Mathematics

Content

Advanced Problem Package	Quadratic Equations

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If $x^2+xy=12$ and $2xy+3y^2+5=0$ then $x+4y$ is equal to :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

2. If 'a' and 'b' are distinct zeroes of the polynomial x^3-2x+c and $a^2\left(2a^2+4ab+3b^2\right)=3$ then $b^2\left(3a^2+4ab+2b^2\right)$ is equal to :

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6

3. Let α and β be the real roots of the equation $x^2-x(k-2)+\left(k^2+3k+5\right)=0.$ The maximum value of $\alpha^2+\beta^2$ is :

(A) 18 **(B)** 19 **(C)** 50/9 **(D)** 50/19

4. For $x\in R,$ the maximum value of $\sqrt{x^4-3x^2-6x+13}-\sqrt{x^4-x^2+1}$ is :

(A) 3 **(B)** $\sqrt{10}$ **(C)** $\sqrt{13}-\sqrt{3}$ **(D)** $2\sqrt{3}$

5. Suppose $A=\{x\,;5x-a\le 0\}, B=\{x\,;6x-b>0\}$, $a,b\in N$ and $A\cap B\cap N=\{2,3,4\}$. The number of such pairs (a, b) is:

(A) 20 **(B)** 25 **(C)** 30 **(D)** 35

6. The number of real solutions to the equation $\sqrt{3x^2-18x+52}+\sqrt{2x^2-12x+162} = \sqrt{-x^2+6x+280}$ is(are) :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

7. a, b, c, d are distinct integers such that $(x-a)(x-b)(x-c)(x-d)=4$ has an integral root r. Then $a+b+c+d$ is equal to :

(A) r **(B)** $2r$ **(C)** $3r$ **(D)** $4r$

8. Let the n real roots of the equation $x^n-2nx^{n-1}+2n(n-1)x^{n-2}+ax^{n-3}+bx^{n-4}+........+c=0$ be $\alpha_1,\alpha_2,\alpha_3......\alpha_n$ then $\sum_{k=1}^{n}(-1)^{k-1}\alpha_k$ is :

(A) Zero **(B)** One **(C)** Two **(D)** Three

9. The number of monic quadratic polynomials of the form x^2+ax+b with integer roots, where 1, a, b are in AP is(are) :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 4

10. Let $A=[-2,4), B=\{x\,;x^2-ax-4\le 0\}$. If $B\subseteq A,$ then the range of real a is :

(A) $[-1,2)$ **(B)** $[-1,2]$ **(C)** [0, 3] **(D)** [0, 3)

11. The sum of all real x such that, $\dfrac{4x^2+15x+17}{x^2+4x+12}=\dfrac{5x^2+16x+18}{2x^2+5x+13}$ is :

(A) 0 **(B)** $\dfrac{-11}{3}$ **(C)** $-\dfrac{20}{3}$ **(D)** $\dfrac{23}{3}$

12. The number of solutions to the equation $2\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{1+(x+3)(x+5)}}}}=x$ is :

(A) 0 **(B)** 2 **(C)** 4 **(D)** 16

PARAGRAPH FOR QUESTIONS 13 - 15

Given that $a>0, \left|ax^2+bx+c\right|\le 1$, if $-1\le x\le 1,\ a,b,c\in R$ and $ax+b$ has its maximum value 2, when $-1\le x\le 1$. Then :

13. $a=$

(A) 3 **(B)** 1 **(C)** 2 **(D)** 4

14. $b=$

(A) -1 **(B)** 2 **(C)** 1 **(D)** 0

15. $c=$

(A) -1 **(B)** 0 **(C)** 1 **(D)** 2

PARAGRAPH FOR QUESTIONS 16 - 18

Consider the equation $x^4-(k-1)x^2+(2-k)=0$. The complete set of possible values of real k for which the equation has:

16. Four distinct real roots is :

(A) $(-\infty,2)$ **(B)** $(2\sqrt{2},-1,2)$

(C) $(\sqrt{2}-1,2\sqrt{2}-1)$ **(D)** $(2,\infty)$

17. 3 distinct real roots is :

(A) $\{2\}$ **(B)** $\{\sqrt{2}-1,2\}$ **(C)** $\{\sqrt{5}-1\}$ **(D)** $\{2\sqrt{2},\sqrt{3}-\sqrt{2}\}$

18. 2 distinct real roots is :

(A) (0, 2) **(B)** $(-\infty,2\sqrt{2}-1)$ **(C)** $(2,\infty)$ **(D)** $\{2\sqrt{2}-1\}\cup(2,\infty)$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

19. The function $f(x)=ax^2-c$ satisfies $-4\le f(1)\le -1$ and $-1\le f(2)\le 5$. Which of the following statements is true?

(A) $-1\le f(3)\le 20$ **(B)** $2\le f(3)\le 18$ **(C)** $-\frac{1}{2}\le f(3)\le 20$ **(D)** $0\le f(3)\le 20$

20. Let $a\neq 0, b, c$ be integers and $\sin\theta, \cos\theta$ be the rational roots of the equation $ax^2+bx+c=0$. Then:

(A) a is a perfect square **(B)** $a+2c$ is a perfect square
(C) $a-2c$ is a perfect square **(D)** b is a perfect square

21. If all roots of the polynomials $6x^2-24x-4a$ and x^3+ax^2+bx-8 are non-negative real numbers, then:

(A) $a=-6$ **(B)** $a=2$ **(C)** $b=10$ **(D)** $b=12$

22. Let $P(x)=x^4+ax^3+bx^2+cx+1$ and $Q(x)=x^4+cx^3+bx^2+ax+1$ with $a,b,c\in R$ and $a\neq c$. If $P(x)=0$ and $Q(x)=0$ have two common roots then :

(A) $b=-2$ **(B)** $b=2$ **(C)** $a+c=0$ **(D)** $a-2c=0$

23. All the roots of x^3+ax^2+bx+c are positive integers greater than 2 and the coefficient satisfy $a+b+c=-46$:

(A) $a=-14$ **(B)** $a=14$
(C) Number of distinct roots of the equation=3 **(D)** Number of distinct roots of the equation=2

24. If the equations $ax^3+(-a+b)x^2+(-b+c)x-c=0$ and $2x^3+x^2+2x-5=0$ have a common root $(a\neq 0, a,b,c\in R)$ then $a+b+c$ is equal to :

(A) 0 **(B)** $5a$ **(C)** $3b$ **(D)** $2c$

25. Let $f(x)=ax^2+bx+c,\ \ a,b,c\in R$. Suppose $|f(x)|\le 1,\ \forall\ x\in[0,1]$ then :

(A) $|a|\le 8$ **(B)** $|a+2b+4c|\le 4$ **(C)** $|a|+|b|+|c|\le 17$ **(D)** $|3a+2b|\le 8$

26. Consider the equation $\sqrt{x+\sqrt{2x-1}}+\sqrt{x-\sqrt{2x-1}}=A$

(A) For A = $\sqrt{2}$, $x\in\left[\frac{1}{2},1\right]$ **(B)** For A = $\sqrt{2}$, $x\in\left[0,\frac{1}{2}\right]$

(C) For A = 1, $x\in\phi$ **(D)** For A = 2, $x=\frac{3}{2}$

27. Suppose $f(x)=-x^2+bx+1$ and $g(x)=x^2+2x+c,\ \ b,c\in R,$ are such that maximum $f(x)\le$ minimum $g(x)$ as x varies over R. Then possible values that c can take is(are) :

(A) 1 **(B)** $\sqrt{2}$ **(C)** 2 **(D)** $\sqrt{5}$

28. The greatest value of the function $f(x)=\frac{1}{2bx^2-x^4-3b^2}$ on the interval $[-2, 1]$ depending on the parameter b is(are) :

(A) $-\frac{1}{3b^2}$ if $b\in[0, 2]$ **(B)** $\frac{1}{4b-4-3b^2}$ if $b\in[0, 4]$

(C) $\frac{1}{8b-16-3b^2}$ if $b\le 2$ **(D)** $-\frac{1}{3b^2}$ if $b\ge 2$

29. Given that a, b, c are positive distinct real numbers such that quadratic expressions ax^2+bx+c, bx^2+cx+a and cx^2+ax+b are always non-negative. Then the expression $\frac{a^2+b^2+c^2}{ab+bc+ca}$ can never lie in :

(A) $(-\infty, 2]$ **(B)** $(-\infty, 1]$ **(C)** $(2, 4)$ **(D)** $[4, \infty)$

30. The equation $8x^4-16x^3+16x^2-8x+a=0,\ a\in R$ has :

(A) Atleast two real roots $\forall\ a\in R$

(B) Atleast two imaginary roots $\forall\ a\in R$

(C) The sum of all non-real roots equal to 2, if $a>\frac{3}{2}$

(D) The sum of all non-real roots equal to 1, if $a\le\frac{3}{2}$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column I are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

31. **MATCH THE COLUMN :**

	Column 1		Column 2
(A)	If a, b, c are length of sides of a triangle, then the roots of the equation $a^2x^2+(b^2+a^2-c^2)x+b^2=0$ are	**(p)**	of opposite signs
(B)	If a, b, c are unequal positive numbers and b is A.M. of a and c, then the roots of the equation $ax^2+2bx+c=0$ are	**(q)**	both positive
(C)	If $a\in R$, then roots of the equation $x^2-(a+1)x-a^2-4=0$ are	**(r)**	both negative
(D)	If a, b, c are unequal positive numbers and b is H.M. of a and c, then the roots of the equation $ax^2+2bx+c=0$ are	**(s)**	real and distinct
		(t)	imaginary

32. Let α, β, γ be three numbers such that $\alpha+\beta+\gamma=2$, $\alpha^2+\beta^2+\gamma^2=6$ and $\alpha^3+\beta^3+\gamma^3=11$, then :

	Column 1		Column 2
(A)	$\alpha^4+\beta^4+\gamma^4$ is equal to	**(p)**	13
(B)	$\alpha^5+\beta^5+\gamma^5$ is equal to	**(q)**	26
(C)	$(\alpha^2-4)(\beta^2-4)(\gamma^2-4)$ is equal to	**(r)**	57
(D)	$\alpha^6+\beta^6+\gamma^6$ is equal to	**(s)**	119
		(t)	129

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

33. The solution of the equation $\frac{8}{\{x\}}=\frac{9}{x}+\frac{10}{[x]}$ is of the form $\frac{k+1}{k}$, $k \in N$ then $k =$ ___________1. ([x] denotes largest integer less than or equal to x, and $\{x\}$ denotes fractional part of x)

34. The value of 'a' so that the equation $x^3-6x^2+11x+a-6=0$ has exactly three integer solutions is _______.

35. Remainder when $P\left(x^5\right)$ is divided by $P(x)=x^4+x^3+x^2+x+1$ is ________.

36. If $a, b, c \in I, a>10$ and $(x-a)(x-12)+2=(x+b)(x+c)$ for all $x \in R$ then $|b-c|=$_________.

37. For real $a, b, c,\ a+b+c=2,\ a^2+b^2+c^2=6$ and $a^3+b^3+c^3=8$ then $(1-a)(1-b)(1-c)=$_______.

38. Let $f(x)=x^2+bx+c,\ b, c \in R$. If $f(x)$ is a factor of both x^4+6x^2+25 and $3x^4+4x^2+28x+5,$ then the minimum value of $f(x)$ is ________.

39. Given that m is a real number not less than –1, such that equation $x^2+2(m-2)x+m^2-3m+3=0$ has two distinct real roots x_1 and x_2. Find the maximum value of $\frac{1}{2}\left(\frac{mx_1^2}{1-x_1}+\frac{mx_2^2}{1-x_2}\right)$.

40. Let p be an integer such that both roots of the equation $5x^2-5px+(66p-1)=0$ are positive integers. Then the value of $\left[\frac{p}{10}\right]$ is equal to ([.] denotes greatest integer function)

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Two rays are drawn through a point A at an angle of $30°$. A point B is taken on one of them at a distance a from the point A. A perpendicular is drawn from the point B to the other ray and another perpendicular is drawn from its foot to AB to meet AB at another point from where the similar process is repeated indefinitely. The length of the resulting infinite polygon line is :

(A) $a\left(2-\sqrt{3}\right)$ **(B)** $a\left(2+\sqrt{3}\right)$ **(C)** a **(D)** None of these

2. The least value of $sin^2\frac{A}{2}+sin^2\frac{B}{2}+sin^2\frac{C}{2}$ is : (Where A, B, C are interior angles of a triangle)

(A) 3/2 **(B)** 3/4 **(C)** 1 **(D)** None of these

3. If A, B, C, D are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k, then the value of $4\,sin\frac{A}{2}+3\,sin\frac{B}{2}+2\,sin\frac{C}{2}+sin\frac{D}{2}$ is equal to :

(A) $2\sqrt{1-k}$ **(B)** $\sqrt{1+k}$ **(C)** $2\sqrt{k}$ **(D)** None of these

4. If $x\,sin\theta = y\,sin\left(\theta+\frac{2\pi}{3}\right) = z\,sin\left(\theta+\frac{4\pi}{3}\right)$, then $\sum xy =$

(A) 1/2 **(B)** $-1/2$ **(C)** 0 **(D)** None of these

5. If $u=\sqrt{a^2\,cos^2\theta+b^2\,sin^2\theta}+\sqrt{a^2\,sin^2\theta+b^2\,cos^2\theta}$, then the difference between maximum and minimum values of u^2 is given by :

(A) $(a-b)^2$ **(B)** $(a+b)^2$ **(C)** a^2+b^2 **(D)** None of these

6. The number of pairs $(x,\,y)$ satisfying the equations $sin\,x+sin\,y=sin(x+y)$ and $|x|+|y|=1$ is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** None of these

7. The value of $cot^{-1}\left(2^2+\frac{1}{2}\right)+cot^{-1}\left(2^3+\frac{1}{2^2}\right)+cot^{-1}\left(2^4+\frac{1}{2^3}\right)+\ldots\infty$ is :

(A) $tan^{-1}\frac{1}{3}$ **(B)** $tan^{-1}\frac{1}{2}$ **(C)** 1 **(D)** None of these

8. In a ΔABC, $\angle B=\frac{\pi}{3}$ and $\angle C=\frac{\pi}{4}$, also D divides BC internally in the ratio 1 : 3, then $\frac{sin\angle BAD}{sin\angle CAD}$ is equal to :

(A) $\frac{1}{\sqrt{6}}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{1}{\sqrt{3}}$ **(D)** None of these

9. The two adjacent sides of a cyclic quadrilateral are 2, 5 and the angle between them is $60°$. If the area of the quadrilateral is $4\sqrt{3}$, then the remaining two sides are :

(A) 1, 2 **(B)** 2, 2 **(C)** 2, 3 **(D)** None of these

10. In a ΔABC, $\angle A > \angle B$. Let $\angle A, \angle B$ satisfy the equation $3\sin x - 4\sin^3 x - k = 0$, where $0 < k < 1$, then $\angle C$ is equal to :

(A) $\frac{\pi}{3}$ **(B)** $\frac{\pi}{2}$ **(C)** $\frac{2\pi}{3}$ **(D)** None of these

11. Points D, E are taken on the side BC of ΔABC, such that $BD = DE = EC$ and let $\angle BAD = x, \angle DAE = y, \angle EAC = z$; then $\frac{\sin(x+y)\sin(y+z)}{\sin x \sin z} =$

(A) 4 **(B)** 6 **(C)** 8 **(D)** None of these

12. If P be any interior point of the equilateral ΔABC of side length 2 units and also x_a, x_b, x_c be the distances of P from the sides BC, CA, AB respectively, then $x_a + x_b + x_c =$

(A) $\sqrt{3}$ **(B)** $3\sqrt{2}$ **(C)** 4 **(D)** None of these

13. If $(x-a)\cos\theta + y\sin\theta = (x-a)\cos\phi + y\sin\phi = a,\ \tan\frac{\theta}{2} - \tan\frac{\phi}{2} = 2e$ and θ, ϕ are unequal angles less than $360°$, then y^2 is equal to :

(A) $2ax-(1+e^2)x^2$ **(B)** $2ax-(1-e^2)x^2$ **(C)** $2ax+(1-e^2)x^2$ **(D)** $2ax+(1+e^2)x^2$

14. The number of solutions of the equation : $x^2+(x+1)\sin\frac{\pi x}{6} = \frac{3+x}{2}$; $-2 \le x \le 0$

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

15. The radius of the circle passing through the incentre $\Delta\, ABC$ and through the end points of BC is given by:

(A) $\frac{a}{2}$ **(B)** $\frac{a}{2}\sec\frac{A}{2}$ **(C)** $\frac{a}{2}\sin A$ **(D)** $a\sec\frac{A}{2}$

16. The median AD of a triangle ABC is bisected at E and BE meets AC in F ; then $AF : AC =$

(A) 3/4 **(B)** 1/3 **(C)** 1/2 **(D)** 1/4

17. If in a $\Delta ABC, \sum \sin 3A = 0$, then at least one angle of ΔABC is :

(A) $60°$ **(B)** $30°$ **(C)** $90°$ **(D)** $45°$

18. If $\left[\sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x\right] = 1$ whose [.] denotes the greatest integer function, then x belongs to :

(A) $[\tan\sin\cos 1, \tan\sin\cos\sin 1]$ **(B)** $[\tan\cos\sin 1, \tan\cos\sin\cos 1]$

(C) $[-1, 1]$ **(D)** $[\sin\cos\tan 1, \sin\cos\sin\tan 1]$

19. If $\sin x + \operatorname{cosec} x + \tan y + \cot y = 4$, where $x, y \in \left[0, \frac{\pi}{2}\right]$, then $\tan\left(\frac{y}{2}\right)$ is a root of the equation :

(A) $\alpha^2+2\alpha+1=0$ **(B)** $\alpha^2+2\alpha-1=0$ **(C)** $2\alpha^2-2\alpha-1=0$ **(D)** None of these

20. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (tan\,\theta)^{tan\,\theta}$, $t_2 = (tan\,\theta)^{cot\,\theta}$, $t_3 = (cot\,\theta)^{tan\,\theta}$ and $t_4 = (cot\,\theta)^{cot\,\theta}$, then :

(A) $t_3 > t_4 > t_1 > t_2$ **(B)** $t_4 > t_3 > t_1 > t_2$ **(C)** $t_4 > t_3 > t_2 > t_1$ **(D)** None of these

21. The period of the function $f(x) = e^{sin^2 x + sin^2\left(x+\frac{\pi}{3}\right) + cos\,x\,cos\left(x+\frac{\pi}{3}\right)}$ is :

(A) 1 **(B)** $\pi/2$ **(C)** π **(D)** Cannot be determined

22. If $t = x + y + z$, then $sin\,x + sin\,y + sin\,z - sin\,t$ equals :

(A) $4\,tan\left(\frac{y+z}{2}\right)tan\left(\frac{z+x}{2}\right)tan\left(\frac{x+y}{2}\right)$ **(B)** $4\,cot\left(\frac{y+z}{2}\right)cot\left(\frac{z+x}{2}\right)cot\left(\frac{x+y}{2}\right)$

(C) $4\,sin\left(\frac{y+z}{2}\right)sin\left(\frac{z+x}{2}\right)sin\left(\frac{x+y}{2}\right)$ **(D)** $4\,cos\left(\frac{y+z}{2}\right)cos\left(\frac{z+x}{2}\right)cos\left(\frac{x+y}{2}\right)$

23. The value of $cos^{-1}x + cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$ is equal to : $\left(\frac{1}{2} \le x \le 1\right)$

(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{3}$ **(C)** π **(D)** 0

24. A quadrilateral *ABCD* in which *AB* = *a*, *BC* = *b*, *CD* = *c* and *DA* = *d* is such that one circle can be inscribed in it and another circle can be circumscribed about it. *cos A* =

(A) $\frac{ad+bc}{ad-bc}$ **(B)** $\frac{ad-bc}{ad+bc}$ **(C)** $\frac{ac+bd}{ac-bd}$ **(D)** $\frac{ac-bd}{ac+bd}$

25. If the equation $a_1 + a_2\,cos\,2x + a_3\,sin^2 x = 1$ is satisfied by every real value of *x*, then the number of possible values of the triplet (a_1, a_2, a_3) is :

(A) 0 **(B)** 1 **(C)** 3 **(D)** Infinite

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

26. Let sides of a ΔABC are in A.P. and $a < min.\{b, c\}$, then *cos A* is equal to :

(A) $\frac{1}{2c}(4b-3c)$ **(B)** $\frac{1}{2c}(4c-3b)$ **(C)** $\frac{1}{2b}(4b-3c)$ **(D)** $\frac{1}{2b}(4c-3b)$

27. If points *D*, *E* and *F* divide sides *BC*, *CA* and *AB* respectively in ratio $\lambda : 1$ (in order) and or $ar(\Delta DEF) = 0.4$ $ar(\Delta ABC)$, then λ is equal to :

(A) $\frac{2-\sqrt{3}}{2}$ **(B)** $\frac{3-\sqrt{5}}{2}$ **(C)** $\frac{2+\sqrt{3}}{2}$ **(D)** $\frac{3+\sqrt{5}}{2}$

28. If in a right angled triangle the greatest side is *a*, then $tan\left(\frac{C}{2}\right) =$

(A) $\frac{a-b}{c}$ **(B)** $\frac{a+b}{c}$ **(C)** $\frac{a-c}{b}$ **(D)** $\frac{a+b}{b}$

29. Which of the following is true ?

(A) $\tan\left|\tan^{-1}x\right|=|x|$ **(B)** $\cot\left|\cot^{-1}x\right|=x$

(C) $\tan^{-1}|\tan x|=|x|$ **(D)** $\sin\left|\sin^{-1}x\right|=|x|$

30. If $f(x)=\left(\sin^{-1}x\right)^3+\left(\cos^{-1}x\right)^3$, then :

(A) Minimum value of $f(x)=-\dfrac{\pi^3}{8}$ **(B)** Minimum value of $f(x)=\dfrac{\pi^3}{32}$

(C) Maximum value of $f(x)=-\dfrac{\pi^3}{8}$ **(D)** Maximum value of $f(x)=\dfrac{7\pi^3}{8}$

31. If $f(x)=\sec^{-1}\left[1+\cos^2 x\right]$, where [.] denotes the greatest integer function, then :

(A) The domain of f is R **(B)** The domain of f is [1, 2]

(C) The range of f is [1. 2] **(D)** The range of f is $\left\{\sec^{-1}1, \sec^{-1}2\right\}$

32. If $(\sin\alpha)x^2-2x+b\geq 2$ for all real values of $x\leq 1$ and $\alpha\in\left(0,\dfrac{\pi}{2}\right)\cup\left(\dfrac{\pi}{2},\pi\right)$, then b can be equal to :

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

33. If $\sin\alpha+\sin\beta=\dfrac{3\sqrt{2}}{5}$ and $\cos\alpha+\cos\beta=\dfrac{4\sqrt{2}}{5}$, then :

(A) $\sin(\alpha+\beta)=\dfrac{12}{13}$ **(B)** $\sin(\alpha+\beta)=\dfrac{24}{25}$ **(C)** $\cos(\alpha+\beta)=\dfrac{5}{13}$ **(D)** $\cos(\alpha+\beta)=\dfrac{7}{25}$

34. If in a triangle *ABC*, atleast one of the following points, orthocenter, centroid, incentre and circumcentre lie outside the triangle, then :

(A) Triangle is obtuse angled

(B) Exactly 2 of these centres will lie outside the triangle

(C) Incentre may be collinear with other three centres

(D) Atleast one of the ex-radii is smaller than the inradius of the triangle

35. The value of θ lying between $\theta=0$ and $\theta=\dfrac{\pi}{2}$ and satisfying the equation :

$$\begin{vmatrix} 1+\cos^2\theta & \sin^2\theta & 4\sin 4\theta \\ \cos^2\theta & 1+\sin^2\theta & 4\sin 4\theta \\ \cos^2\theta & \sin^2\theta & 1+4\sin 4\theta \end{vmatrix}=0 \text{ is :}$$

(A) $\dfrac{11\pi}{24}$ **(B)** $\dfrac{7\pi}{24}$ **(C)** $\dfrac{5\pi}{24}$ **(D)** $\dfrac{\pi}{24}$

36. If $x=\sin\left(2\tan^{-1}2\right)$ and $y=\sin\left(\dfrac{1}{2}\tan^{-1}\dfrac{4}{3}\right)$, then which of the following options is(are) correct ?

(A) $x=y^2$ **(B)** $y^2=1-x$ **(C)** $x^2=\dfrac{y}{2}$ **(D)** $x>y$

37. The equation $(1-tan\theta)(1+tan\theta)sec^2\theta+2^{tan^2\theta}=0$ has :

(A) No solution in the interval $\left(-\frac{\pi}{2},0\right)$ **(B)** Two solutions in the interval $\left(-\frac{\pi}{2},0\right)$

(C) No solution in the interval $\left(0,\frac{\pi}{2}\right)$ **(D)** Two solutions in the interval $\left(0,\frac{\pi}{2}\right)$

38. Which of the following is not a possible value of $f(x)=tan3x\,cot\,x$?

(A) 1 **(B)** 2 **(C)** 4 **(D)** 5

39. In the ΔABC, $b:c=2:1$ and $sin(B-C)=\frac{3}{5}$. Then :

(A) ΔABC is right-angled **(B)** ΔABC is obtuse angled

(C) $a:c=3:1$ **(D)** $a:c=\sqrt{5}:1$

40. Which of the following is/are true?

(A) $\tan 1>\tan^{-1}1$ **(B)** $\sin 1>\cos 1$ **(C)** $\tan 1<\sin 1$ **(D)** $\cos(\cos 1)>\frac{1}{\sqrt{2}}$

41. Which of the following is/are positive?

(A) $\log_{\sin 1}\tan 1$ **(B)** $\log_{\cos 1}(1+\tan 3)$

(C) $\log_{\log_{10}5}(\cos\theta+\sec\theta)$ **(D)** $\log_{\tan 15^\circ}(2\sin 18^\circ)$

42. If $2(\cos(x-y)+\cos(y-z)+\cos(z-x))=-3,$ then:

(A) $\cos x\cos y\cos z=1$ **(B)** $\cos x+\cos y+\cos z=0$

(C) $\sin x+\sin y+\sin z=1$ **(D)** $\cos 3x+\cos 3y+\cos 3z=12\cos x\cos y\cos z$

43. If $2a=2\tan 10^\circ+\tan 50^\circ;\ 2b=\tan 20^\circ+\tan 50^\circ$

$2c=2\tan 10^\circ+\tan 70^\circ;\ 2d=\tan 20^\circ+\tan 70^\circ$

Then which of the following is / are correct ?

(A) $a+d=b+c$ **(B)** $a+b=c$ **(C)** $a>b<c>d$ **(D)** $a<b<c<d$

44. The value of $\frac{\sin x-\cos x}{\sin^3 x}$ is equal to :

(A) $\text{cosec}^2x(1-\cot x)$ **(B)** $1-\cot x+\cot^2 x-\cot^3 x$

(C) $\text{cosec}^2x-\cot x-\cot^3 x$ **(D)** $\frac{1-\cot x}{\sin^2 x}$

45. The inequality $4\sin 3x+5\ge 4\cos 2x+5\sin x$ is true for $x\in$:

(A) $\left[-\pi,\frac{3\pi}{2}\right]$ **(B)** $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ **(C)** $\left[\frac{5\pi}{8},\frac{13\pi}{8}\right]$ **(D)** $\left[\frac{23\pi}{14},\frac{41\pi}{14}\right]$

46. The equation $\cos x\cos 6x=-1$:

(A) has 50 solutions in $[0,100\pi]$ **(B)** has 3 solutions in $[0,3\pi]$

(C) has even number of solutions in $(3\pi,13\pi)$ **(D)** has one solution in $\left[\frac{\pi}{2},\pi\right]$

47. Identify the correct options:

(A) $\dfrac{\sin 3\alpha}{\cos 2\alpha} > 0$ for $\alpha \in \left(\dfrac{3\pi}{8}, \dfrac{23\pi}{48}\right)$ **(B)** $\dfrac{\sin 3\alpha}{\cos 2\alpha} < 0$ for $\alpha \in \left(\dfrac{13\pi}{48}, \dfrac{14\pi}{48}\right)$

(C) $\dfrac{\sin 2\alpha}{\cos \alpha} < 0$ for $\alpha \in \left(-\dfrac{\pi}{2}, 0\right)$ **(D)** $\dfrac{\sin 2\alpha}{\cos \alpha} > 0$ for $\alpha \in \left(\dfrac{13\pi}{48}, \dfrac{14\pi}{48}\right)$

48. The equation $\sin^4 x + \cos^4 x + \sin 2x + k = 0$ must have real solutions if:

(A) $k = 0$ **(B)** $|k| \le \dfrac{1}{2}$ **(C)** $-\dfrac{3}{2} \le k \le \dfrac{1}{2}$ **(D)** $-\dfrac{1}{2} \le k \le \dfrac{3}{2}$

49. Let $f(\theta) = \left(\cos\theta - \cos\dfrac{\pi}{8}\right)\left(\cos\theta - \cos\dfrac{3\pi}{8}\right)\left(\cos\theta - \cos\dfrac{5\pi}{8}\right)\left(\cos\theta - \cos\dfrac{7\pi}{8}\right)$ then:

(A) maximum value of $f(\theta)\forall\theta \in R$ is $\dfrac{1}{4}$ **(B)** maximum value of $f(\theta)\forall\theta \in R$ is $\dfrac{1}{8}$

(C) $f(0) = \dfrac{1}{8}$ **(D)** Number of principle solutions of $f(\theta) = 0$ is 8

50. If r_1, r_2, r_3 are radii of the escribed circles of a triangle ABC and r is the radius of its incircle, then the root(s) of the equation $x^2 - r(r_1r_2 + r_2r_3 + r_3r_1)x + (r_1r_2r_3 - 1) = 0$ is/are:

(A) r_1 **(B)** $r_2 + r_3$ **(C)** 1 **(D)** $r_1r_2r_3 - 1$

51. Let A, B, C be angles of a triangle ABC and let $D = \dfrac{5\pi + A}{32}$, $E = \dfrac{5\pi + B}{32}$, $F = \dfrac{5\pi + C}{32}$, then: (where $D, E, F \ne \dfrac{n\pi}{2}, n \in I, I$ denote set of integers)

(A) $\cot D \cot E + \cot E \cot F + \cot D \cot F = 1$ **(B)** $\cot D + \cot E + \cot F = \cot D \cot E \cot F$

(C) $\tan D \tan E + \tan E \tan F + \tan F \tan D = 1$ **(D)** $\tan D + \tan E + \tan F = \tan D \tan E \tan F$

52. In a ΔABC if $\dfrac{r}{r_1} = \dfrac{r_2}{r_3}$, then which of the following is/are true? (where symbols used have usual meanings)

(A) $a^2 + b^2 + c^2 = 8R^2$ **(B)** $\sin^2 A + \sin^2 B + \sin^2 C = 2$

(C) $a^2 + b^2 = c^2$ **(D)** $\Delta = s(s + c)$

53. ABC is a triangle whose circumcentre, incentre and orthocentre are O, I and H respectively which lie inside the triangle, then :

(A) $\angle BOC = A$ **(B)** $\angle BIC = \dfrac{\pi}{2} + \dfrac{A}{2}$ **(C)** $\angle BHC = \pi - A$ **(D)** $\angle BHC = \pi - \dfrac{A}{2}$

54. In a triangle ABC, $\tan A$ and $\tan \mathrm{B}$ satisfy the inequality $\sqrt{3}x^2 - 4x + \sqrt{3} < 0$, then which of the following must be correct?
(where symbols used have usual meanings)

(A) $a^2 + b^2 - ab < c^2$ **(B)** $a^2 + b^2 > c^2$

(C) $a^2 + b^2 + ab > c^2$ **(D)** $a^2 + b^2 < c^2$

55. $f(x) = \sin^{-1}(\sin x), g(x) = \cos^{-1}(\cos x)$, then:

(A) $f(x) = g(x)$ if $x \in \left(0, \dfrac{\pi}{4}\right)$ **(B)** $f(x) < g(x)$ if $x \in \left(\dfrac{\pi}{2}, \dfrac{3\pi}{4}\right)$

(C) $f(x) < g(x)$ if $x \in \left(\pi, \dfrac{5\pi}{4}\right)$ **(D)** $f(x) > g(x)$ if $x \in \left(\pi, \dfrac{5\pi}{4}\right)$

56. The solution(s) of the equation $\cos^{-1}x=\tan^{-1}x$ satisfy

(A) $x^2=\frac{\sqrt{5}-1}{2}$ **(B)** $x^2=\frac{\sqrt{5}+1}{2}$

(C) $\sin(\cos^{-1}x)=\frac{\sqrt{5}-1}{2}$ **(D)** $\tan(\cos^{-1}x)=\frac{\sqrt{5}-1}{2}$

57. A solution of the equation $\cot^{-1}2=\cot^{-1}x+\cot^{-1}(10-x)$ where $1<x<9$ is:

(A) 7 **(B)** 3 **(C)** 2 **(D)** 5

58. Consider the equation $\sin^{-1}\left(x^2-6x+\frac{17}{2}\right)+\cos^{-1}k=\frac{\pi}{2}$, then:

(A) the largest value of k for which equation has 2 distinct solution is 1

(B) the equation must have real root if $k\in\left(-\frac{1}{2},1\right)$

(C) the equation must have real root if $k\in\left(-1,\frac{1}{2}\right)$

(D) the equation has unique solution if $k=-\frac{1}{2}$

59. The value of x satisfying the equation $(\sin^{-1}x)^3-(\cos^{-1}x)^3+(\sin^{-1}x)(\cos^{-1}x)(\sin^{-1}x-\cos^{-1}x)=\frac{\pi^3}{16}$ cannot be equal to:

(A) $\cos\frac{\pi}{5}$ **(B)** $\cos\frac{\pi}{4}$ **(C)** $\cos\frac{\pi}{8}$ **(D)** $\cos\frac{\pi}{12}$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

60. If $a\sin\theta-b\cos\theta=-\sin 4\theta$ and $a\cos\theta+b\sin\theta=\frac{5}{2}-\frac{3}{2}\cos 4\theta$, then $(a+b)^{2/5}+(a-b)^{2/5}$ is _______.

61. Let incircle of radius 4 units of a triangle ABC touches the side BC at D. If $BD=6,\ DC=8$ and Δ be the area of triangle, then $\sqrt{\sqrt{\Delta}-3}=$__________.

62. The total number of solutions of $\tan\{x\}=\cot\{x\}$; where $\{x\}$ denotes the fractional part of x in $[0,\ 2\pi]$ is ________.

63. Consider the equation $\tan^{-1}x+\cos^{-1}\left(\frac{y}{\sqrt{1+y^2}}\right)=\sin^{-1}\left(\frac{3}{\sqrt{10}}\right)$. Let α = sum of positive integral solutions of x and β = sum of positive integral solutions of y. Then $\beta-\alpha=$__________.

64. If $\sin x+\sin^2x+\sin^3x=1$, then $\cos^6x-4\cos^4x+8\cos^2x=$_________.

65. The number of solutions of $sin^{-1}\left(\frac{1+x^2}{2x}\right)=\frac{\pi}{2}sec(x-1)$ is _________.

66. If the square of the diameter of a circle circumscribing a ΔABC is equal to half the sum of the squares of its sides then $\sum \sin^2 A$ is ________.

67. If $tan\left(142\frac{1}{2}^{\circ}\right)=2+\sqrt{2}-\sqrt{\mu}-\sqrt{\lambda}$, then $\mu+\lambda$ = __________.

68. If $tan\left(\frac{2\pi}{3}-x\right)=\frac{sin\frac{2\pi}{3}-sin\,x}{cos\frac{2\pi}{3}-cos\,x}$ where $0<x<\frac{3\pi}{2}$, and the values of x are x_1 and x_2, then the value of $\frac{12}{\pi}|x_2-x_1|$ is ___________.

69. If in a ΔABC, $a=5,\ b=4$ and $cos(A-B)=\frac{31}{32}$, then the third side c is equal to ___________.

70. If $10\sin^4\alpha+15\cos^4\alpha=6$ and the value of $9\text{cosec}^4\alpha+8\sec^4\alpha$ is S, then find the value of $\frac{S}{25}$.

71. Given that for $a, b, c, d \in R$, if $a\sec(200°)-c\tan(200°)=d$ and $b\sec(200°)+d\tan(200°)=c$, then find the value of $\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20°$.

72. If $\sum_{r=1}^{n}\left(\frac{\tan 2^{r-1}}{\cos 2^r}\right)=\tan p^n-\tan q$, then find the value of $(p+q)$.

73. If $x=\alpha$ satisfy the equation $3^{\sin 2x+2\cos^2 x}+3^{1-\sin 2x+2\sin^2 x}=28$, then $(\sin 2\alpha-\cos 2\alpha)^2+8\sin 4\alpha$ is equal to:

74. If the value of $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}+\cos\frac{7\pi}{7}=-\frac{l}{2}$. Find the value of l.

75. If $x+\sin y=2014$ and $x+2014\cos y=2013, 0\le y\le\frac{\pi}{2}$, then find the value of $[x+y]-2005$ (where [.] denotes greatest integer function)

76. The complete set of values of x satisfying $\frac{2\sin 6x}{\sin x-1}<0$ and $\sec^2 x-2\sqrt{2}\tan x\le 0$ in $\left(0,\frac{\pi}{2}\right)$ is $[a,b)\cup(c,d]$, then find the value of $\left(\frac{cd}{ab}\right)$.

77. The range of value's of k for which the equation $2\cos^4 x-\sin^4 x+k=0$ has atleast one solution is $[\lambda, \mu]$. Find the value of $(9\mu+\lambda)$.

78. The number of solutions of the system of equations:

$$2\sin^2 x+\sin^2 2x=2$$
$$\sin 2x+\cos 2x=\tan x$$

in $[0, 4\pi]$ satisfying $2\cos^2 x+\sin x\le 2 is$:

79. If the sum of all values of $\theta, 0 \le \theta \le 2\pi$ satisfying the equation $(8\cos 4\theta - 3)(\cot\theta + \tan\theta - 2)(\cot\theta + \tan\theta + 2) = 12$ is $k\pi$, then k is equal to:

80. In a ΔABC; inscribed circle with centre I touches sides AB, AC and BC at D, E, F respectively. Let area of quadrilateral $ADIE$ is 5 square units and area of quadrilateral $BFID$ is 10 square units. Find the value of $\dfrac{\cos\left(\dfrac{C}{2}\right)}{\sin\left(\dfrac{A-B}{2}\right)}$.

81. If Δ be area of incircle of a triangle ABC and $\Delta_1, \Delta_2, \Delta_3$ be the area of excircles then find the least value of $\dfrac{\Delta_1\Delta_2\Delta_3}{729\Delta^3}$.

82. In an acute angled triangle $ABC, \angle A = 20°$, let DEF be the feet of altitudes through A, B, C respectively and H is the orthocentre of ΔABC. Find $\dfrac{AH}{AD} + \dfrac{BH}{BE} + \dfrac{CH}{CF}$.

83. Let ΔABC be inscribed in a circle having radius unity. The three internal bisectors of the angles A, B and C are extended to intersect the circumcircle of ΔABC at A_1, B_1 and C_1 respectively. find

$$\frac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C}$$

84. In ΔABC, if circumradius `$R'$ and inradius `r' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then the greatest integer which is less than the semiperimeter of ΔABC is:

85. The complete set of values of x satisfying the inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ is $(2 - \sqrt{\lambda - 2\pi}, 2 + \sqrt{\lambda - 2\pi})$, then $\lambda =$

86. In ΔABC; if $(II_1)^2 + (I_2I_3)^2 = \lambda R^2$, where I denotes incentre; I_1, I_2 and I_3 denote centres of the circles escribed to the sides BC, CA and AB respectively and R be the radius of the circum circle of ΔABC. Find λ.

87. If $2\tan^{-1}\dfrac{1}{5} - \sin^{-1}\dfrac{3}{5} = -\cos^{-1}\dfrac{63}{\lambda}$, then $\lambda =$

88. If $\displaystyle\sum_{n=0}^{\infty} 2\cot^{-1}\left(\frac{n^2+n+4}{2}\right) = k\pi$, then find the value of k.

89. Find number of solutions of the equation $\sin^{-1}(|\log_6^2(\cos x) - 1|) + \cos^{-1}(|3\log_6^2(\cos x) - 7|) = \dfrac{\pi}{2}$, if $x \in [0, 4\pi]$.

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. if p^{th}, q^{th} and r^{th} terms of an H.P. be respectively x, y, z, then $(p-q)xy+(q-r)yz+(r-p)xz=$

(A) $xyz+pqr$ **(B)** pqr **(C)** xyz **(D)** 0

2. If the $(m+1)^{th}, (n+1)^{th}$ and $(r+1)^{th}$ terms of an A.P. are n G.P. and m, n, r are in H.P., then the ratio of the common difference to the first term in the A.P. is equal to :

(A) $\frac{2}{n}$ **(B)** $\frac{1}{n}$ **(C)** $-\frac{1}{n}$ **(D)** $-\frac{2}{n}$

3. $\sum_{r=1}^{99} r!\left(r^2+r+1\right)$ is equal to:

(A) $102!-100!$ **(B)** $100(100!)-1$ **(C)** $99(100!)-1$ **(D)** $100(99!)-1$

4. The sum $\sum_{k=1}^{n} \frac{k^2-\frac{1}{2}}{k^4+\frac{1}{4}}$ is equal to:

(A) $\frac{2n^2-2n+1}{2n^2+2n+1}$ **(B)** $\frac{2n^2-n}{2n^2+2n+1}$ **(C)** $\frac{n^2}{2n^2+2n+1}$ **(D)** $\frac{2n^2}{2n^2+2n+1}$

5. The value of $\frac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right)....\left((2n-1)^4+\frac{1}{4}\right)}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right)....\left((2n)^4+\frac{1}{4}\right)}$ is equal to:

(A) $\frac{1}{4n^2+2n+1}$ **(B)** $\frac{1}{8n^2+4n+1}$ **(C)** $\frac{1}{4\left(2n^2+n+1\right)}$ **(D)** $\frac{n}{8n^2-4n+1}$

6. The sum $\sum_{k=1}^{n} \frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}}$ is equal to:

(A) $\frac{\sqrt{n+1}-2}{\sqrt{n+1}}$ **(B)** $\frac{\sqrt{n+1}-1}{\sqrt{n+1}}$ **(C)** $\frac{\sqrt{n+1}+1}{\sqrt{n+1}}$ **(D)** $\frac{n+1}{n\sqrt{n+1}}$

7. Let S_n, S_{2n}, S_{3n} are respectively the sums of first n, 2n, 3n terms of an arithmetic progression, then $S_{3n}=$

(A) $2\left(S_{2n}-S_n\right)$ **(B)** $\frac{3}{2}\left(S_{2n}-S_n\right)$ **(C)** $3\left(S_{2n}-S_n\right)$ **(D)** $6\left(S_{2n}-S_n\right)$

8. The sum $\frac{19}{1\cdot2\cdot3}\frac{1}{4}+\frac{28}{2\cdot3\cdot4}\frac{1}{8}+\frac{39}{3\cdot4\cdot5}\frac{1}{16}+\frac{52}{4\cdot5\cdot6}\frac{1}{32}+...+$ upto infinite terms is equal to:

(A) $\frac{1}{2}$ **(B)** 1 **(C)** 2 **(D)** 4

9. The sum of infinite series $1^2-\frac{2^2}{5}+\frac{3^2}{5^2}-\frac{4^2}{5^3}+\frac{5^2}{5^4}-\frac{6^2}{5^2}+....\infty$ is equal to:

(A) $\frac{5}{27}$ **(B)** $\frac{25}{27}$ **(C)** $\frac{25}{108}$ **(D)** $\frac{25}{54}$

10. $\frac{n}{1\cdot2\cdot3}+\frac{n-1}{2\cdot3\cdot4}+\frac{n-2}{3\cdot4\cdot5}+...$ upto n terms is equal to:

(A) $\frac{1}{2(n+2)}+\frac{n+1}{4}-\frac{1}{2}$ **(B)** $\frac{1}{2(n+2)}+\frac{n+1}{4}+\frac{1}{2}$

(C) $\frac{1}{n+2}+\frac{n+1}{4}-\frac{1}{2}$ **(D)** $\frac{1}{2(n+2)}+\frac{n+1}{2}+\frac{1}{2}$

For Questions 11 - 13

If $\phi(r)=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+...+\frac{1}{r}$ and $\sum_{r=1}^{n}(2r+1)\phi(r)=P(n)\phi(n+1)-Q(n)$, where $P(n)$ and $Q(n)$ are polynomial function of 'n', then

11. $\sum_{r=0}^{10}P(r)$ is equal to:

(A) 235 **(B)** 506 **(C)** 285 **(D)** 385

12. $\sum_{r=0}^{\infty}\frac{1}{Q(r)}$ is equal to:

(A) 1 **(B)** 2 **(D)** 4 **(D)** 8

13. $P(13)-Q(13)$ is equal to:

(A) 81 **(B)** 78 **(C)** 91 **(D)** 65

For Questions 14 - 16

Let $a_m\,(m=1,2,3....,p)$ be the possible integral values of a for which the graphs of $f(x)=ax^2+2bx+b$ and $g(x)=5x^2-3bx-a$ meets at some point for all real values of b. let $t_r=\prod_{m=1}^{p}(r-a_m)$ and $S_n=\prod_{r=1}^{n}t_r, n\in N$.

14. The minimum possible value of a is:

(A) $\frac{1}{5}$ **(B)** $\frac{5}{26}$ **(C)** $\frac{3}{38}$ **(D)** $\frac{2}{43}$

15. The sum of values of n for which S_n vanishes is:

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11

16. The value of $\sum_{r=5}^{\infty}\frac{1}{t_r}$ is equal to:

(A) $\frac{1}{3}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{1}{15}$ **(D)** $\frac{1}{18}$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

17. Let $a_1, a_2, a_3, ..., a_n$ be the first 'n' terms of an A.P. having common difference 'd' $(d \neq 0)$, then the greatest value of product of two terms equidistant from the extreme terms is:

(A) $a_1 a_n + \frac{d^2(n-1)^2}{4}$ if n is odd **(B)** $a_1 a_n + \frac{d^2(n+1)^2}{4}$ if n is odd

(C) $a_1 a_n + \frac{d^2 n(n+2)}{4}$ if n is even **(D)** $a_1 a_n + \frac{d^2}{4} n(n-2)$ is n is even

18. For all permissible value of x, consider $y = \frac{\sin 3x(\cos 6x + \cos 4x)}{\sin x(\cos 8x + \cos 2x)}$ and range of y is $(-\infty, a) \cup (b, \infty)$. If 2b is the first terms of G.P. and 'a' is its common ratio, then: (S_∞ denotes the sum of infinite terms of G.P.)

(A) $b - a = \frac{10}{3}$ **(B)** $3a + b = 4$ **(C)** $S_\infty = 9$ **(D)** $S_\infty = \frac{27}{10}(a+b)$

19. Let $\{a_n\}$ consists of positive numbers and for any positive integer n, $\frac{a_n + 2}{2} = \sqrt{2s_n}$, where $s_n = \sum_{i=1}^{n} a_1$. Then :

(A) $a_{21} = 82$ **(B)** $a_{12} = 48$ **(C)** $a_{13} = 50$ **(D)** $a_{14} = 54$

20. If x, y, z are three distinct positive real numbers and are in H.P., then $\frac{3x+2y}{2x-y} + \frac{3z+2y}{2z-y}$ is greater then:

(A) 9 **(B)** 10 **(C)** 12 **(D)** 15

21. The sequence $\{a_n\}, n \in N$ satisfies $a_1 = 1$ and $5^{a_{n+1} - a_n} = 1 + \frac{1}{n + \frac{2}{3}}$. Then: (where $[\cdot]$ denotes greatest integer function)

(A) $[a_{501}] = 3$ **(B)** $[a_{207}] = 3$ **(C)** $[a_{223}] = 4$ **(D)** $[a_{625}] = 4$

22. If a, b, c are three positive real numbers, then $\frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b}$ can be never be equal to:

(A) 1 **(B)** 2 **(C)** $\frac{8}{3}$ **(D)** 3

23. Let 'p' be the first of 'n' arithmetic means between two positive numbers and 'q' be first of 'n' harmonic means between same two numbers. The $\frac{q}{p}$ can lie in interval(s):

(A) $(-\infty, 1]$ **(B)** $\left(1, \left(\frac{n+1}{n-1}\right)^2\right)$

(C) $\left(\left(\frac{n-1}{n+1}\right)^2, \left(\frac{n+1}{n-1}\right)^2\right)$ **(D)** $\left[\left(\frac{n+1}{n-1}\right)^2, \infty\right)$

24. Let x, y, z are distinct positive integers and $m = \left(\frac{x^2+y^2+z^2}{x+y+z}\right)^{(x+y+z)}$, $n = x^x y^y z^z$, $P = \left(\frac{x+y+z}{3}\right)^{(x+y+z)}$, then:

(A) $m > n$ **(B)** $n > p$ **(C)** $m < n$ **(D)** $n < p$

25. Let $f(x) = \lim_{n\to\infty}\left(\frac{1}{x+1}+\frac{2}{x^2+1}+\frac{4}{x^4+1}+\ldots+\frac{2^n}{x^{2^n}+1}\right)$ for $x > 1$, then:

(A) $\sum_{r=2}^{6}\frac{1}{f(r)} = 20$ **(B)** $f(5) = \frac{1}{6}$ **(C)** $f(5) = \frac{1}{4}$ **(D)** $\sum_{r=2}^{6}\frac{1}{f(r)} = 15$

26. For a positive integer 'n', let $S(n) = 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots+\frac{1}{2^n-1}$. Then:

(A) $S(200) > 100$ **(B)** $S(200) < 100$ **(C)** $S(100) < 100$ **(D)** $S(100) > 100$

27. Let $a_1, a_2, a_3, \ldots. a_n$ be first 'n' terms of a G.P. with first term 'a' and common ratio 'r'. Then:

(A) $\frac{1}{a_1^2-a_2^2}+\frac{1}{a_2^2-a_3^2}+\ldots+\frac{1}{a_{n-1}^2-a_n^2} = \frac{\left(1-r^{2n}\right)}{a^2 r^{2n-4}\left(1-r^2\right)^2}$

(B) $\frac{1}{a_1^2-a_2^2}+\frac{1}{a_2^2-a_3^2}+\ldots+\frac{1}{a_{n-1}^2-a_n^2} = \frac{\left(1-r^{2n-2}\right)}{a^2 r^{2n-4}\left(1-r^2\right)^2}$

(C) $\frac{1}{a_1^m+a_2^m}+\frac{1}{a_2^m+a_3^m}+..+\frac{1}{a_{n-1}^m+a_n^m} = \frac{\left(r^{mn-m}-1\right)}{a^m\left(1+r^m\right)\left(r^{mn-m}-r^{mn-2m}\right)}$

(D) $\frac{1}{a_1^m+a_2^m}+\frac{1}{a_2^m+a_3^m}+\ldots+\frac{1}{a_{n-1}^m+a_n^m} = \frac{\left(r^{mn-m}-1\right)}{a^m\left(1-r^m\right)\left(r^{mn-m}-r^{mn-2m}\right)}$

28. Let the equation $x^3+px^2+qx-q=0$, where $p, q \in R, q \neq 0$ has 3 real roots α, β, γ in H.P., then:

(A) $\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2} \geq \frac{1}{3}$ **(B)** $9p+2q+27=0$

(C) Maximum value of $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}$ is $\frac{1}{3}$ **(D)** $\frac{p}{q} \geq -\frac{1}{3}$

29. The sum $\sum_{n=1}^{\infty}\sum_{k=1}^{n-1}\frac{k}{2^{n+k}}$ is equal to

(A) $\sum_{k=1}^{\infty}\frac{k}{2^k}$ **(B)** $\sum_{k=1}^{\infty}\frac{k}{4^k}$

(C) $\sum_{m=1}^{\infty}\left(\frac{m}{2^m}\sum_{n=m+1}^{\infty}\frac{1}{2^n}\right)$ **(D)** $\frac{4}{27}$

30. For any natural number $n>1$, consider the sum S =
$\dfrac{1}{n^2+1}+\dfrac{2}{n^2+2}+\dfrac{3}{n^2+3}+\ldots+\dfrac{n}{n^2+n}$, then

(A) $S<\dfrac{1}{2}+\dfrac{1}{2n}$ **(B)** $S>\dfrac{1}{2}+\dfrac{1}{2n}$ **(C)** $S>\dfrac{1}{2}$ **(D)** $S<1$

31. If $n\in N, n>5$ then which of the following holds true ?

(A) $n^n>1\cdot3\cdot5\ldots(2n-1)$ **(B)** $2^n>1+n\sqrt{2^{n-1}}$

(C) $\dfrac{1}{n+1}+\dfrac{1}{n+1}+\dfrac{1}{n+3}+\ldots\dfrac{1}{2n}>\dfrac{1}{2}$ **(D)** $2^n<1+n\sqrt{2^{n-1}}$

32. Let $\{a_n\}$ be a sequence of real numbers such that $a_1=2, a_{n+1}=a_n^2-a_n+1$ for $n=1,2,3,\ldots$ Let
$S=\dfrac{1}{a_1}+\dfrac{1}{a_2}+\ldots+\dfrac{1}{a_{2018}}$, then

(A) $S<1-\dfrac{1}{(2018)^{2018}}$ **(B)** $S>1-\dfrac{1}{(2018)^{2018}}$

(C) $S<1$ **(D)** $S>1-\dfrac{1}{(2017)^{2017}}$

33. Let $a_k=\dfrac{k}{(k-1)^{4/3}+k^{4/3}+(k+1)^{4/3}}$ and $S_n=\sum\limits_{k=1}^{n}a_k$, then

(A) $S_{26}>\dfrac{17}{4}$ **(B)** $S_{26}<\dfrac{17}{4}$ **(C)** $S_{999}<50$ **(D)** $S_{999}>50$

34. Let $S=\dfrac{1}{\sqrt{1}+\sqrt{3}}+\dfrac{1}{\sqrt{5}+\sqrt{7}}+\dfrac{1}{\sqrt{9}+\sqrt{11}}+\dfrac{1}{\sqrt{9997}+\sqrt{9999}}$, then

(A) $S<24$ **(B)** $S>24$ **(C)** $S>18$ **(D)** $S<18$

35. Define $f_n(x)=(1+x)(1+2x)(1+4x)\ldots(1+2^n x)=a_{n,0}+a_{n,1}{}^{x}+a_{n,2}{}^{x^2}+\ldots+a_{n,n}x^2$, where n is a positive integer, then

(A) $a_{100,2}=\dfrac{(2^{100}-1)(2^{102}-4)}{3}$ **(B)** $a_{100,2}=\dfrac{(2^{101}-1)(2^{101}-2)}{3}$

(C) $a_{100,2}-a_{99,2}=2^{201}-2^{101}$ **(D)** $a_{100,2}-a_{99,2}=2^{200}-2^{100}$

36. Let a, b, c be positive integers such that $a+b+c=n$, then

(A) $\left(a^a b^b c^c\right)^{1/n}\le\dfrac{a^2+b^2+c^2}{n}$ **(B)** $\left(a^b b^c c^a\right)^{1/n}\le\dfrac{ab+bc+ca}{n}$

(C) $\left(a^b b^c c^a\right)^{1/n}\le\dfrac{a^2+b^2+c^2}{n}$ **(D)** $\left(a^a b^b c^c\right)^{1/n}+\left(a^b b^c c^a\right)^{1/n}+\left(a^c b^a c^b\right)^{1/n}\le n$

37. Let $S=2016^2+2015^2+2014^2-2013^2-2012^2-2011^2+2010^2+2009^2+2008^2-2007^2+-2006^2-$
$2005^2+\ldots6^2+5^2+4^2-3^2-2^2-1^2$, then S is divisible by

(A) 8 **(B)** 27 **(C)** 112 **(D)** 2017

38. Let $f(x)=\frac{1}{x+1}+\frac{2x}{(x+1)(x+2)}+\frac{3x^2}{(x+1)(x+2)(x+3)}+...+\frac{nx^{n-1}}{(x+1)(x+2)(x+3)...(x+n)}$ then:

(A) $f(x)=\frac{x}{1+x}-\frac{x^n}{(x+1)(x+2)...(x+n)}$ **(B)** $1-\frac{x^n}{(x+1)(x+2)...(x+n)}$

(C) $f'(x)=\left(-\frac{x^n}{(x+1)(x+2)...(x+n)}\right)\left(\sum_{r=1}^{n}\frac{r}{x+r}\right)$

(D) $f'(x)=\left(-\frac{x^{n-1}}{(x+1)(x+2)..(x+n)}\right)\left(\sum_{r=1}^{n}\frac{r}{x+r}\right)$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column I are labelled as (A), (B), (C) & (D) whereas statements in Column II are labeled as P, Q, R, S & T. More than one choice from Column II can be matched with Column I.

39.

	Column –I	Column –II	
(A)	If $A=\sum_{r=1}^{n}r^2, B=\sum_{m=1}^{n}\sum_{r=1}^{m}r-\frac{1}{2}\sum_{r=1}^{n}r$, then $\frac{A}{B}$ is equal to	(P)	1
(B)	For positive numbers a, b, c the minimum value of $\frac{a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2)}{abc}$ is equal to	(Q)	2
(C)	If $x+y+z=1, x, y, z>0$, then the minimum value of $\frac{2x^2}{y+z}+\frac{2y^2}{z+x}+\frac{2z^2}{x+y}$ is equal to	(R)	3
(D)	If $\frac{1}{xy}+\frac{1}{yz}+\frac{1}{zx}=3$ where $x, y, z\in N$, then $x+y+z$ is equal to	(S)	4
		(T)	6

40.

	Column –I	Column –II	
(A)	Let a, b, c are positive real numbers such that $a^3b^2c=12$, then the minimum value of $49a+3b+c$ is equal to	(P)	1
(B)	The minimum value of $\left\|2x^3-\frac{3}{x^2}\right\|$ for $x<0$ is equal to	(Q)	5
(C)	The maximum value of $\frac{x^5(8-x^3)}{\sqrt[3]{25}}$ for $0<x<2$ is equal to	(R)	7
(D)	If $x^7 y^5=a$ and $7x+5y\geq 12\forall x, y>0$, then the minimum value of 'a' is equal to	(S)	15
		(T)	42

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

41. If $\sum_{r=1}^{n} a_r = \sum_{k=1}^{n}\sum_{j=1}^{k}\sum_{i=1}^{j} 2$ and $\lambda = \lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{1}{a_r}\right)^n$ then $\left[\frac{1}{\lambda}\right]$ is equal to (where $[\cdot]$ denotes greatest integer function).

42. Let $a_i + b_i = 1 \forall i = 1,2,....,6$ and $a = \frac{1}{6}(a_1 + a_2 + ...+ a_6), b = \frac{1}{6}(b_1 + b_2 + ...+ b_6)$. Then

$a_1b_1 + a_2b_2 + ...+ a_6b_6 = nab - (a_1 - a_2)^2 - (a_2 - a)^2 - ... - (a_6 - a)^2$ where n is equal to

43. The value of expression $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+...\left[\sqrt{100^2}\right]$, (where[.] denotes greatest integer function) is equal to___.

44. If the value of $\sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{1}{3^i 3^j 3^k}(i \ne j \ne k)$ is equal to $\frac{m}{n}$, where m, n are coprime natural numbers, then $m+n$ is equal to_____.

45. Integers 1, 2, 3,…, n where $n > 2$ are written on a board. Two numbers m, k such that $1 < m < n, 1 < k < n$ are removed and the average of the remaining numbers is found to be 17. What is the maximum sum of the two removed numbers ?

46. Let the equation $x^4 - 16x^3 + px^2 - 256x + q = 0$ has four positive real roots in G.P., then $p+q$ is equal to

47. Let $x_1, x_2, x_3,..., x_{2018}$ be real numbers different from 1, such that $x_1 + x_2 +,,, + x_{2018} = 1$ and

$\frac{x_1}{1-x_1} + \frac{x_2}{1-x_2} + ... + \frac{x_{2018}}{1-x_{2018}} = 1$ then the value of $\frac{x_1^2}{1-x_1} + \frac{x_2^2}{1-x_2} + ... + \frac{x_{2018}^2}{1-x_{2018}}$ is equal to_____.

48. Let $x_1, x_2,...., x_{2018}$ be positive real numbers such that $x_1 + x_2 + ... + x_{2018} = 1$. Determine the smallest constant k such that $k\sum_{i=1}^{2018}\frac{x_i^2}{1-x_i} \ge 1$

49. Let x, y, z are positive real numbers satisfy $2x - 2y + \frac{1}{z} = \frac{1}{2018}, 2y - 2z + \frac{1}{x} = \frac{1}{2018}$, $2z - 2x + \frac{1}{y} = \frac{1}{2018}$ then $x + y - z$ is equal to_____.

50. Let a, b, c be positive real number such that $a+b+c \ge 4$, then find the minimum value of

$$\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)}.$$

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If z_1, z_2, z_3 be three complex numbers such that $|z_1+1|\le 1, |z_2+2|\le 2$ and $|z_3+4|\le 4$, then the maximum value of $|z_1|+|z_2|+|z_3|$ is :

(A) 7 **(B)** 10 **(C)** 12 **(D)** 14

2. The value of $i\log(x-i)+i^2\pi+i^3\log(x+i)+i^4(2\tan^{-1}x)$, (where, x > 0 and $i=\sqrt{-1}$), is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

3. The complex number satisfying arg (z + i) = $\frac{\pi}{4}$ and $\arg(2z+3-2i)=\frac{3\pi}{4}$ simultaneously, is :

(A) $\frac{1}{4}-\frac{3}{4}i$ **(B)** $\frac{1}{4}+\frac{3}{4}i$ **(C)** $-\frac{1}{4}-\frac{3}{4}i$ **(D)** None of these

4. Equation of tangent drawn to the circle $|z| = r$ at the point A (z_0), is :

(A) $\operatorname{Re}\left(\frac{z}{z_0}\right)=1$ **(B)** $\operatorname{Re}\left(\frac{z_0}{z}\right)=1$ **(C)** $\operatorname{Im}\left(\frac{z}{z_0}\right)=1$ **(D)** $\operatorname{Im}\left(\frac{z_0}{z}\right)=1$

5. Consider a square *OABC* in the Argand plane, where '*O*' is origin and $A\equiv A(z_0)$. Then the equation of the circle that can be inscribed in this square is : (vertices of square are given in anti-clockwise order)

(A) $|z-z_0(1+i)|=|z_0|$ **(B)** $2\left|z-\frac{z_0(1+i)}{2}\right|=|z_0|$

(C) $\left|z-\frac{z_0(1+i)}{2}\right|=|z_0|$ **(D)** None of the above

6. If $\left|\frac{z-z_1}{z-z_2}\right|=3$, where z_1 and z_2 are fixed complex numbers and *z* is a variable complex number, then *z* lies on a :

(A) Circle with z_1 as its interior point **(B)** Circle with z_2 as its interior point

(C) Circle with z_1 and z_2 as its interior points **(D)** Circle with z_1 and z_2 as its exterior points

7. Let z_1 and z_2 be the non-real roots of the equation $3z^2 + 3z + b = 0$. If the origin together with the points represented by z_1 and z_2 form an equilateral triangle, then the value of *b* is :

(A) 1 **(B)** 2 **(C)** 3 **(D)** None of the above

8. The equation $(1+a)x^2+2a^2x+a^2+b^2-1=0$ has roots of opposite sign, if *a* + *ib* lies, $(a>-1)$:

(A) On straight line $x + y = 1$

(B) Inside a circle of centre (0, 0) and radius '1'

(C) On a parabola of vertex (0, 0) and focal length '1'

(D) None of the above

9. The roots of $z^n = (z+1)^n$

(A) Lie on the vertices of a regular n-polygon **(B)** Lie on a circle

(C) Are collinear **(D)** None of the above

10. If $|z-4+3i| \le 1$ and α and β be the least and greatest values of $|z|$ and K be the least value of $\frac{x^4+x^2+4}{x}$ on the interval $(0,\infty)$, then K is equal to :

(A) α **(B)** β **(C)** $\alpha+\beta$ **(D)** None of the above

11. The complex numbers $\sin x + i\cos 2x$ and $\cos x - i\sin 2x$ are conjugate to each other for :

(A) $x = n\pi$ **(B)** $x = \left(n+\frac{1}{2}\right)\pi$ **(C)** $x = 0$ **(D)** No value of x

12. If $\left(\frac{3-z_1}{2-z_1}\right)\left(\frac{2-z_2}{3-z_2}\right) = K$, then points $A(z_1), B(z_2), C(3,0)$ and $D(2,0)$ (taken clockwise) will :

(A) lie on a circle only for $K > 0$ **(B)** lie on a circle only for $K < 0$

(C) lie on a circle $\forall K \in R$ **(D)** be the vertices of a square $\forall K \in (0,1)$

13. Let 'z' be a complex number and 'a' be a real parameter such that $z^2 + az + a^2 = 0$, then:

(A) locus of z is a pair of straight lines **(B)** locus of z is a circle

(C) $\arg(z) = \pm\frac{5\pi}{3}$ **(D)** $|z| = 2|a|$

14. The locus represented by the equation $|z-1|+|z+1| = 2$ is :

(A) an ellipse with foci (1, 0) and (–1, 0)

(B) one of the family of circles passing through the points of intersection of the circles $|z-1| = 1$ and $|z+1| = 1$.

(C) the radical axis of the circles $|z-1| = 1$ and $|z+1| = 1$.

(D) the portion of the real axis between the points (1, 0) (-1, 0) including both.

15. If $|z-2| = \min\{|z-1|, |z-5|\}$, where z is a complex number, then :

(A) $\text{Re}(z) = \frac{3}{2}$ **(B)** $\text{Re}(z) = \frac{7}{2}$ **(C)** $\text{Re}(z) \in \left\{\frac{3}{2}, \frac{7}{2}\right\}$ **(D)** None of the above

16. If $z_1, z_2, z_3, \ldots, z_{n-1}$ are the roots of the equation $z^{n-1}+z^{n-2}+z^{n-3}+\ldots+z+1 = 0$, where $n \in N$, $n > 2$ and ω is the cube root of unity, then :

(A) ω^n, ω^{2n} are also the roots of the given equation

(B) $\omega^{1/n}, \omega^{2/n}$ are also the roots of the given equation

(C) $z_1, z_2, z_3, \ldots, z_{n-1}$ form a geometric progression

(D) $a^{\frac{z_r+1}{z_r}}$ is variable for $a > 0$ and $r = 1, 2, \ldots, n-2$

17. If $A(z_1)$, $B(z_2)$, $C(z_3)$ are the vertices of a triangle ABC inscribed in the circle $|z| = 2$. Internal angle bisector of the angle A, meet the circumcircle again at $D(z_4)$, then :

(A) $z_4^2 = z_2z_3$ **(B)** $z_4 = \frac{z_2z_3}{z_1}$ **(C)** $z_4 = \frac{z_1z_2}{z_3}$ **(D)** $z_4 = \frac{z_1z_3}{z_2}$

18. If $|z-2-i| = |z|\sin\left(\frac{\pi}{4} - \arg z\right)$, then locus of z is/an :

(A) ellipse **(B)** circle **(C)** parabola **(D)** pair of straight lines

19. If $|z|=2$ and $\frac{z_1-z_3}{z_2-z_3}=\frac{z-2}{z+2}$, then z_1, z_2, z_3 will be vertices of a/an :

(A) equilateral triangle **(B)** acute angled triangle
(C) right angled triangle **(D)** None of the above

20. For the complex number z, the minimum value of $|z|+|z-\cos\alpha-i\sin\alpha|$ is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** None of the above

Paragraph for Questions 21 - 23

Let $f(x)=x^4-6x^3+26x^2-46x+65$. All the roots of $f(x)=0$ are of the form a_k+ib_k for $k=1, 2, 3, 4$, where $i=\sqrt{-1}$. Further a_k and b_k are all integers. Also $\lambda=|b_1|+|b_2|+|b_3|+|b_4|$ and $\mu=a_1+a_2+a_3+a_4$. If set S is formed whose elements are all $a_i's$ and $b_i's$, then :

21. The value of $\lambda+\mu$ is equal to :

(A) 16 **(B)** 4 **(C)** 10 **(D)** 6.

22. Roots of the equation are :

(A) $-1\pm 2i, -2\pm 3i$ **(B)** $2\pm 2i, 1\pm 3i$ **(C)** $-2\pm 2i, -1\pm 3i$ **(D)** $1\pm 2i, 2\pm 3i$.

23. Number of functions from set $S\to C$, where C has 8 distinct element is :

(A) 8^4 **(B)** 8^8 **(C)** 8^5 **(D)** 8^6.

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

24. Let $\alpha=e^{i2\pi/11}, \lambda=\alpha^6, \mu=\alpha^7, \beta=\alpha^2$. Then :

(A) $\mathrm{Re}\left(\lambda+\lambda^2+\lambda^3+\lambda^4+\lambda^5\right)=-\frac{1}{2}$ **(B)** $(\mu-\beta)(\mu-\beta^2)(\mu-\beta^3).....(\mu-\beta^9)(\mu-\beta^{10})=0$

(C) $(i-\beta)(i-\beta^2)(i-\beta^3).....(i-\beta^{10})=i$ **(D)** None of these

25. $\alpha_1,\alpha_2,\alpha_3,....\alpha_{100}$ are all the 100th roots of unity. The numerical value of $\sum\sum_{1\le i<j\le 100}(\alpha_i\alpha_j)^5$, is :

(A) 20 **(B)** 0 **(C)** $(20)^{1/20}$ **(D)** $\sum_{i=1}^{100}\alpha_i$

26. If A and B respresent complex numbers z_1 and z_2. $P(z)$ is any complex number satisfying $\left|z-\frac{z_1+z_2}{2}\right|=k, (k>0)$, then:

(A) Maximum area of $(\Delta PAB)=\frac{1}{2}k\,|z_1-z_2|$

(B) There are two possible positions of P on argand plane when area of ΔPAB is maximum

(C) Area of ΔPAB = constant $\left(<\frac{1}{2}k\,|z_1-z_2|\right)$, for 4 possible positions of P.

(D) ΔPAB is equilateral triangle of maximum area if $4k^2=3|z_1-z_2|^2$

27. Locus of z, if $\arg[z-(1+i)] = \begin{cases} \dfrac{3\pi}{4}, & \text{when } |z| \le |z-2| \\ \dfrac{-\pi}{4}, & \text{when } |z| > |z-2| \end{cases}$, is :

(A) a pair of straight lines passing through (2, 0)

(B) a pair of straight lines passing through (2, 0), (1, 1)

(C) a line segment **(D)** a set of two rays

28. Let $z \in C$ and if $A = \left\{z : \arg(z) = \dfrac{\pi}{4}\right\}$ and $B = \left\{z : \arg(z-3-3i) = \dfrac{2\pi}{3}\right\}$. Then $n(A \cap B)$ is equal to :

(A) 1 **(B)** $\sum\limits_{r=0}^{99} i^r$ **(C)** 3 **(D)** 0

29. If z_1, z_2, z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ then $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ cannot exceed :

(A) 6 **(B)** 9 **(C)** 12 **(D)** 5

30. Let $f(z)$ be a polynomial function of a complex number z. On division by $z-i, z+i$ and z^2+1 we obtain remainder as α, β and $g(z)$, $(\alpha, \beta \in C)$. Then :

(A) $\alpha = i$ and $\beta = 1+i \Rightarrow g(z) = i\left(\dfrac{z}{2}+1\right)+\dfrac{1}{2}$ **(B)** $\alpha = i$ and $\beta = 1-i \Rightarrow g(z) = \left(z-\dfrac{1}{2}\right)-\dfrac{iz}{2}$

(C) $\alpha = i$ and $\beta = 1+i \Rightarrow g(z) = i\left(z+\dfrac{1}{2}\right)+\dfrac{1}{2}$ **(D)** $\alpha = i$ and $\beta = 1-i \Rightarrow g(z) = \left(z+\dfrac{1}{2}\right)+\dfrac{iz}{2}$

31. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are complex numbers such that $|z_1| = 1, |z_2| = 2$ and $\text{Re}(z_1 z_2) = 0$, then the pair of complex numbers $\omega_1 = a_1 + \dfrac{ia_2}{2}$ and $\omega_2 = 2b_1 + ib_2$ satisfy :

(A) $|\omega_1| = 1$ **(B)** $|\omega_2| = 2$ **(C)** $\text{Re}(\omega_1\omega_2) = 0$ **(D)** $\text{Im}(\omega_1\omega_2) = 0$

32. If from a point P representing the complex number z_1 on the curve $|z| = 2$, pair of tangents are drawn to the curve $|z| = 1$, meeting at point $Q(z_2)$ and $R(z_3)$, then :

(A) complex number $\dfrac{z_1+z_2+z_3}{3}$ will lie on the curve $|z| = 1$

(B) $\left(\dfrac{4}{z_1}+\dfrac{1}{z_2}+\dfrac{1}{z_3}\right)\left(\dfrac{4}{\bar{z}_1}+\dfrac{1}{\bar{z}_2}+\dfrac{1}{\bar{z}_3}\right) = 9$

(C) $\arg\left(\dfrac{z_2}{z_3}\right) = \dfrac{2\pi}{3}$

(D) orthocentre and circumcentre of ΔPQR will coincide

33. The complex number satisfying $|z+\bar{z}| + |z-\bar{z}| = 2$ and $|z+i| + |z-i| = 2$ is / are :

(A) i **(B)** $-i$ **(C)** $1+i$ **(D)** $1-i$

34. If z is a complex number such that $\arg\left(\frac{z-3\sqrt{3}}{z+3\sqrt{3}}\right)=\frac{\pi}{3}$, then the locus of z is :

(A) $|z-3i|=6$ **(B)** $|z-3i|=6$, $Im\, z>0$
(C) $|z-3i|=6$, $Im z<0$ **(D)** Major arc of a circle

35. Value of $\left(\sin\left(\log i^i\right)\right)^3+\left(\cos\left(\log i^i\right)\right)^3$ is :

(A) 1 **(B)** –1 **(C)** $\sum_{k=1}^{8} e^{i\frac{2\pi k}{9}}$ **(D)** 2i

36. If $x^6=(4-3i)^5$, then the product of all of its roots is : (where $\theta=-\tan^{-1}(3/4)$)

(A) $5^5(\cos(\pi+5\theta)+\sin(\pi+5\theta))$ **(B)** $-5^5(\cos 5\theta+i\sin 5\theta)$

(C) $5^5(\cos 5\theta-i\sin 5\theta)$ **(D)** $-5^5(\cos 5\theta-i\sin 5\theta)$

37. If $|z-1|+|z+3|\le 8$, then the possible values of $|z-4|$ belongs to :

(A) (–9, 0) **(B)** [–9, –1] **(C)** [1, 9] **(D)** [5, 9]

38. The possible values of parameter α for which $|z-(\alpha^2-7\alpha+11+i)|=1$ and $\arg z\ge\frac{\pi}{2}$ is satisfied for at least one z are :

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6

39. If $\alpha\ne 1$, $\alpha^5=1$, then $\log_{\sqrt{3}}\left|1+\alpha+\alpha^2+\alpha^3-\frac{2}{\alpha}\right|$ is equal to :

(A) 2 **(B)** 3 **(C)** 4 **(D)** $\min\left(x+\frac{1}{x}\right)$, $(x>0)$

40. If $\alpha=\cos\frac{2\pi}{7}+i\sin\frac{2\pi}{7}$ and $\left|\sum_{r=0}^{3n-1}\alpha^{2^r}\right|^2=32$ then n is :

(A) 8 **(B)** 4 **(C)** $\log_e e^8$ **(D)** $-4\sum_{k=1}^{6} e^{i\frac{2\pi k}{7}}$

41. The number of roots of the equation $z^{15}=1$ satisfying $|\arg z|<\pi/2$ are :

(A) 6 **(B)** 7 **(C)** 8 **(D)** $\frac{7}{i}\sum_{r=1}^{97} i^r$

42. If z is a complex number satisfying $|z|^2-|z|-2<0$, then the possible value(s) of $|z^2+z\sin\theta|$. for all values of θ, is(are) :

(A) 4 **(B)** 5 **(C)** 6 **(D)** 7

43. If $z_n=\cos\frac{\pi}{n(n+1)(n+2)}+i\sin\frac{\pi}{n(n+1)(n+2)}$ for n = 1, 2, 3, …,k, then the value of $\left|\lim_{k\to\infty}(z_1z_2....z_k)\right|$ is :

(A) 1 **(B)** 2 **(C)** $-\sum_{k=1}^{98} e^{i\frac{2\pi k}{99}}$ **(D)** $\sum_{k=0}^{98} e^{i\frac{2\pi k}{99}}$

44. Complex number whose real and imaginary parts x and y are integers and satisfy the equation $3x^2 - |xy| - 2y^2 + 7 = 0$

(A) do not exist **(B)** exist and have equal modulus

(C) form two conjugate pairs **(D)** do not form conjugate pairs

45. If all the three roots of $az^3 + bz^2 + cz + d = 0$ have negative real parts, $(a, b, c \in R)$ then

(A) $ab > 0$ **(B)** $bc > 0$ **(C)** $ad > 0$ **(D)** $bc - ad > 0$

46. If z_1, z_2 and z_3 are three complex numbers such that

$|z_1| = |z_2| = |z_3| = 1$, then $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ is strictly less than

(A) 6 **(B)** 9 **(C)** 12 **(D)** 18

47. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are complex numbers such that $|z_1| = 1, |z_2| = 2$ and Re $(z_1 z_2) = 0$ then the pair of complex numbers $w_1 = a_1 + \frac{ia_2}{2}$ and $w_2 = 2b_1 + ib_2$ satisfy

(A) $|w_1| = 1$ **(B)** $|w_2| = 2$ **(C)** Re $(w_1 w_2) = 0$ **(D)** Im $(w_1 w_2) = 2$

48. For complex number z and w, if $|z|^2 w - |w|^2 z = z - w$ then

(A) $z = w$ **(B)** $z\bar{w} = 1$ **(C)** $z = -w$ **(D)** $\bar{z}w = 1$

49. If $z^3 + (3 + 2i)z + (-1 + ia) = 0$ has one real root, then the value of a lies in the interval $(a \in R)$

(A) (–2, 1) **(B)** (–1, 0) **(C)** (0, 1) **(D)** (–2, 3)

50. If $p(x)$, $q(x)$, $r(x)$ and $s(x)$ are polynomials such that $p(x^3) + xq(x^3) + x^2 r(x^3) = (1 + x + x^2)s(x)$ then

(A) $p(1) = s(1)$ **(B)** $p(1) = r(1)$ **(C)** $p(1) = 3s(1)$ **(D)** $p(1) = 2r(1)$

51. If the roots of the equation $z^4 + \lambda z^3 + (-36 + 15i)z^2 + mz = 0$ are the vertices of a square then $(\lambda + m)$ can be equal to

(A) $35 + 45i$ **(B)** $-35 - 45i$ **(C)** $35 - 45i$ **(D)** $-35 + 45i$

52. Complex numbers z_1, z_2, z_3 and z_4 correspond to the points A, B, C and D, respectively, on a circle $|z| = 1$. If $z_1 + z_2 + z_3 + z_4 = 0$, Then $ABCD$ is necessarily

(A) a triangle **(B)** a square **(C)** a rhombus **(D)** a parallelogram

53. Two triangles having vertices as z_1, z_2, z_3 and a, b, c are similar. Then

(A) $az_1 + bz_2 + cz_3 = 0$ **(B)** $\frac{z_1}{a} + \frac{z_2}{b} + \frac{z_3}{c} = 0$

(C) $z_1(b - c) + z_2(c - a) + z_3(a - b) = 0$ **(D)** $a(z_2 - z_3) + b(z_3 - z_1) + c(z_1 - z_2) = 0$

54. If $k \in R - \{0\}, z$ is a complex number and $k + |k + z^2| = |z^2|$ then the value (s) of arg z is/are

(A) $-\pi$ **(B)** $-\frac{\pi}{2}$ **(C)** $\frac{\pi}{2}$ **(D)** π

55. Let z_1 and z_2 be two complex numbers represented by points on the circle $|z_1| = 1$ and $|z_2| = 2$, respectively. Then

(A) $\max |2z_1 + z_2| = 4$ **(B)** $\max |z_1 - z_2| = 1$

(C) $\left|z_2 + \frac{1}{z_1}\right| \le 3$ **(D)** None of these

56. Equation of line through a and ib such that $a, b \in R$ and $a, b \neq 0$ is

(A) $z\left(\frac{1}{2a}+\frac{1}{2ib}\right)+\bar{z}\left(\frac{1}{2a}-\frac{1}{2ib}\right)=1$ **(B)** $z\left(\frac{1}{2a}+\frac{i}{2b}\right)+\bar{z}\left(\frac{1}{2a}+\frac{i}{2b}\right)=1$

(C) $1\,z\left(\frac{1}{2a}-\frac{i}{2b}\right)+\bar{z}\left(\frac{1}{2a}-\frac{i}{2b}\right)=1$ **(D)** $z\left(\frac{1}{2a}-\frac{i}{2b}\right)+\bar{z}\left(\frac{1}{2a}+\frac{i}{2b}\right)=1$

57. w_1, w_2 be roots of $(a+\bar{c})z^2+(b+\bar{b})z+(\bar{a}+c)=0$ If $|z_1|<|z_2|<1$, then

(A) $|w_1|<1$ **(B)** $|w_1|=1$ **(C)** $|w_2|<1$ **(D)** $|w_2|=1$

58. A complex number z satisfies the equation $|Z^2-9|+|Z^2|=41$, then the true statements among the following are

(A) $|Z+3|+|Z-3|=10$ **(B)** $|Z+3|+|Z-3|=8$

(C) Maximum value of $|Z|$ is 5 **(D)** Maximum value of $|Z|$ is 6

59. Let a, b, c be distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 be the roots of the equation $az^2+bz+c=0$ with $|z_1|=1$. Let P and Q represent the complex numbers z_1 and z_2 in the Argand plane with $\angle POQ = \theta$, $0° < \theta < 180°$ (where O being the origin). Then

(A) $b^2=ac;\ \theta=\frac{2\pi}{3}$ **(B)** $\theta=\frac{2\pi}{3};\ PQ=\sqrt{3}$

(C) $PQ=2\sqrt{3};\ b^2=ac$ **(D)** $\theta=\frac{\pi}{3};\ b^2=ac$

60. Let $Z_1=x_1+iy_1, Z_2=x_2+iy_2$ be complex numbers in fourth quadrant of argand plane and $|Z_1|=|Z_2|=1$, $\text{Re}(Z_1Z_2)=0$. The complex number

$Z_3=x_1+ix_2, Z_4=y_1+iy_2, Z_5=x_1+iy_2, Z_6=x_2+iy_1$, will always satisfy

(A) $|Z_4|=1$ **(B)** $\arg(Z_1Z_4)=-\frac{\pi}{2}$

(C) $\frac{Z_5}{\cos(\arg Z_1)}+\frac{Z_6}{\sin(\arg Z_1)}$ is purely real **(D)** $Z_3^2+(\bar{Z}_6)^2$ is purely imaginary

61. If the imaginary part of $\frac{z-3}{e^{i\theta}}+\frac{e^{i\theta}}{z-3}$ is zero, then z can lie on

(A) a circle with unit radius **(B)** a circle with radius 3 units

(C) a straight line through the point (3, 0) **(D)** a parabola with the vertex (3, 0)

62. If z_1, z_2, z_3 are any three roots of the equation $z^6=(z+1)^6$ then $\arg\left(\frac{z_1-z_3}{z_2-z_3}\right)$ can be equal to

(A) 0 **(B)** π **(C)** $\frac{\pi}{4}$ **(D)** $-\frac{\pi}{4}$

63. Let z_1, z_2, z_3 are the vertices of ΔABC, respectively, such that $\frac{z_3-z_2}{z_1-z_2}$ is purely imaginary number. A square on side AC is drawn outwardly. $P(z_4)$ is the centre of square, then

(A) $|z_1-z_2|=|z_2-z_4|$ **(B)** $\arg\left(\frac{z_1-z_2}{z_4-z_2}\right)+\arg\left(\frac{z_3-z_2}{z_4-z_2}\right)=\frac{\pi}{2}$

(C) $\arg\left(\frac{z_1-z_2}{z_4-z_2}\right)+\arg\left(\frac{z_3-z_2}{z_4-z_2}\right)=0$ **(D)** z_1, z_2, z_3 and z_4 lie on a circle.

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

64. If $|z-1|+|z+1|=k$ then locus of z is :

	Column 1		Column 2
(A)	If $k = 2$	(p)	Ellipse of eccentricity 2/3
(B)	If $k = 5$	(q)	No locus
(C)	if $0 < k < 2$	(r)	Line segment
(D)	If $k = 3$	(s)	Ellipse of eccentricity 2/5

65. If $\arg\left(\frac{z-(1+i)}{z-(3+4i)}\right)=\theta$ Then locus of z is :

	Column 1		Column 2
(A)	Line segment	(p)	If $\theta=\frac{2\pi}{3}$
(B)	Line ray	(q)	If $\theta=\pi$
(C)	Major arc of a circle	(r)	If $\theta=\frac{\pi}{3}$
(D)	Minor arc of a circle	(s)	If $\theta=0$

66. If an equilateral triangle ABC with vertices at z_1, z_2 and z_3 be inscribed in the circle $|z| = 2$ and again a circle is inscribed in the triangle ABC touching the sides AB, BC and CA at $D(z_4)$, $E(z_5)$ and $F(z_6)$ respectively :

	Column 1		Column 2
(A)	Then value of $\mathrm{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1)$ is equal to	(p)	2
(B)	If $\frac{4z_1}{z_3}$ is equal to $a(-1+i\sqrt{3})$, then a is	(q)	– 6
(C)	The value of $|z_1+z_2|^2+|z_2+z_3|^2+|z_3+z_1|^2$ is	(r)	12
(D)	If P is any point on incircle the value of $DP^2+EP^2+FP^2$ is	(s)	6
		(t)	– 2

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/ INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

67. The number of solution(s) of the equation $z^2 - z - |z|^2 - \frac{64}{|z|^5} = 0$ is ____________.

68. If arg(z) < 0, then $-\frac{10}{\pi}$ arg $\left(\frac{z-\bar{z}}{2}\right)$ is equal to ____________.

69. Let w, $\overline{w}$ is complex cube root of unity and P(z) is point on a circle $|z| = 4$ such that $|z - 1|$ is maximum and centroid of triangle formed by $z, -w, -\overline{w}$ is α then -7 Re(α) is____________.

70. If z is a complex number and the minimum value of $|z|+|z-1|+|2z-3|$ is λ and if $y = 2[x]+3 = 3[x-\lambda]$ then find the value of $\frac{1}{5}([x+y])$. (where [.] denotes the greatest integer function).

71. The area of the region bounded by curves.

(i) $|z - z_1| = |z - z_3|$

(ii) $|\text{Re}(z) - \text{Re}(z_1)| = |\text{Re}(z) - \text{Re}(z_3)|$

(iii) $|z - z_2| - |z - z_1| = |z_1 - z_2|$

(where $z_1 = 1+i,\ z_2 = 2+i,\ z_3 = -3+3i$) is $\frac{p}{q}$, (p, q are co-prime) then find $p+q$.

72. Let $\alpha = \cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ and let $A_k = x + y\alpha^k + p\alpha^{2k} + w\alpha^{3k} + f\alpha^{4k}$ where x, y, p, w, f are points on the circle $|z| = 1$, then $\frac{|A_0|^2 + |A_1|^2 + |A_2|^2 + |A_3|^2 + |A_4|^2}{5}$ is equal to _______.

73. Let A, B, C be the set of complex numbers defined as $A = \{z: ||z + 2| - |z - 2|| = 2\}$, $B = \left\{z : \arg\left(\frac{z-1}{z}\right) = \frac{\pi}{2}\right\}$ and $C = \{z: \arg(z - 1) = \pi\}$, then $n(A \cap B \cap C)$ is ______.

74. Let $z = (\cos 12° + i\sin 12° + \cos 48° + i\sin 48°)^6$, then $Im(z)$ is equal to ____________.

75. If $8iz^3 + 12z^2 - 18z + 27i = 0$, then $2|z|$ is equal to ____________.

76. Given a, b, c are cube roots of $q(q < 0)$, then for any value of x, y, z given by $\left|\frac{a^2x^2+b^2y^2+c^2z^2}{b^2x^2+c^2y^2+a^2z^2}\right| + \left(x_1^2 - 2y_1^2\right)\omega + \left(\left[x^2\right]+\left[y^2\right]+\left[z^2\right]\right)\omega^2 = 0$, (where [.] denotes the greatest integer function, ω is cube root of unity, x_1, y_1 are positive integers and y_1, is a prime number) then value of $[x^2 + x_1] + [y^2 + y_1] + [z^2 + x_1^2 + y_1^2] - 10$ is ____________.

77. If $\omega \neq 1$ is a cube root of unity and z is a complex number such that $|z| = 1$ then $\left|\dfrac{2+3\omega+4z\omega^2}{4\omega+3\omega^2 z+2z}\right| =$ _____.

78. If z is a complex number such that $\left|z+\dfrac{1}{z}\right| = 2$ then minimum value of $|z|$ is____

79. If $|z| = 1$ and $z^{2n}+1 \neq 0$ then $\dfrac{z^n}{z^{2n}+1} - \dfrac{(\bar{z})^n}{(\bar{z})^{2n}+1}$ is equal to _____.

80. Let $A(z)$ and $B(z_1)$ be two variable points such that $zz_1 = |z|^2$. If the area enclosed by $|z-\bar{z}|+|z_1+\bar{z}_1| = 10$ is A then the value of $A/8$ is _____.

81. Sum of all the solutions of $z = |z| + z^2$ is ____.

82. Let $z = x+iy$ and $\arg\left(e^{z^2}\right) = \arg\left(e^{(z+i)}\right)$. If $y = f(x)$ is a function, then f(3) is equal to _____.

83. Let z_i, i = 1, 2, …, 6 be the roots of $z^6 + z^4 = 2$ then $\sum_{i=1}^{6} |z_i|^4$ is equal to____.

84. Difference between the square of the least and the square of the greatest values of $|z|$, where $z = e^{i2\phi}\sin\phi + \cos\phi\,(\phi \in R)$, is____.

85. The value of $4\alpha\left(\beta^4 - \alpha^4\right)$, if $\alpha + i\beta$, $\beta \neq 0$ is a root of $z^5 = 1$, is____.

86. If $z\bar{z} = 1$, then the value of $\left[\left|2+\dfrac{1}{z}\right| + |2-z|^2\right]$ is ____.

87. If $1+2|z^2| = |z^2+1| + 2|z+1|^2$, then the value of $\dfrac{|z(z+1)|}{2}$ is ____.

88. Sum of all the solutions of $z^2 + |z| = (\bar{z})^2$ is ____.

89. If the roots Z_1, Z_2, Z_3 of the equation $Z^3 - Z^2 + mZ - 1 = 0$ lie on $|Z| = 1$ and $|(Z_1+3)(Z_2+3)(Z_3+3)| = 10\lambda$ then $\lambda =$ ____.

90. Given $|3z_1 - 2z_2 - 4|^2 = |3z_1 - 1|^2 + |2z_2 + 3|^2$ $\left(z_2 \neq -\dfrac{3}{2}\right)$ If cube roots of $w = \dfrac{3z_1 - 1}{2z_2 + 3}$ are w_1, w_2, w_3; where ($\arg w_1 < \arg w_2 < \arg w_3$), then the value of $\dfrac{w_2^2}{w_1 w_3}$ is ____.

91. Number of complex number z satisfying $z^3 = \bar{z}$ is ____.

92. Let α and β be two complex numbers satisfying $|\alpha+1+i| = 1$ and $|\beta-2-3i| = 6$. Then the value of $6|\alpha|_{\max} - |\beta|_{\max}$ is ____.

93. Let z be a complex number such that $\left|2z+\dfrac{1}{z}\right| = 1$ and $\arg(z) = \theta$. Then minimum value of $\sin^2\theta$ is____.

94. Let AB and CD be parallel chords of the circle $|z| = r$. If z_1, z_2, z_3 and z_4 represent A, B, C and D, respectively, and $z_1 z_2 = k z_3 z_4$ then $\dfrac{75k}{4}$ equals ____.

95. The number of complex numbers which are conjugate of their own cube, is____.

96. If $A(z_1)$; $B(z_2)$; $C(z_3)$ are the vertices of triangle such that

$z_3 = \dfrac{z_2 - iz_1}{1-i}$ & $|z_1| = 3$; $|z_2| = 4$ & $|z_2 + iz_1| = |z_1| + |z_2|$ then area of ΔABC is ___.

Advanced Problem Package	Permutations & Combinations

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Number of ways in which four different toys and five indistinguishable marbles can be distributed between Amar, Akbar and Anthony, if each child receives atleast one toy and one marble, is :

(A) 42 **(B)** 100 **(C)** 150 **(D)** 216

2. You are given an unlimited supply of each of the digits 1, 2, 3 or 4. Using only these four digits, you construct n digit numbers. Such n digit numbers will be called LEGITIMATE if it contains the digit 1 either an even number times or not at all. Number of n digit legitimate numbers are :

(A) $2^n + 1$ **(B)** $2^{n+1} + 2$ **(C)** $2^{n+2} + 4$ **(D)** $2^{n-1}(2^n + 1)$

3. It 5 letters are put in the 5 envelopes. Find the no. of ways so that atleast 2 letters are in wrong envelope:

(A) 120 **(B)** 119 **(C)** 118 **(D)** 117

4. Number of positive integral solutions satisfying the equation $(x_1 + x_2 + x_3)(y_1 + y_2) = 77$, is :

(A) 150 **(B)** 270 **(C)** 420 **(D)** 1024

5. There are counters available in 3 different colours (atleast four of each colour). Counters are all alike except for the colour. If 'm' denotes the number of arrangements of four counters if no arrangement consists of counters of same colour and 'n' denotes the corresponding figure when every arrangement consists of counters of each colour, then :

(A) $m = 2n$ **(B)** $6m = 13n$ **(C)** $3m = 5n$ **(D)** $5m = 3n$

6. An ice cream parlour has ice creams in eight different varieties. Number of ways of choosing 3 ice creams taking atleast two ice creams of the same variety, is :

(Assume that ice creams of the same variety to be identical & available in unlimited supply)

(A) 56 **(B)** 64 **(C)** 100 **(D)** None of these

7. Three digit numbers in which the middle one is a perfect square are formed using the digits 1 to 9. Their sum is :

(A) 134055 **(B)** 270540 **(C)** 170055 **(D)** None of these

8. A guardian with 6 wards wishes everyone of them to study either Law or Medicine or Engineering, Number of ways in which he can make up his mind with regard to the education of his wards if every one of them be fit for any of the branches to study, and atleast one child is to be sent in each discipline is:

(A) 120 **(B)** 216 **(C)** 729 **(D)** 540

9. There are $(p + q)$ different books on different topics in Mathematics. $(p \neq q)$

If L = The number of ways in which these books are distributed between two students X and Y such that X get p books and Y gets q books.

M = The number of ways in which these books are distributed between two student X and Y such that one of them gets p books and another gets q books.

N = The number of ways in which these books are divided into groups of p books and q books then :

(A) $L = M = N$ **(B)** $L = 2M = 2N$ **(C)** $2L = M = 2N$ **(D)** $L = M = 2N$

10. The number 916238457 is an example of nine digit number which contains each of the digit 1 to 9 exactly once. It also has the property that the digits 1 to 5 occur in their natural order, while the digits 1 to 6 do not. Number of such numbers are :

(A) 2268 **(B)** 2520 **(C)** 2975 **(D)** 1560

11. Number of functions defined from $f:\{1,2,3,4,5,6\}\rightarrow\{7,8,9,10\}$ such that the sum $f(1)+f(2)+f(3)+f(4)+f(5)+f(6)$ is odd, is :

(A) 2^{10} **(B)** 2^{11} **(C)** 2^{12} **(D)** $2^{12}-1$

12 The number of non-negative integral solutions of $x+y+z\le n$ where $n\in N$ is :

(A) ${}^{n+4}C_4$ **(B)** ${}^{n+5}C_5$ **(C)** ${}^{n+3}C_3$ **(D)** None of these

13. From 4 men and 6 ladies a committee of 5 is to be selected. The number of ways in which the committee can be formed so that men are in majority, is :

(A) 66 **(B)** 156 **(C)** 60 **(D)** None of these

14. Six persons *A*, *B*, *C*, *D*, *E* and *F* are to be seated at a circular table. The number of ways this can be done if *A* must have either *B* or *C* on his right and *B* must have either *C* or *D* on his right is :

(A) 36 **(B)** 12 **(C)** 24 **(D)** 18

15. The total number of integral solution for x, y , z such that $xyz = 24$ is :

(A) 30 **(B)** 60 **(C)** 90 **(D)** 120

16. How many five digit numbers can be formed from 1, 2, 3, 4, 5 (without repetition), when the digit at the unit place must be greater than that in the tenth place ?

(A) 54 **(B)** 60 **(C)** 5!/3 **(D)** $2\times 4!$

17. Number of ways in which 7 green bottles and 8 blue tube bottles can be arranged in a row if exactly 1 pair of green bottles is side by side, is (Assume all bottles to be alike except for the colour).

(A) 84 **(B)** 360 **(C)** 504 **(D)** None of these

18. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways it which the division may be made is

(A) 420 **(B)** 630 **(C)** 710 **(D)** None of these

19. The number of ways in which 8 distinguishable apples can be distributed among 3 boys such that every boy should get atleast 1 apple and atmost 4 apples is $K.\,{}^{7}P_{3}$ where K has the value equal to :

(A) 14 **(B)** 66 **(C)** 44 **(D)** 22

20. A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it, so that there will be no complete pair is :

(A) 1920 **(B)** 200 **(C)** 110 **(D)** 80

21. There are 10 seats in a double decker bus, 6 in the lower deck and 4 on the upper deck. Ten passenger board the bus, of them 3 refuse to go to the upper deck and 2 insist on going up. The number of way in which the passengers can be accommodated is : (Assume all seats to be duly numbered)

(A) 172800 **(B)** 162800 **(C)** 152800 **(D)** 182800

22. An old man while dialing a 7 digit telephone number remembers that the first four digits consists of one 1's, one 2's and two 3's. He also remembers that the fifth digit is either a 4 or 5 while has no memory of the sixth digit, he remembers that the seventh digit is 9 minus the sixth digit. Maximum number of distinct trials he has to try to make sure that he dials the correct telephone number, is

(A) 360 **(B)** 240 **(C)** 216 **(D)** None of these

23. The total number of different ways a grandfather along with two of his grandsons and four grand daughters can be seated in a line for a photograph so that he is always in the middle and the two grandsons are never adjacent to each other :

(A) 728 **(B)** 600 **(C)** 528 **(D)** 328

24. Number of ways in which 9 different toys be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is:

(A) $\frac{(5!)}{8}$ **(B)** $\frac{9!}{2}$ **(C)** $\frac{9!}{3!(2!)^3}$ **(D)** None of these

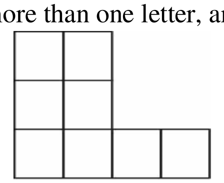
25. In an election three districts are to be canvassed by 2, 3 and 5 men respectively. If 10 men volunteer, the number of ways they can be allotted to the different districts is :

(A) $\frac{10!}{2!3!5!}$ **(B)** $\frac{10!}{2!5!}$ **(C)** $\frac{10!}{(2!)^2 5!}$ **(D)** $\frac{10!}{(2!)^2 3!\, 5!}$

26. The number of ordered pairs (m, n), $m,\ n \in \{1, 2, ..., 50\}$ such that $6^n + 9^m$ is a multiple of 5:

(A) 2500 **(B)** 1250 **(C)** 625 **(D)** 500

27. Number of ways in which the letters of the word "NATION" can be filled in the given figure such that no row remains empty and each box contains not more than one letter, are:



(A) $11\underline{|6}$ **(B)** $12\underline{|6}$ **(C)** $13\underline{|6}$ **(D)** $14\underline{|6}$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

28. The combinatorial $C(n, r)$ is equal to :

(A) number of possible subsets of r members from a set of n distinct members

(B) number of possible binary messages of length n with exactly r 1's

(C) number of non decreasing 2-D paths from the lattice point (0,0) to (r, n)

(D) number of ways of selecting r things out of n different things when a particular thing is always included plus the number of ways of selecting 'r' things out of n, when a particular thing is always excluded

29. Identify the correct statement(s).

(A) Number of naughts standing at the end of $\underline{|125}$ is 30

(B) A telegraph has 10 arms an each arm is capable of 9 distinct positions excluding the position of rest. The number of signals that can be transmitted is $10^{10} - 1$

(C) Number greater than 4 lacs which can be formed by using only the digits 0, 2, 2, 4, 4 and 5 is 90

(D) In a table tennis tournament, each player plays with every other player. If the number of games played is 5050 then the number of players in the tournament is 100

30. There are 10 questions, each question is either True or False. Number of different sequences of not all correct answers is also equal to :

(A) Number of ways in which a normal coin tossed 10 times would fall in a definite order if both Heads and Tails are present.

(B) Number of ways in which a multiple choice question containing 10 alternatives with one or more than one correct alternatives, can be attempted

(C) Number of ways in which it is possible to draw coins from 10 coins of different denominations taken some or all at a time.

(D) Number of different selections of 10 indistinguishable things taken some or all at a time.

31. The continued product, 2. 6. 10. 14….. to n factors is equal to :

(A) ${}^{2n}C_n$ **(B)** ${}^{2n}P_n$

(C) $(n + 1)(n + 2)(n + 3)\ldots.(n + n)$ **(D)** None of these

32. The number of ways in which five different books to be distributed among 3 persons so that each person gets at least one book, is equal to the number of ways in which:

(A) 5 persons are allotted 3 different residential flats so that each person is allotted at most one flat and no two persons are allotted the same flat

(B) number of parallelograms (some of which may be overlapping) formed by one set of 6 parallel lines and other set of 5 parallel lines that goes in other direction

(C) 5 different toys are to be distributed among 3 children, so that each child gets at least one toy

(D) 3 mathematics professors are assigned five different lectures to be delivered, so that each professor gets at least one lecture.

33. The maximum number of permutations of $2n$ letters in which there are only a's and b's, taken all at a time is given by :

(A) ${}^{2n}C_n$ **(B)** $\frac{2}{1}.\frac{6}{2}.\frac{10}{3}.......\frac{4n-6}{n-1}.\frac{4n-2}{n}$

(C) $\frac{n+1}{1}.\frac{n+2}{2}.\frac{n+3}{3}.\frac{n+4}{4}.......\frac{2n-1}{n-1}.\frac{2n}{n}$ **(D)** $\frac{2^n.[1.3.5.........(2n-3)(2n-1)]}{n!}$

34. Number of ways in which 3 different numbers in $A.P.$ can be selected from 1, 2, 3,…...n is :

(A) $\left(\frac{n-1}{2}\right)^2$ if n is even **(B)** $\frac{n(n-2)}{4}$ if n is odd

(C) $\frac{(n-1)^2}{4}$ if n is odd **(D)** $\frac{n(n-2)}{4}$ if n is even

35. The combinatorial coefficient ${}^{n-1}C_p$ denotes :

(A) The number of ways in which n things of which p are alike and rest different can be arranged in a circle

(B) The number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded

(C) Number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty and each box can hold any number of balls

(D) The number of ways in which $(n - 2)$ white balls and p black balls can be arranged in a line if no two black balls are together, balls are all alike except for the colour

36. Triplet (x, y, z) is chosen from the set $\{1, 2, 3, \ldots, n\}$, such that $x \le y < z$. The number of such triplets is :

(A) n^3 **(B)** ${}^{n+1}C_3$ **(C)** ${}^{n}C_2$ **(D)** ${}^{n}C_2 + {}^{n}C_3$

37. Which of the following statements are correct?

(A) Number of 6 letter words that can be formed using letters of the word "CENTRIFUGAL" if each word must contain all the vowels is 3.7 !

(B) There are 15 balls of which some are white and the rest black. If the number of ways in which the balls can be arranged in row, is maximum then the number of white balls must be equal to 7 or 8. Assume balls of the same colour to be alike

(C) There are 12 things, 4 alike of one kind, 5 alike and of another kind and the rest are all different. The total number of combinations is 240

(D) Number of selections that can be made of 6 letters from the word "COMMITIEE" is 35

38. In a plane, there are two families of lines $y = x + r$, $y = -x + r$, where $r \in \{0, 1, 2, 3, 4\}$. The number of squares of diagonals of length 2 formed by the lines is :

(A) $\left(\frac{2}{3}\right)(4!)$ **(B)** $\left(\frac{3}{2}\right)(3!)$ **(C)** 16 **(D)** 9

39. Number of ways in which the letters of the word 'B U L B U L' can be arranged in a line in any order is also equal to the

(A) number of ways in which 2 alike Apples and 4 alike Mangoes can be distributed in 3 children so that each child receives any number of fruits.

(B) Number of ways in which 6 different books can be tied up into 3 bundles, if each bundle is to have equal number of books.

(C) coefficient of $x^2y^2z^2$ in the expansion of $(x+y+z)^6$

(D) number of ways in which 6 different prizes can be distributed equally in three children.

40. Let $p = 2520$, $x =$ number of divisors of p which are multiples of 6, y = number of divisors of p which are multiples of 9, then :

(A) $x = 24$ **(B)** $x = 12$ **(C)** $y = 16$ **(D)** $y = 12$

41. A person wants to invite one or more of his friends for a dinner party. In how many ways can he do so if he has eight friends?

(A) 2^8 **(B)** $2^8 - 1$ **(C)** 8^2 **(D)** ${}^8C_1 + {}^8C_2 + {}^8C_3 + \ldots + {}^8C_8$

42. In an examination, a candidate is required to pass in all four subjects he is studying. The number of ways in which he can fail is :

(A) ${}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4$ **(B)** $4^4 - 1$

(C) $2^4 - 1$ **(D)** ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

43. The number of ways to select 2 ordered numbers from {0, 1, 2, 3, 4} such that the sum of the squares of the selected numbers is divisible by 5 are (repetition of digits is allowed)

(A) ${}^9C_1$ **(B)** ${}^9C_8$ **(C)** 9 **(D)** 7

44. If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is :

(A) $\dfrac{(n-2)(n-3)(n-4)}{6}$ **(B)** ${}^{n-2}C_3$

(C) ${}^{n-3}C_3 + {}^{n-3}C_2$ **(D)** None of these

45. m points on one straight line are joined to n points on another straight line. The number of points of intersection of the line segments thus formed (not lying on given two lines) is :

(A) ${}^{m}C_2.{}^{n}C_2$ **(B)** $\frac{mn(m-1)(n-1)}{4}$ **(C)** $\frac{{}^{m}C_2.{}^{n}C_2}{2}$ **(D)** ${}^{m}C_2+{}^{n}C_2$

46. Consider seven digit number $x_1, x_2, \ldots, x_7$, where $x_1, x_2, \ldots, x_7 \neq 0$ having the property that x_4 is the greatest digit and digits towards the left and right of x_4 are in decreasing order (from left to right). Then total number of such numbers in which all digits are distinct is :

(A) ${}^{9}C_7.{}^{6}C_3$ **(B)** ${}^{9}C_6.{}^{5}C_3$ **(C)** ${}^{10}C_7.{}^{6}C_3$ **(D)** ${}^{9}C_2.{}^{6}C_3$

47. A contest consists of ranking 10 songs of which 6 are Indian classic and 4 are western songs. Number of ways of ranking so that. (mention correct statements)

(A) There are exactly 3 Indian classic songs in top 5 is $(5!)^3$
(B) Top rank goes to Indian classic song is 6(9!)
(C) The rank of all western songs are consecutive is 4!7!
(D) The 6 Indian classic songs are in a specified order is ${}^{10}P_4$

48. $P = n(n^2-1)(n^2-4)(n^2-9)....(n^2-100)$ is always divisible by; ($n \in I$)

(A) 2! 3! 4! 5! 6! **(B)** $(5!)^4$ **(C)** $(10!)^2$ **(D)** 10! 11!

49. A fair coin is tossed n times. Let a_n denote the number of cases in which no two heads occur consecutively. Then which of the following is true?

(A) $a_1 = 2$ **(B)** $a_2 = 3$ **(C)** $a_5 = 14$ **(D)** $a_8 = 55$

50. All the five-digit numbers in which each successive digit exceeds its predecessor are arranged in the increasing order. The 105th number does not contain the digit:

(A) 1 **(B)** 3 **(C)** 4 **(D)** 5

51. The number of ways of selecting two 1×1 squares from a chess board such that they:

(A) Have a common vertex is 98
(B) Have a common side is 112
(C) Neither have a common vertex nor have a common side is 1806
(D) None of these

52. The kindergarten teacher has 25 kids in her class. She takes 5 of them at a time to zoological garden as often as she can without taking the same 5 kids more than once. Then the number of visits, the teacher makes to the garden exceeds that of a kid by:

(A) ${}^{25}C_5-{}^{24}C_4$ **(B)** ${}^{24}C_5$ **(C)** ${}^{25}C_5-{}^{24}C_5$ **(D)** ${}^{24}C_4$

53. Suppose a lot contains 2n objects of which n are identical. The number of ways to select n objects out of these $2n$ objects must be:

(A) 2^n
(B) $({}^{2n+1}C_0+{}^{2n+1}C_1+\ldots+{}^{2n+1}C_n)^{1/2}$
(C) The number of possible subsets of the set $\{a_1, a_2 \ldots, a_n\}$
(D) None of these

54. The number of selections of 4 letters taken from the word "COLLEGE" must be:
(A) 18
(B) 22
(C) Coefficient of x^4 in the expansion of $(1+x)^3\left(1+x+x^2\right)^2$
(D) Coefficient of x^4 in the expansion of $(1+x)^2\left(1+x+x^2\right)^3$

55. The integers from 1 to 1000 are written in order around a circle. Starting at 1, every fifteenth numbers is marked (i.e., 1, 16, 31 etc.). This process is continued until a number is reached which has already been marked, then unmarked numbers are:
(A) 200 (B) 400 (C) 600 (D) 800

56. The number of ways in which 10 students can be divided into three teams, one containing 4 and others 3 each, is:
(A) $\frac{10!}{4!\,3!\,3!}$ (B) 2100 (C) ${}^{10}C_4 \times {}^5C_2$ (D) $\frac{10!}{6!\,3!\,3!}.\frac{1}{2}$

57. The number of isosceles triangles with integer sides if no side exceeding 2008 is:
(A) $(1004)^2$ if equal sides do not exceed 1004
(B) $2(1004)^2$ if equal sides exceed 1004
(C) $3(1004)^2$ if equal sides have any length $\le$ 2008
(D) $(2008)^2$ if equal sides have any length $\le$ 2008

58. The number of ways of distributing 10 different books among 4 students (S_1, S_2, S_3 and S_4) such that S_1 and S_2 get 2 books each and S_3 and S_4 gets 3 books each is:
(A) 12600 (B) 25200 (C) ${}^{10}C_4$ (D) $\frac{10!}{2!\,2!\,3!\,3!}$

59. Given that the divisors of $n = 3^p.\,5^q.\,7^r$ are of the form $4\lambda+1,\ \lambda \ge 0$. Then
(A) $p+r$ is even (B) $p+q+r$ is even or odd
(C) q can be any integer (D) if p is odd, then r is odd

60. For the equation $x+y+z+w=19$, the number of positive integral solutions is equal to:
(A) The number of ways in which 15 identical things can be distributed among 4 persons
(B) The number of ways in which 19 identical things can be distributed among 4 persons
(C) Coefficient of x^{19} in $(x^0+x^1+\ldots+x^{19})^4$
(D) Coefficient of x^{19} in $(x+x^2+x^3+\ldots+x^{19})^4$

61. Let $\vec{a} = \hat{i}+\hat{j}+\hat{k}$ and let $\vec{r}$ be a variable vector such that $\vec{r}.\hat{i},\ \vec{r}.\hat{j}$ and $\vec{r}.\hat{k}$ are positive integers. If $\vec{r}.\vec{a} \le 12$ then the number of values of $\vec{r}$ is:
(A) ${}^{12}C_9 - 1$ (B) ${}^{12}C_3$ (C) ${}^{12}C_9$ (D) ${}^{12}C_3 - 1$

62. The number of 5 letter words formed with the letters of the word Calculus is divisible by:
(A) 2 (B) 3 (C) 5 (D) 7

63. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k (x-2)^{100-k} 3^k$ is also equal to:

(A) Number of ways in which 50 identical books can be distributed in 100 students, if each student can get almost one book

(B) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together

(C) Number of dissimilar terms in $(x_1 + x_2 + x_3 + ... + x_{50})^{51}$

(D) $\dfrac{2.6.10.14.....98}{51!}$

64. Let a, b, c, d be non-zero distinct digits. The number of 4-digit numbers $abcd$ such that ab + cd is even is divisible by:

(A) 3 **(B)** 4 **(C)** 7 **(D)** 11

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

65. In how many ways 7 digit numbers can be formed by using the digits 1, 2, 3, 4, 5 such that

	Column 1		**Column 2**
(A)	Repetition is allowed	**(p)**	78, 120
(B)	Exactly 3 digits will appear	**(q)**	76, 860
(C)	Atleast 2 digit will appear	**(r)**	78, 125
(D)	Atleast 3 digit will appear	**(s)**	18060

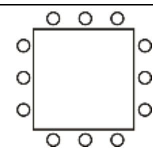
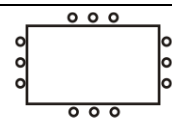
66. In how many ways 7 distinct hats can be arranged among 7 persons such that :

	Column 1		**Column 2**
(A)	No person will get its own hat	**(p)**	1331
(B)	Exactly 3 person will get their own hats	**(q)**	407
(C)	Atleast 2 person will get their own hat	**(r)**	1854
(D)	Atleast 3 person will get their own hat	**(s)**	315

67. A dice is thrown 7 times. Find the number of possible outcomes if :

	Column 1		**Column 2**
(A)	All digits will appear	**(p)**	279930
(B)	Exactly 3 digits will appear	**(q)**	15120
(C)	Atleast 2 digits will appear	**(r)**	264816
(D)	Atmost 5 digits will appear	**(s)**	36, 120

68. In how many ways 12 persons can be seated :

	Column 1		Column 2
(A)	Linearly	**(p)**	$3\times\lvert\underline{11}$
(B)	Circularly	**(q)**	$6\times\lvert\underline{11}$
(C)	Around a square table 	**(r)**	$\dfrac{\lvert\underline{12}}{12}$
(D)	Around a rectangular table 	**(s)**	$\lvert\underline{12}$

69. If p is the total number of ways of arranging 36 girls around a table and q is the exponent of 2 in p then the value of q, if :

	Column 1		Column 2
(A)	The table is circular is	**(p)**	Total number of zeros in $(10^{33}+10^{10})$
(B)	The table is square and 9 girls each side is	**(q)**	Total number of trailing zeros in 138!
(C)	The table is rectangular and 10 girls along length and 8 around along breadth is	**(r)**	The exponent of 5 in 132!
(D)	The table is hexagonal and 6 girls each side is	**(s)**	Exponent of 3 in 70!
		(t)	Total number of trailing zeros in $10^{100}+10^{33}$

SUBJECTIVE INTEGER TYPE

Each of the following question has an integer answer between 0 and 9.

70. A staircase has 10 steps. A person can go up the steps one at a time, two at a time, or any combination of 1's and 2's. If the number of ways in which the person can go up the stairs is p, then find $\dfrac{p}{89}$.

71. $A_1A_2,......A_{2n}$ is regular $2n$ sided polygon $(n\ge 3)$. Find the ratio of number of obtuse angled triangles to the number of acute angled triangles formed by joining the vertices of the polygon.

72. There are n persons sitting around a circular table. They start singing a 2 minute song in pairs such that no two persons sitting together will sing together. This process is continued for 28 minutes. Find n.

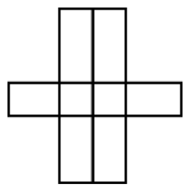
73. If the total number of strictly increasing functions defined from $f:\{1,2,3,4,5,6\}\to\{1,2,3,....9\}$ is k then $\dfrac{k}{12}$ is equal to ________.

74. If the total number of non-decreasing functions defined from $f:\{1,2,3,4,5\}\to\{1,2,3,4,5,6,7,8,9\}$ is m then $\frac{m}{143}$ is equal to _______.

75. If total number of ways of distributing 5 distinct objects and 4 identical objects among 3 persons is (blank group is allowed) $3^{p}\times q$ then $q-2p$ is equal to _______. (HCF of p and q is 5)

76. If the number of ways in which 8 people can be arranged in a line if A and B must be next to each other and C must be somewhere behind D is equal to 'm' then sum of all the digits of m is equal to _______.

77. Six X's have to be placed in the squares of the figure given below, such that each row contains at least one X. If the total no. of different ways this can be done is 'm' then $\frac{m}{13}$ is equal to _________.



78. A conference attended by 200 delegates is held in a hall. The hall has 7 doors, marked A, B, …….., G. At each door, an entry book is kept and the delegates entering through that door sign it in the order in which they enter. If each delegate is free to enter any time and through any door he likes, if the total no. of different sets of seven lists would arise in all is equal to ${}^{n}P_{r}$ then 'n-r' is equal to (Assume that every person signs only at his first entry).

79. If u_r denoted the number of one-one functions from $(x_1, x_2, \ldots\ldots\ldots, x_r)$ to $(y_1, y_2, \ldots\ldots\ldots, y_r)$ such that $f(x_i)\neq y_i$, for $i=1,2,3,\ldots\ldots\ldots,r$ then $u_4=$

80. Three friends went to a shopping mall with \$ 6, 7 and 8 with them in how many ways they can pay a bill of \$ 10 if they have note of only one denomination.

81. In how many rotationally distinct ways can the vertices of a cube be coloured with black or white colour?

82. Fifteen coupons are numbered 1, 2, 3, … 15. Seven coupons are selected such that the largest number appearing on the selected coupon is 9, if total number of ways is ${}^{n}C_{8}$, then n is______.

83. Ramesh has $2n$ number of fruits out of which n of them are identical and remaining n are distinct, If the total number of ways he can distribute these fruits to his two children Bhavesh and Sanjesh such that both of them will receive equal number of fruits is 16 then n is equal to______.

84. In an Ice cream parlor at South City Mall Kolkata, 4 different varieties of ice creams namely Vanila, Strawberry, Chocolate and Butter Scotch were available. On a particular day it was noticed that each customer bought at least one ice cream and at max 10 ice creams, on further investigation it was noticed that no two customer bought same set of ice creams then if the number of customers visited the ice cream shop on that particular day is k then $\frac{k}{100}$ is______.

85. Mr. Anshuman has thrown a dice 6 times in k ways we can get a sum greater than 17 then $\frac{k}{10000}$ is______.

86. If k is the number of positive integral solutions of the in equality $a+b+3c \le 30$? Then $\frac{k}{5}$ is______.

87. Each set has 'm' parallel lines. If the total number of parallelograms thus formed is 225 then m is equal to______.

88. If λ be the number of 3-digit numbers are of the form xyz with $x < y$, $z < y$ and $x \ne 0$, the value of λ is______.

89. In k ways can you place 2 rooks on a chessboard such that they are not in attacking positions, if rooks can attack only in a same row or in a same column? Then $\frac{k}{100}$ is______.

90. Consider a polygon of k sides. If the number of triangles that can be drawn taking vertices of these polygons as vertices of triangles and no sides of triangles is common with any sides of the polygon is 50 then k is______.

91. In a class of 10 students if two prizes (1^{st} and 2^{nd}) has to be given in three subjects Physics, Chemistry & Mathematics and this can be done in k ways, then $\frac{k}{1000}$ is______.

92. Consider 5 points in a plane are situated so that no two of the straight lines joining them are parallel, perpendicular, or coincident. From each point perpendiculars are drawn to all the lines joining the other four points. Determine the maximum number of intersections that these perpendiculars can have?

93. Consider a 6×6 square which is dissected into 9 rectangles by lines parallel to its sides such that all the rectangles have integral sides. What is the minimum number of congruent rectangles?

94. Consider a set $X = \{1, 2, 3, \ldots, 9, 10\}$. If the number of pairs $\{A, B\}$ such that $A \subseteq X$ and $B \subseteq X$ also $A \ne B$ and $A \cap B = \{2, 3, 5, 7\}$ is $3\lambda - 4$ then $\lambda - \mu$ is______.

95. At IITD, roll number of N students are given from 1 to N. Three students are selected from these N students such that their roll numbers are not consecutive, the total number of ways this selection can be done is 10, then N is equal to______.

96. Consider a set $S = \{1, 2, \ldots, 100\}$ two elements p and q are selected from this set S such that $7^p + 7^q$ is divisible by 5, How many ways this selection can be made?

97. In a particular batch of VIDYAMANDIR CLASSES Boston, there are 4 boys and certain number of girls. In every mock test only 5 students including at least 3 boys can appear. If different group of students write the Mock exam every time and number of times test conducted is 66 then find the total number of students in the class.

98. The total number of ways in which 10 Men and 10 Women can form 10 mixed complex (a mixed couple contain a Man and a Woman) is N, then $\frac{N}{9!}$ is equal to______.

99. Find the minimum value of k such that (k!!) is completely divisible by all two-digit prime numbers.

Advanced Problem Package	Binomial Theorem

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Let k and n be the positive integers and $S_k = 1^k + 2^k + 3^k + ... + n^k$. Then ${}^{m+1}C_1 S_1 + {}^{m+1}C_2 S_2 + {}^{m+1}C_3 S_3 + ... {}^{m+1}C_m S_m$ is equal to :

(A) $(n+1)^{m+1}$ **(B)** $(n+1)^m - n$ **(C)** $(n+1)^{m+1} - 1$ **(D)** $(n+1)^{m+1} - n - 1$

2. The sum of the series ${}^nC_1^2 + \frac{1+2}{2}\,{}^nC_2^2 + \frac{1+2+3}{3}\,{}^nC_3^2 ..... + \frac{1+2+3+....n}{n}\,{}^nC_n^2$ is equal to :

(A) $\frac{1}{2}(n\,{}^{2n-1}C_n + {}^{2n}C_n)$ **(B)** $\frac{1}{2}(n\,{}^{2n-1}C_n + {}^{2n}C_n - 1)$

(C) $\frac{1}{2}((n+1)\,{}^{2n-1}C_n - 1)$ **(D)** $\frac{1}{2}(n\,{}^{2n-1}C_n + {}^{2n}C_n - 2)$

3. $\frac{{}^nC_0}{2} - \frac{{}^nC_1}{6} + \frac{{}^nC_2}{10} - \frac{{}^nC_3}{14} + ... + \frac{(-1)^n\,{}^nC_n}{(4n+2)}$, $n \in N$ is equal to :

(A) $\frac{2^{n-1} n!}{3\cdot5\cdot7.....(2n-1)(2n+1)}$ **(B)** $\frac{2^n n!}{1\cdot3\cdot5\cdot7.....(2n-3)(2n-1)}$

(C) $\frac{2^n n!}{3\cdot5.....(2n-1)(2n+1)}$ **(D)** $\frac{2^n (n-1)!}{1\cdot3\cdot5.....(2n-3)(2n-1)}$

4. ${}^nC_1 - \left(1+\frac{1}{2}\right){}^nC_2 + \left(1+\frac{1}{2}+\frac{1}{3}\right){}^nC_3 - \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right){}^nC_4 + ... + (-1)^{n-1}\left(1+\frac{1}{2}+\frac{1}{3}+....+\frac{1}{n}\right){}^nC_n =$

(A) $\frac{n-1}{n}$ **(B)** $\frac{1}{n}$ **(C)** $\frac{1}{n+1}$ **(D)** $\frac{2^n}{n+1}$

5. $\sum_{r=1}^{n} \frac{(-1)^{r-1}\,{}^nC_r (1-x)^r}{r} =$

(A) $x + \frac{x^2}{2} + \frac{x^3}{3} + ... + \frac{x^n}{n}$ **(B)** $\frac{1-x}{1} + \frac{1-x^2}{2} + \frac{1-x^3}{3} + ... + \frac{1-x^n}{n}$

(C) $(x-1) + \frac{x^2-1}{2} + \frac{x^3-1}{3} + ... + \frac{x^n-1}{n}$ **(D)** $n - \frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} ... \frac{x^n}{n}$

6. ${}^nC_0\, x^{2n} + \frac{{}^nC_1}{2} x^{2n-2}(2-x^2) + \frac{{}^nC_2}{3} x^{2n-4}(2-x^2)^2 + ... + \frac{{}^nC_n (2-x^2)^n}{(n+1)} =$

(A) $\frac{2^n - x^{2n+2}}{(n+1)(2-x^2)}$ **(B)** $\frac{2^n - x^{2n}}{(n+1)(2-x^2)}$ **(C)** $\frac{2^{n+1} - x^{2n+2}}{(n+1)(2-x^2)}$ **(D)** $\frac{2^{n+1} - x^{2n}}{(n+1)(2-x^2)}$

7. If $p + q = 1$, then $\sum_{r=0}^{n} r^3\,{}^nC_r\, p^r q^{n-r} =$

(A) $np(n^2 p + 3(n-1)p + 1)$ **(B)** $np((n^2 - n)p^2 + 2(n-1)p + 1)$

(C) $np((n^2 - 3n + 2)p^2 + 2(n-1)p + 1)$ **(D)** $np((n^2 - 3n + 2)p^2 + 3(n-1)p + 1)$

For Questions 8 - 10

If $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n} \frac{r^2}{{}^nC_r} = P(n)a_{n+2} + Q(n)a_{n+1} + a_n + R(n)$, where $P(n), Q(n), R(n)$ are the polynomial functions of n, then :

8. $P(5) =$

(A) 56 **(B)** 42 **(C)** 36 **(D)** 30

9. $Q(5) =$

(A) -5 **(B)** -10 **(C)** -15 **(D)** -18

10. $R(5) =$

(A) -24 **(B)** -20 **(C)** -30 **(D)** -42

For Questions 11 - 13

Let $f_1(x) = (x-2)^2$, $f_2(x) = ((x-2)^2-2)^2$, $f_3(x) = \left((x-2)^2-2)^2-2\right)^2$,..... and so on ; so that

$$f_k(x) = \underbrace{\left(.....\left((x-2)^2-2\right)^2 .....-2\right)^2}_{k \text{ times}} = A_k + B_k x + C_k x^2 + D_k x^3 + ...$$

11. B_5 is equal to :

(A) -2048 **(B)** -32 **(C)** -1024 **(D)** -512

12. C_3 is equal to :

(A) 256 **(B)** 352 **(C)** 320 **(D)** 336

13. C_k is equal to :

(A) $\frac{4^{2k-1}-4^{k-1}}{3}$ **(B)** 4^{2k-2} **(C)** $\frac{4^{2k-1}+4^{k-1}}{5}$ **(D)** 4^{k+1}

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

14. $\sum_{r=1}^{n} r(n-r)\,{}^nC_r^2$ is equal to :

(A) $n^2\,{}^{2n-2}C_{n-2}$ **(B)** $(n-1)^2\,{}^{2n-2}C_{n-1}$

(C) $n(n-1)\,{}^{2n-2}C_{n-2}$ **(D)** $n(n-1)\,{}^{2n-2}C_{n-1}$

15. The value of the sum ${}^nC_1^2 - 2\cdot{}^nC_2^2 + 3\cdot{}^nC_3^2 - 4\cdot{}^nC_4^2 + ... + (-1)^n n\cdot{}^nC_n^2$ where $n \in N$, $n > 3$ will be equal to :

(A) $-n\,{}^{n-1}C_{\frac{n-2}{2}}$ if $n = 4k, k \in I$ **(B)** $n\,{}^{n-1}C_{\frac{n-1}{2}}$ if $n = 4k+1, k \in I$

(C) $n\,{}^{n-1}C_{\frac{n-2}{2}}$ if $n = 4k+2, k \in I$ **(D)** $-n\,{}^{n-1}C_{\frac{n-1}{2}}$ if $n = 4k+3, k \in I$

16. If $n \in N$ and $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then $\sum_{r=0}^{n} (-1)^r a_r \, {}^nC_r$ is equal to :

(A) 0 if $n = 57$ (B) 0 if $n = 77$ (C) ${}^{24}C_8$ if $n = 24$ (D) ${}^{39}C_{13}$ if $n = 39$

17. In the expansion of $(x+a)^n, n \in N,$ if the sum of odd numbered terms be α and the sum of even numbered terms be β, then :

(A) $4\alpha\beta = (x+a)^{2n} - (x-a)^{2n}$ (B) $2(\alpha^2+\beta^2) = (x+a)^{2n} + (x-a)^{2n}$

(C) $\alpha^2 - \beta^2 = (x^2-a^2)^n$ (D) $\alpha^2+\beta^2 = (x+a)^{2n} + (x-a)^{2n}$

18. If the middle term of the expression $(1+x)^{24}, x > 0,$ is the only greatest term of the expansion, then :

(A) $x < 1$ (B) $x < \frac{13}{12}$ (C) $x > \frac{12}{13}$ (D) $x > 1$

19. ${}^nC_m + 3\,{}^{n-1}C_m + 5\,{}^{n-2}C_m + 7\,{}^{n-3}C_m + ... + (2(n-m)+1)\,{}^mC_m$ is equal to :

(A) ${}^{n+2}C_{m+3} + {}^{n+3}C_{m+3}$ (B) ${}^{n+2}C_{m+2} + 2\,{}^{n+2}C_{m+3}$

(C) ${}^{n+1}C_{m+2} + {}^{n+2}C_{m+2}$ (D) ${}^{n+1}C_{m+1} + 2\,{}^{n+1}C_{m+2}$

20. Let n be a positive integer and $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n-1}x^{2n-1} + a_{2n}x^{2n}$, then :

(A) $\sum_{r=0}^{n-1} a_r = \frac{1}{2}(3^n - a_n)$ (B) $\sum_{r=0}^{2n} (-1)^r a_r^2 = a_n$

(C) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + (-1)^{n-1} a_{n-1}^2 = \frac{a_n}{2}(1-(-1)^n a_n)$

(D) $(r+1)a_{r+1} = (n-r)a_r + (2n-r+1)a_{r-1}, 1 \le r \le 2n-1$

21. Let $n \in N, n \ge 4$ and $P = \prod_{r=0}^{n} {}^nC_r$, then :

(A) $P > \left(\frac{2^n}{n+1}\right)^{n+1}$ (B) $P < \left(\frac{2^n}{n+1}\right)^{n+1}$ (C) $P < \left(\frac{2^n-2}{n-1}\right)^{n-1}$ (D) $P < \left(\frac{2^n-2}{n-1}\right)^{n}$

22. Let $S_1 = \sum_{r=0}^{n} ({}^{2n+1}C_{2r})^2$ and $S_2 = \sum_{r=0}^{n} ({}^{2n+1}C_{2r+1})^2$, then :

(A) $S_1 = \frac{1}{2}({}^{4n+2}C_{2n} + (-1)^n\,{}^{2n+1}C_n)$ (B) $S_2 = \frac{1}{2}({}^{4n+2}C_{2n} - (-1)^n\,{}^{2n+1}C_n)$

(C) $S_1 = \frac{1}{2}({}^{4n+2}C_{2n+1})$ (D) $S_2 = \frac{1}{2}({}^{4n+1}C_{2n+1})$

23. Let the coefficient of x^{20} in the expressions $(1+x^2-x^3)^{1000}$, $(1-x^2+x^3)^{1000}$, $(1-x^2-x^3)^{1000}$ and $(1+x^2+x^3)^{1000}$ be respectively a, b, c and d, then :

(A) $a = d$ (B) $a > b$ (C) $a > c$ (D) $b < c$

24. Let $\sum_{r=0}^{200} \alpha_r(1+x)^r = \sum_{r=0}^{200} \beta_r x^r$, where $\alpha_r = 1 \ \forall\ r \geq 98$, then the greatest coefficient in the expansion of $(1+x)^{201}$ is :

(A) ${}^{201}C_{100}$ **(B)** β_{98} **(C)** β_{99} **(D)** β_{100}

25. Let $x = (5\sqrt{3}+8)^{2n+1}$, $n \in N$, then :

(A) $[x]$ is even **(B)** $[x]$ is odd **(C)** $x\{x\} = (11)^{2n+1}$ **(D)** $x\{x\} = (13)^{2n+1}$

where $[\cdot]$ denotes greatest integer and $\{\cdot\}$ denote fraction part function

26. If $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, then the value of $a_1 - a_3 + a_5 - a_7 + ...$ is equal to :

(A) 1 if $n = 4k$ **(B)** -1 if $n = 4k+1$ **(C)** 0 if $n = 4k+2$ **(D)** -1 if $n = 4k+3$

27. If the unit digit of $13^n + 7^n - 3^n$, $n \in N$, is 3 then possible value(s) of n is/are :

(A) 27 **(B)** 103 **(C)** 11 **(D)** 101

28. If the coefficient of x^t and x^{t+1} in $\sum_{r=0}^{n}(1+x)^r$ where $t < n-2$ are equal, then :

(A) n is odd **(B)** n is even

(C) The sum of coefficients of x^t and $x^{t+1} = {}^{n+1}C_{t+2}$

(D) The sum of coefficients of x^t and $x^{t+1} = {}^{n+2}C_{t+2}$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column I are labelled as (A), (B), (C) & (D) whereas statements in Column II are labeled as P, Q, R, S & T. More than one choice from Column II can be matched with Column I.

29. Match the column :

	Column-I		Column-II
(A)	If the fourth term in the expansion of $\left(\frac{x}{a}+\frac{1}{x}\right)^n$ is $\frac{5}{2}$, then a^2 is divisible by	**(P)**	2
(B)	$\sum_{p=1}^{4}\sum_{r=p}^{4} {}^4C_r\, {}^rC_p$ is divisible by	**(Q)**	4
(C)	The coefficient of x^{13} in $(1-x)^5(1+x+x^2+x^3)$ is divisible by	**(R)**	5
(D)	$\sum_{r=0}^{4} {}^4C_r(r-2)^2$ is divisible by	**(S)**	8
		(T)	13

30. Match the column:

	Column-I		Column-II
(A)	The coefficient of x^4 in $(2-x+3x^2)^6$ is	**(P)**	1024
(B)	${}^{11}C_0\,{}^{22}C_{11}-{}^{11}C_1\,{}^{20}C_{11}+{}^{11}C_2\,{}^{18}C_{11}-{}^{11}C_3\,{}^{16}C_{11}+.....=$	**(Q)**	2048
(C)	${}^5C_1\,{}^5C_5-{}^5C_2\,{}^{10}C_5+{}^5C_3\,{}^{15}C_5-{}^5C_4\,{}^{20}C_5+{}^5C_5\,{}^{25}C_5=$	**(R)**	990
(D)	The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^{11}$ is	**(S)**	3125
		(T)	3660

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

31. The value of ${}^{50}C_6-{}^5C_1\,{}^{40}C_6+{}^5C_2\,{}^{30}C_6-{}^5C_3\,{}^{20}C_6+{}^5C_4\,{}^{10}C_6$ is equal to_____.

32. The remainder obtained when $6^{2007}+8^{2007}$ is divided by 49 is equal to_____.

33. ${}^{10}C_0\,{}^{20}C_{10}-{}^{10}C_1\,{}^{18}C_{10}+{}^{10}C_2\,{}^{16}C_{10}-{}^{10}C_3\,{}^{14}C_{10}+{}^{10}C_4\,{}^{12}C_{10}-{}^{10}C_5\,{}^{10}C_{10}$ is equal to_____.

34. The coefficient of x^5y^5 in the expression of $((1+x+y+xy)(x+y))^5$ is equal to_____.

35. The coefficient of $x^{\frac{n^2+n-14}{2}}$ in $(x-1)(x^2-2)(x^3-3)(x^4-4)....(x^n-n),\ n\ge 30$ is equal to_____.

36. ${}^nC_0\,{}^{2n}C_n-{}^nC_1\,{}^{2n-1}C_n+{}^nC_2\,{}^{2n-2}C_n-{}^nC_3\,{}^{2n-3}C_n+...+...+(-1)^n\,{}^nC_n\,{}^nC_n=$

37. The value of $\dfrac{(18^3+7^3+3\cdot 18\cdot 7\cdot 25)}{(3^6+6\cdot 243\cdot 2+15\cdot 81\cdot 4+20\cdot 27\cdot 8+15\cdot 9\cdot 16+6\cdot 3\cdot 32+64)}$ is equal to_____.

38. For what x is the 4th term in the expansion of $\left[\left(5^{1/3}\right)^{-1/2\log_{10}(6-\sqrt{8x})}+\left(\dfrac{5^{\log_{10}(x-1)}}{25^{\log_{10}5}}\right)^{1/6}\right]^m$ is equal to $\dfrac{84}{5}$, if it is known that $\dfrac{14}{9}$ of binomial coefficient of 3rd term, binomial coefficient of 4th term and binomial coefficient of 5th term in the expansion constitute a G.P.

39. Let $x=(5+2\sqrt{6})^n,\ n\in N,$ then find the value of $x-x^2+x[x],$ where $[\cdot]$ denotes greatest integer function.

40. If the coefficient of x^2 + coefficient of x in the expansion of $(1+x)^m(1-x)^n,\ (m\ne n)$ is equal to $-m,$ then the value of $n-m$ is equal to_____.

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1 If $L \equiv \left(\frac{1}{x_l}, l\right); M = \left(\frac{1}{x_m}, m\right); N = \left(\frac{1}{x_n}, n\right)$ where $x_k \neq 0$, denotes the k^{th} terms of a *H.P.* for $k \in N$, then :

(A) $ar(\Delta LMN) = \frac{l^2m^2n}{2}\sqrt{(l-m)^2+(m-n)^2+(n-l)^2}$ (B) ΔLMN is a right angled triangle

(C) The points *L*, *M*, *N* are collinear (D) ΔLMN is equilateral

2. Two points P_1 and P_2 are at distances r_1 and r_2 respectively from the origin *O* and OP_1 and OP_2 makes angle θ_1 and θ_2 respectively with the *x*-axis. Let there be a point *P* on P_1P_2 such that *OP* makes an angle $\frac{\theta_2+\theta_1}{2}$ with the *x*-axis. Then *OP* is :

(A) $\frac{2r_1r_2}{r_1+r_2}\cos\frac{\theta_2-\theta_1}{2}$ (B) $\frac{2r_1r_2}{r_1+r_2}\sin\frac{\theta_2-\theta_1}{2}$

(C) $\frac{r_1r_2}{r_1+r_2}\cos\frac{\theta_2+\theta_1}{2}$ (D) $\frac{r_1r_2}{r_1+r_2}\sin\frac{\theta_2+\theta_1}{2}$

3. If *A* (3, 0) and *B*(6, 0) are two fixed points and $U(\alpha, \beta)$ is a variable point in the plane. *AU* and *BU* meet the *y*-axis at *C* and *D* respectively and *AD* meet *OU* at *V*. Then the coordinate of the point through which *CV* always passes is :

(A) (3, 4) (B) (4, 0) (C) (0, 2) (D) (2, 0)

4. If $\frac{a}{\sqrt{bc}} - 2 = \sqrt{\frac{b}{c}} + \sqrt{\frac{c}{b}}$ where *a*, *b*, *c* > 0, then family of lines $\sqrt{a}x + \sqrt{b}y + \sqrt{c} = 0$ passes through the point:

(A) (1, 1) (B) (1, –2) (C) (–1, 2) (D) (–1, 1)

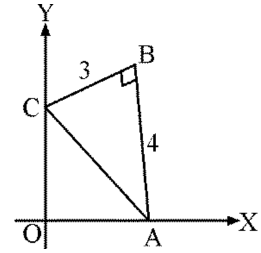
5. *I*(1,0) is the centre of incircle of triangle *ABC*, the equation of *BI* is $x - 1 = 0$ and equation of *CI* is $x - y - 1 = 0$, then angle *BAC* is:

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{2\pi}{3}$

6. If the points where the lines $3x - 2y - 12 = 0$ and $x + ky + 3 = 0$ intersect both the coordinate axes are concyclic, then the number of possible real values of *k* is:

(A) 1 (B) 2 (C) 3 (D) 4

7. In the adjacent figure ΔABC is right angled at *B*. If *AB* = 4 and *BC* = 3 and side *AC* slides along the coordinate axes in such a way that '*B*' always remains in the first quadrant, then *B* always lie on straight line :



(A) $y = x$ (B) $3y = 4x$

(C) $3x = 4y$ (D) $3y + 4x = 0$

8. Consider the triangle OAB where $O \equiv (0,0), B(3,4)$. If orthocenter of triangle is $H(1, 4)$, then coordinates of 'A' is :

(A) $\left(0, \frac{15}{4}\right)$ **(B)** $\left(0, \frac{17}{4}\right)$ **(C)** $\left(0, \frac{21}{4}\right)$ **(D)** $\left(0, \frac{19}{4}\right)$

9. Two vertices of a triangle are (5, –1) and (–2, 3). If orthocenter of the triangle is origin, then the co-ordinates of third vertex is :

(A) (4, 7) **(B)** (3, 7) **(C)** (–4, –7) **(D)** (4, –7)

10. It is desired to construct a right angled triangle $ABC(\angle C = \pi/2)$ in xy-plane so that its sides are parallel to co-ordinates axes and the medians through A and B lie on the lines $y = 3x + 1$ and $y = mx + 2$ respectively. The values of 'm' for which such a triangle is possible is/are :

(A) –12 **(B)** 12 **(C)** 4/3 **(D)** 1/12

11. m, n are integer with $0 < n < m$. A is the point (m, n) on the Cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon $ABCDE$ is :

(A) $2m(m+n)$ **(B)** $m(m+3n)$ **(C)** $m(2m+3n)$ **(D)** $2m(m+3n)$

12. The ends of the base of an isosceles triangle are at (2, 0) and (0, 1) and the equation of one side is $x = 2$ then the orthocenter of the triangle is :

(A) $\left(\frac{3}{4}, \frac{3}{2}\right)$ **(B)** $\left(\frac{5}{4}, 1\right)$ **(C)** $\left(\frac{3}{4}, 1\right)$ **(D)** $\left(\frac{4}{3}, \frac{7}{12}\right)$

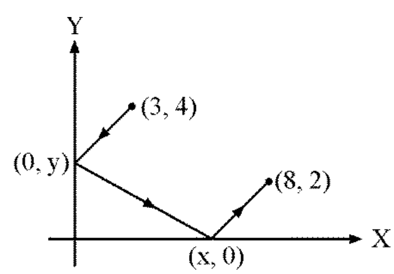
13. Given $A \equiv (1,1)$ and AB is any line through it cutting the x-axis in B. If AC is perpendicular to AB and meets the y-axis in C, then the equation of locus of mid-point P of BC is :

(A) $x + y = 1$ **(B)** $x + y = 2$ **(C)** $x + y = 2xy$ **(D)** $2x + 2y = 1$

14. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, –2) and is running up the line $y = -5x + 18$. At the point (a, b), the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is :

(A) 6 **(B)** 10 **(C)** 18 **(D)** 14

15. Suppose that a ray of light leaves the point (3, 4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8, 2). The value of x is :



(A) $x = 4\frac{1}{2}$ **(B)** $x = 4\frac{1}{3}$

(C) $x = 4\frac{2}{3}$ **(D)** $x = 5\frac{1}{3}$

16. In a triangle ABC, if A (2, –1) and $7x - 10y + 1 = 0$ and $3x - 2y + 5 = 0$ are equations of an altitude and an angle bisector respectively drawn from B, then equation of BC is :

(A) $x + y + 1 = 0$ **(B)** $5x + y + 17 = 0$ **(C)** $4x + 9y + 30 = 0$ **(D)** $x - 5y - 7 = 0$

17. The distance between the two parallel lines is 1 unit. A point 'A' is chosen to lie between the lines at a distance 'd' from one of them. Triangle ABC is equilateral with B on one line and C on the other parallel line. The length of the side of the equilateral triangle is :

(A) $\frac{2}{3}\sqrt{d^2+d+1}$ **(B)** $2\sqrt{\frac{d^2-d+1}{3}}$ **(C)** $2\sqrt{d^2-d+1}$ **(D)** $\sqrt{d^2-d+1}$

Paragraph for Questions 18 - 20

Consider 3 non-collinear points $A(9, 3)$, $B(7, -1)$ and $C(1, -1)$. Let $P(a, b)$ be the centre and R is the radius of circle 'S' passing through points A, B, C. Also $H(\bar{x}, \bar{y})$ are the coordinates of the orthocenter of triangle ABC whose area be denoted by Δ.

18. If D, E and F are the middle points of BC, CA and AB respectively then the area of the triangle DEF is :

(A) 12 **(B)** 6 **(C)** 4 **(D)** 3

19. The value of $a + b + R$ equals :

(A) 3 **(B)** 12 **(C)** 13 **(D)** None of these

20. The ordered pair $(\bar{x}, \bar{y})$ is :

(A) (9, 6) **(B)** (–9, 6) **(C)** (9, –5) **(D)** (9, 5)

Paragraph for Questions 21 - 23

The equation of an altitude of an equilateral triangle is $\sqrt{3}x + y = 2\sqrt{3}$ and one of its vertices is $(3, \sqrt{3})$ then

21. The possible number of triangle is :

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

22. Which of the following can't be the vertex of the triangle :

(A) (0, 0) **(B)** $(0, 2\sqrt{3})$ **(C)** $(3, -\sqrt{3})$ **(D)** None of these

23. Which of the following can be possible orthocenter of the triangle :

(A) $(1, \sqrt{3})$ **(B)** $(0, \sqrt{3})$ **(C)** (0, 2) **(D)** None of these

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

24. $ABCD$ is rectangle with $A(-1, 2)$, $B(3, 7)$ and $AB : BC = 4:3$, if d is the distance of origin from the intersection point of diagonals of rectangle, then possible values of [d] is/are (where [.] denote greatest integer function) :

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6

25. Two straight lines $u = 0$ and $v = 0$ passes through the origin and the angle between them is $\tan^{-1}\left(\frac{7}{9}\right)$. If the ratio of slopes of $v = 0$ and $u = 0$ is $\frac{9}{2}$, then their equations are :

(A) $y = 3x$ and $3y = 2x$ **(B)** $2y = 3x$ and $3y = x$

(C) $y + 3x = 0$ and $3y + 2x = 0$ **(D)** $2y + 3x = 0$ and $3y + x = 0$

26. Let $B(1,-3)$ and $D(0,4)$ represent two vertices of rhombus $ABCD$ in (x, y) plane, then coordinates of vertex A is $\angle BAD = 60°$ can be equal to :

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ **(B)** $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

(C) $\left(\frac{-1+7\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right)$ **(D)** $\left(\frac{-1-7\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right)$

27. The equations of two equal sides AB and AC of an isosceles triangle ABC are $x + y = 5$ and $7x - y = 3$ respectively. If the area of the triangle ABC is 5 square unit, then the possible equations of the side BC is(are) :

(A) $x - 3y + 1 = 0$ **(B)** $3x + y + 2 = 0$ **(C)** $x - 3y - 21 = 0$ **(D)** $3x + y - 12 = 0$

28. A line 'L' is drawn from (4, 3) to meet the lines $L_1 \equiv 3x + 4y + 5 = 0$ and $L_2 \equiv 3x + 4y + 15 = 0$ at points A and B respectively. From point 'A', a line perpendicular to L is drawn meeting the line 'L_2' at A_1. Similarly, from point 'B' a line perpendicular to L, is drawn meeting the line L_1 at B_1. Thus a parallelogram AA_1BB_1 is formed. If the area of the parallelogram AA_1BB_1 is least, the equation of the line L is/are :

(A) $x - 7y + 17 = 0$ **(B)** $x + 7y + 1\ 7 = 0$ **(C)** $3x + y - 31 = 0$ **(D)** $7x + 2y - 31 = 0$

29. If the angle bisector AD of the angle A of the triangle ABC divides the side BC into two segments $BD = 4$, $DC = 2$, then:

(A) $2 < b < 6$ **(B)** $4 < c < 12$ **(C)** $3 < b < 6$

(D) Maximum value of altitude through A is 4

30. If the bisectors of the interior angle A of $\Delta\, ABC$ divides BC into segments $BD = 4$, $DC = 2$. If the length of the altitude $AE > \sqrt{10}$ and if AB and AC are integer. Then the possible length of the side AC is(are) :

(A) 3 **(B)** 4 **(C)** 6 **(D)** 5

31. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angle between them is θ. Which of following relations hold good :

(A) $m_1 + m_2 = \frac{5}{4}$ **(B)** $m_1 m_2 = \frac{3}{8}$ **(C)** $\theta = \sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$

(D) Sum of the abscissa and ordinate of point P is -1.

32. Line $\frac{x}{a} + \frac{y}{b} = 1$ cuts the coordinate axes at $A(a, 0)$ and $B(0, b)$ and the line $\frac{x}{a'} + \frac{y}{b'} = -1$ at $A'(-a', 0)$ and $B'(0, -b')$. If the points A, B, A', B' are concyclic then the orthocenter of the triangle ABA' is :

(A) $(0,0)$ **(B)** $(0,b')$ **(C)** $\left(0,\frac{aa'}{b}\right)$ **(D)** $\left(0,\frac{bb'}{a}\right)$

33. The bisector of angle between the straight lines $y - b = \frac{2m}{1-m^2}(x-a)$ and $y - b = \frac{2m'}{1-m'^2}(x-a)$ are :

(A) $(y-b)(m+m') + (x-a)(1-mm') = 0$ **(B)** $(y-b)(m+m') - (x-a)(1-mm') = 0$

(C) $(y-b)(m-m') + (x-a)(1+mm') = 0$ **(D)** $(y-b)(1-mm') - (x-a)(m+m') = 0$

34. All the points lying inside the triangle formed by the points (1, 3), (5, 6), and (–1, 2) satisfy :

(A) $3x + 2y \geq 0$ **(B)** $2x + y + 1 \geq 0$ **(C)** $-2x + 11 \geq 0$ **(D)** $2x + 3y - 12 \geq 0$

35. Possible values of θ for which the point $(\cos\theta, \sin\theta)$ lies inside the triangle formed by lines $x + y = 2; x - y = 1$ and $6x + 2y = \sqrt{10}$ are :

(A) $\pi/8$ **(B)** $\pi/4$ **(C)** $3\pi/8$ **(D)** $\pi/2$

36. In a $\Delta ABC, A \equiv (\alpha, \beta), B(1,2), C(2,3)$ and point A lies on line $y = 2x + 3$, where $\alpha, \beta \in I$. If the area of ΔABC be such that area of triangle lies in interval [2, 3), then the possible value of $\alpha + \beta$ is/are :

(A) –18 **(B)** –15 **(C)** 9 **(D)** 12

37. The point $(a^2, a+1)$ lies in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin then the possible integral values of a is/are :

(A) $a = 2$ **(B)** $a = -2$ **(C)** $a = 0$ **(D)** $a = -1$

38. The medians AD and BE of a triangle ABC with vertices $A(0, b)$, $B(0, 0)$ and $c(a, 0)$ are perpendicular to each other if :

(A) $a = \sqrt{2}\, b$ **(B)** $a = -\sqrt{2}\, b$ **(C)** $b = \sqrt{3}\, a$ **(D)** $b = -\sqrt{3}\, a$

39. Two sides of a triangle have the joint equation $(x - 3y + 2)(x + y - 2) = 0$, the third side which is variable always passes through the point (–5, –1), then possible values of slope of third side such that origin is an interior point of triangle is/are :

(A) $\frac{-4}{3}$ **(B)** $\frac{-2}{3}$ **(C)** $\frac{-1}{3}$ **(D)** $\frac{1}{6}$

40. Let x_1 and y_1 be the roots of $x^2 + 8x - 2009 = 0$; x_2 and y_2 be the roots of $3x^2 + 24x - 2010 = 0$ and x_3 and y_3 be the roots of $9x^2 + 72x - 2011 = 0$. The points $A(x_1, y_1)\, B(x_2, y_2)$ and $C(x_3, y_3)$:

(A) can not lie on a circle
(B) form a triangle of area 2 sq. units
(C) form a right-angled triangle
(D) are collinear

41. Consider a variable line 'L' which passes through the point of intersection 'P' of the lines $3x + 4y - 12 = 0$ and $x + 2y - 5 = 0$ meeting the coordinate axes at points A and B :

(A) then the locus of middle point of the segment AB has the equation $3x + 4y = 4xy$

(B) then the locus of the feet of the perpendicular from the origin on the variable line 'L' has the equation $2(x^2 + y^2) - 4x - 3y = 0$

(C) Locus of the centroid of the variable triangle OAB has the equation (where 'O' is the origin) $3x + 4y - 6xy = 0$

(D) Locus of the centroid of the variable triangle OAB has the equation (where 'O' is the origin) $3x + 4y + 6xy = 0$

42. A variable line 'L' is drawn through $O(0, 0)$ to meet the lines $L_1 : y - x - 10 = 0$ and $L_2 : y - x - 20 = 0$ at points A and B respectively. A point P is taken on line 'L' :

(A) If $\frac{2}{OP} = \frac{1}{OA} + \frac{1}{OB}$, then locus of P is $3y - 3x = 40$

(B) If $OP^2 = (OA)(OB)$, then locus of P is $(y - x)^2 = 200$

(C) If $\frac{1}{OP^2} = \frac{1}{(OA)^2} - \frac{1}{(OB)^2}$, then locus of P is $(y - x)^2 = 80$

(D) If $\frac{1}{OP^2} = \frac{1}{(OA)^2} + \frac{1}{(OB)^2}$ then locus of P is $(y - x)^2 = 80$

43. Given ΔABC whose vertices are $A(x_1, y_1); B(x_2, y_2); C(x_3, y_3)$. Let there exists a point $P(a, b)$ such that $6a = 2x_1 + x_2 + 3x_3; 6b = 2y_1 + y_2 + 3y_3$:

(A) $P(a,b)$ lies inside the ΔABC

(B) Area of triangle PBC is less than area of ΔABC

(C) $P(a,b)$ lies outside the ΔABC

(D) Area of triangle PBC is greater than area of ΔABC

44. If one vertex of an equilateral triangle of side 'a' is at $(1,0)$, also another one lies on the line $\sqrt{3}x - y - \sqrt{3} = 0$, then co-ordinates of the third vertex may be

(A) $\left(1-\frac{a}{2}, \frac{\sqrt{3}a}{2}\right)$ **(B)** $\left(\frac{a}{2}+1, \frac{-\sqrt{3}a}{2}\right)$ **(C)** $(a+1,0)$ **(D)** $(a-1,0)$

45. A line passing through the origin (O) has point A and B in the same direction such that $OA = AB = r$. Through points A and B two lines are drawn making equal angle $tan^{-1}(\sqrt{3})$ with the line AB. Then points which lies on the locus of point of intersection of the lines is/are:

(A) $(r, \sqrt{2}r)$ **(B)** $(\sqrt{2}r, r)$ **(C)** (r,r) **(D)** $(-\sqrt{2}r, r)$

46. A line 'L' passes through the point $(2,3)$ and making intercept of length 3 units between the lines $2x + y - 2 = 0$ and $4x + 2y - 10 = 0$ then which of the following may be true about the line L:

(A) Parallel to x axis

(B) Parallel to y-axis

(C) having slope $-\frac{3}{4}$

(D) Perpendicular to $2x + y - 2 = 0$

47. For all values of θ, the lines represented by the equation
$(2cos\theta + 3sin\theta)x + (3cos\theta - 5sin\theta)y - (5cos\theta - 2sin\theta) = 0$

(A) pass through a fixed point

(B) Pass through the point $(1,1)$

(C) pass through a fixed point whose reflection in the line $x + y = \sqrt{2}$ is $(\sqrt{2}-1, \sqrt{2}-1)$

(D) pass through the origin

48. The centroid of an equilateral triangle is $(0,0)$. If two vertices of the triangle lies on $x + y - 2 = 0$, then:

(A) Area of triangle is $6\sqrt{3}$ square units

(B) vertex not lying on the line is $(-2,-2)$

(C) foot of the perpendicular from $(0,0)$ to the line is $(1,1)$

(D) vertices on the given line are $(1+\sqrt{3}, 1-\sqrt{3})$ and $(1-\sqrt{3}, 1+\sqrt{3})$

49. In a ΔABC equation of median and altitude from vertex C and B are $x + 2 = 0$ and $x + y - 2 = 0$ respectively, vertex A is at the origin, then:

(A) co-ordinate of point B is $(-4,6)$

(B) equation of AC is $x - y = 0$

(C) Area of ΔABC is 10 square units

(D) vertex C is $(-2,-3)$

50. A ray of light is sent along the line $x-2y=8$. After refracting across the line $x+y=1$ it enters the opposite side after turning by $15°$ away from the line $x+y=1$. Then the equation of line along which refracted ray travels will:

(A) have slope $\frac{5\sqrt{3}-6}{3}$ **(B)** have slope $\frac{5\sqrt{3}-6}{13}$

(C) pass through $\left(0,\frac{13\sqrt{3}-50}{3\sqrt{3}}\right)$ **(D)** pass through $\left(0,\frac{-50\sqrt{3}-31}{39}\right)$

51. The four lines given by $12x^2+7xy-12y^2=0$ and $12x^2+7xy-12y^2-x+7y-1=0$ will make:

(A) Rectangle **(B)** Square **(C)** Rhombus **(D)** Parallelogram

52. Equations of the sides of the triangle having $(3,-1)$ as a vertex, $x-4y+10=0$ *and* $6x+10y-59=0$ being the equations of an angle bisector and a median respectively drawn from different vertices, is/are:

(A) $6x-7y=25$ **(B)** $18x+13y=41$ **(C)** $2x+9y=65$ **(D)** $13x+4y=8$

53. The equation of a pair of straight lines is $ax^2+2hxy+by^2=0$. If the angle by which the axes be rotated so the term containing xy in the equation may be removed is ϕ, then:

(A) $\phi=\frac{\pi}{4}$ if $a=b, h>0$ **(B)** $\phi=\frac{\pi}{8}$, *if* $2h=a-b$

(C) $\phi=\frac{\pi}{8}$, *if* $a=b, h\neq 0$ **(D)** $\phi=\frac{\pi}{4}$, *if* $2h=a-b$

54. The equation of straight lines passing through ordered pairs (a,b) satisfying equation $sec^2((a+1)b)+a^2-1=0$, and having slope $\frac{1}{2}$, is (are):

(A) $x-2y=0$ **(B)** $x+2y=1$ **(C)** $x-2y=2\pi$ **(D)** $x-2y+2\pi=0$

55. $A(1,2)$ and $B(7,10)$ are two points. If $P(x,y)$ is a point such that the angle APB is $60°$ and the area of the ΔAPB is maximum, then which of given is (are) true?

(A) P lies on any line perpendicular of AB **(B)** P lies on the right bisector of AB

(C) P lies on the straight line $3x+4y=36$

(D) P lies on the circle passing through the points $(1,2)$ and $(7,10)$ and having a radius of 10 units

56. A line which makes an acute angle θ with the positive direction of x-axis is drawn through the point $P(3,4)$ to meet the line $x=6$ at R and $y=8$ at S, then:

(A) $PR=3\sec\theta$ **(B)** $PS=4\,\text{cosec}\theta$

(C) $PR+PS=\frac{2(3\sin\theta+4\cos\theta)}{\sin 2\theta}$ **(D)** $\frac{9}{(PR)^2}+\frac{16}{(PS)^2}=1$

57. The equation of the bisectors of the angles between the two intersecting lines $\frac{x-3}{\cos\theta}=\frac{y+5}{\sin\theta}$ and $\frac{x-3}{\cos\phi}=\frac{y+5}{\sin\phi}$ are $\frac{x-3}{\cos\alpha}=\frac{y+5}{\sin\alpha}$ and $\frac{x-3}{\beta}=\frac{y+5}{\gamma}$, then:

(A) $\alpha=\frac{\theta+\phi}{2}$ **(B)** $\beta=-\sin\alpha$ **(C)** $\gamma=\cos\alpha$ **(D)** $\beta=\sin\alpha$

58. Two roads are represented by the equation $y-x=6$ and $x+y=8$. An inspection bunglow has to be so constructed that it is at a distance of 100 from each of the roads. Possible location of the bunglow is given by:

(A) $\left(100\sqrt{2}+1,7\right)$ **(B)** $\left(1-100\sqrt{2},7\right)$ **(C)** $\left(1,7+100\sqrt{2}\right)$ **(D)** $\left(1,7-100\sqrt{2}\right)$

59. Let $L_1: 3x+4y=1$ *and* $L_2: 5x-12y+2=0$ be two given lines. Let image of every point on L_1 with respect to a line L lies on L_2 then possible equation of L can be:

(A) $14x+112y-23=0$ **(B)** $64x-8y-3=0$
(C) $11x-4y=0$ **(D)** $52y-45x=7$

60. Let $A(1,1)$ *and* $B(3,3)$ be two fixed points and *P* be a variable point such that area of ΔPAB remains constant equal to 1 for all position of *P*, then locus of *P* is given by:

(A) $2y=2x+1$ **(B)** $2y=2x-1$ **(C)** $y=x+1$ **(D)** $y=x-1$

61. The vertices of a triangle are $(1,3),(5,0)$ *and* $(-1,2)$. Which of the following inequalities will be satisfied by all points lying inside the triangle?

(A) $3x+2y\ge 0$ **(B)** $2x-3y-12\le 0$ **(C)** $x+y-6\le 0$ **(D)** $2x-y\ge 0$

62. $A(x_1,y_1),B(x_2,y_2),(y_1<y_2)$ are two points on the line $x+y=4$ from which perpendicular *AQ* and *BP* are drawn on line $4x+3y=10$ where *P* and *Q* are the feet of perpendicular such that $AQ=BP=1$. Now considering *AB* as diameter, a circle is drawn which meets the line $4x+3y=10$ at *C* and *D* such that *C* is closer to *P*.
Then which of the following statement(s) is correct?

(A) the value of $\dfrac{y_1+y_2}{x_1+x_2}$ is equal to -3 **(B)** the length PQ is equal to 14
(C) length QD is equal to $5\sqrt{2}-7$ **(D)** radius of circle obtained is $5\sqrt{2}$ units

63. If the two lines represented by $x^2\left(\tan^2\theta+\cos^2\theta\right)-2xy\tan\theta+y^2\sin^2\theta=0$ make angles α,β with the x-axis, then:

(A) $\tan\alpha+\tan\beta=4\operatorname{cosec}2\theta$ **(B)** $\tan\alpha\tan\beta=\sec^2\theta+\tan^2\theta$
(C) $\tan\alpha-\tan\beta=2$ **(D)** $\dfrac{\tan\alpha}{\tan\beta}=\dfrac{2+\sin 2\theta}{2-\sin 2\theta}$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column I are labelled as (A), (B), (C) & (D) whereas statements in Column II are labeled as p, q, r, s & t. More than one choice from Column II can be matched with Column I.

64. MATCH THE FOLLOWING :

	Column 1		**Column 2**
(A)	The number of integral values of 'a' for which point (a, a^2) lies completely inside the triangle formed by lines $x = 0, y = 0, 2y + x = 3$	**(p)**	0
(B)	A point on the line $x + y = 4$ which lies at a unit distance from the line $4x + 3y = 10$ has the coordinates (α, β) then possible value of $\alpha + \beta$	**(q)**	1
(C)	If (α, β) be the Orthocenter of triangle made by lines $x + y = 1, x - y + 3 = 0, 2x + y = 7$ then the value of $\alpha + \beta$ is	**(r)**	2
(D)	In a triangle ABC, the bisector of angles B and C lie along the lines $y = x$ and $y = 0$. If A is (1, 2) then $\sqrt{10}\, d(A, BC)$ equals (where $d(A, BC)$ denotes the perpendicular distance of A from BC.)	**(s)**	4

65. MATCH THE FOLLOWING :

	Column 1		**Column 2**
(A)	Two perpendicular straight lines are drawn from the origin to make an iscosceles triangle together with the line $2x + y = 5$ Then the area of triangle is	**(p)**	$\sqrt{5}$
(B)	Let the line $2x + y = 4$ meet x-axis at A and y-axis at B, and the perpendicular bisector of AB meets the horizontal line through (0, –1) at C. Let G be the centroid of the triangle ABC. Then perpendicular distance from G to AB equals	**(q)**	5
(C)	The number of integral points inside the triangle made by the line $3x + 4y - 12 = 0$ with the coordinate axes which are equidistant from at least two sides is/are (an intergral point is a point both of whose coordinates are integers)	**(r)**	3
(D)	The line $x = c$ cuts the triangle with corners (0, 0), (1, 1) and (9, 1) into two regions. For the area of the two regions to be the same c must be equal to :	**(s)**	1

66. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	A straight line with negative slope through (1, 4) meets the co-ordinate axes at *A* and *B*. The minimum length of $OA+OB$, *O* being the origin is :	**(p)**	$5\sqrt{2}$
(B)	If the point *P* is symmetric to the point $Q(4,\ -1)$ with respect to the bisector of the first quadrant, then the length of *PQ* is :	**(q)**	$3\sqrt{2}$
(C)	On the portion of the straight line $x+y=2$ between the axes a square is constructed away from the origin, with this portion as one of its side. If '*d*' denotes the perpendicular distances of a side of this square from the origin, then the maximum value of '*d*' is :	**(r)**	$\frac{9}{2}$
(D)	If the parametric equation of a line is given by $x=4+\frac{\lambda}{\sqrt{2}}$ and $y=-1+\sqrt{2}\ \lambda$, where λ is the parameter, then the intercept made by the line on the *x*-axis is :	**(s)**	9

67. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The pair of lines $6x^2-\alpha xy-3y^2-24x+3y+\beta=0$ intersect on *x*-axis, then the value of $20\alpha-\beta$.	**(p)**	5
(B)	Let *P* be any point on the line $x-y+3=0$ and *A* be a fixed point (3, 4). If the family of lines given by the equation $(3\sec\theta+5\operatorname{cosec}\theta)x+(7\sec\theta-3\operatorname{cosec}\theta)y+11(\sec\theta-\operatorname{cosec}\theta)=0$ are concurrent at a point *B* for all permissible values of θ and maximum value of $\lvert PA-PB\rvert=2\sqrt{2n}\,(n\in N)$, then find the value of *n*.	**(q)**	8
(C)	A square with centre at (3, 7) and side length 4 units has one of its diagonal parallel to the line *y* = *x*. If the vertices of the square be $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ and (x_4,y_4) then the value of max (y_1, y_2, y_3, y_4) $-\min(x_1,x_2,x_3,x_4)$.	**(r)**	1
(D)	The equation of a line through the mid point of the sides *AB* and *AD* of rhombus *ABCD*, whose one diagonal is $3x-4y+5=0$ and one vertex is *A*(3, 1) is $ax+by+c=0$. Then the absolute value of (*a* + *b*+ *c*) where *a*, *b*, *c* are integers expressed in lowest form:	**(s)**	6

68. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	If $P\left(1+\frac{t}{\sqrt{2}}, 2+\frac{t}{\sqrt{2}}\right)$ be any point on a line then value of t for which the point P lies between parallel lines $x+2y=1$ and $2x+4y=15$ is	**(p)**	(1, 2)
(B)	If the point $(2x_1-x_2+t(x_2-x_1), 2y_1-y_2+t(y_2-y_1))$ divides the join of (x_1,y_1) and (x_2,y_2) internally, then t lies in	**(q)**	$\left(\frac{-\sqrt{13}-1}{2},-1\right)\cup\left(1,\frac{\sqrt{13}-1}{2}\right)$
(C)	If the point (1, t) always remains in the interior of the triangle formed by the lines $y = x$, $y = 0$ and $x + y = 4$, then t lies in	**(r)**	$\left(\frac{-4\sqrt{2}}{3},\frac{5\sqrt{2}}{6}\right)$
(D)	Set of values of 't' for which the point $P(t,t^2-2)$ lies inside the triangle formed by lines $x+y=1$, $y=x+1$ and $y=-1$ is	**(s)**	(0, 1)

69. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$P(3, 1)$, $Q(6, 5)$ and $R(x, y)$ are three points such that angle PRQ is right angle and the area of ΔPRQ is 7, then number of such points R is.	**(p)**	2
(B)	Let $ABCD$ is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that $AE = AF$. The maximum possible area of quadrilateral $CDFE$ is	**(q)**	$\frac{21}{4}$
(C)	Let $A\equiv(0,0), B\equiv(5,0), C\equiv(5,3)$ and $D\equiv(0,3)$ are the vertices of rectangle $ABCD$. If P is a variable point lying inside the rectangle $ABCD$ and $d(P, L)$ denote perpendicular distance of point P from line L. If $d(P,AB)\le\min\{d(P,BC), d(P,AD), d(P,CD)\}$, then area of the region in which P lies is:	**(r)**	0
(D)	The slope of one of lines given by $ax^2+2hxy+by^2=0$ be the square of the slope of the other, if $ab(a+b)+\alpha abh+\beta h^3=0$, then $\alpha+\beta$ is equals:	**(s)**	$\frac{5}{8}$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

70. Let ABC be a triangle. Let A be the point (1, 2), $y = x$ is the perpendicular bisector of AB and $x-2y+1=0$ is the angle bisector of angle C. If the equation of BC is given by $ax+by-5=0$ then the value of $a + b$ is _________.

71. In a ΔABC, the equations of right bisectors of sides AB and CA are $3x+4y=20$ and $8x+6y=65$ respectively. If the vertex A be (10, 10), then the value of $\frac{1}{7}(ar\,\Delta\,ABC)$ is _________.

72. In a ΔABC, the vertex A is (1, 1) and orthocenter is (2, 4). If the sides AB and BC are members of the family of straight lines $ax+by+c=0$. Where a, b, c are in $A.P.$ then the coordinates of vertex C are (h, k). The value of $2h+9k$ is _________.

73. The slopes of three sides of a triangle ABC are $-1,-2,3$ respectively. If the orthocenter of triangle ABC is origin, then the locus of its centroid is $y=\frac{a}{b}x$ where a, b are relatively prime than $b – a$ is equal to _________.

74. The lines $x + y = 0$, $x – 4y = 0$ and $2x – y = 0$ are the altitudes of a triangle. If one of the vertices has coordinates of the form $(\lambda,\ -\lambda)$, if the locus of the centroid of such a triangle is $ax + by = 0$, then the value of $a + b$ is _______.

75. Two equal sides OA and OB of an isosceles triangle lie in the first quadrant. If the slopes of OA and OB are $\frac{7}{17}$ and 1, respectively and the length of perpendicular from O to AB *is* $\sqrt{13}$, if the equation of the side AB is $ax + by = c$ then the value of $(c – a – b)$ is ____________.

76. A line intersects the x-axis at $A(7, 0)$ and y-axis $B(0, – 5)$. A variable line PQ perpendicular to AB intersects the x-axis at P and the y-axis at Q. If AQ and BP intersect at R, show that the locus of R is $x^2 + y^2 – ax + by = 0$. Then the value of $(a – b)$ is :

77. In triangle ABC if the median to side BC has length $\left(11-6\sqrt{3}\right)^{\frac{-1}{2}}$ and it divides angle $\angle A$ into angles 30º and 45º. Then length of side BC is ____________.

78. In a triangle ABC if $AC = 3$, $BC = 4$ and median AD and BE are perpendicular to each other, Δ be the area of the triangle ABC. Then the value of $[\,\Delta\,]$ is ______. (where [.] denotes the greatest integer) :

79. In a triangle ABC, the coordinates of A is (1, 2) and the equations to the medians through B and C are $x+y=5$ and $x=4$. If coordinate of B is (x_1, y_1) and co-ordinate of C is (x_2, y_2) then $x_1y_2+x_2y_1$ equals_____.

80. A straight line through the point $A(-2,-3)$ cuts the lines $x+3y=9$ *and* $x+y+1=0$ at B and C respectively. If $AB.AC=20$, then product of slopes of line is_____.

81. A right angled triangle $ABC\left(\angle C=\frac{\pi}{2}\right)$ is constructed so that its sides are parallel to coordinate axes and the medians through A and B lie on the lines $y=3x+1\,and\ y=mx+2$ respectively. Then product of values of m for which such a triangle is possible is____.

82. If the distance of any point $P(x,y)$ from the origin is defined as $d(x,y)=\max\{|x|,|y|\}, d(x,y)=2$ then the area of curve represented by the locus of point P is S then $[S]$ is____.

83. Two equal sides AB and AC of an acute angle triangle ABC are formed by the equations $7x-y+3=0$ and $x+y-3=0$, side BC is $ax+by-31=0$ where $a-10b=31$. Value of $2a+10b+31=0$ is____.

84. A variable line 'L' of the form $y=mx$ is drawn to meet the lines $L_1:2x+3y-5=0;\ L_2:x+2y-5=0$ and $L_3:6x+4y-5=0$ at points A, B and C. A point $P(a,b)$ is taken on the line 'L'also $\frac{k(a+b)}{OP}=\frac{1}{OA}+\frac{1}{OB}+\frac{1}{OC}$ then value of k is______.

85. ABC is a triangle, whose vertex A is $(3,4)$ $L_1=0, L_2=0$ are the angle bisectors of angle B and C respectively where $L_1=x+2y-5=0$, $L_2=x-2y-3=0$ also $AB=KAI$ where I is the incentre then K is

86. If the equal sides PQ and PR (each equal to 2) of a right angled isosceles ΔPQR be produced to A and B so that $QA.RB=PR^2$ then the line AB passes through a fixed point which also satisfies the line $ax+by-6=0$ then $a+b$ is______.

87. If from point P (4, 4) perpendiculars to the straight lines $3x+4y+5=0$ and $y=mx+7$ meet at Q and R respectively and area of triangle PQR is maximum. Then the value of $12m$ must be____.

88. Let $(3,4)$ be a fixed point. A straight line passing through this point cuts the positive direction of the coordinate axes at the points P and Q. If λ sq unit be the minimum area of the triangle OPQ, O being the origin. Then the value of λ must be____.

89. If the slope of one of the line represented by $ax^2+2hxy+by^2=0$ is the square of the other, then the value of $\frac{a+b}{h}+\frac{8h^2}{ab}$ is______.

90. In a triangle ABC, the bisector of angles B and C lies along the lines $y=x\,and\ y=0$. If A is $(1,2)$ then $\sqrt{10}d(A,BC)$ equal (where $d(A,BC)$ denotes the perpendicular distance of A from BC).

91. Let P be any point on the line $x-y+3=0$ and A be a fixed point $(3,4)$. If the family of lines given by the equation $(3\sec\theta+5\operatorname{cosec}\theta)x+(7\sec\theta-3\operatorname{cosec}\theta)y+11(\sec\theta-\operatorname{cosec}\theta)=0$ are concurrent at a point B for all permissible values of θ and maximum value of $|PA-PB|=2\sqrt{2n}\,(n\in N)$, then find the value of n.

92. The slopes of three sides of a triangle ABC are $-1,-2,3$ respectively. If the orthocenter of triangle ABC is origin, then the locus of its centroid is $y=\frac{a}{b}x$ where a, b are relatively prime than $b-a$ is equal to______.

93. A straight line through the origin O meets the parallel lines $3x-4y=6$ and $6x-8y+c=0$ at points Q and P respectively such that $\left|\frac{OP}{OQ}\right|=\frac{4}{3}$. If $c=2^k$, then k is equal to______.

94. The vertices B and C of a triangle ABC lie on the lines $3y=4x$ *and* $y=0$ respectively and the side BC passes through the point $\left(\frac{2}{3},\frac{2}{3}\right)$. If $ABOC$ is a rhombus, O being the origin and the coordinates of A are (h,k), then $\frac{h}{k}$ is equal to______.

95. If A line is passing through the point $P(1,2)$ cutting the lines $x+y-5=0$ and $2x-y=7$ at A and B respectively such that the harmonic mean of PA and PB is 10. If equation of line is $(y-2)=\tan\left[\pi-\sin^{-1}\left[\frac{a}{\sqrt{146}}\right]-\sin^{-1}\left[\frac{b}{\sqrt{146}}\right]\right](x-1)$. The $|a-b|\times\frac{25}{41}$ is ______.

96. If the straight line through the point $P(3,4)$ makes an angle $\frac{\pi}{6}$ with the x-axis and meets the line $12x+5y+10=0$ at Q then the value of $\frac{(12\sqrt{3}+5)}{11}$ PQ is______.

97. A line through $A(-5,-4)$ meets the lines $x+3y+2=0$, $2x+y+4=0$ and $x-y-5=0$ at B, C and D respectively. If $\left(\frac{15}{AB}\right)^2+\left(\frac{10}{AC}\right)^2=\left(\frac{6}{AD}\right)^2$ equation of line is $2x+by+c=0$ then value of $2+b+c$ is______.

98. If $6a^2-3b^2-c^2+7ab-ac+4bc=0$, then the family of lines $ax+by+c=0$ is concurrent at ordered pairs (A,B) and (C,D), $A>0$ then the value of $A-B-C-D$ is______.

Advanced Problem Package	Circle

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Two Circles of radii 36 units and 9 units touch each other externally, a third circle of radius r touches the two given circles externally and also their common tangent, then the value of r is :

(A) 4 **(B)** 5 **(C)** $\sqrt{17}$ **(D)** $\sqrt{18}$

2. If $A(0,\alpha)$ and $B(0,\beta)$, α, $\beta > 0$ are two vertices of a variable triangle ABC, where the vertex C $(x, 0)$ is variable. The value of x for which $\angle ACB$ is maximum is :

(A) $\frac{\alpha+\beta}{2}$ **(B)** $\sqrt{\alpha\beta}$ **(C)** $\frac{2\alpha\beta}{\alpha+\beta}$ **(D)** $\frac{\alpha\beta}{\alpha+\beta}$

3. Two tangents are drawn from a point P to the circle $x^2+y^2=1$. If the tangents make an intercept of 2 units on the line $x=1$, then locus of P is :

(A) parabola **(B)** pair of lines **(C)** circle **(D)** straight line

4. Let $ABCD$ be a quadrilateral in which $AB \parallel CD$, $AB \perp AD$ and $AB = 3CD$. If the area of the quadrilateral $ABCD$ is 4, then the radius of the circle touching all the four sides of the quadrilateral is

(A) $\sin\frac{\pi}{12}$ **(B)** $\sin\frac{\pi}{3}$ **(C)** $\sin\frac{\pi}{4}$ **(D)** $\sin\frac{\pi}{6}$

5. Three concentric circles of which the biggest is $x^2+y^2=1$, have their radii in $A.P.$ If the line $y=x+1$ cuts all the circles in real and distinct points. The interval in which the common difference of the $A.P.$ will lie is:

(A) $\left(0,\frac{1}{4}\right)$ **(B)** $\left(0,\frac{2-\sqrt{2}}{4}\right)$ **(C)** $\left(0,\frac{1}{2\sqrt{2}}\right)$ **(D)** None of these

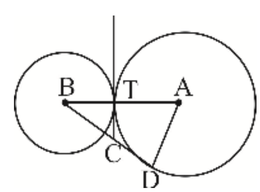
6. One circle has a radius of 5 units and its center is at (0, 5). A second circle has a radius of 12 and its centre is at (12, 0). The radius of a third circle which passes through the center of the second circle and both points of intersection of the first two circles, is equal to:

(A) 13/2 **(B)** 15/2 **(C)** 17/2 **(D)** None of these

7. A circle is inscribed in an equilateral triangle with side lengths 6 units. Another circle is drawn inside the triangle (but outside the first circle), tangent to the first circle and two of the sides of the triangle. The radius of the smaller circle is:

(A) $1/\sqrt{3}$ **(B)** $2/3$ **(C)** $1/2$ **(D)** 1

8. Two circles with centre at A and B, touch at T. BD is the tangent at D and TC is a common tangent. AT has length 3 and BT has length 2. The length of CD is:



(A) 4/3 **(B)** 3/2
(C) 5/3 **(D)** 7/4

9. In a triangle ABC, right angled at A, on the leg AC as diameter, a semicircle is described. The chord joining A with the point of intersection D of the hypotenuse and the semicircle, then the length AC equals to:

(A) $\frac{AB\cdot AD}{\sqrt{AB^2+AD^2}}$ **(B)** $\frac{AB\cdot AD}{AB+AD}$ **(C)** $\sqrt{AB\cdot AD}$ **(D)** $\frac{AB\cdot AD}{\sqrt{AB^2-AD^2}}$

10. In a circle with centre 'O' PA and PB are two chords. PC is the chord that bisects the angle APB. The tangent to the circle at C is drawn meeting PA and PB extended at Q and R respectively. If $QC = 3$, $QA = 2$ and $RC = 4$, then length of RB equals:

(A) 2 **(B)** 8/3 **(C)** 10/3 **(D)** 11/3

11. A circle is inscribed in a rhombus $ABCD$ with one angle 60°. The distance from the centre of the circle to the nearest vertex is equal to 1. If P is any point on the circle, then $|PA|^2+|PB|^2+|PC|^2+|PD|^2$ is equal to :

(A) 12 **(B)** 11 **(C)** 9 **(D)** None of these

12. Considering the circles, $x^2+y^2=25$ and $x^2+y^2=9$. From the point $A(0, 5)$ two segments are drawn touching the inner circle at the points B and C while intersecting the outer circle at the points D and E. If 'O' is the centre of both the circles then the length of the segment OF that is perpendicular to DE, is:

(A) 7/5 **(B)** 7/2 **(C)** 5/2 **(D)** 3

13. Points P and Q are 3 units apart. A circle centered at P with a radius of 3 units intersects a circle centered at Q with radius $\sqrt{3}$ units at point A and B. The area of the quadrilateral $APBQ$ is:

(A) $\sqrt{99}$ **(B)** $\frac{\sqrt{99}}{2}$ **(C)** $\sqrt{\frac{99}{2}}$ **(D)** $\sqrt{\frac{99}{16}}$

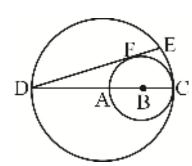
14. Circle $x^2+y^2+16x+12y+c=0$ is touched by a straight line with slope 2 and y-intercept 5 units at a point Q. Then the coordinates of Q are

(A) (–6, –7) **(B)** (–9, –13) **(C)** (–10, –15) **(D)** (–6, 11)

15. Tangents are drawn at the point of intersections of the circles $x^2+y^2=1$ and $x^2+y^2-(\lambda+6)x+(8-2\lambda)y-3=0$. ($\lambda$ being the variable). Then the locus of the point of intersection of these tangents is :

(A) $2x-y+10=0$ **(B)** $x+2y-10=0$ **(C)** $x-2y+10=0$ **(D)** $2x+y-10=0$

16. In the diagram, DC is a diameter of the large circle centered at A, and AC is a diameter of the smaller circle centered at B. If DE is tangent to the smaller circle at F and $DC = 12$ units then the length of DE is:



(A) 13 **(B)** 16 **(C)** $8\sqrt{2}$ **(D)** $10\sqrt{2}$

17. Let C be a circle $x^2+y^2=1$. The line $y=mx+m$ intersects C at the point P other than $(-1,0)$, the number of rational choices for m for which both the coordinates of P are rational, is:

(A) 3 **(B)** 4 **(C)** 5 **(D)** infinitely many

18. The locus of the mid points of the chords of the circle $x^2+y^2-ax-by=0$ which subtend a right angle at $(a/2, b/2)$ is:

(A) $ax+by=0$ **(B)** $ax+by=a^2+b^2$

(C) $x^2+y^2-ax-by+\frac{a^2+b^2}{8}=0$ **(D)** $x^2+y^2-ax-by-\frac{a^2+b^2}{8}=0$

19. If the two circles $C_1: x^2+y^2=16$ and circle C_2 of radius 5 units intersect in such a manner that the common chord of maximum length has a slope equal to 3/4, then the coordinates of the centre of C_2 are:

(A) $\left(\pm\frac{9}{5},\pm\frac{12}{5}\right)$ **(B)** $\left(\pm\frac{9}{5},\mp\frac{12}{5}\right)$ **(C)** $\left(\pm\frac{12}{5},\pm\frac{9}{5}\right)$ **(D)** $\left(\pm\frac{12}{5},\mp\frac{9}{5}\right)$

Paragraph for Questions 20 - 22

Let S_1 and S_2 be two fixed circles touching each other externally with radius 2 and 3 respectively. Let S_3 be a variable circle touching internally both S_1 and S_2 at points A and B respectively. The tangents to S_3 at A and B meet at T, and $TA = 4$.

20. The radius of circle S_3 is :

(A) 2 **(B)** 4 **(C)** 6 **(D)** 8

21. The area of circle circumscribing $x = \frac{-3h}{2}$ is:

(A) 10π **(B)** 20π **(C)** 40π **(D)** 80π

22. Let C_1, C_2, C_3 be centres of circles S_1, S_2, S_3 respectively, then which of the following must be true:

(A) $C_3C_1 + C_3C_2 = 5$ **(B)** $C_3C_1 - C_3C_2 = 3$ **(C)** $C_3C_1 + C_3C_2 = 3$ **(D)** $C_3C_1 - C_3C_2 = 1$

Paragraph for Questions 23 - 25

Let $f(x, y) = 0$ be the equation of a circle such that $f(0, y) = 0$ has equal real roots and $f(x, 0) = 0$ has two distinct real roots. Let $g(x, y) = 0$ be the locus of points 'p' from where tangents to circle $f(x, y) = 0$ make angle $\frac{\pi}{3}$ between them and $g(x, y) = x^2 + y^2 - 5x - 4y + c, c \in R$

23. Let Q be a point from where tangents drawn to circle $g(x, y) = 0$ are mutually perpendicular. If A, B are the points of contact of tangent drawn from Q to circle $g(x, y) = 0$, then area of triangle QAB is:

(A) 25/12 **(B)** 25/8 **(C)** 25/4 **(D)** 25/2

24. The area of region bounded by circle $f(x, y) = 0$ with x-axis in the first quadrant is:

(A) $3 + \frac{25}{8}\left(\pi - \tan^{-1}\frac{1}{2}\right)$ **(B)** $3 + \frac{25}{8}\left(\tan^{-1}\frac{24}{11}\right)$

(C) $3 + \frac{25}{8}\left(2\pi - \tan^{-1}\frac{3}{4}\right)$ **(D)** $3 + \frac{25}{8}\left(2\pi - \tan^{-1}\frac{24}{7}\right)$

25. The number of points with positive integral coordinates satisfying $f(x, y) > 0, g(x, y) < 0; y > 3$ and $x < 6$ is:

(A) 7 **(B)** 8 **(C)** 10 **(D)** 11

Paragraph for Questions 26 - 28

If a circle C_0, with radius 1 unit touches both the axes and as well as line (L_1) through $P(0,4)$, L_1 cut the x-axis at $(x_1, 0)$. Again a circle C_1 is drawn touching x-axis, line L_1 and another line L_2 through point P. L_2 intersects x-axis at $(x_2, 0)$ and this process is repeated n times.

26. The value of x_2 is

(A) 3 (B) 4 (C) 15/2 (D) 17/2

27. The centre of circle C_2 is

(A) (3, 1) (B) $\left(\frac{15}{2}, 1\right)$ (C) $\left(\frac{31}{4}, 1\right)$ (D) none of these

28. The $\lim\limits_{n\to\infty} \frac{x_n}{2^n}$ is

(A) 4 (B) 0 (c) 1 (D) 2

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

29. ABC is a right angled triangle right angled at A, with side $AC = 1$ and $AB = a$, a circle having AC as diameter cuts the side CB at D if $CD = b$ then :

(A) $ab > 1$ **(B)** $ab < 1$ **(C)** $\frac{b}{a} > \frac{1}{a^2 + \frac{1}{2}}$ **(D)** $\frac{b}{a} < \frac{1}{a^2 + \frac{1}{2}}$

30. The equation of the circle which touches the axes of coordinates and the line $\frac{x}{3} + \frac{y}{4} = 1$ and whose centre lies in the first quadrant is $x^2 + y^2 + 2rx + 2ry + r^2 = 0$, then r can be equal to:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 6

31. The equation of the largest circle passing through the points (1, 1) and (2, 2) and lying completely in the first quadrant is :

(A) $x^2 + y^2 - 4x - 2y + 4 = 0$ **(B)** $x^2 + y^2 - 2x - 4y + 4 = 0$

(C) $x^2 + y^2 - 3x - 3y + 4 = 0$ **(D)** $x^2 + y^2 + 3x + 3y - 4 = 0$

32. If $4l^2 - 5m^2 + 6l + 1 = 0$ and the line $lx + my + 1 = 0$ touches a fixed circle, then:

(A) centre of circle is at (3, 0) **(B)** the radius of circle is $\sqrt{5}$

(C) The radius of circle is $\sqrt{3}$ **(D)** the circle passes through (1, 1)

33. The equation of a circle in which the chord joining the points (1, 2) and (2, –1) subtends an angle of $\frac{\pi}{4}$ at any point on the circumference is

(A) $x^2 + y^2 - 5 = 0$ **(B)** $x^2 + y^2 - 6x - 2y + 5 = 0$

(C) $x^2 + y^2 + 6x + 2y - 15 = 0$ **(D)** $x^2 + y^2 + 7x - 2y + 14 = 0$

34. Equations of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$, then:

(A) The radius of the greatest circle touching all the four circles is $(\sqrt{2} + 1)a$

(B) The radius of the smallest circle touching all the four circles is $(\sqrt{2} - 1)a$

(C) Area of region enclosed by four given circles with co-ordinate axes is $(4 - \pi)a^2$ sq.units

(D) The centres of four circles are the vertices of a square

35. If the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ is inscribed in a triangle whose two sides are co-ordinate axes and one side has negative slope cutting intercepts a and b on positive x and positive y axis, then :

(A) $1 - \frac{1}{a} - \frac{1}{b} = \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$ **(B)** $\frac{1}{a} + \frac{1}{b} < 1$

(C) $\frac{1}{a} + \frac{1}{b} > 1$ **(D)** $\frac{1}{a} + \frac{1}{b} + 1 = \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

36. The locus of points of intersection of the tangents to $x^2+y^2=a^2$ at the extremities of a chord of circle $x^2+y^2=a^2$ which touches the circle $x^2+y^2-2ax=0$ is(are):

(A) $y^2=a(a-2x)$ **(B)** $x^2=a(a-2y)$ **(C)** $x^2+y^2=(x-a)^2$ **(D)** $x^2+y^2=(y-a)^2$

36. A circle touches the line $x+y-2=0$ at $(1,1)$ and cuts the circle $x^2+y^2+4x+5y-6=0$ at P and Q. Then :

(A) PQ can never be parallel to the given line $x+y-2=0$

(B) PQ can never be perpendicular to the given line $x+y-2=0$

(C) PQ always passes through (6, – 4) **(D)** PQ always passes through (– 6, 4)

38. Let x, y be real variable satisfy the $x^2+y^2+8x-10y-40=0$. Let $a=\max\left(\sqrt{(x+2)^2+(y-3)^2}\right)$ and $b=\min\left(\sqrt{(x+2)^2+(y-3)^2}\right)$, then:

(A) $a+b=18$ **(B)** $a+b=4\sqrt{2}$ **(C)** $a-b=4\sqrt{2}$ **(D)** $ab=73$

39. A square is inscribed in the circle $x^2+y^2-2x+4y-93=0$ with the sides parallel to the coordinate axes. The coordinates of the vertices are :

(A) (8, 5) **(B)** (8, 9) **(C)** (– 6, 5) **(D)** (– 6, – 9)

40. Let A, B, C, D lie on a line such that $AB = BC = CD = 1$. The points A and C are also joined by a semicircle with AC as diameter and P is a variable point on this semicircle such that $\angle PBD=\theta, 0\le\theta\le\pi$. Let R is the region bounded by arc AP, the straight line PD and line AD

(A) The maximum possible area of region R is $\dfrac{2\pi+3\sqrt{3}}{6}$

(B) If 'L' is the perimeter of region 'R', then L is equal to $3+\pi-\theta+\sqrt{5-4\cos\theta}$

(C) The maximum possible area of region R is $\dfrac{2\pi-3\sqrt{3}}{6}$

(D) If 'L' is the perimeter of region 'R', then L is equal to $3+\pi-\theta+\sqrt{5+4\cos\theta}$

41. Consider two circles S_1 and S_2 (externally touching) having centres at points A and B whose radii are 1 and 2 respectively. A tangent to circle S_1 from point B touches the circle S_1 at point C. D is chosen on circle S_2 so that AC is parallel to BD and two segments BC and AD do not intersect. Segment AD intersect the circle S_1 at E. The line through B and E intersects the circle S_1 at another point F.

(A) The length of segment EF is $\dfrac{2\sqrt{3}}{3}$ **(B)** The area of triangle ABD is $2\sqrt{2}$

(C) The length of the segment DE is 2 **(D)** ABD is a triangle of perimeter $2\sqrt{3}$

42. The circle 'S' touches the sides AB and AD of the rectangle $ABCD$ and cuts the side DC at single point F and the side BC at a single point E. If $|AB|=32, |AD|=40$ and $|BE|=1$

(A) The angle between pair of tangents drawn form the point D to the circle 'S' is $\pi-\tan^{-1}\left(\dfrac{15}{8}\right)$

(B) The Area of trapezium AFCB is 1180 sq.units

(C) The radius of circle is 25 units

(D) The angle between pair of tangents drawn form the point D to the circle 'S' is $\pi-2\tan^{-1}\left(\dfrac{15}{8}\right)$

43. In the triangle ABC, the angle bisector AK is perpendicular to the median BM and $\angle ABC = 120°$, then :

(A) The value of ratio $\frac{BC}{AB}$ is equal to $\frac{\sqrt{13}-1}{2}$

(B) The value of ratio of radius of the circle circumscribing the triangle ABC to the side length AB is equal to $\frac{2}{\sqrt{3}}$

(C) The ratio of the area of ΔABC to the area of the circle circumscribing ΔABC is equal to $\frac{3\sqrt{3}}{32\pi}\left(\sqrt{13}-1\right)$

(D) The value of ratio of the sides AB to AC is equal to 1/2

44. There are two circles in a parallelogram. One of them of radius 3units is inscribed in the parallelogram, and the other touches two sides of the parallelogram and the first circle. The distance between the points of tangency which lie on the same side of the parallelogram is equal to 3 units.

(A) The radius of the other circle is ¾ units

(B) Area of the parallelogram is equal to 75/2 units

(C) Let d_1, d_2 denote the lengths of the diagonals of parallelogram, then the product $d_1.d_2$ is equal to 75

(D) Let d_1, d_2 denote the lengths of the diagonals of parallelogram, then the product $d_1.d_2$ is equal to 95

45. Point M moved on the circle $(x-4)^2 + (y-8)^2 = 20$. Then it broke away from it and move along a tangent to the circle, cuts the x-axis at the point $(-2, 0)$. The coordinates of a point on the circle at which the moving point broke away is:

(A) $\left(-\frac{3}{5},\frac{46}{5}\right)$ **(B)** $\left(-\frac{2}{5},\frac{44}{5}\right)$ **(C)** (6, 4) **(D)** (3, 5)

46. Two chords are drawn from the point $P(h, k)$ on the circle $x^2+y^2=hx+ky$. If the y-axis divides both the chords in the ratio 2:3, then which of the following may be correct?

(A) $k^2>15h^2$ **(B)** $15k^2>h^2$ **(C)** $h^2=15k^2$ **(D)** $k^2>5h^2$

47. The equation(s) of the tangent at the point (0, 0) to the circle, making intercepts of lengths $2a$ and $2b$ units on the coordinates axes, is(are):

(A) $ax+by=0$ **(B)** $ax-by=0$ **(C)** $x-y=0$ **(D)** $bx+ay=0$

48. If the area of the quadrilateral formed by the tangents from the origin to the circle $x^2+y^2+6x-10y+c=0$ and the radii corresponding to the points of contact is 15, then a value of c is :

(A) 9 **(B)** 4 **(C)** 5 **(D)** 25

49. Let C be a circle with centre 'O' and HK is the chord of contact of tangents drawn from a point A. OA intersects the circle 'C' at P and Q and B is the midpoint of HK, then :

(A) AB is the harmonic mean of AP and AQ

(B) OA is the arithmetic mean of AP and AQ

(C) $(AK)^2=(OA)(AB)$

(D) AB is the geometric mean of AP and AQ

50. Chords of the circle $x^2+y^2=9$ are drawn such that segments intercepted from the chords by the curve $y^2-4x-4y=0$ subtend right angle at the origin. If the locus of the middle points of the chords with respect to circle is a curve S, then:

(A) S is a pair of line

(B) S is a circle

(C) S passes through the origin

(D) S meets the given circle $x^2+y^2=9$ at A and B and the tangents at A and B to the circle $x^2+y^2=9$ intersect at (4, 4)

51. A and B are two points in xy plane, which are $2\sqrt{2}$ unit distance apart and subtend an angle of 90° at $C(1,2)$ on the line $x-y+1=0$ which is larger than any angle subtend by the line segment AB at any other point on the line. The equation of the circle through the points A, B and C is:

(A) $x^2+y^2-6x+7=0$ **(B)** $x^2+y^2-4x+2y+3=0$

(C) $x^2+y^2-6y+7=0$ **(D)** $x^2+y^2-4x-2y+3=0$

52. A variable circle passes through the origin O and cuts off portions OP and OQ from X-axis and Y-axis respectively such that $m(OP)+n(OQ)$ is equal to unity. If the circle passes through a fixed point (x_1, y_1) other than O, then:

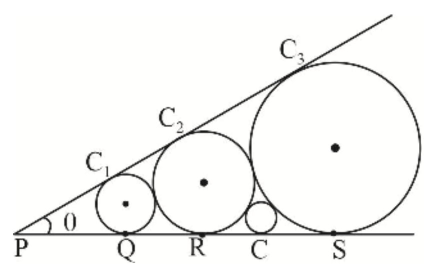
(A) $x_1^2+y_1^2=m^2+n^2$ **(B)** $x_1+y_1=m+n$ **(C)** $mx_1+ny_1=1$ **(D)** $nx_1-my_1=0$

53. If two points $A(-2,\alpha)$ *and* $B(4,\beta)$ are such that the triangle AOB is the right-angle triangle right angle at O. If $S=0$ be the equation of the locus of the foot of the perpendicular P drawn to AB from the point O :

(A) (1, 3) lies on $S=0$ **(B)** Minimum possible area of triangle AOB is 9

(C) Locus is a parabola **(D)** Locus is a circle

54. If the C_1, C_2, C_3 and C are four circles of radius r_1, r_2, r_3, r respectively as shown in figure:



(A) radius of the circle C is $2\sqrt{r_2 r_3}$

(B) $\tan\frac{\theta}{2}=\frac{r_2-r_1}{2\sqrt{r_1 r_2}}$

(C) $PQ=\frac{2r_1^{3/2}r_2^{1/2}}{r_2-r_1}$

(D) If $r_1=2$ and $r_1+r_2+r_3=14$ then $\frac{r_3}{r_2}=2$

55. Let A, B, C and D be four distinct point on a line in that order. The circles with diameter AC is $x^2+y^2+ax+c=0$ and BD is $x^2+y^2-by=0$ intersect at X and Y the line XY meets BC at Z. Let P be a point on XY other than Z, the line CP intersects the circle with diameter AC at C and M, and line BP intersects the circle with diameter BD at B and N and the equation of line AM and DN are $bx+cy+a=0$ and $cx+ay+b=0$ respectively, then which of the following is true (where ω is a cube root of unity)

(A) $a+b+c=1$ **(B)** $a+b\omega+c\omega^2=0$ **(C)** $a+b\omega^2+c\omega=0$ **(D)** $a+b+c=0$

56. An isosceles right angled triangle ABC is such that $\angle B = 90°, AC = \sqrt{2}$ and A and C moves on positive coordinate axis, then

(A) locus of the point B is $y - x = 0$

(B) locus of the circum centre of ΔABC $x^2 + y^2 = \frac{1}{2}$

(C) centre of the circle circumscribing ΔABC will lie on the line $y - 3x = 0$

(D) centre of the circle circumscribing ΔOAC will lie on the $y - 4x = 0$

57. If the circle passing through the distinct points $(a,t),(t,a)$ and (t,t) for the all values of 't' passes through a fixed point then

(A) possible values of a is 4 **(B)** possible values of a is 8

(C) possible values of a is 12 **(D)** possible values of a is 15

58. A line L_1 intersect x and y axes at P and Q respectively. Another line L_2, perpendicular to L_1, cuts x and y axes at T and S respectively. The locus of the point of intersection of the lines PS and QT is a circle passing through the

(A) origin **(B)** point P **(C)** point Q **(D)** point T

59. The equation of a circle of radius 1 touching the circles $x^2 + y^2 - 2|x| = 0$ is :

(A) $x^2 + y^2 + 2\sqrt{3}y - 2 = 0$ **(B)** $x^2 + y^2 - 2\sqrt{3}y + 2 = 0$

(C) $x^2 + y^2 + 2\sqrt{3}y + 2 = 0$ **(D)** $x^2 + y^2 - 2\sqrt{3}y - 2 = 0$

60. A straight line through the vertex P of a triangle PQR intersect the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then:

(A) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{SQ \times SR}}$ **(B)** $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{SQ \times SR}}$

(C) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$ **(D)** $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$

61. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, which of the following equations can represent L_1.

(A) $x + y = 0$ **(B)** $x - y = 0$ **(C)** $x + 7y = 0$ **(D)** $x - 7y = 0$

62. Two circles have centres at $(a,0)$ and $(-a,0)$ and radii r_1 and r_2 $(a > r_1 > r_2)$. Then the points of contact of the common tangents to two circles lies on the

(A) $x^2 + y^2 = a^2 + r_1r_2$ **(B)** $x^2 + y^2 = a^2 - r_1r_2$

(C) $x^2 + y^2 = a^2 - 2r_1r_2$ **(D)** $x^2 + y^2 = a^2 + 2r_1r_2$

63. A circle S of radius unity touches a line L at P. A point A lies on S and N is the foot of the perpendicular from A to L. The area of ΔPAN as A varies cannot be equal to:

(A) 1 **(B)** $\sqrt{3}$ **(C)** $\frac{3\sqrt{3}}{8}$ **(D)** $\frac{\sqrt{3}}{2}$

64. Consider the circle $x^2+y^2-10x-6y+30=0$. Let O be the centre of the circle and tangent at $A(7, 3)$ and $B(5, 1)$ meet at C. Let $S = 0$ represents family of circles passing through A and B, then :

(A) area of quadrilateral $OACB = 4$

(B) the radical axis for the family of circles $S = 0$ is $x + y = 10$

(C) the smallest possible circle of the family $S = 0$ is $x^2+y^2-12x-4y+38=0$

(D) the coordinates of point C are (7, 1)

65. Let C_1 *and* C_2 be centres of two circles whose radii are 2 and 4 respectively. Also $C_1C_2 = 10$ and direct common tangents of these circles touch them at P, Q, R, S. Another circle of radius 'λ' is drawn passing through *P, Q, R, S.* Then

(A) Midpoint of C_1C_2 is centre of the circle passing through *P, Q, R, S*

(B) Centre of the circle passing through *P, Q, R, S* divides $C_1.C_2$ in the ratio $1:2$

(C) $\lambda^2 = 33$

(D) $\lambda^2 = 35$

66. If a circle passes through the point $\left(3, \sqrt{\frac{7}{2}}\right)$ and touches $x + y = 1$ and $x - y = 1$, then the centre of the circle is:

(A) (4, 0) **(B)** (4, 2) **(C)** (6, 0) **(D)** (7, 9)

67. A point M divides A and B in the ratio $1:2$ where A and B diametrically opposite ends of a circle $x^2+y^2-5x-9y+22=0$ square AMCD and BMEF on the length AM and MB are constructed on the same side of line AB if co-ordinates of A is (1, 3) then find the orthocenter of ΔABE.

(A) (1, 6) (B) (1, 5) (C) (3, 3) (D) (4, 6)

68. If the conics equations are $S \equiv \sin^2\theta\, x^2 + 2h\tan\theta xy + \cos^2\theta\, y^2 + 32x + 16y + 19 = 0$,

$S' \equiv \cos^2\theta\, x^2 + 2h'\cot\theta xy + \sin^2\theta y^2 + 16x + 32y + 19 = 0$ intersect at four concyclic points, then: (where $\theta \in [0, \pi/2]$)

(A) $h + h' = 0$ **(B)** $h - h' = 0$ **(C)** $\theta = \pi/4$ **(D)** None of these

69. If largest and smallest value of $\frac{y-4}{x-3}$ is p and q where (x, y) satisfy $x^2+y^2-2x-6y+9=0$ then which of the following is true:

(A) $p+q=\frac{4}{3}$ **(B)** $q=1$ **(C)** $p=\frac{4}{3}$ **(D)** $pq=\frac{4}{3}$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

70. MATCH THE FOLLOWING :

	Column 1		**Column 2**
(A)	If the point $(c, c+2)$ is an interior point of smaller segment of the curve $x^2 + y^2 - 4 = 0$ made by the chord of the curve whose equation is $3x+4y+12=0$,then the value of c is	**(p)**	-1
(B)	If the set represented by $\{(x,y) \mid x^2+y^2+2x \le 1\} \cap \{(x,y) \mid x-y+c \ge 0\}$ contains only one point then the value of c is	**(q)**	ϕ
(C)	*ABCD* is a square of unit area, if a circle touches two sides of *ABCD* and passes through exactly one of its vertices. Then the radius of this circle is	**(r)**	$3\sqrt{2}$
(D)	If tangents are drawn from(4, 4) to the circle $x^2+y^2-2x-2y-7=0$ to meet the circle at A and B, then length of the chord AB is	**(s)**	2
		(t)	$2-\sqrt{2}$

71. MATCH THE FOLLOWING :

	Column 1		**Column 2**
(A)	If the line $2x-y+1=0$ is tangent to the circle at the point (2, 5) whose centre lies on the line $x-2y=4$, then radius of this circle is	**(p)**	$6\sqrt{26}$
(B)	Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If BD = 20 and DC = 16 then the length AC equals	**(q)**	1
(C)	Let C be the circle of radius unity centred at the origin. If two positive numbers x_1 and x_2 are such that the line passing through $(x_1, -1)$ and $(x_2, 1)$ is tangent to C then x_1x_2 is:	**(r)**	$3\sqrt{5}$
(D)	If $\left(a,\frac{1}{a}\right),\left(b,\frac{1}{b}\right),\left(c,\frac{1}{c}\right)$ and $\left(d,\frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units then, $abcd$ is equal to	**(s)**	2
		(t)	$2-\sqrt{2}$

72. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Consider 3 non-collinear points A, B, C with coordinates (0, 6), (5, 5) and (–1, 1) respectively. If the equation of a line tangent to the circle circumscribing the triangle ABC and passing through the origin is $ax+by=0$ then $b-a$ is	**(p)**	–1
(B)	From (3, 4) chords are drawn to the circle $x^2+y^2-4x=0$. The locus of the mid points of the chords is $(x-a)(x-b)+y(y-c)=0$, then the value of $a+b+c$ is	**(q)**	9
(C)	A foot of the normal from the point (4, 3) to a circle is (2, 1) and a diameter of the circle has the equation $2x-y-2=0$. Then the equation of the circle is $x^2+y^2-ax-b=0$, then $a-b$ is	**(r)**	14
(D)	The equation of the circle symmetric to the circle $x^2+y^2-2x-4y+4=0$ about the line $x-y=3$ is $x^2+y^2-ax+by+28=0$, then $a+b$ is	**(s)**	2
		(t)	1

73. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	A circle of constant radius 'a' passes through origin 'O' and cuts the axes of coordinates in points P and Q, then the equation of the locus of the foot of perpendicular from O to PQ is $(x^2+y^2)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=ka^2$,then k is	**(p)**	7
(B)	Tangents are drawn from any point on the circle $x^2+y^2=R^2$ to the circle $x^2+y^2=r^2$. The line joining the points of intersection of these tangents with circle also touch the second. If R equals $k\,r$,then k is	**(q)**	0
(C)	A ray of light incident at the point (–2, –1) gets reflected from the tangent at (0, –1) to the circle $x^2+y^2=1$. The reflected ray touches the circle. If the equation of the line along which the incident ray moves is $ay+bx=11$,then the value of $a+b$ is	**(r)**	4
(D)	The equation of a line inclined at an angle $\frac{\pi}{4}$ to the axis X, such that the two circle $x^2+y^2=4, x^2+y^2-10x-14y+65=0$ intercept equal lengths on it is $ax+by-3=0$,then the value of $a+b$ is	**(s)**	2
		(t)	$2-\sqrt{2}$

74. MATCH THE FOLLOWING :

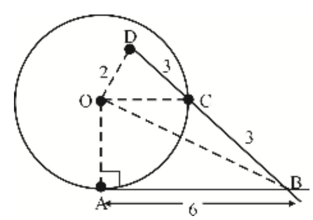
	Column 1		**Column 2**
(A)	P is a point (a, b) in the first quadrant. If the two circles which pass through P and touch both the coordinate axes cut at right angles, then the relation between a and b is given by $a^2 + b^2 - kab = 0$,then the value of k is	**(p)**	3
(B)	If two chords of the circle $x^2 + y^2 - ax - by = 0$, drawn from the point (a, b) is divided by the x-axis in the ratio 2:1 if $a^2 - kb^2 > 0$,then the value of k is	**(q)**	1
(C)	AB is a diameter of a circle. CD is a chord parallel to AB and $2CD = AB$. The tangent at B meets the line AC product at E then AE is equal to (kAB), then the value of k is	**(r)**	4
(D)	The lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to the same circle whose radius is r, then $4r$ is equal to	**(s)**	2
		(t)	$2-\sqrt{2}$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

75. From a point $A(2, 2)$ two chords AB and AC of 1 unit length are drawn to the circle $x^2 + y^2 = 8$. If the equation of the chord BC is given by $ax + by = 15$, then the value of $a + b$ is__________.

76. In the given figure AB is tangent at A to the circle with centre at O; point D is interior to circle and DB intersects the circle at C. If $BC = DC = 3$, $OD = 2$ and $AB = 6$, then find the value of $[r]$ (where r is the radius of circle and [.] represent G.I.F.)



77. If r_1 and r_2 are the radius of two circles passing through (–1, 1) and touching the lines $x + y = 2$, $x - y = 2$, and $r_1 + r_2 = a\sqrt{2}$, then a is equal to _________.

78. If a circle touches the hypotenuse of a right-angled triangle at its middle point and passes through the middle point of shorter side. If a and b ($a < b$) be the length of the sides and the radius of the circle is $\frac{b}{ka}\sqrt{a^2 + b^2}$, then the value of k is __________.

79. On the side AC of an acute angled triangle ABC a point D is taken, such that $AD = 1$, $DC = 2$ and BD is an altitude of ΔABC . A circle of radius 2, which passes through points A and D and touches a circle at the point D circumscribed about the ΔBDC. If the area of ΔABC is Δ then the value of $\frac{1}{11}[\Delta]$ is equal to __________. (Where [.] represents G.I.F.)

80. The centre of *a* circle *C* lies on the line $2x - 2y + 9 = 0$ and this circle cuts $x^2 + y^2 = 4$ orthogonally. If this circle passes through two fixed points (a, b) and (c, d), then the value of $a + b + c + d$ is __________.

81. Let $P\ (a,\ b)$ be a variable point satisfying $4 \le a^2 + b^2 \le 9$ and $b^2 - 4ab + a^2 \le 0$. Let R be the complete region represented in *x-y* plane in which P can lie, if m be the minimum value of $|a+b|$ for all position of P lying in region R. Then $[m]$ is __________. (Where [.] represents G.I.F.)

82. *ABCD* is rectangle a circle passing through *C* touches *AB* and *AD* at *M* and *N* respectively. If the perpendicular distance of *MN* from *C* is 5 then the area of rectangle is___.

83. Through the point of intersection *P* of the circle $x^2 + y^2 = 1$ and $x^2 + y^2 + 2x + 4y + 1 = 0$ a common chord *APB* is drawn terminating on the two circles such that the chords *AP* and *BP* of the given circles subtend equal angles at the respective centres. If the coordinates of *P* are integral and the equation of the chord is $y = 2mx + 1$ then the value of m is _____.

84. A circle *S*, whose radius is 1 unit, touches the *X*-axis at point *A*. The centre *Q* of *S* lies in the first quadrant. The tangent from the origin *O* to the circle touches it at *T* and a point *P* lies on it such that the triangle *OAP* is a right-angled triangle at *A* and its perimeter is 8 unit. The length of *QP* is _____.

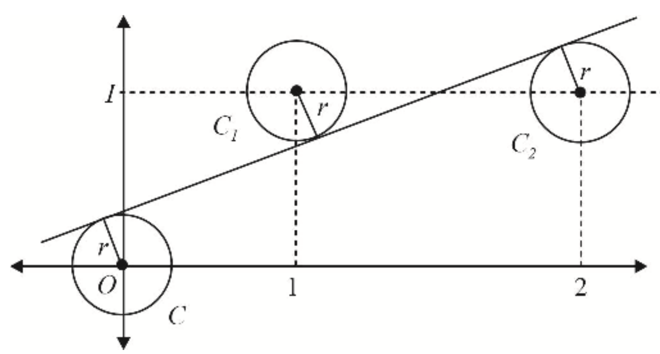
85. *CD* is the common chord of the two circles of equal radii touching a line *L* at *A* and *B*. Let C be closer to the line *L* than *D*. The ratio of the radii of circumcircles of the triangle *ACB* and *ADB* is _____.

86. The centres of two circles C_1 and C_2 each of unit radius are at a distance 6 unit from each other. Let P be the mid-point of the line segment joining the centres of C_1 and C_2. If a common tangent to C_1 and C_2 passing through P is also a common tangent to C_2 and C, then the radius of the circle C is _____.

87. The circle $x^2 + y^2 + 6x - 24y + 72 = 0$ and $x^2 - y^2 + 6x + 16y - 46 = 0$ intersect at four points. The sum of distances from these four points to the point $(-3, 2)$ is $10\,k$. Then value of k is equal to _____.

88. Consider a series of '*n*' concentric circles $C_1, C_2, C_3, ..., C_n$ with radii $r_1, r_2, r_3, ..., r_n$ respectively, such that $r_1 > r_2 ...... > r_n$ and $r_1 = 20$. If the tangents drawn from any point on to C_{i+1} are such that the chord of contact is a tangent to $C_{i+2}\ (i = 1, 2, 3, ...)$ and the angle between the tangents from any point on C_1 to C_2 is $\dfrac{\pi}{3}$, then find the values of $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} r_i$.

89. As shown in the figure, three circles which have the same radius r have centers at $(0,0), (1,1), and\ (2,1)$. If they have a common tangent line, as shown, then the value of $10\sqrt{5}r$ is___.



90. If the circles $x^2 + y^2 + (3 + \sin\beta)x + (2\cos\alpha)y = 0$ and $x^2 + y^2 + (2\cos\alpha)x + 2cy = 0$ touch each other, then the maximum value of c is ______.

91. Let BD be the internal angle bisector of angle B in triangle ABC with D on side AC. The circumcircle of triangle BDC meets AB at E, while the circumcircle of triangle ABD meets BC at F., if $AE = 3$, then CF is equal to ______.

92. Six points $(x_i, y_i); i = 1,2,3,4,5,6$ are taken on the circle $x^2 + y^2 = 4$ such that $\sum_{i=1}^{6} x_i = 8$ and $\sum_{i=1}^{6} y_i = 4$. The line segment joining orthocenter of a triangle made by any three points and the centroid of the triangle made by other three points passes through a fixed point (h, k). The value of $h+k$ is ______.

93. The number of points of intersection of curve $\sin x = \cos y$ and circle $x^2 + y^2 = 1$.

94. Three circles touch one another externally. The tangents at their points of contact meet at a point whose distance from point of contact is 4. Find the ratio of the product of the radii to the double of the sum of the radii of the circles.

95. A circle passes through the point (3, 4) and cuts the circle $x^2 + y^2 = a^2$ orthogonally. The locus of its centre is a straight line. If the distance of the straight line from the origin is 817, then find the value of $a^2 - 8140$.

96. Let S_1 and S_2 denote the circles $x^2 + y^2 + 10x - 24y - 87 = 0$ and $x^2 + y^2 - 10x - 24y + 153 = 0$ respectively. (Let m be the smallest positive value of 'a' for which the line $y = ax$ contains the centre of a circle which touches S_2 externally and S_1 internally). Given that $m^2 = \frac{p}{q}$, where p and q are relatively prime integers, if $(p+q)$ is equal to 13^k, then the value of k is equal to ______.

97. A circle touches the hypotenuse of a right-angled triangle at its middle point and passes through the middle point of the shorter side. If 3 units and 4 units be the length of the sides and 'r' be the radius of the circle, then find the value of '$3r$'.

98. From a point 'P' on the normal $y = x + c$ of the circle $x^2 + y^2 - 2x - 4y + 5 - \lambda^2 = 0$, two tangents are drawn to the same circle touching it at points B and C. If the area of the quadrilateral $OBPC$ (where O is the centre of the circle), is 36 sq. units, the possible positive value of λ, (if it is given that the point P is at a distance of $|\lambda|(\sqrt{2}-1)$ from the circle) is ______.

99. The number of possible integral values of m for which the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y + m^2 = 0$ have exactly two common tangents is ___________.

100. Let PT be a tangent from the point $P(5, 3+\sqrt{3})$ to the circle $x^2 + y^2 + 4x - 6y - 3 = 0$, with centre C, at T and AB is secant which passes through P such that BT is the normal at T. If $Ar(\Delta CAB) + Ar(\Delta CAT) = \frac{\lambda}{25}$, then find the value of $\left(\left[\sqrt{\lambda}\right] - 15\right)$ ([.] denotes G.I.F).

101. If the two circles $(x-1)^2 + (y-3)^2 = r^2$ $(r > 0)$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points, then the number of odd positive integral values of r is____ .

Advanced Problem Package	Conic Section

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The locus of a point on the variable parabola $y^2 = 4ax$, whose distance from focus is constant k, is equal to: (a is parameter)

(A) $4x^2 + y^2 - 4kx = 0$ **(B)** $x^2 + y^2 - 4kx = 0$

(C) $x^2 + 2y^2 - 4kx = 0$ **(D)** $4x^2 - y^2 + 4kx = 0$

2. Let S be the focus of $y^2 = 4x$ and a point P is moving on the curve such that its abscissa is increasing at the rate of 4 units/sec, then the rate of increase of projection of SP on $x + y = 1$ when P is at (4, 4) is :

(A) $-\sqrt{2}$ **(B)** -1 **(C)** $\sqrt{2}$ **(D)** $-\dfrac{3}{\sqrt{2}}$

3. An equilateral triangle SAB is inscribed in the parabola $y^2 = 4ax$ having its focus at 'S'. If chord AB lies towards the left of S, then side length of this triangle is :

(A) $2a(2-\sqrt{3})$ **(B)** $4a(2-\sqrt{3})$ **(C)** $a(2-\sqrt{3})$ **(D)** $8a(2-\sqrt{3})$

4. $\min\left[(x_1 - x_2)^2 + \left(12 - \sqrt{1-x_1^2} - \sqrt{4x_2}\right)^2\right] \forall x_1, x_2 \in R$ is :

(A) $4\sqrt{5}+1$ **(B)** $4\sqrt{5}-1$ **(C)** $\sqrt{5}+1$ **(D)** $\sqrt{5}-1$

5. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is :

(A) $x^2 + 2y^2 - ax = 0$ **(B)** $2x^2 + y^2 - 2ax = 0$

(C) $2x^2 + 2y^2 - ay = 0$ **(D)** $2x^2 + y^2 - 2ay = 0$

6. If the ellipse $\dfrac{x^2}{4} + y^2 = 1$ meets the ellipse $x^2 + \dfrac{y^2}{a^2} = 1$ in four distinct points and $a = b^2 - 5b + 7$, then b does not lie in

(A) [4, 5] **(B)** $(-\infty, 2) \cup (3, \infty)$ **(C)** $(-\infty, 0)$ **(D)** [2, 3]

7. There are exactly two points on the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ whose distance from its centre is same and is equal to $\sqrt{\dfrac{a^2 + 2b^2}{2}}$. Then the eccentricity of the ellipse is :

(A) $\dfrac{1}{2}$ **(B)** $\dfrac{1}{\sqrt{2}}$ **(C)** $\dfrac{1}{3}$ **(D)** $\dfrac{1}{3\sqrt{2}}$

8. Let S and S' be two foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. If a circle described on *SS'* as diameter intersects the ellipse in real and distinct points, then the eccentricity *e* of the ellipse satisfies.

(A) $e=\frac{1}{\sqrt{2}}$ **(B)** $e\in(1/\sqrt{2},1)$ **(C)** $e\in(0,1/\sqrt{2})$ **(D)** None of these

9. From any point *P* lying in first quadrant on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, *PN* is drawn perpendicular to the major axis and produced to *Q* so that *NQ* equals to *PS*, where *S* is the focus (–3, 0). Then the locus of *Q* is :

(A) $5y-3x-25=0$ **(B)** $3x+5y+25=0$ **(C)** $3x-5y-25=0$ **(D)** None of these

10. Tangents are drawn to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a>b)$ and the circle $x^2+y^2=a^2$ at the points where a common ordinate cuts them (on the same side of the x-axis). Then the greatest acute angle between these tangents is given by :

(A) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ **(B)** $\tan^{-1}\left(\frac{a+b}{2\sqrt{ab}}\right)$ **(C)** $\tan^{-1}\left(\frac{2ab}{\sqrt{a-b}}\right)$ **(D)** $\tan^{-1}\left(\frac{2ab}{\sqrt{a+b}}\right)$

11. If the eccentricity of the hyperbola $x^2-y^2\sec^2\alpha=5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\alpha+y^2=25$, then a value of α is :

(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{3}$ **(D)** $\frac{\pi}{2}$

12. If a ray of light incident along the line $3x+(5-4\sqrt{2})y=15$ gets reflected from the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ at the point $(4\sqrt{2},3)$, then its reflected ray goes along the line :

(A) $\sqrt{2}x-y+5=0$ **(B)** $\sqrt{2}y-x+5=0$

(C) $\sqrt{2}y-x-5=0$ **(D)** None of these

13. If two distinct tangents can be drawn from the point $(\alpha,2)$ on different branches of the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$, then

(A) $|\alpha|<\frac{3}{2}$ **(B)** $|\alpha|>\frac{2}{3}$ **(C)** $|\alpha|>3$ **(D)** None of these

14. Area of the triangle formed by the asymptotes of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and any tangent to the hyperbola is $a^2\tan\lambda$ in magnitude then its eccentricity is :

(A) $\sec\lambda$ **(B)** $\operatorname{cosec}\lambda$ **(C)** $\sec^2\lambda$ **(D)** $\operatorname{cosec}^2\lambda$

15. $(x-1)(y-2)=5$ and $(x-1)^2+(y+2)^2=r^2$ intersect at four points *A, B, C, D* and if centroid of ΔABC lies on line $y=3x-4$, then locus of *D* is :

(A) $y=3x$ **(B)** $x^2+y^2+3x+1=0$ **(C)** $3y=x+1$ **(D)** $y=3x+1$

16. If S_1 and S_2 are the foci of the hyperbola whose transverse axis length is 4 and conjugate axis length is 6, S_3 and S_4 are the foci of the conjugate hyperbola, then the area of the quadrilateral $S_1S_2S_3S_4$ is :

(A) 24 **(B)** 26 **(C)** 22 **(D)** None of these

For Questions 17 - 19

Two tangents to a parabola are $x-y=0$ and $x+y=0$. If (2, 3) is focus of the parabola, then :

17. The equation of tangent at vertex is :

(A) $4x-6y+5=0$ **(B)** $4x-6y+3=0$ **(C)** $4x-6y+1=0$ **(D)** $4x-6y+3/2=0$

18. Length of latus rectum of the parabola is :

(A) $\frac{6}{\sqrt{13}}$ **(B)** $\frac{10}{\sqrt{13}}$ **(C)** $\frac{2}{\sqrt{13}}$ **(D)** None of these

19. If P, Q are ends of focal chord of the parabola, then $\frac{1}{SP}+\frac{1}{SQ}=$

(A) $\frac{2\sqrt{13}}{3}$ **(B)** $2\sqrt{13}$ **(C)** $\frac{2\sqrt{13}}{5}$ **(D)** None of these

For Questions 20 - 22

A curve is represented by $C \equiv 21x^2-6xy+29y^2+6x-58y-151=0$.

20. Eccentricity of curve is :

(A) 1/3 **(B)** $1/\sqrt{3}$ **(C)** 2/3 **(D)** $2/\sqrt{5}$

21. The lengths of axes are :

(A) $6, 2\sqrt{6}$ **(B)** $5, 2\sqrt{5}$ **(C)** $4, 4\sqrt{5}$ **(D)** None of these

22. The centre of the conic C is :

(A) (1, 0) **(B)** (0, 0) **(C)** (0, 1) **(D)** None of these

For Questions 23 - 25

Let $P(x, y)$ is a variable point such that $\left|\sqrt{(x-1)^2+(y-2)^2}-\sqrt{(x-5)^2+(y-5)^2}\right|=3$ which represents hyperbola.

23. The eccentricity e' of the corresponding conjugate hyperbola is :

(A) 5/3 **(B)** 4/3 **(C)** 5/4 **(D)** $3/\sqrt{7}$

24. Locus of intersection of two perpendicular tangents to the hyperbola is :

(A) $(x-3)^2+\left(y-\frac{7}{2}\right)^2=\frac{55}{4}$ **(B)** $(x-3)^2+\left(y-\frac{7}{2}\right)^2=\frac{25}{4}$

(C) $(x-3)^2+\left(y-\frac{7}{2}\right)^2=\frac{7}{4}$ **(D)** None of these

25. If origin is shifted to point $\left(3, \frac{7}{2}\right)$ and the axes are rotated through an angle θ in clockwise sense so that equation of given hyperbola changes to the standard form $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then θ is :

(A) $\tan^{-1}\left(\frac{4}{3}\right)$ **(B)** $\tan^{-1}\left(\frac{3}{4}\right)$ **(C)** $\tan^{-1}\left(\frac{5}{3}\right)$ **(D)** $\tan^{-1}\left(\frac{3}{5}\right)$

For Questions 26 - 28

Let $A\left(\frac{1}{2},0\right),B\left(\frac{3}{2},0\right),C\left(\frac{5}{2},0\right)$ be the given points and P be a point satisfying max $\left(PA+PB,PB+PC\right)<2$

26. All points P are points common to:

(A) two ellipse (B) two hyperbola

(C) a circle and an ellipse (D) a circle and an hyperbola

27. The locus of P is symmetric about:

(A) origin **(B)** the line $y=x$ **(C)** y-axis **(D)** x-axis

28. The area of region of the point P is:

(A) $\sqrt{2}\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$ **(B)** $\sqrt{3}\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$ **(C)** $2\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$ **(D)** $3\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$

For Questions 29 - 31

Consider a point P, such that 2 of the normal drawn from it to the parabola are at right angles, then:

29. If the equation of parabola is $y^2=8x$, then locus of P is:

(A) $x^2=4\left(y-6\right)$ **(B)** $y^2=2\left(x-6\right)$ **(C)** $y^2=8\left(x-6\right)$ **(D)** $2x^2=\left(y-6\right)$

30. The ratio of latus rectum of given parabola and that of made by locus of point P is:

(A) 4:1 **(B)** 2:1 **(C)** 16:1 **(D)** 1:1

31. If $P\equiv\left(x_1,y_1\right)$ the slope of third normal is:

(A) $\frac{y_1}{8}$ **(B)** $\frac{y_1}{2}$ **(C)** $-\frac{y_1}{8}$ **(D)** $-\frac{y_1}{2}$

For Questions 32 - 34

Let a hyperbola whose centre is at origin. A line $x+y=2$ touches this hyperbola at $P\left(1,1\right)$ and intersects the asymptotes at A and B such that $AB=6\sqrt{2}$ units. (you can use the concept that incase of hyperbola portion of tangent intercepted between asymptotes is bisected at the point of contact).

32. Equation of asymptotes are

(A) $5xy+2x^2+2y^2=0$ **(B)** $3x^2+2y^2+6xy=0$

(C) $2x^2+2y^2-5xy=0$ **(D)** none of these

33. Angle subtended by AB at centre of the hyperbola is

(A) $sin^{-1}\frac{4}{5}$ **(B)** $sin^{-1}\frac{2}{5}$ **(C)** $sin^{-1}\frac{3}{5}$ **(D)** none of these

34. Equation of the tangent to the hyperbola at $\left(-1,\frac{7}{2}\right)$ is

(A) $5x+2y=2$ **(B)** $3x+2y=4$ **(C)** $3x+4y=11$ **(D)** none of these

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

35. The equation of the directrix of the parabola with vertex at the origin and having the axis along the x-axis and a common tangent of slope 2 with the circle $x^2 + y^2 = 5$ is/are :

(A) $x = 10$ **(B)** $x = 20$ **(C)** $x = -10$ **(D)** $x = -20$

36. Tangent is drawn at any point (x_1, y_1) other than vertex on the parabola $y^2 = 4ax$. If tangents are drawn from any point on this tangent to the circle $x^2 + y^2 = a^2$ such that all the chords of contact pass through a fixed point (x_2, y_2), then :

(A) x_1, a, x_2 are in G.P. **(B)** $\frac{y_1}{2}, a, y_2$ are in G.P.

(C) $-4, \frac{y_1}{y_2}, \frac{x_1}{x_2}$ are in G.P. **(D)** $x_1x_2 + y_1y_2 = a^2$

37. If the focus of the parabola $x^2 - ky + 3 = 0$ is (0, 2), then a value of k is/are:

(A) 4 **(B)** 6 **(C)** 3 **(D)** 2

38. If $y = 2$ be the directrix and (0, 1) be the vertex of the parabola $x^2 + \lambda y + \mu = 0$ then :

(A) $\lambda = 4$ **(B)** $\mu = 8$ **(C)** $\lambda = -8$ **(D)** $\mu = -4$

39. The extremities of latus rectum of a parabola are (1, 1) and (1, –1), then the equation of the parabola can be :

(A) $y^2 = 2x - 1$ **(B)** $y^2 = 1 - 2x$ **(C)** $y^2 = -2x + 3$ **(D)** $y^2 = 2x - 3$

40. Parabola $y^2 = 4x$ and the circle having its centre at (6, 5) intersect at right angle. Possible point of intersection of these curves can be :

(A) (9, 6) **(B)** $(2, \sqrt{8})$ **(C)** (4, 4) **(D)** $(3, 2\sqrt{3})$

41. A normal drawn to parabola $y^2 = 4ax$ meet the curve again at Q such that angle subtended by PQ at vertex is 90º, then coordinates of P can be

(A) $(8a, 4\sqrt{2}a)$ **(B)** $(8a, 4a)$ **(C)** $(2a, -2\sqrt{2}a)$ **(D)** $(2a, 2\sqrt{2}a)$

42. The locus of the midpoint of the focal distance of a variable point moving on the parabola, $y^2 = 4ax$ is a parabola whose

(A) Latus rectum is half the latus rectum of the original parabola **(B)** Vertex is $(a/2, 0)$

(C) Directrix is y-axis **(D)** Focus is $(a, 0)$

43. $\frac{x^2}{r^2 - r - 6} + \frac{y^2}{r^2 - 6r + 5} = 1$ will represents the ellipse, if r lies in the interval

(A) $(-\infty, -2)$ **(B)** $(3, \infty)$ **(C)** $(5, \infty)$ **(D)** $(1, \infty)$

44. The co-ordinates (2, 3) and (1, 5) are the foci of an ellipse which passes through the origin, then the equation of

(A) Tangent at the origin is $(3\sqrt{2} - 5)x + (1 - 2\sqrt{2})y = 0$

(B) Tangent at the origin is $(3\sqrt{2} + 5)x - (1 + 2\sqrt{2})y = 0$

(C) Normal at the origin is $(3\sqrt{2} + 5)x - (2\sqrt{2} + 1)y = 0$

(D) Normal at the origin is $(3\sqrt{2} - 5)x + (1 - 2\sqrt{2})y = 0$

45. If a pair of variable straight lines $x^2+4y^2+\alpha xy=0$ (where α is a real parameter) cut the ellipse $x^2+4y^2=4$ at two points A and B, then the locus of the point of intersection of tangents at A and B is :

(A) $x-2y=0$ **(B)** $2x-y=0$ **(C)** $x+2y=0$ **(D)** $2x+y=0$

46. Which of the following is/are true?

(A) There are infinite positive integral values of a for which $(13x-1)^2+(13y-2)^2=\left(\frac{5x+12y-1}{a}\right)^2$ represents an ellipse

(B) The minimum distance of a point (1, 2) from the ellipse $4x^2+9y^2+8x-36y+4=0$ is 1

(C) If from a point $P(0,\alpha)$ (P is not the origin) two normals other than axes are drawn to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, then $|\alpha|<\frac{9}{4}$

(D) If the length of latus rectum of an ellipse is one-third of its major axis, then its eccentricity is equal to $\frac{1}{\sqrt{3}}$

47. Let E_1 and E_2 be two ellipses $\frac{x^2}{a^2}+y^2=1$ and $x^2+\frac{y^2}{a^2}=1$ (where a is a parameter). Then the locus of the points of intersection of the ellipses E_1 and E_2 is a set of curves comprising

(A) Two straight lines **(B)** One straight line

(C) One circle **(D)** One parabola

48. Consider the ellipse $\frac{x^2}{f(k^2+2k+5)}+\frac{y^2}{f(k+11)}=1$ and $f(x)$ is a positive decreasing function, then :

(A) The set of values of k, for which the major axis is x-axis is $(-3, 2)$

(B) The set of values of k, for which the major axis is y-axis is $(-\infty, 2)$

(C) The set of values of k, for which the major axis is y-axis is $(-\infty,-3)\cup(2,\infty)$

(D) The set of values of k, for which the major axis is y-axis is $(-3,\infty)$

49. If two concentric ellipses are such that the foci of each one are on the other and their major axes are equal. Let e and e' be their eccentricities, then

(A) The quadrilateral formed by joining the foci of the two ellipses is a parallelogram

(B) The angle θ between their axes is given by $\theta=\cos^{-1}\sqrt{\frac{1}{e^2}+\frac{1}{e'^2}-\frac{1}{e^2e'^2}}$

(C) If $e^2+e'^2=1$, then the angle between the axis of the two ellipses is 90º

(D) If $e+e'=1$, then the angle between the axis of the two ellipses is 90º

50. If the tangent drawn at point $(t^2, 2t)$ on the parabola $y^2=4x$ is same as the normal drawn at point $(\sqrt{5}\cos\theta, 2\sin\theta)$ on the ellipse $4x^2+5y^2=20$. Then :

(A) $\theta=\cos^{-1}\left(-\frac{1}{\sqrt{5}}\right)$ **(B)** $\theta=\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ **(C)** $t=-\frac{2}{\sqrt{5}}$ **(D)** $t=-\frac{1}{\sqrt{5}}$

51. The equation $\left|\sqrt{x^2+(y-1)^2}-\sqrt{x^2+(y+1)^2}\right|=K$ will represent a hyperbola for

(A) $K\in(0,2)$ **(B)** $K\in(-2,1)$ **(C)** $K\in(1,\infty)$ **(D)** $K\in(2,\infty)$

52. If $x,y\in R$ then the equation $3x^4-2(19y+8)x^2+(361y^2+2(100+y^4)+64)=2(190y+2y^2)$ represents in rectangular Cartesian system:

(A) parabola **(B)** hyperbola **(C)** circle **(D)** ellipse

53. For which of the following hyperbolas, we can have more than one pair of perpendicular tangents?

(A) $\frac{x^2}{4}-\frac{y^2}{9}=1$ **(B)** $\frac{x^2}{4}-\frac{y^2}{9}=-1$ **(C)** $x^2-y^2=4$ **(D)** $xy=44$

54. For the hyperbola $9x^2-16y^2-18x+32y-151=0$

(A) one of the directrix is $x = 21/5$ **(B)** length of latus rectum = 9/2

(C) foci are (6, 1) and (–4, 1) **(D)** eccentricity is 5/4

55. Circles are drawn on chords of the rectangular hyperbola $xy=4$ parallel to the line $y=x$ as diameters. All such circles pass through two fixed points whose coordinates are

(A) (2, 2) **(B)** (2, –2) **(C)** (–2, 2) **(D)** (–2, –2)

56. The equation $(x-\alpha)^2+(y-\beta)^2=k(lx+my+n)^2$ represents

(A) a parabola for $k<(l^2+m^2)^{-1}$ **(B)** an ellipse for $0<k<(l^2+m^2)^{-1}$

(C) a hyperbola for $k>(l^2+m^2)^{-1}$ **(D)** a point circle for $k=0$

57. If P is a point on a hyperbola, then

(A) Locus of excentre of the circle described opposite to $\angle P$ for $\Delta PSS'(S, S'$ are foci), is tangent at vertex

(B) Locus of excentre of the circle described opposite to $\angle S'$ is hyperbola

(C) Locus of excentre of the circle described opposite to $\angle P$ for $\Delta PSS'(S, S'$ are foci), is hyperbola

(D) Locus of excentre of the circle described opposite to $\angle S'$, is tangent at vertex

58. From point (2, 2) tangents are drawn to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ then point of contact lies in

(A) I quadrant **(B)** II quadrant **(C)** III quadrant **(D)** IV quadrant

59. For hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, let n be the number of points on the plane through which perpendicular tangents are drawn.

(A) if $n = 1$, then $e=\sqrt{2}$ **(B)** if n> 1, then $1<e<\sqrt{2}$

(C) if n = 0, then $e>\sqrt{2}$ **(D)** if $n>1$, then $e>\sqrt{2}$

60. From the points (x_1, y_1) and (x_2, y_2) tangents are drawn to the hyperbola $xy=c^2$, such that a circle passes through these points and the four points of contact, then:

(A) $x_1y_1=x_2y_2$ **(B)** $x_1x_2=y_2y_2$

(C) $x_1y_2+x_2y_1=4c^2$ **(D)** $x_1y_1+x_2y_2=4c^2$

61. Equations of the asymptotes of the hyperbola whose equation is given by $x=a\tan(\theta+\alpha)$ and $y=b\tan(\theta+\beta)$, θ being a parameter, is/are:

(A) $by=ax\tan(\alpha-\beta)$ **(B)** $y=ax\tan(\alpha-\beta)$

(C) $x+a\cot(\alpha-\beta)=0$ **(D)** $y-b\cot(\alpha-\beta)=0$

62. If equation of tangent at *P*, *Q* and vertex *A* of a parabola are $3x+4y-7=0, 2x+3y-10=0$ and $x-y=0$ respectively, then:

(A) Focus is (4, 5) **(B)** Length of latus rectum is $2\sqrt{2}$

(C) Axis is $x+y-9=0$ **(D)** Vertex is $\left(\frac{9}{2},\frac{9}{2}\right)$

63. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If area $(\Delta PVQ) = 20\,unit^2$ then the coordinates of P is/are

(A) $(16,8)$ **(B)** $(16,-8)$ **(C)** $(-16,8)$ **(D)** $(-16,-8)$

64. The chord AB of the parabola $y^2 = 4ax$ cuts the axis of the parabola at C (C is internal to AB). If $A \equiv (at_1^2, 2at_1)$ and $B \equiv (at_2^2, 2at_2)$ and $AC : AB = 1:3$, then:

(A) $t_2 = 2t_1$ **(B)** $t_2 + 2t_1 = 0$ **(C)** $t_1 + 2t_2 = 0$ **(D)** $6t_1^2 = t_2(t_1 + 2t_2)$

65. Variable circle is described to pass through point $(1,0)$ and tangent to the curve $y = tan(tan^{-1}x)$. The locus of the centre of the circle is a parabola whose:

(A) length of the latus rectum is $2\sqrt{2}$ **(B)** axis of symmetry has the equation $x + y = 1$

(C) vertex has the co-ordinates $\left(\frac{3}{4}, \frac{1}{4}\right)$ **(D)** length of the latus rectum is $\sqrt{2}$

66. The range of α for which the points $(\alpha, 2+\alpha)$ and $\left(\frac{3}{2}\alpha, \alpha^2\right)$ lie on opposite sides of the line $2x + 3y = 6$ can lie in intervals:

(A) $(-\infty,-2)$ **(B)** $(-2,0)$ **(C)** $(0,1)$ **(D)** $(2,4)$

67. Let $y^2 = 4ax$ be a parabola and PQ be a focal chord. Let R be the point of intersection of the tangents at P and Q, then:

(A) area of circumcircle of ΔPQR is $\frac{\pi (PQ)^2}{4}$ **(B)** orthocenter of ΔPQR lies at the directrix

(C) incentre of ΔPQR lies at the vertex **(D)** minimum area of the circumcircle a ΔPQR is $4\pi a^2$

68. The normal drawn at the extermities P and Q of a focal chord meet the parabola again in P' and Q' respectively. Then**:**

(A) PQ and $P'Q'$ are perpendicular **(B)** PQ and $P'Q'$ are parallel

(C) $P'Q' = 3PQ$ **(D)** $P'Q' = 2\sqrt{3}PQ$

69. The parabolas $y^2 = 4ax$ and $y^2 = 4c(x-d)$ have a common normal other than the X-axis if and only if:

(A) $c > a$ and $2a > d + 2c$ **(B)** $c < a$ and $2a > d + 2c$

(C) $c > a$ and $2a < d + 2c$ **(D)** $c < a$ and $2a < d + 2c$

70. Let O be the vertex of a parabola and Q be any point on the axis of the parabola. If PQR be any chord passing through Q and PM and RN be the ordinates of P and R, then:

(A) $OM.ON = OQ^2$ **(B)** $OM.ON = 4aOQ$

(C) $PM.RN = 4aOQ$ **(D)** $PM.RN = OQ^2$

71. Coordinates of the feet of normal drawn from the point (7, 14) to the parabola $x^2 - 8x - 16y = 0$ is/are:

(A) $(0,0)$ **(B)** $(-4,3)$ **(C)** $(4,-1)$ **(D)** $(16,8)$

72. The values of a for which $y = ax^2 + ax + \frac{1}{24}$, $x = ay^2 + ay + \frac{1}{24}$ touch each other is/are

(A) $\frac{2}{3}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{13+\sqrt{601}}{12}$ **(D)** $\frac{13-\sqrt{601}}{12}$

73. If P is any point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose foci are S_1 and S_2. Let $\angle PS_1S_2=\alpha$ and $\angle PS_2S_1=\beta$ then,

(A) $PS_1+PS_2=2a$, if $a>b$ **(B)** $PS_1+PS_2=2b$, if $a<b$

(C) $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2}=\frac{1-e}{1+e}$ **(D)** $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2}=\frac{\sqrt{a^2-b^2}}{b^2}\left[a-\sqrt{a^2-b^2}\right]$

74. A line $3x+y=8$ touches a hyperbola $H=0$ at $P(1,5)$ meets its asymptotes at A and B . If $AB=2\sqrt{10}, C(1,1)$ be the centre of hyperbola, e and l are eccentricity and latus rectum of hyperbola then

(A) $e=\frac{\sqrt{7}}{2}$ **(B)** $e=\frac{\sqrt{5}}{2}$ **(C)** $l=\sqrt{2}$ **(D)** $l=2\sqrt{2}$

75. Two tangents $2x+y=2$ and $x-2y=3$ to a parabola touching it at $A(2,-2)$ and $B(5,1)$ respectively. If focus of parabola is $S(\alpha,\beta)$ and latus rectum length is L, then:

(A) $\alpha-\beta=3$ **(B)** $\alpha-\beta=4$ **(C)** $L=\frac{27\sqrt{3}}{25}$ **(D)** $L=\frac{27\sqrt{2}}{25}$

76. Let $P(x_1,y_1)$ and $Q(x_2,y_2), y_1<0, y_2<0$ be the end points of the latus rectum of the ellipse $x^2+4y^2=4$. The equation of the parabolas with latus rectum PQ are:

(A) $x^2+2\sqrt{3}y=3+\sqrt{3}$ **(B)** $x^2-2\sqrt{3}y=3+\sqrt{3}$

(C) $x^2+2\sqrt{3}y=3-\sqrt{3}$ **(D)** $x^2-2\sqrt{3}y=3-\sqrt{3}$

77. Tangents are drawn from the point $(-2,0)$ to the parabola $y^2=8x$, radius of circle (s) that would touch these tangents and the corresponding chord of contact, can be equal to

(A) $4(\sqrt{2}+1)$ (B) $4(\sqrt{2}-1)$ (C) $8\sqrt{2}$ (D) none of these

78. If two distinct chords of the parabola $y^2=4ax$, passing through $(a,2a)$ are bisected on the line $x+y=1$, then length of the latus-rectum can be

(A) 2 **(B)** 1 **(C)** 4 (D) 5

79. The locus of point of intersection of any tangent to the parabola $y^2=4a(x-2)$ with a line perpendicular to it and passing through the focus, is

(A) the tangent to the parabola at the vertex **(B)** $x=2$

(C) $x=0$ **(D)** none of these

80. The tangent to the circle $x^2+y^2=4$ at a point P intersect the parabola $y^2=4x$ at points Q and R. Tangents to the parabola at Q and R intersect at S. If Q lies in the first quadrant such that $PQ=1$ units, then:

(A) $PR=8\,units$ **(B)** $SR=\frac{35}{4}units$

(C) $SQ=\frac{7}{2}units$ **(D)** area of $\Delta SQR=\frac{343}{16}sq.units$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column I are labelled as (A), (B), (C) & (D) whereas statements in Column II are labeled as p, q, r, s & t. More than one choice from Column II can be matched with Column I.

81. Consider the parabola $(x-1)^2+(y-2)^2=\dfrac{(12x-5y+3)^2}{169}$:

Column 1		Column 2	
(A)	Locus of point of intersection of perpendicular tangent	**(p)**	$12x-5y-2=0$
(B)	Locus of foot of perpendicular from focus upon any tangent	**(q)**	$5x+12y-29=0$
(C)	Line along which minimum length of focal chord occurs	**(r)**	$12x-5y+3=0$
(D)	Line about which parabola is symmetrical	**(s)**	$24x-10y+1=0$

82. **MATCH THE FOLLOWING:**

Column 1		Column 2	
(A)	Points from which perpendicular tangents can be drawn to parabola $y^2=4x$	**(p)**	(–1, 2)
(B)	Points from which only one normal can be drawn to parabola $y^2=4x$	**(q)**	(3, 2)
(C)	Points at which chord $x-y-1=0$ of parabola $y^2=4x$ is bisected	**(r)**	(–1,–5)
(D)	Points from which tangents cannot be drawn to parabola $y^2=4x$	**(s)**	(5, –2)

83. **MATCH THE FOLLOWING:**

Column 1		Column 2	
(A)	Distance between the points on the curve $4x^2+9y^2=1$, where tangent is parallel to the line $8x=9y$, is less than	**(p)**	1
(B)	Sum of distances of the foci of the curve $25(x+1)^2+9(y+2)^2=225$ from $(-1,0)$ is more than	**(q)**	4
(C)	Sum of distances from the x-axis of the points on the ellipse $\dfrac{x^2}{9}+\dfrac{y^2}{4}=1$, where the normal is parallel to the line $2x+y=1$, is less than	**(r)**	7
(D)	Tangents are drawn from points on the line $x-y+2=0$ to the ellipse $x^2+2y^2=2$, then all the chords of contact pass through the point whose distance from (2, ½) is more than	**(s)**	5

84. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	If vertices of a rectangle of maximum area inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are extremities of latus rectum. Then eccentricity of ellipse is	**(p)**	$\frac{2}{\sqrt{5}}$
(B)	If extremities of diameter of the circle $x^2+y^2=16$ are foci of a ellipse, then eccentricity of the ellipse, if its size is just sufficient to contain the circle, is	**(q)**	$\frac{1}{\sqrt{2}}$
(C)	If normal at point (6, 2) to the ellipse passes through its nearest focus (5, 2), having centre at (4, 2) then its eccentricity is	**(r)**	$\frac{1}{3}$
(D)	If extremities of latus rectum of the parabola $y^2=24x$ are foci of ellipse and if ellipse passes through the vertex of the parabola, then its eccentricity is	**(s)**	$\frac{1}{2}$

85. If e_1 and e_2 are the roots of the equation $x^2-ax+2=0$, then match the following.

	Column 1		Column 2
(A)	If e_1 and e_2 are the eccentricities of the ellipse and hyperbola, respectively then the values of a are	**(p)**	6
(B)	If both e_1 and e_2 are the eccentricities of the hyperbolas, then values of a are	**(q)**	$2\sqrt{2}+10^{-3}$
(C)	If e_1 and e_2 are eccentricities of hyperbola and conjugate hyperbola, then values of a are	**(r)**	$2\sqrt{2}$
(D)	If e_1 is the eccentricity of the hyperbola for which no such points exist from which perpendicular tangents can be drawn, then the values of a are	**(s)**	5

86. Match the following:

	List 1		List 2
P.	The normal at an end of a latusrectum of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through an end of the minor axis if e^4 is equal to	**1.**	$\frac{1}{9}$
Q.	PQ is a double ordinate of a parabola $y^2=4ax$. If the locus of its point of trisection is another parabola length of whose latus rectum is k times the length of the latus rectum of the given parabola then k is equal to	**2.**	$\frac{1}{a}$
R.	If e and e' are the distances of the extremities of any focal chord from the focus f of the parabola $y^2=4ax$, then $\frac{1}{e}+\frac{1}{e'}$ is equal to	**3.**	1
S	If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e}+\frac{1}{e'^2}$ is equal to	**4.**	$1-e^2$

Codes:

	P	Q	R	S		P	Q	R	S
(A)	4	1	2	3	**(B)**	2	3	4	1
(C)	1	2	3	4	**(D)**	2	4	3	1

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

87. If the normals to the curve $y = x^2$ at the points P, Q & R passes through the point $(0, 3/2)$, find the radius of the circle circumscribing ΔPQR.

88. At any point P on the parabola $y^2 - 2y - 4x + 5 = 0$, a tangent is drawn which meets the directrix at Q. If the locus of R which divides QP externally in the ratio 1 : 2 is $(y-1)^2(x+1)+\lambda = 0$, then find λ.

89. The chord AC of the parabola $y^2 = 4ax$ subtends an angle of 90° at points B & D on the parabola. If A, B, C and D are represented by $t_1, t_2, t_3 \,\&\, t_4$, then find the value of $\left|\dfrac{t_2+t_4}{t_1+t_3}\right|$.

90. The straight line $ax+by+c=0$ cuts the locus of point of intersection of the lines $\frac{tx}{4}-\frac{y}{3}+t=0, \frac{x}{4}+\frac{ty}{3}-t=0$ at A & B such that line AB subtends a right angle at the origin, then $\left[\dfrac{3a-4b}{c}\right]$ is _____. ([.] represents greatest integer function)

91. Let P, Q be two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose eccentric angles differ by a right angle. The tangents at P and Q meet at R. If the chord PQ divides the line segment CR in m : n, then find *m*/*n* (where C is the centre of ellipse).

92. The length of the major axis of the ellipse $(5x-10)^2+(5y+15)^2 = \frac{(3x-4y+7)^2}{4}$ is A. Find [A]. ([.] represents greatest integer function).

93. Find the number of distinct normals that can be drawn to the ellipse $\frac{x^2}{169}+\frac{y^2}{25}=1$ from the point $P(0, 6)$.

94. If e be the eccentricity of a hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola, then the value of $\int_1^3 \frac{f\,f\,f\,..............f(e)}{n\ times}\,de$ is (n is even)

95. A normal to the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$ has equal intercepts on positive x and y-axes. If this normal touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then find $[a^2+b^2]$ ([.] represents greatest integer function).

96. If k be the length of the latus rectum of the hyperbola $16x^2-9y^2+32x+36y-164=0$, then find 3k/8.

97. From a point P three normal are drawn to the parabola $y^2 = 4ax$, such that the product of slopes of two of the normal is p. If the locus of P is a part of the parabola, then $|p|$ equal to

98. Tangent is drawn at any fixed point (x_1, y_1) on the parabola $y^2 = 4ax$. Now tangents are drawn from any point on this tangent to the circle $x^2 + y^2 = a^2$ so that all the chords of contact pass through a fixed point (x_2, y_2).

If $4\left(\frac{x_1}{x_2}\right)+\left(\frac{y_1}{y_2}\right)^2 = ka^2$, then k equals to

99. A chord cut the same branch of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ in P, P' and the asymptotes in Q, Q', then the value of $(PQ+PQ')-(P'Q'+P'Q)$ is_____ .

100. A normal to the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$, has equal intercepts on positive x and positive y axis. If this normal touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then the value of $\frac{9}{25}\left(a^2+b^2\right)$ is____ .

101. Variable pairs of chords at right angles are drawn through any point P (with eccentric angle $\frac{\pi}{4}$) on the ellipse $\frac{x^2}{4}+y^2=1$, to meet the ellipse at two points say A and B. If the line joining A and B passes through a fixed point $Q(a,b)$ such that a^2+b^2 has the value equal to $\frac{m}{n}$, where m, n are relatively prime positive integers, then the value of $\frac{m+n}{3}$ is____ .

102. A normal is drawn to the ellipse $\frac{x^2}{\left(a^2+2a+2\right)^2}+\frac{y^2}{\left(a^2+1\right)^2}=1$, $a>0$ whose centre is at O. If maximum radius of the circle, centered at the origin and touching the normal, is 5 then the positive value of 'a' is….

103. If the normal at the points where the straight line $lx+my=1$ meet the parabola $y^2=4ax$, meet at the point (h, k) on the parabola $y^2=4ax$, then $\frac{kl}{am}$ is equal to____ .

104. 'O' is the vertex of parabola $y^2=4x$ and L is the upper end of latus rectum. If LH is drawn perpendicular to OL meeting x-axis in H, then length of double ordinate through H is $\sqrt{N}$, then $N =$

105. The straight line $\frac{lx}{a}+\frac{my}{b}=n$ meet the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at in the points P and Q. If OP and OQ are along a pair of semi-conjugate diameters, O being the centre of the ellipse, then $\frac{l^2}{n^2}+\frac{m^2}{n^2}$ equals____.

106. From a point P, perpendicular tangents are drawn to the ellipse $x^2+2y^2=2$. If the chords of contact are tangents to a family of concentric circles, having the centres same as that of the ellipse then the ratio of the areas of the largest circle to the smallest circle is____.

107. The tangent to the hyperbola $xy=1$ at the point P intersects the X-axis in T and the Y–axis in T'. The normal to the hyperbola at P intersects the X-axis in N and the Y-axis in N'. Let the areas of the triangles PNT and $PN'T'$ are Δ and Δ' respectively, If $\frac{1}{\Delta}+\frac{1}{\Delta'}$ is constant for all positions of P then the value of constant is _____.

108. The tangents at a point P to the rectangular hyperbola $xy=1$ meets the lines $x-y=0$ *and* $x+y=0$ at Q and R respectively and Δ_1 is the area of the triangle OQR, where O is the origin. The normal at P meets X-axis at M and the Y-axis at N and Δ_2 is the area of the triangle OMN, then the value of $\Delta_1^2\Delta_2$ is ….

109. Transverse and conjugate axes of a rectangular hyperbola are along X-axis and Y-axis respectively and the distance between the foci is $10\sqrt{14}$. Number of the points (x, y) on the curve such that x and y are positive integers, is equal to _____.

110. The normal at four points A, B, C and D on the rectangular hyperbola $xy=c^2$ meet in $P(h,k)$ and $PA^2+PB^2+PC^2+PD^2=n\left(h^2+k^2\right)$, where '$n$' is equal to

111. If the point $\left(\lambda^2,\lambda-2\right)$ is a point lying interior of the region bounded by the parabola $y^2=2x$ and the chord joining the point (2, 2) and (8, –4), then the number of the integral values of λ is ____.

112. The straight line $\frac{x}{4}+\frac{y}{3}=1$ intersects the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at two points A and B, there is a point P on this ellipse such that the area of ΔPAB is equal to $6\left(\sqrt{2}-1\right)$. Then the number of such points P is____.

113. The length of the sub-tangent to the hyperbola $x^2-4y^2=4$ corresponding to the normal having slope unity is $\frac{1}{\sqrt{k}}$, then k is equal to _____.

114. If the normals to curve $y=x^2$ at the points P, Q and R pass through the point $\left(0,\frac{3}{2}\right)$, then the radius of the circle circumscribing ΔPQR is___.

115. Consider a parabola $4y=x^2$ and point $B(0,1)$. Let $A_1\left(x_1,y_1\right),A_2\left(x_2,y_2\right),..............,A_n\left(x_n,y_n\right)$ are n points on the parabola such that $x_r>0$ and $\angle OBA_r=\frac{r\pi}{2n}(r=1,2,3,.........,n)$ then $\pi\left(\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}BA_r\right)$ is equal to ____.

116. If the equation on reflection of $\frac{(x-4)^2}{16}+\frac{(y-3)^2}{9}=1$ about the line $x-y-2=0$ is $16x^2+9y^2+k_1x-36y+k_2=0$ then $\frac{k_1+k_2}{100}$ is___ .

Advanced Problem Package	Functions

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The reflection about the line $x + y = 0$ of the inverse function $f^{-1}(x)$ of a function $f(x)$ is :

(A) $-f(x)$ **(B)** $-f(-x)$ **(C)** $-f^{-1}(x)$ **(D)** $-f^{-1}(-x)$

2. The range of $f(x) = tan^{-1}x + \frac{1}{2}sin^{-1}x$ is :

(A) $(-\pi, \pi)$ **(B)** $\left[-\frac{3\pi}{4}, \frac{3\pi}{4}\right]$ **(C)** $\left(-\frac{3\pi}{4}, \frac{3\pi}{4}\right)$ **(D)** $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

3. Let $f(x)$ be any function. The graphs of $y = f(x-1)$ and $y = f(-x+1)$ are symmetric about the line :

(A) $y = 0$ **(B)** $x = 0$ **(C)** $x = 1$ **(D)** $x = -1$

4. Let $f(x)$ be such that $f(x+2) = f(x)$ and $f(-x) = f(x)$ for any real number x. On the interval $[2, 3]$, $f(x) = x$. Then the formula of $f(x)$ given on $[-2, 0]$ is :

(A) $x + 4$ **(B)** $2 - x$ **(C)** $3 - |x+1|$ **(D)** $2 + |x+1|$

5. Suppose $f(x) = x^3 + log_2\left(x + \sqrt{x^2+1}\right)$. For any $a, b \in R$, to satisfy $f(a) + f(b) \geq 0$, the condition $a + b \geq 0$ is:

(A) Necessary and sufficient **(B)** Necessary but not sufficient
(C) Not necessary but sufficient **(D)** Neither necessary nor sufficient

6. Let $f(x) = sin^4 x - sin x cos x + cos^4 x$, then range of $f(x)$ is :

(A) $\left[0, \frac{9}{8}\right]$ **(B)** $\left[-\frac{9}{8}, 0\right]$ **(C)** $\left(0, \frac{9}{8}\right)$ **(D)** $\left(-\frac{9}{8}, 0\right)$

7. Let $f(x) = 1 + 2cos x + 3 sin x$. If real numbers a, b, c are such that $a\,f(x) + b\,f(c-x) = 1$ holds for any $x \in R$ then $\frac{b cos c}{a} =$

(A) 1 **(B)** – 1 **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

8. A function $f : R \to R$ has property $f(x+y) = f(x) \cdot e^{f(y)-1}$, for every $x, y \in R$ then positive value of $f(4)$ is :

(A) 1 **(B)** 2 **(C)** 4 **(D)** 8

9. The number of positive integers x that satisfy $3^x = x^3 + 3x^2 + 2x + 1$ is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 4

10. It is given that the polynomial $P(x) = x^3 + ax^2 + bx + c$ has three distinct positive integer roots and $P(22) = 21$. Let $Q(x) = x^2 - 2x + 22$. It is also given that $P(Q(x))$ has no real roots then a is equal to :

(A) – 45 **(B)** – 55 **(C)** 45 **(D)** 60

11. The graph of the function $f(x)=\dfrac{9x+7}{3x+12}$ is symmetric to the point :

(A) $(-4,3)$ **(B)** $(-3,4)$ **(C)** $(3,4)$ **(D)** $(-3,-4)$

12. The only real solution to the equation $\left(x^2+100\right)^2=\left(x^3-100\right)^3$ has how many digits in base 10 representation?

(A) one **(B)** two **(C)** three **(D)** four

13. The equation $\sin(\cos x)=x$ has only one root x_1 in $(0,\pi/2)$ and the equation $\cos(\sin x)=x$ has also only one root x_2 in $(0,\pi/2)$. Then :

(A) $x_1>x_2$ **(B)** $x_1<x_2$ **(C)** $x_1=x_2$ **(D)** $x_1=2x_2$

14. Least value of the expression $\dfrac{1}{2bx-\left(x^2+b^2+\sin^2 x\right)}$, $x\in[-1,0]$, $b\in[2,3]$ is :

(A) $\dfrac{1}{4}$ **(B)** $-\dfrac{1}{4}$ **(C)** $\dfrac{-1}{8+\sin^2 1}$ **(D)** None of these

15. Let $f(x)=\left([a]^2-5[a]+4\right)x^3-\left(6[a]^2-5[a]+1\right)x-(\tan x)\operatorname{sgn} x$, be an even function for all $x\in-\left\{(2n+1)\dfrac{\pi}{2};n\in Z\right\}$, then sum of all possible values of a is : (where [.] {.} denotes greatest integer function and fractional part functions, respectively)

(A) $\dfrac{17}{6}$ **(B)** $\dfrac{53}{6}$ **(C)** $\dfrac{35}{3}$ **(D)** None of these

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

16. Define functions $f,g:R\to R$, $f(x)=3x-1+|2x+1|$ and $g(x)=\dfrac{1}{5}(3x+5-|2x+5|)$ then

(A) $f(g(x))=g(f(x))$ **(B)** $(f(f(x)))=g(g(x))$

(C) $f(g(x))=x$ **(D)** for $x<-\dfrac{5}{2}, f(g(x))=x$

17. The function $f(x)$ satisfies $f(10+x)=f(10-x)$ and $f(20-x)=-f(20+x)$. Which of the following statements about $f(x)$ is true?

(A) Periodic function **(B)** Not periodic **(C)** odd function **(D)** Even function

18. Given the system of equations $[3x]+\{y\}+x-y=1$ and $[-y]-\{x\}-x+y=1$, [x] denotes greatest integer $\le$ x, and $\{x\}$ denotes fractional part of x then

(A) $x=-1$ **(B)** $y=-5$ **(C)** $\{x\}=\{y\}$ **(D)** $x+y=4$

19. Consider the function $f(x)=\dfrac{\log x}{x}$

(A) $f(x)$ has horizontal tangent at $x=e$ **(B)** $f(x)$ cuts the x-axis at only one point

(C) $f(x)$ is many-one function **(D)** $f(x)$ has one vertical tangent

20. Which of the following is periodic?

(A) $\operatorname{sgn}\left(e^{-x}\right)$ **(B)** $\sin x+|\sin x|$

(C) minimum $(|x|, \sin x)$ **(D)** $2[-x]+\left[x+\frac{1}{2}\right]+\left[x-\frac{1}{2}\right]$

21. The function $f(x)=\frac{|x-1|}{x^2}$ is one – one in

(A) $(2, \infty)$ **(B)** $(1, 2)$ **(C)** $(0, 1)$ **(D)** $(-\infty, 0)$

22. Consider $f(x)=x\left[x^2\right]+\frac{1}{\sqrt{1-x^2}}$

(A) $f(x)$ is an even function **(B)** $f(x)$ is an odd function

(C) $f(x)$ is periodic **(D)** Range of $f(x)$ has only positive values

23. $f: R \to R,\ f(x)=a_1x+a_3x^3+a_5x^5+..........+a_{2n+1}x^{2n+1}-\cot x$ where $0<a_1<a_3<.........<a_{2n+1}$ then $f(x)$ is

(A) one-one **(B)** many-one **(C)** onto **(D)** into

24. Consider $f(x)=\log_2\left(\frac{1-x}{1+x}\right)-\log_2\left(x+\sqrt{x^2+1}\right)\cdot$ Then :

(A) $f'(0)=0$ **(B)** $f''(0)=0$ **(C)** $f'''(0)=0$ **(D)** $f^{IV}(0)=0$

25. The range of the function $f(x)=x\{x\}-x[-x]$ does not contain, [x] denotes greatest integer $\le$ x, and {x} denotes fractional part of x :

(A) – 1 **(B)** – 2 **(C)** 1 **(D)** 2

26. Let $f(x)=\frac{\sin \pi x}{x}, x \in (0,1)$ and $g(x)=f(x)+f(1-x)$ then :

(A) $g(x)=\frac{1}{2}f\left(\frac{x}{2}\right)f\left(\frac{1-x}{2}\right)$ **(B)** $g(x)=f\left(\frac{x}{2}\right)f\left(\frac{1-x}{2}\right)$

(C) $g(x)$ is symmetric about the y-axis **(D)** $g(x)$ is symmetric about the line $2x-1=0$

27. Let $f: R-\{0\} \to R, f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$

(A) $f(x)$ is even **(B)** $f(x)$ is many-one

(C) $f(x)$ is into **(D)** $\lim\limits_{x\to 0} f(x)=2$

28. Let $f(x)=2+\sqrt{x}$ and $g(x)=\frac{2x}{x^2+1}$, then :

(A) Domain $(f+g+2)=(-1, \infty)$ **(B)** Domain $(f+g+2)=[0, \infty)$

(C) Range $f \cap$ range $(g+2)=[2, 3]$ **(D)** Range $f \cup$ range $(g+2)=[1, \infty)$

29. Let $f(x)=\sin x+\sin\left(x\sqrt{3}\right)$. Then, which of the following are false ?

(A) Maximum value of $f(x)$ cannot be 2 **(B)** Maximum value of $f(x)$ cannot be -2

(C) $f(x)$ is periodic function with period $2\sqrt{3}\pi$ **(D)** $f(x)>0\ \forall\ x \in R$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

30. Equation $e^x = x^n,\ n \in I^+$

	Column 1		**Column 2 (Number of real roots)**
(A)	$n = 1$	**(p)**	3
(B)	$n = 2$	**(q)**	2
(C)	odd $n \geq 3$	**(r)**	1
(D)	even $n \geq 4$	**(s)**	0

31. MATCH THE FOLLOWING

	Column 1		**Column 2 (Range of f (x))**
(A)	$f(x) = \dfrac{(x+3)^2}{x^2+1}$	**(p)**	[0, 3]
(B)	$A = \{(x, y); x, y \in R, x^2 + y^2 \leq 25\}$ $B = \left\{(x, y); x, y \in R, y \geq \dfrac{4x^2}{9}\right\}$ and let $(x, f(x)) = A \cap B$	**(q)**	[3, 9]
(C)	$f(x) = \dfrac{9}{2 - \cos 3x}$	**(r)**	[0, 10]
(D)	$f(x) = 3\sqrt{2} \cdot \sin\left(\sqrt{\dfrac{\pi^2}{16} - x^2}\right)$	**(s)**	[0, 5]

32. Match the following columns :

	Column I		Column II
(i)	Range of $sgn\{x\}$ is: (where {.} represents fractional part function	**(a)**	{1}
(ii)	Domain of $sin^{-1} x + sin^{-1}(1-x)$ is:	**(b)**	[0, 1)
(iii)	Range of $\sqrt{\dfrac{2 tan^{-1} x}{\pi}}$ is:	**(c)**	0, 1}
(iv)	Range of $\dfrac{2}{\pi} sin^{-1}\left[x^2 + x + 1\right]$ is: (where [.] represent greatest integer function)	**(d)**	[0, 1]

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

33. Let 'a' be the real root of the equation $x^3-3x^2+5x-17=0$ and 'b' be the real root of the equation $x^3-3x^2+5x+11=0$. Then $a+b=$ ____ .

34. The number of real numbers x such that $\frac{x}{x+4}=\frac{5[x]-7}{7[x]-5}$ is ______. [x] denotes greatest integer $\leq x$,

35. Let $f(x)=144^{\sin^2 x}+144^{\cos^2 x}$. The number of integral values that $f(x)$ can take is ______.

36. The number of real solutions to the equation $3x-7=\left[x^2-3x+2\right]$ is ___. [x] denotes greatest integer $\leq x$,

37. Compute : $\left[\frac{2106^3}{2104\times 2105}-\frac{2104^3}{2105\times 2106}\right]$. [$x$] denotes greatest integer $\leq x$,

38. The range of the function $f(x)=\frac{x+m}{x^2+1},(m\in R)$ contains the interval [0, 1]. If $m\geq\frac{3}{k}$, then find k.

39. Let $f(x)=(x+1)(x+2)(x+3)(x+4)+5$; where $x\in[-6,6]$. If the range of the function is [a, b]; where $a, b\in N$, then find the value of ($a+b$).

40. Find the minimum number of roots of $f(x)=f\left(\frac{x+4}{x-2}\right)$.

41. If $f(2x+1)=4x^2+14x$, then find the sum of the squares of roots of the equation $f(x)=0$.

42. If $\alpha=e^{2\pi i/13}$ and $f(x)=7\sum_{k=1}^{50}A_i x^k$, then find the value of $\left(\frac{1}{13}\right)\sum_{r=0}^{12}f\left(a^r x\right)$.

43. Let f (x, y) be a periodic function satisfying $f(x, y)=f(2x+2y, 2y-2x)$ for all x, y. Define $g(x)=f\left(2^x, 0\right)$, the find the period of function g.

44. Let $g(x)=\frac{e^x-e^{-x}}{2}$ and $g(f(x))=x$, then evaluate $f\left(\frac{e^{22}-1}{2e^{11}}\right)$.

Advanced Problem Package	Differential Calculus-1

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If $f(x)=\begin{cases}\dfrac{((a-n)nx-\tan x)\sin nx}{x^2} & \text{at } x\neq 0\\ 0 & \text{at } x=0\end{cases}$, where n is a non-zero real number, and f is continuous at $x = 0$, then a is equal to :

(A) 0 **(B)** $\dfrac{n}{n+1}$ **(C)** n **(D)** $n+\dfrac{1}{n}$

2. Let f be a function defined on $(-\pi/2,\pi/2)$ as follows : $f(x)=\begin{cases}\dfrac{2^{|x|}e^{|x|}-|x|-|x|\ln 2-1}{x\tan x} & ,x\neq 0\\ k & ,x=0\end{cases}$. The value of k so that f is continuous at x = 0 is :

(A) $\frac{1}{2}(\ln 2)^2+\frac{1}{2}\ln 2+1$ **(B)** $(\ln 2)^2+\frac{1}{2}(\ln 2)+1$

(C) $(\ln 2)^2+(\ln 2)+\frac{1}{2}$ **(D)** $\frac{1}{2}(\ln 2)^2+\ln 2+\frac{1}{2}$

3. The function $f(x)=[x]+\sqrt{\{x\}}$, where [.] denotes the greatest integer function and {.} denotes the fractional part function respectively, is discontinuous at

(A) all x **(B)** all integer points

(C) no x **(D)** x which is not an integer.

4. Define $f:[0,\pi]\to R$ by $f(x)=\begin{cases}\tan^2 x\left[\sqrt{2\sin^2 x+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right] & ,\ x\neq \pi/2\\ k & ,\ x=\pi/2\end{cases}$ is continuous at $x=\dfrac{\pi}{2}$, then k is equal to :

(A) 1/12 **(B)** 1/6 **(C)** 1/24 **(D)** 1/32

5. $\lim\limits_{x\to\infty}\left(\sqrt[3]{(x+a)(x+b)(x+c)}-x\right)=$

(A) $\sqrt{abc}$ **(B)** $\dfrac{a+b+c}{3}$ **(C)** abc **(D)** $(abc)^{1/3}$

6. Given $f(x) = \dfrac{e^x - \cos 2x - x}{x^2}$ for $x \in R - \{0\}$, $\{x\}$ is fractional part function

$$g(x) = \begin{cases} f\{x\} & n < x < n + \frac{1}{2} \\ f(1-\{x\}) & n + \frac{1}{2} < x \le n+1 \\ \frac{5}{2} & \text{otherwise} \end{cases}$$

Then $g(x)$ is :

(A) Discontinuous at all integral values of x only **(B)** Continuous everywhere except for $x = 0$

(C) Discontinuous at $x = n + \dfrac{1}{2}$; $n \in I$ **(D)** Continuous everywhere

7. $\displaystyle\lim_{n\to\infty}\left(\left(\frac{n}{n+1}\right)^{\alpha} + \sin\frac{1}{n}\right)^n$ when $\alpha \in Q$ is equal to :

(A) $e^{-\alpha}$ **(B)** $-\alpha$ **(C)** $e^{1-\alpha}$ **(D)** $e^{1+\alpha}$

8. $\displaystyle\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r}{n^2+n+r}$ equals :

(A) 0 **(B)** 1/3 **(C)** 1/2 **(D)** 1

9. $\displaystyle\lim_{n\to\infty}\left(\sqrt{n^2+n+1} - \left[\sqrt{n^2+n+1}\right]\right)$ $(n \in I)$ where [ ] denotes the greatest integer function is :

(A) 0 **(B)** 1/2 **(C)** 2/3 **(D)** 1/4

10. If $f(x+y) = f(x) + f(y) + |x|y + xy^2$, $\forall\ x, y \in R$ and $f'(0) = 0$, then :

(A) f need not be differentiable at every non zero x **(B)** f is differentiable for all $x \in R$

(C) f is twice differentiable at $x = 0$ **(D)** None

11. $\displaystyle\lim_{n\to\infty}\frac{1^2\,n + 2^2\,(n-1) + 3^2\,(n-2) + ... + n^2 . 1}{1^3 + 2^3 + 3^3 + ... + n^3}$ is equal to :

(A) 1/3 **(B)** 2/3 **(C)** 1/2 **(D)** 1/6

12. The value of $\displaystyle\lim_{x\to\infty}\frac{\cot^{-1}\left(x^{-a}\log_a x\right)}{\sec^{-1}\left(a^x\log_x a\right)}$ $(a > 1)$ is equal to :

(A) 1 **(B)** 0 **(C)** $\pi/2$ **(D)** Does not exist

13. $\displaystyle\lim_{n\to\infty}\left(\frac{\sqrt[n]{p}+\sqrt[n]{q}}{2}\right)^n$, p, q > 0 equals :

(A) 1 **(B)** $\sqrt{pq}$ **(C)** pq **(D)** $\dfrac{pq}{2}$

14. Let $a = \min(x^2 + 2x + 3, x \in R)$ and $b = \displaystyle\lim_{x\to 0}\frac{\sin x\cos x}{e^x - e^{-x}}$. Then the value of $\displaystyle\sum_{r=0}^{n} a^r b^{n-r}$ is :

(A) $\dfrac{2^{n+1}+1}{3\cdot 2^n}$ **(B)** $\dfrac{2^{n+1}-1}{3\cdot 2^n}$ **(C)** $\dfrac{2^n-1}{3\cdot 2^n}$ **(D)** $\dfrac{4^{n+1}-1}{3\cdot 2^n}$

15. $\underset{x\to\infty}{Limit}\dfrac{\cot^{-1}\left(\sqrt{x+1}-\sqrt{x}\right)}{\sec^{-1}\left\{\left(\dfrac{2x+1}{x-1}\right)^{x}\right\}}$ is equal to :

(A) 1 **(B)** 0 **(C)** $\pi/2$ **(D)** Non existent

16. If f (x) = $\dfrac{e^{2x}-(1+4x)^{1/2}}{\ln(1-x^2)}$ for $x \neq 0$, then f has :

(A) An irremovable discontinuity at x = 0
(B) A removable discontinuity at x = 0 and f (0) = – 4
(C) A removable discontinuity at x = 0 and f (0) = – 1/4
(D) A removable discontinuity at x = 0 and f (0) = 4

Paragraph for Question 17 - 19

Let $f(x)$ is a function continuous for all $x \in R$ except at $x = 0$. Such that $f'(x) < 0$ $x \in (-\infty, 0)$ and $f'(x) > 0$ $x \in (0, \infty)$. Let $\underset{x\to0^+}{Lim} f(x) = 2$, $\underset{x\to0^-}{Lim} f(x) = 3$ and $f(0) = 4$.

17. The value of λ for which $2\left(\underset{x\to0}{Lim} f(x^3-x^2)\right) = \lambda\left(\underset{x\to0}{Lim} f(2x^4-x^5)\right)$ is :

(A) 4/3 **(B)** 2 **(C)** 3 **(D)** 5

18. The values of $\underset{x\to0^+}{Lim}\dfrac{f(-x)\,x^2}{\left\{\dfrac{1-\cos x}{[f(x)]}\right\}}$ where $[\cdot]$ denote greatest integer function and $\{\cdot\}$ denote fraction part function.

(A) 6 **(B)** 12 **(C)** 18 **(D)** 24

19. $\underset{x\to0^-}{Lim}\left(\left[3f\left(\dfrac{x^3-\sin^3 x}{x^4}\right)\right]-f\left(\left[\dfrac{\sin x^3}{x}\right]\right)\right)$ where $[\cdot]$ denote greatest integer function.

(A) 3 **(B)** 5 **(C)** 7 **(D)** 9

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

20. Let f be a function defined on (–1, 1) by $f(x)=\dfrac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}-\{x\}^3}, x\neq 0$.{.} is the fractional part function. Which of the following statements is correct?

(A) $\underset{x\to0+}{lim} f(x)$ exists and equals $\dfrac{\pi}{\sqrt{2}}$
(B) $\underset{x\to0-}{lim} f(x)$ exists and equals $\pi/4$
(C) f is continuous
(D) $\underset{x\to0^-}{lim} f(x)$ exists and equals $\underset{x\to0+}{lim} f(x)$

21. Let $f(x)=\begin{cases}\dfrac{x^{2^{32}}-2^{32}x+4^{16}-1}{(x-1)^2} & ,x\neq1\\ =k & ,x=1\end{cases}$, the value of k so that the function is continuous at $x = 1$ is :

(A) $2^{63}-2^{31}$
(B) $2^{65}-2^{33}$
(C) $\left(2^{16}+1\right)\left(2^{8}+1\right)\left(2^{4}+1\right)\left(2^{2}+1\right)\left(2^{32}+2^{31}\right)$
(D) $\left(2^{32}+1\right)\left(2^{16}+1\right)\left(2^{8}+1\right)\left(2^{4}+1\right)\left(2^{2}+1\right)\left(2^{33}+2^{31}\right)$

22. Which of the following statements are true?

(A) If f is differentiable at x = c, then $\lim\limits_{h\to 0}\dfrac{f(c+h)-f(c-h)}{2h}$ exists and equals $f'(c)$.

(B) Given a function f and a point c in the domain of f, if the $\lim\limits_{h\to 0}\dfrac{f(c+h)-f(c-h)}{h}$ exists, then the function is differentiable at x = c

(C) Let $g(x)=\begin{cases} x^2\sin\dfrac{1}{x^2}, & x\neq 0 \\ 0, & x=0 \end{cases}$, then g' exists

(D) Let $g(x)=\begin{cases} x^2\sin\dfrac{1}{x^2}, & x\neq 0 \\ 0, & x=0 \end{cases}$, then g' exists and is continuous.

23. Let f be a function given by $f(x)=\begin{cases} \dfrac{1}{x\ell n2}-\dfrac{1}{2^x-1} & ;x\neq 0 \\ \dfrac{1}{2} & ;x=0 \end{cases}$, Then :

(A) f is continuous on R

(B) f is differentiable on R and $f'(0)$ equals $\dfrac{-\ell n2}{12}$

(C) f is not differentiable at $x = 0$

(D) f is differentiable on R and $f'(0)$ equals $\dfrac{-\ell n2}{6}$

24. Let f be a function with two continuous derivatives and $f(0)=0$, $f'(0)=0$, $f''(0)=0$. Function g is defined by

$$g(x)=\begin{cases} \dfrac{f(x)}{x}, & x\neq 0 \\ 0, & x=0 \end{cases}$$

Then which of the following statements are correct?

(A) g has a continuous first derivative

(B) $g'(x)$ exist at $x=0$

(C) $g(x)$ is continuous but $g'(x)$ do not exist

(D) $g(x)$ is continuous, but the first derivative of g is not continuous

25. The function $\left.\begin{aligned} f(x) &= \frac{x^2}{a} &&, \quad 0\le x<1 \\ &= a &&, \quad 1\le x<\sqrt{2} \\ &= \frac{2b^2-4b}{x^2} &&, \quad \sqrt{2}\le x<\infty \end{aligned}\right\}$ is continuous for $0\le x<\infty$. Then which of the following statements is correct?

(A) The number of all possible ordered pairs (a, b) is 3

(B) The number of all possible order pairs (a, b) is 4

(C) The product of all possible values of b is – 1

(D) The product of all possible values of b is 1.

26. Let $f: R \to R$ be defined by $f(x)=\begin{cases} x+2x^2 \sin\frac{1}{x} & for \quad x \neq 0 \\ 0 & for \quad x=0 \end{cases}$ then

(A) $f(x)$ is differentiable for all $'x'$ but $f'(x)$ is not continuous at $x=0$

(B) $f'(0)=1$

(C) $f(x)$ is increasing at $x=0$

(D) Both $f(x)$ and $f'(x)$ are differentiable for all $'x'$

27. Let $f(x) = \min(x^3, x^2)$ and $g(x) = [x]^2 + \sqrt{\{x\}^2}$, where $[x]$ denotes the greatest integer and $\{x\}$ denotes the fractional part function. Then which of the following holds?

(A) f is continuous for all x.

(B) g is discontinuous for all $x \in I$.

(C) f is differentiable for all $x \in (1, \infty)$

(D) g is not differentiable for all $x \in I$

28. Let $f(x) = \lim\limits_{n\to\infty} \dfrac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$ then which of the following alternative(s) is/are correct?

(A) $\lim\limits_{x\to\infty} x f(x) = 2$

(B) $\lim\limits_{x\to 1} f(x)$ does not exist.

(C) $\lim\limits_{x\to 0} f(x)$ does not exist

(D) $\lim\limits_{x\to -\infty} f(x)$ is equal to zero.

29. Assume that $\lim\limits_{\theta\to -1} f(\theta)$ exists and $\dfrac{\theta^2+\theta-2}{\theta+3} \le \dfrac{f(\theta)}{\theta^2} \le \dfrac{\theta^2+2\theta-1}{\theta+3}$ holds for certain interval containing the point $\theta = -1$ then $\lim\limits_{\theta\to -1} f(\theta)$ and $\lim\limits_{\theta\to -1} \dfrac{f(\theta)}{\theta^2}$ is :

(A) is equal to $f(-1)$

(B) is equal to 1

(C) is non existent

(D) is equal to – 1

30. Let $f(x) = \begin{cases} \dfrac{\tan^2\{x\}}{x^2-[x]^2} & for \quad x>0 \\ 1 & for \quad x=0 \\ \sqrt{\{x\}}\cot\{x\} & for \quad x<0 \end{cases}$ where $[x]$ is the step up function and $\{x\}$ is the fractional part function of x, then :

(A) $\lim\limits_{x\to 0^+} f(x) = 1$

(B) $\lim\limits_{x\to 0^-} f(x) = 1$

(C) $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2 = 1$

(D) f is continuous at $x = 1$

31. The function, $f(x)=[|x|]-[|x|]$, where [ ] denotes greatest integer function:

(A) is continuous for all positive integers

(B) is discontinuous for all non-positive integers

(C) has finite number of elements in its range

(D) is such that its graph does not lie above the x-axis

32. The function $f(x) = \sqrt{1-\sqrt{1-x^2}}$

(A) has its domain $-1 \le x \le 1$

(B) has finite one sided derivates at the point $x = 0$

(C) is continuous and differentiable at $x = 0$

(D) is continuous but not differentiable at $x = 0$

33. f is a continuous function in [a, b]; g is a continuous function in [b, c]

A function h (x) is defined as :

$$h(x) = f(x) \quad \text{for } x \in [a, b)$$
$$= g(x) \quad \text{for } x \in (b, c]$$

if f (b) = g (b), then

(A) h(x) has a removable discontinuity at x = b
(B) h(x) may or may not be continuous in [a, c]
(C) $h(b^-) = g(b^+)$ and $h(b^+) = f(b^-)$
(D) $h(b^+) = g(b^-)$ and $h(b^-) = f(b^+)$

34. In which of the following cases the given equations has atleast one root in the indicated interval?

(A) $x - \cos x = 0$ in $(0, \pi/2)$
(B) $x + \sin x = 1$ in $(0, \pi/6)$
(C) $\dfrac{a}{x-1} + \dfrac{b}{x-3} = 0$, a, b > 0 in (1, 3)
(D) f (x) – g (x) = 0 in (a, b) where f and g are continuous on [a, b] and f (a) > g (a) and f (b) < g (b)

35. If $f(x) = \begin{cases} \dfrac{x \,.\, \ell n(\cos x)}{\ell n\left(1+x^2\right)} & x \neq 0 \\ 0 & x = 0 \end{cases}$ then :

(A) f is continuous at $x = 0$
(B) f is continuous at $x = 0$ but not differentiable at $x = 0$
(C) f is differentiable at $x = 0$
(D) f is not continuous at $x = 0$

36. $\underset{x \to c}{Lim} f(x)$ does not exist when :

(A) f(x) = [[x]] – [2x – 1], c = 3
(B) f(x) = [x] – x, c = 1
(C) $f(x) = \{x\}^2 - \{-x\}^2$, c = 0
(D) $f(x) = \dfrac{\tan(\text{sgn}\, x)}{\text{sgn}\, x}$, c = 0 .

where [x] denotes step up function and {x} fractional part function.

37. Which of the following limits vanish?

(A) $\underset{x \to \infty}{Limit} \sin \dfrac{1}{\sqrt{x}}$
(B) $\underset{x \to \pi/2}{Limit} (1 - \sin x) \,.\, \tan x$
(C) $\underset{x \to \infty}{Limit} \dfrac{2x^2 + 3}{x^2 + x - 5} \,.\, \text{sgn}(x)$
(D) $\lim\limits_{x \to 3} \dfrac{[x]^2 - 9}{x^2 - 9}$

where [ ] denotes greatest integer function

38. Let $f(x) = |x-1|([x]-[-x])$, then which of the following statement(s) is/are correct. (where [.] denotes greatest integer function.)

(A) $f(x)$ is continuous at $x = 1$
(B) $f(x)$ is derivable at $x = 1$
(C) $f(x)$ is non-derivable at $x = 1$
(D) $f(x)$ is discontinuous at $x = 1$

39. If $y = f(x)$ defined parametrically by $x = 2t - |t-1|$ and $y = 2t^2 + t|t|$, then:

(A) $f(x)$ is continuous for all $x \in R$
(B) $f(x)$ is continuous for all $x \in R - \{2\}$
(C) $f(x)$ is differentiable for all $x \in R$
(D) $f(x)$ is differentiable for all $x \in R - \{2\}$

40. $f:R\to R$ is one-one, onto and differentiable function and $f(4+x)+f(4-x)=0\forall x\in R$ then:

(A) $f^{-1}(2010)+f^{-1}(-2010)=8$

(B) $\int_{-2010}^{2018} f(x)dx=0$

(C) If $f'(-100)>0$, then roots of $x^2-f'(10)x-f'(10)=0$ are non-real

(D) If $f'(10)=20$, then $f'(-2)=20$

41. Let f be a real valued function defined on the interval $(0,\infty)$, by $f(x)=\ln x+\int_0^x\sqrt{1+\sin t}\,dt$. Then, which of the following statement(s) is/are true?

(A) $f''(x)$ exists for all $x\in(0,\infty)$

(B) $f'(x)$ exists for all $x\in(0,\infty)$ and f' is continuous on $(0,\infty)$ but not differentiable on $(0,\infty)$

(C) There exists $\alpha>1$ such that $|f'(x)|<|f(x)|$ for all $x\in(0,\infty)$

(D) There exists $\beta>0$ such that $|f(x)|<|f'(x)|\le\beta$ for all $x\in(0,\infty)$

42. If $f(x)=\begin{cases}3-\left[\cot^{-1}\left(\dfrac{2x^3-3}{x^2}\right)\right], & for\ x>0\\ \{x^2\}\cos\left(e^{1/x}\right), & for\ x<0\end{cases}$, where $\{x\}$ and $[x]$ denote fractional part and the greatest integer function respectively, then which of the following statements does not hold good?

(A) $f(0^-)=0$

(B) $f(0^+)=0$

(C) $f(0)=0\Rightarrow$ continuous at $x=0$

(D) Irremovable discontinuity at $x=0$

43. If $f(x)=\begin{cases}b([x]^2+[x])+1, & for\ x\ge-1\\ \sin(\pi(x+a)), & for\ x<-1\end{cases}$, where $[x]$ denotes the integral part of x, then for what values of a and b, the function is continuous at $x=-1$?

(A) $a=2n+\dfrac{3}{2};b\in R,n\in I$

(B) $a=4n+2;b\in R,n\in I$

(C) $a=4n+\dfrac{3}{2};b\in R^+,n\in I$

(D) $a=4n+1;b\in R^+,n\in I$

44. Let $[x]$ be the greatest integer function, then $f(x)=\dfrac{\sin\frac{1}{4}\pi[x]}{[x]}$ is:

(A) not continuous at any point

(B) continuous at $x=\dfrac{3}{2}$

(C) discontinuous at $x=2$

(D) differentiable at $x=\dfrac{4}{3}$

45. Let $f(x)=\cos x$ and $H(x)=\begin{cases}\min(f(t):0\le t<x), & \text{for } 0\le x\le \pi/2\\ \dfrac{\pi}{2}-x, & \text{for } \dfrac{\pi}{2}<x\le 3\end{cases}$, then:

(A) $H(x)$ is continuous and derivable in [0, 3]

(B) $H(x)$ is continuous but not derivable at $x=\dfrac{\pi}{2}$

(C) $H(x)$ is neither continuous nor derivable at $x=\dfrac{\pi}{2}$

(D) Maximum value of $H(x)$ in [0, 3] is 1

46. Let $f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{x^2-[x]^2}, & \text{for } x>0\\ 1, & \text{for } x=0\\ \sqrt{\{x\}\cot\{x\}}, & \text{for } x<0\end{cases}$, where $[x]$ is greatest integer function and $\{x\}$ is the fractional part function of x, then:

(A) $\lim\limits_{x\to 0^+} f(x)=1$ **(B)** $\lim\limits_{x\to 0^-} f(x)=1$ **(C)** $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2=1$ **(D)** None of these

47. Let $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\dfrac{x}{(rx+1)\{(r+1)x+1\}}$. Then:

(A) $f(0)=0$ **(B)** $f(0)=1$ **(C)** $f(2)=1$ **(D)** $f(3)=1$

48. For $\lim\limits_{x\to 0}\dfrac{\cot^{-1}\left(\frac{1}{x}\right)}{x}$.

(A) RHL exists

(B) LHL does not exist

(C) Limit does not exist as RHL is 1 and LHL is -1

(D) Limit does not exist as RHL and LHL both are non-existent

49. For $a>0$, let $l=\lim\limits_{x\to\frac{\pi}{2}}\dfrac{a^{\cot x}-a^{\cos x}}{\cot x-\cos x}$ and $m=\lim\limits_{x\to-\infty}\left(\sqrt{x^2+ax}-\sqrt{x^2-ax}\right)$, then:

(A) $l>m$, for all $a>0$ **(B)** $l>m$, when $a\ge 1$

(C) $l>m$, for all $a>e^{-a}$ **(D)** $e^l+m=0$

50. Consider the function $f(x)=\left(\dfrac{ax+1}{bx+2}\right)^x$, where $a,b>0$ the $\lim\limits_{x\to\infty} f(x)$ is:

(A) exists for all values of a and b **(B)** zero for $a<b$

(C) non-existent for $a>b$ **(D)** $e^{-(1/a)}$ or $e^{-(1/b)}$, if $a=b$

51. $\lim_{x \to c} f(x)$ does not exist when (where $[x]$ denotes the greatest integer less than or equal to x)

(A) $f(x) = [[x]] - [2x-1], c = 3$ **(B)** $f(x) = [x] - x, c = 1$

(C) $f(x) = \{x\}^2 - \{-x\}^2, c = 0$ **(D)** $f(x) = \dfrac{\tan(\operatorname{sgn} x)}{(\operatorname{sgn} x)}, c = 0$

52. The function $f(x) = \left[x^2\left[\dfrac{1}{x^2}\right]\right], x \neq 0$ is ($[x]$ represents the greatest integer $\leq x$)

(A) continuous at $x = 1$ **(B)** discontinuous at $x = -1$

(C) discontinuous at infinitely many points **(D)** discontinuous at finite number of points

53. A function is defined as $f(x) = [\tan x] + \sqrt{\tan x - [\tan x]}$ $0 \leq x < \dfrac{\pi}{2}$ (where [.] denotes the greatest integer function) then:

(A) $f(x)$ is continuous in $\left[0, \dfrac{\pi}{2}\right)$ **(B)** $f(x)$ is not continuous at $x = 0$

(C) $f(x)$ is continuous at $x = 0, \dfrac{\pi}{4}$ **(D)** $f(x)$ has infinite points of discontinuities

54. Let $f(x) = \sec^{-1}\left(\left[1 + \sin^2 x\right]\right)$ ([.]) denotes the greatest-integer function). Then set of points, where $f(x)$ is not continuous is:

(A) $\left\{\dfrac{n\pi}{2}, n \in I\right\}$ **(B)** $\left\{\{2n-1\}\dfrac{\pi}{2}, n \in I\right\}$ **(C)** $\left\{(2n+1)\dfrac{\pi}{2}, n \in I\right\}$ **(D)** $\{n\pi, n \in I\}$

55. Let $f(x) = \begin{cases} \int_0^x \{1 + |1-t|\}dt & if\ x > 2 \\ 5x - 7 & if\ x \leq 2 \end{cases}$, then:

(A) f is not continuous at $x = 2$

(B) f is continuous but not differentiable at $x = 2$

(C) f is differentiable every where

(D) $\lim_{x \to 2^+} f'(x) = 2$

56. If $F(x) = f(x)g(x)$ and $f'(x)g'(x) = c$, then (where f and g are thrice differentiable)

(A) $F' = c\left[\dfrac{f}{f'} + \dfrac{g}{g'}\right]$ **(B)** $\dfrac{F''}{F} = \dfrac{f''}{f} + \dfrac{g''}{g} + \dfrac{2c}{fg}$

(C) $\dfrac{F'''}{F} = \dfrac{f'''}{f} + \dfrac{g'''}{g}$ **(D)** $\dfrac{F'''}{F''} = \dfrac{f'''}{f''} + \dfrac{g'''}{g''}$

57. $f(x)$ is defined for $x \geq 0$ and has a continuous derivative. It satisfies $f(0) = 1, f'(0) = 0$ and $(1 + f(x))f''(x) = 1 + x$. The values $f(1)$ can't take is/are:

(A) 2 **(B)** 1.75 **(C)** 150 **(D)** 1.35

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

58. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$f(x)=\begin{cases} e^{-1/x^2}, & x\neq 0 \\ 0, & x=0 \end{cases}$	**(p)**	First derivative exists
(B)	$f(x)=\begin{cases} e^{-1/x}, & x>0 \\ 0, & x\le 0 \end{cases}$	**(q)**	First derivative is continuous
(C)	$f(x)=\begin{cases} e^{\frac{-1}{x-e}+\frac{1}{x-\pi}}, & x\in(e,\pi) \\ 0, & x\notin(e,\pi) \end{cases}$	**(r)**	Second derivative exists
(D)	$f(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\neq 0 \\ 0, & x=0 \end{cases}$	**(s)**	Second derivative is continuous

59. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$f(x)=\begin{cases} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+........\left[\frac{8}{x}\right]\right), & x\neq 0 \\ 9k, & x=0 \end{cases}$ The value of k such that f is continuous at x = 0 is ([.] denotes the greatest integer function)	**(p)**	1
(B)	$f(x)=\begin{cases} \left(1+xe^{-1/x^2}\sin\frac{1}{x^4}\right)^{e^{1/x^2}}, & x\neq 0 \\ k, & x=0 \end{cases}$ The value of k such that f is continuous at x = 0 is	**(q)**	2
(C)	f: $[0,\infty)\to R$; $f(x)=\begin{cases} \left(2\sin\sqrt{x}+\sqrt{x}\sin\frac{1}{x}\right)^{x}, & x>0 \\ k, & x=0 \end{cases}$ The value of k such that f is continuous at x = 0 is	**(r)**	3
(D)	$f:(0,\pi)\to R$; $f(x)=\begin{cases} \frac{1-\sin x}{(\pi-2x)^2}\cdot\frac{\ell n\sin x}{\ell n\left(1+\pi^2-4\pi x+4x^2\right)}; & x\neq\frac{\pi}{2} \\ k & ; x=\frac{\pi}{2} \end{cases}$ The value of $8\sqrt{\lvert k\rvert}$ such that f is continuous at x = $\frac{\pi}{2}$ is	**(s)**	4

60. Let f be a polynomial of degree 4 having real coefficients satisfying

$$f'(0)=f'(1)=f'(-1)=0 \text{ and } f(0)=4, f''\left(\frac{1}{2}\right)=-1$$

MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$f(x)=0$ has	**(p)**	root at $x = 2$
(B)	$4-f(x)=0$ has	**(q)**	root at $x = 1$
(C)	$f'(x)+x-1=0$ has	**(r)**	2 equal real roots
(D)	$x f'(x)-4f(x)=0$ has	**(s)**	no real roots

61. Let $f(x)=x^2+ax+b, \forall x\in R$, $f(0)>0$ & $f(x)$ has integral roots. Tangent at $\left(\frac{5}{2}, p\right)$ to $y=f(x)$ is parallel to x-axis & $g(x)=f(x+1)$.

	Column 1		Column 2
(A)	$(a+b)$ can be	**(p)**	–1
(B)	Value of [p] can be (where [.] represents greatest integer function)	**(q)**	1
(C)	Number of points where $g(\|x\|)$ is non differentiable can be	**(r)**	3
(D)	Number of points where $\|g(\|x\|)\|$ is non differentiable can be	**(s)**	–3
		(t)	5

SUBJECTIVE INTEGER TYPE

Each of the following question has an integer answer between 0 and 9.

62. $f(x)=\begin{cases}\dfrac{\sin\left(\pi\cos^2\left(\tan(\sin x)\right)\right)}{\pi x^2} & ,x\neq 0\\ k & ,x=0\end{cases}$. The value of k such that f is continuous at $x = 0$, is

63. If the independent variable x is changed to y, then the expression $x\dfrac{d^2y}{dx^2}+\left(\dfrac{dy}{dx}\right)^2-\dfrac{dy}{dx}=0$ is transformed to $x\dfrac{d^2x}{dy^2}+\left(\dfrac{dx}{dy}\right)^2=\lambda\dfrac{dx}{dy}$, then λ equals.

64. Let f and g be continuously differentiable functions such that $f(0) = 0$, $f'(0) = 2$ and $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ equals.

65. Let $f(x)=\begin{cases}x^2\displaystyle\sum_{r=0}^{\left[\frac{1}{|x|}\right]} r & ;x\neq 0\\ \dfrac{k}{2}; & otherwise\end{cases}$ ([.] denotes the greatest integer function)

The value of k such that f become continuous at $x = 0$ is ________.

66. Let $f: R \rightarrow R$ be a continuous function such that $f(x+y)=f(x)+f(y)+f(x).f(y), \forall\, x, y \in R$. Also $f'(0)=1$. Then $\left[\frac{f(4)}{f(2)}\right]$ equals ([•] represents greatest integer function)

67. Let $f(x) = tan^{-1} x, |x| \le 1$

$$= \frac{\pi}{4} sgn\, x + \frac{x-1}{2}, |x| > 1$$, (where sgn$x$ denotes signum function)

Then the value of $4f'(1^+)$ equals.

68. Let $K>0$ and $\lambda = \lim\limits_{x \to 0} \frac{K\left(1-4\sqrt{K^2-x^2}\right)}{x^2\sqrt{K^2-x^2}}$ is finite then the value of λK is______.

69. Let $f(x)$ be a differentiable function, $f(1)=0,\ f'(1)=2$ then the value of $\lim\limits_{x \to 1} \int_1^{x^2} \frac{x^2 \sin(f(t))dt}{(x-1)^2}$ is____.

70. Let $f:[-1,1] \rightarrow \left[\frac{-\pi}{4} - \tan 1, \frac{\pi}{4} + \tan 1\right]$ defined by $f(x)=\tan x + \tan^{-1} x$ and the derivative of $f^{-1}(x)$ at $x=0$ is 'k' then the value of $\frac{4}{k}$ is ____.

71. The number of point where $|xf(x)| + ||x-2|-1|$ is non-differentiable in $x \in (0, 3\pi)$, where

$$f(x) = \prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right)$$, is_______.

72. If $f\left(\frac{xy}{2}\right) = \frac{f(x).f(y)}{2}; x, y \in R, f(1) = f'(1)$. Then, $\frac{f(3)}{f'(3)}$ is_______.

73. Let $f: R \rightarrow R$ be a differentiable function satisfying $f(x) = f(y) f(x-y), \forall x, y \in R$ and $f'(0) = \int_0^4 \{2x\} dx$, where $\{.\}$ denotes the fractional part function and $f'(-3) = \alpha e^{\beta}$. Then, $|\alpha + \beta|$ is equal to_______.

74. Let $f(x)$ is a polynomial function and $(f(\alpha))^2 + (f'(\alpha))^2 = 0$, then find $\lim\limits_{x \to \alpha} \frac{f(x)}{f'(x)}\left[\frac{f'(x)}{f(x)}\right]$, (where $[.]$ denotes greatest integer function) is_______.

75. Let $f: R \rightarrow R$ is a function satisfying $f(10-x) = f(x)$ and $f(2-x) = f(2+x), \forall x \in R$. If $f(0) = 101$. Then, the minimum possible number of values of x satisfying $f(x) = 101, x \in [0, 25]$ is_______.

76. If $f(x)=\begin{cases}\dfrac{\left(\dfrac{\pi}{2}-\sin^{-1}\left(1-\{x\}^2\right)\right)\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2}\left(\{x\}-\{x\}^3\right)} & , \quad x>0\\ k & , \quad x=0\\ \dfrac{A\sin^{-1}\left(1-\{x\}\right)\cos^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}\left(1-\{x\}\right)} & , \quad x<0\end{cases}$

is continuous at $x=0$, then the value of A is_______. (where $\{.\}$ denotes fractional part of x).

77. In a ΔABC, angles A,B,C are in AP. If $f(C)=\lim\limits_{A\to C}\dfrac{\sqrt{3-4\sin A\sin C}}{|A-C|}$, then $f'\left(\dfrac{\pi}{12}\right)$ is equal to_______.

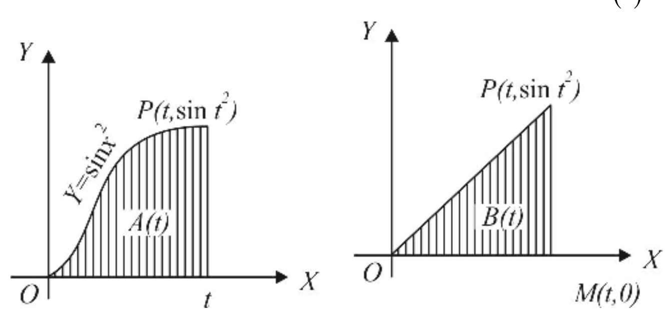
78. Let $f_1(x)$ and $f_2(x)$ be twice differentiable function, Where $F(x)=f_1(x)+f_2(x)$ and $G(x)=f_1(x)-f_2(x), \forall x\in R, f_1(0)=2$ and $f_2(0)=1$. If $f_1'(x)=f_2(x)$ and $f'_2(x)=f_1(x), \forall x\in R$, then the number of solutions of the equation $(F(x))^2=\dfrac{9x^4}{G(x)}$ is_______.

79. Let $f(x)=x\tan^{-1}\left(x^2\right)+x^4$. Let $f^k(x)$ denotes kth derivative of $f(x)\, w.r.t. x, k\in N$, . If $f^{2m}(0)\neq 0, m\in N$, then m equals to_______.

80. Let $f(n)=\left[\sqrt{n}+\dfrac{1}{2}\right]$, where [.] denotes greatest integer function, $\forall n\in N$. Then $\sum\limits_{n=1}^{\infty}\dfrac{2^{f(n)}+2^{-f(n)}}{2^n}$ is equal to_______.

81. The value of $\lim\limits_{x\to\frac{\pi}{2}}\sqrt{\dfrac{\tan x-\sin\left\{\tan^{-1}(\tan x)\right\}}{\tan x+\cos^2(\tan x)}}$ is_______.

82. The figure shows two regions in the first quadrant. $A(t)$ is the area under the curve $y=\sin x^2$ from 0 to t and $B(t)$ is the area of the triangle with vertices 0, P and $M(t,0)$. If $\lim\limits_{t\to 0}\dfrac{A(t)}{B(t)}=\dfrac{1}{k}$, then k is_______.



83. Consider a parabola $y=\frac{x^2}{4}$ and the point $F(0,1)$.

Let $A_1(x_1,y_1),A_2(x_2,y_2),A_3(x_3,y_3),..............,A_N(x_n,y_n)$, are '$n$' points on the parabola such that $x_k>0$ and $\angle OFA_k=\frac{k\pi}{2n}(k=1,2,.....,n)$. If the value of $\lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}FA_k=\frac{m}{\pi}$, then m is_______.

84. The value of $\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+....+\frac{1}{\sqrt{n^2+2n}}\right)$ is_______.

85. Suppose $x_1=\tan^{-1}2>x_2>x_3>.......$ are the real numbers satisfying $\sin(x_{n+1}-x_n)+2^{-(n+1)}.\sin x_n.\sin x_{n+1}=0$ for all $n>1$ and the sequence is convergent and , $l=\lim_{n\to\infty}x_n$, the value of $4l$ is_______.

86. A function $f(x)$ satisfies the relation $f(x+y)=f(x)+f(y)+xy(x+y),\forall x,y\in R$. If $f'(0)=-1$, then $f'(3)=$

87. If $\lim_{x\to\infty}4x\left(\frac{\pi}{4}-\tan^{-1}\frac{x+1}{x+2}\right)=y^2+4y+5$, then the product of all possible value of y is_______.

88. If $\lim_{x\to 0}\frac{a\sin x-bx+cx^2+x^3}{2x^2\ln(1+x)-2x^3+x^4}$ exists and is equal to l then $a+b+c+l=$

89. If $\lim_{n\to\infty}\left(\frac{(n^3+1)(n^3+2^3)(n^3+3^3)..........(n^3+n^3)}{n^{3n}}\right)^{1/n}=4e^{\frac{\pi}{\sqrt{a}}}.e^{-b}$ (where $a,b\in N$) then $a+b$ is_______.

90. Let H_n denotes the harmonic mean of n positive integers $n+1,n+2,n+3,.....,n+n$, if $\lim_{n\to\infty}\left(\frac{H_n}{n}\right)=\frac{1}{k}$ then the value of e^k is_______.

Advanced Problem Package	Differential Calculus-2

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The equation $(x^2+3x+4)^2+3(x^2+3x+4)+4=x$ has :

(A) All its solutions real but not all positive **(B)** Only two of its solutions real

(C) Two of its solutions positive, two negative **(D)** None of its solution real

2. Let $f(x)$ be a differentiable function in the interval (0, 2), then the value of $\int_0^2 f(x)dx$ is :

(A) $f(c)$ where $c \in (0, 2)$ **(B)** $2f(c)$ where $c \in (0, 2)$

(C) $f'(c)$ where $c \in (0, 2)$ **(D)** None of these

For Questions 3 - 5

Consider $f(x)=x+\cos x-a$

3. Which of the following holds good for the above $f(x)$

(A) $a > 1$for which $f(x)$ has exactly one positive root

(B) $a > 2$for which $f(x)$ has exactly one –ve root

(C) $1<a<2\, f(x)$ will have an imaginary set of roots

(D) None of these

4. Which of the following is true for $f(x)$

(A) $\sin\alpha > \dfrac{\cos\alpha-\cos\beta}{\beta-\alpha} > \cos\beta$; For $0<\alpha<\beta<\dfrac{\pi}{2}$

(B) $\sin\alpha > \dfrac{\cos\alpha-\cos\beta}{\beta-\alpha} > \sin\beta$; For $0<\alpha<\beta<\dfrac{\pi}{2}$

(C) $\tan\alpha < \dfrac{\cos\alpha-\cos\beta}{\beta-\alpha} < \tan\beta$; For $0<\alpha<\beta<\dfrac{\pi}{2}$

(D) None of these

5. The equation of tangent to $f(x)$ at $x=\dfrac{\pi}{2}$ is :

(A) $y=\dfrac{\pi}{2}-a$ **(B)** $y=\dfrac{\pi}{4}+a$ **(C)** $x+y=\dfrac{\pi}{4}$ **(D)** $x-y=\dfrac{\pi}{2}$

6. Point 'A' lies on the curve $y=e^{-x^2}$ and has the coordinate $(x,\ e^{-x^2})$ where $x > 0$. Point B has the coordinates $(x, 0)$. If 'O' is the origin then the maximum area of the triangle AOB is :

(A) $\dfrac{1}{\sqrt{2e}}$ **(B)** $\dfrac{1}{\sqrt{4e}}$ **(C)** $\dfrac{1}{\sqrt{e}}$ **(D)** $\dfrac{1}{\sqrt{8e}}$

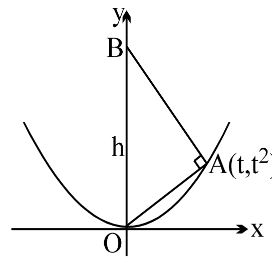
7. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is :

(A) $27\,c^3$ **(B)** $\dfrac{4}{27}c^3$ **(C)** $\dfrac{27}{4}c^3$ **(D)** $\dfrac{4}{9}c^3$

8. The function $f : [a, \infty) \to R$ where R denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided :

(A) $a \ge 1$ **(B)** $a \ge 0$ **(C)** $a \le 0$ **(D)** $a \le 1$

9. The figure shows a right triangle with its hypotenuse OB along the y-axis and its vertex A on the parabola $y = x^2$. Let h represents the length of the hypotenuse which depends on the x-coordinate of the point A. The value of $\lim\limits_{x \to 0} (h)$ equals



(A) 0 **(B)** 1/2

(C) 1 **(D)** 2

10. The least value of 'a' for which the equation, $\dfrac{4}{\sin x} + \dfrac{1}{1 - \sin x} = a$ has atleast one solution on the interval $(0, \pi/2)$ is:

(A) 3 **(B)** 5 **(C)** 7 **(D)** 9

11. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} Min\,\{f(t) : 0 \le t \le x\} & ;\ 0 \le x \le 1 \\ 3 - x & ;\ 1 < x \le 2 \end{cases}$ then $g\left(\dfrac{1}{4}\right) + g\left(\dfrac{3}{4}\right) + g\left(\dfrac{5}{4}\right)$ has the value equal to :

(A) 7/4 **(B)** 9/4 **(C)** 13/4 **(D)** 5/2

12. Given: $f(x) = 4 - \left(\dfrac{1}{2} - x\right)^{2/3}$, $g(x) = \begin{cases} \dfrac{\tan[x]}{x} & , x \ne 0 \\ 1 & , x = 0 \end{cases}$, $h(x) = \{x\}$

$k(x) = 5^{\log_2(x+3)}$, then in [0, 1], Lagranges Mean Value Theorem is NOT applicable to :

(A) f, g, h **(B)** h, k **(C)** f, g **(D)** g, h, k

where $[x]$ and $\{x\}$ denotes the greatest integer and fraction part function.

13. A curve is represented by the equations, $x = \sec^2 t$ and $y = \cot t$ where t is a parameter. If the tangent at the point P on the curve where $t = \pi/4$ meets the curve again at the point Q then | PQ | is equal to :

(A) $\dfrac{5\sqrt{3}}{2}$ **(B)** $\dfrac{5\sqrt{5}}{2}$ **(C)** $\dfrac{2\sqrt{5}}{3}$ **(D)** $\dfrac{3\sqrt{5}}{2}$

14. Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals

(A) $-\dfrac{2\sqrt{3}}{9}$ **(B)** $-\dfrac{2}{9}$ **(C)** $\dfrac{2}{3\sqrt{3}}$ **(D)** $\dfrac{1}{3\sqrt{3}}$

15. Let h be a continuous twice differentiable positive function on an open interval J. Let $g(x) = ln\left(h(x)\right)$ for each $x \in J$

Suppose $\left(h'(x)\right)^2 > h''(x)\, h(x)$ for each $x \in J$. Then :

(A) g is increasing on J **(B)** g is decreasing on J

(C) g is concave up on J **(D)** g is concave down on J

16. Let $F(x) = \displaystyle\int_{\sin x}^{\cos x} e^{(1+\arcsin t)^2} dt$ in $\left[0, \dfrac{\pi}{2}\right]$ then :

(A) $F''(c) = 0$ for all $c \in \left(0, \dfrac{\pi}{2}\right)$ **(B)** $F''(c) = 0$ for some $c \in \left(0, \dfrac{\pi}{2}\right)$

(C) $F'(c) = 0$ for some $c \in \left(0, \dfrac{\pi}{2}\right)$ **(D)** $F(c) \ne 0$ for all $c \in \left(0, \dfrac{\pi}{2}\right)$

17. P and Q are two points on a circle of centre C and radius α, the angle PCQ being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when

(A) $\sin\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$ **(B)** $\sin\theta = \frac{\sqrt{5}-1}{2}$ **(C)** $\sin\theta = \frac{\sqrt{5}+1}{2}$ **(D)** $\sin\theta = \frac{\sqrt{5}-1}{4}$

18. Let a function f be defined as $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & if \quad x > -1 \\ x^2 & if \quad x \le -1 \end{cases}$

Then the number of critical point(s) on the graph of this function is/are :

(A) 4 **(B)** 3 **(C)** 2 **(D)** 1

19. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at x = 0, f (x) has :

(A) a maxima whenever a > 0, b > 0 **(B)** a maxima whenever a > 0, b < 0

(C) minima whenever a > 0, b > 0 **(D)** neither a maxima nor minima whenever a > 0, b < 0

20. Consider $f(x) = \int_1^x \left(t + \frac{1}{t}\right) dt$ and $g(x) = f'(x)$ for $x \in \left[\frac{1}{2}, 3\right]$

If P is a point on the curve y = g(x) such that the tangent to this curve at P is parallel to a chord joining the points $\left(\frac{1}{2}, g\left(\frac{1}{2}\right)\right)$ and (3, g(3)) of the curve, then the coordinates of the point P

(A) can't be found out **(B)** $\left(\frac{7}{4}, \frac{65}{28}\right)$ **(C)** (1, 2) **(D)** $\left(\sqrt{\frac{3}{2}}, \frac{5}{\sqrt{6}}\right)$

21. $f : R \to R$

Statement 1 : $f(x) = 12x^5 - 15x^4 + 20x^3 - 30x^2 + 60x + 1$ is monotonic and surjective on R.

Statement 2 : A continuous function defined on R, if strictly monotonic has its range R.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true

For Questions 22 - 24

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where a, b $\in$ R.

22. For a = 1 if $y = f(x)$ is strictly increasing $\forall$ x $\in$ R then maximum range of values of b is :

(A) $\left(-\infty, \frac{1}{3}\right]$ **(B)** $\left(\frac{1}{3}, \infty\right)$ **(C)** $\left[\frac{1}{3}, \infty\right)$ **(D)** $(-\infty, \infty)$

23. For b = 1, if $y = f(x)$ is non monotonic then the sum of all the integral values of a $\in$ [1, 100], is :

(A) 4950 **(B)** 5049 **(C)** 5050 **(D)** 5047

24. If the sum of the base 2 logarithms of the roots of the cubic $f(x) = 0$ is 5 then the value of 'a' is :

(A) – 64 **(B)** – 8 **(C)** – 128 **(D)** – 256

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

25. The points on the ellipse $4x^2+9y^2=13$ such that the rate of decrease of ordinate is equal to rate of increase in abscissa are :

(A) $\left(\frac{3}{2},\frac{2}{3}\right)$ **(B)** $\left(-\frac{3}{2},-\frac{2}{3}\right)$ **(C)** $\left(\frac{3}{2},-\frac{2}{3}\right)$ **(D)** $\left(-\frac{3}{2},\frac{2}{3}\right)$

26. If Rolle's theorem is applicable to the function *f* defined by $f(x)=\begin{cases} ax^2+b, & |x|<1 \\ 1, & |x|=1 \\ \dfrac{\lambda}{|x|} & |x|>1 \end{cases}$ $for\ x\in[-2,2]$ then :

(A) $a+b=1$ **(B)** $\lambda=a+b$ **(C)** $b=\frac{3}{2}$ **(D)** $3a+b=0$

27. $f(x)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ decreases in the region :

(A) $(-1, 0)$ **(B)** $(0, 1)$ **(C)** $(-\infty, -1)$ **(D)** $(-\infty, 1)$

28. The function $f(x)=x^{1/3}(x-1)$

(A) has 2 inflection points

(B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$

(C) is concave down in $(-1/2, 0)$

(D) Area enclosed by the curve lying in the fourth quadrant is 9/28

29. If $\frac{x}{a}+\frac{y}{b}=1$ is a tangent to the curve $x = Kt,\ y=\frac{K}{t}, K > 0$ then :

(A) $a > 0,\ b > 0$ **(B)** $a > 0, b < 0$ **(C)** $a < 0, b > 0$ **(D)** $a < 0, b < 0$

30. The function $f(x)=\int_0^x \sqrt{1-t^4}\,dt$ is such that :

(A) it is defined on the interval $[-1, 1]$ **(B)** it is an increasing function

(C) it is an odd function **(D)** the point $(0, 0)$ is the point of inflection

31. The co-ordinates of the point(s) on the graph of the function, $f(x)=\frac{x^3}{3}-\frac{5x^2}{2}+7x-4$ where the tangent drawn cut off intercepts from the co-ordinate axes which are equal in magnitude but opposite in sign, is :

(A) $(2, 8/3)$ **(B)** $(3, 7/2)$ **(C)** $(1, 5/6)$ **(D)** None of these

32. If $f(x)=a^{\{a^{|x|}\operatorname{sgn} x\}}$; $g(x)=a^{[a^{|x|}\operatorname{sgn} x]}$ for $a>0,\ a\neq 1$ and $x\in R$, where { } and [ ] denote the fractional part and integral part functions respectively, then which of the following statements can hold good for the function $h(x)$, where $(\ln a)h(x)=(\ln f(x)+\ln g(x))$.

(A) 'h' is even and increasing **(B)** 'h' is odd and decreasing

(C) 'h' is even and decreasing **(D)** 'h' is odd and increasing.

33. On which of the following intervals, the function $x^{100}+\sin x-1$ is strictly increasing.

(A) $(-1, 1)$ **(B)** $(0, 1)$ **(C)** $(\pi/2, \pi)$ **(D)** $(0, \pi/2)$

34. If $f(x)=\begin{bmatrix} 3x^2+12x-1 & , & -1\le x\le 2 \\ 37-x & , & 2<x\le 3\end{bmatrix}$ then :

(A) $f(x)$ is increasing on [– 1, 2]
(B) $f(x)$ is continuous on [– 1 , 3]
(C) $f'(2)$ does not exist
(D) $f(x)$ has the maximum value at $x=2$

35. Consider the function $f(x) = x^2 - x\sin x - \cos x$ then the statements which holds good, are
(A) $f(x) = k$ has no solution for $k < -1$
(B) f is increasing for $x < 0$ and decreasing for $x > 0$
(C) $\lim_{x\to\pm\infty} f(x)\to\infty$
(D) The zeros of $f(x) = 0$ lie on the same side of the origin

36. Assume that inverse of the function f is denoted by g. Then which of the following statement hold good?
(A) If f is increasing then g is also increasing
(B) If f is decreasing then g is increasing.
(C) The function f is injective
(D) The function g is onto

37. For the function $f(x) = \ln(1 - \ln x)$ which of the following do not hold good?
(A) Increasing in (0, 1) and decreasing in (1, e)
(B) Decreasing in (0, 1) and increasing in (1, e)
(C) $x = 1$ is the critical point for $f(x)$.
(D) f has two asymptotes

38. Consider the function $f(x)=\left[\cos\left(\tan^{-1}\left(\sin(\cot^{-1}x)\right)\right)\right]^2$. Which of the following is correct?

(A) Range of f is (0, 1)
(B) f is even
(C) $f'(0) = 0$
(D) The line y = 1 is asymptotes to the graph y = $f(x)$

39. Equation of a line which is tangent to both the curves $y=x^2+1$ and $y=-x^2$ is :

(A) $y=\sqrt{2}x+\frac{1}{2}$
(B) $y=\sqrt{2}x-\frac{1}{2}$
(C) $y=-\sqrt{2}x+\frac{1}{2}$
(D) $y=-\sqrt{2}x-\frac{1}{2}$

40. A function f is defined by $f(x)=\int_0^{\pi}\cos t\cos(x-t)\,dt$, $0\le x\le 2\pi$ then which of the following hold(s) good?

(A) $f(x)$ is continuous but not differentiable in (0, 2π)
(B) Maximum value of f is π
(C) There exists atleast one c ∈ (0, 2π) s.t. $f'(c) = 0$
(D) Minimum value of f is $-\frac{\pi}{2}$

41. Let $f(x)=\frac{x-1}{x^2}$ then which of the following is correct.

(A) $f(x)$ has minima but no maxima
(B) $f(x)$ increases in the interval (0, 2) and decreases in the interval $(-\infty, 0)\cup(2,\infty)$
(C) $f(x)$ is concave down in $(-\infty, 0)\cup(0, 3)$
(D) $x = 3$ is the point of inflection

42. $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true?
(A) $f(x)$ has a relative minimum on (–3, 4)
(B) $f(x)$ has a minimum on [–3, 4]
(C) $f(x)$ is concave upwards on [–3, 4]
(D) If $f(3) = f(4)$ then $f(x)$ has a critical point on [–3, 4]

43. Which of the following statements is/are **TRUE**?
(A) If f has domain [0, ∞) and has no horizontal asymptotes $\lim_{x\to\infty} f(x)\to\infty$ or $\lim_{x\to\infty} f(x)\to-\infty$
(B) If f is continuous on [–1, 1], $f(-1) = 4$ and $f(1) = 3$ then there exist a number r such that $|r| < 1$ and $f(r) = \pi$
(C) $\lim_{x\to\infty} \arcsin\left(\frac{x+1}{x}\right)$ does not exist
(D) For all values of $m \in R$ the line $y - mx + m - 1 = 0$ cuts the circle $x^2+y^2-2x-2y+1=0$ orthogonally

44. If a function f is continuous $\forall x$ and if f has a relative maximum at $(-1,4)$ and a relative minimum at $(3,-2)$, then which of the following statements can be incorrect ?

(A) The graph of f has a point of inflection somewhere between $x=-1$ and $x=3$.

(B) $f'(-1)=0$

(C) The graph of f has a horizontal tangent line at $x=3$

(D) The graph of f intersect both co-ordinate axes.

45. Which of the following function(s) cannot exist?

(A) $f''(x)>0$ for all $x\in R, f'(0)=1$ and $f'(1)=1$

(B) $f''(x)>0$ for all $x\in R, f'(0)=1$ and $f'(1)=2$

(C) $f''(x)\ge 0$ for all $x\in R, f'(0)=1$ and $f(x)\le 100$ for all $x>0$.

(D) $f''(x)>0$ for all $x\in R, f'(0)=1$ and $f(x)\le 1$ for all $x<0$

46. Possible integral value (s) of 'a' for which $x+\dfrac{a}{x^2}>2$ for every $x>0$, is (are)

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

47. Set $f(x)=ax^4+bx^3+cx^2+dx+e, a\ne 0,$ is a function such that $x=1, x=2$ and $x=3$ are normals to the curve $y=f(x)$ such that $f(2)$ is always greater than $f(0)$, then which of the following are true for $f(x)$?

(i) $f(x)$ has 2 local maxima

(ii) there exist only one value of k such that Rolle's theorem is applicable to $f(x)$ on the interval $[0,k]$

(iii) $f(x)=0$ has two imaginary roots.

(A) only (iii) and (i) are true **(B)** (ii) is true and (iii) is false

(C) Only (i) & (ii) are true **(D)** All are true

48. Given a natural number n. Consider the function $f(x)=\dfrac{1}{(1-x)^n}-\dfrac{1}{(1+x)^n}-2nx$, then for all $0<x<1$, which of the following is / are correct

(A) $f''(x)>0$ **(B)** $f'(x)>0$ **(C)** $f(x)>0$ **(D)** None of these

49. Which of the following statement(s) is/are correct ?

(A) If $f'(x)$ and $g'(x)$ exist, $f'(x) > g'(x)$ for all real x, then the graph of $y=f(x)$ and the graph of $y=g(x)$ intersect atmost once.

(B) Let $f:(-2,2)\to R$ be defined as $f(x)=\dfrac{x^3}{4}-\sin \pi x+3,$ then $f(c)=\dfrac{\pi}{2}$ for some $c\in(-2,2)$.

(C) If $f(x)=(x-\alpha)^p(x-\beta)^q, p,q>0$ the $x=c$ in Rolle's theorem divides the segment $\alpha\le x\le\beta$ in the ratio $p:q$.

(D) The equation $3^{x-1}+x=105,$ where x is an integer, has exactly one solution.

50. For any real valued function satisfying $f'(x)-\sin x(f(x)-1)\le 0\ \forall x\in R^+$ and $f(0)=1$ then range of $f(x)$ is $\left(-\infty,\dfrac{a}{2}\right]$ then a is :

(A) prime number **(B)** even number **(C)** odd number **(D)** a perfect square

51. Which of the following are true?

(A) There is no cubic curve for which the tangent lines at two distinct points coincide.

(B) If a is the number of horizontal tangents and b is the number of vertical tangents to the curve $y^2-3xy+2=0$, then $a+b=1$

(C) Number of integral values of a for which cubic $f(x)=x^3+ax+2$ is non monotonic and has exactly one real root is 2.

(D) If $f(x)$ is strictly increasing, then it is one-one.

52. If $\frac{f(x+y)+f(x-y)}{2}=f(x)f(y)$ for $\forall x, y \in R$ and $f(0)\neq 0$ then

(A) $f(2)=f(-2)$ **(B)** $f(3)+f(-3)=0$

(C) $f'(2)+f'(-2)=0$ **(D)** $f'(3)=f'(-3)$

53. If $f(x)=\left(\lim_{x\to\infty}\frac{\log x^n-[x]}{[x]}\right)^{[x^5+2]}$ Where [ ] represents greater integer function, then

(A) $f(x)$ is not differentiable for $x=n^{1/5}$, $n \in I$ **(B)** $f'(x)\neq 0$ for $-1<x<1$

(C) $f(2)=1$ **(D)** $f'(-2)=-1$

54. If P is a point on the curve $5x^2+3xy+y^2=2$ and O is the origin, then OP has

(A) minimum value $\frac{1}{2}$ **(B)** minimum value $\frac{2}{\sqrt{11}}$

(C) maximum value $\sqrt{11}$ **(D)** maximum value 2

55. The interval to which b may belong so that the derivative of function $f(x)=\left(1-\sqrt{\frac{21-4b-b^2}{b+1}}\right)x^3+5x+\sqrt{6}$ is always positive :

(A) $[-7,-1)$ **(B)** $[-6,-2]$ **(C)** $[2,5/2]$ **(D)** $[2,3]$

56. Let $g:R\to R$ be a differentiable function satisfying $g(x)=g(y)g(x-y)\forall x, y\in R$ and $g'(0)=a$ and $g'(3)=b$ then $g'(-3)$ is

(A) $\frac{a^2}{b}$ **(B)** $\frac{g'(3)a}{g'(0)}$ **(C)** $\frac{b}{a}$ **(D)** $\frac{g'(0)a}{g'(3)}$

57. If $1\geq a\geq b\geq c\geq 0$ and 'λ' is a real root of the equation $x^3+ax^2+bx+c=0$ then the value of $|\lambda|$ can be

(A) 1 **(B)** $\frac{3}{2}$ **(C)** 2 **(D)** $\frac{1}{2}$

58. If $x^2+y^2=1, |x|, |y|\leq 1$ then minimum value of $\frac{kx^2}{y^2}+\frac{1}{k}\left(\frac{y^2+kx^2}{x^2}\right)$, where $k>0$ is

(A) 2 **(B)** 3 **(C)** 4 **(D)** None of these

59. If $|f''(x)| \le 1, \forall x \in R$ and $f(0) = 0 = f'(0)$, then which of the following can be true ?

(A) $f\left(-\frac{1}{2}\right) = -\frac{1}{5}$ **(B)** $f(-2) = -5$ **(C)** $f\left(\frac{1}{2}\right) = -\frac{1}{5}$ **(D)** $f(1) = 0$

60. If $f(x) = \begin{cases} x^2 + 3 + \log_{0.5} \log_2 [k+3], & -1 \le x < 0 \\ x^2 + 3x + 2, & 0 \le x \le 1 \end{cases}$, (where $[.]$ denotes the greatest integer function) has minimum value at $x = 0$, then

(A) $k \in [2,5)$ **(B)** $k \in [-2,1)$ **(C)** $k \in [-1, 2]$ **(D)** $k \in [-1,2)$

61. $f(x)$ and $g(x)$ are quadratic polynomials and $|f(x)| \ge |g(x)|, \forall x \in R$. Also $f(x) = 0$ have real roots. Then the equation $h(x)h''(x) + (h'(x))^2 = 0$ (where $\left(\text{where } h(x) = f(x)g(x)\right)$.

(A) Has 6 roots
(B) Has 4 distinct roots
(C) Has exactly one pair of equal roots
(D) Has two pair of equal roots

62. If x_1 and x_2 are positive numbers between 0 and 1, then which of the following is/are true?

(A) $\sin\left(\frac{x_1 + x_2}{2}\right) \le \frac{\sin x_1 + \sin x_2}{2}$
(B) $\tan\left(\frac{x_1 + x_2}{2}\right) \le \frac{\tan x_1 + \tan x_2}{2}$
(C) $\log\left(\frac{x_1 + x_2}{2}\right) \le \frac{\log x_1 + \log x_2}{2}$
(D) $\left(\frac{x_1 + x_2}{2}\right)^2 \le \frac{x_1^2 + x_2^2}{2}$

63. If $f'(x) > f(x)$ for all $x \ge 1$ and $f(1) = 0$, then

(A) $e^x f(x)$ is a decreasing function
(B) $e^{-x} f(x)$ is an increasing function
(C) $f(x) > 0$ for all $x \in (1, \infty)$
(D) $f(x) < 0$ for all $x \in [1, \infty)$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

64. Inequalities with intervals given such that inequalities are valid :

	Column 1		Column 2
(A)	$\frac{x}{1+x} < ln(1+x)$	**(p)**	$(0, \infty)$
(B)	$x - \frac{x^2}{2} < ln(1+x)$	**(q)**	$(-1, 0)$
(C)	$ln(1+x) < x$	**(r)**	$(1, \infty)$
(D)	$\frac{1}{ln(1+x)} - \frac{1}{x} < 1$	**(s)**	$(-1, 0) \cup (0, 1)$

65. $h(x)=3f\left(\frac{x^2}{3}\right)+f(3-x^2)\forall\ x\in(-3,4)$ where $f''(x)>0\ \forall\ x\in(-3,4)$ then

	Column 1		Column 2
(A)	$h(x)$ is increasing	**(p)**	$x\in\left(\frac{3}{2},4\right)$
(B)	$h(x)$ is decreasing	**(q)**	$x\in\left(0,\frac{3}{2}\right)$
(C)	The least and greatest value of x^2+y^2-xy where x, y connected by $x^2+4y^2=4$	**(r)**	$\frac{5-\sqrt{13}}{2},\frac{5+\sqrt{13}}{2}$
(D)	$f(x)=2x^3-9x^2+12x+6$. The global minima of $f(x)$ in $(1,3)$.	**(s)**	$x=2$

66. MATCH THE FOLLOWING:

	Column 1		Column 2
(A)	The value of a for which $x^3+3(a-7)x^2+3(a^2-9)x-2$ has a +ve point of maxima	**(p)**	9/8
(B)	The minimum value of μ for which $x^3-\lambda x^2+\mu x-6=0$ has real and positive roots	**(q)**	$(-\infty,-3)\cup\left(3,\frac{29}{7}\right)$
(C)	The maximum value of $f(x)=sin\,x+cos\,2x$ for $x\in[0,2\pi]$	**(r)**	$\left(3(6)^{\frac{2}{3}},\infty\right)$
(D)	The set of values of x for which $\log(1+x)\le x$	**(s)**	$(-3,3)$
		(t)	$(-1,\infty)$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/ INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

67. If the difference between the greatest and least values of the function $f(x)=\left(3-\sqrt{4-x^2}\right)^2+\left(1+\sqrt{4-x^2}\right)^3$ is p, then find $\left[\frac{p}{2}\right]$. (where [.] represents greatest integer function).

68. $f(x)$ is a fifth order polynomial in 'x' with every root of $f(x)=0$ is real and distinct. Find the number of real roots of $f''(x)f(x)-(f'(x))^2=0$.

69. Let $f(x)=30-2x-x^3$, then find the number of positive integral values of x which satisfies $f(f(f(x)))>f(f(-x))$.

70. If $|\ln x| = px$ has exactly three distinct solutions, then find $[p]$ (where [.] denote greater integer function).

71. If $f : R \to R$ is a monotonic, differentiable real valued function, a, b are two real numbers and $\int_a^b (f(x)+f(a))(f(x)-f(a))\,dx = k\int_{f(a)}^{f(b)} x(b-f^{-1}(x))\,dx$, then the value of k is _________.

72. A line passing through $(21,30)$ and normal to the curve $y = 2\sqrt{x}$. If m is slope of the normal then $m+6=$

73. Let $P = x^3 - \frac{1}{x^3}, Q = x - \frac{1}{x}$ and $'a'$ is the minimum value of $\frac{P}{Q^2}$. Then the value of $[a]$ is _________. (where $[x]=$ the greatest integer $\le x$).

74. If θ is the angle of intersection of curves $y = \left[|\sin x| + |\cos x|\right]$ and $x^2 + y^2 = 5$.
Then the value of $|\tan\theta|$ is _________. (where [.] denotes G.I.F.)

75. Let $f : [0,\infty) \to R$ be a continuous, strictly increasing function such that $f^3(x) = \int_0^x t f^2(t)dt$. If a normal is drawn to the curve. $y = f(x)$ with gradient $\frac{-1}{2}$, then find the intercept made by it on the y-axis.

76. The smallest positive integral value of p for which the function $f(x) = 6px - p\sin 4x - 5x - \sin 3x$ is monotonic increasing and has no critical points on R is :

77. l_1 and l_2 are lengths of side of two variable squares S_1 and S_2 respectively for $l_1 = l_2 + l_2^3 + 6$ at $l_2 = 1$. If rate of change of area of S_2 with respect to area of S_1 is equal to $\frac{1}{8m}$, then $m =$.

78. If the function $\int_0^x f(t)dt \to 5$ as $|x| \to 1$, where f is continuous then the number of integers in the range of p so that the equation $2x + \int_0^x f(t)dt = p$ has roots of opposite sign in $(-1,1)$.

79. Let $f(x)$ be a function such that its derivative $f'(x)$ is continuous in $[a,b]$ and derivable in (a,b). Consider a function $\phi(x) = f(b) - f(x) - (b-x)f'(x) - (b-x)^2 A$. If Rolle's theorem is applicable to $\phi(x)$ on $[a,b]$ and for some $c \in (a,b), \phi'(c) = 0$ and $f(b) = f(a) + (b-a)f'(a) + \lambda f''(c)(b-a)^2$ then 6λ is equals to :

80. The values of parameter a such that the line $\left(\log_2\left(1+5a-a^2\right)\right)x - 5y - \left(a^2-5\right) = 0$ is a normal to the curve $xy = 1$, may lie in the interval (c,d) then $c-d$ is equals to :

81. A polynomial function $P(x)$ of degree 5 with leading coefficient one, increases in the interval $(-\infty,1)$ and $(3,\infty)$ and decreases in the interval $(1,3)$. Given that $P(0) = 4$ and $P'(2) = 0$. Find the value $P'(6)$.

82. The circle $x^2+y^2=1$ cuts the x-axis at P and Q. Another circle with center at Q and variable radius intersect the first circle at R above the x-axis and the line segment PQ at S, the maximum area of the triangle QSR is A, thens $[2A]$ is ________([.]is $G.I.F$)

83. A wire $20cm$ long be divided into two parts, if one part is to be bent into a circle, the other part is to be bent into a square and the two plane figures are to have areas the sum of which is maximum, then side length of square is____.

84. If $\int_e^x t\, f(t)\,dt = \sin x - \cos x - \frac{x^2}{2}$ for all $x \in R$ then value of $\left(-2f\left(\frac{\pi}{6}\right)\right)$ is ____.

85. If $f(x+h) = f(x) + hf'(x+\theta h), 0<\theta<1$, the value of 4θ, when $f(x) = Ax^2+Bx+C$ is______________

86. In the interval (a,b) there exists at least one point c, for any two differentiable function f and g such that $\begin{vmatrix} f(a) & f(b) \\ \phi(a) & \phi(b) \end{vmatrix} = \lambda^2(b-a)\begin{vmatrix} f(a) & f'(c) \\ \phi(a) & \phi'(c) \end{vmatrix}$, then sum of absolute value of λ is______.

87. $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, $\frac{\pi}{2} < x < \pi$, then $\left(\frac{-dy}{dx}\right)$ is equal to_______.

88. The third derivative of a function $f(x)$ vanishes for all x. If $f(0)=1$, $f'(1)=2$ and $f''(1)=-1$, then find $f'(x)$ at $x=3$.

89. A lane of width $27m$ runs at right angle out of a road of $64m$. The maximum length of a pole which can be carried from the road to the lane keeping it horizontal is L, then $\sqrt[3]{L}$ equals to __________

90. If $a^2+b^2=1$ and u is the minimum value of $\frac{b+1}{a+b-2}$ then find the value of u^2.

91. Let f be a differentiable function on R and satisfying $f(x) = -\left(x^2-x+1\right)e^x + \int_0^x e^{x-y}.f'(y)\,dy$.

If $f(1)+f'(1)+f''(1) = ke$, where $k \in N$, then find k.

92. Let $P(x_0, y_0)$ be a point on the curve $C:\left(x^2-11\right)(y+1)+4=0$ where $x_0, y_0 \in N$. If area of the triangle formed by the normal drawn to the curve 'C' at P and the co-ordinate axes is $\left(\frac{a}{b}\right), a,b \in N$ then the least value of $(a-6b)$.

93. Let f be a twice differentiable function defined in $[-3,3]$ such that $f(0)=-4, f'(3)=0, f'(-3)=12$ and $f''(x) \ge -2 \forall x \in [-3,3]$. If $g(x) = \int_0^x f(t)dt$ then find maximum value of $g(x)$, in [–3, 3]

94. If $9 + f''(x) + f'(x) = x^2 + f^2(x)$, where $f(x)$ is twice differentiable function such that $f''(x) \ne 0 \forall x \in R$ and let P be the point of maxima of $f(x)$ then find the number of tangents which can be drawn from P to the circle $x^2+y^2=9$.

Advanced Problem Package	Integral Calculus-1

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If $f(x)=\lim\limits_{n\to\infty} e^{x\tan(1/n)\log(1/n)}$, and $\int \dfrac{f(x)}{\sqrt[3]{\sin^{11}x\cos x}}dx = g(x)+C$ (C being the constant of integration). Then :

(A) $g\left(\dfrac{\pi}{4}\right)=\dfrac{3}{2}$ **(B)** $g(x)$ is continuous for all

(C) $g\left(\dfrac{\pi}{4}\right)=-\dfrac{15}{8}$ **(D)** $g\left(\dfrac{\pi}{4}\right)=-\dfrac{1}{2}$

2. If $f(x)=\lim\limits_{n\to\infty}\left[2x+4x^3+........+2nx^{2n-1}\right](0<x<1)$ then $\int f(x)dx$ is equal to :

(A) $-\sqrt{1-x^2}+c$ **(B)** $\dfrac{1}{\sqrt{1-x^2}}+c$ **(C)** $\dfrac{1}{x^2-1}+c$ **(D)** $\dfrac{1}{1-x^2}+c$

3. The integral $\int \dfrac{\sec^{3/2}\theta-\sec^{1/2}\theta}{2+\tan^2\theta}\tan\theta d\theta$ is :

(A) $\sqrt{2}\tan^{-1}\left(\dfrac{\sec\theta+1}{\sqrt{2\sec\theta}}\right)+C$ **(B)** $\dfrac{1}{\sqrt{2}}\log_e\left|\dfrac{\sec\theta-\sqrt{2\sec\theta}+1}{\sec\theta+\sqrt{2\sec\theta}+1}\right|+C$

(C) $\dfrac{1}{\sqrt{2}}\tan^{-1}\left(\dfrac{\sec\theta+1}{\sqrt{2\sec\theta}}\right)+C$ **(D)** $\sqrt{2}\log_e\left|\dfrac{\sec\theta-\sqrt{2\sec\theta}+1}{\sec\theta+\sqrt{2\sec\theta}+1}\right|+C$

4. If $f(x)=\lim\limits_{n\to\infty}\dfrac{x^n-x^{-n}}{x^n+x^{-n}}, 0<x<1,\ n\in N$ then $\int\left(\sin^{-1}x\right)f(x)dx$ is equal to :

(A) $-\left[x\sin^{-1}x+\sqrt{1-x^2}\right]+C$ **(B)** $x\sin^{-1}x+\sqrt{1-x^2}+C$

(C) $\dfrac{x^2}{2}+C$ **(D)** $\dfrac{1}{2}\left(\sin^{-1}x\right)^2+C$

5. If $I_n=\int\cot^n x dx$, then $I_0+I_1+2\left(I_2+I_3+..........+I_8\right)+I_9+I_{10}=$

(A) $-\sum\limits_{k=1}^{9}\dfrac{\cot^k x}{k}$ **(B)** $\sum\limits_{k=1}^{9}\dfrac{\cot^k x}{k!}$ **(C)** $\sum\limits_{k=1}^{10}\dfrac{\cot^k x}{10}$ **(D)** $-\sum\limits_{k=1}^{10}k\cot^k x$

6. If $\int\dfrac{\left(sin^{3/2}\theta+cos^{3/2}\theta\right)d\theta}{\sqrt{sin^3\theta\, cos^3\theta\, sin(\theta+\alpha)}}=a\sqrt{cos\,\alpha\tan\theta+sin\,\alpha}+b\sqrt{cos\,\alpha+sin\,\alpha\cot\theta}+c$ then :

(A) $a=2\sec\alpha, b=2\cos ec\alpha, c\in R$ **(B)** $a=2\sec\alpha, b=-2\cos ec\alpha, c\in R$

(C) $a=-2\sec\alpha, b=2\cos ec\alpha, c\in R$ **(D)** $a=2\cos ec\alpha, b=2s ec\alpha,\ \ c\in R$

7. If $\int \frac{\cos^2 x+\sin 2x}{(2\cos x-\sin x)^2}dx=\frac{\cos x}{2\cos x-\sin x}+ax+b\ln|2\cos x-\sin x|+c$, then :

(A) $a=\frac{1}{5}, b=\frac{2}{5}$ **(B)** $a=\frac{1}{5}, b=-\frac{2}{5}$ **(C)** $a=-\frac{1}{5}, b=\frac{2}{5}$ **(D)** $a=-\frac{1}{5}, b=-\frac{2}{5}$

Paragraph for Questions 8 - 10

Let us consider the integrals of the following forms $f\left(x,\sqrt{mx^2+nx+p}\right)^{1/2}$

Case I: If $m>0$, then put $\sqrt{mx^2+nx+c}=u\pm x\sqrt{m}$ Case II: If p > 0, then put $\sqrt{mx^2+nx+c}=ux\pm\sqrt{p}$

Case III: If quadratic equation $mx^2+nx+p=0$ has real roots α and β then put $mx^2+nx+p=(x-\alpha)u\,or\,(x-\beta)u$

8. If $\int \frac{dx}{x-\sqrt{9x^2+4x+6}}$ to evaluate I, one of the most proper substitution could be :

(A) $\sqrt{9x^2+4x+6}=u\pm 3x$ **(B)** $\sqrt{9x^2+4x+6}=3u\pm x$

(C) $x=\frac{1}{t}$ **(D)** $9x^2+4x+6=\frac{1}{t}$

9. $\int \frac{\left(x+\sqrt{1+x^2}\right)^{15}}{\sqrt{1+x^2}}dx$ is equal to :

(A) $\frac{\left(x+\sqrt{1+x^2}\right)^{16}}{10}+c$ **(B)** $\frac{1}{15\left(\sqrt{1+x^2}+x\right)}+c$

(C) $\frac{15}{\left(\sqrt{1+x^2}-x\right)}+c$ **(D)** $\frac{\left(\sqrt{1+x^2}+x\right)^{15}}{15}+c$

10. To evaluate $\int \frac{dx}{(x-1)\sqrt{-x^2+3x-2}}$ one of the most suitable substitution could be :

(A) $\sqrt{-x^2+3x-2}=u$ **(B)** $\sqrt{-x^2+3x-2}=\left(ux\sqrt{2}\right)$

(C) $\sqrt{-x^2+3x-2}=u(1-u)$ **(D)** $\sqrt{-x^2+3x-2}=u(x-2)$

11. If $I_n=\int \frac{dx}{\left(x^2+a^2\right)^n}$, where $n\in N$ and n > 1. If I_n and I_{n-1} are related by the relation $PI_n=\frac{x}{\left(x^2+a^2\right)^{n-1}}+QI_{n-1}$. Then P and Q are respectively given by :

(A) $(2n-1)a^2, 2n-3$ **(B)** $2a^2(n-1), 2n-1$

(C) $a^2(n+1), 2n+3$ **(D)** $a^2, a^2(n+1)$

12. $\int \frac{(ax^2-b)dx}{x\sqrt{c^2x^2-(ax^2+b)^2}} =$

(A) $\sin^{-1}\left(\frac{ax+bx^2}{c}\right)+k$ **(B)** $\tan^{-1}\left(\frac{a+bx^2}{cx}\right)+k$

(C) $\sin^{-1}\left(\frac{ax^2+b}{cx}\right)+k$ **(D)** $\tan^{-1}\left(ax^2+bx+c\right)+k$

13. Let $f:\left[0,\frac{\pi}{2}\right]\to R$ be such that $f(0)=3$ and $f'(x)=\frac{1}{1+\cos x}$. If $a<f\left(\frac{\pi}{2}\right)<b$, then a and b can be

(A) $\frac{\pi}{2},\pi$ **(B)** 3, 4

(C) $3+\frac{\pi}{4},3+\frac{\pi}{2}$ **(D)** $3+\frac{\pi}{2},3+\frac{3\pi}{4}$

Paragraph for Questions 14 - 18

Evaluation of indefinite integral with the help of specific substitution:

In general if we have an integral of type $\int f\left(g(x)\right)g'(x)dx$, we substitute $g(x)=t\Rightarrow g'(x)dx=dt$ and the integral becomes $\int f(t)dt$. Some of the substitution can be guessed by keen observation of the nature of given integrand. For example, we have $\frac{d}{dx}\left(x+\frac{1}{x}\right)=1-\frac{1}{x^2}$. So if the integrand is of the type $f\left(x+\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)$, we can substitute $x+\frac{1}{x}=t$.

Some more similar forms are given below

For integral $\int f\left(x-\frac{a}{x}\right)\left(1+\frac{a}{x^2}\right)dx$, put $x-\frac{a}{x}=t$ For integral $\int f\left(x+\frac{a}{x}\right)\left(1-\frac{a}{x^2}\right)dx$, put $x+\frac{a}{x}=t$

For integral $\int f\left(x^2-\frac{a}{x^2}\right)\left(x+\frac{a}{x^3}\right)dx$, put $x^2-\frac{a}{x^2}=t$ For integral $\int f\left(x^2+\frac{a}{x^2}\right)\left(x-\frac{a}{x^3}\right)dx$, put $x^2+\frac{a}{x^2}=t$

Many integrands can be brought into above forms by suitable reductions or transformations

14. $\int \frac{x^2+1}{x^4+1}dx =$

(A) $\frac{1}{\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}x}+C$ **(B)** $\sin^{-1}\frac{x^2+1}{\sqrt{2}x}+C$

(C) $\frac{1}{2}\log\frac{\sqrt{2}x+1}{\sqrt{2}x-1}+C$ **(D)** $x^2+\frac{1}{x^2}+C$

15. $\int \frac{x^2-1}{\left(x^4+3x^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)}dx =$

(A) $\tan^{-1}\left(x+\frac{1}{x}\right)+C$ **(B)** $\left(x+\frac{1}{x}\right)\tan^{-1}\left(x+\frac{1}{x}\right)+C$

(C) $\ln\left|\tan^{-1}\left(x+\frac{1}{x}\right)\right|+C$ **(D)** $\frac{1}{2}\ln\left|x+\frac{1}{x}\right|+C$

16. $\int \frac{x^4-2}{x^2\sqrt{x^4+x^2+2}}dx =$

(A) $\sqrt{x^2+1+\frac{1}{x^2}}+C$ **(B)** $\sqrt{x^2+1+\frac{2}{x^2}}+C$

(C) $\sqrt{x^2+\frac{1}{x^2}}+C$ **(D)** $\sqrt{x^2+\frac{2}{x^2}}+C$

17. Anti-derivative of $\frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}$ is :

(A) $tan^{-1}\left(x+\frac{1}{x}+1\right)$ **(B)** $tan^{-1}\sqrt{x+\frac{1}{x}+1}$ **(C)** $2tan^{-1}\sqrt{x+\frac{1}{x}+1}$ **(D)** $\sqrt{x+\frac{1}{x}+1}$

18. The derivative of $x^{-4}+x^{-5}$ is $-\left(4x^{-5}+5x^{-6}\right)$. So, $\int \frac{5x^4+4x^5}{\left(x^5+x+1\right)^2}dx =$

(A) $x^5+x+1+C$ **(B)** $\frac{1}{x^5+x+1}+C$ **(C)** $x^{-4}+x^{-5}+C$ **(D)** $\frac{x^5}{x^5+x+1}+C$

19. Let $f(xy)=f(x).f(y), \forall x>0, y>0$ and $f(1+x)=1+x\{1+g(x)\}$, where $\lim_{x\to 0} g(x)=0$, then $\int \frac{f(x)}{f'(x)}dx$ is :

(A) $\frac{x^2}{2}+c$ **(B)** $\frac{x^3}{3}+c$ **(C)** $\frac{x^2}{3}+c$ **(D)** None of these

20. If $\int \frac{dx}{x^2\left(x^n+1\right)^{(n-1)/n}} = -\left[f(x)\right]^{1/n}+C$, then $f(x)$ is

(A) $\left(1+x^n\right)$ **(B)** $1+x^{-n}$ **(C)** x^n+x^{-n} **(D)** None of these

21. If $I_{m,n}=\int \cos^m x\sin nx dx$, then $7I_{4,3}-4I_{3,2}=$

(A) Constant **(B)** $-cos^2 x+C$ **(C)** $-cos^4 x cos 3x+C$ **(D)** $cos 7x-cos 4x+C$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

22. The value of the integral $\int e^{\sin^2 x}\left(\cos x+\cos^3 x\right)\sin x dx$ is :

(A) $\frac{1}{2}e^{\sin^2 x}\left(3-\sin^2 x\right)+c$ **(B)** $e^{\sin^2 x}\left(1+\frac{1}{2}\cos^2 x\right)+c$

(C) $e^{\sin^2 x}\left(3\cos^2 x+2\sin^2 x\right)+c$ **(D)** $e^{\sin^2 x}\left(2\cos^2 x+3\sin^2 x\right)+c$

23. $I=\int \frac{dx}{(\sin x-2\cos x)(2\cos x+\sin x)}$ is equal to

(A) $\log_e \sqrt[4]{\frac{\tan x-2}{\tan x+2}}+c$ **(B)** $\frac{1}{4}\log_e\left|\frac{\sin x-2\cos x}{\sin x+2\cos x}\right|+c$

(C) $-\frac{1}{4}\log_e\left|\frac{2\sin x+\cos x}{\sin x+2\cos x}\right|+c$ **(D)** None of these

24. If $\int \sqrt{\cos ecx+1}\ dx = kfog(x)+c$, where k is a real constant, then :

(A) $k=-2, f(x)=cot^{-1}x, g(x)=\sqrt{cos\,ecx-1}$ **(B)** $k=-2, f(x)=tan^{-1}x, g(x)=\sqrt{cos\,ecx-1}$

(C) $k=2, f(x)=tan^{-1}x, g(x)=\frac{cot\,x}{\sqrt{cos\,ecx-1}}$ **(D)** $k=2, f(x)=cot^{-1}x, g(x)=\frac{cot\,x}{\sqrt{cos\,ecx+1}}$

25. If $f'(x)=\frac{1}{-x+\sqrt{x^2+1}}$ and $f(0)=\frac{-(1+\sqrt{2})}{2}$, then the value of $f(5)$ will be :

(A) More than 12 **(B)** Loss than 5 **(C)** More than 5 **(D)** None of these

26. Let $f(x)=\frac{1}{4-3\cos^2 x+5\sin^2 x}$ and its anti-derivative $F(x)=\frac{1}{3}\tan^{-1}(g(x))+c$, then :

(A) $g(x)$ is equal to 3tanx **(B)** $g\left(\frac{\pi}{4}\right)$ is equal to 3

(C) $g'\left(\frac{\pi}{3}\right)$ is equal to 6 **(D)** $g'\left(\frac{\pi}{3}\right)$ is equal to 12

27. If $\int \frac{\sin x-\cos x}{(\sin x+\cos x)\sqrt{\sin x\cos x+\sin^2 x\cos^2 x}}dx=\cos ec^{-1}(g(x))+c\,\forall\, x\in R$, then

(A) $g(x)=1+\sin 2x$ **(B)** $g(x)=1-\sin 2x$

(C) $g(x)\ge 0$ **(D)** $-1\le g(x)\le 1$

28. If $x^2\ne(n\pi-1)\forall n\in N$, then $I=\int \frac{x\sqrt{2\,sin(x^2+1)-sin\,2(x^2+1)}}{2\,sin(x^2+1)+sin\,2(x^2+1)}dx$ is equal to :

(A) $log\left|sec\frac{(x^2+1)}{2}\right|+c$ **(B)** $-log\left|cos\frac{(x^2+1)}{2}\right|+c$

(C) $log\left|tan\frac{(x^2+1)}{2}\right|+c$ **(D)** $-log\left|cot\frac{(x^2+1)}{2}\right|+c$

29. Let $f(x)=\{b^2+(a-1)b+2\}x-\int(\sin^2 x+\cos^4 x)dx$ be an increasing function of $x\in R$ and $b\in R$, then a can take value(s) :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 4

30. If $\int \cos ec^2 x\left(\cos x+\sqrt{\cos 2x}\right)dx=\cot x\log\left(\cos x+\sqrt{\cos 2x}\right)+P\left(\cos ec^2 x-2\right)^t+Q\left(x+\cot x\right)+c$, , then

(A) $t=P+\dfrac{Q}{2}$ **(B)** $P+Q=0$ **(C)** $P\neq Q\neq t$ **(D)** $P-\dfrac{Q}{2}=t$

31. $\int \dfrac{\sin 2x}{8\sin^2 x+17\cos^2 x}dx$ is equal to :

(A) $-\dfrac{1}{9}\log_e\left(8\sin^2 x+17\cos^2 x\right)+c$ **(B)** $\log_e \dfrac{1}{\sqrt[9]{8+9\cos^2 x}}+c$

(C) $\log_e \dfrac{1}{\sqrt[9]{17-9\sin^2 x}}+c$ **(D)** None of these

32. $I=\int \sec^3(2\theta)d\theta$ is equal to :

(A) $\dfrac{1}{2}(\sec\theta\tan\theta)+\log_e\sqrt{\sec\theta+\tan\theta}+c$ **(B)** $\dfrac{1}{4}(\sec 2\theta\tan 2\theta)+\dfrac{1}{2}\log_e\sqrt{\sec 2\theta+\tan 2\theta}+c$

(C) $\dfrac{1}{4}(\sec 2\theta\tan 2\theta)+\log_e\sqrt[4]{\sec 2\theta+\tan 2\theta}+c$ **(D)** None of these

33. $I_1=\int 2^x dx=p(x)+c_1$ and $I_2=\int\left(\dfrac{1}{2}\right)^x dx=m(x)+c_1$ then $p(x)-m(x)$ is equal to

(A) $\{\log_e(2)\}\left(2^x-2^{-x}\right)$ **(B)** $\{\log_e 2\}\left(2^x+2^{-x}\right)$

(C) $\dfrac{1}{\log_e 2}\left(2^x+2^{-x}\right)$ **(D)** $\log_2 e\left(2^x+2^{-x}\right)$

34. If $\int \dfrac{dx}{p^2\sin^2 x+r^2\cos^2 x}=\dfrac{1}{12}\tan^{-1}(3\tan x)+c$, then the value of psinx + rcosx can be :

(A) $\dfrac{6}{\sqrt{5}}$ **(B)** $\sqrt{5}$ **(C)** $6\sqrt{3}$ **(D)** –4

35. $I=\int \dfrac{dx}{\left(1+\sqrt{x}\right)^8}$ is equal to:

(A) $\dfrac{-1}{21\left(1+\sqrt{x}\right)^6}\left(\dfrac{6\sqrt{x}}{1+\sqrt{x}}+1\right)+c$ **(B)** $\dfrac{-1}{21\left(1+\sqrt{x}\right)^7}\left(7\sqrt{x}+1\right)+c$

(C) Either (A) and (B) **(D)** None of these

36. If $\int \dfrac{\sin x}{\sin(x-\alpha)}dx=Ax+B\log\sin(x-\alpha)+c$, then

(A) $A=\sin\alpha$ **(B)** $B=\sin\alpha$ **(C)** $A=\cos\alpha$ **(D)** $B=\cos\alpha$

37. If $\int(\sin 3\theta+\sin\theta)e^{\sin\theta}\cos\theta d\theta=\left(A\sin^3\theta+B\cos^2\theta+C\sin\theta+D\cos\theta+E\right)e^{\sin\theta}+F$ then :

(A) B = 12 **(B)** D = 0 **(C)** B = –12 **(D)** None of these

38. $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx \ \forall\ x>0$ is equal to :

(A) $2\tan^{-1}\sqrt{x^2+\frac{1}{x^2}+1}+c$ **(B)** $\tan^{-1}\sqrt{x+\frac{1}{x}+1}+c$

(C) $2\tan^{-1}\sqrt{x+\frac{1}{x}+1}+c$ **(D)** $2\sec^{-1}\sqrt{x+\frac{1}{x}+2}+c$

39. If $\int \frac{x+\left(cos^{-1}3x\right)^2}{\sqrt{1-9x^2}}dx = P\sqrt{1-9x^2}+Q\left(cos^{-1}3x\right)^3+c$ then :

(A) $P=-\frac{1}{9}$ **(B)** $Q=-\frac{3}{8}$ **(C)** $P=\frac{3}{8}$ **(D)** $Q=-\frac{1}{9}$

40. If $\int \frac{\sec x(2+\sec x)}{(1+2\sec x)^2}$ is $\left(\frac{\cos x+2}{\lambda \sin x}\right)^p$

(A) $\lambda=1$ **(B)** $\lambda=-1$ **(C)** $p=1$ **(D)** $p=-1$

41. A function $f(x)$ continuous on R and periodic with 2π satisfies $f(x)+(\sin x)f(x+\pi)=\sin^2 x$ then,

(A) $f(x)=\frac{\sin^2 x\left(1+\sin^2 x\right)}{(1-\sin x)}$

(B) $f(x)=\frac{\sin^2 x(1-\sin x)}{\left(1+\sin^2 x\right)}$

(C) $\int f(x)dx = x+\cos x-\frac{1}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\tan x\right)+\frac{1}{2\sqrt{2}}\ln\frac{\left(\sqrt{2}-\cos x\right)}{\left(\sqrt{2}+\cos x\right)}+c$

(D) None of these

42. $\int \frac{dx}{\left(\prod_{r=0}^{n}(x+r)\right)}$ is equal to :

(A) $\frac{1}{n!}\left[\sum_{r=0}^{n}(-1)^r.{}^nC_r \ln(x+r)\right]+c$ **(B)** $\frac{1}{n!}\left[\sum_{r=0}^{n-1}(-1)^{r-1}.\ln(x+r-1)\right]+c$

(C) $\frac{1}{n!}\ln\left(\prod_{r=0}^{n}(x+r)^{(-1)^r.{}^nC_r}\right)+c$ **(D)** $\frac{1}{n!}\left[\sum_{r=0}^{n-1}\ln(x+r-1)^{(-1)^{r-1}}\right]$

43. In a certain problem the differentiation of product $(f(x).g(x))$ appears. One student commits mistake and differentiates as $\left(\frac{d(f(x))}{dx}.\frac{d(g(x))}{dx}\right)$ but he gets correct result if $f(x)=x^3$ & $g(4)=9, g(2)=-9$ & $g(0)=\frac{-1}{3}$ then :

(A) $g(x)=\frac{9}{(x-3)^3}$

(B) $\left(\frac{d}{dx}(f(x-3).g(x))\right)_{at\ x=100}=0$

(C) $\lim\limits_{x\to 0}\frac{f(x).g(x)}{x(1+g(x))}=0$

(D) None of these

44. If 'A' is a square matrix and e^A is defined as $e^A=I+A+\frac{A^2}{2!}+\frac{A^3}{3!}+........=\frac{1}{2}\begin{bmatrix} f(x) & g(x) \\ g(x) & f(x)\end{bmatrix}$ where $A=\begin{bmatrix} x & x \\ x & x\end{bmatrix}$ and $0<x<1$, I is an identity matrix, then:

(A) $\int\frac{g(x)}{f(x)}dx=\ln\left(e^x+e^{-x}\right)+c.$

(B) $f(x)\in R^+$ & $g(x)\in R^+$

(C) $f(x)\in R^+$ & $g(x)\in R$

(D) $\int(g(x)+1)\sin x dx=\frac{e^{2x}}{5}(2\sin x-\cos x)+c$

45. $I=\int\frac{(x^2-1)\sqrt{x^4+2x^3-x^2+2x+1}}{x^2(x+1)^2}dx$ is equal to (where $t=x+\frac{1}{x}$)

(A) $\sqrt{t^2+2t-3}-\ln\left(t+1+\sqrt{t^2+2t-3}\right)-\sqrt{3}\sin^{-1}\left(\frac{t+5}{2t+4}\right)+c$

(B) $\sqrt{t^2+t-3}-\ln\left(t+1+\sqrt{t^2+t-3}\right)-\sqrt{3}\sin^{-1}\left(\frac{t+5}{2t+4}\right)+c$

(C) $\sqrt{t^2+2t-3}-\ln\left(t+1+\sqrt{t^2+2t-3}\right)+\sqrt{3}\cos^{-1}\left(\frac{t+5}{2t+4}\right)+c$

(D) None of these

46. Let $f:R\to R$ be a function as $f(x)=(x-1)(x+2)(x-3)(x-6)-100$ If $g(x)$ is a polynomial of degree ≤ 3 such that $\int\frac{g(x)}{f(x)}dx$ does not contain any logarithm function and $g(-2)=10$, then:

(A) $f(x)=0$ has two real & two imaginary roots

(B) $(f(x))_{\min}=-84$

(C) $\int\frac{g(x)}{f(x)}dx=\tan^{-1}\left(\frac{x-2}{2}\right)+c$

(D) $g(2)=-42$

47. $I_1 = \int f(x)dx$ and $I_2 = \int_0^1 f(x)dx$ where $f(x) = x^2 \ln(1-x^2)$, then:

(A) $I_1 = -\left\{\frac{x^3}{1.3} + \frac{x^5}{3.5} + \frac{x^7}{5.7} + \ldots\ldots\ldots\ldots\right\}$ (B) $I_2 = \frac{2}{3}\ln 2 - \frac{8}{9}$

(C) $I_1 = -\left\{\frac{x^5}{1.5} + \frac{x^7}{2.7} + \frac{x^9}{3.9} + \ldots\ldots\right\}$ (D) $I_2 = \frac{2}{3}\ln 2 - \frac{5}{9}$

48. Let $f : R \to R$ be a function satisfying

$f(x+2y) = f(x)e^{2y} + f(2y)e^x + x^2(1-e^{2y}) + 4y^2(1-e^x) + 4xy \forall x, y \in R$ and $f'(0) = 1$, then:

(A) $f(x) = xe^x + x^2$. (B) $f(x) = xe^x - x^2$.

(C) $\int f(x)dx = e^x(x-1) + \frac{x^3}{3} + c$ (D) $\int f(x)dx = e^x(x-1) - \frac{x^3}{3} + c$

49. If $\int\left(\frac{1}{1-x^8}\right)\left[\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)\right]dx$. has $p\tan^{-1} f(x)$ & $q\tan^{-1} g(x)$ terms and $x \in (-1, 1)$, then: (where p and q are constant)

(A) $f(x).g(x) = \frac{x^2-1}{\sqrt{2}}$ (B) $p.q = \frac{\pi^2}{64\sqrt{2}}$

(C) $p.q = \frac{\pi^2}{\sqrt{2}}$ (D) None of these

50. If $A = \int e^{ax}\cos bx\, dx$ and $B = \int e^{ax}\sin bx\, dx$, then which of the following may be correct?

(A) $(A^2+B^2)(a^2+b^2) = e^{2ax}$ (B) $\tan^{-1}\frac{B}{A} + \tan^{-1}\frac{b}{a} = bx$

(C) (A) and (B) both (D) None of these

51. If $\int\sqrt{\frac{\cos x - \cos^3 x}{(1-\cos^3 x)}}dx = f(x) + c$, then $f(x)$ is equal to:

(A) $\frac{2}{3}\sin^{-1}\left(\cos^{\frac{3}{2}} x\right)$ (B) $\frac{3}{2}\sin^{-1}\left(\cos^{\frac{3}{2}} x\right)$ (C) $\frac{2}{3}\cos^{-1}\left(\cos^{\frac{3}{2}} x\right)$ (D) $-\frac{2}{3}\sin^{-1}\left(\cos^{\frac{3}{2}} x\right)$

52. If $\int\frac{dx}{x^{22}(x^7-6)} = A\left\{\ln(p)^6 + 9p^2 - 2p^3 - 18p\right\} + c$, then:

(A) $A = \frac{1}{9072}$, (B) $p = \left(\frac{x^7-6}{x^7}\right)$ (C) $A = \frac{1}{54432}$, (D) $p = \left(\frac{x^7-6}{x^7}\right)^{-1}$

53. $f(x) = \int e^{\tan^{-1}x}(1+x+x^2)d(\cot^{-1}x)$ is equal to:

(A) $-e^{\tan^{-1}x} + C$ (B) $f(x)$ is decreasing (C) $-xe^{\tan^{-1}x} + C$ (D) $f(x)$ is increasing

54. If the anti-derivative of $\dfrac{x^3}{\sqrt{1+2x^2}}$ which passes through (1, 2) is $\dfrac{1}{m}\left(1+2x^2\right)^{1/2}\left(x^2-1\right)+c$. Then :

(A) $c=6$ **(B)** $m=6$ **(C)** $m+c=12$ **(D)** $m+c=8$

55. If $I=\int \dfrac{\sin x+\sin^3 x}{\cos 2x}dx = P\cos x+Q\ln\,|\,\mathrm{f}\,(x)|+R$, then :

(A) $P=1/2,\ Q=-\dfrac{3}{4\sqrt{2}}$ $P=1/2,\ Q=\dfrac{3}{4\sqrt{2}},\ f(x)=\dfrac{\sqrt{2}\cos x+1}{\sqrt{2}\cos x-1}$

(B) $P=1/4, Q=-\dfrac{1}{\sqrt{2}}, f(x)=\dfrac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}$

(C) $P=1/2, Q=-\dfrac{3}{4\sqrt{2}}, f(x)=\dfrac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}$

(D) $P=-1/2, Q=-\dfrac{3}{4\sqrt{2}}, f(x)=\dfrac{\sqrt{2}\cos x+1}{\sqrt{2}\cos x-1}$

56. If $f(x), g(x)$ and $h(x)$ are continuous and positive functions such that:
$f(x)+g(x)+h(x)=\sqrt{f(x)g(x)}+\sqrt{g(x)h(x)}+\sqrt{h(x)f(x)}$, then $\int\left(f(x)+g(x)-2h(x)\right)dx$ is/are

(A) Independent of $f(x)$ **(B)** Independent of $g(x)$

(C) Independent of $h(x)$ **(D)** only A and B

57. Let f is a differentiable function such that $f(x)=x^2+\int_0^x e^{-t}f(x-t)dt$, then:

(A) $\int e^x f(x)dx=\dfrac{e^x x^3}{3}+c$ **(B)** $f(x)$ is neither even nor odd

(C) $\int e^x f(x)dx=\dfrac{e^x x^2}{2}+c$ **(D)** $f(3)=18$

58. Let f is a differentiable function such that $f'(x)=f(x)+\int_0^2 f(x)dx, f(0)=\dfrac{4-e^2}{3}$, then:

(A) $f(x)=e^{-x}-\dfrac{\left(e^2+1\right)}{3}$ **(B)** $f(x)=e^x-\dfrac{\left(e^2-1\right)}{3}$

(C) $f(x)$ is increasing **(D)** $f(x)$ is decreasing

59. Let f be a function such that $f(x).f(y)+2=f(x)+f(y)+f(xy), \forall x,y\in R-\{0\}$ and $f(0)=1, f'(1)=2$, then:

(A) $3\left(\int f(x)dx\right)-x\left(f(x)+2\right)$ is constant **(B)** $3\left(\int f(x)dx\right)-x\left(f(x)-2\right)$ is constant

(C) $\int f(x)dx-x\left(f(x)-2\right)$ is constant **(D)** the only possible value of $f(1)$ is 2

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

60. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	If $\int \frac{2^x}{\sqrt{1-4^x}}dx = k\sin^{-1}(f(x))+C$, then k is greater than	(p)	0
(B)	If $\int \frac{(\sqrt{x})^5}{(\sqrt{x})^7+x^6}dx = a\ln\frac{x^k}{x^k+1}+c$, then ak is less than	(q)	1
(C)	$\int \frac{x^4+1}{x\left(x^2+1\right)^2}dx = k\ln\|x\|+\frac{m}{1+x^2}+n$, where n is the constant of integration, then mk is greater than	(r)	3
(D)	$\int \frac{dx}{5+4\cos x} = k\tan^{-1}\left(m\tan\frac{x}{2}\right)+C$, then k/m is greater than	(s)	4

61. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	$\int \frac{e^{2x}-1}{e^{2x}+1}dx$ is equal to	(p)	$x-\log\left[1+\sqrt{1-e^{2x}}\right]+c$
(B)	$\int \frac{1}{(e^x+e^{-x})^2}dx$ is equal to	(q)	$\log(e^x+1)-1-e^{-x}+c,$
(C)	$\int \frac{e^{-x}}{1+e^x}dx$ is equal to	(r)	$\log(e^{2x}+1)-x+c$
(D)	$\int \frac{1}{\sqrt{1-e^{2x}}}dx$ is equal to	(s)	$-\frac{1}{2\left(e^{2x}+1\right)}+c$

62. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	$\int \frac{dx}{\sqrt{x}(x+9)} =$	(p)	$\log\left\|1-\tan\left(\frac{x}{2}\right)\right\|+c$
(B)	$\int e^x\left(1-\cot x+\cot^2 x\right)dx =$	(q)	$\log\left\|1-\cot\left(\frac{x}{2}\right)\right\|+c$
(C)	$\int \frac{\sin^3 x+\cos^3 x}{\cos^2 x\sin^2 x}dx =$	(r)	$\sec x-\operatorname{cosec} x+c$
(D)	$\int \frac{dx}{1-\cos x-\sin x} =$	(s)	$\frac{2}{3}\tan^{-1}\frac{\sqrt{x}}{3}+c$
		(t)	$-e^x.\cot x+c$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

63. Let $f(x)=\int x^{\sin x}(1+x\cos x\cdot \ln x+\sin x)dx$ and $f\left(\frac{\pi}{2}\right)=\frac{\pi^2}{4}$, then the value of $|\cos(f(\pi))|$ is

64. Let $g(x)=\int \frac{1+2\cos x}{(\cos x+2)^2}dx$ and $g(0)=0$, then the value of $8g(\pi/2)$ is

65. Let $k(x)=\int \frac{(x^2+1)dx}{\sqrt[3]{x^3+3x+6}}$ and $k(-1)=\frac{1}{\sqrt[3]{2}}$, then the value of $k(-2)$ is

66. If $\int \frac{2\cos x-\sin x+\lambda}{\cos x+\sin x-2}dx = A\ln|\cos x+\sin x-2|+Bx+C$. Then the value of $A+B+|\lambda|$ is

67. If $\int\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\ln x dx = A\left(\frac{x}{e}\right)^x+B\left(\frac{e}{x}\right)^x+C$, then the value of A + B is

68. If $\int\left(x^9+x^6+x^3\right)\left(2x^6+3x^3+6\right)^{1/3}dx = a\left(2x^9+3x^6+6x^3\right)^{4/3}+c$, then the value of 48a must be.........

69. If $\int \frac{dx}{1+\sin x}=\tan\left(\frac{x}{2}+a\right)+b$, then the value of $-\frac{4a}{\pi}$ must be............

70. If $\int \frac{dx}{\sqrt{x+7}-\sqrt[4]{x+7}} = P\sqrt{x+7}+Q\sqrt[4]{x+7}+R\,ln\left|(x+7)^{1/4}-1\right|+c$, Then find the value of P + Q + R.

71. If $\int \frac{\left(x^2+2\right)dx}{\left(x^2+1\right)\left(x^2+4\right)} = k\,tan^{-1}\left(\frac{mx}{c-x^2}\right)$, then find the value of 3k + m + c.

72. If $\int \frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}} = k\sqrt[4]{\frac{x-1}{x+2}}+c$, then $3k$ is equal to____________.

73. If the primitive of the function $f(x)=\frac{x^{2009}}{\left(1+x^2\right)^{1006}}$ w.r.t x is equal to $\frac{1}{n}\left(\frac{x^2}{1+x^2}\right)^m+C$ then find the value of $(m+n)$ (where $m,n\in N$)

74. Let $f(x)$ is a quadratic function such that $f(0)=1$ & $f(-1)=4$. If $\int \frac{f(x)dx}{x^2(x+1)^2}$ is a rational function, then $f(10)=$.

75. Let $\int \frac{\left(f'(x)g(x)-g'(x)f(x)\right)dx}{\left(f(x)+g(x)\right)\sqrt{f(x)g(x)-g^2(x)}} = \sqrt{m}\tan^{-1}\left(\frac{f(x)-g(x)}{ng(x)}\right)+c$ where $m,n\in N$ and C is constant of integration $\left(g(x)>0\right)$ then the value of m^2+n^2 is

76. Let a matrix $'A'$ be denoted as $A = diag.\left(5^x, 5^{5^x}, 5^{5^{5^x}}\right)$ If the value of $\int(\det(A))dx = \frac{5^{5^{5^x}}}{(\ln 5)^k}+c$ then k is

77. If $\int\left\{\ln\left(\frac{\cos 2\theta}{1+\sin 2\theta}\right)+\ln\left(\frac{1+\sin 2\theta}{1-\sin 2\theta}\right)^{\cos^2\theta}\right\}d\theta$ is equal to $\frac{1}{a}\ \sin 2\theta \ln.\left|\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right|+b\ln|\cos 2\theta|+c$ where $a,b\in R-\{0\}$ & c is integration constant such that $\cos\theta>\sin\theta>0$ then $(a+b)$ is

78. Let f & g be differentiable function for all $x\in R$ & have the following properties

(i) $f'(x)=f(x)-g(x)$ (ii) $g'(x)=g(x)-f(x)$

(iii) $f(0)=5$ (iv) $g(0)=1$

Then the value of $|f(\ln 2)+g(\ln 3)|$ is equal to

79. If $\int\frac{x^3+x+1}{x^4+x^2+1}dx = A_1\ln\left(x^2+x+1\right)+A_2\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+A_3\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)+A_4\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right)+c$ then the value of $(A_1+A_2+A_3+A_4)$ is

80. Let $\int\frac{\ln\left(x+\sqrt{1+x^2}\right)dx}{\sqrt{1+x^2}} = fog(x)+c,$ where $f(x)=\frac{x^2}{2}$ and g are some functions and c is an arbitrary constant. If $\int f(x).g(x)dx = ax^3g(x)+b\left(1+x^2\right)^{3/2}+c\left(1+x^2\right)^{1/2}+d.$ then $\left(\frac{1}{a+b+c}\right)$ is equal to

81. $\int\sqrt{x}\tan\left\{2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right\}dx$ is equal to $ax^b+K\tan^{-1}\left(\frac{\sqrt{x}}{2}\right)+\frac{\alpha}{\sqrt{1+\sqrt{x}}}+c$ then $a+b$ is equal to (where $a,b,k,\alpha\in R$)

82. If $\int\frac{(\cos x-\sin x+1-x)}{e^x+\sin x+x}dx = \ln(f(x))+g(x)+c$ where c is the constant of integration & $f(x)$ is positive, , then $\frac{f(x)+g(x)}{e^x+\sin x}$ is

83. If $\int\left(x^{2010}+x^{804}+x^{402}\right)\left(2x^{1608}+5x^{402}+10\right)^{\frac{1}{402}}dx = \frac{1}{10a}\left(2x^{2010}+5x^{804}+10x^{402}\right)^{b/402}+c,$ where c is constant Then a is equal to

84. Let $u(x)\ \&\ v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)}=7.$ If $\frac{u'(x)}{v'(x)}=P\ \&\left(\frac{u(x)}{v(x)}\right)'=q$

then $\frac{p+q}{p-q}$ has the value equal to

85. If $\int \frac{\sqrt{2-x-x^2}}{x^2}dx = \frac{A\sqrt{2-x-x^2}}{x} + \frac{B}{4\sqrt{2}}\ln\left|\frac{4-x+4\sqrt{2-x-x^2}}{x}\right| - \sin^{-1}\left(\frac{2x+1}{3}\right) + c$ then $|A+B|$ is equal to_____.

86. If $\int\left\{\left(\frac{x^{-6}-64}{4+2x^{-1}+x^{-2}}\right).\left(\frac{x^2}{4-4x^{-1}+x^{-2}}\right)-\frac{4x^2(2x+1)}{(1-2x)}\right\}dx$ is equal to $f(x)$ where $f(1)=2$ then $f(3)=$

87. Let $f(x)=x+\sin x$. Suppose g denotes the inverse function of f. If the value of $g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)$ is l then $2l=$

88. If $\int(\sin(2020x))(\sin^{2018}x)dx$ is equal to $\frac{(\sin(ax)).(\sin x)^b}{c}+k$ (where k is integration constant) then $\frac{a+b+c}{3}=$

89. If $I=\int\frac{dx}{(x-2)\left(1+\sqrt{7x-10-x^2}\right)}=f(t)+c$ (Where $t=\sqrt{\frac{5-x}{x-2}}$) and $f(0)=k\ln\frac{3-\sqrt{5}}{3+\sqrt{5}}$, ($k>0$) then k^2 is equal to _____.

90. $I=\int e^{x\sin x+\cos x}.\left(\frac{x^4\cos^3 x-x\sin x+\cos x}{x^2\cos^2 x}\right)dx=f(x)+c$, where c is constant and $f(\pi)=e^a\left(b+\frac{1}{b}\right)$ then $a+b$ equal to _____.

91. Suppose $\begin{vmatrix} f'(x) & f(x) \\ f''(x) & f'(x) \end{vmatrix}=0$ where $f(x)$ is differentiable function with $f'(x)\neq 0$ & satisfies $f(0)=1, f'(0)=2$ If $f(x)=e^{\lambda x}+\mu$ then $\lambda+\mu$ is _____.

92. If $\int\sqrt{\frac{1+x^{2n}}{x^{2n}}}\frac{\ln(1+x^{2n})-2n\ln x}{x^{2n+1}}dx=\frac{\alpha p^3}{\beta n}[1-3\ln p]+c$ (where p = $\sqrt{1+\frac{1}{x^{2n}}}$, $\alpha,\beta\in N$) then $\alpha+\beta=$

Advanced Problem Package	Integral Calculus-2

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If p, q, r, s are in arithmetic progression and $f(x)=\begin{vmatrix} p+\sin x & q+\sin x & p-r+\sin x \\ q+\sin x & r+\sin x & -1+\sin x \\ r+\sin x & s+\sin x & s-q+\sin x \end{vmatrix}$ such that $\int_0^2 f(x)dx=-4,$ then the common difference of the progression is :

(A) ± 1 (B) $\frac{1}{2}$ (C) ± 2 (D) None of these

2. If $A=\int_1^{\sin\theta}\frac{t\,dt}{1+t^2}$ and $B=\int_1^{\text{cosec}\,\theta}\frac{dt}{t(1+t^2)}$, then the value of $\begin{vmatrix} A & A^2 & B \\ e^Ae^B & B^2 & -1 \\ 1 & A^2+B^2 & -1 \end{vmatrix}$ is :

(A) $\sin\theta$ (B) $\text{cosec}\,\theta$ (C) 0 (D) 1

3. Area bounded by the curves $y=\left[\frac{x^2}{64}+2\right]$ ([.] denotes the greatest integer function), $y=x-1$ and $x=0$ above the x-axis is :

(A) 2 (B) 3 (C) 4 (D) $2\sqrt{3}$

Paragraph for Questions 4 - 7

Evaluating Integrals Dependent on a parameter

Differentiate I with respect to the parameter within the sign of integrals taking variable of the integrand as constant. Now, evaluate the integral so obtained as a function of the parameter and then integrate the result to get I. Constant of integration can be computed by giving some arbitrary values to the parameter and the corresponding value of I.

4. The value of $\int_0^1\frac{x^a-1}{\log x}dx$ is :

(A) $\log(a-1)$ (B) $\log(a+1)$ (C) $a\log(a+1)$ (D) None of these

5. The value of $\int_0^{\pi/2}\log\left(\sin^2\theta+k^2\cos^2\theta\right)d\theta$, where $k\ge 0$, is :

(A) $\pi\log(1+k)+\pi\log 2$ (B) $\pi\log(1+k)$

(C) $\pi\log(1+k)-\pi\log 2$ (D) $\log(1+k)-\log 2$

6. The value of $\frac{dI}{da}$ when $I=\int_0^{\pi/2}\log\left(\frac{1+a\sin x}{1-a\sin x}\right)\frac{dx}{\sin x}$ (where $|a|<1$) is :

(A) $\frac{\pi}{\sqrt{1-a^2}}$ (B) $-\pi\sqrt{1-a^2}$ (C) $\sqrt{1-a^2}$ (D) $\frac{\sqrt{1-a^2}}{\pi}$

7. If $\int_0^{\pi}\frac{dx}{(a-\cos x)}=\frac{\pi}{\sqrt{a^2-1}}$, then the value of $\int_0^{\pi}\frac{dx}{\left(\sqrt{10}-\cos x\right)^3}$ is :

(A) $\frac{\pi}{81}$ (B) $\frac{7\pi}{162}$ (C) $\frac{7\pi}{81}$ (D) None of these

8. The smaller area enclosed by $y=f(x)$, when $f(x)$ is a polynomial of least degree satisfying $\lim_{x\to 0}\left[1+\frac{f(x)}{x^3}\right]^{1/x}=e$ and the circle $x^2+y^2=2$ above the x-axis is :

(A) $\frac{\pi}{2}$ (B) $\frac{3}{5}$ (C) $\frac{\pi}{2}-\frac{3}{5}$ (D) $\frac{\pi}{2}+\frac{3}{5}$

9. The area bounded by the curves $y=\ln x, y=\ln|x|, y=|\ln x|$ and $y=|\ln|x||$ is :

(A) 5 sq. units (B) 2 sq. units (C) 4 sq. units (D) None of these

10. If the line $y=mx+2$ cuts the parabola $2y=x^2$ at points (x_1,y_1) and $(x_2,y_2)(x_1<x_2)$, then value of m for which $\int_{x_1}^{x_2}\left(mx+2-\frac{x^2}{2}\right)dx$ is minimum is :

(A) $\sqrt{2}$ (B) $\frac{8}{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) 0

11. Maximum area of rectangle whose two sides are $x=x_0, x=\pi-x_0$ and which is inscribed in a region bounded by $y=\sin x$ and x-axis is obtained when $x_0\in$

(A) $\left(\frac{\pi}{4},\frac{\pi}{3}\right)$ (B) $\left(\frac{\pi}{6},\frac{\pi}{4}\right)$ (C) $\left(0,\frac{\pi}{6}\right)$ (D) None of these

12. If $f(x)=a+bx+cx^2$, where $c>0$ and $b^2-4ac<0$, then the area enclosed by the co-ordinate axes, the line $x=2$ and the curve $y=f(x)$ is given by :

(A) $\frac{1}{3}[4f(1)+f(2)]$ (B) $\frac{1}{2}[f(0)+4f(1)]$

(C) $\frac{1}{2}[f(0)+4f(1)+f(2)]$ (D) $\frac{1}{3}[f(0)+4f(1)+f(2)]$

Paragraph for Questions 13 - 15

$f(x)$ satisfies the relation $f(x)-\lambda\int_0^{\pi/2}\sin x\cos t f(t)dt=\sin x$.

13. If $\lambda>2$, then $f(x)$ decreases in which of the following interval?

(A) $(0,\pi)$ (B) $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ (C) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (D) None of these

14. If $f(x)=2$ has at least one real root, then :

(A) $\lambda\in[1,4]$ (B) $\lambda\in[-1,2]$ (C) $\lambda\in[0,1]$ (D) $\lambda\in[1,3]$

15. If $\int_0^{\pi/2}f(x)dx=3$, then the value of λ is :

(A) 1 (B) 3/2 (C) 4/3 (D) None of these

16. If a is a positive integer, then the number of values of a satisfying

$$\int_0^{\pi/2}\left\{a^2\left(\frac{\cos 3x}{4}+\frac{3}{4}\cos x\right)+a\sin x-20\cos x\right\}dx\le -\frac{a^2}{3} \text{ is :}$$

(A) Only one **(B)** Two **(C)** Three **(D)** Four

17. If I is the greatest of the definite integrals $I_1=\int_0^1 e^{-x}\cos^2 x\,dx, I_2=\int_0^1 e^{-x^2}\cos^2 x\,dx$

$$I_3=\int_0^1 e^{-x^2}dx,\ I_4=\int_0^1 e^{-x^2/2}dx \text{ then :}$$

(A) $I=I_1$ **(B)** $I=I_2$ **(C)** $I=I_3$ **(D)** $I=I_4$

18. The value of the integral $\int_0^{\pi}\frac{\sin(n+1/2)x}{\sin x/2}dx\,(n\in N)$ is :

(A) π **(B)** 2π **(C)** 3π **(D)** none of these

19. The value of $\int_0^1 \lim\limits_{n\to\infty}\sum_{k=0}^{n}\frac{x^{k+2}2^k}{k!}dx$ is :

(A) e^2-1 **(B)** 2 **(C)** $\frac{e^2-1}{2}$ **(D)** $\frac{e^2-1}{4}$

20. Let $P(x)$ be a polynomial of least degree whose graph has three points of inflection $(-1,-1)$, $(1, 1)$ and a point with abscissa 0 at which the curve is inclined to the axis of abscissa at an angle of 60°. Then $\int_0^1 P(x)dx$ equals to :

(A) $\frac{3\sqrt{3}+4}{14}$ **(B)** $\frac{3\sqrt{3}}{7}$ **(C)** $\frac{\sqrt{3}+\sqrt{7}}{14}$ **(D)** $\frac{\sqrt{3}+2}{7}$

21. The value of a for which the equation $\int_0^x \sin^2(\frac{t}{2})dt=a^2x^2-\frac{1}{2}(3x-1)+\frac{1}{a^2}$ possess a solution are :

(A) $\pm\frac{1}{\sqrt{2n\pi}}, n\in N$ **(B)** $\pm\frac{1}{\sqrt{2n\pi-\frac{\pi}{2}}}, n\in N$ **(C)** $\pm\frac{1}{\sqrt{n\pi+\frac{\pi}{2}}}, n\in N$ **(D)** None of these

22. Let f, g and h be continuous functions on $[0, a]$ such that $f(x)=f(a-x), g(x)=-g(a-x)$ and $3h(x)-4h(a-x)=5$. Then $\int_0^a f(x)g(x)h(x)\,dx=$

(A) 5/4 **(B)** 3/4 **(C)** 1 **(D)** 0

Paragraph for Questions 23 - 26

Let $f(x)$ and $\phi(x)$ are two continuous functions on R satisfying $\phi(x)=\int_a^x f(t)dt,\ a\ne 0$ and another continuous function $g(x)$ satisfying $g(x+\alpha)+g(x)=0\ \forall\ x\in R,\ \alpha>0$ and $\int_b^{2k} g(t)dt$ is independent of b.

23. If $f(x)$ is an odd function, then :

(A) $\phi(x)$ is also an odd function **(B)** $\phi(x)$ is an even function

(C) $\phi(x)$ is neither an even nor an odd function

(D) For $\phi(x)$ to be an even function, it must satisfy $\int_0^a f(x)\,dx=0$

24. If $f(x)$ is an even function, then :

(A) $\phi(x)$ is also an even function **(B)** $\phi(x)$ is an odd function

(C) If $f(a-x)=-f(x)$, then $\phi(x)$ is an even function

(D) If $f(a-x)=-f(x)$, then $\phi(x)$ is an odd function

25. Least positive value of c if c, k , b are in A.P. is :

(A) 0 **(B)** 1 **(C)** α **(D)** 2α

26. If m, n are even integers and p, q $\in R$, then $\int_{p+m\alpha}^{q+n\alpha} g(t)dt$ is equal to :

(A) $\int_p^q g(x)dx$ **(B)** $(n-m)\int_0^\alpha g(x)dx$

(C) $\int_p^q g(x)dx+(n-m)\int_0^\alpha g(2x)dx$ **(D)** $\int_p^q g(x)dx+(n-m)\int_0^\alpha g(x)dx$

27. If $G(x,t)=\begin{cases} x(t-1), & \text{when } x \le t \\ t(x-1), & \text{when } t < x \end{cases}$ and if t is continuous function of x in [0, 1]

Let $g(x)=\int_0^1 f(t)G(x,t)dt$. Then which is incorrect :

(A) $g(0)+g(1)=0$ **(B)** $g(0)=0$ **(C)** $g(1)=1$ **(D)** $g''(x)=f(x)$

28. If $\int_0^1 \frac{\sin t}{1+t}dt=\alpha$, then the value of $\int_{4\pi-2}^{4\pi} \frac{\sin\frac{t}{2}}{4\pi+2-t}dt$ is :

(A) α **(B)** $-\alpha$ **(C)** $\pi\alpha$ **(D)** 2α

29. The value of $\int_{-\pi}^{\pi} |x\sin[x^2-\pi]|\,dx$, where [.] denotes the greatest integer function is :

(A) $\sum_{r=1}^{6} r\sin r$ **(B)** $\sum_{r=1}^{6}(-1)^r r\sin r$ **(C)** $\sum_{r=1}^{6} r^2\sin r$ **(D)** None of these

30. If $a \le \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \le b$, then $(a,b)=$

(A) $\left(\frac{\pi}{6},\frac{\pi}{4}\right)$ **(B)** $\left(\frac{\pi}{6},\frac{\pi}{4\sqrt{2}}\right)$ **(C)** $\left(\frac{\pi}{4\sqrt{2}},\frac{\pi}{2\sqrt{2}}\right)$ **(D)** None of these

31. Let $I=\int_{\pi/4}^{\pi/3}\frac{\sin x}{x}dx$, then I belongs to :

(A) $\left(\frac{\sqrt{3}}{8},\frac{\sqrt{2}}{6}\right)$ **(B)** $\left(\frac{\sqrt{2}}{2},\frac{\sqrt{3}}{2}\right)$ **(C)** $\left(\frac{1}{2},\frac{\sqrt{2}}{2}\right)$ **(D)** None of these

32. The sum of the series as $n\to\infty$ $\frac{\sqrt{n}}{(3+4\sqrt{n})^2}+\frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2}+\frac{\sqrt{n}}{\sqrt{3}(3\sqrt{3}+4\sqrt{n})^2}+....\frac{1}{49n}$ is :

(A) 1/14 **(B)** 3/28 **(C)** 2/3 **(D)** $\pi/4$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

33. A function $f(x)$ which satisfies the relation $f(x)=e^x+\int_0^1 e^x f(t)\,dt$, then :

(A) $f(0)<0$

(B) $f(x)$ is a decreased function

(C) $f(x)$ is an increasing function

(D) $\int_0^1 f(x)dx>0$

34. Let $f(x)=\int_1^x \frac{3^t}{1+t^2}dt$, where $x>0$, then :

(A) For $0<\alpha<\beta,\ f(\alpha)<f(\beta)$

(B) For $0<\alpha<\beta,\ f(\alpha)>f(\beta)$

(C) $f(x)+\frac{\pi}{4}<\tan^{-1}x, \forall x\ge 1$

(D) $f(x)+\frac{\pi}{4}>\tan^{-1}x, \forall x\ge 1$

35. The values of a for which the integral $\int_0^2 |x-a|dx\ge 1$ is satisfied are :

(A) $[2,\infty)$ **(B)** $(-\infty,0]$ **(C)** (0, 2) **(D)** None of these

36. If $\int_a^b |sin\,x|dx=8$ and $\int_0^{a+b} |cos\,x|dx=9$, then :

(A) $a+b=\frac{9\pi}{2}$ **(B)** $|a-b|=4\pi$ **(C)** $\frac{a}{b}=15$ **(D)** $\int_a^b \sec^2 x dx=0$

37. If $g(x)=\int_0^x 2|t|\,dt$, , then :

(A) $g(x)=x|x|$

(B) g(x) is monotonic

(C) g(x) is differentiable at x = 0

(D) $g'(x)$ is differentiable at x = 0

38. Let $f:[1,\infty)\to R$ and $f(x)=x\int_1^x \frac{e^t}{t}dt-e^x$, then

(A) $f(x)$ is an increasing function

(B) $\lim_{x\to\infty} f(x)\to\infty$

(C) $f'(x)$ has a maxima at x = e

(D) $f(x)$ is a decreasing function

39. The value of $\int_0^1 \frac{2x^2+3x+3}{(x+1)(x^2+2x+2)}dx$ is :

(A) $\frac{\pi}{4}+2\,log\,2-tan^{-1}2$

(B) $\frac{\pi}{4}+2\,log\,2-tan^{-1}\frac{1}{3}$

(C) $2\,log\,2-cot^{-1}3$

(D) $-\frac{\pi}{4}+log\,4+cot^{-1}2$

40. If $A_n = \int_0^{\pi/2} \frac{\sin(2n-1)x}{\sin x} dx$; $B_n = \int_0^{\pi/2} \left(\frac{\sin nx}{\sin x}\right)^2 dx$, for $n \in N$, then :

(A) $A_{n+1} = A_n$ **(B)** $B_{n+1} = B_n$ **(C)** $A_{n+1} - A_n = B_{n+1}$ **(D)** $B_{n+1} - B_n = A_{n+1}$

41. If $f(x) = \int_0^x [f(x)]^{-1} dx$ and $\int_0^1 [f(x)]^{-1} dx = \sqrt{2}$, then :

(A) $f(2) = 2$ **(B)** $f'(2) = 1/2$ **(C)** $f^{-1}(2) = 2$ **(D)** $\int_0^1 f(x) dx = \sqrt{2}$

42. The value of $\int_0^{\infty} \frac{dx}{1+x^4}$ is :

(A) Same as that of $\int_0^{\infty} \frac{x^2+1}{1+x^4} dx$ **(B)** $\frac{\pi}{2\sqrt{2}}$

(C) Same as that of $\int_0^{\infty} \frac{x^2 dx}{1+x^4}$ **(D)** $\frac{\pi}{\sqrt{2}}$

43. If $f(x) = \int_0^x |t-1| dt$, where $0 \le x \le 2$, then :

(A) Range of $f(x)$ is [0, 1] **(B)** $f(x)$ is differentiable at $x = 1$

(C) $f(x) = \cos^{-1} x$ has two real roots **(D)** $f'\left(\frac{1}{2}\right) = \frac{1}{2}$

44. If $\int_a^b \frac{f(x)}{f(a) + f(a+b-x)} dx = 10$, then :

(A) $b = 22, a = 2$ **(B)** $b = 15, a = -5$ **(C)** $b = 10, a = -10$ **(D)** $b = 10, a = -2$

45. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, where $n \in N$, which of the following statements hold good?

(A) $2nI_{n+1} = 2^{-n} + (2n-1)I_n$ **(B)** $I_2 = \frac{\pi}{8} + \frac{1}{4}$

(C) $I_2 = \frac{\pi}{8} - \frac{1}{4}$ **(D)** $I_3 = \frac{3\pi}{32} + \frac{1}{4}$

46. If $f(x)$ is integrable over [1, 2], then $\int_1^2 f(x) dx$ is equal to :

(A) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r}{n}\right)$ **(B)** $\lim_{n\to\infty} \frac{1}{n} \sum_{r=n+1}^{2n} f\left(\frac{r}{n}\right)$

(C) $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r+n}{n}\right)$ **(D)** $\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{2n} f\left(\frac{r}{n}\right)$

47. If $f(2-x)=f(2+x)$ and $f(4-x)=f(4+x)$ for all x and $f(x)$ is a function for which $\int_0^2 f(x)dx=5$, then $\int_0^{50} f(x)dx$ is equal to :

(A) 125 (B) $\int_{-4}^{46} f(x)dx$ (C) $\int_1^{51} f(x)dx$ (D) $\int_2^{52} f(x)dx$

48. $\int_0^x \left\{\int_0^u f(t)dt\right\}du$ is equal to :

(A) $\int_0^x (x-u)f(u)du$ (B) $\int_0^x uf(x-u)du$ (C) $x\int_0^x f(u)du$ (D) $x\int_0^x uf(u-x)du$

49. Which of the following statements(s) is(are) true?

(A) If function $y=f(x)$ is continuous at $x = c$ such that $f(c)\neq 0$, then $f(x)f(c)>0\,\forall x\in(c-h,c+h)$ where h is sufficiently small positive quantity

(B) $\lim\limits_{n\to\infty} \ln\left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right).....\left(1+\frac{n}{n}\right)\right)=1+2\ln 2.$

(C) Let f be a continuous and non-negative function defined on [a, b]. If $\int_a^b f(x)dx=0$, then $f(x)=0\ \forall\ x\in[a,b]$.

(D) Let f be a continuous function defined on [a, b]. If $\int_a^b f(x)dx=0$, then there exists at least one $c\in(a,b)$ for which $f(c) = 0$.

50. The value of $\int_0^1 e^{x^2-x}dx$ is :

(A) < 1 (B) > 1 (C) $> e^{-1/4}$ (D) $< e^{-1/4}$

51. If $f(x)=\int_0^x (\cos(\sin t)+\cos(\cos t))dt$, then $f(x+\pi)$ is :

(A) $f(x)+f(\pi)$ (B) $f(x)+2f(\pi)$ (C) $f(x)+f\left(\frac{\pi}{2}\right)$ (D) $f(x)+2f\left(\frac{\pi}{2}\right)$

52. If x satisfies the equation $x^2\left(\int_0^1 \frac{dt}{t^2+2t\cos\alpha+1}\right)-x\left(\int_{-3}^3 \frac{t^2\sin 2t}{t^2+1}dt\right)-2=0$ $(0<\alpha<\pi)$, then the value of x is :

(A) $2\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$ (B) $-2\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$ (C) $4\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$ (D) $-4\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$

53. If $G(x,t)=\begin{cases} x(t-1), & where\ x\le t \\ t(x-1), & where\ t<x \end{cases}$ and if t is continuous function of x in [0, 1]. Let $g(x)=\int_0^1 f(t)G(x,t)dt$, then :

(A) $g(0)=1$ (B) $g(0)=0$ (C) $g(1)=1$ (D) $g^{11}(x)=f(x)$

54. $\int_{-1/2}^{1/2}\sqrt{\left\{\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right\}}dx$ is :

(A) $4ln\left(\frac{4}{3}\right)$ (B) $4ln\left(\frac{3}{4}\right)$ (C) $-ln\left(\frac{81}{256}\right)$ (D) $ln\left(\frac{256}{81}\right)$

55. Let T > 0 be a fixed real number. Suppose $f\ (x)$ is a continuous function for all $x\in R, f\left(x+T\right)=f\left(x\right)$. If $I=\int_0^T f\left(x\right)dx$ then :

(A) $\int_5^{5+5T} f\left(x\right)dx=5I$ (B) $\int_5^{5+5T} f(2x)dx=10I$

(C) $\int_5^{5+5T} f(3x)dx=5I$ (D) $\int_5^{5+5T} f(3x)dx=15I$

56. Let $f(x)=\int_{\pi^2/4}^{x^2}\frac{\sin x}{1+\cos^2\sqrt{t}}dt$ then

(A) $f'\left(\frac{\pi}{2}\right)=\pi$ (B) $f'\left(-\frac{\pi}{2}\right)=\pi$ (C) $f'\left(\frac{3\pi}{2}\right)=-3\pi$ (D) $f'\left(\pi\right)=\int_{\pi^2}^{\pi^2/4}\frac{dx}{1+cos^2\sqrt{x}}$

57. Let $I_n=\int_{-\pi}^{\pi}\frac{sin\,nx}{sin\,x}dx; n\in N$ then :

(A) $I_{n+2}=I_n$ (B) $\sum_{m=1}^{20}I_{2m+1}=20\pi$

(C) $I_{2m}=0$ where $m=1,2,3,.....$ (D) $I_{n+1}=I_n$

58. If y is a function of x, satisfying $x.\int_0^x y(t)\,dt=(x+1)\int_0^x t\cdot y(t)\,dt$, where $x>0$, given $y\,(1)=1$. Then:

(A) $y=x^3.e^{-\frac{1}{x}}$ (B) $y=y=\frac{e}{x^3}e^{-1/x}$ (C) $y\left(2\right)=\frac{8}{\sqrt{e}}$ (D) $y\left(2\right)=\frac{\sqrt{e}}{8}$

59. If $P=\int_0^{\infty}\frac{x^2}{1+x^4}dx;\ Q=\int_0^{\infty}\frac{x\,dx}{1+x^4}$ and $R=\int_0^{\infty}\frac{dx}{1+x^4}$, then:

(A) $Q=\frac{\pi}{4}$, (B) $P=R$ (C) $P-\sqrt{2}\ Q+R=\frac{\pi}{2\sqrt{2}}$ (D) $P-2\sqrt{2}\ \ Q+R=\frac{\pi}{\sqrt{2}}$

60. Let $u=\int_0^{\pi/4}\left(\frac{\cos x}{\sin x+\cos x}\right)^2dx$ and $v=\int_0^{\pi/4}\left(\frac{\sin x+\cos x}{\cos x}\right)^2dx$, then:

(A) $v=1+\ln 2$ (B) $u=\frac{1+\ln 2}{4}$ (C) $\frac{v}{u}=6$ (D) $\frac{v}{u}=\frac{1}{6}$

61. If $\int_0^2 \frac{\ln(1+2x)}{1+x^2}dx = \left(\tan^{-1}a\right)\left(\ln\sqrt{b}\right)$ where $a,b \in N$, then:

(A) $a=2$ (B) $b=5$ (C) $a^2+b^2=29$ (D) $b-a=4$

62. Comment upon the nature of roots of the quadratic equation $x^2+2x=k+\int_0^1 |t+k|\,dt$ depending on the value of $k \in R$.

(A) real and distinct if $-1<k<0$ (B) real and distinct if $k<-1$

(C) imaginary if $-1<k<0$ (D) real and distinct $k>0$

63. If $\underset{n\to\infty}{Lim}\frac{1}{n^2}\sum_{k=0}^{n-1}\left[k\int_k^{k+1}\sqrt{(x-k)(k+1-x)}\,dx\right] = \frac{\pi}{m^n}$, then:

(A) $m=2,\ n=4$ (B) $m=4,\ n=2$ (C) $m=\sqrt{2},\ n=4$ (D) $m=2^{1/4},\ n=8$

64. Let $g(x)=x^c.e^{2x}$ & let $f(x)=\int_0^x e^{2t}.\left(3t^2+1\right)^{1/2}dt$. For a certain value of 'c', the limit of $\frac{f'(x)}{g'(x)}$ as $x\to\infty$ is finite and non-zero, then:

(A) $c=1$ (B) $\lim\limits_{x\to\infty}\frac{f'(x)}{g'(x)}=\frac{\sqrt{3}}{2}$ (C) $\lim\limits_{x\to\infty}\frac{f'(x)}{g'(x)}=\frac{2}{\sqrt{3}}$ (D) None of these

65. If $I=\int_3^4 \frac{1}{\sqrt[3]{\ln x}}dx$, then:

(A) $I>0.92$ (B) $I<1$ (C) $I>.8$ (D) (B) and (C) only

66. Consider the integrals $I_1=\int_0^1 e^{-x}\cos^2 x\,dx, I_2=\int_0^1 e^{-x^2}\cos^2 x\,dx,\ I_3=\int_0^1 e^{-x^2}dx$ and $I_4=\int_0^1 e^{-x^2/2}dx$, Then:

(A) $I_4>I_2$ (B) $I_2>I_1$ (C) $I_3<I_4$ (D) None of these

67. Which of the following is/are true?

(A) $\frac{\pi}{3\sqrt{3}} \le \int_0^1 \frac{dx}{1+x^2+2x^5}$ (B) $\int_0^1 \frac{dx}{1+x^2+2x^5} \le \frac{\pi}{4}$

(C) $\int_0^{\pi/2}\sqrt{1-\sin^3 x}\,dx \le \frac{1}{2}\left(\sqrt{2}+\ln\left(1+\sqrt{2}\right)\right)$. (D) $1 \le \int_0^{\pi/2}\sqrt{1-\sin^3 x}\,dx$

68. Let $f(x)$ be a continuous function and 'c' is a constant satisfying $\int_0^x f(t)\,dt = e^x - ce^{2x}\int_0^1 f(t)e^{-t}dt$, then:

(A) $f(x)=e^{2x}-2e^x$ (B) $f(x)=e^x-2e^{2x}$ (C) $c=\frac{1}{3-2e}$ (D) $c=\frac{1}{3+2e}$

69. If $f(x)=x+\int_0^1\left(xy^2+x^2y\right)\left(f(y)\right)dy$, then:

(A) $f(x)=\frac{260}{119}$ (B) $f(-1)=\frac{-100}{119}$

(C) $f(x)$ have positive point of local minimum (D) $f(x)$ have negative point of local minimum

70. If $I=\int_{1/2}^{2}\frac{\ln t}{1+t^n}dt,\ (t\neq 1)$

(A) $I>0$ if $n=1$ **(B)** $I<0\ \forall n\geq 3$ **(C)** $I=0$ if $n=2$ **(D)** $I<0$ if $n=1$

71. If $\lim_{x\to 0}\frac{\int_0^x \frac{t^2dt}{\left(a+t^r\right)^{1/p}}}{bx-\sin x}=l$, (where $p\in N, p\geq 2, a>0, r>0$ and $b\neq 0$), then:

(A) If l exists and is non-zero, then $b=1$

(B) If $p=3$ and $l=1$, then $a=8$

(C) If $p=2$ and $a=9$ and l exists and non-zero, then $l=\frac{2}{3}$

(D) If $P=2$ & $a=9$ & l exists, then $l=\frac{1}{3}$

72. Suppose $f(x)$ and $g(x)$ are two continuous functions defined for $0\leq x\leq 1$. Given, $f(x)=\int_0^1 e^{x+t}.f(t)dt$ and $g(x)=\int_0^1 e^{x+t}.g(t)dt+x$. Then:

(A) $f(1)=0$

(B) $g(0)-f(0)=\frac{2}{3-e^2}$

(C) $\frac{g(0)}{g(2)}=\frac{1}{3}$

(D) $g(0)-f(0)=\frac{2}{3+e^2}$

73. We are given the curves $y=\int_{-\infty}^{x}f(t)dt$ through the point $\left(0,\frac{1}{2}\right)$ and $y=f(x)$ where $f(x)>0$ and $f(x)$ is differentiable, $\forall x\in R$ passes through $(0,1)$. Tangents drawn to both the curves at the points with equal abscissae intersect on the same point on the X-axis. Then:

(A) The number of solutions $f(x)=2ex$ is 2

(B) $\lim_{x\to\infty}\left(f(x)\right)^{f(-x)}=e$

(C) $\lim_{x\to\infty}\left(f(x)\right)^{f(-x)}=1$

(D) The number of solutions $f(x)=2ex$ is 1

74. If $f(x)=\int_0^x\left(4t^4-at^3\right)dt$ and $g(x)$ is quadratic satisfying $g(0)+6=g'(0)-c=g''(0)+2b=0$. $y=h(x)$ and $y=g(x)$ intersect in 4 distinct points with abscissae $x_i; i=1,2,3,4$ such that $\sum\frac{i}{x_i}=8, a,b,c\in R^+$ and $h(x)=f'(x)$. Then :

(A) Abscissae of point of intersection are in AP

(B) $a=20$

(C) $c=25$

(D) $c-a=6$

75. Let $f(x)$ and $g(x)$ be differentiable functions such that $f(x)+\int_0^x g(t)dt = \sin x(\cos x - \sin x)$ and $(f'(x))^2 + (g(x))^2 = 1,$ then $f(x)$ and $g(x)$ respectively, can be:

(A) $\frac{1}{2}\sin 2x, \sin 2x$ **(B)** $\frac{\cos 2x}{2}, \cos 2x$ **(C)** $\frac{1}{2}\sin 2x, -\sin 2x$ **(D)** $\cos^2 x, \cos 2x$

76. The function $f:[0, 1] \to [0, 1]$ is continuous and has the property $f(f(x)) = 1 - x$ for all $x \in [0,1]$ and $\alpha = \int_0^1 f(x)dx,$ then:

(A) $f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) = 1$

(B) the value of α equals to $1/2$

(C) $f\left(\frac{1}{3}\right).f\left(\frac{2}{3}\right) = 1$

(D) $\int_0^{\pi/2} \frac{\sin x dx}{(\sin x + \cos x)^3}$ has the same value as α

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

77. If $[.]$ denotes the greatest integer function, then match the following columns:

	Column 1		Column 2
(A)	$\int_{-1}^{1} [x + [x + [x]]] dx$	**(p)**	3
(B)	$\int_{2}^{5} ([x] + [-x]) dx$	**(q)**	5
(C)	$\int_{-1}^{3} \text{sgn}(x - [x]) dx$	**(r)**	4
(D)	$25\int_{0}^{\frac{\pi}{4}} (\tan^6(x-[x]) + \tan^4(x-[x])) dx$	**(s)**	–3

78. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	$\lim\limits_{n\to\infty}\left[\int_0^2 \frac{\left(1+\frac{t}{n+1}\right)^n}{n+1}dt\right]$	**(p)**	$e-\frac{1}{2}e^2-\frac{3}{2}$
(B)	Let $f(x)$ be a function satisfying $f'(x)=f(x)$ with $f(0)=1$ and g be the function satisfying $f(x)+g(x)=x^2$, then the value of the integral $\int_0^1 f(x)g(x)dx$	**(q)**	e^2
(C)	$\int_0^1 e^{e^x}(1+xe^x)dx$ is equal to	**(r)**	e^2-1
(D)	$\lim\limits_{k\to 0}\frac{1}{k}\int_0^k (1+\sin 2x)^{\frac{1}{x}}dx$ is equal to	**(s)**	e^e

79. **MATCH THE FOLLOWING:**

	Column 1		Column 2
(A)	If $f(x)$ is an integrable function for $x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$ and $I_1=\int_{\pi/6}^{\pi/3}\sec^2\theta f(2\sin 2\theta)d\theta$ and $I_2=\int_{\pi/6}^{\pi/3}\operatorname{cosec}^2\theta f(2\sin 2\theta)d\theta$, then I_1/I_2	**(p)**	3
(B)	If $f(x+1)=f(3+x)$ for $\forall x$, and the value of $\int_a^{a+b} f(x)dx$ is independent of a then the value of b can be	**(q)**	1
(C)	The value of $\int_1^4 \frac{\tan^{-1}\left[x^2\right]}{\tan^{-1}\left[x^2\right]+\tan^{-1}\left[25+x^2-10x\right]}$ Where [.] denotes G.I.F.	**(r)**	2
(D)	If $I=\int_0^2\sqrt{x+\sqrt{x+\sqrt{x+\dots\dots\infty}}}\,dx$ (where x >0), then [I] is equal to (where [.] denotes the greatest integer function	**(s)**	4

80. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	If $I=\int_{-2}^{2}\left(\alpha x^3+\beta x+\gamma\right)dx$, then I is	**(p)**	Independent of α
(B)	Let α,β be the distinct positive roots of the equation $\tan x=2x$, then $\gamma\int_0^1(\sin\alpha x.\sin\beta x)dx$ where $\gamma\neq 0$) is	**(q)**	Independent of β
(C)	If $f(x+\alpha)+f(x)=0$, where $\alpha>0$, then $\int_{\beta}^{\beta+2\gamma\alpha} f(x)dx$, where $\gamma\in N$ is	**(r)**	Independent of γ
(D)	$\gamma\int_0^{\alpha}[\sin x]dx$ is, where $\gamma\neq 0$, $\alpha\in\left[(2\beta+1)\pi,(2\beta+2)\pi\right]n\in N$, and where [.] denotes the greatest integer function	**(s)**	Depends on α

81. Let $\mathrm{I(n)}=\int_i^e x^3(\log x)^n dx$, where n is a whole number.

	Column 1		Column 2
(A)	$\frac{64}{5e^4-1}I(2)=$	**(p)**	1
(B)	$\frac{4I(n)+nI(n-1)}{e^4}, for\, n\geq 1=$	**(q)**	3
(C)	The least value of 'n', for which $I(n)<\frac{e^4-4}{4}$, is	**(r)**	2
(D)	$\lim\limits_{n\to\infty} \mathrm{I(n)}=$	**(s)**	0

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

82. Consider a real valued continuous function f such that $f(x)=\sin x+\int_{-\pi/2}^{\pi/2}\left(\sin x+tf(t)\right)dt$. If M and m are maximum and minimum value of the function f, then the value of M/m is _______.

83. Let $f(x)=x^3-\frac{3x^2}{2}+x+\frac{1}{4}$. Then the value of $\left(\int_{1/4}^{3/4} f(f(x))dx\right)^{-1}$ is _______.

84. Let $f:[0,\infty)\to R$ be a continuous strictly increasing function, such that $f^3(x)=\int_0^x t.f^2(t)dt$ for every $x\geq 0$, then value of $f(6)$ is _______.

85. If the value of the definite integral $\int_0^1 {}^{207}C_7 x^{200}.(1-x)^7\,dx$ is equal to 1/k where $k \in N$, then the value of k/26 is____.

86. If the value of $\lim_{n\to\infty}\left(n^{-3/2}\right)\sum_{j=1}^{6n}\sqrt{j}$ is equal to $\sqrt{N}$, then the value of N/12 is ______.

87. The value of $2^{2010}\dfrac{\int_0^1 x^{1004}(1-x)^{1004}dx}{\int_0^1 x^{1004}\left(1-x^{2010}\right)^{1004}dx}$ is _______.

88. Let $J=\int_{-5}^{-4}(3-x^2)\tan(3-x^2)dx$ and $K=\int_{-2}^{-1}(6-6x+x^2)\tan(6x-x^2-6)dx$ Then J + K is _______.

89. Let $I_n=\int_0^1 x^n\sqrt{1-x^2}\,dx$ then find the value of $\lim_{n\to\infty}\dfrac{I_n}{I_{n-2}}$.

90. For differentiable function $f(x)$, if $\int_0^n f'(x)\left([x]-x+\frac{1}{2}\right)dx = A_1\int_0^n f(x)dx + A_2 f(0)+A_3 f(n)+A_4\sum_{r=0}^{n} f(r)$,

(where [.] Denotes the G.I.F and A_1, A_2, A_3, A_4 are constant $n\in N$) then $A_1+A_2+A_3+A_4$ is equal to_____.

91. For positive integers $k=1,2,3,......,n,$ let S_k denotes the area of ΔAOB_k (where 'O' is origin)

such that $\angle AOB_k=\dfrac{k\pi}{2n}, OA=1$ and $OB_k=k.$ If the value of $\lim_{n\to\infty}\dfrac{1}{n^2}\sum_{k=1}^{n}S_k=\dfrac{\alpha}{\pi^2},$ then 'α' is equal to]

92. If $\lim_{n\to\infty}\sum_{k=0}^{n}\dfrac{{}^nC_k}{n^k(k+3)}=p$ then p is_____.

93. Let $f(x)$ be a continuous function with continuous first derivative on (a,b), where $b>a,$ and let

$\lim_{x\to a^+} f(x)=\infty, \lim_{x\to b^-} f(x)=-\infty$ and $f'(x)+f^2(x)\ge -1,$ for all x in (a,b), if the minimum value of $(b-a)$ equals to k then k is_____.

94. Let $f(x)$ be a continuous function such that $f(x)>0$ for all $x\ge 0$ and $(f(x))^{101}=1+\int_0^x f(t)dt.$ then

$(f(101))^{100}$ is equal to_____.

95. If $\int_x^{xy} f(t)dt$ is independent of x and $f(2)=2,$ if the value of $\int_1^x f(t)dt=k.\ln x$ then k is_____.

96. Let $y=f(x)$ be a quadratic function with $f'(2)=1.$ Then the value of the integral $\int_{2-\pi}^{2+\pi} f(x)\cdot\sin\left(\dfrac{x-2}{2}\right)dx$ is__.

97. If a_1, a_2 and a_3 are the three values of a which satisfy the equation $\int_0^{\pi/2}(\sin x+a\cos x)^3dx$

$-\dfrac{4a}{\pi-2}\int_0^{\pi/2}x\cos x\,dx=2$ then $\left(a_1^2+a_2^2+a_3^2\right)$ is equal to_____.

98. If $\int\limits_0^{\infty} \frac{\ell n\, t}{x^2+t^2}\, dt = \frac{\pi\, \ell n\, 2}{4}$ $(x>0)$ then the number of integral values of 'x' satisfying this equation is_____.

99. Let $F(x)=\int\limits_{-1}^{x}\sqrt{4+t^2}\, dt$ and $G(x)=\int\limits_{x}^{1}\sqrt{4+t^2}\, dt$ then the value of $(FG)'(0)$ is___. (where dash denotes the derivative).

100. Let 'x' be a real valued differentiable function satisfying $f\left(\frac{x}{y}\right)=f(x)-f(y)$ and $\lim\limits_{x\to 0}\frac{f(1+x)}{x}=3.$ If the area bounded by the curve $y=f(x)$, the Y-axis and the line $y=3$, where $x, y\in R^+$ is K. then K is_____.

101. Let $S=\left\{(x,y):\frac{y(3x-1)}{x(3x-2)}<0\right\}$, $S'=\{(x,y)\in A\times B: -1\le A\le 1, -1\le B\le 1\}$, then the area of the region enclosed by all points in $S\cap S'$ is_____.

102. A positive real valued continuously differentiable functions f on the real line such that for all x

$f^2(x)=\int\limits_0^x\left((f(t))^2+(f'(t))^2\right)dt+e^2$ then $f(-1)=$

103. If $\int\limits_1^2 \frac{(x^2-1)\,dx}{x^3\cdot\sqrt{2x^4-2x^2+1}}=\frac{1}{k}$ then k is_____.

104. Let $h(x)=(fog)(x)+K$ where K is any constant. If $\frac{d}{dx}(h(x))=-\frac{\sin x}{\cos^2(\cos x)}$ if $j(x)=\int\limits_{f(x)}^{g(x)}\frac{f(t)}{g(t)}dt,$ where f and g are trigonometric functions then the value of $j(0)$ is equal to $(\cos(1)=.54)$

105. For a ≥ 2, if the value of the definite integral $\int\limits_0^{\infty}\frac{dx}{a^2+\left(x-(1/x)\right)^2}$ equals to $\frac{\pi}{5050}$ then $\frac{a}{25}$ is_____.

106. If $\int\limits_0^{\pi}\sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\, dx$ has the value equal to $\left(\frac{\pi}{k}+\sqrt{w}\right)$ where k and w are positive integers then $k^2+w^2=$

107. If the absolute value of the integral $I=\int\limits_{\pi/4}^{\pi/2}\frac{x\cdot\cos 2x\cdot\cos x}{\sin^7 x}dx$ in the lowest form is $\frac{-a}{b}$ where $a, b\in N$, then $(a+b)=$

108. If $\int\limits_0^{\pi}\frac{x\sin^3 x}{4-\cos^2 x}dx=\pi\left(1-\frac{a\ln b}{c}\right)$ where a and b are prime and $c\in N$, then $a+b+c=$

109. Consider a real valued continuous function f such that $f(x)=\sin x+\int_{-\pi/2}^{\pi/2}(\sin x+tf(t))dt.$ then minimum value of $f(x)$ is_____.

Advanced Problem Package	Differential Equations

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The solution of $\dfrac{xdy}{x^2+y^2}=\left(\dfrac{y}{x^2+y^2}-1\right)dx$ is :

(A) $y=x\cot(c-x)$ **(B)** $cos^{-1}\left(\dfrac{y}{x}\right)=-x+c$ **(C)** $y=x\tan(c-x)$ **(D)** $\dfrac{y^2}{x^2}=x\tan(c-x)$

2. The solution of $(y(1+x^{-1})+\sin y)dx+(x+\log_e x+x\cos y)dy=0$ is :

(A) $(1+y^{-1}\sin y)+x^{-1}\log_e x=C$ **(B)** $(y+\sin y)+xy\log_e x=C$

(C) $xy+y\log_e x+x\sin y=C$ **(D)** None of these

3. The solution of $x^3\dfrac{dy}{dx}+4x^2\tan y=e^x\sec y$ satisfying $y(1)=0$ is :

(A) $\tan y=(x-2)e^x\log x$ **(B)** $\sin y=e^x(x-1)x^{-4}$

(C) $\tan y=(x-1)e^x x^{-3}$ **(D)** $\sin y=e^x(x-1)x^{-3}$

4. The solution of $\dfrac{xdx+ydy}{xdy-ydx}=\sqrt{\dfrac{a^2-x^2-y^2}{x^2+y^2}}$ is :

(A) $\sqrt{x^2+y^2}=a\left\{\sin\left(\tan^{-1}\dfrac{y}{x}+C\right)\right\}$ **(B)** $\sqrt{x^2+y^2}=a\cos\left\{\tan^{-1}\dfrac{y}{x}+C\right\}$

(C) $\sqrt{x^2+y^2}=a\{\tan(\sin^{-1}y/x)+C\}$ **(D)** None of these

5. The curve $y=f(x)$ is such that the area of the trapezium formed by the coordinate axes ordinate of an arbitrary point and the tangent at this point equals half the square of its abscissa. The curve is

(A) $y=cx^2\pm x$ **(B)** $y=cx^2\pm 1$ **(C)** $y=cx\pm x^2$ **(D)** $y=cx^2\pm x\pm 1$

6. The real value of m for which the substitution $y=u^m$ will transform the differential equation $2x^4y\dfrac{dy}{dx}+y^4=4x^6$ into a homogeneous equation is

(A) $m=0$ **(B)** $m=1$ **(C)** $m=\dfrac{3}{2}$ **(D)** $m=\dfrac{2}{3}$

7. Solution of the differential equation $x=1+xy\dfrac{dy}{dx}+\dfrac{x^2y^2}{2!}\left(\dfrac{dy}{dx}\right)^2+\dfrac{x^3y^3}{3!}\left(\dfrac{dy}{dx}\right)^3+........$ is :

(A) $y=ln(x)+c$ **(B)** $y=(ln\,x)^2+c$

(C) $y=\pm\sqrt{\{(ln\,x)^2+c}$ **(D)** $xy=x^y+c$

8. The solution of the differential equation $\frac{dy}{dx}=\frac{1}{xy[x^2\sin y^2+1]}$ is :

(A) $x^2(\cos y^2-\sin y^2-2Ce^{-y^2})=2$ **(B)** $y^2(\cos x^2-\sin y^2-2Ce^{-y^2})=2$

(C) $x^2(\cos y^2-\sin y^2-e^{-y^2})=4C$ **(D)** None of these

Paragraph for Questions 9 - 13

Let us represent the derivative dy/dx by p. An equation of the form $y=px+f(p)$ **....(i)**

Is known as Clairut's equation where $f(p)$ is a function of p. To solve equation (1), we differentiate the equation with respect to x, we get $p=p+x\frac{dp}{dx}+f'(p)\frac{dp}{dx}$ $\Rightarrow[x+f'(p)]\frac{dp}{dx}=0\Rightarrow\frac{dp}{dx}=0$ **....(ii)** Or, $x+f'(p)=0$ **....(iii)**

Now, (ii) gives p = constant = c, say

Then eliminating p from (i) we get y = cx + f(c) **....(iv)**

Which is a solution of equation (i). If we eliminate p between (i) and (iii) we will obtain another solution not contained in the general solution (iv). This solution is known as the singular solution.

9. The general equation of the equation y = px + log p which does not contain the singular solution, is

(A) y = cx + logc **(B)** $y=cx+\frac{1}{c}$ **(C)** y = logx + c **(D)** y = –logx + c

10. The singular solution of the differential equation given in previous problem is :

(A) y = –x + 1 **(B)** y = x + 1 **(C)** y + 1 = logx **(D)** y + 1 = –log(–x)

11. Non-singular solution of the differential equation $x\frac{dy}{dx}=y-\left(\frac{dy}{dx}\right)^2$ is :

(A) $y^2=cx+c$ **(B)** $y=cx+c^2$ **(C)** $cy=x^2+c$ **(D)** $y=cx^2+c$

12. Singular solution of the differential equation given in the previous question is :

(A) $y=\frac{x}{4}$ **(B)** $y=\frac{x^2}{4}$ **(C)** $y=-\frac{x^2}{4}$ **(D)** y = x

13. Solution of the differential equation $x^2\left(y-x\frac{dy}{dx}\right)=y\left(\frac{dy}{dx}\right)^2$ which does not contain singular solution is :

(A) $x^2(y-xc)=yc^2$ **(B)** $y=cx+c^2$ **(C)** $y^2=cx^2+c^2$ **(D)** $xy=cx^2+c$

14. The solution of $y=2x\left(\frac{dy}{dx}\right)+x^2\left(\frac{dy}{dx}\right)^4$ is :

(A) $y=2c^{1/2}x^{1/4}+c$ **(B)** $y=2\sqrt{c}\,x^2+c^2$ **(C)** $y=2\sqrt{c}\,(x+1)$ **(D)** $y=2\sqrt{cx}+c^2$

15. Solution of the differential equation $x\cos\left(\frac{y}{x}\right)(ydx+xdy)=y\sin\left(\frac{y}{x}\right)(xdy-ydx)$ is :

(A) $y=cx\cos\left(\frac{x}{y}\right)$ **(B)** $\sec\left(\frac{y}{x}\right)=cxy$ **(C)** $\left(\frac{y}{x}\right)\sec\left(\frac{y}{x}\right)=c$ **(D)** none of these

16. Solution of the differential equation $\left\{\frac{1}{x}-\frac{y^2}{(x-y)^2}\right\}dx+\left\{\frac{x^2}{(x-y)^2}-\frac{1}{y}\right\}dy=0$ is :

(A) $ln\left|\frac{x}{y}\right|+\frac{xy}{x-y}=c$ **(B)** $\frac{xy}{x-y}=ce^{x/y}$ **(C)** $ln|xy|=c+\frac{xy}{x-y}$ **(D)** None of these

17. If the solution of the differential equation $\frac{xdx-ydy}{xdy-ydx}=\sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$ be $\sqrt{f(x,y)}+\sqrt{1+f(x,y)}=c\left(\frac{x+y}{\sqrt{f(x,y)}}\right)$, then $f(x,y)$ is :

(A) x^2+y^2 **(B)** $1+x^2-y^2$ **(C)** x^2-y^2 **(D)** $\frac{x^2-y^2}{x^2+y^2}$

18. The solution of the equation $x\int_0^x y(t)dt=(x+1)\int_0^x ty(t)dt, x>0$ as $y=f(x)$ is :

(A) $y=ce^{-1/x}$ **(B)** $y=\frac{c}{x^3}$ **(C)** $y=-\frac{1}{2}-clnx$ **(D)** $y=\frac{ce^{-1/x}}{x^3}$

19. Solution of the equation $\frac{xdx+ydy}{xdy-ydx}=\sqrt{\frac{a^2-x^2-y^2}{x^2+y^2}}$ is :

(A) $\sqrt{x^2-y^2}=a\tan\left\{\tan^{-1}\left(\frac{y}{x}\right)+c\right\}$ **(B)** $\sqrt{x^2+y^2}=a\sin\left\{\tan^{-1}\left(\frac{y}{x}\right)+c\right\}$

(C) $\sqrt{x^2+y^2}=a\tan\left\{\tan^{-1}\left(\frac{y}{x}\right)+c\right\}$ **(D)** $\sqrt{x^2-y^2}=a\cos\left\{\tan^{-1}\left(\frac{y}{x}\right)+c\right\}$

20. Through any point (*x*, *y*) of a curve which passes through the origin, lines are drawn parallel to the co-ordinate axes. The curve, given that it divides the rectangle formed by the two lines and the axes into two areas, one of which is twice the other, represents a family of :

(A) circles **(B)** pair of straight lines

(C) parabolas **(D)** rectangular hyperbolas

21. The orthogonal trajectories of the family of coaxial circles $x^2+y^2+2gx+C=0$, where *g* is a parameter are

(A) family of circles with centre on *y*-axis **(B)** system of coaxial parabolas

(C) $x^2+y^2-C'x-Cy=0$, where *C'* is an arbitrary constant **(D)** system of circles with centre on *x*-axis

22. A curve $f(x)$ passes through the point $P(1,1)$. The normal to the curve at point *P* is $a(y-1)+(x-1)=0$. If the slope of the tangent at any point on the curve is proportional to the ordinate at that point, then the equation of the curve is

(A) $y=e^{ax}-1$ **(B)** $y-1=e^{ax}$ **(C)** $y=e^{a(x-1)}$ **(D)** $y-a=e^{ax}$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

23. The solution of $\left(\frac{dy}{dx}\right)^2+2y\cot x\frac{dy}{dx}=y^2$ is :

(A) $y-\frac{c}{1+\cos x}=0$ **(B)** $y=\frac{c}{1-\cos x}$ **(C)** $x=2\sin^{-1}\sqrt{\frac{c}{2y}}$ **(D)** $x=2\cos^{-1}\sqrt{\frac{c}{2y}}$

24. The solution of $\frac{dy}{dx}+x=xe^{(n-1)y}$ is :

(A) $\frac{1}{n-1}\log\left(\frac{e^{(n-1)y}-1}{e^{(n-1)y}}\right)=\frac{x^2}{2}+C$ **(B)** $e^{(n-1)y}=Ce^{(n-1)y+(n-1)x^2/2}+1$

(C) $\log\left(\frac{e^{(n-1)y}-1}{(n-1)e^{(n-1)y}}\right)=x^2+C$ **(D)** $e^{(n-1)y}=ce^{(n-1)x^2/2+x}+1$

25. A tangent drawn to the curve $y = f(x)$ at P(x, y) cuts the x-axis and y-axis at A and B respectively such that BP: AP = 3:1, given that f(1) = 1, then :

(A) Equation of curve is $x\frac{dy}{dx} - 3y = 0$ **(B)** Normal at (1,1) is $x + 3y = 4$

(C) Curve passes through (2, 1/8) **(D)** Equation of curve is $x\frac{dy}{dx} + 3y = 0$

26. If $f(x), g(x)$ be twice differentiable functions on [0, 2] satisfying $f''(x) = g''(x)$, $f'(1) = 2g'(1) = 4$ and $f(2) = 3g(2) = 9$, then :

(A) $f(4) - g(4) = 10$ **(B)** $|f(x) - g(x)| < 2 \Rightarrow -2 < x < 0$

(C) $f(2) = g(2) \Rightarrow x = -1$ **(D)** $f(x) - g(x) = 2x$ has real root

27. Given the differential equation $\frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$; $y(1) = \pi$ and the following statements

(A) Solution is $y^2 - \sin y = -2x^3 + c$ **(B)** Solution is $y^2 + \sin y = 2x^3 + c$

(C) $c = \pi^2 - 2$ **(D)** $c = \pi^2 + 2$

28. The equation of the curve satisfying the differential equation $y_2(x^2 + 1) = 2xy_1$ passing through the point (0, 1) and having slope of tangent at x = 0 as 3(where y_2 and y_1 represents 2nd and 1st order derivative), then :

(A) $y = f(x)$ is strictly increasing function **(B)** $y = f(x)$ is non-monotonic function

(C) $y = f(x)$ has three distinct real roots **(D)** $y = f(x)$ has only one negative root

29. For the central conics having their axes along the coordinates :

(A) Differential equation is $y\frac{dy}{dx} = x\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2}$ **(B)** Order is 2 and degree is1

(C) Differential equation is $xy\frac{dy}{dx} = x\left(\frac{dy}{dx}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^3$ **(D)** Order is 2 and degree is 3

30. For the differential equation $(3x + 2y^2)ydx + 2x(2x + 3y^2)dy = 0$

(A) on simplification it reduces to $2(xy^3)d(xy^3) + d(x^3y^4) = 0$ **(B)** solution is $x^2y^6 + x^3y^4 = c$

(C) on simplification it reduces to $2(xy^2)d(xy^2) + d(x^2y^3) = 0$ **(D)** solution is $x^2y^4 + x^2y^3 = 0$

31. If y_1, y_2 are the solution of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, then :

(A) $y = y_1 + c(y_1 - y_2)$ is the general solution of equation

(B) $y = y_1 + c(y_1 + y_2)$ is the general solution of equation

(C) $\alpha y_1 + \beta y_2$ is a solution of $\alpha + \beta = 1$ **(D)** $\alpha y_1 + \beta y_2$ is a solution of $\alpha - \beta = 1$

32. If the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of air is 30°C and the substance cools from 37°C to 34°C in 15 min then:

(A) Temperature of substance will be 32°C at t = $15\log_{7/4}\frac{7}{2}$ min

(B) Temperature of substance will be 31°C at t = $15\log_{7/4} 7$ min

(C) Proportional constant $k = \frac{1}{15}\log\frac{7}{4}$

(D) All of these

33. A right circular cylinder with radius R and height H contains a liquid which evaporates at a rate proportional to its surface area in contact with air (proportionality constant = K > 0). If T is time after which cylinder will be empty, then

(A) T is dependent on R **(B)** T is independent of R

(C) T is dependent on H **(D)** T is dependent on K

34. Let $S_1 = x^2 + y^2 - kx = 0$ and $S_2 = x^2 - y^2 - cx = 0$, then

(A) S_1 and S_2 intersect at an angle of $\pi/4$

(B) S_1 and S_2 intersect orthogonally

(C) abscissa of the point of intersection of S_1 and S_2 is A.M. of c and k.

(D) point of intersection of S_1 and S_2 is origin.

35. If a curve $y = f(x)$, passing through the point (2, 1) satisfies the condition that length of subtangent is equal to slope of tangent in 1st quadrant given that $\frac{dy}{dx} > 0$, then :

(A) Curve $y = f(x)$ is a parabola **(B)** $y = f(x)$ is $y^2 = \frac{x}{2}$

(C) $y = f(x)$ is $y^2 = x - 1$ **(D)** Area bounded by $y = f(x)$, y-axis and y = 0, y = 3 is 18 sq. units

36. Consider the differential equation $\cos^2 x \frac{dy}{dx} - (\tan 2x) y = \cos^4 x; |x| < \frac{\pi}{4}$ and $y\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{8}$ then :

(A) Integrating factor of the differential equation is $\frac{\cos 2x}{1+\cos 2x}$

(B) Solution is $y = \frac{1}{2}\tan 2x \cos^2 x$

(C) Solution is $y = \tan 2x \cos^2 x - \frac{3\sqrt{3}}{8}$

(D) $y\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{8}$

37. If the length of perpendicular from origin to any normal to the curve y = $f(x)$ is equal to its y intercept, then

(A) Curve is $x^2 = 4y + c$ **(B)** Curve is y = c

(C) Curve is x = c **(D)** Equation of normal to the curve y = k

38. If $f(x) = \int_1^x \frac{\log t}{1+t+t^2} dt,\ x \ge 1$, then :

(A) $f(x) = \int_1^x \frac{t \log t}{1+t+t^2} dt,\ x \ge 1$ **(B)** $f(x) = \int_1^{1/x} \frac{\log t}{1+t+t^2} dt,\ x \ge 1$

(C) $f(x) = f\left(\frac{1}{x}\right)$ **(D)** All of these

39. If $f(x)$ is a function such that $x\int_0^x (1-t) f(t) dt = \int_0^x t f(t) dt; f(1) = 1$, then :

(A) $f(x) = -\frac{1}{x} - 3\ln x + 1$ **(B)** $x^2 + y^2 + 2ax + 2by + c = 0$

(C) Degree of the differential equation is 1 **(D)** All of these

40. The solutions of $y = x\left(\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3\right)$ are given by (where $p = \frac{dy}{dx}$ and k is constant)

(A) The constant function y = 0 **(B)** $y = kp^{-3}e^{1/2p^2}(p + p^3)$

(C) $y = kp^3e^{-1/2p^2}(p + p^3)$ **(D)** $ye^{-1/2p^2} = p^{-2} + 1$

41. If $|y| = f(x)$ is solution of $\frac{d^2y}{dx^2} = \frac{x^3}{y^3}\frac{d^2x}{dy^2}$ such that $f(0) = 2$ and $y = g(x)$ is solution of $\frac{d^2y}{dx^2} + \frac{8y^3}{x^3} \cdot \frac{d^2x}{dy^2} = 0$ such that $g(1) = \frac{1}{3}$, then:

(A) Domain of region $f(x) \cap g(x)$ is $[-2, 2]$ **(B)** Domain of region $f(x) \cap g(x)$ is $\left[-\sqrt{3}, \sqrt{3}\right]$

(C) Range of region $f(x) \cap g(x)$ is $[0, 2]$ **(D)** Range of region $f(x) \cap g(x)$ is $\left[0, \sqrt{3}\right]$

42. If $y = e^{-x}\sin x$ and $y_n + a_n y = 0$ where a_n is constant for $n \in N$ & $y_n = \frac{d^ny}{dx^n}$ (nth derivative of y), then:

(A) $a_4 = 4$ **(B)** $a_8 = -16$ **(C)** $a_{12} = 64$ **(D)** $a_{16} = 256$

43. If $y = f(x); f(x) \geq 0$ & $f(0) = 0$ bounding a curvilinear trapezoid with base $[0, x]$ whose area is proportional to 3^{rd} power of $f(x)$. If $f(1) = 3$, then:

(A) Range of $f\left(\sin^2 x\right)$ is $[-3, 3]$

(B) Domain of $f\left(\ln\left(2^x - 3\right)\right)$ is $[2, \infty)$

(C) Range of $f\left(\sec^2 x\right)$ is $[3, \infty)$

(D) Area bounded by $y = f(x)$ line $x = 0, y = 1$ & $y = 2$ is $\frac{7}{9}$

44. If differential equation of the curve $y = ae^{3x} + be^{2x} + ce^{x}$ is $\frac{d^3y}{dx^3} + m\frac{d^2y}{dx^2} + n\frac{dy}{dx} + py = 0$, then:

(A) $(m + n)$ is a prime number **(B)** $(m - n)$ is always divisible by any odd number

(C) $m + n + p$ is a negative integer **(D)** $|m|$ is divisible by two prime numbers

45. If right circular cone with radius 18 & height 27 contains a liquid which evaporates at a rate proportional to its surface area in contact with air (proportionality constant $= k > 0$). If volume of liquid is V & r is radius of surface of liquid left, then:

(A) $\frac{1}{r^2}\frac{dV}{dr} = \frac{3\pi}{2}$ **(B)** $\frac{1}{r^2}\frac{dr}{dt} = -k\pi$

(C) Radius as function of time $r(t) = \frac{-2k}{3}t + c$ **(D)** Total time taken to empty the cone is $\frac{3}{2}$ unit

46. If A tangent drawn to the curve $y=f(x)$ at (x,y) cuts the $x-$ axis and $y-$ axis at A and B respectively such that $\frac{BP}{AP}=\frac{3}{1}$ given $f(1)=1$, then:

(A) differential equation of curve may be $x\frac{dy}{dx}+3y=0$

(B) differential equation of curve may be $x\frac{dy}{dx}=3y$

(C) If tangent at $R(\alpha,\beta)$ intersect again at $S(m,n)$ then $m+2\alpha=0$

(D) equation of normal at $(1,1)$ is $3y=x+2$

47. If $\frac{dy}{dx}=\frac{x^2-y}{x+y}$ such that $y=f(x)$ is a solution of differential equation & $f(0)=0$, then:

(A) $(f(1)+1)^2=\frac{5}{3}$ **(B)** $(f(-1)-1)^2=\frac{1}{3}$ **(C)** $(f(3)+3)^{2/3}=3$ **(D)** $|f(-3)-3|=3$

48. Let C be a curve such that the normal at any point P on it meets x-axis and y-axis at A and Y respectively. If $BP:PA=1:2$ (internally) and the curve passes through the point $(0,4)$ then which of the following alternative(s) is/are correct?

(A) The curves passes through $(\sqrt{10},-6)$

(B) The equation of tangent at $(4,4\sqrt{3})$ is $2x+\sqrt{3}y=20$

(C) The differential equation for the curve is $yy'+2x=0$

(D) The curve represent a hyperbola

49. A differentiable function satisfies $f(x)=\int_0^x\{f(t)\cos t-\cos(t-x)\}dt$. which is of the following hold good?

(A) $f(x)$ has a minimum value $1-e$

(B) $f(x)$ has a maximum value $1-e^{-1}$

(C) $f''\left(\frac{\pi}{2}\right)=e$

(D) $f'(0)=1$

50. Let $\frac{dy}{dx}+y=f(x)$ where y is a continuous function of x with $y(0)=1$ and $f(x)=\begin{cases}e^{-x}, & \text{if } x\le 2\\ e^{-2}, & \text{if } x>2\end{cases}$. Which is of the following hold(s) good?

(A) $y(1)=2e^{-1}$ **(B)** $y'(1)=-e^{-1}$ **(C)** $y(3)=-2e^{-3}$ **(D)** $y'(3)=-2e^{-3}$

51. The function $f(x)$ satisfying the equation $f^2(x)+4f'(x).f(x)+[f'(x)]^2=0$

(A) $f(x)=C.e^{(2-\sqrt{3})x}$

(B) $f(x)=C.e^{(2+\sqrt{3})x}$

(C) $f(x)=C.e^{(\sqrt{3}-2)x}$

(D) $f(x)=C.e^{-(2+\sqrt{3})x}$

52. Which of the following pair(s) is/are orthogonal?

(A) $16x^2+y^2=C$ and $y^{16}=kx$

(B) $y=x+Ce^{-x}$ and $x+2=y+ke^{-y}$

(C) $y=Cx^2$ and $x^2+2y^2=k$

(D) $x^2-y^2=C$ and $xy=k$

53. The general solution of the differential equation , $x\left(\frac{dy}{dx}\right) = y.\log\left(\frac{y}{x}\right)$ is:

(A) $y = xe^{1-Cx}$ **(B)** $y = xe^{1+Cx}$ **(C)** $y = ex.e^{Cx}$ **(D)** $y = xe^{Cx}$

54. Identify the statement(s) which is/are true?

(A) $f(x, y) = e^{y/x} + \tan\frac{y}{x}$ is homogeneous of degree zero.

(B) $x.\log\frac{y}{x}dx + \frac{y^2}{x}\ \sin^{-1}\frac{y}{x}dy = 0$ is homogeneous

(C) $f(x, y) = x^2 + \sin x.\cos y$ is not homogeneous.

(D) $\left(x^2 + y^2\right)dx - \left(xy^2 - y^3\right)dy = 0$ is a homogeneous differential equation.

55. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}.\sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$, then the statement which is correct?

(A) $\lim_{x\to 0} f(x) = 1$ **(B)** $\int_0^{\pi/2} f(x)dx$ is less then $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)dx$ is greater than unity **(D)** $f(x)$ is an odd function

56. Identify the statement(s) which is/are true?

(A) The order of differential equation $\sqrt{1+\frac{d^2y}{dx^2}} = x$ is 1.

(B) solution of the differential equation

$xdy - ydx = \sqrt{x^2+y^2}dx$ **is** $y + \sqrt{x^2+y^2} = Cx^2$.

(C) $\frac{d^2y}{dx^2} = 2\left(\frac{dy}{dx} - y\right)$ is differential equation of family of curves $y = e^x\left(A\cos x + B\sin x\right)$.

(D) The solution of differential equation

$\left(1+y^2\right) + \left(x - 2e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0$ is $xe^{\tan^{-1}y} = e^{3\tan^{-1}y} + k$.

57. Let $y = f(x)$ be a curve in the first quadrant such that the triangle formed by the co-ordinate axis and the tangent at any point on the curve has area 2. If $y(1) = 1$, then $y(2) =$

(A) 0 **(B)** 1 **(C)** 2 **(D)** $\frac{1}{2}$

58. If $f(x) = x + \int_1^x \frac{f(t)}{t}dt$, then $\int_0^{\pi}\frac{\left(f(\sin\theta) - \sin\theta\right)}{\sin\theta}d\theta$ is equal to:

(A) $-\frac{\pi}{2}\ln 2$ **(B)** $\frac{\pi}{2}\ln 2$ **(C)** $-\pi\ln 2$ **(D)** $\pi\ln 2$

59. If $\frac{dy}{dx} + y\frac{dx}{dy} = x, y(-2) = 1,$ then :

(A) $x + 3y^2 = 1$ **(B)** $2x + y + 3 = 0$ **(C)** $x + y + 1 = 0$ **(D)** $x^2 - 4y = 0$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

60. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The value of $\int_{-2}^{2}\left(ax^3+bx+c\right)$ dx depends on	**(p)**	$\frac{1}{n-1}$
(B)	$\int_{-1}^{1}\left(ax^3+bx\right)dx=0$ is true for all real values of	**(q)**	0
(C)	$\left(\sum_{n=1}^{10}\int_{-2n-1}^{-2n}\sin^{27}(x)dx+\sum_{n=1}^{10}\int_{2n}^{2n+1}\sin^{27}(x)dx\right)$	**(r)**	$\frac{1}{n+1}$
(D)	$\int_{0}^{\pi/4}\left(\tan^{n}(x)+\tan^{n-2}(x)\right)d\left(x-[x]\right)$ (where [.] is G.I.F.)	**(s)**	a, b
		(t)	a, c
		(u)	c

61. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	If the curve satisfy the equation $\left(e^x+1\right)ydy=\left(y+1\right)e^x dx$ passes through (0, 0) and (k, 1) then k is	**(p)**	1
(B)	If the curve satisfy the equation $x\frac{dy}{dx}+y=xy^3$ passes, through (1,1) and $\left(\frac{3}{2},p\right)$ then p is	**(q)**	Not defined
(C)	If $\frac{dy}{dx}=\frac{xy+y}{xy+x}$, then the solution of the differential equation always passes through the point origin and (k, 1) then k is	**(r)**	$\ln\left(e-1\right)$
(D)	The solution of the equation $\frac{dy}{dx}=\frac{3x-4y-2}{3x-4y-3}$ passes through origin the distance of it from (–1, 1) is	**(s)**	ϕ

62. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Order of differential equation whose general solution is given by $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}+c_5\sin x$, where c_1,c_2,c_3,c_4,c_5 are arbitrary constants, is	**(p)**	1
(B)	Order of differential equation formed by eliminating the constants from $y=a\sin^2 x+b\cos^2 x+c\sin 2x+d\cos 2x+e\sin x,$, where a, b, c , d are arbitrary constants, is	**(q)**	2
(C)	The degree of equation $\frac{d^2y}{dx^2}-\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=0$	**(r)**	3
(D)	Order of differential equation whose solution is $y=cx+c^2-3c^{3/2}+2$, where c is a parameter is	**(s)**	4

63. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Order 1	**(p)**	Of all parabolas whose axis is the x-axis
(B)	Order 2	**(q)**	Of family of curve $y=a(x+a)^2$, where a is an arbitrary constant
(C)	Degree 1	**(r)**	$\left(1+3\frac{dy}{dx}\right)^{2/3}=\frac{4d^3y}{dx^3}$
(D)	Degree 3	**(s)**	Of family of curve $y^2=2c(x+\sqrt{c})$, where c > 0

64. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	If the function $y=e^{4x}+2e^{-x}$ is a solution of the differential equation $\dfrac{\frac{d^3y}{dx^3}-13\frac{dy}{dx}}{y}=K$, then the value of K/3 is	**(p)**	3
(B)	Number of solutions which satisfy the differential equation $\frac{dy}{dx}+x\left(\frac{dy}{dx}\right)^2-y=0$ is	**(q)**	4
(C)	If real value of m for which the substitution, $y=u^m$ will transform the differential equation, $2x^4y\frac{dy}{dx}+y^4=4x^6$ in to a homogenous equation, then the value of 2m is	**(r)**	2
(D)	If the solution of differential equation $x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=12y$ is $y=Ax^m+Bx^{-n}$, then $\lvert m-n\rvert$ is	**(s)**	1

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/ INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

65. Let $\frac{d}{dx}F(x)=\frac{e^{\sin x}}{x};x>0$. If $\int_1^4 \frac{2e^{\sin x^2}}{x}dx=F(k)-F(1)$, then find one of the possible value of k/4.

66. If the dependent variable y is changed to 'z' by the substitution $y=\tan z$ and the differential equation $\frac{d^2y}{dx^2}=1+\frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$ is changed to $\frac{d^2y}{dx^2}=\cos^2 z+k\left(\frac{dz}{dx}\right)^2$, then the value of k equals ____________.

67. Let $y=y(t)$ be a solution to the differential equation $y'+2ty=t^2$, then $16\lim\limits_{t\to\infty}\frac{y}{t}$ is ____________.

68. If the solution of the differential equation $\frac{dy}{dx}=\frac{1}{x\cos y+\sin 2y}$ is $x=ce^{\sin y}-k(1+\sin y)$, then the value of k is

69. If the independent variable x is changed to y, then the differential equation $x\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3-\frac{dy}{dx}=0$ is changed to $x\frac{d^2x}{dy^2}+\left(\frac{dx}{dy}\right)^2=k$ where k equals ____________.

70. The curve passing through the point (1, 1) satisfies the differential equation $\frac{dy}{dx}+\frac{\sqrt{(x^2-1)(y^2-1)}}{xy}=0$. If the curves passes through the point $(\sqrt{2},k)$ then the value of $[k]$ is (where [.] represents greatest integer function.

71. Tangent is drawn at the point (x_i,y_i) on the curve y $=f(x)$, which intersects the x-axis at $(x_{i+1},0)$. Now, again a tangent is drawn at (x_{i+1},y_{i+1}) on the curve which intersects the x-axis at $(x_{i+2},0)$ and the process is repeated n times, i.e., $i=1,2,3,.........,n$. If $x_1,x_2,x_3,........,x_n$ form an arithmetic progression with common difference equal to $\log_2 e$ and curve passes through (0, 2). Now if curve passes through the point (–2, k), then the value of k is ____________.

72. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Also curve passes through the point (1, 1). Then the length of intercept of the curve on the x-axis is ____________.

73. If the solution of the differential equation $\frac{dy}{dx}-y=1-e^{-x}$ and $y(0)=y_0$ has a finite value, when $x\to\infty$, then the value of $|2/y_0|$ is ____________.

74. Find the constant of integration by the general solution of the differential equation $(2x^2y-2y^4)dx+(2x^3+3xy^3)dy=0$ if curve passes through (1, 1).

75. A tank initially contains 50 gallons of fresh water. Brine contains 2 pounds per gallon of salt, flows into the tank at the rate of 2 gallons per minutes and the mixture kept uniform by stirring, runs out at the same rate. If it will take for the quantity of slat in the tank to increase from 40 to 80 pounds (in seconds) is 206λ, then find λ. (given $\ln 3 = 1.0986$)

76. If $f: R-\{-1\} \to R$ and f is differentiable function which satisfies:

$f(x+f(y)+xf(y)) = y+f(x)+yf(x) \forall x, y \in R-\{-1\}$, $f(1) \neq 1$ then find the value of $2019\left[1+f(2018)\right]$.

77. If $\phi(x)$ is a differential real-valued function satisfying $\phi'(x)+2\phi(x) \le 1$, then the maximum value of $2\phi(x)$, equal to ____.

78. The degree of the differential equation satisfied by the curves $\sqrt{1+x} - a\sqrt{1+y} = 1$, is ____.

79. Let $f(x)$ be a twice differentiable bounded function satisfy $2f^5(x).f'(x)+2(f'(x))^3.f^5(x) = f''(x)$. If $f(x)$ is bounded in between $y=k_1$, and $y=k_2$, Then the number of integers between k_1 and k_2 is/are (where $f(0) = f'(0) = 0$)

80. Let $y(x)$ be a function satisfying $d^2y/dx^2 - dy/dx + e^{2x} = 0$, $y(0) = 2$ and $y'(0) = 1$. If maximum value of $y(x)$ is $y(\alpha)$, Then integral part of (2α) is____.

81. $y = f(x)$ is a particular solution of differential equation $\left(\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right) = e^x$ such that $f(0) = 1 = f'(0)$, then the value of $f^2(\ln 2)$ is ____.

82. The differential equation $\frac{dy}{dx} = \frac{(x+y+2)^2 - (y-1)^2}{(x+3)^2}$ determines a curve $y = f(x)$ & $f(0) = 1$ then $f(3)$ is equal to ____.

83. If $y = f(x)$ is solution of differential equation $\frac{dy}{2x\,dx} + \frac{y}{x^2} = \frac{\sin^{-1}x}{2x^2}$, if $f(0) = 0$, then, $\frac{\pi}{f(1)}$ is equal to ____.

84. If $f'(x) < 2f(x)$ where $f:\left[\frac{1}{2}, 1\right] \to R$ such that $f\left(\frac{1}{2}\right) = 2e$ then maximum value of $f(\ln 2)$ is____.

85. If differential equation of the curve $x^2 - y^2 = c\left(x^2+y^2\right)^2$ is $\frac{dy}{dx} = \frac{x\left(my^2 - x^2\right)}{y\left(3x^2+ny^2\right)}$, then $(m+n)$ is ____.

86. If $x\int_0^x f(t)\,dt = (x+1)\int_0^x t\,f(t)\,dt$ for $x \in R^+$. $f(1) = \frac{1}{e}$ and $g(x) = e^{\frac{1}{x}}.f(x)$ then, $|g'(1)|$ (where g' denotes $\frac{dg}{dx}$) is ____.

87. If differential equation of first degree of a curve is given by $x^2\left(\frac{dy}{dx}\right)^2 - x(2y-1)\frac{dy}{dx} + \left(y^2-y-2\right) = 0$, then $(y - 2015\,.\,x)$ is a positive prime number $'P'$ then the value of P is ____.

88. The order of differential equation of family of circles in a plane is m and highest power of second differential $\left(\frac{d^2y}{dx^2}\right)$ is n then $(m+n)$ ____.

89. Let $y=f(x)$ be a curve C_1 passing through $(2,2)$ and $\left(8,\frac{1}{2}\right)$ and satisfying a differential equation $y\left(\frac{d^2y}{dx^2}\right)=2\left(\frac{dy}{dx}\right)^2$. Curve C_2 is the director circle of the circle $x^2+y^2=2$. If the shortest distance between the curves C_1 and C_2 is $\left(\sqrt{p}-q\right)$ where $p,q\in N$, then find the value of $\left(p^2-q\right)$.

90. A function $y=f(x)$ satisfies $xf'(x)-2f(x)=x^4f^2(x), \forall x>0$ and $f(1)=-6$. Find the value of $f'\left(3^{1/5}\right)$.

91. For $y<0$, if y is a differentiable function of x such that $y(x+y)=x$ and $\int \frac{dx}{x+2y}=-\ln(k-y)+c$ where $k\in N$ then $k=$

92. Let C be the curve passing through the point (1, 1) has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x-axis, If the area bounded by the curve C and x – axis in the first quadrant is $\frac{k\pi}{2}$ square units, then the value of 'k' is____.

93. Let $y=f(x)$ be a curve passing through $\left(e,e^e\right)$ which satisfy the differential equation $(2ny+xy\log_e x)dx-x\log_e x\,dy=0,\ x>0,\ y>0$. If $g(x)=\lim\limits_{n\to\infty} f(x)$, then $\int_{1/e}^{e} g(x)dx=$

Advanced Problem Package	Vector

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

★1. If $\hat{a}, \hat{b}, \hat{c}$ are three non-coplanar, mutually perpendicular unit vectors, then $[\hat{a}\ \vec{p}\ \vec{q}]\hat{a}+[\hat{b}\ \vec{p}\ \vec{q}]\hat{b}+[\hat{c}\ \vec{p}\ \vec{q}]\hat{c}$ is equal to :

(A) $(\hat{a}+\hat{b}+\hat{c})\times(\vec{p}\times\vec{q})$ **(B)** $\hat{a}+\hat{b}+\hat{c}+\vec{p}+\vec{q}$

(C) $\vec{p}+\vec{q}$ **(D)** $\vec{p}\times\vec{q}$

2. If vectors $\vec{a}=\frac{\vec{i}+\vec{j}}{\sqrt{2}}$, $\vec{b}=\frac{-\vec{i}+\vec{j}}{\sqrt{2}}$ and $\vec{c}=\vec{k}$ then the value of $(\vec{r}\cdot\vec{a})^2+(\vec{r}\cdot\vec{b})^2+(\vec{r}\cdot\vec{c})^2$ is equal to :

(A) $|\vec{r}|^2$ **(B)** $2\vec{r}$ **(C)** 0 **(D)** None of these

★3. If $\vec{a}$ is a unit vector and projection of $\vec{x}$ along $\vec{a}$ is 2 units and $(\vec{a}\times\vec{x})+\vec{b}=\vec{x}$, then $\vec{x}$ is given by :

(A) $\frac{1}{2}[\vec{a}-\vec{b}+(\vec{a}\times\vec{b})]$ **(B)** $\frac{1}{2}[2\vec{a}+\vec{b}+(\vec{a}\times\vec{b})]$

(C) $[\vec{a}+(\vec{a}\times\vec{b})]$ **(D)** None of these

★4. A vector $\vec{\alpha}=a\hat{i}+b\hat{j}+c\hat{k}$ is said to be rational vector, if a, b, c are all rational. If a rational vector with magnitude as positive integer makes an angle $\pi/4$ with vector $\vec{\beta}=\sqrt{2}\hat{i}+3\sqrt{2}\hat{j}+4\hat{k}$, then $\vec{\alpha}$:

(A) Surely lies in xy plane **(B)** Surely lies in xz plane

(C) Surely lies in yz plane **(D)** Nothing can be said

★5. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector, which is perpendicular to $(\vec{a}+\vec{b}+\vec{c})$. Now if $\vec{d}=\sin x(\vec{a}\times\vec{b})+\cos y(\vec{b}\times\vec{c})+2(\vec{c}\times\vec{a})$, then minimum value of x^2+y^2 is equal to :

(A) π^2 **(B)** 0 **(C)** $\frac{\pi^2}{4}$ **(D)** $\frac{5\pi^2}{4}$

★6. Let $ABCD$ be a tetrahedron in which position vectors of A, B, C and D are $\vec{i}+\vec{j}+\vec{k}$, $2\vec{i}+\vec{j}+2\vec{k}$, $3\vec{i}+2\vec{j}+\vec{k}$ and $2\vec{i}+3\vec{j}+2\vec{k}$. If ABC be the base of tetrahedron then height of tetrahedron is :

(A) $\sqrt{\frac{3}{2}}$ **(B)** $\sqrt{\frac{3}{5}}$ **(C)** $\frac{1}{3}\sqrt{\frac{2}{3}}$ **(D)** None of these

7. The position vectors of the vertices A, B and C of a triangle are three unit vectors $\hat{a}$, $\hat{b}$ and $\hat{c}$. A vector $\vec{d}$ is such that $\vec{d}\cdot\hat{a}=\vec{d}\cdot\hat{b}=\vec{d}\cdot\hat{c}$ and $\vec{d}=\lambda(\hat{b}+\hat{c})$, then triangle ABC is :

(A) Acute angled **(B)** Obtuse angled **(C)** Right angled **(D)** None of these

8. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are four non-coplanar unit vectors, $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular, such that $\vec{d}$ makes equal angles with all the three vectors $\vec{a}, \vec{b}, \vec{c}$, then :

(A) $\vec{a}+\vec{b}=\vec{b}+\vec{c}=\vec{c}+\vec{a}$ **(B)** $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

(C) $[\vec{d}\ \vec{a}\ \vec{b}]=[\vec{d}\ \vec{b}\ \vec{c}]=[\vec{d}\ \vec{a}\ \vec{c}]$ **(D)** $[\vec{d}\ \vec{a}\ \vec{b}]=[\vec{d}\ \vec{b}\ \vec{c}]=[\vec{d}\ \vec{c}\ \vec{a}]$

9. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be such that $\vec{c}-3\vec{d}=-\vec{b}-\vec{a}$. Then the points with position vectors as $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are :

(A) coplanar **(B)** collinear **(C)** non-coplanar **(D)** None of these

10. Let $\vec{\alpha}=a\hat{i}+b\hat{j}+c\hat{k}$ and $\vec{\beta}=b\hat{i}+c\hat{j}+a\hat{k}$, where $a, b, c \in R$. If 'θ' be the angle between $\vec{\alpha}$ and $\vec{\beta}$ then :

(A) $\theta \in (0, \pi/2)$ **(B)** $\theta \in [0, 2\pi/3]$ **(C)** $\theta \in (2\pi/3, \pi]$ **(D)** None of these

11. If the lines $\vec{r}=\vec{a}+t\left(\vec{b}\times\vec{c}\right)$, and $\vec{r}=\vec{b}+s\left(\vec{c}\times\vec{a}\right)$ intersect (t and s are scalars) then :

(A) $\vec{a}\cdot\vec{c}=0$ **(B)** $\vec{a}\cdot\vec{c}=\vec{b}\cdot\vec{c}$ **(C)** $\vec{b}\cdot\vec{c}=0$ **(D)** None of these

12. If $4\overline{a}+5\overline{b}+9\overline{c}=0$, then $(\overline{a}\times\overline{b})\times[(\overline{b}\times\overline{c})\times(\overline{c}\times\overline{a})]$ is equal to :

(A) A vector perpendicular to plane of $\overline{a}, \overline{b}$ and $\overline{c}$ **(B)** A scalar quantity

(C) $\vec{0}$ **(D)** None of these

★13. Given that $\vec{a}$ is perpendicular to $\vec{b}$ and p is a non-zero scalar, if $p\,\vec{r}+\left(\vec{r}\,.\,\vec{b}\right)\vec{a}=\vec{c}$ then $\vec{r}=$:

(A) $\vec{r}=\dfrac{\vec{c}}{p^2}-\dfrac{\vec{c}.\vec{b}}{p}\vec{a}$ **(B)** $\vec{r}=\dfrac{\vec{c}}{p}+\dfrac{\vec{c}.\vec{b}}{p^2}\vec{a}$ **(C)** $\vec{r}=\dfrac{\vec{c}}{p}-\dfrac{\vec{c}.\vec{b}}{p^2}\vec{a}$ **(D)** None of these

★14. If the plane faces of a tetrahedron are represented by the equations $\overline{r}\cdot(l\hat{i}+m\hat{j})=0,\ \overline{r}\cdot(n\hat{k}+m\hat{j})=0,\ \overline{r}\cdot(n\hat{k}+l\hat{i})=0$ and $\overline{r}\cdot(l\hat{i}+m\hat{j}+n\hat{k})=p$, then the volume of the tetrahedron is :

(A) $\dfrac{lmn}{6p^3}$ **(B)** $\dfrac{3lmn}{2p^3}$ **(C)** $\dfrac{lmn}{p^3}$ **(D)** None of these

15. Let $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=2\hat{i}-\hat{k}$. The point of intersection of the lines $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}$ and $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ is :

(A) $\hat{i}+\hat{j}-3\hat{k}$ **(B)** $3\hat{i}+\hat{j}+\hat{k}$ **(C)** $3\hat{i}+\hat{j}-\hat{k}$ **(D)** None of these

16. If the non-zero vectors $\vec{a}$ and $\vec{b}$ are perpendiculars to each other then the solution of the equation $\vec{r}\times\vec{a}=\vec{b}$ is given by:

(A) $\vec{r}=x\vec{a}+\dfrac{1}{\vec{a}.\vec{a}}(\vec{a}\times\vec{b})\,;\,x\in R$ **(B)** $\vec{r}=x\vec{b}+\dfrac{1}{\vec{b}.\vec{b}}(\vec{a}\times\vec{b})\,;\,x\in R$

(C) $\vec{r}=x\,\vec{a}\times\vec{b}\,;\,x\in R$ **(D)** None of these

17. Let $OPQR$ is a tetrahedron such that O is origin and $\vec{p}, \vec{q}, \vec{r}$ are position vectors of P, Q, R respectively and α is the angle which OP makes with face PQR then :

(A) $|\sin\alpha|=\dfrac{\left|[\vec{p}\;\vec{q}\;\vec{r}]\right|}{|\vec{q}\times\vec{r}+\vec{r}\times\vec{p}+\vec{p}\times\vec{q}\,|\cdot|\vec{p}|}$ **(B)** $\sin^2\alpha - \cos^2\alpha = \sqrt{3}$

(C) $\tan\alpha=\dfrac{\left|[\vec{p}\vec{q}\vec{r}]\right|}{|\vec{q}\times\vec{r}+\vec{r}\times\vec{p}+\vec{p}\times\vec{r}|\cdot|\vec{p}|}$ **(D)** None of these

18. If $\overline{r}=l(\overline{b}\times\overline{c})+m(\overline{c}\times\overline{a})+n(\overline{a}\times\overline{b})$ and $[\overline{a}\;\overline{b}\;\overline{c}]=2$, then $l+m+n$ is equal to :

(A) $\overline{r}\cdot(\overline{b}\times\overline{c}+\overline{c}\times\overline{a}+\overline{a}\times\overline{b})$ **(B)** $1/2\ \overline{r}\cdot(\overline{a}+\overline{b}+\overline{c})$

(C) $(\overline{a}\;\overline{b}\;\overline{c})$ **(D)** None of these

19. If vectors $\overline{b}=(\tan\alpha, -1, 2\sqrt{\sin\frac{\alpha}{2}})$ and $\overline{c}=(\tan\alpha, \tan\alpha, -\dfrac{3}{\sqrt{\sin\alpha/2}})$ are orthogonal and vector $\overline{a}=(1, 3, \sin2\alpha)$ makes an obtuse angle with the z-axis, then :

(A) $\alpha=\tan^{-1}(-2)$ **(B)** $\alpha=\tan^{-1}(-3)$ **(C)** $\alpha=\tan^{-1}(2)$ **(D)** None of these

20. Let position vector of point A be $\hat{i}+\hat{j}+\hat{k}$ and that of point B be $-\hat{i}+\hat{k}$, then the position vector of point $R(\vec{r})$ such that AR is perpendicular to BR and $\vec{r}$ is not perpendicular to $\left(\vec{r}-\left(\hat{j}+2\hat{k}\right)\right)$ is :

(A) $\vec{r}=\hat{i}+2\hat{j}$ **(B)** $\vec{r}=2\hat{i}+\hat{j}-\hat{k}$ **(C)** $\vec{r}=\hat{k}+2\hat{i}$ **(D)** None of these

***21.** If three coterminous edges of a tetrahedron are $\vec{a}$, $\vec{b}$, $\vec{c}$ such that $|\vec{a}|=2$, $|\vec{b}|=3$, $|\vec{c}|=4$, angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{4}$ and $\vec{c}$ and $\vec{a}$ is $\frac{\pi}{6}$. The area of the base is 2 sq. units, then the height of the tetrahedron is :

(A) $3\sqrt{\left(\sqrt{3}-2\right)}$ **(B)** $3\sqrt{\left(\sqrt{6}-2\right)}$ **(C)** $\frac{3\sqrt{\left(\sqrt{6}-2\right)}}{2}$ **(D)** None of these

***22.** If $\left[(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})\right].(\bar{c}\times\bar{b})=0$ then which of the following is always true :

(A) $\bar{a},\bar{b},\bar{c}$ and $\bar{d}$ are necessary coplanar **(B)** at least one of $\bar{a}$ or $\bar{d}$ must lie in plane of $\bar{b}$ and $\bar{c}$

(C) at least one of $\bar{b}$ or $\bar{c}$ must lie in plane of $\bar{a}$ and $\bar{d}$

(D) at least one of $\bar{a}$ or $\bar{b}$ must lie in plane of $\bar{c}$ and $\bar{d}$

***23.** If $\hat{\alpha}$ and $\hat{\beta}$ be two perpendicular unit vectors such that $\vec{x}=\hat{\beta}-(\hat{\alpha}\times\vec{x})$, then $|\vec{x}|$ is equal to :

(A) 1 **(B)** $\sqrt{2}$ **(C)** $\frac{1}{\sqrt{2}}$ **(D)** None of these

24. If $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are on a circle of radius R whose centre is at origin and $\vec{c}-\vec{a}$ is perpendicular to $\vec{d}-\vec{b}$, then $|\vec{d}-\vec{a}|^2+|\vec{b}-\vec{c}|^2=$ (AC is diameter)

(A) R^2 **(B)** $R^2/2$ **(C)** $2R^2$ **(D)** $4R^2$

25. The vector, directed along the internal bisector of the angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ & $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$ with $|\vec{c}|=5\sqrt{6}$ is:

(A) $\frac{5}{3}\left(\hat{i}-7\hat{j}+2\hat{k}\right)$ **(B)** $\frac{5}{3}\left(\hat{i}+7\hat{j}-2\hat{k}\right)$ **(C)** $\frac{5}{3}\left(-\hat{i}-7\hat{j}+2\hat{k}\right)$ **(D)** None of these

Passage for Questions 26 - 30

Unit vectors $\hat{i}$ and $\hat{j}$ are parallel to the adjacent edges of a large square table. The directions $\hat{i}$ and $\hat{j}$ are referred to as East and North. An ant creeping on the table makes the following movements successively :

(i) 4 *cm* 30° East of North **(ii)** 12 *cm* South West

(ii) 6 *cm* East and **(iv)** 9 *cm* West North

26. The resultant displacement of the ant after first two steps is :

(A) $\left(2+12\sqrt{2}\right)\ \hat{i}+\sqrt{3}\hat{j}$ **(B)** $\left(2-6\sqrt{2}\right)\ \hat{i}+\left(2\sqrt{3}+6\sqrt{2}\right)\hat{j}$

(C) $\left(2-6\sqrt{2}\right)\ \hat{i}+\left(2\sqrt{3}-6\sqrt{2}\right)\hat{j}$ **(D)** None of these

27. The final resultant displacement of the ant is :

(A) $\left(\frac{8\sqrt{2}+21}{\sqrt{2}}\right)\hat{i}+\left(\frac{2\sqrt{6}-3}{\sqrt{2}}\right)\hat{j}$ **(B)** $\left(\frac{8\sqrt{2}+21}{\sqrt{2}}\right)\hat{i}+\left(\frac{2\sqrt{6}+3}{\sqrt{2}}\right)\hat{j}$

(C) $\left(\frac{8\sqrt{2}-21}{\sqrt{2}}\right)\hat{i}+\left(\frac{2\sqrt{6}+3}{\sqrt{2}}\right)\hat{j}$ **(D)** $\left(\frac{8\sqrt{2}-21}{\sqrt{2}}\right)\hat{i}+\left(\frac{2\sqrt{6}-3}{\sqrt{2}}\right)\hat{j}$

28. Magnitude of the resultant displacement is given by :

(A) $\sqrt{301+186\sqrt{2}}$ **(B)** $\sqrt{301-168\sqrt{2}-6\sqrt{6}}$

(C) $\sqrt{301+6\sqrt{6}}$ **(D)** None of these

29. Direction of the ant's resultant displacement is :

(A) $\tan\theta=\dfrac{2\sqrt{6}+3}{8\sqrt{2}+21}$ **(B)** $\tan\theta=\dfrac{2\sqrt{6}-3}{8\sqrt{2}-21}$ **(C)** $\tan\theta=\dfrac{2\sqrt{6}-3}{8\sqrt{2}+21}$ **(D)** None of these

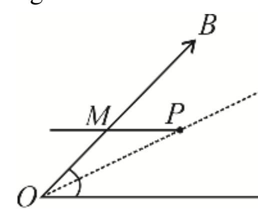
30. Direction of the ant's resultant displacement after first three steps is :

(A) $\tan\theta=\dfrac{\sqrt{3}-3\sqrt{2}}{4-3\sqrt{2}}$ **(B)** $\tan\theta=\dfrac{\sqrt{3}+3\sqrt{2}}{4+3\sqrt{2}}$ **(C)** $\tan\theta=\dfrac{\sqrt{3}-3\sqrt{2}}{4+3\sqrt{2}}$ **(D)** None of these

*Paragraph for Questions 31 - 35

Let two unit vectors along two lines $\overrightarrow{OA}$ and $\overrightarrow{OB}$ be $\hat{a}$ and $\hat{b}$ respectively. Take their point of intersection as the origin and let P be any point on the bisector of angle between the lines OA and OB. Draw PM parallel to AO cutting OB at M.

$$\angle AOP=\angle POM=\angle OPM \text{ and hence } OM=PM.$$



But $\overrightarrow{OM}=t\hat{b}$ and $\overrightarrow{MP}=t\hat{a}$

(Since $\overrightarrow{OM}\parallel\hat{b}$ and $\overrightarrow{MP}\parallel\hat{a}$ and their magnitudes are same).

Then $\overrightarrow{OP}=\vec{r}=\overrightarrow{OM}+\overrightarrow{MP}=t(\hat{b}+\hat{a})$ **...(i)**

For external bisector OP', the angle between OB and OA is the same as the internal bisector of the angle between the unit vectors along them being $-\hat{b}$ and $\hat{a}$ and hence the equation of $\overrightarrow{OP'}$ be

$\overrightarrow{OP'}=\vec{r}=t(\hat{a}-\hat{b})$ **...(ii)**

For any two vectors $\vec{a}$ and $\vec{b}$ the equations (i) and (ii) reduce to $\vec{r}=t\left(\dfrac{\vec{a}}{|\vec{a}|}\pm\dfrac{\vec{b}}{|\vec{b}|}\right)$

31. A vector $\vec{c}$, directed along the internal bisector of the angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ and $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$ with $|\vec{c}|=5\sqrt{6}$, is :

(A) $\frac{5}{3}(\hat{i}-7\hat{j}+2\hat{k})$ **(B)** $\frac{5}{3}(5\hat{i}+5\hat{j}+2\hat{k})$ **(C)** $\frac{5}{3}(\hat{i}+7\hat{j}+2\hat{k})$ **(D)** $\frac{5}{3}(-5\hat{i}+5\hat{j}+2\hat{k})$

32. Let ABC be a triangle and $\vec{a},\vec{b},\vec{c}$ be the position vectors of the point A, B, C respectively. External bisectors of $\angle B$ and $\angle C$ meet at P with the sides of the triangle as $a, b, c,$ the position vector of P becomes :

(A) $\dfrac{(-b)\vec{b}+(-c)\vec{c}}{(b+c)}$ **(B)** $\dfrac{a\vec{a}+(-b)\vec{b}+(-c)\vec{c}}{(a-b-c)}$

(C) $\left(\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}\right)(abc)$ **(D)** $\dfrac{a\vec{a}+b\vec{b}+c\vec{c}}{(\vec{a}+\vec{b}+\vec{c})}$

33. If the interior and exterior bisectors of the angle A of a triangle ABC meet the base BC at D and E, then :

(A) $2BC=BD+BE$ **(B)** $BC^2=BD\times BE$

(C) $\dfrac{2}{BC}=\dfrac{1}{BD}+\dfrac{1}{BE}$ **(D)** None of these

34. If ABC be a triangle of sides a, b, c with position vectors of A, B, C as $\vec{a},\vec{b}$ and $\vec{c}$ respectively, then the position vector of its incentre is :

(A) $\dfrac{(\vec{a}+\vec{b}+\vec{c})}{3}$ **(B)** $\left(\dfrac{\vec{a}\times\vec{b}+\vec{b}+\vec{c}+\vec{c}\times\vec{a}}{a^2+b^2+c^2}\right)$

(C) $\left(\dfrac{a\vec{a}+b\vec{b}+c\vec{c}}{a+b+c}\right)$ **(D)** None of these

35. ABC is a triangle. AD, AD' are internal and external bisectors of angle A, meeting BC at D and D' respectively. A' is the mid-point of DD' and B', C' are similar points on CA and AB. Then A', B', C'

(A) Lie on a plane **(B)** Form an equilateral triangle.

(C) Form an isosceles triangle **(D)** None of these

Paragraph for Questions 36 - 38

If $\vec{a},\vec{b},\vec{c}$ are cyclically permuted, the volume of the scalar triple product remains same. The change in scalar triple product changes the sign of scalar triple product but not the magnitude. In scalar triple product, the position of the dot and cross can be interchanged provided its cyclic nature is preserved. Also, the scalar triple product is ZERO if any two of them are equal.

36. $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$ is $\vec{a}$ and $\vec{c}$ are :

(A) perpendicular **(B)** collinear **(C)** parallel **(D)** None of these

37. The vector $OP = 5i + 12j + 13\hat{k}$ turns through an angle of $\frac{\pi}{2}$ about O passing through the positive side of $\hat{j}$ axis an iff way. The vector in the new position is:

(A) $\dfrac{2}{\sqrt{97}}\left(-30\hat{i}+97\hat{j}-78\hat{k}\right)$ **(B)** $\dfrac{2}{\sqrt{98}}\left(-30\hat{i}+97\hat{j}-78\hat{k}\right)$

(C) $\dfrac{3}{\sqrt{97}}\left(-30\hat{i}+97\hat{j}-78\hat{k}\right)$ **(D)** None of these

38. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three non-parallel unit vectors such that $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{2}\vec{b}$, the angle between $\vec{b}$ and $\vec{c}$ is:

(A) $\pi/2$ **(B)** $\pi/3$ **(C)** $\pi/4$ **(D)** $\pi/6$

Paragraph for Questions 39 - 40

Let $\vec{r}$ is a position vector of a variable point in Cartesian OXY plane such that $\vec{r}\cdot(10\hat{j}-8\hat{i}-\vec{r})=40$ and

$p_1=\max\left\{|\vec{r}+2\hat{i}-3\hat{j}|^2\right\}$, $p_2=\min\left\{|\vec{r}+2\hat{i}-3\hat{j}|^2\right\}$. A tangent line is drawn to the curve $y=\dfrac{8}{x^2}$ at the point A with abscissa 2. The drawn line cuts x-axis at a point B.

39. p_2 is equal to :

(A) 9 **(B)** $2\sqrt{2}-1$ **(C)** $6\sqrt{2}+3$ **(D)** $9-4\sqrt{2}$

40. p_1+p_2 is equal to :

(A) 2 **(B)** 10 **(C)** 18 **(D)** 5

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

41. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angles with $\vec{a}$, $\vec{b}$ and $\vec{c}$, then $|\vec{a}+\vec{b}+\vec{c}+\vec{d}|^2$ is equal to :

(A) $4+\sqrt{3}$ **(B)** $4-\sqrt{3}$ **(C)** $4+2\sqrt{3}$ **(D)** $4-2\sqrt{3}$

★42. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector which is perpendicular to $(\vec{a}+\vec{b}+\vec{c})$ and is represented as $\vec{d}=x(\vec{a}\times\vec{b})+y(\vec{b}\times\vec{c})+z(\vec{c}\times\vec{a})$. Then :

(A) $x^3+y^3+z^3=3xyz$ **(B)** $xy+yz+xz\le 0$

(C) $x=y=z$ **(D)** $x^2+y^2+z^2=xy+yz+zx$

43. Let $\vec{a}$ and $\vec{b}$ be two units vectors if $\vec{u}=\vec{a}-(\vec{a}\cdot\vec{b})\vec{b}$ and $\vec{v}=\vec{a}\times\vec{b}$ then $|\vec{v}|$ is :

(A) $|\vec{u}|$ **(B)** $|\vec{u}|+|\vec{u}\cdot\vec{a}|$ **(C)** $|\vec{u}|+|\vec{u}\cdot\vec{b}|$ **(D)** $|\vec{u}|+\vec{u}.(\vec{a}+\vec{b})$

44. If $\vec{b}$ and $\vec{c}$ are two non-collinear vectors such that $\vec{a}.(\vec{b}+\vec{c})=4$ and $\vec{a}\times(\vec{b}\times\vec{c})=(x^2-2x+6)\vec{b}+(\sin y)\vec{c}$, then :

(A) $x=1$ **(B)** $y=-1$ **(C)** $y=\dfrac{\pi}{2}$ **(D)** $x+y=0$

★45. Let $\vec{\alpha}=a\hat{i}+b\hat{j}+c\hat{k}$, $\vec{\beta}=b\hat{i}+c\hat{j}+a\hat{k}$ and $\vec{\gamma}=c\hat{i}+a\hat{j}+b\hat{k}$ be three coplanar vectors, and $\vec{v}=\hat{i}+\hat{j}+\hat{k}$. Then $\vec{v}$ may be :

(A) $\perp$ to $\vec{\alpha},\vec{\beta},\vec{\gamma}$ **(B)** $\parallel$ to $\vec{\alpha},\vec{\beta},\vec{\gamma}$ **(C)** anti $\parallel$ to $\vec{\alpha},\vec{\beta},\vec{\gamma}$ **(D)** None of these

46. Let $\hat{\alpha}, \hat{\beta}$ and $\hat{\gamma}$ be the unit vectors such that $\hat{\alpha}$ and $\hat{\beta}$ are mutually perpendicular and $\hat{\gamma}$ is equally inclined to $\hat{\alpha}$ and $\hat{\beta}$ at an angle θ. If $\hat{\gamma}=x\hat{\alpha}+y\hat{\beta}+z(\hat{\alpha}\times\hat{\beta})$, then :

(A) $z^2=1-2x^2$ **(B)** $z^2=1-2y^2$ **(C)** $z^2=1-x^2-y^2$ **(D)** $x^2=y^2$

47. Unit vectors $\hat{a}$ and $\hat{b}$ are inclined at an angle 2θ and $|\hat{a}-\hat{b}|<1$. If $0\le\theta<\pi$ then θ may belong to :

(A) $[0, \pi/6)$ **(B)** $(5\pi/6, \pi)$ **(C)** $[\pi/6, \pi/2]$ **(D)** $[\pi/6, 5\pi/6]$

48. The position vectors of the vertices *A, B* and *C* of a triangle are $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$ and $\hat{i}+\hat{k}$ respectively. A unit vector $\hat{r}$ lying in the plane of ΔABC and perpendicular to *IA*, where *I* is the incentre of the triangle is :

(A) $\dfrac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$ **(B)** $\dfrac{\hat{i}-\hat{j}}{\sqrt{2}}$ **(C)** $\dfrac{\hat{j}-\hat{i}}{\sqrt{2}}$ **(D)** $\dfrac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}$

49. *A* and *B* are two points in space with position vector $\bar{a}$ and $\bar{b}$ respectively. Then the value of λ such that the system of equations $|3\bar{r}-2\bar{a}-\bar{b}|=|\bar{a}-\bar{b}|$ and $[\bar{r}-\lambda\bar{a}-(1-\lambda)\bar{b}]\cdot(\bar{a}-\bar{b})=0$ does not have any solution :

(A) 2 **(B)** 3 **(C)** –2 **(D)** None of these

50. The position vectors of the vertices A, B, C of a triangle are $\bar{a}, \bar{b}$ and $\bar{c}$ respectively, where $\bar{c} = \bar{a} \times \bar{b}$ and $\bar{a}$ and $\bar{b}$ are non–collinear vectors. If $\bar{d}$, the position vector of the centroid of the triangle ABC, makes equal angles 'α' with the vectors $\bar{a}, \bar{b}$ and $\bar{c}$, then :

(A) $|\bar{a}| = |\bar{b}|$. **(B)** $|\bar{a}| \neq |\bar{b}|$.

(C) the value of α is $\cos^{-1}\frac{1}{\sqrt{3}}$ if $\bar{a}.\bar{b} = 0$. **(D)** the value of α is $\cos^{-1}\sqrt{\frac{2}{5}}$ if $\bar{a}.\bar{b} = 0$.

51. The vector sum of $\vec{a}$ and $\vec{b}$ trisects the angle θ between them. If $|\vec{a}| = a$; $|\vec{b}| = b$; $a > b$, then:

(A) $\theta = 3\cos^{-1}\left(\frac{2b}{a}\right)$ **(B)** $\theta = 3\cos^{-1}\left(\frac{a}{2b}\right)$ **(C)** $|\vec{a}+\vec{b}| = \frac{a^2+b^2}{b}$ **(D)** $|\vec{a}+\vec{b}| = \frac{a^2-b^2}{b}$

52. The position vectors of the points A, B, C are respectively (1,1,1), (1,–1,2), (0,2,–1). The unit vector parallel to the plane determined by A, B, C and perpendicular to the vector (1,0,1) is/are :

(A) $\frac{5i+j-\hat{k}}{3\sqrt{3}}$ **(B)** $\frac{i+5j-\hat{k}}{3\sqrt{3}}$ **(C)** $\frac{-5i-j+\hat{k}}{3\sqrt{3}}$ **(D)** $\frac{-i-5j+\hat{k}}{3\sqrt{3}}$

53. A line passes through the points whose position vectors are $\vec{i}+\vec{j}-2\vec{k}$ and $\vec{i}-3\vec{j}+\vec{k}$. The position vector of a point on it at a unit distance from the first point is :

(A) $\frac{1}{5}(5\vec{i}+\vec{j}-7\vec{k})$ **(B)** $\frac{1}{5}(5\vec{i}+9\vec{j}-13\vec{k})$ **(C)** $\vec{i}-4\vec{j}+3\vec{k}$ **(D)** $\frac{1}{\sqrt{29}}(2\vec{i}-4\vec{j}+3\vec{k})$

54. The volume of a right triangular prism $ABC\,A_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are $A(1, 0, 1)$; $B(2,0, 0)$ and $C(0, 1, 0)$ the position vectors of the vertex A_1 can be :

(A) (2, 2, 2) **(B)** (0, 2, 0) **(C)** (0, –2, 2) **(D)** (0, –2, 0)

55. For non-zero vectors $\vec{a}, \vec{b}, \vec{c}$, $|(\vec{a}\times\vec{b}).\vec{c}| = |\vec{a}||\vec{b}||\vec{c}|$ holds then :

(A) $\vec{a}.\vec{b}=0, \vec{b}.\vec{c}=0$ **(B)** $\vec{b}.\vec{c}=0, \vec{c}.\vec{a}=0$ **(C)** $\vec{c}.\vec{a}=0, \vec{a}.\vec{b}=0$ **(D)** $\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=0$

56. Three points having position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ will be collinear if :

(A) $\lambda\vec{a}+\mu\vec{b}=(\lambda+\mu)\vec{c}$ **(B)** $[\vec{a}\ \vec{b}\ \vec{c}] = 0$

(C) $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=0$ **(D)** $\vec{a}\times\vec{c}=\vec{b}$

57. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are any four vectors, then $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})$ is a vector:

(A) Perpendicular to $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$

(B) Along the line of intersection of two planes, one containing $\vec{a}, \vec{b}$ other containing $\vec{c}, \vec{d}$

(C) Equally inclined to both $\vec{a}\times\vec{b}$ and $\vec{c}\times\vec{d}$

(D) None of these

58. $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an angle $\alpha(\alpha\in[0,\pi])$ to each other and $|\vec{a}+\vec{b}|<1$ then α can lie in :

(A) $\alpha\in\left(\frac{\pi}{3},\frac{2\pi}{3}\right)$ **(B)** $\alpha\in\left(\frac{\pi}{3},\frac{\pi}{2}\right)$ **(C)** $\alpha\in\left(\frac{2\pi}{3},\frac{5\pi}{6}\right)$ **(D)** $\alpha\in\left(\frac{2\pi}{3},\frac{5\pi}{7}\right)$

59. If $\vec{a}=2\hat{i}-\hat{j}+\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-2\hat{k}$ be three vectors. A vector in the plane of the $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is :

(A) $2\hat{i}+3\hat{j}-3\hat{k}$ **(B)** $2\hat{i}+3\hat{j}+3\hat{k}$ **(C)** $-2\hat{i}-\hat{j}+5\hat{k}$ **(D)** $2\hat{i}+\hat{j}+5\hat{k}$

60. The vector along the bisector of the angle between the two vectors $2\hat{i}-2\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}-2\hat{k}$ having magnitude of 2 units is:

(A) $\frac{2}{\sqrt{10}}\left(3\hat{i}-\hat{k}\right)$ **(B)** $\frac{1}{\sqrt{26}}\left(\hat{i}-4\hat{i}+3\hat{k}\right)$ **(C)** $\frac{2}{\sqrt{26}}\left(\hat{i}-4\hat{i}+3\hat{k}\right)$ **(D)** $\frac{1}{\sqrt{26}}\left(\hat{i}-4\hat{i}-3\hat{k}\right)$

61. Let $\vec{\lambda}=\vec{a}\times(\vec{b}+\vec{c}),\ \vec{\mu}=\vec{b}\times(\vec{c}+\vec{a})$ and $\vec{v}=\vec{c}\times(\vec{a}+\vec{b})$. Then :

(A) $\vec{\lambda}+\vec{\mu}=\vec{v}$ **(B)** $\vec{\lambda},\ \vec{\mu}$ and $\vec{v}$ are coplanar

(C) $\vec{\lambda}+\vec{\mu}+\vec{v}=\vec{0}$ **(D)** $\vec{\lambda}+\vec{v}=\vec{\mu}$

62. Let the position vectors of the points A, B, C be $\vec{a}=\hat{i}+2\hat{j}+3\hat{k},\ \vec{b}=-\hat{i}-\hat{j}+8\hat{k}$ and $\vec{c}=-4\hat{i}+4\hat{j}+6\hat{k}$ respectively, then:

(A) ΔABC is equilateral **(B)** ΔABC is right angled

(C) $|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=2|\vec{a}-\vec{b}|^2$ **(D)** $|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=|\vec{a}-\vec{b}|^2$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

63. If $\vec{a},\vec{b}$ and $\vec{c}$ are the position vectors of the points *A, B* and *C,* where $\vec{c}=x\vec{a}+y\vec{b}$, then match the position of the point *C* according to the given conditions.

	Column 1		Column 2
(A)	$x>0,\ y>0,\ x+y<1$	**(p)**	Outside ΔOAB
(B)	$x>0,\ y>0,\ x+y>1$	**(q)**	Inside $\angle OAB$
(C)	$x>0,\ y<0,\ x+y<1$	**(r)**	Inside $\angle OBA$
(D)	$x<0,\ y>0,\ x+y<1$	**(s)**	Inside $\angle AOB$

64. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	If $\vec{a},\vec{b},\vec{c}$ are three mutually perpendicular vectors where $\lvert\vec{a}\rvert=\lvert\vec{b}\rvert=2,\lvert\vec{c}\rvert=1$, then $1/12[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]$ is	**(p)**	–3/4
(B)	If $\vec{a},\vec{b}$ are two unit vectors inclined at $\pi/3$, then $[\vec{a}\ \ \vec{b}+\vec{a}\times\vec{b}\ \ \vec{b}]$ is	**(q)**	0
(C)	If $\vec{b},\vec{c}$ are orthogonal unit vectors and $\vec{b}\times\vec{c}=\vec{a}$, then $[\vec{a}+\vec{b}+\vec{c}\ \ \vec{a}+\vec{b}\ \ \vec{b}+\vec{c}]$ is	**(r)**	4/3
(D)	If $[\vec{x}\ \vec{y}\ \vec{a}]=[\vec{x}\ \vec{y}\ \vec{b}]=[\vec{a}\ \vec{b}\ \vec{c}]=0$ each vector being a non-zero vector, then $[\vec{x}\ \vec{y}\ \vec{c}]$ is	**(s)**	1

65. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The possible value of a if $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+2\hat{j})+\mu(-\hat{i}+\hat{j}+a\hat{k})$ are two skew lines	(p)	–4
(B)	The angle between the vectors $\vec{a}=\lambda\hat{i}-3\hat{j}-\hat{k}$ and $\vec{b}=2\lambda\hat{i}+\lambda\hat{j}-\hat{k}$ is acute, whereas the vector $\vec{b}$ makes with axes of coordinates an obtuse angle, then λ may be	(q)	–2
(C)	The possible value of a such that $\left(2\hat{i}-\hat{j}+\hat{k}\right),\left(\hat{i}+2\hat{j}+(1+a)\hat{k}\right)$ and $\left(3\hat{i}+a\hat{j}+5\hat{k}\right)$ are coplanar is	(r)	2
(D)	If $\vec{A}=2\hat{i}+\lambda\hat{j}+3\hat{k}$, $\vec{B}=2\hat{i}+\lambda\hat{j}+\hat{k}$, $\vec{C}=3\hat{i}+\hat{j}$ and $\vec{A}+\lambda\vec{B}$ is perpendicular to $\vec{C}$, then $\lvert 2\lambda \rvert$ is	(s)	3

66. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$\vec{a}$ and $\vec{c}$ are unit vectors and $\left\lvert\vec{b}\right\rvert=4$ with $\vec{a}\times\vec{b}=2\vec{a}\times\vec{c}$. The angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{b}-2\vec{c}=\lambda\vec{a}$, then λ belongs to	(p)	$\left(0, \frac{\pi}{2}\right)$
(B)	If Vector $\vec{a}=(x, y, z)$ makes equal angles with the vectors $\vec{b}=(y, -2z, 3x)$ and $\vec{c}=(2z, 3x, -y)$ and is perpendicular to the vector $\vec{d}=(1, -1, 2)$ with $\lvert\vec{a}\rvert=2\sqrt{3}$ and the angle between $\vec{a}$ and the unit vector $\hat{j}$ is obtuse then $x+y-z$ belongs to	(q)	$(0, e)$
(C)	Let ABC be a triangle whose centroid is G orthocentre is H and circumcentre is origin O. If D is any point in the plane of the triangle such that no three of O, A, B, C, D are collinear satisfying the relation $\overrightarrow{AD}+\overrightarrow{BD}+\overrightarrow{CH}+3\overrightarrow{HG}=\lambda\overrightarrow{HD}$, then scalar λ belongs to	(r)	$[0, \pi)$
(D)	If $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\lvert\vec{a}\rvert=\lvert\vec{c}\rvert=1, \lvert\vec{b}\rvert=4$ and $\left\lvert\vec{b}\times\vec{c}\right\rvert=\sqrt{15}$ if $\vec{b}-2\vec{c}=4\lambda\vec{a}$, then λ belongs to	(s)	(e, π)

67. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	If $\vec{a}, \vec{b}, \vec{c}$ represents the sides of the triangle ABC, then	(p)	$\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}$
(B)	If $\vec{a}, \vec{b}, \vec{c}$ represents three co-terminus edges of regular tetrahedron, then	(q)	$\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=0$
(C)	If $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$, then	(r)	$\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$
(D)	$\vec{a}+\vec{b}+\vec{c}=\vec{0}$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, then	(s)	$\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}=-\frac{3}{2}$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/ INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

★68. Two given points P and Q in the rectangular cartesian coordinates lie on $y = 2^{x+2}$ such that $\overrightarrow{OP}.\hat{i} = -1$ and $\overrightarrow{OQ}.\hat{i} = +2$ where $\hat{i}$ is a unit vector along the x-axis . The magnitude of $\dfrac{\overrightarrow{OQ} - 4\overrightarrow{OP}}{2}$ is _______.

★69. Line L_1 is parallel to a vector $\vec{\alpha} = 3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through a point A (7,6,2) and the line L_2 is parallel to a vector $\vec{\beta} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through a point B (5, 3,4). Now a line L_3 parallel to a vector $\vec{r} = 2\hat{i} + 2\hat{j} + \hat{k}$ intersects the lines L_1 and L_2 at points C and D respectively. then $\left|\overline{CD}\right|$ is_______.

70. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\left(\vec{a} \times \vec{b}\right)$ and $\vec{c}$ is 30^0, then the value of $\left|\left(\vec{a} \times \vec{b}\right) \times 2\vec{c}\right|$ is_______.

71. If $\vec{a} \times \left(\vec{b} \times \vec{c}\right) + \left(\vec{a}.\vec{b}\right)\vec{b} = (4 - 2\beta - \sin\alpha)\vec{b} + \left(\beta^2 - 1\right)\vec{c}$ and $\left(\vec{c}.\vec{c}\right)\vec{a} = \vec{c}$ where $\vec{b}$ and $\vec{c}$ are non-collinear and α, β are scalars then $\beta =$ _______.

★72. Let $\vec{u}$ and $\vec{v}$ be unit vectors. if $\vec{\omega}$ is a vector such that $\vec{\omega} + \left(\vec{\omega} \times \vec{u}\right) = \vec{v}$, if the maximum volume of the parallelepiped formed by $\vec{u}, \vec{v}$ and $\vec{\omega}$ is p then $12p =$ _______.

73. Let $\bar{u}, \bar{v}, \bar{w}$ be such that $|\bar{u}| = 1,\ |\bar{v}| = 2,\ |\bar{w}| = 3$. If the projection $\bar{v}$ along $\bar{u}$ is equal to that of $\bar{w}$ along $\bar{u}$ and $\bar{v}$, $\bar{w}$ are perpendicular to each other, then $\dfrac{|\bar{u} - \bar{v} + \bar{w}|^2}{2}$ equals _______.

74. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, where $x_1, x_2, x_3 \in \{-3, -2, -1, 0, 1, 2\}$. Number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular, is p then $\dfrac{p}{5} =$

75. If in a triangle ABC, $\overrightarrow{BC} = \dfrac{\vec{u}}{|\vec{u}|} - \dfrac{\vec{v}}{|\vec{v}|}$ and $\overrightarrow{AC} = \dfrac{2\vec{u}}{|\vec{u}|}$ where $|\vec{u}| \neq |\vec{v}|$, then 1 + cos 2A + cos 2B + cos 2C =

76. Let $\bar{u}$ and $\bar{v}$ are unit vectors and $\bar{w}$ is a vector such that $\bar{u} \times \bar{v} + \bar{u} = \bar{w}$ and $\bar{w} \times \bar{u} = \bar{v}$, Then the value of $[uvw]$ =_______.

77. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 1$, then $\sqrt{[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}]}$ is_______.

78. $\vec{a}$ and $\vec{b}$ are two non-collinear vectors then the points with position vectors $l_1\vec{a} + m_1\vec{b}, l_2\vec{a} + m_2\vec{b}, l_3\vec{a} + m_3\vec{b}$ are collinear then find the value of $\begin{vmatrix} 1 & 1 & 1 \\ l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{vmatrix}$.

79. Let $A(2\hat{i}+3\hat{j}+5\hat{k}), B(-\hat{i}+3\hat{j}+2\hat{k})$ and $C(\lambda\hat{i}+5\hat{j}+\mu\hat{k})$ are vertices of a triangle and its median through A is equally inclined to the positive directions of the axes. Find the value of $2\lambda-\mu$

80. Let A, B, C be points with position vectors $r_1=2\hat{i}-\hat{j}+\hat{k}$, $r_2=\hat{i}+2\hat{j}+3\hat{k}$ and $r_3=3\hat{i}+\hat{j}+2\hat{k}$ relative to the origin 'O'. Find the shortest distance between point B and plane OAC.

81. Volume of tetrahedron whose vertices are the points with position vectors $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+7\hat{k}$, $5\hat{i}-\hat{j}+h\hat{k}$ and $7\hat{i}-4\hat{j}+7\hat{k}$ is 11 cubic units then the value of h is _______ $(h>1)$

82. If $\vec{a},\vec{b},\vec{c}$ are unit vectors such that $\vec{a}\cdot\vec{b}=0=\vec{a}\cdot\vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. Then the value of $\left|\vec{a}\times\vec{b}-\vec{a}\times\vec{c}\right|$ is ______.

83. Given that the vectors $\vec{a},\vec{b}$ and $\vec{c}$ (no two of them are collinear). Further if $(\vec{a}+\vec{b})$ is collinear with $\vec{c}$, $(\vec{b}+\vec{c})$ is collinear with $\vec{a}$ and $|\vec{a}|=|\vec{b}|=|\vec{c}|=\sqrt{2}$. Then the value of $|\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}|$ is _____.

84. Let $\vec{a}=\hat{i}-\hat{j}, \vec{b}=\hat{i}+2\hat{j}+2\hat{k}, \vec{c}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{d}=2\hat{i}-\hat{j}+\hat{k}$, then the shortest distance between the lines $\vec{r}=\vec{a}+t\vec{b}$ and $\vec{r}=\vec{c}+p\vec{d}$ is k, then the value of $\frac{1}{k^2}$ is _____.

Advanced Problem Package	Three Dimensional Geometry

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The equation of line intersecting and perpendicular to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and passing through (–2, –5, 7) is :

(A) $\frac{x+2}{14}=\frac{y+5}{123}=\frac{z-7}{104}$ **(B)** $\frac{x+2}{14}=\frac{y+5}{137}=\frac{z-7}{-204}$

(C) $\frac{x+2}{76}=\frac{y+5}{137}=\frac{z-7}{-254}$ **(D)** None of these

★2. The direction cosines of the projection of the line $\frac{1}{2}(x-1)=-y=z+2$ on the plane $2x + y - 3z = 4$ are :

(A) $\left(\frac{2}{\sqrt{6}},\frac{-1}{\sqrt{6}},\frac{1}{\sqrt{6}}\right)$ **(B)** $\left(\frac{-2}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}}\right)$ **(C)** $\left(\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{-1}{\sqrt{6}}\right)$ **(D)** None of these

3. A mirror and a source of light are situated at the origin O and at a point on the line OX respectively. A ray of light from the source strikes the mirror at O and is reflected. The direction ratios of the normal to the plane of the mirror are (1, –1, 1); then the direction cosines of the reflected ray are :

(A) $\frac{1}{3},-\frac{2}{3},\frac{2}{3}$ **(B)** $-\frac{1}{3},\frac{2}{3},\frac{2}{3}$ **(C)** $-\frac{1}{3},-\frac{2}{3},-\frac{2}{3}$ **(D)** $\frac{-1}{3},\frac{-2}{3},\frac{2}{3}$

4. The planes $x + y - z = 0$, $y + z - x = 0$, $z + x - y = 0$ meet :

(A) in a line **(B)** taken two at a time in parallel lines

(C) in a unique point **(D)** None of these

5. If the line $x = y = z$ intersect the line $\sin A\ x + \sin B\ y + \sin C\ z = 2d^2$, $\sin 2A\ x + \sin 2B\ y + \sin 2C\ z = d^2$ then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ is equal to : (where $A + B + C = \pi$)

(A) 1/16 **(B)** 1/8 **(C)** 1/32 **(D)** 1/12

6. In a three dimensional co-ordinate system P, Q and R are images of a point A (a, b, c) in the x-y, the y-z and the z-x planes respectively. If G is the centroid of triangle PQR then area of triangle AOG is : (O is the origin)

(A) $\frac{3}{2}(a^2+b^2+c^2)$ **(B)** $a^2 + b^2 + c^2$ **(C)** $\frac{2}{3}(a^2+b^2+c^2)$ **(D)** 0

7. A variable plane passes through a fixed point (a, b, c) and meets the coordinate axes in A, B, C. The locus of the point common to the plane and also planes through A, B, C parallel to coordinate planes is:

(A) $ayz + bzx + cxy = xyz$ **(B)** $axy + byz + czx = xyz$

(C) $axy + byz + czx = abc$ **(D)** $bcx + acy + abz = abc$

8. The equation of the plane bisecting the acute angle between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is :

(A) $23x - 13y + 32z + 45 = 0$ **(B)** $5x - y - 4z = 3$

(C) $5x - y - 4z + 45 = 0$ **(D)** $23x - 13y + 32z + 3 = 0$

9. Direction cosines of normal to the plane containing lines $x = y = z$ and $x - 1 = y - 1 = \frac{z-1}{d}$ (where $d \in R - \{1\}$), are :

(A) $\left\{\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}},0\right\}$ **(B)** $\left\{\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}}\right\}$ **(C)** $\left\{0,-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right\}$ **(D)** None of these

10. The equation of the straight line through the origin parallel to the line $(b + c)x + (c + a)y + (a + b)z = k = (b - c)x + (c - a)y + (a - b)z$ is :

(A) $\frac{x}{b^2-c^2}=\frac{y}{c^2-a^2}=\frac{z}{a^2-b^2}$ (B) $\frac{x}{b}=\frac{y}{c}=\frac{z}{a}$

(C) $\frac{x}{a^2-bc}=\frac{y}{b^2-ca}=\frac{z}{c^2-ab}$ (D) None of these

11. A variable plane makes intercepts on the co-ordinate axes the sum of whose squares is constant and equal to k^2. Then the locus of the foot of the perpendicular from the origin to the plane is

(A) $(x^{-2} + y^{-2} + z^{-2})(x^2 + y^2 + z^2) = k^2$ (B) $k^2(x^{-2} + y^{-2} + z^{-2})(x^2 + y^2 + z^2) = 1$

(C) $(x^2 + y^2 + z^2)^2 = \frac{1}{k}$ (D) None of these

12. P is a point on the plane $lx + my + nz = p$. A point Q is taken on the line OP such that $OP.OQ = p^2$. Then the locus of Q is :

(A) $p(lx + my + nz) = x^2 + y^2 + z^2$ (B) $(lx + my + nz)(x^2 + y^2 + z^2) = p^2$

(C) $lx + my + nz = (x^2 + y^2 + z^2)p$ (D) None of these

13. The orthogonal projection of the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ on the plane $3x + 4y + 5z = 0$ is :

(A) $\frac{x}{7}=\frac{y}{1}=\frac{z}{-5}$ (B) $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ (C) $\frac{x}{1}=\frac{y}{-2}=\frac{z}{1}$ (D) None of these

14. Shortest distance between the lines $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{1}$ is equal to :

(A) $\sqrt{14}$ (B) $\sqrt{7}$ (C) $\sqrt{2}$ (D) None of these

***15.** The point of intersection of the line, passing through (0, 0, 1) and intersecting the lines $x + 2y + z = 1$, $-x + y - 2z = 2$ and $x + y = 2$, $x + z = 2$ with xy plane is :

(A) $\left(\frac{5}{3}, -\frac{1}{3}, 0\right)$ (B) (1, 1, 0) (C) $\left(\frac{2}{3}, -\frac{1}{3}, 0\right)$ (D) $\left(-\frac{5}{3}, \frac{1}{3}, 0\right)$

16. If $abc \neq 0$ and let (p_1, q_1, r_1) be the image of (p, q, r) in the plane $ax + by + cz + d = 0$, then :

(A) $\frac{p_1-p}{a}=\frac{q_1-q}{b}=\frac{r_1-r}{c}$ (B) $a(p + p_1) + b(q + q_1) + c(r + r_1) + 2d = 0$

(C) Both **(A)** and **(B)** (D) None of these

17. A perpendicular is drawn from a point (1, 6, 3) to the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. What will be coordinates of the foot of perpendicular :

(A) (1, 3, 5) (B) (0, 3, –2) (C) (2, 4, –5) (D) (1, 3, 4)

18. The foot of the perpendicular from the point $O(0, 0, 0)$ to the line of intersection of the planes $x + y + z = 4$ and $2x + y + 3z = 1$ is point A. Then the equation of line OA is :

(A) $\frac{x}{2}=\frac{y}{1}=\frac{z}{3}$ (B) $\frac{x}{8}=\frac{y}{29}=\frac{z}{-13}$ (C) $\frac{x}{-2}=\frac{y}{1}=\frac{z}{1}$ (D) None of these

19. A variable line passing through the point $P(0, 0, 2)$ always makes angle 60^0 with z–axis, intersects the plane $x + y + z = 1$. Then the locus of point of intersection of the line and the plane is :

(A) $x^2 + y^2 + 3(z-2)^2 = 0$ **(B)** $x^2 + y^2 = 3(z+2)^2$

(C) $x^2 + y^2 = 2(z-3)^2$ **(D)** $x^2 + y^2 = 3(z-2)^2$

★20. A ray of light is coming along the line $\frac{x-2}{3}=\frac{y-1}{4}=\frac{z-6}{5}$ and strikes the plane mirror kept along the plane through the points (2, 1, 0), (5, 0, 1) and (4, 1, 1). Then the equation of reflected ray is :

(A) $\frac{x+10}{4}=\frac{y+15}{5}=\frac{z+14}{3}$ **(B)** $\frac{x-10}{4}=\frac{y-15}{5}=\frac{z-14}{3}$

(C) $\frac{x-15}{-4}=\frac{y-14}{-5}=\frac{z-10}{3}$ **(D)** None of these

★21. The equation of a line on the plane $x + y + z = 1$ such that the line $\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-1}{1}$ and the required line form a plane which is perpendicular to the plane $x + y + z = 1$ is :

(A) $\frac{3x+1}{2}=\frac{3y+1}{-1}=\frac{3z+1}{-1}$ **(B)** $\frac{3x-1}{-2}=\frac{3y-1}{1}=\frac{3z-1}{1}$

(C) $\frac{3x-1}{2}=\frac{3y-1}{-1}=\frac{3z-1}{-1}$ **(D)** None of these

22. The image of the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ in the plane $x + 2y + z = 12$ is :

(A) $\frac{x-6}{2}=\frac{y+\frac{7}{2}}{-2}=\frac{z-13}{2}$ **(B)** $\frac{x-6}{4}=\frac{y+\frac{7}{2}}{-7}=\frac{z-13}{10}$

(C) $\frac{x+6}{2}=\frac{y-\frac{7}{2}}{-3}=\frac{z-13}{6}$ **(D)** None of these

23. If a plane passes through the intersection of $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2 (a_1, a_2 \neq 0)$ and contains the x-axis, then :

(A) $d_1a_2 = a_1d_2$ **(B)** $a_1a_2 = d_1d_2$ **(C)** $d_1a_1 = d_2a_2$ **(D)** None of these

24. The plane $x = 0$ is rotated through an angle α about its line of intersection with the plane $z = 0$. Then equation of the plane in new position is (are) :

(A) $x \pm \sqrt{\sec\alpha - 1}\, z = 0$ **(B)** $x \pm \sqrt{\cos\alpha + 1}\, z = 0$

(C) $x \pm \sqrt{\sec\alpha + 1}\, z = 0$ **(D)** None of these

★Passage for Questions 25 - 29

A conic section is obtained by the intersection of two inverted cones (having same vertex) with a plane. If a plane passes through the vertex, then its intersection with the cones either represents a pair of straight lines or a point depending upon whether it intersects the cones or not. If plane does not pass through vertex, then section may be hyperbola or circle depending upon whether plane is parallel to axis of the cone or perpendicular to it. If it has any other inclination, then intersection of plane and cone gives an ellipse.

25. If a variable point P moves such that the line passing through P and Q (0, 0, 2) makes an angle 60º with z-axis, then locus of P is :

(A) $x^2 - y^2 - 3(z-2)^2 = 0$ **(B)** $x^2 - y^2 + 3(z-2)^2 = 0$

(C) $x^2 + y^2 - 3(z-2)^2 = 0$ **(D)** $x^2 + y^2 + 3(z-2)^2 = 0$

26. The locus of intersection of locus of P with the plane $x + y + z = 1$ is :

(A) a pair of straight lines **(B)** a hyperbola

(C) a circle **(D)** an ellipse

27. The locus of intersection of locus of P and the plane $x + y + z = 2$

(A) a pair of straight lines **(B)** a hyperbola

(C) a circle **(D)** a point

28. The locus of intersection of locus of P with $x + y = 2$

(A) a straight lines **(B)** a hyperbola **(C)** a circle **(D)** an ellipse

29. The locus of intersection of locus of P with $2x + y + z = 2$ is :

(A) a straight lines **(B)** a hyperbola **(C)** a circle **(D)** a point

***Passage for Questions 30 - 33**

Consider a three dimensional Cartesian system with origin at O and three rectangular coordinates axes x, y and z-axis. Suppose that the distance between two points P and Q in the space having their coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively be defined by the following formula $d(P,Q) = |x_2 - x_1| + |y_2 - y_1| + |z_2 - z_1|$.

Although the formula of distance between two points has been defined in a new way, yet the other definition remain same (like section formula, direction cosines etc.). So, in general equations of straight line in space remain unchanged. Now answer the following questions.

30. If l, m, n represent direction cosines (if we can call it) of a vector $\overrightarrow{OP}$, then which of the following relations holds?

(A) $l^2 + m^2 + n^2 = 1$ **(B)** $l + m + n = 1$ **(C)** $|l + m + n| = 1$ **(D)** $|l| + |m| + |n| = 1$

31. Locus of point P if $d(O,P) = k$, where k is a positive constant number, represents :

(A) a sphere of radius k **(B)** a set of eight planes forming an octahedron

(C) a set of eight planes forming hexagonal prism **(D)** an infinite cylinder of radius k

32. Let A be a point (1, 2, 3) in the given reference system. Then locus of the point P in the first octant satisfying the equation $d(O,P) = d(A,P)$ does not contain :

(A) any of the coordinates axes **(B)** any of the coordinates planes

(C) any plane parallel to coordinates axes **(D)** any plane parallel to coordinate planes

33. An equilateral triangle has its vertices on the axes of coordinates and area $\sqrt{3}$ square units. The coordinates of the orthocenter of the triangle are :

(A) (1, 1, 1) **(B)** $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ **(C)** $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ **(D)** $\left(\frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}\right)$

Passage for Questions 34 - 36

Two lines whose equations are $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{\lambda}$ and $\frac{x-2}{3}=\frac{y-3}{2}=\frac{z-2}{3}$ lie in the same plane, then:

34. The value of $\sin^{-1}\sin\lambda$ is equal to:

(A) 3 **(B)** $\pi-3$ **(C)** 4 **(D)** $\pi-4$

35. Point of intersection of the line lies on:

(A) $3x+y+z=20$ **(B)** $3x+y+z=25$ **(C)** $3x+2y+z=24$ **(D)** None of these

36. Equation of plane containing both lines is:

(A) $x+5y-3z=10$ **(B)** $x+6y+5z=20$ **(C)** $x+6y-5z=10$ **(D)** None of these

Passage for Questions 37 - 38

Consider the tetrahedron formed by the planes $y+z=0,\ z+x=0,\ x+y=0,\ x+y+z=a$.

37. The direction cosines of the shortest distance lie between the planes $y+z=0$ and $z+x=0$ is:

(A) $-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}$ **(B)** $-\frac{1}{\sqrt{6}},-\frac{1}{\sqrt{6}},-\frac{1}{\sqrt{6}}$

(C) $-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$ **(D)** None of these

38. The shortest distance between any two opposite edges of the tetrahedron is:

(A) $\frac{2}{\sqrt{6}}a$ **(B)** $\frac{1}{\sqrt{6}}a$ **(C)** $\frac{1}{\sqrt{3}}a$ **(D)** None of these

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

***39.** The locus of the point equidistant from the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ and $\frac{x}{2}=\frac{y}{3}=\frac{z}{1}$ can be:

(A) $x+y+2z=0$ **(B)** $x+y-2z=0$ **(C)** $3x+5y+4z=0$ **(D)** $3x+5y+4z+1=0$

40. The equation of a plane passing through the line $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-2}{-2}$ and making an angle of 30^0 with the plane $x+y+z=5$ is :

(A) $x+(3+2\sqrt{2})y+(2+\sqrt{2})z-11-6\sqrt{2}=0$ **(B)** $x+(3-2\sqrt{2})y+(2-\sqrt{2})z-11+6\sqrt{2}=0$

(C) $x+(3-\sqrt{2})y+(2+\sqrt{2})z-11-6\sqrt{2}=0$ **(D)** $x+(3+\sqrt{2})y+(2-\sqrt{2})z-11+6\sqrt{2}=0$

***41.** A plane meets a set of three mutually perpendicular planes in the sides of a triangle whose angles are A, B and C respectively. The squares of cosines of angles which first plane makes with the other planes are:

(A) $\cot B\cot C,\cot C\cot A,\cot A\cot B$

(B) $\tan B\tan C,\tan C\tan A,\tan A\tan B$

(C) $\operatorname{cosec} B\operatorname{cosec} C,\operatorname{cosec} C\operatorname{cosec} A,\operatorname{cosec} A\operatorname{cosec} B$

(D) None of these

★42. The straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$ are perpendicular if :

(A) $\frac{f}{a}+\frac{g}{b}+\frac{h}{c}=0$ (B) $\frac{a}{f}+\frac{b}{g}+\frac{c}{h}$ (C) $\frac{h}{a}+\frac{g}{b}+\frac{f}{c}$ (D) None of these

★43. The coordinates of points, whose perpendicular distances from yz, zx and xy-planes are in $A.P.$ and whose distances from x, y and z axes are $\sqrt{13}$, $\sqrt{10}$ and $\sqrt{5}$ respectively is :

(A) (1, 2, 3) (B) (–1, 2, 3) (C) (1, –2, 3) (D) (–1, –2, –3)

44. The plane $3y + 4z = 0$ is rotated about its line of intersection with the plane $x = 0$ through an angle 60^0. The equation of the plane in its new position is :

(A) $3y+4z+5\sqrt{3}x=0$ (B) $3y+4z-5\sqrt{3}x=0$

(C) $3y-4z-5\sqrt{3}x=0$ (D) $3y-4z+5\sqrt{3}x=0$

45. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy = c^2$, $z = 0$ if c is equal to :

(A) 1/3 (B) –1/3 (C) $\sqrt{5}$ (D) $-\sqrt{5}$

46. Consider the lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ the equation of the line which :

(A) Bisects the angle between the lines is $\frac{x}{3}=\frac{y}{3}=\frac{z}{6}$

(B) Bisects the angle between the lines is $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) Passes through origin and is perpendicular to the given lines is $x = y = -z$

(D) None of these

47. The equations of the planes through the origin which are parallel to the line $\frac{x-1}{2}=\frac{y+3}{-1}=\frac{z+1}{-2}$ and at a distance 5/3 from it are :

(A) $2x + 2y + z = 0$ (B) $x + 2y + 2z = 0$ (C) $2x - 2y + 3z = 0$ (D) $x - 2y + 2z = 0$

★48. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). The equations of the remaining sides are :

(A) $\frac{x-7}{5}=\frac{y-2}{2}=\frac{z-4}{6}$ (B) $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

(C) $\frac{x-7}{2}=\frac{y-2}{-2}=\frac{z-4}{6}$ (D) $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

★49. A ray M is sent along the line $\frac{x-0}{2}=\frac{y-2}{2}=\frac{z-1}{0}$ and is reflected by the plane $x = 0$ at point A. The reflected ray is again reflected by the plane $x+2y=0$ at point B. The initial ray and final reflected ray meets at point J. Then :

(A) The co-ordinates of point B is $(4,-2,1)$ (B) The co-ordinates of point J is $(-3,-1,1)$

(C) The centroid of ΔABJ is (0, 0, 0) (D) The co-ordinates of point J is $(2,-1,1)$

50. The equation of a line in xz plane equally inclined with x and z axes which is at a unit distance from the line $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z}{1}$ is :

(A) $\frac{x-\sqrt{6}+1}{1}=\frac{y}{0}=\frac{z}{-1}$ (B) $\frac{x-1-\sqrt{2}}{1}=\frac{y}{0}=\frac{z}{1}$

(C) $\frac{x-1}{1}=\frac{y}{0}=\frac{z}{1}$ (D) None of these

*51. Projection of line $\frac{x+1}{2}=\frac{y+1}{-1}=\frac{z+3}{4}$ on the plane $x + 2y + z = 6$; has equation :

(A) $x + 2y + z - 6 = 0 = 9x - 2y - 5z - 8$ (B) $x + 2y + z + 6 = 0,\ 9x - 2y + 5z = 4$

(C) $\frac{x-1}{4}=\frac{y-3}{-7}=\frac{z+1}{10}$ (D) $\frac{x+3}{4}=\frac{y-2}{7}=\frac{z-7}{-10}$

52. If three planes $P_1 \equiv 2x+y+z-1=0$, $P_2 \equiv x-y+z-2=0$ and $P_3 = \alpha x-y+3z-5=0$ intersect each other at point P on XOY plane and at point Q on YOZ plane, where O is the origin then identify the correct statement(s)

(A) the value of α is 4

(B) straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4}=\frac{y+1}{-1}=\frac{z}{3}$

(C) the length of projection of $\overrightarrow{PQ}$ on x-axis is 1

(D) centroid of the triangle OPQ is $\left(\frac{1}{3},-\frac{1}{2},\frac{1}{2}\right)$

53. The coordinate of the points on the line $\frac{x+2}{3}=\frac{y+1}{2}=\frac{z-3}{2}$ which are at a distance $3\sqrt{2}$ from the point (1, 2, 3)

(A) (–2, –1, 3) (B) (2, 2, 4) (C) $\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ (D) $\left(\frac{47}{11},\frac{42}{11},\frac{56}{11}\right)$

54. The equations of the lines of shortest distance between the lines $\frac{x}{2}=\frac{y}{-3}=\frac{z}{1}$ and $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ are:

(A) $3(x-21)=3y+92=3z-32$ (B) $\frac{x-62/3}{1/3}=\frac{y+31}{1/3}=\frac{z-31/3}{1/3}$

(C) $\frac{x-21}{1/3}=\frac{y+92/3}{1/3}=\frac{z-32/3}{1/3}$ (D) $\frac{x-2}{1/3}=\frac{y+3}{1/3}=\frac{z-1}{1/3}$

55. The equation of planes bisecting the angle between the planes $2x-y+2z+3=0$ and $3x-2y+6z+8=0$ is / are:

(A) $5x-y-4z-45=0$ (B) $5x-y-4z-3=0$

(C) $23x-13y+32z+45=0$ (D) $23x-13y+32z+5=0$

56. The equation of the plane parallel to plane $x+y+2z=5$ at a distance $\sqrt{6}$ units from this plane is / are:

(A) $x+y+2z+11=0$ (B) $x+y+2z=11$

(C) $x+y+2z+1=0$ (D) $x+y+2z-1=0$

57. If the line $\frac{x-2}{-1}=\frac{y+2}{1}=\frac{z+k^2-1}{4}$ is one of the angle bisector of the lines $\frac{x}{1}=\frac{y}{-2}=\frac{z}{3}$ and $\frac{x}{-2}=\frac{y}{3}=\frac{z}{1}$, then the value of k is /are:

(A) – 3 (B) 2 (C) 3 (D) – 2

58. Let PM be perpendicular from the point $P(1, 2, 3)$ to x-y plane. If $\overrightarrow{OP}$ makes an angle θ with the positive direction of z-axis and $\overrightarrow{OM}$ makes an angle ϕ with the positive direction of x-axis, where O is the origin and θ and ϕ are acute angles, then:

(A) $\tan\theta = \frac{\sqrt{5}}{3}$ **(B)** $\sin\theta\sin\phi = \frac{2}{\sqrt{14}}$ **(C)** $\tan\phi = 2$ **(D)** $\cos\theta\cos\theta = \frac{1}{\sqrt{14}}$

59. If lines $x = y = z$, $x = \frac{y}{2} = \frac{z}{3}$ and the third line passing through (1, 1, 1) form a triangle of area $\sqrt{6}$ units, then point of intersection of third line with second line will lie on:

(A) $\frac{x-2}{3} = \frac{y-4}{8} = \frac{z-6}{9}$ **(B)** $x + 2y + z = 16$

(C) $x - 2y - z = 16$ **(D)** $\frac{x-3}{3} = \frac{y-4}{8} = \frac{z-3}{5}$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labelled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

60. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	The point in which the line $\frac{x+1}{-1} = \frac{y-12}{5} = \frac{z-7}{2}$ cuts the surface $11x^2 - 5y^2 + z^2 = 0$ is(are)	**(p)**	$(2, -3, 1)$
(B)	A line with direction cosines proportional to 1, –5, –2 meets each of the lines $x = y + 5 = z + 1$ and $x + 5 = 3y = 2z$. The coordinates of each of the points of intersection are given by	**(q)**	$(1, 2, 3)$
(C)	Let $P = 0$ is the equation of the plane passing through the line of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z = 8$. Then the points which lies on the plane $P = 0$ is(are)	**(r)**	$(0, 9, 17)$
(D)	The image of the point (1, 3, 4) in the plane $y + z - 6 = 0$ is(are)	**(s)**	$\left(\frac{1}{2}, 1, \frac{1}{3}\right)$
		(t)	None of these

61. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The plane $x-2y+7z+21=0$ contains the line	**(p)**	$\frac{x-3}{3}=\frac{y+5}{-4}=\frac{z-7}{5}$
(B)	An equation of the line passing through $3\hat{i}-5\hat{j}+7\hat{k}$ and perpendicular to the plane $3x-4y+5z=8$ is (λ, μ are parameters)	**(q)**	$\frac{x+1}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$
(C)	Equation of the line of shortest distance between the lines $\frac{x}{2}=\frac{y}{-3}=\frac{z}{1}$ and $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ is	**(r)**	$\frac{x-3}{-2}=\frac{y-1}{7}=\frac{z-4}{13}$
(D)	The line of intersection of the planes $\vec{r}.\left(3\hat{i}-\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}+4\hat{j}-2\hat{k}\right)=2$ is parallel to the line given by	**(s)**	$3(x-21)=3y+92=3z-32$
		(t)	None of these

62. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The coordinates of a point on the line $x=4y+5,\ z=3y-6$ at a distance 3 from the point (5, 0, –6) is(are)	**(p)**	(0, 0, 0)
(B)	The plane containing the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{4}=\frac{z-6}{7}$ passes through	**(q)**	(17, 3, 3)
(C)	A line passes through two points A(2, –3, –1) and B(8, –1, 2). The coordinates of a point on this line nearer to the origin at a distance of 14 units from A is(are)	**(r)**	(2, 5, 7)
(D)	The coordinates of the foot of the perpendicular from the point (3, –1, 11) on the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is(are)	**(s)**	(14, 1, 5)
		(t)	None of these

63. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	The D.R.s of the line which is perpendicular to lines $\frac{x-7}{2}=\frac{y+17}{-3}=\frac{z-6}{-1}$ and $\frac{x+5}{-1}=\frac{y+3}{2}=\frac{z-4}{-2}$ is	**(p)**	(0, 4, 1)
(B)	Image of (2, 1, 1); in the plane $x+y-z=3$ is	**(q)**	(8, 5, 1)
(C)	Foot of perpendicular from (0, 2, 7) to the line $\frac{x+2}{-1}=\frac{y-1}{3}=\frac{z-3}{-2}$ is	**(r)**	$\left(-\frac{3}{2},-\frac{1}{2},4\right)$
(D)	The point where the line joining (2, 1, 3) and (4, –2, 5) meets the plane $2x+y-z=3$ is	**(s)**	$\left(\frac{8}{3},\frac{5}{3},\frac{1}{3}\right)$
		(t)	None of these

64. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Let image of the line $\frac{x-1}{3}=\frac{y-3}{5}=\frac{z-4}{2}$ in the plane $2x - y + z + 3 = 0$ be L. A plane $7x + By + Cz + D = 0$ is such that it contains the line L and perpendicular to the plane $2x - y + z + 3 = 0$ then value of $\frac{B+C+D}{10}$ is	**(p)**	4
(B)	ABC is a triangle where $A(2,3,5)$; $B(-1,3,2)$ and $C(\lambda,5,\mu)$. If the median through A is equally inclined to the axes then value of $\mu-\lambda+1$ is	**(q)**	3
(C)	Length of projection of line segment joining $P(-1,2,0)$ and $Q(1,-1,2)$ on $2x-y-2z=4$ is	**(r)**	$\frac{7}{2}$
(D)	If $A(a,b,c)$ is any point on plane $3x+2y+z=7$, then the least value of $a^2+b^2+c^2$ is	**(s)**	$\frac{6}{11}$
		(t)	None of these

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

65. The shortest distance between the lines $2x + y + z = 1$, $3x + y + 2z = 2$ and $x = y = z$ is d then $1/d^2$ = _____.

66. Three lines $y–z–1 = 0$, $x = 0$; $x + z + 1 = 0$, $y = 0$; $x–z–1 = 0$, $y = 0$ intersect the xy plane at A, B and C. If the orthocentre of ΔABC is (p, q, r) then $3p + q + r$ = ____________.

67. The minimum distance of the point (1, 1, 1) from the plane $x + y + z = 1$ measured perpendicular to the line $\frac{x-x_1}{1}=\frac{y-y_1}{2}=\frac{z-z_1}{3}$ is d then $\frac{3d^2}{7}$ = ____________.

68. The maximum distance between the point $P(0, 0, 3)$ and the circle $x^2 + y^2 - 2\sqrt{5}\,x - 4y + 8 = 0;\ z = 0$ is ____________.

69. The equation of the plane which is equidistant from lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{1}=\frac{z-1}{2}$ is $Ax+By+Cz+D=0$ then $|A+B+C+D|$ = ____________.

70. Plane $2x + 3y + 4z = 5$ is rotated about the line where it cuts the xy-plane by an angle α. In the new position the plane contains the point (1, 1, 1). If the angle $\alpha=\cos^{-1}\sqrt{\frac{p}{q}}$, ($\frac{p}{q}$ is rational number in its simplest form) then $q-2p=$ ____________.

71. A variable plane is at a constant distance p from the origin and meets the axes in A, B and C. If the locus of the centroid of the tetrahedron $OABC$ is $x^{-2}+y^{-2}+z^{-2}=\lambda p^{-2}$ then the value of $\sqrt{\lambda}$ is______.

72. If the planes $x=cy+bz,\ y=az+cx,\ z=bx+ay$ pass through one line then the value of $a^2+b^2+c^2+2abc$ is________.

73. If the planes $x-y+z+1=0,\ \lambda x+3y+2z-3=0,\ 3x+\lambda y+z-2=0$ form a triangular prism then λ is _____.

74. If a, b, c be the lengths of the intercepts of the plane passing through the intersection of the planes $2x+y+2z=9,\ 4x-5y-4z=1$ and the point (3, 2, 1) on the coordinate axes, then $(5a+b+c)/2=$ _____.

75. If Q is the foot of perpendicular from the point $P(4,-5,3)$ on the line $\frac{x-5}{3}=\frac{y+2}{-4}=\frac{z-6}{5}$, then $[PQ]=$ _____. (where [.] denote greatest integer function)

76. The projection of the line $\frac{x}{2}=\frac{y-1}{2}=\frac{z-1}{1}$ on a plane P is $\frac{x}{1}=\frac{y-1}{1}=\frac{z-1}{-1}$. The plane P passes through $(k,-2,0)$ then $k=$ ______.

77. If $10y-8x-(x^2+y^2+z^2)=40,\ P_1=\max\left\{\sqrt{(x+2)^2+(y-3)^2+z^2}\right\},\ P_2=\min\left\{\sqrt{(x+2)^2+(y-3)^2+z^2}\right\}$, then P_1-P_2 is ____.

78. Let for $\lambda\in[0,\infty)$ such that $(x,\ y,\ z)\neq(0,0,0)$ and $(\hat{i}+\hat{j}+3\hat{k})x+(3\hat{i}-3\hat{j}+\hat{k})y+(4\hat{i}+5\hat{j})z=\lambda(x\hat{i}+y\hat{j}+3\hat{k})$, then the value of $\frac{x-y-z}{x}$ is equal to ______.

79. If the distance between the plane $Ax-2y+z=d$ and the plane containing the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is equal to _____.

80. If the distance of point of intersection of lines $\frac{x-4}{1}=\frac{y+3}{-4}=\frac{z+1}{7}$ and $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z+10}{8}$ from $(1,-4,7)$ is α, then $\frac{\alpha^2}{13}$ is equal to ______.

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Entries of a 2×2 determinant are chosen from the set $\{-1, 1\}$. The probability that determinant has zero value is :

(A) 1/4 **(B)** 1/3 **(C)** 1/2 **(D)** None of these

2. A number is chosen at random from the numbers 10 to 99. By seeing the number a man will laugh if product of the digits is 12. If he choose three numbers with replacement then the probability that he will laugh at least once is :

(A) $1-\left(\frac{3}{5}\right)^3$ **(B)** $\left(\frac{43}{45}\right)^3$ **(C)** $1-\left(\frac{4}{25}\right)^3$ **(D)** $1-\left(\frac{43}{45}\right)^3$

★3. 8n players $P_1, P_2, P_3.......P_{8n}$ play a knock out tournament. It is known that all the players are of equal strength. The tournament is held in 3 rounds where the players are paired at random in each round. If it is given that P_1 wins in the third round. The probability that P_2 looses in the second round is :

(A) $\frac{n}{8n-1}$ **(B)** $\frac{n}{8n+1}$ **(C)** $\frac{2n}{4n-1}$ **(D)** None of these

4. Two subsets A and B of a set containing n elements are chosen at random. The probability that $A \subseteq B$ is :

(A) $\frac{1}{2}$ **(B)** $\frac{2^n}{n!}$ **(C)** $\left(\frac{2}{3}\right)^n$ **(D)** $\left(\frac{3}{4}\right)^n$

5. Three different dice are rolled three times. The Probability that they show different numbers only two times is :

(A) $\frac{1}{3}$ **(B)** $\left(\frac{{}^6P_3}{6^3}\right)^2$ **(C)** $\frac{107}{(54)^3}$ **(D)** $\frac{100}{243}$

★6. 2^n players of equal strength are playing a knock out tournament. If they are paired randomly in all rounds, the probability that out of two particular players S_1 and S_2 exactly one will reach in semi final is $(n \in N, n \geq 2)$:

(A) $\frac{8 \times (2^n - 4)}{2^n(2^n - 1)}$ **(B)** $\frac{(2^n - 4)}{2^n(2^n - 1)}$ **(C)** $\frac{(2^{n-1} - 4)}{2^n(2^n - 1)}$ **(D)** None of these

7. A speaks truth in 60% cases and B speaks truth in 70% cases. The probability that they will say the same thing while describing single event is :

(A) 0.56 **(B)** 0.54 **(C)** 0.38 **(D)** 0.94

★8. Three smallest squares are chosen randomly on a chess board the probability that these squares have exactly two corners, but no side common is :

(A) $\frac{80}{{}^{64}C_3}$ **(B)** $\frac{72}{{}^{64}C_3}$ **(C)** $\frac{6^3}{{}^{64}C_3}$ **(D)** None of these

★9. Four die are thrown simultaneously. The probability that 4 and 3 appear on two of the die given that 5 and 6 have appeared on other two die is :

(A) 1/6 **(B)** 1/36 **(C)** 12/151 **(D)** None of these

★10. A fair coin is tossed 5 times then the probability that no two consecutive heads occur, is :

(A) $\frac{11}{32}$ **(B)** $\frac{15}{32}$ **(C)** $\frac{13}{32}$ **(D)** None of these

★**11.** The probability that a randomly chosen 3 digit number has exactly 3 factors :

(A) $\frac{2}{225}$ **(B)** $\frac{7}{900}$ **(C)** $\frac{1}{800}$ **(D)** None of these

12. Out of 6 pairs of distinct gloves 8 gloves are randomly selected, then the probability that there exist exactly 2 pairs in it is :

(A) $\frac{16}{33}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{12}{33}$ **(D)** $\frac{10}{33}$

13. A cube painted red on all sides is cut into 125 equal small cubes. A small cube when picked is found to show red color on one of its faces. The probability that two more faces also show red color is :

(A) $\frac{4}{75}$ **(B)** $\frac{4}{49}$ **(C)** $\frac{1}{8}$ **(D)** $\frac{8}{125}$

★**14.** 2n balls (all distinct in size) are arranged in a row. First few of these balls are black rest all white, both odd in number. The probability that there is exactly, one black ball in one of all possible arrangements is:

(A) $\frac{n}{2^{2n-1}}$ **(B)** $\frac{n}{2^{n-1}}$ **(C)** $\frac{n}{2^{2n-2}}$ **(D)** $\frac{n}{2^{n-2}}$

15. Each of 10 passengers board any of the three buses randomly which had no passenger initially. The probability that each bus has got at least one passenger is :

(A) $1-\frac{2^{10}}{3^{10}}$ **(B)** $1-\frac{{}^{10}C_3\times 3^7}{3^{10}}$ **(C)** $\frac{{}^{10}P_3 3^7}{3^{10}}$ **(D)** $\frac{3^{10}-3.2^{10}+3}{3^{10}}$

★**16.** *A* and *B* play a game of tennis. The situation of the game is as follows; if one scores two consecutive points after a deuce he wins; if loss of a point is followed by win of a point, it is deuce. The chance of a server to win a point is 2/3. The game is at deuce and *A* is serving. Probability that *A* will win the match is : (Serves are changed after each game)

(A) 3/5 **(B)** 2/5 **(C)** 1/2 **(D)** 4/5

17. Six different balls are put in three different boxes, no box being empty. The probability of putting balls in the boxes in equal numbers is :

(A) 3/10 **(B)** 1/6 **(C)** 1/5 **(D)** None of these

★**18.** The probability so that '*r*' 1 × 1 squares which are selected from a m × n chess board such that no two of them share the same row or same column is :

(A) $\frac{{}^mC_r\,{}^nC_r}{{}^{mn}C_r}$ **(B)** $\frac{{}^mC_r\,{}^nC_{r-1}}{{}^{mn}C_r}$ **(C)** $\frac{r\,{}^mC_r\,{}^nC_r}{{}^{mn}C_r}$ **(D)** None of these

19. A business man is expecting two telephone calls. Mr Walia may call any time between 2 p.m and 4 p.m. while Mr Sharma is equally likely to call any time between 2.30 p.m. and 3.15 p.m. The probability that Mr Walia calls before Mr Sharma is :

(A) 1/18 **(B)** 1/6 **(C)** 1/16 **(D)** None of these

20. A student appears for test I, II and III. The student is successful if he passes either in test I, II or I, III. The probability of the student passing in test I, II and III are respectively *p*, *q* and 1/2. If the probability of the student to be successful is 1/2 then :

(A) $p=q=1$ **(B)** $p=q=\frac{1}{2}$ **(C)** $p=1, q=0$ **(D)** $p=1, q=\frac{1}{2}$

21. Three of six vertices of a regular hexagon are chosen at random. The probability that the triangle with three vertices is equilateral equal to :

(A) 1/2 **(B)** 1/5 **(C)** 1/10 **(D)** 1/20

★**22.** A machine containing n different balls, when switched on, can throw up any number of balls one by one. The probability of throwing r balls is directly proportional to r. Given that a particular ball is the first ball to pop up, the probability that machine has thrown up all the balls is :

(A) $\frac{2}{n+1}$ **(B)** $\frac{1}{n+1}$ **(C)** $\frac{2}{n}$ **(D)** None of these

★**23.** From $4m + 1$ tickets numbered as 1, 2, … $4m + 1$. Three tickets are chosen at random. The probability that the numbers are in A.P. with even common difference is

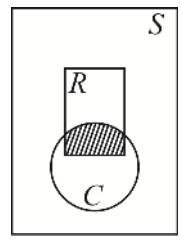
(A) $\frac{2(2m-1)}{3(16m^2-1)}$ **(B)** $\frac{3(2m+1)}{2(16m^2+1)}$ **(C)** $\frac{3(2m-1)}{2(4m^2-1)}$ **(D)** None of these

24. Let A and B be two events such that $P(A \cap B') = 0.20$, $P(A' \cap B) = 0.15$, $P(A' \cap B') = 0.1$, then $p(A/B)$ is equal to,

(A) 11/14 **(B)** 2/11 **(C)** 2/7 **(D)** 1/7

Paragraph for Questions 25 - 28

Consider a random experiment in which the outcomes cannot be identified discretely. Then the sample space of such an experiment will not contain distinguishable elements. An example of such a sample space can be an interval in the set of real numbers. Consider the following experiment:



Let your pen drop, tip downwards, into one of the pages of your notebook and note the point on the paper that the pen first touches. Here the sample space S consists of all the points on the paper. Let R and C be the events that the pencil drops into a rectangle and a circle as illustrated in the adjacent figure.

Clearly the sample space S and event sets R and C contain a uniform distribution of points. We consider only those sample spaces which have some finite geometrical measurement such as length area, and in which a point is selected at random. The probability of an event R, i.e. the selected point belongs to R will be given by $P(R) = \frac{\text{area of } R}{\text{area of } S}$.

Such a probability space is said to be uniform or continuous.

25. A point is selected at random inside a circle. The probability that the point is closer to the centre of the circle than to its circumference :

(A) 1/2 **(B)** 1/3 **(C)** 1/4 **(D)** $1/\sqrt{2}$

26. A point is selected at random inside an equilateral triangle whose side length is 3. The probability its distance to any corner is greater that 1 is

(A) $\frac{2\pi}{9\sqrt{3}}$ **(B)** $1-\frac{2\pi}{9\sqrt{3}}$ **(C)** $\frac{\sqrt{3}\pi}{9}$ **(D)** $1-\frac{\sqrt{3}\pi}{9}$

27. A point X is selected at random from a line segment AB with mid point O. The probability that the line segments AX, XB and AO can form a triangle is :

(A) 1/2 **(B)** 1/3 **(C)** 1/4 **(D)** 3/4

★**28.** A coin of diameter 1/2 is tossed randomly onto the rectangular cartesian plane. The probability that the coin does not intersect any line whose equation is of the form $x = k$, or $y = k, k$ is integer, is :

(A) 1/2 **(B)** 1/3 **(C)** 1/4 **(D)** 2/3

Paragraph for Questions 29 - 31

A player 'A' plays a game against a machine. At reach round he deposits one rupee in a slot and then flips a coin which has a probability p of showing a head. If a head shows, he gets back the rupee he deposited and one more rupee from the machine. If a tail shows, he loses his rupee. Let A starts with 10 rupee and $q = 1 - p$

29. The probability that he will be drained out of his rupee exactly at the eleventh round is

(A) q^{11} **(B)** $1-p^{11}$ **(C)** $pq^{10}+q^{11}$ **(D)** 0

30. The probability that all his money will be finished exactly at the twelfth round is :

(A) q^{12} **(B)** $1-p^{12}$ **(C)** ${}^{10}C_1\, pq^{11}$ **(D)** None of these

31. The probability that he is left with no money by the 14th round or earlier is :

(A) $q^{10}(1+10pq+65p^2q^2)$ **(B)** $q^{14}(p^2q+36pq+7)$

(C) $q^{12}+3pq^{13}+3p^{13}q+p^{12}$ **(D)** $1-{}^{10}C_1\, pq^{11}-{}^{10}C_2 p^2q^{12}$

Paragraph for Questions 32 - 34

Let B_n denotes the event that n fair dice are rolled once with $P(B_n)=\frac{1}{2^n}, n\in N$, e.g. $P(B_1)=\frac{1}{2},\ldots,P(B_n)=\frac{1}{2^n}$.

Hence $B_1,B_2,B_3,\ldots,B_n$ are pairwise mutually exclusive events as $n\to\infty$. The event A occurs with atleast one of the event $B_1,B_2,B_3,\ldots,B_n$ and denotes that the numbers appearing on the dice is S.

32. If even number of dice has been rolled, the probability that $S = 4$, is :

(A) very closed to $\frac{1}{2}$ **(B)** very closed to $\frac{1}{4}$ **(C)** very closed to $\frac{1}{16}$ **(D)** very closed to $\frac{1}{32}$

33. Probability that greatest number on the dice is 4 if three dice are known to have been rolled, is :

(A) $\frac{37}{216}$ **(B)** $\frac{64}{216}$ **(C)** $\frac{27}{216}$ **(D)** $\frac{31}{216}$

34. If $S=3$, $P(B_2/S)$ has the value equal to :

(A) $\frac{8}{169}$ **(B)** $\frac{24}{169}$ **(C)** $\frac{72}{169}$ **(D)** $\frac{16}{169}$

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

35. Suppose X is a random variable which takes values 0, 1, 2, 3, …… and $P(X=r)=pq^r$. Where $0<p<1, q=1-p$ and $r=0,1,2,3$ ……. Then :

(A) $P(X\ge a)=q^a$ **(B)** $P(X\ge a+b\,|\,X\ge a)=P(X\ge b)$

(C) $P(X=a+b\,|\,X\ge a)=P(X=b)$ **(D)** $P(X\ge a+b\,|\,X\ge b)=P(X\ge a)$

36. Consider the cartesian plane R^2 and let X denotes the subset of points for which both coordinates are integers. A coin of diameter 1/2 is tossed randomly into the plane. The probability p that the coin covers a point of X satisfies:

(A) $p=\frac{\pi}{16}$ **(B)** $p<\frac{\pi}{3}$ **(C)** $p>\frac{\pi}{30}$ **(D)** $p=\frac{1}{4}$

37. A square in inscribed in a circle. If p_1 is the probability that a randomly chosen point of the circle lies within the square and p_2 is the probability that the point lies outside the square, then :

(A) $p_1=p_2$ **(B)** $p_1>p_2$ **(C)** $p_1<p_2$ **(D)** $p_1^2-p_2^2<1/3$

38. If $\frac{1+4p}{4}$, $\frac{1-p}{3}$ and $\frac{1-2p}{2}$ are the probabilities of three mutually exclusive events then p may be :

(A) 1/8 **(B)** 1/2 **(C)** 3/8 **(D)** 5/8

★39. Players $P_1, P_2, P_3, \ldots, P_m$, of equal skill, play a game consecutively in pairs as $P_1P_2, P_2P_3, P_3P_4, \ldots\ P_{m-1}P_m, P_mP_1, \ldots$, and any player who wins two consecutive games (i.e k and $(k+1)$th game) wins the match. If the chance that the match is won at the rth game is k then :

(A) $k=\frac{1}{8}, \text{if } r=5$ **(B)** $k=\frac{5}{32}, \text{if } r=5$ **(C)** $k=\frac{3}{32}, \text{if } r=6$ **(D)** $k=\frac{5}{64}, \text{if } r=6$

★40. Two persons A, and B, have respectively $n+1$ and n coins, which they toss simultaneously. Then probability P that A will have more heads then B belongs :

(A) $\frac{1}{4}<P<\frac{3}{4}$ **(B)** $\frac{1}{2}<P<\frac{3}{4}$ **(C)** $\frac{1}{4}<P<\frac{1}{2}$ **(D)** $\frac{1}{3}<P<\frac{3}{4}$

41. If A and B are two events such that $P(\boldsymbol{A}) = 1/2$ and $P(\boldsymbol{B}) = 2/3$, then :

(A) $P(A\cup B) \geq 2/3$ **(B)** $P(A\cap B') \leq 1/3$

(C) $1/6 \leq P(A\cap B) \leq 1/2$ **(D)** $1/6 \leq P(A'\cap B) \leq 1/2$

42. A bag contains n (white and black) balls. It is given that the probability that the bag contains exactly r white balls is directly proportional to r $(0 \leq r \leq n)$. A ball is drawn at random and is found to be white. The probability that there is only one white ball in the bag is p then :

(A) $p=\frac{1}{55}$ if $n=5$ **(B)** $p=\frac{6}{55}$ if $n=5$ **(C)** $p=\frac{1}{91}$ if $n=6$ **(D)** $p=\frac{13}{93}$ if $n=6$

43. Let X be a set containing n elements. If two subsets A and B of X are picked at random, the probability that A and B have the same number of elements is :

(A) $\frac{^{2n}C_n}{2^{2n}}$ **(B)** $\frac{1}{^{2n}C_n}$ **(C)** $\frac{1.3.5......(2n-1)}{2^n.(n!)}$ **(D)** $\frac{3^n}{4^n}$

44. A parent particle can be divided into 0, 1 or 2 particles with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ respectively, after splitting. Beginning with one particle, the progenitor, let us denote by X_i, the number of particles in the i generation. Then :

(A) $P(X_2>0)=\frac{39}{64}$ **(B)** $P(X_2>0)=\frac{37}{64}$ **(C)** $P\left(\frac{(X_1=2)}{(X_2=1)}\right)=\frac{1}{5}$ **(D)** $P\left(\frac{(X_1=2)}{(X_2=1)}\right)=\frac{3}{64}$

45. If A and B are two events such that $P(A)=\frac{3}{4}$ and $P(B)=\frac{5}{8}$ then:

(A) $P(A\cup B)\geq\frac{3}{4}$ **(B)** $P(A'\cap B)\leq\frac{1}{4}$ **(C)** $\frac{3}{8}\leq P(A\cap B)\leq\frac{5}{8}$ **(D)** $\frac{1}{8}\leq P(A\cap B')\leq\frac{3}{8}$

46. A student has to match historical levents viz., Dandi march, Quit India Movement and Mahatma Gandhi's assassination with the years 1948, 1930 and 1942. The student has no knowledge of the correct answer decides to match the events and years randomly. Let $E_i\ (0\leq i\leq 3)$ denote the event that the student gets exactly i correct answers, then which of the following is/are NOT correct?

(A) $P(E_0)+P(E_3)=P(E_1)$ **(B)** $P(E_0).P(E_1)=P(E_3)$

(C) $P(E_0\cap E_1)=P(E_2)$ **(D)** $P(E_0)+P(E_1)+P(E_3)=1$

47. A bag contains 20 blue marbles, 12 red marbles and some other number of green marbles. If the probability of drawing green marble in one try is $\frac{1}{y}$ then which of the following statements is/are correct?

(A) Number of possible integral values of y is 5 **(B)** Number of possible integral values of y is 6
(C) Sum of possible integral values of y is 69 **(D)** Sum of possible integral values of y is 96

48. A dice is rolled three times. Let E_1 denote the event of getting a number larger than the previous number each time and E_2 denote the event that the numbers (in order) form an increasing AP then:

(A) $P(E_1) \geq P(E_2)$ **(B)** $P(E_2 \cap E_1) = \frac{1}{36}$ **(C)** $P(E_2 / E_1) = \frac{3}{10}$ **(D)** $P(E_1) = \frac{10}{3}P(E_2)$

49. One card is missing from a pack of cards. Let A be the event that missing card is a spade. Then two cards are drawn, and S be the event that they are spades then:

(A) $P(A') = \frac{3}{4}$ **(B)** $P(S/A) = \frac{{}^{13}C_2}{{}^{51}C_2}$ **(C)** $P(A/S) = \frac{11}{50}$ **(D)** $P(A) = P(A/S)$

50. A square is inscribed in a circle. If p_1 is the probability that a randomly chosen point of the circle lies within the square and p_2 is the probability that the point lies outside the square, then:

(A) $p_1 = p_2$ **(B)** $p_1 > p_2$ **(C)** $p_1 < p_2$ **(D)** $p_1^2 - p_2^2 < \frac{1}{3}$

51. A student appears for tests I, II, and III. The student is successful if he passes either in tests I and II or tests I and III. The probabilities of the student passing in tests I, II, III are p, q and $\frac{1}{2}$, respectively. If the probability that the student is successful is $\frac{1}{2}$, then: (Assuming his performance in tests are independent).

(A) $p = 1, q = 0$ **(B)** $p = 2/3, q = 1/2$
(C) $p = 3/5, q = 2/3$ **(D)** There are infinitely many values of p and q

52. The letters of the word PROBABILITY are written down at random in a row. Let E_1 denote the event that two I, s are together and E_2 denote the event that two B's are together, then:

(A) $P(E_1) = P(E_2) = \frac{3}{11}$ **(B)** $P(E_1 \cap E_2) = \frac{2}{55}$ **(C)** $P(E_1 \cup E_2) = \frac{18}{55}$ **(D)** $P(E_1 / E_2) = \frac{1}{5}$

53. Let a, b, c be three integers such that $a^2 + b^2 + c^2 = 2$. Then for the system of simultaneous equations $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, where $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$, which of the following statements is/are true?

(A) The probability that the system of equations has unique solution is 1/2
(B) The number of triplets (a, b, c) for which of the system of equations has infinitely many solutions is 6
(C) If $a = 0$, the number of ordered pairs (b, c) for which the system of equations has no solution is 2
(D) The number of elements in the range of $ab + bc + ca$ is 2

54. Sixteen players S_1, S_2, …., S_{16} play in a tournament. They are divided into eight pairs at random. From each pair, a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength then:

(A) The probability that the player S_1 is among the eight winners is 1/2

(B) The probability that the player S_1 is among the eight winners is 1/4

(C) The probability that exactly one of the two players S_1 and S_2 is among the eight winners is 8/15

(D) The probability that exactly one of the two players S_1 and S_2 is among the eight winners is 7/15

55. *A* and *B* are two independent events. The probability that both *A* and *B* occur is 1/6 and the probability that neither of them occurs is 1/3. Then the probability of the occurrence of *A* may be:

(A) 1/2 **(B)** 1/3 **(C)** 1/4 **(D)** 2/3

56. The probabilities of events $A \cap B$, A, B and $A \cup B$ are respectively in A.P. with second term equal to the common difference. Therefore, *A* and *B* are:

(A) Mutually exclusive **(B)** Independent

(C) Such that one of them must occur **(D)** Such that one is twice as likely as the other

57. 5 players of equal strength play one each with each other. *P*(*A*) = probability that at least one player wins all matches he(they) play. *P*(*B*) = probability that at least one player loses all his (their) matches.

(A) $P(A)=\frac{5}{16}$ **(B)** $P(B)=\frac{7}{16}$ **(C)** $P(A \cap B)=\frac{5}{32}$ **(D)** $P(A \cup B)=\frac{15}{32}$

58. If *A* and *B* are exhaustive events in a sample space such that probabilities of the events $A \cap B$, A, B and $A \cup B$ are in A.P. If $P(A)=K$, where $0<K\le 1$, then:

(A) $P(B)=\frac{K+1}{2}$ **(B)** $P(A\cap B)=\frac{3K-1}{2}$ **(C)** $P(A\cup B)=1$ **(D)** $P(A'\cup B')=\frac{3(1-K)}{2}$

59. A boy has a collection of the blue and green marbles. The number of blue marbles belong to the set {2, 3, 4, …,13}. If two marbles are chosen simultaneously and random from his collection, then the probability that they have different colours is 1/2. Possible number of blue marbles is:

(A) 3 **(B)** 6 **(C)** 10 **(D)** 12

60. A fair coin is tossed *n* times. Let *X* denote the number of times head occurs. If $P(X=4), P(X=5)$ and $P(X=6)$ are in arithmetic progression, then the value of *n* can be:

(A) 14 **(B)** 12 **(C)** 10 **(D)** 7

61. If A_1, A_2, …., A_n be any events of the same sample space then:

(A) $\sum_{i=1}^{n} P(A_i)=1$ **(B)** $\sum P(A_i)\le 1$ if $A_1, A_2, ..., A_n$ are disjoint

(C) $\sum P(A_i)\ge 1$ if $A_1, A_2, ..., A_n$ are exhaustive events

(D) None of these

62. A family has three children. Event '*A*' is that family has at most one boy, Event '*B*' is that family has at least one boy and one girl, Event '*C*' is that the family has at most one girl. Then:

(A) Events '*A*' and '*B*' are independent **(B)** Events '*A*' and '*B*' are not independent

(C) Events *A*, *B*, *C* are not independent **(D)** Events *A*, *B*, *C* are not independent

63. A certain coin is tossed with probability of showing head being '*p*' Let '*q*' denote the probability that when the coin is tossed four times the number of heads obtained is even. Then:

(A) There is no value of *p*, if $q=\frac{1}{4}$ **(B)** There is exactly one value of *p* if $q=\frac{3}{4}$

(C) There are exactly two values of *p* if $q=\frac{3}{5}$ **(D)** There are exactly four values of *p* if $q=\frac{4}{5}$

64. A bag contains four tickets marked with numbers 112, 121, 211, and 222. One ticket is drawn at random from the bag. Let $E_i\left(i=1, 2, 3\right)$ denote the event that i^{th} digit on the ticket is 2. Then:

(A) E_1 and E_2 are independent **(B)** E_2 and E_3 are independent

(C) E_3 and E_1 are independent **(D)** E_1, E_2, E_3 are independent

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

65. Observe the following columns:

	Column 1		Column 2
(A)	The probability that A, B, C solve a problem is $\frac{1}{2},\frac{1}{3}$ and $\frac{1}{4}$. If the probability that the problem will be solved is λ and that the problem is solved by only one of them is μ, then	**(p)**	$\lambda+\mu=\frac{13}{24}$
(B)	The probability of hitting a target by three men is $\frac{1}{2},\frac{1}{3}$ and $\frac{1}{4}$ respectively. If the probability that exactly two of them will hit the target is λ and that at least two of them hit the target is μ, then	**(q)**	$\lambda+\mu=\frac{29}{24}$
(C)	A bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. One ball is drawn from each bag. If the probability that both are black is λ and that both are white is μ, then	**(r)**	$\lambda+\mu=\frac{11}{24}$
		(s)	$\lambda-\mu=7/24$
		(t)	$\mu-\lambda=1/24$

66. **MATCH THE FOLLOWING :**

	Column 1		Column 2
(A)	A pack of cards contain 51 cards. Cards are drawn from the pack without replacement. If 1^{st} 13 cards drawn are all red, then the probability that the missing card is black	**(p)**	$\frac{1}{2}$
(B)	A pack of cards is counted with face downwards. It is found that one card is missing. One card is drawn and is found to be red. The probability that the missing card is red.	**(q)**	$\frac{25}{51}$
(C)	A box has 2 white, 4 black and 6 green balls. Person A, draws a ball from it. Then from the remaining balls person B draws two balls which are found to be green. The probability that A has drawn a black ball.	**(r)**	$\frac{8}{9}$
(D)	Let p, q be chosen one by one from the set $\{1, \sqrt{2}, \sqrt{3}, 2, e, \pi\}$ with replacement. Now a circle is drawn taking (p, q) as its centre. The probability that at the most two rational points exist on the circle. (Rational points are those points whose both the coordinates are rational).	**(s)**	$\frac{2}{5}$
		(t)	None of these

★67. In a tournament there are twelve players $S_1, S_2, ..., S_{12}$ and divided into six pairs at random. From each game a winner is decided on the basis of a game played between the two players of the pair. Assuming all the pairs are of equal strength, then match the following.

	Column 1		**Column 2**
(A)	Probability that S_2 is among the losers is	**(p)**	5/22
(B)	Probability that exactly one S_3 and S_4 is among the losers, is	**(q)**	10/11
(C)	Probability that both S_2 and S_4 are among the winners is	**(r)**	1/2
(D)	Probability of S_4 and S_5 not playing against each other is	**(s)**	6/11
		(t)	None of these

★68. 'n' whole numbers are randomly chosen and multiplied:

	Column 1		**Column 2**
(A)	The probability that the last digit is 1, 3, 7 or 9 is	**(p)**	$\frac{8^n - 4^n}{10^n}$
(B)	The probability that the last digit is 2, 4, 6, 8 is	**(q)**	$\frac{5^n - 4^n}{10^n}$
(C)	The probability that last digit is 5 is	**(r)**	$\frac{4^n}{10^n}$
(D)	The probability that the last digit is zero is	**(s)**	$\frac{10^n - 8^n - 5^n + 4^n}{10^n}$
		(t)	None of these

69. A bag contains some white and some black balls, all combinations being equally likely. The total number of balls in the bag is 12. Four balls are drawn at random without replacement. Match the entries in the following two lists.

	Column 1		**Column 2**
(A)	The probability that all the four balls are black is equal to	**(p)**	$\frac{14}{33}$
(B)	If the bag contains 10 black and 2 white balls, then the probability that all four balls are black is equal to	**(q)**	$\frac{1}{5}$
(C)	If all the four balls are black, then the probability that the bag contains 10 black balls is equal to	**(r)**	$\frac{70}{429}$
(D)	The probability that two balls are black and two are white is equal to	**(s)**	$\frac{13}{165}$
		(t)	None of these

SUBJECTIVE INTEGER TYPE

Each of the following question has an integer answer between 0 and 9.

*70. Two integer 'a' and 'b' are randomly selected from the set {1, 2, $\cdots$} (with replacement) then if the probability of $\frac{1}{5}\left(a^2+b^2\right)$ being positive integer is $\frac{p}{q}$ (where $H.C.F$ (p, q) = 1) then $q-2p=$ ....

71. If a and b are chosen randomly from the set consisting of numbers 1, 2, 3, 4, 5, 6 with replacement. If the probability that $\underset{x\to 0}{Lt}\left(\frac{a^x+b^x}{2}\right)^{2/x}=6$ is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-p=$ ....

72. A bag contained 3 maths book and 2 physics books. A book is drawn at random if it is of math, 2 more books of maths together with this book put back in the bag and if it is of physics it is not replaced in the bag. This experiment is repeated 3 time. If third draw gives math book, The probability that first two drawn books were of physics is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-8(p+1)=$ __________.

73. Two friends decide to meet at a spot between 2 p.m. and 3 p.m. whosoever arrives first agrees to wait for 15 minutes for the other. The probability that they meet is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-p=$ _______.

74. There are two bags, bag I contains 4 red and 5 white balls, while bag II contains 5 red and 4 white balls. Two balls are drawn from bag I. If they are of the same colour, another ball is drawn from bag I. If the first two drawn balls are of different colours, one ball is drawn from bag II. If the third drawn ball is red, then the probability that the first two drawn balls were white is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-3(p+1)=$ ________.

75. There are two purses. The first contains 9 fifty paise coins and a one-rupee coin, while the second purse has 10 fifty-paise coins. Nine coins are transferred from the first purse to the second randomly. Then nine coins are transferred from the second purse to the first randomly. The probability of finding a one rupee coin in the first purse after these transfers is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-p=$ _________.

*76. The probability that two queens, placed at random on a chess board, do not take on each other is $\frac{p}{q}$ (where H.C.F (p, q) = 1) then $q-p-5=$ __________.

*77. In an organization number of women are μ times that of men. If α things are to be distributed among them than the probability that the number of things received by men are odd is $\left(\frac{1}{2}-\left(\frac{1}{2}\right)^{\alpha+1}\right)$. Then $\mu=$ _________.

78. Six fair dice are thrown independently. The probability that there are exactly 2 different pairs (A pair is an ordered combination like 2, 2, 1, 3, 5, 6) is p, then $4p$ is______.

79. Raj and Sanchita are playing game in which they throw a die alternately till one of them gets a six. The probability that Sanchita win the game is p then the value of $5p$?

80. Two dice are thrown simultaneously what is the probability that sum of the two numbers will be 5 before 7?

81. Rajesh doesn't like to study. Probability that he will study is 1/3. If he studied, then probability that he will fail is 1/2 and if he didn't study then probability that he will fail is 3/4. If in result Rajesh failed, then what is the probability that he didn't study.

82. On a rod of length 6 units, lengths 1, 2 units are measured at random, the probability that no points of the measured lines will coincide is______.

83. In a board exam there are two sections each section has 5 questions. As per the given condition a candidate has to answer any 6 questions out of of 10 questions. What is the probability that a student answered 6 questions such that not more than 4 questions selected from one section?

84. Assume that the birth of a boy or girl to a couple to be equally likely, and exhaustive. For a couple having 6 children, the probability that their "three oldest are boys" is p then the value of $10p$.

85. Two fair dice are thrown till outcome is 12. The probability that one has to do 20 throws for this is______.

86. The probability of randomly drawing five cards from a deck that has exactly one Ace is______.

87. An ordinary deck of 52 playing cards is randomly divided into 4 groups of 13 cards each. The probability that each group has exactly 1 jack?

88. In a poker game, the probability of getting a "pair" in five cards is___ [A pair consists of 2 cards of the same kind (eg, 2 kings) and 3 cards that are different from the kind of the pair (eg, different from kings) and that are all different from each other.]

89. Letters of the word MATHEMATICS are arranged in all the possible ways, and a word is selected randomly then the probability that letter C is exactly between S and H is______.

90. A group of students comprising 3 girls and 5 boys went for a picnic. During a game they were arranged in a circle then the probability that each boy has one girl on at least one side is______.

91. The probabilities that a student passes in Mathematics, physics and chemistry are m, p and c, respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in exactly two. Then $p + m + c$ is______.

92. Eight players $P_1, P_2, \ldots P_8$ play a knock out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if $i < j$. Assuming that the players are paired at random in each round, then the probability that the player P_4 reaches the final is______.

93. If A and B are two events such that $P(A) = 0.3$, $P(B) = 0.25$, $P(A \cap B) = 0.2$, then $10P\left(\left(\frac{A^C}{B^C}\right)^C\right)$ is equal to______.

94. A number is selected at random from the first twenty-five natural numbers. If it is a composite number, then it is divided by 5. But if it is not a composite number, it is divided by 2. The probability that there will be no remainder in the division is______.

95. There are three bags each containing 5 white balls and 2 black balls and 2 bags each containing 1 white ball and 4 black balls: a black ball having been drawn, the probability that it came from the first group is______.

96. A speaks truth 3 times out of of 4, and B 7 times out of 10, they both assert that a white ball has been drawn from a bag containing 6 balls all of different colours; the probability of truth of the assertion is______.

97. A bag contains 10 different balls. Five balls are drawn simultaneously and then replaced and then seven balls are drawn. The probability that exactly three balls are common to the two drawn is p then the value of $10p$ is______.

Advanced Problem Package	Matrices & Determinants

SINGLE CORRECT ANSWER TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. If matrix $A=\begin{pmatrix}1 & -1\\ 1 & -2\end{pmatrix}$, satisfies $A^n=5I-8A$, then $n=$

(A) 4 **(B)** 5 **(C)** 6 **(D)** 7

2. Let $A=[a_{rs}]_{n\times n}$ be a $n\times n$ matrix such that $a_{rs}=(r-s)2^{i(r-s)}$ where $i=\sqrt{-1}$, then $A=$ ($\bar{A}$ denotes $[\overline{a_{rs}}]$ complex conjugate)

(A) $\bar{A}]$ **(B)** $-\bar{A}$ **(C)** $(\bar{A})^T$ **(D)** $-(\bar{A})^T$

3. A_1 is a matrix formed by replacing all the elements in $A_{n\times n}$ by corresponding cofactors, A_2 is matrix formed by replacing all elements of A_1 by corresponding cofactors and A_3, A_4 … formed so on, if $|A_n|=K$, then $|A|=$

(A) K^{n^n} **(B)** $K^{(n-1)^n}$ **(C)** $K^{\frac{1}{(n-1)^n}}$ **(D)** $\frac{1}{K^{(n-1)^n}}$

4. If A is an idempotent matrix i.e. $A^2=A$, and $B=I-A$, then which of the following is incorrect:

(A) $B^2=B$ **(B)** $B^2=I$ **(C)** $AB=0$ **(D)** $BA=0$

5. $A=[a_{ij}]_{n\times n}$ be a square matrix, n is odd such that $a_{ij}=(-1)^i\ {}^nC_i\times{}^nC_j$, then trace $(A)=$ (Trace (A) denotes sum of diagonal elements of A)

(A) 0 **(B)** 1 **(C)** –1 **(D)** 2

6. Let $A=[a_{ij}]_{3\times3}, B=[b_{ij}]_{3\times3}$ where $b_{ij}=3^{i-j}a_{ij}$, $C=[c_{ij}]_{3\times3}$, where $c_{ij}=4^{i-j}b_{ij}$ be any three matrices. If $|A|=4$, then $|B|+|C|=$ ($|X|$ denotes determinant of matrix X)

(A) 8 **(B)** 48 **(C)** 192 **(D)** 4

7. If $f(x)=\begin{vmatrix}x & 1+x^2 & x^3\\ \ln(1+x^2) & e^x & \sin x\\ \cos x & \tan x & \sin^2 x\end{vmatrix}$, then :

(A) $f(x)$ is divisible by x **(B)** $f(x)=0$

(C) $f'(x)=0$ **(D)** $\lim\limits_{x\to0}\frac{f(x)}{x}=1$

8. System of equations $ax+4y+z=0$, $2y+3z=1, 3x-bz=-2$ then which of the following is not true

(A) Unique solution if $ab\neq15$ **(B)** Infinitely many solutions if $a=3, b=5$

(C) No solution if $ab=15, a\neq3$ **(D)** No solution if $ab=15, a\neq5$

9. If $1, \omega, \omega^2$ are cube roots of unity then system of equations : $x+2\omega y+3\omega^2 z=1-\omega^2$, $2x+3\omega y+\omega^2 z=\omega^2-\omega$

$3x+\omega y+2\omega^2 z=\omega-1$ has

(A) unique solution **(B)** infinitely many solutions

(C) no solution **(D)** exactly 3 different solutions

10. Let S be the set of all 3×3 symmetric matrices whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0. A matrix is selected from set S, what is the probability that the selected matrix is non singular

(A) 1/4 **(B)** 1/3 **(C)** 1/2 **(D)** 2/3

11. Let $f(x)=\begin{vmatrix} x\cos x & 2x\sin x & x\tan x \\ 1 & x & 1 \\ 1 & 2x & 1 \end{vmatrix}$, then $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}=$

(A) 0 **(B)** 1 **(C)** –1 **(D)** Does not exist

12. If a,b,c are the roots of the equation $x^3+2x^2+1=0$, then $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=$

(A) 8 **(B)** –8 **(C)** 0 **(D)** 2

13. Matrix A is such that $A^2=2A-I$, where I is identity matrix, then for $n\ge 2, A^n=$

(A) $nA-(n-1)I$ **(B)** $nA-I$ **(C)** $2^{n-1}A-(n-1)I$ **(D)** $2^{n-1}A-I$

14. The system of homogeneous equations $\lambda x+(\lambda+1)y+(\lambda-1)z=0$, $(\lambda+1)x+\lambda y+(\lambda+2)z=0$,

$(\lambda-1)x+(\lambda+2)y+\lambda z=0$ Has non trivial solution for :

(A) Exactly 3 real values of λ **(B)** Exactly 2 real values of λ

(C) Exactly 1 real value of λ **(D)** Infinitely many real values of λ

15. For any real values of X,Y,Z,L,M,N value of $\begin{vmatrix} \cos(X-L) & \cos(X-M) & \cos(X-N) \\ \cos(Y-L) & \cos(Y-M) & \cos(Y-N) \\ \cos(Z-L) & \cos(Z-M) & \cos(Z-N) \end{vmatrix}=$

(A) 0 **(B)** 1 **(C)** $\cos X\cos Y\cos Z+\cos L\cos M\cos N$

(D) $(\cos X-\cos Y)(\cos Y-\cos Z)(\cos Z-\cos X)(\cos L-\cos M)(\cos M-\cos N)(\cos N-\cos M)$

16. If $A=\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, P=\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, $Q=P^TAP$ then $PQ^{2014}P^T=$

(A) $\begin{pmatrix} 1 & 2^{2014} \\ 0 & 1 \end{pmatrix}$ **(B)** $\begin{pmatrix} 1 & 4028 \\ 0 & 1 \end{pmatrix}$ **(C)** $(P^T)^{2013}A^{2014}P^{2013}$ **(D)** $P^TA^{2014}P$

17. Let A be a 3×3 non-singular matrix then which of the following is not true

(A) $|-A|=-|A|$ **(B)** $|Adj(A)|=|A|^2$

(C) $Adj(Adj\ A)=|A|^2\ A$ **(D)** $A\cdot Adj(A)=|A|I_3$

18. If A be 3×3 non-singular matrix, $|A|=K$, then $|(xA)^{-1}|=$ $(where\ \ x\ne 0)$

(A) xK **(B)** $\dfrac{1}{xK}$ **(C)** $\dfrac{1}{x^3K^3}$ **(D)** $\dfrac{1}{x^3K}$

19. Number of distinct real values of K, such that the system of equations $x+2y+z=1$, $x+3y+4z=K$, $x+5y+10z=K^2$ has infinitely many solutions is:

(A) zero **(B)** one **(C)** two **(D)** three

Paragraph for Questions 20 - 22

Let $A=\begin{bmatrix}4 & 1\\-9 & -2\end{bmatrix}$, and $A^{100}=\begin{bmatrix}a & b\\c & d\end{bmatrix}$, then:

20. $a+d=$

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

21. $\dfrac{c}{b}=$

(A) 9 **(B)** –9 **(C)** $\dfrac{-1}{9}$ **(D)** Not defined

22. Which of the following is true:

(A) $A^{200}+2A^{100}-I=0$ **(B)** $A^{200}-2A^{100}-I=0$

(C) $A^{200}-2A^{100}+I=0$ **(D)** $A^{200}+2A^{100}+I=0$

Paragraph for Questions 23 - 25

Let A be a square matrix and I be identity matrix of same order then $A-\lambda I$ is called characteristic matrix of A, λ is some complex number. $|A-\lambda I|=0$ is known as characteristic equation of matrix A; and its roots are called Characteristic roots or Eigen values of A. Every matrix satisfies its characteristic equation.

23. Eigen values of matrix $\begin{bmatrix}2 & 1 & 1\\2 & 3 & 4\\-1 & -1 & -2\end{bmatrix}$ are :

(A) $1,-1,3$ **(B)** $2,\dfrac{1\pm\sqrt{7}}{2}$ **(C)** $-3,3\pm2\sqrt{2}$ **(D)** $2,-2,3$

24. Which of the following matrices do not have eigen values 1 *and* -1

(A) $\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ **(B)** $\begin{bmatrix}0 & -i\\i & 0\end{bmatrix}$ **(C)** $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ **(D)** $\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$

25. If $A=\begin{bmatrix}1 & 0 & 0\\0 & 1 & 2\\0 & -2 & 3\end{bmatrix}$ & $I=\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$, then if $A^7-4A^6+6A^5=\alpha A^2+\beta A+\gamma I$ then (α,β,γ) is :

(A) $(5,-11,7)$ **(B)** $(-21,115,-91)$ **(C)** $(-13,44,-28)$ **(D)** None of these

MULTIPLE CORRECT ANSWERS TYPE

Each of the following Question has 4 choices A, B, C & D, out of which ONE or MORE Choices may be Correct:

26. A and B be two 3×3 matrices such that $A^5 = B^5$ and $A^4B = B^4A$, $A \neq B$ then :

(A) $A^4 = B^4$ **(B)** $|A^4 + B^4| = 0$

(C) $(A^4 - B^4)\cdot(A+B) = 0$ **(D)** $(A^4 + B^4)\cdot(A-B) = 0$

27. Let X and Y be two matrices different from I, such that $XY = YX$ and $X^n - Y^n$ is invertible for some natural number n. If $X^n - Y^n = X^{n+1} - Y^{n+1} = X^{n+2} - Y^{n+2}$, then:

(A) $I - X$ is singular **(B)** $I - Y$ is singular

(C) $X + Y = XY + I$ **(D)** $(I-X)(I-Y)$ is non singular

28. Let matrices be $X = \begin{bmatrix} 2 & 2 \\ 5 & 1 \end{bmatrix}$, $Y = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$ and $Z = \begin{bmatrix} 4 & -5 \\ -3 & 4 \end{bmatrix}$, then (tr($A$) denotes trace of A)

(A) $\sum_{k=0}^{\infty} \frac{tr(X(YZ)^k)}{2^k} = 6$ **(B)** $\sum_{k=1}^{\infty} \frac{tr(X(YZ)^k)}{2^k} = 3$

(C) $\sum_{k=1}^{\infty} \frac{tr(X(YZ)^k)}{2^k} = 6$ **(D)** $\sum_{k=0}^{\infty} \frac{tr(X(YZ)^k)}{2^k} = 3$

29. $Adj(A) = \begin{pmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{pmatrix}$, then $|A| =$

(A) 1 **(B)** –1 **(C)** 2 **(D)** –2

30. Let $f(x) = \begin{vmatrix} (2^x - 2^{-x})^2 & (2^x + 2^{-x})^2 & 1 \\ (3^x - 3^{-x})^2 & (3^x + 3^{-x})^2 & 1 \\ (4^x - 4^{-x})^2 & (4^x + 4^{-x})^2 & 1 \end{vmatrix}$ & $g(x) = \begin{vmatrix} 2x-2 & x-1 & x-1 \\ 3x-4 & 2x-3 & x-1 \\ 3x-5 & 2x-4 & 2x-4 \end{vmatrix}$ then :

(A) $f(4) = 0$ **(B)** $f(4) = 1020$

(C) $f(x) = g(x)$ has one solution **(D)** $f(x) = g(x)$ has three solutions

31. Let $f(x) = \begin{vmatrix} 7 & 2 & x^2-12 \\ 6 & x^2-12 & 3 \\ x^2-12 & 2 & 7 \end{vmatrix}$ then :

(A) $f(x) = 0$ has 6 real roots **(B)** $f(x) = 0$ has 4 real roots

(C) Sum of real roots of $f(x) = 0$ is 0 **(D)** Sum of real roots of $f(x) = 0$ is 9

32. Let X, Y, Z be 2×2 matrices with real entries. Define $\odot$ as follows $X \odot Y = \frac{1}{2}(XY + YX)$ then :

(A) $X \odot Y = Y \odot X$ **(B)** $X \odot I = X$

(C) $X \odot X = X^2$ **(D)** $X \odot (Y \odot Z) = X \odot Y + X \odot Z$

33. If $A \,\&\, B$ are two invertible matrices of same order, then $Adj(AB) =$

(A) $|A||B|A^{-1}B^{-1}$ **(B)** $adj(A)\cdot adj(B)$ **(C)** $|A||B|(AB)^{-1}$ **(D)** $adj(B)\cdot adj(A)$

34. If $f(x) = \begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^a \end{vmatrix}$, a, b being natural numbers, then:

(A) constant term of $f(x)$ is 0 **(B)** coefficient of x in $f(x)$ is 0

(C) constant term of $f(x)$ is $a-b$ **(D)** constant term of $f(x)$ is $a+b$

35. Atleast one root of the equation $\begin{vmatrix} x^2 + \sin x \cos x & x(1+\sin x) \\ x + \cos x & x+1 \end{vmatrix} = 0$ lies in:

(A) $\left(0, \frac{\pi}{6}\right)$ **(B)** $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ **(C)** $\left(\frac{-\pi}{3}, \frac{-\pi}{4}\right)$ **(D)** $\left(\frac{-\pi}{4}, \frac{\pi}{6}\right)$

36. Let in a skew symmetric matrix of order n the maximum number of non-zero elements is M_1 & let in an upper triangular matrix of order n the minimum number of zero elements is M_2 then :

(A) $\frac{M_1}{M_2} = 2$ **(B)** $M_1 = 2\sum_{k=1}^{n-1} k$ **(C)** $M_2 = \sum_{k=1}^{n} k$ **(D)** $\frac{M_1}{M_2} = \frac{2(n-1)}{n+1}$

37. Let A be a square matrix of order n, $A = [a_{ij}]_{n\times n}, a_{ij} = i^n - j^n$, then :

(A) $|A| = 0, n$ is odd

(B) $|A|$ is perfect square if n is even

(C) $|A| = 0 \ \forall n$

(D) A is skew symmetric when n is odd & symmetric when n is even

38. If $f(x) = \begin{vmatrix} a^{-x} & e^{x\ln a} & x^2 \\ a^{-3x} & e^{3x\ln a} & x^4 \\ a^{-5x} & e^{5x\ln a} & 1 \end{vmatrix}$ then :

(A) Graph of $f(x)$ is symmetric about origin **(B)** Graph of $f(x)$ is symmetric about y axis

(C) $f^{iv}(0) = 0$ **(D)** $f(x)\ln\left(\frac{a-x}{a+x}\right)$ is an even function

39. If A and B are square matrices of same order such that they commute then :

(A) $A^m \,\&\, B^n$ commute $m, n \in N$ **(B)** $(A+B)^n = {}^nC_0A^n + {}^nC_1A^{n-1}B + ... + {}^nC_nB^n, n \in N$

(C) $A - \lambda I, B + \mu I$ commute $\forall \lambda, \mu \in R$ **(D)** $A + \lambda I,\ \mu I - B$ commute $\forall \lambda, \mu \in R$

40. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 & 5 \\ 2 & 3 & 8 \\ 7 & 2 & 9 \end{bmatrix}$, then :

(A) $adj(adjA) = -A$ **(B)** $|adj(AB)| = 576$

(C) $|adj(adj(adj(adjA)))| = 1$ **(D)** $|adj(B^{-1})| = \frac{1}{24}$

41. If $ax+by+cz=0,\ bx+cy+az=0,\ cx+ay+bz=0,\ a,b,c\in R^{+}$ then :

(A) System have only trivial solution if $a^3+b^3+c^3>3abc$

(B) System will have non trivial solution only if $a=b=c$

(C) System have no solution if $a^3+b^3+c^3=3abc$

(D) If system have non trivial solution then minimum value of $(x-1)^2+(y-2)^2+(z-3)^2$ is 12

42. A be the set of all square matrices of order 3 with elements either 0, 1, or –1, then:

(A) $O(A)=3^9$

(B) Number of symmetric matrices in A whose trace is 0 = 7 × Number of skew symmetric matrices in A

(C) Number of matrices in A such that each of 0, 1, and –1 occurs atleast once at any position is 18150

(D) All skew symmetric matrices in A are singular

43. Let A be set of all 2×2 matrices of the form $\begin{pmatrix} a & b \\ c & a \end{pmatrix}$, such that $a,b,c\in\{0,1,2,3,4\}$ then **:**

(A) Number of matrices in A which are symmetric or skew symmetric is 25

(B) Number of matrices in A which are symmetric or skew symmetric & determinant divisible by 5 is 9

(C) Number of matrices in A for which trace is not divisible by 5 & determinant divisible by 5 is 16

(D) Number of matrices in A for which determinant is not divisible by 5 is 109

44. If $x^a y^b=e^m, x^c y^d=e^n,\ P=\begin{vmatrix} m & b \\ n & d \end{vmatrix}, Q=\begin{vmatrix} a & m \\ c & n \end{vmatrix},\ R=\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then :

(A) $P\log_x e=Q\log_y e$ **(B)** $x=e^{P/R}, y=e^{Q/R}$

(C) $P\log_e x=Q\log_e y$ **(D)** $x=e^{R/P}, y=e^{R/Q}$

45. $A=\begin{pmatrix} -3 & -1 & 2 \\ 3 & 1 & -1 \\ 4 & 2 & 5 \end{pmatrix}, A\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}=\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, A\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},\ A\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, B=\begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix}$, then :

(A) Trace $(B)=-8$ **(B)** $|AdjB|=4$

(C) $|Adj(B)|=\frac{1}{4}$ **(D)** Sum of all elements of B is –10

46. Let $f(n)=\begin{vmatrix} n & n+1 & n+2 \\ {}^{n}P_n & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^{n}C_n & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$, where the symbols have their usual meanings. Then $f(n)$ is divisible by

(A) n^2+n+1 **(B)** $(n+1)!$ **(C)** $n!$ **(D)** None of these

47. If $A(\theta)=\begin{bmatrix} \sin\theta & i\cos\theta \\ i\cos\theta & \sin\theta \end{bmatrix}$, then which of the following is not true

(A) $A(\theta)^{-1}=A(\pi-\theta)$ **(B)** $A(\theta)+A(\pi+\theta)$ is a null matrix

(C) $A(\theta)$ is invertible for all $\theta\in R$ **(D)** $A(\theta)^{-1}=A(-\theta)$

48. If $\Delta = \begin{vmatrix} \sin\theta\cos\theta & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\theta\sin\phi & \sin\theta\sin\phi & 0 \end{vmatrix}$ then

(A) Δ is independent of θ **(B)** Δ is independent of ϕ

(C) Δ is a constant **(D)** $\left.\frac{d\Delta}{d\theta}\right]_{\theta=\frac{\pi}{2}} = 0$

49. If $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} = \begin{vmatrix} \alpha^2 & \beta^2 & \beta^2 \\ \beta^2 & \alpha^2 & \beta^2 \\ \beta^2 & \beta^2 & \alpha^2 \end{vmatrix}$, then

(A) $\alpha^2 = a^2+b^2+c^2$ **(B)** $\beta^2 = ab+bc+ca$ **(C)** $\alpha^2 = ab+bc+ca$ **(D)** $\beta^2 = a^2+b^2+c^2$

50. Let $a, \lambda, \mu \in R$ consider the system of linear equations $ax+2y=\lambda$, $3x-2y=\mu$

Which of the following statement(s) is (are) correct?

(A) If $a=-3$ then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda+\mu=0$ the system has infinitely many solutions for $a=-3$

(D) If $\lambda+\mu \neq 0$, then the system has no solutions for $a=-3$

51. $\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & a+b \\ 0 & 1 & 2a+3b \end{vmatrix}$ is divisible by

(A) $a+b$ **(B)** $a+2b$ **(C)** $2a+3b-1$ **(D)** a^2

52. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -684\alpha$$

(A) –4 **(B)** 9 **(C)** –9 **(D)** 4

53. If $A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is an orthogonal matrix of order 3, then:

(A) $a=-2$ **(B)** $a=2, b=1$ **(C)** $b=-1$ **(D)** $b=1$

54. How many 3×3 matrices M with entries from {0, 1, 2} are there for which the sum of the diagonal entries of M^TM is 5.

(A) 198 **(B)** 162 **(C)** 126 **(D)** 135

55. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [P_{ij}]$ be a $n\times n$ matrix with $P_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when n =

(A) 57 **(B)** 55 **(C)** 58 **(D)** 56

56. The value of θ lying between $\theta=0$ and $\theta=\frac{\pi}{2}$ and satisfying the equation $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}=0$ is:

(A) $\frac{7\pi}{24}$ **(B)** $\frac{5\pi}{24}$ **(C)** $\frac{11\pi}{24}$ **(D)** $\frac{\pi}{24}$

57. If maximum and minimum values of the determinant $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$ are α and β, then:

(A) $\alpha+\beta^{99}=4$

(B) $\alpha^3-\beta^{17}=26$

(C) $\left(\alpha^{2n}-\beta^{2n}\right)$ is always an even integer for $n\in N$

(D) A triangle can be constructed having it sides as $\alpha-\beta$, $\alpha+\beta$ and $\alpha+3\beta$

58. Let X and Y be two arbitrary, 3×3, non-zero skew-symmetric matrices and Z be an arbitrary 3×3, non-zero symmetric matrix. Then which of the following is (are) skew symmetric:

(A) $Y^3Z^4-Z^4Y^3$ **(B)** $X^{44}+Y^{44}$ **(C)** $X^4Z^3-Z^3X^4$ **(D)** $X^{23}+Y^{23}$

59. Which of the following is (are) not the square of a 3×3 matrix with real entries :

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **(B)** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **(C)** $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **(D)** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

60. Let M be a 2×2 symmetrix matrix with integer entries. Then M is invertible, if:

(A) The first column of M is the transpose of the second row of M.

(B) The second row of M is the transpose of first column of M

(C) M is a diagonal matrix with non-zero entries in the main diagonal

(D) The product of entries in the main diagonal of M is not the square of an integer

61. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to zero, if:

(A) a, b, c are in A.P **(B)** a, b, c are in G.P

(C) a, b, c are in H.P **(D)** $(x-\alpha)$ is a factor of $ax^2+2bx+c$

62. If a, b, c are non-zero real numbers such that $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}=0$, then:

(A) $\frac{1}{a}+\frac{1}{b\omega}+\frac{1}{c\omega^2}=0$ **(B)** $\frac{1}{a}+\frac{1}{b\omega^2}+\frac{1}{c\omega}=0$ **(C)** $\frac{1}{a\omega}+\frac{1}{b\omega^2}+\frac{1}{c}=0$ **(D)** None of these

63. Let $\{\Delta_1, \Delta_2, \Delta_3, .. \Delta_k\}$ be the set of third order determinants that can be made with the distinct non-zero real numbers $a_1, a_2, ...a_9$, Then:

(A) $k = 9!$

(B) $\sum_{i=1}^{k} \Delta_i = 0$

(C) At least one $\Delta_i = 0$

(D) None of these

64. For 3×3 Matrices M and N, which of the following statement(s) is (are) not correct

(A) $N^T MN$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric

(B) $MN - NM$ is skew symmetric for all symmetric matrices M and N

(C) MN is symmetric for all symmetric matrices M and N

(D) (adj M) (adj N) = adj(MN) for all invertible matrices M and N.

65. The system of equations $6x+5y+\lambda z=0$, $3x-y+4z=0$, $x+2y-3z=0$, has:

(A) Only a trivial solution for $\lambda \in R$

(B) Exactly one non-trivial solution for some real λ

(C) Infinite number of non-trivial solutions for one value of λ

(D) Only one solution for $\lambda \neq -5$

MATRIX MATCH TYPE

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) & (D) whereas statements in Column 2 are labeled as p, q, r, s & t. More than one choice from Column 2 can be matched with Column 1.

66. Consider the 2×2 matrix $A = \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}$

	Column 1		Column 2
(A)	A is idempotent	**(p)**	Either $x=1, y=0$ or $x=-1, y\in R$
(B)	A is involutory	**(q)**	Either $x=0, y\in R$ or $x=1, y=0$
(C)	A is orthogonal	**(r)**	$x=0, y\in R$
(D)	A is singular	**(s)**	$x=\pm1, y=0$

67. If $f(x)=\begin{vmatrix}(\alpha x+1)\cos^2 x & x & 1-x\\ \beta\sin x & x^2 & 2x\\ (\gamma x^2+1)\tan x & x & 1-x^2\end{vmatrix}$

	Column 1		Column 2
(A)	$\lim\limits_{x\to 0}\frac{f(x)}{x}+\lim\limits_{x\to 0}\frac{f(x)}{x^2}$	(p)	0
(B)	$\lim\limits_{x\to 0}\frac{f'(x)}{x}+f'(0)$	(q)	–1
(C)	$\lim\limits_{x\to 0}\frac{1}{x^6}\int_{x^2}^{x^3} f(x)dx=A,[A]$ is ([.] denotes greatest integer function)	(r)	–2
(D)	If $\alpha=\beta=\gamma=0, g(x)=\frac{f(x)}{x^2}$ then $\left[g\left(\frac{\pi}{4}\right)\right]$ is ([.] denotes greatest integer function)	(s)	$f''(0)$

68. The elements of 3×3 matrix A are either 1 or –1, then

	Column 1		Column 2
(A)	Total number of such A which are symmetric	(p)	3
(B)	Maximum value of determinant $\|A\|$	(q)	4
(C)	Minimum value of determinant $\|A\|$	(r)	–4
(D)	Maximum value of trace of A	(s)	64

69. A is non singular matrix of order n.

	Column 1		Column 2
(A)	$(adj(A))^{-1}$	(p)	$\frac{A}{\|A\|}$
(B)	$adj(kA)$	(q)	$\frac{adj(adj\,A)}{\|A\|^{n-1}}$
(C)	$adj(adj(kA))$	(r)	$k^{n-1}(adj\ A)$
(D)	$adj(A^{-1})$	(s)	$k^{(n-1)^2}\|A\|^{n-2}A$

70. $x+y+z=6,\ x+2y+3z=10,\ x+2y+\alpha z=\beta$

	Column 1		Column 2
(A)	Unique solution	(p)	$\alpha=3$
(B)	No solution	(q)	$\alpha\neq 3$
(C)	Infinitely many solutions	(r)	$\beta=10$
(D)	Atleast 2 solutions	(s)	$\beta\neq 10$

NUMERICAL VALUE TYPE

This section has Numerical Value Type Questions. The answer to each question is a NUMERICAL/INTEGER VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places.

71. Let $A=[a_{ij}]_{n\times n}$, n is odd natural number. Then determinant of matrix $(A-A^T)^{2015}$ is _______.

72. If x, y, z distinct common roots of $z^6-1=0$ and $z^{21}-1=0$ then $\begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-x-z & 2y \\ 2z & 2z & z-x-y \end{vmatrix}$ is equal to_____.

73. Let $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ & $P=\begin{bmatrix} \cos\frac{\pi}{12} & \sin\frac{\pi}{12} \\ -\sin\frac{\pi}{12} & \cos\frac{\pi}{12} \end{bmatrix}$ and $Q=P^TAP$, then if $PQ^{2014}P^T=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then sum of digits of b is____.

74. Let ω be complex cube root of unity. Let $S=\begin{bmatrix} 1 & a & b \\ \omega^4 & 1 & c \\ \omega^2 & \omega^7 & 1 \end{bmatrix}$, where each of a,b,c are either ω or ω^2. Then number of distinct non singular possible such matrices S is________.

75. Let $\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}=ax^4+bx^3+cx^2+dx+e$ be an identity in x, then $-\left[\left(\frac{a+b+c+d+e}{a+e}\right)\right]$ is ____. ([.] denotes greatest integer function).

76. $\begin{vmatrix} {}^5C_1 & {}^5C_2 & {}^5C_3 \\ {}^4C_1 & {}^4C_2 & {}^4C_3 \\ {}^3C_1 & {}^3C_2 & {}^3C_3 \end{vmatrix}-\begin{vmatrix} {}^5C_1 & {}^6C_2 & {}^7C_3 \\ {}^4C_1 & {}^5C_2 & {}^6C_3 \\ {}^3C_1 & {}^4C_2 & {}^5C_3 \end{vmatrix}=$ ______.

77. $A=\begin{bmatrix} \frac{1}{2}|[x]| & |\sin y| \\ \cos z & 1 \end{bmatrix}, B=\begin{bmatrix} \{x\} & \{y\} \\ \{z\} & 1 \end{bmatrix}$, if $x\in[-2,2]$, $y,z\in(-\pi,\pi)$ if number of triplets (x,y,z) such that $A=B$ is k, then value of $k/7$ is _______.

78. $A=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then $A^{2014}=\lambda A^{2013}+\mu A^{2012}, \lambda+\mu=$ ___________.

79. Let A be a 3×3 matrix which contains five 'a' & four 'b' then number of symmetric matrices possible is k, number of zeros at the end of $k!$ is _________.

80. Matrix A satisfies $A^2=3A-2I$, and $A^{-1}=\frac{\lambda I+kA^3}{\mu}$, then $\lambda+k\mu$ is ________.

81. In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix}=0$, then $\sin^2 A+\sin^2 B+\sin^2 C=$_____.

82. For all values of $\theta \in \left[0, \frac{\pi}{2}\right]$, the determinant of the matrix $\Delta = \begin{bmatrix} -2 & \tan\theta + \sec^2\theta & 3 \\ -\sin\theta & \cos\theta & \sin\theta \\ -3 & -4 & 3 \end{bmatrix}$ is always greater than or equal to _____.

83. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = \left[q_{ij}\right]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals _____.

84. Number of positive integral solutions of the equation $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 30$ are _____.

85. Let $A = \begin{bmatrix} 1 & \frac{-1-i\sqrt{3}}{2} \\ \frac{-1+i\sqrt{3}}{2} & 1 \end{bmatrix}$, Then $A^{100} = 2^k . A$ where k is ____.

86. If the system of linear equations $x + ky + 3z = 0$, $3x + ky - 2z = 0$, $2x + 4y - 3z = 0$ has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to ____.

87. If $S_r = \alpha^r + \beta^r + \gamma^r$ the value of $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix}$ is equal to $(\alpha - \beta)^k (\beta - \gamma)^{2k-2} (\gamma - \alpha)^{k^2-2}$. Then k is ____.

88. Consider three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then the value of the sum

$tr(A) + tr\left(\frac{ABC}{2}\right) + tr\left(\frac{A(BC)^2}{4}\right) + tr\left(\frac{A(BC)^3}{8}\right) + ... + \infty$ is ____

89. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$ then $A + 2B$ equals ____

90. Let $\{\Delta_1, \Delta_2, \Delta_3, ... \Delta_k\}$ be the set of third order determinants that can be made with the distinct nonzero real numbers $a_1, a_2, a_3, ... a_9$ then $k = (a+b)!$ where $a + b$ equals ____

91. Let k be a positive real number and let $A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$

If det (adj A) + det (adj B) = 10^6, then greatest integer of k is equal to ____

92. For a real number α, if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ of linear equations, has infinitely many solutions, then $1+\alpha+\alpha^2 =$

93. If $\begin{vmatrix} x & x+y & x+y+z \\ 2x & 3x+2y & 4x+3y+2z \\ 3x & 6x+3y & 10x+6y+3z \end{vmatrix}=64$, then the real value of x is____

94. Let $\Delta_1=\begin{vmatrix} a & b & c \\ c & d & c+d \\ a & b & a+b+c \end{vmatrix}$ and $\Delta_2=\begin{vmatrix} a & b & a+c \\ b & d & b+d \\ a & c & a+d+c \end{vmatrix}$ then the value of $\left|\dfrac{\Delta_1}{\Delta_2}\right|$, where $b \neq 0$ and $ad \neq bc$ is

95. Let M be a 3×3 matrix satisfying $M\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}=\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $M\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then the sum of diagonal entries of M is ____.

96. Let ω be the complex number $\cos\dfrac{2\pi}{3}+i\sin\dfrac{2\pi}{3}$. Then, the number of distinct complex numbers z satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}=0$ is equal to____.

97. The characteristic equation of a matrix A is $\lambda^3-5\lambda^2-3\lambda+2=0$ then $|\text{adj } A| =$

98. If $a_i^2+b_i^2+c_i^2=1, (i=1, 2, 3)$ and $a_ia_j+b_ib_j+c_ic_j=0 \ (i\neq j;\ i, j=1, 2, 3)$ then the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is

99. If $x_1, x_2, x_3, ..., x_{13}$ are in A.P then the value of $\begin{vmatrix} e^{x_1} & e^{x_4} & e^{x_7} \\ e^{x_4} & e^{x_7} & e^{x_{10}} \\ e^{x_7} & e^{x_{10}} & e^{x_{13}} \end{vmatrix}$ is

100. For $a, b, c, x, y, z \in R$, if $\Delta_1=\begin{vmatrix} (a-x)^2 & (b-x)^2 & (c-x)^2 \\ (a-y)^2 & (b-y)^2 & (c-y)^2 \\ (a-z)^2 & (b-z)^2 & (c-z)^2 \end{vmatrix}$ and $\Delta_2=\begin{vmatrix} (1+ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix}$ then $|\Delta_1/\Delta_2|=$_____.

MATHEMATICS

ANSWER KEY & SOLUTION

QUADRATIC EQUATIONS

1	2	3	4	5	6	7	8	9	10
A	C	A	B	C	B	D	A	C	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	A	C	D	A	B	A	D	ABCD	ABC
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
AD	AC	AD	ABD	ABCD	ACD	CD	CD	BD	BCD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
[A-T] [B-R, S] [C-P, S] [D-T]	[A-Q] [B-R] [C-P] [D-T]	2	0	5	1	0	4	5	7

TRIGONOMETRY

1	2	3	4	5	6	7	8	9	10
B	B	D	C	A	D	B	A	C	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	A	B	B	B	B	A	A	B	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	C	B	B	D	BC	BD	A	ABD	BD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
AD	CD	BD	ABC	AB	BD	BD	AB	AD	ABD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
BD	BD	ABD	ABCD	ABCD	ACD	ABCD	ABC	BCD	CD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
BC	ABC	BC	AC	ABC	AC	AB	ABD	ABD	2
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
3	6	6	4	1	2	9	8	6	3
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
2	3	1	3	9	6	7	8	8	3
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	
1	2	2	7	9	16	65	1	8	

SEQUENCE AND SERIES

1	2	3	4	5	6	7	8	9	10
D	D	B	D	B	B	C	B	D	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	B	C	B	C	D	AD	BCD	CD	AB
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
ABD	ABC	AD	AB	CD	AC	BC	ABCD	BC	ACD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
ABC	BCD	BC	BC	BD	ABCD	**ABCD**	**BD**	[A-Q] [B-T] [C-P] [D-R]	[A-T] [B-Q] [C-S] [D-P]
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
2	6	661750	289	69	352	0	2017	2018	4

COMPLEX NUMBERS

1	2	3	4	5	6	7	8	9	10
D	A	D	A	B	B	A	B	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	A	D	C	C	A	C	C	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	D	C	ABC	BD	ABCD	D	BD	BC	AD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
ABCD	ABCD	AB	BD	BC	AB	CD	ABC	AD	BD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
BD	AB	AC	BC	ABC	CD	ABCD	AB	ABD	AB
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
AB	AD	CD	BC	ABC	AD	BD	AC	AB	ABCD
61	**62**	**63**	**64**		**65**		66		
AC	AB	CD	[A-r] [B-s] [C-q] [D-p]		[A-q] [B-s] [C-r] [D-p]		[A – q] [B-p] [C-r] [D-s]		
67	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**	**76**
1	5	7	6	5	5	0	0	3	9
77	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**	**86**
1	0.414	0	6.25	1	0.20	10	1	1	10
87	**88**	**89**	**90**	**91**	**92**	**93**	**94**	**95**	**96**
0.50	0	4	1	5	4.88	0.875	18.75	5	6.25

PERMUTATIONS & COMBINATIONS

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	D	B	C	B	B	A	D	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	C	A	D	D	B	C	B	D	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	B	C	C	A	B	C	ABD	BC	BC
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
BC	BCD	ABCD	CD	BD	BD	ABCD	BD	ACD	AC
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
BD	CD	ABC	ABC	AB	AD	ABCD	ABCD	ABD	ABCD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
ABC	AB	ABC	AC	ABCD	BC	ABC	BD	ABC	AD
61	**62**	**63**	**64**	65			66		
BC	ABC	AD	ABD	[A-r [B-s] [C-p] [D-q]]			[A-r] [B-s] [C-p] [D-q]		
67		68		69				**70**	**71**
[A-q] [B-s] [C-p] [D-r]		[A-s] [B-r] [C-p] [D-q]		[A-p, r, s] [B-p, r, s] [C-q, t] [D-q, t]				1	3
72	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**	**81**
7	7	9	5	9	2	6	9	47	23
82	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**	**91**
14	4	10	3.150	243	6	240	15.68	10	729
92	**93**	**94**	**95**	**96**	**97**	**98**	**99**		
315	2	7	7	2500	10	4	5		

BINOMIAL THEOREM

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	A	B	B	C	D	B	D	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	A	AD	ABCD	BCD	ABC	BC	CD	ABCD
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	
BCD	CD	ABC	ACD	AC	CD	ABC	BD	[A-P, Q] [B-R, T] [C-P, Q] [D-P, Q, S]	
30		**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**
[A-T] [B-Q][C-S] [D-R]		2250000	21	1024	2252	13	1	1	2
39	**40**								
1	3								

STRAIGHT LINE

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	D	D	C	B	B	D	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	B	A	B	B	B	B	D	B	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	D	A	BD	ABCD	AB	ABCD	AC	ABD	BD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
BC	BC	AD	ABC	ABC	ABCD	BD	AB	BCD	AD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
ABC	ABD	AB	ABC	ABD	BC	ABC	ABCD	ABC	AC
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
ABCD	ABC	AB	ACD	BC	ABCD	ABC	ABCD	AB	CD
61	**62**	**63**	**64**		**65**		**66**		
ABC	ABCD	ACD	[A-p] [B-s] [C-q] [D-s]		[A-q] [B-p] [C-s] [D-r]		[A-s] [B-p] [C-q] [D-r]		
67		**68**		**69**			**70**	**71**	**72**
[A-s] [B-p] [C-q] [D-r]		[A-r] [B-p] [C-s] [D-q]		[A-r] [B-s] [C-q] [D-p]			2	6	2
73	**74**	**75**	**76**	**77**	**78**	**79**	**80**	**81**	**82**
7	6	8	2	2	3	13	3	9	16
83	**84**	**85**	**86**	**87**	**88**	**89**	**90**	**91**	**92**
3	1.80	4	3	16	24	6	4	5	7
93	**94**	**95**	**96**	**97**	**98**				
4	2	5	12	27	9				

CIRCLE

1	2	3	4	5	6	7	8	9	10
A	B	A	B	B	A	A	B	D	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	A	B	A	A	C	D	C	B	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	D	d	d	d	c	c	d	BC	AD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
AB	ABD	AB	ABCD	AB	AC	AC	ACD	ACD	AB
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
ABC	ABC	ABCD	ABC	BC	ABD	ABD	AD	ABC	BCD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
CD	CD	AD	BC	BCD	AB	ABCD	ABC	BC	BD
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	
BC	AB	ABD	ACD	AC	AC	BC	ABC	AC	
70		**71**		**72**		**73**		**74**	
[A-q] [B-p][C-t] [D-r]		[A-r] [B-p][C-q] [D-q]		[A-s] [B-q][C-t] [D-r]		[A-r, B-s, C-p, D-q]		[A-r, B-p, C-s, D-p]	
75	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**
8	4	6	4	1	0	2	25	0.5	1.66
85	**86**	**87**	**88**	**89**	**90**	**91**	**92**	**93**	**94**
1	8	4	40	5	1	3	3	0	8
95	**96**	**97**	**98**	**99**	**100**	**101**			
5	2	5	6	7	4	3			

CONIC SECTION

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	B	B	B	D	B	B	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	D	A	A	A	B	A	B	C	B
21	**22**	**23**	**24**	**25**	26	27	28	**29**	**30**
A	C	C	D	B	A	D	B	B	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	A	C	B	AC	BCD	BD	AD	AC	AC
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
CD	ABCD	AC	AC	AC	ABC	AC	AC	ABC	AD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
A	ABC	B	ABCD	AD	BCD	AB	CD	ABC	BC
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
CD	ABCD	AB	BD	BCD	AC	ABD	**BC**	**AD**	**AC**
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
ABD	ABCD	ABC	BC	BD	BC	AB	AB	AB	D
81		**82**		**83**		**84**		**85**	
[A-r] [B-s] [C-p] [D-q]		[A-p, r] [B-p, q, r] [C-q] [D-q, s]		[A-p,q,r,s] [B-p,q,r,s] [C-q,r,s] [D-p]		[A-q] [B-q] [C-s] [D-p]		[A-p, s] [B-q] [C-r] [D-p, q, s]	
86	**87**	**88**	**89**	**90**	**91**	**92**	**93**	**94**	**95**
A	1	4	1	2	1	6	3	4	8
96	**97**	**98**	**99**	**100**	**101**	**102**	**103**	**104**	**105**
4	2	0	0	3	6.33	2	4	80	2
106	**107**	**108**	**109**	**110**	**111**	**112**	**113**	**114**	**115**
4	2	8	3	3	1	3	3	1	4
116									
1.32									

FUNCTIONS

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	D	C	C	A	A	B	A	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	A	B	B	C	ACD	AC	ABC	ABC	ABCD
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	
ABCD	AD	AC	BD	AB	AD	ABCD	BCD	CD	
30		**31**		**32**		**33**	**34**	**35**	**36**
[A-s] [B-r] [C-q] [D-p]		[A-r] [B-s] [C-p] [D-p]		[i-c] [ii-d][iii-b] [iv-c]		2	3	22	5
37	**38**	**39**	**40**	**41**	**42**	**43**	**44**		
8	4	2049	2	37	7	12	1		

DIFFERENTIAL CALCULUS-1

1	2	3	4	5	6	7	8	9	10
D	D	C	A	B	D	C	C	B	B
11	12	13	14	15	16	17	18	19	20
A	A	B	D	A	B	C	B	B	AB
21	22	23	24	25	26	27	28	29	30
AC	AC	AB	AB	AC	AB	AC	ABD	AD	AC
31	32	33	34	35	36	37	38	39	40
ABCD	ABD	AC	ABCD	AC	BC	ABC	AC	AD	ABD
41	42	43	44	45	46	47	48	49	50
B, C	ABC	AC	BCD	AD	AC	ACD	AB	BCD	BCD
51	52	53	54	55	56	57	58		
BC	BC	AC	BC	BD	ABC	ABCD	[A-p, q, r, s] [B-p, q, r, s] [C-p, q, r, s] [D-r]		
59		60		61			62	63	64
[A-s] [B-p] [C-p] [D-p]		[A-s] [B-r] [C-q, r] [D-p]		[A-p, q] [B-p, s] [C-q] [D-r, t]			1	1	6
65	66	67	68	69	70	71	72	73	74
1	8	2	2	4	8	5	3	4	1
75	76	77	78	79	80	81	82	83	84
9	1.41	0	3	2	3	1	1.5	4	2
85	86	87	88	89	90				
3.14	8	3	12.07	6	2				

DIFFERENTIAL CALCULUS-2

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	A	B	A	D	C	A	C	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	D	A	D	B	B	A	A	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	C	B	D	AB	ABCD	AC	ABCD	AD	ABCD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
AB	BD	BCD	ABCD	ABC	ACD	ABC	BCD	AC	CD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
BCD	BCD	BCD	ABC	AC	BCD	BC	ABC	ABC	ACD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
ABC	AC	AC	BD	ABCD	AD	AD	**B**	D	D
61	**62**	**63**	**64**			**65**		**66**	
ABD	BD	BC	[A-p, q, r, s] [B-p, r] [C-p, q, r, s] [D-p, q, r, s]			[A-p] [B-q] [C-r] [D-s]		[A-q] [B-r] [C-p] [D-t]	
67	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**	**76**
9	0	2	0	2	1	3	2	9	5
77	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**	**86**
4	2	3	5	1200	1	0	1	2	2
87	**88**	**89**	**90**	**91**	**92**	**93**	**94**		
2	0	5	9	9	3	48	0		

INTEGRAL CALCULUS-1

1	2	3	4	5	6	7	8	9	10
C	D	B	A	A	B	D	A	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	C	C	A	C	B	C	D	A	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	AB	AB	BD	AC	ABD	AC	AB	ABC	BCD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
ABC	BC	CD	ABD	ABC	BC	BC	CD	AD	AD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B,C	AC	ABC	ACD	AC	ABCD	BC	AC	AB	ABC
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	
CD	CB	BC	BD	AC	ABC	ABD	BC	AD	
60		**61**		**62**		**63**	**64**	**65**	**66**
[A-p, q] [B-r, s] [C-p] [D-p, q]		[A-r] [B-s] [C-q] [D-p]		[A-s] [B-t] [C-r] [D-q]		1	4	2	3
67	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**	**76**
0	2	1	10	6	4	3015	521	8	3
77	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**	**86**
2.5	4	0.5	3.6	2.05	1	403	1	0	12
87	**88**	**89**	**90**	**91**	**92**				
1.59	2019	1.8	2.14	2	11				

INTEGRAL CALCULUS-2

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	C	B	C	A	C	D	C	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	D	C	D	C	D	D	A	D	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	D	B	D	D	D	C	B	D	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	A	AB	A	ABC	ABD	ABC	AB	ACD	AD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
ABC	BC	ABD	ABC	AB	BC	ABD	AB	ACD	AC
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
AD	AB	BD	ACD	AC	ABCD	ABC	BD	A,B,C,D	A,B
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
A,B,C	A,B,D	A,B	A,B	A,B,C	A,B,C	A,B,C,D	B,C	BD	A,B,C
71	**72**	**73**	**74**	**75**	**76**	**77**		**78**	
ABC	A,B,C	C,D	A,B,C	CD	A,B,D	[A-s] [B-s] [C-r] [D-q]		[A-r] [B-p] [C-s] [D-q]	
79		**80**		**81**		**82**	**83**	**84**	**85**
[A-q] [B-r, s] [C-p] [D-p]		[A-p, q] [B-p, q, r] [C-q, s] [D- s]		[A-r] [B-p] [C-q] [D-s]		3	4	6	8
86	**87**	**88**	**89**	**90**	**91**	**92**	**93**	**94**	**95**
8	4	0	1	1	2	0.72	3.14	101	4
96	**97**	**98**	**99**	**100**	**101**	**102**	**103**	**104**	**105**
8	10.50	2	0	8.15	2	1	8	85	101
106	**107**	**108**	**109**						
153	61	10	4.14						

DIFFERENTIAL EQUATIONS

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	B	A	A	C	C	A	A	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	C	C	D	B	A	C	D	B	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	C	ABCD	AB	CD	AB	BC	AD	AB	AB
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
AC	ABCD	BCD	CD	ABD	ABD	CD	BC	AC	AB
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
BC	ABC	BCD	ACD	AC	ABCD	A,B,C	AD	A,B,C	A,B,D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	
C,D	A,B,C,D	B,C	A,B	A,B, C	B,C	A,D	C	C,D	
60		**61**		**62**		**63**		**64**	
[A-u] [B-s] [C-q] [D-p]		[A-r] [B-s] [C-p] [D-q]		[A-r] [B-s] [C-q] [D-p]		[A-q, s] [B-p] [C-p] [D-q, r, s]		[A-q] [B-r] [C-p] [D-s]	
65	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**
4	2	8	2	1	1	8	2	4	1
75	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**
8	1	1	1	3	1	3	7	8	8
85	**86**	**87**	**88**	**89**	**90**	**91**	**92**	**93**	
2	3	2	5	62	8	1	1	0	

VECTORS

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	A	B	A	D	C	C	D	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	C	C	D	C	A	A	B	A	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	C	D	A	C	D	B	B	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	B	C	C	A	B	A	B	D	C
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
CD	AB	AC	AC	ABC	ABCD	AB	BC	ABC	AC
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
BD	BD	AB	AD	ABCD	AC	BC	AB	CD	AC
61	**62**	63		64		**65**		**66**	
BC	AC	[A-q, r, s] [B-p, s] [C-p, r] [D-p, q]		[A-r] [B-p] [C-s] [D-q]		[A-p, q, r, s] [B-p, q] [C-p, r] [D-r]		[A-r, s] [B-q, r] [C-q, r] [D-p, q, r]	
67		**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**
[A-r] [B-p][C-p, q] [D-s]		5	9	3	1	6	7	5	**0**
76	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**	
1	4	0	2	$2\sqrt{5/7}$	7	1	3	2	

THREE DIMENSIONAL GEOMETRY

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	A	D	C	A	D	A	A	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	A	C	A	ABC	A	B	D	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	B	A	D	C	B	A	B	A	D
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	D	D	D	B	C	B	A	BC	AB
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
A	A	ABCD	AB	CD	C	AD	BD	AB	AB
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	
AC	ABCD	AC	ABC	BC	BC	AC	ABC	AB	
60		61		62		**63**		**64**	
[A-p,q][B-p,q] [C-r,s][D-q]		[A-q] [B-p][C-s] [D-r]		[A-t][B-p][C-s][D-r]		[A-q,s][B-s][C-r][D-p]		[A-q] [B-p][C-p] [D-r]	
65	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**
2	1	4	5	6	3	4	1	4	7
75	**76**	**77**	**78**	**79**	**80**				
4	5	2	4	6	2				

PROBABILITY

1	2	3	4	5	6	7	8	9	10
C	D	D	D	D	A	B	C	C	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	A	B	C	D	C	B	D	D	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	A	D	A	C	B	A	C	D	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	C	A	B	ABCD	ABC	BD	BC	AD	AD
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
ABCD	AC	AC	AC	ABCD	ABCD	BC	ABCD	AC	BD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
ABCD	BCD	ACD	AC	AB	AD	ACD	ABCD	ABC	AD
61	**62**	**63**	**64**	**65**		**66**		**67**	
BC	AD	AC	ABC	[A-q,s][B-p,t][C-r,t]		[A-t][B-q] [C-s][D-r]		[A-r][B-s][C-p][D-q]	
68		**69**		**70**	**71**	**72**	**73**	**74**	**75**
[A-r] [B-p][C-q][D-s]		[A-q][B-p][C-r][D-q]		7	8	3	9	8	9
76	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**
8	3	1.44	2.27	0.40	0.75	0.45	0.9523	1.25	0.0130
86	**87**	**88**	**89**	**90**	**91**	**92**	**93**	**94**	**95**
0.299	0.1054	0.422	0.018181	0.142857	1.35	0.1142	1.33	0.20	0.3488
96	**97**								
0.9722	4.17								

MATRICES & DETERMINANTS

1	2	3	4	5	6	7	8	9	10
C	D	C	B	C	A	D	D	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	A	A	C	A	B	C	D	C	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	A	C	B	BCD	ABC	AB	CD	AD
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
AC	ABC	CD	AB	BCD	AB	AB	ACD	ABCD	ABC
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
ABD	ABCD	ABC	AB	ACD	AC	ABC	BD	AB	BCD
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
AC	BC	AC	A	BCD	AC	ABC	CD	AC	CD
61	**62**	**63**	**64**	**65**	**66**		**67**		
BD	ABC	AB	CD	CD	[A-q, r] [B-p, s] [C-s] [D-r]		[A-q] [B-r, s] [C-p] [D-p]		
68		**69**		**70**		**71**	**72**	**73**	**74**
[A-s] [B-q] [C-r] [D-p]		[A-p, q] [B-r] [C-s] [D-p, q]		[A-q] [B-p, s] [C-p, r] [D-p, r]		0	0	7	2
75	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**
9	0	6	7	2	1	2.25	3	103	3
85	**86**	**87**	**88**	**89**	**90**	**91**	**92**	**93**	**94**
99	10	2	6	6	9	4	1	4	2
95	**96**	**97**	**98**	**99**	**100**				
9	1	4	1	0	1				

SOLUTIONS

Mathematics

Content

QUADRATIC EQUATION

1.(A) $5x^2+5xy=60,\ 24xy+36y^2=-60$. Add and factorise to get $(x+4y)(5x+9y)=0$

2.(C) $\left(a^3-2a+c\right)-\left(b^3-2b+c\right)=0 \Rightarrow a^2+ab+b^2=2$

Let $x=a^2\left(2a^2+4ab+3b^2\right),\ y=b^2\left(3a^2+4ab+2b^2\right)$

Then $x+y-8=2(ab+2)\left(a^2+ab+b^2-2\right)=0 \Rightarrow 3+y-8=0$

3.(A) $D\ge 0$ gives $K\in\left[-4,\frac{-4}{3}\right]$ and $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=19-(K+5)^2$

So, maximum $\alpha^2+\beta^2=19-(-4+5)^2=18$

4.(B) The given expression may be written as $\sqrt{(x-3)^2+\left(x^2-2\right)^2}-\sqrt{x^2+\left(x^2-1\right)^2}$, P (x, y) on parabola $y=x^2$ and $A(3,2), B(0,1)$

$(PA-PB)_{\max}=AB$

5.(C) $x\le\frac{a}{5},\ x>\frac{b}{6}$ So, $4\le\frac{a}{5}<5$ and $1\le\frac{b}{6}<2$

So, $a\in[20,25)$ and $b\in[6,12)$

So, number of pairs (a, b) is ${}^6C_1\times{}^5C_1$

6.(B) Complete the squares of each expression to get LHS least value = $\sqrt{25}+\sqrt{144}=17$ and RHS maximum value $=\sqrt{289}=17$ So, $x=3$ is the only solution.

7.(D) $(r-a)(r-b)(r-c)(r-d)=4$. Assume $a>b>c>d$ then $r-a<r-b<r-c<r-d$.

So, we must have $r-a=-2,\ r-b=-1,\ r-c=1,\ r-d=2$. (integers)

8.(A) $\sum\alpha_i=2n$ and $\sum\alpha_i\alpha_j=2n(n-1)$

So, $\sum\alpha_i^2=\left(\sum\alpha_i\right)^2-2\sum\alpha_i\alpha_j=4n$

So, $\sum(\alpha_i-2)^2=0 \Rightarrow \alpha_i=2$

9.(C) $\alpha+\beta=-a,\ \alpha\beta=2a-1,\ 2a=1+b$

So, $(\alpha+2)(\beta+2)=3 \Rightarrow \alpha+2=3,\ \beta+2=1$ or $\alpha+2=-1,\ \beta+2=-3$

10.(D) Roots $x_1=\frac{a}{2}-\sqrt{4+\frac{a^2}{4}},\ x_2=\frac{a}{2}+\sqrt{4+\frac{a^2}{4}}$

$B\subseteq A \Rightarrow x_1\ge -2$ and $x_2<4$

Simplify to get $a\in[0,3)$

11.(B) Let $f(x)=4x^2+5x+17,\ \ g(x)=x^2+4x+12,\ h(x)=x^2+x+1$

$\frac{f(x)}{g(x)}=\frac{f(x)+h(x)}{g(x)+h(x)}$; Simplify $f(x)=g(x)\ \ (h(x)>0) \Rightarrow 3x^2+11x+5=0$

12.(A) Notice that RHS x is always positive and the inner square root term gets simplified to $\sqrt{1+(x+3)(x+5)}=|x+4|$. This way simplifying the whole LHS side amounts to $2(x+1)=x$

13.(C) 14.(D) 15.(A)

Let $f(x)=ax^2+bx+c$

$$f(0)=c, f(1)=a+b+c, f(-1)=a-b+c$$

$$b=\frac{f(1)-f(-1)}{2}, a=\frac{f(1)+f(-1)}{2}-f(0)$$

Maximum value of $ax+b=a+b=2$

$\Rightarrow \quad f(1)-f(0)=2;\ f(1)-f(0)\le 1-(-1) \quad \Rightarrow\ f(1)=1$ and $f(0)=-1 \ \Rightarrow\ c=-1$

Least value of $f(x)=f(0)=-1$

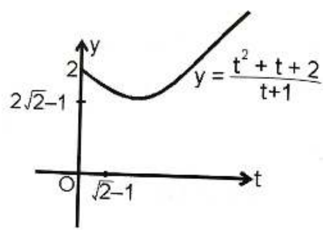
$\Rightarrow \quad -\frac{b}{2a}=0 \ \Rightarrow\ b=0 \ \Rightarrow\ f(1)=f(-1)=1 \quad \therefore\ a=2$

16.(B) 17.(A) 18.(D)

Put $x^2=t$

$$t^2-(k-1)t+2-k=0$$

$$k=\frac{t^2+t+2}{t+1} \qquad t\ge 0$$



From graph it is clear that

$k\in\{2\sqrt{2}-1\}\cup(2,\infty), 2$ real and distinct roots

$k\in\{2\}, 3$ real and distinct roots

$k\in(2\sqrt{2}-1, 2), 4$ real and distinct roots.

19.(ABCD) $a=\frac{1}{3}[f(2)-f(1)],\ c=\frac{1}{3}[f(2)-4f(1)]$, So, $f(3)=9a-c=\frac{1}{3}[8f(2)-5f(1)]$

20.(ABC) $\sin\theta, \cos\theta=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$\sin^2\theta+\cos^2\theta=1\Rightarrow b^2-2ac=a^2\Rightarrow b^2=a(a+2c)$ and $b^2-4ac=a(a-2c)$

So, $a, a-2c, \mathrm{a}+2\mathrm{c}$ are all perfect squares.

21.(AD) Let α,β,γ be the roots of the cubic then $AM\ge GM$ gives $-a\ge 6$ and real roots of $6x^2-24x-4a=0$ gives the condition $24^2+24\times 4a\ge 0\Rightarrow a=-6$ So, $\alpha=\beta=\gamma=2$

22.(AC) $P(x)-Q(x)=x(a-c)(x^2-1)$

As $P(0)=Q(0)=1$ So, $x=-1$ and $x=1$ are the common roots.

23.(AD) Let the roots be α,β,γ then $\Sigma\alpha=-a,\ \Sigma\alpha\beta=b,\ \alpha\beta\gamma=-c$

Now, $(a+b+c+1)=-45\Rightarrow -a-b-c-1=45$

i.e., $\Sigma\alpha-\Sigma\alpha\beta+\alpha\beta\gamma-1=45=(\alpha-1)(\beta-1)(\gamma-1)$ or $(\alpha-1)(\beta-1)(\gamma-1)=3\times3\times5$

So, $\alpha=4,\ \beta=4,\ \gamma=6$

24.(ABD) $(x-1)\underbrace{\left(2x^2+3x+5\right)}_{\text{Img. roots}}=0$ and $(x-1)\left(ax^2+bx+c\right)=0$ So, $a+b+c=0$ or $\frac{a}{2}=\frac{b}{3}=\frac{c}{5}$

25.(ABCD) Putting $x=0,1,\frac{1}{2}$ one by one in $f(x)$, we have $|c|\le 1$, $|a+b+c|\le 1$, $\left|\frac{a}{4}+\frac{b}{2}+c\right|\le 1$

So, $-1\le c\le 1$, $-1\le a+b+c\le 1$ and $-4\le a+2b+4c\le 4$

Rearranging these inequalities, we obtain $-8\le 3a+2b\le 8$, $|a|\le 8$, $|b|\le 8$, $|c|\le 1$

26.(ACD) Let $\sqrt{2x-1}=t$ then the given equation becomes $|t+1|+|t-1|=A\sqrt{2}$

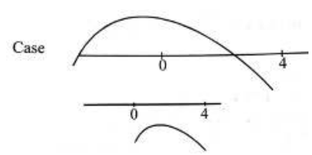
Now consider different cases, $t\le -1$, $-1<t<1$ and $t\ge 1$ etc.

27.(CD) According to given condition, $\frac{\left(b^2+4\right)}{2}\le\frac{-(4-4c)}{2}$; $c\ge\frac{b^2+8}{4}$; $c_{\min}=\frac{8}{4}=2$

28.(CD) $f(x)=\frac{1}{2bx^2-x^4-3b^2}$ $x\in[-2,1]$

Let $g(t)=2bt-t^2-3b^2$ $t\in[0,4]$

$g(t)<0\,\forall t\in[0,4]$



Case-I if $b\le 0$, then $g(t)$ will decrease in $[0,4]$

Maximum value of $f(t)=f(4)=\frac{1}{8b-16-3b^2}$

29.(BD) $a>0, b^2\le 4ac$

$b>0, c^2\le 4ab$

$c>0, a^2\le 4bc$

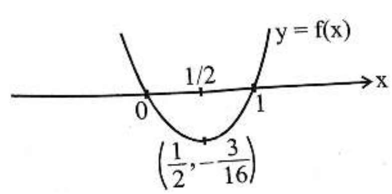
$\Rightarrow$ $a^2+b^2+c^2<4(ab+bc+ca)$ [$\because$ all equal is impossible]

Also $a^2+b^2+c^2>ab+bc+ca$ ($\because$ a, b, c are distinct)

$\Rightarrow$ $\frac{a^2+b^2+c^2}{ab+bc+ca}\in(1,4)$

30.(BCD) $x^4-2x^3+2x^2-x=-\frac{a}{8}$

Let $f(x)=x^4-2x^3+2x^2-x=x(x-1)\left(x^2-x+1\right)$



$f'(x)=4x^3-6x^2+4x-1$

$f''(x)=12x^2-12x+4>0$ $\forall x\in R$

$f'\left(\frac{1}{2}\right)=0$

Also $f(1-x)=f(x)$

If $-\frac{a}{8}\ge-\frac{3}{16}$ $\Rightarrow$ $a\le\frac{3}{2}$

Then equation has two real roots $\alpha, 1-\alpha$

$\Rightarrow$ sum of all no real roots $= 2-1=1$

If $a > \dfrac{3}{2}$, then equation has 4 non real roots

$\Rightarrow$ sum of all no real roots = 2.

31. $A-T; B-R,S; C-P,S; D-T$

(A) D=$\left(2ab\cos C\right)^2 - 4a^2b^2 = -4a^2b^2\sin^2 C < 0$

(B) $D = 4\left(b^2-ax\right) > 0$ $\left[\because \dfrac{a+c}{2} = b\sqrt{ac} \Rightarrow b^2 > ac\right]$

And $f(x) = ax^2 + abx + c > 0 \quad \forall \quad x \geq 0$

Hence $f(x)=0$ has both negative roots.

(C) $f(x) = x^2 - (a+1)x - \left(a^2+4\right)$

$f(0) < 0$

(D) $\sqrt{ac} > b \Rightarrow b^2 - ac < 0$; $D = 4\left(b^2-ac\right) < 0$

32. $A-Q; B-R; C-P; D-T$

$\sum \alpha\beta = \dfrac{2^2-(6)}{2} = -1$

$11 - 3\alpha\beta\gamma = 2\left(6-(1)\right) = 14 \Rightarrow \alpha\beta\gamma = -1$

$\therefore$ $x^2 - 2x^2 - x + 1 = 0 \begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}$

$\therefore$ $x^3 = 2x^2 + x - 1$ for $x = \alpha, \beta, \gamma$

$\Rightarrow$ $x^4 = 2x^3 + x^2 - x = 5x^2 + x - 2 \quad \Rightarrow \quad \alpha^4 + \beta^4 + \gamma^5 = 5(6) + 2 - 6 = 26$

$x^5 = 5x^3 + x^2 - 2x = 11x^2 + 3x - 5$ for $x = \alpha, \beta, \gamma$

$\Rightarrow$ $\alpha^5 + \beta^5 + \gamma^5 = 11(6) + 3(2) - 15 = 57$

$x^6 = 11x^3 + 3x^2 - 5x = 25x^2 + 6x - 11$ for $x = \alpha, \beta, \gamma$

$\Rightarrow$ $\alpha^6 + \beta^6 + \gamma^6 = 25(6) + 6(2) - 33 = 129$

$-\left(4-\alpha^2\right) + \left(4-\beta^2\right)\left(4-\gamma^2\right) = -2(2-\alpha)(2-\beta)(2-\gamma)(2+\alpha)(2+\beta)(2+\gamma)$

$= \left(8 - 2(4) - 2 + 1\right)\left((-8-8+2+1)\right) = (-1)(-13) = 13$

33.(2) Let $[x] = t, \{x\} = y, 0 \leq y < 1, x = t + y, x \neq 0$

Simplifying, $8t^2 = 11ty + 10y^2 < 11t + 10$ or $8t^2 - 11t - 10 < 0 \Rightarrow -\dfrac{5}{8} < t < 2$

So, $t = 1$ and $y = \dfrac{1}{2}$

34.(0) (Hint: Roots are 1, 2, 3)

35.(5) $P\left(x^5\right)=\left(x^{20}-1\right)+\left(x^{15}-1\right)+\left(x^{10}-1\right)+\left(x^5-1\right)+5$

$P(x)=\dfrac{x^5-1}{x-1}$. Hence, remainder = 5

36.(1) $(x-a)(x-12)+2=0$ should have integral roots.

So, $(x-a)(x-12)=-2=1\times-2$ or -1×2 or -2×1 ; So, $a=9,15$

37.(0) Notice that a, b, c are roots of the equation $x^3-2x^2-x+2=0$

Hence, $x^3-2x^2-x+2=(x-a)(x-b)(x-c)$ *and* Take $\lim\limits_{x\to1}$ on both sides.

38.(4) $f(x)$ will also be a factor of $3\left(x^4+6x^2+25\right)-\left(3x^4+4x^2+28x+5\right)=14\left(x^2-2x+5\right)$

So, $f(x)=x^2-2x+5=(x-1)^2+4$

39.(5) $\Delta>0 \Rightarrow (m-2)^2-\left(m^2-3m+3\right)>0$

$\Rightarrow \quad m<1 \quad \Rightarrow \quad m\in[-1,1)$

$$m\left(\frac{x_1^2}{1-x_1}+\frac{x_2^2}{1-x_2}\right)=m\left[\frac{x_1^2+x_2^2-x_1x_2\left(x_1+x_2\right)}{\left(1-x_1\right)\left(1-x_2\right)}\right]$$

$$=m\left[\frac{\left(x_1+x_2\right)^2-2x_1x_2-x_1x_2\left(x_1+x_2\right)}{\left(1-x_1\right)\left(1-x_2\right)}\right]=m\left[\frac{4(m-2)^2-(2-2(m-2))\left(m^2-3m+3\right)}{1+2(m-2)+m^2-3m+3}\right]$$

$$=\left[\frac{2m\left[2\left(m^2-4m+4\right)+(m-3)\left(m^2-3m+3\right)\right]}{m^2-m}\right]=2\left(m^2-3m+1\right)\in(-2,10]$$

40.(7) $p^2-20(66p-1)=k^2$

$\Rightarrow \quad (5p-132)^2-k^2=17404$

$\Rightarrow \quad (5p-132-k)(5p-132+k)=2^2\times10\times229$

$5p-132-k=2\times19$

$5p-132+k=2\times229$

$\Rightarrow \quad p=76$

And $\quad 5p-132-k=2$

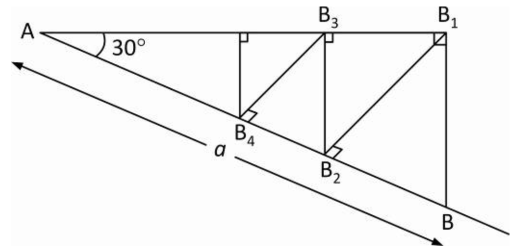
$5p-132+k=2\times19\times229$

Gives on integer solution for p

Hence, $\quad p=76$

TRIGONOMETRY

1.(B) Suppose $B_1, B_2, B_3, B_4 \ldots$ are the feet of perpendiculars respectively.



$\therefore \quad BB_1 = a\sin 30°,\ AB_1 = a\cos 30°$.

$B_1B_2 = AB_1 \sin 30° = a\cos 30° \sin 30°$

$AB_2 = AB_1 \cos 30° = a\cos 30° \cos 30°$

$B_2B_3 = a\cos^2 30° \sin 30°,\ AB_3 = a\cos^3 30°$ and so on.

$\therefore$ Length of the infinite polygonal line

$= BB_1 + B_1B_2 + \ldots = a \sin 30° + a \sin 30° \cos 30° + a \sin 30° \cos^2 30° + \ldots$

$$= \frac{a}{2}\left\{1 + \frac{\sqrt{3}}{2} + \left(\frac{\sqrt{3}}{2}\right)^2 + \ldots\right\} = a\left(2+\sqrt{3}\right)$$

2.(B) $\frac{1}{2}\left(1-\cos A + 1 - \cos B + 1 - \cos C\right) = \frac{3}{2} - \frac{1}{2}\left(\cos A + \cos B + \cos C\right) \qquad \ldots\text{(i)}$

Let $\cos A + \cos B + \cos C = \lambda$, $\quad \therefore \quad 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + 1 - 2\sin^2\frac{C}{2} = \lambda$,

$\Rightarrow \quad 2\sin^2\frac{C}{2} - 2\cos\frac{A-B}{2}\sin\frac{C}{2} + \lambda - 1 = 0$.

$\therefore \quad 4\cos^2\frac{A-B}{2} - 8(\lambda - 1) \geq 0 \qquad \left(\because \ \sin\frac{C}{2} \text{ is real}\right), \qquad \Rightarrow \qquad \lambda \leq \frac{2+\cos^2\frac{A-B}{2}}{2} = \frac{2+1}{2} = \frac{3}{2}$

$\therefore$ From (i) : The least value of $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = \frac{3}{2} - \frac{1}{2}.\frac{3}{2} = \frac{3}{4}$

3.(D) Given : $A < B < C < D \quad \ldots\text{(i)}$ and

$\sin A = \sin B = \sin C = \sin D = k = (+)ve \qquad \ldots\text{(ii)}$

Now, $\quad B = \pi - A,\ C = 2\pi + A,\ D = 3\pi - A$

We have chosen the values so as to satisfy the conditions (i) and (ii).

$\therefore \quad 4\sin\frac{A}{2} + 3\sin\frac{1}{2}(\pi - A) + 2\sin\frac{1}{2}(2\pi + A) + \sin\frac{1}{2}(3\pi - A)$

$= 4\sin\frac{A}{2} + 3\cos\frac{A}{2} - 2\sin\frac{A}{2} - \cos\frac{A}{2} = 2\left(\sin\frac{A}{2} + \cos\frac{A}{2}\right)$

$= 2\sqrt{\sin^2\frac{A}{2} + \cos^2\frac{A}{2} + 2\sin\frac{A}{2}\cos\frac{A}{2}} = 2\sqrt{1+\sin A} = 2\sqrt{1+k}$

4.(C) Let each ratio is equal to $\frac{1}{k}$

$\therefore \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = k\left\{\sin\theta + \sin\left(\theta + \frac{2\pi}{3}\right) + \sin\left(\theta + \frac{4\pi}{3}\right)\right\}$

$= k\left\{2\sin\left(\theta + \frac{2\pi}{3}\right)\cos\frac{2\pi}{3} + \sin\left(\theta + \frac{2\pi}{3}\right)\right\} = 0 \qquad \left(\text{As } \cos\frac{2\pi}{3} = \frac{-1}{2}\right)$

$\therefore \quad \sum xy = xy + yz + zx = 0$.

5.(A) $u^2 = (a^2+b^2) + 2\left\{a^2b^2(1-2\sin^2\theta\cos^2\theta) + \sin^2\theta\cos^2\theta(a^4+b^4)\right\}^{1/2}$

$= (a^2+b^2) + 2\left\{a^2b^2 + \sin^2\theta\cos^2\theta(a^2-b^2)^2\right\}^{1/2} = (a^2+b^2) + \left\{4a^2b^2 + \sin^2 2\theta(a^2-b^2)^2\right\}^{1/2}$

When $\sin 2\theta = 0$ or 1, then u^2 is min. or maximum respectively.

$\therefore$ Maximum $u^2 = (a^2+b^2)+(a^2+b^2) = 2(a^2+b^2)$ and Minimum $u^2 = a^2+b^2+2ab$.

$\therefore$ Difference is $a^2+b^2-2ab = (a-b)^2$

6.(D) Given : $\sin x + \sin y = \sin(x+y)$

$\Rightarrow \quad 2\sin\frac{x+y}{2}\left(\cos\frac{x-y}{2} - \cos\frac{x+y}{2}\right) = 0 \Rightarrow 4\sin\frac{x+y}{2}\sin\frac{x}{2}\sin\frac{y}{2} = 0$

$\Rightarrow \quad \sin\frac{x+y}{2} = 0$ or $\sin\frac{x}{2} = 0$ or $\sin\frac{y}{2} = 0$ **....(i)**

Given : $|x|+|y| = 1$ **....(ii)** $|x| \le 1$ and $|y| \le 1$

From (i) : $x+y=0,\ x=0,\ y=0$

Putting $x=0$ in (ii) : $y=\pm 1$ and also putting $y=0$ in (ii) : $x=\pm 1$

Again, putting $x+y=0 \Rightarrow y=-x$ in (ii) : $2|x|=1 \Rightarrow x=\pm\frac{1}{2}$

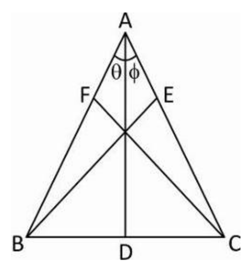
We get 6 pairs of (x, y) such as $(0,\pm 1), (\pm 1, 0), \left(\frac{1}{2}, -\frac{1}{2}\right), \left(-\frac{1}{2}, \frac{1}{2}\right)$

7.(B) $T_n = \cot^{-1}\left(2^{n+1} + \frac{1}{2^n}\right) = \cot^{-1}\frac{2^{n+1}.2^n+1}{2^n} = \tan^{-1}\frac{2^n(2-1)}{1+2^{n+1}.2^n} = \tan^{-1}\frac{2^{n+1}-2^n}{1+2^{n+1}.2^n} = \tan^{-1}2^{n+1} - \tan^{-1}2^n$.

Putting $n = 1, 2, 3, \ldots, n$ and then adding, $S_n = \tan^{-1}2^{n+1} - \tan^{-1}2$

$\therefore \quad S_\infty = \tan^{-1}\infty - \tan^{-1}2 = \frac{\pi}{2} - \tan^{-1}2 = \cot^{-1}2 = \tan^{-1}\frac{1}{2}$

8.(A) $\frac{AD}{BD} = \frac{\sin 60°}{\sin\theta}$ and $\frac{AD}{DC} = \frac{\sin 45°}{\sin\phi}$

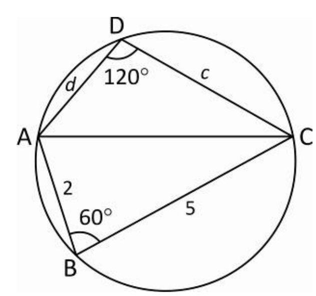


$\therefore \quad \frac{DC}{BD} = \frac{AD}{BD} \times \frac{DC}{AD} = \frac{\sin 60°}{\sin 45°} \cdot \frac{\sin\phi}{\sin\theta}$

$\Rightarrow \quad \frac{3}{1} = \frac{\sqrt{3}/2}{1/\sqrt{2}} \cdot \frac{\sin\phi}{\sin\theta} \qquad \therefore \qquad \frac{\sin\theta}{\sin\phi} = \frac{1}{\sqrt{6}}$

9.(C) Given : $AB = 2,\ BC = 5$ and $\angle ABC = 60°$

$\angle CDA = 180° - 60° = 120°$ ($\because$ $ABCD$ is cyclic quadrilateral).



Let $CD = c$ and $DA = d$.

$\therefore$ Area of quadrilateral ABCD = Area of ΔABC

+ Area of $\Delta ACD = \frac{1}{2}AB.BC\sin 60° + \frac{1}{2}CD.DA\sin 120°$

$= \frac{1}{2}.2.5.\frac{\sqrt{3}}{2} + \frac{1}{2}.cd.\frac{\sqrt{3}}{2} = 4\sqrt{3}$ (Given)

Thus $\dfrac{\sqrt{3}}{4}cd = 4\sqrt{3} - \dfrac{5\sqrt{3}}{2} = \dfrac{3\sqrt{3}}{2} \Rightarrow cd = 6$ ....**(i)**

Now, $AB^2 + BC^2 - 2AB \,.\, BC \cos 60° = AC^2 = CD^2 + DA^2 - 2CD \,.\, DA \cos 120°$.

$\Rightarrow \quad 4 + 25 - 2.2\,.\,5\,.\,\dfrac{1}{2} = c^2 + d^2 + cd \Rightarrow c^2 + d^2 + cd = 19$ ....**(ii)**

From (i) and (ii) : $c^2 + d^2 = 13$ and $c^2 d^2 = 36$

$\therefore \quad c^2$ and d^2 are the roots of $t^2 - 13t + 36 = 0$

$\therefore \quad t = 9, 4 \Rightarrow c^2 = 9, d^2 = 4$ or $c^2 = 4, d^2 = 9$

Hence, $c = 3, d = 2$ or $c = 2, d = 3$. Thus the other two sides are 2 and 3.

10.(C) $3\sin x - 4\sin^3 x - k = 0 \Rightarrow \sin 3x - k = 0$ ....**(i)**

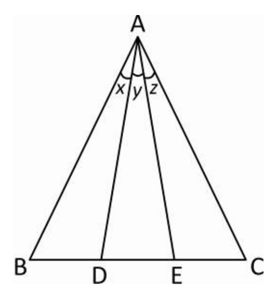
$\because \quad A$ and B satisfy (i),

$\therefore \quad \sin 3A - k = 0$ and $\sin 3B - k = 0$, i.e.,

$\sin 3A = \sin 3B \Rightarrow 3A = 180° - 3B (\because \ A > B) \Rightarrow A + B = 60°$

$\therefore \quad C = 180° - 60° = \dfrac{2\pi}{3}$

11.(A) From $\Delta ADC, \dfrac{\sin(y+z)}{DC} = \dfrac{\sin C}{AD}$



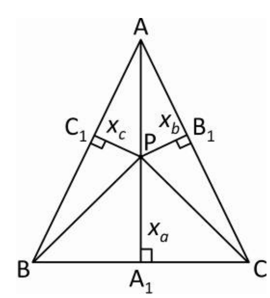
From $\Delta ABD, \dfrac{\sin x}{BD} = \dfrac{\sin B}{AD}$

From $\Delta AEC, \dfrac{\sin z}{EC} = \dfrac{\sin C}{AE}$

And from $\Delta ABE, \dfrac{\sin(x+y)}{BE} = \dfrac{\sin B}{AE}$

$\therefore \quad \dfrac{\sin(x+y)\sin(y+z)}{\sin x \sin z} = \dfrac{BE}{AE} \times \dfrac{DC}{AD} \times \dfrac{AD}{BD} \times \dfrac{AE}{EC} = \dfrac{2BD \times 2EC}{BD \times EC} = 4$

12.(A) Let A_1, B_1, C_1 be the feet of altitudes drawn from P to sides BC, CA, AB respectively.



Given : $PA_1 = x_a, PB_1 = x_b$ and $PC_1 = x_c$

$\therefore \quad \Delta ABC = \Delta BPC + \Delta CPA + \Delta APB$

$\Rightarrow \quad \dfrac{\sqrt{3}}{4}\,.\,4 = \dfrac{1}{2}\,.\,2(x_a + x_b + x_c)$

$\therefore \quad x_a + x_b + x_c = \sqrt{3}$

13.(B) We know that $\cos\theta = \dfrac{1 - \tan^2\frac{\theta}{2}}{1 + \tan^2\frac{\theta}{2}}$ and $\sin\theta = \dfrac{2\tan\frac{\theta}{2}}{1 + \tan^2\frac{\theta}{2}}$

$\therefore \quad \tan\dfrac{\theta}{2}, \tan\dfrac{\phi}{2}$ (are unequal) are the roots of $(x - a)\dfrac{1 - t^2}{1 + t^2} + y\,.\,\dfrac{2t}{1 + t^2} = a \Rightarrow xt^2 - 2yt + 2a - x = 0$

$\therefore \quad \tan\dfrac{\phi}{2}\tan\dfrac{\theta}{2} = \dfrac{2a - x}{x}$ ....**(i)** and $\quad tan\dfrac{\theta}{2}+tan\dfrac{\phi}{2}=\dfrac{2y}{x}$ $\quad$ **....(ii)**

Now $\quad \left(tan\dfrac{\theta}{2}-tan\dfrac{\phi}{2}\right)^2=\left(tan\dfrac{\theta}{2}+tan\dfrac{\phi}{2}\right)^2-4tan\dfrac{\theta}{2}tan\dfrac{\phi}{2}$. But $tan\dfrac{\theta}{2}-tan\dfrac{\phi}{2}=2e$

$\therefore \quad (2e)^2=\left(\dfrac{2y}{x}\right)^2-4\left(\dfrac{2a-x}{x}\right) \Rightarrow 4e^2=\dfrac{4y^2}{x^2}-4\left(\dfrac{2a-x}{x}\right)$

$\Rightarrow \quad e^2=\dfrac{y^2}{x^2}-\dfrac{2a-x}{x}$, i.e., $y^2=2ax-\left(1-e^2\right)x^2$

14.(B) $\left(x^2-\dfrac{3+x}{2}\right)+(x+1)sin\dfrac{\pi x}{6}=0 \Rightarrow \left(2x^2-x-3\right)+2(x+1)sin\dfrac{\pi x}{6}=0$

$\Rightarrow \quad (x+1)\left\{(2x-3)+2\,sin\dfrac{\pi x}{6}\right\}=0 \Rightarrow x+1=0$ $\quad$ **....(i)**

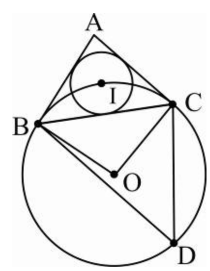
or $\quad (2x-3)+2\,sin\dfrac{\pi x}{6}=0$ $\quad$ **....(ii)**

From (i) : $x=-1$ and from (ii) : $2\,sin\dfrac{\pi x}{6}=3-2x \Rightarrow sin\dfrac{\pi x}{6}=\dfrac{3-2x}{2}$

On drawing the graphs of $y=sin\dfrac{\pi x}{6}$ and $y=\dfrac{3-2x}{2}$, we see that they intersect at the point $x=1,\ y=\dfrac{1}{2}$. But x = 1 does not lie in the given domain $-2\le x\le 0$

Hence $x=-1$ is the only solution.

15.(B) Let O be the centre of the required circle and R_1 is the radius and $BC=a$



Then $\angle BOC=2\angle BDC=(180°-A) \Rightarrow \angle BDC=90^o-\dfrac{A}{2}$

Now using Sine Law in ΔBOC

$$\frac{a}{\sin\angle BDC}=2R_1$$

$$R_1=\frac{a}{2}\sec\frac{A}{2}$$

16.(B) Let $\dfrac{AF}{FC}=\dfrac{\lambda}{1}$

And let $D(0, 0)$, $B(-a,0), C(a,0), A(h,k)$, then

$$E=\left(\frac{h}{2},\frac{k}{2}\right) \text{ and } F=\left(\frac{\lambda a+h}{\lambda+1},\frac{k}{\lambda+1}\right)$$

Now, $\quad B, E, F$ are collinear

$$\Rightarrow \quad \begin{vmatrix} \dfrac{\lambda a+h}{\lambda+1} & \dfrac{k}{\lambda+1} & 1 \\ \dfrac{h}{2} & \dfrac{k}{2} & 1 \\ -a & 0 & 1 \end{vmatrix}=0 \Rightarrow \lambda=\frac{1}{2} \Rightarrow \frac{AF}{AC}=\frac{1}{3}$$

17.(A) $\sum sin\,3A = 0$

$\Rightarrow \quad sin\,3A + sin\,3B + sin\,3C = 0 \quad \Rightarrow \quad 2\sin\left[\frac{3A+3B}{2}\right]\cos\left[\frac{3A-3B}{2}\right] + 2\sin\frac{3C}{2}\cos\frac{3C}{2} = 0$

$\Rightarrow \quad 2\sin\frac{3}{2}(\pi - C)\cos\frac{3}{2}(A-B) + 2\sin\frac{3C}{2}\cos\frac{3C}{2} = 0$

$\Rightarrow \quad 2\sin\left[\frac{3\pi}{2} - \frac{3C}{2}\right]\cos\left(\frac{3A}{2} - \frac{3B}{2}\right) + 2\sin\frac{3C}{2}\cos\frac{3C}{2} = 0$

$\Rightarrow \quad -2\cos\frac{3C}{2}\cos\left(\frac{3A}{2} - \frac{3B}{2}\right) + 2\sin\frac{3C}{2}\cos\frac{3C}{2} = 0$

$\Rightarrow \quad 2\cos\frac{3C}{2}\left[\sin\frac{3C}{2} - \cos\left(\frac{3A}{2} - \frac{3B}{2}\right)\right] = 0 \quad \Rightarrow \quad 2\cos\frac{3C}{2}\left[\sin 3\left[\frac{\pi}{2} - \frac{(A+B)}{2}\right] - \cos\left(\frac{3A}{2} - \frac{3B}{2}\right)\right] = 0$

$\Rightarrow \quad 2\cos\frac{3C}{2}\left[-\cos\left(\frac{3A}{2} + \frac{3B}{2}\right) - \cos\left(\frac{3A}{2} - \frac{3B}{2}\right)\right] = 0$

$\Rightarrow \quad -2\cos\frac{3C}{2}\left(2\cos\frac{3A}{2}\cos\frac{3B}{2}\right) = 0 \quad \Rightarrow \quad 4\,cos\frac{3A}{2}\,cos\frac{3B}{2}\,cos\frac{3C}{2} = 0$

$\Rightarrow \quad cos\frac{3A}{2}$ or $cos\frac{3B}{2}$ or $cos\frac{3C}{2} = 0 \quad \Rightarrow \quad \frac{3A}{2} = \frac{\pi}{2}$ or $\frac{3B}{2} = \frac{\pi}{2}$ or $\frac{3C}{2} = \frac{\pi}{2}$

$\Rightarrow \quad A = \frac{\pi}{3}$ or $B = \frac{\pi}{3}$ or $C = \frac{\pi}{3} \quad \Rightarrow$ At least one angle of ΔABC is $60°$

18.(A) We have $\left[sin^{-1}\,cos^{-1}\,sin^{-1}\,tan^{-1}\,x\right] = 1$

$\Rightarrow \quad 1 \le sin^{-1}\,cos^{-1}\,sin^{-1}\,tan^{-1}\,x \le \frac{\pi}{2} \quad \Rightarrow \quad sin\,1 \le cos^{-1}\,sin^{-1}\,tan^{-1}\,x \le 1$

$\Rightarrow \quad cos\;sin1 \ge sin^{-1}\,tan^{-1}\,x \ge cos1 \quad \Rightarrow \quad sin\;cos\;sin1 \ge tan^{-1}\,x \ge sin\;cos1$

$\Rightarrow \quad tan\;sin\;cos\;sin1 \ge x \ge tan\;sin\;cos1$ Hence $x \in [tan\;sin\;cos1,\; tan\;sin\;cos\;sin1]$

19.(B) $sin\,x + cosec\;x \ge 2$ when $x \in \left[0, \frac{\pi}{2}\right]$ and $tan\,y + cot\,y \ge 2$ when $x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow$ Solution of the equation possible only when $sin\,x + cosec\;x = 2$ and $tan\,y + cot\,y = 2$

$\Rightarrow \quad sin\,x = 1$ and $tan\,y = 1 \quad \Rightarrow \quad x = \frac{\pi}{2},\; y = \frac{\pi}{4} \quad \Rightarrow \quad tan\left(\frac{y}{2}\right) = tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1$

Note that $\left(\sqrt{2} - 1\right)$ satisfies $\alpha^2 + 2\alpha - 1 = 0$

20.(B) $\theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow tan\;\theta \in (0,1)$ and $cot\,\theta \in (1, \infty) \quad \Rightarrow \quad t_1 = (tan\,\theta)^{tan\,\theta} \in (0,1)$ and $t_2 = (tan\,\theta)^{cot\,\theta} \in (0,1)$

Also $t_2 < t_1$ since numbers between 0 and 1 decrease with increasing power.

$t_3 = (cot\,\theta)^{tan\,\theta} \in (1, \infty)$ and $t_4 = (cot\,\theta)^{cot\,\theta} \in (1, \infty)$

Also $t_4 > t_3$ since numbers greater than 1 increase with increasing power. So, $t_4 > t_3 > t_1 > t_2$

21.(D) $sin^2\,x + sin^2\left(x + \frac{\pi}{3}\right) + cos\,x\,cos\left(x + \frac{\pi}{3}\right) = sin^2\,x + \left(\frac{sin\,x}{2} + \frac{\sqrt{3}}{2}cos\,x\right)^2 + cos\,x\left(\frac{cos\,x}{2} - \frac{\sqrt{3}}{2}sin\,x\right)$

$= sin^2\,x + \frac{sin^2\,x}{4} + \frac{3}{4}cos^2\,x + \frac{\sqrt{3}}{2}sin\,x\,cos\,x + \frac{cos^2\,x}{2} - \frac{\sqrt{3}}{2}sin\,x\,cos\,x = \frac{5}{4}\left(sin^2\,x + cos^2\,x\right) = \frac{5}{4}$

$\Rightarrow \quad f(x)$ is a constant. So, its period is undefined.

22.(C) $\sin x+\sin y+\sin z-\sin t=(\sin y+\sin z)+(\sin x-\sin t)$

$$=2\sin\left(\frac{y+z}{2}\right)\cos\left(\frac{y-z}{2}\right)+2\sin\left(\frac{x-t}{2}\right)\cos\left(\frac{x+t}{2}\right)$$

$$=2\sin\left(\frac{y+z}{2}\right)\cos\left(\frac{y-z}{2}\right)+2\sin\left(\frac{-(y+z)}{2}\right)\cos\left(\frac{x+t}{2}\right)=2\sin\left(\frac{y+z}{2}\right)\left[\cos\left(\frac{y-z}{2}\right)-\cos\left(\frac{x+t}{2}\right)\right]$$

$$=2\sin\left(\frac{y+z}{2}\right)\left[2\sin\left(\frac{y-z+x+t}{4}\right)\sin\left(\frac{x+t-y+z}{4}\right)\right]=4\sin\left(\frac{y+z}{2}\right)\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x+z}{2}\right)$$

23.(B) Let $\cos^{-1}x=y \Rightarrow x=\cos y, \frac{1}{2}\le x\le 1 \Rightarrow 0\le y\le\frac{\pi}{3}$

$$\Rightarrow \quad \frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}=\frac{x}{2}+\frac{\sqrt{3}}{2}\sqrt{1-x^2}=\frac{\cos y}{2}+\frac{\sqrt{3}}{2}\sin y=\cos\left(\frac{\pi}{3}-y\right) \Rightarrow \cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)=\frac{\pi}{3}-y$$

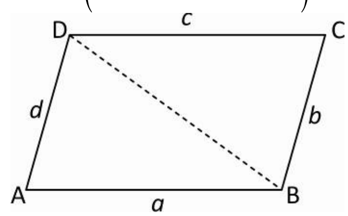
$$\Rightarrow \quad \cos^{-1}x+\cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)=y+\frac{\pi}{3}-y=\frac{\pi}{3}$$

24.(B) Since a circle can be inscribed in the quadrilateral, we have $a+c=b+d$

Since quadrilateral is cyclic, $C=\pi-A$

$$\cos A=\frac{a^2+d^2-BD^2}{2ad} \quad\Rightarrow\quad 2ad\cos A=a^2+d^2-\left(b^2+c^2-2bc\cos C\right)$$

$$\Rightarrow \quad 2(ad+bc)\cos A=a^2+d^2-b^2-c^2$$



$$\Rightarrow \quad \cos A=\frac{a^2+d^2-b^2-c^2}{2(ad+bc)}$$

$a+c=b+d \Rightarrow (a-d)=(b-c) \Rightarrow (a-d)^2=(b-c)^2$

$$\Rightarrow \quad a^2+d^2-b^2-c^2=2ad-2bc \quad\Rightarrow\quad \cos A=\frac{2(ad-bc)}{2(ad+bc)}=\frac{ad-bc}{ad+bc}$$

25.(D) $a_1+a_2\cos 2x+a_3\sin^2 x=1$

$$\Rightarrow \quad a_1+a_2\left(1-2\sin^2 x\right)+a_3\sin^2 x=1 \quad\Rightarrow\quad \sin^2 x(a_3-2a_2)+(a_1+a_2-1)=0$$

Since the above equation is satisfied by every real value of *x*, we have

$a_3-2a_2=0$ and $a_1+a_2-1=0$

3 unknowns and 2 equations $\Rightarrow$ infinite number of solutions

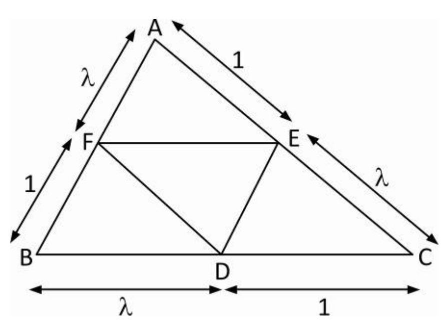
26.(BC) Case 1 : When $2c=a+b$ $\therefore$ $\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{b^2+c^2-(2c-b)^2}{2bc}=\frac{4bc-3c^2}{2bc}=\frac{4b-3c}{2b}$

Case 2 : When $2b=a+c$ $\therefore$ $\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{b^2+c^2-(2b-c)^2}{2bc}=\frac{4bc-3b^2}{2bc}=\frac{4c-3b}{2c}$

27.(BD) $ar(\Delta AEF)=\frac{1}{2}\times AF\times AE\sin A=\frac{1}{2}\left(\frac{\lambda c}{\lambda+1}\right)\left(\frac{b}{\lambda+1}\right)\sin A$

$$=\frac{\lambda}{(\lambda+1)^2}\left(\frac{1}{2}bc\sin A\right)=\frac{\lambda\Delta}{(\lambda+1)^2}$$

Similarly, $ar(\Delta BDF)=ar(\Delta CDE)=\frac{\lambda\Delta}{(\lambda+1)^2}$



Now, $\frac{ar(\Delta DEF)}{ar(\Delta ABC)}=\frac{2}{5} \Rightarrow \frac{\Delta-\frac{3\lambda\Delta}{(\lambda+1)^2}}{\Delta}=\frac{2}{5}$

or $1-\frac{3\lambda}{(\lambda+1)^2}=\frac{2}{5}$ or $\frac{\lambda}{(\lambda+1)^2}=\frac{1}{5}$ or $5\lambda=\lambda^2+2\lambda+1$ or $\lambda^2-3\lambda+1=0 \Rightarrow \lambda=\frac{3\pm\sqrt{5}}{2}$

28.(A) Greatest side is $a \Rightarrow \angle A=90° \Rightarrow b^2+c^2=a^2 \Rightarrow \sin C=\frac{c}{a} \Rightarrow \frac{2\tan\left(\frac{C}{2}\right)}{1+\tan^2\left(\frac{C}{2}\right)}=\frac{c}{a}$

Put $\tan\left(\frac{C}{2}\right)=t$ to get : $2at=c+ct^2$ or $ct^2-2at+c=0 \Rightarrow t=\frac{2a\pm\sqrt{4a^2-4c^2}}{2c}=\frac{2a\pm\sqrt{4b^2}}{2c}=\frac{a\pm b}{c}$

$\frac{a+b}{c}>\frac{c}{c}=1$ (Due to triangle inequality)

$\Rightarrow$ If $\tan\left(\frac{C}{2}\right)=\frac{a+b}{c}$, then $\tan\left(\frac{C}{2}\right)>1$ or $\frac{C}{2}>\frac{\pi}{4}$ or $C>\frac{\pi}{2}$ which is not possible.

$\Rightarrow \tan\left(\frac{C}{2}\right)=\frac{a-b}{c}$

29.(ABD) Since $\left|\tan^{-1}x\right|=\begin{cases}\tan^{-1}x & \text{if } 0\le\tan^{-1}x<\frac{\pi}{2}\\ -\tan^{-1}x & \text{if } -\frac{\pi}{2}<\tan^{-1}x<0\end{cases}=\begin{cases}\tan^{-1}x & \text{if } x\ge 0\\ -\tan^{-1}x & \text{if } x<0\end{cases}$

$\Rightarrow \left|\tan^{-1}x\right|=\tan^{-1}|x|$ for all $x\in R \Rightarrow \tan\left|\tan^{-1}x\right|=\tan\tan^{-1}|x|=|x|$ Hence, (A) is correct.

Likewise $\sin\left|\sin^{-1}x\right|=\sin\sin^{-1}|x|=|x|$ for all $|x|\le 1$. Hence (D) is correct.

$\cot\left|\cot^{-1}x\right|=\cot\cot^{-1}x=x$ Hence (B) is correct.

Since $|\tan x|$ is not necessarily always equal to $\tan|x|$ Hence $\tan^{-1}|\tan x|\ne|x|$

30.(BD) $\left(\sin^{-1}x\right)^3+\left(\cos^{-1}x\right)^3=$

$\left(\sin^{-1}x+\cos^{-1}x\right)^3-3\sin^{-1}x\cos^{-1}x\left(\sin^{-1}x+\cos^{-1}x\right)$

$=\left(\frac{\pi}{2}\right)^3-3\sin^{-1}x\cos^{-1}x\left(\frac{\pi}{2}\right)=\frac{\pi^3}{8}-\frac{3\pi}{2}\sin^{-1}x\left(\frac{\pi}{2}-\sin^{-1}x\right)$

$=\frac{\pi^3}{8}-\frac{3\pi^2}{4}\sin^{-1}x+\frac{3\pi}{2}\left(\sin^{-1}x\right)^2=\frac{\pi^3}{8}+\frac{3\pi}{2}\left[\left(\sin^{-1}x\right)^2-\frac{\pi}{2}\sin^{-1}x\right]$

$=\frac{\pi^3}{8}+\frac{3\pi}{2}\left[\left(\sin^{-1}x-\frac{\pi}{4}\right)^2\right]-\frac{3\pi^3}{32}=\frac{\pi^3}{32}+\frac{3\pi}{2}\left(\sin^{-1}x-\frac{\pi}{4}\right)^2$

$\Rightarrow$ The least value is $\frac{\pi^3}{32}$ and since $\left(\sin^{-1}x-\frac{\pi}{4}\right)^2\le\left(\frac{3\pi}{4}\right)^2$

$\Rightarrow$ The greatest value is $\frac{\pi^3}{32}+\frac{9\pi^2}{16}\times\frac{3\pi}{2}=\frac{7\pi^3}{8}$

31.(AD) For $f(x)$ to be defined, $\left[1+\cos^2 x\right] \in (-\infty, -1] \cup [1, \infty)$

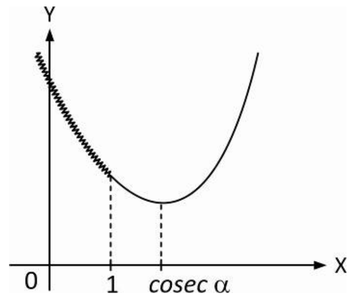
$\cos^2 x \in [0, 1] \Rightarrow 1+\cos^2 x \in [1, 2] \Rightarrow \left[1+\cos^2 x\right] \in \{1, 2\}$

As $\left[1+\cos^2 x\right] \in (-\infty, 1] \cup [1, \infty)\ \forall\ x \in R$, the domain of f is R.

As we have seen above $\left[1+\cos^2 x\right] \in \{1, 2\}\ \forall\ x \in R \quad \Rightarrow \quad$ The range of f is $\left\{\sec^{-1} 1, \sec^{-1} 2\right\}$

32.(CD) x-coordinate corresponding to the vertex is given by

$$x = \frac{1}{\sin\alpha} = \operatorname{cosec}\alpha > 1.$$



The graph of $f(x) = (\sin\alpha)x^2 - 2x + (b-2),\ \forall\ x \le 1$ is shaded above.

Minimum of $f(x)$ must be greater than 0.

Minimum occurs at $x = 1$.

$\Rightarrow \quad \sin\alpha - 2 + b - 2 \ge 0 \Rightarrow b \ge 4 - \sin\alpha,\ \alpha \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$

$\Rightarrow \quad b > 3$

33.(BD) $\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{3\sqrt{2}}{5}$ and $\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{4\sqrt{2}}{5}$

Dividing the above two equations we get : $\tan\left(\frac{\alpha+\beta}{2}\right) = \frac{3}{4}$

$$\Rightarrow \quad \sin(\alpha+\beta) = \frac{2\tan\left(\frac{\alpha+\beta}{2}\right)}{1+\tan^2\left(\frac{\alpha+\beta}{2}\right)} = \frac{2\times\frac{3}{4}}{1+\frac{9}{16}} = \frac{24}{25} \text{ and } \cos(\alpha+\beta) = \frac{1-\tan^2\left(\frac{\alpha+\beta}{2}\right)}{1+\tan^2\left(\frac{\alpha+\beta}{2}\right)} = \frac{1-\frac{9}{16}}{1+\frac{9}{16}} = \frac{7}{25}$$

34.(ABC) Triangle must be obtuse because in an acute angled triangle all the four centres lie inside the triangle and in a right angled triangle two lie inside and other two lie on the triangle.

In an obtuse angled triangle circumcentre and orthocentre lie outside the triangle.

If the triangle is isosceles obtuse angled, then incentre may be collinear with the other three centres.

$$r = \frac{\Delta}{s},\ r_1 = \frac{\Delta}{s-a},\ r_2 - \frac{\Delta}{s-b},\ r_3 = \frac{\Delta}{s-c} \qquad \Rightarrow \qquad r < r_1, r_2 \text{ and } r_3.$$

35.(AB) Applying the operations $R_1 \to R_1 - R_3$ and $R_2 \to R_2 - R_3$, we have $\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ \cos^2\theta & \sin^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$

$$\Rightarrow \quad \left(1+4\sin 4\theta + \sin^2\theta\right) - \left(-\cos^2\theta\right) = 0 \text{ or } 1+4\sin 4\theta + 1 = 0 \text{ or } \sin 4\theta = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right)$$

$$\Rightarrow \quad 4\theta = n\pi + (-1)^n\left(-\frac{\pi}{6}\right) \text{ or } \theta = \frac{n\pi}{4} + \frac{1}{4}(-1)^n\left(-\frac{\pi}{6}\right)$$

$$0 \le \theta \le \frac{\pi}{2} \Rightarrow \theta = \frac{7\pi}{24}, \frac{11\pi}{24}$$

36.(BD) $x = \sin\left(2\tan^{-1} 2\right)$

Assume that $\tan^{-1} 2 = \theta \Rightarrow x = \sin(2\theta) = \dfrac{2\tan\theta}{1+\tan^2\theta} = \dfrac{2(2)}{1+2^2} = \dfrac{4}{5} \quad \Rightarrow \quad y = \sin\left(\dfrac{1}{2}\tan^{-1}\dfrac{4}{3}\right)$

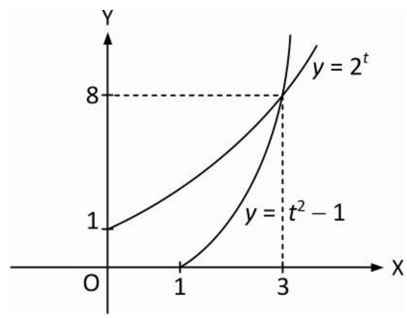
Assume that $\tan^{-1}\dfrac{4}{3} = \alpha \Rightarrow y = \sin\left(\dfrac{\alpha}{2}\right)$

Now $\quad \tan\alpha = \dfrac{4}{3} \Rightarrow \sin\alpha = \dfrac{4}{5}$ and $\cos\alpha = \dfrac{3}{5}$

$\cos\alpha = 1 - 2\sin^2\dfrac{\alpha}{2} \Rightarrow \dfrac{3}{5} = 1 - 2\sin^2\dfrac{\alpha}{2} \Rightarrow \sin\dfrac{\alpha}{2} = \pm\dfrac{1}{\sqrt{5}}$

Since $\quad \alpha = \tan^{-1}\dfrac{4}{3}, 0 < \alpha < \dfrac{\pi}{2} \Rightarrow \sin\left(\dfrac{\alpha}{2}\right) > 0 \Rightarrow y = \sin\dfrac{\alpha}{2} = \dfrac{1}{\sqrt{5}}$

37.(BD) $\left(1-\tan^2\theta\right)\left(1+\tan^2\theta\right)\sec^2\theta + 2^{\tan^2\theta} = 0$



$\Rightarrow \qquad \left(1-\tan^2\theta\right)\left(1+\tan^2\theta\right) + 2^{\tan^2\theta} = 0$

$\Rightarrow \qquad \left(1-\tan^4\theta\right) + 2^{\tan^2\theta} = 0$

Put $\tan^2\theta = t$ to get :

$2^t = t^2 - 1,\ t > 0$

Solving the above equation graphically $t = 3$ is a solution

$\Rightarrow \qquad \tan^2\theta = 3$ or $\tan\theta = \pm\sqrt{3} \qquad \Rightarrow \qquad \theta = \pm\dfrac{\pi}{3}$ are solutions.

Also after $t = 3$ at some value of t, 2^t becomes above $t^2 - 1$

$\Rightarrow$ One root also exist after $t = 3$

Hence two solution in $\left(0, \dfrac{\pi}{2}\right)$ and in $\left(-\dfrac{\pi}{2}, 0\right)$

38.(AB) $f(x) = \tan 3x \cdot \cot x = \dfrac{3\tan x - \tan^3 x}{1 - 3\tan^2 x} \cdot \dfrac{1}{\tan x} = \dfrac{3 - \tan^2 x}{1 - 3\tan^2 x}$

Let $\quad y = \dfrac{3-\tan^2 x}{1-3\tan^2 x} \Rightarrow \tan^2 x = \dfrac{y-3}{3y-1} \qquad$ As $\quad \tan^2 x > 0, \dfrac{y-3}{3y-1} > 0 \Rightarrow y \in \left(-\infty, \dfrac{1}{3}\right) \cup (3, \infty)$

39.(AD) $\sin(B-C) = \dfrac{3}{5} \Rightarrow \cos(B-C) = \dfrac{4}{5} = \dfrac{1-t^2}{1+t^2}$, where $t = \tan\left(\dfrac{B-C}{2}\right)$

$\Rightarrow \qquad t^2 = \dfrac{1}{9} \Rightarrow t = \pm\dfrac{1}{3} = \tan\left(\dfrac{B-C}{2}\right) = \dfrac{b-c}{b+c}\cot\left(\dfrac{A}{2}\right)$

Put $\quad b = 2c$ to get $\pm\dfrac{1}{3} = \dfrac{1}{3}\cot\left(\dfrac{A}{2}\right) \Rightarrow \cot\left(\dfrac{A}{2}\right) = \pm 1$

$\cot\left(\dfrac{A}{2}\right)$ can't be negative $\qquad \Rightarrow \qquad \cot\left(\dfrac{A}{2}\right) = 1 = \cot\left(\dfrac{\pi}{4}\right) \Rightarrow \dfrac{A}{2} = \dfrac{\pi}{4}$ or $A = \dfrac{\pi}{2}$

$a = \sqrt{b^2 + c^2} = \sqrt{4c^2 + c^2} = \sqrt{5}c \Rightarrow a : c = \sqrt{5} : 1.$

40.(ABD) **(A)** $\tan 1 > \tan^{-1} \Rightarrow \tan 1 > \frac{\pi}{4}$

(B) $\sin 1 > \cos 1$

$\sin 57.3° > \cos 57.3°$

(C) $\tan 1 > \sin 1$ (not possible)

Because $\tan 57.3° > 1 > \sin 57.3°$

(D) $\cos 1 < \frac{\pi}{4} \Rightarrow \cos(\cos 1) > \cos\left(\frac{\pi}{4}\right)$

41.(BD) **(A)** $\tan 1 > 1$ and $\sin 1 < 1$, then $\log_{\sin 1} \tan 1 < 0$

(B) $1 + \tan 3 < 1$ and $\cos 1 < 1$, then $\log_{\cos 1}(1 + \tan 3) > 0$

(C) $\cos\theta + \sec\theta > 2$ and $\log_{10} 5 < 1$, then $\log_{\log_{10} 5}(\cos\theta + \sec\theta) < 0$

(D) $2\sin 18° < 1$ and $\tan 15° < 1$, then $\log_{\tan 15°} 2\sin 18° > 0$

42.(BD) $2\Sigma(\cos x \cos y) + 2\Sigma(\sin x \sin y) + 3 = 0$

$(\Sigma\cos x)^2 + (\Sigma\sin x)^2 = 0 \Rightarrow \Sigma\cos x = 0$ and $\Sigma\sin x = 0$

$\cos 3x + \cos 3y + \cos 3z = 4(\cos^3 x + \cos^3 y + \cos^3 z) - 3(\cos x + \cos y + \cos z) = 12\cos x \cos y \cos z$

43.(ABD) **(A)** $2(a + d) = 2(b + c)$

(B) $\tan 50° = \frac{1}{\tan 40°} = \frac{1 - \tan^2 20°}{2\tan 20°}$

$\Rightarrow \tan 20° + 2\tan 50° = \tan 70° \Rightarrow 2a + 2b = 2c$

(D) $\tan 20° - 2\tan 10° = \tan 20° \tan^2 10° > 0 \Rightarrow \tan 20° > 2\tan 10° \Rightarrow b > a$ and $d > c$

44.(ABCD) $\frac{(1 - \cot x)}{\sin^2 x} = (1 - \cot x)\cdot \operatorname{cosec}^2 x = (1 - \cot x)(1 + \cot^2 x)$

45.(ABCD) $4\sin 3x + 5 \geq 4\cos 2x + 5\sin x \Rightarrow (\sin x - 1)(4\sin x + 1)^2 \leq 0 \ \forall\ x \in R$

46.(ACD) $\cos x \cos 6x = -1$

Case-I : $\cos x = 1$ and $\cos 6x = -1$ Not possible

Case-2 : $\cos 6x = 1$ and $\cos x = -1 \Rightarrow x = (2n - 1)\pi, (n \in I)$

47.(ABCD) Verify in given intervals

48.(ABC) $2k = \sin^2 2x - 2\sin 2x - 2$

Let $\sin 2x = t$ $t \in [-1, 1]$

$2k = t^2 - 2t - 2 \Rightarrow k \in \left[-\frac{3}{2}, \frac{1}{2}\right]$

49.(BCD) $f(\theta) = \left(\cos\theta - \cos\frac{\pi}{8}\right)\left(\cos\theta - \cos\frac{3\pi}{8}\right)\left(\cos\theta - \cos\frac{5\pi}{8}\right)\left(\cos\theta - \cos\frac{7\pi}{8}\right) = \cos^4\theta - \cos^2\theta + \frac{1}{8}$

50.(CD) $x^2 - r(r_1r_2 + r_2r_3 + r_1r_3)x + (r_1r_2r_3 - 1) = 0$

$x^2 - (r_1r_2r_3)x + (r_1r_2r_3 - 1) = 0 \Rightarrow$ Roots are 1 and $r_1r_2r_3 - 1$

51.(BC) $D+E+F=\frac{\pi}{2}$

52.(ABC) If $\frac{r}{r_1}=\frac{r_2}{r_3} \Rightarrow \frac{s-a}{s}=\frac{s-c}{s-b} \Rightarrow a^2+b^2=c^2 \Rightarrow \angle C=90°$

53.(BC) $\angle BOC = 2\angle A$

$\angle BIC = \pi/2 + A/2$

$\angle BHC = \pi - A$

54.(AC) $\sqrt{3}x^2-4x+\sqrt{3}<0$

$\Rightarrow \quad (\sqrt{3}x-1)(x-\sqrt{3})<0 \quad \Rightarrow \quad \frac{1}{\sqrt{3}}<x<\sqrt{3}$

$30° < A,\ B < 60° \quad \Rightarrow \quad 60° < C < 120°$

55.(ABC) Use graphs of $\sin^{-1}(\sin x),\ \cos^{-1}(\cos x)$

56.(AC) $\cos^{-1}x=\tan^{-1}x \quad \Rightarrow \quad x\in[0,1]$

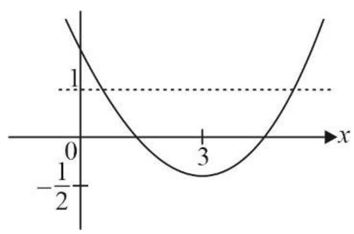
$\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)=\tan^{-1}x \quad \Rightarrow \quad x^2=\sqrt{1-x^2} \quad \Rightarrow \quad x^4+x^2-1=0 \ ; \ x^2=\frac{\sqrt{5}-1}{2}$

57.(AB) Check with the options

58.(ABD) $\sin^{-1}\left(x^2-6x+\frac{17}{2}\right)=\sin^{-1}k$

where $-1\le k\le 1$

$y=x^2-6x+\frac{17}{2}$



59.(ABD) $(\sin^{-1}x-\cos^{-1}x)((\sin^{-1}x)^2+(\cos^{-1}x)^2+2\sin^{-1}x\cos^{-1}x)=\frac{\pi^3}{16}$

$\Rightarrow \quad \sin^{-1}x-\cos^{-1}x=\frac{\pi}{4} \quad \Rightarrow \quad \cos^{-1}x=\frac{\pi}{8} \quad \Rightarrow \quad x=\cos\frac{\pi}{8}$

60.(2) $a\sin\theta-b\cos\theta=-\sin 4\theta=-2\sin 2\theta\cos 2\theta=-4\sin\theta\cos\theta\left(\cos^2\theta-\sin^2\theta\right)=-4\sin\theta\cos^3\theta+4\sin^3\theta\cos\theta$ ....(i)

$a\cos\theta+b\sin\theta=\frac{5-3\cos 4\theta}{2}=\frac{2+3-3\cos 4\theta}{2}=1+3\sin^2 2\theta$

$=\left(\sin^2\theta+\cos^2\theta\right)^2+3.\left(2\sin\theta\cos\theta\right)^2=\sin^4\theta+\cos^4\theta+14\sin^2\theta\cos^2\theta$ ....(ii)

Now solving, (i) and (ii) we get : $a=\cos^5\theta+10\cos^3\theta\sin^2\theta+5\cos\theta\sin^4\theta$ and

$b=5\sin\theta\cos^4\theta+10\cos^2\theta\sin^3\theta+\sin^5\theta$

Thus $\quad a+b=(\cos\theta+\sin\theta)^5$ and $a-b=(\cos\theta-\sin\theta)^5$

$\therefore \quad (a+b)^{2/5}+(a-b)^{2/5}=(\cos\theta+\sin\theta)^2+(\cos\theta-\sin\theta)^2=2$

61.(3) $s=\frac{(x+6)+(x+8)+14}{2}=x+14$

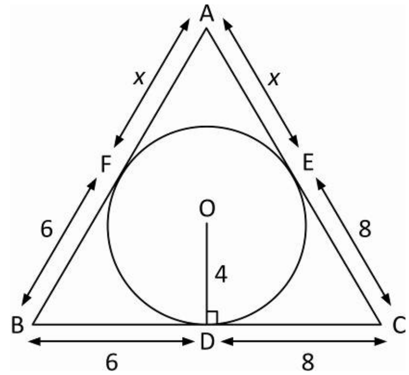
$\Rightarrow \quad \Delta=\sqrt{s(s-a)(s-b)(s-c)}=\sqrt{(x+14)(8)(6)(x)}$

Now $\quad r=\frac{\Delta}{s} \Rightarrow 4=\sqrt{\frac{48x}{x+14}}$

$\Rightarrow \quad 16(x+14)=48x$ or $x=7$

$\Rightarrow \quad \Delta=\sqrt{48\times 7\times 21}=84$

$\Rightarrow \quad \sqrt{\sqrt{\Delta-3}}=3$



62.(6) $tan\{x\}=cot\{x\}$

$\Rightarrow \quad \frac{tan\{x\}}{cot\{x\}}=1 \quad \Rightarrow \quad tan^2\{x\}=1 \quad \Rightarrow \quad tan\{x\}=\pm 1$

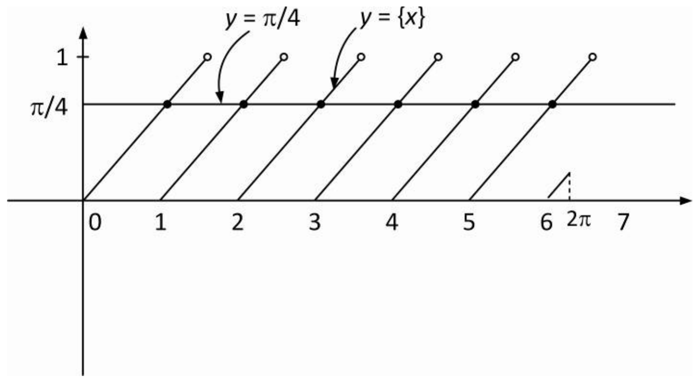
But $\{x\}\in(0,1) \Rightarrow tan\{x\} \to \in[0, tan 1) \quad \Rightarrow \quad tan\{x\}\neq -1$

$\therefore \quad tan\{x\}=1$

$\Rightarrow \quad \{x\}=tan^{-1}1=\frac{\pi}{4}$ $\quad$ ....(i)

But $x\in[0, 2\pi]$

$\therefore$ Clearly, (i) holds where graphs of $y=\{x\}$ and $y=\frac{\pi}{4}$ intersect each other in $[0, 2\pi]$ which are 6 points as shown below in diagram.



63.(6) $tan^{-1}x+cos^{-1}\frac{y}{\sqrt{1+y^2}}=sin^{-1}\left(\frac{3}{\sqrt{10}}\right) \quad \Rightarrow \quad tan^{-1}x+tan^{-1}\left(\frac{1}{y}\right)=tan^{-1}(3)$

$\Rightarrow \quad \frac{xy+1}{y-x}=3$ or $xy+1=3y-3x$ or $y=\frac{3x+1}{3-x}$

The positive integral solutions are $(1, 2), (2, 7) \quad \Rightarrow \quad \alpha=1+2=3$ and $\beta=2+7=9 \Rightarrow \beta-\alpha=9-3=6$

64.(4) $sin\,x+sin^2x+sin^3x=1$

$\Rightarrow \quad sin\,x(1+sin^2x)=1-sin^2x=cos^2x \quad \Rightarrow \quad sin^2x(1+sin^2x)^2=cos^4x$

$\Rightarrow \quad (1-cos^2x)(2-cos^2x)^2=cos^4x \quad \Rightarrow \quad (1-cos^2x)(4+cos^4x-4cos^2x)=cos^4x$

$\Rightarrow \quad 4-8cos^2x+5cos^4x-cos^6x=cos^4x \quad \Rightarrow \quad cos^6x-4cos^4x+8cos^2x=4$

65.(1) For $sin^{-1}\left(\frac{1+x^2}{2x}\right)$ to exist, $\left|\frac{1+x^2}{2x}\right|\le 1$

$\Rightarrow \quad |1+x^2|\le|2x| \Rightarrow 1+x^2\le 2|x| \Rightarrow |x|^2-2|x|+1\le 0 \Rightarrow (|x|-1)^2\le 0 \quad \Rightarrow \quad |x|=1$ or $x=\pm 1$

For x = 1, the equation is satisfied. For $x=-1$, the equation is not satisfied. So, only one solution.

66.(2) For a triangle inscribed in a circle, we have

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R \Rightarrow \sin^2 A+\sin^2 B+\sin^2 C=\frac{a^2+b^2+c^2}{4R^2}$$

According to the question $(2R)^2=\frac{a^2+b^2+c^2}{2}$ or $a^2+b^2+c^2=8R^2 \Rightarrow \sin^2 A+\sin^2 B+\sin^2 C=\frac{8R^2}{4R^2}=2$

67.(9) $\tan\left(142\frac{1}{2}^\circ\right)=-\tan\left(37\frac{1}{2}^\circ\right)=-\tan\left(60^\circ-22\frac{1}{2}^\circ\right)=\tan\left(22\frac{1}{2}^\circ-60^\circ\right)=\frac{\left(\sqrt{2}-1\right)-\sqrt{3}}{1+\left(\sqrt{2}-1\right)\left(\sqrt{3}\right)}=\frac{\sqrt{2}-\sqrt{3}-1}{1+\sqrt{6}-\sqrt{3}}$

$$=\frac{\left(\sqrt{2}-\sqrt{3}-1\right)\left(1-\sqrt{6}+\sqrt{3}\right)}{\left(1+\sqrt{6}-\sqrt{3}\right)\left(1-\sqrt{6}+\sqrt{3}\right)}=\frac{4\sqrt{2}-4\sqrt{3}+2\sqrt{6}-4}{6\sqrt{2}-8}=\frac{\left(2\sqrt{2}-2\sqrt{3}+\sqrt{6}-2\right)\left(3\sqrt{2}+4\right)}{\left(3\sqrt{2}-4\right)\left(3\sqrt{2}+4\right)}=2+\sqrt{2}-\sqrt{3}-\sqrt{6}$$

$\Rightarrow \quad \mu=3, \lambda=6 \Rightarrow \mu+\lambda=9$

68.(8) $\tan\left(\frac{2\pi}{3}-x\right)=\frac{2\sin\left(\frac{\pi}{3}-\frac{x}{2}\right)\cos\left(\frac{\pi}{3}+\frac{x}{2}\right)}{-2\sin\left(\frac{\pi}{3}-\frac{x}{2}\right)\sin\left(\frac{\pi}{3}+\frac{x}{2}\right)}=-\cot\left(\frac{\pi}{3}+\frac{x}{2}\right)=\tan\left(\frac{\pi}{2}+\frac{\pi}{3}+\frac{x}{2}\right)=\tan\left(\frac{5\pi}{6}+\frac{x}{2}\right)$

$\Rightarrow \quad \frac{2\pi}{3}-x=n\pi+\frac{5\pi}{6}+\frac{x}{2} \Rightarrow \frac{3x}{2}=-n\pi-\frac{\pi}{6} \Rightarrow \quad x=\frac{-2n\pi}{3}-\frac{\pi}{9}$

$\Rightarrow \quad x=\frac{5\pi}{9},\frac{11\pi}{9} \Rightarrow \quad \frac{12}{\pi}\left|x_2-x_1\right|=\frac{12}{\pi}\times\frac{6\pi}{9}=8$

69.(6) $\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\left(\frac{C}{2}\right)=\frac{1}{9}\cot\left(\frac{C}{2}\right)$

$$\cos(A-B)=\frac{1-\tan^2\left(\frac{A-B}{2}\right)}{1+\tan^2\left(\frac{A-B}{2}\right)}=\frac{1-\frac{1}{81}\cot^2\left(\frac{C}{2}\right)}{1+\frac{1}{81}\cot^2\left(\frac{C}{2}\right)}=\frac{31}{32} \Rightarrow \cot^2\left(\frac{C}{2}\right)=\frac{9}{7}$$

$$\cos(C)=\frac{1-\tan^2\left(\frac{C}{2}\right)}{1+\tan^2\left(\frac{C}{2}\right)}=\frac{1-\frac{7}{9}}{1+\frac{7}{9}}=\frac{2}{16}=\frac{1}{8} \Rightarrow \cos(C)=\frac{a^2+b^2-c^2}{2ab}=\frac{25+16-c^2}{40}=\frac{1}{8} \Rightarrow c=6$$

70.(3) $10(1-\cos 2\alpha)^2+15(1+\cos 2\alpha)^2=24 \Rightarrow (5\cos 2\alpha+1)^2=0 \Rightarrow \cos 2\alpha=-\frac{1}{5} \Rightarrow \tan^2\alpha=\frac{3}{2}$

71.(2) $a^2\sec^2 200^\circ=c^2\tan^2 200^\circ+d^2+2cd\tan 200^\circ$

$b^2\sec^2 200^\circ=c^2+d^2\tan^2 200^\circ-2cd\tan 200^\circ$

$\Rightarrow \quad a^2+b^2=c^2+d^2$

$(a\sec 200^\circ-c\tan 200^\circ)^2=d^2$

$(b\sec 200^\circ+d\tan 200^\circ)^2=c^2$

$\Rightarrow \quad (c^2+d^2)(\sec^2 200^\circ+\tan^2 200^\circ)+(2bd-2ac)\sec 200^\circ\tan 200^\circ=c^2+d^2$

$\Rightarrow \quad (c^2+d^2)(2\tan^2 200^\circ)=(2ac-2bd)\sec 200^\circ\tan 200^\circ \Rightarrow \frac{2(c^2+d^2)}{ac-bd}=\frac{2\sec 200^\circ}{\tan 200^\circ}=\frac{2}{\sin 200^\circ}=\frac{-2}{\sin 20^\circ}$

72.(3) $\sum_{r=1}^{n} \frac{\sin(2^r - 2^{r-1})}{\cos 2^r \cos 2^{r-1}} = \sum_{r=1}^{n} (\tan 2^r - \tan 2^{r-1}) = \tan 2^n - \tan 1$

73.(1) $3^{\sin 2x + 2\cos^2 x} + \frac{3^3}{3^{\sin 2x + 2\cos^2 x}} = 28$

Let $3^{\sin 2x+2\cos^2 x} = t,\ t^2 - 28t + 27 = 0 \quad \Rightarrow \quad t = 1, 27$

If $t = 1 \quad \Rightarrow \quad \sin 2x + 2\cos^2 x = 0$

$2\cos x(\sin x + \cos x) = 0 \quad \Rightarrow \quad x = \frac{\pi}{2}, \frac{3\pi}{4}, \frac{7\pi}{4}$

If $t = 27 \quad \Rightarrow \quad \sin 2x + 2\cos^2 x = 3$ (Not possible)

$(\sin 2\alpha - \cos 2\alpha)^2 + 8\sin 4\alpha = 1 + 7\sin 4\alpha = 1$ (at $\alpha = \frac{\pi}{2}, \frac{3\pi}{4}, \frac{7\pi}{4}$)

74.(3) $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} - 1 = \frac{\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}} \cdot \cos\frac{4\pi}{7} - 1 = \frac{-3}{2}$

75.(9) $\sin y - 2014\cos y = 1 \quad \Rightarrow \quad y = \frac{\pi}{2}$

76.(6) $\frac{2\sin 6x}{\sin x - 1} < 0 \quad \Rightarrow \quad \sin 6x > 0 \quad \Rightarrow \quad x \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

77.(7) $\sin^4 x - 4\sin^2 x + (2+k) = 0$

Let $\sin^2 x = t \qquad t \in [0, 1]$

$t^2 - 4t + (2+k) = 0$

$f(0)f(1) \le 0$

$(k+2)(k-1) \le 0 \quad \Rightarrow \quad -2 \le k \le 1$

78.(8) $2\sin^2 x + \sin^2 2x = 2$

$2\sin^4 x - 3\sin^2 x + 1 = 0 \quad \Rightarrow \quad (2\sin^2 x - 1)(\sin^2 x - 1) = 0$

$\sin 2x + \cos 2x = \tan x$

$2\tan x + 1 - \tan^2 x = \tan x(1 + \tan^2 x)$

$\Rightarrow \quad \tan^3 x + \tan^2 x - \tan x - 1 = 0 \quad \Rightarrow \quad (1+\tan x)(\tan^2 x - 1) = 0$

$2\cos^2 x + \sin x \le 2$

$2\sin^2 x - \sin x \ge 0$

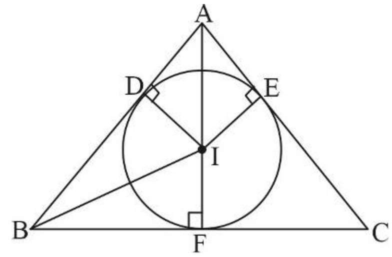
$\sin x(2\sin x - 1) \ge 0$

79.(8) $(8\cos 4\theta - 3)(\cot\theta - \tan\theta)^2 = 12$

$8(2\cos^2 2\theta - 1) - 3\left(\frac{4\cos^2 2\theta}{\sin^2 2\theta}\right) = 12$

$16\cos^4 2\theta - 8\cos^2 2\theta - 3 = 0 \quad \Rightarrow \quad (4\cos^2 2\theta - 3)(4\cos^2 2\theta + 1) = 0 \quad \Rightarrow \quad \cos 2\theta = \pm\frac{\sqrt{3}}{2}$

80.(3) $\frac{1}{2}r(AD+AE)=5$; $\frac{1}{2}r(BF+BD)=10$



$$\Rightarrow \quad \frac{BF+BD}{AD+AE}=2 \quad \Rightarrow \quad \frac{r\cot\frac{B}{2}+r\cot\frac{B}{2}}{r\cot\frac{A}{2}+r\cot\frac{A}{2}}=2$$

Applying C and D, $\dfrac{\cos\frac{C}{2}}{\sin\frac{A-B}{2}}=3$

81.(1) $\dfrac{\Delta_1\Delta_2\Delta_3}{\Delta^3}=\dfrac{(r_1r_2r_3)^2}{r^6}=\left(\dfrac{s}{s-a}\times\dfrac{s}{s-b}\times\dfrac{s}{s-c}\right)^2$

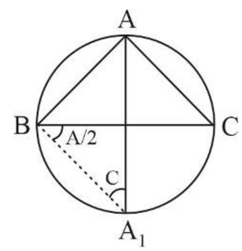
$$\frac{(s-a)+(s-b)+(s-c)}{3}\geq[(s-a)(s-b)(s-c)]^{1/3} \quad \Rightarrow \quad \frac{s^3}{(s-a)(s-b)(s-c)}\geq 27$$

Minimum value $=1$

82.(2) $\dfrac{AH}{AD}+\dfrac{BH}{BE}+\dfrac{CH}{CF}=\dfrac{R}{\Delta}(a\cos A+b\cos B+c\cos C)=\dfrac{R^2}{\Delta}(\sin 2A+\sin 2B+\sin 2C)$

$$=\frac{4R^2}{\Delta}\sin A\sin B\sin C=\frac{bc\sin A}{\Delta}=2$$

83.(2) $\dfrac{c}{\sin C}=\dfrac{AA_1}{\sin\left(B+\frac{A}{2}\right)}$



$AA_1\cos\dfrac{A}{2}=\sin B+\sin C \qquad (\because R=1)$

$$\Rightarrow \quad \frac{AA_1\cos\frac{A}{2}+BB_1\cos\frac{B}{2}+CC_1\cos\frac{C}{2}}{\sin A+\sin B+\sin C}=2$$

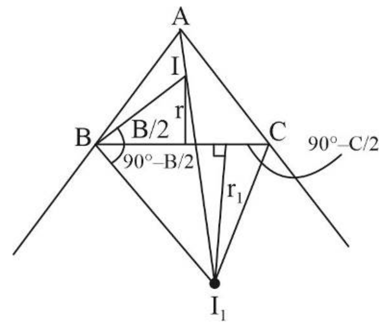
84.(7) $(R^2-4Rr+4r^2)+(4r^2-12r+9)=0$

$$(R-2r)^2+(2r-3)^2=0 \quad \Rightarrow \quad r=\frac{3}{2};\ R=2r$$

ΔABC is an equilateral triangle.

85.(9) $5-2\pi>x^2-4x$, $x^2-4x+(2\pi-5)<0$, $2-\sqrt{9-2\pi}<x<2+\sqrt{9-2\pi} \Rightarrow \lambda=9$

86.(16) $\sin\frac{B}{2} = \frac{r}{IB}$



$IB = 4R\sin\frac{A}{2}\sin\frac{C}{2}$

$\sin\left(90° - \frac{B}{2}\right) = \frac{r_1}{BI_1} \quad \Rightarrow \quad BI_1 = 4R\sin\frac{A}{2}\cos\frac{C}{2}$

$(II_1)^2 = (BI)^2 + (BI_1)^2 = 16R^2\sin^2\frac{A}{2}$ .... **(1)**

$I_2I_3\cos\left(90° - \frac{A}{2}\right) = a$ (by using pedal triangle)

$I_2I_3 = 4R\cos\frac{A}{2}$

$(I_2I_3)^2 = 16R^2\cos^2\frac{A}{2}$ .... **(2)**

From (1) & (2) we get $\lambda = 16$

87.(65) $2\tan^{-1}\left(\frac{1}{5}\right) - \sin^{-1}\left(\frac{3}{5}\right)$

$\tan^{-1}\left(\frac{5}{12}\right) - \sin^{-1}\left(\frac{3}{5}\right)$

$\tan^{-1}\left(\frac{5}{12}\right) - \tan^{-1}\left(\frac{3}{4}\right) = -\left(\tan^{-1}\left(\frac{3}{4}\right) - \tan^{-1}\left(\frac{5}{12}\right)\right) = -\tan^{-1}\left(\frac{16}{63}\right) = -\cos^{-1}\left(\frac{63}{65}\right) \Rightarrow \lambda = 65$

88.(1) $\sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{2}{n^2+n+4}\right) = \sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{\frac{1}{2}}{\frac{n^2}{4}+\frac{n}{4}+1}\right) = \sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{\left(\frac{n}{2}+\frac{1}{2}\right)-\frac{n}{2}}{\frac{n}{2}\left(\frac{n}{2}+\frac{1}{2}\right)+1}\right) = \sum_{n=0}^{\infty} 2\left(\tan^{-1}\left(\frac{n}{2}+\frac{1}{2}\right) - \tan^{-1}\left(\frac{n}{2}\right)\right)$

89.(8) $\cos^{-1}(|3\log_6^2(\cos x) - 7|) = \cos^{-1}(|\log_6^2(\cos x) - 1|)$

$|3\log_6^2(\cos x) - 7| = |\log_6^2(\cos x) - 1|$

Let $|3t - 7| = |t - 1| \Rightarrow t = 3$ and $t = 2$

$\Rightarrow \cos x = 6^{-\sqrt{3}}$ and $6^{-\sqrt{2}}$

SEQUENCE AND SERIES

1.(D) $\frac{1}{x}=a+(p-1)d \quad \frac{1}{y}=a+(q-1)d \quad \frac{1}{z}=a+(r-1)d$

$$\frac{1}{x}-\frac{1}{y}=(p-q)d$$

$$\Rightarrow \quad xy(p-q)=\frac{y-x}{d};\, yz(q-r)=\frac{z-y}{d};\, zx(r-p)=\frac{x-z}{d}$$

$$\Rightarrow \quad (p-q)xy+yz(q-r)+zx(r-p)=\frac{1}{d}(y-x+z-y+x-z)=0$$

2.(D) $(a+nd)^2=(a+md)(a+rd)$

$$\Rightarrow \quad \left(n^2-mr\right)d^2=ad(r+m-2n) \quad \Rightarrow \quad \frac{d}{a}=\frac{\left(\frac{2mr}{n}-2n\right)}{n^2-mr} \qquad \left[\because\ n=\frac{2mr}{m+r}\right]$$

$$\Rightarrow \quad \frac{d}{a}=-\frac{2}{n}$$

3.(B) $\sum_{r=1}^{99} r!\left((r+1)^2-r\right)=\sum_{r=1}^{99}\left((r+1)(r+1)!-rr!\right)=100(100!)-1$

4.(D) $\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{\left(k^2-k+\frac{1}{2}\right)\left(k^2+k+\frac{1}{2}\right)}=\sum_{k=1}^{n}\left(\frac{k-\frac{1}{2}}{k^2-k+\frac{1}{2}}-\frac{k+\frac{1}{2}}{k^2+k+\frac{1}{2}}\right)=1-\frac{n+\frac{1}{2}}{n^2+n+\frac{1}{2}}=1-\frac{2n+1}{2n^2+2n+1}$

5.(B) $k^4+\frac{1}{4}=\left(k^2-k+\frac{1}{2}\right)\left(k^2+k+\frac{1}{2}\right)$

$$\prod_{k=1}^{n}\frac{\left((2k-1)^2-(2k-1)+\frac{1}{2}\right)\left((2k-1)^2+(2k-1)+\frac{1}{2}\right)}{\left((2k)^2-2k+\frac{1}{2}\right)\left((2k)^2+2k+\frac{1}{2}\right)}=\prod_{k=1}^{n}\frac{(2k-1)^2-(2k-1)+\frac{1}{2}}{(2k)^2+2k+\frac{1}{2}}=\frac{\frac{1}{2}}{4n^2+2n+\frac{1}{2}}=\frac{1}{8n^2+4n+1}$$

6.(B) $\sum_{k=1}^{n}\frac{1}{\sqrt{k}\sqrt{k+1}\left(\sqrt{k+1}+\sqrt{k}\right)}=\sum_{k=1}^{n}\frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k}\sqrt{k+1}}=\sum_{k=1}^{n}\left(\frac{1}{\sqrt{k}}-\frac{1}{\sqrt{k+1}}\right)=1-\frac{1}{\sqrt{n+1}}$

7.(C) $S_{2n}-S_n=\frac{2n}{2}(2a+(2n-1)d)-\frac{n}{2}(2a+(n-1)d)=\frac{n}{2}(2a+(3n-1)d)=\frac{1}{3}S_{3n}$

8.(B) $S_n=\sum_{r=1}^{n}\frac{\left(r^2+6r+12\right)}{(r)(r+1)(r+2)}\frac{1}{2^{r+1}}=\sum_{r=1}^{n}\frac{2(r+2)(r+3)-r(r+4)}{(r)(r+1)(r+2)^{r+1}}$

$$=\sum_{r=1}^{n}\left(\frac{r+3}{r(r+1)2^r}-\frac{r+4}{(r+1)(r+2)2^{r+1}}\right)=1-\frac{n+4}{(n+1)(n+2)2^{n+1}}$$

9.(D) $S = \sum_{r=1}^{\infty} r^2 \left(-\frac{1}{5}\right)^{r-1}$

$$\sum_{r=0}^{\infty} x^r = \frac{1}{1-x}$$

$$\Rightarrow \quad \sum_{r=1}^{\infty} r x^r = \frac{x}{(1-x)^2} = \frac{1}{x-1} + \frac{1}{(x-1)^2} \Rightarrow \sum_{r=1}^{\infty} r^2 x^{r-1} = -\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \Rightarrow \sum_{r=1}^{\infty} r^2 \left(-\frac{1}{5}\right)^{r-1} = \frac{25}{54}$$

10.(A) $\sum_{r=1}^{n} \frac{n-(r-1)}{r(r+1)(r+2)} = \frac{(n+1)}{2} \sum_{r=1}^{n} \frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} - \sum_{r=1}^{n} \left(\frac{1}{r+1} - \frac{1}{r+2}\right)$

$$= \frac{(n+1)}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right) - \left(\frac{1}{2} - \frac{1}{n+2}\right) = \frac{1}{2(n+2)} + \frac{n+1}{4} - \frac{1}{2}$$

11.(D) **12.(B)** **13.(C)**

$$\sum_{r=1}^{n}\left((r+1)^2 - r^2\right)\phi(r) = \sum_{r=1}^{n} (r+1)^2 \left(\phi(r) - \phi(r+1)\right) + \sum_{r=1}^{n}\left((r+1)^2 \phi(r+1) - r^2 \phi(r)\right)$$

$$= -\sum_{r=1}^{n} (r+1) + (n+1)^2 \phi(n+1) - 1^2 \phi(1) = -\left(1+2+3+\ldots+(n+1)\right) + (n+1)^2 \phi(n+1)$$

$$= -\left(1+2+3+\ldots+(n+1)\right) + (n+1)^2 \phi(n+1) = -\frac{(n+1)(n+2)}{2} + (n+1)^2 \phi(n+1)$$

$$\sum_{r=0}^{9} P(r) = 1^2 + 2^2 + \ldots 10^2 = \frac{10 \times 11 \times 21}{6} = 385$$

$$\sum_{r=0}^{\infty} \frac{1}{Q(r)} = \sum_{r=0}^{\infty} \frac{2}{(r+1)(r+2)} = 2\sum_{r=0}^{\infty}\left(\frac{1}{r+1} - \frac{1}{r+2}\right) = 2$$

$$P(n) - Q(n) = (n+1)^2 - \frac{(n+1)(n+2)}{2} = \frac{n(n+1)}{2}$$

14.(B)

15.(C)

16.(D)

17.(AD) $a_k a_{n-(k-1)} = \left(a_1 + (k-1)d\right)\left(a_n - (k-1)d\right)$

$$= a_1 a_n + d(k-1)(a_n - a_1) - (k-1)^2 d^2$$

$$= a_1 a_n + d^2 (k-1)(n-k)$$

$$\le \begin{cases} a_1 a_n + d^2\left(\frac{n+1}{2} - 1\right)\left(n - \frac{n+1}{2}\right) = a_1 a_n + \frac{(n-1)^2}{4} d^2 & \text{if } n \text{ is odd} \\ a_1 a_n + \frac{d^2}{4} n(n-2) & \text{if } n \text{ is even} \end{cases}$$

18.(BCD) $y = \frac{\tan 3x}{\tan x} = \frac{3-\tan^2 x}{1-3\tan^2 x} = \frac{8}{3\left(1-3\tan^2 x\right)} + \frac{1}{3}$

$$\Rightarrow \quad y \in \left(-\infty, \frac{1}{3}\right) \cup (3, \infty)$$

$$a = \frac{1}{3}, b = 3, s_\infty = \frac{6}{1-\frac{1}{3}} = 9$$

19.(CD) $\frac{a_1+2}{2} = \sqrt{2a_1} \Rightarrow (a_1-2)^2 = 0 \Rightarrow a_1 = 2$

$$(a_n+2)^2 = 8s_n$$

$$(a_{n-1}+2)^2 = 8s_{n-1}$$

$$\Rightarrow \quad (a_n+2)^2 - (a_{n-1}+2)^2 = 8a_n \Rightarrow (a_n - a_{n-1})(a_n + a_{n-1}+4) = 8a_n$$

$$\Rightarrow \quad a_n^2 - a_{n-1}^2 = 4(a_n + a_{n-1}) \Rightarrow a_n - a_{n-1} = 4$$

20.(AB) $\frac{3x+\frac{4xz}{x+z}}{2x-\frac{2xz}{x+z}} + \frac{3z+\frac{4xz}{x+z}}{2z-\frac{2xz}{z+x}} = \frac{3x^2+7xz}{2x^2} + \frac{7xz+3z^2}{2z^2} = 3 + \frac{7}{2}\left(\frac{z}{x}+\frac{x}{z}\right) \in (10, \infty)$

21.(ABD) $\frac{5^{a_{n+1}}}{5^{a_n}} = \frac{3n+5}{3n+2}$

$$\frac{5^{a_2}}{5^{a_1}} \times \frac{5^{a_3}}{5^{a_2}} \times ..... \frac{5^{a_n}}{5^{a_{n-1}}} = \frac{8}{5} \times \frac{11}{8} \times \frac{14}{11} \times ..... \frac{3n+2}{3n-1}$$

$$\frac{5^{a_n}}{5^{a_1}} = \frac{3n+2}{5} \quad \Rightarrow \quad 5^{a_n} = 3n+2 \quad \Rightarrow \quad a_n = \log_5(3n+2)$$

$$[a_n] = 3 \quad \Rightarrow \quad \log_5(3n+2) \in [3, 4)$$

$$\Rightarrow \quad n \in \{41, 42, ..., ...207\}$$

$$[a_n] = 4 \Rightarrow \frac{623}{3} \le n < \frac{3123}{3} \Rightarrow n \in \{208, 209, ..., 1041\}$$

22. (ABC) $\frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b} \ge 2\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right)$

$$= 2\left[(a+b+c)\left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right) - 3\right] \ge 2\left(\frac{9}{2} - 3\right) = 3$$

$$\frac{3}{\left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right)} \le \frac{(b+c)+(c+a)+(a+b)}{3} \quad \Rightarrow \quad (a+b+c)\left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right) \ge \frac{9}{2}$$

23.(AD) $p = a+n\left(\dfrac{b-a}{n+1}\right) = \dfrac{nb+a}{n+1}$

$$\frac{1}{q} = \frac{1}{a}+n\left(\frac{\frac{1}{b}-\frac{1}{a}}{n+1}\right) = \frac{na+b}{(n+1)ab}$$

$$q = \frac{(n+1)ab}{na+b}$$

$$\therefore \quad q\left(na+\frac{p(n+1)-a}{n}\right) = (n+1)a\left(\frac{(n+1)p-a}{n}\right)$$

$$\Rightarrow \quad q\left(\left(n^2-1\right)a+p(n+1)\right) = (n+1)a\left((n+1)p-a\right)$$

$$\Rightarrow \quad a^2+a\left(q(n-1)-(n+1)p\right)+pq = 0$$

$$D \ge 0 \quad \Rightarrow \quad \left(q(n-1)-(n+1)p\right)^2-4pq \ge 0$$

$$\Rightarrow \quad (q-p)^2 n^2+(q+p)^2-2n\left(q^2-p^2\right)-4pq \ge 0$$

$$\Rightarrow \quad (q-p)^2 n^2+(q+p)^2-2n\left(q^2+p^2\right) \ge 0$$

$$\Rightarrow \quad (q-p)\left((q-p)n^2+(q-p)-2n(q+p)\right) \ge 0$$

$$\Rightarrow \quad (q-p)\left(q(n-1)^2-p(n+1)^2\right) \ge 0$$

$$q \in (-\infty, p] \cup \left[\left(\frac{n+1}{n-1}\right)^2 p, \infty\right)$$

24.(AB) $\dfrac{\underbrace{(x+x+.....x)}_{\text{x times}}+\underbrace{y+y+.....y}_{\text{y times}}+\underbrace{z+z+.....z}_{\text{z times}}}{(x+y+z)} \ge \left(x^x y^y z^z\right)^{\frac{1}{x+y+z}}$

$$\ge \frac{x+y+z}{\underbrace{\left(\frac{1}{x}+\frac{1}{x}+.....\frac{1}{x}\right)}_{\text{x times}}+\underbrace{\left(\frac{1}{y}+\frac{1}{y}+....\frac{1}{y}\right)}_{\text{y times}}+\underbrace{\left(\frac{1}{z}+\frac{1}{z}+.....\frac{1}{z}\right)}_{\text{z times}}}$$

25.(CD) $\dfrac{1}{x-1}+\left(\dfrac{1}{x+1}-\dfrac{1}{x-1}\right)+\dfrac{2}{x^2+1}+\dfrac{4}{x^4+1}+...+\dfrac{2^n}{x^{2n}+1}$

$$= \frac{1}{x-1}+2\left(\frac{1}{x^2+1}-\frac{1}{x^2-1}\right)+\frac{4}{x^4+1}+....+\frac{2^n}{x^{2n}+1}$$

$$= \frac{1}{x-1}+4\left(\frac{1}{x^4+1}-\frac{1}{x^4-1}\right)+\frac{8}{x^8+1}+.....+\frac{2^n}{x^{2^n}+1} = \frac{1}{x-1}-\frac{2^{n+1}}{x^{2^{n+1}}-1} \quad \Rightarrow \quad f(x) = \frac{1}{x-1} \quad \forall\, x>1$$

26.(AC) $S(n)=1+\underbrace{\left(\frac{1}{2}+\frac{1}{3}\right)}_{2\text{ terms}}+\underbrace{\left(\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}\right)}_{4terms}+\underbrace{\left(\frac{1}{8}+\frac{1}{9}++....+\frac{1}{15}\right)}_{8\text{terms}}+.....\underbrace{\left(\frac{1}{2^n-1}+\frac{1}{2^{n-1}+1}....\frac{1}{2^n-1}\right)}_{2^{n-1}\text{terms}}$

$<+\left(\frac{1}{2}+\frac{1}{2}\right)+\left(\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+...+\frac{1}{8}\right)+...+\left(\frac{1}{2^{n-1}}+\frac{1}{2^{n-1}}+...\frac{1}{2^{n-1}}\right)=1+1+1+1+1+...+1=n$

$S(n)>\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+\left(\frac{1}{9}+\frac{1}{10}+...+\frac{1}{26}\right)+.....+\left(\frac{1}{2^{n-1}+1}+\frac{1}{2^{n-1}+2}+\frac{1}{2^n}\right)$

$>\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\left(\frac{1}{16}+\frac{1}{16}+.....+\frac{1}{16}\right)+....\left(\frac{1}{2^n}+\frac{1}{2^n}+...+\frac{1}{2^n}\right)$

$=\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+...+\frac{1}{2}=\frac{n}{2}\quad\Rightarrow\quad S(2n)>n$

27.(BC) $\frac{1}{a^2\left(1-r^2\right)}\left[1+\frac{1}{r^2}+\frac{1}{r^4}+...\frac{1}{r^{2(n-2)}}\right]=\frac{1-\frac{1}{r^{2(n-1)}}}{\left(1-\frac{1}{r^2}\right)a^2\left(1-r^2\right)}=\frac{\left(1-r^{2n}\right)}{a^2\left(1-r^2\right)^2r^{2n-4}}$

$\frac{1}{a^m\left(1+r^m\right)}\left[1+\frac{1}{r^m}+\frac{1}{r^{2m}}+...+\frac{1}{r^{(n-2)m}}\right]=\frac{\left(1-\frac{1}{r^{(n-1)m}}\right)}{\left(1-\frac{1}{r^m}\right)a^m\left(1+r^m\right)}=\frac{\left(1-r^{mn-m}\right)}{\left(1-r^m\right)\left(r^{mn-n}-r^{mn-2m}\right)}$

28.(ABCD) $\frac{1}{\alpha}+\frac{1}{\gamma}=\frac{2}{\beta}\Rightarrow\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{3}{\beta}=\frac{q}{q}=1\quad\Rightarrow\quad\beta=3$

$\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\geq\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}\quad\Rightarrow\quad 3\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\right)\geq\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)^2=1\quad\Rightarrow\quad\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\geq\frac{1}{3}$

Again $\quad\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\geq\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}\Rightarrow 3\left(\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}\right)\leq\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)^2=1$

Put $\beta=3$ in the equation $27+9p+3q-q=0\Rightarrow 9p+2q+27=0$

$\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}\leq\frac{1}{3}\quad\Rightarrow\quad\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}=\frac{-p}{q}\leq\frac{1}{3}\Rightarrow\frac{p}{q}\geq-\frac{1}{3}$

29.(BC) $S=\sum_{n=1}^{\infty}\sum_{k=1}^{n-1}\frac{k}{2^{n+k}}=\sum_{k=1}^{\infty}\left(\frac{k}{2^k}\sum_{n=k+1}^{\infty}\frac{1}{2^n}\right)=\sum_{k=1}^{\infty}\frac{k}{2^k}\times\frac{2}{2^{k+1}}\times 2$

$\sum_{k=1}^{\infty}\frac{k}{4^k}=\frac{4}{9}$

30.(ACD) $\quad n^2<n^2+1<n^2+2<....<n^2+n$

$\Rightarrow\quad\frac{1}{n^2+1}+\frac{2}{n^2+2}+....+\frac{n}{n^2+n}<\frac{1}{n^2}+\frac{2}{n^2}+...\frac{n}{n^2}=\frac{1}{2}+\frac{1}{2n}$

and $\quad\frac{1}{n^2+1}+\frac{2}{n^2+2}+....+\frac{n}{n^2+n}>\frac{1}{n^2+n}+...\frac{n}{n^2+n}=\frac{1}{2}$

31.(ABC) (a) $\left(1\cdot3\cdot5...(2n-1)\right)^{1/n}<\dfrac{1+3+5+...(2n-1)}{n}=n \Rightarrow 1\cdot3\cdot5....(2n-1)<n^n$

(b) $\dfrac{1+2+2^2+....+2^{n-1}}{n}>\left(2^{1+2+...(n-1)}\right)^{1/n}=2^{\frac{n-1}{2}} \Rightarrow 2^n\cdot>1+n2^{\frac{n-1}{2}}$

(c) $\dfrac{1}{n+1}+\dfrac{1}{n+2}+...+\dfrac{1}{n+n}>\dfrac{1}{2n}+\dfrac{1}{2n}+...+\dfrac{1}{2n}=\dfrac{1}{2}$

32.(BCD) $\dfrac{1}{a_{n+1}-1}=\dfrac{1}{a_n(a_n-1)}=\dfrac{1}{a_n-1}-\dfrac{1}{a_n}$

$\dfrac{1}{a_n-1}-\dfrac{1}{a_{n+1}-1}=\dfrac{1}{a_n}$

$\Rightarrow \displaystyle\sum_{r=1}^{2018}\frac{1}{a_r}=\frac{1}{a_1-1}-\frac{1}{a_{2019}-1}=1-\frac{1}{a_{2019}-1}<1$

$a_{2019}-1=a_{2018}(a_{2018}-1)=a_{2018}\,a_{2017}(a_{2017}-1)$

$\Rightarrow a_{2019}-1=a_{2018}\,a_{2017}\,a_{2016}...a_2\,a_1(a_1-1)=a_1a_2a_3...a_{2018}$

$2018<\dfrac{2018}{\frac{1}{a_1}+\frac{1}{a_2}+...+\frac{1}{a_{2018}}}<(a_1a_2....a_{2018})^{\frac{1}{2018}}$

$\Rightarrow a_{2019}-1>(2018)^{2018} \Rightarrow 1-\dfrac{1}{a_{2019}-1}>1-\dfrac{1}{(2018)^{2018}}$

33.(BC) $\dfrac{k}{(k-1)^{4/3}+k^{4/3}+(k+1)^{4/3}}<\dfrac{k}{(k-1)^{4/3}+((k-1)(k+1))^{2/3}+(k+1)^{4/3}}$

$=\dfrac{\left((k+1)^{2/3}-(k-1)^{2/3}\right)k}{(k+1)^2-(k-1)^2}=\dfrac{1}{4}\left((k+1)^{2/3}-(k-1)^{2/3}\right)$

$\Rightarrow \quad S_n<\dfrac{1}{4}\displaystyle\sum_{k=1}^{n}\left((k+1)^{2/3}-(k-1)^{2/3}\right)=\frac{1}{4}\left((n+1)^{2/3}+n^{2/3}-1\right)$

$\Rightarrow \quad S_{999}<\dfrac{1}{4}\left((1000)^{2/3}+(999)^{2/3}-1\right)<\dfrac{1}{4}(100+100-1)<50$

$S_{26}<\dfrac{1}{4}\left((27)^{2/3}+(26)^{2/3}-1\right)<\dfrac{1}{4}(9+9-1)=\dfrac{17}{4}$

34.(BC) $S=\dfrac{1}{\sqrt{1}+\sqrt{3}}+\dfrac{1}{\sqrt{5}+\sqrt{7}}+....+\dfrac{1}{\sqrt{9997}+\sqrt{9999}}$

$>\dfrac{1}{\sqrt{3}+\sqrt{5}}+\dfrac{1}{\sqrt{7}+\sqrt{9}}+\dfrac{1}{\sqrt{9999}+\sqrt{10001}}$

$\Rightarrow \quad 2s>\dfrac{1}{\sqrt{1}+\sqrt{3}}+\dfrac{1}{\sqrt{3}+\sqrt{5}}+\dfrac{1}{\sqrt{5}+\sqrt{7}}+....+\dfrac{1}{\sqrt{9999}+\sqrt{10001}}$

$=\dfrac{1}{2}\left(\sqrt{3}-\sqrt{1}+\sqrt{5}-\sqrt{3}+\sqrt{7}-\sqrt{5}+....+\sqrt{10001}-\sqrt{9999}\right)=\dfrac{1}{2}\left(\sqrt{10001}-\sqrt{1}\right)>\dfrac{1}{2}(100-1) \Rightarrow S>\dfrac{99}{4}>24$

35.(BD) $a_n, 2 = \sum_{i=0}^{n-1} \sum_{j=i+1}^{n} 2^i 2^j = \frac{1}{2}\left[\left(1+2+2^n+...+2^n\right)^2 - \left(1^2+2^2+2^4+....+2^{2n}\right)\right]$

$$= \frac{\left(2^{n+1}-1\right)\left(2^{n+1}-2\right)}{3}$$

36.(ABCD) $\dfrac{\underbrace{(a+a+...a)}_{\text{a times}}+\underbrace{(b+b+...b)}_{\text{b times}}+\underbrace{(c+c+....c)}_{\text{c times}}}{a+b+c} \geq \left(a^a b^b c^c\right)^{\frac{1}{a+b+c}}$

$$\Rightarrow \quad \left(a^a b^b c^c\right)^{1/n} \leq \frac{a^2+b^2+c^2}{n}$$

$$\frac{\underbrace{(a+a+...a)}_{\text{b times}}+\underbrace{(b+b+...b)}_{\text{c times}}+\underbrace{(c+c+...c)}_{\text{a times}}}{a+b+c} \geq \left(a^b b^c c^a\right)^{\frac{1}{a+b+c}}$$

$$\frac{a^2+b^2+c^2}{n} \geq \frac{ab+bc+ca}{n} \geq \left(a^b b^c c^a\right)^{\frac{1}{n}}$$

$$\left(a^a b^b c^c\right)^{\frac{1}{n}} + \left(a^b b^c c^a\right)^{\frac{1}{n}} + \left(a^c b^a c^b\right)^{\frac{1}{n}} \leq \frac{a^2+b^2+c^2}{n} + \frac{ab+bc+ca}{n} + \frac{ac+ba+cb}{b} = n$$

37.(ABCD) $S = 3\left(1+2+3+4++...+2016\right) = 2^4 \times 3^3 \times 7 \times 2017$

$$\left[\because\ 6^2+5^2+4^2-3^2-2^2-1^2 = 3\left[(6+3)+(5+2)+(4+1)\right]\right.$$

$$\left.= 3\left(1+2+3+4+5+6\right)\right]$$

38.(BD) $f(x) = \sum_{r=1}^{n} \frac{rx^{r-1}}{(x+1)(x+2)...(x+r)}$

$$f(x) = \left(1-\frac{x}{1+x}\right) + \sum_{r=2}^{n} \frac{(x+r)x^{r-1}-x^r}{(x+1)(x+2)....(x+r)}$$

$$= \left(1-\frac{x}{1+x}\right) + \sum_{r-2}^{n}\left(\frac{x^{r-1}}{(x+1)(x+2)...(x+r-1)} - \frac{x^r}{(x+1)(x+2)...(x+r)}\right) = 1 - \frac{x^n}{(x+1)(x+2)...(x+n)}$$

$$1-f(x) = \prod_{r=1}^{n} \frac{x}{r+x}$$

$$\Rightarrow \quad \frac{-f'(x)}{1-f(x)} = \frac{1}{x}\sum_{r=1}^{n}\frac{r}{x+r}$$

$$\Rightarrow \quad f'(x) = -\frac{x^{n-1}}{(x+1)(x+2)....(x+n)}\sum_{r=1}^{n}\frac{r}{x+r}$$

39. **[A-Q] [B-T] [C-P] [D-R]**

(A) $$A=\frac{n(n+1)(2n+1)}{6}$$

$$B=\sum_{m=1}^{n}\frac{m(m+1)}{2}-\frac{1}{2}\frac{n(n+1)}{2}$$

$$\frac{1}{6}\sum_{m=1}^{n}(m+2)m(m+1)-m(m+1)(m-1)-\frac{n(n+1)}{4}$$

$$=\frac{n(n+1)(n+2)}{6}-\frac{n(n+1)}{4}=\frac{n(n+1)(2n+1)}{12}$$

(B) $$\frac{\sum a\left(b^2+c^2\right)}{abc}\ge\frac{a(2bc)+b(2ca)+c(2ab)}{abc}=6$$

(C) $$2\sum\frac{x^2-1+1}{1-x}=2\sum-(x+1)+\frac{1}{1-x}=-8+2\sum\frac{1}{1-x}\ge-8+2\left(\frac{9}{2}\right)=1$$

$$\left[\because\ \frac{3}{\sum\frac{1}{1-x}}\le\frac{\sum(1-x)}{3}=\frac{2}{3}\ \Rightarrow\ \sum\frac{1}{1-x}\ge\frac{9}{2}\right]$$

(D) $\because\ x,y,z\in N\quad\Rightarrow\quad\frac{1}{xy}\le1\quad\Rightarrow\quad\frac{1}{xy}+\frac{1}{yz}+\frac{1}{zx}\le3\qquad\therefore\qquad x=y=z=1$

40. **[A-T] [B-Q] [C-S] [D-P]**

(A) $$\frac{\left(\frac{49a}{3}\right)3+\left(\frac{3b}{2}\right)2+c}{6}\ge\left(\left(\frac{49a}{3}\right)^3\left(\frac{3b}{2}\right)^2c\right)^{1/6}=\left(\frac{7^6a^3b^2c}{12}\right)^{1/6}\Rightarrow\ 49a+3b+c\ge6\times7=42$$

(B) $x=-t,t>0$

$$2x^3-\frac{3}{x^2}=-\left(2t^3+\frac{3}{t^2}\right)\le-5$$

$$\because\qquad\frac{t^3+t^3+\frac{1}{t^2}+\frac{1}{t^2}+\frac{1}{t^2}}{5}\ge\left(\left(t^3\right)^2\frac{1}{\left(t^2\right)^3}\right)^{1/5}=1\ \Rightarrow\qquad2t^3+\frac{3}{t^2}\ge5$$

(C) $$\left(\frac{\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{8-x^3}{3}+\frac{8-x^3}{3}}{8}\right)^8\ge\left(\frac{8-x^3}{3}\right)\left(\frac{x^3}{5}\right)\ \Rightarrow\ \left(8-x^3\right)x^5\le\left(3^35^5\right)^{1/3}=3\times5^{5/3}$$

(D) $$\frac{(x+x+x+x+x+x+x)+(y+y+y+y+y)}{12}\ge\left(x^7y^5\right)^{1/12}\Rightarrow(7x+5y)\ge12a^{1/2}$$

$\therefore\qquad12a^{1/12}\ge12\qquad\Rightarrow\qquad a\ge1$

41.(2) $S_n = \sum_{r=1}^{n} a_r = \sum_{k=1}^{k}\sum_{j=1}^{k} 2j = \sum_{k=1}^{n} k(k+1) = \frac{1}{3}\sum_{k=1}^{n}\left((k(k+1)(k+2)-(k-1)k(k+1)\right)$

$$S_n = \frac{1}{3}n(n+1)(n+2)$$

$$a_n = S_n - S_{n-1} = \frac{1}{3}(n+1)(n)\left((n+2)-(n-1)\right) = n(n+1)$$

$$\sum_{r=1}^{n}\frac{1}{a_r} = \sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+1}\right) = 1-\frac{1}{n+1} = \frac{n}{n+1}$$

$$\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n = e^{\lim_{n\to\infty} -\frac{n}{n+1}} = e^{-1}$$

$$\frac{1}{\lambda} = e \quad \Rightarrow \quad \left[\frac{1}{\lambda}\right] = 2$$

42.(6) $a+b = \frac{1}{6}\sum_{i=1}^{6}(a_i+b_i) = 1$

$$\sum_{i=1}^{6}(a_i-a)^2 + \sum_{i=1}^{6}a_ib_i = 6a^2 - 2a\sum a_i + \sum a_i^2 + \sum a_ib_i$$

$$= 6a^2 - 2a(6a) + \sum_{i=1}^{6}a_i(a_i+b_i) = -6a^2+6a = 6a(1-a) = 6ab$$

43.(661750) $S = \sum_{k=1}^{99} k\left((k+1)^2-k^2\right)+100 = \frac{100\times 99\times 401}{6}+100 = 661750$

44.(289) $\sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{1}{3^i3^j3^k} = \left(\frac{1}{1-\frac{1}{3}}\right)^3 - 3\left(\sum_{i=0}^{\infty}\frac{1}{3^{2i}}\frac{1}{\left(1-\frac{1}{3}\right)} - \sum_{i=0}^{\infty}\frac{1}{3^{3i}}\right) - \sum_{i=0}^{\infty}\frac{1}{3^i}$

$(i \neq j \neq k)$

$$= \frac{27}{8} - \frac{3}{2}\times 3\left(\frac{1}{1-\frac{1}{9}}\right) + 2\frac{1}{1-\frac{1}{27}} = \frac{27}{8}-\frac{81}{16}+\frac{27}{13} = \frac{81}{208}$$

45.(69) $\frac{n(n+1)}{2}-(m+k) = 17(n-2) 3 \le \frac{n^2-33n+68}{2} = m+k \le 2n-1$

$\Rightarrow \quad n \in [31,35] \quad n = \{31,32,33,34,35\} \quad \therefore \quad$ Maximum value of $m+k = 35+34 = 69$

46.(352) $x^4 - 16x^3 + px^2 - 256x + q = 0$

x_1, x_2, x_3, x_4 are in G.P. $\Rightarrow \quad x_1x_4 = x_2x_3$

$$(x_1+x_4)+(x_2+x_3) = 16$$

$$x_1x_4(x_2+x_3)+x_2x_3(x_1+x_4)=256$$

$$\Rightarrow \quad x_1x_4(16)=256 \quad \Rightarrow \quad x_1x_4=x_2x_3=16$$

$$P=x_1x_4+x_2x_3+(x_1+x_4)(x_2+x_3)$$

$$x_1x_2x_3x_4=(16)^2 \quad \Rightarrow \quad (x_1x_2x_3x_4)^{1/4}=2^2=4$$

$$\frac{x_1+x_2+x_3+x_4}{4}=4$$

$$\Rightarrow \quad x_1=x_2=x_3=x_4=4 \quad \Rightarrow \quad P=6\times16=96, q=256$$

47.(0) $\displaystyle\sum_{r=1}^{2018}\frac{x_r^2}{1-x_r}=\sum_{r=1}^{2018}\left(-(x_r+1)+\frac{1}{1-x_r}\right)$

$$=\sum_{r=1}^{2018}\left[-(x_r+1)+\frac{1-x_r+x_r}{1-x_r}\right]=\sum_{r=1}^{2018}\left[-(x_r+1)+1+\frac{x_r}{1-x_r}\right]=-1+1=0$$

48.(2017) $\displaystyle\sum_{i=1}^{2018}\frac{x_i^2}{1-x_i}=\sum_{i=1}^{2018}-(x_i+1)+\frac{1}{1-x_i}=-2019+\sum_{i=1}^{2018}\frac{1}{1-x_i}$

$$\frac{(1-x_1)+(1+x_2)+....(1-x_{2018})}{2018}\geq\frac{2018}{\frac{1}{1-x_1}+\frac{1}{1-x_2}+...+\frac{1}{1-x_{2018}}} \quad \Rightarrow \quad \sum_{i=1}^{2018}\frac{1}{1-x_i}\geq\frac{(2018)^2}{2017}$$

$$\therefore \quad \sum_{i=1}^{2018}\frac{x_i^2}{1-x_i}\geq\frac{(2018)^2}{2017}-2019=\frac{(2018)^2-\left((2018)^2-1\right)}{2017}=\frac{1}{2017}$$

$\Rightarrow$ the least value of k satisfying $\displaystyle k\sum_{i=1}^{2018}\frac{x_i^2}{1-x_i}\geq 1 \;=2017$

49.(2018) Adding, $\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}=\dfrac{3}{2018}$

$$2xz-2yz+1=\frac{z}{2018}\,;\quad 2yz-2zx+1=\frac{x}{2018}\,;\quad 2yz-2xy+1=\frac{y}{2018}$$

Adding, $x+y+z=2018\times3$

$$\because \quad \frac{3}{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}}\leq\frac{x+y+z}{3} \quad \Rightarrow \quad x=y=z=2018$$

50.(4) $\displaystyle -\frac{a^3}{(a-b)(c-a)}-\frac{b^3}{(b-c)(a-b)}-\frac{c^3}{(c-a)(b-c)}$

$$=-\frac{a^3(b-c)+b^3(c-a)+c^3(a-b)}{(a-b)(b-c)(c-a)}=-\frac{(b-c)(c-a)(a-b)(-a-b-c)}{(a-b)(b-c)(c-a)}=a+b+c\geq4$$

COMPLEX NUMBER

1.(D) Since $|z + a| \le a$ implies z lies on or inside a circle with centre (-a, 0) and radius a, we have $|z_1|+|z_2|+|z_3|\le 14$.

2.(A) Hence, $i\left\{\log\left(\frac{x-i}{x+i}\right)\right\}-\pi+2\tan^{-1}x=k(\text{say})$

$\therefore \quad \log\left(\frac{x+i}{x-i}\right)=i\left(k+\pi-2\tan^{-1}x\right)$ Or $\frac{x+i}{x-i}=e^{i\theta}$, where $\theta=k+\pi-2\tan^{-1}x$

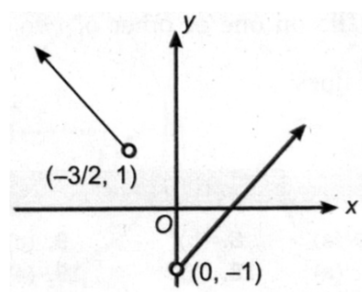
$\Rightarrow \quad x+i=(x\cos\theta+\sin\theta)+i(x\sin\theta-\cos\theta) \quad \Rightarrow \quad x=x\cos\theta+\sin\theta$

And $\quad 1=x\sin\theta-\cos\theta$

$\Rightarrow \quad x=\cot\frac{\theta}{2} \quad \Rightarrow \quad \theta=2\cot^{-1}x$ Or $k+\pi-2\tan^{-1}x=2\cot^{-1}x$

$\Rightarrow \quad k+\pi=2\left(\cot^{-1}x+\tan^{-1}x\right)=2\left(\frac{\pi}{2}\right) \quad \Rightarrow \quad k+\pi=\pi$ or $k=0$.

3.(D) $\arg(z+i)=\frac{\pi}{4}$



$\Rightarrow \quad \arg\{x+iy+i\}=\frac{\pi}{4} \qquad \{x>0, y+1>0\}$

$\Rightarrow \quad \frac{y+1}{x}=1, x>0, y>-1$

Now, $\quad x = y + 1, x > 0, y > -1 \qquad \ldots(1)$

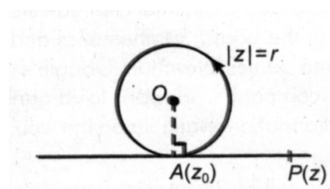
Also, $\quad \arg\{(2z+3-2i)\}=\frac{3\pi}{4}, \{2x+3<0, 2y-2>0\}$

$\Rightarrow \quad \frac{2y-2}{2x+3}=-1, x<-\frac{3}{2}, y>1 \quad \Rightarrow \quad 2y - 2 = -2x - 3$

$\Rightarrow \quad 2x + 2y = -1$

$\Rightarrow \quad x+y=-\frac{1}{2}, x<-\frac{3}{2}, y>1 \qquad \ldots(2)$ Hence, (1) and (2) have no solutions.

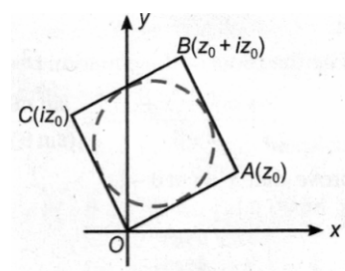
4.(A) $\angle OAP=\frac{\pi}{2}$



$\Rightarrow \quad \frac{z-z_0}{z_0}$ is purely imaginary. $\quad \Rightarrow \quad \frac{z-z_0}{z_0}+\frac{\bar{z}-\bar{z}_0}{\bar{z}_0}=0$

$\Rightarrow \quad \frac{z}{z_0}+\frac{\bar{z}}{\bar{z}_0}=2 \quad \Rightarrow \quad 2\,\text{Re}\left(\frac{z}{z_0}\right)=2$

$\Rightarrow \quad \text{Re}\left(\frac{z}{z_0}\right)=1.$



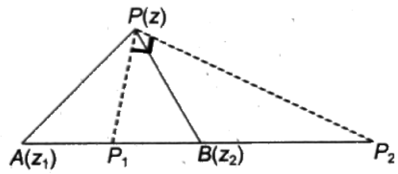
5.(B) Clearly, mid-point of OB is on centre of the circle whose radius is equal to $\frac{|z_0|}{2}$.

$\Rightarrow \quad$ Required equation of circle is $\left|z-\frac{z_0(1+i)}{2}\right|=\frac{|z_0|}{2}$.

6.(B) Let internal and external bisectors of $\angle APB$ meet the line joining A and B at P_1 and P_2, respectively.

Now, $\frac{AP_1}{P_1B} = \frac{PA}{PB} = \frac{3}{1}$ (internal division)

And $\frac{AP_2}{P_2B} = \frac{PA}{PB} = \frac{3}{1}$ (external division)



Thus, P_1 and P_2 are fixed points

Also, $\angle P_1PP_2 = \frac{\pi}{2}$

Thus, P lies on a circle having P_1P_2 as its diameter. Clearly, $B(z_2)$ lies inside this circle.

7.(A) $z_1 + z_2 = -1$ and $z_1z_2 = \frac{b}{3}$.

As z_1, z_2 and origin form an equilateral triangle,

We have $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

$\Rightarrow \quad z_1^2 + z_2^2 + 0^2 = z_1z_2 + 0 + 0 \quad \Rightarrow \quad (z_1 + z_2)^2 = 3z_1z_2 \quad \Rightarrow \quad 1 = b$

8.(B) As the roots are of opposite signs, product of roots < 0.

$\Rightarrow \quad a^2 + b^2 - 1 < 0 \quad \Rightarrow \quad a^2 + b^2 < 1 \quad \Rightarrow \quad |a + ib| < 1$

So, the point a + ib lies inside a circle of centre (0, 0) and radius 1.

9.(C) $\left|\frac{z+1}{z}\right| = 1^{1/n} = 1$

$\Rightarrow \quad |z + 1| = |z| \quad \Rightarrow$ z lies on the right bisector of the line joining the points (-1, 0) and (0, 0).

$\Rightarrow$ Roots are collinear.

10.(B) Given, $1 \geq |z - (4 - 3i)| \geq \begin{cases} |z| - |4 - 3i| \\ |4 - 3i| - |z| \end{cases}$

$\Rightarrow \quad |z| \leq 6$ and $|z| \geq 4 \quad \Rightarrow \quad 4 \leq |z| \leq 6 \quad \Rightarrow \alpha = 4, \beta = 6$

Let $\quad y = \frac{x^4 + x^2 + 4}{x} = x^3 + x + \frac{4}{x} = x^3 + x \frac{1}{x} + \frac{1}{x} + \frac{1}{x} + \frac{1}{x}$

Since $x \in (0, \infty)$, therefore $x^3, x, \frac{1}{x}, \frac{1}{x}, \frac{1}{x}, \frac{1}{x}$ are positive. Sum will be the least when $x^2 = x = \frac{1}{x} \Rightarrow x = 1$.

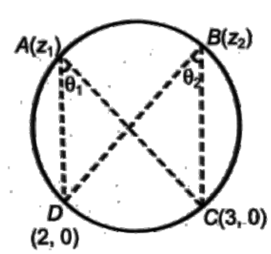
$\therefore \quad K = 6 \qquad$ Hence, $K = \beta$.

11.(D) Let $(\cos x - i \sin 2x) = \sin x + i \cos 2x \quad \Rightarrow \quad \cos x + i \sin 2x = \sin x + i \cos 2x$

$\therefore \quad \cos x = \sin x$ and $\sin 2x = \cot 2x \quad \Rightarrow \quad \tan x = 1$ and $\tan 2x = 1$

Which is impossible.

12.(A) $arg\left(\frac{3 - z_1}{2 - z_1}\right) + arg\left(\frac{2 - z_2}{3 - z_2}\right) = arg\left(\frac{3 - z_1}{2 - z_1} \cdot \frac{2 - z_2}{3 - z_2}\right)$



Now if, $\left(\frac{3 - z_1}{2 - z_1} \cdot \frac{2 - z_2}{3 - z_2}\right)$ is a positive real number, then its argument will be zero.

So, angle θ_1 and θ_2 are equal in magnitude but opposite in signs.

So, chord DC subtends equal angles at A and B. So, points are concyclic for K > 0.

13.(A) $z^2 + ax + a^2 = 0 \Rightarrow z = a\omega, a\omega^2$ (where 'ω' is non-real root of unity)

$\Rightarrow$ Locus of z is a pair of straight lines and arg(z) = arg(a) + arg(ω) or arg(z) = arg(a) + arg(ω^2)

$\Rightarrow$ $\arg(z) = \pm\dfrac{2\pi}{3}$ Also, $|z| = |a||\omega|$ or $|z| = |a||\omega^2|$ $\Rightarrow$ $|z| = |a|$

14.(D) Required locus is the set of all points, sum of whose distance from (-1, 0) and (1, 0) is equal to 2. Again distance between (-1, 0) and (1, 0) is also 2.

$\therefore$ The required locus is the portion of the real axis between the points (-1, 0) and (1, 0).

15.(C) $|z - 2| = \min\{|z - 1|, |z - 5|\}$

i.e., $|z - 2| = |z - 1|$, where $|z - 1| < |z - 5|$ $\Rightarrow$ Re(z) = $\dfrac{3}{2}$ which satisfy $|z - 1| < |z - 5|$

Also, $|z - 2| = |z - 5|$, where $|z - 5| < |z - 1|$ $\Rightarrow$ Re(z) = $\dfrac{7}{2}$ which satisfy $|z-5| < |z-1|$

16.(C) $\therefore$ $z^{n-1} + z^{n-2} + z^{n-3} + ... + z + 1 = 0$

$\Rightarrow$ $(z-1)\left(z^{n-1} + z^{n-2} + ... + z + 1\right) = 0$ $\Rightarrow$ $z^n - 1 = 0$ $\Rightarrow$ $z^n = e^{f2\pi r}, (r \in N)$ $\Rightarrow$ $z = e^{f2\pi r/n}$

$\Rightarrow$ The roots are $e^{i2\pi/n}, e^{i4\pi/n}, ..., e^{i(2n-n)\pi/n}$

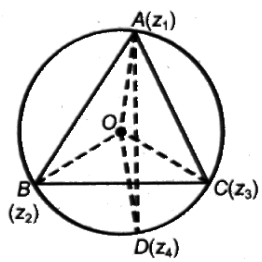
Which is a GP with a common ratio $e^{i\frac{2\pi}{n}}$. Also, $a^{\frac{z_r+1}{z_r}} = a^{e^{i2\pi/n}}$ which is a constant.

17.(A) Now, $\angle BOD = 2\angle BAD = A$

And $\angle COD = 2\angle CAD = A$

From the rotation about the point O, we get,

$$\frac{z_4}{z_2} = e^{iA}, \frac{z_3}{z_4} = e^{iA} \Rightarrow z_4^2 = z_2 z_3.$$



18.(C) $|z - 2 - i| = |z|\sin\left(\dfrac{\pi}{4} - \arg z\right)$

$\Rightarrow$ $|(x-2) + i(y-1)| = |z|\left(\dfrac{1}{\sqrt{2}}\cos\theta - \dfrac{1}{\sqrt{2}}\sin\theta\right)$ (where $\theta = \arg z$)

$\Rightarrow$ $\sqrt{(x-2)^2 + (y-1)^2} = \dfrac{1}{\sqrt{2}}|z|\cos\theta - \dfrac{1}{\sqrt{2}}|z|\sin\theta$ $\Rightarrow$ $\sqrt{(x-2)^2 + (y-1)^2} = \dfrac{1}{\sqrt{2}}(x - y)$

Which is a parabola with vertex (2, 1) and directrix x – y = 0.

Alternate approach

$\alpha = \dfrac{\pi}{4} - \theta$

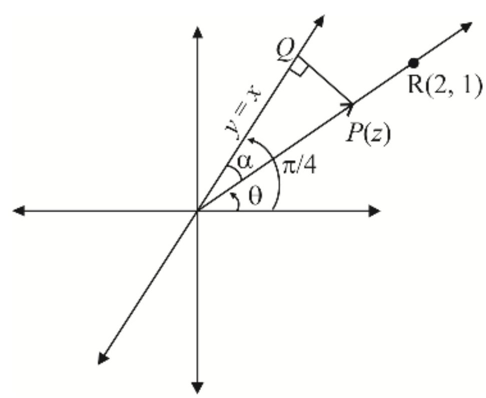
$PQ = |z|\sin\left(\dfrac{\pi}{4} - \theta\right)$ is distance of $P(z)$ from $y = x$ line

on argand plane which is equal to PR.

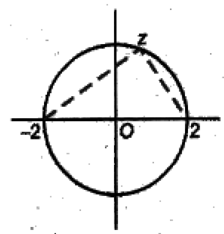
As distance of a point from a fixed line is equal to its distance from

a fixed point so locus of given point is a parabola

whose directrix is $y = x$ and focus is (2, 1)



19.(C) Clearly, $\arg\left(\frac{z-2}{z+2}\right)=\pm\frac{\pi}{2}$



$\Rightarrow$ $\arg\left(\frac{z_1-z_2}{z_1-z_3}\right)=\pm\frac{\pi}{2}$

$\Rightarrow$ z_1, z_2, z_3 will be the vertices of a right-angled triangle.

20.(B) We are finding out sum of distances of complex number z from origin and $(\cos\alpha,\sin\alpha)$. This sum will be minimum if z lies on the line joining the two points and minimum value of sum will be distance between those two points.

i.e., $\sqrt{(\cos\alpha-0)^2+(\sin\alpha-0)^2}=1$.

21-23. 21.(A) 22.(D) 23.(C)

$f(x)=x^4-6x^3+26x^2-46x+65$.

Since complex roots occur in conjugate pairs.

$a_1=a_2,\ a_3=a_4,\ b_1=-b_2\ \ \&\ \ b_3=-b_4$

Product $=\left(a_1^2+b_1^2\right)\left(a_3^2+b_3^2\right)=65$ and sum of roots $=2(a_1+a_3)=6\Rightarrow a_1+a_3=3$

Without loss of generality let $a_1^2+b_1^2<a_3^2+b_3^2$

Case I $a_1^2+b_1^2=1,\ a_3^2+b_3^2=65\Rightarrow$ $a_1=0,\ b_1=1$ $\Rightarrow$ $a_3=3$ $(\because a_1+a_3=3)$

Not possible $\because$ $b_3=\sqrt{65-9}$ not an integer

Case II $a_1^2+b_1^2=5,\ a_3^2+b_3^2=13$

Sub-case I $a_1=2,\ b_1=1$

$\Rightarrow$ $a_3=1$, not possible $\because$ $b_3=\sqrt{12}$ not an integer

Sub-case II $a_1=1,\ b_1=2$

$\Rightarrow$ $a_3=2,\ b_3=3$ integer (only possible solution)

$\Rightarrow$ roots of equation are $1\pm 2i,\ 2\pm 3i$.

$\Rightarrow$ $\mu=1+1+2+2=6,\ \lambda=2+2+3+3=10$ and $S=\{1,\ 2,\ -2,\ 3,\ -3\}$

$\Rightarrow$ Number of functions from S to C are 8^5

24.(ABC) $\lambda=\alpha^6$ is a root of

$1+x+x^2+x^3+\ldots+x^{10}=0$ .....(1)

$\therefore$ $1+\lambda+\lambda^2+\ldots+\lambda^{10}=0$

$\Rightarrow$ $1+\lambda+\lambda^2+\lambda^3+\lambda^4+\lambda^5+\bar{\lambda}+\overline{\lambda^2}+\overline{\lambda^3}+\overline{\lambda^4}+\overline{\lambda^5}=0$ $\Rightarrow$ $\operatorname{Re}\left(\lambda+\lambda^2+\ldots.+\lambda^5\right)=-\frac{1}{2}$

$\beta,\beta^2,\beta^3,\ldots..,\beta^{10}$ are roots of (1)

$\therefore$ $(x-\beta)\left(x-\beta^2\right)\left(x-\beta^3\right)\ldots\left(x-\beta^{10}\right)=1+x+x^2+\ldots.+x^{10}$ ..... (2)

Now put x = μ

$(\mu-\beta)\left(\mu-\beta^2\right)\ldots\left(\mu-\beta^{10}\right)=1+\mu+\ldots+\mu^{10}=0$ $[\because \mu$ is a root of Eq. (1)$]$

Put x = i in Eq. (2), $(i-\beta)\left(i-\beta^2\right)\ldots.\left(i-\beta^{10}\right)=i$.

25.(BD) $\sum\limits_{1\le i<j\le 100}\sum \alpha_i^5\alpha_j^5 = \left(\alpha_1^{5}+\alpha_2^{5}+\alpha_3^{5}+\alpha_4^{5}+\alpha_5^{5}+....\right)^2-\left(\alpha_1^{10}+\alpha_2^{10}+\alpha_3^{10}+\alpha_4^{10}+\alpha_5^{10}+...\right)=0-0=0$

Because $\left(\alpha_1^{r}+\alpha_2^{r}+....\alpha_{100}^{r}\right)=\begin{cases}100, & \text{if } r=100k\\ 0, & \text{if } r\ne 100k\end{cases}$

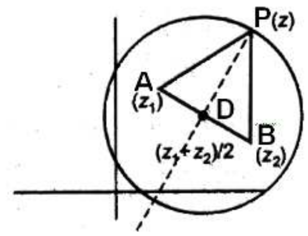
26.(ABCD) Maximum area of $(\Delta PAB)=\frac{1}{2}(PD)(AB)=\frac{1}{2}k\,|z_1-z_2|$

When ΔPAB is equilateral triangle of maximum area, then

$PD=\frac{\sqrt{3}}{2}AB \Rightarrow 4k^2\,|z_1-z_2|^2$

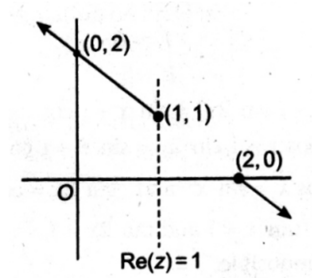
For maximum area of ΔPAB, P has two possible positions on the circle.

For the given area of triangle less than the maximum area there are four possible positions of P.



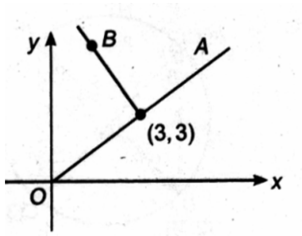
27.(D) The given equation is written is $\arg\left(z-(1+i)\right)=\begin{cases}\frac{3\pi}{4} & \text{when }, x\le 2\\ \frac{-\pi}{4} & \text{when, } x>2\end{cases}$

$\Rightarrow$ The locus is a set of two rays.



28.(BD) We can observe that $3+3i\in A$ but $\notin B$

$\therefore$ $n(A\cap B)=0$.



29.(BC) Let $y=|z_1-z_2|^2+|z_2-z_3|^2+|z_3-z_1|^2=(z_1-z_2)(\overline{z}_1-\overline{z}_2)+(z_2-z_3)(\overline{z}_2-\overline{z}_3)+(z_3-z_1)(\overline{z}_3-\overline{z}_1)$

$= 6-(z_1\overline{z}_2+z_2\overline{z}_1+z_2\overline{z}_3+z_3\overline{z}_2+z_3\overline{z}_1+z_1\overline{z}_3)....(1)$

Now, we know $|z_1+z_2+z_3|^2\ge 0 \quad\Rightarrow\quad 3+(z_1\overline{z}_2+z_2\overline{z}_1+z_1\overline{z}_3+z_3\overline{z}_1+z_2\overline{z}_3+z_3\overline{z}_2)\ge 0....(2)$

From Eqs. (1) and (2), we get : $y\le 9$

30.(AD) $f(i)=\alpha,\ f(-i)=\beta$

Now $f(z)=\text{Quotient}(z^2+1)+(az+b)$ where $g(z)=az+b$ is remainder and $a,b\in C$

Now $f(i)=ai+b$ and $f(-i)=-ai+b \quad\Rightarrow\quad a=\frac{\alpha-\beta}{2i}$ and $b=\frac{\alpha+\beta}{2}$

Then, $g(z)=i\left(\frac{z}{2}+1\right)+\frac{1}{2}$ when $\alpha=i$ and $\beta=1+i$

and $\alpha=i$ and $\beta=1-i\Rightarrow g(z)=\left(z+\frac{1}{2}\right)+\frac{iz}{2}$

31.(ABCD) Now, $a_1^2 + b_1^2 = 1$,

$a_2^2 + b_2^2 = 4$ and $a_1a_2 = b_1b_2$

$\therefore$ $a_2^2 + b_2^2 = 4a_1^2 + 4b_1^2 \Rightarrow (a_2 + 2ia_1)^2 = (2b_1 + ib_2)^2 \Rightarrow a_2 = \pm 2b_1$

$2a_1 = \pm b_2$

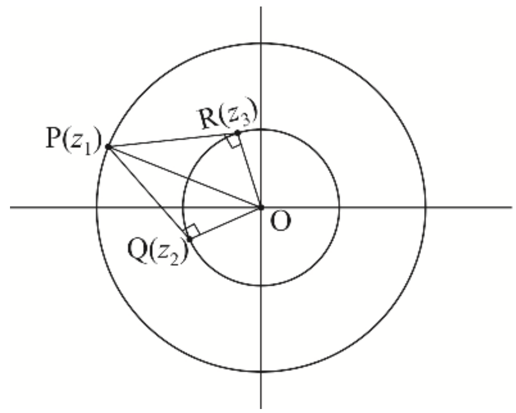
$$|\omega_1|^2 = a_1 + a_1^2 + \frac{a_2^2}{4} = a_1^2 + b_1^2 = 1 \Rightarrow |\omega_1| = 1$$

And $|\omega_2|^2 = 4b_1^2 + b_2^2 = 4 \Rightarrow |\omega_2| = 2$

$$\text{Re}(\omega_1\omega_2) = 2a_1b_1 - \frac{a_2b_2}{2} = 0$$

$$\text{Im}(\omega_1\omega_2) = a_1b_2 + a_2b_1 = 0$$

32.(ABCD) Option (A), (D) are true as PQR is an equilateral triangle. So orthocentre, circumcentre and centroid, all will coincide.



(B) $\left|\frac{z_1 + z_2 + z_3}{3}\right| = 1 \Rightarrow |z_1 + z_2 + z_3|^2 = 9$

$$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$$

$$\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right)\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right) = 9.$$

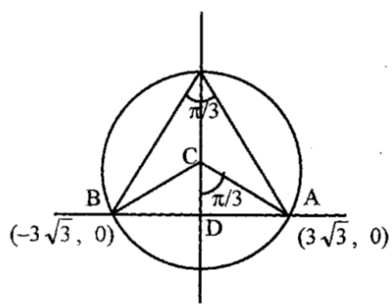
(C) $\angle QOR = 120^\circ$.

33.(AB) $|z + \bar{z}| + |z - \bar{z}| = 2 \Rightarrow |2\text{Re}(z)| + |2i\,\text{Im}(z)| = 2 \Rightarrow |x| + |y| = 1$

Which is the locus of a square.

Also, $|z + i| + |z - i| = 2$ represents a line. Hence, (a), (b) are the correct answer.

34.(BD) $\tan 60^\circ = \frac{3\sqrt{3}}{y} \Rightarrow y = 3$



$$|z - 3i| = \sqrt{9 + 27} = 6, I_m(Z) > 0$$

35.(BC) $i^i = \left(e^{\frac{i\pi}{2}}\right)^i = e^{-\frac{\pi}{2}}$

$$\log i^i = -\frac{\pi}{2}$$

36.(AB) $x^6 = (4 - 3i)^5 \qquad = -5^5(\cos(5\theta) + i\sin(5\theta))$

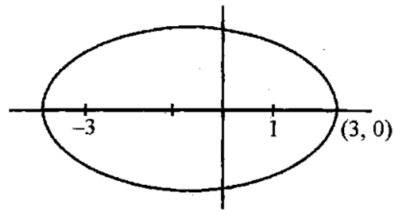
Hence $x_1x_2....x_6 = -(4 - 3i)^5$

37.(CD) $2ae = 4$

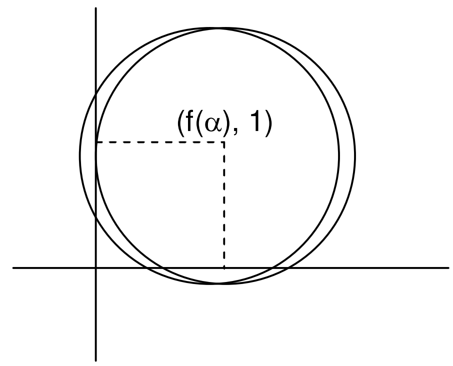
$8\ e = 4$

$e = 1/2$

$|Z - 4| \in [1, 9]$



38.(ABC) From the figure it is clear that for $\arg z \ge \frac{\pi}{2}$ circle should either lie in second quadrant or touch the positive imaginary axis.



$\Rightarrow \quad f(\alpha) \le 1$

$\Rightarrow \quad \alpha^2 - 7\alpha + 11 \le 1$

$\Rightarrow \quad \alpha^2 - 7\alpha + 10 \le 0$

$\Rightarrow \quad (\alpha - 2)(\alpha - 5) \le 0$

$$2 \le \alpha \le 5$$

39.(AD) $\log_{\sqrt{3}}\left|1 + \alpha + \alpha^2 + \alpha^3 - \frac{2}{\alpha}\right| = \log_{\sqrt{3}}\left|-\alpha^4 - \frac{2}{\alpha}\right| = \log_{\sqrt{3}} 3 = 2$

40.(BD) $\sum_{r=0}^{3n-1} \alpha^{2^r} = n(\alpha + \alpha^2 + \alpha^4)$

$1 + (\alpha + \alpha^2 + \alpha^4) + (\alpha^3 + \alpha^5 + \alpha^6) = 0$

$(\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) = 3 + (\alpha + \alpha^2 + ...\alpha^6) = 2$

Hence, $(\alpha + \alpha^2 + \alpha^4)$ is root of $t^2 + t + 2 = 0$

$t = \frac{-1 \pm \sqrt{7}i}{2}$

$\Rightarrow \quad \left|\sum_{r=0}^{3n-1} \alpha^{2^r}\right|^2 = n^2 |\alpha + \alpha^2 + \alpha^4|^2 = 2n^2 = 32$

$n = 4$

41.(BD) $z^{15} = 1 \Rightarrow z = \left(\cos\frac{2\aleph k}{15} + i\sin\frac{2\aleph k}{15}\right), k = 0, \pm 1, \pm 2, ..., \pm 7$

$\Rightarrow \quad |\arg(z)| = \left|\frac{2\pi k}{15}\right| < \frac{\pi}{2} \quad \Rightarrow \quad |k| < \frac{15}{4} \quad \Rightarrow \quad k = 0, \pm 1, \pm 2, \pm 3.$

42.(AB) $(|z| + 1)(|z| - 2) < 0 \Rightarrow -1 < |z| < 2$

$|z^2 + z\sin\theta| \le |z|^2 + |z|\sin\theta < 6$

43.(AC) $z_1.z_2.....z_n = e^{i\pi\left\{\frac{1}{1.2.3.} + \frac{1}{2.3.4} + ... + \frac{1}{k(k+1)(k+2)}\right\}} \quad (1)$

Let $S_k = \frac{1}{1.2.3} + \frac{1}{2.3.4} + ... + \frac{1}{k(k+1)(k+2)}$

Consider $t_r = \frac{1}{2}\left\{\frac{r+2-r}{r(r+1)(r+2)}\right\} = \frac{1}{2}\left\{\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}\right\}$

$S_k = \sum t_r = \frac{1}{2}\left\{\frac{1}{1.2} - \frac{1}{(k+1)(k+2)}\right\} = \frac{1}{4} - \frac{1}{2(k+1)(k+2)}$

$\therefore \quad \underset{k\to\infty}{\text{Lt}}\ S_k = \frac{1}{4}$ and hence from (1) $\underset{k\to\infty}{\text{Lt}}\left(z_1 \ldots z_k\right) = e^{i\pi/4} = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$

44.(BC) (i) $xy > 0$, $3x^2 - xy - 2y^2 = -7$ or $(3x+2y)(x-y) = -7$ x and y being integers, we can take

$3x+2y = 7$ and $x - y = -1$

$x = 1$, $y = 2$

If x and y are charged to $-x, -y$, equation remains same.

$x = -1,\ y = -2$ is also a solution pair.

$3x+2y = -1$ and $x+y = 7$ do not give integral solutions.

(ii) $xy = 0$ will not give any integral solution.

(iii) $xy < 0$ $3x^2 + xy - 2y^2 = -7$

$(3x-2y)(x-y) = -7$

$3x - 2y = -7$ and $x + y = 1$ leads to $x = -1$ $y = 2$

$3x - 2y = 7$ and $x + y = -1$ lead to x = 1 and $y = -2$.

Required complex numbers are $1+2i,\ 1-2i,\ -1+2i,\ -1-2i$ which form two conjugate pairs.

45.(ABC) Let $z_1 = x_1$, z_2, $z_3 = x_2 \pm iy_2$

$\Rightarrow \quad z_1 + z_2 + z_3 = -\frac{b}{a} \quad \Rightarrow \quad x_1 + 2x_2 = -\frac{b}{a} < 0 \Rightarrow ab < 0$

Also $z_1 z_2 z_3 = x_1\left[x_2^2 + y_2^2\right] = -\frac{d}{a} < 0 \Rightarrow ad > 0$

Also $z_1 z_2 + z_2 z_3 + z_1 z_3 = \frac{c}{a}$

$\Rightarrow \quad x_1(x_2 + iy_2) + x_1(x_2 - iy_2) + x_2^2 + y_2^2 = 2x_1 x_2 + x_2^2 + y_2^2 > 0 \quad \Rightarrow \quad \frac{c}{a} > 0 \quad \Rightarrow \quad \frac{b}{a}\frac{c}{a} > 0 \Rightarrow bc > 0$

46.(CD) $y = |z_1 - z_2|^2 + |z_1 - z_3|^2 + |z_3 - z_1|^2$

$= (z_1 - z_2)(\overline{z}_1 - \overline{z}_2) + (z_2 - z_3)(\overline{z}_2 - \overline{z}_3) + (z_3 - z_1)(\overline{z}_3 - \overline{z}_1) = 6 - \left(z_1\overline{z}_2 + \overline{z}_2\,\overline{z}_1 + z_2\,\overline{z}_3 + z_3\,\overline{z}_2 + z_3\,\overline{z}_1 + z_1\,\overline{z}_3\right)$

Also $|z_1 + z_2 + z_3|^2 \geq 0 \quad \Rightarrow \quad 3 + \left(z_1 + \overline{z}_2 + z_2\overline{z}_1 + z_1\,\overline{z}_3 + z_3\,\overline{z}_1 + z_2\,\overline{z}_3 + z_3\,\overline{z}_2\right) \geq 0 \quad \Rightarrow \quad y \leq 9$

47.(ABCD) Re $(z_1 z_2) = 0$

$\Rightarrow \quad a_1 a_2 = b_1 b_2$

$|z_1| = 1, |z_2| = 2$

$\Rightarrow \quad a_1^2 + b_1^2 = 1,\ a_2^2 + b_2^2 = 4$

$w_1 w_2 = \left(2a_1 b_1 - (a_2 b_2)/2\right) + i\left(a_2 b_1 + a_1 b_2\right)$

$a_2^2 + b_2^2 = 4a_1^2 + 4b_1^2$

$$\Rightarrow \ (a_2+2ia_1)^2=(2b_1+ib_2)^2 \ \Rightarrow \ a_2=\pm 2b_1 \text{ and } b_2=\pm 2a_1$$

$$|w_1|^2=a_1^2+\frac{a_2^2}{4}=a_1^2+b_1^2=1$$

$$|w_1|^2=4b_1^2+b_2^2=4\left(a_1^2+b_1^2\right)=4$$

$$\operatorname{Re}(w_1w_2)=\left(2a_1b_1-\frac{1}{2}(2b_1)(2a_1)\right)=0$$

$$\operatorname{Im}(w_1w_2)=\left(2b_1^2+2a_1^2\right)=2\left(a_1^2+b_1^2\right)=2$$

48.(AB) For complex number z and w, we are given that

$$|z|^2 w-|w|^2 z=z-w \qquad \ldots\text{(i)}$$

$$\Rightarrow \quad z\left(1+|w|^2\right)=w\left(1+|z|^2\right) \ \Rightarrow \ \frac{z}{w}=\frac{1+|z|^2}{1+|w|^2}=a \text{ real number}$$

$$\Rightarrow \quad \overline{\left(\frac{z}{w}\right)}=\frac{z}{w} \ \Rightarrow \ \frac{\bar{z}}{\bar{w}}=\frac{z}{w}$$

$$\Rightarrow \quad \bar{z}w=z\bar{w} \qquad \ldots\text{(ii)}$$

Again from equation (i), we get $z\bar{z}w=w\bar{w}z=z-w$

$$z(\bar{z}w-1)-w(\bar{w}z-1)=0$$

$$z(\bar{z}w-1)-w(z\bar{w}-1)=0 \qquad \text{(using equation (ii))}$$

$$\Rightarrow \quad (z\bar{w}-1)(z-w)=0 \ \Rightarrow \ z\bar{w}=1 \text{ or } z=w$$

49.(ABD) Let $z=\alpha$ be the real root of the given equation. Then $\alpha^3(3+2i)\alpha+(-1+i\alpha)=0$

$$\Rightarrow \ \left(\alpha^3+3\alpha-1\right)+i(\alpha+2\alpha)=0 \ \Rightarrow \ \alpha^3+3\alpha-1=0 \text{ and } a=-\frac{\alpha}{2} \ \Rightarrow \ -\frac{a}{8}-\frac{3a}{2}-1=0 \ \Rightarrow \ a^3+12a+8=0$$

Let $f(a)=a^3+12a+8$

$$f(-1)<0,\ f(0)>0,\ f(-2)<0,\ f(3)>0,\ f(1)>0 \quad \Rightarrow \quad a\in(-2,1) \text{ or } (-1,0)\ (-2,3)$$

50.(AB) $p\left(x^3\right)+xq\left(x^3\right)+x^2r\left(x^3\right)=\left(1+x+x^2\right)s(x)$

Putting $x = 1$, we get

$$\Rightarrow \quad p(1)+q(1)+r(1)=3s(1) \qquad \ldots\text{(i)}$$

Putting $x=\omega$, where ω is cube root of unity, we get

$$\Rightarrow \quad p(1)+\omega q(1)+\omega^2 r(1)=0$$

Putting $x=\omega^2$, we get

$$\Rightarrow \quad p(1)+\omega^2 q(1)+\omega r(1)=0$$

Adding (i), (ii), (iii) we get

$$3p(1)=3s(1) \ \Rightarrow p(1)=s(1)$$

Multiplying (ii) by $(-\omega)$ and (iii) by $\left(-\omega^2\right)$, we get

$$-\omega p(1)-\omega^2 q(1)-r(1)=0$$

$$-\omega^2 p(1)-\omega q(1)-r(1)=0$$

Adding (ii), (iii), (iv) and (v) we get $\left(2-\omega-\omega^2\right)p(1)+\left(\omega+\omega^2-2\right)r(1)=0 \quad \Rightarrow \quad p(1)=r(1)$

51.(AB) We have $z^4+\lambda z^3+(-36+15i)z^2+mz=0$

Clearly, one root is $z=0$

Let the other roots be z, iz and $z+iz$ which are the roots of the equation

$$z^3+\lambda z^2+(-36+15i)z+m=0$$

From the equation, $z+iz+z+iz=-\lambda \quad \Rightarrow \quad 2(1+i)z=\lambda$

Sum of the product of roots taken two at a time.

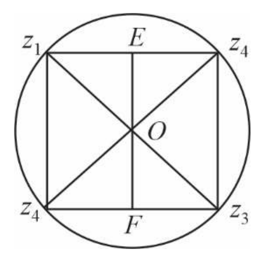
$iz^2+z^2+iz^2+iz^2-z^2=-36+15i$

$\Rightarrow \quad 3iz^2=-36+15i \quad \Rightarrow \quad z^2=5+12i \quad \Rightarrow \quad z=3+2i$ or $-3-2i$

Product of all roots, $z^3 i(1+i)=-m$

So, (λ, m) can be $(-2-10i,\ 37+55i)$ or $(2+10i,\ -37-55i)$

52.(AD) $z_1+z_2+z_3+z_4=0 \quad \Rightarrow \quad \dfrac{z_1+z_2}{2}=-\dfrac{z_3+z_4}{2}$



$\dfrac{z_3+z_4}{2}$ represents mid-point F of DC.

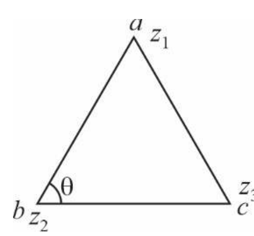
From (1), OE and OF are in opposite direction and $OE = OF$.

Thus, ΔOAB is isosceles, since $OA = OB = 1$ [A, B lie on $|z|=1$].

Hence, OE is perpendicular to AB. Also, OF is perpendicular to CD.

Therefore, $ABCD$ is a parallelogram inscribed in a circle and hence, it is a rectangle.

53.(CD) Triangles are similar



$\Rightarrow \quad -z_1z_2z_3=-abc$

$\Rightarrow \quad \arg\left(\dfrac{z_1-z_2}{z_3-z_2}\right)=\arg\left(\dfrac{a-b}{c-b}\right)$ and $\left|\dfrac{z_1-z_2}{z_3-z_2}\right|=\left|\dfrac{a-b}{c-b}\right|$

$\therefore \quad \dfrac{z_1-z_2}{z_3-z_2}=\dfrac{a-b}{c-b}$

$\therefore \quad (z_1-z_2)(c-b)=(a-b)(z_3-z_2)$

$\Rightarrow \quad z_1(b-c)+z_2(c-a)+z_3(a-b)=0$ and $a(z_2-z_3)+b(z_3-z_1)+c(z_1-z_2)=0$

54.(BC) $k+\left|k+z^2\right|=|z|^2$

$\Rightarrow \quad \left|k+z^2\right|^2=\left(|z|^2-k\right)^2 \quad \Rightarrow \quad \left(k+z^2\right)\left(k+\bar{z}^2\right)=|z|^4+k^2-2k|z|^2$

$\Rightarrow \quad k^2+(z\bar{z})^2+k\left(z^2+\bar{z}^2\right)=(z\bar{z})^2+k^2-2k(z\bar{z}) \quad \Rightarrow \quad k\left(z^2+\bar{z}^2+2z\bar{z}\right)=0 \quad \Rightarrow \quad k(z+\bar{z})^2=0$

As $k\neq 0$, $z+\bar{z}=0$ i.e., Re(z) = 0 So, z is a purely imaginary number. $\Rightarrow$ $\arg z=\pm\dfrac{\pi}{2}$

55.(ABC) Since z_1 and z_2 lie on $|z|=1$ and $|z|=2$, respectively, then $|z_1|=1$ and $|z_2|=2$

Now, $|2z_1+z_2| \le 2|z_1|+|z_2| \le 4 \Rightarrow \max|2z_1+z_2|=4$

Also, $|z_1-z_2| \ge ||z_1|-|z_2|| = |1-2| = 1 \Rightarrow \min|z_1-z_2|=1$

And, $\left|z_2+\frac{1}{z_1}\right| \le |z_2|+\frac{1}{|z_1|}=2+1=3 \Rightarrow \left|z_2+\frac{1}{z_1}\right| \le 3$

56.(AD) If z lies on the line joining a and ib, then

$$\arg\left(\frac{z-ib}{z-a}\right)=0 \text{ or } \pi$$

$$\Rightarrow \frac{z-ib}{z-a}=\frac{\bar{z}+ib}{\bar{z}-a} \Rightarrow z(-a-ib)+\bar{z}(-ib+a)+2iab=0$$

$$\Rightarrow z\left(-\frac{1}{2ib}-\frac{1}{2a}\right)+\bar{z}\left(-\frac{1}{2a}+\frac{1}{2ib}\right)+1=0 \Rightarrow z\left(\frac{1}{2a}+\frac{1}{2ib}\right)+\bar{z}\left(\frac{1}{2a}-\frac{1}{2ib}\right)=1 \Rightarrow z\left(\frac{1}{2a}-\frac{i}{2ib}\right)+\bar{z}\left(\frac{1}{2a}-\frac{i}{2b}\right)=1$$

57.(BD) $a+\bar{c} \ne 0$

Now, $(a+\bar{c})w_1^2+(b+\bar{b})w_1+(\bar{a}+c)=0 \Rightarrow (\bar{a}+c)\bar{w}_1^2+(b+\bar{b})w_1+(\bar{a}+c)=0$

Since, $\bar{w}_1 \ne 0$, we get $(a+\bar{c})\frac{1}{\bar{w}_1^2}+(b+\bar{b})\frac{1}{\bar{w}_1}+(\bar{a}+c)=0$

Hence, $\frac{1}{\bar{w}_1}$ is also a root, but $\frac{1}{\bar{w}_1}$, so $w_1=\frac{1}{\bar{w}_1} \Rightarrow |w_1|=1$ Similarly $|w_2|=1$

58.(AC) $|Z^2-9|+|Z^2|=41$ N $\Rightarrow |Z+3|+|Z-3|=10 \Rightarrow 10=|Z+3|+|Z-3| \ge |Z+3+Z-3| \Rightarrow |Z| \le 5$

59.(AB) $|z_1+z_2|=\left|-\frac{b}{a}\right|=1$

$|z_1z_2|=\left|\frac{c}{a}\right|=1$

$\therefore \quad |z_2|=1$

$\quad |z_1+z_2|^2=1$

$\therefore \quad 2+z_1\bar{z}_2+z_2\bar{z}_1=1$

Now $z_2=z_1e^{i\theta}$

$\therefore \quad |z_1+z_1e^{i\theta}|=|z_1||1+e^{i\theta}|=1$

$\therefore \quad 2\cos\frac{\theta}{2}=1$

$\therefore \quad \theta=\frac{2\pi}{3}$

Now, $\frac{(z_1+z_2)^2}{z_1z_2}=1 \Rightarrow \frac{b^{2a}}{a^2}=\frac{c}{a} \Rightarrow b^2=ac$

$PQ=|z_1-z_2|=\sqrt{3}$

60.(ABCD) $Z_1 = e^{i\theta_1}$, $Z_2 = e^{i\theta_2}$, $\text{Re}(Z_1 Z_2) = 0 \Rightarrow \theta_1 + \theta_2 = -\frac{\pi}{2}$, (as z_1, z_2 lie in fourth quadrant)

$Z_3 = e^{-i\theta_1}$, $Z_4 = -ie^{i\theta_1}$, $Z_5 = \cos\theta_1(1-i)$, $Z_6 = \sin\theta_1(-1+i)$

61.(AC) Let $z-3 = re^{i\phi}$ $\quad\therefore\quad$ $\frac{z-3}{e^{i\theta}} + \frac{e^{i\theta}}{z-3} = re^{i(\phi-\theta)} + \frac{1}{r}e^{i(\phi-\theta)}$

Imaginary part of the above = $r\sin(\phi-\theta) - \frac{1}{r}\sin(\phi-\theta) - \frac{1}{r}\sin(\phi-\theta)$

Given that r sin $(\phi-\theta) - \frac{1}{r}\sin(\phi-\theta) = 0$

$\Rightarrow$ $r - \frac{1}{r} = 0$ or $\sin(\phi-\theta) = 0$

$r - \frac{1}{r} = 0 \Rightarrow r = 1$

$\Rightarrow$ $|z-3| = 1$

$\Rightarrow$ z lies on a circle with unit radius.

$\sin(\phi-\theta) = 0 \Rightarrow \phi = \theta$

$\therefore$ $z-3 = re^{i\theta}$

$z-3 = r\cos\theta,\ y = r\sin\theta \quad\Rightarrow\quad x-3 = r\cos\theta,\ y = r\sin\theta$

$\frac{x-3}{\cos\theta} = \frac{y}{\sin\theta} = r$

The represents a straight line through (3, 0)

62.(AB) Given equation $z^6 = (z+1)^6$

$\Rightarrow$ $|z^6| = |(z+1)^6| \quad\Rightarrow\quad |z| = |z+1| \quad\Rightarrow$ Roots are collinear $\quad\Rightarrow\quad \arg\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = 0 \text{ or } \pi$

63.(CD) P is centre of square $\Rightarrow$ $AP = PC$ and $-APC = \frac{\pi}{2} - ABC$ is also $\frac{\pi}{2}$.

$\Rightarrow$ $ABCP$ is a cyclic quadrilateral.

$\Rightarrow$ A, B, C, P lie on a circle.

$\Rightarrow$ In circle chord, $AP = PC$

$\Rightarrow$ arc AP = arc PC

$\Rightarrow$ $-ABP = -PBC$

$\Rightarrow$ BP is angular bisector.

Let $\arg\left(\frac{z_1 - z_2}{z_4 - z_2}\right) = -\alpha$

$\Rightarrow$ $\arg\left(\frac{z_3 - z_2}{z_4 - z_2}\right) = \alpha$

Add (i) and (ii), $\arg\left(\frac{z_1 - z_2}{z_4 - z_2}\right) + \arg\left(\frac{z_3 - z_2}{z_4 - z_2}\right) = 0$

64. **[A-r] [B-s] [C-q] [D-p]**

If k = 2, line segment

If 0 < k < 2, No locus

If k = 5

Let $z_1 = 1$ $\quad z_2 = -1$

$|z_1 - z_2| = 2 = 2ae$

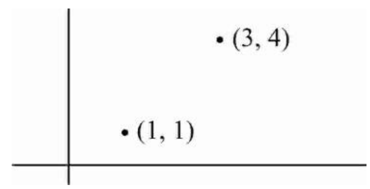
$5 = 2a \quad \Rightarrow \quad e = \dfrac{2}{5}$

65. **[A-q] [B-s] [C-r] [D-p]**

If $\theta = 0$, outside portion of line $\quad \therefore \quad$ line ray

If $\theta = \pi$, middle portion of line $\quad \therefore \quad$ line segment

If $\theta = \dfrac{2\pi}{3}$ (Obtuse) $\quad \therefore \quad$ minor arc



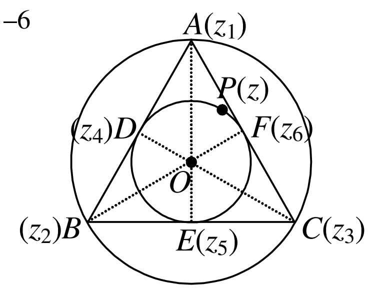
66. **[A-q] [B-p] [C-r] [D-s]**

(A) As we know,

$$|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\,\mathrm{Re}\, z_1\bar{z}_2$$

$\Rightarrow \quad AB^2 = OA^2 + OB^2 - 2\mathrm{Re}\; z_1\bar{z}_2 \quad \Rightarrow \quad \mathrm{RE}(z_1\bar{z}_2) = -2$

Similarly, $\mathrm{Re}(z_1\bar{z}_3) = \mathrm{Re}(z_3\bar{z}_1) = -2 \quad \Rightarrow \quad \mathrm{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = -6$



(B) Since $\angle AOC = \dfrac{2\pi}{3}$

$$\therefore \quad \frac{z_1 - 0}{z_3 - 0} = \frac{2}{2} e^{\frac{2\pi i}{3}}$$

$$\Rightarrow \quad \frac{4z_1}{z_3} = \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \times 4 = 2(-1 + i\sqrt{3}) \quad \Rightarrow \quad a = 2$$

(C) We have $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$

$$= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) + 2\,\mathrm{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = 12$$

(D) $DP^2 + EP^2 + FP^2 = |z - z_4|^2 + |z - z_5|^2 + |z - z_6|^2 = 3|z|^2 + |z_4|^2 + |z_5|^2 + |z_6|^2 = 6$

67.(1) Here $z^2 - z = |z|^2 + \dfrac{64}{|z|^5}$ $\quad \ldots\ldots\ldots$(i)

$\Rightarrow \quad z^2 - z = \bar{z}^2 - \bar{z}$ $\quad (\because z^2 - z$ is purely real number)

$\Rightarrow \quad (z - \bar{z})(z + \bar{z} - 1) = 0$

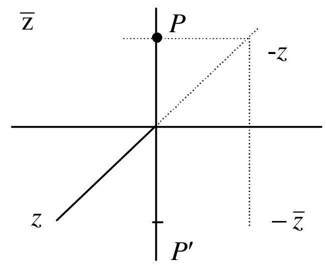
$\Rightarrow \quad z = \bar{z}$ as $z + \bar{z} = 1$ is not possible $\left(\because x \neq \dfrac{1}{2}\right)$

$\Rightarrow \quad z = x$

$\therefore$ Equation (i), given as $x^2 - x - |x|^2 - \dfrac{64}{|x|^5} = 0 \quad \Rightarrow \quad x = -2$

$\therefore$ Only one solution

68.(5) Clearly $P\left(\frac{\bar{z}-z}{2}\right)$ will lie on imaginary axis



$P'\left(\frac{z-\bar{z}}{2}\right) \Rightarrow \arg\left(\frac{z-\bar{z}}{2}\right)$ is $-\frac{\pi}{2}$

69.(7) For $|z-1|$ maximum $z = -4$ $\Rightarrow$ centroid $(\alpha) = \frac{z-w-\bar{w}}{3} = \frac{-4+1}{3} = -1$ $\Rightarrow$ Re $(\alpha) = -1$

70.(6) $|z|+|z-1|+|2z-3| = |z|+|z-1|+|3-2z| \ge |z+z-1+3-2z| = 2$

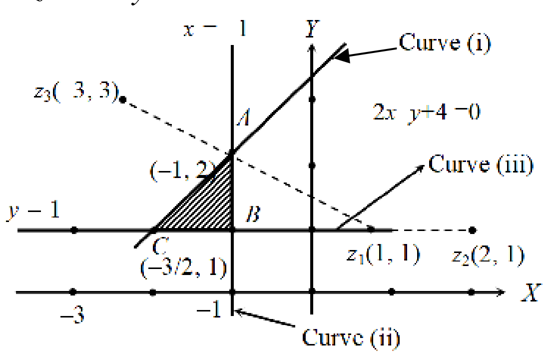
$\therefore \quad \lambda = 2$

$2[x]+3 = 3([x]-2)$

$[x] = 9 \;\therefore\; y = 2\times 9+3 = 21$

$\therefore \quad [x+y] = [x+21] = [x]+21 = 30$

71.(5) If $z = x+iy$



Required area = area of ΔABC

$= \frac{1}{2}\times(AB)(BC)$

$= \frac{1}{2}\times(2-1)\times\left(-1-\left(-\frac{3}{2}\right)\right)$

$= \frac{1}{4} = \frac{P}{Q} \Rightarrow P+Q = 5$

72.(5) $|A_k|^2 = A_k.\bar{A}_k = (x + y\alpha^k + p\alpha^{2k} + w\alpha^{3k} + f\alpha^{4k})\ (\bar{x} + \bar{y}\alpha^{-k} + \bar{p}\alpha^{-2k} + \bar{w}\alpha^{-3k} + \bar{f}\alpha^{-4k})$

$= x\bar{x} + y\bar{y} + p\bar{p} + w\bar{w} + f\bar{f} + x(\bar{y}\alpha^{-k} + \bar{p}\alpha^{-2k} + \omega\alpha^{-3k} + \bar{f}\alpha^{-4k})$

$+y\left(x\alpha^{k} + \bar{p}\alpha^{-k} + \bar{w}\alpha^{-2k} + \bar{f}\alpha^{-3k}\right) + \ldots\ldots + f(\bar{x}\alpha^{4k} + \bar{y}\alpha^{3k} + \bar{p}\alpha^{2k} + \bar{w}\alpha^{k})$

$\therefore \quad |A_0|^2 + |A_1|^2 + |A_2|^2 + |A_3|^2 + |A_4|^2 = \sum_{k=0}^{4}|A_n|^2 = 5(|x|^2 + |y|^2 + |p|^2 + |w|^2 + |f|^2) + x\sum_{k=0}^{4}(\bar{y}\alpha^{-k} + \ldots + \bar{f}\alpha^{-4k})$

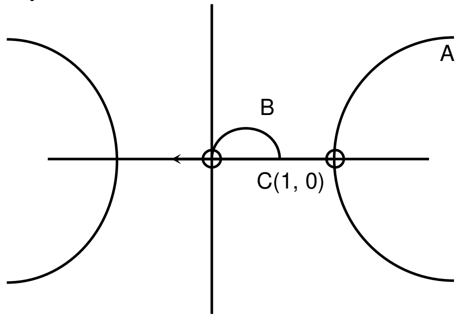
$+y\sum_{k=0}^{4}(\bar{x}\alpha^{k} + \bar{p}\alpha^{-k} + \ldots + \bar{f}\alpha^{-3k}) + \ldots + f\sum_{k=0}^{4}\left(\bar{x}\alpha^{4k} + \ldots + \bar{w}\alpha^{k}\right)$

But $\sum_{k=0}^{4}\alpha^{ak} = 0$, if a is not divisible by 5

$\Rightarrow \quad |A_0|^2 + |A_1|^2 + \ldots\ldots + |A_4|^2 = 5(|x|^2 + |y|^2 + |p|^2 + |w|^2 + |f|^2)$

$= 5(5)$ ($\because$ x, y, p, w, f are points on $z\bar{z} = 1$ i.e. $|z|^2 = 1$).

73.(0) A, B, C can be shown geometrically as below.



Clearly $A \cap B \cap C = \phi$.

74.(0) $\omega = \cos 12° + \cos 48° + i(\sin 48° + \sin 12°) = 2\cos 30° \cos 18° + 2i \sin 30° \cos 18° = 2\cos 18°\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

$z = \omega^6 = 2^6 \cos^6 18°(\cos\pi + i\sin\pi)$

$z = -2^6(\cos^6 18°) \qquad \therefore \qquad I_m(z) = 0$

75.(3) $8iz^3 + 12z^2 - 18z + 27i = 0 \quad \Rightarrow \quad 4iz^2(2z-3i) - 9(2z-3i) = 0$

$(4iz^2 - 9)(2z-3i) = 0 \quad \Rightarrow \quad |2z| = |3i|, \ |4iz^2| = 9$

$|z| = \frac{3}{2}, \ 4|z|^2 = 9 \Rightarrow |z| = \frac{3}{2} \Rightarrow 2|z| = 3$

76.(9) $\left|\frac{a^2x^2 + b^2y^2 + c^2z^2}{b^2x^2 + c^2y^2 + a^2z^2}\right| + \left(x_1^2 - 2y_1^2\right)\omega + \left(\left[x^2\right] + \left[y^2\right] + \left[z^2\right]\right)\omega^2 = 0 \qquad \ldots\text{(i)}$

$1 + \omega + \omega^2 = 0$ $\quad$ ($\because$ ω is cube root of unity) $\qquad \ldots$(ii)

Comparing (i) and (ii)

$$\left|\frac{a^2x^2 + b^2y^2 + c^2z^2}{b^2x^2 + c^2y^2 + a^2z^2}\right| = \left(x_1^2 - 2y_1^2\right) = \left[x^2\right] + \left[y^2\right] + \left[z^2\right]$$

a, b, c are cube roots of $q \ \Rightarrow \quad a = \lambda,\ b = \lambda\omega,\ c = \lambda\omega^2$

$$\left|\frac{a^2x^2 + b^2y^2 + c^2z^2}{b^2x^2 + c^2y^2 + a^2z^2}\right| = \left|\frac{\lambda^2x^2 + \lambda^2\omega^2y^2 + \lambda^2 \ \ \omega^4 \ \ z^2}{\lambda^2\omega^2x^2 + \lambda^2\omega^4y^2 + \lambda^2z^2}\right| = \left|\frac{1}{\omega^2}\right| = |\omega| = 1$$

$\therefore \quad x_1^2 - 2y_1^2 = 1 \ \Rightarrow \ x_1^2 = 1 + 2y_1^2 \ \Rightarrow \ x_1$ is odd.

Let $\quad x_1 = 2n + 1$, then $y_1^2 = 2n(n+1) \ \Rightarrow y_1$ is even

$\therefore \quad y_1$ is prime (given) $\Rightarrow \ y_1 = 2$

$\therefore \quad x_1 = 3 \quad \therefore [x^2] + [y^2] + [z^2] = 1$

$\therefore \quad [x^2 + x_1] + [y^2 + y_1] + [z^2 + x_1^2 + y_1^2] - 10$

$= [x^2] + [x_1] + [y^2] + [y_1] + [z^2] + [x_1^2] + [y_1^2] - 10 = 1 + 3 + 2 + 9 + 4 - 10 = 9.$

77.(1) $\left|\dfrac{2+3\omega+4\omega^2 z}{4\omega+3\omega^2 z+2z}\right| = \left|\dfrac{1}{z}\cdot\dfrac{2+3\omega+4\omega^2 z}{2+3\omega^2+4\omega z^{-1}}\right| = \left|\dfrac{1}{z}\right|\left|\dfrac{2+3\omega+4\omega^2 z}{2+3\omega^2+4\omega\bar{z}}\right| = \left|\dfrac{1}{z}\right|\left|\dfrac{2+3\omega+4\omega^2 z}{2+3\omega+4\omega^2 z}\right| = \left|\dfrac{1}{z}\right| = 1$

78.(0.414) $\left||z|-\dfrac{1}{|z|}\right| \le \left|z+\dfrac{1}{z}\right| = 2$

$\Rightarrow \quad -2 \le |z|-\dfrac{1}{|z|} \le 2 \quad \Rightarrow \quad |z|^2+2|z|-1\ge 0$ and $|z|^2-2|z|-1\le 0$

$\Rightarrow \quad |z| \ge \sqrt{2}-1, |z| \le \sqrt{2}+1 \quad \Rightarrow \quad |z|_{\min} = \sqrt{2}-1$

79.(0) $|z|=1$

Let $z\cos\theta + i\sin\theta$

$$\Rightarrow \quad \frac{z^n}{z^{2n}+1}-\frac{\bar{z}^n}{\bar{z}^{2n}+1}$$

$$= \frac{\cos n\theta + i\sin n\theta}{1+\cos 2n\theta + i\sin 2n\theta} - \frac{\cos n\theta - i\sin n\theta}{1+\cos 2n\theta - i\sin 2n\theta}$$

$$= \frac{\cos n\theta + i\sin n\theta}{2\cos n\theta(\cos n\theta + i\sin n\theta)} - \frac{\cos n\theta - i\sin n\theta}{2\cos n\theta(\cos n\theta - i\sin n\theta)} = \frac{1}{2\cos n\theta} - \frac{1}{2\cos n\theta} = 0$$

80.(6.25) $zz_1 = |z|^2 \Rightarrow z_1 = \bar{z}$

Now $|z-\bar{z}|+|z_1+\bar{z}_1|=10$

$\Rightarrow \quad |z-\bar{z}|+|z+\bar{z}|=10 \Rightarrow \quad |2iy|+|2x|=10 \quad \Rightarrow \quad |x|+|y|=5$

This represents a square and its area is $2\times 5^2 = 50$ units.

$\therefore \quad A = 50 \Rightarrow \dfrac{A}{8} = \dfrac{50}{8} = 6.25$

81.(1) $z=|z|+z^2$

Let $z=re^{i\theta}, r>0 \quad \Rightarrow \quad re^{i\theta} = r + r^2 e^{2i\theta}$

Comparing real and imaginary parts, we get

$r\cos\theta = r + r^2\cos 2\theta$ and $r\sin\theta = r^2\sin 2\theta$

If $r=0$; $z=0$

If $\cos\theta = 1 + r\cos 2\theta$ and $\sin\theta = r\sin 2\theta$

$\Rightarrow \quad \sin\theta = 2r\sin\theta\cos\theta$

$\Rightarrow \quad \sin\theta = 0$ or $r\cos\theta = \dfrac{1}{2}$

$\Rightarrow \quad \theta = 0, \pi$ or $\cos\theta = 1 + \dfrac{2\cos^2\theta - 1}{2\cos\theta}$

$\Rightarrow \quad z = 0$ or $2\cos^2\theta = 2\cos\theta + 2\cos^2\theta - 1$

$\Rightarrow \quad z=0$ or $\cos\theta = \dfrac{1}{2} \quad \Rightarrow \quad z = 0$ or $\theta = \pm\dfrac{\pi}{3}$

Thus, solutions are $z = 0, e^{ir/3}, e^{-ir/3}$

82.(0.20) $\left(e^{z^2}\right)=e^{\left(x^2-y^2\right)+2ixy} \Rightarrow \arg\left(e^{z^2}\right)=2xy$

Similarly, $\arg\left(e^{(z+i)}\right)=(y+1)$

Given $2xy=y+1 \Rightarrow y=\frac{1}{2x-1} \Rightarrow f(3)=\frac{1}{5}$

83.(10) Let $z^2=t$

$\Rightarrow t^3+t^2-2=0 \Rightarrow (t-1)\left(t^2+2t+2\right)=0 \Rightarrow t=1$ or $t^2+2t+2=0 \Rightarrow z=\pm 1$ or $z^2=-1\pm i$

$\Rightarrow |z|^2=\sqrt{2} \Rightarrow |z|^4=2 \Rightarrow \sum_{i=1}^{6}|z_i|^4=1+1+2+2+2+2=10$

84.(1) $z=e^{i2\theta}\sin\phi+\cos\phi$

$\Rightarrow z=\cos\phi+\sin\phi(\cos 2\phi+i\sin 2\phi)$

$\Rightarrow z=\cos\phi+\cos 2\phi\sin\phi+i\sin 2\phi\sin\phi$

$\Rightarrow |z|=\sqrt{\cos^2\phi+\cos^2 2\phi\sin^2\phi+2\cos\phi\sin\phi\cos 2\phi+\sin^2 2\phi\sin^2\phi}$

$\Rightarrow |z|=\sqrt{\cos^2\phi+\sin^2\phi+\sin 2\phi\cos 2\phi} \Rightarrow |z|=\sqrt{1+\frac{\sin 4\phi}{2}} \Rightarrow |z|_{\min}=\sqrt{\frac{1}{2}}$ and $|z|_{\max}=\sqrt{\frac{3}{2}}$

85.(1) Given $z=\alpha+i\beta$ is root of $z^5=1$

Also $|z|^5=1$

$\Rightarrow |z|=1 \Rightarrow \bar{z}=\frac{1}{z}$

Now $4\alpha\left(\beta^4-\alpha^4\right)$

$=4\alpha\left(\beta^2-\alpha^2\right)|z|^2=-\frac{1}{2}(z+\bar{z})\left[(z-\bar{z})^2+(z+\bar{z})^2\right]=-(z+\bar{z})\left(z^2+\bar{z}^2\right)$

$=-\left(z^3+z+\frac{1}{z}+\frac{1}{z^3}\right)=-\frac{1}{z^3}\left(\frac{z^8-1}{z^2-1}\right)=-\frac{z^8-1}{z^5-z^3}=\frac{1-z^3}{1-z^3}=1 \qquad (\because z^5=1)$

86.(10) $\left|2+\frac{1}{z}\right|^2+|2-z|^2=(2+\bar{z})^2+|2-z|^2=2\left[2^2+|z|^2\right]=10$

87.(0.50) $1+2|z|^2=|z^2+1|^2+2|z+1|^2$

$\Rightarrow 1+2z\bar{z}=\left(z^2+1\right)\left(\bar{z}^2+1\right)+2(z+1)(\bar{z}+1) \Rightarrow (z\bar{z})^2+2(z+\bar{z})+z^2+\bar{z}^2+2=0$

$\Rightarrow (z\bar{z})^2+2(z+\bar{z})+z^2+\bar{z}^2+2=0 \Rightarrow (z\bar{z}-1)^2+(z+\bar{z}+1)^2=0$

$\Rightarrow z\bar{z}=1$ and $z+\bar{z}+1=0 \Rightarrow z+\frac{1}{z}+1=0$

$\Rightarrow z^2+z=-1 \Rightarrow |z^2+z|=1$

88.(0) $z^2+|z|=\bar{z}^2$

Taking conjugate, we get $\bar{z}^2+|z|=z^2$

Adding (i) and (ii), we get

$\Rightarrow \quad 2|z|=0 \quad \Rightarrow \quad z=0$

89.(4) Z_1, Z_2, Z_3 are the roots of the equation $Z^3-Z^2+mZ-1=0$.

$\Rightarrow \quad Z_1+Z_2+Z_3=1$ and $Z_1Z_2Z_3=1$

Also, $|Z_1|=|Z_2|=|Z_3|=1$

Now, $|(Z_1+3)(Z_2+3)(Z_3+3)|$

$$=\left|Z_1Z_2Z_3+9(Z_1+Z_2+Z_3)+27+3\sum Z_1Z_2\right| = \left|1+9+27+3Z_1Z_2Z_3\left(\frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)\right|$$

$$=\left|1+9+27+3Z_1Z_2Z_3\left(\bar{Z}_1+\bar{Z}_2+\bar{Z}_3\right)\right| = \left|1+9+27+3Z_1Z_2Z_3\left(Z_1+Z_2+Z_3\right)\right| = \left|37+3(1)(1)\right|=40$$

90.(1) $|3z_1-2z_2-4|^2=|3z_1-1|^2+|2z_2+3|^2$

$\Rightarrow \quad |(3z_1-1)-(3z_2+3)|^2=|3z_1-1|^2+|2z_2+3|^2$

$\Rightarrow \quad A(3z_1-1),\ B(2z_2+3)$ and O(0) are situated as shown in the following figure.

Thus, OA and OB are perpendicular to each other.

$\Rightarrow \quad \arg\left(\frac{3z_1-1}{2z_2+3}\right)=\pm\frac{\pi}{2}$

So, $w=\frac{3z_1-1}{2z_2+3}$ is purely imaginary number.

$\Rightarrow \quad w=ki$

Hence, cube roots of w are $(ki)^{\frac{1}{3}}$

$\Rightarrow \quad w_1=k^{1/3}e^{i\pi/6},\ w_2=k^{1/3}e^{i5\pi/6}$ and $w_3=k^{1/3}e^{i3\pi/2} \quad \Rightarrow \quad \frac{w_2^2}{w_1w_3}=1$

91.(5) $z^3=\bar{z}$

$\Rightarrow \quad |z|^3=|\bar{z}|=|z| \quad \Rightarrow \quad |z|=0$ or $|z|^2=1 \quad \Rightarrow \quad z=0$ or $z^4=z\bar{z}=|z|^2=1$

Now $z^4=1$ has four roots. Thus, total number of roots is 5.

92.(4.88) $|\alpha+1+i|=1$

i.e., α lies on the circle having centre at $(-1,-i)$ and radius 1.

$\Rightarrow \quad |\alpha|_{max}=\sqrt{2}+1$

Similarly for $|\beta-2-3i|=6$, we have

$|\beta|_{max}=6+\sqrt{13}$

$\Rightarrow \quad 6|\alpha|_{max}-|\beta|_{max}=6\sqrt{2}-\sqrt{13}=\sqrt{72}-\sqrt{13}$

93.(0.875) Let $z = r(\cos\theta + i\sin\theta)$

$$\left|2z + \frac{1}{z}\right| = 1$$

$$\Rightarrow \left|\left(2r + \frac{1}{r}\right)\cos\theta + i\left(2r - \frac{1}{r}\right)\sin\theta\right| = 1$$

$$\Rightarrow \left(2r + \frac{1}{r}\right)^2 \cos^2\theta + \left(2r - \frac{1}{r}\right)^2 \sin^2\theta = 1$$

$$\Rightarrow \left(2r + \frac{1}{r}\right)^2 \left(2r + \frac{1}{r}\right)^2 \sin^2\theta + \left(2r - \frac{1}{r}\right)^2 \sin^2\theta = 1 \Rightarrow 4r^2 + \frac{1}{r^2} + 4 - 8\sin^2\theta = 1$$

$$\Rightarrow 4r^2 + \frac{1}{r^2} + 3 = 8\sin^2\theta$$

Now, $\dfrac{4r^2 + \frac{1}{r^2}}{2} \geq 2$ (using A.M. $\geq$ G.M)

$$\Rightarrow 4r^2 + \frac{1}{r^2} \geq 4 \Rightarrow 8\sin^2\theta \geq 7$$

94.(18.75) Let PQ be the diameter perpendicular to AB.

Let P and Q be represented by z and $-z$.

$\therefore$ $z_1 = ze^{i\alpha}, ze^{-i\alpha}$ and $z_3 = (-z)e^{i\beta}, z_4 = (-z)e^{i\beta}$

$\therefore$ $z_1 z_2 = z_3 z_4 \Rightarrow k = 1 \Rightarrow \dfrac{75k}{4} = \dfrac{75}{4} = 18.75$

95.(5) Let z be such a complex number, then according to the question

$z^3 = \bar{z}$ …(i)

$\Rightarrow$ $|z| = 0$ or $|z| = 1 \Rightarrow z = 0$ or $z = \cos\theta + i\sin\theta$

Putting $z = \cos\theta + i\sin\theta$ in (i), we get

$\cos 3\theta = \cos\theta$ and $\sin 3\theta = -\sin\theta$

$\Rightarrow$ $\sin\theta = 0$ or $\sin\theta = 1$ or $\sin\theta = -1$

$\Rightarrow$ $\theta = 0$ or π, or $\dfrac{\pi}{2}$ or $-\dfrac{\pi}{2}$

$\Rightarrow$ $z = 1, -1, i, -i$ or $z = 0$

96.(6.25) Let $z_2 = 4$ $z_1 = -3i$

$$z_3 = \frac{1}{1-i} = \frac{1+i}{2}$$

1.(D) No. of ways $= \left(3^4 - {}^3C_1 2^4 + {}^3C_2 1^4\right) \times {}^4C_2 = 36 \times 6 = 216$

2.(D) No. of ways $= 3^n + {}^nC_2 . 3^{n-2} + {}^nC_4 3^{n-4} + ...$

$$= \frac{1}{2}\left[(3+1)^n + (3-1)^n\right] = \frac{4^n + 2^n}{2} = 2^{n-1}\left(2^n + 1\right)$$

3.(B) Total number of ways $= 5! - 1 = 119$

4.(C) $(x_1 + x_2 + x_3)(y_1 + y_2) = 77$

case I : $x_1 + x_2 + x_3 = 7$ and $y_1 + y_2 = 11$ case II: $x_1 + x_2 + x_3 = 11$ and $y_1 + y_2 = 7$

$= {}^6C_2 \times {}^{10}C_1 + {}^{10}C_2 \times {}^6C_1 = 150 + 270 = 420.$

5.(B) Case I : 3 alike, 1 distinct $= 3 \times 2 \times \frac{\underline{|4}}{\underline{|3}} = 24$ Case II :2 alike, 2 distinct $= 3 \times 1 \times \frac{\underline{|4}}{\underline{|2}} = 36$

Case III: 2 alike, 2 alike $= {}^3C_2 \times \frac{\underline{|4}}{\underline{|2}\,\underline{|2}} = 18$

$m = 24 + 36 + 18 = 78$ and $n = 36$

$\frac{m}{n} = \frac{78}{36} = \frac{13}{6}$

6.(B) Case I : 2 alike, 1 distinct $= 8 \times 7 = 56$ Case II :3 alike = 8

Total no. of ways = 56 + 8 = 64

7.(A) Sum of unit place $(S_1) = 27 \times 45 = 1215$ Sum of 10^{th} place $(S_2) = (1+4+9) \times 81 = 1134$

Sum of 100^{th} place $(S_3) = 27 \times 45 = 1215$ Total sum $= 1215 + 1134 \times 10 + 1215 \times 100 = 134055$

8.(D) No. of ways $= 3^6 - {}^3C_1 (2)^6 + {}^3C_2 (1)^6 = 540.$

9.(C) $L = \frac{\underline{|p+q}}{\underline{|p}\,\underline{|q}},\ M = \frac{\underline{|p+q}}{\underline{|p}\,\underline{|q}} \times \underline{|2},\ N = \frac{\underline{|p+q}}{\underline{|p}\,\underline{|q}}$

$L = N = \frac{M}{2} \Rightarrow 2L = 2N = M$

10.(B) Case I : $\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{5}\boxed{\ }\boxed{\ }\boxed{\ } \rightarrow {}^5C_1 \times \underline{|3} = 30$

Case II: $\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{5}\boxed{\ }\boxed{\ } \rightarrow {}^6C_1 \times {}^5C_4 \times \underline{|3} = 180$

Case III : $\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{5}\boxed{\ } \rightarrow {}^7C_1 \times {}^6C_4 \times \underline{|3} = 630$

Case IV : $\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\boxed{5} \rightarrow {}^8C_1 \times {}^7C_4 \times \underline{|3} = 1680$

Total such numbers = 30 + 180 + 630 + 1680 = 2520

11.(B) Number of functions = $\dfrac{\text{total no. of functions}}{2} = \dfrac{2^{12}}{2} = 2^{11}$

12.(C) Total no. of non- negative integral solution of $x + y + z \le n$

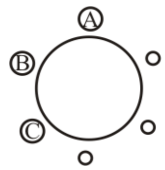
= total no. of non-negative integral solution of $x + y + z + w = n = {}^{n+4-1}C_{4-1} = {}^{n+3}C_3$

13.(A) Case I - 3 men and 2 women : No. of ways = ${}^{4}C_3 \times {}^{6}C_2 = 60$

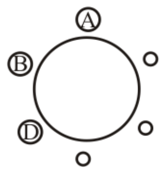
Case II – 4 men and 1 women : no. of ways = ${}^{4}C_4 \times {}^{6}C_1 = 6$

Total no. of ways = 60 + 6 = 66.

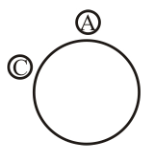
14.(D)



$\longrightarrow \lfloor 3 = 6$



$\longrightarrow \lfloor 3 = 6$



$\longrightarrow {}^{3}C_1 \times \lfloor 2 = 6$

= 18

15.(D) $xyz = 2^3 \times 3^1$

Number of non-negative integral solution $= {}^{3+3-1}C_{3-1} \times {}^{1+3-1}C_{3-1} = 30$

$\therefore$ Number of negative integral solutions = ${}^{3}C_2 \times 30 = 90$

Hence, total number of integral solutions = 120.

16.(B) Number of five-digit numbers = $\lfloor 3(1+2+3+4) = 60$

17.(C) $\_T\_T\_T\_T\_T\_T\_T\_T\_$

No. of ways = ${}^{9}C_6 \times \dfrac{\lfloor 6}{\lfloor 5} = 504$

18.(B)

Case	P_1	P_2	P_3	No. of ways
1.	1	2	4	$\dfrac{\lfloor 7}{\lfloor 1 \lfloor 2 \lfloor 4} \times \lfloor 3 = 630$

19.(D)

	P_1	P_2	P_3	No. of ways
	1	3	4	$\dfrac{\lfloor 8}{\lfloor 1 \lfloor 3 \lfloor 4} \times \lfloor 3 = 1680$
	2	3	3	$\dfrac{\lfloor 8}{\lfloor 2 \lfloor 3 \lfloor 3} \times \dfrac{\lfloor 3}{\lfloor 2} = 1680$
	2	2	4	$\dfrac{\lfloor 8}{\lfloor 2 \lfloor 2 \lfloor 4} \times \dfrac{\lfloor 3}{\lfloor 2} = 1260$

$= 4620$

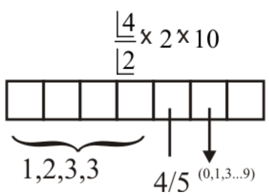
$K.\,{}^7P_3 = 4620$

$K = \dfrac{4620}{210} = 22$

20.(D) No of ways = ${}^{10}C_1.{}^{8}C_1.{}^{6}C_1.{}^{4}C_1 \times \dfrac{1}{\lfloor 4} = 80$

21.(A) No. of ways = ${}^4P_2 \times {}^6P_3 \times \lfloor 5 = 172800$

22.(B) No. of ways = $12 \times 2 \times 10 = 240$



23.(C) $\underline{S_1}\,\_\,\underline{S_2}\,\underline{G}\,\_\,\_\,\_\,\_ \longrightarrow$ No. of ways $= 2 \times 2 \times \lfloor 4 = 96$ (both grandsons on same side of grandfather)

$\_\,\underline{S_1}\,\_\,\underline{G}\,\_\,\underline{S_2}\,\_ \rightarrow$ No.of ways $= 2 \times 3 \times 3 \times \lfloor 4 = 432$ (opposite side)

Total ways = 96 + 432 = 528

24.(C)

C_1	C_2	C_3	C_4	No. of ways
2	2	2	3	$\dfrac{\lfloor 9}{(\lfloor 2)^3 \lfloor 3}$

25.(A) Let D_1, D_2 D_3 be the districts that require 2, 3, 5 men respectively. Number of ways $= \dfrac{\lfloor 10}{\lfloor 3 \lfloor 2 \lfloor 5}$.

26.(B) Digit in units place in $6^n = 6$

Digit in units place in $9^m = \begin{cases} 1 & m = EVEN \\ 9 & m = ODD \end{cases}$

$\therefore$ Digit in units place in $6^n + 9^m = \begin{cases} 7 & m = EVEN \\ 5 & m = ODD \end{cases}$

Hence, for m being an ODD integer, the required expression $(6^n + 9^m)$ will be a multiple of 5.

$\Rightarrow$ The number of ordered pairs $= 50 \times 25 = 1250$

27.(C)

28.(ABD) **(A)** $^{n}C_r$

(B) $\dfrac{\lfloor n}{\lfloor r \lfloor n-r} = {}^{n}C_r \left(\underbrace{0,0,0,...0}_{n-r},\underbrace{1,1,...1}_{r}\right)$

(C) $^{n+r}C_r$

(D) $^{n-1}C_{r-1} + {}^{n-1}C_r = {}^{n}C_r$

29.(BC) **(A)** $E_2(\lfloor 125) = \left[\dfrac{125}{2}\right] + \left[\dfrac{125}{2^2}\right] + \left[\dfrac{125}{2^3}\right] + ... + \left[\dfrac{125}{2^7}\right] = 119$

$E_5(\lfloor 125) = \left[\dfrac{125}{5}\right] + \left[\dfrac{125}{5^2}\right] + \left[\dfrac{125}{5^3}\right] = 31$

(B) $10^{10} - 1$

(C) $\boxed{4}\boxed{}\boxed{}\boxed{}\boxed{}$ Case I : 2 alike, 2 different $\rightarrow 1 \times {}^{3}C_1 \times \dfrac{\lfloor 4}{\lfloor 2}$

Case II : 4 distinct $\rightarrow \lfloor 4$

$\boxed{5}\boxed{}\boxed{}\boxed{}\boxed{}$ Case I : 2 alike, 2 alike - $\dfrac{\lfloor 4}{\lfloor 2 \lfloor 2}$

Case II : 2 alike, 2 distinct - $2 \times \dfrac{\lfloor 4}{\lfloor 2}$

Total numbers = 60 + 30 = 90

(D) $^{n}C_2 = 5050$

30.(BC) **(A)** $2^{10} - 2$; **(B)** $2^{10} - 1$; **(C)** $2^{10} - 1$; **(D)** 10

31.(BC) $2^n .1.3.5.7....$

$$^{2n}P_n = \frac{\lfloor 2n}{\lfloor n} = \frac{1.2.3.4.5.6....2n}{\lfloor n}$$

$$= \frac{2^n .1.2.3....n.1.3.5....}{\lfloor n}$$

$$= 2^n . \frac{\lfloor n .1.3.5....}{\lfloor n} = 2^n .1.3.5.....$$

32.(BCD) Number of ways = $3^5 - {}^{3}C_1 2^5 + {}^{3}C_2 = 150$

(A) No. of ways = $^{5}P_3 = 60$

(B) No. of parallelograms = $^{6}C_2 \times {}^{5}C_2 = 150$

(C) No. of ways = $3^5 - {}^{3}C_1 2^5 + {}^{3}C_2 = 150$

(D) No. of ways = $3^5 - {}^{3}C_1 2^5 + {}^{3}C_2 = 150$

33.(ABCD) No. of permutation = $\dfrac{\lfloor 2n}{\lfloor n \lfloor n} = {}^{2n}C_n$ (where n a's and n b's)

34.(CD) Case I: n = 2m

If common diff = 1 ; no. of triplets = $2m-2$

common diff = 2 ; no. of triplets = $2m-4$

common diff = 3 ; no. of triplets = $2m-6$

.

.

.

common diff = m – 1 ; no. of triplets = 2

Total number of triplets = 2 + 4 + 6 + ...+ (2m – 2) $= m(m-1) = \dfrac{n(n-2)}{4}$

Case II : n = 2m + 1 ;

If common difference = 1; no. of triplets = 2m – 1

common difference = 2; no. of triplets = 2m – 3

.

.

.

common diff = m ; no. of triplets = 1

Total number of triplets = 1 + 3 + 5 + ...+ (2m – 1) = $\dfrac{m}{2}[1+(2m-1)] = m^2 = \left(\dfrac{n-1}{2}\right)^2$

35.(BD) (A) Number of ways = $\dfrac{\lfloor n-1}{\lfloor p}$ **(B)** Number of ways = ${}^{n-1}C_p$

(C) Number of ways = ${}^{n-1}C_{p-1}$ **(D)** Number of ways = ${}^{n-1}C_p$

36.(BD) Number of selections when x < y < z = ${}^nC_3$

Number of selections when x = y < z = ${}^nC_2$ (we have to select only two numbers out of n numbers)

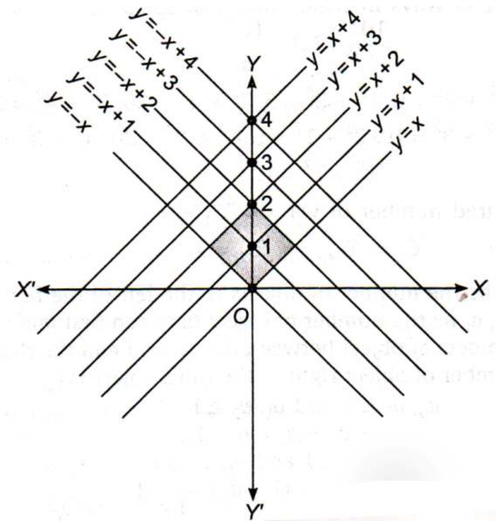
$\therefore$ Required number = ${}^nC_3 + {}^nC_2 = {}^{n+1}C_3$

37.(ABCD) **(A)** $3\times 7!$ **(B)** Total no. of arrangements of '*r*' white and 15 – *r* black balls = $\dfrac{15!}{r!(15-r)!}$

(C) Total no. of combinations = $5\times 6\times 2^3 = 240$

(D) Total no. of selections = 35

38.(BD) Hence, number of squares whose diagonals length is 2 unit



$= 3 \times 3 = 9$

39.(ACD) Number of ways = $\frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2}$

(A) No. of ways = ${}^{2+3-1}C_{3-1} \times {}^{4+3-1}C_{3-1} = {}^{4}C_{2} \times {}^{6}C_{2} = \frac{\lfloor 4}{\lfloor 2 \lfloor 2} \times \frac{\lfloor 6}{\lfloor 4 \lfloor 2} = \frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2}$

(B) No. of ways = $\frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2} \times \frac{1}{\lfloor 3}$ **(C)** Coeff of $x^2 y^2 z^2 = \frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2}$ **(D)** No. of ways = $\frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2}$

40.(AC) $p = 2520 = 2^3 \times 3^2 \times 5^1 \times 7^1$

$x = 3 \times 2 \times 2 \times 2 = 24$

$y = 4 \times 1 \times 2 \times 2 = 16$

41.(BD) Number of ways = $2^8 - 1$

42.(CD) Number of ways = $2^4 - 1$

43.(ABC) i.e., $0^2 + 0^2 = 0, 1^2 + 2^2 = 5, 2^2 + 4^2 = 20, 3^2 + 4^2 = 25$

$1^2 + 3^2 = 10, 2^2 + 1^2 = 5, 4^2 + 2^2 = 20, 4^2 + 3^2 = 25,$

$3^2 + 1^2 = 10$

$\therefore$ Required number of ways = ${}^{9}C_{1} = 9$ Also, ${}^{9}C_{8} = {}^{9}C_{9-8} = {}^{9}C_{1} = 9$

44.(ABC) Let a_0 be the number of objects to the left of the first object chosen, a_1 be the number of object between first and second, the numbers of object between the second and the third is a_2 and number of object right to the third object is a_3.

$\therefore$ $a_0, a_3 \geq 0$ and $a_1, a_2 \geq 1$

Also, $a_0 + a_1 + a_3 = n - 3$ ……(i)

$\because$ $1 + a_0, 1 + a_3 \geq 1$ and $a_1, a_2 \geq 1$ $\Rightarrow$ $(1 + a_0) + a_1 + a_2 + (1 + a_3) = n - 1$

$\therefore$ Required number of ways = ${}^{n-1-1}C_{4-1} = {}^{n-2}C_{3} = \frac{(n-2)(n-3)(n-4)}{6}$

Also, ${}^{n-2}C_{3} = {}^{n-3}C_{3} + {}^{n-3}C_{2}$

45.(AB) Each selection of 4 points, two on one line and two on the other will give one point of intersection.

$\therefore$ Required number of intersection = ${}^{m}C_2.{}^{n}C_2 = \frac{m(m-1)}{2}\frac{n(n-1)}{2} = \frac{mn(m-1)(n-1)}{4}$

46.(AD) Number of selections of 7 digits out of the digit 1, 2, 3, …., 9 = ${}^{9}C_7$

Number of digits out of these 7 selected digits excluding the greatest digit = 6

These 6 digits can be divided in two groups each having 3 digits = $\frac{6!}{3!3!2!} = {}^{6}C_3 \times \frac{1}{2!}$

But the 3 digits on one side can go on the other side

$\therefore$ Required number of ways = ${}^{9}C_7.{}^{6}C_3.\frac{1}{2!}.2! = {}^{9}C_7.{}^{6}C_3 = {}^{9}C_2.{}^{6}C_3$

47.(ABCD) (A) ${}^{6}C_3\,{}^{4}C_2\,5!\,5! = (5!)^3$ (B) ${}^{6}C_1\,9!$ (C) $(6+1)!\,4!$ (D) ${}^{10}P_4$

48.(ABCD) We know that product of r consecutive integer is divisible by $r!$ we have $P = (n-10)(n-9)(n-8)...(n+10)$ is product of 21 consecutive integers.

Which is divisible by 21!

$[(n-10)(n-9)(n-8)...n][(n+1)(n+2)(n+3)...(n+10)]$

Which is divisible by 11! 10!

$[(n-10)...(n-6)]\times[(n-5)...(n-1)]\times[n(n+1)...(n+4)]\times[(n+5)...(n+9)]\times(n+10)$

Which is divisible by $(5!)^4$

Also, it can be shown that P is divisible by 2! 3! 4! 5! 6!

49.(ABD) The cases for $a_1\{H,T\}$ i.e., $a_1 = 2$

The case for $a_2 : \{HT, TH, TT\}, a_2 = 3$

For $n \geq 3$, if the first outcome is *H*, then next just *T* and then a_{n-2}. If the first outcome is *T*, then a_{n-1} should follow.

So, $a_n = a_{n-1} + a_{n-2}$

So, $a_3 = a_1 + a_2 = 5$, $a_4 = 3 + 5 = 8$ and so on.

50.(ABCD) Number of five-digit numbers starting with $1 = {}^{8}C_4 = 70$

Number of five-digit numbers starting with $2 = {}^{7}C_4 = 35$

Total numbers = $70 + 35 = 105$

Thus, 105^th five-digit number is 26789

51.(ABC) Consider square of 2×2, in which we have 2 pairs of squares which have common vertex.

We have such 7×7 squares of size 2×2.

So, number of ways of selecting two squares having one vertex common is $7\times 7\times 2 = 98$.

In each row or column, we have 7 pairs of squares having one side common.

So, number of ways of selecting two squares having one side common is $7\times 8\times 2 = 112$.

Therefore, number of ways of selecting two squares such that they neither have a common vertex nor have a common side is ${}^{64}C_2 - 98 - 112 = 1806$

52.(AB) Number of visits teacher makes = Number of ways of selecting 5 students from 25 = ${}^{25}C_5$

Number visits a particular kid makes = Number of ways of selecting 4 students accompanying him/her $= {}^{24}C_4$

So, difference between the number of trips made by teacher and kid $= {}^{25}C_5 - {}^{24}C_4 = {}^{24}C_5$

53.(ABC) We can take the objects as:
0 identical and n distinct;
1 identical and $n-1$ distinct;
2 identical and $n-2$ distinct;
And so on…

$\therefore$ Total number of ways = $\sum_{r=0}^{n} {}^nC_r = 2^n$

Thus, (A) is true
Number of possible subsets of a set containing n elements is 2^n.
Also, ${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + ... + {}^{2n+1}C_n = 2^{2n}$
Thus, (B) and (C) are also true.

54.(AC) The required number of selections of 4 letters

= coeff. of x^4 in the expansion of $(x^0+x^1)(x^0+x^1)(x^0+x^1+x^2)(x^0+x^1+x^2)(x^0+x^1)$

= coeff. of x^4 in $(1+x)^3(1+x+x^2)^2 = 18$

55.(ABCD) In 1st round all the integers, which leaves the remainder 1 when divided by 15, will be marked. Last number of this category is 991. Next number to be marked is $(991+15-1000)=6$.

So, second round of integers which leave the remainder 6 when divided by 15 will be marked.
Last number of this category is 996
Next number to be marked is $(996+15-1000)=11$

Thus, third round of integers which leave the remainder 11 when divided by 15 will be marked.
Last number of this category is 986.
Next number to be marked is $(986+15-100)=1$; which has already been marked.

56.(BC) Required number of teams $= \dfrac{{}^{10}C_4 \,.\, {}^6C_3 \,.\, {}^3C_3}{2!} = \dfrac{10!}{4!\times 6!} \times \dfrac{6!}{3!\times 3!\times 3!} \times 3 = {}^{10}C_4 \times {}^5C_2 = 2100$

57.(ABC) If the sides are a, a, b then the triangle will be formed only when $2a > b$. So, for any $a \in N, b$ can change from 1 to $2a-1$.
When $a \le 1004$ then

Number of triangles = $1+3+5+...+(2(1004)-1) = (1004)^2$

And when $1005 \le a \le 2008, b$ can take any value from 1 to 2008.

Since a has 1004 possibilities, Number of triangles = $1004 \times 2008 = 2(1004)^2$

$\therefore$ Total number of isosceles triangles = $3(1004)^2$

58.(BD) Since, S_1 and S_2 get 2 books each and S_3 and S_4 get 3 books each.

$\therefore$ Required No. of ways = $\dfrac{10!}{2!\,2!\,3!\,3!}$

59.(ABC) $\because \quad 3^p = (4-1)^p = 4\lambda_1 + (-1)^p$

$5^q = (4+1)^q = 4\lambda_2 + 1 \quad$ and $\quad 7^r = (8-1)^r = 8\lambda_3 + (-1)^r$

Hence, both p and r must be odd, or both must be even. Thus, $p+r$ is always even. Also, $p+q+r$ can be odd or even.

60.(AD)

61.(BC) Let $x = \vec{r}.\hat{i}$, $y = \vec{r}.\hat{j}$ and $z = \vec{r}.\hat{k}$

where x, y and z are the components of $\vec{r}$ along x, y and z-axis resp. and they are given to be positive.

$\therefore \quad \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Also $\vec{r}.\vec{a} \le 12$

$\Rightarrow \quad x + y + z \le 12$

$\therefore$ (the number of values of $\vec{r}(I)$) = (the number of positive integral solutions of ($x+y+z \le 12$))

$\Rightarrow \quad I = \sum_{n=3}^{12} {}^{n-1}C_2 = {}^2C_2 + {}^3C_2 + ... + {}^{11}C_2 \quad \Rightarrow \quad I = {}^3C_0 + {}^3C_1 + {}^4C_2 + ... + {}^{11}C_9 \;\; \{\because {}^nC_r = {}^nC_{n-r}\}$

$\therefore \quad I = {}^4C_1 + {}^4C_2 + ... + {}^{11}C_9 = ... = {}^{12}C_9$

62.(ABC) **63.(AD)** **64.(ABD)**

65. **[A-r] [B-s] [C-p] [D-q]**

(A) $5^7 = 78{,}125$

(B) ${}^5C_3 \times \left[3^7 - {}^3C_1 2^7 + {}^3C_2\right] = 10 \times [2187 - 384 + 3] = 18060$

(C) $5^7 - {}^5C_1 = 78{,}120$

(D) $5^7 - {}^5C_1 - {}^5C_2\left[2^7 - {}^2C_1\right] = 76{,}860$

66. **[A-r] [B-s] [C-p] [D-q]**

(A) $\underline{|7}\left(1 - \frac{1}{\underline{|1}} + \frac{1}{\underline{|2}} - \frac{1}{\underline{|3}} + - - - \frac{1}{\underline{|7}}\right) = 1854$

(B) ${}^7C_3 \times D_4 = 315$

(C) $\underline{|7} - D_7 - {}^7C_1.D_6 = \underline{|7} - 1854 - 265 \times 7 = 1331$

(D) $\underline{|7} - D_7 - {}^7C_1D_6 - {}^7C_2D_5 = 407.$

67. **[A-q] [B-s] [C-p] [D-r]**

(A) $(6)^7 - {}^6C_1(5)^7 + {}^6C_2(4)^7 - {}^6C_3(3)^7 + {}^6C_4(2)^7 - {}^6C_5(1)^7$

$= 279936 - 468750 + 245760 - 43740 + 1920 - 6 = 15,\ 120$

(B) ${}^6C_3.\left[(3)^7 - {}^3C_1(2)^7 + {}^3C_2\right] = 20 \times [2187 - 384 + 3] = 36,\ 120$

(C) $6^7 - {}^6C_1 = 279930$

(D) ${}^6C_1 \times 5^7 - {}^6C_2(4)^7 + {}^6C_3(3)^7 - {}^6C_4(2)^7 + {}^6C_5(1)^7 = 264816$

68. **[A-s] [B-r] [C-p] [D-q]**

(A) $\underline{|12}$ **(B)** $\underline{|11}$ **(C)** $3 \times \underline{|11}$ **(D)** $6 \times \underline{|11}$

69. **[A-p, r, s]** **[B-p, r, s]** **[C-q, t]** **[D-q, t]**

Answers to the column-II as **p.** 32, **q.** 33, **r.**32, **s.** 32, **t.** 33

(A) Total number of ways of arrangement = 35!. So exponent of 2 in 35! = 32

(B) Total number of ways of arrangement = 9 × 35!. So exponent of 2 in 9 × 35! = 32

(C) Total number of ways of arrangement = 18 × 35!. So exponent of 2 in 18 × 35! = 33

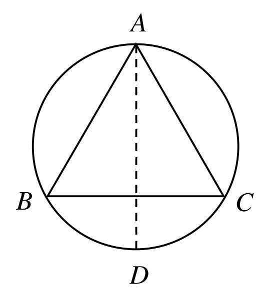
(D) Total number of ways of arrangement = 6 × 35!. So exponent of 2 in 6 × 35! = 33

70.(1) $x+2y=10,$ where x is the number of times he takes single steps, and y is the number of times he takes two steps

Cases	Total number of ways
1. $x=0,\ y=5$	$5!/5!=1$
2. $x=2,\ y=4$	$6!/2!4!=15$
3. $x=4,\ y=3$	$7!/4!3!=35$
4. $x=6,\ y=2$	$8!/2!6!=28$
5. $x=8,\ y=1$	$9!/8!=9$
6. $x=10,\ y=0$	10!/10! = 1

$\therefore \quad p=89$

71.(3) For every acute angle $\Delta ABC,$ we will



Get 3 obtuse angled triangles by shifting one of the vertices to the diametrically opposite point on the circum circle of the polygon.

72.(7) ${}^{n}C_2-n=14 \quad \Rightarrow \quad$ n = 7

73.(7) $\mathrm{f}(1)<\mathrm{f}(2)<\mathrm{f}(3)<\mathrm{f}(4)<\mathrm{f}(5)<\mathrm{f}(6)$

Total no. of strictly increasing functions = ${}^{9}\mathrm{C}_6=\dfrac{9\times8\times7}{6}=84$

74.(9) $\mathrm{f}(1)\le\mathrm{f}(2)\le\mathrm{f}(3)\le\mathrm{f}(4)\le\mathrm{f}(5)$

Total no. of non-decreasing functions = ${}^{13}\mathrm{C}_5=1287$

$$\frac{\mathrm{m}}{143}=9$$

75.(5) Number of ways = $3^5\times{}^{4+3-1}\mathrm{C}_{3-1}=3^5\times{}^{6}\mathrm{C}_2$

p = 5, q = 15; q – 2p = 15 – 10 = 5

76.(9) ____AB____

$\mathrm{m}=\dfrac{7!\times2!}{2}=5040$ Sum of digits = 9

77.(2) **Case I**: First row contains 1x, second row contains 4x and third row contain 1x

No. of ways = $2\times1\times2=4$

Case II: First row contains 2x, second row contains 3x, and third row contain 1x

No. of ways = $1\times{}^4C_3\times2=8$

Case III: First row contains 1x, second row contains 3x, and third row contain 2x

No. of ways = $2\times{}^4C_3\times1=8$

Case IV : First row contains 2x, second row contains 2x, and third row contain 2x

No. of ways = $1\times{}^4C_2\times1=6$

Total number of ways = 4 + 8 + 8 + 6 = 26

78.(6) Let $x_1, x_2, x_3, \ldots, x_7$ persons are entering into Gate A, B, C, D,…..G gates respectively

Now, $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 200$

Total no. of ways to enter = ${}^{200+7-1}C_{7-1} = {}^{206}C_6 = {}^{206}C_{200}$

Now the arrangement of these 200 persons will also count so

Total no. of ways = ${}^{206}C_{200}\times\underline{|200} = {}^{206}P_{200}$

n = 206, r = 200

n-r = 6 No. of different list = ${}^{206}P_6$

79.(9) $U_4=\underline{|4}\left(1-\frac{1}{\underline{|1}}+\frac{1}{\underline{|2}}-\frac{1}{\underline{|3}}+\frac{1}{\underline{|4}}\right)=9$

80.(47) They have note of same denomination hence that denomination must be HCF of 6, 7 and 8 which is 1, so each of them have notes of $1.

Now let us assume that the number of notes that they have is x_1, x_2, and x_3. So, the numbers of ways in which they can denote Rs. 10 is the same as the number of solutions to the question

$x_1+x_2+x_3=10$

Subject to conditions

$0\le x_1\le6,\ 0\le x_2\le7,\ 0\le x_3\le8$

Hence the required number of ways = coefficient of x^{10} in

$\left(1+x+x^2+\ldots+x^6\right)\left(1+x+x^2+\ldots x^7\right)\left(1+x+x^2+\ldots+x^8\right)$

= coefficient of x^{10} in $\left(1-x^7\right)\left(1-x^8\right)\left(1-x^9\right)\left(1-x\right)^{-3}$

= coefficient of x^{10} in $\left(1-x^7-x^8-x^9\right)\left(1+{}^3C_1x+{}^4C_2x^2+{}^5C_3x^3+\ldots{}^{12}C_{10}x^{10}\right)$

$={}^{12}C_2-{}^5C_3-{}^4C_2-{}^3C_1=47$

81.(23) Each vertex can be coloured in 2 ways, so there are $2^8 = 256$ ways (when vertices are fixed under the action of the trivial symmetry) now consider symmetry under different rotations. There are $2^4 = 4$ (out of 256) that are fixed under a 90-degree rotation about the axis joining the mid-points of two opposite faces. There are $2^4 = 16$ (out of 256) that are fixed under a 120° rotation about the axis joining two diametrically opposite corners. (In this case rotation splits up the 8 vertices as $1+3+3+1=8$). There are $2^4 = 16$ (out of 256) that are fixed under a 180° rotation about the axis joining the mid-points of two opposite faces. (In this case rotation splits up the 8 vertices as $2+2+2+2=8$). There are $2^4 = 16$ (out of 256) that are fixed under a 180° rotation about the axis joining the mid-points of two opposite edges. (In this case rotation splits up the 8 vertices as $2+2+2+2=8$).

Hence the number of orbits is $\frac{1}{24}(1\times 256+6\times 4+8\times 16+3\times 16+6\times 16)=23$

82.(14)

83.(4) Ramesh has to 1st select in fruits out of $2n$ fruits, for this following case may arise.

Case (i) Number of distinct fruits 0 and number of identical fruits n then number of ways is ${}^nC_0$.

Case (ii) Number of distinct fruits 1 and number of identical fruits $n-1$ then number of ways is ${}^nC_1$.

Case (iii) Number of distinct fruits 2 and number of identical fruits $n-2$ then number of ways is ${}^nC_2$. Ans so on

Last case number of distinct fruits n and number of identical fruits 0 then number of ways in ${}^nC_n$

So total number of ways is ${}^nC_0+{}^nC_1+{}^nC_2+\ldots+{}^nC_n = 2^n\ \left(2^n\right)=16 \quad \Rightarrow \quad n=4$.

84.(10) Let a particular customer buys ice creams Vanila, Strawberry, Chocolate and Butter Scotch of a, b, c and d numbers then from the given condition $0 < a+b+c+d \le 10$. Let us assume a dummy variable k such that $a+b+c+d+k=10$ then number of integral solution of this equation is ${}^{10+5-1}C_{5-1} = {}^{14}C_4$

$${}^{10+5-1}C_{5-1} = {}^{14}C_4 = \frac{14!}{(4!)(10!)} = \frac{14\times 13\times 12\times 11}{24} = 1001.$$

But there is one case when k=10 then $a+b+c+d=10$, this case is not applicable case hence required answer is $1001-1=1000 \Rightarrow \frac{k}{100}=10$

85.(3.150) Let $x_1, x_2, \ldots, x_6$ be the number that appears on the six dies. As per the given condition $x_1+x_2+x_3+\ldots+x_6 \le 17$. Introducing a dummy variable $x_7\ (0 \le x_7)$ the inequality becomes an equation $x_1+x_2+\ldots+x_6+x_7=17$. Here $1 \le x_i \le 6$ where i = 1, 2m … , 6 and $0 \le x_7$. So that No of solution =

coefficient of x^{17} in $\left(x+x^2+\ldots+x^6\right)^6\left(1+x+x^2+x^3+\ldots\right)$ = coefficient of x^{17} in $\left(1-x^6\right)^6(1-x)^{-7}$

= coefficient of x^{17} in $\left(1-6x^6\right)(1-x)^{-7}$ = coefficient of x^{17} in $\left(1-6x^6\right)$

$\left(1+{}^7C_1x+\ldots+{}^{10}C_4x^4+{}^{11}C_5x^5+\ldots{}^{17}C_{11}x^{11}+\ldots\right)={}^{17}C_{11}-6.{}^{11}C_5$

Since total number of outcomes is 6^6 Required number of ways to get a sum greater than

$17 = 6^6-\left({}^{17}C_{11}-6\,{}^{11}C_5\right)=46656-12376-6\times 642=31508 \Rightarrow 3.150$

86.(243) Since it is given that $a+b+3c \le 30$, so assume a number $d \ge 0$ such that $a+b+3c+d=30$, a, b and c are the positive integers so $a=a_1+1$, $b=b_1+1$, and $c=c_1+1$ then the given equation we can write as $(a_1+1)+(b_1+1)+3(c_1+1)+d=30$. Or $a_1+b_1+3c_1+d=25$. Since coefficient of c_1 is 3 hence it can-not takes a value more than 8 if we form cases then we will have 9 cases:

Case (i) $c_1=0$ then $a_1+b_1+d=25$. Since coefficient of c_1 is 3 hence it can-not takes a value more than 8 if we form cases then we will have 9 cases:

Case (i) $c_1=0$ then $a_1+b_1+d=25$ and number of non-negative integral solution is

${}^{25+3-1}C_{3-1} = {}^{27}C_2 = 351$.

Case (ii) $c_1=1$ then $a_1+b_1+d=22$ and number of non-negative integral solution is

${}^{22+3-1}C_{3-1} = {}^{24}C_2 = 276$.

Case (iii) $c_1=1$ then $a_1+b_1+d=19$ and number of non-negative integral solution is

${}^{19+3-1}C_{3-1} = {}^{21}C_2 = 210$.

Case (iv) $c_1=3$ then $a_1+b_1+d=16$ and number of non-negative integral solution is ${}^{16+3-1}C_{3-1} = {}^{18}C_2 = 153$.

Case (v) $c_1=4$ then $a_1+b_1+d=13$ and number of non-negative integral solution is

${}^{13+3-1}C_{3-1} = {}^{15}C_2 = 105$.

Case (vi) $c_1=5$ then $a_1+b_1+d=10$ and number of non-negative integral solution is

${}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$.

Case (vii) $c_1=9$ then $a_1+b_1+d=7$ and number of non-negative integral solution is ${}^{7+3-1}C_{3-1} = {}^{9}C_2 = 36$.

Case (viii) $c_1=7$ then $a_1+b_1+d=4$ and number of non-negative integral solutions is

${}^{4+3-1}C_{3-1} = {}^{6}C_2 = 15$.

Case (ix) $c_1=8$ then $a_1+b_1+d=1$ and number of non-negative integral solutions is ${}^{1+3-1}C_{3-1} = {}^{3}C_2 = 3$.

So total number of integral solutions is $51+276+210+153+105+66+36+15+3 = \frac{1215}{5} = 243$.

87.(6) In the system now total m and another set of m parallel lines. From m lines two lines can be selected in ${}^{m}C_2$ ways. So total number of parallelograms is $({}^{m}C_2)^2 = 225$

88.(240) Since, $x \ge 1$ then $y \ge 2$ $[\because x < y]$

If $y=n$, then x takes values form 1 to $n-1$ and z can take the values from 0 to $n-1$ (i.e., n values)

Thus, for each values of $y(2 \le y \le 9)$, x and z take $n(n-1)$ values.

Hence, the 3-digit numbers are of the form xyz

$$= \sum_{n=2}^{9} n(n-1) = \sum_{n=1}^{9} n(n-1) \quad [\because \text{at } n=1, (n-1)=0]$$

$$= \sum_{n=1}^{9} n^2 - \sum_{n=1}^{9} n = \frac{9(9+1)(18+1)}{6} - \frac{9(9+1)}{2} = 240 = \lambda$$

89.(15.68) The first rook can be placed in 64 ways, and the second rook cannot be placed in the same row or the same column. So, it has 7 rows and 7 columns left for it. It can be placed in $7\times7=49$ ways.

But the order in which the rooks are placed is not important. So, it will be divided by 2!

Total number of ways = $64\times49/2=1568$. $\Rightarrow$ 15.68

90.(10) Total number of triangles formed is ${}^kC_3$, now consider the following cases

Case (i): Number of triangles having three sides common with the sides of polygon is 0.

Case (ii): Number of triangles having two sides common with the sides of polygon is k.

Case (iii): Number of triangles having one sides common with the sides of polygon = number of ways of selecting 3 points (vertices) out of which only two are consecutive is $K\left({}^{K-4}C_1\right)=K(K-4)$. So, number of required triangles is ${}^kC_3-k-k(k-4)=k(k-1)(k-2)/6-k-k(k-4)=\dfrac{k(k-4)(k-5)}{6}=50$

91.(729) In Physics 1st and 2nd prize can be given in $10\times9=90$ ways and similarly prizes in other 2 subjects can be given in 90 ways, Hence total number of ways is $90\times90\times90=729000$. $\Rightarrow$ 729.

92.(315) Lets consider 5 points *A*, *B*, *C*, I and *E*. Consider a point *A*, number of straight lines that can be drawn through *B*, *C*, *D* and *E* is $\left({}^4C_2\right)=6$ straight lines, so from *A* we can draw perpendicular to these 6 straight lines, similarly we will get 6 perpendiculars from other points as well so total number of perpendiculars is $6\times5=30$.

The maximum number of intersections of these 30 straight lines is $\left({}^{30}C_2\right)=435$, but these 435 points are not distinct means out of 30 straight lines not all of them are non-concurrent. We have to consider the following cases.

Case (i) Consider one of the ten lines drawn from original 5 points, say this line *AB*, there are three perpendiculars are drawn on *AB* from points *C*, *D* and *E*, these 3 perpendiculars are parallel to each other hence they will not intersect each other these three point if not parallel could have intersected at 3 points hence we lost $3\times10=30$ points.

Case (ii) Since three altitudes of a triangle intersect each other at a point (Called Ortho centre). From the original 5 points we will have $\left({}^{10}C_3\right)=10$ triangles. Consider one of these 10 triangles, perpendicular from vertex to opposite side (altitude) will intersect at one point instead of 3 points, hence from each triangle we lost 2 points so total number of points that we have to subtract is $2\times10=20$ points.

Case (iii) Consider one of the original 5 points (say A) since from A we have drawn 6 perpendiculars and these 6 perpendiculars are concurrent hence they intersect each other at one point instead of $\left({}^6C_2\right)=15$ points so from one of this point we lost $15-1=14$ points, hence total we lost $14\times5=70$ points.

So total number of points is $435-30-20-70=315$ points.

93.(2) Let's start with the minimum number of non-congruent rectangles and for that purpose with minimum area. If area is 1 sq. unit the we will get only 1 rectangle of 1×1

If area is 2 sq. unit then we will get only 1 rectangle of 1×2

If area is 3 sq. unit then we will get only 1 rectangle of 1×3

If area is 4 sq. unit then we will get only 2 rectangle of 1×4 or 2×2

If area is 5 sq. unit then we will get only 1 rectangle of 1×5

If area is 6 sq. unit then we will get 2 rectangle of 1×6 or 2×3

If area is 7 sq. unit then we will get only 1 rectangle of 1×7 but it is not possible in 6×6 square

Last rectangle will be of area 8 square unit

Here we have considered all the non-congruent rectangles and summation of these areas is

$1+2+3+4+4+5+6+6+8=39>36$ hence, we must have at least 2 congruent rectangles = 2

94.(7) Since it is given that $A\cap B=\{2, 3, 5, 7\}$ so remaining numbers $\{1, 4, 6, 8, 9, 10\}$ can be a member of either set A or set B or neither in set A nor in set B, so each member of set $\{1, 4, 6, 8, 9, 10\}$ has 3 options so total number of ways/pairs is 3^6 but since $A\neq B$ hence required number of ways is 3^6-1.

95.(7) ${}^{N-2}C_3=10$

96.(2500) Since unit digit of 7^k ends in 7, 9, 3 or 1 (corresponding to $k=4x+1,\ 4x+2,\ 4x+3$ *and* $4x$ respectively.)

Thus, 7^p+7^q cannot end in 5 for any values of p, q. So, for 7^p+7^q to be divisible by 5, it should end in 0.

For 7^p+7^q to end in 0, the forms of p and q should be as follows:

	P	Q
1	$4x$ (Unit digit) 1	$4y+2$ (Unit digit 9)
2	$4x+1$ (Unit digit 4)	$4y+3$ (Unit digit 3)
3	$4x+2$ (Unit digit 9)	$4y$ (Unit digit 1)
4	$4x+3$ (Unit digit 3)	$4y+1$ (Unit digit 7)

Thus, for a given value of m there are just 25 values of n for which 7^p+7^q ends in 0.

[For instance, if $p=4x$, then $q=2, 6, 10, \ldots, 98$]

Thus there are $100\times25=2500$ ordered pairs (m, n) for which 7^p+7^q is divisible by 5.

97.(10) Let number of girls is 'g' then number of group having 4 boys and 1 girl = $\left({}^4C_4\right)\times\left({}^gC_1\right)=g$ and number 4 of groups having 3 boys and 2 girls = $\left({}^4C_3\right)\times\left({}^gC_2\right)=2g(g-1)$ thus, total number of tests is $g+2g(g-1)=66$

or $2g^2-g-66=0$ only integral value of g is 6

Hence total number of students is $4+6=10$

98.(4) Obviously N = 10!, therefore $\dfrac{N}{9!}=10$.

99.(5)

Binomial Theorem

1.(D) $(1+n)^{m+1}-n^{m+1}=1+{}^{m+1}C_1 n+{}^{m+1}C_2 n^2+{}^{m+1}C_3 n^3+....+{}^{m+1}C_m n^m$

Put $n=1, 2, 3, ...n$ and add

$(1+n)^{m+1}-1^{m+1}=n+{}^{m+1}C_1 S_1+{}^{m+1}C_2 S_2+{}^{m+1}C_3 S_3+....+{}^{m+1}C_m S_m$

$\Rightarrow \quad {}^{m+1}C_1 S_1+{}^{m+1}C_2 S_2+{}^{m+1}C_3 S_3+....+{}^{m+1}C_m S_m=(n+1)^{m+1}-(n+1)$

2.(B) $\sum_{r=1}^{n}\frac{r(r+1)}{2r}\,{}^nC_r^2=\frac{1}{2}\sum_{r=1}^{n}(r\,{}^nC_r^2+{}^nC_r^2)=\frac{1}{2}\sum_{r=1}^{n}(n\,{}^{n-1}C_{r-1}\,{}^nC_{n-r}+{}^nC_r^2)=\frac{1}{2}(n\;{}^{2n-1}C_{n-1}+{}^{2n}C_n-1)$

3.(A) $S=\sum_{r=0}^{n}\frac{(-1)^r\,{}^nC_r}{(4r+2)}=\int_0^1 x(1-x^4)^n\,dx=\frac{1}{2}\int_0^1(1-x^2)^n dx$

$I_n=\int_0^1(1-x^2)^n\,dx=x(1-x^2)^n\Big|_0^1+2n\int_0^1 x^2(1-x^2)^{n-1}dx=2nI_{n-1}-2nI_n$

$I_n=\frac{2n}{(2n+1)}I_{n-1}=\frac{2^2n(n-1)}{(2n+1)(2n-1)}I_{n-2}=.....$

$=\frac{2^n n(n-1)(n-2)....1}{(2n+1)(2n-1)(2n-3).....3}I_0 \quad =\frac{2^n n!}{3\cdot5\cdot7.......(2n-1)(2n+1)}$

4.(B) $S=\sum_{r=1}^{n}(-1)^{r-1}\left(1+\frac{1}{2}+\frac{1}{3}+....+\frac{1}{r}\right)\;{}^nC_r$

$=\sum_{r=1}^{n}\left({}^nC_r(-1)^{r-1}\int_0^1(1+x+x^2+....+x^{r-1})\;dx\right)=\sum_{r=0}^{n}\int_0^1\left(\frac{1-x^r}{1-x}\right)\;{}^nC_r(-1)^{r-1}dx$

$=\int_0^1\left(\sum_{r-0}^{n}\frac{{}^nC_r(1-x^r)(-1)^r}{1-x}\right)dx=\int_0^1\left(\frac{0-(1-x)^n}{(1-x)}\;dx=\int_0^1(1-x)^{n-1}dx=\int_0^1 x^{n-1}dx=\frac{1}{n}\right)dx$

5.(B) $\sum_{r=0}^{n}{}^nC_r(1-x)^r(-1)^r=x^n$

$\Rightarrow \quad \sum_{r=1}^{n}{}^nC_r(-1)^{r-1}(1-x)^{r-1}=\frac{1-x^n}{1-x}$

$\Rightarrow \quad \int_1^x\left(\sum_{r=1}^{n}{}^nC_r(-1)^{r-1}(1-x)^{r-1}\right)dx=\int_1^x(1+x+x^2+x^3+....+x^{n-1})\;dx$

$\Rightarrow \quad -\sum_{r=1}^{n}\frac{{}^nC_r(-1)^{r-1}(1-x)^r}{r}=\frac{x-1}{1}+\frac{x^2-1}{2}+\frac{x^3-1}{3}+....+\frac{x^n-1}{n}$

$\Rightarrow \quad \sum_{r=1}^{n}\frac{{}^nC_r(1-x)^r(-1)^{r-1}}{r}=\frac{1-x}{1}+\frac{1-x^2}{2}+\frac{1-x^3}{3}+....+\left(\frac{1-x^n}{n}\right)$

6.(C) $\sum\limits_{r=0}^{n}\frac{{}^nC_r}{r+1}a^{n-r}b^r = \frac{1}{n+1}\sum\limits_{r=0}^{n}{}^{n+1}C_{r+1}a^{n-r}b^r = \frac{1}{(n+1)b}[(a+b)^{n+1}-a^{n+1}]$

Put $a = x^2,\, b = 2-x^2$

$\Rightarrow \quad \sum\limits_{r=1}^{n}\frac{{}^nC_r}{r+1}x^{2n-2r}(2-x^2)^r = \frac{2^{n+1}-x^{2n+2}}{(n+1)(2-x^2)}$

7.(D) $n\sum\limits_{r=1}^{n} r^2\,{}^{n-1}C_{r-1}p^rq^{n-r} = n\sum\limits_{r=2}^{n}(r+1)(r-1)\,{}^{n-1}C_{r-1}\,p^rq^{n-r} + n\sum\limits_{r=1}^{n}{}^{n-1}C_r p^r q^{n-r}$

$= n(n-1)\sum\limits_{r=3}^{n}(r-2)^{n-2}C_{r-2}p^rq^{n-r} + 3n(n-1)\sum\limits_{r=2}^{n}{}^{n-2}C_{r-2}p^rq^{n-r} + n\sum\limits_{r=1}^{n}{}^{n-1}C_{r-1}p^rq^{n-r}$

$= n(n-1)(n-2)\sum\limits_{r=3}^{n}{}^{n-3}C_{r-3}p^rq^{n-r} + 3n(n-1)p^2(q+p)^{n-2} + np(q+p)^{n-1}$

$= n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np = np((n-1)(n-2)p^2 + 3(n-1)p + 1$

8-10

8.(B) **9.(D)** **10.(C)**

$\sum\limits_{r=0}^{n}\frac{r^2}{{}^nC_r} = a_n + \sum\limits_{r=0}^{n}\frac{(r-1)(r+1)}{{}^nC_r} = a_n + (n+1)\sum\limits_{r=0}^{n}\frac{r+2-3}{{}^{n+1}C_{r+1}}$

$= a_n + (n+1)(n+2)\sum\limits_{r=0}^{n}\frac{1}{{}^{n+2}C_{r+2}} - 3(n+1)\sum\limits_{r=0}^{n}\frac{1}{{}^{n+1}C_{r+1}}$

$= a_n + (n+1)(n+2)\left(a_{n+2} - 1 - \frac{1}{n+2}\right) - 3(n+1)(a_{n+1}-1)$

$= a_n + (n+1)(n+2)a_{n+2} - 3(n+1)a_{n+1} - n(n+1)$

11-13

11.(C) **12.(D)** **13.(A)**

Clearly, $A_k = 4$

$\therefore \quad 4 + B_kx + C_kx^2 + D_kx^3 = \ldots\ldots = (4 + B_{k-1}x + C_{k-1}x^2 + \ldots - 2)^2 = 4 + 4B_{k-1}x + (4C_{k-1} + B_{k-1}^2)x^2 + \ldots\ldots$

$\Rightarrow \quad B_k = 4B_{k-1}$ and $C_k = 4C_{k-1} + B_{k-1}^2$

$\therefore \quad B_k = 4B_{k-1} = 4^2B_{k-2} = \ldots\ldots = 4^{k-1}B_1 = -4^k$

$C_k = B_{k-1}^2 + 4C_{k-1} = 4^{2k-2} + 4C_{k-1}$

$= 4^{2k-2} + 4(4^{2k-4}) + 4^2C_{k-2} = 4^{2k-2} + 4^{2k-3} + 4^2(4^{2k-6} + 4C_{k-3}) = 4^{2k-2} + 4^{2k-3} + 4^{2k-4} + 4^3C_{k-3}$

$= 4^{2k-2} + 4^{2k-3} + 4^{2k-4} + \ldots + 4^k + 4^{k-1}C_1 = 4^{2k-2} + 4^{2k-3} + 4^{2k-4} + \ldots + 4^k + 4^{k-1} = \frac{4^{k-1}(4^k-1)}{(4-1)} = \frac{4^{2k-1}-4^{k-1}}{3}$

14.(AD) $\sum\limits_{r=1}^{n}(r\,{}^nC_r)[(n-r)^nC_r] = n\sum\limits_{r=1}^{n}\left[{}^{n-1}C_{r-1}\left((n-r)\,{}^nC_{n-r}\right)\right] = n^2\sum\limits_{r=1}^{n}{}^{n-1}C_{r-1}\,{}^{n-1}C_{n-r-1} = n^2\,{}^{2n-2}C_{n-2} = n(n-1)\,{}^{2n-2}C_{n-1}$

$\left[\because n^2\,{}^{2n-2}C_{n-2} = n^2\frac{(2n-2)!}{n!(n-2)!} = \frac{n(n-1)(2n-2)!}{(n-1)!(n-1)!} = n(n-1)\,{}^{2n-2}C_{n-1}\right]$

15.(ABCD) $-\sum_{r=1}^{n}(-1)^r r\,{}^nC_r^2=-n\sum_{r=1}^{n}(-1)^r\,{}^{n-1}C_{r-1}{}^nC_r=-n$ coefficient of x^n in $(1-x^2)^{n-1}(1-x)$

16.(BCD) $(x^3-1)^n=(x-1)^n(1+x+x^2)^n$

Coefficient of x^n in $(x^3-1)^n={}^nC_0a_0-{}^nC_1a_1+{}^nC_2a_2-{}^nC_3a_3+...+(-1)\,{}^nC_na_n$

$\Rightarrow\quad \sum_{r=0}^{n}{}^nC_r(-1)^r a_r=$coefficient of x^n in $(x^3-1)^n$

$$=\begin{cases}{}^nC_{\frac{2n}{3}}(-1)^{\frac{2n}{3}}={}^nC_{\frac{n}{3}} & \text{if } n \text{ is multiple of } 3\\ 0 & \text{if } n \text{ is not multiple of } 3\end{cases}$$

17.(ABC) $\alpha={}^nC_0\,x^n+{}^nC_2x^{n-2}a^2+{}^nC_4x^{n-4}a^4+...$

$\beta={}^nC_1x^{n-1}a+{}^nC_3x^{n-3}a^3+{}^nC_5x^{n-5}a^5+...$

$\alpha+\beta=(x+a)^n$ and $\alpha-\beta=(x-a)^n$

$(x+a)^{2n}-(x-a)^{2n}=(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$

$(x+a)^{2n}+(x-a)^{2n}=(\alpha+\beta)^2+(\alpha-\beta)^2=2(\alpha^2+\beta^2)$

$(x+a)^n(x-a)^n=(x^2-a^2)^n=(\alpha+\beta)\,(\alpha-\beta)=\alpha^2-\beta^2$

18.(BC) $T_{r+1}>T_r\quad\Rightarrow\quad {}^{24}C_rx^r>{}^{24}C_{r-1}x^{r-1}$

$\Rightarrow\quad \frac{(25-r)x}{r}>1\quad\Rightarrow\quad x>\frac{r}{25-r}\quad\Rightarrow\quad T_{r+1}>T_{r+2}\quad\Rightarrow\quad x<\frac{r+1}{24-r}$

Put $r=12$

$\therefore\quad \frac{12}{13}<x<\frac{13}{12}$

19.(CD) $({}^nC_m+{}^{n-1}C_m+{}^{n-2}C_m+...+{}^mC_m)+2({}^{n-1}C_m+{}^{n-2}C_m+...+{}^mC_m)$

$+2({}^{n-2}C_m+{}^{n-3}C_m+...+{}^mC_m)+...+2({}^{m+1}C_m+{}^mC_m)+2({}^mC_m)$

$={}^{n+1}C_{m+1}+2({}^nC_{m+1}+{}^{n-1}C_{m+1}+{}^{n-2}C_{m+1}+...+{}^{m+2}C_{m+1}+{}^{m+1}C_{m+1})$

$={}^{n+1}C_{m+1}+2\,{}^{n+1}C_{m+2}={}^{n+1}C_{m+2}+{}^{n+2}C_{m+2}$

20.(ABCD) $a_r=a_{2n-r}\ \forall r\in\{0,1,2,...,n\}$

$a_0+a_1+a_2+...+a_{n-1}+a_n+a_{n+1}+...+a_{2n}=3^n$

$\Rightarrow\quad a_0+a_1+a_2+...+a_{n-1}=\frac{(3^n-a_n)}{2}$

$(1-x+x^2)^n=a_0x^{2n}-a_1x^{2n-1}+a_2x^{2n-2}-a_3x^{2n-3}+...+a_{2n}$

$\Rightarrow \quad a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 =$ coefficient of x^{2n} in $(1+x^2+x^4)^n = a_n$

$\Rightarrow \quad a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + (-1)^{n-1} a_{n-1}^2 = \dfrac{a_n - (-1)^n a_n^2}{2}$

$$n(1+2x)(1+x+x^2)^{n-1} = \sum_{r=1}^{2n} r\, a_r\, x^{r-1}$$

$$n(1+2x)(1+x+x^2)^n = (1+x+x^2)\sum_{r=1}^{2n} r\, a_r\, x^{r-1}$$

Equating coefficient of x^r from both the sides

$n(a_r + 2a_{r-1}) = (r+1)a_{r+1} + ra_r + (r-1)a_{r-1} \Rightarrow (r+1)a_{r+1} = (n-r)a_r + (2n-r+1)a_{r-1}$

21.(BCD) $({}^nC_1\, {}^nC_2 ... {}^nC_{n-1})^{\frac{1}{n-1}} < \dfrac{{}^nC_1 + {}^nC_2 + ... + {}^nC_{n-1}}{n-1} \Rightarrow P < \left(\dfrac{2^n - 2}{n-1}\right)^{n-1}$

Also, $({}^nC_0\,{}^nC_1\,{}^nC_2 ... {}^nC_{n-1}\,{}^nC_n)^{\frac{1}{n+1}} < \dfrac{{}^nC_0 + {}^nC_1 + C_2 + ... + {}^nC_n}{n+1} \Rightarrow P < \left(\dfrac{2^n}{n+1}\right)^{n+1}$

22.(CD) ${}^{2n+1}C_0^2 + {}^{2n+1}C_1^2 + {}^{2n+1}C_2^2 + .. + {}^{2n+1}C_{2n+1}^2 = {}^{4n+2}C_{2n+1}$

${}^{2n+1}C_0^2 - {}^{2n+1}C_1^2 + {}^{2n+1}C_2^2 - {}^{2n+1}C_3^2 + ... - {}^{2n+1}C_{2n+1}^2 = 0$

$\Rightarrow \quad \displaystyle\sum_{r=0}^{n} {}^{2n+1}C_{2r}^2 = \sum_{r=0}^{n} {}^{2n+1}C_{2r+1} = \frac{1}{2}({}^{4n+2}C_{2n+1})$

23.(ABC) $f(x) = (1+x^2-x^3)^{1000} = a_0 + a_1x + ... + a_{20}x^{20} + ...$

$f(-x) = (1+x^2+x^3)^{1000} = a_0 - a_1x + ... + a_{20}x^{20} + ...$

$a = d$

$g(x) = (1-x^2+x^3)^{1000} = b_0 + b_1x + ... + b_{20}x^{20} + ...$

$g(-x) = (1-x^2-x^3) = b_0 - b_1x + ... + b_{20}x^{20} + ...$

$\Rightarrow \quad b = c \qquad$ Clearly d is the largest.

24.(ACD) $\beta_{100} = {}^{100}C_{100} + {}^{101}C_{100} + {}^{102}C_{100} + ... {}^{200}C_{100} = {}^{201}C_{100}$

$\beta_{99} = {}^{99}C_{99} + {}^{100}C_{99} + {}^{101}C_{99} + ... {}^{200}C_{99} = {}^{201}C_{100}$

25.(AC) $\left(5\sqrt{3}+8\right)^{2n+1} - \left(5\sqrt{3}-8\right)^{2n+1} = 2\left({}^{2n+1}C_1(5\sqrt{3})^{2n}8 + {}^{2n+1}C_3(5\sqrt{3})^{2n-2}8^3 + ...\right)$

$\Rightarrow \quad [x] + \{x\} - N = 2k,\ k \in I$

$0 < \{x\} < 1$ and $0 < N < I$

$\Rightarrow \quad -1 < \{x\} - N < 1$ and $\{x\} - N \in I \quad \Rightarrow \quad \{x\} - N = 0 \quad \Rightarrow \quad xN = x\{x\} = (11)^{2n+1}$

26.(CD) Put $x = i \quad \Rightarrow \quad i^n = (a_0 - a_2 + a_4 - a_6 + ...) + i(a_1 - a_3 + a_5 - a_7 + ...)$

$$\Rightarrow \quad a_1 - a_3 + a_5 - a_7 + ... = \begin{cases} 0 & \text{if } n \text{ is even} \\ 1 & \text{if } n = 4k+1,\ k \in I \\ -1 & \text{if } n = 4k+3 \end{cases}$$

27.(ABC) $(13^n - 3^n) + 7^n = 10k + 7^n \Rightarrow$ last digit is 3 if $n = 4k+3,\ k \in I$

28.(BD) Coefficient of $x^t = {}^tC_t + {}^{t+1}C_t + {}^{t+2}C_t + \ldots.. + {}^nC_t = {}^{n+1}C_{t+1}$

Coefficient of $x^{t+1} = {}^{t+1}C_{t+1} + {}^{t+2}C_{t+1} + {}^{t+3}C_{t+1} + \ldots.. + {}^nC_{t+1} = {}^{n+1}C_{t+2}$

${}^{n+1}C_{t+1} = {}^{n+1}C_{t+2} \Rightarrow n - t = t + 2 \Rightarrow t = \dfrac{n-2}{2}$ their sum $= {}^{n+2}C_{t+2}$

29. **A-P, Q ; B-R, T ; C-P, Q ; D-P, Q, S**

(A) ${}^nC_3\left(\dfrac{x}{a}\right)^{n-3}\left(\dfrac{1}{x}\right)^3 = \dfrac{5}{2} \Rightarrow n - 6 = 0$

$\Rightarrow a^3 = 8 \Rightarrow a = 2$

(B) $S = \displaystyle\sum_{p=1}^{4}\sum_{r=p}^{4}\frac{4!}{r!(4-r)!}\frac{r!}{(r-p)!p!} = \sum_{p=1}^{4}{}^4C_p\sum_{r=p}^{4}\frac{(4-p)!}{(r-p)!(4-r)!}$

$= \displaystyle\sum_{p=1}^{4}{}^4C_p 2^{4-p} = 3^4 - 2^4 = 65$

(C) Coefficient of x^{13} in $(1-x)(1-x^4)^4 = -(-{}^4C_3) = 4$

(D) $\displaystyle\sum_{r=0}^{4}{}^4C_r(r-2)^2 = 16$

30. **A-T ; B-Q ; C-S ; D-R**

(A) Coefficient of $x^4 = \dfrac{6!}{2!\,4!}2^2 + \dfrac{6!}{3!\,2!}2^3\,3 + \dfrac{6!}{2!\,4!}2^4\,3^2 = 3660$

(B) ${}^{11}C_0\,{}^{22}C_{11} - {}^{11}C_1\,{}^{20}C_{11} + {}^{11}C_2\,{}^{18}C_{11} - {}^{11}C_3\,{}^{16}C_{11} + \ldots.$

$=$ coefficient of x^{11} in ${}^{11}C_0(1+x)^{22} - {}^{11}C_1(1+x)^{20} + \ldots..$

$=$ coefficient of x^{11} in $((1+x)^2 - 1)^{11} =$ coefficient of x^{11} in $x^{11}(2+x)^{11} = 2^{11} = 2048$

(C) ${}^5C_1\,{}^5C_5 - {}^5C_2\,{}^{10}C_5 + \ldots. + {}^5C_5\,{}^{25}C_5$

$=$ coefficient of x^5 in $[1 - (1 - (1+x)^5)^5]$

$=$ coefficient of x^5 in $[1 - (-x(1 + (1+x) + (1+x)^2 + (1+x)^3 + (1+x)^4))^5]$

$=$ coefficient of x^5 in $[1 + x^5(1 + (1+x) + (1+x)^2 + (1+x)^3 + (1+x)^4)^5] = 5^5 = 3125$

(D) Coefficient of x^4 in $= \dfrac{11!}{4!\,7!} + \dfrac{11!}{2!\,8!} + \dfrac{11!}{9!} + \dfrac{11!}{2!\,9!} = 990$

31.(2250000)

$S = {}^{50}C_6 - {}^5C_1\,{}^{40}C_6 + \ldots + {}^5C_4\,{}^{10}C_6$

$=$ Coefficient of x^6 in ${}^5C_0(1+x)^{50} - {}^5C_1(1+x)^{40} + \ldots + {}^5C_4(1+x)^{10} - {}^5C_5(1+x)^0$

$=$ Coefficient of x^6 in $((1+x)^{10} - 1)^5$

$=$ Coefficient of x^6 in $x^5\ ({}^{10}C_1 + {}^{10}C_2x + \ldots + {}^{10}C_{10}x^9)^5$

$= {}^5C_4({}^{10}C_1)^4\left({}^{10}C_2\right) = 5 \times 10^4 \times 5 \times 9 = 2250000$

32.(21) $(7-1)^{2007}+(7+1)^{2007}=49k+2007\times7\times2=49\lambda+21$

33.(1024) ${}^{10}C_0{}^{20}C_{10}-{}^{10}C_1{}^{18}C_{10}+....{}^{10}C_5{}^{10}C_{10}$

$=$ Coefficient of x^{10} in ${}^{10}C_0(1+x)^{20}-{}^{10}C_1(1+x)^{18}+{}^{10}C_2(1+x)^{16}-{}^{10}C_3(1+x)^{14}+....+{}^{10}C_9(1+x)^2+{}^{10}C_{10}$

$=$ Coefficient of x^{10} in $((1+x)^2-1)^{10}$ $=$ Coefficient of x^{10} in x^{10} $(2+x)^{10}=2^{10}$

34.(2252) Coefficient of x^5y^5 in

$$(1+x)^5(1+y)^5(x+y)^5=\sum_{r=0}^{5}{}^5C_r{}^5C_{5-r}{}^5C_{5-r}=\sum_{r=0}^{5}{}^5C_r^3$$

35.(13) Coefficient of $x^{\frac{n(n+1)}{2}-7}=-7+(1\cdot6+2\cdot5+3\cdot4)-1\cdot2\cdot4)=13$

36.(1) Coefficient of x^n in ${}^nC_0(1+x)^{2n}-{}^nC_1(1+x)^{2n-1}+{}^nC_2(1+x)^{2n-2}-{}^nC_3(1+x)^{2n-3}+...+(-1)^n\,{}^nC_n(1+x)^n$

$=$ Coefficient of x^n in $(1+x)^n((1+x)-1)^n=1$

37.(1) $N=\dfrac{(7+18)^3}{(3+2)^6}=1$

38.(2) $\dfrac{14}{9}{}^mC_2{}^mC_4=\left({}^mC_3\right)^2\ \Rightarrow m=9$

$${}^9C_3 5^{-\log_{10}(6-\sqrt{8x})}\cdot5^{\frac{\log_{10}(x-1)-2\log_{10}5}{2}}=\frac{84}{5}$$

$\Rightarrow\quad 6\sqrt{2x}-x-10=0$

$\Rightarrow\quad x-6\sqrt{2x}+10=0$

$\Rightarrow\quad \left(\sqrt{x}-5\sqrt{2}\right)\left(\sqrt{x}-\sqrt{2}\right)=0$

$\Rightarrow\quad x=2,\ 50$ (rejected)

Hence, $x=2$

39.(1) $(5+\sqrt{2})^n+(5-2\sqrt{6})^n=2k, k\in I$

$\Rightarrow\quad [x]+\{x\}+N=2k,\ 0<N<1,\ 0<\{x\}<1,\ \{x\}+N\in I$

$\Rightarrow\quad \{x\}+N=1$

$\therefore\quad x-x^2+x[x]=x-x(x-[x])=x(1-\{x\})$

$=xN=(5+2\sqrt{6})^n(5-2\sqrt{6})^n=1$

40.(3) ${}^mC_2+{}^nC_2-{}^mC_1\,{}^nC_1+{}^mC_1-{}^nC_1=-m$

$\Rightarrow\quad (m-n)^2+3(m-n)=0$

$\Rightarrow\quad n-m=3$

STRAIGHT LINE

1.(C) $\frac{1}{x_l}=a+(l-1)d$;

$$\frac{1}{x_m}=a+(m-1)d;\frac{1}{x_n}=a+(n-1)d \quad \Rightarrow \quad \begin{vmatrix} a+(l-1)d & l & 1 \\ a+(m-1)d & m & 1 \\ a+(n-1)d & n & 1 \end{vmatrix}=0$$

2.(A) Area of ΔOPP_1 + area of ΔOPP_2 = area of ΔOP_1P_2

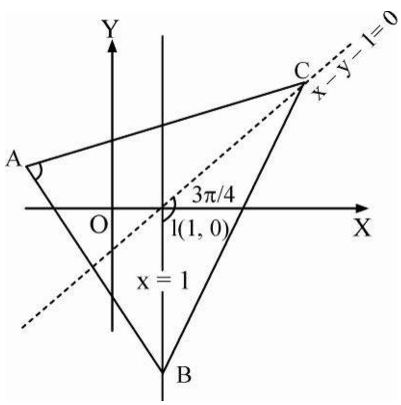
3.(D) Equation of CV is $y=\frac{3\beta}{(3-\alpha)}\left(1-\frac{x}{2}\right)$. Hence fixed point is (2, 0).

4.(D) $a-2\sqrt{bc}=b+c \quad \Rightarrow \quad \left(\sqrt{b}+\sqrt{c}\right)^2-\left(\sqrt{a}\right)^2=0 \quad \Rightarrow \quad \sqrt{b}+\sqrt{c}-\sqrt{a}=0$

Or $\quad \sqrt{b}+\sqrt{c}+\sqrt{a}=0$ (rejected)

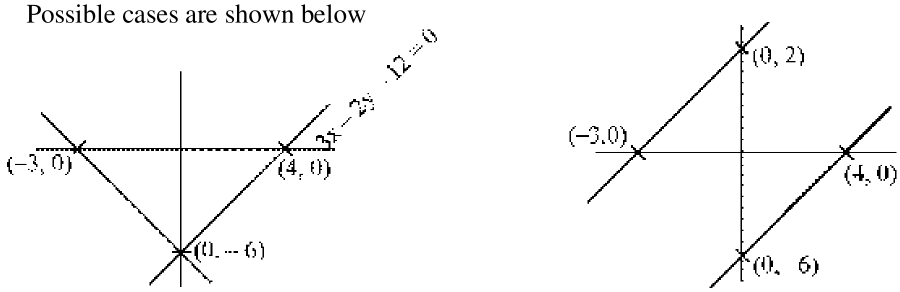
$\sqrt{b}+\sqrt{c}-\sqrt{a}=0 \quad \Rightarrow \quad \sqrt{a}x+\sqrt{b}y+\sqrt{c}=0$ Passes through fixed point (–1, 1)

5.(C) $\angle BIC=\frac{3\pi}{4}=\frac{\pi}{2}+\frac{A}{2} \quad \Rightarrow \quad A=\frac{\pi}{2} \quad \left[\because \angle BIC=\frac{\pi}{2}+\frac{A}{2}\right]$

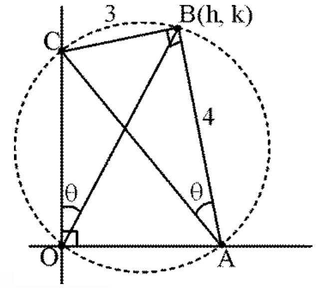


Hence, $\angle BAC=\frac{\pi}{2}$

6.(B) Possible cases are shown below



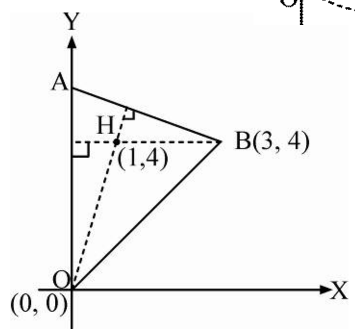
7.(B) $\tan\theta = \dfrac{3}{4}$ (from Fig.)



$\Rightarrow$ Slope of $OB = \tan\left(\dfrac{\pi}{2}-\theta\right) = \cot\theta$

$\Rightarrow$ Locus of (h, k) is $\dfrac{k}{h} = \dfrac{4}{3}$ $\Rightarrow$ $3y - 4x = 0$

8.(D) $BH \perp OA$ $\Rightarrow$ A lies on y-axis equation on AH is:

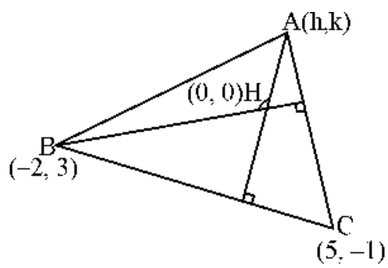


$$y-4 = -\frac{3}{4}(x-1)$$

Put $x = 0$

$\Rightarrow$ $y = 4 + \dfrac{3}{4} = \dfrac{19}{4}$ $\Rightarrow$ $A = \left(0, \dfrac{19}{4}\right)$

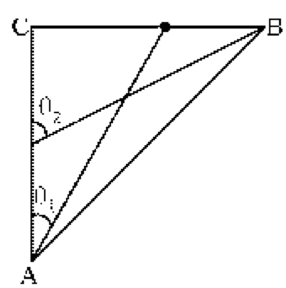
9.(C) Slope of $AH = \dfrac{k}{h} = \dfrac{7}{4}$



Slope of $AC = \dfrac{k+1}{h-5} = \dfrac{2}{3}$

So, $3\left(\dfrac{7}{4}h+1\right) = 2(h-5)$

10.(B) $\tan\theta_1 = \dfrac{BC}{2AC}$



$\tan\theta_2 = \dfrac{BC}{(AC/2)} = \dfrac{2BC}{AC}$

$\dfrac{\tan\theta_1}{\tan\theta_2} = \dfrac{1}{4}$

Case-I: If $m < 3$

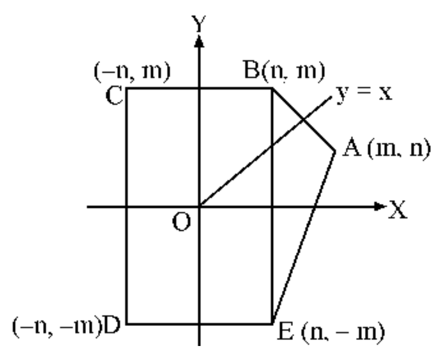
$\tan\theta_1 = \dfrac{1}{3}$ and $\tan\theta_2 = \dfrac{1}{m}$

$\dfrac{m}{3} = \dfrac{1}{4}$ $\Rightarrow$ $m = \dfrac{3}{4}$

Case-II: If $m > 3$,

$\tan\theta_1 = \dfrac{1}{m}$ and $\tan\theta_2 = \dfrac{1}{3}$ $\Rightarrow$ $\dfrac{3}{m} = \dfrac{1}{4}$ $\Rightarrow$ $m = 12$

11.(B) Area of rectangle $BCDE = 4mn$



Area of $\Delta ABE = \dfrac{2m(m-n)}{2}$

$= m^2 - mn$

$\therefore$ area of pentagon $= 4mn + m^2 - mn = m^2 + 3mn$

Aliter :

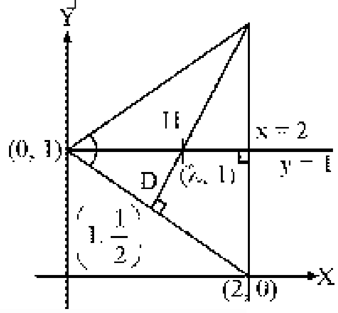
$$ar(ABCDE) = \frac{1}{2}\left\| \begin{matrix} m & n \\ n & m \end{matrix} \right| + \left| \begin{matrix} n & m \\ -n & m \end{matrix} \right| + \left| \begin{matrix} -n & m \\ -n & -m \end{matrix} \right| + \left| \begin{matrix} -n & -m \\ n & -m \end{matrix} \right| + \left| \begin{matrix} n & -m \\ m & n \end{matrix} \right\|$$

$$= \frac{1}{2}\left[m^2 - n^2 + mn + mn + mn + mn + mn + n^2 + m^2\right] = m^2 + 3mn$$

12.(B) Slope of $HD = 2$

$$\Rightarrow \quad \frac{1-\frac{1}{2}}{\lambda - 1} = 2 \quad \Rightarrow \quad \lambda = \frac{5}{4}$$

$$\therefore \quad H \equiv \left(\frac{5}{4}, 1\right)$$



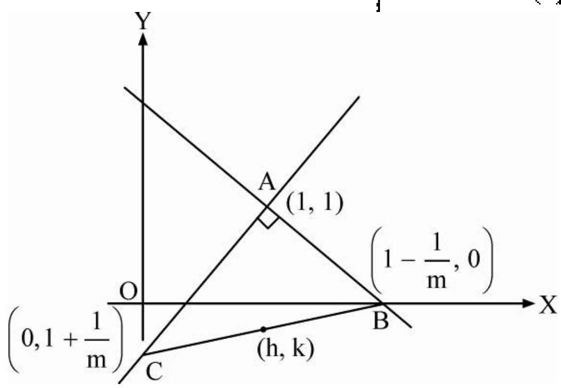
13.(A) $y - 1 = m(x - 1)$

$$y - 1 = -\frac{1}{m}(x - 1)$$

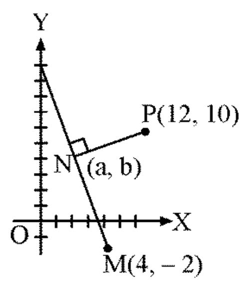
$$2h = 1 - \frac{1}{m}$$

$$2k = 1 + \frac{1}{m}$$

Locus is $x + y = 1$



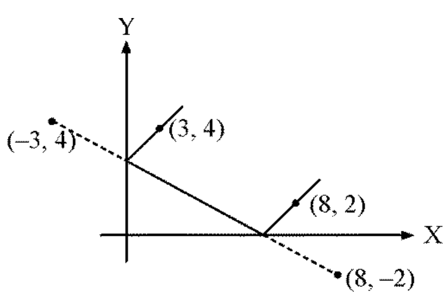
14.(B) (a, b) is the foot of $\perp$ ar of (12, 10) on the line $y + 5x = 18$



$$\Rightarrow \quad \frac{a-12}{5} = \frac{b-10}{1} = -\left(\frac{10 + 5(12) - 18}{26}\right) \quad \Rightarrow \quad (a, b) = (2, 8)$$

15.(B) $\dfrac{0+2}{x-8} = \dfrac{4+2}{-3-8} = -\dfrac{6}{11}$

$$x = \frac{13}{3}$$



16.(B) Image of $A(2,-1)$ w.r.t.

$3x-2y+5=0, A'$ is given by

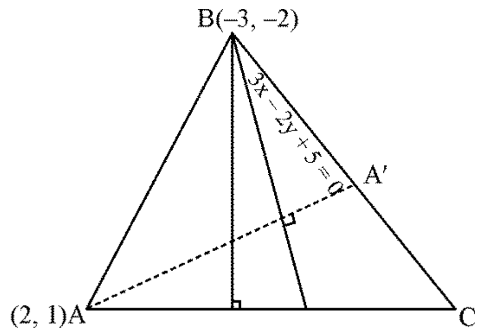
$$\frac{x-2}{3}=\frac{y+1}{-2}=-2\cdot\frac{(6+2+5)}{13}=-2$$

$$A'=(-4,3)$$

Equation of BC is

$$y-3=\frac{3+2}{-4+3}(x+4)$$

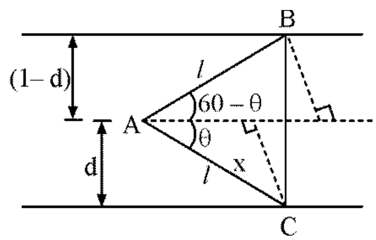
$$5x+y+17=0$$



17.(B) $l\sin(60-\theta)=l\frac{\sqrt{3}}{2}\cos\theta-\frac{l}{2}\sin\theta$

$=1-d$

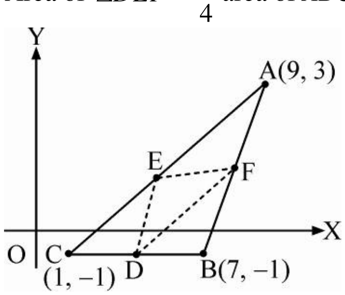
$\Rightarrow \quad l\cos\theta=\frac{2-d}{\sqrt{3}}$ & $\ell\sin\theta=d$



Squaring and adding Eqns. (1) and (2)

$\Rightarrow \quad l^2=d^2+\frac{4+d^2-4d}{3}=\frac{4}{3}(d^2-d+1) \quad \Rightarrow \quad l=2\sqrt{\frac{d^2-d+1}{3}}$

18.(D) Area of $\Delta DEF=\frac{1}{4}$ area of ABC



$$=\frac{1}{4}\times\frac{1}{2}\times 6\times 4=3$$

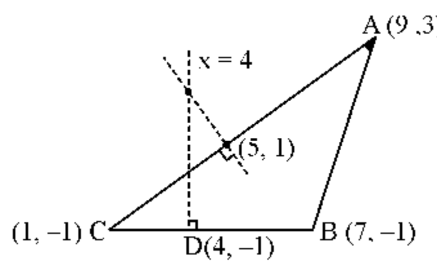
19.(B) Equation of perpendicular bisector of AC is

$y-1=-2(x-5)$

$\Rightarrow \quad y+2x=11$

Perpendicular bisector of BC is

$\therefore \quad$ Circumcentre $\equiv(4,3)$

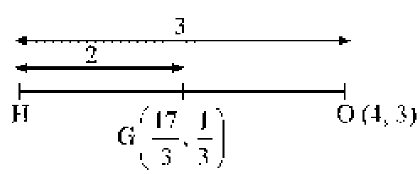


$\Rightarrow \quad R=\sqrt{(4-1)^2+(3+1)^2}=5 \quad \Rightarrow \quad a+b+R=4+3+5=12$

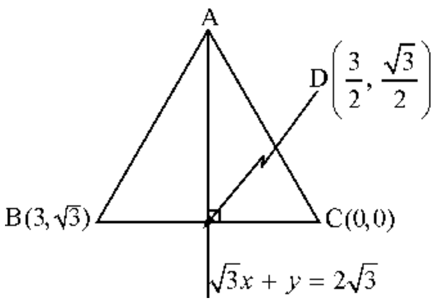
20.(C) $O\equiv$ circumcentre, $G\equiv$ centroid

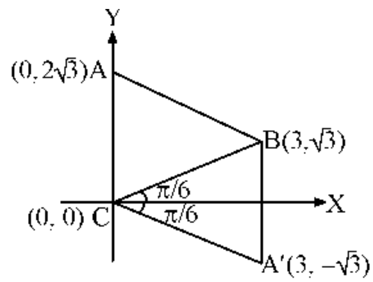
$\Rightarrow \quad H\equiv\left(3\left(\frac{17}{3}\right)-2(4), 3\left(\frac{1}{3}\right)-2(3)\right)$

$H\equiv(9,-5)$



21-23.





Equation of BD is : $x - \sqrt{3}y = 0$

$$\Rightarrow \quad D \equiv \left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right) \text{ and } C \equiv (0, 0)$$

$AC = BC = 2\sqrt{3}$

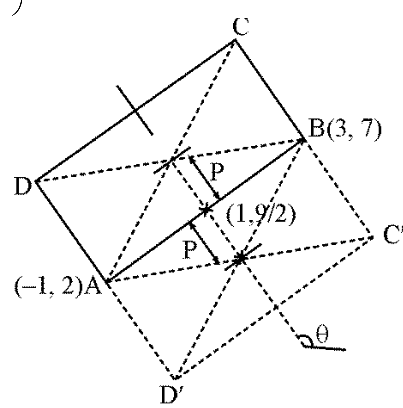
21.(B) Two triangles are possible

22.(D) $A \equiv \left(0, 2\sqrt{3}\right)$ and $A'\left(3, -\sqrt{3}\right)$

23.(A) Orthocentre $\equiv$ centroid of Δ

$$H_1 \equiv \left(\frac{0+3+0}{3}, \frac{0+\sqrt{3}+2\sqrt{3}}{3}\right) \equiv \left(1, \sqrt{3}\right); \quad H_2 \equiv \left(\frac{0+3+3}{2}, \frac{0+\sqrt{3}-\sqrt{3}}{3}\right) \equiv (2, 0)$$

24.(BD) $BC = \frac{3}{4}AB = \frac{3}{4}\sqrt{41}$



$P = \frac{1}{2}BC = \frac{3}{8}\sqrt{41}$; $\tan\div = -\frac{4}{5}$

Intersection point of diagonals can be

$$\left(1 + \frac{3}{8}\sqrt{41}\left(-\frac{5}{\sqrt{41}}\right), \frac{9}{2} + \frac{3}{8}\sqrt{41}\left(\frac{4}{\sqrt{41}}\right)\right)$$

or $\left(1 - \frac{3}{8}\sqrt{41}\left(-\frac{5}{\sqrt{41}}\right), \frac{9}{2} - \frac{3}{8}\sqrt{41}\left(\frac{4}{\sqrt{41}}\right)\right)$ i.e., $\left(-\frac{7}{8}, 6\right) or \left(\frac{23}{8}, 3\right)$

25.(ABCD) $\frac{m_1}{m_2} = \frac{9}{2}, \frac{m_1 - m_2}{1 + m_1 m_2} = \pm\frac{7}{9} \quad \Rightarrow \quad 9m_2^2 - 9m_2 + 2 = 0$

$9m_2^2 + 9m_2 + 2 = 0 \qquad m_2 = \frac{2}{3}, \frac{1}{3} m_2 = \frac{-2}{3}, \frac{-1}{3}$

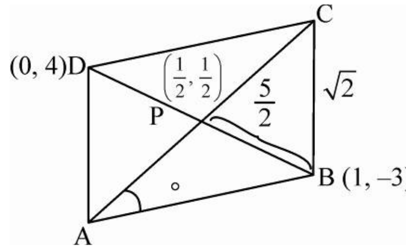
$m_2 = \frac{2}{3}, m_1 = 3 \quad \Rightarrow \quad y = 3x, 3y = 2x$

$m_2 = \frac{1}{3}, m_1 = \frac{3}{2} \quad \Rightarrow \quad 3y = x, 2y = 3x$

$m_2 = \frac{-2}{3}, m_1 = -3 \quad \Rightarrow \quad 3x + y = 0, 2x + 3y = 0$

$m_2 = -\frac{1}{3}, m_1 = \frac{-3}{2} \quad \Rightarrow \quad x + 3y = 0, \ 3x + 2y = 0$

26.(AB) $\tan\theta = \frac{1}{7}, \sin\theta \frac{1}{5\sqrt{2}}, \cos\theta = \frac{7}{5\sqrt{2}}$



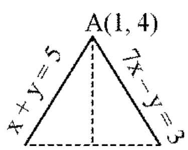
$$AP \equiv \frac{5}{2}\sqrt{2}\cot 30° = \frac{5}{2}\sqrt{2}\sqrt{3}$$

$$A \equiv \left(\frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$$

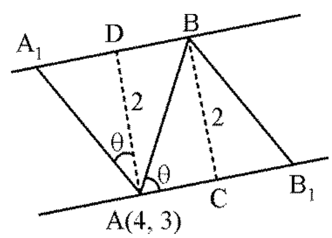
Or, $\left(\frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$

$\Rightarrow \quad A = \left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ Or $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

27.(ABCD)



28.(AC) Area of parallelogram $AA_1BB_1 = \frac{8}{sin 2\theta}$



Which in least for $\theta = \frac{\pi}{4}$

So $1 = \left|\frac{m + \frac{3}{4}}{1 - \frac{3m}{4}}\right|$

29.(ABD) Taking B as origin

BC as x axis and let A (h, k)

Now $b + c > a$

$\Rightarrow \quad b + c > 6 \quad \Rightarrow \quad b + 2b > 6 \quad \Rightarrow \quad b > 2$.

Again $cos B = \frac{b^2 + c^2 - 36}{2bc} \Rightarrow \frac{b^2 + 4b^2 - 36}{4b^2} < 1 \quad \Rightarrow \quad b < 6 \quad \Rightarrow \quad 2 < b < 6 \;\; so \;\; 4 < c < 12.$

Since $b = 2c$.

Now $h^2 + k^2 = c^2$ and $(h-6)^2 + k^2 = b^2$

On eliminating $b^2 \quad \Rightarrow \quad k^2 = 16 - (h-8)^2 \quad \Rightarrow \quad max\, k = 4$.

30.(BD) $\frac{c}{b} = 2 \quad \Rightarrow \quad c^2 = 4b^2$

$c^2 = h^2 + k^2$

$b^2 = (h-6)^2 + k^2 \quad \Rightarrow \quad h = \frac{b^2 + 12}{4}$

Now it in given that $k^2 > 10$

$\Rightarrow \quad c^2 - h^2 > 10 \quad \Rightarrow \quad 4b^2 - \left(\frac{b^2+12}{4}\right)^2 > 10$

$\Rightarrow \quad b^2 \in \left(20 - \sqrt{96}, 20 + \sqrt{96}\right) \quad \Rightarrow \quad 11 < b^2 < 30$

31.(BC) $(3x+4y+5)(x+2y+3)=0$

$$m_1=\frac{-3}{4}, m_2=-\frac{1}{2} \Rightarrow m_1m_2=\frac{3}{8}$$

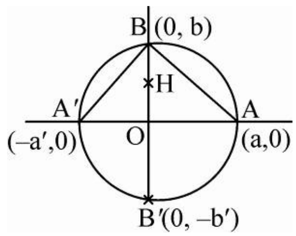
$$p=(1,-2) \quad ; \quad \tan\theta=\frac{-\frac{1}{2}+\frac{3}{4}}{1+\frac{3}{8}}=\frac{2}{11} \quad \Rightarrow \quad \sin\theta=\frac{2}{5\sqrt{5}}$$

32.(BC) H is the image of B'

$\Rightarrow \quad H\equiv(0,b')$

Applying power of 'O'

$$aa'=bb' \Rightarrow b'=\frac{aa'}{b}$$



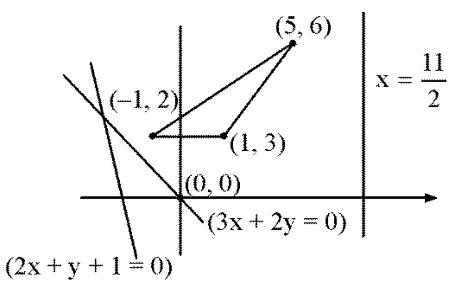
33.(AD) $m=\tan\theta_1, m'=\tan\theta_2$ inclination of given lines are $2\theta_1$ and $2\theta_2$

$\therefore$ Inclination of angle bisectors $\theta_1+\theta_2$ or $\theta_1+\theta_2+\frac{\pi}{2}$

$\therefore$ Slopes will be $\frac{m+m'}{1-mm'}, \frac{mm'-1}{m+m'}$

Eqn. of bisectors are $y-b=\frac{(m+m')}{(1-mm')}(x-a)$, $\quad y-b=\frac{(mm'-1)}{(m+m')}(x-a)$

34.(ABC)



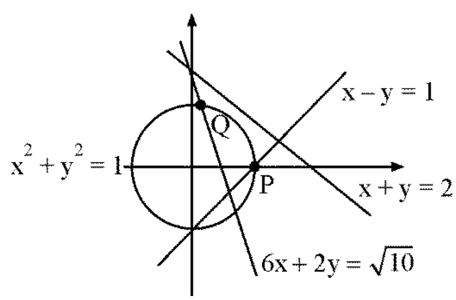
35.(ABC) For Q:

$6\cos\theta+2\sin\theta=\sqrt{10}$

$\tan^{-1}3=\alpha, \sin(\theta+\alpha)=\frac{1}{2}$

$\theta+\alpha=\frac{5\pi}{6}$

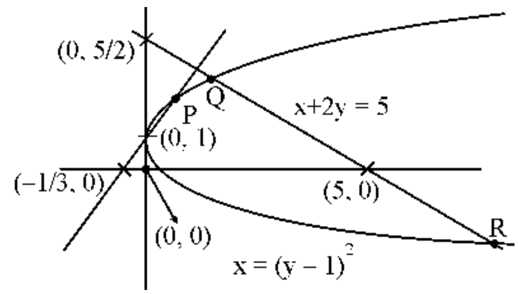
$\theta=\frac{5\pi}{6}-\tan^{-1}3$



36.(ABCD) Area = Mod of $\frac{1}{2}\begin{vmatrix}\alpha & 2\alpha+3 & 1\\ 1 & 2 & 1\\ 2 & 3 & 1\end{vmatrix}=\frac{1}{2}|-\alpha+2\alpha+3-1|$

$$\Rightarrow \frac{|\alpha+2|}{2}\in[2,3) \quad \Rightarrow \quad \alpha\in(-8,-6]\cup[2,4) \quad \Rightarrow \quad (\alpha,\beta)=(-7,-11),(-6,-9),(2,7),(3,9)$$

37.(BD) Intersection of $x=(y-1)^2$ with lines are

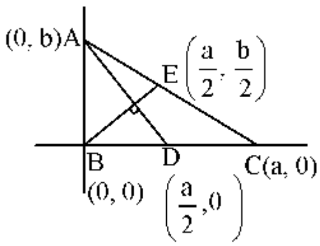


$$(y-1)^2=\frac{y-1}{3} \quad \Rightarrow \quad y=1,\ y=\frac{4}{3} \quad \Rightarrow \quad P\equiv\left(\frac{1}{9},\frac{4}{3}\right)$$

$$(y-1)^2=5-2y \Rightarrow y^2=4 \Rightarrow y=\pm 2 \qquad Q\equiv(1,2), R\equiv(9,-2)$$

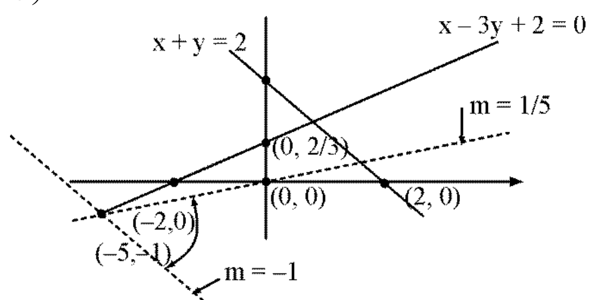
$$\Rightarrow \qquad a+1\in(-2,1)\cup\left(\frac{4}{3},2\right); \qquad a\in(-3,0)\cup\left(\frac{1}{3},1\right)$$

38.(AB) $AD \perp BE$



$$\Rightarrow \qquad \left(\frac{b}{a}\right)\frac{-b}{(a/2)}=-1 \quad \Rightarrow \quad 2b^2=a^2 \quad \Rightarrow \quad a=\pm\sqrt{2}b$$

39.(BCD) $m\in\left(-1,\frac{1}{5}\right)$ for origin to lie inside the triangle

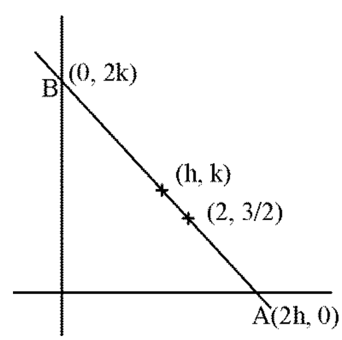


40.(AD) $D=\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Applying $C_1 \to C_1+C_2$, we get, $D=\begin{vmatrix} x_1+y_1 & y_1 & 1 \\ x_2+y_2 & y_2 & 1 \\ x_3+y_3 & y_3 & 1 \end{vmatrix}=\begin{vmatrix} -8 & y_1 & 1 \\ -8 & y_2 & 1 \\ -8 & y_3 & 1 \end{vmatrix}=0$

41.(ABC)

(A) Intersection point of lines is $\left(2,\frac{3}{2}\right)$,



Equation, of AB is $\frac{x}{2h}+\frac{y}{2k}=1$

Put $\left(2,\frac{3}{2}\right) \Rightarrow \frac{1}{h}+\frac{3}{4k}=1$

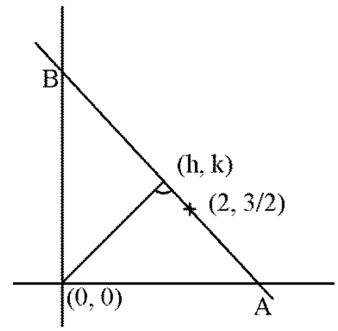
$\Rightarrow \qquad 4k+3h=4hk \qquad \Rightarrow \qquad 3x+4y=4xy$

(B) Equation of AB is $hx+ky=h^2+k^2$

Put $\left(2,\frac{3}{2}\right)$

$\Rightarrow \quad 2h+\frac{3k}{2}=h^2+k^2$

$\Rightarrow \quad 2\left(x^2+y^2\right)-4x-3y=0$



(C) Equation of AB is $\frac{x}{3h}+\frac{y}{3k}=1$

Put $\left(2,\frac{3}{2}\right)$

42.(ABD)

Let $OA=r_1$, $OB=r_2$, $OP=r$

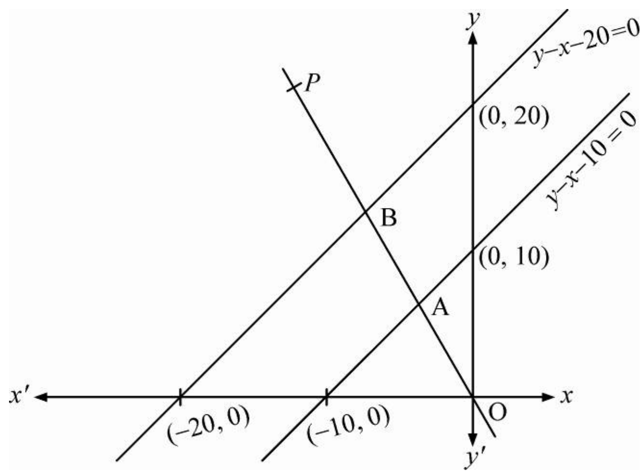
$\therefore \quad A \equiv \left(r_1\cos\theta,\ r_1\sin\theta\right)$

$B \equiv \left(r_2\cos\theta,\ r_2\sin\theta\right)$

$P \equiv \left(r\cos\theta,\ r\sin\theta\right)=\left(h,\ k\right)$

$\because \quad A$ lies on the line $y-x-10=0$

$\Rightarrow \quad r_1\sin\theta-r_1\cos\theta=10 \quad \Rightarrow \quad OA=\frac{10}{\sin\theta-\cos\theta}$



Similarly $OB=\frac{20}{\sin\theta-\cos\theta}$

(A) $\because \quad \frac{2}{OP}=\frac{1}{OA}+\frac{1}{OB} \quad \Rightarrow \quad \frac{2}{r}=\frac{1}{r_1}+\frac{1}{r_2} \quad \Rightarrow \quad \frac{2}{r}=\frac{\sin\theta-\cos\theta}{10}+\frac{\sin\theta-\cos\theta}{20}$

$\Rightarrow \quad \frac{40}{r}=3\sin\theta-3\cos\theta \quad \Rightarrow \quad 3\left(r\sin\theta\right)-3\left(r\cos\theta\right)=40 \quad \therefore$ Locus of P is $3y-3x=40$

(B) $OP^2=\left(OA\right)\left(OB\right) \quad \Rightarrow \quad r^2=\frac{10}{\sin\theta-\cos\theta}\times\frac{20}{\sin\theta-\cos\theta}$

$\Rightarrow \quad \left(r\sin\theta-r\cos\theta\right)^2=200 \quad \therefore$ Locus of P is $\left(y-x\right)^2=200$

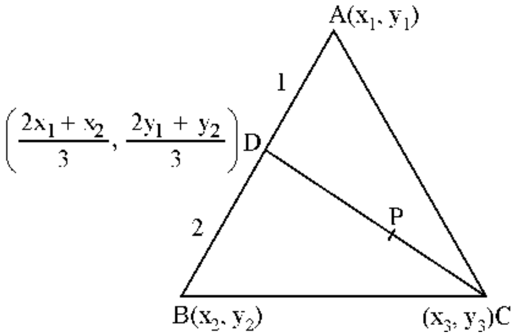
(C) $\therefore \quad \frac{1}{OP^2}=\frac{1}{OA^2}-\frac{1}{OB^2} \quad \Rightarrow \quad \frac{1}{r^2}=\frac{\left(\sin\theta-\cos\theta\right)^2}{10^2}-\frac{\left(\sin\theta-\cos\theta\right)^2}{20^2}$

$\because \quad$ Locus of P is $\left(y-x\right)^2=\frac{400}{3}$

(D) $\because \quad \frac{1}{OP^2}=\frac{1}{OA^2}+\frac{1}{OB^2} \quad \Rightarrow \quad \frac{1}{r^2}=\frac{\left(\sin\theta-\cos\theta\right)^2}{10^2}+\frac{\left(\sin\theta-\cos\theta\right)}{20^2}$

$\therefore \quad$ Locus of P is $\left(y-x\right)^2=80$

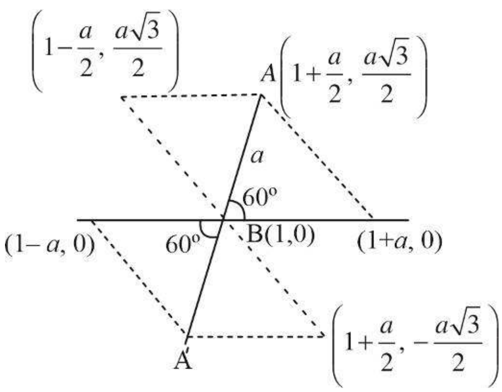
43.(AB)



$$P \equiv \left(\frac{\frac{2x_1+x_2}{3}+x_3}{2}, \frac{\frac{2y_1+y_2}{3}+y_3}{2}\right) \equiv \left(\frac{2x_1+x_2+3x_3}{6}, \frac{2y_1+y_2+3y_3}{6}\right)$$

$\therefore$ P lies inside the ΔABC $\Rightarrow$ Area of ΔPBC < area of ΔABC

44.(ABC)



45.(ABD)

$\alpha = tan^{-1}\sqrt{3}$

$tan\,\alpha = \sqrt{3}$

$tan\,\alpha = \dfrac{PC}{r/2}$

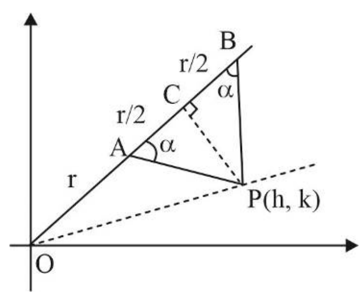
$PC = \dfrac{r}{2}tan\,\alpha$

Given $OA = AB = r$ in ΔOCP

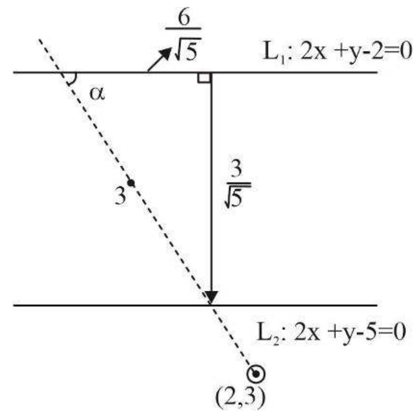
$(OP)^2 = (OC)^2 + (CP)^2$

$h^2 + k^2 = \left(\dfrac{3r}{2}\right)^2 + \left(\dfrac{r}{2}tan\,\alpha\right)^2$

Locus is $x^2 + y^2 = \left(\dfrac{9+tan^2\alpha}{4}\right)r^2$



46.(BC) Let slope of line 'L' is m



$$\tan\alpha = \left|\frac{m-(-2)}{1+m(-2)}\right| = \frac{1}{2}$$

$$\frac{m+2}{1-2m} = \pm\frac{1}{2}$$

$$2(m+2) = \pm(1-2m)$$

$+ \qquad\qquad -$

$4m = -3$ or $4 = -1$ (*Not possible*)

$m = -\frac{3}{4}$ or line is perpendicular to the x-axis

47.(ABC) The equation $(2x+3y-5)\cos\theta + (3x-5y+2)\sin\theta = 0$

$$(2x+3y-5) + \tan\theta(3x-5y+2) = 0$$

$$L_1 + \lambda L_2 = 0$$

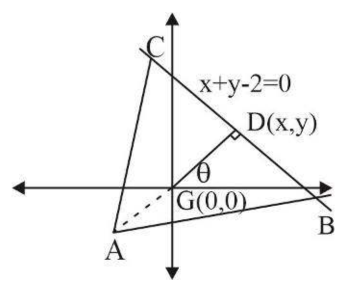
It represents the family of lines passing through the point of intersection of L_1 *and* L_2

Point of intersection of L_1 *and* L_2 is $(1,1)$

By reflection formula $\frac{x-1}{1} = \frac{y-1}{1} = \frac{-2\left(2-\sqrt{2}\right)}{2}$

Mirror image of $(1,1)$ is $\left(\sqrt{2}-1, \sqrt{2}-1\right)$

48.(ABCD) distance of $Point(0,0)$ from $x+y-2=0$; $GD = \sqrt{2}$

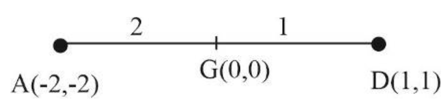


$$\tan\theta = -\frac{1}{m_{CB}} = 1$$

$$\cos\theta = \frac{1}{\sqrt{2}}, \sin\theta = \frac{1}{\sqrt{2}}$$

$$x = 0 + \sqrt{2}\times\frac{1}{\sqrt{2}} = 1$$

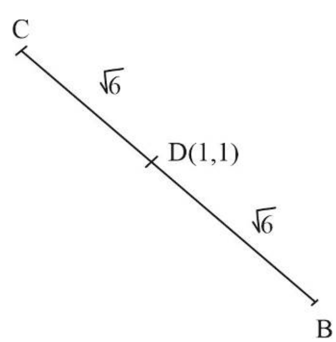
$$y = 0 + \sqrt{2}\times\frac{1}{\sqrt{2}} = 1$$



Point $D(1,1)$ and Point $A(-2,-2)$

$$AD = 3\sqrt{2}$$

As $AD = \frac{\sqrt{3}}{2}(side\ of\ \Delta ABC) = \frac{\sqrt{3}}{2}a = 3\sqrt{2}$



$$a = 2\sqrt{6}$$

$$m_{BC} = -1$$

$$\tan\alpha = -1$$

$$\cos\alpha = -\frac{1}{\sqrt{2}}, \sin\alpha = \frac{1}{\sqrt{2}}$$

$$Pt\ B\left(1+\frac{\sqrt{6}}{\sqrt{2}},1-\frac{\sqrt{6}}{\sqrt{2}}\right);\qquad B\left(1+\sqrt{3},1-\sqrt{3}\right);\qquad C\left(1-\sqrt{3},1+\sqrt{3}\right)$$

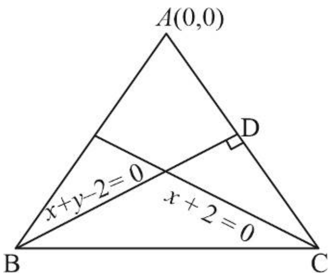
Area of $\Delta ABC=\frac{\sqrt{3}}{4}\times 4\times 6=6\sqrt{3}$ square units

49.(ABC) As C lies on $x+2=0$

x-coordinate of C is -2

$m_{AC}.m_{BD}=-1$

$$m_{AC}=\frac{-1}{.m_{BD}}=1$$



Equation of AC is $y=x$

Coordinate of C is $(-2,-2)$

D is $(1,1)$

If slope of AB is m then equation of AB is $y=mx$

$B(-4,-4m)$

As it lies on $x+y-2=0$

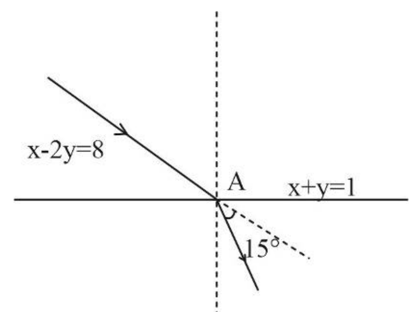
$-4-4m-2=0\ ;\quad m=\frac{-3}{2}$ Coordinate of B is $(-4,6)$

50.(AC) $x-2y=8$ ……(i)

$x+y=1$..(ii)

Solving (i), (ii) we get

$y=-7/3, x=\frac{10}{3};\ A\left(\frac{10}{3},-\frac{7}{3}\right)$



Let the refracted ray have the slope $=m$

$$\text{Then } tan15^\circ=\left|\frac{m-\frac{1}{2}}{1+m.\frac{1}{2}}\right|=\left|\frac{2m-1}{m+2}\right| \text{ Or } tan\left(45^\circ-30^\circ\right)=\frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}}=\left|\frac{2m-1}{m+2}\right|$$

$$\text{Or } \left|\frac{2m-1}{m+2}\right|=\frac{\sqrt{3}-1}{\sqrt{3}+1}=2-\sqrt{3}$$

$$\frac{2m-1}{m+2}=2-\sqrt{3},-2+\sqrt{3}$$

$$2m-1=\pm\left(2-\sqrt{3}\right)(m+2)$$

$$\sqrt{3}m=5-2\sqrt{3},\ \left(4-\sqrt{3}\right)m=2\sqrt{3}-3$$

$$m=\frac{5\sqrt{3}-6}{3},\frac{5\sqrt{3}-6}{13}$$

Let the angle between $x+y=1$ and the line through $A\left(\frac{10}{3},-\frac{7}{3}\right)$ with the slope $\frac{5\sqrt{3}-6}{3}$ be ϕ. Then

$$\tan\phi = \left|\frac{\frac{5\sqrt{3}-6}{3}-(-1)}{1+\frac{5\sqrt{3}-6}{3}(-1)}\right| = 5\sqrt{3}+8$$

If the angle between $x+y=1$ and the line through $A(10/3,-7/3)$ with the slope $\frac{5\sqrt{3}-6}{13}$ be Ψ then

$$\tan\Psi = \left|\frac{\frac{5\sqrt{3}-6}{13}-(-1)}{1+\frac{5\sqrt{3}-6}{13}(-1)}\right| = \frac{7+5\sqrt{3}}{19-5\sqrt{3}}$$

$$= \frac{(7+5\sqrt{3})(19+5\sqrt{3})}{19^2-(5\sqrt{3})^2} = \frac{80+50\sqrt{3}}{11} > 0$$

$\tan\phi > \tan\Psi ; \phi > \Psi$

The slope of the refracted ray $= \frac{5\sqrt{3}-6}{3}$

The equation of the refracted ray is $y+\frac{7}{3} = \frac{5\sqrt{3}-6}{3}\left(x-\frac{10}{3}\right)$

$3(5-2\sqrt{3})x-3\sqrt{3}y+13\sqrt{3}-50=0$

51.(ABCD) $12x^2+7xy-12y^2 = 12x^2+16xy-9xy-12y^2 = 4x(3x+4y)-3y(3x+4y) = (3x+4y)(4x-3y)$

The pair of lines given by the first equation has their separate equations $3x+4y=0, 4x-3y=0$ which are at right angle.

Again $12x^2+7xy-12y^2-x+7y-1=0=(3x+4y)(4x-3y)-x+7y-1 = (3x+4y-1)(4x-3y+1)$

Thus the four lines are $3x+4y=0,\ 4x-3y=0,\ 3x+4y-1=0$ and $4x-3y+1=0$ of which the first and the third are parallel while the second and the fourth are also parallel. Moreover, the first and the second are at right angle. Hence the lines form a rectangle.

The distance between $3x+4y=0$ *and* $3x+4y-1=0$ is given by

$d =$ distance of $(0,0)$ *from* $3x+4y-1=0$

$\{(0,0)$ *being a point on* $3x+4y=0\}$ $= \left|\frac{-1}{\sqrt{3^2+4^2}}\right| = \frac{1}{5}$

The distance between $4x-3y=0$ *and* $4x-3y+1=0$ is given by $d' = \left|\frac{1}{\sqrt{3^2+4^2}}\right| = \frac{1}{5}\ \therefore d=d'$

Hence lines form a square.

52.(ABC) vertex $A \equiv (3,-1), B(x_1, y_1), C(x_2, y_2)$

Equation of median through B be $6x+10y-59=0$

Equation of angle bisector through C be $x-4y+10=0$

$x_2-4y_2+10=0$, $D=\left\{\frac{x_2+3}{2}, \frac{y_2-1}{2}\right\}$ the mid-point of AC is on median through B.

$x_2=10, y_2=5$ i.e., coordinate of $C \equiv (10,5)$

equation of $AC \equiv 6x-7y=25$

Let slope of BC be m, then BC and AC equally inclined to $x-4y+10=0$ we have

$$\frac{\frac{1}{4}-m}{1+\frac{1}{4}m}=\frac{\frac{6}{7}-\frac{1}{4}}{1+\frac{6}{7}\times\frac{1}{4}} \Rightarrow m=-2/9$$

So, equation of $BC = 2x+9y=65$

Also, B is on $2x+9y=65$ *and* $6x+10y-59=0$

By solving if $B \equiv \left\{-\frac{7}{2}, 8\right\}$ equation of $AB \equiv 18x+13y=41$

53.(AB) Here the origin remains fixed. Let the axes be turned about the fixed origin through an angle ϕ in the anticlockwise sense and new coordinates of the point (x, y) becomes (x', y').

Then the equation of transformation will be $x = x'\cos\phi - y'\sin\phi$; $y = x'\sin\phi + y'\cos\phi$

The changed equation will be

$$a(x'\cos\phi - y'\sin\phi)^2 + 2h(x'\cos\phi - y'\sin\phi)(x'\sin\phi + y'\cos\phi) + b(x'\sin\phi + y'\cos\phi)^2 = 0$$

$$\left(a\cos^2\phi + 2h\sin\phi\cos\phi + b\sin^2\phi\right)x^2 + (-a\sin 2\phi + 2h\cos 2\phi + b\sin 2\phi)x'y'$$

$$+\left(a\sin^2\phi - 2h\sin\phi\cos\phi + b\cos^2\phi\right)y^2 = 0$$

The transformed equation of the pair of lines for the new axes will be

$$\left(a\cos^2\phi + h\sin 2\phi + b\sin^2\phi\right)x^2 + \{(b-a)\sin 2\phi + 2h\cos 2\phi\}xy + \left(a\sin^2\phi - h\sin 2\phi + b\cos^2\phi\right)y^2 = 0$$

This equation will not contain xy if $(b-a)\sin 2\phi + 2h\cos 2\phi = 0$ *or* $\tan 2\phi = \frac{2h}{a-b}$

$\phi = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}$ This is the required angle.

54. (ACD) $\sec^2(a+1)b + a^2 - 1 = 0$

$1+\tan^2(a+1)b + a^2 - 1 = 0$; $\tan^2(a+1)b + a^2 = 0$

$a = 0$ *and* $\tan(a+1)b = 0$; $a = 0$ *and* $b = n\pi,\ n \in I$

Equation of line passing through $(0, n\pi)$ and having slope $\frac{1}{2}$

$y - n\pi = \frac{1}{2}(x-0)$; $x - 2y + 2n\pi = 0$

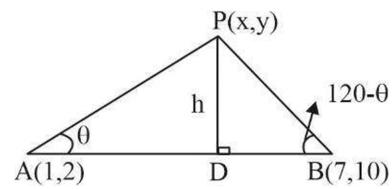
At $n=0$ $\quad x-2y=0$

$n=1$ $\quad x-2y+2\pi=0$

$n=-1$ $\quad x-2y-2\pi=0$

55. (BC) As $\angle APB = 60°$

Area will be $\frac{1}{2} \times AP \times PB \times sin(60°) = \frac{1}{2} AB \times h$



$\frac{1}{2} \times (h\, cosec\theta)(h\, cosec(120-\theta)) \times \frac{\sqrt{3}}{2} = \frac{1}{2} \times 10 \times h$

$h = \frac{20}{\sqrt{3}} sin\theta\, sin(120-\theta) = \frac{10}{\sqrt{3}}[cos(2\theta-120) - cos(120)]$

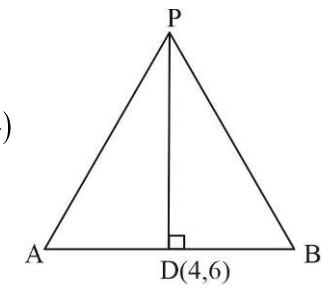
For max area h should be max.

So $2\theta - 120 = 0 \Rightarrow \theta = 60° \Rightarrow$ triangle is equilateral. So,

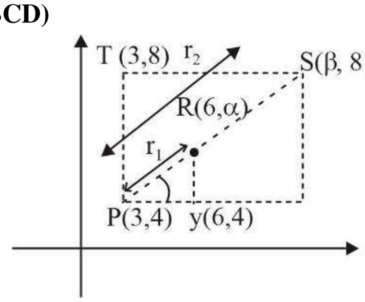
PD will be $\perp$ bisector as well as median equation of PD will be $(y-6) = \frac{-6}{8}(x-4)$

$4y - 24 = -3x + 12$

$3x + 4y = 36$ and P lies on right bisector of AB

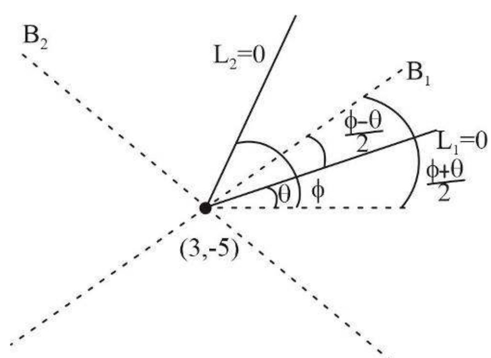


56. (ABCD)



$PU = (PR)cos\theta \Rightarrow \frac{3}{cos\theta} = PR \text{ or } PR = 3sec\theta$; $PT = (PS)sin\theta \Rightarrow \frac{4}{sin\theta} = PS \text{ or } PS = 4cosec\theta$

57. (ABC)



Lines given will pass through $(3,-5)$ making angle θ and ϕ with x-axis.

Let B_2, B_1 be the angle bisector of lines L_1 and L_2.

Then B_1 will be inclined at an angle of $\frac{\theta+\phi}{2}$. So, its equation will be $\frac{x-3}{cos\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{sin\left(\frac{\theta+\phi}{2}\right)}$

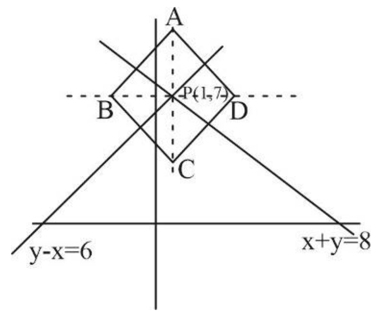
So, $\alpha = \frac{\theta+\phi}{2}$

Also, B_2 will be at 90° from B_1 so its angle of inclination will be $90+\left(\frac{\theta+\phi}{2}\right) or\, 90+\alpha$

So its equation will be $\frac{x-3}{cos(90+\alpha)}=\frac{y+5}{sin(90+\alpha)}$ or $\frac{x-3}{-sin\,\alpha}=\frac{y+5}{cos\,\alpha}$

$\beta=-sin\,\alpha\; and\; \gamma=cos\,\alpha$

58. (ABCD) There will be 4 points which will lie equidistant to both line and will be vertices of square will sides 200 and diagonal length $200\sqrt{2}$.



Length of $PC=100\sqrt{2}=AP=BP=PD$

So coordinate of $D\equiv\left[1+100\sqrt{2},7\right]$ $A\equiv\left[1,7+100\sqrt{2}\right]$

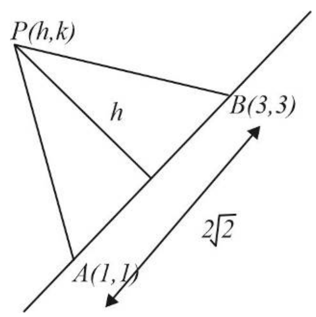
$B\equiv\left[1-100\sqrt{2},7\right]$ $C\equiv\left[1,7-100\sqrt{2}\right]$

59.(AB) L must be angle bisector of $L_1\; and\; L_2$

L is given by $\frac{3x+4y-1}{5}=\pm\frac{5x-12y+2}{13}$

$14x+112y-23=0$ (+ sign) ; $64x-8y-3=0$ $(-sign)$

60.(CD)



p will be on line parallel to $y=x$ at a perpendicular distance of $\frac{1}{\sqrt{2}}$

locus of P will be $y-x=1$ or $y-x=-1$

61. (ABC) Let $f(x,y)=3x+2y$

$f(1,3)>0, f(5,0)>0\, and\; f(-1,2)>0$

All three vertices satisfy the inequality $3x+2y\geq 0$, thus inequality holds for all point inside the triangle

Let $f(x,y)=2x-3y-12$

$f(1,3)<0, f(5,0)<0\; and\; f(-1,2)<0$

All three vertices satisfy the inequality $2x-3y-12\leq 0$, thus the inequality holds for all points lying inside the triangle.

Let $f(x,y)=x+y-6, then\; f(1,3)<0, f(5,0)<0\, and\; f(-1,2)<0$

So, the vertices $(1,3)\, and\, (-1,2)$ do not satisfy the inequality $2x-y\geq 0$. Hence some points lying inside the triangle do not satisfy the inequality

62.(ABCD) $x_1 + y_1 = 4,\ x_2 + y_2 = 4$

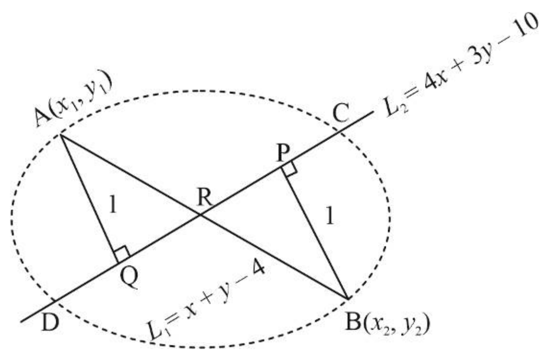
Also, $\left|\dfrac{4x_1+3y_1-10}{5}\right|=1$, $\left|\dfrac{4x_2+3y_2-10}{5}\right|=1$

$4x_1+3y_1-10=\pm 5$

$4x_1+3y_1-10=5$ or $4x_2+3y_2-10=-5$ (as (x_1,y_1) and (x_2,y_2) are similar)

$x_1=3;\ \ x_2=-7;\ \ y_1=1;\ \ y_2=11$

So, $\dfrac{y_1+y_2}{x_1+x_2}=-3$



Also L_2 divides L_1 in ratio $-\left\{\dfrac{4(3)+3(1)-10}{4(-7)+3(11)-10}\right\}=1$

$R\equiv\{-2,6\}$

$AR=5\sqrt{2}\Rightarrow QR=7=PR$

$PQ=14$

$RD=5\sqrt{2};\ QR=7\Rightarrow QD=5\sqrt{2}-7$

63.(ACD) Let the lines represented by the given equation be $y=x\tan\alpha$ and $y=x\tan\beta$

Then $\tan\alpha+\tan\beta=\dfrac{2\tan\theta}{\sin^2\theta}=\dfrac{2}{\sin\theta\cos\theta}=4\operatorname{cosec}2\theta$

$$\tan\alpha\tan\beta=\frac{\tan^2\theta+\cos^2\theta}{\sin^2\theta}=\sec^2\theta+\cot^2\theta$$

$$\tan\alpha-\tan\beta=\pm\sqrt{\frac{4}{\sin^2\theta\cos^2\theta}-4\left(\sec^2\theta+\cot^2\theta\right)}=\pm2\sqrt{\frac{1-\left(\sin^2\theta+\cos^4\theta\right)}{\sin^2\theta\cos^2\theta}}$$

$$=\pm2\sqrt{\frac{\left(\cos^2\theta-\cos^4\theta\right)}{\sin^2\theta\cos^2\theta}}=\pm2$$

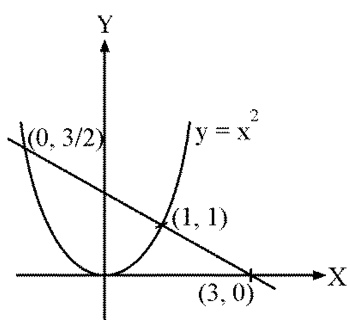
And $\dfrac{\tan\alpha}{\tan\beta}=\dfrac{4\operatorname{cosec}2\theta+2}{4\operatorname{cosec}2\theta-2}=\dfrac{2+\sin2\theta}{2-\sin2\theta}$

64. **[A-p] [B-s] [C-q] [D-s]**

(A) $2a^2+a-3<0$

$(2a+3)(a-1)<0$

$\Rightarrow\quad a\in\left(\dfrac{-3}{2},1\right)$



But $a > 0$

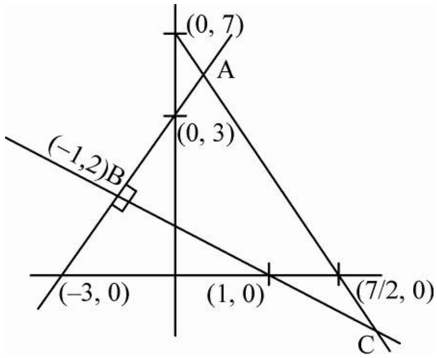
Hence no. of integral values of $a = 0$

(B) Perpendicular distance of $P(\lambda,4-\lambda)$ from $4x + 3y = 10$

$=\dfrac{|4\lambda+3(4-\lambda)-10|}{5}=1\ \Rightarrow\quad |\lambda+2|=5\Rightarrow\lambda=-2\pm5=3,-7$

$$P \equiv (3,1),(-7,+11)$$

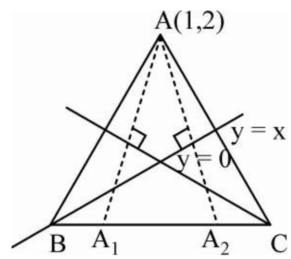
(C) Point $B \equiv (-1,2)$



Aliter : The given triangle is right angled.

$\therefore$ Co-ordinates of orthocentre is $(-1,\ 2)$, by solving the equation $x+y=1$ and $x-y+3=0$

(D) Images of A w.r.t. $y=x$ and $y=0$ lies on BC which are (2, 1), (1, –2)



$\therefore$ Equation of BC is $y=3x-5$

Perpendicular distance of A from

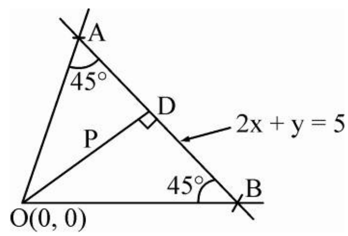
$$BC = \frac{|3-2-5|}{\sqrt{10}}$$

$$d(A,BC)=\frac{4}{\sqrt{10}} \quad \Rightarrow \quad \sqrt{10}d(A,BC)=4$$

65. **[A-q] [B-p] [C-s] [D-r]**

(A) Since the triangle is isosceles, so

$OD = AD = BD$

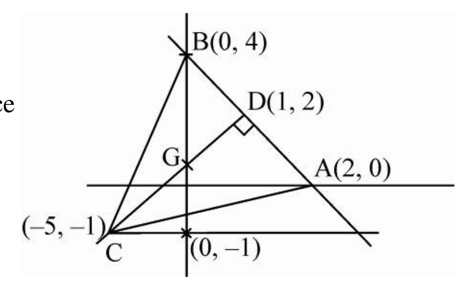


$$OD = P = \frac{|5|}{\sqrt{5}}$$

Area of $\Delta OAB = \frac{1}{2}(2P)P = P^2$

Area = 5

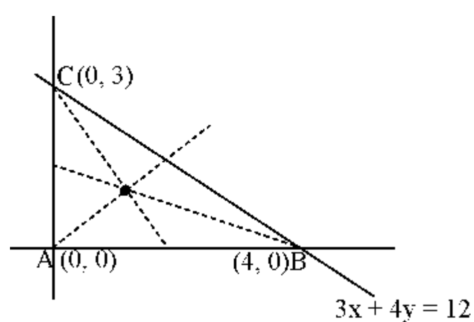
(B) Slope of CD is $\frac{1}{2}, C \equiv (-5,-1)$



Since G is centroid of triangle ABC,so Perpendicular distance from G to AB = $\frac{1}{3}$ (Perpendicular distance from C to AB)

$$=\frac{1}{3}\left(\frac{|-10-1-4|}{\sqrt{4}+1}\right)=\sqrt{5}$$

(C) Point P must lie on at least one of the angle bisectors.
Incentre,



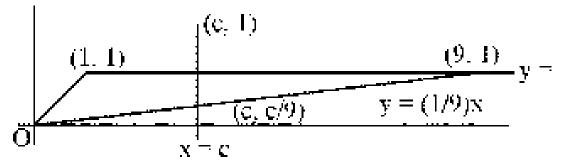
$$I \equiv \left(\frac{4(0)+3(4)+5(0)}{4+3+5}, \frac{4(3)+3(0)+5(0)}{4+3+5}\right)$$

$I \equiv (1,\ 1)$

$P \equiv (1,\ 1)$ only

(D) $2\times\frac{1}{2}\left(1-\frac{c}{9}\right)(9-c)=\frac{1}{2}\times 8\times 1 \Rightarrow c=3$

Or $c = 15$ which is not possible



66. **[A-s] [B-p] [C-q] [D-r]**

(A) Let equation of line is $y-4=m(x-1)(m<0)$

$\therefore \quad OA=1+\frac{4}{(-m)}$ and $OB=4+(-m) \Rightarrow \quad OA+OB=5+\frac{4}{(-m)}+(-m)$

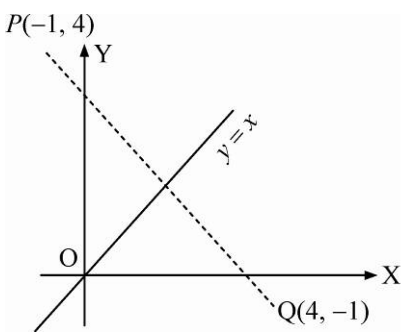
Apply $\text{AM}\geq\text{GM}$ on $\frac{4}{(-m)}$ & $(-m)$

$$\therefore \quad \frac{\frac{4}{(-m)}+(-m)}{2}\geq\sqrt{\frac{4}{(-m)}\times(-m)}$$

$$\frac{4}{(-m)}+(-m)\geq 4 \quad\Rightarrow\quad 5+\frac{4}{(-m)}+(-m)\geq 9 \quad\Rightarrow\quad OA+OB\geq 9$$

(B) Image of point $Q(4,-1)$, w.r.t. to line mirror $y=x$ is $P(-1,4)$

$\therefore \quad PQ=\sqrt{50}=4\sqrt{2}$



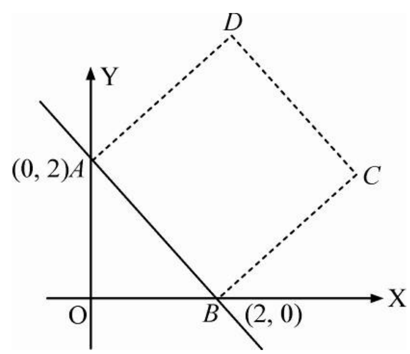
(C) From figure,

Perpendicular distance of O from $AB=\sqrt{2}$

Perpendicular distance of O from $CD=3\sqrt{2}$

Perpendicular distance of O from $AD=\sqrt{2}$

Perpendicular distance of O from $BC=\sqrt{2}$



(D) Equation of line

$$x=4+\frac{\lambda}{\sqrt{2}} \ \& \ y=-1+\sqrt{2}\lambda \text{ can be written as } 2x=8+\sqrt{2}\lambda \ \& \ y=-1+\sqrt{2}\lambda$$

$\therefore \quad$ Equation of line is $2x-y=9$

$\therefore \quad$ Length of x-intercept $=\frac{9}{2}$

67. **[A-s] [B-p] [C-q] [D-r]**

(A) $6x^2 - axy - 3y^2 - 24x + 3y + \beta = 0$

$\therefore$ Intersection point of lines lies on x-axis.

Put $y = 0$.

$\Rightarrow$ $6x^2 - 24x + \beta = 0$ must have equal real roots

$\Rightarrow$ $D = 0 \Rightarrow (24)^2 - 24\beta = 0 \Rightarrow \beta = 24$

Apply condition of pair of lines, i.e, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

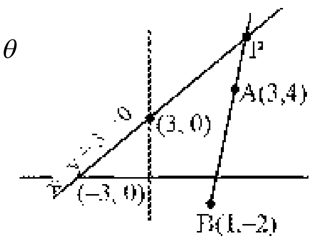
$\Rightarrow$ $6(-3)\beta + (-\alpha)(-12)\frac{3}{2} - 6\left(\frac{3}{2}\right)^2 - (-3)(12)^2 - \beta\left(\frac{\alpha}{2}\right)^2 = 0$

$\Rightarrow$ $-18(24) + 18\alpha - \frac{27}{2} + 3(144) - 6\alpha^2 = 0 \Rightarrow 6\alpha^2 - 18\alpha + \frac{27}{2} = 0$

$\Rightarrow$ $4\alpha^2 - 12\alpha + 9 = 0$

$\alpha = 3/2$ Hence, $20\alpha - \beta = 6$

(B) $(3x + 7y + 11)\sec\theta + (5x - 3y - 11)\operatorname{cosec}\theta = 0$ passes through $(1, -2)$ $\forall$ permissible θ

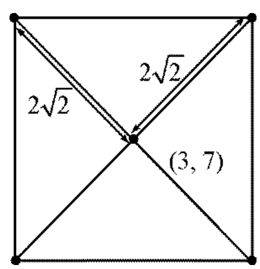


$B \equiv (1, -2)$

It is clear that $|PA - PB| \le AB$

$\Rightarrow$ $|PA - PB|_{\max} = AB = \sqrt{6^2 + 2^2} = 2\sqrt{10} \Rightarrow n = 5$

(C) Vertices of square are



$$\left(3 \pm 2\sqrt{2}\left(\frac{1}{\sqrt{2}}\right), 7 \pm 2\sqrt{2}\frac{1}{\sqrt{2}}\right),$$

$$\left(3 \pm 2\sqrt{2}\left(-\frac{1}{\sqrt{2}}\right), 7 \pm 2\sqrt{2}\frac{1}{\sqrt{2}}\right)$$

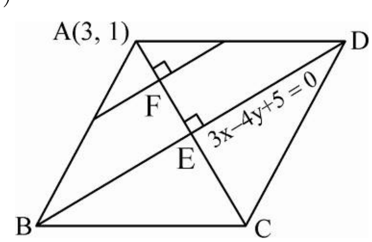
i.e., (5, 9), (1, 5), (1, 9), (5, 5)

$\therefore$ $\max(y_1, y_2, y_3, y_4) - \min(x_1, x_2, x_3, x_4) = 9 - 1 = 8$

(D) Point E

$$\frac{x-3}{3} = \frac{y-1}{-4} = -\frac{9-4+5}{25} = -\frac{2}{5} \Rightarrow E \equiv \left(\frac{9}{5}, \frac{13}{5}\right)$$

Line, L passing through midpoint of sides AB and AD passes through F.



$$F \equiv \left(\frac{3 + \frac{9}{5}}{2}, \frac{\frac{13}{5} + 1}{2}\right) \equiv \left(\frac{12}{5}, \frac{9}{5}\right)$$

Equation of line L is $3x - 4y = \frac{36}{5} - \frac{36}{5} = 0$

$\Rightarrow$ $3x - 4y = 0 \Rightarrow |a + b + c| = |3 - 4| = 1$

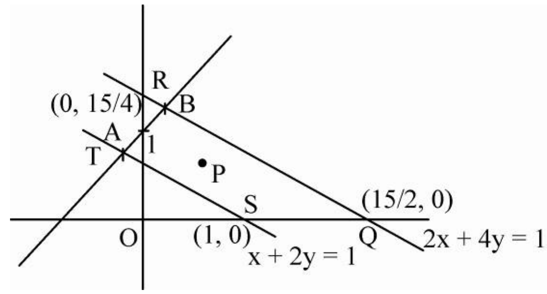
68. **[A-r]** **[B-p]** **[C-s]** **[D-q]**

(A) $P\left(1+\frac{t}{\sqrt{2}}, 2+\frac{t}{\sqrt{2}}\right)$ point P lies on the line $y = x+1$

$$A = \left(\frac{-1}{3}, \frac{2}{3}\right), B = \left(\frac{11}{6}, \frac{17}{6}\right)$$

$$1+\frac{t}{\sqrt{2}} \in \left(\frac{-1}{3}, \frac{11}{6}\right)$$

$$t \in \left(\frac{-4\sqrt{2}}{3}, \frac{5\sqrt{2}}{6}\right)$$



Aliter : For point P lies in between parallel lies

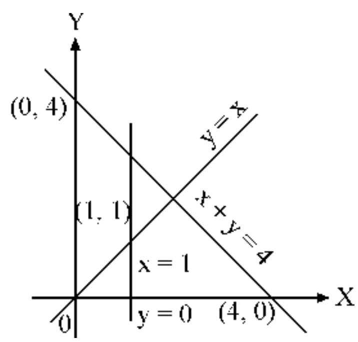
$\Rightarrow$ With respect to line QR, point P and origin are on same side and with respect to line ST, point P and origin are on opposite side

$$\Rightarrow \quad 2\left(1+\frac{t}{\sqrt{2}}\right)+4\left(2+\frac{t}{\sqrt{2}}\right)-15<0 \text{ and } \left(1+\frac{t}{\sqrt{2}}\right)+2\left(2+\frac{t}{\sqrt{2}}\right)-1>0 \Rightarrow t \in \left(\frac{-4\sqrt{2}}{3}, \frac{5\sqrt{2}}{6}\right)$$

(B) $P\left((2-t)x_1+(t-1)x_2, (2-t)y_1+(t-1)y_2\right)$ divides $(x_1, y_1), (x_2, y_2)$ internally in ratio $(t-1):(2-t)$

$\therefore \quad (t-1)(2-t) > 0 \Rightarrow t \in (1,2)$

(C) $t \in (0,1)$ (See from figure)



(D) For A :

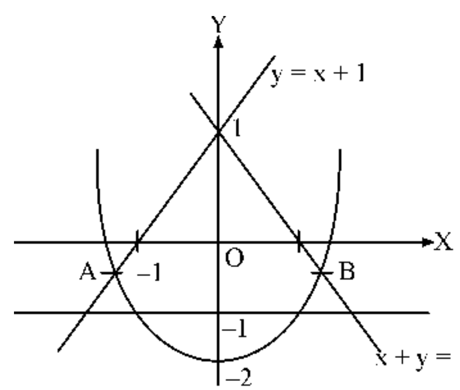
$x^2 - 2 = x + 1$

$$x^2 - x - 3 = 0$$

$$x = \frac{1 \pm \sqrt{13}}{2}; \quad x = \frac{1-\sqrt{3}}{2}$$

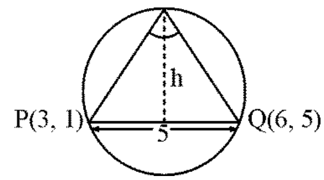
Similarly for B, $\quad x = \frac{\sqrt{13}-1}{2}$

$$\therefore \quad t \in \left(\frac{1-\sqrt{13}}{2}, -1\right) \cup \left(1, \frac{\sqrt{13}-1}{2}\right)$$



69. **[A-r] [B-s] [C-q] [D-p]**

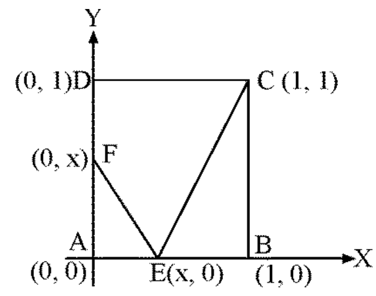
(A) $\Delta = \frac{1}{2} \times 5 \times h = 7 \Rightarrow h = \frac{14}{5}$



But h max = Radius of circle with diameter $PQ = \frac{5}{2} < \frac{14}{5}$

$\Rightarrow$ No such triangle exists.

(B) Area of $CDEF$,



$$A(x) = 1 - \frac{1}{2}x^2 - \frac{1}{2}(1-x) = \frac{1+x-x^2}{2}$$

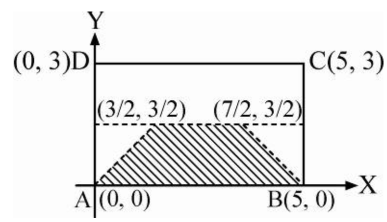
$$A_{max} = A\left(\frac{1}{2}\right) = \frac{1+\frac{1}{2}-\frac{1}{4}}{2} = \frac{5}{8}$$

At $x = \frac{1}{2}$

(C) $d(P, AB) \le d(P < BC)$

$d(P, AB) \le d(P < CD)$

$d(P, AB) \le d(P < AD)$



$\Rightarrow$ P lies in region as shown $\quad Area = \frac{1}{2} \times \frac{3}{5} \times (5+2) = \frac{21}{4}$

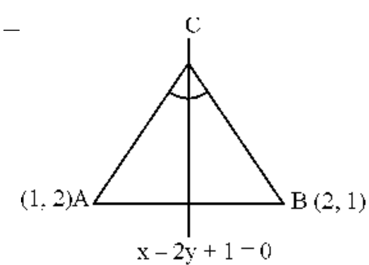
(D) $a + 2hm + bm^2 = 0$

Where $m = \frac{y}{x} \quad \Rightarrow \quad m_1 + m_1^2 = -\frac{2h}{b}; \quad m_1^3 = \frac{a}{b}$

$$\Rightarrow -\frac{8h^3}{b^3} = \left(m_1 + m_1^2\right)^3 = m_1^3 + m_1^6 + 3m_1^3\left(m_1 + m_1^2\right) = \frac{a}{b} + \frac{a^2}{b^2} + \frac{3a}{b}\left(-\frac{2h}{b}\right)$$

$$\Rightarrow -\frac{8h^3}{b^3} = \frac{ab + a^2 - 6ah}{b^2} \Rightarrow ab(a+b) - 6abh + 8h^3 = 0 \Rightarrow \alpha + \beta = 8 -$$

70.(2) Image of A say A' w.r.t. $x - 2y + 1 = 0$ lies on BC.



$$\frac{x-1}{1} = \frac{y-2}{-2} = -2\frac{(1-4+1)}{1+2^2} = \frac{4}{5}$$

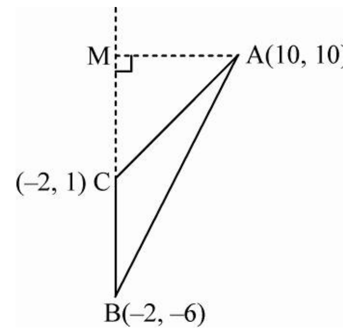
$$A' \equiv \left(\frac{9}{5}, \frac{2}{5}\right)$$

Equation of BC joining $A'\left(\frac{9}{5}, \frac{2}{5}\right)$ and $B(2, 1)$ is $y - 1 = \frac{1-\frac{2}{5}}{2-\frac{9}{5}}(x-2) = \frac{3}{1}(x-2)$

$3x - y - 5 = 0 \Rightarrow a + b = 3 - 1 = 2$

71.(6) Let perpendicular bisector of AB is $3x+4y-20=0$ and perpendicular bisector of AC is $8x+6y-65=0$

$\Rightarrow$ Image of A w.r.t. $3x+4y-20=0$ is B and image of A w.r.t $8x+6y-65=0$ is C.

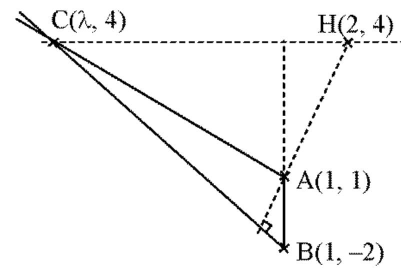


For B, $\dfrac{x-10}{3}=\dfrac{y-10}{4}=-2\left(\dfrac{30+40-20}{25}\right) \quad \Rightarrow \quad B\equiv(-2,-6)$

For C, $\dfrac{x-10}{8}=\dfrac{y-10}{6}=-2\left(\dfrac{80+60-65}{100}\right) \quad \Rightarrow \quad C\equiv(-2,1)$

Area of $\Delta ABC=\dfrac{1}{2}(10+2)(1+6)=42 \quad \Rightarrow \quad$ Required value $=6$

72.(2) $a+c-2b=0$



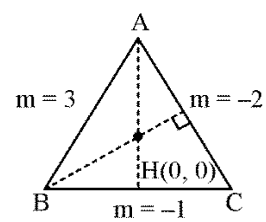
$\Rightarrow$ $ax+by+c=0$ passes through fixed point $(1,-2)$ $\Rightarrow$ $B\equiv(1,-2)$

Slope of $BC=-\dfrac{2-1}{4-1}=\dfrac{4+2}{\lambda-1}$

$\lambda-1=-18 \quad \Rightarrow \quad \lambda=-17$

$C\equiv(-17,4)\equiv(h,k) \qquad \therefore \quad 2h+9k=-34+36=2$

73.(7) Let slopes of BC, CA, AB be -1, -2, 3 respectively.



$\because$ Slope of BC is $-1 \quad \Rightarrow \quad$ Slope of $AD=1$

$\therefore$ Equation of AD is $y=x \quad \Rightarrow \quad$ co-ordinate of $A(x_1,x_1)$

Similarly co-ordinate of $B\left(x_2,\dfrac{x_2}{3}\right)$ & $C\left(x_3,-\dfrac{x_3}{3}\right)$

$\because$ Slope of $BC=\dfrac{\dfrac{x_2}{2}+\dfrac{x_3}{3}}{x_2-x_3}=\dfrac{3x_2+2x_3}{6(x_2-x_3)}=-1 \quad \Rightarrow \quad 9x_2=4x_3$

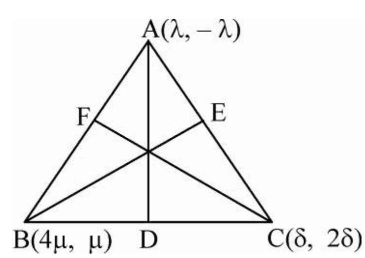
$\because$ Slope of $CA=\dfrac{x_1+\dfrac{x_3}{3}}{x_1-x_3}=-2=\dfrac{3x_1+x_3}{3(x_1-x_3)}\Rightarrow 9x_1=5x_3$

$$G\equiv(h,k)=\left(\frac{x_1+x_2+x_3}{3},\frac{x_1+\dfrac{x_2}{2}-\dfrac{x_3}{3}}{3}\right)$$

$3h=x_3+\dfrac{5x_3}{9}+\dfrac{4x_3}{9}=2x_3 \qquad$ **...(1)** $\quad$ and $\quad 3k=\dfrac{5x_3}{9}+\dfrac{2x_3}{9}-\dfrac{x_3}{9}=\dfrac{4x_3}{9} \qquad$ **...(2)**

From eqs. (1) and (2), we get $\dfrac{k}{h}=\dfrac{2}{9}\Rightarrow y=\dfrac{2}{9}x;\ b-a=9-2=7$

74.(6) Equation of AD, $x+y=0$



Equation of line BE, $x-4y=0$

Equation of line CF, $2x-y=0$

Let centroid is $(x,\ y)$

$x=\dfrac{\lambda+4\mu+\delta}{3}$ ...(i) and $y=\dfrac{\mu-\lambda+2\delta}{3}$ ...(ii)

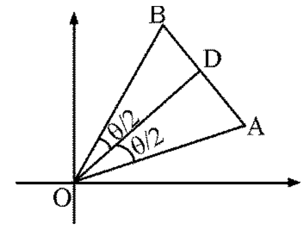
$BC \perp AD$; $\dfrac{\mu - 2\delta}{4\mu - \delta} = 1 \quad \Rightarrow \quad \mu - 2\delta = 4\mu - \delta \Rightarrow 3\mu = -\delta$

$AC \perp BE$; $\dfrac{2\delta + \lambda}{\delta - \lambda} = -4 \quad \Rightarrow \quad 2\delta = \lambda$

So we can say $\dfrac{\lambda}{6} = \dfrac{\delta}{3} = \dfrac{\mu}{-1}$ by using equation (i) and (ii),

$\dfrac{x}{y} = \dfrac{6-4+3}{-1-6+6} = -5 \quad \Rightarrow \quad x = -5y \quad \Rightarrow \quad x + 5y = 0$

75.(8) Slope of OA and OB are respectively $\dfrac{7}{17}$ and 1, OD is $\perp$ to AB.



Let equation of AB is $x\cos\alpha + y\sin\alpha = \sqrt{13}$,

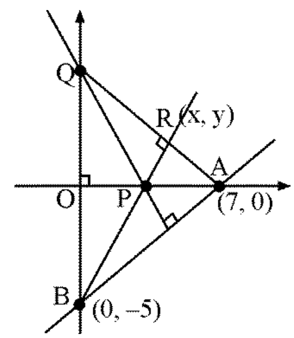
where OD makes α angle with x-axis.

So $\quad \tan\theta = \dfrac{1 - \frac{7}{17}}{1 + \frac{7}{17}} = \dfrac{10}{24} = \dfrac{5}{12} \Rightarrow \tan\dfrac{\theta}{2} = \dfrac{1}{5}$

$\tan\alpha = \tan\left(\dfrac{\theta}{2} + \tan^{-1}\left(\dfrac{7}{17}\right)\right) = \dfrac{\frac{1}{5} + \frac{7}{17}}{1 - \frac{7}{5\,.\,17}} = \dfrac{2}{3}$

$\sin\alpha = \dfrac{2}{\sqrt{13}}, \quad \cos\alpha = \dfrac{3}{\sqrt{13}}$

So equation of AB is $3x + 2y = 13$.



76.(2) As x and y-axis perpendicular the point P becomes orthocenter of triangle ABQ, so R is foot of 3rd altitude.

$\Rightarrow \quad \left(\dfrac{y+5}{x}\right)\left(\dfrac{y}{x-7}\right) = -1 \quad \Rightarrow \quad x^2 + y^2 - 7x + 5y = 0$

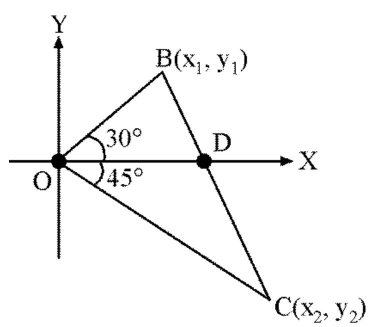
77.(2) Taking O as origin

And x –axis as the median

$\dfrac{y_1}{x_1} = \tan 30^o = \dfrac{1}{\sqrt{3}}$

$\dfrac{y_2}{x_2} = \tan 135^o = -1$

$y_1 = \dfrac{x_1}{\sqrt{3}}, y_2 = -x_2$



Also $\quad \dfrac{x_1 + x_2}{2} = \left(11 - 6\sqrt{3}\right)^{\frac{-1}{2}} = p$

$\dfrac{y_1 + y_2}{2} = 0$

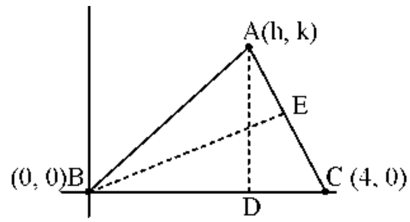
$x_1 = \dfrac{2\sqrt{3}p}{\sqrt{3}+1}, x_2 \dfrac{2p}{\sqrt{3}+1}, y_1 = \dfrac{2p}{\sqrt{3}+1}, y_2 = -\dfrac{2p}{\sqrt{3}+1}$

$BC = 2$.

78.(3) Area of $\Delta\, ABC = 2k$

Since $AD \perp BE$, So

$$k^2 + (h+4)(h-2) = 0$$



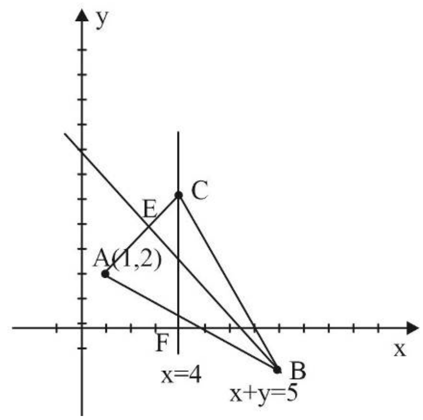
And $AC = 3$, so

$$(h-4)^2 + k^2 = 9$$

Hence $\quad k^2 = \dfrac{11}{4}$

$$[\Delta] = 3$$

79.(13) Any point B on $x+y=5$ is $(h, 5-h)$ and C on $x=4$ is $(4,k)$



Midpoint of AB is $\left(\dfrac{1+h}{2}, \dfrac{7-h}{2}\right)$ which lies on $x=4$, so $\dfrac{1+h}{2} = 4 \Rightarrow h = 7$

Thus, the coordinates of B are $(7,-2)$

Again, the midpoint of AC is $\left(\dfrac{5}{2}, \dfrac{2+k}{2}\right)$ which lies on $x+y=5$, so $k=3$.

Thus, the coordinates of C are $(4,3)$

80.(3) Let the slope of the line be $\tan\theta$. Any point on the line is $(-2 + r\cos\theta, -3 + r\sin\theta)$. If $AB = r_1$, then B has coordinates $(-2 + r_1\cos\theta, -3 + r_1\sin\theta)$, which lies ion $x + 3y = 9$

$$-2 + r_1\cos\theta - 9 + 3r_1\sin\theta - 9 = 0$$

$$r_1 = \frac{20}{\cos\theta + 3\sin\theta}$$

If $AC = r_2$ then C has coordinates $(-2 + r_2\cos\theta, -3 + r_2\sin\theta)$ which lies on $x + y + 1 = 0$

$$-2 + r_2\cos\theta - 3 + r_2\sin\theta + 1 = 0 \Rightarrow r_2 = \frac{4}{\cos\theta + \sin\theta}$$

Given $r_1 r_2 = 20$

$$\left|\left(\frac{20}{\cos\theta + 3\sin\theta}\right)\left(\frac{4}{\cos\theta + \sin\theta}\right)\right| = 20$$

$$\left|(\cos\theta + 3\sin\theta)(\cos\theta + \sin\theta)\right| = 4$$

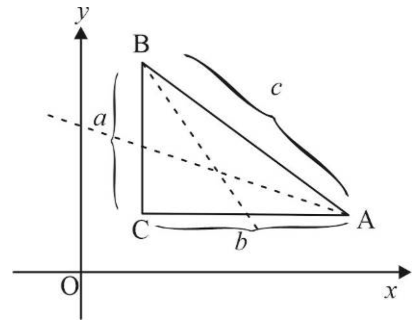
$$\cos^2\theta + 3\sin^2\theta + 4\cos\theta\sin\theta = \pm 4$$

$$1 + 3\tan^2\theta + 4\tan\theta = \pm 4\left(1 + \tan^2\theta\right)$$

$$7\tan^2\theta + 4\tan\theta + 5 = 0 \ or \ \tan^2\theta - 4\tan\theta + 3 = 0$$

The first equation has no real roots, so $\tan^2\theta - 4\tan\theta + 3 = 0 \Rightarrow \tan\theta = 1 \, or \, 3$

81. (9) If the slopes of the medians be $m_1 \, and \, m_2$, then

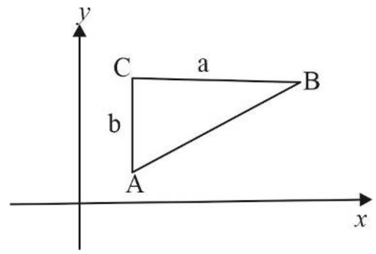


$$m_1 = \frac{a/2}{b} = 3, \; m_2 = \frac{a}{b/2}$$

(a and b are the length of the sides)

$$\frac{m_1}{m_2} = \frac{a \times b}{2b \times a \times 2} = \frac{1}{4}$$

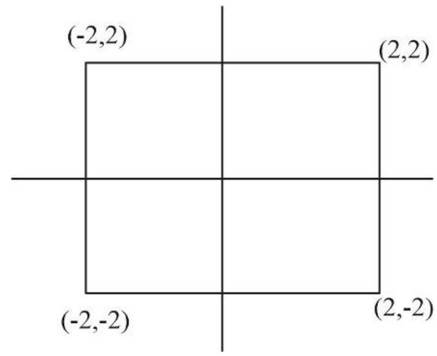
$$m_2 = 12$$



$$m_1 = \frac{b}{a/2} = 3; \; m_2 = \frac{b/2}{a}$$

$$\frac{m_1}{m_2} = \frac{2b \times 2a}{a \times b} = 4; \; m_2 = 3/4$$

82.(16) Point P is (x, y)



Given $max\{|x|,|y|\} = 2$

Case I: $|x| = 2$

$x = +2$, and $x = -2$

Then $|y| \le 2$; $-2 \le y \le 2$

Case II: $|y| = 2$

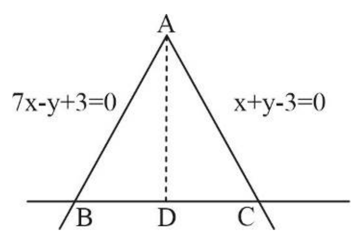
$y = 2 \; and \; y = -2$

Then $|x| \le 2$; $-2 \le x \le 2$

Such curve will be a square of side 4 units and area will be 16 square units.

83. (3) $L_1 : 7x - y + 3 = 0$

$L_2 : -x - y + 3 = 0$



$a_1 a_2 + b_1 b_2 < 0$

Bisector with positive sign will give acute angle bisector

$$\frac{7x - y + 3}{\sqrt{50}} = +\left(\frac{-x - y + 3}{\sqrt{2}}\right) = 3x + y - 3 = 0$$

equation of line BC with slope $-\dfrac{1}{m_{AD}}$ and passing through $(1, -10)$ (from given relation)

is $y + 10 = \dfrac{1}{3}(x - 1)$

$x - 3y - 31 = 0$

$a = 1, b = -3, c = -31$

$2a + 10b - c = 2 - 30 + 31 = 3$

84.(1.80) Let $OA = r_1, OB = r_2, OC = r_3$

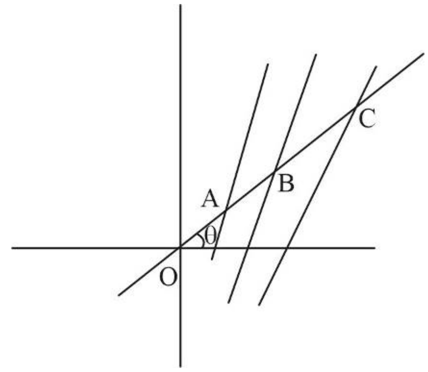
And line 'L' makes angle θ with passing through origin $(y = mx\ form)$

Point $A(r_1 \cos\theta, r_1 \sin\theta); B(r_2 \cos\theta, r_2 \sin\theta); C(r_3 \cos\theta, r_3 \sin\theta)$

Now as point *A*, *B* and *C* lies on the lines L_1, L_2 and L_3 respectively,

$2r_1 \cos\theta + 3r_1 \sin\theta - 5 = 0$

$\frac{1}{r_1} = \frac{2\cos\theta + 3\sin\theta}{5}, \ \frac{1}{r_2} = \frac{\cos\theta + 2\sin\theta}{5}, \frac{1}{r_3} = \frac{6\cos\theta + 4\sin\theta}{5},$

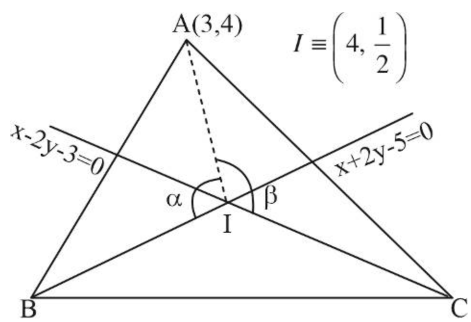


Now, $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{9}{5}(\cos\theta + \sin\theta) = \frac{k(a+b)}{OP}$ $[(a, b) \equiv (r\cos\theta, r\sin\theta)$ where $OP = r]$

$k = \frac{9}{5}$

85.(4) $AI = \frac{\sqrt{53}}{2}, \ m_{AI} = \frac{4 - \frac{1}{2}}{-1} = \frac{-7}{2}$

$\tan\alpha = \left|\frac{\frac{-7}{2} + \frac{1}{2}}{1 + \frac{7}{4}}\right|$



$\beta = 90 + \frac{B}{2}; \quad \tan\beta = \left|\frac{\frac{-7}{2} - \frac{1}{2}}{1 - \frac{7}{4}}\right| = \left|\frac{\frac{8}{2}}{\frac{3}{4}}\right| = 16/3$

$\tan\beta = \tan\left(90 + \frac{B}{2}\right) = \frac{16}{3} \Rightarrow \sin\frac{B}{2} = \frac{3}{\sqrt{265}}$

Now, $\frac{AI}{\sin\frac{B}{2}} = \frac{AB}{\sin\alpha} \Rightarrow \frac{\sqrt{53} \times \sqrt{265}}{2 \times 3} = \frac{AB}{12} \times \sqrt{265} \Rightarrow AB = 2\sqrt{53}$

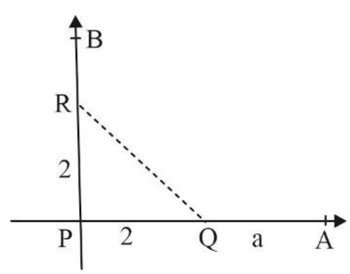
Given $AB = k.AI \Rightarrow 2\sqrt{53} = k.\frac{\sqrt{53}}{2} \Rightarrow k = 4$

86.(3) Let $QA = a$

Given $QA.RB = PR^2 = 4$

$RB = \frac{4}{a}$

Point $A(2 + a, 0); B\left(0, 2 + \frac{4}{a}\right)$



Now equation of line *AB*; $y = \frac{\frac{4}{a} + 2}{-(2+a)}(x - 2 - a)$

$-a(2+a)y = (4 + 2a)(x - 2 - a)$

$-ay = 2(x-2-a)$

$2x-4+ay-2a=0$

$(x-2)+\frac{a}{2}(y-2)=0$

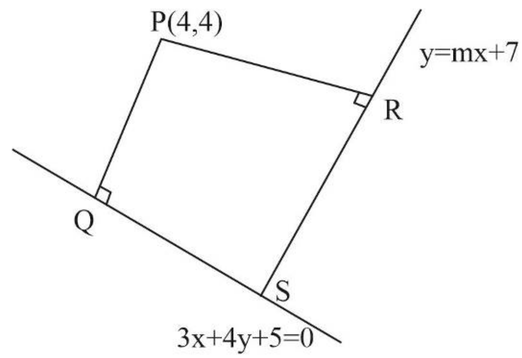
Always passes through $(2,2)$

If it satisfies $ax+by-6=0$ then $2a+2b-6=0$

$a+b=3$

87.(16) As rectangle $PQSR$ is a cyclic quadrilateral.

ΔPQR will have maximum area if $\angle QPR = \angle QSR = 90$



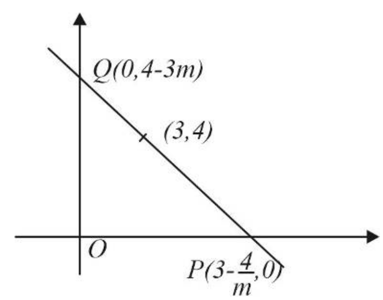
$m_{QS}.m_{SR}=-1$

$\left(-\frac{3}{4}\right)m=-1$

$m=\frac{4}{3}$

$12m=12\times\frac{4}{3}=16$

88.(24) Line passing through the point $(3,4)$ is $y-4=m(x-3)$



As slope m is negative here $ar(\Delta OPQ)$ $\Delta=\frac{1}{2}(4-3m)\left(3-\frac{4}{m}\right)$

$\Delta=\frac{1}{2}\left[24+\left(-9m-\frac{16}{m}\right)\right]$

By AM-GM inequality $\frac{-9m-\frac{16}{m}}{2}\geq\sqrt{9\times16}$

$\left(-9m-\frac{16}{m}\right)_{min}=24$, $\Delta_{min}=\frac{1}{2}\times48=24$

89.(6) Let m and m^2 be the slopes of the lines $m+m^2=-\frac{2h}{b}$ ....(i) and $m^3=\frac{a}{b}$ ....(ii)

Cubing equation (i)

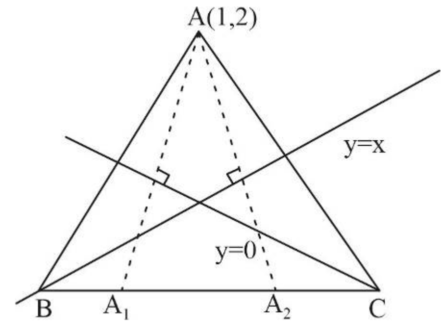
$\left(m+m^2\right)^3=-\frac{8h^3}{b^3}$

$\Rightarrow\ m^3+m^6+3mm^2\left(m+m^2\right)=-\frac{8h^3}{b^3}$

$\Rightarrow\ \frac{a}{b}+\frac{a^2}{b^2}+\frac{3a}{b}\left(-\frac{2h}{b}\right)=-\frac{8h^3}{b^3}$

$\frac{a(a+b)}{b^2}+\frac{8h^3}{b^3}=\frac{6ah}{b^2}\ \Rightarrow\ \frac{(a+b)}{h}+\frac{8h^3}{ab}=6$

90.(4) Images of A w.r.t $y = x$ and $y = 0$ lies on BC which are $(2,1),(1,-2)$



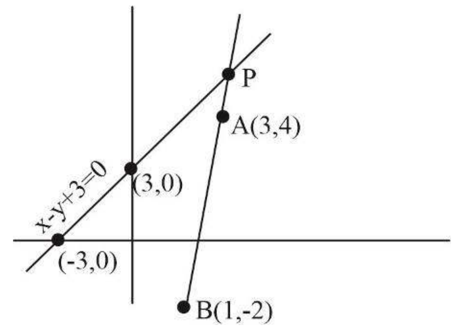
Equation of BC is $y = 3x - 5$

Perpendicular distance of A from $BC = \frac{|3-2-5|}{\sqrt{10}}$

$d(A, BC) = \frac{4}{\sqrt{10}}$

$\sqrt{10}d(A, BC) = 4$

91.(5) $(3x+7y+11)\sec\theta + (5x-3y-11)\operatorname{cosec}\theta = 0$ passes through intersection of $3x+7y+11=0$ and $5x-3y-11=0$ for permissible values of θ given by $(1,-2)$

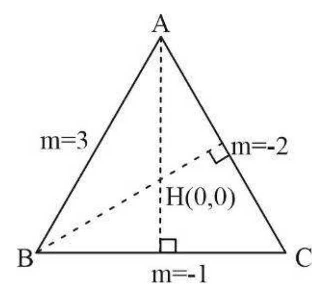


$B \equiv (1,-2)$

It is clear that $|PA - PB| \le AB$ (Triangle inequality)

$|PA - PB|_{max} = AB = \sqrt{6^2 + 2^2} = 2\sqrt{10} \Rightarrow n = 5$

92.(7) Let the slopes of BC, CA and AB be $-1, -2, 3$ respectively and orthocenter be H.



Slope of $AH = 1 \Rightarrow A = (x_1, x_1)$

Slope of $BH = \frac{1}{2} \Rightarrow B = \left(x_2, \frac{1}{2}x_2\right)$

Slope of $CH = -\frac{1}{3} \Rightarrow C = \left(x_3, -\frac{x_3}{3}\right)$

Slope of $BC = \dfrac{\frac{x_2}{2} + \frac{x_3}{3}}{x_2 - x_3} = \dfrac{3x_2 + 2x_3}{6(x_2 - x_3)} = -1 \Rightarrow 9x_2 = 4x_3$

Slope of $CA = \dfrac{x_1 + \frac{x_3}{3}}{x_1 - x_3} = -2 = \dfrac{3x_1 + x_3}{3(x_1 - x_3)} \Rightarrow 9x_1 = 5x_3$

$$G \equiv (h,k) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{x_1 + \frac{x_2}{2} - \frac{x_3}{3}}{3}\right)$$

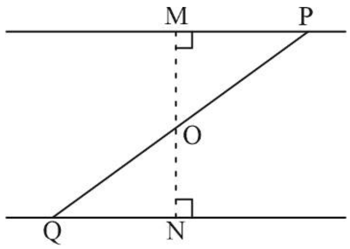
$3h = x_3 + \frac{5x_3}{9} + \frac{4x_3}{9} = 2x_3$ ....(i)

$3k = \frac{5x_3}{9} + \frac{2x_3}{9} - \frac{x_3}{9} = \frac{4x_3}{9}$ ....(ii)

From equation (i) and (ii), we get

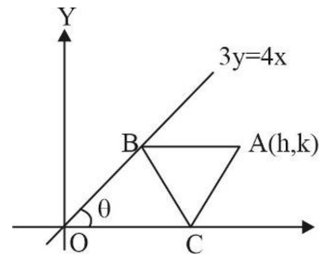
$\frac{k}{h} = \frac{2}{9} \Rightarrow y = \frac{2}{9}x; b - a = 9 - 2 = 7$

93.(4)



$$\frac{OQ}{OP}=\frac{OM}{ON}=\frac{6/5}{|c|/10}=\frac{3}{4} \Rightarrow |c|=16 \Rightarrow c=\pm 16 \Rightarrow k=4$$

94.(2) Let $\angle BOC=\theta$ then $\tan\theta=\frac{4}{3}$



$\sin\theta=\frac{4}{5}$ and $\cos\theta=\frac{3}{5}$

If $OC=t$, then $OB=BA=AC=t$

Coordinates of C are $(t,0)$, coordinates of B are $(OB\cos\theta, OB\sin\theta)=\left(\frac{3}{5}t,\frac{4}{5}t\right)$

Equation of BC is $y-0=\frac{\frac{4}{5}t}{\frac{3}{5}t-t}(x-t)\Rightarrow y=-2(x-t)$

It passes through $\left(\frac{2}{3},\frac{2}{3}\right)\Rightarrow\frac{2}{3}=-2\left(\frac{2}{3}-t\right)\Rightarrow t=1$

Coordinates (h,k) of A are given by $h=\frac{3}{5}t+t=\frac{8}{5}t=\frac{8}{5}$ and $k=\frac{4}{5}t=\frac{4}{5}\Rightarrow\frac{h}{k}=2$

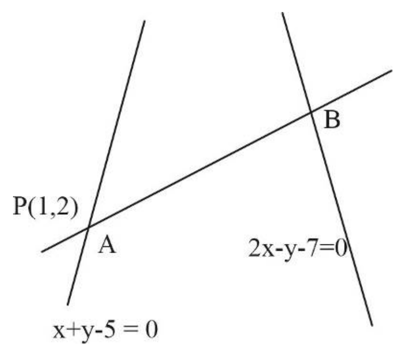
95.(5) Let the equation of the line passing through $P(1,2)$ be $y-2=m(x-1)$ ....(i)

Let $PA=r_1, PB=r_2$ then $A=(1+r_1\cos\theta, 2+r_1\sin\theta)$; $B=(1+r_2\cos\theta, 2+r_2\sin\theta)$

Where $\tan\theta=m$

As A is on the line $x+y-5=0, 1+r_1\cos\theta+2+r_1\sin\theta-5=0$ Or $r_1(\cos\theta+\sin\theta)=2$

$\frac{1}{r_1}=\frac{\cos\theta+\sin\theta}{2}$ ....(ii)



As B is on the line $2x-y-7=0$

$2(1+r_2\cos\theta)-(2+r_2\sin\theta)-7=0$

$r_2(2\cos\theta-\sin\theta)=7$

$\frac{1}{r_2}=\frac{(2\cos\theta-\sin\theta)}{7}$ ....(iii)

Now, HM of PA, PB is 10

$10=\frac{2PA.PB}{PA+PB}$; $\therefore 10=\frac{2r_1r_2}{r_1+r_2}$

$\frac{r_1+r_2}{r_1r_2}=\frac{1}{5}$; $\frac{1}{r_1}+\frac{1}{r_2}=\frac{1}{5}$ Or $\frac{2\cos\theta-\sin\theta}{7}+\frac{(\sin\theta+\cos\theta)}{2}=\frac{1}{5}$

From (ii), (iii) Or $10(2\cos\theta-\sin\theta)+35(\cos\theta+\sin\theta)=14$

$55\cos\theta+25\sin\theta=14$

$\sqrt{55^2+25^2}\cos(\theta-\alpha)=14$ where $\cos\alpha=\frac{55}{\sqrt{55^2+25^2}}$ or $\cos(\theta-\alpha)=\frac{14}{5\sqrt{146}}$

$\theta = \alpha + cos^{-1}\dfrac{14}{5\sqrt{146}} = cos^{-1}\dfrac{11}{\sqrt{146}} + cos^{-1}\dfrac{14}{5\sqrt{146}}$

From (i) the equation of the line is $(y-2) = tan\left[cos^{-1}\left[\dfrac{11}{\sqrt{146}}\right] + cos^{-1}\left[\dfrac{14}{5\sqrt{146}}\right]\right](x-1)$

Also $cos^{-1}x + sin^{-1}x = \dfrac{\pi}{2}$

96.(12) The equation of any line passing through the given point $P(3,4)$ and making an angle $\pi/6$ with x-axis is

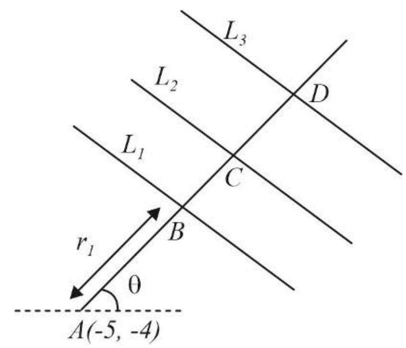
$\dfrac{x-3}{cos\,30°} = \dfrac{y-4}{sin\,30°} = r$ where r represents the distance of any point Q on this line from the given point $P(3,4)$.

Coordinates of any point Q on line are $(3 + r\,cos\,30°, 4 + r\,sin\,30°)$ which lies on the line $12x + 5y + 10 = 0$ then

$12\left(3 + \dfrac{r\sqrt{3}}{2}\right) + 5\left(4 + \dfrac{r}{2}\right) + 10 = 0 \Rightarrow r = \dfrac{132}{12\sqrt{3}+5}$

97.(27) Co-ordinate of B is $(-5 + r_1\,cos\,\theta, -4 + r_1\,sin\,\theta)$; $C(-5 + r_2\,cos\,\theta, -4 + r_2\,sin\,\theta)$; $D(-5 + r_3\,cos\,\theta, -4 + r_3\,sin\,\theta)$

As point B, C, D lies on L_1, L_2, L_3



$(-5 + r_1\,cos\,\theta) + 3(-4 + r_1\,sin\,\theta) + 2 = 0$

$r_1 = \dfrac{15}{cos\,\theta + 3\,sin\,\theta}; \dfrac{15}{r_1} = cos\,\theta + 3\,sin\,\theta = \dfrac{15}{AB}$ ....(i)

Similarly; $\dfrac{10}{r_2} = 2\,cos\,\theta + sin\,\theta = \dfrac{10}{AC}$ ....(ii)

$\dfrac{6}{r_3} = cos\,\theta - sin\,\theta = \dfrac{6}{AD}$ ......(iii)

Put in given expression $(2\,cos\,\theta + 3\,sin\,\theta)^2 = 0 \Rightarrow tan\,\theta = -\dfrac{2}{3}$

$y + 4 = -\dfrac{2}{3}(x+5)$, $2x + 3y + 22 = 0$, $2 + b + c = 27$

98. (9) $6a^2 - 3b^2 - c + 7ab - ac + 4bc = 0$

$6a^2 + a[7b - c] + 4bc - 3b^2 - c^2 = 0$

$a = \dfrac{c - 7b \pm \sqrt{49b^2 + c^2 - 14bc - 24(4bc - 3b^2 - c^2)}}{12}$; $a = \dfrac{c - 7b \pm \sqrt{(11b)^2 + (5c)^2 - 110bc}}{12}$

$a = \dfrac{c - 7b \pm (11b - 5c)}{12}$; $a = \dfrac{c - 7b + (11b - 5c)}{12}$ or $a = \dfrac{c - 7b - (11b - 5c)}{12}$

$a = \dfrac{4b - 4c}{12}$ or $a = \dfrac{6c - 18b}{12}$; $a = \dfrac{b-c}{3}$ or $a = \dfrac{c - 3b}{2}$

So $3a - b + c = 0$ and $-2a - 3b + c = 0$ will concurrent at $(3,-1)$ and $(-2,-3)$

$\Rightarrow A = 3, B = -1, C = -2, D = -3$

CIRCLE

1.(A) $2\sqrt{rr_1}+2\sqrt{rr_2}=2\sqrt{r_1r_2} \Rightarrow \sqrt{r}(6+3)=6\times3 \Rightarrow$ r = 4 [Length of direct common tangent $\sqrt{d^2-(r_1-r_2)^2}$]

2.(B) For $\angle ACB$ to be maximum, circle passing through *A, B* will touch *x*-axis at *C*.
By power of point *O*

$$OC^2=(OA)(OB)$$

$$x^2=\alpha\beta \qquad x=\sqrt{\alpha\beta}$$

3.(A) Combined equation of pair of tangents is $(xh+yk-1)^2=(x^2+y^2-1)(h^2+k^2-1)$

Put $x=1, (h-1)^2+2ky(h-1) = y^2(h^2-1)$

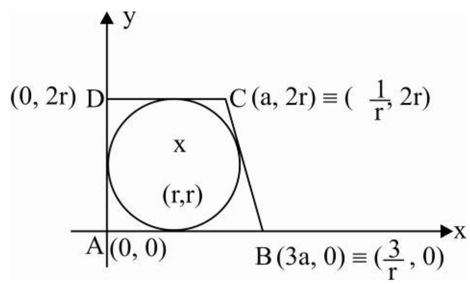
$$y^2(h+1)-2ky-(h-1)=0$$

$$AB=|y_1-y_2|=2 \Rightarrow 4=\frac{4k^2}{(h+1)^2}+\frac{4(h-1)}{(h+1)}$$

$$(h+1)^2=k^2+(h^2-1) \Rightarrow k^2=2(h+1)$$

$$y^2=2(x+1)$$

4.(B) Area of trapezium $ABCD=\frac{1}{2}(a+3a)(2r)=4 \Rightarrow ar=1$



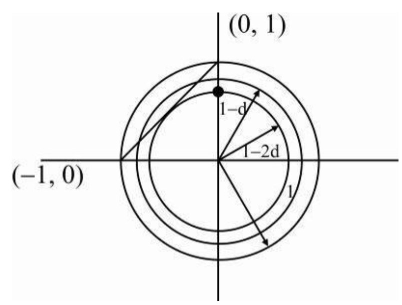
Equation of *BC* is $y=-r^2\left(x-\frac{3}{r}\right)$

$$y+r^2x-3r=0$$

$\because$ *BC* is tangent to circle

$\Rightarrow \frac{|r+r^3-3r|}{\sqrt{1+r^4}}=r \Rightarrow r^4+4-4r^2=1+r^4 \Rightarrow r=\frac{\sqrt{3}}{2}$

5.(B) For max common difference, the smaller circle just touches the line

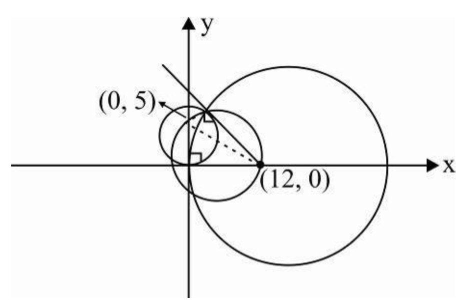


$\Rightarrow 1-2d=\frac{1}{\sqrt{2}}$

$$d=\frac{\sqrt{2}-1}{2\sqrt{2}}$$

$\therefore d\in\left(0,\frac{\sqrt{2}-1}{2\sqrt{2}}\right)$ i.e., $\left(0,\frac{2-\sqrt{2}}{4}\right)$

6.(A) Equation of required circle is $(x^2+y^2-10y)+\lambda(x^2+y^2-24x)=0$



The circle process through the point (12, 0)
i.e. $144+\lambda(144-24\times12)=0 \quad \therefore \quad \lambda=1$

$\therefore$ Equation required circle is $x^2+y^2-12x-5y=0$

$\therefore$ Radius $=\frac{13}{2}$

7.(A) $R + r + x = GB$ when $x = PB$

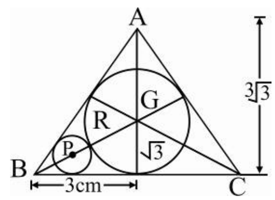
$\sqrt{3} + r + x = 2\sqrt{3}$

$\left[R = 3\sqrt{3}\cdot\frac{1}{3} = \sqrt{3}\right]$

$x = \sqrt{3} - r = PB$

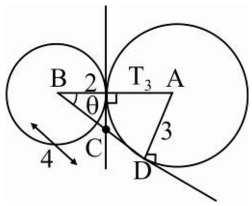
Now $\sin 30° = \frac{r}{\sqrt{3}-r} = \frac{1}{2}$

$2r = \sqrt{3} - r \Rightarrow r = \frac{1}{\sqrt{3}}$



8.(B) $BC = \frac{2}{\cos\theta} = \frac{2}{\frac{4}{5}} = \frac{5}{2}$

$CD = 4 - \frac{5}{2} = \frac{3}{2}$



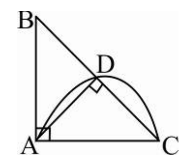
9.(D) $(AC)^2 = (AD)^2 + (CD)^2$

$(AB)^2 = (BD)(BC) = BD(BD + DC)$

$\Rightarrow \quad (BD)(DC) = (AB)^2 - (BD)^2 = (AD)^2 \quad \Rightarrow \quad DC = \frac{AD^2}{BC} = \frac{(AD)^2}{\sqrt{(AB)^2 - (AD)^2}}$

$\therefore \quad (AC)^2 = (AD)^2 + \frac{(AD)^4}{(AB)^2 - (AD)^2} = \frac{(AB)^2(AD)^2}{(AB)^2 - (AD)^2}$

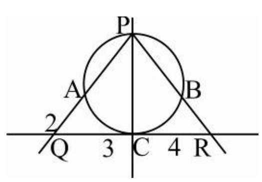
$AC = \frac{(AB)(AD)}{\sqrt{(AB)^2 - (AD)^2}}$



10.(B) $(PQ)(AQ) = (QC)^2 \quad \Rightarrow \quad PQ = \frac{9}{2}$

$\frac{QC}{RC} = \frac{PQ}{PR} \Rightarrow PR = 6$

$(RC)^2 = (RB)(RP) \Rightarrow RB = \frac{8}{3}$

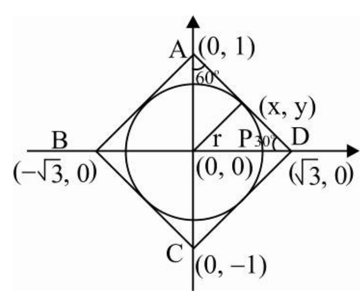


11.(B) $r = \sqrt{3}\sin 30° = \frac{\sqrt{3}}{2}$

$(PA)^2 + (PB)^2 + (PC)^2 + (PD)^2$

$= \left(x - \sqrt{3}\right)^2 + y^2 + x^2 + (y-1)^2 + \left(x + \sqrt{3}\right)^2 + y^2 + x^2 + (y+1)^2$

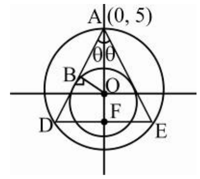
$= 4\left(x^2 + y^2 + 2\right) = 4\left(\frac{3}{4} + 2\right) = 11$



12.(A) $\sin\theta = \frac{3}{5}$

Slope of AB, AE are $\tan\left(\frac{\pi}{2} - \theta\right), \tan\left(\frac{\pi}{2} + \theta\right) = \frac{4}{3}, -\frac{4}{3}$

Equation of OB is $y=-\frac{3}{4}x$



$\Rightarrow \quad 4y+3x=0$

D is image of A *w.r.t.* OB

$$\frac{x-0}{3}=\frac{y-5}{4}=-2\frac{4(5)}{25}=-\frac{8}{5}$$

$$D\equiv\left(\frac{-24}{5},\frac{-7}{5}\right)\Rightarrow OF=\frac{7}{5}$$

Aliter : $\sin\theta=\frac{3}{5}$

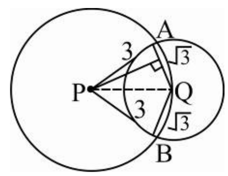
$\therefore \quad OB\perp AD \quad \Rightarrow \quad AD=8$

Let $OF=x$

In ΔADF

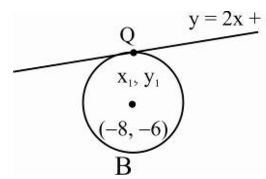
$$\cos\theta=\frac{5+2}{8} \quad \Rightarrow \quad \frac{4}{5}=\frac{5+x}{8}; \qquad x=\frac{7}{5}$$

13.(B) Area of $APBQ=2(\text{area } \Delta APQ)$



$$=2\left(\frac{1}{2}\times\sqrt{3}\times\sqrt{3^2-\left(\frac{\sqrt{3}}{2}\right)^2}\right)=\sqrt{3}\cdot\frac{\sqrt{33}}{2}=\frac{\sqrt{99}}{2}$$

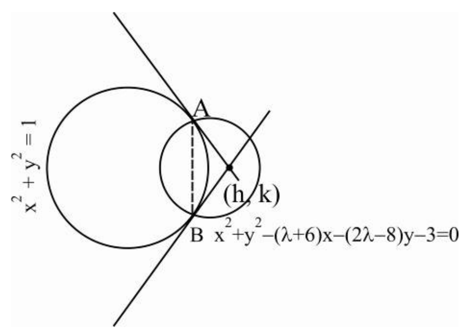
14.(A) Let the line touches the given circle at (x_1, y_1), then $y_1=2x_1+5$ and $\frac{(y_1+6)}{x_1+8}\times 2=-1$



$\Rightarrow \quad x_1=-6$ and $y_1=-7$

15.(A) Let both the circles intersects at A and B, then the equation of the line AB is

$$xh+yk=1 \qquad \ldots(1)$$



$$AB=S_1-S_2=0$$

$\Rightarrow \quad (\lambda+6)x+(2\lambda-8)y+2=0 \qquad \ldots(2)$

(1) & (2) represent the same line, so

$\Rightarrow \quad \frac{h}{\lambda+6}=\frac{k}{2\lambda-8}=\frac{-1}{2}$

$\Rightarrow \quad \frac{2h}{2\lambda+12}=\frac{k}{2\lambda-8}=\frac{2h-k}{20}=-\frac{1}{2}\left(as\,\frac{a}{b}=\frac{c}{b}=\frac{a-c}{b-d}\right)$

$\Rightarrow \quad 2x-y=-10$

$\Rightarrow \quad 2x-y+10=0$

16.(C) $\sin\theta=\frac{r}{3r}=\frac{1}{3}$ where angle EDC is θ

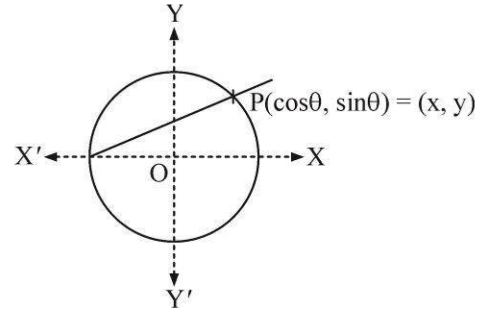
$$DE=2(2r\cos\theta)=4\times 3\left(\frac{2\sqrt{2}}{3}\right)$$

$$DE=8\sqrt{2}$$

17.(D) $m=\dfrac{\sin\theta}{\cos\theta+1}=\tan\left(\dfrac{\theta}{2}\right)$

$$P(x,y)=\left(\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}},\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)$$

$\therefore$ $x,y\in Q$, if $m\in Q\Rightarrow$ Infinite points are possible.



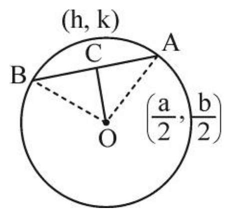
18.(C) In triangle OAC

$$OA^2=AC^2+OC^2$$

$$\frac{a^2}{4}+\frac{b^2}{4}=2\left[\left(h-\frac{a}{2}\right)^2+\left(k-\frac{b}{2}\right)^2\right]$$

$\Rightarrow$ $h^2+k^2-ah-bk+\dfrac{a^2}{8}+\dfrac{b^2}{8}=0$

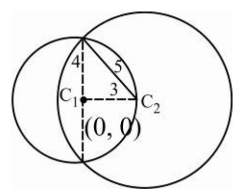
Required Locus is $x^2+y^2-ax-by+\dfrac{a^2+b^2}{8}=0$



19.(B) Slope of $C_1C_2=-\dfrac{4}{3}=\tan\theta$

$\Rightarrow$ $C_2\equiv\left(0+3\left(-\dfrac{3}{5}\right),0+3\left(\dfrac{4}{5}\right)\right)$ Or $\left(0-3\left(-\dfrac{3}{5}\right),0-3\left(\dfrac{4}{5}\right)\right)$

$C_2\equiv\left(\dfrac{9}{5},-\dfrac{12}{5}\right)$ or $\left(-\dfrac{9}{5},\dfrac{12}{5}\right)$

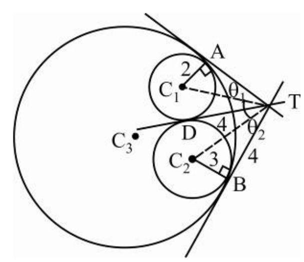


20-22. *AT* and *BT* are radical axis to C_3 and C_1 and C_3 & C_2 respectively.

$\therefore$ T is radical centre $\Rightarrow$ radical axis of C_1 and C_2 i.e., common tangent passes through *T*.

TA = *TB* = *TD* = 4

$$\tan\frac{\theta_1}{2}=\frac{2}{4}=\frac{1}{2},\tan\frac{\theta_2}{2}=\frac{3}{4},(\angle ATD=\theta_1,\angle BTD=\theta_2)$$



20.(D) $r_3=TA\tan\left(\dfrac{\theta_1+\theta_2}{2}\right)=4\left(\dfrac{\frac{1}{2}+\frac{3}{4}}{1-\frac{1}{2}\cdot\frac{3}{4}}\right)=8$

21.(B) Circumcircle of ΔTAB will pass through C_3 has TC_3 as diameter

$\Rightarrow$ Area $=\pi\left(\dfrac{TC_3}{2}\right)^2, TC_3=\dfrac{TA}{\cos\left(\frac{\theta_1+\theta_2}{2}\right)}=\dfrac{4}{1/\sqrt{5}}=4\sqrt{5}$

Area = $\pi\left(2\sqrt{5}\right)^2=20\pi$

22.(D) $C_3C_1=r_3-r_1$

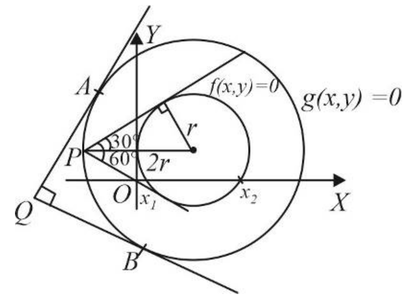
$C_3C_2=r_3-r_2$ $\Rightarrow$ $C_3C_1-C_3C_2=r_2-r_1=1$

23.-25 $g(x,y)=0$ represent a circle concentric with circle $f(x,y)=0$ with radius twice the radius of

$f(x,y)=0, centre=\left(\frac{5}{2},2\right)$

$f(x,y)=x^2+y^2-5x-4y+4=0$

$g(x,y)=x^2+y^2-5x-4y-\frac{59}{4}=0$



23.(D) Area of $\Delta QAB=\frac{1}{2}\times5\times5=\frac{25}{2}$

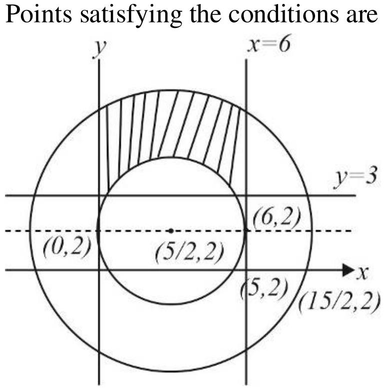
24.(D) $\theta=tan^{-1}\frac{3}{4}$

$$2\theta=tan^{-1}\left(\frac{2\left(\frac{3}{4}\right)}{1-\frac{9}{16}}\right)=tan^{-1}\left(\frac{24}{7}\right)$$

Area of region inside $f(x,y)=0$ above x-axis.

$$=\frac{1}{2}\left(\frac{5}{2}\right)^2\left(2\pi-tan^{-1}\left(\frac{24}{7}\right)\right)+\frac{1}{2}\times3\times2=3+\frac{25}{8}\left(2\pi-tan^{-1}\left(\frac{24}{7}\right)\right)$$

25.(D) Points satisfying the conditions are



$(1,5)(1,6)(2,5)(2,6)(3,5)(3,6)(4,5)(4,6)(5,4)(5,5)(5,6)$

26.(C) **27. (C) 28. (D)**

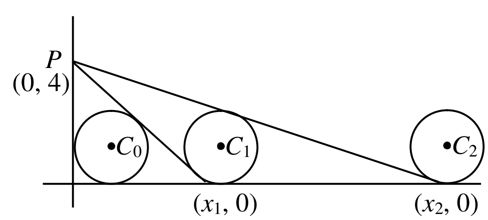
26.28 **Explanation:**

For C_n circle let centre be $(\alpha_n,1)$ and it touches line L_n and L_{n+1}.

i.e. $\frac{x}{x_n}+\frac{y}{4}=1,\ \frac{x}{x_{n+1}}+\frac{y}{4}=1$

$$\Rightarrow \frac{4\alpha_n+x_n-4x_n}{\sqrt{16+x_n^{\,2}}}=1 \qquad \ldots(i)$$

$$\Rightarrow \frac{4\alpha_n+x_{n+1}-4x_{n+1}}{\sqrt{16+x_{n+1}^2}}=-1 \qquad \ldots(ii)$$



$\Rightarrow \quad 3(x_{n+1}-x_n)=\sqrt{16+x_n^2}+\sqrt{16+x_{n+1}^2}$

Let $\quad x_n = 4cot\,\theta_n\, 0<\theta_n<\dfrac{\pi}{2} \qquad \Rightarrow \qquad tan\left(\dfrac{\theta_{n+1}}{2}\right)=\dfrac{1}{2}tan\left(\dfrac{\theta_n}{2}\right)$

As $\quad x_1=3 \quad \Rightarrow \quad tan\dfrac{\theta_1}{2}=\dfrac{1}{2} \qquad \Rightarrow \quad tan\left(\dfrac{\theta_n}{2}\right)=\dfrac{1}{2^n} \quad \Rightarrow \quad x_n=\left(\dfrac{2^{2n}-1}{2^{n-1}}\right) \qquad x_2=\dfrac{15}{2}$

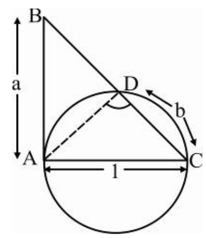
Putting is (i), we get $\alpha_2=\dfrac{31}{4}$.

$$\lim_{n\to\infty}\frac{x_n}{2^n}=\lim_{n\to\infty}\frac{2^{2n}-1}{2^{n-1}2^n}=2$$

29.(BC) In triangle ADC

$\tan C=\dfrac{AD}{b}=\dfrac{a}{1} \quad \Rightarrow \quad AD=ab<AC=1\Rightarrow ab<1$

In triangle *ADC*

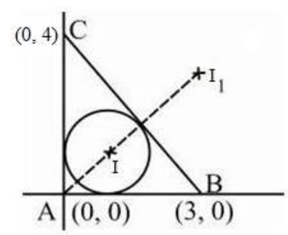


$(AC)^2=(AD)^2+(CD)^2=a^2b^2+b^2=1$, then $b^2=\dfrac{1}{a^2+1}$

$$\frac{b^2}{a^2}=\frac{1}{a^4+a^2}>\frac{1}{a^4+a^2+\frac{1}{4}}$$

$$\frac{b}{a}>\frac{1}{a^2+\frac{1}{2}}$$

30.(AD) Circles satisfying the given condition are incircle and excircle of the ΔABC as shown



$$I\equiv\left(\frac{4(3)}{4+3+5},\frac{3(4)}{4+3+5}\right)\equiv(1,1)$$

$$I_1\equiv\left(\frac{4(3)}{-5+4+3},\frac{4(3)}{-5+4+3}\right)\equiv(6,6)$$

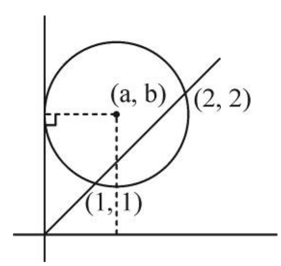
31.(AB) Equation of the circle passing through (1, 1) and (2, 2) is $\{(x-1)(x-2)+(y-1)(y-2)\}+\lambda(x-y)=0$

$\Rightarrow \qquad x^2+y^2-(3-\lambda)x-(3+\lambda)y+4=0 \quad \Rightarrow \quad$ Radius $=\sqrt{\left(\dfrac{3-\lambda}{2}\right)^2+\left(\dfrac{3+\lambda}{2}\right)^2-4}$

Now let (a,b) be the centre of the circle.

Then we have (i) $|a|=r, |b|>r$ or (ii) $|b|=r, |a|>r$

Case (i) $|a|=r \quad \Rightarrow \quad a^2=r^2$



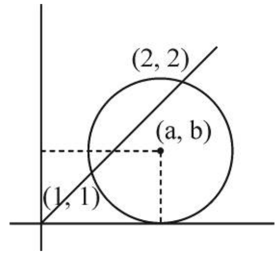
$\Rightarrow \qquad \left(\dfrac{3-\lambda}{2}\right)^2=\left(\dfrac{3-\lambda}{2}\right)^2+\left(\dfrac{3+\lambda}{2}\right)^2-4 \quad \Rightarrow \qquad \lambda=1$ or -7

$\lambda=1 \qquad \Rightarrow \qquad |a|=1$ and $|b|=2 \quad \Rightarrow \quad r=1$

$\lambda=-7 \qquad \Rightarrow \qquad |a|=5$ and $|b|=2 \quad \Rightarrow \quad r=5$ (rejected)

$\therefore \qquad$ Equation of circle is $x^2+y^2-2x-4y+4=0$

Case(ii) $|b| = r \Rightarrow \left(\frac{3-\lambda}{2}\right)^2 = 4 \Rightarrow \lambda = -1$ or 7



If $\lambda = -1 \Rightarrow |a| = 2,\ r = 1$ so $|a| > r$

$\lambda = 7 \Rightarrow |a| = 2,\ r = 5$ so $|a| < r$

$\therefore \lambda = 7$, is invalid

$\therefore$ Equation of circle is $x^2 + y^2 - 4x - 2y + 4 = 0$

32.(ABD) $9l^2 + 6l + 1 = 5\left(l^2 + m^2\right) \Rightarrow \frac{(3l+1)}{\sqrt{l^2+m^2}} = \sqrt{5}$

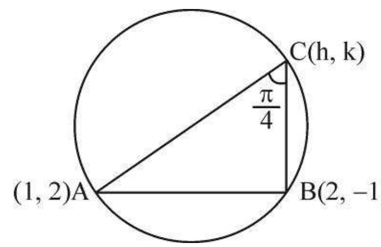
$\Rightarrow$ Perpendicular distance of (3, 0)

From $lx + my + 1 = 0$ is $\sqrt{5}$

$\Rightarrow$ Centre of circle = (3, 0)

Radius = $\sqrt{5}$ & Equation of circle is $(x-3)^2 + y^2 = 5$

33.(AB) $m(AC) = m_1 = \frac{k-2}{h-1}$ & $m(BC) = m_2 = \frac{k+1}{h-2}$

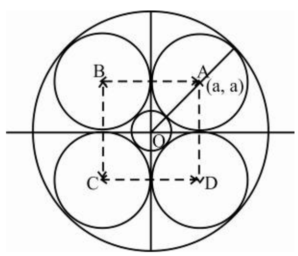


$\therefore \quad \tan\frac{\pi}{4} = \left|\frac{\frac{k-2}{h-1} - \frac{k+1}{h-2}}{1 + \frac{k-2}{h-1}\cdot\frac{k+1}{h-2}}\right| \Rightarrow \pm 1 = \frac{(k-2)(h-2)-(k+1)(h-1)}{(h-1)(h-2)+(k-2)(k+1)}$

Required equation of circle by taking and $-ve$ sign,

We get $x^2 + y^2 = 5$ & $x^2 + y^2 - 6x - 2y + 5 = 0$

34.(ABCD) $OA = \sqrt{2}a$



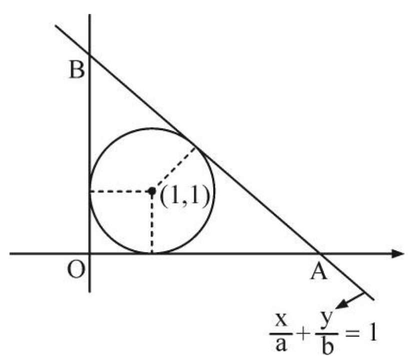
Largest radius $= \sqrt{2}a + a$

Smallest radius $= \sqrt{2}a - a$

Area enclosed by circles

$= (2a)^2 - 4\left(\frac{\pi}{4}a^2\right) = (4-\pi)a^2$

35.(AB) point (1, 1) & (0, 0) are of the same side of the line $\frac{x}{a} + \frac{y}{b} - 1 = 0$

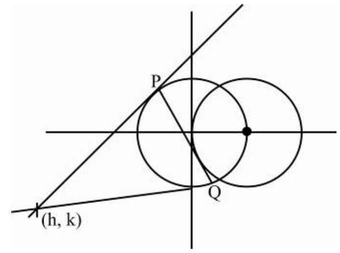


$\Rightarrow \quad \frac{1}{a} + \frac{1}{b} < 1$

Also, perpendicular distance from centre to the line is equal to radius

$$\frac{-\frac{1}{a} - \frac{1}{b} + 1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = 1 \Rightarrow 1 - \frac{1}{a} - \frac{1}{b} = \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

36.(AC) PQ is chord of contact of (h, k) *w.r.t.* $x^2+y^2=a^2$



$\Rightarrow$ Equation of PQ is $xh+yk=a^2$

$\because$ PQ is tangent to $(x-a)^2+y^2=a^2$

$\Rightarrow$ $\dfrac{|ah-a^2|}{\sqrt{h^2+k^2}}=a \quad \Rightarrow \quad (h-a)^2=h^2+k^2$

Locus is $(x-a)^2=x^2+y^2$ i.e., $y^2=a(a-2x)$

37.(AC) Equation of the circle touching the line $x+y-2=0$ at $(1,1)$ is $(x-1)^2+(y-1)^2+\lambda(x+y-2)=0$

i.e. $x^2+y^2+x(\lambda-2)+y(\lambda-2)+(2-2\lambda)=0$

Now equation of PQ is $x(\lambda-6)+y(\lambda-7)+(8-2\lambda)=0$

$\Rightarrow$ PQ can never be parallel to the line $x+y+20$ & PQ always passes through $(6,-4)$

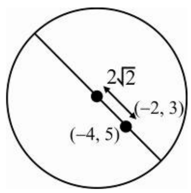
38.(ACD) $a=9+2\sqrt{2}$

$b=9-2\sqrt{2}$

$ab=81-8=73$

$a+b=18$

$a-b=4\sqrt{2}$



39.(ACD)

$\because$ Sides of the square are parallel to co-ordinate axes

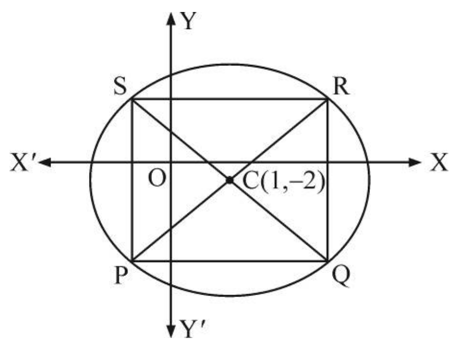
$\therefore$ Slope of PR and SQ are 1 and -1 respectively.

Co-ordinate of P & Q are

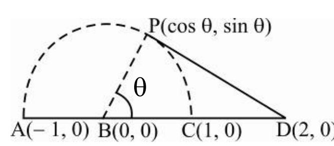
$\left(1\pm 7\sqrt{2}\cdot\dfrac{1}{\sqrt{2}},\,-2\pm 7\sqrt{2}\cdot\dfrac{1}{\sqrt{2}}\right)$ i.e. $(8,5)$ and $(-6,-9)$

Similarly co-ordinate of S & Q are

$\left(1\pm 7\sqrt{2}\left(-\dfrac{1}{\sqrt{2}}\right),\,-2\pm 7\sqrt{2}\left(\dfrac{1}{\sqrt{2}}\right)\right)$ i.e. $(-6,5)$ & $(8,-9)$



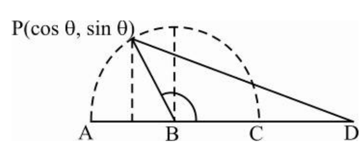
40.(AB) Area $=\dfrac{1}{2}(1)^2(\pi-\theta)+\dfrac{1}{2}(2)\sin\theta \;=\dfrac{\pi}{2}+\sin\theta-\dfrac{\theta}{2}$



$A=\dfrac{\pi}{2}+\sin\theta-\dfrac{\theta}{2}$

$\dfrac{dA}{d\theta}=\cos\theta-\dfrac{1}{2} \quad \Rightarrow \quad \theta=\dfrac{\pi}{3}$ (for maximum or minimum)

A is max. for $\theta=\dfrac{\pi}{3}$



$\Rightarrow$ $A_{\max}=\dfrac{\pi}{3}+\dfrac{\sqrt{3}}{2}=\dfrac{2\pi+3\sqrt{3}}{6}$

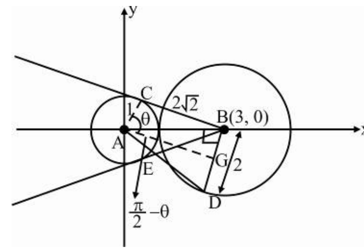
$L=3+(\pi-\theta)+\sqrt{(2-\cos\theta)^2+\sin^2\theta}\;=3+(\pi-\theta)+\sqrt{5-4\cos\theta}$

41.(ABC) In triangle ABC

$$\tan\theta = 2\sqrt{2}$$

$$AG = 2\sqrt{2}$$

$m_{BD} = 2\sqrt{2}$ where m_{BD} is the slope of BD



Then the coordinate of point $D \equiv \left(3-2\left(\frac{1}{3}\right), 0-2\left(\frac{2\sqrt{2}}{3}\right)\right) \equiv \left(\frac{7}{3}, \frac{-4\sqrt{2}}{3}\right)$

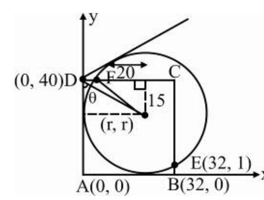
slope of line AE= Slope of line AD $= \frac{-4\sqrt{2}}{7}$

So coordinate of point $E = \left(0-1\left(-\frac{7}{9}\right), 0-1\left(\frac{4\sqrt{2}}{9}\right)\right) = \left(\frac{7}{9}, \frac{-4\sqrt{2}}{9}\right)$

Eqn. of BE is $y = \frac{\sqrt{2}}{5}(x-3)$, Solving intersection of line BE with circle S_1,

$$x^2 + \frac{2}{25}(x-3)^2 = 1 \Rightarrow \quad 27x^2 - 12x - 7 = 0 \quad \Rightarrow \quad (9x-7)(3x+1) = 0 \quad \Rightarrow \quad F \equiv \left(-\frac{1}{3}, \frac{-2\sqrt{2}}{3}\right)$$

42.(ABC) Equation of circle is $x^2 + y^2 - 2rx - 2ry + r^2 = 0$



Put (32, 1)

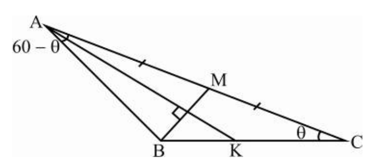
$\Rightarrow \quad r^2 - 66r + 1025 = 0 \quad \Rightarrow \quad r = 25,\ r = 41,\ r = 41$, is rejected

$$\theta = \tan^{-1}\left(\frac{25}{15}\right) \quad \Rightarrow \quad 2\theta = 2\tan^{-1}\left(\frac{5}{3}\right) = \pi - \tan^{-1}\left(\frac{15}{8}\right)$$

Area of $AFCB = \frac{1}{2}(40)(32+27) = 1180$

43.(ABCD) ΔAMB is isosceles

$\because$ angle bisector of $\angle A$ is $\perp$ to MB



$$\frac{c}{\sin\theta} = \frac{2c}{\sin 60°} \Rightarrow \sin\theta = \frac{\sqrt{3}}{4}$$

$$a^2 = c^2 + 4c^2 - 4c^2\cos(60° - \theta)$$

$$= 5c^2 - 4c^2\left(\frac{\sqrt{13}}{4}\frac{1}{2} + \frac{\sqrt{3}}{2}\frac{\sqrt{3}}{4}\right)$$

$$a = \frac{\sqrt{7-\sqrt{13}}}{\sqrt{2}}c = \frac{\sqrt{13}-1}{2}c$$

(A) $\frac{a}{c} = \frac{\sqrt{13}-1}{2}$ **(B)** $R = \frac{abc}{4\Delta} = \frac{ac(2c)}{2ac\frac{\sqrt{3}}{2}} = \frac{2c}{\sqrt{3}}$

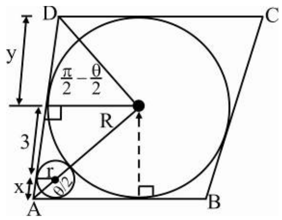
(C) $\frac{\Delta}{\pi R^2} = \frac{\frac{1}{2}\left(ac\frac{\sqrt{3}}{2}\right)}{\pi\frac{4}{3}c^2} = \frac{3\sqrt{3}}{16\pi}\frac{a}{c} = \frac{3\sqrt{3}}{32\pi}\left(\sqrt{13}-1\right)$ **(D)** $AB = c, AC = 2c, \frac{AB}{AC} = \frac{1}{2}$.

44.(ABC) $2\sqrt{Rr}=3\Rightarrow r=\frac{9}{4\times 3}=\frac{3}{4}$

$\frac{r}{R}=\frac{x}{x+3}=\frac{1}{4}\Rightarrow x=1$

$\tan\frac{\theta}{2}=\frac{3}{4}$

$y=R\cot\left(\frac{\pi}{2}-\frac{\theta}{2}\right)=3\tan\frac{\theta}{2}=\frac{9}{4}$



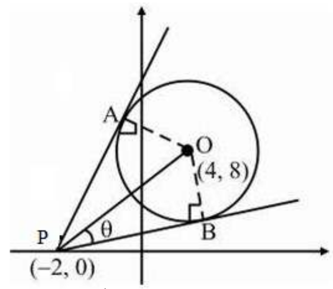
$ABCD$ is a rhombus with side $=4+\frac{9}{4}=\frac{25}{4}$ Area of $ABCD=\frac{25}{4}\times\frac{25}{4}\sin\theta=\frac{25}{4}\times\frac{25}{4}\frac{2\left(\frac{3}{4}\right)}{1+\frac{9}{16}}$

$A=\frac{75}{2}=\frac{1}{2}d_1d_2\ \Rightarrow\ d_1d_2=75$

45.(BC) $\sin\theta=\frac{2\sqrt{5}}{10}=\frac{1}{\sqrt{5}}$

Slopes of PA and PB are $\tan(\alpha\pm\theta)$

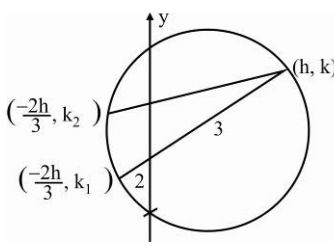
Where $\tan\alpha=\frac{8}{6}=\frac{4}{3}=\frac{\frac{4}{3}+\frac{1}{2}}{1-\frac{4}{3},\frac{1}{2}},\frac{\frac{4}{3}-\frac{1}{2}}{1+\frac{4}{3},\frac{1}{2}}=\frac{11}{2},\frac{5}{10}$

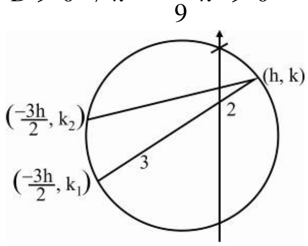


$\therefore\quad A,B\equiv\left(4+2\sqrt{5}\left(\frac{-11}{5\sqrt{5}}\right),8+2\sqrt{5}\left(\frac{2}{5\sqrt{5}}\right)\right)\ \left(4-2\sqrt{5}\left(\frac{-1}{\sqrt{5}}\right)8-2\sqrt{5}\left(\frac{2}{\sqrt{5}}\right)\right)\equiv\left(\frac{-2}{5},\frac{44}{5}\right),(6,4)$

46.(ABD) Put $x=\frac{-2h}{3}, y^2-ky+\frac{10h^2}{9}=0$

$D>0\Rightarrow k^2-\frac{40}{9}h^2>0$





Put $x=\frac{-3h}{2}\quad\Rightarrow\quad y^2-ky+\frac{15h^2}{4}=0$

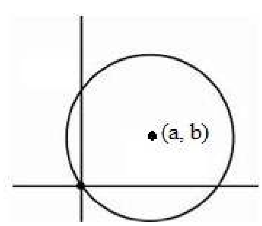
$D>0\Rightarrow k^2-15h^2>0$

47.(ABD) Eqn. of circle is $(x\pm a^2)+(y\pm b)^2=a^2+b^2$

or $(x\pm b)^2+(y\pm a)^2=a^2+b^2$

$x^2+y^2\pm 2ax\pm 2by=0$ or $x^2+y^2\pm 2bx\pm 2ay=0$

Equation of tangent origin is $ax\pm by=0$ Or $bx\pm ay=0$



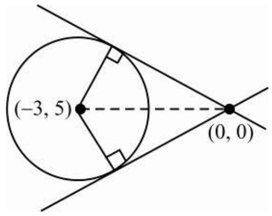
48.(AD) Area of quadrilateral

$$15=\sqrt{34-C}\sqrt{C}$$

$C^2-34C+225=0$

$(C-25)(C-9)=0$

$C=9,25$

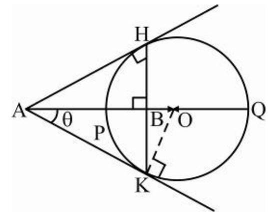


49.(ABC) $\dfrac{AP+AQ}{2}=\dfrac{(OA-r)+(OA+r)}{2}=OA$

$\cos\theta=\dfrac{AK}{OA}=\dfrac{AB}{AK} \quad \Rightarrow \quad (AK)^2=(OA)(AB)$

Also $(AK)^2=(AP)(AQ)$

$\therefore \quad (OA)(AB)=(AP)(AQ) \quad \Rightarrow \left(\dfrac{AP+AQ}{2}\right)AB=(AP)(AQ) \quad \Rightarrow \quad AB=\dfrac{2(AP)(AQ)}{(AP+AQ)}$



50.(BCD) If the mid-point of the chord be $P(h,k)$, then the equation of the chord is

$hx+ky=h^2+k^2$ ....(i)

Homogenise the equation $y^2-4x-4y=0$ with the help of (i), Thus,

$y^2=4(x+y)\left(\dfrac{hx+ky}{h^2+k^2}\right)$ or $4hx^2+\left(4k-h^2-k^2\right)y^2+4(h+k)xy=0$ ....(ii)

It represents a pair of perpendicular lines if $4h+4k-h^2-k^2=0$ or $h^2+k^2-4h-4k=0$

Hence, the locus of P is $x^2+y^2-4x-4y=0$ ....(iii)

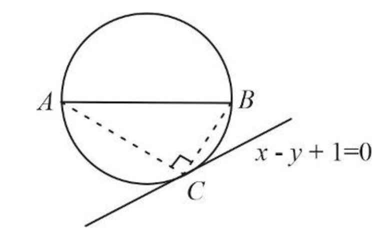
The chord of contact of circle (iii) with the given circle is $4x+4y=9$, which is also the chord of contact of tangents drawn from the point (4, 4)

51.(CD) The line segment *AB* subtend the greatest angle at *C* on the line $x-y+1=0$, so the circle drawn through *A* and *B* must touch the line $x-y+1=0$ at *C*. The equation of the circle can be assumed as

$(x-1)^2+(y-2)^2+\lambda(x-y+1)=0$

$x^2+y^2+(\lambda-2)x-(\lambda+4)y+5+\lambda=0$ ....(i)

Again $\angle ACB=\dfrac{\pi}{2}$, so the line *AB* is diameter of circle.



Thus the radius of the circle (i) is $\sqrt{2}$. $\left(\dfrac{\lambda-2}{2}\right)^2+\left(\dfrac{\lambda+4}{2}\right)^2-(5+\lambda)=2\Rightarrow\lambda=\pm2$

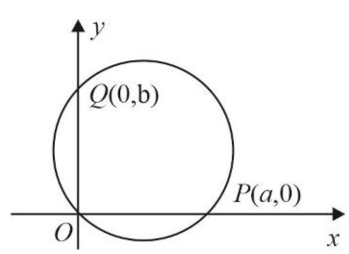
52.(CD) Le the equation of the variable circle be $x^2+y^2-ax-by=0$ ....(i)

It cuts X-axis at $P(a,0)$ and Y-axis at $Q(0,b)$. Thus,

$OP=a \ and \ OQ=b$

Given, $\quad m.OP+n.OQ=1$

$ma+nb=1 \ or \ b=\dfrac{1-ma}{n}$



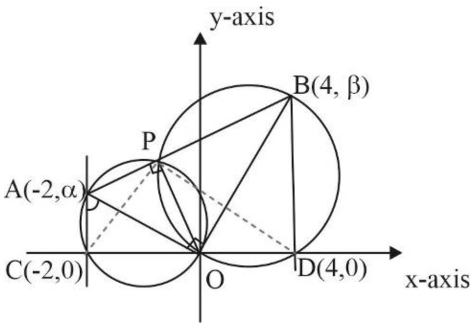
Putting the value of b in the equation of the circle (i), we have

$$\left(x^2+y^2-\frac{1}{n}y\right)-a\left(x-\frac{m}{n}y\right)=0 \;\ldots\ldots\ldots\text{(ii)}$$

Since, 'm' and 'n' are constants, so for all values of 'a'. The above circle given by (ii) passes through the points of intersection of the circle $x^2+y^2-\frac{1}{n}y=0$ and line $x-\frac{m}{n}y=0$

The circle always passes through the point $\left(\frac{m}{m^2+n^2},\frac{n}{m^2+n^2}\right)$.

53.(AD)



Locus is the circle $(x+2)(x-4)+y^2=0$

54.(BC) $QR=2\sqrt{r_1r_2}$

Similarly, $RC=2\sqrt{rr_2}$, $CS=2\sqrt{rr_3}$, $RS=2\sqrt{r_3r_2}$, $QS=2\sqrt{r_1r_3}$ and $QC=2\sqrt{r_1r}$

55. (BCD) Since XY, AM and ND are concurrent $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$

$$a\left(bc-a^2\right)-b\left(b^2-ac\right)+c\left(ab-c^2\right)=0$$

$$abc-a^3-b^3+abc+abc-c^3$$

$$3abc-a^3-b^3-c^3=0$$

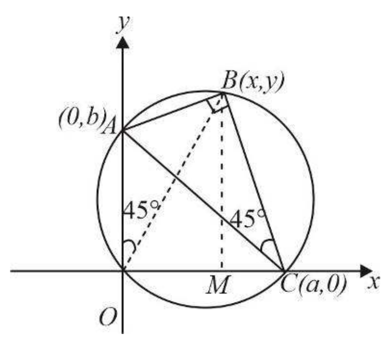
$$a^3+b^3+c^3=3abc=(a+b+c)\left(a+b\omega+c\omega^2\right)\left(a+b\omega^2+c\omega\right)=0$$

56.(AB) From the figure is obvious that $y-x=0$

$a^2+b^2=2$

Circumcentre of the triangle ABC $(a/2, b/2)$ so the locus of the circumcnetre of the triangle ABC is $x^2+y^2=\frac{1}{2}$

Since triangle is isosceles



57.(ABCD) $x^2+y^2+2gx+2fy+c=0 \;\ldots\ldots\ldots\text{(i)}$

be the equation of the circle

Point $(a,t),(t,a)$ and (t,t) lie on the given circle

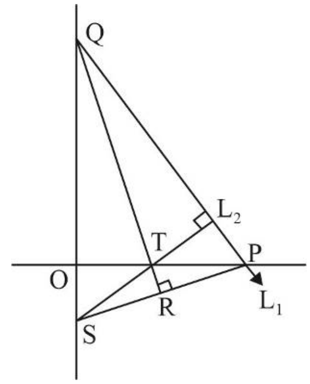
After getting the value of g, f and c from equation (i) $\left(x^2+y^2-ax-ay\right)-t(x+y-2a)=0$

The circle $x^2+y^2-ax-ay=0$ always touches the line $x+y-2a=0$ at (a, a)

58.(ABC) T is the orthocenter of the triangle QPS so the $\angle QRP$ *and* $\angle QOP$ are 90° and the points Q and P are fixed.

So locus of R is a circle having PQ as diameter.

Point O will also lie on the circle



59.(BC) The given circles are $x^2 + y^2 - 2x = 0,\ x > 0$

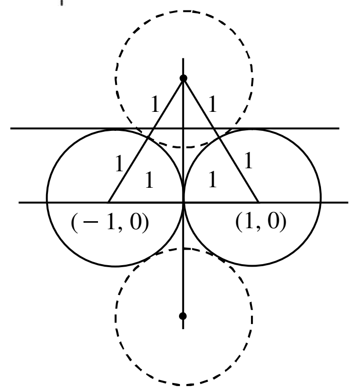
and $\quad x^2 + y^2 + 2x = 0,\ x < 0$

From the figure, the centres of the required circles will be

$\left(0, \sqrt{3}\right)$ and $\left(0, -\sqrt{3}\right)$

$\therefore$ Equations of circles will be

$$(x-0)^2 + \left(y \mp \sqrt{3}\right)^2 = 1^2$$



60.(BD) For two chords PT and QR of the circle, which intersect at S, we have

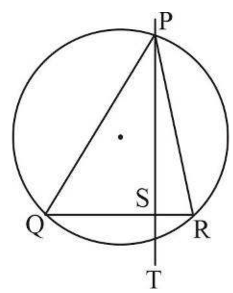
$PS \times ST = QS \times SR$

Now, $\dfrac{\dfrac{1}{PS} + \dfrac{1}{ST}}{2} > \sqrt{\dfrac{1}{PS} \times \dfrac{1}{ST}}$

$\dfrac{1}{PS} + \dfrac{1}{ST} > \dfrac{2}{\sqrt{QS \times SR}}$

Again, $\dfrac{QS + SR}{2} > \sqrt{QS \times SR}$ Or $\dfrac{QR}{2} > \sqrt{QS \times SR}$ *or* $\dfrac{1}{\sqrt{QS \times SR}} > \dfrac{2}{QR}$

$\dfrac{1}{PS} + \dfrac{1}{ST} > \dfrac{4}{QR}$



61.(BC) Given circle is $x^2 + y^2 - x + 3y = 0$ **....(i)**

Given line is $x + y - 1 = 0$ **....(ii)**

Equation of any line through the origin is $y - mx = 0$ **....(iii)**

Let H be the centre of circle (i), then $H = \left(\dfrac{1}{2}, -\dfrac{3}{2}\right)$

As equal chords of a circle are equidistant from the centre, we have L_1 *and* L_2 are equidistant from H.

$$\Rightarrow \left|\frac{\frac{1}{2} - \frac{3}{2} - 1}{\sqrt{2}}\right| = \left|\frac{-\frac{3}{2} - \frac{m}{2}}{\sqrt{1+m^2}}\right| \Rightarrow \sqrt{2} = \frac{3+m}{2\sqrt{1+m^2}} \Rightarrow 7m^2 - 6m - 1 = 0 \Rightarrow m = 1, -\frac{1}{7}$$

Therefore, equation of L_1, $\ x - y = 0$ or $\ x + 7y = 0$

62.(AB) The equations of two given circles are $S_1 : (x-a)^2 + y^2 = r_1^2$ and $S_2 : (x+a)^2 + y^2 = r_2^2$

Any point on the circles $S_1\ is\ (a + r_1 \cos\theta, r_1 \sin\theta)$. Tangent to S_1 at this point is

$x\cos\theta + y\sin\theta = r_1 + a\cos\theta$ ......(i)

The line (i) touches the circle S_2 if $|r_1 + a\cos\theta + a\cos\theta| = r_2$

Or $2a\cos\theta + r_1 = \pm r_2$ ....(ii)

Now, if the point of contact be represented by (h,k), then $h = a + r_1\cos\theta\ and\ k = r_1\sin\theta$

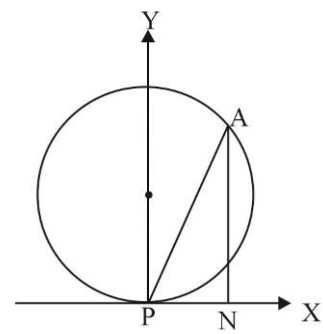
$h^2 + k^2 = a^2 + r_1^2 + 2ar_1\cos\theta = a^2 + r_1(\pm r_2)$

$h^2 + k^2 = a^2 \pm r_1 r_2$

Thus, (h,k) lies on the circle $x^2 + y^2 = a^2 \pm r_1 r_2$

63.(ABD) Let the lines L be the X-axis and the circle S be $x^2 + (y-1)^2 = 1$, which touches the line L at $P(0,0)$.

Let any point on the circle S be $A(\cos\theta, 1+\sin\theta)$



Area of $\Delta PAN = \frac{1}{2}|\cos\theta(1+\sin\theta)| = \frac{1}{2}|f(\theta)|$

Where, $f(\theta) = \cos\theta(1+\sin\theta)$

$f'(\theta) = \cos\theta(\cos\theta) - \sin\theta(1+\sin\theta) = \cos 2\theta - \sin\theta$

$f'(\theta) = 0 \Rightarrow \cos 2\theta - \sin\theta = 0$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

$f''(\theta) = -2\sin 2\theta - \cos\theta$

$f''(\theta) < 0\ if\ \theta = \frac{\pi}{6}\ and\ \frac{5\pi}{6}$

$f\left(\frac{\pi}{6}\right) = \cos\frac{\pi}{6}\left(1+\sin\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{4}$ and $f\left(\frac{5\pi}{6}\right) = \cos\frac{5\pi}{6}\left(1+\sin\frac{5\pi}{6}\right) = -\frac{3\sqrt{3}}{4}$

Maximum area of $\Delta PAN = \frac{3\sqrt{3}}{8}$ $\qquad ar(\Delta PAN) \le \frac{3\sqrt{3}}{8}$

64.(ACD) Coordinates of O are (5, 3) and radius = 2

Equation of tangent at A(7, 3) is $7x + 3y - 5(x+7) - 3(y+3) + 30 = 0$

i.e. $2x - 14 = 0$ i.e. $x = 7$

Equation of tangent at B(5, 1) is $5x + y - 5(x+5) - 3(y+1) + 30 = 0$

i.e. $-2y + 2 = 0$ i.e. $y = 1$

$\therefore$ coordinates of C are (7, 1)

$\therefore$ area of OACB = 4 . . . $\because$ angle between AC & BC $90°$, therefore quadrilateral OACB is a square.

Equation of AB is $x - y = 4$ (radical axis)

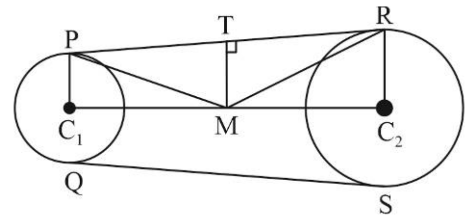
$\Rightarrow$ equation of smallest circle is $(x-7)(x-5) + (y-3)(y-1) = 0$

$x^2 + y^2 - 12x - 4y + 38 = 0$

65.(AC) Let T be the midpoint of PR, perpendicular from T to PR meets C_1C_2 at M

Also $MP^2 = MR^2 = MQ^2 = MS^2 = MT^2 + TR^2$

Hence M is centre of required circle.



$$PR = \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$$

$$= \sqrt{100-4} = \sqrt{96}$$

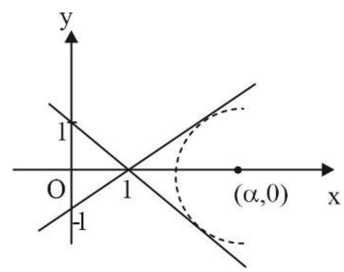
$$\lambda^2 = MT^2 + TR^2 = \left(\frac{2+4}{2}\right)^2 + TR^2 = 9 + \frac{96}{4} = 9 + 24 = 33$$

66.(AC) We know, angle bisector of angle between two tangents from a point (outside the circle) should pass through the centre of circle.

As, angle bisector of given two tangents is x-axis.

$\Rightarrow$ centre of circle will lie on x-axis.

So, Let centre $\equiv (\alpha, 0)$

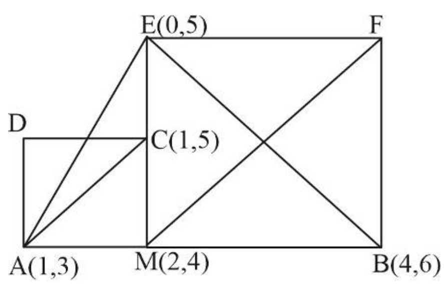


$$\Rightarrow \quad \frac{|\alpha + 0 - 1|}{\sqrt{2}} = \sqrt{(\alpha-3)^2 + \left(0 - \sqrt{7/2}\right)^2}$$

$$\Rightarrow \quad (\alpha - 1)^2 = 2\left[(\alpha-3)^2 + \frac{7}{2}\right] \Rightarrow \alpha = 6, 4$$

$\Rightarrow$ centre of circle $\equiv (6, 0)$ or $(4, 0)$

67.(BC)



Shown in the figure since $C(1,5)$ is the orthocentre of triangle of triangle ΔAEB .

Similarly for the other side the coordinates of $C(3,3)$

68.(ABC) The curve through the intersection of S_1 and S_2 is given by $S_1 = \lambda S_2 = 0$

Or $$x^2(\sin^2\theta + \lambda\cos^2\theta) + 2(h\tan\theta + 2h'\cot\theta)\,xy - (\cos^2\theta + \lambda\sin^2\theta)y^2$$
$$+(32+16\lambda)\,x + (16+32\lambda)y + 19(1+\lambda) = 0$$

The above equation will represent a circle if $\sin^2\theta + \lambda\cos^2\theta = \cos^2\theta + \lambda\sin^2\theta$

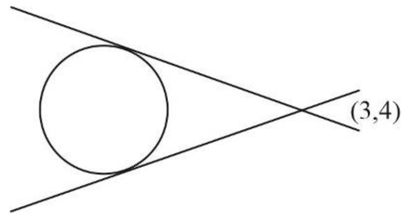
$\Rightarrow \quad (1-\lambda)\sin^2\theta = (1-\lambda)\cos^2\theta \quad \Rightarrow \quad (1-\lambda)(\sin^2\theta - \cos^2\theta) = 0$ i.e., $\lambda = 1$ or $\theta = \frac{\pi}{4}$

Also, $h\tan\theta - \lambda h'\cot\theta = 0$ [$\because$ Coefficient of xy is zero]

Or $h\tan\theta = \lambda h'\cot\theta$

Which is satisfied if $\lambda = 1$ and $\theta = \frac{\pi}{4}$ i.e., $h = h'$ or $\lambda = -1, \theta = \frac{\pi}{4}$ and, hence, $h + h' = 0$

69.(AC) Given circle $(x-1)^2+(y-3)^2=1$



Let of tangent to circle is $y-3=m(x-1)+1\sqrt{1+m^2}$

$(3,4)$ lies on axis $1=2m+\sqrt{1+m^2}$

$(1-2m)^2=1+m^2$

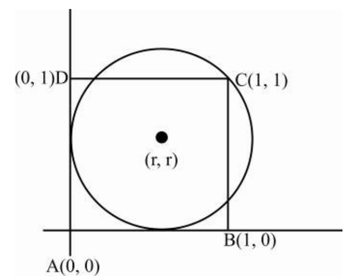
$4m^2-4m+1=1+m^2$

$3m^2-4m=0$; $m=0,\frac{4}{3}$; $\frac{y-4}{n-3}=m=0 \; or \; \frac{4}{3}$

Smallest is 0 and largest $=\frac{4}{3}$

70. **[A-q] [B-p] [C-t] [D-r]**

(A) Line does not intersect the circle



(B) Line is tangent to circle, then $\frac{1-c}{\sqrt{2}}=\sqrt{2}$

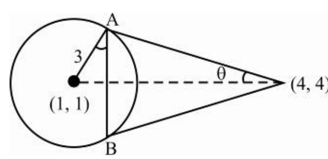
$[\because (-1,0)$ lies above the line $y-x-c=0]$ $\Rightarrow$ $c=-1$

(C) Equation of given circle is $x^2+y^2-2rx-2ry+r^2=0$

It passes through (1, 1), so $r^2-4r+2=0, r=2\pm\sqrt{2}$

$r=2-\sqrt{2}$ $\quad (\because r<1)$

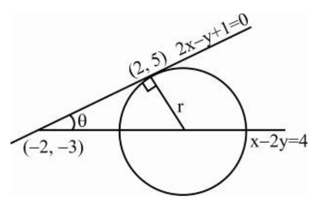
(D) $\sin\theta=\frac{3}{3\sqrt{2}}=\frac{1}{\sqrt{2}}$



$\theta=\frac{\pi}{4}$ $\quad AB=2r\cos\theta=2(3)\frac{1}{\sqrt{2}}=3\sqrt{2}$

71. **[A-r] [B-p] [C-q] [D-q]**

(A) $\tan\theta=\frac{2-\frac{1}{2}}{1+2\left(\frac{1}{2}\right)}=\frac{3}{4}$



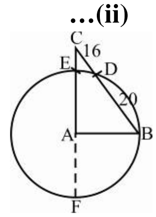
$\frac{r}{\sqrt{8^2+4^2}}=\tan\theta=\frac{3}{4}=\frac{r}{4\sqrt{5}}$ $\quad\Rightarrow\quad$ $r=3\sqrt{5}$

(B) $(AC)^2+r^2=(36)^2$ **...(i)**

$(CE)(CA)=(CD)(BC)$

$\Rightarrow$ $(AC-r)(AC+r)=16\times36$ **...(ii)**

On adding (1) and (2), we get :



$\Rightarrow$ $2(AC)^2=36\times52$

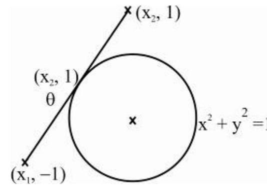
$\Rightarrow$ $AC=6\sqrt{26}$

(C) Let the equation of tangent is $x\cos\theta + y\sin\theta = 1$

$(x_1, -1)$ lies on tangent

$\Rightarrow \quad x_1\cos\theta - \sin\theta = 1$

$x_1\cos\theta = 1 + \sin\theta$ **...(i)**



Now $(x_2, 1)$ lies on tangent

$\Rightarrow \quad x_2\cos\theta = 1 - \sin\theta$ **...(ii)**

$(1)\times(2)$, $x_1x_2\cos^2\theta = 1 - \sin^2\theta = \cos^2\theta \quad \Rightarrow \quad x_1x_2 = 1$

(D) Let circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ Put $\left(t, \frac{1}{t}\right)$

$t^4 + 2gt^3 + ct^2 + 2ft + 1 = 0$ *a, b, c, d* are the roots of this equation. So, $abcd = 1$

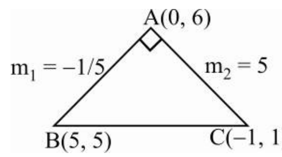
72. **[A-s] [B-q] [C-t] [D-r]**

(A) Note that the *ABC* is right angled at *A*

$\therefore$ Equation of circle $(x+1)(x-5)+(y-1)(y+5)=0$

$x^2 + y^2 - 4x - 6y - 5 + 5 = 0$

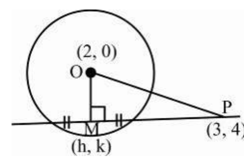
$x^2 + y^2 - 4x - 6y = 0$



Hence the circle passes through the origin $\quad \therefore \quad$ tangent at (0, 0) is $4x + 6y = 0$

(B) Locus is Arc of the circle with *OP* as diameter intercepted by the given circle, *i.e.*,

$(x-2)(x-3) + y(y-4) = 0$



(C) Diameter of the Circle Is Given As

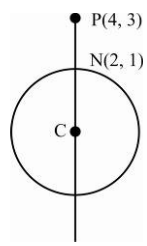
$2x - y - 2 = 0$ **...(1)**

Slope of *PN*: $\dfrac{3-1}{4-2} = 1$

Equation of normal through *PN* is

$y - 1 = (x - 2)$

$x - y - 1 = 0$ **...(2)**



Solving (1) and (2), centre is (1, 0)

Hence equation of the circle is $(x-1)^2 + y^2 = (2-1)^2 + 1$

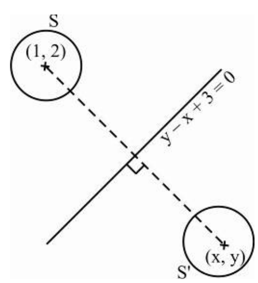
$x^2 + y^2 - 2x - 1 = 0$

(D) $\dfrac{x-1}{-1} = \dfrac{y-2}{1} = -2\dfrac{(2-1+3)}{2}$

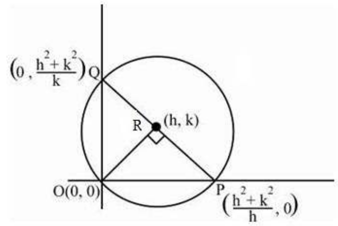
$\Rightarrow \quad (x, y) = (5, -2)$

Eqn. of *S*' is $(x-5)^2 + (4+2)^2 = 1$

$x^2 + y^2 - 10x + 4y + 28 = 0$



73. [A-r, B-s, C-p, D-q]

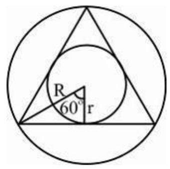


(A) Equation of PQ is

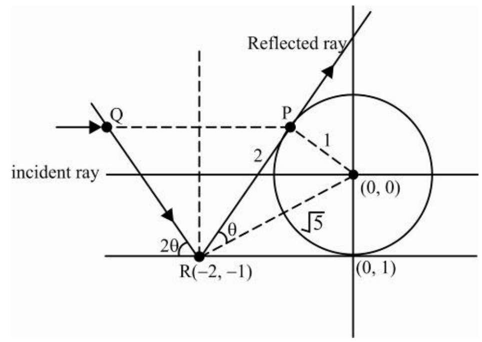
$hx+ky=h^2+k^2$

$PQ^2=4a^2$; $\left(h^2+k^2\right)^2\left(\frac{1}{h^2}+\frac{1}{k^2}\right)=4a^2$

(B) $r=R\cos 60^\circ=\frac{R}{2}$



(C) $\tan 2\theta=\frac{2\left(\frac{1}{2}\right)}{1-\frac{1}{4}}=\frac{4}{3}$

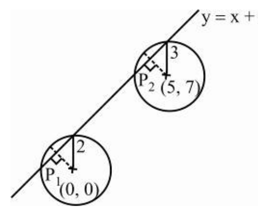


Slope of $QR=-\frac{4}{3}$

$\therefore$ Equation of incident ray QR is $y+1=-\frac{4}{3}(x+2)$

$3y+4x+11=0$

(D) $r_1^2-P_1^2=r_2^2-P_2^2$



$\Rightarrow \quad 4-\frac{C^2}{2}=9-\frac{(2-C)^2}{2} \quad \Rightarrow \quad C=-\frac{3}{2}$

$\Rightarrow$ Equation of line $y=x-\frac{3}{2}$

74. [A-r, B-p, C-s, D-p]

(A) Let the circle be $(x-r)^2+(y-r)^2=r^2$

$x^2+y^2-2rx-2ry+r^2=0.......(r_1,r_2)$

Orthogonally $\Rightarrow 2r_1r_2+2r_1r_2=r_1^2+r_2^2 \Rightarrow 6r_1r_2=(r_1+r_2)^2$

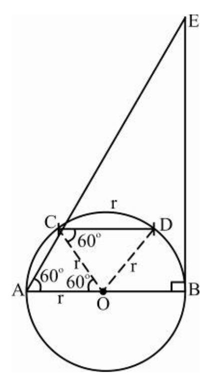
Circle passes through (a, b)

$\Rightarrow \quad r^2-2(a+b)r+\left(a^2+b^2\right)=0$

From (1), $6\left(a^2+b^2\right)=4(a+b)^2 \quad \Rightarrow a^2+b^2-4ab=0$

(B) Put $\left(h,-\frac{b}{2}\right)$ to equation of circle $h^2+\frac{b^2}{4}-ah+\frac{b^2}{2}=0$

$h^2-ah+\frac{3b^2}{4}=0$ must get two distinct real roots $\Rightarrow D>0\Rightarrow a^2-3b^2>0$



(C) From figure it is clear that

$\Rightarrow \quad AE=\frac{AB}{\cos 60^\circ}=2(AB)$

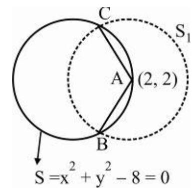
(D) 2r = perpendicular distance between two parallel tangents

$\Rightarrow \quad 2r=\left|\frac{4+\frac{7}{2}}{5}\right|=\frac{15}{2\times 5}=\frac{3}{2}$

75.(8) So equation of circle having centre (2, 2) and radius 1 unit is $S_1 \equiv (x-2)^2 + (y-2)^2 = 1$

Equation of chord AB is $S_1 - S_2 = 0$

$$4x + 4y = 15$$



76.(4) $r^2 = \dfrac{2\left(OD^2 + OB^2\right) - BD^2}{4}$

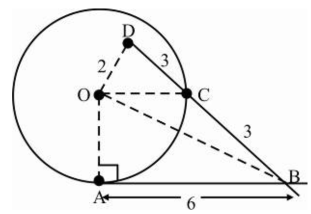
$$r^2 = \frac{2\left(4 + (OB)^2\right) - 6^2}{4} \qquad \text{...(1)}$$

Also, $(OB)^2 = (OA)^2 + (AB)^2 = r^2 + 36$ **...(2)**

From (1) and (2), we get

$\therefore \quad 4r^2 = 2\left(r^2 + 36\right) - 28$

$r^2 = 22 \Rightarrow r = \sqrt{22} \qquad \Rightarrow \qquad [r] = 4$



77.(6) It is obvious that the centre of the circle will lie on x axis let it is $(\lambda, 0)$, then

$$r^2 = (\lambda + 1)^2 + 1 = \left(\frac{|\lambda - 2|}{\sqrt{2}}\right)^2$$

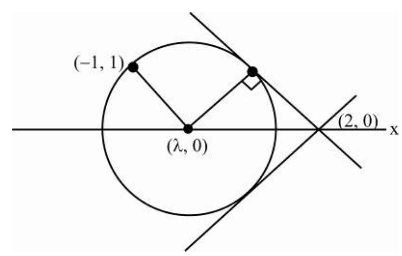
$\lambda^2 + 8\lambda = 0 \qquad \Rightarrow \qquad \lambda = 0, -8$

$\Rightarrow \qquad$ Equation of circle is $x^2 + y^2 = 2$

and $\quad (x+8)^2 + y^2 = 50$ i.e., $x^2 + y^2 - 2 = 0$

and $\quad x^2 + y^2 + 16x + 14 = 0$

$$r_1 + r_2 = \sqrt{2} + \sqrt{50} = 6\sqrt{2}$$



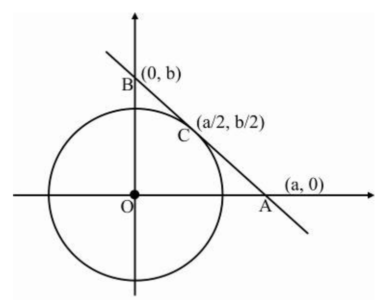
78.(4) Let equation of circle is

$\left(x - \dfrac{a}{2}\right)^2 + \left(y - \dfrac{b}{2}\right)^2 + \lambda\left(bx + ay - ab\right) = 0$ it passes through the point $\left(\dfrac{a}{2}, 0\right)$, then we get $\lambda = \dfrac{b}{2a}$

So, equation of circle will be

$$x^2 + \frac{a^2}{4} + y^2 + \frac{b^2}{4} - ax - by + \frac{b^2}{2a}x + \frac{b}{2}y - \frac{b^2}{2} = 0$$

$x^2 + y^2 + x\left(-a + \dfrac{b^2}{2a}\right) + y\left(\dfrac{b}{2} - b\right) + \dfrac{a^2}{4} - \dfrac{b^2}{4} = 0$. Radius is $\dfrac{b}{4a}\sqrt{a^2 + b^2}$



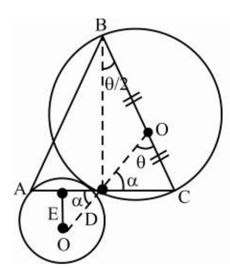
79.(1) $\cos\alpha = \dfrac{1/2}{2} = \dfrac{1}{4}$ [from ΔOED]

Then $\quad \theta = \pi - 2\alpha$

$$BD = 2\cot\frac{\theta}{2} = 2\cot\left(\frac{\pi}{2} - \alpha\right) = 2\tan\alpha \quad \text{[from } \Delta BDC\text{]}$$

$$= 2\sqrt{15}$$

$\therefore \quad$ Area of $\Delta ABC = \dfrac{1}{2} \times (AC)(BD) = \dfrac{1}{2} \times 3 \times 2\sqrt{15} = 3\sqrt{15} = \sqrt{135}$



80.(0) Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ as centre lies an $2x - 2y + 9 = 0 \Rightarrow -2g + 2f + 9 = 0$

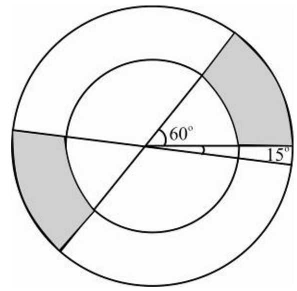
As it cuts $x^2 + y^2 - 4 = 0$ orthogonally $2g + 0 + 2f \times 0 = c - 4 = 0 \quad \Rightarrow c = 4$

So, the equation circle is $x^2 + y^2 + (2f + 9)x + 2fy + 4 = 0$

$x^2 + y^2 + 9x + 4 + 2f(x + y) = 0$ it passes through the through the intersection of

$x^2 + y^2 + 9x + 4 = 0$ and $x + y = 0 \Rightarrow$ Point as $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(-4, 4)$

81.(2) $4 \le a^2 + b^2 \le 9$ represents the region lying between the two concentric circles of radius of radius 2 and 3 having centre at $(0,0)$



For $b^2 - 4ab + a^2 \le 0$ Let $\frac{b}{a} = \tan\theta$

$$\left(\frac{b}{a}\right)^2 - \frac{4b}{a} + 1 \le 0$$

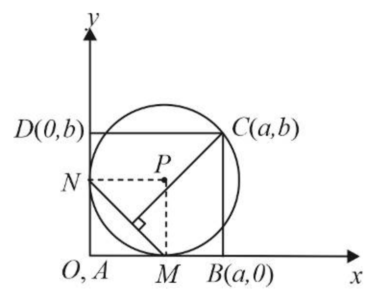
$\Rightarrow \quad \tan\theta \in \left[2-\sqrt{3}, 2+\sqrt{3}\right] \quad \Rightarrow \quad \theta \in [15°, 75°]$

So the shaded region in the given figure is the region R

Hence $|a+b|$ is $|r(\cos\theta + \sin\theta)| = r\sqrt{2}\left|\sin(\theta + 45°)\right| \ge 2\sqrt{2}\sin(15° + 45°)$ because $r \ge 2$

Minimum = $2\sqrt{2}\frac{\sqrt{3}}{2} = \sqrt{6}$

82.(25) Let r be the radius of circle



Equation of line MN is $x + y = r$

Now, $\frac{|a+b-r|}{\sqrt{2}} = 5 \Rightarrow a + b = 5\sqrt{2} + r$ ....(i)

Equation of the circle $x^2 + y^2 - 2rx - 2ry + r^2 = 0$; $\quad C(a,b)$ lies on the circle

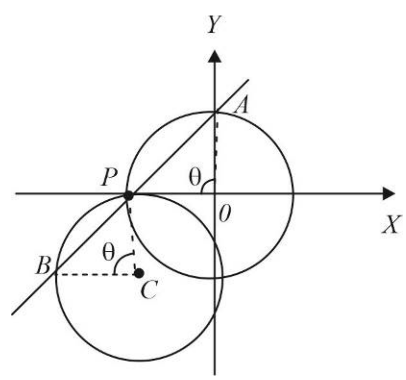
$a^2 + b^2 - 2ar - 2br + r^2 = 0$

By using (i)

$a^2 + b^2 = 2\left(5\sqrt{2} + r\right)r - r^2 = 10\sqrt{2}r + r^2$ ....(ii)

The required area of the rectangle $= ab = \frac{1}{2}\left[(a+b)^2 - \left(a^2 + b^2\right)\right] = \frac{1}{2}\left[\left(5\sqrt{2} + r\right)^2 - \left(10\sqrt{2}r + r^2\right)\right] = 25$

83.(0.5) Solving $x^2 + y^2 = 1$



And $x^2 + y^2 + 2x + 4y + 1 = 0$

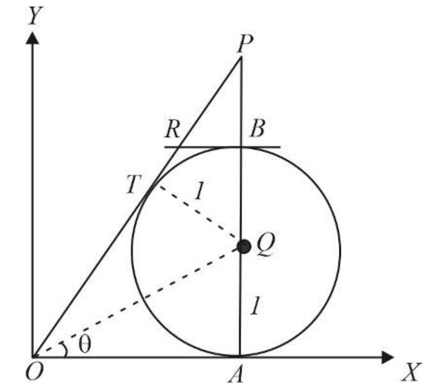
We get, $y = 0$ or $-\frac{4}{5}$; $\quad x = -1$ or $x = \frac{3}{5}$

The circles intersect at $(-1, 0)$ and $\left(\frac{3}{5}, -\frac{4}{5}\right)$

The coordinates of P are integral, so P is $(-1, 0) \Rightarrow 0 = -2m + 1 \Rightarrow m = 0.5$

84.(1.66) Let $OA = h$ and $PQ = x$,

Triangles OAP and QTP are similar



$\Rightarrow \dfrac{AP}{PT} = \dfrac{OA}{QT} = \dfrac{OP}{PQ} \quad \Rightarrow \quad \dfrac{x+1}{PT} = \dfrac{h}{1} = \dfrac{OP}{x}$ ....(i)

From equation (i), we get $OP = hx$

Perimeter of $\Delta OAP = 8$

$OA + AP + OP = 8$

$(h+1)(x+1) = 8$ ....(ii)

From equation (i) $PT = \dfrac{x+1}{h}$

Again, from equation (i), $OP = hx \Rightarrow OT + PT = hx$ or $PT = hx - h$

Thus, $\dfrac{x+1}{h} = hx - h$ or $h^2(x-1) = x+1$

$h^2\left[\dfrac{8}{h+1} - 2\right] = \dfrac{8}{h+1}$ ....from (ii)

$h^3 - 3h^2 + 4 = 0 \Rightarrow (h-2)^2(h+1) = 0$; $h = 2$

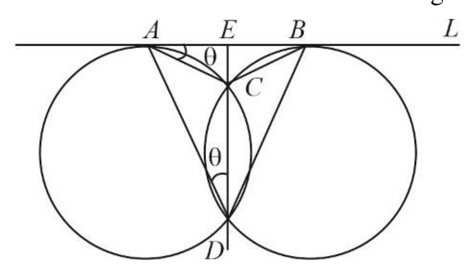
From the equation (ii) we have $(h+1)(x+1) = 8$; $x = \dfrac{5}{3} \Rightarrow PQ = \dfrac{5}{3}$

85.(1) Let R_1 be the radius of the circumcircle of ΔACB, then $R_1 = \dfrac{AB}{2\sin(\angle ACB)}$

Let R_2 be the radius of the circumcircle of ΔADB, then $R_2 = \dfrac{AB}{2\sin(\angle ADB)}$

Let the common chord CD meets the common tangent L at E. since the common chord bisects the common tangent,

So, $AE = BE$



Also $\angle EAC = \angle ADE = \theta$

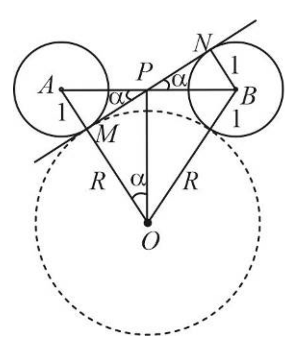
$\angle ACE = 90 - \theta$

$\angle ACB = 180 - 2\theta$

And $\angle ADB = 2\theta$

$R_1 = \dfrac{AB}{2\sin(180 - 2\theta)} = \dfrac{AB}{2\sin 2\theta} = R_2$; $\dfrac{R_1}{R_2} = 1$

86.(8) From the figure, we have $PB = 3, BN = 1$



$\tan\alpha = \dfrac{1}{2\sqrt{2}}$

And ΔABO is isosceles

OP is perpendicular to AB, so $\angle AOP = \alpha \Rightarrow \tan\alpha = \dfrac{2\sqrt{2}}{R}$

Hence, $\dfrac{1}{2\sqrt{2}} = \dfrac{2\sqrt{2}}{R} \Rightarrow R = 8$

87.(4) On adding and subtracting the curves we get $(x+3)^2 = 4(y-1),\ y^2 - 20y + 59 = 0$ respectively

Let point of intersection be $(x_1,\ y_1)(x_2,\ y_1)(x_3,\ y_2)(x_4,\ y_2) \quad \Rightarrow\ y_1 + y_2 = 20$ and $y_1,\ y_2 > 0$

The required sum of distances $= \sum\sqrt{(x_1+3)^2 + (y_1-2)^2} = \sum\sqrt{4(y_1-1)+(y_1-2)^2} = \sum y_1 = 2(y_1+y_2) = 40$

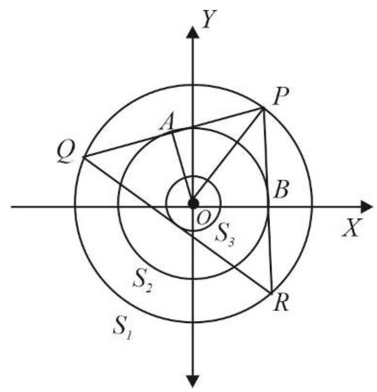
88.(40) Let us select the centre of the series of circles as origin O,

Let the tangents drawn from any point on the circle $S_1 : x^2 + y^2 = a^2$ to $S_2 : x^2 + y^2 = b^2$ be such that the chord of contact touches the circle $S_3 : x^2 + y^2 = c^2$.

Any point, P on the circle S_1 is $(a\cos\theta, a\sin\theta)$

The chord of contact of tangents drawn from P to the circle S_2 is $(a\cos\theta)x + (a\sin\theta)y = b^2$

Which touches S_3 if $b^2 = ac$



a, b, and *c* are in geometric progression.

Thus, the conditions of the problem hold if $r_1, r_2, r_3, ...$ are in G.P.

Again, the angle between the tangents PA and PB is $\dfrac{\pi}{3}$

$\angle OPQ = \dfrac{\pi}{6}$

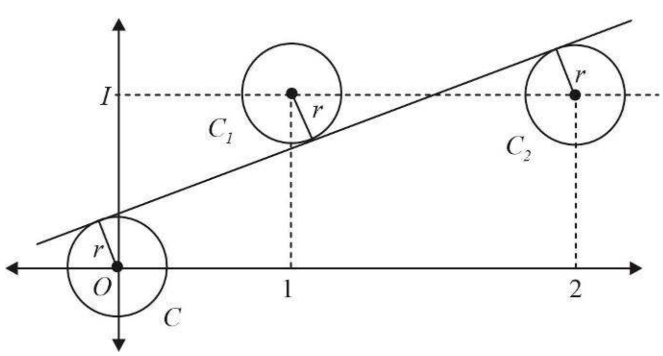
$\sin\dfrac{\pi}{6} = \dfrac{OA}{OP} = \dfrac{b}{a} \Rightarrow \dfrac{b}{a} = \dfrac{1}{2}$

Thus, the common ratio of G.P. is $\dfrac{1}{2}$

Hence, the radii $r_1, r_2, r_3, ...$ form a G.P. of common ratio $\dfrac{1}{2}$

Thus, $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n} r_i = \lim\limits_{n\to\infty}(r_1 + r_2 + ... + r_n) = \lim\limits_{n\to\infty}\dfrac{r_1\left[1-\left(\frac{1}{2}\right)^n\right]}{1-\frac{1}{2}} = \lim\limits_{n\to\infty} 2r_1\left[1-\left(\frac{1}{2}\right)^n\right] = 2r_1 = 2(20) = 40$

89. (5)



The equation of the line joining the origin and the centre of circle $C_2(2,1)$ is

$y = \dfrac{x}{2}$ or $x - 2y = 0$

Let the equation of common tangent be $x-2y+c=0$ .....(i)

Perpendicular distance from (0, 0) on this line $=$ Perpendicular distance from (1, 1)

Or $\left|\frac{c}{\sqrt{5}}\right|=\left|\frac{c-1}{\sqrt{5}}\right|$ or $c=1-c \; or \; c=\frac{1}{2}$

The equation of common tangent is $x-2y+\frac{1}{2}=0 \; or \; 2x-4y+1=0$ ....(ii)

The length of perpendicular from $(2,1)$ on the line (ii) is $r=\left|\frac{4-4+1}{\sqrt{20}}\right|=\frac{1}{2\sqrt{5}}=\frac{\sqrt{5}}{10}$

90.(1) $x^2+y^2+(3+\sin\beta)x+(2\cos\alpha)y=0$ ....(i)

$x^2+y^2+(2\cos\alpha)x+2cy=0$ ....(ii)

Since both the circles are passing through the origin $(0,0)$, the equation of tangent at $(0,0)$

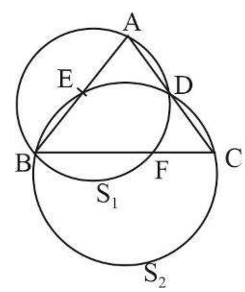
Tangent at $(0,0)$ to circle (ii) is therefore, $(3+\sin\beta)x+(2\cos\alpha)y=0$

Tangent at $(0,0)$ to circle (ii) is $(2\cos\alpha)x+2cy=0$

Therefore, (i) and (ii) must be identical

Comparing (i) and (ii), we get $\frac{3+\sin\beta}{2\cos\alpha}=\frac{2\cos\alpha}{2c}$ or $c=\frac{2\cos^2\alpha}{3+\sin\beta}$ or $c_{max}=1 \; when \; \sin\beta=-1 \; and \; \alpha=0$

91.(3) Let circumcircle of triangle BDC be S_1 and the circumcircle of triangle ABD be S_2



So, $\frac{AD}{CD}=\frac{AB}{BC}$ ....(i)

Now, for circle S_1; $AE\times AB=AD\times AC$

$AE=\frac{AD\times AC}{AB}$ ....(ii)

For circle S_2; $CF\times CB=CD\times CA$

$CF=\frac{CD\times CA}{AB}$ ....(iii)

From equation (ii) and (iii), we get $\frac{AE}{CF}=\frac{AD\times CB}{AB\times CD}$ ....(iv)

From equation (i) and (iv), $AE=CF$ So, $CF=3$

92.(3) The circumcenter of the triangle formed by any three points lies at the origin. The orthocenter of the triangle formed by the point $A_1(x_1,y_1), A_2(x_2,y_2) \; and \; A_3(x_3,y_3)$ is given by $H(x_1+x_2+x_3, y_1+y_2+y_3)=(a,b)$

The centroid of the triangle formed by the other three points is given by

$G\left(\frac{x_4+x_5+x_6}{3},\frac{y_4+y_5+y_6}{3}\right) or \; G\left(\frac{8-a}{3},\frac{4-b}{3}\right)$

Equation of the line joining H and G, is $y-b=\frac{\frac{4-b}{3}-b}{\frac{8-a}{3}-a}(x-a)$ or $(2-a)(y-b)=(1-b)(x-a)$ ....(i)

The above equation holds for all values of 'a' and 'b' if $x=2$ and $y=1$

Thus, the line (i) always passes through the fixed point (2, 1).

93.(0) Let point on circle be $(\cos\theta, \sin\theta)$

If this point lies on the curve $\sin x = \cos y$, we have

$$\sin(\cos\theta) = \cos(\sin\theta) \Rightarrow \frac{\pi}{2} - \cos\theta = 2n\pi \pm \sin\theta,\ n \in Z$$

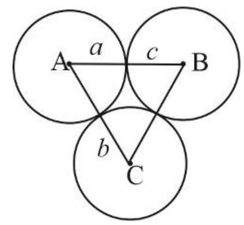
$$\cos\theta \pm \sin\theta = \frac{\pi}{2} - 2n\pi$$

But $\left|\frac{\pi}{2} - 2n\pi\right| > \sqrt{2}$ for $\forall\ n \in Z$

$\Rightarrow$ No solution $\Rightarrow$ Number of points of intersection = 0.

94.(8) In radius of ΔABC is 4

We have to find $\frac{abc}{2(a+b+c)}$



$r = \frac{\Delta}{S}$

$S = a+b+c,\ \Delta = \sqrt{abc(a+b+c)} \Rightarrow 4 = \sqrt{\frac{abc}{a+b+c}} \Rightarrow \frac{abc}{2(a+b+c)} = 8$

95.(5) Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

Since, it passes through (3, 4), $6g + 8f + c = -25$. As it cuts the circle $x^2 + y^2 = a^2$ orthogonally

$$2g \times 0 + 2f \times 0 = c - a^2 \Rightarrow c = a^2$$

$\Rightarrow \quad 6g + 8f + a^2 + 25 = 0$

$\therefore$ Locus of the centre $(-g, -f)$ is $6x + 8y - (a^2 + 25) = 0$.

$\therefore$ Distance of the line from the origin is $817 = \frac{a^2 + 25}{\sqrt{36+64}} \Rightarrow a^2 + 25 = 8170 \Rightarrow a^2 = 8145$

96.(2) $S_1 : C_1(-5, 12)$

$S_2 : C_2(5, 12)$

$r_1 = 16,\ r_2 = 4$

$CC_2 = r + 4$ (Where C is the centre of circle touching C_2 externally and C_1 internally)

$CC_1 = 16 - r$ (Not $r - 16, \because S_2$ is contained by S_1)

$\Rightarrow \quad CC_1 + CC_2 = 20$

$\therefore$ Locus of C is the ellipse with foci at C_1 and C_2 and length of major axis = 20

$\therefore$ Locus of C is

$$\frac{x^2}{100} + \frac{(y-12)^2}{75} = 1 \qquad \text{........(i)}$$

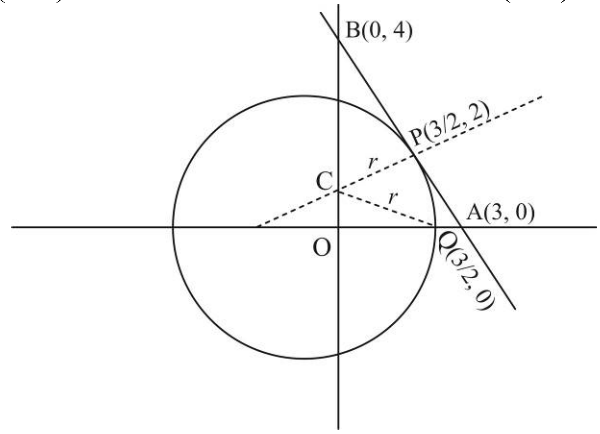
According to question $y = ax$ is tangent to this ellipse from (0, 0)

Equation of tangent to (i) is $y - 12 = mx + \sqrt{100m^2 + 75}$

But this is passing through (0, 0)

$\Rightarrow \quad -12 = \sqrt{100m^2 + 75} \Rightarrow m^2 = \frac{69}{100} \Rightarrow p + q = 169 \Rightarrow k = 2.$

97.(5) Let the centre be C(h, k). And P be the mid point of $AB \Rightarrow P=\left(\frac{3}{2}, 2\right)$ and also Q is mid point of $OQ \Rightarrow Q=\left(\frac{3}{2}, 0\right)$



$\because \quad CP \perp AB \Rightarrow \dfrac{2-k}{\frac{3}{2}-h}=\dfrac{3}{4}$

$\Rightarrow \quad 6h-8k=-7 \qquad$ **....(i)**

$\because \quad$ CP = CQ

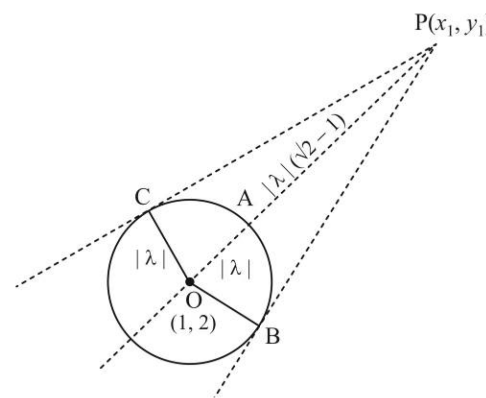
$\Rightarrow \quad \left(h-\frac{3}{2}\right)^2+(k-2)^2=\left(h-\frac{3}{2}\right)^2+k^2$

$\Rightarrow \quad k = 1$, putting in equation (i), we get :

$$6h=1 \Rightarrow h=\frac{1}{6}$$

$\therefore \quad$ Radius $(r) = CQ=\sqrt{\left(\frac{1}{6}-\frac{3}{2}\right)^2+1} = \sqrt{\left(\frac{1-9}{6}\right)^2+1}=\frac{5}{3} \qquad \therefore \qquad 3r = 5.$

98.(6) Since the line $y = x + c$ is normal to the given circle, $c = 1$.
So the equation of line is $y = x + 1 \qquad$ **....(i)**
Also the radius of the circle is $|\lambda|$.



Given $AP=|\lambda|\left(\sqrt{2}-1\right)$

$\Rightarrow \quad OP=\sqrt{2}|\lambda| \qquad \Rightarrow \qquad PC=|\lambda|$

$\Rightarrow \quad$ Area of quadrilateral OBPC

$$=2\times\frac{1}{2}|\lambda|^2=36 \Rightarrow \lambda=\pm 6$$

99.(7) Given circles are $x^2+y^2=4 \qquad$ **....(i)**

and $\quad x^2+y^2-6x-8y+m^2=0 \qquad$ **....(ii)**

Centre of circle (i) is $A(0,0)$ and that of circle (ii) is $B(3,4)$

Radius of circle (i), $r_1=2$ and radius of circle (ii), $r_2=\sqrt{25-m^2}$

For $\sqrt{25-m^2}$ to be defined $25-m^2 \ge 0 \Rightarrow -5 \le m \le 5 \qquad$ **....(A)**

For circles (i) and (ii) to have exactly two common tangents $|r_1-r_2|<AB<r_1+r_2$

(i) $\quad$ Now, $AB<r_1+r_2 \Rightarrow 5<2+\sqrt{25-m^2}$

$\Rightarrow \quad 9<25-m^2 \Rightarrow m^2-16<0 \Rightarrow -4<m<4 \qquad$ **....(I)**

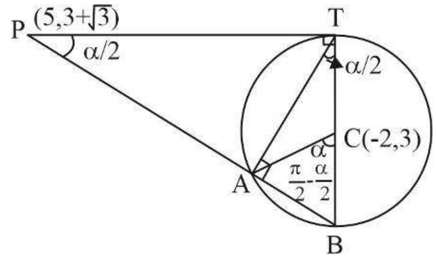
$\therefore \quad \sqrt{25-m^2}>3 \Rightarrow r_2>r_1$

(ii) $\quad AB>|r_1-r_2|$

$$5>r_2-r_1 \Rightarrow 5>\sqrt{25-m^2}-2 \Rightarrow 49>25-m^2 \Rightarrow m^2+24>0 \Rightarrow -\infty<m<\infty \quad \textbf{....(II)}$$

From (I) and (II), $-4<m<4$
Therefore, possible integral values of m are : $-3, -2, -1, 0, 1, 2, 3$
Hence number of possible integral values of m is 7

100.(4) $CT = CB = r = 4$



$$PT = \sqrt{5^2 + \left(3+\sqrt{3}\right)^2 + 4\times5 - 6\left(3+\sqrt{3}\right) - 3}$$

$PT = 6$

Let $\angle ACB = \alpha$

$$Ar(\Delta CAB) = \frac{1}{2} r.r \sin\alpha = 8\sin\alpha$$

$$Ar(\Delta CAT) = \frac{1}{2} r.r \sin(\pi - \alpha) = 8\sin\alpha$$

In ΔPBT $\tan\frac{\alpha}{2} = \frac{BT}{PT} = \frac{8}{6}$

$$\sin\alpha = \frac{2\tan\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}} = \frac{2\times\frac{4}{3}}{1+\frac{16}{9}} = \frac{24}{25}$$

$$Ar(\Delta CAB) + Ar(\Delta CAT) = 16\sin\alpha = 16\times\frac{24}{25} = \frac{384}{25}$$

$\lambda = 384$ hence $\left[\sqrt{384}\right] - 15 = 19 - 15 = 4$

101.(3) The equation of first circle $(x-1)^2 + (y-3)^2 = r^2$

whose centre is $C_1(1, 3)$ and radius $r_1 = r$ and equation of second circle $x^2 + y^2 - 8x + 2y + 8 = 0$

whose centre is $C_2(4, -1)$ and radius $r_2 = \sqrt{4^2 + 1^2 - 8} = \sqrt{17-8} = \sqrt{9} = 3$

Two circles intersect in two distinct points, then:

$|r_1 - r_2| < C_1C_2 < r_1 + r_2 \quad \Rightarrow \quad |r-3| < \sqrt{(4-1)^2 + (-1-3)^2} < r+3$

$\Rightarrow \quad |r-3| < \sqrt{9+16} < r+3 \quad \Rightarrow \quad |r-3| < 5 < r+3$

$\Rightarrow \quad |r-3| < 5$ and $5 < r+3 \quad \Rightarrow \quad 2 < r < 8 \quad \Rightarrow \quad r = 3, 5, 7$

CONIC SECTION

1.(A) Let the point be $P(at^2, 2at)$.

Then according to question, $SP = at^2 + a = k$ ....(i)

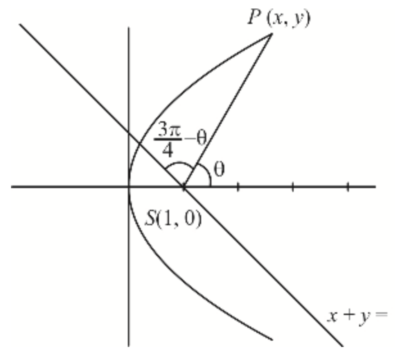
Let (α, β) is the moving point, then $\alpha = at^2, \beta = 2at$

$\Rightarrow \quad \frac{\alpha}{\beta} = \frac{t}{2}$ and $a = \frac{\beta^2}{4\alpha}$ $\quad$ ($\because$ point (α,β) lies on $y^2 = 4ax$)

On substituting these values in equation (i),

$$\frac{\beta^2}{4\alpha}\left(1+\frac{4\alpha^2}{\beta^2}\right) = k \Rightarrow \beta^2 + 4\alpha^2 = 4k\alpha \Rightarrow 4x^2 + y^2 - 4kx = 0 \text{ is the required locus.}$$

2.(A) Projection on $x + y = 1$ is $L = SP\cos\left(\frac{3\pi}{4}-\theta\right)$ where $\tan\theta$ is slope of SP



$SP = x+1$

$$\frac{dL}{dt} = \frac{d}{dt}\left[(x+1)\cos\left(\frac{3\pi}{4}-\theta\right)\right]$$

$$\tan\theta = \frac{y}{x-1}$$

$$\sec^2\theta\frac{d\theta}{dt} = \frac{1}{(x-1)}\frac{dy}{dt} - \frac{y}{(x-1)^2}\frac{dx}{dt}$$

P lies on $y^2 = 4x$

Hence, $2y\frac{dy}{dt} = \frac{4dx}{dt}, \quad \frac{dy}{dt} = \frac{8}{y}$

So, at $x = y = 4$, we have $\sec^2\theta = 1 + \left(\frac{4}{3}\right)^2, \quad \frac{dy}{dt} = 2$

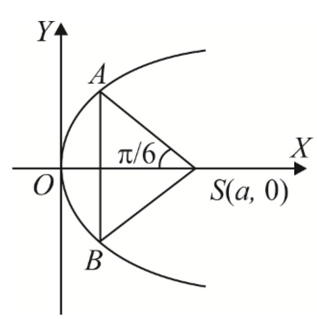
Hence, $\frac{25}{9}\frac{d\theta}{dt} = \frac{2}{3} - \frac{16}{9}$

$\frac{25}{9}\frac{d\theta}{dt} = -\frac{10}{9} \Rightarrow \frac{d\theta}{dt} = -\frac{2}{5}$

Hence, $\frac{dL}{dt} = \cos\left(\frac{3\pi}{4}-\theta\right)\times 4 + (x+1)\sin\left(\frac{3\pi}{4}-\theta\right)\times\frac{d\theta}{dt}$

Also, $\sin\theta = \frac{4}{5}$ and $\cos\theta = \frac{3}{5} \Rightarrow \frac{dL}{dt} = -\sqrt{2}$

3.(B) Let $A = (at_1^2, 2at_1), B \equiv (at_1^2, -2at_1)$.



We have $m_{AS} = \tan\left(\frac{5\pi}{6}\right)$

$\Rightarrow \frac{2at_1}{at_1^2 - a} = -\frac{1}{\sqrt{3}} \Rightarrow t_1^2 + 2\sqrt{3}t_1 - 1 = 0 \Rightarrow t_1 = -\sqrt{3} \pm 2$

Clearly $t_1 = -\sqrt{3} - 2$ is rejected.

Thus, $t_1 = (2-\sqrt{3})$

Hence, $AB = 4at_1 = 4a(2-\sqrt{3})$

4.(B) Let $y_1 = 12 - \sqrt{1 - x_1^2}$ and $y_2 = \sqrt{4x_2}$ or $x_1^2 + (y_1 - 12)^2 = 1$ and $y_2^2 = 4x_2$

Thus (x_1, y_1) lies on the circle $x^2 + (y-12)^2 = 1$ and (x_2, y_2) lies on the parabola $y^2 = 4x$.

Thus, given expression is the shortest distance between the curves $x^2 + (y-12)^2 = 1$ and $y^2 = 4x$.

Now the shortest distance always occurs along common normal to the curves and normal to circle passes through the centre of the circle.

Normal to the parabola $y^2 = 4x$ is $y = mx - 2m - m^3$ passes through (0, 12) gives $m^3 + 2m + 12 = 0$, which has only one real root $m = -2$.

Hence, corresponding point on the parabola is (4, 4).

Thus, required minimum distance $= \sqrt{4^2 + 8^2} - 1 = 4\sqrt{5} - 1$.

5.(B) Tangent at point P is $ty = x + at^2$. . . . .(i)

Line perpendicular to equation (i) passes through $(a, 0)$

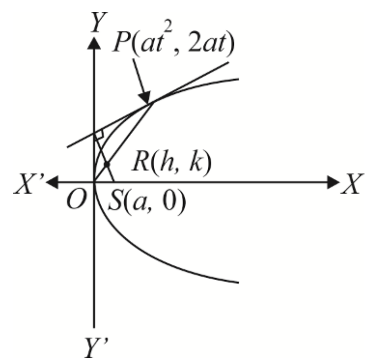
$\therefore \quad y - 0 = -t(x-a)$ or $tx + y = ta$ or $y = t(a-x)$ . . . .(ii)

Equation of OP

$$y - \frac{2}{t}x = 0 \text{ or } y = \frac{2}{t}x \quad \text{. . . .(iii)}$$

From equations (ii) and (iii), eliminating t, we get

$$y^2 = 2x(a-x) \quad \text{or} \quad 2x^2 + y^2 - 2ax = 0$$



6.(D) For the two ellipses to intersect at four distinct points, $a > 1$

$\Rightarrow b^2 - 5b + 7 > 1 \Rightarrow b^2 - 5b + 6 > 0 \Rightarrow b \in (-\infty, 2) \cup (3, \infty) \Rightarrow b$ does not lie in [2, 3]

7.(B) Since there are exactly two points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose distance from centre is same, the points would be either end points of the major axis or of the minor axis.

But $\sqrt{\frac{a^2 + 2b^2}{2}} > b$, so the points are the vertices of major axis.

Hence, $a = \sqrt{\frac{a^2 + 2b^2}{2}} \Rightarrow a^2 = 2b^2 \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{1}{\sqrt{2}}$

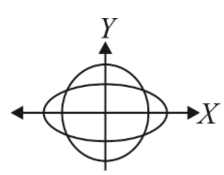
8.(B) Radius of the circle having SS' as diameter is $r = ae$

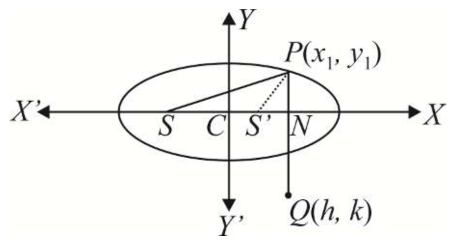
If it cuts an ellipse, then $r > b$

$$\Rightarrow ae > b \quad \Rightarrow e^2 > \frac{b^2}{a^2}$$

$$\Rightarrow e^2 > 1 - e^2 \quad \Rightarrow e^2 > \frac{1}{2}$$

$$\Rightarrow e > \frac{1}{\sqrt{2}} \quad \Rightarrow e \in \left(\frac{1}{\sqrt{2}}, 1\right)$$





9.(B) $a^2 = 25$ and $b^2 = 16$

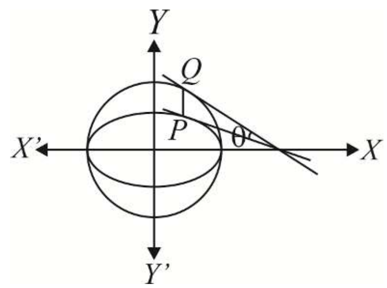
$\Rightarrow$ $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

Let point Q be (h, k), where $k < 0$

Given that $|k| = SP = a + ex_1$, where $P(x_1, y_1)$ lies on the ellipse

$\Rightarrow$ $|k| = a + eh$ (as $x_1 = h$)

$\Rightarrow$ $-y = a + ex \Rightarrow 3x + 5y + 25 = 0$



10.(A) Tangent to the ellipse at $P(a\cos\alpha, b\sin\alpha)$ is $\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$

Tangent to the circle at $Q(a\cos\alpha, a\sin\alpha)$ is $\cos\alpha x + \sin\alpha y = a$

Now angle between tangents is θ,

Then $\tan\theta = \left|\dfrac{-\frac{b}{a}\cot\alpha - (-\cot\alpha)}{1 + \left(-\frac{b}{a}\cot\alpha\right)(-\cot\alpha)}\right|$

$$= \left|\frac{\cot\alpha(1 - \frac{b}{a})}{1 + \frac{b}{a}\cot^2\alpha}\right| = \left|\frac{a-b}{a\tan\alpha + b\cot\alpha}\right| = \left|\frac{a-b}{(\sqrt{a\tan\alpha} - \sqrt{b\cot\alpha})^2 + 2\sqrt{ab}}\right|$$

Now the greatest value of the above expression is $\left|\frac{a-b}{2\sqrt{ab}}\right|$ when $\sqrt{a\tan\alpha} = \sqrt{b\cot\alpha}$

$\Rightarrow$ $\theta_{\text{maximum}} = \tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$

11.(B) For hyperbola $\frac{x^2}{5} - \frac{y^2}{5\cos^2\alpha} = 1$

We have $e_1^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{5\cos^2\alpha}{5} = 1 + \cos^2\alpha$

For ellipse $\frac{x^2}{25\cos^2\alpha} + \frac{y^2}{25} = 1$

We have $e_2^2 = 1 - \frac{25\cos^2\alpha}{25} = \sin^2\alpha$

Given that $e_1 = \sqrt{3}e_2$

$\Rightarrow$ $e_1^2 = 3e_2^2$ $\Rightarrow$ $1 + \cos^2\alpha = 3\sin^2\alpha$

$\Rightarrow$ $2 = 4\sin^2\alpha$ $\Rightarrow$ $\sin\alpha = \frac{1}{\sqrt{2}}$

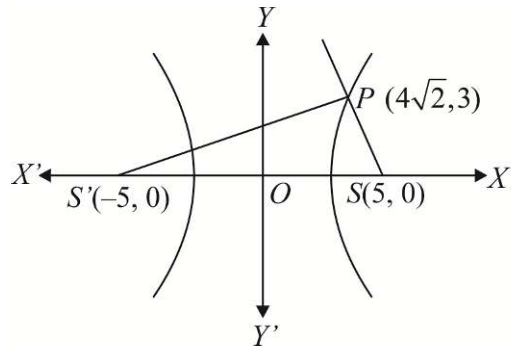
12.(D) The given hyperbola is $\dfrac{x^2}{16}-\dfrac{y^2}{9}=1 \;\Rightarrow\; e=\dfrac{5}{4}$

Its foci are $(\pm 5,0)$.

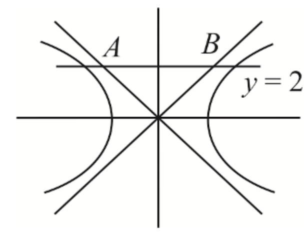
The ray is incident at $P(4\sqrt{2},3)$.

The incident ray passes through $(5,0)$; so the reflected ray must pass through $(-5,0)$.

Its equation is $\dfrac{y-0}{x+5}=\dfrac{3}{4\sqrt{2}+5}$ or $3x-y(4\sqrt{2}+5)+15=0$



13.(A) For two distinct tangents on different branches the point should lie on the line $y=2$ and between A and B (where A and B the points on the asymptotes).



Equations of asymptotes are $4x=\pm 3y$

Solving with $y=2$, we have

$$x=\pm\frac{3}{2} \qquad \therefore \qquad -\frac{3}{2}<\alpha<\frac{3}{2}$$

14.(A) Any tangent to hyperbola forms a triangle with the asymptotes which has constant area ab.

Given $\quad ab=a^2\tan\lambda$

$$\Rightarrow \quad \frac{b}{a}=\tan\lambda \;\Rightarrow\; e^2-1=\tan^2\lambda \;\Rightarrow\; e^2=1+\tan^2\lambda=\sec^2\lambda \;\Rightarrow\; e=\sec\lambda$$

15.(A) If (x_t,y_t) is the point of intersection of given curves, then $\dfrac{\sum_{i=1}^{4}x_i}{4}=\dfrac{1+1}{2}$ and $\dfrac{\sum_{i=1}^{4}y_i}{4}=0$

Now $\quad \dfrac{\sum_{i=1}^{3}x_i}{3}=\dfrac{4-x_4}{3}$ and $\dfrac{\sum_{i=1}^{3}y_i}{3}=-\dfrac{y_4}{3}$

Centroid $\left(\dfrac{\sum_{i=1}^{3}x_i}{3},\dfrac{\sum_{i=1}^{3}y_i}{3}\right)$ lies on the line $y=3x-4$. Hence, $\dfrac{-y_4}{3}=\dfrac{3(4-x_4)}{3}-4 \;\Rightarrow\; y_4=3x_4$

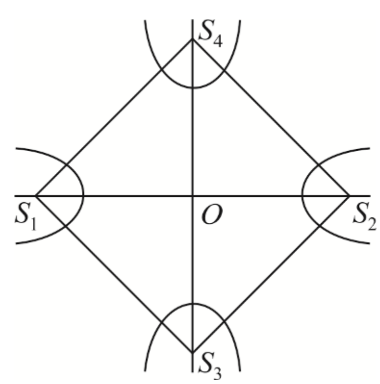
16.(B) Required area $=4$ area $\Delta S_2OS_4=4\times\dfrac{1}{2}ae\times be_1$

$$=4\times\frac{1}{2}\times 2\times 3\times ee_1 \qquad \ldots.(i)$$

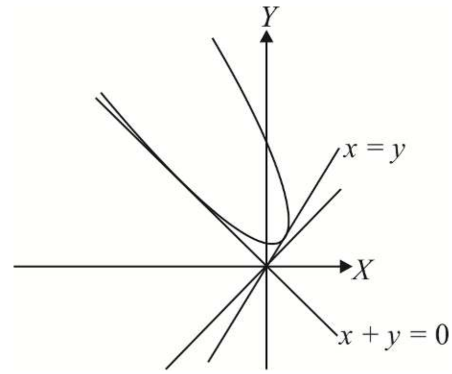
$$b^2=a^2(e^2-1)\Rightarrow e^2=\frac{9}{4}+1=\frac{13}{4}$$

Also $\quad \dfrac{1}{e_1^2}=1-\dfrac{1}{e^2}=1-\dfrac{4}{13}=\dfrac{9}{13}$

$e_1^2=\dfrac{13}{9} \qquad$ Required area $=12\times\dfrac{\sqrt{13}}{2}\times\dfrac{\sqrt{13}}{3}=2\times 13=26$



17-19. **17.(A)** **18.(B)** **19.(C)**



We know that foot of perpendicular from focus upon any tangent lies on the tangent at vertex of the parabola.

Now, if foot of perpendicular of (2, 3) on the line $x-y=0$ is (x_1, y_1),

then $\frac{x_1-2}{1}=\frac{y_1-3}{-1}=-\frac{2-3}{2} \Rightarrow x_1=\frac{5}{2}$ and $y_1=\frac{5}{2}$

If foot of perpendicular of $(2,3)$ on the line $x+y=0$ is (x_2, y_2),

then $\frac{x_2-2}{1}=\frac{y_2-3}{1}=-\frac{2+3}{2} \Rightarrow x_2=-\frac{1}{2}$ and $y_2=\frac{1}{2}$

Now tangent at vertex passes through the points $\left(\frac{5}{2},\frac{5}{2}\right)$ and $\left(-\frac{1}{2},\frac{1}{2}\right)$.

Then, its equation is $y-\frac{1}{2}=\frac{2}{3}\left(x+\frac{1}{2}\right)$ or $4x-6y+5=0$

Latus rectum of the parabola

$$=4\times\text{(Distance of focus from tangent at vertex)} =4\times\left|\frac{8-18+5}{\sqrt{52}}\right|=\frac{10}{\sqrt{13}}$$

We know that $\frac{1}{SP}+\frac{1}{SQ}=\frac{1}{a}$, where a is $\left(\frac{1}{4}\right)$th of latus rectum.

$$\Rightarrow \quad \frac{1}{SP}+\frac{1}{SQ}=\frac{2\sqrt{13}}{5}$$

20-22. **20.(B)** **21.(A)** **22.(C)**

$21x^2-6xy+29y^2+6x-58y-151=0$

$3(x-3y+3)^2+2(3x+y-1)^2=180$

$$\Rightarrow \frac{(x-3y+3)^2}{60}+\frac{(3x+y-1)^2}{90}=1 \Rightarrow \left(\frac{x-3y+3}{\sqrt{1+3^2}\sqrt{6}}\right)^2+\left(\frac{3x+y-1}{3\sqrt{1+3^2}}\right)^2=1$$

Thus C is an ellipse whose lengths of axes are $6, 2\sqrt{6}$.

Hence e = $1/\sqrt{3}$

The major and the minor axes are $x-3y+3=0$ and $3x+y-1=0$, respectively.

Their point of intersection gives the centre of the conic. $\therefore$ Centre $\equiv(0,1)$

23-25.

23.(C) $2a=3$

Distance between the foci (1, 2) and (5, 5) is 5.

$\therefore \quad 2ae=5 \qquad \therefore \quad e=\frac{5}{3}$

Now if e' is eccentricity of the corresponding conjugate hyperbola, then

$$\frac{1}{e^2}+\frac{1}{e'^2}=1 \Rightarrow e'=\frac{5}{4}$$

24.(D) Director circle is given by $(x-h)^2+(y-k)^2=a^2-b^2$

Where (h,k) is centre, i.e. the midpoint of foci $\left(\frac{1+5}{2},\frac{2+5}{2}\right)\equiv\left(3,\frac{7}{2}\right)$.

$$b^2=a^2(e^2-1)=\left(\frac{3}{2}\right)^2\left(\left(\frac{5}{3}\right)^2-1\right)=4$$

Therefore, the director circle is $(x-3)^2+\left(y-\frac{7}{2}\right)^2=\frac{9}{4}-4 \quad\Rightarrow\quad (x-3)^2+\left(y-\frac{7}{2}\right)^2=-\frac{7}{4}$

This does not represent any real point.

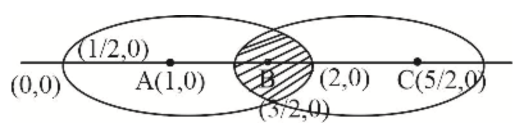
25.(B) Slope of transverse axis is $\frac{3}{4}$.

Therefore, the angle of rotation is $\theta=\tan^{-1}\frac{3}{4}$.

26-28 **26.(A)** **27.(D)** **28.(B)**

$PA+PB<2 \text{ and } PB+PC<2$

Region inside ellipse with foci A and B $2a=2, a=1$



$2ae=\frac{3}{2}-\frac{1}{2}=1$

$b^2=a^2-a^2e^2=1-\frac{1}{4}=\frac{3}{4}$

Equation of ellipse is $\frac{(x-1)^2}{1}+\frac{y^2}{3/4}=1$

$PB+PC<2 \Rightarrow P$ lies inside ellipse with foci B and C

Whose equation is $\frac{(x-2)^2}{1}+\frac{y^2}{3/4}=1$

Locus of P is shown by shaded region which is symmetric about x-axis

Area of region $=4\int_1^{3/2}\frac{\sqrt{3}}{2}\sqrt{1-(x-2)^2}\,dx=\sqrt{3}\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$

29.(B) Any normal to $y^2=8x \text{ is } mx-4m-2m^3$ and this passes through the point P if

$2m^3+(4-h)m=0$ ....(i)

Let m_1,m_2,m_3 be roots, then $m_1+m_2+m_3=0$

$m_1m_3+m_3m_1+m_1m_2=\frac{4-h}{2} \text{ and } m_1m_2m_3=-\frac{k}{2}$

$m_1m_2=-1 \text{ and } m_3=\frac{k}{2}$ ....(ii)

m_3 is the root of equation (i) $\frac{k^3}{4}+(4-h)\frac{k}{2}+k=0$ i.e. $k^2=2(h-6)$

$y^2=2(x-6)$

30.(A) Latus rectum of $y^2 = 8x \, is \, 8$

latus rectum of $y^2 = 2(x-6) \, is \, 2 \;\; ratio = 4:1$

31.(B) From (ii), $m_3 = \frac{k}{2}$ it passes through (x_1, y_1)

$m_3 = \frac{y_1}{2}$

32.(A) Equation of tangent in parametric form $\frac{x-1}{-1/\sqrt{2}} = \frac{y-1}{1/\sqrt{2}} = \pm 3\sqrt{2}$

$A \equiv (4,-2), B \equiv (-2,4)$

Equation of asymptotes (OA and OB)

$y+2 = \frac{-2}{4}(x-4) \Rightarrow 2y+4+x-4=0$

$2y+x=0 \, and \; y-4 = \frac{4}{-2}(x+2) \Rightarrow y-4=-2x-4$

$2y+x=0, \; (2x+y)(x+2y)=0 \Rightarrow 2x^2+2y^2+5xy=0$

33.(C) $m_{OQ} = -\frac{1}{2}, m_{OB} = -2$

$tan\,\theta = \left|\frac{-1/2+2}{1+1}\right| = \frac{3}{4}; \;\; sin\,\theta = \frac{3}{5}, \theta = sin^{-1}\frac{3}{5}$

34.(B) Equation of the hyperbola is $2x^2+2y^2+5xy+\lambda=0$. It passes through $(1,1)$

So, $2+2+5+\lambda=0 \Rightarrow \lambda=-9$

Hyperbola $2x^2+2y^2+5xy=9$

Tangent at $\left(-1, \frac{7}{2}\right)$

$2x(-1)+2y\left(\frac{7}{2}\right)+5\frac{x\left(\frac{7}{2}\right)+(-1)y}{2}=9$

$3x+2y=4$

35.(AC) The line $y=2x+c$ is a tangent to $x^2+y^2=5.$

If $\quad c^2=25$

$\Rightarrow \quad c=\pm 5$

Let the equation of parabola be $y^2=4ax.$ Then

$\frac{a}{2}=\pm 5 \qquad \Rightarrow \qquad a=\pm 10$

$\Rightarrow$ Equation of the parabola is $y^2=\pm 40x$

$\Rightarrow$ Equation of the directrix are $x=\pm 10.$

36.(BCD) Let $(x_1, y_1) \equiv (at^2, 2at)$

Tangent at this point is $ty = x + at^2$.

Any point on this tangent is $\left(h, \left(\frac{h+at^2}{t}\right)\right)$.

Chord of contact of this point with respect to the circle $x^2 + y^2 = a^2$ is

$$hx + \left(\frac{h+at^2}{t}\right)y = a^2 \quad \text{or} \quad (aty - a^2) + h\left(x + \frac{y}{t}\right) = 0$$

Which is a family of straight lines passing through point of intersection of

$$ty - a = 0 \text{ and } x + \frac{y}{t} = 0$$

So, the fixed point is $\left(-\frac{a}{t^2}, \frac{a}{t}\right)$ $\quad \therefore \quad x_2 = -\frac{a}{t^2}, y_2 = \frac{a}{t}$

Clearly, $\quad x_1x_2 = -a^2, y_1y_2 = 2a^2$

Also, $\quad \frac{x_1}{x_2} = -t^4$

$$\frac{y_1}{y_2} = 2t^2 \quad \Rightarrow \quad 4\frac{x_1}{x_2} + \left(\frac{y_1}{y_2}\right)^2 = 0$$

37.(BD) Given parabola is $x^2 - ky + 3 = 0$ $\quad$ or $\quad x^2 = k\left(y - \frac{3}{k}\right)$

Let $\quad x = X, y - \frac{3}{k} = Y$

Then the parabola is $X^2 = kY$

Whose focus is $\left(0, \frac{k}{4}\right)$.

Therefore, the focus of $x^2 = k\left(y - \frac{3}{k}\right)$ is $\left(0, \frac{3}{k} + \frac{k}{4}\right) \equiv (0, 2)$

$\therefore \quad \frac{3}{k} + \frac{k}{4} = 2 \quad \Rightarrow \quad 12 + k^2 = 8k$

$\Rightarrow \quad k^2 - 8k + 12 = 0 \quad \Rightarrow \quad (k-6)(k-2) = 0 \quad \Rightarrow \quad k = 2, 6$

38.(AD) Here $x^2 = -\lambda\left(y + \frac{\mu}{\lambda}\right)$

Therefore, vertex $= \left(0, -\frac{\mu}{\lambda}\right)$ and the directrix is $\left(y + \frac{\mu}{\lambda}\right) - \frac{\lambda}{4} = 0$.

Comparing with the given data, $-\frac{\mu}{\lambda} = 1$ and $\frac{\mu}{\lambda} - \frac{\lambda}{4} = -2$

$\therefore \quad -1 - \frac{\lambda}{4} = -2 \quad$ or $\quad \lambda = 4 \Rightarrow \mu = -4$.

39.(AC) Given that the extremities of the latus rectum are (1, 1) and (1, –1)

$\Rightarrow \quad 4a = 2 \Rightarrow a = \dfrac{1}{2} \quad \Rightarrow$ The focus of the parabola is (1, 0).

$\Rightarrow$ The vertex can be $\left(\dfrac{1}{2}, 0\right)$ and $\left(\dfrac{3}{2}, 0\right)$.

$\Rightarrow$ The equations of the parabola can be $y^2 = 2\left(x - \dfrac{1}{2}\right)$

or $\quad y^2 = -2\left(x - \dfrac{3}{2}\right) \quad \Rightarrow \quad y^2 = 2x - 1 \quad$ or $\quad y^2 = -2x + 3$

40.(AC) Let the possible point be $(t^2, 2t)$. Equation of tangent at this point is $yt = x + t^2$.

It must pass through $(6,5)$. (Since normal to circle always passes through its centre)

$\Rightarrow \quad t^2 - 5t + 6 = 0 \quad \Rightarrow \quad t = 2,3 \quad \Rightarrow$ Possible points are (4, 4), (9, 6).

41.(CD) $t_2 = -t_1 - \dfrac{2}{t_1}$

Also, $\dfrac{2at_1}{at_1^2} \times \dfrac{2at_2}{at_2^2} = -1 \quad \Rightarrow \quad t_1 t_2 = -4$

$\therefore \quad \dfrac{-4}{t_1} = -t_1 - \dfrac{2}{t_1} \quad \Rightarrow \quad t_1^2 + 2 = 4$ and $t_1 = \pm\sqrt{2}$ So, point can be $(2a, \pm 2\sqrt{2}a)$.

42.(ABCD) Any point on the parabola is $P(at^2, 2at)$.

Therefore, midpoint of $S(a,0)$ and $P(at^2, 2at)$ is $R\left(\dfrac{a + at^2}{2}, at\right) \equiv (h,k)$.

$\therefore \quad h = \dfrac{a + at^2}{2}, k = at$

Eliminate 't'

i.e., $\quad 2x = a\left(1 + \dfrac{y^2}{a^2}\right) = a + \dfrac{y^2}{a}$ i.e., $\quad 2ax = a^2 + y^2 \quad$ i.e., $\quad y^2 = 2a\left(x - \dfrac{a}{2}\right)$

It's a parabola with vertex at $\left(\dfrac{a}{2}, 0\right)$, latus rectum $= 2a$

Directrix is $x - \dfrac{a}{2} = -\dfrac{a}{2}$, $\quad$ i.e., $\quad x = 0$

Focus is $x - \dfrac{a}{2} = \dfrac{a}{2}$, $\quad$ i.e., $\quad x = a \quad$ i.e., $\quad (a, 0)$

43.(AC) $r^2 - r - 6 > 0$ and $r^2 - 6r + 5 > 0$

$\Rightarrow \quad (r-3)(r+2) > 0$ and $(r-1)(r-5) > 0$

$\Rightarrow \quad (r < -2 \text{ or } r > 3)$ and $(r < 1$ or $r > 5)$

$\Rightarrow \quad r < -2$ or $r > 5$

44.(AC) Tangent and normal are bisectors of $\angle SPS'$, P(0,0)

Now equation of SP is $y = 3x/2$ and that of $S'P$ is $y = 5x$

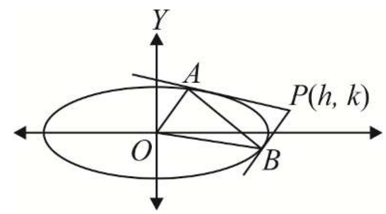
Then equations of their bisectors are $\dfrac{3x-2y}{\sqrt{13}} = \pm\dfrac{5x-y}{\sqrt{26}}$

or $\quad 3x-2y = \pm\dfrac{5x-y}{\sqrt{2}} \Rightarrow$ Lines are $(3\sqrt{2}-5)x+(1-2\sqrt{2})y=0$ and $(3\sqrt{2}+5)x - (2\sqrt{2}+1)y=0$

Now (2, 3) and (1, 5) lie on the same side of $(3\sqrt{2}-5)x+(1-2\sqrt{2})y=0,$ which is equation of tangent.

Points (2, 3) and (1, 5) lie on the different sides of $(3\sqrt{2}+5)x-(2\sqrt{2}+1)y=0,$ which is equation of normal.

45.(AC) Let the point of intersection of tangents at A and B be $P(h,k)$, then equation of AB is



$$\frac{xh}{4}+\frac{yk}{1}=1 \qquad \textbf{....(i)}$$

Homogenizing the equation of ellipse using Equation (i), we get :

$$\frac{x^2}{4}+\frac{y^2}{1}=\left(\frac{xh}{4}+\frac{yk}{1}\right)^2$$

$$\Rightarrow \quad x^2\left(\frac{h^2-4}{16}\right)+y^2(k^2-1)+\frac{2hk}{4}xy=0 \qquad \textbf{....(ii)}$$

Given equation of OA and OB is $x^2+4y^2+\alpha xy=0$ **....(iii)**

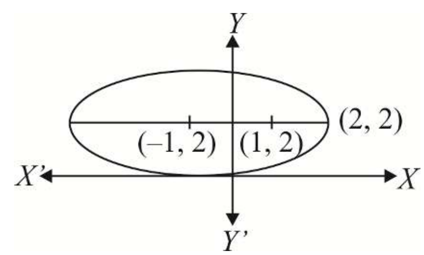
$\because$ Equation (ii) and (iii) represent same lines,

Hence, $\dfrac{h^2-4}{16}=\dfrac{k^2-1}{4}=\dfrac{hk}{2\alpha} \Rightarrow h^2-4=4(k^2-1) \Rightarrow h^2-4k^2=0 \Rightarrow$ Locus $(x-2y)(x+2y)=0$

46.(ABC)

(A) The given equation is $\left(x-\dfrac{1}{13}\right)^2+\left(y-\dfrac{2}{13}\right)^2=\dfrac{1}{a^2}\left(\dfrac{5x+12y-1}{13}\right)^2$

It represents ellipse, if $\dfrac{1}{a^2}<1 \Rightarrow a^2>1 \Rightarrow a>1$



(B) $4x^2+8x+9y^2-36y=-4$

$\Rightarrow \quad 4(x^2+2x+1)+9(y^2-4y+4)=36$

$\Rightarrow \quad \dfrac{(x+1)^2}{9}+\dfrac{(y-2)^2}{4}=1$

Hence, (–1, 2) is centre and (1, 2) lies on the major axis.

Then required minimum distance is 1.

(C) Equation of normal at $P(\theta)$ is $5\sec\theta\, x-4\operatorname{cosec}\theta\, y=25-16,$ and it passes through $P(0,\alpha)$

$\therefore \quad \alpha=\dfrac{-9}{4\operatorname{cosec}\theta} \Rightarrow \alpha=\dfrac{-9}{4}\sin\theta \Rightarrow |\alpha|<\dfrac{9}{4}$

(D) $\dfrac{2b^2}{a}=\dfrac{2a}{3}\Rightarrow 3b^2=a^2 \Rightarrow$ From $b^2=a^2(1-e^2), 1=3(1-e^2)\Rightarrow e=\sqrt{2/3}$

47.(AC) Let $P(h,k)$ be the point of intersection of E_1 and E_2

$$\Rightarrow \quad \frac{h^2}{a^2}+k^2=1 \quad \Rightarrow \quad h^2=a^2(1-k^2) \quad \text{and} \quad \frac{h^2}{1}+\frac{k^2}{a^2}=1 \quad \Rightarrow \quad k^2=a^2(1-h^2)$$

Eliminating a from equations, we get :

$$\frac{h^2}{1-k^2}=\frac{k^2}{1-h^2} \quad \Rightarrow \quad h^2(1-h^2)=k^2(1-k^2) \quad \Rightarrow \quad (h-k)(h+k)(h^2+k^2-1)=0$$

Hence, the locus is a set of curves consisting of the straight lines $y=x, y=-x$ and circle $x^2+y^2=1$.

48.(AC) $f(x)$ is a decreasing function and for major axis to be x-axis

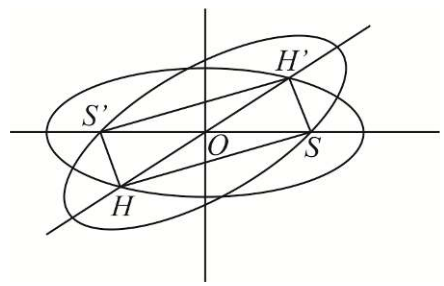
$$f(k^2+2k+5)>f(k+11) \quad \Rightarrow \quad k^2+2k+5<k+11 \quad \Rightarrow \quad k\in(-3,2)$$

Then for the remaining values of k, i.e., $k\in(-\infty,-3)\cup(2,\infty)$, major axis is y-axis.

49.(ABC) Clearly O is the midpoint of SS' and HH'

$\Rightarrow$ Diagonals of quadrilateral $HSH'S'$ bisect each other, so it is a parallelogram.

Let $H'OH=2r \Rightarrow OH=r=ae'$



H lies on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (suppose)

$$\therefore \quad \frac{r^2\cos^2\theta}{a^2}+\frac{r^2\sin^2\theta}{b^2}=1$$

$$e'^2\cos^2\theta+\frac{e'^2\sin^2\theta}{1-e^2}=1 \qquad [\because b^2=a^2(1-e^2)]$$

$$\Rightarrow \quad e'^2\cos^2\theta-\frac{e'^2\cos^2\theta}{1-e^2}=1-\frac{e'^2}{1-e^2} \quad \Rightarrow \quad \cos^2\theta=\frac{1}{e^2}+\frac{1}{e'^2}-\frac{1}{e^2e'^2} \quad \Rightarrow \quad \theta=\cos^{-1}\sqrt{\frac{1}{e^2}+\frac{1}{e'^2}-\frac{1}{e^2e'^2}}$$

$$\text{For } \theta=90^\circ, \frac{e^2+e'^2}{e^2e'^2}=\frac{1}{e^2e'^2} \quad \Rightarrow \quad e^2+e'^2=1$$

50.(AD) The equation of the tangent at $(t^2,2t)$ to the parabola $y^2=4x$ is

$$2ty=2(x+t^2) \quad \Rightarrow \quad ty=x+t^2$$

$$\Rightarrow \quad x-ty+t^2=0 \qquad \textbf{....(i)}$$

The equation of the normal at point $(\sqrt{5}\cos\theta, 2\sin\theta)$ on the ellipse $4x^2+5y^2=20$ is

$$\Rightarrow \quad (\sqrt{5}\sec\theta)x-(2\operatorname{cosec}\theta)y=5-4$$

$$\Rightarrow \quad (\sqrt{5}\sec\theta)x-(2\operatorname{cosec}\theta)y=1 \qquad \textbf{....(ii)}$$

Given that equations (i) and (ii) represent the same line.

$$\Rightarrow \quad \frac{\sqrt{5}\sec\theta}{1}=\frac{-2\operatorname{cosec}\theta}{-t}=\frac{-1}{t^2} \quad \Rightarrow \quad t=\frac{2}{\sqrt{5}}\cot\theta \text{ and } t=-\frac{1}{2}\sin\theta$$

$$\Rightarrow \quad \frac{2}{\sqrt{5}}\cot\theta=-\frac{1}{2}\sin\theta \quad \Rightarrow \quad 4\cos\theta=-\sqrt{5}\sin^2\theta$$

$$\Rightarrow \quad 4\cos\theta=-\sqrt{5}(1-\cos^2\theta) \quad \Rightarrow \quad \sqrt{5}\cos^2\theta-4\cos\theta-\sqrt{5}=0$$

$\Rightarrow \quad (\cos\theta-\sqrt{5})(\sqrt{5}\cos\theta+1)=0 \quad \Rightarrow \quad \cos\theta=-\frac{1}{\sqrt{5}} \quad [\because \cos\theta\neq\sqrt{5}]$

$\Rightarrow \quad \theta=\cos^{-1}\left(-\frac{1}{\sqrt{5}}\right)$

Putting $\cos\theta=-\frac{1}{\sqrt{5}}$ in $t=-\frac{1}{2}\sin\theta$, we get : $t=-\frac{1}{2}\sqrt{1-\frac{1}{5}}=-\frac{1}{\sqrt{5}}$

Hence, $\theta=\cos^{-1}\left(-\frac{1}{\sqrt{5}}\right)$ and $t=-\frac{1}{\sqrt{5}}$.

51.(A) We have, $\left|\sqrt{x^2+(y-1)^2}-\sqrt{x^2+(y+1)^2}\right|=K$

Which is equivalent to $|S_1P-S_2P|=$ constant, where $S_1\equiv(0,1), S_2\equiv(0,-1)$ and $P\equiv(x,y)$.

The above equation represents a hyperbola. So, we have $2a=K$

And $\quad 2ae=S_1S_2=2$

Where $2a$ is the transverse axis and e is the eccentricity.

Dividing, we have $e=\frac{2}{K}$

Since, $e>1$ for a hyperbola, therefore $K<2$.

Also, K must be a positive quantity. So, we have $K\in(0,2)$.

52.(ABC) $3x^4-2(19y+8)x^2+[(19y^2)+(10)^2+(10)^2+y^4+y^4+8^2]=2(19\times10y+10y^2-8y^2)$

$\Rightarrow \quad 3x^4-2(19y+8)x^2+(19y-10)^2+(10-y^2)^2+(y^2+8)^2=0$

$\Rightarrow \quad 3x^4-2(19y-10+10-y^2+y^2+8)x^2+(19y-10)^2+(10-y^2)^2+(y^2+8)^2=0$

$\Rightarrow \quad [x^2-(19y-10)]^2+[x^2-(10-y^2)]^2+[x^2-(y^2+8)]^2=0$

$\Rightarrow \quad x^2-19y+10=0, x^2-10+y^2=0$ and $x^2-y^2-8=0$

The three curves represented by the given equation are $x^2=19y-10$ (parabola), $x^2+y^2=10$ (circle) and $x^2-y^2=8$ (hyperbola).

53.(B) Locus of point of intersection of perpendicular tangents is director circle of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Equation of director circle is $x^2+y^2=a^2-b^2$ which is real if $a>b$.

54.(ABCD) Given hyperbola can be written as $\frac{(x-1)^2}{16}-\frac{(y-1)^2}{9}=1$

$\Rightarrow \quad \frac{X^2}{16}-\frac{Y^2}{9}=1 \quad$ (where $X=x-1, Y=y-1$)

$$e=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+\frac{9}{16}}=\frac{5}{4}$$

Directrices are $X = \pm\frac{a}{e} \Rightarrow x-1 = \pm\frac{16}{5} \Rightarrow x = \frac{21}{5}$ and $x = -\frac{11}{5}$

Length of latus rectum $= \frac{2b^2}{a} = \frac{9}{2}$

The foci are given by $X = \pm ae, Y = 0 \Rightarrow (6,1),(-4,1)$

55.(AD) Circle with points $P(2t_1, 2/t_1)$ and $Q(2t_2, 2/t_2)$ as diameter is given by

$$(x-2t_1)(x-2t_2)+\left(y-\frac{2}{t_1}\right)\left(y-\frac{2}{t_2}\right)=0 \qquad \textbf{....(i)}$$

Also, slope of PQ is given by $-\frac{1}{t_1t_2} = 1 \Rightarrow t_1t_2 = -1$

Hence, from (1), circle is $(x^2+y^2-8)-2(t_1+t_2)(x-y)=0$

Which is of the form $S+\lambda L=0$.

Hence, circles pass through the points of intersection of the circle $x^2+y^2-8=0$ and the line $x=y$.

The points of intersection are (2, 2) and (–2, –2).

56.(BCD) $(x-\alpha)^2+(\gamma-\beta)^2 = k(lx+my+n)^2$

$$\Rightarrow \quad \sqrt{(x-\alpha)^2+(y-\beta)^2} = \sqrt{k}\sqrt{l^2+m^2}\frac{|lx+my+n|}{\sqrt{l^2+m^2}} \quad \Rightarrow \quad \frac{PS}{PM} = \sqrt{k}\sqrt{l^2+m^2}$$

where point $P(x,y)$ is any point on the curve.

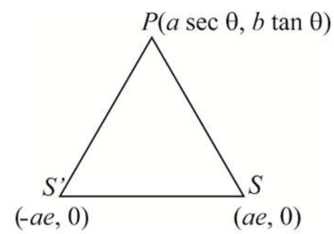
Fixed point $S(\alpha,\beta)$ is focus and fixed line $lx+my+n=0$ is directrix.

If $k(l^2+m^2)=1, P$ lies on parabola. If $k(l^2+m^2)<1, P$ lies on ellipse.

If $k(l^2+m^2)>1, P$ lies on hyperbola. If $k=0, P$ lies on a point circle.

57.(AB) Let (h,k) be excentre opposite $\angle S'$, then

$$h = \frac{+ae(ae\sec\theta+a)+ae(-a+ae\sec\theta)+2ae(a\sec\theta)}{2a(e-1)}$$



$h = ae\sec\theta\left(\frac{e+1}{e-1}\right), k = \frac{be\tan\theta}{e-1}$ (for $a\sec\theta>0$), locus is a hyperbola

Again let (h,k) be excentre opposite $\angle P$

$$\therefore \quad h = \frac{-2a^2e\sec\theta+a^2e^2\sec\theta+a^2e-a^2e^2\sec\theta+a^2e}{2ae\sec\theta-2ae}$$

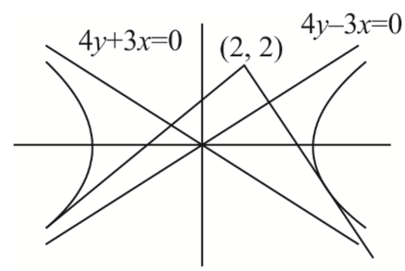
$\Rightarrow \quad h=-a, \text{for } a\sec\theta>0$ or $h=a, \text{for } a\sec\theta<0 \Rightarrow$ locus is tangent at vertex.

58.(CD) Equations of asymptotes are

$$4y-3x=0 \text{ and } 4y+3x=0$$

As point (2, 2) lies above the asymptotes.

Hence, points of contact of the tangents are in III and IV quadrants.



59.(ABC) Locus of point of intersection of perpendicular tangents is director circle $x^2+y^2=a^2-b^2$.

Now, $e^2=1+\frac{b^2}{a^2}$

If $a^2>b^2$, then there are infinite (or more than 1) points on the circle $\Rightarrow e^2<2 \Rightarrow e<\sqrt{2}$.

If $a^2<b^2$, then there does not exist any point on the plane $\Rightarrow e^2>2 \Rightarrow e>\sqrt{2}$.

If $a^2=b^2$, there is exactly one point (centre of the hyperbola) $e=\sqrt{2}$.

60.(BC) Equation of the chords of contact of the given points $P(x_1,y_1)$ *and* $Q(x_2,y_2)$ are respectively

$xy_1+x_1y=2c^2$ ....(i) $\qquad xy_2+x_2y=2c^2$ ....(ii)

Equation of any conic section passing through the points of intersection of the curve $xy=c^2$ and the lines (i) and (ii) is given by $\left(xy_1+x_1y-2c^2\right)\left(xy_2+x_2y-2c_2\right)+\lambda\left(xy-c^2\right)=0$ ....(iii)

It represents a circle if $y_1y_2=x_1x_2$ *and* $x_1y_2+x_2y_1+\lambda=0$ ....(iv)

Also, the circle (iii) passes through the points P and Q, so $\left(2x_1y_1-2c^2\right)\left(x_1y_2+x_2y_1-2c^2\right)+\lambda\left(x_1y_1-c^2\right)=0$

$2\left(x_1y_2+x_2y_1-2c^2\right)+\lambda=0$ ....(v)

From equation (iv) and (v) we have $2\left(-\lambda-2c^2\right)+\lambda=0$

$\lambda=-4c^2$; $x_1y_2+x_2y_1=4c^2$

61.(CD) $\tan(\alpha-\beta)=\tan\{(\theta+\alpha)-(\theta+\beta)\}=\dfrac{\frac{x}{a}-\frac{y}{b}}{1+\frac{x}{a}\frac{y}{b}}$

Or $xy-(bx-ay)\cot(\alpha-\beta)+ab=0$, which represents a rectangular hyperbola

We may rewrite the equation as $[x+a\cot(\alpha-\beta)][y-b\cot(\alpha-\beta)]+ab\,\mathrm{cosec}^2(\alpha-\beta)=0$

Thus, the asymptotes are $x+a\cot(\alpha-\beta)=0$ and $y-b\cot(\alpha-\beta)=0$

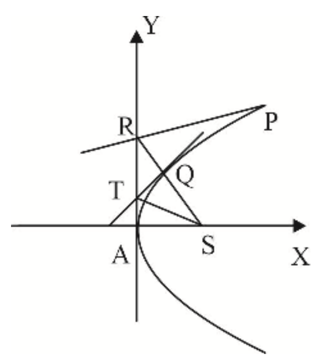
62.(ABCD) $R\equiv(1,1), T\equiv(2,2)$

Equation of RS is $4x-3y-1=0$

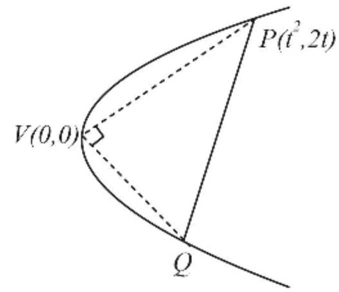
Equation of TS is $3x-2y-2=0$

Length of latus rectum $=4\times 1/\sqrt{2}=2\sqrt{2}$

Axis is $x+y-9=0 \Rightarrow vertex\equiv\left(\frac{9}{2},\frac{9}{2}\right)$



63.(AB) m of $PV=\frac{2t-0}{t^2-0}=\frac{2}{t}$



the equation of QV is $y=-\frac{t}{2}x$

solving it with $y^2=4x, Q=\left(\frac{16}{t^2},\frac{-8}{t}\right)$

Now, $ar(\Delta PVQ)=\frac{1}{2}\times PV.VQ=20(given)$

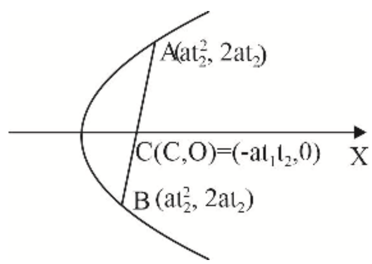
$PV^2.VQ^2=40^2$ Or $\left(\left(t^2\right)^2+(2t)^2\right)\left\{\left(\frac{16}{t^2}\right)^2+\left(\frac{-8}{t}\right)^2\right\}=40^2$

$\left(t^2\right)\left(4+t^2\right)\frac{1}{t^2}\left(\frac{256}{t^2}+64\right)=40^2$

$\left(\frac{256\times 4}{t^2}+256+256+64t^2\right)=40^2$

$\left(t^2-16\right)\left(t^2-1\right)=0 \therefore t=\pm 4,\pm 1$

64.(BD) Let $C(c, 0)$



$t_1t_2=\frac{-c}{a}\Rightarrow c=-at_1t_2$

$\frac{AC^2}{CB^2}=\frac{\left(at_1^2+at_1t_2\right)^2+4a^2t_1^2}{\left(at_2^2+at_1t_2\right)^2+4a^2t_2^2}=\frac{t_1^2}{t_2^2}$

$\frac{CB}{AC}=\pm\frac{t_2}{t_1}\Rightarrow\frac{CB}{AC}+1=\pm\frac{t_2}{t_1}+1$

$\frac{AB}{AC}=\frac{t_1+t_2}{t_1}=3$

$\frac{AB}{AC}=\frac{t_1-t_2}{t_1}=3$

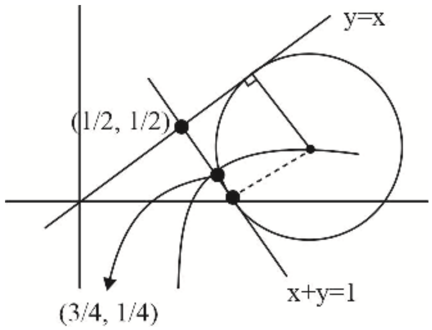
$t_2+t_1=3t_1 \, or \, t_1-t_2=3t_1$

$2t_1-t_2=0 \, or \, 2t_1+t_2=0$

$6t_1^2-2t_2^2-t_1t_2=0=6t_1^2+3t_1t_2-4t_1t_2-2t_2^2$

$\left(3t_1-2t_2\right)\left(2t_1+t_2\right)=0$

65.(BCD)



$$4a = 2\left(\frac{|0-1|}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$$

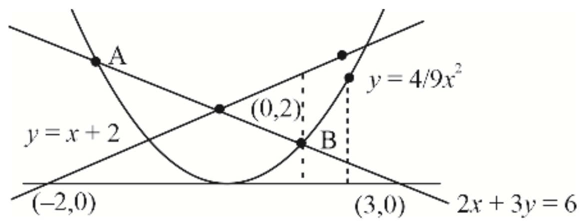
66.(AC) For point A, B

$$2x + 3\left(\frac{4}{9}x^2\right) = 6$$

$$2x^2 + 3x - 9 = 0$$

$$2x^2 + 6x - 3x - 9 = 0 = (2x-3)(x+3) = 0$$

$$A \equiv (-3, 4) \; and \; B \equiv \left(\frac{3}{2}, 1\right)$$

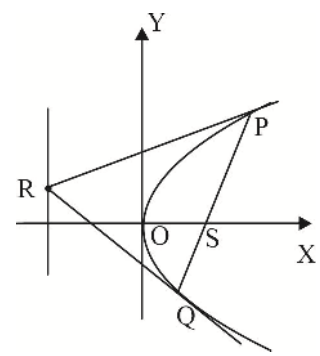


$$\frac{3}{2}\alpha = -3 \Rightarrow \alpha = -2$$

$$\frac{3}{2}\alpha = \frac{3}{2} \Rightarrow \alpha = 1$$

$$\alpha \in (-\infty, -2) \cup (0, 1)$$

67.(ABD) The tangents at the ends of a focal chord of a parabola intersect at the directrix perpendicularly. So, the orthocenter being the vertex of the triangle at which right angle occurs therefore lies on the directrix. Also, the circumcircle of ΔPQR will have PQ as diameter.



Area of circumcircle $= \frac{\pi}{4}(PQ)^2$

Further the minimum length of a focal chord equals to the latus rectum, so the minimum area of the circumcircle.

$$= \frac{\pi}{4}(4a)^2 = 4\pi a^2$$

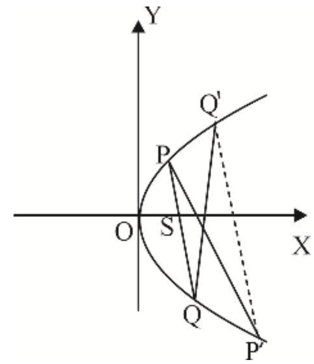
Again the tangents at P and Q are say

$t_1 y = x + at_1^2 \; and \; t_2 y = x + at_2^2$ respectively and $t_1 t_2 = -1$ their distance from the origin are

$$d_1 = \frac{at_1^2}{\sqrt{1+t_1^2}} \; and \; d_2 = \frac{at_2^2}{\sqrt{1+t_2^2}} = \frac{a}{\sqrt{t_1^2 + t_1^4}}$$

Clearly, $d_1 \; and \; d_2$ cannot be equal for all values of t_1 . So, the incentre cannot lie at the vertex.

68.(BC) Let the coordinates of P and Q be $\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$ then $t_1 t_2 = -1$ and $PQ = a\left(t_1 - t_2\right)^2$



If the normal at P meets the parabola again at $P'\left(t_1'\right)$, then $t_1' = -t_1 - \dfrac{2}{t_1}$ ………(i)

And if the normal at Q meets the parabola again at $Q'\left(t_2'\right)$ then $t_2' = -t_2 - \dfrac{2}{t_2}$ …….(ii)

From the equation (i) and (ii) we have,

$$t_1' + t_2' = -\left(t_1 + t_2\right) - \left(\frac{2}{t_1} + \frac{2}{t_2}\right) = -\left(t_1 + t_2\right) - \frac{2}{t_1 t_2}\left(t_1 + t_2\right) = t_1 + t_2$$

$$t_1' - t_2' = t_2 - t_1 + \frac{2}{t_2} - \frac{2}{t_1} = \left(t_2 - t_1\right) + \frac{2}{t_1 t_2}\left(t_1 - t_2\right) = 3\left(t_2 - t_1\right) \text{ …….(iv)}$$

Now the slope of the line PQ is $m_1 = \dfrac{2}{t_1 + t_2}$ and the slope of the line $P'Q'$ is $m_2 = \dfrac{2}{t_1' + t_2'} = \dfrac{2}{t_1 + t_2}$

Thus, the lines PQ and $P'Q'$ are parallel,

Further $P'Q' = a\left|t_1' - t_2'\right|\sqrt{\left(t_1' + t_2'\right)^2 + 4} = 3a\left(t_1 - t_2\right)^2 = 3PQ$

69.(AD) Equation of a normal to $y^2 = 4ax$ is $y = mx - 2am - am^3$ ……….(i)

Equation of the normal to $y^2 = 4c\left(x - d\right)$ with the same slope 'm' is $y = m\left(x - d\right) - 2cm - cm^3$ ……..(ii)

The equation (i) and (ii) represent the same straight lines if $-2am - am^3 = -dm - 2cm - cm^3$

$\left(c - a\right)m^3 = \left(2a - d - 2c\right)m$

Either $m = 0$, which implies that one normal is always X-axis, Or $m^2 = \dfrac{2a - d - 2c}{c - a}$

For non-zero roots $m^2 > 0$

$\dfrac{2a - d - 2c}{c - a} > 0$

$c - a > 0$ and $2a - d - 2c > 0$; $c - a < 0$ and $2a - d - 2c < 0$

70.(AC) Let $y^2 = 4ax$ be the parabola and Q be the point $\left(h, 0\right)$. If P and R are $\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$, then the equation of PR is $\left(t_1 + t_2\right)y = 2x + 2at_1 t_2$

If this line passes through the point $\left(h, 0\right)$, then we have $0 = 2h + 2at_1 t_2 \Rightarrow t_1 t_2 = -\dfrac{h}{a}$

Now $OM.ON = at_1^2 . at_2^2 = a^2\left(t_1 t_2\right)^2 = a^2\left(-\dfrac{h}{a}\right)^2 = h^2 = OQ^2$

And $PM.RN = \left|2at_1\right| . \left|2at_2\right| = 4a^2\left|t_1 t_2\right| = 4a^2\left|\left(-\dfrac{h}{a}\right)\right| = 4ah = 4aOQ$

71.(ABD) Equation of the parabola is $x^2-8x-16y=0 \Rightarrow x^2-8x+16=16+16y$; $(x-4)^2=16(y+1)$

Shifting the origin to $(4,-1)$, above equation becomes $X^2=16Y$ ....(i)

Equation of a normal to (i) is $X=mY-8m-4m^3$ and foot of the normal is $\left(-8m,4m^2\right)$.

It passes through $(7,14)$, which transforms to $(3,15)$ in the new system. So, we have

$3=15m-8m-4m^3 \Rightarrow 4m^3-7m+3=0$

$(m-1)(2m-1)(2m+3)=0 \Rightarrow m=1,\frac{1}{2},-\frac{3}{2}$

The coordinates of the feet in the shifted coordinates system are $(-8,4),(-4,1)$ and $(12,9)$

The required coordinates are therefore (changing to original coordinate system) $(-4,3),(0,0)\,and\,(16,8)$

72. (ABCD) Point of contact will lie an $y=x$ (let say (α,α)) and slope of tangent can be ± 1 and $\alpha = a\,\alpha^2+a\alpha+\frac{1}{24}$

eliminating α we get $\left(\frac{\pm 1-a}{2a}\right)^2=a\left(\frac{\pm 1-a}{2a}\right)^2+a+\frac{1}{24}$

On solving we get $a=\frac{2}{3},\frac{3}{2},\frac{13\pm\sqrt{601}}{12}$

73.(ABC) Facts

74.(BC) By properties $PA=PB=\sqrt{10}$

$\frac{x-1}{-1/\sqrt{10}}=\frac{y-5}{3/\sqrt{10}}=\pm\sqrt{10}$

$A=(0,8);\ B=(2,2)$

Slopes of asymptotes are -7 and 1

If angles between asymptotes be θ then $tan\,\theta=\left|\frac{-7-1}{1-7}\right|=\frac{4}{3}$

Now, $2tan^{-1}\frac{b}{a}=tan^{-1}\frac{4}{3} \Rightarrow \frac{\frac{2b}{a}}{1-\frac{b^2}{a^2}}=\frac{4}{3}$

$\frac{b}{a}=\frac{1}{2}$ ....(i)

$\frac{b^2}{a^2}=e^2-1 \Rightarrow e=\frac{\sqrt{5}}{2}$

Also, area of $\Delta CAB=ab$

$ab=\frac{1}{2}\left|1(8-2)+0(2-1)+2(1-8)\right|=4$ ....(ii)

$b^2=2, a=2b=2\sqrt{2}$, length of latus rectum $=\frac{2b^2}{a}=\frac{4}{2\sqrt{2}}=\sqrt{2}$

75.(BD) As tangents are perpendicular to each other and its points of intersections $P\left(\frac{7}{5},\frac{-4}{5}\right)$ lies on directrix. AB is focal chord

Equation of AB is $x-y=4$, as focus is (α,β) $so\ \alpha-\beta=4 \Rightarrow \beta=\alpha-4$

Now $\angle ASP = 90°$

$$\frac{-2-\alpha+4}{2-\alpha}\times\frac{-\frac{4}{5}-\alpha+4}{\frac{7}{5}-\alpha}=-1\ ;\quad \frac{2-\alpha}{2-\alpha}\times\frac{\frac{16}{5}-\alpha}{\frac{7}{5}-\alpha}=-1$$

$$\frac{16}{5}-\alpha=\alpha-\frac{7}{5};\ \alpha=\frac{23}{10}$$

Focus is $\left(\frac{23}{10},-\frac{17}{10}\right)$

Also, $l_1=AS=\frac{3\sqrt{2}}{10}, l_2=BS=\frac{27\sqrt{2}}{10}$

$$\frac{1}{a}=\frac{1}{l_1}+\frac{1}{l_2}\Rightarrow a=\frac{27\sqrt{2}}{100}\Rightarrow 4a=\frac{27\sqrt{2}}{25}$$

76.(BC) Ellipse $\frac{x^2}{4}+\frac{y^2}{1}=1\Rightarrow$ eccentricity $e=\frac{\sqrt{3}}{2}$; Extremities of latus rectums are $\left(\pm\sqrt{3},\pm\frac{1}{2}\right)$

Given $y_1<0, y_2<0$

$P\left(\sqrt{3},-\frac{1}{2}\right)$ and $Q\left(-\sqrt{3},-\frac{1}{2}\right)$

$PQ=2\sqrt{3}=$ length of latus rectum of parabola

Two parabolas with vertices $\left(0,-\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$ and $\left(0,\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$ are possible

77.(AB) Point 'P' clearly lies on the directrix of $y^2=8x$

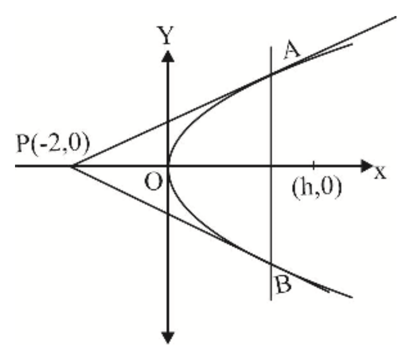
Thus slope of PA and PB are 1 and -1 respectively

Equation of PA $: y=x+2$, equation of $PB: y=-x-2$, equation of $AB: x=2$

Let the centre of the circle be $(h,0)$ and radius be 'r'

$$\Rightarrow\left|\frac{h+2}{\sqrt{2}}\right|=\frac{|h-2|}{1}=r\Rightarrow h^2+4+4h=2\left(h^2+4-4h\right)\Rightarrow h^2-12h+4=0$$

$$h=\frac{12\pm8\sqrt{2}}{2}=6\pm4\sqrt{2}\Rightarrow|h-2|=4\left(\sqrt{2}-1\right),4\left(\sqrt{2}+1\right)$$



78.(AB) Any point on the line $x+y=1$ can be taken $(t,1-t)$

Equation of the chord, with this as mid-point is

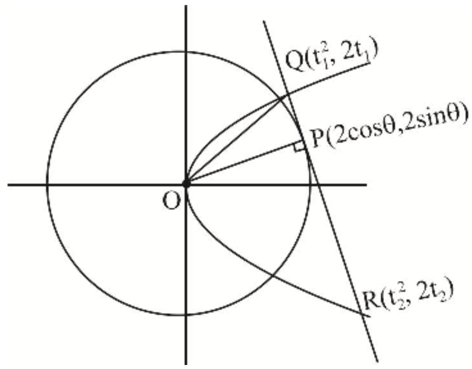
$y(1-t)-2a(x+t)=(1-t)^2-4at$, it passes through $(a,2a)$

So $t^2-2t+2a^2-2a+1=0$, this should have 2 distinct real roots so $a^2-a<0, 0<a<1$,

so length of latus-rectum <4

79.(AB) It is a well-known property of a parabola that a tangent and normal to it from focus intersect at tangent at vertex.

80.(D) $OQ^2 = OP^2 + PQ^2$



$t_1^4 + 4t_1^2 = 5$

$t_1 = 1$

$Q \equiv (1, 2)$

$PQ = 1 \Rightarrow \cos\theta + 2\sin\theta = 2$

$\cos\theta = \frac{4}{5}, \sin\theta = \frac{3}{5}$

$P = \left(\frac{8}{5}, \frac{6}{5}\right)$

Now R lies on tangent $4x + 3y = 10$

$t_2 = -\frac{5}{2}$; $\quad R = \left(\frac{25}{4}, -5\right)$ and $S = \left(-\frac{5}{2}, -\frac{3}{2}\right)$

81. **[A-r] [B-s] [C-p] [D-q]**

Locus of point of intersection of perpendicular tangent is directrix which is $12x - 5y + 3 = 0$.

Parabola is symmetrical about its axis, which is a line passing through the focus (1, 2) and perpendicular to the directrix, which has equation $5x + 12y - 29 = 0$.

Minimum length of focal chord occurs along the latus rectum line, which is a line passing through the focus and parallel to directrix, i.e,. $12x - 5y - 2 = 0$.

Locus of foot of perpendicular from focus upon any tangent is tangent at the vertex, which is parallel to directrix and equidistant from directrix and latus rectum line, i.e., $12x - 5y + \lambda = 0$

where $\frac{|\lambda - 3|}{\sqrt{12^2 + 5^2}} = \frac{|\lambda + 2|}{\sqrt{12^2 + 5^2}} \Rightarrow \lambda = \frac{1}{2}$. Hence, equation of tangent at vertex is $24x - 10y + 1 = 0$.

82. **[A-p, r] [B-p, q, r] [C-q] [D-q, s]**

Points through which perpendicular tangent can be drawn to the parabola $y^2 = 4x$ lie on the directix. Points (–1, 2) and (–1, –5) lie on the directrix. Also from these points only one normal can be drawn.

83. **[A-p, q, r, s] [B-p, q, r, s] [C-q, r, s] [D-p]**

(A) Equation of tangent at $\left(\frac{\cos\theta}{2}, \frac{\sin\theta}{3}\right)$ is $2x\cos\theta + 3y\sin\theta = 1$ which is parallel to the given line $8x = 9y$

$\therefore \quad \cos\theta = \pm\frac{4}{5}, \sin\theta = \mp\frac{3}{5}$

Hence, points are $\left(\frac{2}{5}, -\frac{1}{5}\right)$ and $\left(-\frac{2}{5}, \frac{1}{5}\right)$.

Distance between the points is $\sqrt{\frac{16}{25} + \frac{4}{25}} = \frac{2}{\sqrt{5}}$ which is less than 1.

(B) The given equation is $\frac{(x+1)^2}{9}+\frac{(y+2)^2}{25}=1 \Rightarrow e^2=1-\frac{9}{25}=\frac{16}{25}\Rightarrow e=\frac{4}{5}$

Hence, the foci are $S,S'\equiv(-1,-2\pm 4)\equiv S(-1,2)$ and $S'(-1,-6)$

The required sum of distances $=2+6=8$.

(C) Equation of normal at $(3\cos\theta,2\sin\theta)$ is $3x\sec\theta-2y\mathrm{cosec}\theta=5$

which is parallel to the given line $2x+y=1$. Therefore, $\cos\theta=\mp\frac{3}{5},\sin\theta=\pm\frac{4}{5}$

Hence, points are $\left(\frac{-9}{5},\frac{8}{5}\right)$ and $\left(\frac{9}{5},-\frac{8}{5}\right)$. The required sum of distances $=\frac{16}{5}$.

(D) Consider any point $(t,t+2),t\in R$, on the line $x-y+2=0$.

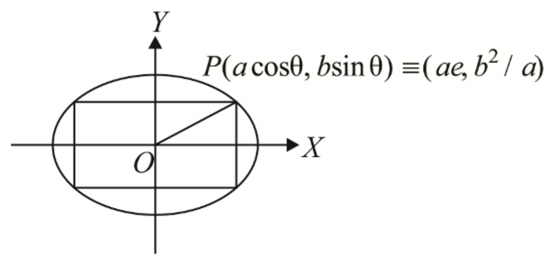
The chord of contact of this point w.r.t. ellipse is $xt+2y(t+2)=2 \Rightarrow (4y-2)+t(x+2y)=0$

These lines passes through point of intersection of lines $4y-2=0$ and $x+2y=0$

$\therefore$ $x=-1$, y = 1/2 Hence, the point is $(-1,1/2)$, whose distance from (2, 1/2) is 3.

84. **[A-q]** **[B-q]** **[C-s]** **[D-p]**

(A) Let one of the vertices of the rectangle be $P(a\cos\theta,b\sin\theta)$.



Then its area $A=2a\cos\theta\times 2b\sin\theta=2ab\sin 2\theta$.

Hence, $A_{\max}=2ab$

Now area of rectangle formed by extremities of

$LR=(2ae)(2b^2/a)=4eb^2$.

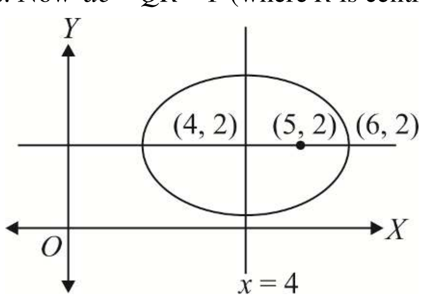
Given that $2ab=4eb^2\Rightarrow\frac{2b}{a}e=1 \Rightarrow \frac{4b^2}{a^2}e^2=1$

$\Rightarrow 4(1-e^2)e^2=1 \Rightarrow 4e^4-4e^2+1=0 \Rightarrow (2e^2-1)^2=0 \Rightarrow e=\frac{1}{\sqrt{2}}$

(B) For ellipse, distance between the foci, $2ae=8$ and length of semi-minor axis, is $b=4$.

Now, $b^2=a^2-a^2e^2 \Rightarrow 16=a^2-16 \Rightarrow a^2=32 \Rightarrow e=\sqrt{1-\frac{16}{32}}=\frac{1}{\sqrt{2}}$

(C) Normal at point $P(6,2)$ to the ellipse passes through its focus $Q(5,2)$. Then P must be extremity of the major axis. Now $ae=QR=1$ (where R is centre) and $a-ae=1$.



$\therefore$ $a=2$

$b^2=a^2-a^2e^2=4-1=3 \Rightarrow e=\sqrt{1-\frac{3}{4}}=\frac{1}{2}$

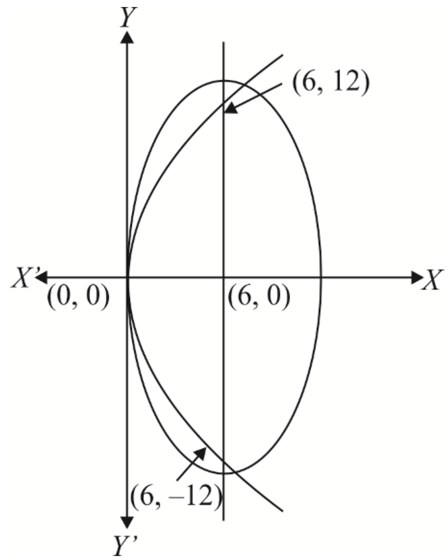
(D) Extremities of LR of parabola $y^2 = 24x$ are $(6, \pm 12)$.

For ellipse, $2be = 24$ and extremity of minor axis is (0, 0). Hence, $a = 6$.

Now, $a^2 = b^2 - b^2e^2$

$\Rightarrow \quad b^2 = 36 + 144 = 180$

$\Rightarrow \quad e = \sqrt{1 - \frac{36}{180}} = \sqrt{1 - \frac{1}{5}} = \frac{2}{\sqrt{5}}$



85. **[A-p, s]** **[B-q]** **[C-r]** **[D-p, q, s]**

(A) We must have

$e_1 < 1 < e_2 \Rightarrow f(1) < 0 \Rightarrow 1 - a + 2 < 0 \Rightarrow a > 3$

(B) We must have

I. $D \ge 0$ or $a^2 - 8 \ge 0$ or $a \in (-\infty, -2\sqrt{2}] \cup [2\sqrt{2}, \infty)$

II. $1 \cdot f(1) > 0$ or $1 - a + 2 > 0$ or $a < 3$ **III.** $\frac{a}{2} > 1 \Rightarrow a > 2$

From equation (i), (ii) and (iii) we have $a \in (2\sqrt{2}, 3)$.

(C) We must have $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$

$\Rightarrow \quad \frac{(e_1 + e_2)^2 - 2e_1e_2}{e_1^2 e_2^2} = 1 \quad \Rightarrow \quad \frac{a^2 - 4}{4} = 1 \quad \Rightarrow \quad a = \pm 2\sqrt{2}$

(D) We must have $\sqrt{2} < e_1$ since product of roots is 2, the other root must lie in $(0, \sqrt{2})$

$\Rightarrow \quad f(\sqrt{2}) < 0 \quad \Rightarrow \quad 2 - a\sqrt{2} + 2 < 0 \quad \Rightarrow \quad a > 2\sqrt{2}$

86.(A) (P) Let an end of latus rectum be $\left(ae, b\sqrt{1-e^2}\right)$, then the equation of normal at this end is $\frac{x - ae}{ae/e^2} = \frac{y - b\sqrt{1-e^2}}{\frac{b\sqrt{1-e^2}}{b^2}}$

It will pass through the end $(0, -b)$ if $\quad -a^2 = \frac{-b^2\left(1 + \sqrt{1-e^2}\right)}{\sqrt{1-e^2}}$ or $\frac{b^2}{a^2} = \frac{\sqrt{1-e^2}}{1 + \sqrt{1-e^2}}$

or $1 + \sqrt{1-e^2} = \frac{\sqrt{1-e^2}}{1-e^2} \Rightarrow \sqrt{1-e^2} + \left(1 - e^2\right) = 1 \Rightarrow e^4 + e^2 = 1$

(Q) Let PQ be the double ordinate of the parabola $y^2 = 4ax$. R and S be the points of trisection of $PQ \Rightarrow PR = RS = SQ$

If $R(h, k) \Rightarrow S(h, -k)$

$RS = 2k, PR = 2k, PL = 3k \Rightarrow P(h,k)$

But P lies on $y^2 = 4ax \Rightarrow (3k)^2 = 4ah \Rightarrow 9k^2 = 4ah$

Locus of $R(h,k) is 9y^2 = 4ax$

Length of the latus rectum $= \frac{4a}{9} \Rightarrow k = \frac{1}{9}$

(R) let $P(a_1^2, 2at_1)\ Q(a_2^2, 2at_2)$ be the extremes $\Rightarrow t_1 t_2 = -1$

$$e = SP = \sqrt{(at_1^2 - a)^2 + 4a^2 t_1^2} = a(1+t_1^2)\ and\ e' = SQ = a(1+t_2^2)$$

$$(e-a)(e'-a) = a^2 t_1^2 t_2^2 = a^2 \Rightarrow ee' - a(e+e') = 0 \Rightarrow \frac{1}{e} + \frac{1}{e'} = \frac{1}{a}$$

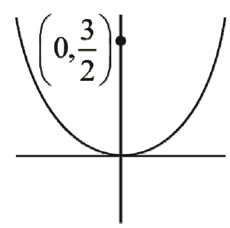
(S) Let $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be the given hyperbola $\Rightarrow b^2 = a^2(e^2 - 1)$ and conjugate is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \Rightarrow a^2 = b^2(e'^2 - 1)$

On eliminating, $\frac{a^2}{b^2}$, we get $e'^2 - 1 = \frac{1}{e^2 - 1} \Rightarrow \frac{1}{e^2} + \frac{1}{e'^2} = 1$

87.(1) $y = x^2$

Normal at (t, t^2) is given by

$x + 2ty = t + 2t^3$ if it passes through $\left(0, \frac{3}{2}\right)$



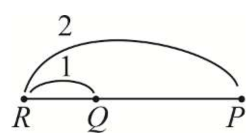
$\Rightarrow \quad t^3 = t \Rightarrow t = 0, 1, -1 \quad \Rightarrow \quad$ points are $P(0,0), Q(1,1), R(-1,1)$

$\angle P = 90° \quad \Rightarrow R = \frac{QR}{2} = 1$

88.(4) $(y-1)^2 = 4x - 4 = 4(x-1)$

Let $P(t^2+1, 2t+1)$, tangent is given by $x - ty + t^2 + t - 1 = 0$. It intersects directrix $x = 0$

at $\quad Q\left(0, \frac{t^2+t-1}{t}\right)$



$\Rightarrow \quad R$ is given by $\left(-t^2 - 1, \frac{t-2}{t}\right)$

$$x = -t^2 - 1, y = 1 - \frac{2}{t} \Rightarrow t = \frac{2}{1-y} \quad \Rightarrow \quad x + 1 = -\frac{4}{(y-1)^2} \Rightarrow (y-1)^2(x-1) + 4 = 0 \quad \Rightarrow \quad \lambda = 4$$

89.(1) $A(at_1^2, 2at_1), B(at_2^2, 2at_2), C(at_3^2, 2at_3), D(at_4^2, 2at_4)$

$\angle ABC = 90° \quad \Rightarrow \frac{2}{t_1+t_2} \times \frac{2}{t_2+t_3} = -1 \Rightarrow (t_1+t_2)(t_2+t_3) = -4$

$\angle ADC = 90° \quad \Rightarrow \frac{2}{t_1+t_4} \times \frac{2}{t_3+t_4} = -1 \Rightarrow (t_1+t_4)(t_3+t_4) = -4$

t_2, t_4 are roots of $(t+t_1)(t+t_3) + 4 = 0 \quad \Rightarrow \quad t^2 + (t_1+t_3)t + t_1 t_3 + 4 = 0 \quad \Rightarrow t_2 + t_4 = -(t_1+t_3)$

90.(2) $\frac{tx}{4}-\frac{y}{3}+t=0$ ... (1) and $\frac{x}{4}+t\frac{y}{3}-t=0$ ... (2)

Locus of point of intersection of (1) and (2) is given by $t=\frac{\frac{y}{3}}{\frac{x}{4}+1}=\frac{\frac{x}{4}}{1-\frac{y}{3}}$

$$\frac{y}{3}-\frac{y^2}{9}=\frac{x^2}{16}+\frac{x}{4} \quad \text{... (3)}$$

Equation of pair of straight lines joining origin to the points of intersection of $ax+by+c=0$ & (3) is given by

$\frac{x^2}{16}+\frac{y^2}{9}+\left(\frac{x}{4}-\frac{y}{3}\right)\left(\frac{ax+by}{-c}\right)=0$. If they are at right angles, then

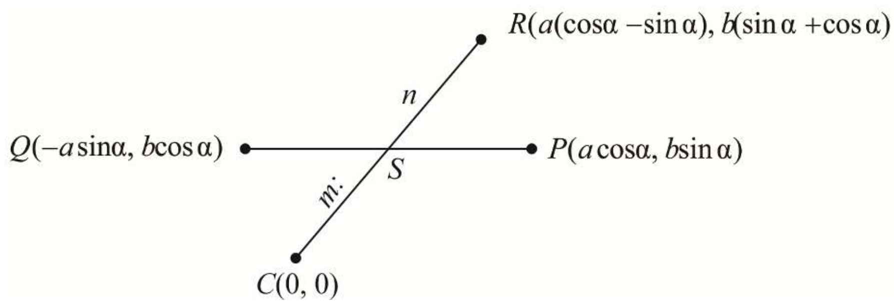
$$\left(\frac{1}{16}-\frac{a}{4c}\right)+\left(\frac{1}{9}+\frac{b}{3c}\right)=0 \quad \Rightarrow \quad \frac{c-4a}{16c}+\frac{c+3b}{9c}=0$$

$$9c-36a+16c+48b=0 \quad \Rightarrow \quad 36a-48b-25c=0 \quad \text{Hence } \left[\frac{3a-4b}{c}\right]=\left[\frac{25}{12}\right]=2$$

91.(1) Let $P(a\cos\alpha, b\sin\alpha)$ & $Q(-a\sin\alpha, b\cos\alpha)$

Tangents at P & Q are given by $\frac{x\cos\alpha}{a}+\frac{y\sin\alpha}{b}=1 \quad \Rightarrow \quad \frac{-x\sin\alpha}{a}+\frac{y\cos\alpha}{b}=1$

Point of intersection R is given by $\left(a(\cos\alpha-\sin\alpha), b(\sin\alpha+\cos\alpha)\right)$



$$S\left(\frac{ma(\cos\alpha-\sin\alpha)}{m+n}, \frac{mb(\sin\alpha+\cos\alpha)}{m+n}\right)$$

As P, Q, S are collinear then $\begin{vmatrix} a\cos\alpha & b\sin\alpha & 1 \\ -a\sin\alpha & b\cos\alpha & 1 \\ \frac{ma(\cos\alpha-\sin\alpha)}{m+n} & \frac{mb(\sin\alpha+\cos\alpha)}{m+n} & 1 \end{vmatrix}=0$

$$mR_1+mR_2-(m+n)R_3 \Rightarrow \begin{vmatrix} 0 & 0 & m-n \\ -a\sin\alpha & b\cos\alpha & 1 \\ \frac{ma(\cos\alpha-\sin\alpha)}{m+n} & \frac{mb(\sin\alpha+\cos\alpha)}{m+n} & 1 \end{vmatrix}=0$$

$m=n \Rightarrow m:n:1:1$

92.(6) $\sqrt{(x-2)^2+(y+3)^2}=\frac{1}{2}\left|\frac{3x-4y+7}{5}\right|$

Focus at $(2,-3)$, directrix at $3x-4y+7=0$ and e = ½

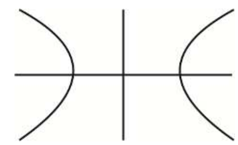
Distance of focus from directrix is 5 which is given by $\frac{a}{e}-ae$ i.e. $\frac{3a}{2}$ Hence $a=\frac{10}{3}$, $A=\frac{20}{3}$

93.(3) Normal is given by $13\sec\theta x-5\operatorname{cosec}\theta y=144$

If it passes through (0, 6), then $\sin\theta=\frac{-5}{24}$

Corresponding to it $\cos\theta$ has 2 values, hence these two normals and one normal is y axis

94.(4) $\frac{1}{e^2}+\frac{1}{e'^2}=1 \quad \Rightarrow \quad e'=f(e)=\frac{e}{\sqrt{e^2-1}} \quad \Rightarrow \quad e''=f(e')=e$



$$f(f(f(......f(e))))=\begin{cases}\frac{e}{\sqrt{e^2-1}} & n\ odd\\ e & n\ even\end{cases} \quad ; \quad \int_1^3 e\,de=4$$

95.(8) Normal to the hyperbola is given by $2\cos\theta x+\cot\theta y=5$, if its intercepts are equal then

$2\cos\theta=\cot\theta$ i.e. $\sin\theta=\frac{1}{2}$, hence normal is given by $x+y=\frac{5}{\sqrt{3}}$

If it touches given ellipse then $c^2=a^2m^2+b^2$ i.e. $a^2+b^2=\frac{25}{3}$

96.(4) $\frac{(x+1)^2}{9}-\frac{(y-2)^2}{16}=1 \quad ; \quad k=\frac{2\times16}{3}=\frac{32}{3}$

97.(2) Let the point is $P(h,k)$, then $k=mh-2am-am^3 \Rightarrow am^3+(2a-h)m+k=0$ ....(i)

The slopes m_1,m_2,m_3 of the normal drawn from P are the roots of the equation (i), hence,

And $m_1m_2m_3=-\frac{k}{a}$

Also, according to the question $m_1m_2=p$

From the last of the above three relations, we get $m_3=-\frac{k}{ap}$

m_3 is a root of the equation (i), so we get $a\left(-\frac{k}{ap}\right)^3+(2a-h)\left(-\frac{k}{ap}\right)+k=0$

$-\frac{k}{a^2p^3}\left[k^2+(2a-h)ap^2-a^2p^3\right]=0$

$k^2+(2a-h)ap^2-a^2p^3=0$

The locus of $P(h,k)$ is $y^2+(2a-x)ap^2-a^2p^3=0$

$y^2=ap^2x+(p-2)a^2p^2$ ....(ii)

Equation (ii) will represent the given parabola $y^2=4ax$ if $p^2=4\,and\ p-2=0 \Rightarrow p=2$

98.(0) Tangents at $\left(x_1, y_1\right)$ on the parabola $y^2=4ax$ is $yy_1=2a\left(x+x_1\right)$ ….(i)

Any point on this line (i) may be selected as $P\left(t, \frac{2a\left(t+x_1\right)}{y_1}\right)$

Equation of chord of contact of the tangents drawn from P to the circle $x^2+y^2=a^2$ is $tx+\frac{2a\left(t+x_1\right)}{y_1}y=a^2$

$$txy_1+2a\left(t+x_1\right)y-a^2y_1=0$$

$$\left(2ax_1y-a^2y_1\right)+t\left(xy_1+2ay\right)=0 \text{ …….(ii)}$$

The line (ii), for all values of t passes through the point of intersection of the lines

$$2ax_1y-a^2y_1=0 \text{ and } xy_1+2ay=0$$

$$y=\frac{ay_1}{2x_1} \text{ and } x=-\frac{a^2}{x_1}$$

According to question the line (ii) passes through $\left(x_2, y_2\right)$, so, $y_2=\frac{ay_1}{2x_1} \text{ and } x_2=-\frac{a^2}{x_1}$

Now, $4\left(\frac{x_1}{x_2}\right)+\left(\frac{y_1}{y_2}\right)^2=4\left(x_1 / \frac{-a^2}{x_1}\right)+\left(y_1 / \frac{ay_1}{2x_1}\right)^2=-\frac{4x_1^2}{a^2}+\frac{4x_1^2}{a^2}=0$

99.(0) Let $y=mx+c$

Equation of pair of asymptotes is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$

For intersection points of chord and pair of asymptotes $\frac{x^2}{a^2}-\frac{(mx+c)^2}{b^2}=0 \begin{smallmatrix}\nearrow x_1\\ \searrow x_2\end{smallmatrix}$ ……….(i)

For intersection points of chord and hyperbola, $\frac{x^2}{a^2}-\frac{(mx+c)^2}{b^2}=1 \begin{smallmatrix}\nearrow x_3\\ \searrow x_4\end{smallmatrix}$ ………..(ii)

$x_1+x_2=x_3+x_4$; $y_1+y_2=y_3+y_4$

Mid points of PP' and QQ' are same

$RQ=RQ'$ ……….(i)

$RP=RP'$ ……..(ii)

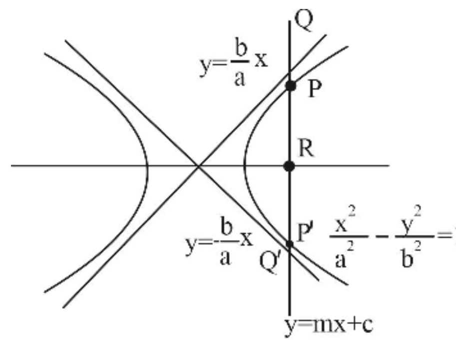
(i) – (ii)

$PQ=P'Q'$

$PQ+PP'=P'Q'+PP'$

$P'Q=PQ'$

$\left(PQ-P'Q'\right)+\left(PQ'-P'Q\right)=0$



100. (3) Equation of normal at $\left(2\sec\theta, \tan\theta\right)$ is $\frac{2x}{\sec\theta}+\frac{y}{\tan\theta}=5$

$Slope=-1 \Rightarrow -\frac{2\tan\theta}{\sec\theta}=-1$; $\theta=\frac{\pi}{6}$

Normal becomes $y=-x+\frac{5}{\sqrt{3}}$

It is tangent to ellipse $\Rightarrow a^2+b^2=\frac{25}{3}$; $\frac{9}{25}\left(a^2+b^2\right)=3$

101.(6.33) Let $X = x-\sqrt{2}, Y = y-\frac{1}{\sqrt{2}}$

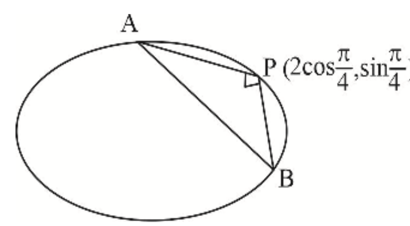
Equation of ellipse $\frac{x^2}{4}+y^2=1$ becomes $\frac{\left(X+\sqrt{2}\right)^2}{4}+\left(Y+\frac{1}{\sqrt{2}}\right)^2=1$

$$\frac{(X)^2}{4}+(Y)^2+\frac{1}{\sqrt{2}}X+\sqrt{2}Y=0$$

Let equation of AB be $aX+bY=1$ Homogenising, we get $\frac{(X)^2}{4}+(Y)^2+\left(\frac{1}{\sqrt{2}}X+\sqrt{2}Y\right)(aX+bY)=0$

$$PA \perp PB \Rightarrow coeff.of\ X^2+coeff.of\ Y^2=0$$

$$\frac{a}{\sqrt{2}}+\sqrt{2}b+\frac{5}{4}=0\ ;\ -\frac{4}{5\sqrt{2}}a-\frac{4\sqrt{2}}{5}b=1$$



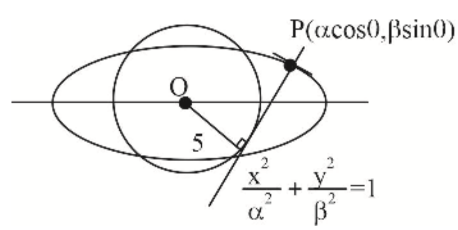
AB passes through fixed point $\left(-\frac{4}{5\sqrt{2}},\frac{-4\sqrt{2}}{5}\right)=\left(-\frac{4}{5\sqrt{2}}+\sqrt{2},\frac{-4\sqrt{2}}{5}+\frac{1}{\sqrt{2}}\right)$

$$\left(\frac{6}{5\sqrt{2}},\frac{-3}{5\sqrt{2}}\right)$$

$$a=\frac{6}{5\sqrt{2}},b=\frac{-3}{5\sqrt{2}}\ ;\ a^2+b^2=\frac{36+9}{50}=\frac{9}{10}=\frac{m}{n}$$

102.(2) Equation of normal at P is $\alpha x\sec\theta-\beta y\operatorname{cosec}\theta=\alpha^2-\beta^2$

$$r=\frac{\left|\alpha^2-\beta^2\right|}{\sqrt{\alpha^2\sec^2\theta+\beta^2\operatorname{cosec}^2\theta}}\ ;\ r=\frac{\left|\alpha^2-\beta^2\right|}{\sqrt{\alpha^2+\beta^2+\alpha^2\tan^2\theta+\beta^2\cot^2\theta}}$$

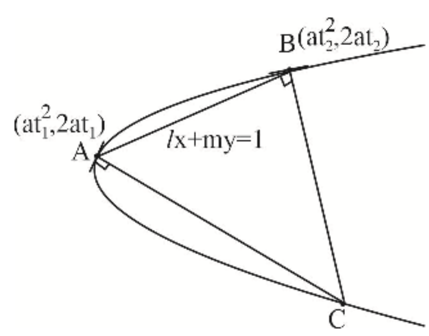


$$\le\frac{\left|\alpha^2-\beta^2\right|}{\sqrt{\alpha^2+\beta^2+2\alpha\beta}}$$

$$r_{max}=|\alpha-\beta|=a^2+2a+2-\left(a^2+1\right)=2a+1\ ;\ \ 5=2a+1\Rightarrow a=2$$

103.(4) Put $\left(at^2,2at\right)$ to equation of AB i.e., $lx+my=1$

$$lat^2+2mat-1=0\begin{smallmatrix}\nearrow t_1\\ \searrow t_2\end{smallmatrix}$$



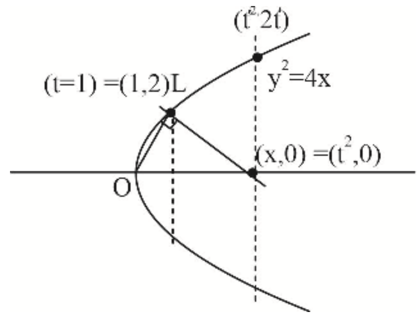
$$t_1+t_2=-\frac{2m}{l}$$

$$t_1+t_2+t_3=0$$

$$t_3=\frac{2m}{l}$$

$$C=\left(at_3^2,2at_3\right)=\left(\frac{4m^2a}{l^2},\frac{4am}{l}\right)$$

104.(80) $\dfrac{0-2}{x-1} = -\dfrac{1}{2} \Rightarrow x-1=4 \Rightarrow x=5$



$t'^2 = 5 \Rightarrow t' = \sqrt{5}$

Length of double ordinate $4t' = 4\sqrt{5} = \sqrt{80}$

$N = 80$

105.(2) If P and Q are α and β, then $|\alpha-\beta| = \dfrac{\pi}{2}$

Equation of PQ is $\dfrac{x}{a}\cos\dfrac{\alpha+\beta}{2} + \dfrac{y}{b}\sin\dfrac{\alpha+\beta}{2} = \cos\dfrac{\alpha-\beta}{2} = \dfrac{1}{\sqrt{2}}$

This line and the given line must be identical.

106.(4) The point P lies on the director circle $x^2+y^2=3$

Let P be $\left(\sqrt{3}\cos\theta, \sqrt{3}\sin\theta\right)$. Chord of contact is $x\sqrt{3}\,cos\,\theta + 2y\sqrt{3}\,sin\,\theta = 2$. If it is a tangent to the circle

$x^2+y^2=a^2$.

Then, $a = \dfrac{2}{\sqrt{3cos^2\theta + 12\,sin^2\theta}} = \dfrac{2}{\sqrt{\dfrac{15}{2} - \dfrac{9}{2}cos\,2\theta}}$

Now, $-\dfrac{9}{2} \le -\dfrac{9}{2}cos\,2\theta \le \dfrac{9}{2} \Rightarrow \dfrac{2}{\sqrt{12}} \le a \le \dfrac{2}{\sqrt{3}}$

Thus, $\dfrac{\max imum\,area}{\min imum\,area} = \dfrac{\left(\dfrac{2}{\sqrt{3}}\right)^2}{\left(\dfrac{2}{\sqrt{12}}\right)^2} = 4$

107.(2) Tangent at $P\left(t,\dfrac{1}{t}\right)$ is $x+t^2y=2t \Rightarrow T(2t,0)$ and $T'\left(0,\dfrac{2}{t}\right)$

Normal at $P\left(t,\dfrac{1}{t}\right) is\, t^3x - ty = t^4 - 1 \Rightarrow N\left(t-\dfrac{1}{t^3},0\right)$ and $P\left(t,\dfrac{1}{t}\right) is\, t^3x - ty = t^4 - 1 \Rightarrow N'\left(0,\dfrac{1}{t}-t^3\right)$

Now the area of $\Delta PTN\ is\ \Delta = \dfrac{1}{2}\left|\dfrac{1}{t}\left(t+\dfrac{1}{t^3}\right)\right| = \dfrac{\left(1+t^4\right)}{2t^4}$

And the area of $\Delta PT'N'\ is\ \Delta = \dfrac{1}{2}\left|t\left(\dfrac{1}{t}+t^3\right)\right| = \dfrac{\left(1+t^4\right)}{2}$

Hence, $\dfrac{1}{\Delta} + \dfrac{1}{\Delta'} = 2\left(\dfrac{t^4+1}{1+t^4}\right) = 2$

$\dfrac{1}{\Delta} + \dfrac{1}{\Delta'}$ is constant for all positions of P.

108.(8) Let the coordinates of P are $\left(t, \frac{1}{t}\right)$.

Equation of the tangent at P is $x + t^2 y = 2t$

It meets $y = x \, and \, y = -x$ at the points $Q\left(\frac{2t}{1+t^2}, \frac{2t}{1+t^2}\right)$ and $R\left(\frac{2t}{1-t^2}, -\frac{2t}{1-t^2}\right)$ respectively. Hence,

$$\Delta_1 = \frac{1}{2}\left[\left(\frac{2t}{1+t^2}\right)\left(\frac{2t}{1-t^2}\right) + \left(\frac{2t}{1-t^2}\right)\left(\frac{2t}{1+t^2}\right)\right] = \left|\frac{4t^2}{1-t^4}\right|$$

Again, the normal at P is $t^3 x - ty = t^4 - 1$

It meets the X-axis and the Y axis at $M\left(\frac{t^4-1}{t^3}, 0\right)$ and $N\left(0, \frac{1-t^4}{t}\right)$ respectively, hence.

$$\Delta_2 = \frac{1}{2}\left|\left(\frac{t^4-1}{t^3}\right)\left(\frac{1-t^4}{t}\right)\right| = \left|\frac{\left(t^4-1\right)^2}{2t^4}\right| \;;\; \Delta_1^2 \Delta_2 = 8$$

109.(3) $ae = 5\sqrt{14} \, and \, e = \sqrt{2} \Rightarrow a^2 = 175$

So, the equation of the hyperbola is $x^2 - y^2 = 175$ Or $(x-y)(x+y) = 175$

$175 = 5^2 .7 \, and \, x > y$ so, we can have $x - y = 1, x + y = 175 \Rightarrow x = 88 \, and \, y = 87$

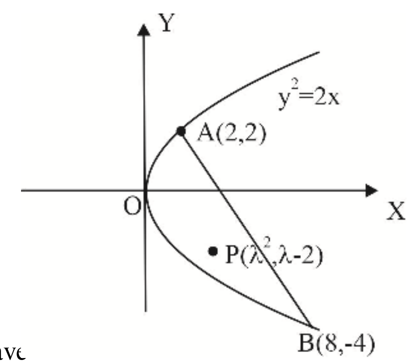
$x - y = 5, x + y = 35 \Rightarrow x = 20 \, and \, y = 15$; $\quad x - y = 7, x + y = 25 \Rightarrow x = 16 \, and \, y = 9$

110.(3) If coordinates of A, B, C and D are $(x_i, y_i); i = 1,2,3,4,$ then we have $\sum_{i=1}^{4} x_i = h, \sum_{i=1}^{4} y_i = k, \sum_{i=1}^{4} x_i^2 = h^2 \, and \sum_{i=1}^{4} y_i^2 = k^2,$

Now $PA^2 + PB^2 + PC^2 + PD^2 = \sum_{i=1}^{4}\left[(h - x_i)^2 + (k - y_i)^2\right]$

$$= 4\left(h^2 + k^2\right) - 2h\sum_{i=1}^{4} x_i - 2k\sum_{i=1}^{4} y_i + \sum_{i=1}^{4} x_i^2 + \sum_{i=1}^{4} y_i^2 \quad = 4\left(h^2 + k^2\right) - 2h^2 - 2k^2 + h^2 + k^2 = 3\left(h^2 + k^2\right)$$

111.(1) The points $P\left(\lambda^2, \lambda - 2\right)$ lies inside the parabola $y^2 = 2x$, so



$(\lambda - 2)^2 - 2\lambda^2 < 0$; $\lambda^2 + 4\lambda - 4 > 0$

The roots of the equation $\lambda^2 + 4\lambda - 4 = 0$ are $\lambda = -2 - 2\sqrt{2}$ and $-2 + 2\sqrt{2}$

So the solution of the above inequality is

$\lambda < -2 - 2\sqrt{2}$ or $\lambda > -2 + 2\sqrt{2}$ ....(i)

Further, the equation of the chord is $y - 2 = \frac{-4-2}{8-2}(x - 2) \Rightarrow x + y - 4 = 0$

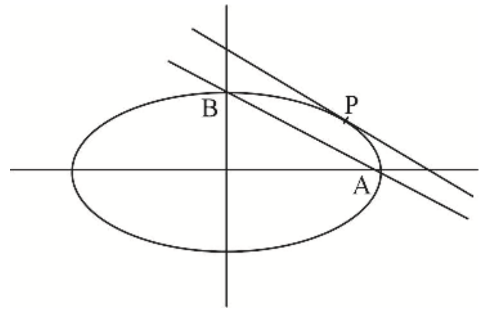
The origin O and the point $P\left(\lambda^2, \lambda - 2\right)$ lies on the same side of the line, so we have

$\lambda^2 + \lambda - 2 - 4 < 0 \Rightarrow \lambda^2 + \lambda - 6 < 0$

$(\lambda + 3)(\lambda - 2) < 0 \Rightarrow -3 < \lambda < 2$ ....(ii)

The common values from (i) and (ii) are $-2 + 2\sqrt{2} < \lambda < 2 \Rightarrow \lambda \in \left(-2 + 2\sqrt{2}, 2\right)$

112.(3) Let perpendicular distance of P from the line be h



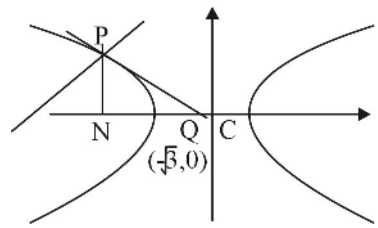
$\frac{1}{2}\times h\times 5=6(\sqrt{2}-1)$ (as $\Delta PAB=6(\sqrt{2}-1)$)

$h=\frac{12\left(\sqrt{2}-1\right)}{5}$

Now distance of tangent parallel to AB i.e. $4y+3x=12\sqrt{2}$,

from line AB is $\frac{12\left(\sqrt{2}-1\right)}{5}$. There are just three such points.

113.(3) The corresponding normals are $y=x\pm\frac{5\sqrt{3}}{3}$.



Considering $y=x+\frac{5\sqrt{3}}{3}$, corresponding tangent is $y=-x-\sqrt{3}$

Point of contact is $\left(\frac{-4}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

Length of sub tangent $=NQ=NC-QC=\frac{4}{\sqrt{3}}-\sqrt{3}=\frac{1}{\sqrt{3}}$ $\Rightarrow$ $k=3$

114(1). equation of normal is $y-t^2=-\frac{1}{2t}(x-t)$ it passes through $\left(0,\frac{3}{2}\right)$

$t=0$ or $t=1,-1$

hence P,Q,R are (0, 0) (1, 1) and (–1, 1), PQR is a right triangle, radius of circum circle is 1.

115.(4) Let $A_r=\left(2t,t^2\right)$

Slope of BA, $BA_r=\frac{t^2-1}{2t}=tan\left(\frac{\pi}{2}+\theta_r\right)$

$tan\left(\theta_r\right)=\frac{2t}{1-t^2}=tan\,2\phi$

$\phi=\frac{\theta_r}{2}=\frac{r\pi}{4n}$ where $tan\,\phi=t$ and $BA_r=\sqrt{\left(t^2-1\right)^2+4t^2}=t^2+1=sec^2\phi=sec^2\left(\frac{r\pi}{4n}\right)$

$\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{n}BA_r=\int\limits_0^1 sec^2\left(\frac{\pi x}{4}\right)dx=\frac{4}{\pi}$

116.(1.32) Image of (h,k) on ellipse about $x-y-2=0$ is say (h',k')

$\frac{h'-h}{1}=\frac{k'-k}{-1}=-2\frac{h-k-2}{1+1}=-h+k+2$

$h'=k+2,k'=h-2$; $(h,k)=(k'+2,h'-2)$

$\frac{(h-4)^2}{16}+\frac{(k-3)^2}{9}=1$; $\frac{(k'+2-4)^2}{16}+\frac{(h'-2-3)^2}{9}=1$; $16h'^2+9k'^2-36k'-160h'+292=0$

Equation of reflection of ellipse is $16x^2+9y^2-160x-36y+292=0$

$\frac{k_1+k_2}{100}=\frac{292-160}{100}=1.32$

FUNCTIONS

1.(B) Check by drawing a rough graph.

2.(D) Domain of $f(x)$ is $[-1,1]$ and $f(x)$ is continuous and increasing. So, $\tan^{-1}x \in \left[\frac{-\pi}{4},\frac{\pi}{4}\right]$ and $\frac{1}{2}\sin^{-1}x \in \left[-\frac{\pi}{4},\frac{\pi}{4}\right]$

3.(C) As $-(x-1)=-x+1,$ the two graphs are symmetrical about the line $x-1=0$

4.(C) $f(x)=x$ on [2, 3], period = 2

$f(x)=x+4$ on $[-2,-1]$ and $f(x)=x+2$ on [0, 1]

So, $f(x)=2-x$ on $[-1,0]$ as it is an even function

Hence, on $[-2,0]$, $f(x)=3-|x+1|$

5.(A) Notice that $f(x)$ is odd and increasing So, if $a+b\ge 0,\ a\ge -b$ then $f(a)\ge f(-b)$ or $f(a)\ge -f(b)$, $f(a)+f(b)\ge 0$ and these steps can be traced back.

6.(A) Simplify $f(x)$ to get $f(x)=1-\frac{1}{2}\sin 2x-\frac{1}{2}\sin^2 2x\ =1-\frac{1}{2}t-\frac{t^2}{2}, t=\sin 2x, t\in[-1,1]$

$$=\frac{9}{8}-\frac{1}{2}\left(t+\frac{1}{2}\right)^2$$

7.(B) Notice that $f(x)+f(\pi-x)=2$. So, $\frac{1}{2}f(x)+\frac{1}{2}f(\pi-x)=1$. Hence, $a=b=\frac{1}{2}$ and $c=\pi$

8.(A) Note that $f(0)=1$ and putting $x = 0$ in the given functional equation, we have $f(y)=e^{f(y)-1}$. Now consider a function $g(t)=t-e^{t-1}$. Here $g'(t)<0$ if $t>1$ and $g'(t)>0$ if $t<1$ and $g(1)=0$. So, $g(t)$ has only one root $t=1$.

9.(A) $3^x=x(x+1)(x+2)+1$ LHS = multiple of 3 and RHS = not divisible by 3.

10.(B) $Q(x)=(x-1)^2+21,\ Q(x)$ takes all values in interval $[21,\infty)$. Since $P(Q(x))$ has no real roots, all three roots of $P(x)$ must be less than 21.

P(22) = 21 gives

$(22-x_1)(22-x_2)(22-x_3)=21\ =1\times 3\times 7$ where x_1, x_2, x_3 are roots of P(x).

So, $\{(22-x_1),(22-x_2),(22-x_3)\}=\{1,3,7\}$

So, $x_1=15,\quad x_2=19,\quad x_3=21$

Hence, $a=-(x_1+x_2+x_3)=-55$

11.(A) The given graph can be rewritten as $(y-3)(x+4)=\frac{-29}{3}$. [A rectangular hyperbola with centre at (– 4, 3)]

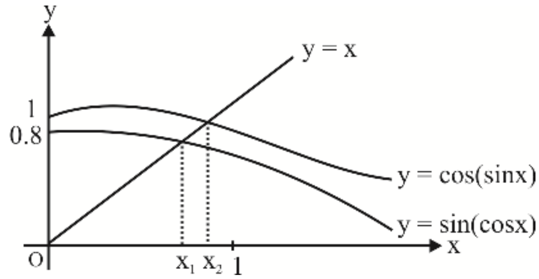
12.(A) $f(x)=\sqrt[3]{x^2+100}$ = increasing function

The given equation can be written as $f(f(x))=x$ so, $f(x)=x \Rightarrow (f(x))^3=x^3$,

$$x^2+100=x^3\quad ;\quad x^3-x^2-100=0\quad ;\quad (x-5)(x^2+4x+20)=0$$

Hence $x=5$

13.(B)



14.(B) Given expression $=\dfrac{1}{-f(x)}$; where $f(x)=x^2+b^2+\sin^2 x-2bx=(x-b)^2+\sin^2 x$

Now, $x\in[-1,0]$ and $b\in[2,3]$ $\Rightarrow$ $|x-b|\in[2,4]\Rightarrow b\in[4,16]$ ad $\sin^2 x\in\left[0,\sin^2 1\right]$.

As x increases from -1 to 0 ; $\sin^2 x$ and $(x-b)^2$ both decreases from $\sin^2 1$ to 0 and $(1+b)^2$ to b^2 respectively and $b\in[2,3]$.

$$\Rightarrow \quad f(x)\in\left[4,16+\sin^2 1\right] \Rightarrow \frac{1}{f(x)}\in\left[\frac{1}{16+\sin^2 1},\frac{1}{4}\right] \Rightarrow \frac{1}{-f(x)}\in\left[-\frac{1}{4},-\frac{1}{16+\sin^2 1}\right]$$

So, least value of expression is $\dfrac{-1}{4}$

15.(C) $\because$ $f(x)$ is an even function

$$\Rightarrow \quad f(x)=f(-x)\ \forall\ x\in R-\left\{(2n+1)\frac{\pi}{2};\ n\in Z\right\}$$

$$\Rightarrow \quad \left([a]^2-5[a]+4\right)x^3-\left(6[a]^2-5\{a\}+1\right)x-(\tan x)\,sgn(x)$$

$$=-\left([a]^2-5[a]+4\right)x^3+\left(6[a]^2-5\{a\}+1\right)x+\tan x\, sgn(-x)$$

$$\Rightarrow \quad 2\left([a]^2-5[a]+4\right)x^3-2\left(6\{a\}^2-5\{a\}+1\right)x+\tan x\, sgn(-x)$$

$$\Rightarrow \quad 2\left([a]^2-5[a]+4\right)x^3-2\left(6\{a\}^2-5\{a\}+1\right)x-\tan x$$

$$\left(sgn(x)+sgn(-x)\right)=0\ \forall\ x\in R-\left\{(2n+1)\frac{\pi}{2};\ n\in Z\right\}$$

$$\Rightarrow \quad \left([a]^2-5[a]+4\right)x^3-\left(6\{a\}^2-5\{a\}+1\right)x=0\ \forall\ x\in R-\left\{(2n+1)\frac{\pi}{2};\ n\in Z\right\}$$

But $sgn\,x+sgn(-x)=0\ \forall\ x\in R$

$$\Rightarrow \quad \left([a]^2-5[a]+4\right)x^3-\left(6\{a\}^2-5\{a\}+1\right)x=0\ \forall\ x\in R-\left\{(2n+1)\frac{\pi}{2};\ n\in Z\right\}$$

$$\Rightarrow \quad [a]^2-5[a]+4=0 \text{ and } 6\{a\}^2-5\{a\}+1=0$$

$$\Rightarrow \quad ([a]-1)([a]-4)=0 \text{ and } (3\{a\}-1)(2\{a\}-1)=0$$

$\Rightarrow \quad [a]=1$ or $[a]=4$ and $\{a\}=\dfrac{1}{3}$ or $\{a\}=\dfrac{1}{2}$

$\therefore \quad [a]=1, \{a\}=\dfrac{1}{3} \Rightarrow \quad a=\dfrac{4}{3}$

$[a]=1, \{a\}=\dfrac{1}{2} \Rightarrow \quad a=\dfrac{3}{2}$

$[a]=4, \{a\}=\dfrac{1}{3} \Rightarrow \quad a=\dfrac{13}{3}$

$[a]=4, \{a\}=\dfrac{1}{2} \Rightarrow \quad a=\dfrac{9}{2}$

$\therefore$ Sum of all possible values of $a=\dfrac{4}{3}+\dfrac{3}{2}+\dfrac{13}{3}+\dfrac{9}{2}=\dfrac{8+9+26+27}{6}=\dfrac{35}{3}$

16.(ACD) $f(x)=5x,\ x\ge\dfrac{-1}{2} \ = x-2,\ x<\dfrac{-1}{2}$

$g(x)=\dfrac{x}{5},\ \ x\ge\dfrac{-5}{2}= x+2,\ \ x<\dfrac{-5}{2}$

For $x<\dfrac{-1}{2},\ g(f(x))=g(x-2)=(x-2)+2=x$ For $x\ge\dfrac{-1}{2},\ g(f(x))=g(5x)=\dfrac{1}{5}(5x)=x$

For $x<\dfrac{-5}{2},\ f(g(x))=f(x+2)=(x+2)-2=x$ For $x\ge\dfrac{-5}{2},\ f(g(x))=f\left(\dfrac{x}{5}\right)=5\left(\dfrac{x}{5}\right)=x$

17.(AC) From the given condition, $f(10+(10-x))=f(10-(10-x))$ i.e., $f(20-x)=f(x)$

Similarly, $-f(20+x)=f(x)$

So, $f(x)=-f(20+x)=f(20+(20+x))$

$f(x)=f(x+40)$. Periodic and $f(-x)=f(20+x)=-f(x)$ odd function.

18.(ABC) Adding the given equations, $[3x]+[-y]+\{y\}-\{x\}=2$

So, $\{y\}-\{x\}$ is an integer and since $-1<\{y\}-\{x\}<1$, so, $\{y\}-\{x\}=0$

19.(ABC) For horizontal tangent $(f'(x)=0)\Rightarrow x=e$

$f'(x)\gtrless 0$for $x\in(e,\infty)$ and $x\in(0,e)$ respectively. So, many one function.

For vertical tangent, $f'(x)=\infty \ \Rightarrow x=0$ which is not in the domain of $f(x)$

20.(ABCD) **(A)** Sgn $\left(e^{-x}\right)=1$ **(B)** LCM of 2π and π

(C) $\min.(|x|, \sin x)=\sin x$ **(D)** Simplify to get $-\left\{x+\dfrac{1}{2}\right\}-\left\{x-\dfrac{1}{2}\right\}-2\{-x\}$

21.(ABCD) $f(x)=\dfrac{1-x}{x^2}\quad x<1,\ x\ne0$ $\qquad f'(x)=\dfrac{x-2}{x^3},\ x<1,\ x\ne0$ and

$=\dfrac{x-1}{x^2}\quad x>1$ $\qquad =\dfrac{2-x}{x^3},\ x>1$

22.(AD) Domain of $f(x)$ is $(-1,1)$. So, $\left[x^2\right]=0$

23.(AC) $f(x)$ = odd degree polynomial + bounded function $\cot^{-1}x\in(0,\pi)$ and $f'(x)>0$

24.(BD) $f(x)$ is odd function

25.(AB) $f(x)=x^2,\ x\in I = x^2+x,\ x\notin I$

26.(AD) $g(x)=\frac{\sin\pi x}{x}+\frac{\sin\pi x}{1-x}=\frac{\sin\pi x}{x(1-x)}=\frac{2\sin\frac{\pi x}{2}\cos\frac{\pi x}{2}}{x(1-x)}$ and $g\left(\frac{1}{2}+x\right)=g\left(\frac{1}{2}-x\right)$

27.(ABCD) Simplify $f(-x)$ to get equal to $f(x)$. So, even function.

$f(x)$ is even so it is not one-one

$$\lim_{x\to 0}\frac{x}{e^x-1}+\frac{x}{2}+1=1+1=2.$$

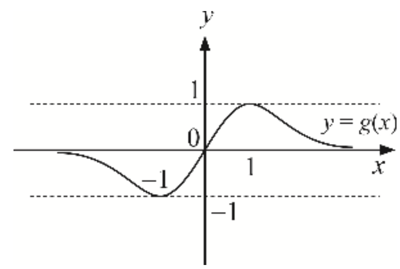
As $\lim_{x\to\infty} f(x)\to\infty$ and $f(x)$ is even. So, $f(x)>0$ and hence not onto.

28.(BCD) Domain of $f(x)=[0,\infty)$

Domain of $g(x)=(-\infty,\infty)$

$\Rightarrow$ Domain of $(g+2)(x)=(-\infty,\infty)$

$\therefore$ Domain of $(f+g+2)(x)=[0,\infty)\cap(-\infty,\infty)=[0,\infty)$



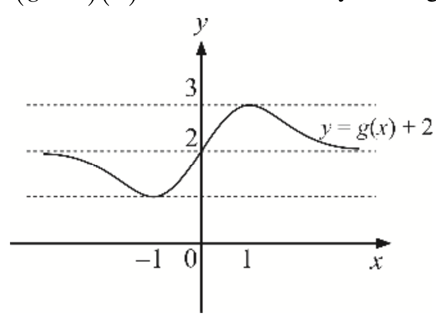
Further range of $f(x)=[2,\infty)$ as $f'(x)=\frac{1}{2\sqrt{x}}>0\ \forall\ x\in(0,\infty)$

$\Rightarrow$ $f(x)$ is increasing and continuous function on $[0,\infty)$

Now $f(x)=\frac{2x}{1+x^2}$. Graph of $g(x)$ is as shown below

Clearly $g(x)$ has its range $[-1,1]$

$\Rightarrow$ $(g+2)(x)$ can be obtained by shifting the graph of $g(x)$ by 2 units above x-axis as shown below



$\therefore$ Range of $(g+2)(x)$ is $[1,3]$

$\therefore$ Range of $f(x)\cap$ range of $(g+2)(x)=[1,3]\cap[2,\infty)=[2,3]$

and range of $f(x)\cup$ range of $(g+2)(x)=[1,3]\cap[2,\infty)=[1,\infty)$

29.(CD) If $f(x)=2$

$\Rightarrow$ $\sin x=1,\ \sin x\sqrt{3}=1\Rightarrow x=(4n+1)\frac{\pi}{2}$ and $x\sqrt{3}=(4m+1)\frac{\pi}{2}$

$\Rightarrow$ $\sqrt{3}=\frac{4m+1}{4n+1}$ which is a contradiction as LHS is irrational whereas RHS is rational

$\therefore$ $f(x)$ cannot attain value 2.

$\therefore$ 2 cannot be the maximum value of $f(x)$.

$\Rightarrow$ $\sin x = \sin x\sqrt{3} = -1$

Which is again impossible by same reason So, (B) is true.

Now period of $f(x) = LCM\left(2\pi, \frac{2\pi}{\sqrt{3}}\right)$ which does not exist as multiples of 2π are $\pm 2\pi, \pm 4\pi, \ldots$

whereas multiples of $\frac{2\pi}{\sqrt{3}}$ are $\pm\frac{2\pi}{\sqrt{3}}, \pm\frac{4\pi}{\sqrt{3}}, \pm\frac{6\pi}{\sqrt{3}}$ Therefore, (C) is false.

Now, $f(0) = 0$ $\therefore$ $f(x) > 0\ \forall\ x \in R$ is false, i.e., (D) is false.

30. **[A-s] [B-r] [C-q] [D-p]**

Consider the ratio $f(x) = \frac{x^n}{e^x}$, $f(0) = 0$

$f'(x) = \frac{x^{n-1}(n-x)}{e^x}$ $\lim_{x\to\infty} f(x) = 0$

f is increasing for $0 < x < a$ and decreasing for $x > a$

So, $f(x) = 1$ has two solutions if $f(a) > 1$, one if $f(a) = 1$ and none if $f(a) < 1$

31. **[A-r] [B-s] [C-p] [D-p]**

(A) $y = \frac{x^2+9+6x}{x^2+1} \Rightarrow (y-1)x^2 - 6x + (y-9) = 0$ $D \ge 0 \Rightarrow 0 \le y \le 10$

(B) Range of $A = [-5, 5]$ and of B is $R^+ \cup \{0\}$. Hence, range of $f(x)$ is $[0, 5]$

(C) $y = \frac{9}{2-\cos 3x} \Rightarrow \cos 3x = \frac{2y-9}{y} \in [-1, 1]$ $3 \le y \le 9$

(D) Domain $\left[\frac{-\pi}{4}, \frac{\pi}{4}\right] \Rightarrow \sqrt{\frac{\pi^2}{16} - x^2} \in \left[0, \frac{\pi}{4}\right]$. So, $0 \le f(x) \le 3$

32. **[(i)-(c)] [(ii)-(d)] [(iii)-(b)] [(iv)-(c)]**

(i) When $\{x\} = 0$, then $sgn\{x\} = 0$ and when $0 < \{x\} < 1$, then $sgn\{x\} = \{0, 1\}$

(ii) For domain $-1 \le x \le 1$ and $-1 \le 1 - x \le 1$

$\Rightarrow$ $-1 \le x \le 1$ and $0 \le x \le 2 \Rightarrow 0 \le x \le 1$ $\Rightarrow$ Domain of function = [0, 1]

(iii) $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}\ \forall\ x \in R$

$\Rightarrow$ $-1 < \frac{2}{\pi}\tan^{-1} x < 1$; for $f(x) = \sqrt{\frac{2\sin^{-1} x}{\pi}}$; $\frac{2\tan^{-1} x}{\pi} \ge 0$

$\Rightarrow$ $0 \le \frac{2\tan^{-1} x}{\pi} < 1$ $\Rightarrow$ $0 \le \sqrt{\frac{2}{\pi}\tan^{-1} x} < 1$ $\Rightarrow$ Range of $f(x) = [0, 1)$

(iv) $\frac{3}{4} \le x^2 + x + 1 < \infty$, but for $\sin^{-1}\left[x^2 + x + 1\right]$

$-1 \le \left[x^2 + x + 1\right] \le 1$ $\Rightarrow$ $\frac{3}{4} \le x^2 + x + 1 < 2$

Case 1: When $\frac{3}{4} \le x^2 + x + 1 < 1$

$$\Rightarrow \quad \left[x^2+x+1\right]=0 \Rightarrow sin^{-1}\left[x^2+x+1\right]=0 \Rightarrow \quad f(x)=\frac{2}{\pi}sin^{-1}\left[x^2+x+1\right]=0$$

Case 2: When $1 \le x^2 + x + 1 < 2$

$$\Rightarrow \quad \left[x^2+x+1\right]=1 \Rightarrow sin^{-1}\left[x^2+x+1\right]=\frac{\pi}{2} \Rightarrow f(x)=\frac{2}{\pi}.\frac{\pi}{2}=1$$

$\therefore$ Range of given function is {0, 1}

33.(2) Use the concept of inflexion point of a cubic i.e., a cubic graph is symmetrical about its inflexion point.

34.(3) Let $[x]=m$ then $\frac{x}{x+4}=\frac{5m-7}{7m-5}$,

Rearranging, $x=\frac{10m-14}{m+1}$

Now, we have $m \le x < m+1 \Rightarrow m \le \frac{10m-14}{m+1} < m+1$

Simplify to get, possible integral values of m to be 2, 6 and 7.

35.(22) Consider the function $f(t)=144^t+144^{1-t}$, $t=\sin^2 x$, $0 \le t \le 1$

$$f'(t)=\log 144\cdot\left(144^t-144^{1-t}\right)$$

So, f is decreasing on $\left(0,\frac{1}{2}\right)$ and increasing on $\left(\frac{1}{2},1\right)$.

So, $\quad f_{\min}=f\left(\frac{1}{2}\right)=24$ and $f_{\max}=f(0)=f(1)=45$

36.(5) Use the concept, $[t] \le t < [t]+1$, $t \in R$

So, $3x-7 \le x^2-3x+2 < (3x-7)+1$ and $3x \in I$

Solving, $x \in \left\{\frac{7}{3},\frac{8}{3},\frac{9}{3},\frac{10}{3},\frac{11}{3}\right\}$

37.(8) Let $t=2105$ then given expression is $\frac{(t+1)^3}{(t-1)t}-\frac{(t-1)^3}{t(t+1)}=8+\frac{16t}{t^3-t}=8+\frac{16}{t^2-1}$

So, integral part = 8, $\frac{16}{t^2-1}<1$

38.(4) Given function is $y=\frac{x+m}{x^2+1}$

$$\Rightarrow \quad yx^2-x+(y-m)=0$$

We must have $D \ge 0$ for $y \in [0,1]$

$$\Rightarrow \quad 1-4y(y-m)\ge 0 \text{ for } y\in[0,1] \Rightarrow g(y)=4y^2-4my-1\le 0 \text{ for } y\in[0,1]$$

$$\Rightarrow \quad g(y)\le 0 \text{ for } y\in[0,1] \Rightarrow g(0)\le 0,\ g(1)\le 0$$

$$\Rightarrow \quad -1\le 0 \text{ and } 3-4m\le 0 \Rightarrow m\ge\frac{3}{4} \Rightarrow k=4$$

39.(2049) Let $f(x)=(x^2+5x+4)(x^2+5x+6)+5$

$$=\left[(x^2+5x+5)-1\right]\left[(x^2+5x+5)+1\right]+5=(x^2+5x+5)^2-1+5$$

$$\Rightarrow \quad f(x)=(x^2+5x+5)^2+4$$

Hence, $f(x)$ has minimum value 4 when $x^2+5x+5=0$, i.e., $x=\frac{-5\pm\sqrt{5}}{2}\in[-6,6]$

Also maximum occurs at $x=6$.

$$\Rightarrow \quad f(x)_{max}=(36+30+5)^2+4=(71)^2+4=5041+4=5045$$

Range is [4, 5045] $\equiv$ [a, b] (Given)

$$\Rightarrow \quad a=4;\ b=5045 \Rightarrow a+b=5049$$

40.(2) Given $f(x)=f\left(\frac{x+4}{x-2}\right)$

One of the possibilities is $x=\frac{x+4}{x-2} \Rightarrow x^2-2x-x-4=0 \Rightarrow x^2-3x-4=0$

Disc. = $9+16=25>0$

$\therefore$ Minimum number of roots = 2

41.(37) Given $f(2x+1)=4x^2+14x$ ....(i)

We are to find the sum of squares of roots of equation $f(x)=0$. Putting $2x+1=y$

$$\Rightarrow \quad x=\frac{y-1}{2} \text{ in (i), we have } f(y)=4\left(\frac{y-1}{2}\right)^2+14\left(\frac{y-1}{2}\right)$$

$$\Rightarrow \quad f(y)=y^2+1-2y+7y-7 \Rightarrow f(y)=y^2+5y-6$$

$$\therefore \quad f(x)=0$$

$$\Rightarrow \quad x^2+5x-6=0 \Rightarrow (x+6)(x-1)=0 \Rightarrow x=-6 \text{ or } x=1 \quad \therefore \quad \alpha^2+\beta^2=36+1=37$$

42.(7) Clearly α is a non-real complex root of equation $x^{13}=1$

But α is non-real complex, α is a root of $1+x+x^2+...+x^{12}=0$

Now, $\sum_{r=0}^{12} r(\alpha^{??}x)=f(x)+f(\alpha x)+f(\alpha^2 x)+...+f(\alpha^{12}x)$

$$=\left(7+\sum_{k=1}^{50}A_k x^k\right)+\left(7+\sum_{k=1}^{50}A_k\alpha^k x^k\right)+...+\left(7+\sum_{k=1}^{50}A_k\alpha^{124}x^k\right)$$

$$=91+\sum_{k=1}^{50}A_k x^k\left[1+\alpha^k+...+\alpha^{12k}\right]=91+\sum_{k=1}^{50}A_k x^k\frac{1-(\alpha^k)^{12}}{1-\alpha^k}=91+\sum_{k=1}^{50}A_k x^k\frac{(1-\alpha^{13k})}{(1-\alpha^k)}$$

$$=91+0 \quad (\because \alpha \text{ is a root of } x^{13}=1 \Rightarrow \alpha^{13}=1 \Rightarrow \alpha^{13}-1=0)$$

$$\therefore \quad \frac{1}{13}\sum_{r=0}^{13}f(\alpha^r x)=7$$

43.(12) Given $g(x)=f\left(2^{x},0\right)=f\left(2,\,2+2.0,\,2.0-2.2^{x}\right)$

$$=f\left(2^{x+1},-2^{x+1}\right)=f\left(2.2^{x+1}+2\left(-2^{x+1}\right),\,2\left(-2^{x+1}\right)-2.2^{x+1}\right)$$

$$=f\left(0,-2^{x+3}\right)=f\left(-2^{x+4},-2^{x+4}\right)=f\left(-2^{x+6},0\right)$$

$$=f\left(-2^{x+7},2^{x+7}\right)=f\left(0,2^{x+9}\right)=f\left(2^{x+10},2^{x+10}\right)$$

$$=f\left(2^{x+12},0\right)=g(x+12)\Rightarrow \text{Period of g is 12.}$$

44.(1) Let $y=g(x)=\dfrac{e^{x}-e^{-x}}{2}\quad\Rightarrow\quad e^{2x}-1=2ye^{x}$

Substituting $e^{x}=t$, we have $t^{2}-2yt-1=0$

$$\Rightarrow\qquad t=\frac{2y\pm\sqrt{4y^{2}+4}}{2}=y\pm\sqrt{y^{2}+1}$$

$$\Rightarrow\qquad t=\frac{2y\pm\sqrt{4y^{2}+4}}{2}=y\pm\sqrt{y^{2}+1}$$

$$\Rightarrow\qquad e^{x}=y+\sqrt{y^{2}+1}\qquad\qquad\left(\because e^{x}>0\right)$$

$$\Rightarrow\qquad x=ln\left(y+\sqrt{y^{2}+1}\right)\Rightarrow g^{-1}(y)=ln\left(y+\sqrt{y^{2}+1}\right)$$

$$\therefore\qquad g^{-1}(x)=f(x)=ln\left(x+\sqrt{x^{2}+1}\right)\qquad\left(\because g(f(x))=x\Rightarrow g^{-1}(x)=f(x)\right)$$

$$\therefore\qquad f\left(\frac{e^{22}-1}{2e^{11}}\right)=ln\left(\frac{e^{22}-1}{2e^{11}}+\sqrt{\left(\frac{e^{22}-1}{2e^{11}}\right)^{2}+1}\right)=ln\left(\frac{e^{22}-1}{2e^{11}}+\sqrt{\left(\frac{e^{22}+1}{2e^{11}}\right)^{2}}\right)=ln\left[e^{11}\right]=11$$

DIFFERENTIAL CALCULUS-1

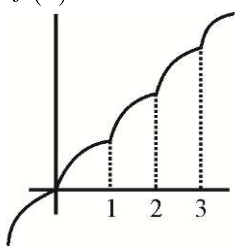
1.(D) $\lim\limits_{x\to 0}\dfrac{((a-n)nx-\tan x)\sin nx}{x^2}=0$

$\lim\limits_{x\to 0}\left(\left((a-n)n-\dfrac{\tan x}{x}\right)\dfrac{\sin nx}{nx}\right)n=0 \;\Rightarrow\; (a-n)n=1$

$a=\dfrac{1}{n}+n$

2.(D) $k=\lim\limits_{x\to 0^+}\dfrac{(2e)^x-x-x\ln 2-1}{x^2\dfrac{\tan x}{x}}=\lim\limits_{x\to 0^+}\dfrac{1+x\ln(2e)+\dfrac{x^2}{2!}(\ln(2e))^2+....-x-x\ln 2-1}{x^2\dfrac{\tan x}{x}}=\dfrac{1}{2}(\ln 2+1)^2=\dfrac{1}{2}(\ln 2)^2+\ln 2+\dfrac{1}{2}$.

3.(C) Graph of $f(x)$



4.(A) Let $\sin x=t$ and evaluate $\lim\limits_{t\to 1}\dfrac{t^2}{1-t^2}\left[\sqrt{2t^2+3t+4}-\sqrt{t^2+6t+2}\right]$ by rationalization

5.(B) $\mathop{Lim}\limits_{x\to\infty}\,[(x+a)\,(x+b)\,(x+c)\,]^{1/3}-x$ (Using $a-b=\dfrac{a^3-b^3}{a^2+ab+b^2}$)

$$=\frac{[x^3+(a+b+c)x^2+(ab+bc+ca)x+abc]-x^3}{[(x+a)(x+b)(x+c)]^{2/3}+x^2+x[(x+a)(x+b)(x+c)]^{1/3}}$$

$$\mathop{Limit}\limits_{x\to\infty}\frac{x^2\left[(a+b+c)+\dfrac{ab+bc+ca}{x}+\dfrac{abc}{x^3}\right]}{x^2\left[\left(1+\dfrac{a+b+c}{x}+\dfrac{ab+bc+ca}{x^2}+\dfrac{abc}{x^2}\right)+1+\left(1+\dfrac{a+b+c}{x}+\dfrac{ab+bc+ca}{x^2}+\dfrac{abc}{x^3}\right)^{1/3}\right]^{1/3}}=\frac{a+b+c}{3}$$

6.(D) $\mathop{Limit}\limits_{h\to 0}\,g(n+h)=f\,[\{n+h\}]=f\,(h)=\mathop{Limit}\limits_{h\to 0}\dfrac{e^h-\cos 2h-h}{h^2}=\dfrac{(e^h-1)+(1-\cos 2h)-h}{h^2}$

$=\mathop{Limit}\limits_{h\to 0}\dfrac{e^h-h-1}{h^2}+\mathop{Limit}\limits_{h\to 0}\dfrac{(1-\cos 2h)}{4h^2}.4=\dfrac{1}{2}+2=\dfrac{5}{2}$

$\mathop{Limit}\limits_{h\to 0}\,g(n-h)=\mathop{Limit}\limits_{h\to 0}\,f\,(1-\{n-h\})=\mathop{Limit}\limits_{h\to 0}\,f\,(h)$

$=\dfrac{e^h-\cos 2h-h}{h^2}\;\left(\{n-h\}=\{-h\}=1-h\right)=g(n)=\dfrac{5}{2}$.

Hence $g(x)$ is continuous at $\forall\, x\in I$. Hence $g(x)$ is continuous $\forall\, x\in R$

7.(C) 1^{∞} form: $l = e^{\underset{n\to\infty}{Lim}\, n\left(\left(\frac{n}{n+1}\right)^{\alpha} + \sin\frac{1}{n} - 1\right)} = e^{\underset{n\to\infty}{Lim}\, n\sin\frac{1}{n} + \underset{n\to\infty}{Lim}\, n\left(\left(\frac{n}{n+1}\right)^{\alpha} - 1\right)} = e \cdot e^{l}$

Consider $l = \underset{n\to\infty}{Lim}\, n\left(\left(\frac{n}{n+1}\right)^{\alpha} - 1\right) = \underset{n\to\infty}{Lim}\, n\left(\left(\frac{1}{1+1/n}\right)^{\alpha} - 1\right)$; put n = $\frac{1}{y}$

$= \underset{y\to 0}{Lim}\, \frac{1}{y}\left(\left(\frac{1}{1+y}\right)^{\alpha} - 1\right) = \underset{y\to 0}{Lim} \frac{1-(1+y)^{\alpha}}{y} = -\alpha$ (using binomial);

8.(C) Let $f(n) = \frac{1}{n^2+n+1} + \frac{2}{n^2+n+2} + \ldots.. + \frac{n}{n^2+n+n}$

Consider g (n) = $\frac{1}{n^2+n+n} + \frac{2}{n^2+n+n} + \ldots.. \frac{n}{n^2+n+n} = \frac{1+2+3+\ldots.+n}{n^2+2n} = \frac{n(n+1)}{2(n^2+2n)}$

$g(n) < f(n)$ ....(1)

ly $h(n) = \frac{1}{n^2+n+1} + \frac{2}{n^2+n+1} + \ldots\ldots. + \frac{n}{n^2+n+1} = \frac{n(n+1)}{2(n^2+n+1)}$

$\therefore$ $f(n) < h(n)$ ....(2)

From (1) and (2) : $g(n) < f(n) < h(n)$

But $\underset{n\to\infty}{Lim}\, g(n) = \underset{n\to\infty}{Lim}\, h(n) = \frac{1}{2}$; Hence using Sandwich theorem $\therefore$ $\underset{n\to\infty}{Lim}\, f(n) = \frac{1}{2}$ Ans.

9.(B) n < $\sqrt{n^2+n+1}$ < n + 1 Hence $\left[\sqrt{n^2+n+1}\right]$ = n

After rationalization $\lim\limits_{h\to\infty}\left[\frac{h+1}{\sqrt{n^2+n+1}+n}\right] = \frac{1}{2}$

10.(B) $f'(x) = \underset{h\to 0}{Lim} \frac{f(x+h)-f(x)}{h} = \underset{h\to 0}{Lim} \frac{f(h)+|x|h+xh^2}{h}$ where $x = h$ and $y = x$

$\because$ $f(0) = 0$; hence $f'(x) = \underset{h\to 0}{Lim}\left(\frac{f(h)-f(0)}{h} + |x| + xh\right)$

$f'(x) = f'(0) + |x| = |x|$

11.(A) $\lim\limits_{x\to\infty} \frac{1^2 n + 2^2(n-1) + 3^2(n-2) + \ldots\ldots\ldots\ldots.. + n^2(n-(n-1))}{\sum r^3}$

N^{r}. = $n(1^2+2^2+\ldots..n^2) - (1.2^2 + 2.3^2 + 3.4^2 + \ldots\ldots\ldots\ldots... + (n-1)n^2)$

$= n\sum r^2 - \sum\limits_{r=2}^{n}(r-1).r^2 = n\sum r^2 - \sum\limits_{r=1}^{n}(r^3 - r^2) = n\sum r^2 - \left[\sum r^3 - \sum r^2\right]$

$= (n+1)\sum r^2 - \sum r^3$ $\therefore \lim\limits_{x\to\infty} \frac{(n+1)\sum r^2 - \sum r^3}{\sum r^3}$

$\lim\limits_{x\to\infty} \frac{(n+1)n(n+1)(2n+1)}{6n(n+1)n(n+1)} - 1$; $\frac{4}{3} - 1 = \frac{1}{3} \Rightarrow A$

12.(A) $\lim\limits_{x\to\infty} \dfrac{\cot^{-1}\left(\dfrac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\dfrac{a^x}{\log_a x}\right)}$; $\underset{x\to\infty}{as}\left(\dfrac{\log_a x}{x^a}\right)\to 0$ and $\left(\dfrac{a^x}{\log_a x}\right)\to\infty$ (using L'Hospital rule) $\therefore$ $\lim\limits_{x\to\infty} = \dfrac{\pi/2}{\pi/2} = 1$

13.(B) $L = \underset{n\to\infty}{Lim}\left(\dfrac{p^{1/n}+q^{1/n}}{2}\right)^n = e^{\underset{n\to\infty}{Lim}\, n\left(\frac{p^{1/n}+q^{1/n}}{2}-1\right)} = e^{\underset{n\to\infty}{Lim}\, n\left(\frac{(p^{1/n}-1)+(q^{1/n}-1)}{2}\right)} = e^{\underset{n\to\infty}{Lim}\frac{1}{2}\left(\frac{(p^{1/n}-1)}{1/n}+\frac{(q^{1/n}-1)}{1/n}\right)}$

$= e^{\left(\frac{\ln p+\ln q}{2}\right)} = e^{\left(\ln\sqrt{pq}\right)} = \sqrt{pq}$

14.(D) $a = [(x+1)^2+2]_{\min} \Rightarrow a = 2$ $\qquad b = \underset{x\to 0}{Lim}\dfrac{\sin 2x}{\dfrac{2(e^{2x}-1).2x}{2x}} = \dfrac{1}{2}$

Now $\displaystyle\sum_{r=0}^{n} a^r b^{n-r} = \sum 2^r\left(\frac{1}{2}\right)^{n-r} = \frac{1}{2^n}\sum_{r=0}^{n} 2^{2r} = \frac{1}{2^n}\sum_{r=0}^{n} 4^r = \frac{1}{2^n}\,[1+4+4^2+......+4^n] = \frac{1}{2^n}\left[\frac{4^{n+1}-1}{3}\right] = \frac{4^{n+1}-1}{3.2^n}$

15.(A) $\underset{x\to\infty}{Lim}\sqrt{x+1}-\sqrt{x} = 0 \quad \Rightarrow \cot^{-1}(0) = \pi/2$

$\underset{x\to\infty}{Lim}\left(\dfrac{2x+1}{x-1}\right)^x \to \infty \quad \Rightarrow sec^{-1}(\infty) = \pi/2 \quad \therefore \quad \lim\limits_{x\to\infty}\dfrac{\pi/2}{\pi/2} = 1$

16.(B) $\underset{x\to 0}{Lim}\dfrac{e^{2x}-(1+4x)^{1/2}}{\ln(1-x^2)}$; $\underset{x\to 0}{Lim}\dfrac{e^{2x}-(1+4x)^{1/2}}{\dfrac{\ln(1-x^2)}{-x^2}\cdot(-x^2)}$; $\underset{x\to 0}{Lim}\dfrac{(1+4x)^{1/2}-e^{2x}}{x^2}$

$\underset{x\to 0}{Lim}\dfrac{\left(1+\frac{1}{2}4x+\frac{1}{2}\cdot\left(\frac{1}{2}-1\right)\frac{1}{2!}16x^2+......\right)-\left(1+\frac{2x}{1!}+\frac{4x^2}{2!}+.......\right)}{x^2} = -2-2 = -4$

17.(C) $x\to 0$, $\quad x^3-x^2 = x^2(x-1)\to 0^-$

$x\to 0$, $\quad 2x^4-x^5 = x^4(2-x)\to 0^+$

$2\,(3) = \lambda\,(2) \Rightarrow \lambda = 3$

18.(B) $\underset{x\to 0^+}{Lim}\dfrac{f(-x)\,x^2}{\left(\dfrac{1-\cos x}{[f(x)]}\right)-\left[\dfrac{1-\cos x}{[f(x)]}\right]} = \dfrac{3x^2}{\dfrac{1-\cos x}{2}-0} = 6\times 2 = 12$

19.(B) For $x\to 0^-$ $\left[f\left(\dfrac{x^3-\sin^3 x}{x^4}\right)\right] = \left[f\left(\dfrac{x-\sin x}{x^3}\right)\left(\dfrac{x^2+\sin^2 x+x\sin x}{x}\right)\right]$

$= \left[f(0^-)\right] = \left[3^+\right] = 3$ for $x\in(-\infty, 0)$ decreasing function

For $x\to 0^-$ $\quad f\left[\dfrac{\sin x^3}{x}\right] = f\left[\dfrac{\sin x^3}{x^3}x^2\right] = f[(0^+)] = f(0) = 4$

$\lim\limits_{x\to 0^-}(9-4) = 5$

20.(AB) For L.H.L. at $x = o^-$

$\{-h\} = 1-h$

L.H.L. $= \lim\limits_{x\to 0^-} \dfrac{\cos^{-1}(1-\{-h\}^2).\sin^{-1}(1-\{-h\})}{\{-h\}(1-\{-h\}^2)} = \lim\limits_{h\to 0} \dfrac{\cos^{-1}(1-(1-h)^2)\cdot\sin^{-1}(1-(1-h))}{(1-h)(1-(1-h)^2)}$

$= \lim\limits_{h\to 0} \dfrac{\cos^{-1}(h(2-h))\cdot\sin^{-1}h}{(1-h)\cdot(h(2-h))} = \lim\limits_{h\to 0} \dfrac{\cos^{-1}(h(2-h))}{(1-h)(2-h)}\cdot\left(\dfrac{\sin^{-1}h}{h}\right) = \dfrac{\pi}{4}$

R.H.L. $= \lim\limits_{x\to 0^+} \dfrac{\cos^{-1}(1-h^2)\cdot\sin^{-1}(1-h)}{h(1-h^2)}$

$= \lim\limits_{h\to 0} \dfrac{\cos^{-1}(1-h^2)}{h(1-h^2)}\cdot\sin^{-1}(1)$ .... (let $\cos^{-1}(1-h^2) = t$)

$= \dfrac{\pi}{2}\lim\limits_{t\to 0}\dfrac{t}{\sqrt{2}\sin t/2} = \dfrac{\pi}{2}\lim\limits_{t\to 0}\left(\dfrac{t/2}{\sin t/2}\right) = \dfrac{\pi}{2}$

Let $\cos^{-1}(1-h^2) = t$

$h^2 = 1-\cos t \quad ; \quad h^2 = 2\sin^2\dfrac{t}{2}$

21.(AC) Let $2^{32} = n$

$k = \lim\limits_{x\to 1}\dfrac{x^n - nx + n - 1}{(x-1)^2} = \lim\limits_{x\to 1}\dfrac{x^n - 1 - n(x-1)}{(x-1)^2}$

$= \lim\limits_{x\to 1}\dfrac{(x^{n-1} + x^{n-2} + ...+1) - n}{x-1} = (n-1)+(n-2)+.........1 = \dfrac{2^{32}(2^{32}-1)}{2} = 2^{63} - 2^{31}$

22.(AC) (A) is true

$\lim\limits_{h\to 0}\dfrac{f(c+h)-f(c)+f(c)-f(c-h)}{h}$

$= \lim\limits_{h\to 0}\dfrac{f(c+h)-f(c)}{h} + \lim\limits_{h\to 0}\dfrac{f(c-h)-f(c)}{-h} = f'(c)+f'(c) = 2f'(c)$ (f is differentiable)

(B) is false. Existence of limit is no guarantee for differentiability

(C) is true

(D) is false

23.(AB) Use expansion of 2^x.

24.(AB) $g(x) = \begin{cases} \dfrac{f(x)}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

$h(x) = g'(x) = \begin{cases} \dfrac{xf'(x)-f(x)}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

For $h(x) = g'(x)$

L.H.S. $= \lim\limits_{x\to 0}\dfrac{-h.f'(-h)-f'(-h)}{h^2} = \lim\limits_{x\to 0}\dfrac{-h.f'(-h)-f'(-h)+f'(-h)}{2h} = \dfrac{f''(0)}{2} = 0$

R.H.L $= \lim\limits_{h\to 0}\dfrac{h.f'(h)-f(h)}{h^2} = \lim\limits_{h\to 0}\dfrac{k.f''(h)+f'(h)-f'(h)}{2h} = \dfrac{f''(0)}{2} = 0$

25.(AC) We get $(a, b) = (-1,1)$, $\left(1,1+\sqrt{2}\right),\left(1,1-\sqrt{2}\right)$

26.(AB) Using first principle method $\quad f'(0) = \lim\limits_{x\to 0}\dfrac{x+2x^2\sin\frac{1}{x}-0}{x-0} = \lim\limits_{x\to 0}\left(1+2x\sin\dfrac{1}{x}\right)=1$

$$f'(x) = 1-2\cos\frac{1}{x}+4x\sin\frac{1}{x}$$

Function f is differentiable for all x, but derivative of $f(x)$ is not continuous at $x=0$

$$f'\left(\frac{1}{2k\pi}\right) = -1 < 0$$

There is not interval around 0 where f is increasing. So it should not make sense to say that f is increasing at $x=0$.

27.(AC) $f(x) = \begin{cases} x^3 & if\ \ x\le 1 \\ x^2 & if\ \ x>1 \end{cases}$

$g(x) = [x]^2 + \sqrt{\{x\}^2}$

$f(x)$ is continuous every where

Continuity of $g(x)$ need to be checked only at integral points $(n \in I)$

$g(n) = n^2$

$\lim\limits_{h\to 0} g(n^+) = \lim\limits_{h\to 0}\left(n^2+\sqrt{h^2}\right) = n^2$

$\lim\limits_{h\to 0} g(n^-) = \lim\limits_{h\to 0}\left((n-1)^2+\sqrt{\{-h\}^2}\right) = \lim\limits_{h\to 0}\left((n-1)^2+\sqrt{(1-h)^2}\right) = (n-1)^2+1$

For continuity

L.H.L. = R.H.L = $g(n)$

$n^2 = (n-1)^2+1$; $n^2 = n^2+1-2n+1$; $n=1$

So $g(x)$ is continuous at only one integral value $x=1$

28.(ABD) $\quad f(x) = \mathop{Lim}\limits_{n\to\infty}\dfrac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}} = \begin{cases} 2\sin\dfrac{1}{x}, & x>1\ or\ x<-1 \\ \dfrac{2(\sin 1)-1}{2}, & x=1 \\ \dfrac{-2(\sin 1)-1}{2}, & x=-1 \\ x, & -1\le x<1 \end{cases}$

$\mathop{Lim}\limits_{x\to\infty} x\,f(x) = \mathop{Lim}\limits_{x\to\infty} 2x\sin\dfrac{1}{x} = 2$

$\mathop{Lim}\limits_{x\to 1^+} 2\sin\dfrac{1}{1+h} = 2\sin 1,\ \lim\limits_{x\to 1^-} x = 1$

$\mathop{Lim}\limits_{x\to 1} f(x)$ does not exist.

$\mathop{Lim}\limits_{x\to 0} f(x)$ exist

$\mathop{Lim}\limits_{x\to -\infty} f(x) = \mathop{Lim}\limits_{x\to -\infty} 2\sin\dfrac{1}{x} = 0$

29.(AD) $\lim\limits_{\theta\to-1}\dfrac{\theta^2+\theta-2}{\theta+3}=\lim\limits_{\theta\to-1}\dfrac{f(\theta)}{\theta^2}\le\lim\limits_{\theta\to-1}\dfrac{\theta^2 2\theta-1}{\theta+3}$

$-1\le\lim\limits_{\theta\to-1}\dfrac{f(\theta)}{\theta^2}\le-1$

Using squeeze theorem or sand with theorem $\lim\limits_{\theta\to-1}\dfrac{f(\theta)}{\theta^2}=-1$

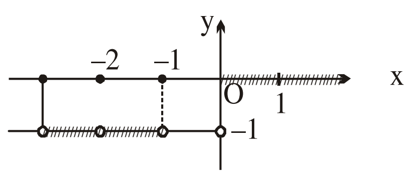
$\lim\limits_{\theta\to-1}f(\theta)=-1$

30.(AC) $\lim\limits_{x\to0^+}\dfrac{\tan^2\{x\}}{x^2-[x]^2}\quad x>0\ =\lim\limits_{x\to0}\dfrac{\tan^2 h}{h^2}=1$

$\lim\limits_{x\to0^-}\sqrt{\{x\}\cot\{x\}}\ =\lim\limits_{x\to\infty}\sqrt{\dfrac{1-h}{\tan(1-h)}}=\sqrt{\cot 1}$

31.(ABCD) $[|\,x\,|\,]-|\,[x]\,|\ =\begin{bmatrix}0 & x=-1\\ -1 & -1<x<0\\ 0 & 0\le x\le1\\ 0 & 1<x\le2\end{bmatrix}\Rightarrow$ range is $\{0,-1\}$

The graph is



32.(ABD) $f(x)=\dfrac{|x|}{\sqrt{1+\sqrt{1-x^2}}}$

$f'(0^+)=\lim\limits_{h\to0}\dfrac{\dfrac{h}{\sqrt{1+\sqrt{1-h^2}}}-0}{h}=\dfrac{1}{\sqrt2}$

$f'(0^-)=\lim\limits_{h\to0}\dfrac{\dfrac{h}{\sqrt{1+\sqrt{1-x^2}}}-0}{-h}=-\dfrac{1}{\sqrt2}$

33.(AC) If $f(b)=g(b)$ then

L.H.L. = $h(b^-)=f(b^-)=f(b)$

R.H.L. = $h(b^+)=g(b^+)=g(b)$

For $h(x)$ to have nonremovable discontinuity

L.H.L. = R.H.L.

$f(b)=g(b)$ (given)

$h(x)$ has a removable discontinuity at $x=b$

$g(b^-)$ and $f(b^+)$ have no existence.

34.(ABCD) **(A)** f (x) = x – cos x ; f (0) < 0, f (π/2) > 0

(B) f (x) = x + sin x – 1

f (0) = – 1 < 0; f (π/6) = $\frac{\pi}{6}+\frac{1}{2}-1>0$

(C) f (x) = a(x – 3) + b(x – 1) in [1, 3]

f (1) = – 2a < 0; f (3) = 2b > 0 ⇒ f (x) = 0 in (1, 3)

(D) h (x) = f (x) – g (x)

h (a) = f (a) – g (a) > 0

h (b) = f (b) – g (b) < 0

Hence using IVT all the four have at least one root in indicated interval.

35.(AC) $$f'(0^+)=\lim_{h\to 0}\frac{h\ln(\cos h)}{h\left(\frac{\ln(1+h^2)}{h^2}\right)\times h^2}=\lim_{h\to 0}\frac{\ln(\cosh)}{h^2}=\lim_{h\to 0}\frac{-\sinh}{(\cosh)\times 2h}=-\frac{1}{2}$$

$$f'(0^-)=\lim_{h\to 0}\frac{-h\ln(\cosh)}{-h\left(\frac{\ln(1+h^2)}{h^2}\right)\times h^2}=-\frac{1}{2}$$

36.(BC) **(A)** $f(3^+)=3-5=-2,\ f(3^-)=2-4=-2$ **(B)** $f(1^+)=1-1=0,\ f(1^-)=-1$

(C) $f(0^+)=-1,\ f(0^-)=1$ **(D)** $f(0^+)=\tan 1,\ f(0^-)=\tan 1$

37.(ABC) $$\lim_{x\to 0}\frac{(1-\sin x)\sin x}{\cos x}\cdot\frac{1+\sin x}{1+\sin x}$$

$$\lim_{x\to\pi/2}\frac{\sin x\cos x}{1+\sin x}=0$$

38.(AC) We have, $f(x)=|x-1|([x]-[-x])$

$$\therefore\quad f(x)=\begin{cases}(x-1)(1+2), & \text{when } 1<x<2\\ 0, & when\ x=1\\ (1-x), & when\ 0<x<1\end{cases}$$

$$\Rightarrow\quad f'(1^+)=\lim_{h\to 0}\frac{3(h)-0}{h}=3 \text{ and } f'(1^-)=\lim_{h\to 0}\frac{(1-(1-h))-0}{-h}=-1$$

⇒ $f(x)$ is continuous at $x=1$ and non-derivable at $x=1$

39.(AD) Here, $x=2t-|t-1|$ and $y=2t^2+t|t|$

Now, when $t<0$,

$x=2t-\{-(t-1)\}=3t-1$ and $y=2t^2-t^2=t^2\Rightarrow y=\frac{1}{9}(x+1)^2$

When $0\le t<1$,

$x=2t-(-(t-1))=3t-1$ and $y=2t^2+t^2=3t^2\Rightarrow y=\frac{1}{3}(x+1)^2$

When $t\ge 1$ $x=2t-(t-1)=t+1$ and $y=2t^2+t^2=3t^2\Rightarrow y=3(x-1)^2$

Thus, $y = f(x) = \begin{cases} \frac{1}{9}(x+1)^2, & x < -1 \\ \frac{1}{3}(x+1)^2, & -1 \le x < 2 \\ 3(x-1)^2, & x \ge 2 \end{cases}$

$f(x)$ is continuous at $x = 2$.

Now, to check differentiability at $x = -1$ and 2.

Differentiability at $x = -1$,

$$\text{LHD} = Lf'(-1) = \lim_{h \to 0} \frac{f(-1-h) - f(-1)}{-h} = \lim_{h \to 0} \frac{\frac{1}{9}(-1-h+1)^2 - 0}{-h} = 0$$

$$\text{RHD} = Rf'(-1) = \lim_{h \to 0} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \to 0} \frac{\frac{1}{3}(-1+h+1)^2 - 0}{h} = 0$$

Hence, $f(x)$ is differentiable at $x = -1$.

Differentiability at x = 2.

$$\text{LHD} = Lf'(2) = \lim_{h \to 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \to 0} \frac{\frac{1}{3}(2-h+1)^2 - 3}{-h} = 2$$

$$\text{RHD} = Rf'(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \to 0} \frac{3(2+h-1)^2 - 3}{h} = 6$$

Hence, $f(x)$ is not differentiable at $x = 2$

40.(ABD) Graph is symmetrical about (4, 0)

$\Rightarrow \quad f(4+x) = -f(4-x) \quad \Rightarrow \quad f(x) = -f(8-x)$

Since f is invertible, replace x by $f^{-1}(x)$

$x = -f(8 - f^{-1}(x))$ or $f^{-1}(-x) + f^{-1}(x) = 8$

$\Rightarrow \quad f^{-1}(2010) + f^{-1}(-2010) = 8 \quad \Rightarrow \quad$ Option (a) is true.

and $\displaystyle\int_{-2010}^{2018} f(x)\,dx = \int_{-2010}^{2018} f(8-x)\,dx = -\int_{2010}^{2018} f(x)\,dx = 0$

$\Rightarrow$ Option (b) is true.

Also, $D = \left(f'(10)\right)^2 + 4f'(10) > 0$

As, $f'(-100) > 0 \Rightarrow f'(10) \ge 0$

$f'(-100) > 0 \Rightarrow f'(10) \ge 0 \qquad cx^2 - f'(10)x - f'(10) = 0$ has real roots.

$\Rightarrow$ Option (c) is false.

As, $f'(4+x) = f'(4-x)$

$f'(x)$ is symmetric about $x = 4 \quad \Rightarrow \quad f'(10) = f'(-2) = 20 \quad \Rightarrow \quad$ Option (d) is true.

41.(B, C) Here, $f'(x)=\frac{1}{x}+\sqrt{1+\sin x}, x>0$ but $f'(x)$ is not differentiable in $(0,\infty)$ as $\sin x$ may be -1 and then

$f''(x)=-\frac{1}{x^2}+\frac{1}{2}\frac{\cos x}{\sqrt{1+\sin x}}$ will not exist.

$\Rightarrow$ $f'(x)$ is continuous for all $x\in(0,\infty)$ but $f'(x)$ is not differentiable on $(0,\infty)$.

$\therefore$ Option (b) is true.

Also, $f'(x)\le 3$, if $x>1$ and $f(x)>3$, if $x>e^3$

$\therefore$ Let $\alpha=e^3$ $\Rightarrow$ Option (c) is true.

(d) is not possible, as $f(x)\to\infty$ when $x\to\infty$.

42.(ABC) RHL $=\lim\limits_{x\to 0^+}\left(3-\left[\cot^{-1}\left(\frac{2x^3-3}{x^2}\right)\right]\right)=3-\left[\cot^{-1}(-\infty)\right]=3-3=0$

LHL $=\lim\limits_{x\to 0}\{x^2\}\cos\left(e^{-\frac{1}{x}}\right)=\lim\limits_{h\to 0}h^2\cos\left(e^{\frac{1}{h}}\right)=0$ and $f(0)=0$

Hence, $f(x)$ is continuous at $x=0$.

43.(AC) $\lim\limits_{x\to -1^+}f(x)=\lim\limits_{h\to 0}\left(b\left([-1+h]^2+[-1+h]\right)+1\right)=\lim\limits_{h\to 0}\left(b\left((-1)^2-1\right)+1\right)=1$

$\Rightarrow$ $b\in R$

$\lim\limits_{x\to -1^-}f(x)=\lim\limits_{x\to -1^-}\sin(\pi(x+a))=\lim\limits_{h\to 0}\sin\left(\pi((-1-h)+a)\right)$

$\therefore$ $\sin\pi a=-1$

$\pi a=2n\pi+\frac{3\pi}{2}\Rightarrow a=2n+\frac{3}{2}$

44.(BCD) $f(x)=\begin{cases}\dfrac{\sin\frac{\pi}{4}}{1}=\dfrac{1}{\sqrt{2}}, & 1\le x<2\\[2ex] \dfrac{\sin\frac{\pi}{2}}{1}=\dfrac{1}{2}, & 2\le x<3\end{cases}$

Hence, $f(x)$ is continuous at $f(x)\ \frac{3}{2}$, differentiable at $\frac{4}{3}$ and discontinuous at 2.

45.(AD) $H(x)=\begin{cases}\cos x, & 0\le x<\frac{\pi}{2}\\ \frac{\pi}{2}-x, & \frac{\pi}{2}<x\le 3\end{cases}$

$H'\left(\frac{\pi^-}{2}\right)=-\sin x=-1$

$H'\left(\frac{\pi^+}{2}\right)=-1$

Hence, $H(x)$ is continuous and derivable in $[0,3]$ and has maximum value 1 in $[0,3]$.

46.(AC) We have, $f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{x^2-[x]^2} & , \quad x>0\\ 1 & , \quad x=0\\ \sqrt{\{x\}\cot\{x\}} & , \quad x<0\end{cases}$

$$\text{RHL} = \lim_{x\to0^+} f(x)=\lim_{h\to0}\frac{\tan^2\{h\}}{h^2-[h]^2}=\lim_{h\to0}\frac{\tan^2 h}{h^2}=1$$

$$\text{LHL} \quad =\lim_{h\to0} f(x)=\lim_{h\to0}\sqrt{\{-h\}\cot\{-h\}}=\lim_{h\to0}\sqrt{(1-h)\cot(1-h)}=\sqrt{\cot 1}$$

$$\therefore \quad \cot^{-1}\left(\lim_{x\to0^-} f(x)\right)^2=\cot^{-1}\left(\sqrt{\cot 1}\right)^2=1$$

47.(ACD) Given, $f(x)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\dfrac{x}{(rx+1)\{(r+1)x+1\}}=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\dfrac{[(r+1)x+1]-(rx+1)}{(rx+1)[(r+1)x+1]}$

$$=\lim_{n\to\infty}\sum_{r=0}^{n}\left(\frac{1}{rx+1}-\frac{1}{(r+1)x+1}\right)=\lim_{n\to\infty}\left[\begin{array}{l}\left(\frac{1}{1}-\frac{1}{x+1}\right)+\left(\frac{1}{x+1}-\frac{1}{2x+1}\right)\\ +\ldots\ldots+\left(\frac{1}{nx+1}-\frac{1}{(n+1)x+1}\right)\end{array}\right]=\lim_{n\to\infty}\left[1-\frac{1}{(n+1)x+1}\right]=\lim_{n\to\infty}\frac{(n+1)x}{(n+1)x+1}$$

48.(AB) $\text{RHL} = \lim\limits_{x\to0^+}\dfrac{\cot^{-1}(1/x)}{x}=\lim\limits_{x\to0^+}\dfrac{\tan^{-1}}{x}=1$

$\text{LHL} \quad =\lim\limits_{x\to0^-}\dfrac{\cot^{-1}(1/x)}{x}=\lim\limits_{x\to0^-}\dfrac{\pi+\tan^{-1}x}{x}=-\infty \quad \therefore$ RHL exists and LHL does not exists.

49.(BCD) Put $x=\dfrac{\pi}{2}-h$

$$l=\lim_{h\to0}\frac{a^{\tanh}-a^{\sinh}}{\tanh-\sin\text{h}}=\lim_{h\to\infty}\frac{a^{\sinh}\left(a^{\tanh-\sinh}-1\right)}{\tanh-\sinh}=\log_e a$$

and $m=\lim\limits_{x\to-\infty}\left(\sqrt{x^2+ax}-\sqrt{x^2-ax}\right)=\lim\limits_{x\to-\infty}\dfrac{(x^2+ax)-(x^2-ax)}{\sqrt{x^2+ax}+\sqrt{x^2-ax}}=\lim\limits_{x\to-\infty}\dfrac{2ax}{|x|\left(\sqrt{1+\frac{a}{x}}+\sqrt{1-\frac{a}{x}}\right)}=-a$

50.(BCD) $f(x)=\lim\limits_{x\to\infty}\left[1+\dfrac{ax+1}{bx+2}-1\right]^x=\lim\limits_{x\to\infty}\left[1+\dfrac{(a-b)x-1}{bx+2}\right]^x=e^{\lim\limits_{x\to\infty}\left[\frac{(a-b)x-1}{bx+2}\right]^x}$

When $a=b, f(x)=e^{\lim\limits_{x\to\infty}\frac{x}{bx+2}}=e^{\frac{1}{b}}$ or $e^{\frac{1}{a}}$.

When $a>b, f(x)=e^{\infty}=\infty,$ does not exist.

When $a<b, f(x)=e^{-\infty}=0.$

51.(BC) (A) $\lim\limits_{x\to 3}[[x]]-[2x-1]$

RHL $=\lim\limits_{h\to 0}[[3+h]]-[6+2h-1]=\lim\limits_{h\to 0}3-5=-2$

LHL $\lim\limits_{h\to 0}[[3-h]]-[6-2h-1]=\lim\limits_{h\to 0}2-4=-2$ $\quad\therefore$ limit exists

(B) $\lim\limits_{x\to 1}[x]-x$

RHL $=\lim\limits_{h\to 0}[1+h]-(1+h)=\lim\limits_{h\to 0}(1-1-h)=0$

LHL $=\lim\limits_{h\to 0}[1-h]-(1+h)=\lim\limits_{h\to 0}0-1+h=-1$ $\quad\therefore$ Limit does not exist.

(C) $\lim\limits_{x\to 0}\{x\}^2-\{-x\}^2$

RHL $=\lim\limits_{x\to 0}\{h\}^2-\{-h\}^2=\lim\limits_{h\to 0}h^2-1(1-h)^2=-1$

LHL $=\lim\limits_{h\to 0}\{-h\}^2-\{h\}^2=\lim\limits_{h\to 0}(1-h)^2-h^2=1$ $\quad\therefore$ Limit does not exist.

(D) $\lim\limits_{x\to 0}\dfrac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x}$; RHL $=\lim\limits_{h\to 0}\dfrac{\tan(1)}{-1}=\tan 1$

LHL $=\lim\limits_{h\to 0}\dfrac{\tan(-1)}{-1}=\tan 1$ $\quad\therefore$ Limit exists.

52.(BC) We have $f(1)=f(-1)=1$

Let $x>1$ then $0<\dfrac{1}{x^2}<1 \Rightarrow \left[\dfrac{1}{x^2}\right]=0\Rightarrow f(x)=0\,\forall x>1$

Also, if $x<-1$, then $x^2>1\Rightarrow 0<\dfrac{1}{x^2}<1 \quad\Rightarrow\quad f(x)=0\,\forall x<-1$

Hence $f(1)=1$, $f(-1)=1$ and $f(x)=0$ if $|x|>1$ $\quad\therefore\quad f(x)$ cannot be continuous at $x=1$ and $x=-1$

Again Let $\dfrac{1}{2}<x^2<1\Rightarrow 1<\dfrac{1}{x^2}<2 \Rightarrow \left[\dfrac{1}{x^2}\right]=1\Rightarrow\dfrac{1}{2}<x^2\left[\dfrac{1}{x^2}\right]<1 \quad\therefore\quad \left[x^2\left[\dfrac{1}{x^2}\right]\right]=0$

$\Rightarrow\quad f(x)=0$ if $x\in\left(-1,\dfrac{1}{\sqrt{2}}\right)\cup\left(\dfrac{1}{\sqrt{2}},1\right)$

Next, Let $\dfrac{1}{3}<x^2<\dfrac{1}{2}\Rightarrow 2<\dfrac{1}{x^2}<3 \Rightarrow \left[\dfrac{1}{x^2}\right]=2\Rightarrow\dfrac{2}{3}<x^2\left(\dfrac{1}{x^2}\right)<1 \quad\therefore\quad \left[x^2\left[\dfrac{1}{x^2}\right]\right]=0\Rightarrow f(x)=0$ if

$x\in\left(-\dfrac{1}{\sqrt{2}},\dfrac{1}{\sqrt{3}}\right)\cup\left(\dfrac{1}{\sqrt{3}},\dfrac{1}{2}\right)$

At $x=\pm\dfrac{1}{\sqrt{2}},x^2=\dfrac{1}{2}.\dfrac{1}{x^2}=2\Rightarrow f(x)=1$

Similarly, at $x=\dfrac{1}{\sqrt{3}},\dfrac{1}{2},\dfrac{1}{\sqrt{5}}$...............

$\therefore\quad f(x)$ is discontinuous at infinite number of points

Given by $x\in\left\{\pm\dfrac{1}{\sqrt{n}},n\in N\right\}$.

53.(AC) $f(x)=[\tan x]+\sqrt{\tan x-[\tan x]}=[t]+\sqrt{t-[t]}$, where $t=\tan x$

Clearly $0\le t<\infty$ as $0\le x<\frac{\pi}{2}$. Possible points of discontinuities may be, at which $t\in N$.

Let $t=k\in N$

L.H.L. (at $t=k$) $=\lim\limits_{t\to k^-}[t]+\sqrt{t-[t]}=\lim\limits_{h\to 0}[k-h]+\sqrt{k-h-[k-h]}=\lim\limits_{h\to 0}\left\{k-1+\sqrt{k-h-k+1}\right\}=k$

R.H.L. (at $t=k$) $=\lim\limits_{t\to k^+}[t]+\sqrt{t-[t]}=\lim\limits_{h\to 0}k+\sqrt{k+h-[k+h]}=\lim\limits_{h\to 0}k+\sqrt{k+h-k}=k$

$\therefore$ The function is continuous if $t=k\in N$ Thus function $f(x)$ is continuous for all $x\in\left[0,\frac{\pi}{2}\right)$.

54.(BC) If $0\le\sin^2 x<1$ then $f(x)=\sec^{-1}(-1)=0$

If $\sin^2 x=1$ then $f(x)=\sec^{-1}(2)=\frac{\pi}{3}$

Thus, $f(x)$ is not continuous if $\sin x=\pm 1$ i.e., $x=$ Odd multiple of $\frac{\pi}{2}$

55.(BD) For $x>2, \int_0^x\{1+|1-t|\}dt=\int_0^1(2-t)dt+\int_1^x dt=1+\frac{x^2}{2}$

Thus, $f(x)=\begin{cases}1+\frac{x^2}{2}, & x>2\\ 5x-7, & x\le 2\end{cases}$

$\lim\limits_{x\to 2+}f(x)=1+\frac{4}{2}=3=f(2)=\lim\limits_{x\to 2-}f(x)$

$f'(2+)=\lim\limits_{x\to 0+}\frac{1+(1/2)(2+h)^2-3}{h}=\lim\limits_{h\to 0}\frac{(1/2)h^2+2h}{h}=2$

$f'(2-)=\lim\limits_{x\to 0-}\frac{5(2+h)-7-3}{h}=5$ Hence f is continuous but not differentiable at $x=2$

56.(ABC) Given $F(x)=f(x).g(x)$.......(1)

Differentiating both sides w.r.t, x we get

$F'(x)=f'(x).g(x)+g'(x).f(x)\quad\Rightarrow\ F'(x)=f'(x)g'(x)\left[\frac{f(x)}{f'(x)}+\frac{g(x)}{g'(x)}\right]\ \Rightarrow\ F'=c\left[\frac{f}{f'}+\frac{g}{g'}\right]\Rightarrow$ (a) is correct

Again, differentiating both sides w.r.t., x we get

$F''(x)=f''(x).g(x)+g''(x).f(x)+2f'(x).g'(x)\ \Rightarrow\ F''(x)=f''(x).g(x)+g''(x).f(x)+2c$......(2)

Dividing both sides by $F(x)=f(x).g(x)$

$(\because f'(x).g'(x)=c)$ then $\frac{F''(x)}{F(x)}=\frac{f''(x)}{f(x)}+\frac{g''(x)}{g(x)}+\frac{2c}{f(x)g(x)}$ or $\frac{F''}{F}=\frac{f''}{f}+\frac{g''}{g}+\frac{2c}{fg}\Rightarrow$ (b) is correct

Again given $f(x)g'(x)=c$

Differentiating both sides w.r.t., x we get $f'(x)g''(x)+g'(x)f''(x)=0$

From (2), $F''(x) = f''(x).g(x) + g''(x).f(x) + 2c$

Differentiating both sides w.r.t., x we get

$F'''(x) = f''(x).g'(x) + f'''(x).g(x) + g''.f'(x) + f(x).g'''(x) + 0 = f'''(x).g(x) + g'''(x).f(x) + 0$ [from (3)]

Now dividing both sides by $F(x) = f(x)g(x)$

Then $\dfrac{F'''(x)}{F(x)} = \dfrac{f'''(x)}{f(x)} + \dfrac{g'''(x)}{g(x)}$ or $\dfrac{F'''}{F} = \dfrac{f'''}{f} + \dfrac{g'''}{g}$

57.(ABCD) $1+x$ is never zero, so $1+f(x)$ is never zero. It is 1 for $x=0$, so it is always positive.

Hence $f''(x)$ is always positive. $f'(x) > 0$, for all $x > 0$ and hence f is strictly increasing.

So, in particular, $1 + f(x) \geq 2$ for all x.

We have $f''(x) \leq \dfrac{(1+x)}{2}$.

Integrating, $f'(x) \leq f'(0) + \dfrac{x}{2} + \dfrac{x^2}{4} = \dfrac{x}{2} + \dfrac{x^2}{4}$.

Integrating, again, $f(x) \leq f(0) + \dfrac{x^2}{4} + \dfrac{x^3}{12}$. Hence $f(1) \leq 1 + \dfrac{1}{4} + \dfrac{1}{12} = \dfrac{4}{3}$.

58. **[A-p, q, r, s]** **[B-p, q, r, s]** **[C-p, q, r, s]** **[D-r]**

(A) $f(x) = \begin{cases} \dfrac{2}{x^3} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

$f'(0) = \lim\limits_{h\to 0} \dfrac{e^{-\frac{1}{h^2}} - 0}{h} = \lim\limits_{h\to 0} \dfrac{1}{h.e^{1/h^2}} = \lim\limits_{t\to\infty} t.e^{-t^2}$ let $\dfrac{1}{h} = t$

Applying L.H. Rule $= \lim\limits_{t\to\infty} \dfrac{1}{2t.e^{t^2}} = 0$

Similarly $= \lim\limits_{t\to-\infty} \dfrac{t}{e^{-t^2}} = 0$

$f''(0) = \lim\limits_{h\to 0} \dfrac{2}{h^4.e^{1/h^2}} = \lim\limits_{t\to\infty} \dfrac{2t^4}{e^{t^2}}$ let $\dfrac{1}{h} = t$

Applying L.H. Rule $= \lim\limits_{t\to\infty} \dfrac{8t^3}{2t.e^{t^2}} == \lim\limits_{t\to\infty} \dfrac{4t^2}{e^{t^2}} == \lim\limits_{t\to\infty} \dfrac{8t}{2te^{t^2}} = 0$

$f(x)$ is infinitely differentiable

$f^{(k)}(0) = 0$ for all k $\qquad$ A $\rightarrow$ p, q, r, s

(B) Similar to (A) $\qquad$ B $\rightarrow$ p, q, r, s

(C) $f_1(x) = g(x-e)$

$f_2(x) = g(\pi - x)$ where $g(x) = e^{-\frac{1}{x}}$ So $f(x) = f_1(x)f_2(x)$

As f_1, f_2 have both 1st and 2nd derivatives existing and continuous, the function f also will.

(D) Only 1st derivative exists and its not continuous.

59. **[A-s]** **[B-p]** **[C-p]** **[D-p]**

(A) Use sandwich theorem $x\left(\frac{1+2+3...+8}{x}-8\right)<\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+......+\left[\frac{8}{x}\right]\le x\left(\frac{1+2+3+...+8}{x}\right)$

Taking limits find that $\lim\limits_{x\to 0} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+....+\left[\frac{8}{x}\right]\right)=36$

(B) 1^{∞} form

$k=e^{\lim\limits_{x\to 0} xe^{-1/x^2}\sin\frac{1}{x^4}.e^{1/x^2}}=e^{\lim\limits_{x\to 0} x\cdot\sin\frac{1}{x^4}}=e^0=1$ B $\to$ p

(C) $u=\left(2\sin\sqrt{x}+\sqrt{x}\sin 1/x\right)^x$

$\ell nu=x\ell n\left(2\sin\sqrt{x}+\sqrt{x}\sin\frac{1}{x}\right)=x\ell n\left(\frac{\left(2\sin\sqrt{x}+\sqrt{x}\sin\frac{1}{x}\right).\sqrt{x}}{\sqrt{x}}\right)$

$=x\ell n\sqrt{x}+x\ell n\left(\frac{2\sin\sqrt{x}}{\sqrt{x}}+\sin\frac{1}{x}\right)=x\ell n\sqrt{x}+xg(x)$

$\lim\limits_{x\to 0} xg(x)=0$ $g(x)$ is bounded

$\lim\limits_{x\to 0}\frac{x}{2}\ell nx=\lim\limits_{x\to 0}\frac{\ell nx}{2/x}$

Using L.H. rule $=\lim\limits_{x\to 0}\frac{1/x}{-2/x^2}=\lim\limits_{x\to 0}-\frac{x}{2}=0$ $=\lim\limits_{x\to 0}(\ell nu)=0$ $u=e^0=1$ $C\to p$

(D) $k=\lim\limits_{x\to\frac{\pi}{2}} f(x)=\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1-\sin x}{4\left(\frac{\pi}{2}-x^2\right)}\right).\frac{\ln\sin x}{\ln(1+\pi^2-4\pi x+4x^2)}$

$=\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1-\sin x}{4(\pi-x)^2}\right)\cdot\lim\limits_{x\to\frac{\pi}{2}}\frac{\ln\sin x}{\ln(1+\pi^2-4\pi x+4x^2)}=\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1-\sin x}{4\left(\frac{\pi}{2}-x\right)^2}\right)\cdot\lim\limits_{x\to\frac{\pi}{2}}\frac{(1+\pi^2-4\pi x+4x^2)\cos x}{\sin x\cdot(8x-4\pi)}$

Let $x-\frac{\pi}{2}=t$

$\lim\limits_{t\to 0}\left(\frac{1-\cos t}{4t^2}\right)\cdot\frac{-\sin t}{-8t}\cdot\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1+\pi^2-4\pi x+4x^2}{\sin x}\right)=\frac{1}{8}\lim\limits_{t\to 0}\frac{2\sin^2 t/2}{16\cdot t^2/4}\cdot 1$; $K=\frac{1}{64}$; $8\sqrt{|k|}=1$ $D\to p$

60. **[A-s]** **[B-r]** **[C-q, r]** **[D-p]**

$F'(x)=kx(x^2-1)$

$F''(x)=k(3x^2-1)\to F''\left(\frac{1}{2}\right)=k\left(\frac{3}{4}-1\right)\Rightarrow k=4$

$F'(x)=4x(x^2-1)$ $F(x)=x^4-2x^2+\mu$

$f(0)=4\Rightarrow\mu=4$ $F(x)=x^4-2x^2+4=(x^2-1)^2+3$

No real roots A $\to s$; B $\to r$; C $\to q, r$; D $\to p$

61. **[A-p, q]** **[B-p, s]** **[C-q]** **[D-r, t]**

$$f(x) = x^2 + ax + b$$

$$f'(x)\Big|_{\left(\frac{5}{2}, p\right)} = 0 \;\Rightarrow 2x + a\Big|_{\left(\frac{5}{2}, p\right)} = 0 \;\Rightarrow 2\times\frac{5}{2} + a = 0$$

$$a = -5 \qquad \therefore \qquad f(x) = x^2 - 5x + b$$

Let its roots are $\alpha \,\&\, \beta,\ \alpha, \beta \in I$

$f(0) > 0 \qquad b > 0$

$\alpha + \beta = 5 \qquad \alpha\beta = b > 0$

Both $\alpha \,\&\, \beta$ are +ve (α, β) can be (1, 4) or (2, 3)

$\therefore \qquad b = 4$ or 6

$$f(x) = x^2 - 5x + 4 \text{ or } f(x) = x^2 - 5x + 6$$

62.(1) $k = \lim\limits_{x\to 0} \dfrac{\sin(\pi\cos^2(\tan(\sin x)))}{\pi x^2} = \lim\limits_{x\to 0} \dfrac{\sin(\pi - \pi\cos^2(\tan(\sin x)))}{\pi x^2} = \lim\limits_{x\to 0} \dfrac{\sin(\pi\sin^2(\tan(\sin x)))}{\pi x^2} = 1$

63.(1) $\dfrac{d^2x}{dy^2} = -\left(\dfrac{dx}{dy}\right)^3 \dfrac{d^2y}{dx^2}$

$\Rightarrow \quad \dfrac{d^2y}{dx^2} = \dfrac{-d^2x}{dy^2}\dfrac{1}{\left(\dfrac{dx}{dy}\right)^3} \qquad \therefore \qquad x\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 - \dfrac{dy}{dx} = 0$

Becomes $-x\dfrac{d^2x}{dy^2}\dfrac{1}{\left(\dfrac{dx}{dy}\right)^3} + \dfrac{1}{\left(\dfrac{dx}{dy}\right)^2} - \dfrac{1}{\dfrac{dx}{dy}} = 0$

$\Rightarrow \quad x\dfrac{d^2x}{dy^2} + \left(\dfrac{dx}{dy}\right)^2 = \dfrac{dx}{dy} \qquad \therefore \qquad \lambda = 1$

64.(6) $g(x) = f(-x + f(f(x)))$

$g'(x) = f'(-x + f(f(x)) \times (-1 + f'(f(x)) \times f'(x))$

$g'(0) = 6$

65.(1). In the vicinity of $x = 0$, we have $x^2 \displaystyle\sum_{r=0}^{\left[\frac{1}{|x|}\right]} r \;= x^2\left(1 + 2 + 3 + \ldots \left[\frac{1}{|x|}\right]\right)$

Use sandwich theorem $P = x^2\left(1 + 2 + 3 + \ldots\ldots + \left[\dfrac{1}{|x|}\right]\right) = \dfrac{x^2\left(1 + \left[\frac{1}{|x|}\right]\right)}{2}\left[\dfrac{1}{|x|}\right]$

So $\quad \dfrac{1}{2}(1 - |x|) < P \le \dfrac{1}{2}(1 + |x|)$

Then the limit is $\dfrac{1}{2}$

66.(8) $1+f(x+y)=f(x)+f(y)+f(x).f(y)+1$

$1+f(x+y)=(1+f(x))(1+f(y))$

Let $1+f(x)=g(x)$

$g(x+y)=g(x).g(y)$

From this functional equation $g(x)=e^{\lambda x}$

$f(x)=e^{\lambda x}-1$, $f'(x)=\lambda.e^{\lambda x}, f'(0)=\lambda=1$, $f(x)=e^{x}-1$

$$\left[\frac{e^4-1}{e^2-1}\right]=\left[e^2+1\right]=[7.29+1]=8$$

67.(2) $$f'(1^+)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}=\lim_{h\to 0}\frac{\frac{\pi}{4}+\frac{1+h-1}{2}-\frac{\pi}{4}}{h}=\frac{1}{2}$$

68.(2) Let $x=K\sin\theta$

$$\lambda=\lim_{\theta\to 0}\frac{1}{K^2}\left(\frac{1-4K\cos\theta}{\sin^2\theta\cos\theta}\right)$$ λ is finite $K=\frac{1}{4}\Rightarrow \lambda=\frac{2}{K}$

69.(4) $$\lim_{x\to 1}\frac{x^2\sin(f(x^2))\times 2x+2x\int_1^{x^2}\sin f(t)dt}{2(x-1)}$$

Applying L – H rule again. We will get $2f'(1)=4$

70.(8) $$\left(f^{-1}(x)\right)'_{at\ x=0}=\frac{1}{f'(0)}$$

71.(5) We know, $\sin 3\theta=3\sin\theta-4\sin^3\theta=\sin\theta(1+2\cos 2\theta)$ $\quad\therefore\quad \sin 3\theta=\sin\theta(1+2\cos 2\theta)$ …(i)

Now, on putting $3\theta=x,\frac{x}{3},\frac{x}{3^2},......$ one by one in Eq. (i), we get $\Rightarrow$ $\sin x=\sin\left(\frac{x}{3}\right).\left(1+2\cos\frac{2x}{3}\right)$

$$\Rightarrow\quad \frac{\sin x}{x}=\lim_{n\to\infty}\frac{\sin x/3^n}{x/3^n}\prod_{k=1}^{n}\frac{\left(1+2\cos\left(\frac{2x}{3^k}\right)\right)}{3^n}\quad\Rightarrow\quad \frac{\sin x}{x}=\prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right)=f(x)$$

Thus, $\left|xf(x)\right|+\left|\left|x-2\right|-1\right|=\frac{|x||\sin x|}{|x|}+\left|\left|x-2\right|-1\right|$ is not

Differentiable at $\{\pi,2\pi,1,2,3\}$ $\quad\therefore$ Number of points are 5.

72.(3) Here, $$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f\left(\frac{2x+2h}{2}\right)-f(x)}{h}=\lim_{h\to 0}\frac{f\left(\frac{2x(1+h/x)}{2}\right)-f\left(\frac{2x.1}{2}\right)}{h}$$

$$=\frac{f(2x)}{2x}.\lim_{h\to 0}\frac{f(1+h/x)-f(1)}{h/x}=\frac{f(2x)}{2x}.f'(1)=\frac{f(2x)}{2x}.f(1),$$ given $f(1)=f'(1)$

$$=\frac{f\left(\frac{2x.1}{2}\right)}{x},$$ using $f\left(\frac{xy}{2}\right)=\frac{f(x).f(y)}{2}$ $\quad\therefore\quad f'(x)=\frac{f(x)}{x}\Rightarrow\frac{f(x)}{f'(x)}=x\Rightarrow\frac{f(3)}{f'(3)}=3$

73.(4) Given, $f(x)=f(y)f(x-y)$

On replacing x by $(x+y)$,

$f(x+y)=f(y).f(x) \quad \Rightarrow \quad f(x)=e^{kx} \Rightarrow f'(x)=ke^{kx}$

But, $f'(0)=\int_0^4 \{2x\}dx=2 \quad \Rightarrow \quad f'(0)=k=2 \Rightarrow f'(x)=2e^{2x}$

$f'(-3)=2e^{-6} \Rightarrow |\alpha+\beta|=4$

74.(1) Here, $(f(\alpha))^2+(f'(\alpha))^2=0 \Rightarrow f(\alpha)=f'(\alpha)=0 \quad \therefore \quad \alpha$ is repeated root of $f(x)$

Now, $\lim_{x\to\alpha}\frac{f(x)}{f'(x)}\left[\frac{f'(x)}{f(x)}\right]=\lim_{x\to\alpha}\frac{f(x)}{f'(x)}\left(\frac{f'(x)}{f(x)}-\left\{\frac{f'(x)}{f(x)}\right\}\right)=\lim_{x\to\alpha}\left(1-\frac{f(x)}{f'(x)}.\left\{\frac{f'(x)}{f(x)}\right\}\right)$

[as, $x=\alpha$ is the repeated root and $\left\{\frac{f(x)}{f'(x)}\right\}$ is bounded; $\therefore \lim_{x\to\alpha}\frac{f(x)}{f'(x)}=0]=1-0=1$

75.(9) Since, $f(10-x)=f(x)=f(4-x) \quad \Rightarrow \quad f(10-x)=f(4-x)$

Say, $4-x=t \Rightarrow f(6+t)=f(t) \quad \Rightarrow \quad f(x)$ is periodic function with period 6.

So, for $x\in[0,25]$

$f(x)=101$ at $x=0,6,12,18,24$

Total number = 5

Since $f(2-x)=f(2+x) \Rightarrow f(x)$ is symmetric about $x=2$ line

Due to symmetry in one period length.

$f(x)=101$ has one solution at $x=4$ other than 0 and 6.

Now, $f(x)=101$ at $x=4,10,16,22$

Total numbers = 4 Hence, at least minimum possible number of values of $x=9$.

76.(1.41) RHL $=\lim_{h\to0}\frac{\left(\frac{\pi}{2}-\sin^{-1}(1-h^2)\right)\sin^{-1}(1-h)}{\sqrt{2}(h-h^2)}=\lim_{h\to0}\frac{\cos^{-1}(1-h^2)\sin^{-1}(1-h)}{\sqrt{2}h(1-h^2)}=\lim_{h\to0}\frac{\sin^{-1}\sqrt{2h^2-h^4}.\sin^{-1}(1-h)}{\sqrt{2}h(1+h)(1-h)}$

$=\lim_{h\to0}\frac{\sin^{-1}\left(h\sqrt{2-h^2}\right)\sin^{-1}(1-h)}{\sqrt{2}h(1+h)(1-h)}=\lim_{h\to0}\frac{1}{\sqrt{2}}.\frac{\sin^{-1}\left(h\sqrt{2-h^2}\right)}{h.\sqrt{2-h^2}}.\frac{\sqrt{2-h^2}}{1+h}.\frac{\sin^{-1}(1-h)}{1-h}$

$=\frac{1}{\sqrt{2}}.\sqrt{2}.\frac{\sin^{-1}(1)}{1}=\frac{\pi}{2}\Rightarrow K=\frac{\pi}{2}$ ....(1)

LHL $=\lim_{h\to0}\frac{A\sin^{-1}(1-(1-h)).\cos^{-1}(1-(1-h))}{\sqrt{2(1-h).(1-(1-h))}}$

$\lim_{h\to0}\frac{A.\sin^{-1}(h).\cos^{-1}(h)}{\sqrt{2}(1-h).h}=\frac{A.\pi/2}{\sqrt{2}}=\frac{A\pi}{2\sqrt{2}} \quad \Rightarrow \quad \frac{A\pi}{2\sqrt{2}}=\frac{\pi}{2}\Rightarrow A=\sqrt{2}$

77.(0) A, B, C are in AP. $\therefore$ $2B = A + C$ and $A + B + C = 180°$ $\therefore$ $B = 60°$

$$\therefore \quad \cos B = \frac{1}{2} = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow a^2 + c^2 = b^2 + ac \Rightarrow (a-c)^2 = b^2 - ac$$

Or $\quad |\sin A - \sin C| = \sqrt{\sin^2 B - \sin A \sin C}$

$$\Rightarrow \quad 2\cos\left(\frac{A+C}{2}\right)\left|\sin\left(\frac{A-C}{2}\right)\right| = \sqrt{\frac{3}{4} - \sin A \sin C} \Rightarrow 2\left|\sin\left(\frac{A-C}{2}\right)\right| = \sqrt{3 - 4\sin A \sin C}$$

$$\therefore \quad \lim_{A \to C} \frac{\sqrt{3 - 4\sin A \sin C}}{|A - C|} = \lim_{A \to C} \frac{2\left|\sin\left(\frac{A-C}{2}\right)\right|}{|A-C|}$$

$$\lim_{A \to C} \left| \frac{\sin\left(\frac{A-C}{2}\right)}{\left(\frac{A-C}{2}\right)} \right| = |1| = 1 \Rightarrow f(x) = 1 \qquad \therefore \qquad f'(x) = 0$$

78.(3) $F(x) = 3e^x$ and $G(x) = e^{-x}$

The equation $9x^4 = (F(x))^2 G(x)$ becomes $x^4 = e^x$

Hence, number of solutions = 3

79.(2) Let $g(x) = x\tan^{-1}(x^2)$. It is an odd function. So, $g^{2m}(0) = 0$.

Let $h(x) = x^4$ So, $f(x) = g(x) + h(x) \Rightarrow f^{2m}(0) = g^{2m}(0) + h^{2m}(0) = h^{2m}(0) \neq 0$

It happens when $2m = 4 \Rightarrow m = 2$

80.(3) Let $\left[\sqrt{n} + \frac{1}{2}\right] = K \Rightarrow K \le \sqrt{n} + \frac{1}{2} < (K+1) \Rightarrow (K - 1/2)^2 \le n < (K + 1/2)^2$

$$\Rightarrow \quad K^2 - K + 1/4 \le n < K^2 + K + 1/4, K \in N, n \in N \quad \Rightarrow \quad K^2 - K + 1 \le n \le K^2 + K$$

$$\text{So, } S = \sum_{n=1}^{\infty} \frac{2^{f(n)} + 2^{-f(n)}}{2^n} = \sum_{n=1}^{\infty} \frac{2^K + 2^{-K}}{2^n} = \left(\frac{2 + 2^{-1}}{2} + \frac{2^1 + 2^{-1}}{2^2}\right) + \left(\frac{2^2 + 2^{-2}}{2^3} + \frac{2^2 + 2^{-2}}{2^4} + \frac{2^2 + 2^{-2}}{2^5} + \frac{2^2 + 2^{-2}}{2^6}\right)$$

$$\text{As, } \begin{cases} K = 1 & \Rightarrow & 1 \le n \le 2 \\ K = 2 & \Rightarrow & 3 \le n \le 6 \\ K = 3 & \Rightarrow & 7 \le n \le 12 \\ \vdots & & \vdots \end{cases}$$

$$\therefore \quad S = \sum_{K=1}^{\infty} \sum_{n=K^2-K+1}^{K^2+K} \frac{2^K + 2^{-K}}{2^n} = \sum_{K=1}^{\infty} (2^K + 2^{-K}) . \sum_{n=K^2-K+1}^{K^2+K} \frac{1}{2^n}$$

$$= \sum_{K=1}^{\infty} (2^K + 2^{-K}) . \left[\frac{1}{2^{K^2-K+1}} + \frac{1}{2^{K^2-K+2}} + ... + \frac{1}{2^{K^2+K}}\right] = \sum_{K=1}^{\infty} (2^K + 2^{-K}) \frac{\frac{1}{2^{K^2-K+1}}\left(1 - \frac{1}{2^{2K}}\right)}{1 - 1/2}$$

$$= (2^1 - 2^{-3}) + (2^0 - 2^{-8}) + (2^{-3} - 2^{-15}) + (2^{-8} - 2^{-24}) + ...\infty = 2 + 1 = 3$$

81.(1) We have, $\lim\limits_{x\to\frac{\pi}{2}}\sqrt{\dfrac{\tan x-\sin\left\{\tan^{-1}(\tan x)\right\}}{\tan x+\cos^2(\tan x)}}$

Now, RHL at $x=\dfrac{\pi}{2}$,

$$=\lim_{x\to\frac{\pi^+}{2}}\sqrt{\frac{\tan x-\sin\left\{\tan^{-1}(\tan x)\right\}}{\tan x+\cos^2(\tan x)}}=\lim_{x\to\frac{\pi^+}{2}}\sqrt{\frac{\tan x-\sin(x-\pi)}{\tan x+\cos^2(\tan x)}}=1 \qquad \left[\because\ \tan^{-1}(\tan x)=x-\pi,\text{when } x>\frac{\pi}{2}\right]$$

Again, LHL at $x=\dfrac{\pi}{2}$,

$$=\lim_{x\to\frac{\pi-}{2}}\sqrt{\frac{\tan x-\sin\left\{\tan^{-1}(\tan x)\right\}}{\tan x+\cos^2(\tan x)}}=\lim_{x\to\frac{\pi-}{2}}\sqrt{\frac{\tan x-\sin(x)}{\tan x+\cos^2(\tan x)}}\left[\because \tan^{-1}(\tan x)\right]=x,\text{when } x<\frac{\pi}{2}$$

$$=\lim_{x\to\frac{\pi-}{2}}\sqrt{\frac{1-\frac{\sin x}{\tan x}}{1+\frac{\cos^2(\tan x)}{\tan x}}}=\sqrt{\frac{1+0}{1+0}}=1=\lim_{x\to\frac{\pi}{2}}\sqrt{\frac{\tan x-\sin\left\{\tan^{-1}(\tan x)\right\}}{\tan x+\cos^2(\tan x)}}=1$$

82.(1.5) We have, $A(t)=\int_0^t \sin x^2.dx; B(t)=\dfrac{t\sin t^2}{2}$ $\quad\therefore\quad \lim\limits_{t\to0}\dfrac{A(t)}{B(t)}=\lim\limits_{t\to0}\dfrac{2\int_0^t\sin x^2.dx}{t\sin t^2}=\lim\limits_{t\to0}\dfrac{2\int_0^t\sin x^2dx}{t^3.\frac{\sin^2 t}{t^2}}$

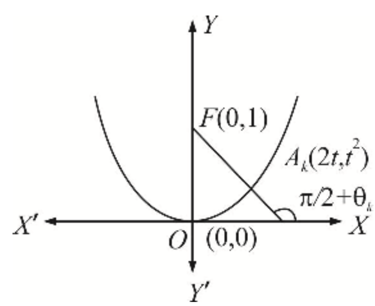
$$\lim_{t\to0}\frac{2\int_0^t\sin x^2.dx}{t^3}=\lim_{t\to0}\frac{2.\sin\left(t^2\right)}{3t^2}$$

[applying Newton –Leibnitz's rule followed by L' Hospital's rule]

$\therefore\quad \lim\limits_{t\to0}\dfrac{A(t)}{B(t)}=\dfrac{2}{3}$

Then, $m+n=2+3=5$

83.(4) Let $A_k=\left(2t,t^2\right)$ $\therefore$ Slope of $FA_k=\dfrac{t^2-1}{2t-0}=\tan\left(\dfrac{\pi}{2}+\theta_k\right)$



$\tan(\theta_k)=\dfrac{2t}{1-t^2}=\tan(2\phi)$, where $t=\tan\phi$

$\therefore\quad \phi=\dfrac{\theta_k}{2}=\dfrac{k\pi}{4n}$, where $\tan\phi=t$.

Also, $FA_k=\sqrt{\left(t^2-1\right)^2+(2t)^2}=t^2+1=1+\tan^2\phi=\sec^2\left(\dfrac{k\pi}{4n}\right)$

$$\therefore \lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}FA_k=\lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}\sec^2\left(\frac{k\pi}{4n}\right)=\int_0^1\sec^2\left(\frac{\pi x}{4}\right)dx=\frac{4}{\pi}.\left[\tan\left(\frac{\pi x}{4}\right)\right]_0^1=\frac{4}{\pi}$$

Hence, $m=4$.

84.(2) Let $f(x)=\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+.....+\frac{1}{\sqrt{n^2+2n}}\right)$

As, $\frac{1}{\sqrt{n^2+2n}}<\frac{1}{\sqrt{n^2}}=\frac{1}{\sqrt{n^2}}$

$\frac{1}{\sqrt{n^2+2n}}<\frac{1}{\sqrt{n^2+1}}<\frac{1}{\sqrt{n^2}}$

On adding, we get $\frac{2n+1}{\sqrt{n^2+2n}}<\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+.....+\frac{1}{\sqrt{n^2+2n}}<\frac{2n+1}{\sqrt{n^2}}$

$2<\lim\limits_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+.....+\frac{1}{\sqrt{n^2+2n}}\right)<2 \quad \therefore \quad \lim\limits_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+....+\frac{1}{\sqrt{n^2+2n}}\right)=2$

85.(3.14) Here, $\sin(x_{n+1}-x_n)+2^{-(n+1)}\sin x_n.\sin x_{n+1}=0$

$\Rightarrow \quad \cot x_{n+1}-\cot x_n=2^{-(n+1)} \quad \Rightarrow \quad \cot x_n-\cot x_{n-1}=2^{-n}$

$\Rightarrow \quad \cot x_{n+1}-\cot x_1=\frac{1}{2^2}+\frac{1}{2^3}+......+\frac{1}{2^{n+1}} \quad \Rightarrow \quad \cot x_{n+1}=\frac{1}{2}+\frac{1}{2^2}+........\frac{1}{2^{n+1}}$

$\Rightarrow \quad \lim\limits_{n\to\infty}\cot x_{n+1}=1 \quad \therefore \quad \lim\limits_{n\to\infty}x_{n+1}=\frac{\pi}{4}$

$l=\frac{\pi}{4}\Rightarrow 4l=\pi$

86.(8) We have, $f(x+y)=f(x)+f(y)+xy(x+y)$

$f(0)=0 \quad \therefore \quad \lim\limits_{h\to 0}\frac{f(h)}{h}=-1 \quad \therefore \quad \lim\limits_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$=\lim\limits_{h\to 0}\frac{f(x)+f(h)+xh(x+h)-f(x)}{h}=\lim\limits_{h\to 0}\frac{f(h)}{h}+\lim\limits_{h\to 0}x(x+h)=-1+x^2$

$\therefore \quad f'(x)=-1+x^2 \quad \therefore \quad f(x)=\frac{x^3}{3}-x+c \quad \therefore \quad f(x)$ is a polynomial function.

$f(x)$ is twice differentiable for all $x\in R$ and

$f'(3)=3^2-1=8$.

87.(3) $\lim\limits_{x\to\infty}4x\frac{\tan^{-1}\left(\frac{1}{2x+3}\right)}{\left(\frac{1}{2x+3}\right)}\times\frac{1}{2x+3}=2 \quad \Rightarrow \quad y^2+4y+5=2 \quad \Rightarrow \quad y=-1,-3.$

88.(12.07) Given limit $=\lim\limits_{x\to 0}\frac{a\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-......\right)-bx+cx^2+x^3}{2x^2\left(x-\frac{x^2}{2}+\frac{x^3}{3}-.....\right)-2x^3+x^4}=\lim\limits_{x\to 0}\frac{(a-b)x+cx^2+\left(1-\frac{a}{6}\right)x^3+\frac{ax^5}{120}....}{2\frac{x^5}{3}-\frac{x^6}{2}+...}$

For this limit to exist, we must have $a=b,c=0,a=6$ and given limit $=\frac{a}{120}\times\frac{3}{2}=\frac{6\times 3}{120\times 2}=\frac{3}{40}$.

89.(6) $\ln P = \frac{1}{n}\left[\left\{\ln\left(n^3+1\right)+\ln\left(n^3+2^3\right)+.........+\ln\left(n^3+n^3\right)\right\}-3n\ln n\right]$

Hence $T_r = \frac{1}{n}\left[\ln\left(n^3+r^3\right)-3\ln n\right] = \frac{1}{n}\left[3\ln n+\ln\left(1+\left(\frac{r}{n^3}\right)^3\right)-3.\ln n\right] = \frac{\ln\left(1+\left(\frac{r}{h}\right)^3\right)}{n}$

Let $S = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(1+\left(\frac{r}{n}\right)^3\right) = \int_0^1 \ln\left(1+x^3\right)dx = \int_0^1 \ln(1+x)dx + \int_0^1 \ln\left(x^2-x+1\right)dx$

$= \ln(1+x).x]_0^1 - \int_0^1 \frac{1+x-1}{1+x}dx + \ln\left(1-x+x^2\right).x]_0^1 - \int_0^1 \frac{x(2x-1)}{x^2-x+1}dx$

$\ln 2-(1-\ln 2)-\int_0^1 \frac{2x^2-x}{x^2-x+1}dx$

$S = \ln 4 - 1 - I_1$

$I_1 = \int_0^1\left(2+\frac{x-2}{x^2-x+1}\right)dx = 2-\frac{\pi}{\sqrt{3}}$ $\Rightarrow$ $S = \ln 4 - 1 - 2 + \frac{\pi}{\sqrt{3}} = \ln 4 + \frac{\pi}{\sqrt{3}} - 3.$ $\therefore$ $\ln P = \ln 4 + \frac{\pi}{\sqrt{3}} - 3$

$\Rightarrow$ $P = 4e^{\frac{\pi}{\sqrt{3}}}.e^{-3}.$

90.(2) $H_n = \frac{n}{\frac{1}{n+1}+\frac{1}{n+2}+.........+\frac{1}{n+n}}$ $\Rightarrow$ $\frac{n}{H_n} = \frac{1}{n+1}+\frac{1}{n+2}+.......+\frac{1}{n+n}$

$\Rightarrow$ $\lim_{n\to\infty}\left(\frac{n}{H_n}\right) = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n+r} = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{\frac{1}{n}}{1+\frac{r}{n}} = \int_0^1\frac{dx}{1+x} = \ln 2$

$\therefore$ $\lim_{n\to\infty}\left(\frac{H_n}{n}\right) = \frac{1}{\ln 2}.$

DIFFERENTIAL CALCULUS-2

1.(D) Let $f(x)=x^2+3x+4$

Then we have to find solutions of $f(f(x))=x$

$f(x)-x>0$

$\therefore$ $f(f(x))-f(x)>0$

$\therefore$ $f(f(x))>x,$ thus $f(f(x))=x$ has no solution.

2.(B) Let us consider $g(t)=\int_0^t f(x)dx$

Applying LMVT in interval (0, 2)

We get $\frac{g(2)-g(0)}{2-0}=g'(c)$

$$\int_0^2 f(x)dx=2f(c) \quad c\in(0,2).$$

3-5. **3.(A)** **4.(B)** **5.(A)**

(3) $f'(x)=1-\sin x$

$\therefore$ $f(x)$ is monotonically increasing

$f(0)=1-a$ (for exactly one positive root)

$f(0)<0\Rightarrow 1-a<0\Rightarrow a>1$

(4) $f(x)=x+\cos x-a$

$f'(x)=1-\sin x$

$f'(c)=1-\sin c$

$\alpha<c<\beta$

Apply L.M.V.T for $x\in(\alpha,\beta)$

$$f'(c)=\frac{(\beta+\cos\beta-a)-(\alpha+\cos\alpha-a)}{\beta-\alpha}=\frac{(\beta-\alpha)+(\cos\beta-\cos\alpha)}{\beta-\alpha}$$

$$=1+\frac{\cos\beta-\cos\alpha}{\beta-\alpha}$$

$\alpha<c<\beta$

$\Rightarrow$ $\sin\alpha<\sin c<\sin\beta \Rightarrow -\sin\beta<-\sin c<-\sin\alpha$ $\Rightarrow$ $1-\sin\Rightarrow<1-\sin c<1-\sin\alpha$

$\Rightarrow$ $1-\sin\beta<1+\frac{\cos\beta-\cos\alpha}{\beta-\alpha}<1-\sin\alpha$

$$\sin\beta>\frac{\cos\alpha-\cos\beta}{\beta-\alpha}>\sin\alpha\Rightarrow\sin\alpha<\frac{\cos\alpha-\cos\beta}{\beta-\alpha}<\sin\beta$$

(5) $f'(x)=1-\sin x$

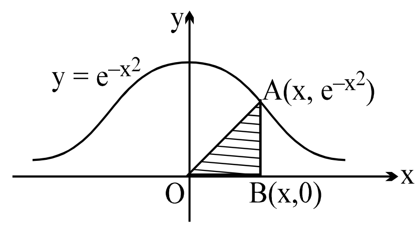
$f'(x)\big|_{x=\frac{\pi}{2}}=0$

$\therefore$ Equation of tangent $y=\frac{\pi}{2}-a$

6.(D) $A = \dfrac{xe^{-x^2}}{2}$; $\quad A' = \dfrac{1}{2}\left[e^{-x^2} - 2x^2 \cdot e^{-x^2}\right]$

$= \dfrac{e^{-x^2}}{2}\left[1-2x^2\right] = 0 \Rightarrow x = \dfrac{1}{\sqrt{2}}$ gives A_{max}.

$\therefore \quad A_{max} = \dfrac{e^{-1/2}}{2\sqrt{2}} = \dfrac{1}{\sqrt{8e}}$



7.(C) $x^2y = c^3$

$x^2\dfrac{dy}{dx} + 2xy \quad \Rightarrow \dfrac{dy}{dx} = 0 = -\dfrac{2y}{x}$

Equation of tangent at (x, y)

$Y - y = -\dfrac{2y}{x}(X - x)$

$Y = 0$, gives, $X = \dfrac{3x}{2} = a$

And $\quad X = 0$, gives, $Y = 3y = b \qquad$ Now $\quad a^2b = \dfrac{9x^2}{4}.3y = \dfrac{27}{4}x^2y = \dfrac{27}{4}c^3 \Rightarrow$ (C)

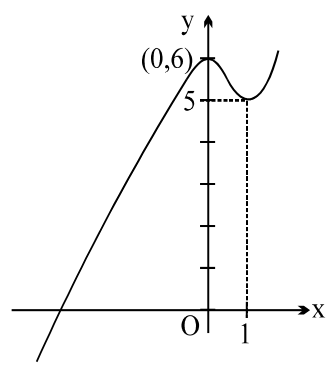
8.(A) $f(x) = 2x^3 - 3x^2 + 6$

$f'(x) = 6x^2 - 6x = 6(x^2 - x) = 0$ gives $x = 0$ or $x = 1$

For inverse to exist function must be Bijective

Hence Domain is $[1, \infty)$ for one-one function

$a \geq 1$



9.(C) Let $\quad A = (t, t^2)$; $\qquad m_{OA} = t$; $\qquad m_{AB} = -\dfrac{1}{t}$

Equation of AB, $y - t^2 = -\dfrac{1}{t}(x - t^2)$

Put $\quad x = 0$

$h = t^2 + 1 \qquad$ (as $x \to 0$ then $t \to 0$)

Now $\quad \lim\limits_{t\to 0}(h) = \lim\limits_{t\to 0}(1+t^2) = 1$

10.(D) $F(x) = \dfrac{4}{\sin x} + \dfrac{1}{1-\sin x}$

$F'(x) = \dfrac{-4\cos x}{\sin^2 x} + \dfrac{\cos x}{(1-\sin x)^2}$

For maxima or minima

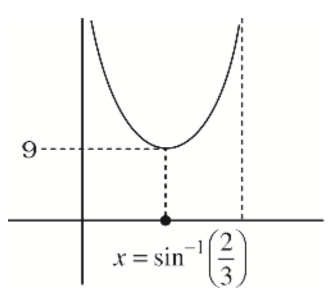
$F'(x) = 0 \Rightarrow \sin x = \dfrac{2}{3}$

$f(x) \to \infty$ for $x \to 0^+$

$f(x) \to \infty$ for $x \to \dfrac{\pi}{2}^-$

$F(x)_{\min}$ when $\sin x = \dfrac{2}{3}$ (Point of minima)

$F(x)_{\min} = a_{\min} = 9$



11.(D) $f'(x) = 12x^2 - 2x - 2 = 2[6x^2 - x - 1] = 2(3x + 1)(2x - 1)$

Hence $g(x) = \begin{cases} f(x) & if\ 0 \le x < \dfrac{1}{2} \\ f\left(\dfrac{1}{2}\right) & if\ \dfrac{1}{2} \le x \le 1 \\ 3 - x & if\ 1 < x \le 2 \end{cases}$

12.(A) f is not differentiable at $x = \dfrac{1}{2}$

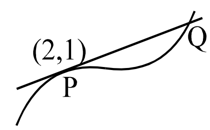
g is not continuous in [0, 1] at x = 0 & 1 ; h is not continuous in [0, 1] at x = 1

$k(x) = (x+3)^{\ln_2 5} = (x + 3)^p$ where $2 < p < 3$

13.(D) Eliminating t gives $y^2 (x - 1) = 1$.

Equation of tangent at P(2, 1) is $x + 2y = 4$.

Solving with curve x = 5 & y = – 1/2 $\Rightarrow$ Q (5, 1/2) $\Rightarrow$ PQ = $\dfrac{3\sqrt{5}}{2}$]



14.(A) $f(x) = x(x^2 - 3x + 2)$, $f(x) = x(3x^2 - 1) = x(\sqrt{3}x - 1)(\sqrt{3}x + 1)$

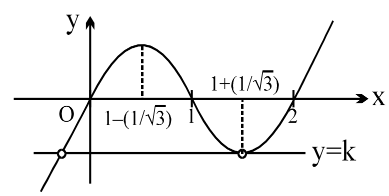
$f(x) = x(x - 2)(x - 1)$

Graph of $y = f(x)$ is as shown

Now $f(x) = k$ to have exactly one positive and negative solution

We have, $k = f\left(1 + \dfrac{1}{\sqrt{3}}\right)$

$\therefore \quad k = \left(1 + \dfrac{1}{\sqrt{3}}\right)\left(\dfrac{1}{\sqrt{3}} - 1\right)\left(\dfrac{1}{\sqrt{3}}\right) = \left(\dfrac{1}{3} - 1\right)\left(\dfrac{1}{\sqrt{3}}\right) = \dfrac{-2}{3\sqrt{3}}$

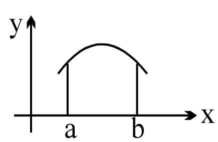


15.(D) Given $g(x) = \ln(h(x))$

$g'(x) = \dfrac{h'(x)}{h(x)}$

$g''(x) = \dfrac{h(x)h''(x) - (h'(x))^2}{h^2(x)} < 0$ (given)

$\therefore \quad g''(x) < 0 \Rightarrow g(x)$ is concave down



16.(B) $F'(x) = e^{(1+\sin^{-1}(\cos x))^2} \cdot (-\sin x) - e^{(1+x)^2} \cdot \cos x$

$F'(0) = 0 - e = -e$

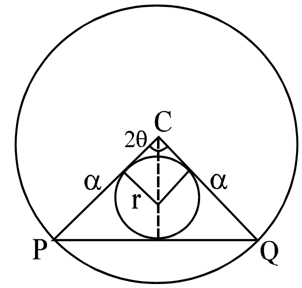
$F'\left(\dfrac{\pi}{2}\right) = -e - 0 = -e$] Hence Rolle's theorem is applicable for the function $F'(x)$

$\Rightarrow$ there is some c in $\left(0, \dfrac{\pi}{2}\right)$ for which F"(c) = 0 as Rolle's theorem is applicable for F'(x) $\left[0, \dfrac{\pi}{2}\right]$ in also F

(0) = $\displaystyle\int_0^1 f(t)dt$ and $F\left(\dfrac{\pi}{2}\right) = \displaystyle\int_1^0 f(t)dt$, hence F (0) & $F\left(\dfrac{\pi}{2}\right)$ have opposite signs

$\Rightarrow$ F (c) = 0 for some $c \in \left(0, \dfrac{\pi}{2}\right)$

17.(B) $r = \dfrac{\Delta}{s}$ where Δ = Area of triangle CPQ and s = semiperimeter of ΔCPQ.



$$r = \frac{\alpha^2 \sin 2\theta}{2s} = \frac{\alpha^2 \sin 2\theta}{2\alpha + 2\alpha \sin\theta} = \frac{\alpha}{2}.\frac{\sin 2\theta}{1+\sin\theta}$$

Consider $f(\theta) = \dfrac{\sin 2\theta}{1+\sin\theta}$

$$f'(\theta) = \frac{(1+\sin\theta)2\cos 2\theta - \sin 2\theta.\cos\theta}{(1+\sin\theta)^2} = 0$$

$2(1 + \sin\theta)(1 - 2\sin^2\theta) - 2\sin\theta\,(1 - \sin^2\theta) = 0$

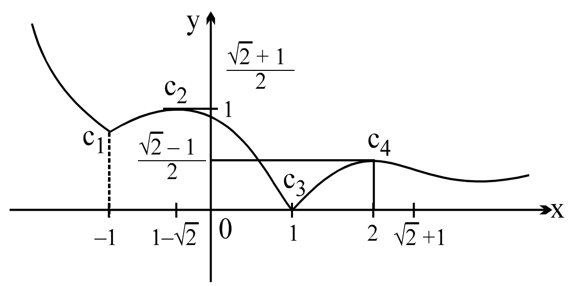
$2(1 - 2\sin^2\theta) = 2\sin\theta\,(1 - \sin\theta)$

$1 - 2\sin^2\theta = \sin\theta - \sin^2\theta$

$\sin^2\theta + \sin\theta - 1 = 0$

$sin\theta = \dfrac{-1 \pm \sqrt{1+4}}{2}$ $\quad\therefore\quad sin\theta = \dfrac{\sqrt{5}-1}{2}$]

18.(A) $f(x) = \begin{cases} \dfrac{x-1}{1+x^2} & if \quad x \ge 1 \\ \dfrac{1-x}{1+x^2} & if \quad -1 < x < 1 \\ x^2 & if \quad x \le -1 \end{cases}$



$f'(x) = 0$ gives $x = \sqrt{2} + 1$ or

$1 - \sqrt{2}$. The function has a continuity at x = – 1.

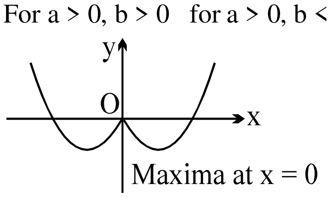
As $x \to \infty$, $f(x) \to 0$.

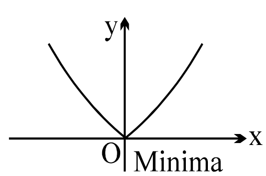
The graph is as shown .

Note that critical points are those where **dy/dx = 0** if it exists or **dy/dx** is non existent .

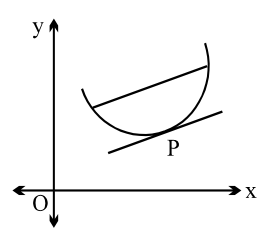
The points $x = -1,\ 1 - \sqrt{2}$, 1 and $1 + \sqrt{2}$ are the four critical points on the graph of this function $\Rightarrow$ A

19.(A) For a > 0, b > 0 for a > 0, b < 0





20.(D) $f(x) = \displaystyle\int_1^x \left(t + \frac{1}{t}\right)dt \quad \Rightarrow \quad f'(x) = x + \frac{1}{x}$



$\therefore\ g(x) = x + \dfrac{1}{x}$ for $x \in \left[\dfrac{1}{2}, 3\right]$

$g\left(\dfrac{1}{2}\right) = 2 + \dfrac{1}{2} = \dfrac{5}{2}$, $\qquad g(3) = 3 + \dfrac{1}{3} = \dfrac{10}{3}$

Let $P \equiv (c, g(c))$, $c \in \left[\dfrac{1}{2}, 3\right]$

By LMVT, $g'(c) = \dfrac{g(3)-g\left(\frac{1}{2}\right)}{3-\frac{1}{2}}$ $\quad\therefore\quad$ $1-\dfrac{1}{c^2} = \dfrac{\frac{10}{3}-\frac{5}{2}}{3-\frac{1}{2}}$

$\Rightarrow \quad c^2 = \dfrac{3}{2} \quad \Rightarrow \quad c = \sqrt{\dfrac{3}{2}}$ $\quad\therefore\quad$ $g(c) = \sqrt{\dfrac{3}{2}} + \dfrac{1}{\sqrt{\frac{3}{2}}} = \dfrac{5}{\sqrt{6}}$ $\quad\therefore\ P \equiv \left(\sqrt{\dfrac{3}{2}}, \dfrac{5}{\sqrt{6}}\right)$

21.(C) $f(x) = 12x^5 - 15x^4 + 20x^3 - 30x^2 + 60x + 1$

$f'(x) = 60x^4 - 60x^3 + 60x^2 - 60x + 60$

$= 60(x^4 - x^3 + x^2 - x + 1) \ = 60(x^3(x-1) + x(x-1) + 1) \ = 60((x^3+x)(x-1)+1) = 60(x(x^2+1)(x-1)+1)$

$f'(x) > 0$ for $x \in R$

So $f(x)$ is monotonically increasing

But is not necessary that if a function is monotonic then it will have range R.

22.(C) a = 1

$f(x) = 8x^3 + 4x^2 + 2bx + 1$

$f'(x) = 24x^2 + 8x + 2b = 2(12x^2 + 4x + b)$

For increasing function, $\quad f'(x) \geq 0 \ \forall \ x \in R \quad \therefore \quad D \leq 0 \Rightarrow 16 - 48b \leq 0 \Rightarrow b \geq \dfrac{1}{3} \Rightarrow$ **(C)**

23.(B) If $b = 1$

$f(x) = 8x^3 + 4ax^2 + 2x + a$

$f'(x) = 24x^2 + 8ax + 2 \quad$ or $\quad f'(x) = 2(12x^2 + 4ax + 1)$

For non monotonic $f'(x) = 0$ must have distinct roots

Hence $\ D > 0 \ $ i.e. $\ 16a^2 - 48 > 0 \quad \Rightarrow \quad a^2 > 3; \quad \therefore \quad a > \sqrt{3}$ or $a < -\sqrt{3}$

$\therefore \quad a \in \{2, 3, 4, ...100\}$

Sum = 5050 – 1 = 5049

24.(D) If x_1, x_2 and x_3 are the roots then $\log_2 x_1 + \log_2 x_2 + \log_2 x_3 = 5$

$\log_2(x_1x_2x_3) = 5$

$x_1x_2x_3 = 32$

$-\dfrac{a}{8} = 32 \quad \Rightarrow \quad a = -256$

25.(AB) $8x\dfrac{dx}{dt} + 18y\dfrac{dy}{dt} = 0 \qquad$ **....(i)**

Given $\dfrac{dx}{dt} = \dfrac{-dy}{dt}$ **....(ii)**

From (1) and (2) $4x_1 = 9y_1$

$4x_1^2 + \dfrac{9.16.x_1^2}{81} = 13 \Rightarrow x_1 = \pm\dfrac{3}{2}$

Points are $\left(\dfrac{3}{2}, \dfrac{2}{3}\right), \left(-\dfrac{3}{2}, -\dfrac{2}{3}\right)$

26.(ABCD) Since $f(x)$ is even, so we check in $[0,2]$

$$f(x)=\begin{cases} ax^2+b, & 0\le x<1 \\ 1, & x=1 \\ \dfrac{\lambda}{x} & 1<x\le 2 \end{cases}$$

For continuity, a + b = 1 = λ (1)

$$f'(x)=\begin{cases} 2ax, 0<x<1 \\ -\dfrac{\lambda}{x^2}, 1<x<2 \end{cases} \Rightarrow \quad \text{For continuity, } 2a=-\lambda \quad ..(2)$$

From (1) and (2), $a=-\dfrac{1}{2}, b=\dfrac{3}{2}, \lambda=1$

27.(AC) $h(x)=\dfrac{1-x^2}{1+x^2}=\dfrac{2}{1+x^2}-1$, $h'(x)=\dfrac{-4x}{(1+x^2)^2}$, $h'(x)>0$ for $x\in(-\infty,0)$

$\cos^{-1}$ is a decreasing function

f decreases when h increases

i.e., when $x\in(-\infty,0)$

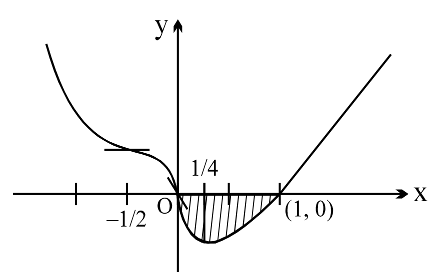
28.(ABCD) $y=x^{1/3}(x-1)$

$$\frac{dy}{dx}=\frac{4}{3}x^{1/3}-\frac{1}{3}\cdot\frac{1}{x^{2/3}}=\frac{1}{3x^{2/3}}[4x-1]$$

Hence f is $\uparrow$ for $x>\left.\dfrac{1}{4}\right]$, ($x^{2/3}$ is always positive) and $x=\dfrac{1}{4}$ the curves has a local minima

And $f\downarrow$ for $x<\dfrac{1}{4}$

Now $f'(x)=\dfrac{4}{3}x^{1/3}-\dfrac{1}{3}\cdot x^{-2/3}$ (non existent at x = 0, vertical tangent)



$$f''(x)=\frac{4}{9}\cdot\frac{1}{x^{2/3}}+\frac{1}{3}\cdot\frac{2}{3}\cdot\frac{1}{x^{5/3}}$$

$$=\frac{2}{9x^{2/3}}\left[2+\frac{1}{x}\right]=\frac{2}{9x^{2/3}}\left[\frac{2x+1}{x}\right]$$

$\therefore$ $f''(x)=0$ at $x=-\dfrac{1}{2}$ (Inflection point), $x=0$ is also point of inflation.

Graph of $f(x)$ is as A = $\left|\int_0^1\left(x^{4/3}-x^{1/3}\right)dx\right| = \left|\left[\dfrac{3}{7}x^{3/7}-\dfrac{3}{4}x^{4/3}\right]_0^1\right| = \left|\dfrac{3}{7}-\dfrac{3}{4}\right| = 3\left|\dfrac{4-7}{28}\right| = \dfrac{9}{28} \Rightarrow$ **(D)**

29.(AD) $\dfrac{dy}{dx}=\dfrac{-b}{a}=\dfrac{-1}{t^2}\Rightarrow ab>0$

30.(ABCD) $f'(x) = \sqrt{1-x^4} > 0$ in $(-1, 1) \Rightarrow f$ is $\uparrow$

Now $f(x) + f(-x) = \int_0^x \sqrt{1-t^4}\, dt + \int_0^{-x} \sqrt{1-t^4}\, dt \Rightarrow \int_0^x \sqrt{1-t^4}\, dt + \left(-\int_0^x \sqrt{1-y^4}\, dy\right)$ $(t = -y)$

$= 0 \Rightarrow f(x)$ is odd

Again $f''(x) = \dfrac{-4x^3}{2\sqrt{1-x^4}}$ which vanished at $x = 0$. (0, 0) is inflection since f is well defined in $[-1, 1] \Rightarrow$ A,B,C,D]

31.(AB) Since intercepts are equal in magnitude but opposite in sign $\Rightarrow \left.\dfrac{dy}{dx}\right|_P = 1$

Now $\dfrac{dy}{dx} = x^2 - 5x + 7 = 1 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2$ or 3

32.(BD) $h(x) = \dfrac{\ell n\left(f(x)\,.\,g(x)\right)}{\ell n\, a} = \dfrac{\ell n\, a^{\left\{a^{|x|}\,.\,\mathrm{sgn}\, x\right\} + \left[a^{|x|}\,.\,\mathrm{sgn}\, x\right]}}{\ell n\, a}$

$\left(\left\{a^{|x|}\,\mathrm{sgn}\, x\right\} + \left[a^{|x|}\,\mathrm{sgn}\, x\right] = a^{|x|}\,\mathrm{sgn}\, x\right)$ $(\because \{y\} + [y] = y)$

$= \begin{bmatrix} a^x & for & x > 0 \\ 0 & for & x = 0 \\ -a^{-x} & for & x < 0 \end{bmatrix} \Rightarrow h(x)$ is an odd function.

33.(BCD) $f'(x) = 100x^{99} + \cos x$

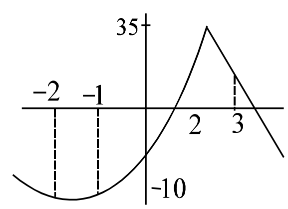
For $x \in (0,1)$ and $\left(0, \dfrac{\pi}{2}\right)$, cos x and x are both +ve $\Rightarrow f'(x)$ is increasing

For $x \in \left(\dfrac{\pi}{2}, \pi\right)$, $x > 1$ hence $100x^{99}$ obviously $> \cos x \Rightarrow \uparrow$]

34.(ABCD) Note that $f(x)$ is continuous at $x = 2$ and f is decreasing for

(2, 3) and increasing for [–1, 2].

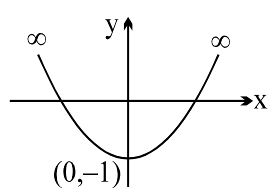
At $x = 2$, f has a maxima



35.(ABC) Graph of y = f (x) $\Rightarrow$ (A) and (C)

$f'(x) = x(2 - \cos x)$

$f(x)$ is even function



36.(ACD) If f and g are inverse then (fog)(x) = x

$f'[g(x)]\, g'(x) = 1$

If f is increasing $\Rightarrow$ $f' > 0$ $\Rightarrow$ sign of g' is also + ve $\Rightarrow$ (A) is correct

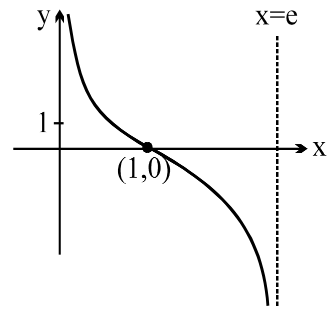
If f is decreasing $\Rightarrow$ $f' < 0$ $\Rightarrow$ sign of g' is – ve $\Rightarrow$ (B) is false

Since f has an inverse $\Rightarrow$ f is bijective $\Rightarrow$ f is injective $\Rightarrow$ (C) is correct

Inverse of a bijective mapping is bijective

$\Rightarrow$ g is also bijective $\Rightarrow$ g is onto $\Rightarrow$ (D) is correct

37.(ABC) $f(x) = \ln(1 - \ln x)$



Domain (0, e)

$f'(x) = -\dfrac{1}{(1-\ln x)} \cdot \dfrac{1}{x} < 0 \;\Rightarrow$ decreasing $\forall$ x in its domain

$\Rightarrow$ (A) & (B) are incorrect

$f'(1) = -1 \quad \Rightarrow$ (C) is also incorrect

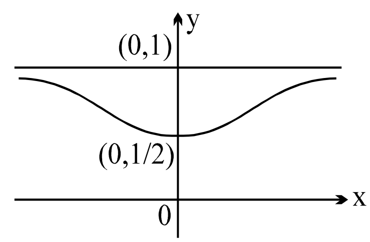
Also $f(1) = 0$; $\lim\limits_{x \to e^{-1}} f(x) \to -\infty$; $\lim\limits_{x \to 0^+} f(x) \to \infty$

$$f''(x) = \frac{-\ln x}{x^2(1-\ln x)^2}$$

$f''(1) = 0,\ x = 1$ which is a point of inflection graph is as shown y axis and x = e are two asymptotes

38.(BCD)

Domain is $x \in R$



Also $f(x) = \left[\cos\left(\tan^{-1}(\sin\theta)\right)\right]^2$ where $\cot\theta = x$

$= \left[\cos\left(\tan^{-1}\left(\dfrac{1}{\sqrt{1+x^2}}\right)\right)\right]^2 = (\cos\phi)^2$ where $\tan\phi = \dfrac{1}{\sqrt{1+x^2}}$

$= \left(\dfrac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)^2$

$g(x) = \dfrac{1+x^2}{2+x^2} = 1 - \dfrac{1}{2+x^2}$

Range is $\left[\dfrac{1}{2}, 1\right)$; $f'(x) = \dfrac{2x}{(2+x^2)^2}$

Hence $f'(0) = 0$

Also $\lim\limits_{x \to \infty} f(x) = 1$. Hence (B), (C), (D)

39.(AC) Let the common tangent is $y = mx + c$

Solving with first curve we get, $x^2 + 1 = mx + c \;\Rightarrow\; x^2 - mx + 1 - c = 0$

D = 0

$\boxed{m^2 = 4(1-c)}$ ....(A)

Solving with second curve

$mx + c = -x^2 \quad \Rightarrow \quad x^2 + mx + c = 0$

D = 0

$\boxed{m^2 = 4c}$ ....(B)

From A and B : $8c = 4 \Rightarrow c = \dfrac{1}{2}$. Hence $m = \pm\sqrt{2}$

The tangent lines are $y = \sqrt{2}x + \dfrac{1}{2}$ and $y = -\sqrt{2}x + \dfrac{1}{2}$

40.(CD) $f(x)=\int_0^{\pi}\cos t\cos(x-t)dt$ ....(1)

$=\int_0^{\pi}-\cos t\cdot\cos(x-\pi+t)dt$

$f(x)=\int_0^{\pi}\cos t\cdot\cos(x+t)dt$ ....(2)

(1) + (2) gives

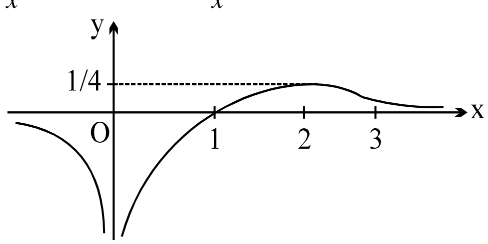
$2f(x)=\int_0^{\pi}\cos t(2\cos x\cdot\cos t)dt$

$\therefore \quad f(x)=\cos x\int_0^{\pi}\cos^2 t\,dt = 2\cos x\int_0^{\pi/2}\cos^2 t\,dt$

$f(x)=\dfrac{\pi\cos x}{2}$

Alternatively: Convert the integrand into sum of two cosine functions.]

41.(BCD) $f'(x)=\dfrac{2-x}{x^3}$ and $f''(x)=\dfrac{x-3}{x^4}$. $x=2$ is point of maxima. $x=3$ is point of inflection



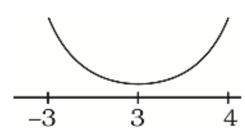
42.(BCD)

(A) $f(x)$ has no relative minimum on (–3, 4)

(B) $f(x)$ is continuous function on [–3, 4]

$\Rightarrow$ $f(x)$ has min. and max. on [–3, 4] by IVT

(C) $f''(x)>0$ $\Rightarrow$ f (x) is concave upwards on [–3, 4]



(D) $f(3)=f(4)$

By Rolle's theorem $\exists\, c\in(3,4)$, where $f'(c)=0$ $\Rightarrow$ $\exists$ critical point on [– 3, 4]

43.(BCD) **(A)** False, e.g. $f(x)=\sin\sqrt{x}$ **(B)** True, from IVT

(C) True as $\lim\limits_{x\to\infty}\sin^{-1}\left(1+\dfrac{1}{x}\right)=\sin^{-1}$ (a quantity greater than one) $\Rightarrow$ not defined

(D) True, as the line passes through the centre of the circle.

44.(ABC) Because f is continuous for all x, the intermediate value theorem implies that the graph of f must intersect the $x-$axis. The graph must also intersect the y axis since f is defined for all x, in particular at $x=0$, as f need not be differentiable. Hence A, B and C need not be correct.

45.(AC) (A) $f''(x)>0\forall\times\in R\Rightarrow f'(x)$ is a strictly increasing function hence $f'(1)>f'(0)$, but given $f'(0)=f'(1)=1$ which is not possible.

(B) $f''(x)>0,\ \forall x\in R\Rightarrow f'(x)$ is increasing $f'(1)>f'(0)$, which is true.

46.(BCD) $a>(2-x)x^2,\ \forall\ x>0$

Let $f(x)=(2-x)x^2$

$\therefore$ $a>$ maximum value of $f(x)$ for $x>0$

$\therefore$ $a>\dfrac{32}{27}\Rightarrow$ Least natural number is 2.

47.(BC) Since $x=1,2,3$ are normal

$\Rightarrow$ $f'(x)=4ax^3+3bx^2+2cx+d$

$\Rightarrow$ $b=-8a,\ C=22a,\ d=-24a$

Also $f(2)>f(0)$

$\therefore$ $x=1\ \&\ 3$ are point of local maxima

Also $f(k)=f(0)$

$\Rightarrow$ $ak^4+bk^3+ck^2+dk=0$

$\Rightarrow$ $ak(k-4)\left(k^2-4k+6\right)=0$ $\Rightarrow$ $k=4$

48.(ABC) $f'(x)=n(1-x)^{-(n+1)}+n(1+x)^{-(n+1)}-2n$

$f''(x)=n(n+1)(1-x)^{-(n+2)}-n(n+1)(1+x)^{-(n+2)}$

for $0<x<1$

$\dfrac{1}{1-x}>\dfrac{1}{1+x}$ $\Rightarrow$ $f''(x)>0$

$\Rightarrow$ $f'(x)$ is increasing $\Rightarrow$ $f'(x)>f'(0)=0$

$\Rightarrow$ $f(x)$ is increasing

$f(x)>f(0)=0.$

49.(ABC)

50.(ACD) $f'(x)-\sin x\ f(x)\le-\sin x$

$\Rightarrow$ $f(x)\le1$ $\Rightarrow$ $(-\infty,1]$

51.(ABC)

52.(AC) $2f(x)f(y)=f(x+y)+f(x-y)$

Inter changing x,y

$2f(y)f(x)=f(y+x)+f(y-x)$

then $f(x)=f(-x)$

$f'(x)=-f'(-x)$

53.(AC) $\lim\limits_{x\to\infty} \dfrac{n\log x}{[x]} - 1 = 0 - 1 = -1$

$f(x) = (-1)^n = \pm 1$

54.(BD) $x = r\cos\theta,\ y = r\sin\theta \to$

$\dfrac{4}{r^2} = 5(1+\cos 2\theta) + 3\sin 2\theta + 1 - \cos 2\theta$

$= 6 + 4\cos 2\theta + 3\sin 2\theta$, which has maximum 11 and minimum 1

$\therefore$ OP has minimum $\dfrac{2}{\sqrt{11}}$ and maximum 2.

55.(ABCD) $f'(x) = 3\left(1 - \sqrt{\dfrac{21-4b-b^2}{b+1}}\right)x^2 + 5 > 0$

$\dfrac{\sqrt{21-4b-b^2}}{b+1} \le 1$

$21 - 4b - b^2 \ge 0 \Rightarrow -7 \le b \le 3$

Case-I

$b + 1 < 0$

then $-7 \le b < -1$ ...(i)

Case-II

If $b + 1 > 0$

$b \in [2, \infty)$ ...(ii)

$\Rightarrow$ $b \in [-7, -1) \cup [2, 3]$

56.(AD) Differentiating partially w.r.t. x

$g'(x) = g(y)[g'(x-y)]$

Put $y = x$

$g'(x) = g(x).g'(0) = a.g(x) \Rightarrow g(x) = e^{ax}$

57.(AD) $\lambda^3 = -a\lambda^2 - b\lambda - c$

$\Rightarrow$ $\lambda^4 = (1-a)\lambda^3 + (a-b)\lambda^2 + (b-c)\lambda + c$

If $|\lambda| > 1 \Rightarrow |\lambda|^4 \le |\lambda|^3$ (not possible)

for $|\lambda| \le 1$

$|\lambda|^4 \le (1-a) + (a-b) + (b+c) + c$ $\Rightarrow$ $|\lambda| \le 1.$

58.(B) Put $x = \sin\theta,\ y = \cos\theta$

$\therefore$ $k\tan^2\theta + \dfrac{1}{k}\cot^2\theta + 1$

$k\tan^2\theta + \dfrac{1}{k\tan^2\theta} \ge 2$

59.(D) $\dfrac{-x^2}{2} \le f(x) \le \dfrac{x^2}{2}$

60.(D) $f'(x) < 0 \forall x \in [-1, 0)$

and $f'(x) > 0 \forall x \in [0, 1]$

hence for $f(x)$ to have a minima at $x = 0$

LHL $\ge$ RHL

$\Rightarrow \quad \lim\limits_{x \to 0^-} f(x) \ge \lim\limits_{x \to 0^+} f(x) \quad \Rightarrow \quad 3 + \log_{0.5} \log_2 [k+3] \ge 2 \quad \therefore \quad k \in [-1, 2).$

61.(ABD) Clearly if $f(x) = a(x-\alpha)(x-\beta)$ then $g(x) = b(x-\alpha)(x-\beta)$

$\Rightarrow \quad h(x) = k(x-\alpha)^2(x-\beta)^2 \Rightarrow h(x)h'(x) = 2k(x-\alpha)^3(x-\beta)^3(2x-\alpha-\beta)$

So distinct roots of $\dfrac{d}{dx}(h(x)h'(x)) = 0,$ are 4.

62.(BD) In sine curve chord joining two points on the curve lies below the curve, hence (A) cannot be true. In log curve also, the same pattern is following but in tan curve and x^2 curves opposite pattern is followed.

63.(BC) (B) $\dfrac{df(x)}{dx} - f(x) > 0$

$\Rightarrow \quad \dfrac{e^{-x} df(x)}{dx} - e^{-x} f(x) > 0 \qquad \left(\because e^{-x} > 0\right)$

$\Rightarrow \quad \dfrac{d}{dx}\left(e^{-x} f(x)\right) > 0 \quad \Rightarrow \quad e^{-x} f(x)$ is an increasing function.

$e^{-x} f(x) > e^{-1} f(1) = 0 \quad \Rightarrow \quad e^{-x} f(x) > 0 \Rightarrow f(x) > 0$ for all $x > 1$.

64. **[A-p, q, r, s]** **[B-p, r]** **[C-p, q, r, s]** **[D-p, q, r, s]**

Let $f(x) = \ln(1+x) - \dfrac{x}{1+x}$, $f(0) = 0$

$f'(x) = \dfrac{x}{(1+x)^2}$

$\therefore$ For $x > 0$ $f(x)$ is $\uparrow \quad \Rightarrow \quad f(x) > f(0)$

$\ln(1+x) > \dfrac{x}{1+x}$

For $x < 0$ $f(x)$ is $\downarrow \quad \Rightarrow \quad f(x) > f(0)$

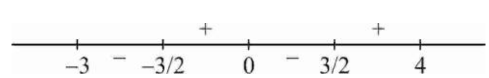
$\ln(1+x) > \dfrac{x}{1+x}$ for $\forall\ x \in (-1, \infty) \quad \therefore \quad$ [A-p, q, r, s]

Similarly other parts can be done.

65. **[A-p]** **[B-q]** **[C-r]** **[D-s]**

$h'(x) = 3f'\left(\dfrac{x^2}{3}\right) \times \dfrac{2x}{3} - 2f'\left(3-x^2\right)x = 2x\left(f'\left(\dfrac{x^2}{3}\right) - f'\left(3-x^2\right)\right) = 0$

$\Rightarrow \quad x = 0$ or $\dfrac{x^2}{3} = 3 - x^2 \Rightarrow x = 0$ or $\pm\dfrac{3}{2}$ as $f'(x)$ is one-one



[A-p] [B-q]

$x^2+4y^2=4 \quad \Rightarrow \quad x=2\cos\theta,\ y=\sin\theta$

$x^2+y^2-xy = 4\cos^2\theta+\sin^2\theta-2\sin\theta\cos\theta$

$$=3\left(\frac{1+\cos 2\theta}{2}\right)+1-\sin 2\theta=\frac{5}{2}+\frac{3}{2}\cos 2\theta-\sin 2\theta$$

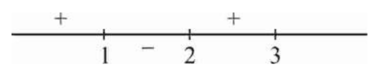
$$\in\left[\frac{5-\sqrt{13}}{2},\frac{5+\sqrt{13}}{2}\right]$$

[C-r]

$f(x)=2x^3-9x^2+12x+6$

$f'(x)=6x^2-18x+12$

$=6\left(x^2-3x+2\right)=6(x-1)(x-2)$



Global minimum of $f(x)$ occur at $x=2$ in (1, 3) [D-s]

66. **[A-q]** **[B-r]** **[C-p]** **[D-t]**

(A) $f'(x)=3x^2+6x(a-7)+3(a^2-9)$

$D>0 \Rightarrow 36(a-7)^2-36(a^2-9)>0$

$\Rightarrow \quad a^2-14a+49-a^2+9>0 \Rightarrow \quad -14a>-58$

$14a<58$

$a<\frac{29}{7} \qquad \therefore \qquad a\in(-\infty,-3)\cup\left(3,\frac{29}{7}\right)$

(B) Let $\alpha,\beta,\gamma=\lambda \Rightarrow \alpha\beta+\beta\gamma+\gamma\alpha=\mu \Rightarrow \alpha\beta\gamma=6$

$\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{3}>(\alpha\beta\gamma)^{\frac{2}{3}} \Rightarrow \mu>3(6)^{\frac{2}{3}}$

(C) $f'(x)=\cos x-2\sin 2x=\cos x(1-4\sin x)=0$

$x=\frac{\pi}{2},\sin x=\frac{1}{4}$

$f''(x)=-\sin x-4\cos 2x=-\sin x-4(1-2\sin^2 x)=8\sin^2 x-\sin x-4$

For $x=\frac{\pi}{2}, f''(x)=3>0$

For $\sin x=\frac{1}{4}, f''(x)=2-\frac{1}{4}-4<0$

Maximum value is $\frac{1}{4}+1-2\frac{1}{16} \Rightarrow \frac{5}{4}-\frac{1}{8}=\frac{9}{8}$

(D) $f(x)=\log(1+x)-x$

$f(x)$ is defined for $x>-1$

$f'(x)=\frac{1}{1+x}-1=\frac{-x}{1+x}$

$f'(x)\le 0$ for $x\ge 0$

$f'(x)\ge 0$ for $-1<x\le 0$

$\therefore \quad \log(1+x)\le x$ for $-1<x\le 0$

67.(9) Let $t=\sqrt{4-x^2}$ $|x|\le 2 \therefore t\in[0,2]$.

Let $g(t)=(3-t)^2+(1+t)^3$ for $t\in[0,2]$.

$g'(t)=(t+3)(3t-1)=0$

$\therefore \quad t=-3,\frac{1}{3}$

Also $g''(t)=2(3t+4)$

$$g(0)=10,\ g(2)=28,\ g\left(\frac{1}{3}\right)=\frac{256}{27}.$$

$$\therefore \quad p=28-\frac{256}{27}=\frac{500}{27}$$

68.(0) Let $f(x)=a(x-x_1)(x-x_2)(x-x_3)(x-x_4)(x-x_5)$

Take log on both sides & Differentiate $\frac{f'}{f}=\frac{1}{x-x_1}+\frac{1}{x-x_2}+\frac{1}{x-x_3}+\frac{1}{x-x_4}+\frac{1}{x-x_5}$

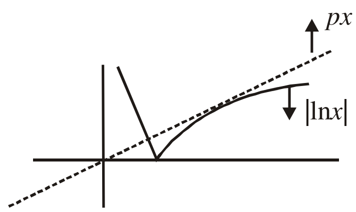
Differentiate again were x

$$\frac{f.f''-(f')^2}{(f)^2}=\frac{1}{(x-x_1)^2}+\ldots+\frac{1}{(x-x_5)^2}<0$$

69.(2) $f'(x)=-2-3x^2<0 \Rightarrow f(x)$ is decreasing function

$\Rightarrow f(x)>-x \Rightarrow (x-3)(x^2+3x+10)<0 \Rightarrow x<3$

70.(0)



$y=px$ touches

$y=|\ln x|$ for $p=\frac{1}{e}$.

$\therefore$ The equation has 3 distinct solution if $p\in\left(0,\frac{1}{e}\right)$, therefore $[p]=0$.

71.(2) Since $f(x)$ is monotonic, therefore, $f^{-1}(x)$ exists.

Let $\quad f^{-1}(x)=z$,then $x=f(z)$

$x=f(a)\Rightarrow z=f^{-1}(f(a))=a, x=f(b) \quad \Rightarrow \quad z=f^{-1}(f(b))=b$ and $dx=f'(z)dz$

$$\therefore \quad \int_{f(b)}^{f(a)} x(b-f^{-1}(x))\,dx=\int_a^b f(z)(b-z)f'(z)dz$$

$$=-\frac{1}{2}(b-a)(f(a))^2+\frac{1}{2}\int_a^b (f(z))^2dz=\frac{1}{2}\int_a^b f^2(z)-f^2(a))dz=\frac{1}{2}\int_a^b (f(x)-f(a))(f(x)+f(a))\,dx$$

$\therefore \quad k=2$

72.(1) Equation of the normal is $y=mx-2m-m^3$ $(a=1)$

If it passes through $(21,30)$ we have $30=21m-2m-m^3 \Rightarrow m^3-19m+30=0$

Then $m=-5,2,3$

But if $m=2$ or 3 the point $(am^2,-2am)$ do not lie on the parabola.

Therefore $m=-5 \qquad \therefore m+6=1$

73.(3) $Q^3 = P - 3Q$

$\frac{P}{Q^2} = Q + \frac{3}{Q} \Rightarrow f(Q) = Q + \frac{3}{Q} \Rightarrow f'(Q) = 1 - \frac{3}{Q^2} \Rightarrow Q = \pm\sqrt{3} \Rightarrow f(Q)$ will be minimum at $Q = \sqrt{3}$

So minimum value of $f(Q)$ is $2\sqrt{3}$ i.e. minimum of $\left[\frac{P}{Q^2}\right] = \left[2\sqrt{3}\right] = 3$

74.(2) $1 \le |\sin x| + |\cos x| \le \sqrt{2}$

So $y = [|\sin x| + |\cos x|] = 1$

Now intersection points are $(2,1)$ and $(-2,1)$

Differentiating the equation $x^2 + y^2 = 5, \frac{dy}{dx} = -\frac{x}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(2,1)} = -2$ and $\left(\frac{dy}{dx}\right)_{(-2,1)} = 2$

75.(9) Given $f^3(x) = \int_0^x t f^2(t)dt$ …. (1)

Differentiating both the sides of equation (1) by using leibnitz, we get

$3f^2(x).f'(x) = xf^2(x)$ $\therefore f(x) = \frac{x^2}{6}, x \ge 0$

Slope of normal $= \frac{-3}{x_1}$ $\Rightarrow x_1 = 6$ and $y = 6$

Hence $y-$ intercepts $= 9$

76.(5) We have $f'(x) = 6p - 4p\cos 4x - 5 - 3\cos 3x$

$= 4p(1 - \cos 4x) + 2(p-4) + 3(1 - \cos 3x)$

Hence $f'(x) \ge 0 \Rightarrow p > 4 \forall x \in R$

Hence least integral value of $p = 5$.

77.(4) $\sqrt{A_1} = \sqrt{A_2} + A_2^{3/2} + 6$

$$\frac{\frac{dA_2}{dt}}{\frac{dA_1}{dt}} = \frac{1}{32}$$

78.(2) Let $F(x) = 2x + \int_0^x f(t)dt - p$ in $(-1, 1)$

$F(0) = -p$ $F(1^-) = 7 - p$

$F(-1^+) = 3 - p$

$F(-1)F(0) < 0$ & $F(0)F(1) < 0 \Rightarrow p \in (0,3)$

79.(3) $\phi'(c) = 0 \Rightarrow f''(c) = 2A$

$\phi(a) = \phi(b) \Rightarrow f(b) = f(a) + (b-a)f'(a) + (b-a)^2 A \Rightarrow \lambda = \frac{1}{2}$

80.(5) Given curve is : $xy = 1 \Rightarrow y = \frac{1}{x} \Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$

$\therefore$ Slope of normal = $x^2 > 0$

$\therefore$ Slope of given line $= \frac{\log_2(1+5a-a^2)}{5} > 0$

$\Rightarrow$ $\log_2\left(1+5a-a^2\right) > 0 \Rightarrow 1+5a-a^2 > 1 \Rightarrow a^2 - 5a < 0$

Hence $a \in (0,5)$. Slope of $y = 1$ is zero, so $\tan\theta = 2 \Rightarrow |\tan\theta| = 2$

81.(1200) $\because$ Degree of $P(x)$ is 5 with leading coefficient one.

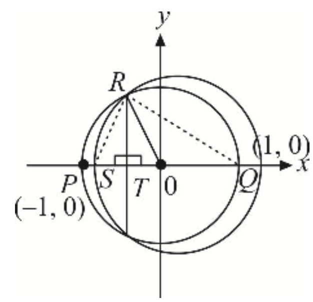
$\therefore$ Degree of $P'(x)$ is 4 with leading coefficient five.

$\therefore$ $P'(x) = 5(x-1)(x-3)(x-2)^2$

$\therefore$ $P'(6) = 5 \times 5 \times 3 \times 4^2 = 1200$

82.(1) Let r the radius of the second circle. The circle are

$S_1 = x^2 + y^2 - 1 = 0$; $S_2 = x^2 + y^2 - 2x + 1 - r^2 = 0$



The common chord is $S_1 - S_2 = 0 \rightarrow x = 1 - \frac{r^2}{2}$

Let it meet x-axis at T

$$RT^2 = OR^2 - OT^2 = 1 - \left(1 - \frac{r^2}{2}\right)^2 = \frac{r^2}{4}\left(4 - r^2\right)$$

$$\Delta QSR = \frac{1}{2}.QS.RT = \frac{r^2}{4}\sqrt{4-r^2} = f(r)$$

$$f'(r) = 0 \rightarrow r^2 = \frac{8}{3}$$

$\therefore$ Maximum area $= \frac{2}{3}\sqrt{4 - \frac{8}{3}} = \frac{4}{3\sqrt{3}} = A$ $\qquad [2A] = \left[\frac{4 \times 2}{3\sqrt{3}}\right] = 1$

83.(0) Let the two parts of the wire be x and z, where x is the circumference of the circle and z is the perimeter of the square.

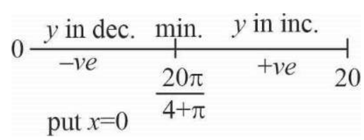
Given, $x + z = 20$ $\qquad \therefore z = 20 - x$ $\qquad$ ...(i)

Let y be the sum of area of the circle and the square.

Then $y = \pi\left(\frac{x}{2\pi}\right)^2 + \left(\frac{z}{4}\right)^2 = \frac{x^2}{4\pi} + \frac{(20-x)^2}{16}$ $\qquad$ ...(ii)

$\therefore$ $\frac{dy}{dx} = \frac{x}{2\pi} - \frac{(20-x)}{8} = \frac{4x - (20-x)\pi}{8\pi} = \frac{(4+\pi)\,x - 20\pi}{8\pi}$ $\qquad$ ...(iii)

Sign scheme for $\frac{dy}{dx}$ i.e. for $\left[(4+\pi)x - 20\pi\right]$ is



Here $0 \le x \le 20$

y has local minimum at $x=\frac{20\pi}{4+\pi}$

y will have maximum (greatest) value at $x=0$ or at $x=20$

From (ii), at $x=0$, $y=\frac{400}{16}=25$ sq. cm. at $x=20$, $y=\frac{400}{4\pi}=\frac{100}{\pi}$ sq. cm

Hence y has greatest value at $x=20$

Thus for greatest area the whole wire should be bent into a circle.

84.(1) $\int_e^x t\,f\,t)\,dt=\sin x-x\cos x-\frac{x^2}{2}$

Differentiating both sides

$xf(x)=\cos x-\cos x+x\sin x-x$

$f(x)=\sin x-1 \qquad \Rightarrow \qquad -2f\left(\frac{\pi}{6}\right)=-2\left(-\frac{1}{2}\right)=1$

85.(2) Given $f(x)=Ax^2+Bx+C$, so $f'(x)=2Ax+B$, ...(i)

$\therefore \quad f(x+h)=A(x+h)^2+B(x+h)+C=Ax^2+2Axh+Ah^2+Bx+Bh+C$

$=\left(Ax^2+Bx+C\right)+\left(2Axh+Ah^2+Bh\right)$.

Also $f'(x+\theta h)=2A(x+\theta h)+B$, from (i)

$=2Ax+2A\theta h+B$

$\therefore$ From $f(x+h)=f(x)+hf'(x+\theta h)$, we get

$\left(Ax^2+Bx+C\right)+\left(2Axh+Ah^2+Bh\right)=\left(Ax^2+Bx+C\right)+h(2Ax+2A\theta h+B)$

or $\quad Ah^2=2A\theta h^2 \quad$ or $\quad 2\theta=1 \quad$ or $\quad \theta=\frac{1}{2}$.

86.(2) From first mean Value Theorem, we know that it $f(x)$ and $\phi(x)$ be functions such that these are continuous in a $\le x\le b$ and $f'(x),\phi'(x)$ exist in $a<x<b$, then there exists at least one point c in (a,b), such that

$f(b)-f(a)=(b-a)f'(c)$ ...(i)

and $\phi(b)-\phi(a)=(b-a)\phi'(c)$. ...(ii)

Multiplying (ii) by $f(a)$ and (i) by $\phi(a)$ we get

$\phi(b)f(a)-f(a)\phi(a)=(b-a)\phi'(c)f(a)$ ...(iii)

$f(b)\phi(a)-f(a)\phi(a)=(b-a)f'(c)\phi(a)$ ...(iv)

Subtracting (iv) from (iii), we get

$\phi(b)f(a)-f(b)\phi(a)=(b-a)[\phi'(c)f(a)-f'(c)\phi(a)]$

or $\quad \begin{vmatrix} f(a) & f(b) \\ \phi(a) & \phi(b) \end{vmatrix}=(b-a)\begin{vmatrix} f(a) & f'(c) \\ \phi(a) & \phi'(c) \end{vmatrix} \quad \Rightarrow \quad \lambda^2=1$ so sum of absolute value λ is 2.

87.(2) $y=\tan^{-1}(\cot x)+\cot^{-1}(\tan x)$

$$y=\frac{\pi}{2}-x+\frac{3\pi}{2}-x,\ \ \frac{\pi}{2}<x<\pi$$

$$\frac{dy}{dx}=-2$$

$$-\frac{dy}{dx}=2$$

88.(0) $f'''(x)=0$

$f''(x)=c=-1$

$f'(x)=-x+d$

$d=3$

$$f(x)=-\frac{x^2}{2}+3x+1$$

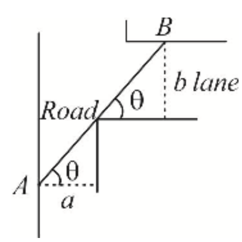
$f'(x)=-x+3$

89.(5) Let AB be the pole $AB=f(\theta)=a\sec\theta+b\cos ec\theta$

$f'(\theta)=0\ \Rightarrow\ a\sec\theta\tan\theta=b\text{cosec}\theta\cot\theta\ \Rightarrow\ a\sin^3\theta=b\cos^3\theta\ \Rightarrow$

$$\frac{\cos^2\theta}{a^{\frac{2}{3}}}=\frac{\sin^2\theta}{b^{\frac{2}{3}}}=\frac{1}{a^{\frac{2}{3}}+b^{\frac{2}{3}}}$$

$$\rightarrow\qquad \frac{a}{\cos\theta}=\frac{a\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{1}{2}}}{a^{\frac{1}{3}}}=a^{\frac{2}{3}}\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{1}{2}}$$



Similarly, $\dfrac{b}{\cos\theta}=b^{\frac{2}{3}}\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{1}{2}}$

Maximum $AB=\dfrac{a}{\cos\theta}+\dfrac{b}{\sin\theta}=\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}\quad\Rightarrow\quad L=\left(64^{\frac{2}{3}}+27^{\frac{2}{3}}\right)^{\frac{3}{2}}=(16+9)^{\frac{3}{2}}=125$

90.(9) Put $a=\cos\theta$ and $b=\sin\theta$

Hence, $E=\dfrac{b+1}{a+b-2}=\dfrac{\sin\theta+1}{\sin\theta+\cos\theta-2}=f(\theta)$

For maxima/minima, $f'(\theta)=0$

$\Rightarrow\qquad (\sin\theta+\cos\theta-2)(\cos\theta)-(\sin\theta+1)(\cos\theta-\sin\theta)=0$

$\Rightarrow\qquad \sin\theta\cos\theta+\cos^2\theta-2\cos\theta-\sin\theta\cos\theta+\sin^2\theta-\cos\theta+\sin\theta=0$

$\Rightarrow\qquad \cos^2\theta-2\cos\theta+\sin^2\theta-\cos\theta+\sin\theta=0$

$\Rightarrow \quad 1-3\cos\theta+\sin\theta=0 \qquad \Rightarrow \quad 1+\sin\theta=3\cos\theta$

$\Rightarrow \quad (1+\sin\theta)^2=9\cos^2\theta=9-9\sin^2\theta \qquad \Rightarrow \quad 10\sin^2\theta+2\sin\theta-8=0$

$\Rightarrow \quad 5\sin^2\theta+\sin\theta-4=0 \qquad \Rightarrow \quad 5\sin^2\theta+5\sin\theta-4\sin\theta-4=0$

$\Rightarrow \quad (\sin\theta+1)(5\sin\theta-4)=0 \qquad \Rightarrow \quad \sin\theta=-1 \text{ or } \sin\theta=\frac{4}{5}$

Now value of E, when $\sin\theta=-1$ is 0.

and value of E when $\sin\theta=\frac{4}{5}$ is $\frac{\sin\theta+1}{\sin\theta+\cos\theta-2}=\frac{\frac{4}{5}+1}{\frac{4}{5}+\frac{3}{5}-2}$ or $\frac{\frac{4}{5}+1}{\frac{4}{5}-\frac{3}{5}-2}$

i.e., $E=\frac{-9}{3}$ or $\frac{9}{-9}$

Hence, $E_{\min imum}=-3$ When $\sin\theta=\frac{4}{5}$ and $\cos\theta=\frac{3}{5}\Rightarrow u=-3$ Hence, $u^2=9$.

91.(9) Given, $f(x)=e^x.\int_0^x e^{-y}.f'(y)dy-\left(x^2-x+1\right)e^x$ ....(i)

Differentiating both the sides

$$f'(x)=e^x.e^{-x}.f'(x)+e^x\int_0^x e^{-y}.f'(y)dy-\left[\left(x^2-x+1\right)e^x+e^x(2x-1)\right]$$

$$f'(x)=f'(x)+\int_0^x e^{x-y}.f'(y)dy-e^x\left(x^2+x\right) \qquad \Rightarrow \qquad 0=f(x)+\left(x^2-x+1\right)e^x-e^x\left(x^2+x\right)$$

[Substituting $\int_0^x e^{x-y}f'(y)dy=f(x)+\left(x^2-x+1\right)e^x$ from (i)]

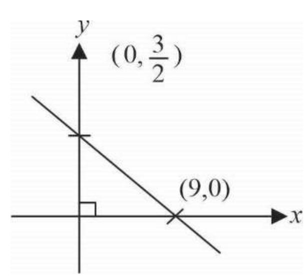
$f(x)=e^x\left(x^2+x-x^2+x-1\right)$; $f(x)=e^x.(2x-1)\Rightarrow f(1)=e.$

Also, $f'(x)=e^x.2+(2x-1)e^x\Rightarrow f'(1)=2e+e=3e$ and $f''(x)=2e^x+(2x-1)e^x+2e^x$

$\Rightarrow \quad f''(1)=2e+e+2e=5e$ Hence, $f(1)+f'(1)+f''(1)=9e\Rightarrow k=9$

92.(3) $\left(x^2-11\right)(y+1)=-4=-2\times 2$

On comparing we get $(x_0,y_0)\equiv(3,1)$



$$y'=\frac{8x}{\left(x^2-11\right)^2}$$

$y'|_{x=3}=6 \qquad \therefore \qquad m_N=\frac{-1}{6}$

Equation of normal $y-1=\frac{-1}{6}(x-3)$

$x+6y=9 \qquad \therefore$ Area $=\frac{1}{2}\times 9\times\frac{3}{2}=\frac{27}{4}$

93.(48) Given, $f''(x) \ge -2 \forall x \in [-3,3]$

$\therefore \quad \int_{-3}^{3} f''(x)dx \ge -2\int_{-3}^{3} dx \Rightarrow f'(x)]_{-3}^{3} \ge -12 \Rightarrow -12 \ge -12$

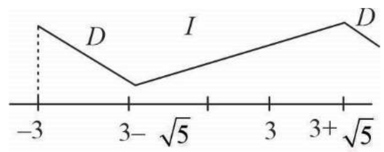
$\therefore \quad f''(x) = -2 \forall x \in [-3,3]$ or apply LMVT in $y = f'(x)$ in $[-3,3]$, we get $f''(x) = -2$

Now, $f'(x) = -2x + C$

$f'(3) = C - 6 = 0 \Rightarrow C = 6$

$f'(x) = 6 - 2x \Rightarrow f(x) = 6x - x^2 + C$

$f(0) = C = -4$



Hence, $f(x) = 6x - x^2 - 4$

Now, $g(x) = \int_0^x f(t)dt = \int_0^x \left(-t^2 + 6t - 4\right)dt \quad \therefore \quad g(x) = \frac{-x^3}{3} + 3x^2 - 4x$ in $[-3, 3]$

$g'(x) = -x^2 + 6x - 4$

$g(-3) = 48$ which is maximum value in $[-3,3]$.

94.(0) Given, $9 + f''(x) + f'(x) = x^2 + f^2(x)$ or $9 + \frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + y^2$

As, P be the point of maxima of $f(x)$ so $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0 \Rightarrow \quad x^2 + y^2 = 9 + \frac{d^2y}{dx^2}$

So, $P(x, y) = 9 + \frac{d^2y}{dx^2}$

So, $P(x, y)$ will lie inside the circle $x^2 + y^2 = 9$. Hence, no tangent is possible.

INTEGRAL CALCULUS-1

1.(C) $f(x) = \lim_{n\to\infty} e^{x\tan(1/n)\log(1/n)}$

But $\lim_{n\to\infty} \tan(1/n)\log 1/n = \lim_{n\to\infty}\left[-\frac{\log n}{n}\frac{\tan 1/n}{1/n}\right] = 0$

So $f(x) = e^0 = 1$. Hence

$$\int \frac{f(x)}{\sqrt[3]{\sin^{11} x\cos x}}dx = \int \frac{dx}{\sqrt[3]{\sin^{11} x\cos x}} = \int \frac{1+t^2}{\left(t^{11/3}\right)}dt\,(t = \tan x)$$

$$\int\left(t^{-11/3}+t^{-5/3}\right)dt = -\frac{3}{8}t^{-8/3} - \frac{3}{2}t^{-2/3} + C = -\frac{3}{8}\frac{(1+4\tan^2 x)}{\tan^2 x\sqrt[3]{\tan^2 x}} + C$$

2.(D) Let $g_n(x) = 1 + x^2 + x^4 + ... + x^{2n} = \frac{x^{2n+2}-1}{\left(x^2-1\right)}$

So $2x + 4x^3 + ... + 2nx^{2n-1} = h_n(x) = g_n'(x) = \frac{2x\left(nx^{2n+2} - (n+1)x^{2n} + 1\right)}{\left(x^2-1\right)^2}$

Now, $f(x) = \lim_{n\to\infty} h_n(x) = \frac{2x}{\left(x^2-1\right)^2}$ as $0 < x < 1$

Thus $\int f(x)dx = \int \frac{2x}{\left(x^2-1\right)^2}dx = -\frac{1}{x^2-1}$ +C

3.(B) The given integral is $I = \int \frac{(\sec\theta - 1)\sqrt{\sec\theta}}{1+\sec^2\theta}\tan\theta d\theta = \int \frac{(\sec\theta-1)\sec\theta\tan\theta}{\left(1+\sec^2\theta\right)\sqrt{\sec\theta}}d\theta$

Put, $\sqrt{\sec\theta} = t \Rightarrow \frac{1}{2\sqrt{\sec\theta}}.\sec\theta\tan\theta d\theta = dt$

$$I = \int \frac{\left(t^2-1\right)}{1+t^4}.2dt = 2\int \frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt = 2\int \frac{\left(1-\frac{1}{t^2}\right)dt}{\left(t+\frac{1}{t}\right)-2}$$

Put $t + \frac{1}{t} = u \Rightarrow \left(1-\frac{1}{t^2}\right)dt = du$

$$I = 2\int \frac{du}{u^2-2} = \frac{2}{2\sqrt{2}}\log_e\left|\frac{u-\sqrt{2}}{u+\sqrt{2}}\right| + C = \frac{1}{\sqrt{2}}\log_e\left|\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right| + C$$

4.(A) $f(x)=\lim\limits_{n\to\infty}\dfrac{x^{2n}-1}{x^{2n}+1}=-1\,(0<x<1)$, so

$$\int \sin^{-1}x\,(f(x))dx=-\int \sin^{-1}x\,dx=-\left[x\sin^{-1}x+\sqrt{1-x^2}\right]+C$$

5.(A) $I_n=\int \cot^n x\,dx=\int \cot^{n-2}x\cot^2 x\,dx=\int \cot^{n-2}x\left(\operatorname{cosec}^2 x-1\right)dx$

Thus $I_n+I_{n-2}=-\dfrac{\cot^{n-1}x}{n-1}$

$$I_0+I_1+2\left(I_2+.......+I_8\right)+I_9+I_{10}=\left(I_0+I_2\right)+\left(I_3+I_1\right)+\left(I_4+I_2\right)+\left(I_5+I_3\right)+\left(I_6+I_4\right)$$
$$+\left(I_7+I_5\right)+\left(I_8+I_6\right)+\left(I_9+I_7\right)+\left(I_{10}+I_8\right)$$
$$=-\left(\frac{\cot x}{1}+\frac{\cot^2 x}{2}+........+\frac{\cot^9 x}{9}\right)(\text{using }(i))$$
$$-\sum_{k=1}^{9}\frac{\cot^k x}{k}$$

6.(B) Let $I=\dfrac{\left(\sin^{3/2}\theta+\cos^{3/2}\theta\right)d\theta}{\sqrt{\sin^3\theta\cos^3\theta\sin(\theta+\alpha)}}=\int\dfrac{\sin^{3/2}\theta d\theta}{\sqrt{\sin^3\theta\cos^3\theta\sin(\theta+\alpha)}}+\int\dfrac{\cos^{3/2}\theta d\theta}{\sqrt{\sin^3\theta\cos^3\theta\sin(\theta+\alpha)}}$

Put $(\cos\alpha\tan\theta)+\sin\alpha=t^2$ in the first integral and $\cos\alpha+\cot\theta\sin\alpha=u^2$ in second integral

$\Rightarrow$ $\sec^2\theta d\theta=\dfrac{2tdt}{\cos\alpha}$ and $\operatorname{cosec}^2\theta d\theta=-\dfrac{2udu}{\sin\alpha}$

$$I=\int\frac{2tdt}{(\cos\alpha)t}-\int\frac{2udu}{(\sin\alpha)u}=\frac{2}{\cos\alpha}t-\frac{2}{\sin\alpha}u+c$$
$$=\frac{2}{\cos\alpha}\sqrt{\cos\alpha\tan\theta+\sin\alpha}-\frac{2}{\sin\alpha}\sqrt{\cos\alpha+\cot\theta\sin\alpha}+c$$

7.(D) Let $I=\int\dfrac{\cos^2 x+\sin 2x}{(2\cos x-\sin x)^2}dx=\int\dfrac{(\cos x+2\sin x)\cos x}{(2\cos x-\sin x)^2}dx$

Integrating by parts, taking *cos x* as the first and $\dfrac{(\cos x+2\sin x)}{(2\cos x-\sin x)^2}$ as the second function,

we have $=\cos x\left\{\dfrac{1}{2\cos x-\sin x}\right\}-\int\dfrac{-\sin x dx}{2\cos x-\sin x}=\cos x\left\{\dfrac{1}{2\cos x-\sin x}\right\}+\int\dfrac{\sin x dx}{2\cos x-\sin x}$

$$=\frac{\cos x}{(2\cos x-\sin x)}+\int\frac{\frac{-1}{5}(2\cos x-\sin x)-\frac{2}{5}(-2\sin x-\cos x)}{2\cos x-\sin x}dx$$

8.(A) As m = 9 > 0, hence we can substitute $\sqrt{9x^2+4x+6}=u\pm 3x$

9.(D) Here as per notations given, we can substitute $\sqrt{1+x^2}=(u-x)$ as $m=1>0$ and p = 4 > 0

$$\Rightarrow \quad I=\int \frac{u^{15}}{u}du=\int u^{14}du=\frac{1}{15}u^{15}+c$$

10.(D) Here m = – 1 < 0

P = – 2<0

Also $-x^2+3x-2=-(x-1)(x-2)$

We can use case 3

$\Rightarrow$ Putting $\sqrt{-x^2+3x-2}=u(x-2)$ or $(x-1)u$ or $u(1-x)$

11.(B) $I_n=\int \frac{dx}{(x^2+a^2)^n}=\frac{1}{(x^2+a^2)^n}.x-\int \frac{(-n)2x}{(x^2+a^2)^{n+1}}x.dx$ [integrating by parts using I as second function]

$$I_n=\frac{x}{(x^2+a^2)^n}+2n\int \frac{x^2+a^2-a^2}{(x^2+a^2)^{n+1}}dx=\frac{x}{(x^2+a^2)^n}+2n\left(I_n-a^2I_{n+1}\right)$$

12.(C) Let $I=\int \frac{(ax^2-b)dx}{x\sqrt{c^2x^2-(ax^2+b)^2}}=\int \frac{(ax^2-b)dx}{x^2\sqrt{c^2-\left(ax+\frac{b}{x}\right)^2}}=\int \frac{\left(a-\frac{b}{x^2}\right)dx}{\sqrt{c^2-\left(ax+\frac{b}{x}\right)^2}}$

Put $ax+\frac{b}{x}=c\sin\theta$

Then $I=\int \frac{c\cos\theta d\theta}{c\cos\theta}=\int d\theta=\theta+k, k$ is constant of integration $=\sin^{-1}\left(\frac{ax^2+b}{cx}\right)+k$

13.(C) $\because f'(x)=\frac{1}{1+\cos x}=\frac{1}{2\cos^2(x/2)}=\frac{1}{2}\sec^2\left(\frac{x}{2}\right)$

Integrating both sides with respect to *x*, we have $f(x)=\tan\left(\frac{x}{2}\right)+c$

$\therefore \quad f(0)=0+c=3$ then $f(x)=\tan\left(\frac{x}{2}\right)+3 \qquad \therefore \quad f\left(\frac{\pi}{2}\right)=\tan\left(\frac{\pi}{4}\right)+3=4$

Now, $3+\frac{\pi}{4}=3+\frac{22}{7\times 4}=3+\frac{11}{14}=\frac{53}{14}$

And $3+\frac{\pi}{4}=3+\frac{22}{14}=3+\frac{11}{7}$

14.(A) $\int \frac{x^2+1}{x^4+1}dx=\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$

Put $x-\frac{1}{x}=t\Rightarrow\left(1+\frac{1}{x^2}\right)dx=dt \qquad \therefore\ I=\int \frac{dt}{t^2+2}=\frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}}+c$

15.(C) $I=\int \frac{x^2-1}{\left(x^4+3x^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)}dx=\int \frac{1-\frac{1}{x^2}}{\left(x^2+\frac{1}{x^2}+3\right)\tan^{-1}\left(x+\frac{1}{x}\right)}dx$

Put $x+\frac{1}{x}=t \Rightarrow \left(1-\frac{1}{x^2}\right)dx=dt$ and $x^2+\frac{1}{x^2}+2=t^2$ $\therefore I=\int \frac{dt}{\left(t^2+1\right)\tan^{-1}t}=\ln\left|\tan^{-1}t\right|+C$

16.(B) $I=\int \frac{x^4-2}{x^2\sqrt{x^4+x^2+2}}dx=\int \frac{x^4-2}{x^3\sqrt{x^2+1+\frac{2}{x^2}}}dx$

Put $x^2+\frac{2}{x^2}+1=t \Rightarrow \left(x-\frac{2}{x^3}\right)dx=\frac{dt}{2}$, we get $I=\int \frac{dt}{2\sqrt{t}}=\sqrt{t}+C$

17.(C) $I=\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx=\int \frac{x^2-1}{(x+1)^2\sqrt{x^3+x^2+x}}dx$

Put $x+\frac{1}{x}+1=t^2 \Rightarrow \left(1-\frac{1}{x^2}\right)dx=2tdt$

$\therefore I=\int \frac{2tdt}{\left(t^2+1\right)t}=2\int \frac{dt}{t^2+1}$

18.(D) Divide numerator and demoninator by x^{10} we get $I=\int \frac{5x^{-6}+4x^{-5}}{\left(1+x^{-4}+x^{-5}\right)^2}dx$

19.(A) $f(xy)=f(x)f(y)$ ....(i)

Put $x=y=1$, we get : $f(1)=f^2(1)\Rightarrow f(1)=1$

Differentiating w.r.t. x (partially), we get : $yf'(xy)=f'(x).f(y)$

Putting $x=1$, $yf'(y)=f'(1).f(y)\Rightarrow \frac{f(y)}{f'(y)}=\frac{y}{f'(1)}$

$\therefore \quad \int \frac{f(x)}{f'(x)}dx=\int \frac{x}{f'(1)}dx=\frac{1}{f'(1)}\left(\frac{x^2}{2}+c\right)$

$f'(1)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}=\lim_{h\to 0}\frac{1+h+hg(h)-1}{h}=1$

20.(B) We have $\int \frac{dx}{x^2\left(x^n+1\right)^{(n-1)/n}}=\int \frac{dx}{x^2x^{n-1}\left(1+\frac{1}{x^n}\right)^{(n-1)/n}}=\int \frac{dx}{x^{n+1}\left(1+x^{-n}\right)^{(n-1)/n}}$

Put $1+x^{-n}=t$

$\therefore \quad -nx^{-n-1}dx=dt$ or $\frac{dx}{x^{n+1}}=-\frac{dt}{n}=-\frac{1}{n}\int t^{1/n-1}dt=-\frac{1}{n}.\frac{t^{1/n-1+1}}{1/n-1+1}+c$

21.(C) $I_{4,3}=\int \cos^4 x\sin 3x dx$

Integrating by parts, we have $I_{4,3}=-\dfrac{\cos 3x\cos^4 x}{3}-\dfrac{4}{3}\int \cos^3 x\sin x\cos 3x dx$

But $\sin x\cos 3x=-\sin 2x+\sin 3x\cos x$ so, $I_{4,3}=-\dfrac{\cos 3x\cos^4 x}{3}+\dfrac{4}{3}\int \cos^3 x\sin 2x dx-\dfrac{4}{3}\int \cos^4 x\sin 3x dx+C$

Therefore, $\dfrac{7}{3}I_{4,3}-\dfrac{4}{3}I_{3,2}=-\dfrac{\cos 3x\cos^4 x}{3}+C$

22.(AB) Put $t=\sin^2 x$

The integral reduces to $I=\dfrac{1}{2}\int e^t(2-t)dt=\dfrac{3}{2}e^t-\dfrac{te^t}{2}+c=\dfrac{1}{2}e^{\sin^2 x}\left(3-\sin^2 x\right)+c=e^{\sin^2 x}\left(1+\dfrac{1}{2}\cos^2 x\right)+c$

23.(AB) $I=\int \dfrac{1}{\sin^2 x-4\cos^2 x}dx=\int \dfrac{\sec^2 x}{\tan^2 x-4}dx$

Let $\tan x=t, \sec^2 x dx=dt$ $=\int \dfrac{dt}{t^2-4}=\int \dfrac{dt}{(t+2)(t-2)}=\int \dfrac{1}{4}\left[\dfrac{1}{t-2}-\dfrac{1}{t+2}\right]dt$

24.(BD) $I=\int \sqrt{\operatorname{cosec} x+1}\,dx=\int \dfrac{\cot x}{\sqrt{\operatorname{cosec} x-1}}dx$

Put $\operatorname{cosec} x=t$

$$I=-\int \frac{dt}{t\sqrt{t-1}}$$

Put $t-1=u^2$

So that $I=-\int \dfrac{2udu}{u\left(u^2+1\right)}=-2\tan^{-1}u+c$

25.(AC) $f(x)=\int\left(x+\sqrt{x^2+1}\right)dx=\dfrac{x^2}{2}+\dfrac{x}{2}\sqrt{x^2+1}+\log\left|x+\sqrt{x^2+1}\right|+c$

Putting x= 0 in above equation, we get $f(0)=c\Rightarrow c=\dfrac{-\left(1+\sqrt{2}\right)}{2}$

26.(ABD) $F(x)=\int \dfrac{1}{4-3\cos^2 x+5\sin^2 x}dx=\int \dfrac{1}{9-8\cos^2 x}dx$

$=\int \dfrac{\sec^2 x}{9\sec^2 x-8}dx=\int \dfrac{\sec^2 x}{1+9\tan^2 x}dx=\dfrac{1}{3}\tan^{-1}(3\tan x)+c \Rightarrow g(x)=3\tan x$

$g\left(\dfrac{\pi}{4}\right)=3$

27.(AC) We have $I=\int \frac{\sin x-\cos x}{(\sin x+\cos x)\sqrt{\sin x\cos x+\sin^2 x\cos^2 x}}dx=\cos ec^{-1}(g(x))+c$

$$I=\int \frac{-\cos 2x}{(1+\sin 2x)\sqrt{\sin x\cos x+\sin^2 x\cos^2 x}}dx$$

Let $1+\sin 2x=y \quad \Rightarrow \quad I=\int \frac{dy}{y\sqrt{y^2-1}} \Rightarrow I=\cos ec^{-1}(y)+c$

28.(AB) We have been given that $I=\int \frac{x\sqrt{2\sin(x^2+1)-\sin 2(x^2+1)}}{2\sin(x^2+1)+\sin 2(x^2+1)}dx \quad \Rightarrow I=\int x\tan\frac{(x^2+1)}{2}dx$

29.(ABC) $f'(x)=b^2+(a-1)b+2-\sin^2 x-\cos^4 x$

$b^2+(a-1)b+2-1$ for minimum value $(a-1)^2-4<0 \Rightarrow a\in(-1,3)$

30.(BCD) Let $I=\int \cos ec^2 x\log(\cos x+\cos 2x)dx$

$$=-\cot x\log\left(\cos x+\sqrt{\cos 2x}\right)+\int \frac{\cot x\left(-\sin x+\frac{-2\sin 2x}{2\sqrt{\cos 2x}}\right)}{\cos x+\sqrt{\cos 2x}}dx$$

$$=-\cot x\log\left(\cos x+\sqrt{\cos 2x}\right)-\int \frac{\cot x\sin x\sqrt{\cos 2x}+\sin 2x}{\sqrt{\cos 2x}\left(\cos x+\sqrt{\cos 2x}\right)}dx$$

Multiplying by $(\cos 2x)^{1/2}\left\{\cos x+(\cos 2x)^{1/2}\right\}$ in numerator and denominator, we get

$$I=-\cot x\log\left(\cos x+\sqrt{\cos 2x}\right)-\int \frac{\cos x}{\sin^2 x\sqrt{\cos 2x}}dx+\int \cot^2 x dx$$

Now put $t=\frac{1}{u}$

$\Rightarrow \quad \int \frac{\cos x}{\sin^2 x\sqrt{\cos 2x}}dx=-\sqrt{\cos ec^2 x-2}+c_1$ (where c_1 is integration constant)

31.(ABC) $\int \frac{\sin 2x}{8\sin^2 x+17\cos^2 x}dx=\int \frac{2\sin x.\cos x}{8\sin^2 x+17-17\sin^2 x}dx=\int \frac{\sin 2x}{17-9\sin^2 x}dx$

Let $\sin^2 x=t \Rightarrow \sin 2x dx=dt$

$$\int \frac{dt}{17-9t}=-\frac{1}{9}\log\left(17-9\sin^2 x\right)+c$$

32.(BC) $I=\int \sec^3 2\theta d\theta=\int \sec^2 2\theta\sec 2\theta d\theta=\frac{\tan 2\theta}{2}\sec 2\theta-\int \frac{2\tan^2 2\theta\sec 2\theta}{2}d\theta$

$$\Rightarrow 2I=\frac{\tan 2\theta}{2}\sec 2\theta+\frac{1}{2}\ln[\sec 2\theta+\tan 2\theta]+c$$

33.(CD) $I_1 = \int 2^x dx = \frac{2^x}{\log 2} = p(x) + c_1$; $I_2 = \int 2^{-x} dx = -\frac{2^{-x}}{\log 2} = m(x) + c_1$

34.(ABD) According to given problem $\int \frac{dx}{p^2 \sin^2 x + r^2 \cos^2 x} = \int \frac{\sec^2 x dx}{r^2 + p^2 \tan^2 x}$

Let $tanx$ = t

$$\Rightarrow \quad \int \frac{dt}{r^2 + p^2 t^2} = \frac{1}{pr} \tan\left(\frac{pt}{r}\right) + c = \frac{1}{pr} = 12$$

$$\frac{p}{r} = 3 \Rightarrow p^2 = 36 \Rightarrow p = \pm 6 \Rightarrow r = \pm 2$$

We know that $-\sqrt{p^2 + r^2} \le p \sin x + r \cos x \le \sqrt{p^2 + r^2}$

$$-40 \le 6 \sin x \pm 2 \cos x \le 40 \ \forall \ x$$

35.(ABC) $I = \int \frac{dx}{\left(1+\sqrt{x}\right)^8}$ Let $1 + \sqrt{x} = t \Rightarrow \frac{1}{2} x^{-1/2} dx = dt \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$

36.(BC) $I = \int \frac{\sin\left(\overline{x-\alpha} + \alpha\right)}{\sin(x-\alpha)} dx \Rightarrow I = \int \{\cos\alpha + \sin\alpha \cot(x-\alpha)\} dx \ = x\cos\alpha + \sin\alpha \log(\sin(x-\alpha)) + c$

37.(BC) $I = \int \left(4\sin\theta - 4\sin^3\theta\right) e^{\sin\theta} \cos\theta d\theta = \left(A\sin^3\theta - B\sin^2\theta + C\sin\theta + D\cos\theta + B + E\right) e^{\sin\theta} + F$

Taking $\sin\theta = t$

$$I = \int \left(4t - 4t^3\right) e^t dt = \underbrace{\left(At^3 - Bt^2 + Ct + D\cos\theta + E\right)}_{f(t)} e^t + F$$

D = 0 as $f(t)$ is a polynomial in t.

$\left(4t - 4t^3\right) e^t = \left(At^3 + Bt^2 + Ct + E\right) e^t + \left(3t^2 - 2Bt + C\right) e^t \Rightarrow A - 4$; 3A = B $\Rightarrow B = -12$

38.(CD) $\int \frac{(x-1)dx}{(x+1)\sqrt{x^3 + x^2 + x}}$

Multiplying numerator and denominator by $(x + 1)$, we get: $\int \frac{\left(x^2 - 1\right)}{\left(x^2 + 2x + 1\right)\sqrt{x^3 + x^2 + x}} dx = \int \frac{1 - \frac{1}{x^2}}{\left(x + 2 + \frac{1}{x}\right)\sqrt{x + 1 + \frac{1}{x}}}$

Either substitute $x + 1 + \frac{1}{x} = t^2$ to get (A) or substituting $x + 2 + \frac{1}{x} = t^2$ to get (C)

39.(AD) Let $3x = \cos\theta \ \Rightarrow -\frac{1}{3}\int \frac{\frac{\cos\theta}{3} + \theta^2}{\sin\theta} \sin\theta d\theta = -\frac{1}{3}\int\left(\frac{1}{3}\cos\theta + \theta^2\right) d\theta = -\frac{1}{9}\sin\theta - \frac{\theta^3}{9} + c$

40.(AD) Let $I = \int \frac{\sec x(2 + \sec x)}{(1 + 2\sec x)^2} dx = \int \frac{2\cos x + 1}{(2 + \cos x)^2} dx \Rightarrow I = \int \frac{\left(\cos x(\cos x + 2) + \sin^2 x\right)}{(2 + \cos x)^2} dx$

41.(BC) $f(x)+(\sin x)f(x+\pi)=\sin^2 x.$ ...(1)

$x \to x+\pi$

$f(x+\pi)+(\sin(x+\pi))f(x+2\pi)=\sin^2(x+\pi)$

$f(x+\pi)-(\sin x)f(x)=\sin^2 x$ ... (2)

from (1) & (2) $f(x)=\dfrac{\sin^2 x(1-\sin x)}{1+\sin^2 x}$

Now $\displaystyle\int \frac{\sin^2 x(1-\sin x)}{(1+\sin^2 x)}dx=\int\frac{(1+\sin^2 x-1)(1-\sin x)dx}{(1+\sin^2 x)}=x-\int\frac{(1-\sin x)+\sin x+\sin^3 x}{(1+\sin^2 x)}dx$

$\displaystyle =x-\int\frac{dx}{1+\sin^2 x}+\int\frac{\sin x dx}{(2-\cos^2 x)}-\int \sin x\ dx$

$\displaystyle =x-\frac{1}{2}\int\frac{\sec^2 x dx}{\tan^2 x+\frac{1}{2}}+\int\frac{\sin x dx}{(2-\cos^2 x)}-\int\sin x\ dx\ =x-\frac{1}{2}.\sqrt{2}\tan^{-1}(\sqrt{2}\tan x)-\frac{1}{2\sqrt{2}}\ln\frac{(\sqrt{2}+\cos x)}{\sqrt{2}-\cos x}+\cos x+C$

42.(AC) $\displaystyle\int\frac{dx}{x(x+1)(x+2).........(x+n)}$

Consider $\displaystyle\frac{1}{x(x+1)(x+2)........(x+n)}=\frac{1}{n!}\left[\frac{1}{x}-\frac{n}{1.(x+1)}+\frac{n(n-1)}{1.2(x+2)}-\frac{n(n-1)(n-2)}{1.2.3(x+3)}.......+\frac{(-1)^n.n(n-1)(n-2)....1}{(1.2.3.....n)(x+n)}\right]$

$\displaystyle\frac{1}{x(x+1)(x+2)........(x+n)}=\frac{1}{n!}\left[\frac{{}^nC_0}{x}-\frac{{}^nC_1}{(x+1)}+\frac{{}^nC_2}{(x+2)}.....+\frac{(-1)^n.{}^nC_n}{(x+n)}\right]$

So, $\displaystyle I=\int\frac{dx}{x(x+1)(x+2)......(x+n)}=\frac{1}{n!}\int\left(\frac{{}^nC_0}{x}-\frac{{}^nC_1}{(x+1)}+\frac{{}^nC_2}{(x+2)}.....+\frac{(-1)^n.{}^nC_n}{(x+n)}\right)dx$

$\displaystyle I=\frac{1}{n!}\left[{}^nC_0(\ln x)-{}^nC_1\ln(x+1)+{}^nC_2\ln(x+2)+..................+(-1)^n.{}^nC_n.\ln(x+n)\right]+c$

$\displaystyle\frac{1}{n!}\left[\sum_{r=0}^{n}(-1)^r.{}^nC_r\ln(x+r)\right]+c$ or $\displaystyle I=\frac{1}{n!}\left[\ln\left(\prod_{r=0}^{n}(x+r)^{(-1)^r.{}^nC_r}\right)\right]$

43.(ABC) Let $h(x)=f(x).g(x)$

Differentiate both sides w.r.t x

$h'(x)=f'(x).g(x)+f(x)g'(x)$

But due to mistake $h'(x)=f'(x).g'(x)$

So, $f'(x).g'(x)=f'(x).g(x)+f(x)g'(x)$

$3x^2.g'(x)=3x^2.g(x)+x^3.g'(x)$

$\displaystyle\int\frac{g'(x)}{g(x)}=\int\frac{-3}{(x-3)}\Rightarrow g(x)=\frac{9}{(x-3)^3}$

44.(ACD)

$$I=\begin{bmatrix}1&0\\0&1\end{bmatrix}, A=\begin{bmatrix}x&x\\x&x\end{bmatrix},\quad A^2=\begin{bmatrix}2x^2&2x^2\\2x^2&2x^2\end{bmatrix}$$

$$A^3=\begin{bmatrix}4x^3&4x^3\\4x^3&4x^3\end{bmatrix}....................$$

So $I+A+\dfrac{A^2}{2!}+\dfrac{A^3}{3!}+............=\begin{bmatrix}1&0\\0&1\end{bmatrix}+\begin{bmatrix}x&x\\x&x\end{bmatrix}+\dfrac{1}{2!}\begin{bmatrix}2x^2&2x^2\\2x^2&2x^2\end{bmatrix}$

$$\begin{bmatrix}\dfrac{f(x)}{2}&\dfrac{g(x)}{2}\\\dfrac{g(x)}{2}&\dfrac{f(x)}{2}\end{bmatrix}=\begin{bmatrix}1+x+\dfrac{2x^2}{2!}+\dfrac{4x^2}{3!}.........&x+\dfrac{2x^2}{2!}+\dfrac{4x^2}{3!}+..........\\x+\dfrac{2x^2}{2!}+\dfrac{4x^3}{3!}+.........&1+x+\dfrac{2x^2}{2!}+\dfrac{4x^3}{3!}+....\end{bmatrix}$$

So $\qquad f(x)=2+(2x)+\dfrac{(2x)^2}{2!}+\dfrac{(2x)^3}{3!}+....................$

$f(x)=1+e^{2x}\Rightarrow\qquad f(x)\in(1,\infty)\in R^+$

Similarly $\qquad g(x)=(2x)+\dfrac{(2x)^2}{2!}+\dfrac{(2x^3)}{3!}+.......$

$\qquad g(x)=e^{2x}-1\Rightarrow g(x)\in(-1,\infty)\in R$

How $\displaystyle\int\frac{g(x)}{f(x)}dx=\int\frac{(e^{2x}-1)}{(e^{2x}+1)}dx=\int\frac{(e^x-e^{-x})}{(e^x+e^{-x})}dx=\ln(e^x+e^{-x})+c$

& $\displaystyle\int(g(x)+1)\sin x dx=\int e^{2x}.\sin x dx=\frac{e^{2x}}{5}(2\sin x-\cos x)+c$

45.(AC) $\displaystyle I=\int\frac{(x^2-1)\sqrt{x^4+2x^3-x^2+2x+1}}{x^2(x+1)^2}dx$

$$I=\int\frac{\left(1-\dfrac{1}{x^2}\right)\sqrt{\left(x^2+\dfrac{1}{x^2}\right)+2\left(x+\dfrac{1}{x}\right)-1}}{\left(x+\dfrac{1}{x}+2\right)}dx$$

Let $x+\dfrac{1}{x}=t\Rightarrow\left(1-\dfrac{1}{x^2}\right)dx=dt$

$$I=\int\frac{\sqrt{(t^2-2)+2t-1}}{(t+2)}dt=\int\frac{\sqrt{t^2+2t-3}}{(t+2)}dt$$

$$I=\int\frac{(t^2+2t-3)dt}{(t+2)\sqrt{t^2+2t-3}}=\int\frac{t(t+2)}{(t+2)\sqrt{t^2+2t-3}}dt-3\int\frac{dt}{(t+2)\sqrt{t^2+2t}}$$

Let $I_1 = \int \frac{tdt}{\sqrt{t^2+2t-3}}, I_2 = \int \frac{dt}{(t+2)\sqrt{t^2+2t-3}}$

$$I_1 = \int \frac{tdt}{\sqrt{(t+1)^2-4}} = \int \frac{(t+1)dt}{(t+1)^2-4} - \int \frac{dt}{\sqrt{(t+1)^2-4}}$$

$I_1 = \sqrt{t^2+2t-3} - \ln\left|(t+1)+\sqrt{t^2+2t-3}\right|$ & $I_2 = \int \frac{dy}{\sqrt{1-2y-3y^2}}$ put $t+2 = \frac{1}{y}$

$I_2 = \frac{1}{\sqrt{3}}\sin^{-1}\left(\frac{y+1/3}{2/3}\right) = \frac{1}{\sqrt{3}}\sin^{-1}\left(\frac{5+t}{2+t}\right)$; Now $I = \sqrt{t^2+2t-3} - \ln\left|(t+1)+\sqrt{t^2+2t-3}\right| - \sqrt{3}\sin^{-1}\left(\frac{5+t}{2+t}\right)+c$

46.(ABCD)

$$f(x) = (x-1)(x+2)(x-3)(x-6)-100$$

$$f(x) = \left(x^2-4x+3\right)\left(x^2-4x-12\right)-100$$

$$f(x) = \left(x^2-4x\right)^2 - 9\left(x^2-4x\right)-136$$

$$f(x) = \left(x^2-4x+8\right)\left(x^2-4x-17\right)$$

So equation $f(x)=0$ has two real & two imaginary roots

Now $f(x) = \left(x^2-4x-17\right)\left(x^2-4x+8\right)$

$$f(x) = \left((x-2)^2-21\right)\left((x-2)^2+4\right)$$

$(f(x))_{\min} = (-21)(4) = -84$ at $x=2$

$$\frac{g(x)}{f(x)} = \frac{g(x)}{\left(x^2-4x-17\right)\left(x^2-4x+8\right)} = \frac{(Ax+B)}{\left(x^2-4x-17\right)} + \frac{(cx+D)}{\left(x^2-4x+8\right)}$$

clearly A, B and C must be zero.

$$\therefore \frac{g(x)}{\left(x^2-4x-17\right)\left(x^2-4x+8\right)} = \frac{D}{\left(x^2-4x+8\right)}$$

$$g(x) = D\left(x^2-4x-17\right)$$

$$g(-2) = D(4+8-17)$$

So $\frac{g(x)}{f(x)} = \frac{2\left(x^2-4x-17\right)}{\left(x^2-4x-17\right)\left(x^2-4x+8\right)}$

$\frac{g(x)}{f(x)} = \frac{2}{x^2-4x+8}$; $\int \frac{g(x)}{f(x)}dx = \int \frac{2}{x^2-4x+8}dx = 2\int \frac{dx}{(x-2)^2+(2)^2}$

$$\int \frac{g(x)}{f(x)}dx = 2.\frac{1}{2}\tan^{-1}\left(\frac{x-2}{2}\right)+c = \tan^{-1}\left(\frac{x-2}{2}\right)+c$$

47.(BC) $\ln(1-x)=-\left\{x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+........\right\}$

Replace x by x^2

$$\ln(1-x^2)=-\left\{x^2+\frac{x^4}{2}+\frac{x^6}{3}+\frac{x^8}{4}+....\right\}$$

$$\ln(1-x^2)=-\left\{x^4+\frac{x^6}{2}+\frac{x^8}{3}+\frac{x^8}{4}+........\right\}$$

$$x^2\ln(1-x^2)=-\left\{x^4+\frac{x^6}{2}+\frac{x^8}{3}+\frac{x^{10}}{4}+.......\right\}$$

$$\int x^2\ln(1-x^2)dx=-\left\{\frac{x^5}{1.5}+\frac{x^7}{2.7}+\frac{x^9}{2.7}+\frac{x^9}{3.9}+\frac{x^{11}}{4.11}........\right\}$$

Applying I.B.P with limits 0 to 1 for LHS

$$\left(\frac{x^3}{3}\ln(1-x^2)\right)_0^1-\int_0^1\frac{x^3}{3}.\frac{(-2x)}{(1-x^2)}dx$$

$$\left[\frac{x^3}{3}\ln(1-x^2)\right]_0^1+\frac{2}{3}\left(\frac{-x^3}{3}-x+\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|\right)_0^1$$

$$\frac{1}{3}\ln 2+\frac{1}{3}\ln 2-\frac{2}{3}-\frac{2}{9}+\lim_{x\to 1}\left(\frac{x^3}{3}-\frac{1}{3}\right)\ln(1-x)$$

$$\therefore\ \lim_{x\to 1}(x^3-1)\ln(1-x)=0$$

So $\frac{2}{3}\ln 2-\frac{8}{9}=RHS$

$$\therefore\ \frac{1}{1.5}+\frac{1}{2.7}+\frac{1}{3.9}+......=\frac{2}{3}\ln 2-\frac{8}{9}$$

48.(AC) $f(x+2y)=f(x)e^{2y}+f(2y)e^x+x^2(1-e^{2y})+4y^2(1-e^x)+4xy$ ......(1)

Putting $x=y=0$ in equation (1), we get $f(0)=2f(0)\Rightarrow f(0)=0$

Now, $f'(x)=\lim_{y\to 0}\frac{f(x+2y)-f(x)}{2y}$

$$=\lim_{y\to 0}\frac{f(x)e^{2y}+e^x f(2y)+x^2(1-e^{2y})+4y^2(1-e^x)+4xy-f(x)}{2y}$$

$$=\lim_{y\to 0}\frac{f(x)(e^{2y}-1)+e^x(f(2y)-f(0))-(e^{2y}-1)x^2+4y^2(1-e^x)+4xy}{2y}$$

$$= f(x)\lim_{y\to 0}\frac{\left(e^{2y}-1\right)}{2y} + e^x \lim_{y\to 0}\frac{f(2y)-f(0)}{2y} - x^2\lim_{y\to 0}\frac{\left(e^{2y}-1\right)}{2y} + 2x + \lim_{y\to 0} 2y\left(1-e^x\right)$$

$$f'(x) = f(x) + e^x - x^2 + 2x$$

$$e^{-x}f'(x) - e^{-x}f(x) = 1 - x^2e^{-x} + 2xe^{-x}$$

$$\frac{d}{dx}\left(e^{-x}f(x)\right) = \frac{d}{dx}\left(x + x^2e^{-x}\right) \Rightarrow e^{-x}f(x) = x + x^2e^{-x} + c$$

put $x = 0$, we get $c = 0$ Hence $f(x) = xe^x + x^2$.

49.(AB) $\int\left(\frac{1}{1-x^8}\right)\left[\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)\right]dx$

$$= \int\left(\frac{1}{1-x^8}\right)\left[\frac{\pi}{2} - \sin^{-1}\frac{2x}{1+x^2} + \tan^{-1}\left(\frac{2x}{1-x^2}\right)\right]dx$$

$$= \frac{\pi}{8}\int\left(\frac{1}{1+x^2} + \frac{1}{1-x^2}\right)dx + \frac{\pi}{4}\int\frac{dx}{x^4+1} \quad = \frac{\pi}{8}\left(\tan^{-1}x + \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|\right) + \frac{\pi}{8}\int\left[\frac{x^2+1}{x^4+1} - \frac{x^2-1}{x^4+1}\right]dx$$

$$= \frac{\pi}{8}\left[\tan^{-1}x + \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right| + \frac{1}{\sqrt{2}}\tan^{-1}\frac{x^2-1}{x\sqrt{2}} - \frac{1}{2\sqrt{2}}\ln\left|\frac{x^2 - x\sqrt{2}+1}{x^2 + x\sqrt{2}+1}\right|\right] + c.$$

50.(ABC) $A + Bi = \int a^{ax}.e^{ibx}.dx = \int e^{(a+ib)x}dx = \frac{e^{(a+ib)x}}{a+ib}$

$$\Rightarrow \quad (A+iB)(a+ib) = e^{(a+ib)x} = e^{ax}.e^{ibx} \qquad \ldots (1)$$

Comparing the modulus of both sides, $\sqrt{A^2+B^2}\sqrt{a^2+b^2} = e^{ax}$

Squaring, we get the result $\left(A^2+B^2\right)\left(a^2+b^2\right) = e^{2ax}$

Comparing the arguments of both sides in equation (1), $\tan^{-1}\frac{B}{A} + \tan^{-1}\frac{b}{a} = bx.$

51.(CD) Let $I = \int\sqrt{\frac{\cos x - \cos^3 x}{\left(1-\cos^3 x\right)}}dx = \int\frac{\sqrt{\cos x}\left(\sqrt{1-\cos^2 x}\right)}{\sqrt{1-\left(\cos^{\frac{3}{2}}x\right)}}dx$

If $\cos^{\frac{3}{2}}x = p$, then $\left(-\frac{3}{2}\cos^{\frac{1}{2}}x\sin x\right)dx = dp$

$$I = -\frac{2}{3}\int\frac{dp}{\sqrt{1-p^2}}, = -\frac{2}{3}\sin^{-1}\left(\cos^{\frac{3}{2}}x\right) + C = \frac{2}{3}\cos^{-1}\left(\cos^{\frac{3}{2}}x\right) + C_1$$

52.(CB) $I=\int \frac{dx}{x^{29}\left(1-\frac{6}{x^7}\right)}$

Let $\left(1-\frac{6}{x^7}\right)=p \Rightarrow \frac{42}{x^8}dx=dp$ and $x^7=\left(\frac{6}{1-p}\right)$

$I=\frac{1}{42}\int \frac{(1-p)^3}{(6)^3\, p}dp=\frac{1}{(42).(216)}\int \frac{1-p^3-3p+3p^2}{p}dp=\frac{1}{54432}\left[\ln p^6+9p^2-2p^3-18p\right]+c$

53.(BC) $\int e^{\tan^{-1}x}\left(1+x+x^2\right)d\left(\cot^{-1}x\right)$

$\int e^{\tan^{-1}x}\left(1+x+x^2\right)\left(\frac{-1}{1+x^2}\right)dx$

$\int e^t\left(1+\tan t+\tan^2 t\right)(-dt)$

Let $\tan^{-1}x=t$

$\frac{dx}{1+x^2}=dt$

$-\int e^t\left(\tan t+\sec^2 t\right)dt \Rightarrow -e^t\tan t+c. \quad \Rightarrow \quad -xe^{\tan^{-1}x}.+c$

54.(BD) $\int \frac{x^3}{\sqrt{1+2x^2}}dx=\frac{1}{m}\left(1+2x^2\right)^{1/2}\left(x^2-1\right)+2$

$\int \frac{x^2.x}{\sqrt{1+2x^2}}dx$ Put $1+2x^2=t^2 \Rightarrow 4xdx=2tdt$

$\frac{1}{2}\int \frac{\left(t^2-1\right)tdt}{2\quad t}$ $2xdx=tdt$

$\frac{1}{4}\int\left(t^2-1\right)dt \Rightarrow \frac{1}{4}\left(\frac{t^3}{3}-t\right)+c$

$\Rightarrow \quad \frac{1}{4}t\left(\frac{t^2}{3}-1\right)+c\ ,\ \frac{1}{4}t\frac{\left(t^2-3\right)}{3}+c$

$\Rightarrow \quad \frac{t}{12}\left(t^2-3\right)+c$

$\Rightarrow \quad \frac{\left(\sqrt{1+2x^2}\right)\left(2x^2+1-3\right)}{12}+c$

$\Rightarrow \quad \frac{\left(\sqrt{1+2x^2}\right)\left(x^2-1\right)}{6}+c$

$m=6$

55.(AC) $I=\int \frac{\sin x+\sin^3 x}{\cos 2x}dx$

$$I=\int \frac{\left(1+\sin^2 x\right)\sin x}{\left(2\cos^2 x-1\right)}dx$$

$$I=\int \frac{\left(2-\cos^2 x\right)\left(-\sin x\right)}{\left(1-2\cos^2 x\right)}dx$$

Let $\cos x=t$

$-\sin x dx=dt$

$$I=\int \frac{\left(2-t^2\right)}{\left(1-2t^2\right)}dt$$

$$I=\int \frac{2}{\left(1-2t^2\right)}dt+\frac{1}{2}\int\left(\frac{\left(-2t^2\right)-1}{\left(1-2t^2\right)}\right)dt$$

$$I=\int \frac{2dt}{\left(1-2t^2\right)}+\frac{1}{2}\int\frac{\left(1-2t^2\right)-1}{\left(1-2t^2\right)}dt$$

$$I=\frac{3}{2}\int\frac{dt}{\left(1-2t^2\right)}+\frac{1}{2}\int dt$$

$$I=\frac{1}{2}t\frac{-3}{4}\int\frac{dt}{\left(t^2-\frac{1}{2}\right)}$$

$$I=\frac{t}{2}-\frac{3}{4\sqrt{2}}\ln\left|\frac{t-\frac{1}{\sqrt{2}}}{t+\frac{1}{\sqrt{2}}}\right|+c$$

$$I=\frac{\cos x}{2}-\frac{3}{4\sqrt{2}}\ln\left|\frac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}\right|+c$$

56.(ABC)

Here, $f(x)+g(x)+h(x)$

$=\sqrt{f(x).g(x)}+\sqrt{g(x).h(x)}+\sqrt{h(x).f(x)}$

$$\Rightarrow \quad \frac{1}{2}\left\{\left(\sqrt{f(x)}-\sqrt{g(x)}\right)^2+\left(\sqrt{g(x)}-\sqrt{h(x)}\right)^2+\left(\sqrt{h(x)}-\sqrt{f(x)}\right)^2\right\}=0$$

$$\Rightarrow \quad f(x)=g(x)=h(x)$$

$$\therefore\ f(x)+g(x)-2h(x)=0$$

57.(ABD) We have $f(x)=x^2+\int_0^x e^{-t}f(x-t)dt$

$\Rightarrow \quad f(x)=x^2+\int_0^x e^{-(x-t)}f(x-(x-t))dt$

$\left[u\sin g \int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$

$\Rightarrow \quad f(x)=x^2+e^{-x}\int_0^x e^t f(t)dt$ ....(i)

On differentiating both the sides, get

$f'(x)=2x+e^{-x}\left[e^x f(x)\right]-e^{-x}\int_0^x e^t f(t)dt$ ....(ii)

[using Leibnitz rule]

$\Rightarrow \quad f(x)+f'(x)=x^2+2x+f(x) \;\Rightarrow\; f'(x)=x^2+2x$ ....(iii)

On integrating both the sides of equation (iii), we get $f(x)=\dfrac{x^3}{3}+x^2+C$

But $f(0)=0\Rightarrow C=0$ Hence, $f(x)=\dfrac{x^3}{3}+x^2$

58.(BC) It is given that $f'(x)=f(x)+\int_0^2 f(x)dx$

$\Rightarrow f'(x)=f(x)+c \quad \left[\therefore c=\int_0^2 f(x)dx\right]$

$\Rightarrow \quad \dfrac{f'(x)}{f(x)+c}=1$

On integrating both the sides of above expression, $\log_e\{f(x)+c\}=k\Rightarrow f(x)+c=ke^x$

[k being constant of integration]

Since, $f(0)=\dfrac{4-e^2}{3}$ and $f(0)+c=k$ $\quad\therefore\; k=\dfrac{4-e^2}{3}+c\Rightarrow c=k-\left(\dfrac{4-e^2}{3}\right)$

From Eqs. (i) and (ii), We get $f(x)+k-\left(\dfrac{4-e^2}{3}\right)=k\left(e^x\right) \;\Rightarrow\; f(x)=k\left(e^k-1\right)+\left(\dfrac{4-e^2}{3}\right)$

Now, to find constant of integration k.

On integrating both the sides form 0 to 2, we get $\int_0^2 f(x)dx=k\int_0^2\left(e^x-1\right)dx+\dfrac{4-e^2}{3}\int_0^2 dx$

$\Rightarrow \quad c=k\left(e^2-2-1\right)+\dfrac{4-e^2}{3}(2) \;\Rightarrow\; c=\dfrac{2}{3}\left(4-e^2\right)+k\left(e^2-3\right)$

From Eqs. (ii) and (iv), we get $k-\left(\dfrac{4-e^2}{3}\right)=\dfrac{2}{3}\left(4-e^2\right)+k\left(e^2-3\right)\Rightarrow k=1$

On putting $k=1$ in Eq. (iii), we get $f(x)=\left(e^x-1\right)+\dfrac{4-e^2}{3}$ Hence, $f(x)=e^x-\dfrac{\left(e^2-1\right)}{3}$

59.(AD) We have, $f(x).f(y)+2=f(x)+f(y)+f(xy)$ ....(i)

On replacing $x,y\to 1$, we get $\left(f(1)\right)^2+2=3f(1)$

$\Rightarrow \quad f^2(1)-3f(1)+2=0 \quad \Rightarrow \quad f(1)=2$ or 1

But $f(1)$ cannot be equal to one as $f(0)=1 \quad \Rightarrow \quad f(1)=2$ ...(ii)

On replacing y by $1/x$ in Eq. (i) we get $f(x).f\left(1/x\right)+2=f(x)+f(1/x)+f(1)$

$\Rightarrow \quad f(x).f(1/x)+2=f(x)+f(1/x)+2$ [using Eq. (ii)

$$\Rightarrow \quad f(x)=\frac{f\left(1/x\right)}{f(1/x)-1} \text{ and } f(1/x)=\frac{f(x)}{f(x)-1} \quad \text{...(iii)}$$

$$\text{Now, } f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f(x+h)+\dfrac{f(1/x)}{1-f(1/x)}}{h}=\lim_{h\to 0}\frac{f(x+h)-f(x+h).f(1/x)+f(1/x)}{h\{1-f(1/x)\}}$$

$$=\lim_{h\to 0}\frac{f(x+h)-f(x+h)-f(1/x)-f\left(\dfrac{x+h}{x}\right)+2+f(1/x)}{h\{1-f(1/x)\}} \quad \text{[using Eq. (i)]}$$

$$=\lim_{h\to 0}\frac{f\left(1+\dfrac{h}{x}\right)-2}{h\{f(1/x)-1\}}=\lim_{h\to 0}\frac{f(1+h/x)-f(1)}{\dfrac{h}{x}.x\{f(1/x)-1\}} \quad \text{[using Eq. (ii)}$$

$$=\frac{f'(1)}{x\{f(1/x)-1\}} \quad \left[\because \text{ from eq. (iii)}, f(x).f\left(\frac{1}{x}\right)=\frac{f(x)f\left(\dfrac{1}{x}\right)}{\left\{f\left(\dfrac{1}{x}\right)-1\right\}\{f(x)-1\}}\right]$$

$$f'(x)=\frac{2\{f(x)-1\}}{x}\Rightarrow xf'(x)=2\{f(x)-1\}$$

On integrating both the sides of the above expression, we get $xf(x)-\int f(x)dx=2\int f(x)dx-2x+\lambda$,

Where λ is the constant of integral

$\Rightarrow \quad 3\int f(x)dx=x\{2+f(x)\}-\lambda$ Hence, $3\int f(x)dx-x\{2+f(x)\}=\lambda$ (constant)

60. **[A-p, q]** **[B-r, s]** **[C-p]** **[D-p, q]**

(A) Let $I=\int\frac{2^x}{\sqrt{1-4^x}}dx=\frac{1}{\log 2}\int\frac{1}{\sqrt{1-t^2}}dt$

Putting $2^x=t, 2^x\log 2dx=dt$

$$I=\frac{1}{\log 2}\sin^{-1}\left(\frac{t}{1}\right)+C=\frac{1}{\log 2}\sin^{-1}\left(2^x\right)+C$$

(B) $\int \frac{dx}{\left(\sqrt{x}\right)^2+\left(\sqrt{x}\right)^7}=\int \frac{dx}{\left(\sqrt{x}\right)^7\left(1+\frac{1}{\left(\sqrt{x}\right)^5}\right)}$

Put $\frac{1}{\left(\sqrt{x}\right)^5}=y, \frac{dy}{dx}=-\frac{5}{2\left(\sqrt{x}\right)^7}$

$\therefore\ I=\int -\frac{2dy}{5(1+y)}=-\frac{2}{5}\ln|1+y|+C=\frac{2}{5}\ln\left(\frac{1}{1+\frac{1}{\left(\sqrt{x}\right)^5}}\right)$

(C) Add and subtract $2x^2$ in the numerator, then k = 1 and m = 1

(D) $I=\int \frac{dx}{5+4\cos x}=\int \frac{dx}{5\left(\sin^2\frac{x}{2}+\cos^2\frac{x}{2}\right)+4\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)}$

Let $t=\tan\frac{x}{2}\Rightarrow 2dt=\sec^2\frac{x}{2}dt \Rightarrow I=\int \frac{2dt}{9+t^2}=\frac{2}{3}\tan^{-1}\left(\frac{t}{3}\right)+C$

61. **[A-r]** **[B-s]** **[C-q]** **[D-p]**

(A) $\int \frac{e^{2x}-1}{e^{2x}+1}dx=\int \frac{e^x-e^{-x}}{e^x+e^{-x}}dx=\log\left(e^x+e^{-x}\right)$

(B) $I=\int \frac{1}{\left(e^x+e^{-x}\right)^2}dx=\int \frac{e^{2x}}{\left(e^{2x}+1\right)^2}dx$

Put $e^{2x}+1=t\Rightarrow 2e^{2x}dx=dt,\ we\ get \Rightarrow I=\frac{1}{2}\int \frac{1}{t^2}dt=-\frac{1}{2}\frac{1}{t}+c$

(C) $I=\int \frac{e^{-x}}{1+e^x}dx=\int \frac{e^{-x}e^{-x}}{e^{-x}+1}dx$

Put $e^{-x}+1=t\Rightarrow -e^{-x}dx=dt$

$\Rightarrow\quad I=-\int \frac{(t-1)}{t}dt=\int\left(\frac{1}{t}-1\right)dt=\log t-t+C=\log\left(e^{-x}+1\right)-\left(e^{-x}+1\right)+C$

(D) $I=\int \frac{1}{\sqrt{1-e^{2x}}}dx=\int \frac{e^{-x}}{\sqrt{e^{-2x}-1}}dx$

Put $e^{-x}=t\Rightarrow -e^{-x}dx=dt$

$\Rightarrow\quad I=-\int \frac{1}{\sqrt{t^2-1}}dt=-\log\left[t+\sqrt{t^2-1}\right]+C=-\log\left[e^{-x}+\sqrt{e^{-2x}-1}+C\right]$

62. **[A-s] [B-t] [C-r] [D-q]**

(A) $\int \frac{dx}{\sqrt{x}(x+9)}$, $x=t^2 = 2\int \frac{dt}{t^2+9} = \frac{2}{3}tan^{-1}\left(\frac{\sqrt{x}}{3}\right)+c$

(B) $\int e^x\left(1-cot\,x+cot^2\,x\right)dx = \int e^x\left(-cot\,x+cosec^2 x\right)dx = -e^x\,cot\,x+c$

(C) $\int \frac{sin^3 x+cos^3 x}{cos^2 x\,sin^2 x}dx = \int (tan\,x\,sec\,x+cot\,x\,cosec\,x)dx = sec\,x-cosec\,x+c$

(D) $\int \frac{dx}{1-cos\,x-sin\,x} = \int \frac{dx}{2\,sin^2\frac{x}{2}-2\,sin\frac{x}{2}cos\frac{x}{2}} = \frac{1}{2}\int \frac{cosec^2\frac{x}{2}}{1-cot\frac{x}{2}}dx = \ell n\left|1-cot\frac{x}{2}\right|+c$

63.(1) $f(x)=\int x^{\sin x}(1+x\cos x.\ln x+\sin x)dx$

If $F(x)=x^{\sin x}=e^{\sin x\ln x}$

$\therefore$ $f(x)=\int(F(x)+xF'(x))=xF(x)+C$

$f(x)=x.x^{\sin x}+C$

64.(4) $g(x)=\int\frac{\cos x(\cos x+2)+\sin^2 x}{(\cos x+2)^2}dx=\int\underbrace{\cos x}_{II}\underbrace{\frac{1}{(\cos x+2)}}_{I}dx+\int\frac{\sin^2 x}{\cos x+2}dx$

$=\frac{1}{\cos x+2}\sin x-\int\frac{\sin^2 x}{(\cos x+2)^2}dx+\int\frac{\sin^2 x}{(\cos x+2)^2}dx$

$g(x)=\frac{\sin x}{\cos x+2}+C$

$g(0)=0\Rightarrow C=0$

65.(2) $k(x)=\int\frac{(x^2+1)dx}{(x^3+3x+6)^{1/3}}$

Put $x^3+3x+6=t^3\Rightarrow 3(x^2+1)dx=3t^2dt$

$k(x)=\int\frac{t^2dt}{t}=\frac{t^2}{2}+C$

$k(x)=\frac{1}{2}(x^3+3x+6)^{2/3}+C$

$k(-1)=\frac{1}{2}(2)^{2/3}+C\Rightarrow C=0$

66.(3) $\frac{d}{dx}(A\ln|\cos x+\sin x-2|+Bx+C)=A\frac{\cos x-\sin x}{\cos x+\sin x-2}+B$

$\therefore 2=A+B,-1=-A+B,\lambda=-2B$ $\qquad\therefore A=3/2,B=1/2,\lambda=-1$

67.(0) $\int\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\ln x dx$

Put $\left(\frac{x}{e}\right)^x=t$ Or $x\ln\left(\frac{x}{e}\right)=\ln t$

$\therefore \left(x.\frac{1}{x/e}\frac{1}{e}+\ln\left(\frac{x}{e}\right)\right)dx=\frac{1}{t}dt$ $\therefore (1+\ln x-\ln e)dx=\frac{1}{t}dt$

$\therefore (\ln e+\ln x-\ln e)dx=\frac{1}{t}dt$

$\therefore (\ln x)dx=\frac{1}{t}dt$ Or $I=\int\left(t+\frac{1}{t}\right)\frac{1}{t}dt=\int 1.dt+\int\frac{1}{t^2}dt=t-\frac{1}{t}+C$

68.(2) $I=\int\left(x^9+x^6+x^3\right)\left(2x^6+3x^3+6\right)^{1/3}dxI=\int x\left(x^8+x^5+x^2\right)\left(2x^6+3x^3+6\right)^{1/3}dx$

Let $t=2x^9+3x^6+6x^3$

$dt=18\left(x^8+x^5+x^2\right)dx=\frac{1}{18}\int t^{1/3}dt=\frac{1}{24}t^{4/3}+c$

69.(1) $I=\int\frac{dx}{(1+\sin x)}=\int\frac{dx}{1+\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}=\int\frac{\sec^2\frac{x}{2}dx}{\tan^2\frac{x}{2}+2\tan\frac{x}{2}+1}$

Let $t=\tan\frac{x}{2}\Rightarrow 2dt=\sec^2\frac{x}{2}dx=2\int\frac{dt}{(t+1)^2}=-\frac{2}{(1+t)}+c=-\frac{2}{\left(1+\tan\frac{x}{2}\right)}+c$

$=1-\frac{2}{\left(1+\tan\frac{x}{2}\right)}+c-1=\left(\frac{\tan\frac{x}{2}-1}{\tan\frac{x}{2}+1}\right)+c-1$

70.(10) $x+7=t^4$

$\int\frac{4t^3dt}{t^2-t}=4\int\frac{t^2-1+1}{t-1}dt=4\int(t+1)+\frac{1}{t-1}dt$

$=4\left(\frac{t^2}{2}+t+\ell n|t-1|\right)+c=2\sqrt{x+7}+4(x+7)^{1/4}+4\ell n\left|\sqrt[4]{x+7}-1\right|+c$

$P=2, Q=4, R=4$

71.(6) $\frac{x^2+2}{\left(x^2+1\right)\left(x^2+4\right)}=\frac{1}{3}\frac{1}{x^2+1}+\frac{2}{3}\frac{1}{x^2+4}$

$$\int \frac{x^2+2}{\left(x^2+1\right)\left(x^2+4\right)}dx = \frac{1}{3}tan^{-1}(x)+\frac{1}{3}tan^{-1}\left(\frac{x}{2}\right)+c$$

$$=\frac{1}{3}tan^{-1}\left(\frac{3x}{2-x^2}\right) \Rightarrow K=\frac{1}{3}, m=3, c=2$$

72.(4) $$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} = \int \left(\frac{x+2}{x-1}\right)^{3/4}\frac{dx}{(x+2)^2}$$

$$\frac{x-1}{x+2}=t \quad =\frac{1}{3}\int \frac{dt}{t^{3/4}} = \frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+c$$

73.(3015) $$I = \int \frac{x^{2009}}{\left(1+x^2\right)^{1006}}dx = \frac{1}{n}\left(\frac{x^2}{1+x^2}\right)^m + c$$

Put $1+x^2=t$

$2xdx=dt$

So $I=\frac{1}{2}\int \frac{(t-1)^{1004}}{t^{1006}}dt = \frac{1}{2}\int\left(1-\frac{1}{t}\right)^{1004}.\left(\frac{1}{t^2}\right)dt$

Put $1-\frac{1}{t}=y \Rightarrow \frac{1}{t^2}dt=dy$

$$I=\frac{1}{2}\int y^{1004}dy = \frac{y^{1005}}{2\times 1005}+c$$

$$I=\frac{1}{2}\int y^{1004}dy = \frac{y^{1005}}{2\times 1005}+c$$

$\Rightarrow \quad m=1005, \qquad n=2010 \qquad$ So $\qquad m+n=3015$

74.(521) Let $g(x)=\int \frac{f(x)dx}{x^2(x+1)^2}$

$$g(x)=\int\left(\frac{A}{x}+\frac{B}{x^2}+\frac{C}{(x+1)}+\frac{D}{(x+1)^2}\right)dx; \quad g(x)=A\ln|x|-\frac{B}{x}+C\ln|x+1|-\frac{D}{x+1}+E$$

$\therefore\ g(x)$ is rational function

$\Rightarrow \quad A=C=0$

Hence $g(x)=\int\left(\frac{B}{x^2}+\frac{D}{(x+1)^2}\right)dx$

Hence $f(x)$ must be of the form of $f(x)=B(x+1)^2+Dx^2$

$f(0)=1 \Rightarrow B=1;\ f(-1)=4 \Rightarrow D=4$

So $f(x)=5x^2+2x+1$

$f(10)=521$

75.(8) $I=\int \frac{f'(x)g(x)-g'(x)f(x)dx}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}}$

$$I=\int \frac{\frac{\{f'(x)g(x)-g'(x)f(x)\}dx}{(g(x))^2}}{\left(\frac{f(x)}{g(x)}+1\right)\sqrt{\frac{f(x)}{g(x)}-1}}dx$$

Put $\frac{f(x)}{g(x)}-1=t^2$

$$\frac{(f'(x)g(x)-g'(x)f(x))}{(g(x))^2}dx=2tdt$$

$$I=\int \frac{2tdt}{(t^2+2)t}=\frac{2}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)+c$$

Hence $m=2,\quad n=2$ So $m^2+n^2=8$

76.(3) $Det(A)=\begin{vmatrix} 5^x & 0 & 0 \\ 0 & 5^{5^x} & 0 \\ 0 & 0 & 5^{5^{5^x}} \end{vmatrix}$

$$Det(A)=5^x.5^{5^x}.5^{5^{5^x}}$$

$$I=\int 5^x.5^{5^x}.5^{5^{5^x}}\,dx$$

Put $5^{5^{5^x}}=t$

$$5^{5^{5^x}}.5^{5^x}.5^x.dx=\frac{dt}{(\ln 5)^3};\quad I=\int \frac{dt}{(\ln 5)^3}\Rightarrow I=\frac{t}{(\ln 5)^3}+c;\qquad I=\frac{5^{5^{5^x}}}{(\ln 5)^3}+c$$

77.(2.5) $I=\int\left\{2\cos^2\theta\ln\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)-\ln\frac{(\cos\theta+\sin\theta)}{(\cos\theta-\sin\theta)}\right\}d\theta$

$$I=\int\left\{(2\cos^2\theta-1)\ln\frac{(\cos\theta+\sin\theta)}{(\cos\theta-\sin\theta)}\right\}d\theta$$

use $I.B.P$

$$I=\left(\ln\left|\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right|\right).\frac{\sin 2\theta}{2}-\int\frac{2}{\cos 2\theta}.\frac{\sin 2\theta}{2}d\theta$$

$$I=\frac{\sin 2\theta}{2}\ln\left|\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right|+\frac{1}{2}\ln|\cos 2\theta|+c$$

Hence $a=2, b=\frac{1}{2}$

78.(4) $f'(x)+g'(x)=0 \Rightarrow \quad f(x)+g(x)=k$

Put $x=0$

$f(0)+g(0)=k \Rightarrow k=6$

$f(x)+g(x)=6 \qquad \ldots(i)$

again

$f'(x)-g'(x)=2(f(x)-g(x))$

$\frac{f'(x)-g'(x)}{f(x)-g(x)}=2$

By integrating both sides w.r.t. x

$\ln|f(x)-g(x)|=2x+c$

Put $x=0$

$\ln|5-1|=c \Rightarrow c=\ln 4$

so $\quad \ln\left|\frac{f(x)-g(x)}{4}\right|=2x.$

$f(x)-g(x)=4e^{2x} \qquad \ldots(ii)$

From (i) & (ii)

$f(x)=3+2e^{2x} \quad \& \quad g(x)=3-2e^{2x}$

$f(\ln 2)=3+2e^{2\ln 2}=3+2e^{\ln 4}=11$

$g(\ln 3)=3-2e^{2\ln 3}=3-2e^{\ln 9}=3-18=-15$

$|f(\ln 2)+g(\ln 3)|=4$

79.(0.5) $I=\int\frac{(x^3+x+1)dx}{(x^2+x+1)(x^2-x+1)}=\int\left(\frac{x(x^2-x+1)+(x^2+x+1)-x}{(x^2+x+1)(x^2-x+1)}\right)dx$

$$I=\underbrace{\int\frac{(xdx)}{(x^2+x+1)}}_{I_1}+\underbrace{\int\frac{dx}{(x^2-x+1)}}_{I_2}-\underbrace{\int\frac{xdx}{(x^4+x^2+1)}}_{I_3}$$

For I_3, put $x^2=t \Rightarrow \; 2xdx=dt$

$$I_3=\frac{1}{2}\int\frac{dt}{t^2+t+1}=\frac{1}{2}\int\frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$I_3=\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) \qquad \ldots(i)$$

Similarly $I_2 = \int \frac{dx}{x^2 - x + 1} = \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$I_2 = \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)$ ...(ii)

& $I_1 = \int \frac{xdx}{x^2 + x + 1} = \frac{1}{2}\int \frac{(2x+1)dx}{(x^2 + x + 1)} - \frac{1}{2}\int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$I_1 = \frac{1}{2}\ln\left(x^2 + x + 1\right) - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)$

Hence $I = I_1 + I_2 - I_3$

$I = \frac{1}{2}\ln\left(x^2 + x + 1\right) - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + c$

so $A_1 = \frac{1}{2}, A_2 = \frac{-1}{\sqrt{3}}, A_3 = \frac{2}{\sqrt{3}}, A_4 = \frac{-1}{\sqrt{3}}$

$A_1 + A_2 + A_3 + A_4 = \frac{1}{2}$

$[A_1 + A_2 + A_3 + A_4] = 0$

80.(3.6) $\int \frac{\ln\left(x + \sqrt{1+x^2}\right)}{\sqrt{1+x^2}}dx = \frac{\left(\ln\left(x+\sqrt{1+x^2}\right)\right)^2}{2} + c^1$

So $f(x) = \frac{x^2}{2}$ & $g(x) = \ln\left(x + \sqrt{1+x^2}\right)$

Now $\int \frac{x^2}{2}.\ln\left(x+\sqrt{1+x^2}\right)dx = \frac{x^3}{6}.\ln\left(x+\sqrt{1+x^2}\right) - \int \frac{1}{\sqrt{1+x^2}}.\frac{x^3}{6}dx$

$\int f(x).g(x) = \frac{1}{6}x^3 g(x) - \frac{1}{6}\int \frac{x^2.x}{\sqrt{1+x^2}}dx$

Put $1 + x^2 = t^2$

$1 + x^2 = t^2$

$\int f(x).g(x) = \frac{1}{6}x^3 g(x) - \frac{1}{6}\int \frac{\left(t^2-1\right)tdt}{t}$

$= \frac{x^3}{6}g(x) - \frac{1}{6}\left(\frac{t^3}{3} - t\right) + d = \frac{x^3}{6}g(x) - \frac{1}{18}\left(1+x^2\right)^{3/2} + \frac{1}{6}\left(1+x^2\right)^{1/2} + d$

So $a = \frac{1}{6}, b = \frac{-1}{18}, c = \frac{1}{6} \Rightarrow \frac{1}{a+b+c} = \frac{18}{5}$

81.(2.05) $$\int \sqrt{x}\tan\left\{2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right\}dx$$

Put $\sqrt{x}=\tan^2\theta$

$$\frac{1}{2\sqrt{x}}dx = 2\tan\theta\sec^2\theta d\theta$$

$$dx = 4\tan^3\theta\sec^2\theta d\theta$$

$$\int \tan^2\theta\tan\left\{2\tan^{-1}\left(\frac{\sqrt{\sqrt{1+\tan^2\theta}+1}-\sqrt{\sqrt{1+\tan^2\theta}-1}}{\sqrt{\sqrt{1+\tan^2\theta}+1}+\sqrt{\sqrt{1+\tan^2\theta}-1}}\right)\right\}4\tan^3\sec^2\theta d\theta$$

$$I = 4\int \tan^5\theta\sec^2\theta.\tan\theta(2\tan^{-1}\left(\frac{\sqrt{1+\cos\theta}-\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}+\sqrt{1-\cos\theta}}\right)d\theta$$

$$I = 4\int \tan^5\theta\sec^2\theta.\tan\theta(2\tan^{-1}\left(\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right)d\theta$$

$$I = 4\int \tan^5\theta\sec^2\theta.\tan\left(\frac{\pi}{2}-\theta\right)d\theta$$

$$I = 4\int \tan^4\theta\sec^2\theta d\theta$$

$$I = \frac{4}{5}\tan^5\theta + c$$

$$I = \frac{4}{5}x^{5/4}+c$$

82.(1) $$I = \int \frac{\left(e^x+\cos x+1\right)-\left(e^x+\sin x+x\right)}{\left(e^x+\sin x+x\right)}dx$$

$$I = \ln\left|e^x+\sin x+x\right| - x + c \ ; \quad f(x) = e^x+\sin x + x$$

$$g(x) = -x \qquad \text{so} \quad f(x)+g(x) = e^x+\sin x \Rightarrow \frac{f(x)+g(x)}{e^x+\sin x} = 1$$

83.(403) $$I = \int x.\left(x^{2009}+x^{803}+x^{401}\right)\left(2x^{1608}+5x^{402}+10\right)^{1/402}dx$$

$$I = \int \left(x^{2009}+x^{508}+x^{401}\right)\left(2x^{2010}+5x^{804}+10x^{10}\right)^{1/402}dx$$

Put $2x^{2010}+5x^{804}+10x^{402}=t$

$$4020\left(x^{2009}+x^{803}+x^{401}\right)dx = dt$$

$$I = \int \frac{1}{4020}(t)^{1/402}dt \ ; \quad I = \frac{1}{4020}.\frac{t^{403/402}}{\left(403/402\right)}+c \ ; \qquad I = \frac{\left(2x^{2010}+5x^{804}+10x^{402}\right)^{403/402}}{4030}+c$$

$$a = 403 \Rightarrow a - 400 = 3$$

84.(1) $u(x)=7v(x)\Rightarrow u'(x)=7v'(x) \quad \Rightarrow \quad p=7$

$$\frac{u(x)}{v(x)}=7\Rightarrow\left(\frac{u(x)}{v(x)}\right)'=0 \quad \Rightarrow \quad q=0 \qquad \text{Now } \frac{p+q}{p-q}=\frac{7+0}{7-0}=1$$

85.(0) $I=\int\frac{dx}{\sqrt{2-x-x^2}}+2\int\frac{dx}{x^2\sqrt{2-x-x^2}}-\int\frac{dx}{x\sqrt{2-x-x^2}}$

$$I=-\int\frac{dx}{\sqrt{\frac{9}{4}-\left(x+\frac{1}{2}\right)}}+2\int\frac{-dt}{\sqrt{2-\frac{1}{t}-\frac{1}{t^2}}}+\int\frac{\frac{1}{t}\,dt}{2-\frac{1}{t}-\frac{1}{t^2}}$$

Put $x=\frac{1}{t} \quad \Rightarrow \quad dx=\frac{-1}{t^2}dt$

$$I=-\sin^{-1}\left(\frac{2x+1}{3}\right)-\frac{1}{2}\int\frac{(4t-1)\,dt}{\sqrt{2t^2-t-1}}+\frac{1}{2}\int\frac{dt}{\sqrt{2t^2-t-1}}$$

$$=-\sin^{-1}\left(\frac{2x+1}{3}\right)-\frac{1}{2}\frac{\sqrt{2t^2-t-1}}{\left(\frac{1}{2}\right)}+\frac{1}{2\sqrt{2}}\int\frac{dt}{\sqrt{t^2-\frac{t}{2}-\frac{1}{2}+\frac{1}{16}-\frac{1}{16}}}$$

$$I=-\sin^{-1}\left(\frac{2x+1}{3}\right)-\frac{\sqrt{2-x-x^2}}{x}+\frac{1}{2\sqrt{2}}\ln\left|\frac{(4-x)+4\sqrt{2-x-x^2}}{4x}\right|+k$$

$$I=\frac{-\sqrt{2-x-x^2}}{x}+\frac{1}{2\sqrt{2}}\ln\left|\frac{(4-x)+\sqrt{2-x-x^2}}{x}\right|-\sin^{-1}\left(\frac{2x+1}{3}\right)+c \quad \Rightarrow \quad A=-1\ \&\ B=1 \quad \Rightarrow A+B=0$$

86.(12) $\int\left(\frac{1-\left(4x^2\right)^3}{x^6.\left(4x^2+2x+1\right)}.\frac{x^6}{\left(4x^2-4x+1\right)}\right)-\frac{4x^2(2x+1)}{(1-2x)}dx$

$$I=\int\left\{\frac{(1-2x)(1+2x)\left(4x^2-2x+1\right)\left(4x^2+2x+1\right)}{\left(4x^2+2x+1\right)(2x-1)^2}-\frac{4x^2(2x+1)}{(1-2x)}\right\}dx$$

$$I=\int\left\{\frac{(1+2x)\left(4x^2-2x+1\right)}{(2x-1)}-\frac{4x^2(2x+1)}{(1-2x)}\right\}dx\,; \quad I=\int(2x+1)\,dx=x^2+x+c$$

87.(1.59) $f(x)=y=x+\sin x$

$$\frac{dy}{dx}=1+\cos x\,; \quad g'(y)=\frac{dx}{dy}=\frac{1}{1+\cos x}$$

$$y=\frac{\pi}{4}+\frac{1}{\sqrt{2}}=x+\sin x\Rightarrow x=\frac{\pi}{4}$$

So $g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)=\frac{1}{1+\cos\frac{\pi}{4}}=\frac{1}{1+\frac{1}{\sqrt{2}}}\,; \quad g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)=\frac{\sqrt{2}\left(\sqrt{2}-1\right)}{\left(\sqrt{2}+1\right)\left(\sqrt{2}-1\right)}=2-\sqrt{2}$

88.(2019) $I=\int \sin(2020x).\sin^{2018}x\ dx.$

$I=\int \{\sin x(2019x+x)\}.\sin^{2018}x\ dx.$

$I=\int \sin^{2018}x\left(\left(\sin(2019x)\right)\cos x+(\cos 2019x).\sin x\right)dx$

$$I=\underset{I}{\int (\sin 2019x)\left(\sin^{2018}x\right)\cos x.dx}+\underset{II}{\int (\cos 2019x)\left(\sin^{2019}x\right)dx}$$

$$I=\frac{(\sin 2019x)\left(\sin^{2019}x\right)}{(2019)}-\int\frac{(2019)\left(\cos(2019x)\right)\left(\sin^{2019}x\right)}{2019}dx+\int(\cos 2019x)\left(\sin^{2019}x\right)dx$$

$$I=\frac{(\sin 2019x)\left(\sin^{2019}x\right)}{(2019)}+c$$

89.(1.8) Let $\sqrt{7x-10-x^2}=(x-2)t$

$(5-x)(x-2)=(x-2)^2t^2$

$(5-x)=(x-2)t^2$

$$x-2=\frac{3}{t^2+1}$$

$$1+\sqrt{7x-10-x^2}=\frac{t^2+3t+1}{t^2+1}$$

$$x=\frac{2t^2+5}{t^2+1}\Rightarrow\ dx=\frac{-6t}{\left(t^2+1\right)}dt$$

$$I=\int\frac{\left(t^2+1\right)}{3}.\frac{\left(t^2+1\right)}{\left(t^2+3t+1\right)}.\left(\frac{-6t}{\left(t^2+1\right)^2}\right)dt$$

$$I=-\int\frac{(2t+3-3)}{\left(t^2+3t+1\right)}dt=-\int\frac{(2t+3)dt}{\left(t^2+3t+1\right)}+3\int\frac{dt}{\left(t+\frac{3}{2}\right)^2-\frac{5}{4}}$$

$$I=-\ln\left|t^2+3t+1\right|+\frac{3}{\sqrt{5}}\int\left[\frac{1}{\left(t+\frac{3}{2}\right)-\frac{\sqrt{5}}{2}}-\frac{1}{\left(t+\frac{3}{2}\right)+\frac{\sqrt{5}}{2}}\right]dt$$

$$I=-\ln\left|t^2+3t+1\right|+\frac{3}{\sqrt{5}}\ln\left|\frac{2t+3-\sqrt{5}}{2t+3+\sqrt{5}}\right|+c$$

90.(2.14) since $\int e^{f(x)}.\left(x\ f'(x)+\dfrac{f''(x)}{(f'(x))^2}\right)dx=e^{f(x)}.\left(x-\dfrac{1}{f'(x)}\right)+c$

Hence $I=\int e^{x\sin x+\cos x}.\left(\dfrac{x^4\cos^3 x-x\sin x+\cos x}{x^2\cos^2 x}\right)dx$

$I=\int e^{x\sin x+\cos x}\left(x.(x\cos x)+\dfrac{(\cos x-x\sin x)}{(x\cos x)^2}\right)dx$

Let $f(x)=x\sin x+\cos x$

$f'(x)=x\cos x$

$f''(x)=\cos x-x\sin x$

So $I=e^{x\sin x+\cos x}\left(x-\dfrac{1}{x\cos x}\right)+c$

91.(2) $(f'(x)f'(x))-f(x).f''(x)=0$

$$\frac{(f'(x))^2-f(x)f''(x)}{(f'(x))^2}=0$$

$$\frac{d}{dx}\left(\frac{f(x)}{f'(x)}\right)=0\Rightarrow\frac{f(x)}{f'(x)}=c$$

at $x=0,\quad \dfrac{f(0)}{f'(0)}=c\Rightarrow c=\dfrac{1}{2}$

$\Rightarrow\quad \dfrac{f'(x)}{f(x)}=2\Rightarrow \ln|f(x)|=2x+k.$

$\therefore\quad f(0)=1\Rightarrow k=0$

$f(x)=e^{2x}\quad ;\quad \lambda=2,\ \mu=0$

92. (11) Let $I=\int\sqrt{\dfrac{1+x^{2n}}{x^{2n}}}\dfrac{\{\ln(1+x^{2n})\}-2n\ln x}{x^{2n+1}}dx$

$$I=\int\sqrt{\frac{1+x^{2n}}{x^{2n}}}\left\{\frac{\ln\left(\frac{1+x^{2n}}{x^{2n}}\right)}{x^{2n+1}}\right\}dx=\frac{-1}{2n}\int p\ln p^2(2p)dp$$

For $\left(\dfrac{1}{x^{2n}}+1\right)=p^2,\dfrac{-2n}{x^{(2n+1)}}dx=2p\ dp$

$$I=\frac{4}{-2n}\int p^2\ln p dp=-\frac{2}{n}\left[\frac{p^3\ln p}{3}-\int\frac{p^3}{3}\left(\frac{1}{p}\right)dp\right]=-\frac{2}{n}\left[\frac{p^3\log_e p}{3}-\frac{p^3}{9}\right]=\frac{2p^3}{9n}[1-3\ln p]+c.$$

INTEGRAL CALCULUS-2

1.(A) Let q = p + d, r = p + 2d, s = p + 3d

$$\therefore \qquad f(x)=\begin{vmatrix} p+\sin x & p+d+\sin x & -2d+\sin x \\ p+d+\sin x & p+2d+\sin x & -1+\sin x \\ p+2d+\sin x & p+3d+\sin x & 2d+\sin x \end{vmatrix}$$

Applying $R_1 \to R_1 + R_3 - 2R_2$, we get :

$$f(x)=\begin{vmatrix} 0 & 0 & 2 \\ p+d+\sin x & p+2d+\sin x & -1+\sin x \\ p+2d+\sin x & p+3d+\sin x & 2d+\sin x \end{vmatrix}$$

Given, $\int_0^2 f(x)\,dx=-4 \Rightarrow \int_0^2 \left(-2d^2\right)dx=-4 \Rightarrow d^2=1 \Rightarrow = \pm 1$

2.(C) $A=\int_1^{\sin\theta}\frac{t\,dt}{1+t^2}$ Let $t=\frac{1}{x}, dt=-\frac{1}{x^2}dx$ $=\int_1^{\cos ec\,\theta}\frac{1}{x}\cdot\frac{-1}{x^2}\cdot\frac{x^2+1}{x^2}dx=-\int_1^{\operatorname{cosec}\theta}\frac{dx}{x\left(1+x^2\right)}=-B$

3.(C) If $-8<x<8$, then y = 2

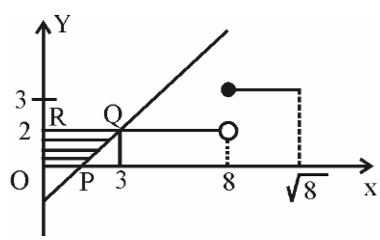
If $x\in\left(-8\sqrt{2},-8\right]\cup[8,\sqrt{2}),$, then y = 3, etc

Intersection of y = x – 1 and y = 2

We get $x=3\in(-8,8)$

Intersection of y = x – 1 and y = 3

We get $x=4\notin(-8\sqrt{2},-8]\cup[8,8\sqrt{2})$



4.(B) Let $I(a)=\int_0^1\frac{x^a-1}{\log x}dx$

Differentiating w.r.t. *a* keeping *x* as constant

$$\therefore \qquad \frac{dI(a)}{da}=\int_0^1\frac{d}{da}\left(\frac{x^a-1}{\log x}\right)dx$$

$$=\int_0^1\frac{x^a\log x}{\log x}dx=\int_0^1 x^a dx=\left.\frac{x^{a+1}}{a+1}\right|_0^1$$

$$=\frac{1}{(a+1)}$$

Integrating both sides w.r.t. *a*, we get

$I(a)=\log(a+1)+c$

For $\quad a=0, I(0)=\log 1+c$

$0=0+c$

$\therefore \quad I=\log(a+1)$

5.(C) Let $F(k)=\int\limits_0^{\pi/2}\ln(\sin^2\theta+k^2\cos^2\theta)d\theta$

$$F'(k)=\int\limits_0^{\pi/2}\frac{1}{\sin^2\theta+k^2\cos^2\theta}2k\cos^2\theta d\theta$$

$$= 2k\int_0^{\pi/2}\frac{\cos^2\theta}{\sin^2\theta + k^2\cos^2\theta}d\theta = 2k\int_0^{\pi/2}\frac{d\theta}{\tan^2\theta + k^2}$$

$$= 2k\int_0^{\pi/2}\frac{\sec^2\theta - \tan^2\theta}{\tan^2\theta + k^2}d\theta = 2k\int_0^{\infty}\frac{dt}{t^2+k^2} - 2k\int_0^{\pi/2}d\theta + 2k^3\int_0^{\pi/2}\frac{d\theta}{\tan^2\theta + k^2}\ \text{(Putting } t = \tan\theta)$$

$$= 2k\frac{1}{k}\tan^{-1}\frac{1}{k}\Big|_0^{\infty} - 2k\frac{\pi}{2} + k^2F'(k)$$

$$\Rightarrow (1-k^2)F'(k) = \pi - k\pi = \pi(1-k) \Rightarrow F'(k) = \frac{\pi}{1+k} \quad \Rightarrow F(k) = \pi\log(1+k) + c$$

For $k = 1,\ F(1) = 0 \Rightarrow c = -\pi\log 2 \Rightarrow F(k) = \pi\log(1+k) - \pi\log 2$

6.(A) Let $I(a) = \int_0^{\pi/2}\log\left(\frac{1+a\sin x}{1-a\sin x}\right)\frac{dx}{\sin x}$

$$\frac{dI}{da} = \int_0^{\pi/2}\frac{2\sin x}{1-a^2\sin^2 x}\frac{dx}{\sin x} = \int_0^{\pi/2}\frac{2\sec^2 x dx}{1+\tan^2 x - a^2\tan^2 x} = \int_0^{\pi/2}\frac{2\sec^2 x dx}{1+(1-a^2)\tan^2 x} = \int_0^{\infty}\frac{2dt}{1+(1-a^2)t^2}\ \text{(Put } \tan x = t)$$

$$= \frac{2}{\sqrt{1-a^2}}\left[\tan^{-1}\left(t\sqrt{1-a^2}\right)\right]_0^{\infty} = \frac{\pi}{\sqrt{1-a^2}} \qquad \Rightarrow I = \pi\sin^{-1}a[\text{as } I(0) = 0]$$

7.(C) $\int_0^{\pi}\frac{dx}{(a-\cos x)} = \frac{\pi}{\sqrt{a^2-1}}$

Differentiating both sides with respect to a, we get : $-\int_0^{\pi}\frac{dx}{(a-\cos x)^2} = \frac{\pi}{(a^2-1)^{3/2}}$

Again differentiating with respect to a, we get : $2\int_0^{\pi}\frac{dx}{(a-\cos x)^3} = \frac{\pi(1+2a^2)}{(a^2-1)^{5/2}}$

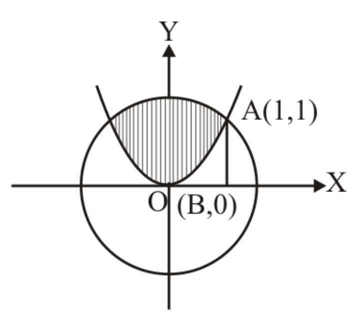
Put $a = \sqrt{10}$, we get $\int_0^{\pi}\frac{dx}{(\sqrt{10}-\cos x)^3} = \frac{7\pi}{81}$

8.(D) Since, $\lim_{x\to 0}\left[1+\frac{f(x)}{x^3}\right]^{1/x}$ exists, so, $\lim_{x\to 0}\frac{f(x)}{x^3} = 0$

$\therefore$ $f(x) = a_4x^4 + a_5x^5 + ... + a_nx^n, a_n \neq 0, n \geq 4$

Since, f(x) is of least degree $\Rightarrow f(x) = a_4x^4$

Again $\lim_{x\to 0}\left[1+\frac{f(x)}{x^3}\right]^{\frac{1}{x}} = e \quad \Rightarrow \quad a_4 = 1 \quad \Rightarrow \quad f(x) = x^4$



The graph of $y = x^4$ and $x^2 + y^2 = 2$ are shown in figure

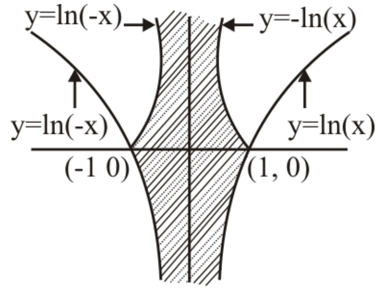
$\therefore$ The required area $= 2\int_0^1\left(\sqrt{2-x^2} - x^4\right)dx = \frac{\pi}{2} - \frac{3}{5}$

9.(C) $y = \ln x$, $y = \ln|x|$, $y = |\ln x|$, $y = |\ln|x||$

Required area $= 4\left|\int_0^1 \ln x dx\right| = 4.\lim\limits_{\varepsilon\to 0}\left|\int_\varepsilon^1 \ln x dx\right|$

$= 4\left|\lim\limits_{\varepsilon\to 0}\left[x(\ln x - 1)\right]_\varepsilon^1\right|$

$= 4$ sq. units



10.(D) Eliminating y from two equations,

We get $2mx + 4 = x^2 \Rightarrow x^2 - 2mx - 4 = 0$

x_1 and x_2 are roots of this quadratic equation

$\therefore \quad x_1 + x_2 = 2m$ and $x_1x_2 = -4$

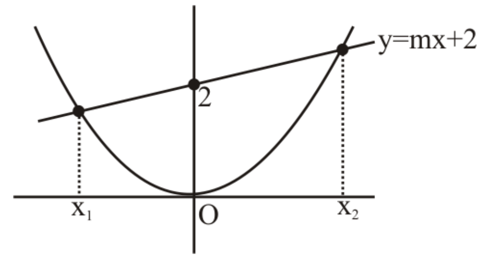
Now,

$$\int_{x_1}^{x_2}\left(mx + 2 - \frac{x^2}{2}\right)dx = \frac{m}{2}\left(x_2^2 - x_1^2\right) + 2\left(x_2 - x_1\right) - \frac{1}{6}\left(x_2^3 - x_1^3\right)$$

$$= \left(x_2 - x_1\right)\left[\frac{m}{2}\left(x_2 + x_1\right) + 2 - \frac{1}{6}\left(x_1^2 + x_1x_2 + x_2^2\right)\right]$$

$$= \sqrt{\left(x_2 - x_1\right)^2 - 4x_2x_1}\left[\frac{m}{2}\left(x_2 + x_1\right) + 2 - \frac{1}{6}\left\{\left(x_2 + x_1\right)^2 - x_2x_1\right\}\right] = \sqrt{4m^2 + 16}\left[\frac{m^2 + 4}{3}\right]$$

Clearly above is minimum if m = 0.



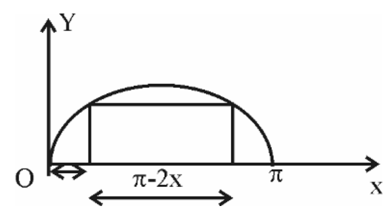
11.(B) A = Area $= \sin x(\pi - 2x)$

$$\frac{dA}{dx} = (\pi - 2x)\cos x - 2\sin x = 0 \Rightarrow \tan x = \frac{\pi}{2} - x$$

Let $\quad f(x) = \tan x + x - \frac{\pi}{2}$

$f\left(\frac{\pi}{6}\right)$ is negative; $f\left(\frac{\pi}{4}\right)$ is positive

So one root lies between $\left(\frac{\pi}{6}, \frac{\pi}{4}\right)$



12.(D) Area of OABL $= \int\limits_0^2 ydx = \int\limits_0^2\left(a + bx + cx^2\right)dx = \left(2a + 2b + \frac{8}{3}c\right) = \frac{1}{3}[6a + 6b + 8c]$

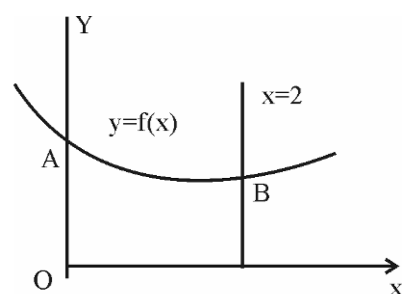
But $\quad f(x) = a + bx + cx^2$

$f(0) = a, f(1) = a + b + c$

$f(2) = a + 2b + 4c$

$\frac{1}{3}\{f(0) + 4f(1) + f(2)\} = \frac{1}{3}\{a + 4(a + b + c) + a(a + 2b + 4c)\}$

$= \frac{1}{3}\{6a + 6b + 8c\}$



13-15. **13.(C)** **14.(D)** **15.(C)**

$$f(x)-\lambda\int_0^{\pi/2}\sin x\cos t\, f(t)dt=\sin x \Rightarrow f(x)-\lambda\sin x\int_0^{\pi/2}\cos t\, f(t)dt=\sin x$$

$\Rightarrow$ $f(x)-A\sin x=\sin x$ or $f(x)=(A+1)\sin x$, where $A=\lambda\int_0^{\pi/2}\cos t\, f(t)dt$

$\Rightarrow$ $A=\lambda\int_0^{\pi/2}(\cos t)(A+1)\sin t\,dt$

$$=\frac{\lambda(A+1)}{2}\int_0^{\pi/2}\sin 2t\,dt=\frac{\lambda(A+1)}{2}\left[\frac{-\cos 2t}{2}\right]_0^{\pi/2}=\frac{\lambda(A+1)}{2}$$

$\Rightarrow$ $A=\dfrac{\lambda}{2-\lambda}\Rightarrow f(x)=\left(\dfrac{\lambda}{2-\lambda}+1\right)\sin x\Rightarrow f(x)=\left(\dfrac{2}{2-\lambda}\right)\sin x$

$$\left(\frac{2}{2-\lambda}\right)\sin x=2$$

$\Rightarrow$ $\sin x=(2-\lambda)\Rightarrow |2-\lambda|\le 1\Rightarrow -1\le\lambda-2\le 1\Rightarrow 1\le\lambda\le 3$

$$\int_0^{\pi/2} f(x)dx=3\Rightarrow\int_0^{\pi/2}\frac{2}{2-\lambda}\sin x\,dx=3\Rightarrow -\left[\frac{2}{2-\lambda}\cos x\right]_0^{\pi/2}=3 \quad\Rightarrow\quad \frac{2}{2-\lambda}=3\Rightarrow\lambda=4/3$$

16.(D) The L.H.S. of given inequality is equal to

$$\left[a^2\left(\frac{\sin 3x}{12}+\frac{3}{4}\sin x\right)-a\cos x-20\sin x\right]_0^{\pi/2}=a^2\left(-\frac{1}{12}+\frac{3}{4}\right)-a(0-1)-20=\frac{2a^2}{3}+a-20$$

Thus the given inequality is $\dfrac{2a^2}{3}+a-20\le-\dfrac{a^2}{3}$

$$a^2+a-20\le 0\Leftrightarrow -5\le a\le 4$$

Since a is a positive integer so a = 1,2,3,4

17.(D) For $0<x<1$, we have $\frac{1}{2}x^2<x^2<x, i.e., -x^2>-x$, so that $e^{-x^2}>e^{-x}$

Hence $\int_0^1 e^{-x^2}\cos^2 x\,dx>\int_0^1 e^{-x}\cos^2 x\,dx$. Also $\cos^2 x\le 1$, therefore $\int_0^1 e^{-x^2}\cos^2 x\,dx\le\int_0^1 e^{-x^2}dx<\int_0^1 e^{-x^2/2}=I_4$

18.(A) We have $2\sin\frac{x}{2}\left(\frac{1}{2}+\cos x+\cos 2x+........+\cos nx\right)$

$$=\sin\frac{x}{2}+2\sin\frac{x}{2}\cos x+2\sin\frac{x}{2}\cos 2x+.......+2\sin\frac{x}{2}\cos nx$$

$$=\sin\frac{x}{2}+\sin\frac{3x}{2}-\sin\frac{x}{2}+\sin\frac{5x}{2}-\sin\frac{3x}{2}+.......+\sin\left(n+\frac{1}{2}\right)x-\sin\left(n-\frac{1}{2}\right)x=\sin\left(n+\frac{1}{2}\right)x$$

$$\frac{1}{2}+\cos x+\cos 2x+...+\cos nx=\frac{\sin\left(n+\frac{1}{2}\right)x}{2\sin\frac{x}{2}}$$

$\Rightarrow$ $$\int_0^{\pi}\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin x/2}dx=2\left(\int_0^{\pi}\frac{1}{2}dx+\int_0^{\pi}\cos x\,dx+...+\int_0^{\pi}\cos nx\,dx\right)=2\left(\frac{\pi}{2}+\sin x\Big|_0^{\pi}+...+\frac{\sin nx}{n}\Big|_0^{\pi}\right)=\pi$$

19.(D) $\int_0^1 \lim_{n\to\infty}\sum_{k=0}^{n}\frac{x^{k+2}2^k}{k!}dx=\int_0^1 x^2\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(2x)^k}{k!}dx=\int_0^1 x^2e^{2x}dx$

20.(A) Since the required function is a polynomial, the absciassae of the points of inflection can only be among the roots of the second derivative. Consequently $p''(x)=ax(x-1)(x+1)=a(x^3-x)$

Since at the point x = 0, $p'(0)=\tan 60^\circ=\sqrt{3}$

$$p'(x)=\int_0^x p''(x)dx+\sqrt{3}=a\left(\frac{x^4}{4}-\frac{x^2}{2}\right)+\sqrt{3}$$

Then, since P(1) = 1, we get : $p(x)=\int_1^x p'(x)dx+1=a\left(\frac{x^5}{20}-\frac{x^3}{6}+\frac{7}{60}\right)+\sqrt{3(x-1)}+1$

Since $\quad p(-1)=-1, so\, a=\frac{60\left(\sqrt{3}-1\right)}{7}$

21.(B) We have $\int_0^x \sin^2\frac{t}{2}dt=\frac{1}{2}\int_0^x(1-\cos t)dt=a^2x^2-\frac{3x}{2}+\frac{1}{2}+\frac{1}{a^2}$

$\Rightarrow\quad \frac{x}{2}-\frac{(\sin x)}{2}=\left(ax-\frac{1}{a}\right)^2+\frac{x}{2}+\frac{1}{2}\quad\Rightarrow\quad 2\left(ax-\frac{1}{a}\right)^2+1=-\sin x$

Since $\left(ax-\frac{1}{a}\right)^2\geq 0$ for all x, so the only possibility is $ax=\frac{1}{a}$ and sin x = -1

i.e., $\quad x=2n\pi-\frac{\pi}{2}$. Thus $a=\frac{1}{\sqrt{x}}=\pm\frac{1}{\sqrt{2n\pi-\frac{\pi}{2}}}n\in N$

22.(D) $\int_0^a f(x)g(x)h(x)dx=\int_0^a f(a-x)g(a-x)h(a-x)dx$

$$=-\int_0^a f(x)g(x)\left(\frac{3h(x)-5}{4}\right)dx=-\frac{3}{4}\int_0^a f(x)g(x)h(x)dx+\frac{5}{4}\int_0^a f(x)g(x)dx$$

Therefore, $\quad \frac{7}{4}\int_0^a f(x)g(x)h(x)dx=\frac{5}{4}\int_0^a f(x)g(x)dx=0$

$\therefore\quad \int_0^a f(x)g(x)dx=\int_0^a f(a-x)g(a-x)dx=\int_0^a f(x)\{-g(x)\}dx$

$\Rightarrow\quad \int_0^a f(x)g(x)dx=0 \qquad$ So, $\int_0^a f(x)g(x)h(x)=0$

23-26.

23.(B) $f(x)$ is an odd function $\Rightarrow f(x)=-f(-x)$

$$\phi(-x)=\int_a^{-x}f(t)dt,\ \text{put } t=-y$$

$$\Rightarrow\quad \phi(-x)=\int_{-a}^{x}f(-t)(-dt)=\int_{-a}^{x}f(t)dt=\int_{-a}^{x}f(t)dt+\int_a^x f(t)dt=0+\int_a^x f(t)dt=\phi(x).$$

24.(D) If $f(x)$ is an even function, then $\phi(-x) = -\int_{-a}^{x} f(t)dt = -\int_{-a}^{a} f(t)dt - \int_{a}^{x} f(t)dt$

$$= -2\int_{0}^{a} f(t)dt - \int_{a}^{x} f(t)dt \qquad \text{(As } f(x) \text{ is an even function)}$$

Now, $\int_{0}^{a} f(t)dt = \int_{0}^{a} f(a-t)dt = \int_{0}^{a} -f(t)dt$ [Using $f(a-x) = -f(x)$]

$\Rightarrow \int_{0}^{a} f(t)dt = 0 \Rightarrow \phi(-x) = -\int_{a}^{x} f(t)dt = -f(x) \Rightarrow \phi(x)$ is an odd function.

25.(D) $g(x+\alpha) + g(x) = 0$

$\Rightarrow g(x+2\alpha) + g(x+\alpha) = 0 \Rightarrow g(x+2\alpha) = g(x) \Rightarrow g(x)$ is periodic with period 2α

$\Rightarrow \int_{b}^{2k} g(t)dt = \int_{a}^{b+c} g(x)dx$ ($\because$ b, k, c are in A.P.)

This is independent of b, then c has least value 2α.

26.(D) $\int_{p+m\alpha}^{q+m\alpha} g(t)dt = \int_{p+m\alpha}^{p} g(x)dx + \int_{p}^{q} g(x)dx + \int_{q}^{q+m\alpha} g(x)dx$

$$= -m\int_{0}^{\alpha} g(x)dx + \int_{p}^{q} g(x)dx + n\int_{0}^{\alpha} g(x)dx = \int_{p}^{q} g(x)dx + (n-m)\int_{0}^{\alpha} g(x)dx$$

27.(C) $G(0,t) = 0$ for $t \ge 0$ so $g(0) = \int_0^1 f(t).\ 0\ dt = 0$

$G(1,t) = t.(1-1) = 0$ for $t < 1$. Hence $g(1) = \int_0^1 f(t).\ 0\ dt = 0,$

Also $g(x) = \int_0^x f(t)t(x-1)dt + \int_x^1 f(t)x(t-1)dt = (x-1)\int_0^x tf(t)dt + x\int_x^1 f(t)(t-1)dt$

Hence, $g'(x) = (x-1)xf(x) + \int_0^x tf(t)dt + \int_x^1 f(t)(t-1)dt - xf(x)(x-1)$

$= \int$ if $(t)dt - \int_0^x f(t)dt +$. Thus $g''(x) = f(x)$

28.(B) Let $I = \int_{4\pi-2}^{4\pi} \frac{\sin(t/2)}{4\pi+2-t}dt$

$$= (4\pi-(4\pi-2))\int_0^1 \frac{\sin\left(\frac{(4\pi-(4\pi-2))t+4\pi-2}{2}\right)}{4\pi+2-((4\pi-(4\pi-2))t+4\pi-2)}dt \quad \left[\therefore \int_a^b f(x)dx = (b-a)\int_0^1 f((b-a)x+a)dx\right]$$

$$= 2\int_0^1 \frac{\sin(t-1)dt}{(4-2t)}$$

29.(D) $\int_{-\pi}^{\pi} \left|x\sin\left[x^2-\pi\right]\right|dx = 2\int_{0}^{\pi}\left|x\sin\left[x^2-\pi\right]\right|dx = 2\int_{0}^{\pi} x\left|\sin\left[x^2-\pi\right]\right|dx$

Since $-\pi \le x^2 - \pi \le \pi^2 \le \pi^2 - \pi \le 7$ so the last integral is equal to

$$2\left[\int_0^{\sqrt{\pi-3}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi-3}}^{\sqrt{\pi-2}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi-2}}^{\sqrt{\pi-1}} x\left|\sin\left[x^2-\pi\right]\right|dx\right.$$

$$+\int_{\sqrt{\pi-1}}^{\sqrt{\pi}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi}}^{\sqrt{1+\pi}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi+1}}^{\sqrt{\pi+2}} x\left|\sin\left[x^2-\pi\right]\right|dx$$

$$+\int_{\sqrt{2+\pi}}^{\sqrt{\pi+3}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi+3}}^{\sqrt{x+4}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{4+\pi}}^{\sqrt{\pi+5}} x\left|\sin\left[x^2-\pi\right]\right|dx$$

$$\left.+\int_{\sqrt{\pi+5}}^{\sqrt{\pi+6}} x\left|\sin\left[x^2-\pi\right]\right|dx+\int_{\sqrt{\pi+6}}^{\pi} x\left|\sin\left[x^2-\pi\right]\right|dx\right]$$

30.(B) Since $4-x^2\ge 4-x^2-x^3\ge 4-2x^2\ \forall x\in[0,1]$

$\Rightarrow\quad \sqrt{4-x^2}\ge\sqrt{\left(4-x^2-x^3\right)}\ge\sqrt{4-2x^2}\ \forall x\in[0,1]\ \Rightarrow\quad \dfrac{1}{\sqrt{4-x^2}}\le\dfrac{1}{\sqrt{\left(4-x^2-x^3\right)}}\le\dfrac{1}{\sqrt{4-2x^2}}\ \forall x\in[0,1]$

$\therefore\quad \displaystyle\int_0^1\frac{dx}{\sqrt{2^2-x^2}}\le\int_0^1\frac{dx}{\sqrt{4-x^2-x^3}}\le\frac{1}{\sqrt2}\int_0^1\frac{dx}{\sqrt{\left(\left(\sqrt2\right)^2-x^2\right)}}$

31.(A) Let $f(x)=\dfrac{\sin x}{x}\quad\therefore f'(x)=\dfrac{x\cos x-\sin x}{x^2}=\dfrac{(x-\tan x)\cos x}{x^2}\ \forall x\in\left[\dfrac{\pi}{4},\dfrac{\pi}{3}\right]$

$f(x)=\dfrac{\sin x}{x}$ is decreased on the interval $\left[\dfrac{\pi}{4},\dfrac{\pi}{3}\right]$

$\Rightarrow\quad$ the least value of the function $m=f\left(\dfrac{\pi}{3}\right)=\dfrac{\sin(\pi/3)}{(\pi/3)}=\dfrac{3\sqrt3}{2\pi}$

And the greatest value of the function M = $f\left(\dfrac{\pi}{4}\right)=\dfrac{\sin(\pi/4)}{(\pi/4)}=\dfrac{2\sqrt2}{\pi}$

32.(A) Let $P=\lim\limits_{n\to\infty}\left\{\dfrac{\sqrt n}{\left(3+4\sqrt n\right)^2}+\dfrac{\sqrt n}{\sqrt2\left(3\sqrt2+4\sqrt n\right)^2}+\dfrac{\sqrt n}{\sqrt3\left(3\sqrt3+4\sqrt n\right)^2}+\ldots+\dfrac{1}{49n}\right\}$

$$P=\lim_{n\to\infty}\left\{\frac{\sqrt n}{\sqrt1\left(3\sqrt1+4\sqrt n\right)^2}+\frac{\sqrt n}{\sqrt2\left(3\sqrt2+4\sqrt n\right)^2}+\frac{\sqrt n}{\sqrt3\left(3\sqrt3+4\sqrt n\right)^2}+\ldots+\frac{\sqrt n}{\sqrt n\left(3\sqrt n+4\sqrt n\right)^2}\right\}$$

$$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\sqrt n}{\sqrt r\left(3\sqrt r+4\sqrt n\right)^2}=\lim_{n\to\infty}\sum_{r=1}^{n}\frac1n\frac{1}{\sqrt{\frac rn}\left\{3\sqrt{\frac rn}+4\right\}^2}=\int_0^1\frac{dx}{\sqrt x\left(3\sqrt x+4\right)^2}$$

Put $\quad 3\sqrt x+4=t;\qquad \dfrac{3}{2\sqrt x}dx=dt$

When x = 0, $\Rightarrow t=4$ and when $x=1\Rightarrow t=7\qquad\therefore\qquad P=\dfrac23\displaystyle\int_4^7\frac{dt}{t^2}=\frac23\left(-\frac1t\right)\Bigg|_4^7$

33.(AB) $f(x)=e^x+\int_0^1 e^x f(t)dt = e^x + k\,e^x$ where $k=\int_0^1 f(t)dt$

$\therefore \quad k=\int_0^1 \left(e^t+ke^t\right)dt = e+ke-1-k \qquad \therefore\ k=\dfrac{e-1}{2-e}$, thus $f(x)=e^x\left(1+\dfrac{e-1}{2-e}\right)=\dfrac{e^x}{2-e}$

Obviously, $f(0)=\dfrac{1}{2-e}<0$

Also, $f'(x)=\dfrac{e^x}{2-e}<0$ for $\forall x\in R.$

Also, $\int_0^1 f(x)dx = \int_0^1 \dfrac{e^x}{2-e}dx = \left[\dfrac{e^x}{2-e}\right]_0^1 = \dfrac{e-1}{2-e}<0$

34.(A) $f'(x)=\dfrac{3^x}{1+x^2}>0\,\forall x>0 \Rightarrow f'(x)=\dfrac{3^x}{1+x^2}>\dfrac{1}{1+x^2}, \forall x\ge 1$

$\Rightarrow \quad \int_1^x f'(x)dx > \int_1^x \dfrac{1}{1+x^2}dx \quad \Rightarrow \quad f(x)>\tan^{-1}x-\tan^{-1}1 \Rightarrow f(x)+\pi/4>\tan^{-1}x$

35.(ABC) For $a\le 0$,

Given equation becomes $\int_0^2 (x-a)dx \ge 1 \Rightarrow a\le \dfrac{1}{2} \Rightarrow a\le 0$

For $0<a<2$,

$\int_0^2 |x-a|\,dx \ge 1 \Rightarrow \int_0^a (a-x)dx + \int_a^2 (x-a)dx \ge 1$

$\Rightarrow \quad \dfrac{a^2}{2}+2-2a+\dfrac{a^2}{2}\ge 1 \Rightarrow a^2-2a+1\ge 0 \Rightarrow (a-1)^2\ge 0$

For $a\ge 2$,

$\int_0^2 |x-a|\,dx \ge 1 \Rightarrow \int_0^2 (a-x)dx \ge 1 \Rightarrow 2a-2\ge 1 \Rightarrow a\ge \dfrac{3}{2}$

$= a\ge 2$

36.(ABD) We know $\int_a^b |\sin x|\,dx$ represents the area under the curve from $x=a$ to $x=b$. We also know that area from $x=a$ to $x=a+\pi$ is 2.

$\therefore \quad \int_a^b |\sin x|\,dx = 8 \Rightarrow b-a=\dfrac{8\pi}{2} \qquad (1)$

Similarly, $\int_0^{a+b} |\cos x|\,dx = 9 \Rightarrow a+b-0=\dfrac{9\pi}{2} \qquad (2)$

From (1) and (2), $a=\dfrac{\pi}{4}$ and $b=\dfrac{17\pi}{4}$

$\Rightarrow \quad |a+b|=\dfrac{9\pi}{2}, |a-b|=4\pi, \dfrac{a}{b}=17$ and $\int_a^b \sec^2 x\,dx = [\tan x]_{\pi/4}^{17\pi/4}=0$.

37.(ABC) $g(x)=\int_0^x 2|t|\,dt=\begin{cases}\int_0^x -2t\,dt, & x<0\\ \int_0^x 2t\,dt, & x\ge 0\end{cases}=\begin{cases}\left[-t^2\right]_0^x, & x<0\\ \left[t^2\right]_0^x, & x\ge 0\end{cases}=\begin{cases}-x^2, & x<0\\ x^2, & x\ge 0\end{cases}=x|x|$

Clearly, continuous and differentiable at $x=0$

Also, $g'(x)=\begin{cases}-2x, & x<0\\ 2x, & x>0\end{cases}$ which is non-differentiable at $x=0$.

38.(AB) $f(x)=x\int_1^x \frac{e^t}{t}dt-e^x \quad\Rightarrow\quad f'(x)=x\frac{e^x}{x}+\int_1^x\frac{e^t}{t}dt-e^x$

$f'(x)=\int_1^x\frac{e^t}{t}dt>0\left[\because x\in[1,\infty)\right]$ $\quad f(x)$ is an increasing function.

39.(ACD) $\int_0^1\frac{2x^2+3x+3}{(x+1)\left(x^2+2x+2\right)}=\int_0^1\frac{2(x^2+2x+2)-(x+1)}{(x+1)\left(x^2+2x+2\right)}dx=\int_0^1\left(\frac{2}{x+1}-\frac{1}{x^2+2x+2}\right)dx$

$\left[2\log(x+1)-\tan^{-1}(x+1)\right]_0^1$

$2\log 2-\tan^{-1}2+\tan^{-1}1$

$2\log 2-\tan^{-1}2+\frac{\pi}{4}$

$\log 4-\left(\frac{\pi}{2}-\cot^{-1}2\right)+\frac{\pi}{4}$

$-\frac{\pi}{4}+\log 4+\cot^{-1}2$

From equation (1), $I=2\log 2-\tan^{-1}\left(\frac{2-1}{1+2\times 1}\right)=2\log 2-\tan^{-1}\frac{1}{3}=2\log 2-\cot^{-1}3$

40.(AD) $A_{n+1}-A_n=\int_0^{\pi/2}\frac{\sin(2n+1)x-\sin(2n-1)x}{\sin x}=\int_0^{\pi/2}2\cos 2nx\,d\,x=0$

$\Rightarrow \quad A_{n+1}=A_n$

$B_{n+1}-B_n=\int_0^{\pi/2}\frac{\sin^2(n+1)x-\sin^2 nx}{\sin^2 x}dx=\int_0^{\pi/2}\frac{\sin(2n+1)x}{\sin x}dx=A_{n+1}$

41.(ABC) $f(x)=\int_0^x\frac{1}{f(x)}dx\Rightarrow f'(x)=\frac{1}{f(x)}\cdot 1-0\Rightarrow f(x)f'(x)=1$

$\Rightarrow \quad \int f(x)f'(x)dx=\int 1dx \quad\Rightarrow\quad \frac{1}{2}[f(x)]^2=x+c \qquad (1)$

Now given that $\int_0^1[f(x)]^{-1}dx=\sqrt{2}\Rightarrow f(1)=\sqrt{2}$

$\Rightarrow$ From (1), $\frac{1}{2}[f(1)]^2=1+c\Rightarrow c=0 \quad\Rightarrow\quad f(x)=\pm\sqrt{2x}$

But $\quad f(1)=\sqrt{2}\Rightarrow f(x)=\sqrt{2x}\Rightarrow f(2)=2$

Also, $f'(x)=\frac{1}{\sqrt{2x}} \Rightarrow f'(2)=1/2$

$$\int_0^1 f(x)dx=\int_0^1 \sqrt{2x}dx=\left[\frac{(2x)^{3/2}}{3}\right]_0^1=\frac{(2)^{3/2}}{3}$$

Also, $f^{-1}(x)=\frac{x^2}{2} \Rightarrow f^{-1}(2)=2$

42.(BC) $I=\int_0^\infty \frac{dx}{1+x^4}$ (1)

$$=\int_0^\infty \frac{x^2+1-x^2}{1+x^4}dx=\int_0^\infty \frac{x^2}{1+x^4}dx+\int_0^\infty \frac{1-x^2}{1+x^4}dx=I_1+I_2$$

$$I_2=\int_0^\infty \frac{\frac{1}{x^2}-1}{\frac{1}{x^2}+x^2}dx.$$

Put $x+\frac{1}{x}=y$

$\Rightarrow$ $I_2=\int_\infty^\infty \frac{-1}{y^2-2}dy=0$ $\Rightarrow$ $I=\int_0^\infty \frac{dx}{1+x^4}=\int_0^\infty \frac{x^2dx}{1+x^4}$ (2)

Adding equations (1) and (2), we get :

$\Rightarrow$ $2I=\int_0^\infty \frac{1+x^2dx}{1+x^4}=\int_0^\infty \frac{\frac{1}{x^2}+1}{\frac{1}{x^2}+x^2}dx$, put $x-\frac{1}{x}=y$

$\Rightarrow$ $2I=\int_{-\infty}^\infty \frac{dy}{y^2+2}=\left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{y}{\sqrt{2}}\right]_{-\infty}^\infty=\frac{\pi}{\sqrt{2}}$ $\Rightarrow$ $I=\frac{\pi}{2\sqrt{2}}$

43.(ABD) Given than $f(x)=\int_0^x |t-1|\,dt$ $\Rightarrow$ $f(x)=\int_0^x (1-t)dt, 0\le x\le 1=x-\frac{x^2}{2}$

Also $f(x)=\int_0^1 (1-t)dt+\int_1^x (t-1)dt$, where $1\le x\le 2=\frac{1}{2}+\frac{x^2}{2}-x+\frac{1}{2}=\frac{x^2}{2}-x+1$

Thus, $f(x)=\begin{cases} x-\frac{x^2}{2}, & 0\le x\le 1 \\ \frac{x^2}{2}-x+1, & 1<x\le 2 \end{cases} \Rightarrow f'(x)=\begin{cases} 1-x, & 0\le x<1 \\ x-1, & 1<x<2 \end{cases}$

Thus, $f(x)$ is continuous as well as differentiable at $x=1$.

Also, $f(x)=\cos^{-1}x$ has one real root, draw the graph and verify.

44.(ABC) Let $I=\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx$ (1)

$=\int_a^b \frac{f(a+b-x)}{f(a+b-x)+f(x)}dx$ (2)

Adding equations (1) and (2), we get

$\Rightarrow \quad 2I=\int_a^b 1dx=b-a \Rightarrow I=\left(\frac{b-a}{2}\right)=10$ (Given) $\quad \therefore \quad b-a=20$

45.(AB) $I_n=\int_0^1 \frac{dx}{\left(1+x^2\right)^n}=\int_0^1\left(1+x^2\right)^{-n}dx=\left.\frac{x}{\left(1+x^2\right)^n}\right|_0^1-\int_0^1(-n)\left(1+x^2\right)^{-n-1}2x\times xdx=\frac{1}{2^n}+2n\int_0^1\frac{x^2dx}{\left(1+x^2\right)^{n+1}}$

$=\frac{1}{2^n}+2n\int_0^1\frac{1+x^2-1}{\left(1+x^2\right)^{n+1}}dx=\frac{1}{2^n}+2nI_n-2nI_{n+1}$

$\Rightarrow \quad 2nI_{n+1}=2^{-n}+(2n-1)I_n \Rightarrow 2I_2=\frac{1}{2}+I_1=\frac{1}{2}+\left.\tan^{-1}x\right|_0^1 \Rightarrow I_2=\frac{1}{4}+\frac{\pi}{8}$

Also $\quad 4I_3=2^{-2}+3I_2=\frac{1}{4}+3\left(\frac{1}{4}+\frac{\pi}{8}\right)=\frac{1}{4}+\frac{3\pi}{32}$

46.(BC) $\lim_{n\to\infty}\frac{1}{n}\sum_{r=n+1}^{2n}f\left(\frac{r}{n}\right)=\int_1^2 f(x)dx$

$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}f\left(\frac{r+n}{n}\right)=\int_0^1 f(1+x)dx=\int_1^2 f(t)dt=\int_1^2 f(x)dx$

$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}f\left(\frac{r}{n}\right)=\int_0^1 f(x)dx$

$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}f\left(\frac{r}{n}\right)=\int_0^2 f(x)dx$

47.(ABD) $f(2-x)=f(2+x), f(4-x)=f(4+x)$

$\Rightarrow \quad f(4+x)=f(4-x)=f(2+2-x)=f(2-(2-x))=f(x) \Rightarrow 4$ is a period of $f(x)$

$\int_0^{50}f(x)dx=\int_0^{48}f(x)dx+\int_{48}^{50}f(x)dx=12\int_0^4 f(x)dx+\int_0^2 f(x)dx$

(In second integral replacing x by $x+48$ and then using $f(x)=f(x+48)$)

$=12\left(\int_0^2 f(x)dx+\int_0^2 f(4-x)dx\right)+5=12\left(\int_0^2 f(x)dx+\int_0^2 f(4+x)dx\right)+5=24\int_0^2 f(x)dx+5=125$

$\int_{-4}^{46}f(x)dx=\int_{-4}^{-2}f(x)dx+\int_{-2}^{-2+48}f(x)dx$

$=\int_0^2 f(x+4)dx+12\int_0^4 f(x)dx=\int_0^2 f(x)dx+24\int_0^4 f(x)dx$

Also $\int_{2}^{52} f(x)dx = \int_{2}^{4} f(x)dx + \int_{4}^{4+48} f(x)dx$

$= \int_{0}^{2} f(4-x)dx + 12\int_{0}^{4} f(x)dx = \int_{0}^{2} f(4+x)dx + 24\int_{0}^{2} f(x)dx = \int_{0}^{2} f(x)dx + 24\int_{0}^{2} f(x)dx$

$\int_{1}^{51} f(x)dx = \int_{1}^{3} f(x)dx + \int_{3}^{3+48} f(x)dx$

$\int_{1}^{3} f(x)dx + 12\int_{0}^{4} f(x)dx$

$\int_{0}^{2} f(x+1)dx + 24\int_{0}^{2} f(x)dx \neq 125$

48.(AB) L.H.S. $= \int_0^x \left\{\int_0^u f(t)dt\right\}du$

Integrating by parts choose '1' as the second function

$= \left\{u\int_0^u f(t)dt\right\}_0^x - \int_0^x f(u)u\,du = x\int_0^x f(t)dt - \int_0^x f(u)u\,du$

$= x\int_0^x f(u)du - \int_0^x f(u)u\,du = \int_0^x f(u)(x-u)du$ = R.H.S.

49.(ACD) The expression $f(x)f(c)\forall x \in (c-h, c+h)$ where $h \to 0^+$ is equivalent to $\lim\limits_{x\to 0} f(x)f(c)$ which equals to $(f(c))^2$ because $f(x)$ is continuous.

(A) We have $I = \lim\limits_{n\to\infty} \frac{1}{n}\ln\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)...\left(1+\frac{n}{n}\right)\right]$

$= \lim\limits_{n\to\infty} \frac{1}{n}\ln\prod_{k=1}^{n}\left(1+\frac{k}{n}\right) = \lim\limits_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n}\right) = \int_1^2 \ln x\,dx = [x(\ln x - 1)]_1^2 = -1 + 2\ln 2$

(C) Given $f(x) \geq 0 \Rightarrow \int_a^b f(x)dx \geq 0.$

But given $\int_a^b f(x)dx = 0,$ so this can be true only when $f(x) = 0.$

(D) $\int_a^b f(x)dx = 0 \Rightarrow y = f(x)$ cuts x axis at least once.

So, there exists at least one $c \in (a,b)$ for which $f(x) = 0.$

50.(AC) $\int_0^1 e^{x^2-x}dx$

For $x \in (0,1), x^2 - x \in (-1/4, 0)$ $\quad \Rightarrow e^{-1/4} < e^{x^2-x} < e^0$ $\quad \Rightarrow e^{\frac{-1}{4}} < \int_0^1 e^{x^2-x}dx < 1$

51.(AD) $f(x+\pi)=\int_0^{x+\pi}(\cos(\sin t)+\cos(\cos t))dt=\int_0^{\pi}(\cos(\sin t)+\cos(\cos t))dt+\int_x^{x+\pi}(\cos(\sin t)+\cos(\cos t))dt$

$=f(x)+\int_0^{\pi}(\cos(\sin t)+\cos(\cot t))dt$ ($\because$ for $g(x)=\cos(\sin x)+\cos(\cos x), f(x+\pi)=f(x))=f(\pi)+f(x)$

$=f(x)+2f\left(\frac{\pi}{2}\right)$ ($\because g(x)$ has period $\pi/2$)

52.(AB) Let $f(x)=\int_0^{x^2}\left(\frac{t^2-5t+4}{2+e^t}\right)dt$; $f^1(x)=\left(\frac{x^4-5x^2+4}{2+e^{x^2}}\right)\times 2x$ For extremum $f^1(x)=0$

53.(BD) For x = 0

$G(0,t)=0, t\geq 0$

$g(1)=\int_0^1 f(t)G(0,t)dt=\int_0^1 0.dt=0$

And for x = 1

$G(1,t)=0, t<1$

$g(1)=\int_0^1 f(t)G(1,t)dt=0$

Also, $g(x)=\int_0^x f(t)G(x,t)dt+\int_x^1 f(t)G(x,t)dt=(x-1)\int_0^x tf(t)dt=x\int_x^1(t-1)f(t)dt$

$\therefore$ $g^{11}(x)=xf(x)+\{0-(x-1)f(x)\}=f(x)$

54.(ACD) Let $I=\int_{-1/2}^{1/2}\sqrt{\left(\frac{x+1}{x-1}-\frac{x-1}{x+1}\right)}dx=\int_{-1/2}^{1/2}\left|\frac{4x}{x^2-1}\right|dx=-2\int_0^{1/2}\frac{4x}{(x^2-1)}dx=-4\left\{ln\left|x^2-1\right|\right\}_0^{1/2}=-4ln\left(\frac{3}{4}\right)$

$=4ln\left(\frac{4}{3}\right)=ln\left(\frac{256}{81}\right)=-ln\left(\frac{81}{256}\right)$

55.(AC) $\int_a^{a+nT}f(x)dx=n\int_0^T f(x)dx$

56.(ABCD) $f'(x)=\sin x\times\frac{2x}{1+\cos^2 x}+\cos x\int_{\pi^2/4}^{x^2}\frac{1}{1+\cos^2\sqrt{t}}dt$

57.(ABC) $I_{n+2}-I_n=\int_{-\pi}^{\pi}\frac{\sin(n+2)x-\sin nx}{\sin x}dx=\int_{-\pi}^{\pi}2\cos(n+1)dx=0\Rightarrow I_1=I_3=I_5=....$

$I_{2m}=2\int_0^{\pi}\frac{sin\,2mx}{sin\,x}dx=0\quad(\because f(\pi-x)=-f(x))$

58.(BD) Differentiating both sides w.r.t. x

$\int_0^x y(t)\,dt+xy(x)=\int_0^x t\,y(t)\,dt+(x+1)x\,y(x)$ or $\int_0^x y(t)\,dt=\int_0^x t\,y(t)\,dt+x^2y(x)$

again differentiating $y(x)=xy(x)+x^2y'(x)+2xy(x)$ or $x^2\frac{dy}{dx}+(3x-1)y=0$

separating variable and integrating $y=\frac{e}{x^3}e^{-1/x}$]

59.(ABCD)

Consider $Q=\int_0^\infty \frac{x\,dx}{1+x^4}$ $\quad x^2=t;\ 2xdx=dt$ $\quad =\frac{1}{2}\int_0^\infty \frac{dt}{1+t^2}$ $\quad =\frac{1}{2}\tan^{-1}t\Big]_0^\infty$ $\quad =\frac{\pi}{4}$

Consider $P=\int_0^\infty \frac{x^2}{1+x^4}dx$ $\quad x=1/t;\ dx=-1/t^2\ dt$

$$=\int_0^\infty \frac{1}{t^2}\cdot\frac{t^4}{1+t^4}\frac{1}{t^2}dt=\int_0^\infty \frac{dt}{1+t^4}$$

$\therefore$ $\quad P=\int_0^\infty \frac{dt}{1+t^4}=R$

Now $P+R=\int_0^\infty \frac{1+x^2}{1+x^4}dx=\int_0^\infty \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx=\int_0^\infty \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}dx$ $\quad$ put $x-\frac{1}{x}=t \Rightarrow 1+\frac{1}{x^2}dx=dt$

$$=\int_{-\infty}^\infty \frac{dt}{t^2+2}=2\int_0^\infty \frac{dt}{t^2+2}=2\cdot\frac{1}{\sqrt{2}}\tan^{-1}\frac{1}{\sqrt{2}}\Big]_0^\infty \quad =\sqrt{2}\cdot\frac{\pi}{2}=\frac{\pi}{\sqrt{2}}$$

60.(AB) $v=\ =\int_0^{\pi/4}\frac{1+\sin 2x}{\cos^2 x}dx=\int_0^{\pi/4}(\sec^2 x+2\tan x)dx=\tan x+2\ln(\sec x)]_0^{\pi/4}=(1+\ln 2)-0=1+\ln 2$

again $u=\int_0^{\pi/4}\frac{\cos^2 x}{1+\sin 2x}dx$ $\qquad$ putting $\sin 2x=\frac{2\tan x}{1+\tan^2 x}$

$$u=\int_0^{\pi/4}\frac{dx}{1+\tan^2 x+2\tan x}=\int_0^{\pi/4}\frac{dx}{(1+\tan x)^2}=\int_0^{\pi/4}\frac{dx}{(1+\tan(\pi/4)-x)^2}$$

$$=\int_0^{\pi/4}\frac{dx}{\left(1+\frac{1-\tan x}{1+\tan x}\right)^2}=\int_0^{\pi/4}\frac{(1+\tan x)^2}{4}dx \quad u=\frac{1}{4}\left[\int_0^{\pi/4}(1+2\tan x+\tan^2 x)dx\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{4}+2\ln\sec x\Big|_0^{\pi/4}+\tan x-x\Big|_0^{\pi/4}\right]=\frac{1}{4}\left[\frac{\pi}{4}+\frac{2}{2}(\ln 2)+1-\frac{\pi}{4}\right]=\frac{1}{4}[(\ln 2)+1]$$

Hence $\quad u=\frac{1}{4}[(\ln 2)+1]$ and $v=(1+\ln 2)$; $\qquad \frac{v}{u}=4$

61.(ABC) Let $I=\int_0^2\frac{\ln(1+2x)}{1+x^2}dx$

put $x=\tan\theta \Rightarrow$ $\quad dx=\sec^2\theta d\theta;$ when $x=2,\ \theta=\tan^{-1}(2)$

$$I=\int_0^{\tan^{-1}2}\ln(1+2\tan\theta)\,d\theta \qquad ....(1)$$

applying king

$$I=\int_0^{\tan^{-1}2}\ln\left(1+2\tan(\tan^{-1}2-\theta)\right)d\theta=\int_0^{\tan^{-1}2}\ln\left(1+2\left(\frac{2-\tan\theta}{1+2\tan\theta}\right)\right)d\theta$$

$$I=\int_0^{\tan^{-1}2}\ln 5\,d\theta-\underbrace{\int_0^{\tan^{-1}2}\ln(1+2\tan\theta)\,d\theta}_{I}$$

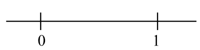
$$2I=\tan^{-1}1.\ln 5$$

$$I=\frac{1}{2}\tan^{-1}2.\ln 5=\tan^{-1}2.\ln\sqrt{5}$$

Hence $a=2$ and $b=5$ $\quad\therefore\ a^2+b^2=29$ $\quad$ Note that: $\displaystyle\int_0^a \frac{\ln(1+ax)}{1+x^2}dx=\tan^{-1}a.\ln\sqrt{(1+a^2)}$

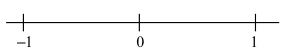
62.(ABD)

$$D=4+4\left(k+\int_0^1|k+t|\,dt\right)=4+4k+4\int_0^1|k+t|\,dt$$



let $\quad k\geq 0$

$$I=\int_0^1|k+t|\,dt=\int_0^1(k+t)dt=k+\frac{1}{2}$$



Hence D becomes $\quad 4+4k+4\left(k+\frac{1}{2}\right)$

$$4+8k+2=8k+6>0=4\left(2k+\frac{3}{2}\right)$$

let $\quad k\leq -1, I=-\displaystyle\int_0^1(k+t)\,dt=-\left[kt+\frac{t^2}{2}\right]_0^1=-\left[k+\frac{1}{2}\right]$

$\therefore \quad D=4+4k-4\left(k+\frac{1}{2}\right)$

$D=2>0$

$-1<k<0 \qquad I=\displaystyle\int_0^1|k+t|dt$

Let $\quad k=-y \quad\Rightarrow\quad 0<y<1$

$$I = \int_0^1 |t-y|\,dt = \int_0^y (y-t)\,dt + \int_y^1 (t-y)\,dt = \left[ty - \frac{t^2}{2}\right]_0^y + \left[\frac{t^2}{2} - yt\right]_y^1$$

$$= \left(y^2 - \frac{y^2}{2}\right) + \left(\frac{1}{2} - y\right) - \left(\frac{y^2}{2} - y^2\right) = y^2 - y + \frac{1}{2} = k^2 + k + \frac{1}{2}$$

$$D = 4 + 4k + 4\left(k^2 + k + \frac{1}{2}\right) = 4\left[1 + k + k^2 + k + \frac{1}{2}\right] = 4\left[(k+1)^2 + \frac{1}{2}\right] > 0$$

Hence $D > 0 \ \forall\ k \in R \Rightarrow$ roots are real and distinct

63.(AB) $x = k\cos^2\theta + (k+1)\sin^2\theta$ [note $\alpha = k$ and $\beta = k+1$]

$x = k + \sin^2\theta$

$dx = \sin 2\theta\, d\theta$

$$I = \int_0^{\pi/2} \sin\theta\cos\theta\, 2\sin\theta\cos\theta\, d\theta = \frac{1}{2}\int_0^{\pi/2} \sin^2 2\theta\, d\theta \quad \text{put } 2\theta = t$$

$$= \frac{1}{4}\int_0^{\pi} \sin^2 t\, dt = \frac{1}{2}\int_0^{\pi/2} \sin^2 t\, dt = \frac{\pi}{8}$$

$$\therefore \quad L = \frac{1}{n^2}\lim_{n\to\infty}\sum_{k=0}^{n-1} k\left(\frac{\pi}{8}\right) = \lim_{n\to\infty}\frac{\pi}{8}\frac{1}{n^2}[1+2+....+n-1] = \lim_{n\to\infty}\frac{\pi}{8}\frac{(n-1)n}{2}\cdot\frac{1}{n^2} = \frac{\pi}{16}$$

64.(AB) **(A)** $g'(x) = cx^{c-1}.e^{2x} + x^c.e^{2x}.2$

$f'(x) = e^{2x}\left(3x^2+1\right)^{1/2}$

$$\lim_{x\to\infty}\frac{f'(x)}{g'(x)} = \lim_{x\to\infty}\frac{e^{2x}\left(3x^2+1\right)^{1/2}}{c\,x^{c-1}.e^{2x} + 2x^c.e^{2x}} = \lim_{x\to\infty}\frac{x\left(3+\frac{1}{x^2}\right)^{1/2}}{x^c\left(\frac{c}{x}+2\right)}$$

If $x\to\infty$ it will be finite if $c = 1$ and $\lim\limits_{x\to\infty}$ will be $\dfrac{\sqrt{3}}{2}$

65.(ABC) For $x > e$, we know that $1 < \ln x < \dfrac{x}{e}$

$$\left(\frac{e}{x}\right)^{1/3} < \frac{1}{\sqrt[3]{\log x}} < 1$$

$$\Rightarrow \int_3^4 e^{1/3}x^{-1/3}dx < I < \int_3^4 dx \Rightarrow \frac{3}{2}e^{1/3}\left(4^{2/3} - 3^{2/3}\right) < I < 1 \Rightarrow 0.92 < I < 1.$$

66.(ABC) For $0 < x < 1$, $x^2 < x$

$$\Rightarrow -x^2 > -x \Rightarrow e^{-x^2} > e^{-x} \Rightarrow \int_0^1 e^{-x^2}\cos^2 x\,dx > \int_0^1 e^{-x}\cos^2 x\,dx \text{ and } \cos^2 x \le 1$$

$$\Rightarrow \int_0^1 e^{-x^2}\cos^2 x\,dx < \int_0^1 e^{-x^2}dx\ ;\ < \int_0^1 e^{-\frac{1}{2}x^2}dx = 1;\ < \int_0^1 e^{-\frac{1}{2}x^2}dx = 1$$

Hence, I_4 is the greatest integral.

67.(A,B,C,D)

We have $1+x^2+2x^5>1+x^2$ and $1+x^2+2x^5<1+x^2+2x^2=1+3x^2$ $\quad\left[\because\ x^5<x^2\ on(0,1)\right]$

Hence, we have $\dfrac{1}{1+3x^2}<\dfrac{1}{1+x^2+2x^5}<\dfrac{1}{1+x^2}$

$$\Rightarrow \quad \int_0^1\frac{dx}{1+3x^2}<\int_0^1\frac{dx}{1+x^2+2x^5}<\int_0^1\frac{dx}{1+x^2} \Rightarrow \left[\frac{\tan^{-1}\sqrt{3}x}{\sqrt{3}}\right]_0^1<\int_0^1\frac{dx}{1+x^2+2x^5}<\left[\tan^{-1}x\right]_0^1$$

$$\Rightarrow \quad \frac{\pi}{3\sqrt{3}}\le\int_0^1\frac{dx}{1+x^2+2x^5}\le\frac{\pi}{4}.$$

Which is the desired result.

$\sin^4 x\le\sin^3 x\le\sin^2 x$

$$\Rightarrow \quad -\sin^2 x\le-\sin^3 x\le-\sin^4 x \Rightarrow \quad 1-\sin^2 x\le1-\sin^3 x\le1-\sin^4 x$$

So, we get $\sqrt{1-\sin^2 x}\le\sqrt{1-\sin^3 x}\le\sqrt{1-\sin^4 x}$

$$\int_0^{\pi/2}\sqrt{1-\sin^2 x}dx=\int_0^{\pi/2}\cos x dx=\left[\sin x\right]_0^{\pi/2}=1,\text{ and } \int_0^{\pi/2}\sqrt{1-\sin^4 x}dx=\int_0^{\pi/2}\sqrt{\left(1+\sin^2 x\right)\cos^2 x}dx$$

$$=\int_0^{\pi/2}\cos x\sqrt{1+\sin^2 x}dx=\int_0^1\sqrt{1+t^2}dt=\left[\frac{1}{2}\left(t\sqrt{t^2+1}+\ln\left(t+\sqrt{t^2+1}\right)\right)\right]=\frac{1}{2}\left(\sqrt{2}+\ln\left(1+\sqrt{2}\right)\right)$$

$$\therefore \quad 1\le\int_0^{\pi/2}\sqrt{1-\sin^3 x}\le\frac{1}{2}\left(\sqrt{2}+\ln\left(1+\sqrt{2}\right)\right).$$

68.(BC) Put $x=0$ in the given equation $0=1-c.\int_0^1 f(t)e^{-t}dt$

Let $c=\dfrac{1}{k}$ where $k=\int_0^1 f(t)e^{-t}dt.$

Differentiating the given equation, we get $f(x)=e^x-2e^{2x}$

Now, $k=\int_0^1\left(e^t-2e^{2t}\right)e^{-t}dt=\int_0^1\left(1-2e^t\right)dt=3-2e$ $\qquad\therefore\qquad c=\dfrac{1}{k}=\dfrac{1}{3-2e}.$

69.(BD) $f(x)=x+x\int_0^1 y^2f(y)dy+x^2\int_0^1 yf(y)dy=x\left(1+\int_0^1 y^2f(y)dy\right)+x^2\left(\int_0^1 yf(y)dy\right)$

$\Rightarrow \quad f(x)$ is a quadratic expression.

Let $f(x)=ax+bx^2$, then $f(y)=ay+by^2,$ $\qquad$ Where $a=1+\int_0^1 y^2f(y)dy$

$$=1+\int_0^1 y^2\left(ay+by^2\right)dy\quad=1+\left(\frac{ay^4}{4}+\frac{by^5}{5}\right)_0^1$$

$$a=1+\frac{a}{4}+\frac{b}{5}\ \&\ b=\int_0^1 y\ f(y)dy$$

$$\frac{3a}{4}+\frac{b}{5}=1$$

70.(ABC)

$$I=\int_{1/2}^{2}\frac{\ln t}{1+t^n}dt \qquad ....(1)$$

put $\quad t=\frac{1}{u} \quad \Rightarrow \quad dt=-\frac{1}{u^2}du$

$$I=\int_{2}^{1/2}\frac{\ln(1/u)}{1+(1/u^n)}\left(-\frac{1}{u^2}\right)du \quad =-\int_{1/2}^{2}\frac{u^n\ln u}{(1+u)^n u^2}du$$

$$I=-\int_{1/2}^{2}\frac{t^{n-2}\ln t}{1+t^n}dt \qquad ....(2)$$

$$2I=\int_{1/2}^{2}\frac{(1-t^{n-2})\ln t}{(1+t^n)}dt$$

for $n=1$, the integrand is $+ve$ in $(1/2,1)\cup(1,2)$ and zero only for $t=1\Rightarrow 1>0$

for $n=2$, integrand is zero $\quad \Rightarrow \quad I=0$

for $n\geq 3$, integrand is always $-ve$ except for $t=1$ hence $I<0$

Note that for $n=0$ $I=\int_{1/2}^{2}\ln t\,dt=\left(t\ln t-t\right)_{1/2}^{2}=\left(2\ln 2-2\right)-\left(\frac{1}{2}\ln\frac{1}{2}-\frac{1}{2}\right)=\frac{5}{2}\ln 2-\frac{3}{2}>0]$

71.(ABC) $\lim\limits_{x\to 0}\frac{\int_0^x\frac{t^2dt}{\left(a+t^r\right)^{1/p}}}{bx-\sin x}=\lim\limits_{x\to 0}\frac{\frac{x^2}{\left(a+x^r\right)^{1/p}}}{b-\cos x}$ using L' Hosp0ital's rule

For existence of limit, $\lim\limits_{x\to 0}$ denominator $=0$

$\therefore \quad b-1=0\Rightarrow b=1$

$$l=\lim_{x\to 0}\frac{x^2}{\left(a+x^r\right)^{1/p}}\cdot\frac{x^2}{\left(1-\cos x\right)}\cdot\frac{1}{x^2}=2\lim_{x\to 0}\frac{1}{\left(a+x^r\right)^{1/p}}=\frac{2}{a^{1/p}}$$

If $p=3$ and $l=1$, then $1=\frac{2}{a^{1/3}}\Rightarrow a=8$ $\qquad$ If, $p=2$ and $a=9$, then $l=\frac{1}{9^{1/2}}=\frac{2}{3}$

72.(ABC)

Here, $f(x)=e^x\underbrace{\int_0^1 e^t.f(t)dt}_{A(say)} \qquad ....(i)$

$f(x)=Ae^x \Rightarrow \quad f(t)=Ae^t$

Where, $A=\int_0^1 e^t.f(t)dt \;\Rightarrow\; A=\int_0^1 e^t.Ae^t dt;\; A=A\int_0^1 e^{2t}dt\neq 0$

Hence, $f(x)=0\Rightarrow f(1)=0$

Again, $g(x)=e^x\int_0^1 e^t g(t)dt+x$

$\Rightarrow \quad g(x)=Be^x+x \quad$ ....(ii)

$\Rightarrow \quad g(t)=Be^t+t$

Where, $B=\int_0^1 e^t g(t)dt \Rightarrow B=\int_0^t e^t\left(Be^t+t\right)dt \Rightarrow B=B\int_0^1 e^{2t}dt+\int_0^1 e^t.t.dt$

But $\int_0^1 e^{2t}dt=\frac{1}{2}\left(e^2-1\right)$ and $\int_0^1 te^t dt=1$

$\therefore \quad B=\frac{B}{2}\left(e^2-1\right)+1 \Rightarrow 2B=B\left(e^2-1\right)+2 \Rightarrow 3B=Be^2+2 \Rightarrow B=\frac{2}{3-e^2}$

From Equation (ii), $g(x)=\left(\frac{2}{3-e^2}\right)e^x+x \Rightarrow g(0)=\frac{2}{3-e^2}$

Also, $f(0)=0$

$\therefore \quad g(0)-f(0)=\frac{2}{3-e^2}-0=\frac{2}{3-e^2}$; $g(2)=\frac{2e^2}{3-e^2}+2=\frac{6}{3-e^2} \quad \therefore \frac{g(0)}{g(2)}=\frac{2}{3-e^2}.\frac{3-e^2}{6}=\frac{1}{3}$

73.(CD) We have the equations of the tangents to the curve $y=\int_{-\infty}^{x} f(t)dt$ and $y=f(x)$ at arbitrary pints on them are

$Y-\int_{-\infty}^{x} f(t)dt=f(x)(X-x) \quad$ ...(i)

And $Y-f(x)=f'(x)(X-x) \quad$ ...(ii)

As Eqs. (i) and (ii) intersect at the same point on the X-axis Putting $Y=0$ and equating x -coordinates, we have

$$x-\frac{f(x)}{f'(x)}=x-\frac{\int_{-\infty}^{x} f(t)dt}{f(x)} \Rightarrow \frac{f(x)}{\int_{-\infty}^{x} f(t)dt}=\frac{f'(x)}{f(x)} \Rightarrow \int_{-\infty}^{x} f(t)dt=cf(x) \quad .....(iii)$$

As $f(0)=1 \Rightarrow \int_{-\infty}^{0} f(t)dt=c\times 1 \Rightarrow c=\frac{1}{2}$

$\Rightarrow \quad \int_{-\infty}^{x} f(t)dt=\frac{1}{2}f(x)$; differentiating both the sides and integrating and using boundary conditions, we get $f(x)=e^{2x}$; $y=2ex$ is tangent to $y=e^{2x}$.

$\therefore$ Number of solutons $=1$.

Clearly, $f(x)$ is increasing for all x. $\quad \therefore \quad \lim_{x\to\infty}\left(e^{2x}\right)^{e^{-2x}}=1 \quad (\infty^\circ$ form)

74.(ABC) We have, $g(x)=g(0)+xg'(0)+\frac{x^2}{2}g''(0)=-bx^2+cx-6$, $h(x)=g(x)=4x^4-ax^3+bx^2-cx+6=0$ has 4 distinct real roots.

Given biquadratic equation has 4 distinct positive roots.

Let the roots be x_1, x_2, x_3 and x_4.

Now, $\dfrac{\dfrac{1}{x_1}+\dfrac{2}{x_2}+\dfrac{3}{x_3}+\dfrac{4}{x_4}}{4} \geq \sqrt[4]{\dfrac{24}{x_1x_2x_3x_4}}$

$\Rightarrow \quad 2 \geq 2 \Rightarrow \dfrac{1}{x_1}=\dfrac{2}{x_2}=\dfrac{3}{x_3}=\dfrac{4}{x_4}=k \;\Rightarrow\; \dfrac{1}{x_1}.\dfrac{2}{x_2}.\dfrac{3}{x_3}.\dfrac{4}{x_4}=k^4 \;\Rightarrow\; \dfrac{24}{3/2}=k^4 \Rightarrow k=2$

$\therefore \quad$ Roots are $\dfrac{1}{2},1.\dfrac{3}{2}$ and 2. Also, $a=20$ and $c=25$

75(CD) Given, $(f'(x))^2+(g(x))^2=1$

$f(x)+\int_0^x g(t)dt=\sin x(\cos x-\sin x)$

Differentiating both the sides, we get

$f'(x)+g(x)=\cos 2x-\sin 2x$ ....(i)

Squaring both the sides of Eq. (i), we get

$(f'(x))^2+(g(x))^2+2f'(x).g(x)=1-\sin 4x$

$\Rightarrow \quad 1+2f'(x).g(x)=1-\sin 4x$

$\therefore \quad 2f'(x)g(x)=-\sin 4x$

Now, substituting $g(x)=-\dfrac{\sin 4x}{2f'(x)}$ in Eq. (i), we get

$f'(x)-\dfrac{\sin 4x}{2f'(x)}=\cos 2x-\sin 2x$

Put $f'(x)=t$

$\Rightarrow \quad 2t^2-2(\cos 2x-\sin 2x)t-\sin 4x=0 \Rightarrow \quad t=\dfrac{2(\cos 2x-\sin 2x)\pm\sqrt{4(1-\sin 4x)+8\sin 4x}}{4}$

$\therefore \quad 4t=2(\cos 2x-\sin 2x)\pm\sqrt{4(1-\sin 4x)+8\sin 4x} \Rightarrow 2t=(\cos 2x-\sin 2x)\pm\sqrt{1+\sin 4x}$

Taking + ve sign, $2t=\cos 2x-\sin 2x+\cos 2x+\sin 2x$

$f(x)=\dfrac{1}{2}\sin 2x+C_1$ or $f(x)=\dfrac{\cos 2x}{2}+C_2$

$f(0)=0$

$\Rightarrow \quad C_1=0$ and $C_2=-1/2$

$\therefore \quad f(x)=\dfrac{1}{2}\sin 2x$ or $f(x)=\dfrac{\cos 2x-1}{2}$

If $f'(x)=\cos 2x$, then $g(x)=-\sin 2x$

If $f'(x)=-\sin 2x$, then $g(x)=\cos 2x$

i.e. $f(x)=\dfrac{1}{2}\sin 2x$ and $g(x)=\cos 2x$

$\Rightarrow \quad f(x)=\dfrac{\cos 2x-1}{2}$ and $g(x)=\cos 2x$

76.(ABD)

Given, $f(f(x)) = -x+1$

Replacing x by $f(x)$, we get

$f(f(f(x))) = -f(x)+1$

$f(1-x) = -f(x)+1$

$f(x)+f(1-x) = 1$

Now, $\alpha = \int_0^1 f(x)dx = \int_0^1 f(1-x)dx$ (using King's property)

$\Rightarrow \quad 2\alpha = \int_0^1 (f(x)+f(1-x))dx \Rightarrow \quad 2\alpha = \int_0^1 dx = 1 \Rightarrow \alpha = \frac{1}{2}$

Put $x = \frac{1}{4}$ in Eq. (i), $f\left(\frac{1}{4}\right)+f\left(1-\frac{1}{4}\right) = 1 \Rightarrow f\left(\frac{1}{4}\right)+f\left(\frac{3}{4}\right) = 1$

Now $I = \int_0^{\pi/2} \frac{\sin x}{(\sin x+\cos x)^3}dx$

$$I = \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2}-x\right)}{\left(\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)\right)^3}dx \Rightarrow I = \int_0^{\pi/2} \frac{\cos x}{(\cos x+\sin x)^3}dx$$

$$\therefore\ 2I = \int_0^{\pi/2} \frac{1}{(\sin x+\cos x)}dx = \frac{1}{2}\int_0^{\pi/2} \frac{dx}{\left(\frac{1}{\sqrt{2}}\sin x+\frac{1}{\sqrt{2}}\cos x\right)^2}$$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{dx}{\sin^2\left(\frac{\pi}{4}+x\right)} = \frac{1}{2}\int_0^{\pi/2} \operatorname{cosec}^2\left(\frac{\pi}{4}+x\right)dx = -\frac{1}{2}\left[\cot\left(\frac{\pi}{4}+x\right)\right]_0^{\pi/2} = -\frac{1}{2}[-1-1] = 1 \quad \Rightarrow I = \frac{1}{2}$$

77. **[A-s] [B-s] [C-r] [D-q]**

(A) $\int_{-1}^{1} [x+[x+[x]]]dx$ (use property $[x+n] = [x]+n$ if n is integer)

$= \int_{-1}^{1} 3[x]dx = 3\int_{-1}^{1} [x]dx = -3$

(B) $\int_2^5 ([x]+[-x])dx = \int_2^5 -1dx = -3$

(C) $\operatorname{sgn}(n-[x]) = \begin{cases} 1, \text{if } x \text{ is not an integer} \\ 0, \text{if } x \text{ is an integer} \end{cases}$ Hence, $\int_{-1}^{3} \operatorname{sgn}(x-[x])dx = 4(1-0) = 4.$

(D) Let $I = 25\int_0^{\pi/4} \left(\tan^6(x-[x])+\tan^4(x-[x])\right)dx$ $\left\{\because\ 0 < x \le \frac{\pi}{4} \Rightarrow [x] = 0\right\}$

$\therefore I = 25\int_0^{\pi/4} \left(\tan^6 x+\tan^4 x\right)dx$

$$= 25\int_0^{\pi/4} \tan^4 x\left(\tan^2 x+1\right)dx = 25\int_0^{\pi/4} \tan^4 x\sec^2 x\,dx = 25\left(\frac{\tan^5 x}{5}\right)_0^{\pi/4} = 25\times\frac{1}{5} = 5$$

78. **[A-r] [B-p] [C-s] [D-q]**

(A) $\lim\limits_{n\to\infty}\left[\dfrac{\int_0^2\left(1+\dfrac{t}{n+1}\right)^n dt}{n+1}\right] \Rightarrow \lim\limits_{n\to\infty}\left[\left(1+\dfrac{t}{n+1}\right)^{n+1}\right]_0^2 = \lim\limits_{n\to\infty}\left(1+\dfrac{2}{n+1}\right)^{n+1}-1$

(B) $f'(x)=f(x)\Rightarrow f(x)=Ce^x$ and since $f(0)=1$

$\therefore \quad 1=f(0)=C \qquad\qquad \therefore \quad f(x)=e^x$ and hence $g(x)=x^2-e^x$.

Thus, $\int_0^1 f(x)g(x)dx = \int_0^1 (x^2e^x-e^{2x})\,dx = x^2e^x\Big| - 2\int_0^1 xe^x dx - \dfrac{e^{2x}}{2}\Big|_0^1 = (e-0)-2xe^x\Big|_0^1 + 2e^x\Big|_0^1 - \dfrac{1}{2}(e^2-1)$

$= (e-0)-2e+2e-2-\dfrac{1}{2}(e^2-1) = e-\dfrac{1}{2}e^2-\dfrac{3}{2}$

(C) $I=\int_0^1 e^{e^x}(1+xe^x)dx \qquad$ Let $e^x=t \Rightarrow \int_1^e e^t(1+t\log t)\dfrac{dt}{t} = \int_1^e e^t\left(\dfrac{1}{t}+\log t\right)dt = \left[e^t\log t\right]_1^e = e^e$

(D) $L=\lim\limits_{k\to 0}\dfrac{\int_0^k (1+\sin 2x)^{1/x}dx}{k}\left(\text{form}\dfrac{0}{0}\right) \qquad \Rightarrow L=\lim\limits_{k\to 0}(1+\sin 2k)^{\frac{1}{k}} = e^{\lim\limits_{k\to 0}\frac{1}{k}(\sin 2k)} = e^2$

79. **[A-q] [B-r, s] [C-p] [D-p]**

(A) $I_1=\int_{\pi/6}^{\pi/3}\sec^2\theta\, f(2\sin 2\theta)d\theta$

Applying property $\int_a^b f(a+b-x)dx=\int_a^b f(x)dx \Rightarrow I_1=\int_{\pi/6}^{\pi/3}\sec^2\left(\dfrac{\pi}{2}-\theta\right)f\left(2\sin 2\left(\dfrac{\pi}{2}-\theta\right)\right)d\theta$

$I_1=\int_{\pi/6}^{\pi/3}\operatorname{cosec}^2\theta\, f\left(2\sin 2\theta\right)d\theta = I_2$.

(B) $f(x+1)=f(x+3)\Rightarrow f(x)=f(x+2) \Rightarrow f(x)$ is periodic with period 2.

Then $\int_a^{a+b} f(x)dx$ is independent of a, for which b is multiple of 2.

$\Rightarrow b=2,4,6...$

(C) Let $I=\int_1^4 \dfrac{\tan^{-1}[x^2]}{\tan^{-1}[x^2]+\tan^{-1}[25+x^2-10x]} \qquad \ldots\ldots(1)$

Applying $\int_a^b f(x)dx=\int_a^b \cdot f(a+b-x)dx$, we get : $I=\int_1^4\dfrac{\tan[(5-x)^2]}{\tan^{-1}[(5-x)^2]+\tan^{-1}[x^2]}dx \qquad \ldots\ldots.(2)$

Adding equations (1) and (2), we get : $2I=\int_1^4 dx \Rightarrow 2I=3\Rightarrow I=3/2$

(D) Let $y=\sqrt{x+\sqrt{x+\sqrt{x+...}}}=\sqrt{x+y}$

$\Rightarrow \quad y^2 - y - x = 0 \qquad \Rightarrow \quad y = \dfrac{1 \pm \sqrt{1+4x}}{2.1}$

$\Rightarrow \quad y = \dfrac{1+\sqrt{1+4x}}{2} \qquad (\because y > 1)$

$$\Rightarrow \quad I = \int_0^2 \frac{1+\sqrt{1+4x}}{2}dx = \left[\frac{x}{2} + \frac{(1+4x)^{3/2}}{\frac{3}{2}.2.4}\right]_0^2 = \left[\left(1+\frac{27}{12}\right) - \left(0+\frac{1}{12}\right)\right] = 1 + \frac{26}{12} = \frac{19}{6}$$

80. **[A-p, q] [B-p, q, r] [C-q, s] [D- s]**

(A) $I = \int_{-2}^{2}(\alpha x^3 + \beta x + \gamma)dx$

$\alpha x^3 + \beta x$ is an odd function

$I = 0 + 2\int_0^2 \gamma dx = 2.2\gamma = 4\gamma$

(B) $I = \dfrac{1}{2}\int_0^1 2\sin\alpha x \sin\beta x\, dx = \dfrac{1}{2}\int_0^1 (\cos(\alpha-\beta)x - \cos(\alpha+\beta)x)dx = \dfrac{1}{2}\left[\dfrac{\sin(\alpha-\beta)x}{\alpha-\beta} - \dfrac{\sin(\alpha+\beta)x}{\alpha+\beta}\right]_0^1$

$= \dfrac{1}{2}\left[\dfrac{\sin(\alpha-\beta)}{\alpha-\beta} - \dfrac{\sin(\alpha+\beta)}{\alpha+\beta}\right]$

Also, $2\alpha = \tan\alpha$ and $2\beta = \tan\beta$ $\Rightarrow 2(\alpha-\beta) = \tan\alpha - \tan\beta$ and $2(\alpha+\beta) = \tan\alpha + \tan\beta$

$2(\alpha-\beta) = \dfrac{\sin(\alpha-\beta)}{\cos\alpha\cos\beta}$ and $2(\alpha+\beta) = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$

Substituting these values, we get : $I = (\cos\alpha\cos\beta) - (\cos\alpha\cos\beta) = 0.$

(C) $f(x+\alpha) + f(x) = 0 \Rightarrow f(x+2\alpha) + f(x+\alpha) = 0 \Rightarrow f(x+2\alpha) = f(x)$

$\Rightarrow f(x)$ is periodic with period 2α $\Rightarrow \int_{\beta}^{\beta+2\gamma\alpha} f(x)dx = \gamma\int_0^{2\alpha} f(x)dx.$

(D) Let $I = \int_0^{\alpha}[\sin x]dx, \alpha \in [(2\beta+1)\pi, (2\beta+2)\pi], \beta \in N,$ [where $[\cdot]$ denotes the greatest integer function.]

$I = \int_0^{2\beta\pi}[\sin x]dx + \int_{2\beta\pi}^{(2\beta+1)\pi}[\sin x]dx + \int_{(2\beta+1)\pi}^{\alpha}[\sin x]dx$

$= \beta\int_0^{2\pi}[\sin x]dx + 0 + \int_{(2\beta+1)\pi}^{\alpha}(-1)dx = -\beta\pi + (2\beta+1)\pi - \alpha = (\beta+1)\pi - \alpha$

$\Rightarrow \gamma\int_0^{\alpha}[\sin x]dx$ depends on α, β and α, β

81. **[A-r] [B-p] [C-q] [D-s]**

To solve I(n) elegantly, take $f(x) = \dfrac{1}{4}x^4(\log x)^n, n \geq 1$

$$f'(x) = x^3(\log x)^n + \frac{n}{4}x^4(\log x)^{n-1}\left(\frac{1}{x}\right) = x^3(\log x)^n + \frac{n}{4}x^3(\log x)^{n-1}$$

Integrating in the interval [1, e], we get $\dfrac{1}{4}e^4 = I(n) + \dfrac{n}{4}I(n-1) \Rightarrow \dfrac{4I(n)+nI(n-1)}{e^4} = 1$

Now $\quad I(0) = \dfrac{e^4-1}{4} \Rightarrow I(1) = \dfrac{3e^4+1}{16} \Rightarrow I(2) = \dfrac{5e^4-1}{32}$

Hence $\frac{64}{5e^4-1}I(2)=2$. $\lim\limits_{n\to\infty} I(n)=\int_1^e x^3\left(\lim\limits_{n\to\infty}(\log x)^n\right)dx$ $\therefore \lim\limits_{n\to\infty} I(n)=0$

Finally, observe that $I(2)=\frac{5e^4-1}{32}>\frac{4}{3}$ (verify)

Now $I(3)=\frac{e^4-3I(2)}{4}<\frac{e^4-4}{4}$ $\therefore n=3$

82.(3) We have $f(x)=\sin x+\int\limits_{-\pi/2}^{\pi/2}(\sin x+t\,f(t))dt=\sin x+\pi\sin x+\int\limits_{-\pi/2}^{\pi/2}t\,f(t)dt$

$\therefore f(x)=(\pi+1)\sin x+A$

Now, $A=\int\limits_{-\pi/2}^{\pi/2}t((\pi+1)\sin t+A)dt=2(\pi+1)\left(\int\limits_0^{\pi/2}\underset{(I)}{t}\,\underset{(II)}{\sin t}\,dt\right)$ (By part)

$\Rightarrow A=2(\pi+1)$

Hence, $f(x)=(\pi+1)\sin x+2(\pi+1)$.

Therefore, $f_{\max.}=3(\pi+1)=M$

and $f_{\min.}=(\pi+1)=m$. $\Rightarrow$ $\frac{M}{m}=3$

83.(4) Given $f(x)=x^3-\frac{3x^2}{2}+x+\frac{1}{4}=\frac{1}{4}(4x^3-6x^2+4x+1)$

$$=\frac{1}{4}(4x^3-6x^2+4x-1+2)$$

$$f(x)=\frac{1}{4}[x^4-(1-x)^4]+\frac{2}{4}$$

$$\therefore \quad f(1-x)=\frac{1}{4}[(1-x)^4-x^4]+\frac{2}{4}$$

$$\therefore \quad f(x)+f(1-x)=\frac{2}{4}+\frac{2}{4}=1 \qquad (1)$$

Replacing x by $f(x)$ we have

$$f[f(x)]+f[1-f(x)]=1 \qquad (2)$$

Now $I=\int\limits_{1/4}^{3/4}f(f(x))dx$ (3)

Also, $I=\int\limits_{1/4}^{3/4}f(f(1-x))dx=\int\limits_{1/4}^{3/4}f(1-f(x))dx$ (4) [using (1)]

Adding (3) and (4), $2I=\int\limits_{1/4}^{3/4}\left[f(f(x))+f(1-f(x))\right]dx=\int\limits_{1/4}^{3/4}dx$

$\Rightarrow 2I=\frac{1}{2}\Rightarrow I=\frac{1}{4}$

$\therefore I=\frac{1}{4}\ \therefore I^{-1}=4$

84.(6) Given $f^3(x)=\int_0^x t.f^2(t)dt$

Differentiating, $3f^2(x)f'(x)=xf^2(x)$

$$f(x)\neq 0 \therefore f'(x)=\frac{x}{3};\ \therefore f(x)=\frac{x^2}{6}+C$$

But $f(0)=0\Rightarrow C=0$

$f(6)=6$

85.(8) Let $I=\int_0^1 {}^{207}C_7.\underbrace{x^{200}}_{II}.\underbrace{(1-x)^7}_{I}dx$

$$I={}^{207}C_7\left[\underbrace{\left.(1-x)^7\cdot\frac{x^{201}}{201}\right|_0^1}_{zero}+\frac{7}{201}\int_0^1(1-x)^6\cdot x^{201}dx\right]={}^{207}C_7\cdot\frac{7}{201}\int_0^1(1-x)^6\cdot x^{201}dx$$

Integrating by parts again 6 more times $={}^{207}C_7\cdot\frac{7!}{201.202.203.204.205.206.207}\int_0^1 x^{207}dx$

$$\frac{(207)!}{7!(200)!}\cdot\frac{7!}{201.202\cdots207}\cdot\frac{1}{208}=\frac{(207)!}{(207)!7!}\cdot\frac{7!}{208}=\frac{1}{208}=\frac{1}{k}\Rightarrow k=208$$

86.(8) $I=\lim_{n\to\infty}\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\cdots+\sqrt{6n}}{n\sqrt{n}}=\lim_{n\to\infty}\frac{1}{n}\sum\sqrt{\frac{r}{n}}=\int_0^6\sqrt{x}\,dx=\left[\frac{2}{3}x^{3/2}\right]_0^6=\frac{2}{3}\cdot 6\sqrt{6}=\sqrt{96}$

87.(4) $I_1=\int_0^1 x^{1004}(1-x)^{1004}dx=2\int_0^{1/2}x^{1004}(1-x)^{1004}dx$ (1)

And $I_2=\int_0^1 x^{1004}(1-x^{2010})^{1004}dx$ Put $x^{1005}=t$ $\Rightarrow 1005x^{1004}dx=dt$

$$\Rightarrow I_2=\frac{1}{1005}\int_0^1(1-t^2)^{1004}dt=\frac{1}{1005}\int_0^1\left(t(2-t)\right)^{1004}dt=\frac{1}{1005}\int_0^1 t^{1004}(2-t)^{2004}dt$$

Now put $t=2y$ $\Rightarrow dt=2dy$

$$\Rightarrow I_2=\frac{1}{1005}\int_0^{1/2}(2y)^{1004}(2-2y)^{1004}dt=\frac{1}{1005}2\cdot2^{1004}\cdot2^{1004}\int_0^{1/2}y^{1004}(1-y)^{1004}dy$$

$$=\frac{1}{1005}2^{2009}\int_0^{1/2}y^{1004}(1-y)^{1004}dy=\frac{1}{1005}2^{2008}I_1\Rightarrow\frac{I_1}{I_2}=\frac{1005}{2^{2008}}\Rightarrow\frac{2^{2010}}{1005}\frac{I_1}{I_2}=4$$

88.(0) We have $J=\int_{-5}^{-4}(3-x^2)\tan(3-x^2)dx.$

Put $(x+5)=t$, we get : $J=\int_0^1(3-(t-5)^2)\tan(3-(t-5)^2)dt=\int_0^1(-22+10t-t^2)\tan(-22+10t-t^2)dt.$

Now, $K=\int_{-2}^{-1}(6-6x+x^2)\tan(6x-x^2-6)dx.$

Put $(x+2)=z,$ we get :

$$K=\int_0^1(6-6(z-2)+(z-2)^2)\tan(6(z-2)-(z-2)^2-6)dz=\int_0^1(22-10z+z^2)\tan(-22+10z-z^2)dz$$

Hence, $(J+K)=0.$

89.(1) $I_n=\int_0^1 x^n(1-x^2)^{1/2}dx$

$$=\int_0^1 x^{n-1}\cdot x(1-x^2)^{1/2}dx=\left(-x^{n-1}\frac{(1-x^2)^{3/2}}{3}\right)_0^1+\int_0^1(n-1)x^{n-2}\frac{(1-x^2)^{3/2}}{3}dx$$

$$=O+\left(\frac{n-1}{3}\right)\int_0^1 x^{n-2}(1-x^2)\sqrt{1-x^2}\,dx$$

$3I_n+(n-1)I_n=(n-1)I_{n-2}$

$$\frac{I_n}{I_{n-2}}=\frac{(n-1)}{(n+2)}\Rightarrow\lim_{n\to\infty}\frac{I_n}{I_{n-2}}=1$$

90.(1) $I=\int_0^n f'(x)[x]dx-\int_0^n xf'(x)dx+\frac{1}{2}\int_0^n f'(x)dx$

$$\sum_{r=1}^{n}\int_{r-1}^{r}f'(x)[x]dx-\left\{(xf(x))_0^n-\int_0^n f(x)dx\right\}+\frac{1}{2}(f(x))_0^n$$

$$=\sum_{r=1}^{n}(r-1)\int_{r-1}^{r}f'(x)dx-nf(n)+\frac{1}{2}f(n)-\frac{1}{2}f(0)+\int_0^n f(x)dx$$

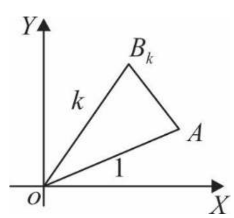
$$=\sum_{r=1}^{n}(r-1)\{f(r)-f(r-1)\}-nf(n)+\frac{1}{2}f(n)+\int_0^n f(x)dx-\frac{1}{2}f(0)$$

$$=-f(1)-f(2)-.......-f(n-1)-f(n)+\frac{1}{2}f(n)+\frac{1}{2}f(0)+\int_0^n f(x)dx$$

$$=-\sum_{r=1}^{n}f(r)+\frac{1}{2}f(n)+\frac{1}{2}f(0)+\int_0^n f(x)dx$$

91.(2) $OB_k=k,\ \angle AOB_k=\frac{k\pi}{2n}$

$S_k=\frac{1}{2}k\sin\frac{k\pi}{2n}$ (using $\Delta=\frac{1}{2}ab\sin\theta$)



$\therefore$ $$L=\lim_{n\to\infty}\frac{1}{n^2}\sum_{k=1}^{n}S_k=\frac{k}{2n^2}\sum_{n=1}^{\infty}\sin\frac{k\pi}{2n}$$

$$=\frac{1}{2n}\sum_{n=1}^{\infty}\frac{k}{n}\sin\frac{k\pi}{2n}=\frac{1}{2}\int_0^1 x.\sin\frac{\pi x}{2}dx=\frac{1}{2}\left[\frac{-2}{\pi}x\cos\frac{\pi x}{2}\bigg|+\frac{2}{\pi}\int_0^1 x.\sin\frac{\pi x}{2}dx\right]=\frac{1}{2}\left[0+\frac{2}{\pi}.\frac{2}{\pi}\left(\sin\frac{\pi x}{2}\right)_0^1\right]=\frac{2}{\pi^2}.$$

92.(0.72) $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{{}^nC_k}{n^k(k+3)}=\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k+3}{}^nC_k.\frac{1}{n^k}$

$=\lim_{n\to\infty}\sum_{k=0}^{n}{}^nC_k.\frac{1}{n^k}\int_0^1 x^{k+2}dx\left\{\because \int_0^1 x^{k+2}dx=\frac{1}{k+3}\right\}$

$=\int_0^1\left(x^2\lim_{n\to\infty}\sum_{k=0}^{n}{}^nC_k.\left(\frac{x}{n}\right)^k\right)dx=\int_0^1 x^2\left\{\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n\right\}dx=\int_0^1 x^2.e^x dx$ $\qquad\left\{\because \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n=e^x\right\}$

$=\left(x^2.e^x\right)_0^1-\int_0^1 2xe^x dx=e-2$

93.(3.14) $f'(x)+f^2(x)\geq -1$

$\therefore$ $f^2(x)+1\geq -f'(x)$ in (a,b)

$1\geq -\frac{f'(x)}{1+f^2(x)}$ in (a,b)

$\therefore$ $\int_a^b dx\geq -\int_a^b\frac{f'(x)}{1+f^2(x)}dx$

$b-a\geq -(\tan^{-1}(f(x))_a^b=-\left[\left(-\frac{\pi}{2}\right)-\frac{\pi}{2}\right]$

$\therefore$ $(b-a)\geq \pi.$

94.(101) Given $(f(x))^{101}=1+\int_0^x f(t)dt$

Differentiating $101.(f(x))^{100}.f'(x)=f(x)$

$\therefore$ $101.(f(x))^{99}.f'(x)=1(as\ f(x)>0)$

Integrating, $\frac{(101)(f(x))^{100}}{100}=x+C$

but $f(0)=1$

$\Rightarrow$ $C=\frac{101}{100}$

$\therefore$ $\frac{101}{100}(f(x))^{100}=x+\frac{101}{100}$

Putting $x=101$,

$\frac{101}{100}(f(101))^{100}+\frac{101}{100}=\frac{(101)(101)}{100}$

$\therefore$ $(f(101))^{100}=101.$

95.(4) We have $\int_x^{xy} f(t)dt$.

Differentiating with respect of x, treating y as constant, we get

$$\frac{d}{dx}\int_x^{xy} f(t)dt = 0 \Rightarrow \quad yf(xy) - f(x) = 0 \qquad \ldots(1)$$

Putting $y = \frac{1}{x}$ in equation (1), we have $\frac{1}{x} f(1) - f(x) = 0$

$$\Rightarrow \quad f(x) = f(1).\frac{1}{x} \quad \Rightarrow \quad \int_1^x f(x)dx = f(1)\int_1^x \frac{1}{x}dx = f(1).\ln x$$

Now, putting $y = \frac{1}{2}$ and $x = 2$ in equation (1),

We have $\frac{1}{2} f(1) - f(2) = 0 \Rightarrow \quad f(1) = 2f(2) = 4$ Hence, we have $\int_1^x f(x)dx = 4\ln x$.

96.(8) Let $f(x) = ax^2 + bx + c$

$f'(x) = 2ax + b$

$f'(2) = 4a + b = 1 \qquad \ldots.(1)$

now $I = \int_{2-\pi}^{2+\pi} f(x)\cdot\sin\left(\frac{x-2}{2}\right)dx$ put $x - 2 = t \Rightarrow dx = dt$

$$I = \int_{-\pi}^{\pi} f(2+t)\cdot\sin\left(\frac{t}{2}\right)dt \qquad \ldots.(2)$$

Also, $I = \int_{-\pi}^{\pi} f(2-t)\cdot\sin\left(-\frac{t}{2}\right)dt$ (using $a+b-x$ property)

$$I = -\int_{-\pi}^{\pi} f(2-t)\cdot\sin\left(\frac{t}{2}\right)dt \qquad \ldots.(3)$$

(2) + (3)

$$2I = \int_{-\pi}^{\pi} [f(2+t) - f(2-t)]\sin\frac{t}{2}dt \qquad \ldots.(4)$$

now $f(2+t) - f(2-t)$

$$\left[a(2+t)^2 + b(2+t) + c\right] - \left[a(2-t)^2 + b(2-t) + c\right]$$

$$a\left[(2+t)^2 - (2-t)^2\right] + b[2+t-2+t]$$

$= 8at + 2bt = 2t(4a+b) = 2t$ (as $4a + b = 1$)

hence $2I = \int_{-\pi}^{\pi} 2t\cdot\sin\frac{t}{2}dt \Rightarrow I = \int_{-\pi}^{\pi} t\cdot\sin\frac{t}{2}dt = 2\int_0^{\pi} t\cdot\sin\frac{t}{2}dt$ (put $\frac{t}{2} = y$)

$$I = 8\int_0^{\pi/2} y\sin y\, dy \Rightarrow I = 8 \text{ as } \int_0^{\pi/2} y\sin y\, dy = 1$$

97.(10.50) $\displaystyle\int_0^{\pi/2} (\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\, dx = 2$

Let $\displaystyle I_1 = \int_0^{\pi/2} (\sin x + a\cos x)^3 dx = \int_0^{\pi/2} (\sin^3 x + a^3\cos^3 x + 3a\sin^2 x\cos x + 3a^2 \sin x\cos^2 x)dx$

$$= \int_0^{\pi/2} \sin^3 x\, dx \ + a^3 \int_0^{\pi/2} \cos^3 x\, dx + \ 3a \int_0^{\pi/2} \sin^2 x\cos x\, dx \ + 3a^2 \int_0^{\pi/2} \sin x\cos^2 x\, dx$$

$$= \frac{2}{3} + \ a^3\left(\frac{2}{3}\right) \ + 3a\int_0^{\pi/2} (1-\cos^2 x)\cos x\, dx \ + 3a^2 \int_0^{\pi/2} \sin x(1-\sin^2 x)\, dx$$

$$= \frac{2}{3}\left(1+a^3\right) + 3a\ \left(1-\frac{2}{3}\right) + 3a^2\left(1-\frac{2}{3}\right) = \frac{2}{3} + \frac{2a^2}{3} + a + a^2$$

$$I_1 = \frac{2a^3}{3} + a^2 + a + \frac{2}{3} \ ; \quad I_2 = \int_0^{\pi/2} \underbrace{x}_{I} \cdot \underbrace{\cos x}_{II}\, dx \ = x\sin x\Big|_0^{\pi/2} - \int_0^{\pi/2} \sin x\, dx$$

$$I_2 = x\sin x + \cos x\Big|_0^{\pi/2} ; \quad I_2 = \frac{\pi}{2} - 1 = \frac{\pi-2}{2}$$

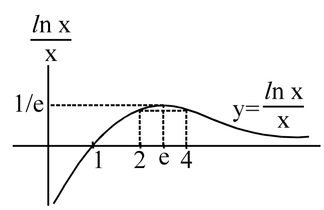
$$I = \frac{2a^3}{3} + a^2 + a + \frac{2}{3} - \frac{4a}{\pi-2}.\frac{\pi-2}{2} \quad \Rightarrow \quad \frac{2a^3}{3} + a^2 - a + \frac{2}{3} = 2$$

$$\Rightarrow \qquad 2a^3 + 3a^2 - 3a + 2 = 6 \qquad \Rightarrow \qquad 2a^3 + 3a^2 - 3a - 4 = 0$$

$$a_1 + a_2 + a_3 = -\frac{3}{2} \qquad \Rightarrow \sum a_1 a_2 \ = -\frac{3}{2} \qquad \therefore \qquad \sum {a_1}^2 = \frac{9}{4} + \frac{6}{2} + \frac{21}{4}$$

98.(2) $\displaystyle I = \int_0^{\pi/2} \frac{\ell n\,(x\tan\theta)\cdot x\sec^2\theta}{x^2(1+\tan^2\theta)}\, d\theta = \frac{1}{x}\int_0^{\pi/2} (\ln x + \ln\tan\theta)\, d\theta$

$$= \frac{\ell n\, x}{x}\int_0^{\pi/2} d\theta + \frac{1}{x}\int_0^{\pi/2} \ln\tan\theta\, d\theta = \frac{\pi}{2}\frac{\ell n\, x}{x} + \ \text{zero}$$



Hence $\displaystyle \frac{\pi}{2}\frac{\ell n\, x}{x} = \frac{\pi}{4}\ell n\, 2 \Rightarrow \frac{\ell n\, x}{x} = \frac{\ell n\, 2}{2} \Rightarrow x = 2 \ \text{ or } \ 4$

Note from the graph of $y = \dfrac{\ell n\, x}{x}$ that for all values of $y = \dfrac{\ell n\, x}{x} \in \left(0, \dfrac{1}{e}\right)$, there can be two values of x on either side of $x = e$ for which $\dfrac{\ell n\, x}{x}$ will have the same value.]

99.(0) $\displaystyle F(x) = \int_{-1}^{x} f(t)\, dt$ and $\displaystyle G(x) = \int_x^1 f(t)\, dt$ where $f(t) = \sqrt{4+t^2}$

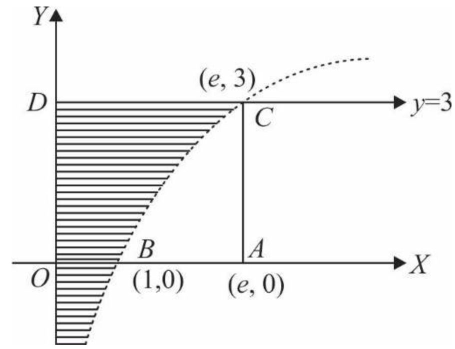
now $\quad H(x) = F(x).G(x)\,; \qquad H'(x) = F(x).G'(x) + G(x).F(x)$

$$H'(x) = \left(\int_{-1}^{x} f(t)\, dt\right)\left(-\sqrt{4+x^2}\right) + \left(\int_x^1 f(t)\, dt\right)\left(\sqrt{4+x^2}\right)$$

$$H'(x)=\sqrt{4+x^2}\left[\int_x^1\sqrt{4+t^2}\,dt-\int_{-1}^{x}\sqrt{4+t^2}\,dt\right]\;;H'(0)=2\left[\int_0^1\sqrt{4+t^2}\,dt-\int_{-1}^{0}\sqrt{4+t^2}\,dt\right]$$

put $\quad t=-y \quad =\left[\int_0^1\sqrt{4+t^2}\,dt+\int_1^0\sqrt{4+y^2}\,dy\right] \quad =\left[\int_0^0\sqrt{4+t^2}\,dt\right]=$ zero

100.(8.15) Given, $f\left(\dfrac{x}{y}\right)=f(x)-f(y)$ Putting, $x=y=1, f(1)=0$



Now, $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}=\lim\limits_{h\to 0}\dfrac{f\left(1+\dfrac{h}{x}\right)}{h}$ [from Eq. (i)]

$$=\lim_{h\to 0}\frac{f\left(1+\dfrac{h}{x}\right)}{\dfrac{h}{x}.x}$$

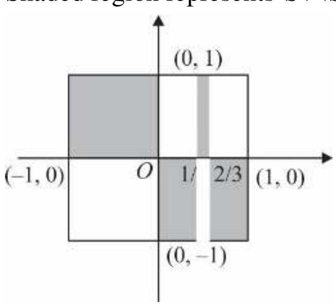
$$\Rightarrow\quad f'(x)=\frac{3}{x} \qquad \left[\sin ce,\ \lim_{x\to 0}\frac{f(1+x)}{x}=3\right]$$

$$\Rightarrow\quad f(x)=3\log x+c$$

Putting $x=1\Rightarrow c=0 \;\Rightarrow\; f(x)=3\log x=y$

$\therefore$ Required area $\displaystyle\int_{-\infty}^{3}x\,dy=\int_{-\infty}^{3}e^{y/3}dy=3\left[e^{y/3}\right]_{-\infty}^{3}=3(e-0)=3e$ sq units

101.(2) Shaded region represents $S\cap S'$ clearly are enclosed is 2 sq units.



102.(1) Differentiating both sides w.r.t. x

$$2f(x).f'(x)=(f(x))^2+(f'(x))^2 \quad\text{or}\quad (f(x)-f'(x))^2=0 \quad\Rightarrow\quad f'(x)=f(x)$$

(from the given relation $f(0)^2=e^2\Rightarrow f(0)=e$ or $-e$ (to be rejected))

now $\quad\dfrac{f'(x)}{f(x)}=1 \quad\Rightarrow\quad \ln(f(x))=x+C;$ but $f(0)=e$

$\therefore\quad \ln(e)=C \quad\Rightarrow\quad C=1 \quad\therefore\quad \ln(f(x))=x+1 \;\Rightarrow\; f(x)=e^{x+1}$

103.(8) $I=\displaystyle\int_1^2\frac{(x^2-1)\,dx}{x^5\cdot\sqrt{2-2x^{-2}+x^{-4}}}$ (taking x^2 out from the radical sign)

$$=\int_1^2\frac{(x^{-3}-x^{-5})\,dx}{\sqrt{2-2x^{-2}+x^{-4}}}$$

put $2-2x^{-2}+x^{-4}=t^2$; when $x=2,\ t\to 5/4$

when $x=1, t\to 1$

$$+4\left(x^{-3}-x^{-5}\right)dx=2tdt$$

$$\left(x^{-3}-x^{-5}\right)dx=\frac{tdt}{2}=\frac{1}{2}\int_1^{5/4}\frac{t\,dt}{t}=\frac{1}{2}[t]_1^{5/4}=\left(\frac{5}{4}-1\right).\frac{1}{2}=\frac{1}{8}$$

104.(85) $\frac{d}{dx}(h(x))=-\frac{\sin x}{\cos^2(\cos x)}\Rightarrow -\int\frac{\sin x}{\cos^2(\cos x)}dx=h(x)$

$$\therefore\quad h(x)=-\int\frac{\sin x}{\cos^2(\cos x)}dx\ ;\qquad h(x)=\tan(\cos x)+C\qquad ....(1)$$

comparing (1) with $h(x)=(fog)(x)+K$

we have $f(x)=\tan x$ and $g(x)=\cos x$

hence $j(0)=\int_{f(0)}^{g(0)}\frac{\tan t}{\cos t}dt=\int_0^1(\tan t\ \sec t)dt=\ \sec t\Big]_0^1=\sec(1)-1$

105.(101) $I=\int_0^\infty\frac{dx}{x^2+\frac{1}{x^2}+(a^2-2)}=\int_0^\infty\frac{x^2\,dx}{x^4+(a^2-2)x^2+1}\qquad\left(a^2-2=k\geq 0\right)$

$$=\int_0^\infty\frac{x^2\,dx}{x^4+kx^2+1}=\frac{1}{2}\int_0^\infty\frac{(x^2+1)+(x^2-1)}{x^4+kx^2+1}dx=\underbrace{\frac{1}{2}\int_0^\infty\frac{1+(1/x^2)}{x^2+(1/x^2)+k}dx}_{I_1}+\underbrace{\frac{1}{2}\int_0^\infty\frac{1-(1/x^2)}{x^2+(1/x^2)+k}dx}_{I_2}$$

now proceed, $I_1=\frac{\pi}{2a}$ and $I_2=0$

$$\therefore\ \boxed{I=\frac{\pi}{2a}}\ ;\frac{\pi}{2a}=\frac{\pi}{5050}\Rightarrow\quad a=2525$$

106.(153) $I=\int_0^\pi\sqrt{3+2(\cos x+\cos x+\cos 2x)}\,dx=\int_0^\pi\sqrt{3+2(2\cos x+2\cos^2 x-1)}\,dx$

$$=\int_0^\pi\sqrt{4\cos^2 x+4\cos x+1}\,dx=\int_0^\pi|1+2\cos x|dx=\int_0^{2\pi/3}(1+2\cos x)\,dx-\int_{2\pi/3}^{\pi}(1+2\cos x)\,dx$$

$$=x+2\sin x\Big|_0^{2\pi/3}-x+2\sin x\Big|_{2\pi/3}^{0}=\left(\frac{2\pi}{3}+\sqrt{3}\right)-\left(\pi-\frac{2\pi}{3}-\sqrt{3}\right)=\frac{\pi}{3}+2\sqrt{3}=\frac{\pi}{3}+\sqrt{12}$$

$$\therefore\quad k=3\ \text{and}\ w=12\ \Rightarrow\ \left(k^2+w^2\right)=9+144=153$$

107.(61) Consider $I_1=\int\frac{\cos 2x\cos x}{\sin^7 x}dx=\int\frac{(1-2\sin^2 x)\cos x}{\sin^7 x}dx$

$x=t\Rightarrow\ \sin xdx=dt$

$$I_1=\int\frac{(1-2t^2)dt}{t^7}=\int t^{-7}\,dt\ -2\int t^{-5}\,dt=\frac{2}{4}\frac{1}{t^4}\ -\frac{1}{6t^6}=\frac{1}{2\sin^4 x}-\frac{1}{6\sin^6 x}=\frac{\operatorname{cosec}^4 x}{2}-\frac{\operatorname{cosec}^6 x}{6}$$

now integrating by parts the integral

$$I=\int_{\pi/4}^{\pi/2}\underset{I}{x}\cdot\underbrace{\frac{\cos 2x\cos x}{\sin^7 x}}_{II}dx=x\left(\frac{\text{cosec}^4 x}{2}-\frac{\text{cosec}^6 x}{6}\right)\Bigg|_{\pi/4}^{\pi/2}-\int_{\pi/4}^{\pi/2}\left(\frac{\text{cosec}^4 x}{2}-\frac{\text{cosec}^6 x}{6}\right)dx$$

$$I=\underbrace{\left[\left\{\frac{\pi}{2}\left(\frac{1}{2}-\frac{1}{6}\right)\right\}-\left\{\frac{\pi}{4}\left(2-\frac{4}{3}\right)\right\}\right]}_{zero}=-J$$

now $\quad -J=\int_{\pi/4}^{\pi/2}\left(\frac{\text{cosec}^2 x}{2}-\frac{\text{cosec}^4 x}{6}\right)\text{cosec}^2 x\,dx=\int_{\pi/4}^{\pi/2}\left(\frac{1+\cot^2 x}{2}-\frac{(1+\cot^2 x)^2}{6}\right)\text{cosec}^2 x\,dx$

$\cot x=y,-\cos ec^2 xdx=dy$

$$-J=+\int_0^1\left(\frac{1+y^2}{2}-\frac{(1+y^2)^2}{6}\right)dy$$

$$-J=\frac{16}{45}\quad\Rightarrow\quad J=-\frac{16}{45}\quad\Rightarrow\quad a+b=16+45=61$$

108.(10) $I=\int_0^{\pi}\frac{x\sin^3 x}{4-\cos^2 x}dx$

$I=\int_0^{\pi}\frac{(\pi-x)\sin^3 x}{4-\cos^2 x}dx$ add $2I=\pi\int_0^{\pi}\frac{\sin^3 x}{4-\cos^2 x}dx=2\pi\int_0^{\pi/2}\frac{\sin^3 x}{4-\cos^2 x}dx$; $I=\pi\int_0^{\pi/2}\frac{\sin^3 x}{4-\cos^2 x}dx$

put $\cos x=t$

$$I=\pi\int_0^1\frac{1-t^2}{4-t^2}dt=\pi\int_0^1\frac{4-t^2-3}{4-t^2}dt=\pi\left[1-\int_0^1\frac{3\,dt}{4-t^2}\right]=\pi\left[1-3\frac{1}{2\cdot 2}\ln\frac{2+t}{2-t}\right]_0^1=\pi\left[1-\frac{3\ln 3}{4}\right]$$

hence $\quad a=3,b=3,c=4\quad\Rightarrow\quad a+b+c=10$

109.(4.14) We have $f(x)=\sin x+\int_{-\pi/2}^{\pi/2}(\sin x+tf(t))dt.$

$=\sin x+\pi\sin x+\int_{-\pi/2}^{\pi/2}tf(t)dt$

$\therefore\quad f(x)=(\pi+1)\sin x+A$

Now, $A=\int_{-\pi/2}^{\pi/2}t((\pi+1)\sin t+A)dt=2(\pi+1)\int_0^{\pi/2}t\sin t\,dt$

$\Rightarrow\quad A=2(\pi+1)$

Hence, $f(x)=(\pi+1)\sin x+2(\pi+1)$

$\therefore\quad f_{\max}=3(\pi+1)$ and $f_{\min}=(\pi+1)$.

DIFFERENTIAL EQUATIONS

1.(C) $\dfrac{xdy}{x^2+y^2}=\left(\dfrac{y}{x^2+y^2}-1\right) \Rightarrow \dfrac{xdy-ydx}{x^2+y^2}=-dx$

$\int d\left(\tan^{-1}\left(\dfrac{y}{x}\right)\right)=-\int dx \Rightarrow \tan^{-1}\left(\dfrac{y}{x}\right)=-x+c \quad \Rightarrow \dfrac{y}{x}=\tan(c-x)$

2.(C) The given equation can be written as $y(1+x^{-1})dx+(x+\log x)dx+\sin ydx+x\cos ydy=0$

$\Rightarrow \quad d(y(x+\log x))+d(x\sin y)=0 \Rightarrow \quad y(x+\log x)+x\sin y=C$

3.(B) Rewriting the given equation the form $x^4\cos y\dfrac{dy}{dx}+4x^3\sin y=xe^x \Rightarrow \dfrac{d}{dx}(x^4\sin y)=xe^x$

$\Rightarrow \quad x^4\sin y=\int xe^x dx+C=(x-1)e^x+C$

Since $y(1)=0$, so $C=0$ Thus $\sin y=x^{-4}(x-1)e^x$

4.(A) Taking, $x=r\cos\theta$ and $y=r\sin\theta$, so that $x^2+y^2=r^2$ and $y/x=\tan\theta$, we have $xdx+ydy=rdr$ and $xdy-ydx=x^2\sec^2\theta d\theta=r^2 d\theta.$

The given equation can be transformed into $\dfrac{rdr}{r^2d\theta}=\sqrt{\dfrac{a^2-r^2}{r^2}} \Rightarrow \dfrac{dr}{d\theta}=\sqrt{a^2-r^2}$

$\Rightarrow \quad C+\sin^{-1}r/a=\theta=\tan^{-1}y/x$

$\Rightarrow \quad y=x\tan\left(C+\sin^{-1}\dfrac{1}{a}\sqrt{x^2+y^2}\right)$ or $\sqrt{x^2+y^2}=a\sin\left(\text{const.}+\tan^{-1}\dfrac{y}{x}\right)$

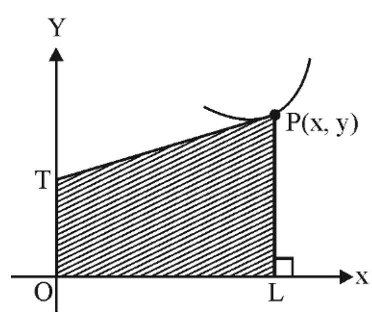
5.(A) Let $P(x,y)$ be any point on the curve. Length of intercept on y-axis by any tangent at

$$P(x,y)=OT=y-x\frac{dy}{dx}$$

$\therefore$ Area of trapezium OLPTO $=\dfrac{1}{2}(PL+OT)OL=\dfrac{1}{2}\left(y+y-x\dfrac{dy}{dx}\right)x=\dfrac{1}{2}\left(2y-x\dfrac{dy}{dx}\right)x$

According to question,

Area of trapezium OLPTO $=\dfrac{1}{2}x^2$



i.e., $\dfrac{1}{2}\left(2y-x\dfrac{dy}{dx}\right)x=\pm\dfrac{1}{2}x^2 \Rightarrow 2y-x\dfrac{dy}{dx}=\pm x$

$\dfrac{xdy}{dx}-2y=\pm x$

$\dfrac{xdy}{dx}-\dfrac{2y}{x}=\pm 1$

I.F., $=e^{-\int\frac{2}{x}}=\dfrac{1}{x^2}$

Now solution is: $\dfrac{y}{x^2}=\int\pm\dfrac{1}{x^2}dx \quad \Rightarrow \quad \dfrac{y}{x^2}=\pm\dfrac{1}{x}+c \quad \Rightarrow \quad y=cx^2\pm x$

6.(C) The substitution $y = u^m \Rightarrow \frac{dy}{dx} = mu^{m-1}\frac{du}{dx}$ changes the equation to, $2.x^4.u^m.mu^{m-1}\frac{du}{dx} + u^{4m} = 4x^6$

$\Rightarrow \frac{du}{dx} = \frac{4x^6 - u^{4m}}{2mx^4x^{2m-1}}$. Since it is homogeneous, the degree of $4x^6 - u^{4m}$ and $2mx^4u^{2m-1}$ must be same

$\Rightarrow 6 = 4m = 4 + 2m - 1 \Rightarrow m = \frac{3}{2}$.

7.(C) The given equation is reduced to $x = e^{xy(dy/dx)} \Rightarrow \log x = xy\frac{dy}{dx}$

$$\Rightarrow \quad \int y\,dy = \int \frac{1}{x}\log_e x\,dx$$

$$\frac{y^2}{2} = \frac{(\log_e x)^2}{2} + C \Rightarrow y = \pm\sqrt{\{(\log x)^2 + C\}}$$

8.(A) The given differential equation can be written as $\frac{dx}{dy} = xy[x^2 \sin y^2 + 1]$

$\Rightarrow \frac{1}{x^3}\frac{dx}{dy} - \frac{1}{x^2}y = y\sin y^2$. This equation is reducible to linear equation, so putting $-1/x^2 = u$, the last equation can be written as $\frac{du}{dy} + 2uy = 2y\sin y^2$.

The integrating factor of this equation is e^{y^2}. So required solution is $ue^{y^2} = \int 2y\sin y^2.e^{y^2}dy + C$

$$= \int(\sin t).e^t\,dt + C = \frac{1}{2}e^{y^2}(\sin y^2 - \cos y^2) + C \qquad \text{where } (t = y^2)$$

$$\Rightarrow \quad 2u = (\sin y^2 - \cos y^2) + Ce^{-y^2} \quad \Rightarrow \quad 2 = x^2[\cos y^2 - \sin y^2 - 2Ce^{-y^2}]$$

9.(A) The general solution will be obtained by replacing p by c, where c is an arbitrary constant. So, the solution is $y = cx + \log c$.

10.(D) Differentiating the given equation we get $\left(x + \frac{1}{p}\right)\frac{dp}{dx} = 0$. For singular solution, we take $x + \frac{1}{p} = 0 \Rightarrow p = -\frac{1}{x}$.

Putting in the given equation, we obtain $y = -1 + \log\left(-\frac{1}{x}\right) \Rightarrow y + 1 = -\log(-x)$.

11.(B) Putting $\frac{dy}{dx} = p$, the equation becomes $y = xp + p^2$, which is the Clairut's equation. So, the solution is obtained by replacing p by c, so, $y = cx + c^2$, where c is the arbitrary constant.

12.(C) Singular form is $x + f'(p) = 0 \Rightarrow x + 2p = 0$

Therefore $p = -\frac{x}{2}$ putting in the given equation,

We get $y = x\left(-\frac{x}{2}\right) + \frac{x^2}{4} \Rightarrow y = -\frac{x^2}{4}$

13.(C) Put $x^2 = u$ and $y^2 = v$, then $\frac{dy}{dx} = \frac{x}{y}\left(\frac{dv}{du}\right)$

The equation then becomes, $x^2\left(y - \frac{x^2}{y}\frac{dv}{du}\right) = y.\frac{x^2}{y^2}\left(\frac{dv}{du}\right)^2 \Rightarrow y^2 - x^2\frac{dv}{du} = \left(\frac{dv}{du}\right)^2$ or $v = u\frac{dv}{du} + \left(\frac{dv}{du}\right)^2$

which is Clairut's equation in variables u and v, so the solution is $v = uc + c^2 \Rightarrow y^2 = cx^2 + c^2$

14.(D) Writing $p = \frac{dy}{dx}$ and differentiating w.r.t. x, we have $p = 2p + 2x\frac{dp}{dx} + 2xp^4 + 4p^3x^2\frac{dp}{dx}$

$\Rightarrow \quad 0 = p(1 + 2xp^3) + 2x\frac{dp}{dx}(1 + 2p^3x) \quad \Rightarrow \quad p + 2x\frac{dp}{dx} = 0 \ \Rightarrow 2\frac{dp}{p} = -\frac{dx}{x}$

$\Rightarrow \quad 2\log p + \log x = \text{const.} \Rightarrow p^2x = c$ or $p = \pm\sqrt{\frac{c}{x}}$ for $p = \sqrt{\frac{c}{x}}$ we get option 'D'

15.(B) $x\left(\cos\frac{y}{x}\right)(ydx + xdy) = y\sin\left(\frac{y}{x}\right)(xdy - ydx)$

Dividing both sides by x^2dx we get : $\left(\cos\frac{y}{x}\right)\left(\frac{y}{x} + \frac{dy}{dx}\right) = \frac{y}{x}\sin\left(\frac{y}{x}\right)\left(\frac{dy}{dx} - \frac{y}{x}\right)$

Which is homogeneous equation

Putting $y = vx$ we get :

$\therefore \quad \frac{dy}{dx} = v + x\frac{dv}{dx} \quad$ Or $\quad \cos v\left(v + v + x\frac{dv}{dx}\right) = v\sin v\left(x\frac{dv}{dx}\right)$

$\Rightarrow \quad 2v\cos v = x\frac{dv}{dx}(v\sin v - \cos v)$

Separating the variables, we get :

$\Rightarrow \quad \frac{2dx}{x} = \left(\frac{v\sin v - \cos v}{v\cos v}\right)dv$

Integrating $2\ln x + \ln c = \ln\sec v - \ln v$

$\Rightarrow \quad cx^2 = \frac{\sec v}{v} \Rightarrow \sec v = cx^2v \quad \Rightarrow \quad \sec\frac{y}{x} = cxy$

Where 'c' is an arbitrary constant.

16.(A) The given equation can be written as $\left(\frac{dx}{x} - \frac{dy}{y}\right) + \left(\frac{x^2dy - y^2dx}{(x-y)^2}\right) = 0$

or $\quad \left(\frac{dx}{x} - \frac{dy}{x}\right) + \left(\frac{\frac{dy}{y^2} - \frac{dx}{x^2}}{\left(\frac{1}{y} - \frac{1}{x}\right)^2}\right) = 0 \quad$ or $\quad \left(\frac{dx}{x} - \frac{dy}{y}\right) + \left(\frac{\frac{dy}{y^2} - \frac{dx}{x^2}}{\left(\frac{1}{x} - \frac{1}{y}\right)^2}\right) = 0$

Integrating, we get $\ln|x| - \ln|y| - \frac{1}{\left(\frac{1}{x} - \frac{1}{y}\right)} = c \quad$ or $\quad \ln\left|\frac{x}{y}\right| + \frac{xy}{(x-y)} = c$

Where 'c' is an arbitrary constant.

17.(C) Put $x = r\sec\theta$ and $y = r\tan\theta$. So, $x^2 - y^2 = r^2$ .....(1)

and $\sin\theta = \dfrac{y}{x}$ .....(2)

Then differentiating (1) we get : $2x\,dx - 2y\,dy = 2r\,dr$

Or $x\,dx - y\,dy = r\,dr$ .....(3)

And differentiating (2) we get : $\dfrac{x\,dy - y\,dx}{x^2} = \cos\theta\,d\theta$

Or $x\,dy - y\,dx = x^2\cos\theta\,d\theta = r^2\sec^2\theta\cos\theta d\theta = r^2\sec\theta d\theta$ .....(4)

Substituting values from (3) and (4) in the given differential equation, we get :

$$\frac{rdr}{r^2\sec^2\theta d\theta} = \sqrt{\left(\frac{1+r^2}{r^2}\right)} = \frac{\sqrt{1+r^2}}{r} \quad \text{or} \quad \frac{dr}{\sqrt{(1+r^2)}} = \sec\theta\,d\theta$$

Integrating both sides $\ln(r + \sqrt{(1+r^2)}) = \ln(\sec\theta + \tan\theta) + \ln c$

Where c is an arbitrary constant.

or $(r + \sqrt{(1+r^2)}) = c(\sec\theta + \tan\theta)$ or $\sqrt{(x^2-y^2)} + \sqrt{(1+x^2-y^2)} = c\left(\dfrac{x+y}{\sqrt{(x^2-y^2)}}\right)$

18.(D) Differentiating both sides w.r.t, x of the given equation

$$x.y(x) + \int_0^x y(t)dt.1 = (x+1)xy(x) + \int_0^x ty(t)dt \quad \text{or} \quad \int_0^x y(t)dt = x^2y(x) + \int_0^x ty(t)dt$$

Again differentiating both sides w.r.t., x

$$y(x) = x^2y(x) + y(x)2x + xy(x)$$

or $(1-3x)y(x) = x^2y'(x)$ or $\dfrac{y'(x)}{y(x)} = \left(\dfrac{1}{x^2} - \dfrac{3}{x}\right)$

Integrating, we get $\ln y(x) = -\dfrac{1}{x} - 3\ln x + \ln c$

or $\ln\left(\dfrac{x^3y(x)}{c}\right) = -\dfrac{1}{x}$ or $\dfrac{x^3y(x)}{c} = e^{-1/x}$ or $y(x) = \dfrac{ce^{-1/x}}{x^3}$

19.(B) Put $x = r\cos\theta$ and $y = r\sin\theta \Rightarrow x^2 + y^2 = r^2$ ....(1)

and $\theta = \tan^{-1}\left(\dfrac{y}{x}\right)$ .... (2)

then differentiating (1) we get :

$$2x\,dx + 2y\,dy = 2r\,dr$$

or $x\,dx + y\,dy = r\,dr$ ....(3)

and differentiating (2) we get :

$\dfrac{x\,dy + y\,dy}{x^2} = \sec^2\theta\,d\theta$ or $x\,dy + y\,dx = x^2\sec^2\theta\,d\theta$ ....(4)

Substituting values from (3) and (4) in the given differential equation, we get :

$$\frac{r\,dr}{r^2\,d\theta} = \sqrt{\left(\frac{a^2 - r^2}{r^2}\right)} = \frac{\sqrt{(a^2 - r^2)}}{r} \qquad \text{or} \qquad \frac{dr}{\sqrt{(a^2 - r^2)}} = d\theta$$

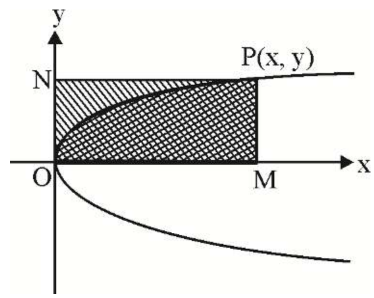
Integrating both sides we gets : $\sin^{-1}\left(\frac{r}{a}\right) = \theta + c \Rightarrow r = a\sin(\theta + c)$

or $\quad \sqrt{(x^2 + y^2)} = a\sin\left(\tan^{-1}\left(\frac{y}{x}\right) + c\right)$ {from (1) and (2)} where c is an arbitrary constant.

20.(C) Let $P(x, y)$ be the point on the curve passing through the origin $O(0,0)$, and let PN and PM be the lines parallel to the x-and y-axes, respectively. If the equation of the curve is $y = y(x)$, then the area POM equals

$\int_0^x y\,dx$ and the area PON equals $xy - \int_0^x y\,dx$.

Assuming that 2(POM) = PON, we there fore have



$$2\int_0^x y\,dx = xy - \int_0^x y\,dx \qquad \Rightarrow 3\int_0^x y\,dx = xy$$

Differentiating both sides of this gives

$$3y = x\frac{dy}{dx} + y \qquad \Rightarrow 2y = x\frac{dy}{dx} \Rightarrow \frac{dy}{y} = 2\frac{dx}{x}$$

$\Rightarrow \qquad \log y = 2\log x + C \Rightarrow y = Cx^2$,

With C being a constant. This solution represents a parabola. We will get a similar result if we has started instead with 2(PON) = POM.

21.(A) Differentiating the given equation, we have $2x + 2y\frac{dy}{dx} + 2g = 0 \quad \Rightarrow g = -\left(x + y\frac{dy}{dx}\right)$

Putting this value in $x^2 + y^2 + 2gx + C = 0$, we have

$$x^2 + y^2 - 2x\left(x + y\frac{dy}{dx}\right) + C = 0$$

Replacing $\frac{dy}{dx}$ by $\frac{-dx}{dy}$, we have the differential equation of orthogonal trajectories as

$$y^2 - x^2 - 2xy\frac{dx}{dy} + C = 0 \qquad \Rightarrow \qquad 2x\frac{dx}{dy} - \frac{1}{y}x^2 = \frac{-C}{y} - y$$

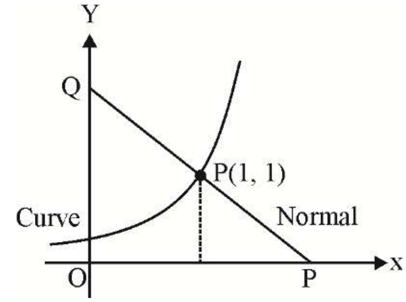
Putting $x^2 = v$, we gave $\frac{dv}{dy} - \frac{1}{y}v = -\frac{C}{y} - y$, which is linear differential equation in v and y whose I.F. is $\frac{1}{y}$.

Hence $\quad \frac{v}{y} = \int\left(-\frac{C}{y^2} - 1\right)dy + C = \frac{C}{y} - y + k \quad$ where 'k' is integration constant

$\Rightarrow \qquad x^2 + y^2 - ky - C = 0$, which represent system of circles with centre on y-axis.

22.(C) Given, equation of normal at $P(1,1)$ is $ay + x = a + 1$ $\therefore$ Slope of tangent at $P = a \Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = a$

Given $\frac{dy}{dx} \propto y \Rightarrow \frac{dy}{dx} = ky \Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = k = a$



$\therefore$ $\frac{dy}{dx} = ay \Rightarrow \frac{dy}{y} = a\,dx$ (Variables being separated)

$\Rightarrow$ $\ln y = ax + c$

It is passing through (1, 1) then $c = -a$

$\Rightarrow$ equation of the curve is $y = e^{a(x-1)}$.

23.(ABCD) Solving for $\frac{dy}{dx}$, we obtain $\frac{dy}{dx} = \frac{-2y\cot x \pm \sqrt{4y^2\cot^2 x + 4y^2}}{2} = y(-\cot x \pm \operatorname{cosec} x)$

Thus, we have $\frac{dy}{y} = (-\cot x + \operatorname{cosec} x)dx$

$\Rightarrow$ $\log y = -\log\sin x + \log\tan\frac{x}{2} + \log c$ $\Rightarrow$ $y = \frac{c\tan\frac{x}{2}}{\sin x} = \frac{c}{2\cos^2\frac{x}{2}} = \frac{c}{1+\cos x}$

Solving $\frac{dy}{y} = (-\cot x + \operatorname{cosec} x)dx$, we get $y = \frac{c}{1-\cos x}$ $\Rightarrow$ $x = 2\sin^{-1}\sqrt{\frac{C}{2y}}$.

24.(AB) Rewriting the given equation, we get $\frac{dy}{dx} = x(e^{n-1)y} - 1)$

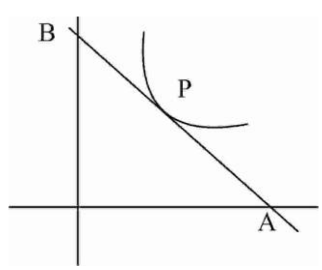
$\Rightarrow$ $\frac{dy}{e^{(n-1)y} - 1} = xdx \Rightarrow \frac{1}{n-1}\int \frac{(n-1)e^{(n-1)y}}{(e^{(n-1)y} - 1)e^{(n-1)y}}dy = \frac{x^4}{2} + C$

$\Rightarrow$ $\frac{1}{n-1}\int \frac{du}{u(u-1)} = \frac{x^2}{2} + C$ (where $u = e^{(n-1)y}$)

$\Rightarrow$ $\frac{1}{n-1}\log\frac{u-1}{u} = \frac{x^2}{2} + C \Rightarrow \frac{1}{n-1}\log\left(\frac{e^{(n-1)y} - 1}{e^{(n-1)y}}\right) = \frac{x^2}{2} + C \Rightarrow e^{(n-1)y} = C\,e^{(n-1)y + (n-1)x^2/2} + 1$

25.(CD) $Y - y = \frac{dy}{dx}(X - x)$

$A\left(x - y\frac{dx}{dy}, 0\right), B\left(0, y - x\frac{dy}{dx}\right)$



$\frac{3\left(x - y\frac{dx}{dy}\right) + 0}{4} = x,$

$x\frac{dy}{dx} + 3y = 0$; $\frac{dy}{y} + 3\frac{dx}{x} = 0$

$\ell ny + 3\ell nx = \ell nc \Rightarrow yx^3 = c = 1,$ as $f(1) = 1$

26.(AB) $f''(x)=g''(x) \Rightarrow f'(x)=g'(x)+c \Rightarrow f(x)=g(x)+cx+d$

$f'(1)=4,\ g'(1)=2 \Rightarrow c=2$

$f(2)=9,\ g(2)=3 \Rightarrow d=2$

$f(x)-g(x)=2x+2$

27.(BC) $\dfrac{dy}{dx}=\dfrac{6x^2}{2y+\cos y} \Rightarrow \int(2y+\cos y)dy=\int 6x^2dx \Rightarrow \left(y^2+\sin y\right)=2x^3+c$ and $c=\pi^2-2$

28.(AD) $\left(x^2+1\right)\dfrac{d^2y}{dx^2}=2x\dfrac{dy}{dx} \Rightarrow \int\dfrac{y_2}{y_1}dx=\int\dfrac{2x}{x^2+1}dx \Rightarrow \ell n|y_1|=\ell n\left|x^2+1\right|+\ell nc$

$y_1=c\left(x^2+1\right)$ and $c=3 \Rightarrow \int dy=\int 3\left(x^2+1\right)dx \Rightarrow y=x^3+3x+1$

29.(AB) Given equation of conic having its axis along the co-ordinate axes is $ax^2+by^2=1$

$\Rightarrow ax+by\dfrac{dy}{dx}=0 \Rightarrow a+b\left(\dfrac{dy}{dx}\right)^2+by\dfrac{d^2y}{dx^2}=0 \Rightarrow y\dfrac{dy}{dx}=x\left(\dfrac{dy}{dx}\right)^2+xy\dfrac{d^2y}{dx^2}$

30.(AB) $3xydx+4x^2dy+2y^3dx+6xy^2dy=0 \Rightarrow 3xydx+4x^2dy+2d\left(xy^3\right)=0$

$\Rightarrow 3x^2y^4dx+4x^3y^3dy+2\left(xy^3\right)d\left(xy^3\right)=0 \Rightarrow d\left(x^3y^4\right)+2\left(xy^3\right)d\left(xy^3\right)=0 \Rightarrow x^2y^6+x^3y^4=c$

31.(AC) $\dfrac{dy}{dx}+Py=Q$ ....(i)

$\dfrac{dy_1}{dx}+Py_1=Q$ ....(ii)

and $\dfrac{dy_2}{dx}+Py_2=Q$ ....(iii)

$\dfrac{d\left(y_1-y_2\right)}{dx}+P\left(y_1-y_2\right)=0$ ....(iv)

And $\dfrac{d\left(y-y_1\right)}{dx}+P\left(y-y_1\right)=0$ $\therefore \dfrac{d\left(y-y_1\right)}{d\left(y_1-y_2\right)}=\dfrac{y-y_1}{y_1-y_2}$

$\Rightarrow \ln\left(y-y_1\right)=\ln\left(y_1-y_2\right)+\ln c \Rightarrow y-y_1=c\left(y_1-y_2\right) \Rightarrow y=y_1+c\left(y_1-y_2\right)$

Also $\alpha y_1+\beta y_2$ is a solution $\Rightarrow \alpha y_1+\beta y_2=y_1(1+c)-cy_2$

$\alpha+\beta=1$

32.(ABCD) $\dfrac{d\theta}{dt}=-k\left(\theta-\theta_a\right)$

$\ln\left|\theta-\theta_a\right|_{37}^{34}=-k\times 15 \Rightarrow k=\dfrac{1}{15}\ln\dfrac{7}{4}$ where $\theta_a=30^\circ C$

$t\left(\theta=31^\circ\right)=\ln\left(\dfrac{37-30}{31-30}\right)\dfrac{15}{\ln 7/4}=15\log_{7/4}7$

$t\left(\theta=32^\circ\right)=\ln\left(\dfrac{37-30}{32-30}\right)\dfrac{15}{\ln 7/4}=15\log_{7/4}\dfrac{7}{2}$

33.(BCD) $\frac{dv}{dt}=-kA \Rightarrow A\frac{dH}{dt}=-k.A \quad \Rightarrow \quad \int_H^0 dH=-k\int_0^T dt$

34.(CD) $x^2+y^2-kx=0$

Diff. w.r.t. x, $2x+2y\frac{dy}{dx}=k=\frac{x^2+y^2}{x}$

For orthogonal curve, $2x-2y\frac{dx}{dy}=\frac{x^2+y^2}{x} \quad \Rightarrow \quad \frac{x^2-y^2}{2xy}=\frac{dx}{dy}$

35.(ABD) $\left|y\frac{dx}{dy}\right|=\frac{dy}{dx}$

$\Rightarrow y\frac{dx}{dy}=\frac{dy}{dx} \quad \because \frac{dy}{dx}>0 \Rightarrow \frac{dy}{dx}=\sqrt{y} \Rightarrow 2\sqrt{y}=x+c$

$\because$ it passes through (2, 1) $\quad \therefore c=0 \quad \therefore 2\sqrt{y}=x$ or $y^2=\frac{x^2}{4}$

Area $=\int 2\sqrt{y}\,dy=4\sqrt{3}\ sq.unit$

36.(ABD) $\frac{dy}{dx}-\frac{\tan 2x}{\cos^2 x}y=\cos^2 x$

I.F. $=e^{-\int\frac{\tan 2x}{\cos^2 x}dx}=e^{-\int\frac{2\sin 2x}{\cos 2x(1+\cos 2x)}dx}=e^{\log\frac{\cos 2x}{1+\cos 2x}}=\frac{\cos 2x}{1+\cos 2x}$

$y.\frac{\cos 2x}{1+\cos 2x}=\int\cos^2 x\times\frac{\cos 2x}{1+\cos 2x}dx+c=\frac{1}{2}\frac{\sin 2x}{2}+c$

$y=\frac{1}{2}\tan 2x\cos^2 x+c$

$\because y\left(\frac{\pi}{6}\right)=\frac{3\sqrt{3}}{8} \therefore c=0 \quad \therefore y=\frac{1}{2}\tan 2x\cos^2 x$

37.(CD) Equation of normal is $Y-y+\frac{dx}{dy}(X-x)=0$

$\therefore$ Length of perpendicular from origin is $\frac{\left|y+x\frac{dx}{dy}\right|}{\sqrt{1+\left(\frac{dx}{dy}\right)^2}}=y+x\frac{dx}{dy}\Rightarrow 1+\left(\frac{dx}{dy}\right)^2=1$

38.(BC) $t=\frac{1}{y}$

$\int_1^{1/x}\frac{-\ell ny}{1+\frac{1}{y}+\frac{1}{y^2}}\times\frac{-1}{y^2}dy=\int_1^{1/x}\frac{\ell ny}{y^2+y+1}dy$

39.(AC) Diff. the equation we get : $x(1-x)f(x)+\int_0^x (1-t)f(t)dt = xf(x)$

$$\int_0^x (1-t)f(t)dt - x^2 f(x) \Rightarrow (1-x)f(x) = x^2 f'(x) + f(x).2x \Rightarrow \int \frac{f'(x)}{f(x)}dx = \int\left(\frac{1}{x^2}-\frac{3}{x}\right)dx$$

$$\ln f(x) = -\frac{1}{x} - 3\ln x + c$$

$\therefore \quad f(1)=1 \quad \therefore c=1$

. $\quad x^2+y^2+2ax+2by+c=0$

$\because \quad a, b, c$ are arbitrary constants. $\quad \therefore$ order of the corresponding differential equation is 3.

Differentiating three times and eliminating the constant we get degree as 1.

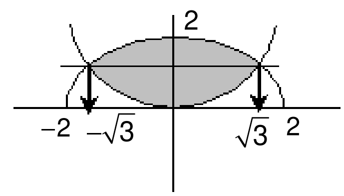
If $a+b=c^2$ and $a-2b=c \Rightarrow a=\frac{2c^2+c}{3}$ and $b=\frac{c^2-c}{3}$

40.(AB) Clearly, the constant function $y = 0$ is a solution. Differentiating the given equation with respect to x, we get.

$$p=\frac{dy}{dx}=\left(p+p^3\right)+x\left(\frac{dp}{dx}+3p^2\frac{dp}{dx}\right)$$

$$\Rightarrow -p^3 - x\left(1+3p^2\right)\frac{dp}{dx} \Rightarrow -\frac{dx}{x}=\left(\frac{1}{p^3}+\frac{3}{p}\right)dp \quad \Rightarrow \log xp^3 = \frac{1}{2p^2}+c \Rightarrow xp^3 = ke^{1/2p^2}$$

41.(BC) $\because \quad \frac{d^2y}{dx^2} = -\left(\frac{dy}{dx}\right)^3 \frac{d^2x}{dy^2}$



$\therefore \quad y=f(x)$ is solution of $\frac{dy}{dx}=\frac{-x}{y}$

i.e. $f(x)=\sqrt{4-x^2}$ & $y=y(x)$ is solution of $\frac{dy}{dx}=\frac{2y}{x}$ $\quad$ i.e. $g(x)=\frac{1}{3}x^2$

42.(ABC) $\quad y=e^{-x}\sin x$

$\Rightarrow \quad y_2 = e^{-x}\cos x - e^{-x}\sin x \quad \Rightarrow \quad y_1 = \sqrt{2}e^{-x}\sin\left(\frac{\pi}{4}-x\right) = -\sqrt{2}e^{-x}\sin\left(x-\frac{\pi}{4}\right)$

$\therefore \quad y_2 = \left(\sqrt{2}\right)^2 e^{-x}\sin\left(x-\frac{\pi}{2}\right)$

$y_3 = -\left(\sqrt{2}\right)^{2\backslash 3} e^{-x}\sin\left(x-\frac{3\pi}{4}\right)$ & $y_4 = 4e^{-x}\sin(x-\pi) \quad \Rightarrow y_4 = -4y$

$\Rightarrow \quad y_4+4y=0 \quad \Rightarrow y_8+4y_4=0 \quad \Rightarrow y_8-16y_4=0$

43.(BCD) $\quad \because \quad \int_0^x f(x)dx = kf^3(x) \Rightarrow \frac{1}{2}f^2(x)=\frac{x}{3k} \quad \Rightarrow f(x)=3\sqrt{x}$

So, $\quad f\left(\sin^2 x\right)=3|\sin x|$

44.(ACD) $\because$ Differential equation is $\dfrac{d^3y}{dx^3}-6\dfrac{d^2y}{dx^2}+11\dfrac{dy}{dx}-6y=0$

45.(AC) $\because V=\dfrac{1}{3}\pi r^2 h$ and $\dfrac{h}{27}=\dfrac{r}{18}$

$\therefore \quad V=\dfrac{1}{3}\pi\dfrac{27}{18}.r^3=\dfrac{\pi}{2}r^3 \quad ; \quad \dfrac{dV}{dr}=\dfrac{3\pi r^2}{2}\,;\ \dfrac{1}{r^2}\dfrac{dV}{dr}=\dfrac{3\pi}{2}$

46.(ABCD) $\because \quad A\equiv\left(x-y\dfrac{dx}{dy},\ 0\right)$ and $B\equiv\left(0,\ y-x\dfrac{dy}{dx}\right)$

$\therefore \quad BP=\sqrt{x^2\left(1+\left(\dfrac{dy}{dx}\right)^2\right)}=|x|\sqrt{1+\left(\dfrac{dy}{dx}\right)^2}$

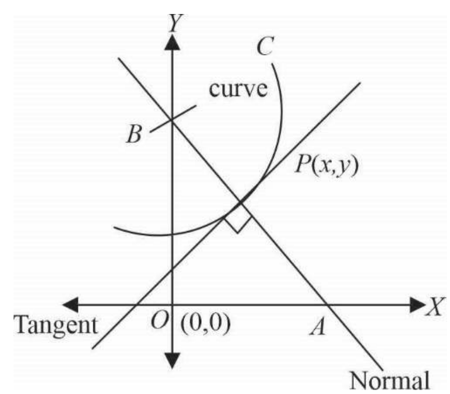
$$AP=\sqrt{y^2\left(1+\left(\dfrac{dx}{dy}\right)^2\right)}=\dfrac{|y|\sqrt{\left(\dfrac{dy}{dx}\right)^2+1}}{\left|\dfrac{dy}{dx}\right|}$$

$\therefore \quad \dfrac{BP}{AP}=\dfrac{|x|}{|y|}.\left|\dfrac{dy}{dx}\right|=\dfrac{3}{1} \quad \Rightarrow \quad \dfrac{dy}{dx}=\pm\dfrac{3y}{x} \quad \Rightarrow \quad \ln y=\pm\ln x^3 \quad \Rightarrow \quad y=x^3$ or $\dfrac{1}{x^3}$

47.(ABC) $\because \quad \dfrac{dy}{dx}+1=\dfrac{x^2-y+x+y}{x+y}$

$\Rightarrow \quad \dfrac{d(x+y)}{dx}=\dfrac{x^2+x}{x+y}\,;\ \dfrac{(x+y)^2}{2}=\dfrac{x^3}{3}+\dfrac{x^2}{2}+c$ for $x=0=y \Rightarrow c=0$

48.(AD) The equation of normal of $P(x,y)$ is $(Y-y)=\dfrac{-1}{\frac{dy}{dx}}(X-x)$



$\therefore A\left(x+y\dfrac{dy}{dx},0\right)$ and $B\left(0,y+\dfrac{x}{\frac{dy}{dx}}\right)$

Now $\dfrac{1\left(x+y\dfrac{dy}{dx}\right)+2(0)}{1+2}=x \Rightarrow x+y\dfrac{dy}{dx}=3x$

$y\dfrac{dy}{dx}=2x$ .....(1)

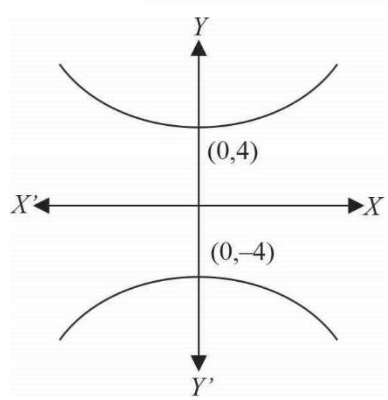
$\Rightarrow \quad \int y\,dy=\int 2x\,dx \Rightarrow \dfrac{y^2}{2}=x^2+C$

Also $(0,4)$ satisfy it, so $C=8$

$\therefore \quad y^2=2x^2+16$ (equation of curve)

Which represent a hyperbola

Also $\left.\dfrac{dy}{dx}\right]_{(4,4\sqrt{3})}=\dfrac{2(4)}{4\sqrt{3}}=\dfrac{2}{\sqrt{3}}$



49.(ABC) $f(x)=\int_0^x \{f(t)\cos t-\cos(t-x)\}dx$

$=\int_0^x f(t)\cos t dt-\int_0^x \cos(-t)dt \qquad \left[\int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$

$f(x)=\int_0^x f(t)\cos t dt-\sin x$

Differentiating both the sides, we get $f'(x)=f(x)\cos x-\cos x$

Let $f(x)=y; f'(x)=\dfrac{dy}{dx}$

$\dfrac{dy}{dx}-y\cos x=-\cos x$

If $=e^{-\int \cos x dx}=e^{-\sin x}$

Therefore, $y.e^{-\sin x}=-\int e^{-\sin x}; y=Ce^{\sin x}+1$

If s $\qquad y=0$ $\qquad$ (from the given relation)

$\Rightarrow \qquad C=-1$

Therefore, $f(x)=1-e^{\sin x}$

Now, minimum value $=1-e$ $\qquad$ (when $x=\pi/2$)

Maximum value $=1-e^{-1}$ $\qquad$ (when $x=-\pi/2$)

$f'(x)=-e^{\sin x}\cos x$

Therefore, $f'(0)=-1$

$f''(x)=-\left[\cos^2 x.e^{\sin x}-e^{\sin x}.\sin x\right]; \quad f''\left(\dfrac{\pi}{2}\right)=e$

50.(ABD) $\dfrac{dy}{dx}+y=f(x)\Rightarrow$ IF $=e^x$

$ye^x=\int e^x f(x)dx+C$

Now, if $0\le x\le 2$, then $ye^x=\int e^x e^{-x}dx+C$

$\Rightarrow \qquad ye^x=x+C$

$x=0, y(0)=1, C=1$

$\therefore \qquad ye^x=x+1$ $\qquad$ ...(i)

$y=\dfrac{x+1}{e^x}; y(1)=\dfrac{2}{e}\Rightarrow y'=\dfrac{e^x-(x+1)e^x}{e^{2x}}$

$y'(1)=\dfrac{e-2e}{e^2}=\dfrac{-e}{e^2}=-\dfrac{1}{e}$

If $x>2$, $ye^x=\int e^{x-2}dx$

$ye^{x} = e^{x-2} + C$

$y = e^{-2} + Ce^{-x}$

As y is continuous.

$\therefore \quad \lim_{x\to 2} \frac{x+1}{e^{x}} = \lim_{x\to 2}\left(e^{-2} + Ce^{-x}\right)$

$3e^{-2} = e^{-2} + Ce^{-2} \Rightarrow C = 2$

$\therefore$ for $x > 2$

$y = e^{-2} + 2e^{-x}$

Hence, $y(3) = 2e^{-3} + e^{-2} = e^{-2}\left(2e^{-1} + 1\right)$

$y' = -2e^{-x}$; $\quad y'(3) = -2e^{-3}$

51.(CD) Here, $\left(f'(x)\right)^2 + 4f'(x).f(x) + \left(f(x)\right)^2 = 0$

$\Rightarrow \quad \left(\frac{f'(x)}{f(x)}\right)^2 + 4\left(\frac{f'(x)}{f(x)}\right) + 1 = 0 \Rightarrow \frac{f'(x)}{f(x)} = \frac{-4 \pm \sqrt{16-4}}{2} \Rightarrow \frac{f'(x)}{f(x)} = -2 \pm \sqrt{3}$

Integrating both the sides $\log|f(x)| = \left(-2 \pm \sqrt{3}\right)x + C_1 \Rightarrow f(x) = e^{\left(-2\pm\sqrt{3}\right)x}.C$

52.(ABCD)

(A) $32x + 2y\frac{dy}{dx} = 0$

$\Rightarrow \quad \frac{dy}{dx} = m_1 = -\frac{16x}{y}$ and $16y^{15}\frac{dy}{dx} = k \Rightarrow \frac{dy}{dx} = m_2 = \frac{k}{16y^{15}}$

$m_1 m_2 = -\frac{16x}{y}.\frac{k}{16y^{15}} = -\frac{x}{y^{16}}.k = -\frac{x}{y^{16}}.\frac{y^{16}}{x} = -1$

(B) $\frac{dy}{dx} = 1 - ce^{-x} = 1 - (y-x) = -(y-x-1)$ [using $ce^{-x} = y - x$]

and $\frac{dy}{dx} - k.\frac{dy}{dx}e^{-y} = 1$

$\frac{dy}{dx}\left[1 - ke^{-y}\right] = 1$ or $\left[1-(x+2-y)\right]\frac{dy}{dx} = 1$ [using $ke^{-y} = x - y + 2$]

$\frac{dy}{dx} = m_2 = \frac{1}{y-x-1} \Rightarrow m_1 m_2 = -1$

(C) $\frac{dy}{dx} = 2cx = 2x.\frac{y}{x^2} = \frac{2y}{x} = m_1$

Also , $2x + 4y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{2y} = m_2$

Hence, $m_1 m_2 = -1$

(D) $x^2 - y^2 = c$

$2x - 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y} = m_1$

$xy = k$

$x\frac{dy}{dx} + y = 0 \qquad \Rightarrow \frac{dy}{dx} = -\frac{y}{x} = m_2$

$\therefore \qquad m_1 m_2 = -1$

53.(BC) $x\frac{dy}{dx} = y\log\left(\frac{y}{x}\right)\frac{dy}{dx} = \frac{y}{x}$, put $y = vx$

$\Rightarrow \qquad \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \qquad v + x\frac{dv}{dx} = v\log v \Rightarrow x\frac{dv}{dx} = v(\log v - 1)$

$\Rightarrow \qquad \int \frac{dv}{v(\log v - 1)} = \int \frac{dx}{x} \Rightarrow \int \frac{dx}{t-1} = \int \frac{dx}{x},$

Let $\log v = t$

$\Rightarrow \qquad \log(t-1) = \log(x) + \log C \Rightarrow \qquad \log(t-1) = \log(xC)$

$\Rightarrow \qquad t = 1 + xC \quad \Rightarrow \quad \log\frac{y}{x} = 1 + xC$

$\Rightarrow \qquad y = x.e^{1+Cx}$ and $\log\frac{y}{x} = \log e + Cx \quad \Rightarrow \qquad y = ex.e^{Cx}$

54.(AB) (A) $f(x,tx) = e^t + \tan^{-1}(t)$, independent of x. $\Rightarrow$ Homogeneous differential equation.

(B) $\log\left(\frac{y}{x}\right)dx + \frac{y^2}{x^2}\sin^{-1}\frac{y}{x}.dy = 0 \Rightarrow \frac{dy}{dx} = \frac{\log\left(\frac{y}{x}\right)}{\frac{y^2}{x^2}\sin^{-1}\left(\frac{y}{x}\right)}$; $f(x,y) = \frac{\log\left(\frac{y}{x}\right)}{\frac{y^2}{x^2}\sin^{-1}\left(\frac{y}{x}\right)}$

$\therefore \qquad f(x,tx) = \frac{\log(t)}{t^2\sin^{-1}(t)}$, independent of $x \Rightarrow$ Homogeneous differential equation.

(C) $f(x,y) = x^2 + \sin x.\cos y$

$f(x,tx) = x^2 + \sin x.\cos(tx)$, not independent of x. $\Rightarrow$ Not homogeneous differential equation.

(D) $f(x,y) = \frac{x^2 + y^2}{xy^2 - y^3}$

$\Rightarrow \qquad f(x,tx)$ is not independent of $f(x,tx)$

$\Rightarrow \qquad$ Not homogeneous differential equation.

55.(A,B, C)

$$\frac{dy}{dx} - y\cot x = -\frac{\sin x}{x^2}$$

Integrating Factor $= e^{\int -\cot x dx} = e^{-\log \sin x} = \frac{1}{\sin x}$

$\therefore$ Required solution is $y.\frac{1}{\sin x} = \int -\frac{\sin x}{x^2}.\frac{1}{\sin x}dx + C$

$$y.\frac{1}{\sin x} = \frac{1}{x} + C$$

As $y\left(\frac{\pi}{2}\right) = \frac{2}{\pi} \Rightarrow C = 0$

$\therefore$ $y = \frac{\sin x}{x} \Rightarrow \lim_{x\to 0} f(x) = 1$

$\therefore$ $I = \int_0^{\pi/2} \frac{\sin x}{x} dx$

Since, $\frac{\sin x}{x}$ is decreasing, when $x > 0$

$\Rightarrow$ $f(x) < f(0) \Rightarrow \int_0^{\pi/2} f(x) < \frac{\pi}{2}$ and $x < \frac{\pi}{2} \Rightarrow f(x) > \int \left(\frac{\pi}{2}\right) \Rightarrow \int_0^{\pi/2} f(x)dx > 1$

56.(BC) **(A)** Order of the differential equation is 2.

(B) $\frac{xdy - ydx}{\sqrt{x^2+y^2}} = dx \Rightarrow \frac{\frac{xdy-ydx}{x^2}}{\sqrt{\left(1+\frac{y^2}{x^2}\right)}} = \frac{dx}{x}$

$\therefore$ $\ln\left|\frac{y}{x} + \sqrt{1+\frac{y^2}{x^2}}\right| = \ln|Cx|$; $\frac{y}{x} + \frac{\sqrt{x^2+y^2}}{x} = Cx$ i.e. $y + \sqrt{x^2+y^2} = Cx^2$

(C) $y = e^x(A\cos x + B\sin x)$

$$\frac{dy}{dx} = e^x(A\cos x + B\sin x) + e^x(-A\sin x + B\cos x) = y + e^x(-A\sin x + B\cos x)$$

$\therefore$ $\frac{d^2y}{dx^2} = \frac{dy}{dx} + e^x(-A\sin x + B\cos x) - y$

$$\frac{dy}{dx} + \left(\frac{dy}{dx} - y\right) - y = 2\left(\frac{dy}{dx} - y\right)$$

(D) $(1+y^2)\frac{dy}{dx} + x = 2e^{\tan^{-1}y} \Rightarrow \frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{e^{\tan^{-1}y}}{1+y^2}$

If $= e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y} \Rightarrow xe^{\tan^{-1}y} = 2\int e^{\tan^{-1}y}.\frac{e^{\tan^{-1}y}}{1+y^2}dy \Rightarrow xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + k$

57.(AD) The tangent at $P(x,y)$ is $Y-y-(X-x)y_1=0$

It meets x-axis at $A\left(x-\frac{y}{y_1},0\right)$ and y-aixs at $B(0,y-xy_1)$

$\Delta OAB = 2 \rightarrow OA.OB = 4$

$\left(x-\frac{y}{p}\right)(y-xp)=4, p=\frac{dy}{dx}$

$(y-xp)^2=-4p$

$y-xp=2\sqrt{-p}$

$y=xp+2\sqrt{-p}$

The general solution is $y=cx+2\sqrt{-c}$

$x=1, y=1 \rightarrow c=-1, y=2-x$

$y(2)=0$

Differentiating (1) w.r.t. c

$0=x-\frac{1}{\sqrt{-c}}$

Eliminating c form (1) and (2)

$y=-\frac{1}{x}+\frac{2}{x}=\frac{1}{x}$

$y(2)=\frac{1}{2}$

58.(C) $f'(x)=1+\frac{f(x)}{x}$

$\frac{dy}{dx}-\frac{y}{x}=1$

$\therefore$ Integrating factor $=e^{\int -\frac{1}{x}dx}=\frac{1}{x}$

$\therefore$ $y\times\frac{1}{x}=\int 1\times\frac{1}{x}dx+c=\ln x+c$

$y=x\ln x+cx \quad \Rightarrow c=1$

$\therefore$ $\frac{f(x)-x}{x}=\ln x \quad \Rightarrow \int_0^{\pi}\ln(\sin\theta)d\theta=-\pi\ln 2$

59.(CD) It is Clairauts's equation, $y=xp-p^2$

General solution $y=cx-c^2 \rightarrow$

$x+y+1=0$ for $c=-1$

Singular solution is $y=\frac{x^2}{4}$.

60. **[A-u]** **[B-s]** **[C-q]** **[D-p]**

$$A=\int_{-2}^{2}\left(ax^3+bx+c\right)dx=4c\ ; \qquad B=\int_{-1}^{1}\left(ax^3+bx\right)dx=0$$

$$C=\sum_{n=1}^{10}\left(\int_{-2n-1}^{-2n}\sin^{27}x\,dx+\int_{2n}^{2n+1}\sin^{27}x\,dx\right)=\sum_{n=1}^{10}\left(\int_{2n+1}^{2n}\sin^{27}x\,dx+\int_{2n}^{2n+1}\sin^{27}x\,dx\right)=0$$

$$D=\left.\frac{\tan^{n-1}x}{n-1}\right|_0^{\pi/4}=\frac{1}{n-1}$$

61. **[A-r]** **[B-s]** **[C-p]** **[D-q]**

(A) $y\left(e^x+1\right)dy=\left(y+1\right)e^x dx\ \Rightarrow\int\frac{e^x}{e^x+1}dx=\int\frac{y}{y+1}dy\Rightarrow\ln\left(e^x+1\right)=y-\ln\left|y+1\right|+\ln c$

$\Rightarrow\left(e^x+1\right)\left|\left(y+1\right)\right|=ce^y$ which can also be written as $\left(e^x+1\right)\left(y+1\right)=\pm ce^y$

(B) $x\frac{dy}{dx}+y=xy^3\Rightarrow\frac{1}{y^3}\frac{dy}{dx}+\frac{1}{y^2x}=1\ \Rightarrow$ Let $\frac{1}{y^2}=t\Rightarrow\frac{dt}{dx}-\frac{2t}{x}=-2\ \Rightarrow\frac{1}{y^3}\frac{dy}{dx}=-\frac{1}{2}\frac{dt}{dx}$

Which is linear in form. Also the equation can be written as $xdy+ydx=xy^3dx$

$$d\left(xy\right)=x^3y^3\frac{dx}{x^2}\Rightarrow\frac{1}{\left(xy\right)^3}d\left(xy\right)=\frac{dx}{x^2}\Rightarrow-\frac{1}{2x^2y^2}+\frac{1}{x}+c=0$$

(C) $\frac{dy}{dx}=\frac{xy+y}{xy+x}\Rightarrow\left(\frac{1+y}{y}\right)dy=\left(\frac{1+x}{x}\right)dx$

On integrating both sides, we get, $\log y+y=\log x+x+\log A$

$\log\left(\frac{y}{Ax}\right)=x-y\Rightarrow y=A\times xe^{x-y}$

(D) Put $3x-4y=X\ \Rightarrow 3-\frac{4dy}{dx}=\frac{dX}{dx}\Rightarrow\frac{dy}{dx}=\frac{1}{4}\left(3-\frac{dX}{dx}\right)$

Therefore the given equation reduced to

$$\frac{3}{4}-\frac{1}{4}\frac{dX}{dx}=\frac{X-2}{X-3}\Rightarrow\frac{-1dX}{4dx}=\frac{4X-8-3\left(X-3\right)}{4\left(X-3\right)}=\frac{X+1}{4\left(X-3\right)}$$

$$-\left(\frac{X-3}{X+1}\right)dX=dx\Rightarrow-\left(1-\frac{4}{X+1}\right)dX=dx=-X+4\log\left(X+1\right)=x+c'$$

62. **[A-r]** **[B-s]** **[C-q]** **[D-p]**

(A) $y=a\cos\left(x+b\right)+ce^x+d\sin x$ **(B)** $y=a\left(\frac{1-\cos 2x}{2}\right)+b\left(\frac{1+\cos 2x}{2}\right)+\sin 2x+d\cos 2x+e\sin x$

(C) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3=\left(\frac{d^2y}{dx^2}\right)^2$ **(D)** differentiating, $\frac{dy}{dx}=c$ Order is 1

63. **[A-q, s]** **[B-p]** **[C-p]** **[D-q, r, s]**

(A) Equation of the required parabola is of the form

$y^2 = 4a(x-h)$. Differentiating, we have $2y\frac{dy}{dx} = 4a \Rightarrow y = \left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 0$

(B) We have $y = a(x+a)^2$ .....(i) $\Rightarrow \frac{dy}{dx} = 2a(x+a)$ ......(ii)

Dividing equation (i) and (ii), we get $\frac{y}{\frac{dy}{dx}} = \frac{x+a}{2} \Rightarrow x+a = \frac{2y}{y_1}$, where $y_1 = \frac{dy}{dx}$

(C) The given equation is $\left(1+3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$

Cubing, we get $\left(1+3\frac{dy}{dx}\right)^2 = 64\left(\frac{d^3y}{dx^2}\right)^3$

(D) We have $y^2 = 2c\left(x+\sqrt{c}\right)$ ...(i)

Diff. w.r.t. x, we get $2y\frac{dy}{dx} = 2c \quad \Rightarrow c = y\frac{dy}{dx}$

Putting in equation, (i), we get $y^2 = 2\left(y\frac{dy}{dx}\right)x + 2\left(y\frac{dy}{dx}\right)^{3/2}$

64. **[A-q]** **[B-r]** **[C-p]** **[D-s]**

(A) $y = e^{4x} + 2e^{-x}; y_1 = 4e^{4x} - 2e^{-x}; y_2 = 16e^{4x} + 2e^{-x}$

$y_3 = 64e^{4x} - 2e^{-x}$

Now, $y_3 - 13y_1 = \left(64e^{4x} - 2e^{-x}\right) - 13\left(4e^{4x} - 2e^{-x}\right) = 12e^{4x} + 24e^{-x}$

$y_3 - 13y_1 = 12\left(e^{4x} + 2e^{-x}\right) = 12y$

65.(4) $\int_1^4 \frac{2e^{\sin x^2}}{x}dx = \int_1^{16} \frac{e^{\sin t}}{t}dt = \left[F(t)\right]_1^{16} = F(16) - F(1) \Rightarrow \frac{k}{4} = 4 \quad \Rightarrow \quad k = 16$

66.(2) Given y = tan$z$

$\frac{dy}{dx} = \sec^2 z.\frac{dz}{dx}$ ....(i)

Now $\frac{d^2y}{dx^2} = \sec^2 z.\frac{d^2z}{dx^2} + \frac{dz}{dx}\frac{d}{dx}\left(\sec^2 z\right)$ [using product rule]

$\frac{d^2y}{dx^2} = \sec^2 z.\frac{d^2z}{dx^2} + \left(\frac{dz}{dx}\right).2\sec^2 z.\tan z$ .....(ii)

Now $1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2 = 1 + \frac{2(1+\tan z)}{\sec^2 z}.\sec^4 z.\left(\frac{dz}{dx}\right)^2$

$= 1 + 2(1+\tan z)\sec^2 z\left(\frac{dz}{dx}\right)^2 = 1 + 2\sec^2 z\left(\frac{dz}{dx}\right)^2 + 2\tan z.\sec^2 z\left(\frac{dz}{dx}\right)^2$ .....(iii)

From (ii) and (iii), we have RHS of (ii) = RHS of (iii)

67.(8) $\frac{dy}{dt}+2ty=t^2$

I.F. $= e^{t^2}$ $\quad\therefore\quad$ Solution is $y\cdot e^{t^2}=\int t\cdot t e^{t^2}dt$

$$y.e^{t^2}=t.\frac{e^{t^2}}{2}-\frac{1}{2}\int e^{t^2}dt+C$$

$$y=\frac{t}{2}-e^{-t^2}\int\frac{e^{t^2}dt}{2}+Ce^{-t^2}$$

$$\lim_{t\to\infty}\frac{y}{t}=\frac{1}{2}-\lim_{t\to\infty}\frac{\int\frac{e^{t^2}}{2}}{te^{t^2}}=\frac{1}{2}$$

68.(2) $\frac{dy}{dx}=\frac{1}{x\cos y+2\sin y\cos y}$

$\therefore\frac{dx}{dy}=x\cos y+2\sin y\cos y$ $\qquad\therefore\frac{dx}{dy}+(-\cos y)x=2\sin y\cos y$

$\therefore$ I. F. $= e^{-\int\cos y dy}e=e^{-\sin y}$

The solution is $x.e^{-\sin y}=2\int e^{-\sin y}.\sin y\cos ydy=-2\sin y.e^{-\sin y}-2\int\left(-e^{-\sin y}\right).\cos ydy$

$$=-2\sin y.e^{-\sin y}+2\int\left(-e^{-\sin y}\right).\cos ydy$$

69.(1) $\frac{dy}{dx}=\frac{1}{dx/dy};\frac{d^2y}{dx^2}=\frac{d}{dy}\left(\frac{1}{dx/dy}\right).\frac{dy}{dx}=-\frac{1}{(dx/dy)^3}\frac{d^2x}{dy^2}$

Hence $x\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3-\frac{dy}{dx}=0$

Become $-x\frac{1}{(dx/dy)^3}\frac{d^2x}{dy^2}+\frac{1}{(dx/dy)^3}-\frac{1}{(dx/dy)}=0$

Or $x\frac{d^2x}{dy^2}-1+\left(\frac{dx}{dy}\right)^2=0\Rightarrow x\frac{d^2x}{dy^2}+\left(\frac{dx}{dy}\right)^2=1$

70.(1) $\frac{dy}{dx}=-\frac{\sqrt{(x^2-1)(y^2-1)}}{xy}$

$$\int\frac{y}{\sqrt{y^2-1}}dy=-\int\frac{\sqrt{x^2-1}}{x}dx$$

Let $\quad y^2-1=t^2\Rightarrow 2ydy=2tdt$ $\qquad\therefore\int\frac{t}{t}dt=-\int\frac{x^2-1}{x\sqrt{x^2-1}}dx$

$$t=-\int\frac{x}{\sqrt{x^2-1}}dx+\int\frac{1}{x\sqrt{x^2-1}}dx$$

$$\sqrt{y^2-1}=-\sqrt{x^2-1}+\sec^{-1}x+c$$

Passing through (1, 1)

$\Rightarrow \quad \sqrt{y^2-1}=-\sqrt{x^2-1}+\sec^{-1}x \quad \Rightarrow \quad \sqrt{k^2-1}=-1+\frac{\pi}{4}$

$\Rightarrow \quad k^2-1=1+\frac{\pi^2}{16}-\frac{\pi}{2} \quad \Rightarrow \quad k^2=2+\frac{\pi^2}{16}-\frac{\pi}{2}=1.046\,; \quad [k]=\sqrt{1.046}=1$

71.(8) Equation the tangent at $P(x_1, y_1)$ of $y=f(x)$

$$y-y_1=\frac{dy}{dx}(x-x_1) \qquad \text{..........(i)}$$

This tangent cuts the x-axis so $x_2=x_1-\frac{y_1}{\left(\frac{dy}{dx}\right)}$

$\because \quad x_1, x_2, x_3, ...x_n$ are in AP

$x_2-x_1=-\frac{y_1}{\frac{dy}{dx}}=\log_2 e$ given

$-y=\log_2 e\frac{dy}{dx}$

$\frac{dy}{y}\log_2 e=-dx$ Integrating both side

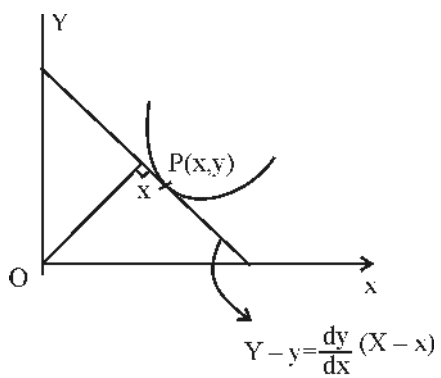
$\log_e y=-x\log_e 2+c$

$y=ke^{-x\log_e 2}$

Put $\quad x=-2$

$y=2e^{+2\log_e^2}=2\times 4=8$

72.(2)



Equation of tangent is $X\frac{dy}{dx}-y-Y\frac{dy}{dx}+y=0$ perpendicular distance from origin is

$\therefore$ $\perp$ from (0, 0) = x

$$\left|\frac{0-0-x\frac{dy}{dx}+y}{\sqrt{\left(\frac{dy}{dx}\right)^2+1}}\right|=x \quad\Rightarrow\quad \left|\frac{x\frac{dy}{dx}-y}{\sqrt{\left(\frac{dy}{dx}\right)^2+1}}\right|=x \;\Rightarrow\; \left(x\frac{dy}{dx}-y\right)^2=x^2\left(1+\left(\frac{dy}{dx}\right)^2\right)$$

$$x^2\left(\frac{dy}{dx}\right)^2+y^2-2xy\frac{dy}{dx}=x^2+x^2\left(\frac{dy}{dx}\right)^2 \Rightarrow \frac{y^2-x^2}{2xy}=\frac{dy}{dx} \quad ......\text{(i) (homogenous)}$$

Put y = vx in (i)

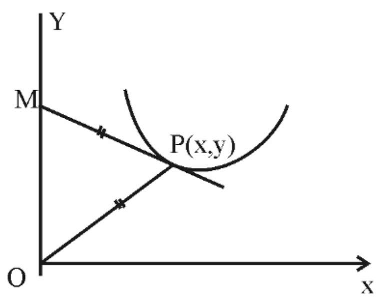
$$v+x\frac{dv}{dx}=\frac{v^2-1}{2v}$$

$$\int\frac{2v}{v^2+1}dv=-\int\frac{dx}{x} \qquad\Rightarrow\qquad \ln\left(v^2+1\right)=-\ln x+\ln c$$

$$v^2+1=\frac{c}{x} \qquad\Rightarrow\qquad \frac{y^2+x^2}{x^2}=\frac{c}{x}\Rightarrow y^2+x^2=cx$$

Passes through (1, 1), then c = 2 $\Rightarrow$ $x^2+y^2-2x=0$

73.(4) $\frac{dy}{dx}-y=1-e^{-x}$



P = –1

$Q=1-e^{-x}$

I.F. = $e^{\int p dx}=e^{\int -1dx}=e^{-x}$

$$\therefore\; y.e^{-x}=\int e^{-x}\left(1-e^{-x}\right)dx+C$$

$$y.e^{-x}=-e^{-x}+\frac{1}{2}e^{-2x}+C$$

$$y=-1+\frac{1}{2}e^{-x}+Ce^{x}$$

x = 0, y = y_0

So $\quad C=y_0+\frac{1}{2}$

$$y=-1+\frac{1}{2}e^{-x}+\left(y_0+\frac{1}{2}\right)e^{x}$$

$$x\to\infty;\; y\to \textit{finite value so } y_0+\frac{1}{2}=0$$

74.(1) $2x^2ydx-2y^4dx+2x^3dy+3xy^3dy=0$

Dividing throughout by x^3y, we get $2\frac{dx}{x}-\frac{2y^3}{x^3}dx+2.\frac{dy}{y}+\frac{3y^2}{x^2}dy=0$

$$\Rightarrow \quad 2\frac{dx}{x}+2\frac{dy}{y}+\frac{3y^2x^4dy-2y^3x^3dx}{x^6}=0 \Rightarrow \quad 2\frac{dx}{x}+2\frac{dy}{y}+\frac{3y^2x^2dy-2y^3xdx}{x^4}=0$$

$$\Rightarrow \quad 2d(\ln x)+2d(\ln y)+d\left(\frac{y^3}{x^2}\right)=0 \Rightarrow \quad 2\ln|x|+2\ln|y|+\frac{y^3}{x^2}=C$$

When $x=1, y=1$. So, $C=1$

75.(8) Let the salt content at time $'t'$ be $u\ \ lb,$ so that its rate of change is du/dt.

$=2gal\times 2lb=4lb/\min$

If c be the concentration of the brine at time t, the rate at which the salt content decreases due to the out flow

$=2\ gal\times c\,1b/\min\ \ =2c1b/\min$

$$\therefore \quad \frac{du}{dt}=4-2c \qquad \text{....(i)}$$

Also, since there is no increase in the volume of the liquid, the concentrations $c=\frac{u}{50}$

$\therefore$ Eq. (i) becomes $\frac{du}{dt}=4-\frac{2u}{50}$

Separating the variables and integrating, we have

$$\int dt=25\int\frac{du}{100-u} \quad \text{or} \quad t=-25\ln(100-u)+K \qquad \text{....(ii)}$$

initially when $t=0, u=0$

$$0=-25\ln\ 100+K \qquad \text{...(iii)}$$

Eliminating $'K'$ from Eqs. (ii) and (iii), we get $t=25\ln\left(\frac{100}{100-u}\right)$

Taking $t=t_1$ when $u=40$ and $t=t_2$ when $u=80$, we have $t_1=25\ln\left(\frac{100}{60}\right)$ and $t_2=25\ln\left(\frac{100}{20}\right)$

$\therefore$ The required time $(t_2-t_1)=25\left(\ln 5-\ln\frac{5}{3}\right)=25\ln 3$

$=25\times 1.0986=26\min 28s\ \ =1648s=206\times 8=206\times\lambda\ ;\ \lambda=8$

76.(1) $f(x+f(y)+xf(y))=y+f(x)+yf(x)$ ....(i)

Differentiating w.r.t. x and y is constant

$f'(x+f(y)+xf(y))(1+f(y))=f'(x)+yf'(x)$ ....(ii)

From Eq. (i) again differentiating w.r.t. y as x is constant

$f'(x+f(y)+xf(y))(1+x)f'(y)=1+f(x)$ ...(iii)

From Eqs. (ii) and (iii)

$$\frac{1+f(y)}{(1+x)f'(y)}=\frac{(1+y)f'(x)}{1+f(x)}$$

$$\frac{(1+y)f'(y)}{1+f(y)}=\frac{1+f(x)}{(1+x)f'(x)}=\lambda$$

$\therefore \quad f'(y)=\dfrac{1+f(y)}{1+y}\lambda$ and $f'(x)=\dfrac{1+f(x)}{\lambda(1+x)}$

$\therefore \quad \lambda=\dfrac{1}{\lambda}\Rightarrow\lambda=\pm1 \quad \therefore \quad \dfrac{f'(x)}{1+f(x)}=\pm\dfrac{1}{1+x}$

Integrating both the sides $f(x)=C(1+x)^{\pm1}-1$

From Eq. (i) put $x=y=0$

$f(f(0))=f(0)$

From Eq. (ii), $f(0)=C-1$

So, $f(C-1)=C-1$

$\therefore \quad C=0,1$ (taking +ve sign)

So, $f(0)=-1$ and $f(x)=(1+x)-1=x$ and $C=1$

$\therefore \quad f(x)=(1+x)^{-1}-1=\dfrac{-x}{1+x}$

$1+f(x)=\dfrac{1}{1+x} \quad \therefore \quad 1+f(2018)=\dfrac{1}{2019} \quad \therefore \quad (2019)(1+f(2018))=1$

77.(1) $\phi'(x)+2\phi(x)\le 1$

$e^{2x}\phi'(x)+2\phi(x)e^{2x}\le e^{2x}$

$\dfrac{d}{dx}\left(e^{2x}\phi(x)-\dfrac{1}{2}e^{2x}\right)\le 0$

$\therefore \quad \left(e^{2x}\phi(x)-\dfrac{1}{2}\right)$ is a non-increasing function of x.

$\therefore$ $2\phi(x)$ is always less than or equal to 1.

78.(1) $\dfrac{1}{2}(1+x)^{-1/2}-\dfrac{a}{2}(1+y)^{-1/2}\dfrac{dy}{dx}=0$

$\Rightarrow \quad \dfrac{1}{\sqrt{1+x}}=\dfrac{a}{\sqrt{1+y}}.\dfrac{dy}{dx}\Rightarrow a=\dfrac{\sqrt{1+y}}{\sqrt{1+x}}.\dfrac{1}{dy/dx}$

Putting this value in the given equation.

$\dfrac{dy}{dx}\sqrt{1+x}-\dfrac{1+y}{\sqrt{1+x}}=\dfrac{dy}{dx}$

$\Rightarrow \quad (1+x)\dfrac{dy}{dx}=1+y+\sqrt{1+x}\dfrac{dy}{dx}$

$\Rightarrow \quad \left(1+x-\sqrt{1+x}\right)\dfrac{dy}{dx}=1+y$

$\therefore$ Degree of the given equation is 1.

79.(3) Here, $2f^5(x).f'(x).\left|1+(f'(x))^2\right| = f''(x)$

$$\Rightarrow \quad f^4(x)d\left(f^2(x)\right) = \frac{f''(x)}{1+(f'(x))^2} \quad \Rightarrow \quad \left|f^4(x)d\left(f^2(x)\right)\right| = \int d\left(\tan^{-1}(f'(x))\right)$$

$$\Rightarrow \quad \frac{f^6(x)}{3} + C = \tan^{-1}(f'(x)), \text{ as } f(0) = f'(0) = 0 \Rightarrow c = 0$$

$$\Rightarrow \quad \frac{f^6(x)}{3} = \tan^{-1}(f'(x)) \quad \Rightarrow \quad 0 \le \frac{f^6(x)}{3} < \frac{\pi}{2}$$

$$\therefore \quad -\left(\frac{3\pi}{2}\right)^{1/6} < f(x) < \left(\frac{3\pi}{2}\right)^{1/6} \quad \Rightarrow \quad \text{Number of integers between } k_1 \text{ and } k_2 \text{ are } 3.$$

80.(1) Put, $dy/dx = t$

$$\therefore \quad dt/dx - t + e^{2x} = 0, I.F. = e^{-\int dx}$$

$$\therefore \quad \text{solution is, } t.e^{-x} = \int -e^{2x}.e^{-x}dx + C$$

$$\Rightarrow \quad te^{-x} = -e^x + C: \quad y'(0) = 1 \Rightarrow C = 2$$

$$\therefore \quad dy/dx = \frac{2-e^x}{e^{-x}} \int dy = \int \left(2e^x - e^{2x}\right)dx$$

$$\Rightarrow \quad y = 2e^x - \frac{e^{2x}}{2} + C^1, y(0) = 2 \Rightarrow C^1 = 1/2$$

$$\therefore \quad y(x) = 2e^x - \frac{e^{2x}}{2} + \frac{1}{2} \Rightarrow \quad y_{max}(x) = \frac{5}{2} \text{ for } x = \log 2 \qquad \therefore [2x] = [2\log 2] = [\log 4] = 1$$

81.(3) $\because \quad \dfrac{d}{dx}\left\{\dfrac{dy}{dx}.y\right\} = e^x$

$$\Rightarrow \quad y\frac{dy}{dx} = e^x + c \quad \Rightarrow \quad 1\times 1 = e^0 + c \quad \Rightarrow c = 0 \quad \Rightarrow \quad ydy = e^x dx$$

$$\frac{1}{2}y^2 = e^x + c \quad \therefore \frac{1}{2} = 1 - c \quad \Rightarrow c = -\frac{1}{2}$$

$$y^2 = 2e^x - 1 \quad \Rightarrow f^2(x) = 2e^x - 1$$

82.(7) Let $x+3 = X$ & $y-1 = Y$

$$\therefore \frac{dY}{dX} = \frac{(X+Y)^2 - Y^2}{X^2} = \frac{X^2 + 2XY}{X^2}$$

$$\Rightarrow \frac{dY}{dX} - \frac{2Y}{X} = 1$$

$$\text{I.F.} = e^{\int -\frac{2}{X}dX} = e^{-2\ln X} = \frac{1}{X^2}$$

$$\therefore \quad Y \times \frac{1}{X^2} = \int \frac{1}{X^2} \times 1\, dX + c$$

$\Rightarrow \quad \dfrac{(y-1)}{(x+3)^2} = -\dfrac{1}{(x+3)} + c \quad \Rightarrow (y-1) = c(x+3)^2 - (x+3)$

$\therefore \quad 0 = 9c - 3 \quad \Rightarrow \quad c = \dfrac{1}{3}$

$y = 1 + \dfrac{1}{3}(x+3)^2 - (x+3)$

$y(3) = 1 + 12 - 6 = 7$

83.(8) $\because \quad \dfrac{dy}{dx^2} = \dfrac{dy}{dx} \times \dfrac{dx}{dx^2} = \dfrac{1}{2x}\dfrac{dy}{dx}$

$\therefore$ equation, $\dfrac{dy}{dx} + \dfrac{2y}{x} = \dfrac{\sin^{-1} x}{x}$

I.F. $= e^{\int \frac{2}{x} dx} = e^{2\ln x} = x^2$

$\therefore \quad y \times x^2 = \int \dfrac{\sin^{-1} x}{x} \times x^2 dx$

$= \sin^{-1} x \,.\, \dfrac{x^2}{2} - \int \dfrac{1}{\sqrt{1-x^2}} \times \dfrac{x^2}{2} dx$

$= \dfrac{1}{2}x^2 \sin^{-1} x + \dfrac{1}{2}\int \dfrac{1-x^2-1}{\sqrt{1-x^2}} dx$

$= \dfrac{1}{2}x^2 \sin^{-1} x + \dfrac{1}{2}\int \sqrt{1-x^2}\, dx - \dfrac{1}{2}\int \dfrac{dx}{\sqrt{1-x^2}}$

$= \dfrac{1}{2}x^2 \sin^{-1} x + \dfrac{1}{4}\left[x\sqrt{1-x^2} + \sin^{-1} x\right] - \dfrac{1}{2}\sin^{-1} x + c$

$y \times x^2 = \dfrac{1}{2}\left(2x^2 - 1\right)\sin^{-1} x + \dfrac{1}{4}x\sqrt{1-x^2} + c \quad \Rightarrow \quad y(1) \times 1^2 = \dfrac{1}{4}(2-1) \times \dfrac{\pi}{2} + 0 = \dfrac{\pi}{8}$

84.(8) If $f'(x) < 2f(x)$

$\dfrac{dy}{dx} - 2y < 0$

$e^{-2x}.\dfrac{dy}{dx} - 2e^{-2x} y < 0$

$\dfrac{d}{dx}\left(y \,.\, e^{-2x}\right) < 0$

$g(x) = f(x)e^{-2x}$ is decreasing function

$\therefore g(x) \le g\left(\dfrac{1}{2}\right) \Rightarrow f(x) \le 2e^{2x}$

85.(2) $\because \quad \dfrac{dy}{dx} = \dfrac{x\left(3y^2 - x^2\right)}{y\left(3x^2 - y^2\right)}$

$\therefore \quad m = 3$ & $n = -1$

86.(3) $\because$ differentiations, $xf(x)+1\times\int_0^x f(t)dt=(x+1)xf(x)+\int_0^x tf(t)dt$

Again differentiate, we have $f(x)=x^2f'(x)+f(x)\;.\;2x+xf(x)$

$\frac{f'(x)}{f(x)}=\frac{1}{x^2}-\frac{3}{x}\Rightarrow$ $\ln f(x)=-\frac{1}{x}-3\ln x+c$ $\because$ $f(1)=\frac{1}{e}$

$-1=-1-0+c$

$c=0$

$\therefore$ $\ln f(x)=\ln e^{-\frac{1}{x}}+\ln x^{-3}$

$f(x)=e^{-\frac{1}{x}}\times\frac{1}{x^3}$

$e^{\frac{1}{x}}f(x)=\frac{1}{x^3};\;\;g(x)=\frac{1}{x^3};\;\;g'(x)=\frac{-3}{x^4};\;\;g'(1)=-3$

87.(2) $\because$ $\left(x\frac{dy}{dx}\right)^2-2xy\frac{dy}{dx}+y^2+\left(x\frac{dy}{dx}-y\right)-2=0$ $\Rightarrow$ $\left(x\frac{dy}{dx}-y\right)^2+\left(x\frac{dy}{dx}-y\right)-2=0$

$\therefore$ $x\frac{dy}{dx}-y=-2$ or 1 $\Rightarrow$ $\frac{dy}{dx}-\frac{y}{x}=-\frac{2}{x}$ or $\frac{1}{x}$

I.F. $=e^{\int -\frac{1}{x}dx}=\frac{1}{x}$

$\therefore$ Solution is $y\times\frac{1}{x}=\int\frac{-2}{x}\times\frac{1}{x}dx$ or $\int\frac{1}{x}\times\frac{1}{x}dx$

$\frac{y}{x}=+\frac{2}{x}+c$ or $-\frac{1}{x}+c$

$y=2+cx$ or $-1+cx$

$\therefore$ Either $(y-cx)=2$ or $(y-cx)=-1$ for $c=2015$

We have two answers

Either $(y-2015\;.\;x)=2$ or $(y-2015\;.\;x)=-1$

88.(5) $\because$ Differential equation is $\left\{1+\left(\frac{dy}{dx}\right)^2\right\}\frac{d^3y}{dx^3}-3\frac{dy}{dx}\left(\frac{d^2y}{dx^2}\right)^2=0$

So, $m=3$ and $n=2$

89.(62) Given $y\left(\frac{d^2y}{dx^2}\right)=2\left(\frac{dy}{dx}\right)^2\Rightarrow\frac{y''}{y'}=\frac{2y'}{y}\Rightarrow\ln\;y'=2\ln y+\ln a$

$\Rightarrow$ $\frac{y'}{y^2}=a\Rightarrow\int\frac{dy}{y^2}=\int adx\Rightarrow\frac{-1}{y}=ax+b$

Since curve is passing through $(2,2)$ and $\left(8,\frac{1}{2}\right)$

So, $2a+b=\frac{-1}{2}$ ...(i)

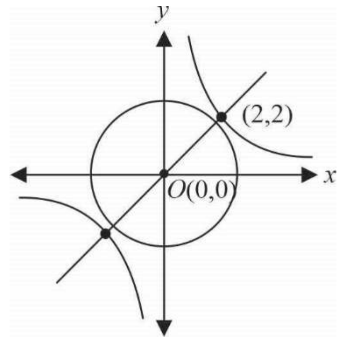
$8a+b=-2$ ...(ii)

On solving (i) and (ii), we get $a=\frac{-1}{4}, b=0$

Hence, $C_1 : xy=4$ and curve C_2 is $x^2+y^2=4$.

$\therefore$ Shortest distance between C_1 and $C_2 = 2\sqrt{2}-2=\sqrt{8}-2$

Hence, $\left(p^2-q\right)=64-2=62$



90.(8) Given, $\frac{xdy}{dx}-2y=x^4y^2$ or $\frac{dy}{dx}-\frac{2}{x}y=x^3y^2$. Now dividing by y^2, we get

$\frac{1}{y^2}\frac{dy}{dx}-\frac{2}{y}\frac{1}{x}=x^3$ ...(i)

This is a Bernouli's differential equation, substituting $\frac{-2}{y}=t$, we get

$\Rightarrow \quad \frac{2}{y^2}\frac{dy}{dx}=\frac{dt}{dx}$. So, equation (i) becomes

$\Rightarrow \quad \frac{1}{2}\frac{dt}{dx}+\frac{t}{x}=x^3 \Rightarrow \frac{dt}{dx}+\frac{2}{x}t=2x^3$

$I.F.=e^{\int \frac{2}{x}dx}=e^{2\ln x}=e^{\ln x^2}=x^2$

So, general solution is given by $x^2t=\frac{2x^6}{6}+C \Rightarrow \frac{-x^2.2}{y}=\frac{x^6}{3}+C \Rightarrow \frac{-2}{y}=\frac{x^4}{3}+\frac{C}{x^2}$

If $x=1, y=-6, \Rightarrow C=0$

$\therefore \quad \frac{-2}{y}=\frac{x^4}{3} \Rightarrow y=\frac{-6}{x^4} i.e.\ t(x)=\frac{-6}{x^4}$

Now, $\frac{dy}{dx}=24x^{-5}=\frac{24}{x^5}$. Hence $\left.\frac{dy}{dx}\right]_{x=3^{1/5}}=\frac{24}{3}=8.$

91.(1) $y(x+y)=x \Rightarrow (x+2y)\frac{dy}{dx}=1-y \Rightarrow \frac{dx}{x+2y}=\frac{dy}{1-y} \Rightarrow \int\frac{dx}{x+2y}=-\ln(1-y)+c \Rightarrow k=1$

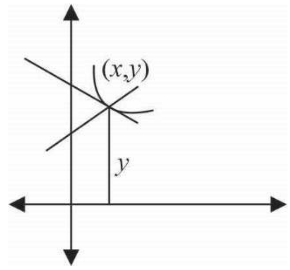
92.(1) Equation of normal is

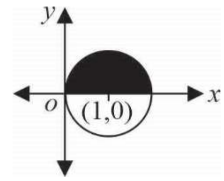
$Y-y=-\frac{1}{m}(X-x)$

$X+mY-(x+my)=0$ ....(1)

Perpendicular distance from (0, 0) to equation (1) is $\left|\frac{x+my}{\sqrt{1+m^2}}\right|=|y|$

$\Rightarrow \quad (x+my)^2 = y^2(1+m^2) \Rightarrow x^2 + 2mxy = y^2 \Rightarrow m = \dfrac{y^2 - x^2}{2xy}$ ... (2)





Put $y^2 = t \Rightarrow 2y\dfrac{dy}{dx} = \dfrac{dt}{dx}$

$\therefore$ Equation (2) becomes

$x\dfrac{dt}{dx} = t - x^2 \Rightarrow \dfrac{dt}{dx} - \dfrac{1}{x}t = -x$

$\therefore \quad I.F. = e^{-\int \frac{1}{x}dx} = e^{-\ln x} = \dfrac{1}{x}$

Now general solution is given by $t\left(\dfrac{1}{x}\right) = -x + C \Rightarrow y^2\left(\dfrac{1}{x}\right) = -x + C$

As $(1,1)$ satisfy it, so $C = 2$

$\Rightarrow \quad y^2 = -x^2 + 2x \Rightarrow x^2 + y^2 - 2x = 0$ Hence required area $= \dfrac{k\pi}{2}$ $\therefore$ $k = 1$

93.(0) $(2n\,y + x\,y\log_e x)dx - x\log_e x\,dy = 0$

$\Rightarrow \quad \dfrac{dy}{y} = \left(\dfrac{2n}{x\log_e x} + 1\right) \Rightarrow \log y = 2n\log|\log x| + x + c$ & $c = 0$ [$\because$ curve passes through (e, e^e)]

$\therefore \quad y = e^{x+\log(\log x)^{2n}} = e^x(\log x)^{2n} \quad \Rightarrow \quad f(x) = e^x(\log x)^{2n}$

$$g(x) = \lim_{x\to\infty} f(x) = \begin{cases} \to \infty \ \text{if} \ x < \dfrac{1}{e} \\ 0 \ \text{if} \ \dfrac{1}{e} < x < e \\ \to \infty \ \text{if} \ x > e \end{cases}$$

Now, $\therefore \displaystyle\int_{1/e}^{e} g(x)\,dx = 0$

VECTORS

1.(D) $\left[\hat{a}\ \vec{p}\ \vec{q}\right]$ = projection of $\vec{p}\times\vec{q}$ in the direction of $\hat{a}$. Hence the given vector is $\vec{p}\times\vec{q}$

2.(A) Any vector $\vec{r}$ can be written as $\vec{r}=\left(\vec{r}\cdot\vec{a}\right)\vec{a}+\left(\vec{r}\cdot\vec{b}\right)\vec{b}+\left(\vec{r}\cdot\vec{c}\right)\vec{c}$ where $\vec{a},\vec{b},\vec{c}$ are three mutually perpendicular unit vectors.

$\Rightarrow$ $\left(\vec{r}\cdot\vec{a}\right)^2+\left(\vec{r}\cdot\vec{b}\right)^2+\left(\vec{r}\cdot\vec{c}\right)^2=\left[\left(\vec{r}\cdot\vec{a}\right)\vec{a}+\left(\vec{r}\cdot\vec{b}\right)\vec{b}+\left(\vec{r}\cdot\vec{c}\right)\vec{c}\right]\cdot\vec{r}\Rightarrow\vec{r}\cdot\vec{r}=\left|\vec{r}\right|^2$

$\Rightarrow$ Clearly **(A)** is the answers.

3.(B) $\left(\bar{a}\times\bar{x}\right)+\bar{b}=\bar{x}\Rightarrow\bar{a}\times\left(\bar{a}\times\bar{x}\right)+\left(\bar{a}\times\bar{b}\right)=\bar{a}\times\bar{x}$

$\left(\bar{a}\cdot\bar{x}\right)\bar{a}-\left(\bar{a}\cdot\bar{a}\right)\bar{x}+\left(\bar{a}\times\bar{b}\right)=\bar{x}-\bar{b}$

Projection of $\bar{x}$ along $\bar{a}$ is 2 units $\Rightarrow\dfrac{\left(\bar{a}\cdot\bar{x}\right)}{|\bar{a}|}=2\quad\Rightarrow\bar{a}\cdot\bar{x}=2$ So $\bar{x}=\dfrac{1}{2}\left[2\bar{a}+\bar{b}+\left(\bar{a}\times\bar{b}\right)\right]$.

4.(A) Given that $\dfrac{1}{\sqrt{2}}=\dfrac{\sqrt{2}a+3\sqrt{2}b+4c}{6.m}$, m = $|\bar{\alpha}|$ $\Rightarrow$2(a + 3b) + 4$\sqrt{2}$ c = 6m

Since a, b, c are rational, so c = 0, therefore vector lies in the xy plane.

5.(D) $\bar{d}.\bar{a}=\cos y[\bar{a}\,\bar{b}\,\bar{c}]=-\bar{d}.(\bar{b}+\bar{c})$ [as $\bar{d}.(\bar{a}+\bar{b}+\bar{c})=0$] $\Rightarrow$ $\cos y==-\dfrac{\bar{d}\cdot(\bar{a}+\bar{b})}{[\bar{a}\,\bar{b}\,\bar{c}]}$

Similarly $\sin x=-\dfrac{\bar{d}\cdot\left(\bar{a}+\bar{b}\right)}{\left[\bar{a}\ \bar{b}\ \bar{c}\right]}$ and $2=-\dfrac{\bar{d}\cdot\left[\bar{a}+\bar{c}\right]}{\left[\bar{a}\ \bar{b}\ \bar{c}\right]}$

$\sin x+\cos y+2=0\Rightarrow\sin x+\cos y=-2\Rightarrow\sin x=-1,\ \cos y=-1$

Since we want minimum value of x^2+y^2, so $x=-\dfrac{\pi}{2}$, $y=\pi$.

6.(C) $\overrightarrow{AB}\times\overrightarrow{AC}=(\vec{i}+\vec{k})\times(2\vec{i}+\vec{j})=\vec{k}+2\vec{j}-\vec{i}$.

Height of tetrahedron = $\dfrac{\left|\overrightarrow{AD}\cdot\overrightarrow{AB}\times\overrightarrow{AC}\right|}{6\left|\overrightarrow{AB}\times\overrightarrow{AC}\right|}=\dfrac{\left|(\vec{i}+2\vec{j}+\vec{k})\cdot(-\vec{i}+2\vec{j}+\vec{k})\right|}{6\left|-\vec{i}+2\vec{j}+\vec{k}\right|}=\left|\dfrac{-1+4+1}{6\sqrt{6}}\right|=\dfrac{4}{6\sqrt{6}}=\dfrac{1}{3}\sqrt{\dfrac{2}{3}}$.

7.(C) $\bar{d}\cdot\hat{a}=\bar{d}\cdot\hat{b}=\bar{d}\cdot\hat{c}\ \Rightarrow\ \lambda(1+\hat{b}\cdot\hat{c})=\lambda(1+\hat{b}\cdot\hat{c})=\lambda(\hat{a}\cdot\hat{b}+\hat{a}\cdot\hat{c})$

$\Rightarrow$ $1+\hat{b}\cdot\hat{c}=\hat{a}\cdot\hat{b}+\hat{a}\cdot\hat{c}$ $\Rightarrow$ $1-\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}-\hat{a}\cdot\hat{c}=0$

$\Rightarrow$ $1-\hat{a}\cdot\hat{b}+(\hat{b}-\hat{a})\cdot\hat{c}=0$ $\Rightarrow$ $\hat{a}\cdot(\hat{a}-\hat{b})+(\hat{b}-\hat{a})\cdot\hat{c}=0$

$\Rightarrow$ $(\hat{a}-\hat{c})\cdot(\hat{a}-\hat{b})=0\Rightarrow\hat{a}-\hat{c}$ is perpendicular to $(\hat{a}-\hat{b})$

$\Rightarrow$ The triangle is right angled.

8.(D) Since $\bar{d}$ makes equal angles with the vectors $\bar{a}$, $\bar{b}$ and $\bar{c}\Rightarrow$ d = $\dfrac{\mu(\bar{a}+\bar{b}+\bar{c})}{3}$

[$\bar{d}$ passes through the centroid of the triangle with position vectors $\bar{a}$, $\bar{b}$, $\bar{c}$] **... (1)**

Again $[\bar{a}\ \bar{b}\ \bar{c}]\bar{d}=[\bar{d}\ \bar{b}\ \bar{c}]\bar{a}+[\bar{d}\ \bar{c}\ \bar{a}]\bar{b}+[\bar{d}\ \bar{a}\ \bar{b}]\bar{c}$ **... (2)**

From (1) and (2), we get $[\bar{d}\ \bar{b}\ \bar{c}]=[\bar{d}\ \bar{c}\ \bar{a}]=[\bar{d}\ \bar{a}\ \bar{b}]$.

9.(A) $\overline{a}+\overline{b}+\overline{c}-3\overline{d}=0$ and $1+1+1-3=0 \Rightarrow \overline{a}, \overline{b}, \overline{c}$ and $\overline{d}$ are coplanar.

10.(B) $\vec{\alpha}\cdot\vec{\beta} = ab + bc + ca = \sqrt{a^2+b^2+c^2}\ \sqrt{b^2+c^2+a^2}\ \cos\theta \quad \Rightarrow \quad \cos\theta = \dfrac{ab+bc+ca}{\left(a^2+b^2+c^2\right)}$

Now $(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$

$a^2 + b^2 + c^2 \geq ab + bc + ca \Rightarrow \dfrac{ab+bc+ca}{a^2+b^2+c^2} \leq 1$

Also $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \geq 0$

$\Rightarrow \quad \dfrac{ab+bc+ca}{a^2+b^2+c^2} \geq -1/2 \quad \Rightarrow \quad -\dfrac{1}{2} \leq \cos\theta \leq 1 \Rightarrow \theta \in [0, 2\pi/3]$.

11.(B) For the point of intersection of the lines

$\vec{a}+t\left(\vec{b}\times\vec{c}\right)=\vec{b}+s\left(\vec{c}\times\vec{a}\right) \Rightarrow \vec{a}.\vec{c}+t\left(\vec{b}\times\vec{c}\right).\vec{c}=\vec{b}.\vec{c}+s\left(\vec{c}\times\vec{a}\right).\vec{c} \quad \Rightarrow \quad \vec{a}.\vec{c}=\vec{b}.\vec{c}$.

Hence **(B)** is the correct answer.

12.(C) $4\overline{a}+5\overline{b}+9\overline{c}=0 \Rightarrow$ Vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar

$\Rightarrow \quad \vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ are collinear $\Rightarrow \left(\vec{b}\times\vec{c}\right)\times\left(\vec{c}\times\vec{a}\right)=\vec{0}$.

Hence **(C)** is the correct answer

13.(C) We have $p\overline{r}+(\overline{r}\cdot\overline{b})\overline{a}=\overline{c} \Rightarrow p\overline{r}\cdot\overline{b}+(\overline{r}\cdot\overline{b})\overline{a}\cdot\overline{b}=\overline{c}\cdot\overline{b}$

$\Rightarrow \quad p\overline{r}\cdot\overline{b}+0=\overline{c}\cdot\overline{b} \Rightarrow \overline{r}.\overline{b}=\dfrac{\overline{c}\cdot\overline{b}}{p} \Rightarrow p\overline{r}+\dfrac{\overline{c}\cdot\overline{b}}{p}\overline{a}=\overline{c} \Rightarrow \overline{r}=\dfrac{\overline{c}}{p}-\dfrac{\overline{c}.\overline{b}}{p^2}\overline{a}$.

14.(D) $\ell x+my=0$ **....(i)**, $my+nz=0$ **....(ii)**, $\ell x+nz=0$ **....(iii)** and $\ell x+my+nz=p$ **....(iv)**

Solving (i), (ii), (iv) $\quad A\left(\dfrac{p}{\ell}, \dfrac{-p}{m}, \dfrac{p}{n}\right)$

Similarly other points are $B\left(\dfrac{p}{\ell}, \dfrac{p}{m}, \dfrac{-p}{n}\right)$, $C\left(\dfrac{-p}{\ell}, \dfrac{p}{m}, \dfrac{p}{n}\right)$ and O(0, 0, 0)

$\overline{OA}=\dfrac{p}{\ell}\hat{i}-\dfrac{p}{m}\hat{j}+\dfrac{p}{n}\hat{k}$, $\overline{OB}=\dfrac{p}{\ell}\hat{i}+\dfrac{p}{m}\hat{j}-\dfrac{p}{n}\hat{k}$ and $\overline{OC}=-\dfrac{p}{\ell}\hat{i}+\dfrac{p}{m}\hat{j}+\dfrac{p}{n}\hat{k}$

Volume of parallelopiped $=\dfrac{p^3}{\ell mn}\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}=\dfrac{4p^3}{\ell mn} \quad \therefore \quad$ Volume of tetrahedron $=\dfrac{1}{6}\times\dfrac{4p^3}{\ell mn}$

15.(C) Given $\vec{r}\times\vec{a}=\vec{b}\times\vec{a} \Rightarrow \vec{r}\times\vec{a}-\vec{b}\times\vec{a}=0 \quad \Rightarrow \quad (\vec{r}-\vec{b})\times\vec{a}=0 \Rightarrow (\vec{r}-\vec{b})$ is parallel/collinear to $\vec{a}$

$\Rightarrow \quad \vec{r}-\vec{b}=p\vec{a} \Rightarrow \vec{r}=b+p\vec{a}$ **... (1)** (Equation of the line)

Similarly, $\vec{r}\times\vec{b}-\vec{a}\times\vec{b}=0 \quad \Rightarrow \quad \vec{r}=a+s\vec{b}$ **... (2)** (Equation of the line)

The point of intersection of (1) and (2) gives $\vec{b}+p\vec{a}=\vec{a}+s\vec{b} \Rightarrow \vec{b}(1-s)+\vec{a}(p-1)=0$

Since $\vec{a}$ and $\vec{b}$ are non collinear $\quad \Rightarrow 1-s=0$ and $p-1=0 \Rightarrow p=s=1$

Hence point of intersection is $\vec{a}+\vec{b}=3\hat{i}+\hat{j}-\hat{k}$.

16.(A) As $\overline{a}$ and $\overline{b}$ are $\perp \Rightarrow \overline{a}\cdot\overline{b}=0$

$$\overline{r}\times\overline{a}=\overline{b} \qquad \textbf{... (i)}$$

Taking cross product with $\overline{a}$ $\Rightarrow$ $\overline{a}\times(\overline{r}\times\overline{a})=\overline{a}\times\overline{b}$ $\Rightarrow$ $\overline{r}=\dfrac{(\overline{a}\cdot\overline{r})}{\overline{a}\cdot\overline{a}}\overline{a}+\dfrac{(\overline{a}\times\overline{b})}{\overline{a}\cdot\overline{a}}$

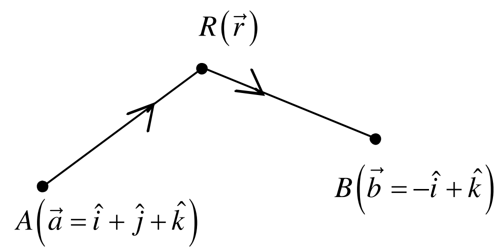
17.(A) Any vector $\perp$ to plane of PQR is $\overline{PQ}\times\overline{PR}$ $=(\overline{q}-\overline{p})\times(\overline{r}-\overline{p})$ $=\overline{q}\times\overline{r}+\overline{p}\times\overline{q}+\overline{r}\times\overline{p}$

If OP makes an angle α with face PQR. Then it will make an angle $\left(\dfrac{\pi}{2}-\alpha\right)$ with its normal.

18.(B) Taking dot product with $\overline{a},\overline{b}\;\&\;\overline{c}$ respectively $\Rightarrow$ $l=\dfrac{\overline{r}\cdot\overline{a}}{[\overline{a}\,\overline{b}\,\overline{c}]}=\dfrac{\overline{r}\cdot\overline{a}}{2}$; $m=\dfrac{\overline{r}\cdot\overline{b}}{2},n=\dfrac{\overline{r}\cdot\overline{c}}{2}$

19.(A) $\overline{b}\cdot\overline{c}=0$ and $\overline{a}\cdot\hat{k}<0$

20.(D) $\overrightarrow{AR}=\left(\vec{r}-\vec{a}\right)$, $\overrightarrow{BR}=\left(\vec{r}-\vec{b}\right)$.



Since $\overrightarrow{AR}$ is perpendicular to $\overrightarrow{BR}$

$\Rightarrow\left(\vec{r}-\vec{a}\right)\cdot\left(\vec{r}-\vec{b}\right)=0$ $\Rightarrow|\vec{r}|^2-\left(\vec{a}+\vec{b}\right)\vec{r}+\vec{a}\cdot\vec{b}=0$

$\Rightarrow\vec{r}\cdot\left(\vec{r}-\left(\vec{a}+\vec{b}\right)\right)+0=0$ $\Rightarrow$ Either $\vec{r}$ is perpendicular to $\vec{r}-\left(\vec{a}+\vec{b}\right)$ or $\vec{r}=\left(\vec{a}+\vec{b}\right)\Rightarrow\vec{r}=\hat{j}+2\hat{k}$.

21.(B) Volume of tetrahedron $V=\dfrac{1}{6}\left[\bar{a}\bar{b}\bar{c}\right]$

Here $\left[\bar{a}\bar{b}\bar{c}\right]^2=\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}=\begin{vmatrix}4 & 3 & 4\sqrt{3}\\ 3 & 9 & 6\sqrt{2}\\ 4\sqrt{3} & 6\sqrt{2} & 16\end{vmatrix}=4[144-72]+3\left[24\sqrt{6}-48\right]+4\sqrt{3}\left[18\sqrt{2}-36\sqrt{3}\right]$

$$=288+72\sqrt{6}-144+72\sqrt{6}-432=144\sqrt{6}-288=144\left(\sqrt{6}-2\right)$$

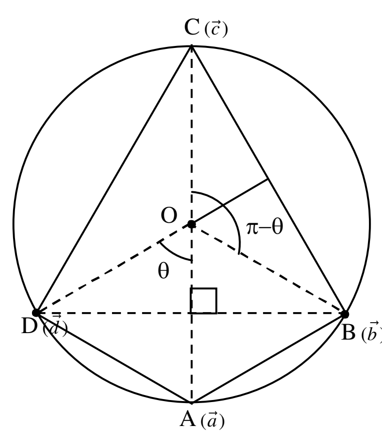
$$V=\frac{1}{6}\times12\sqrt{\left(\sqrt{6}-2\right)}=2\sqrt{\sqrt{6}-2}\quad\text{Since } V=\frac{1}{3}\times\text{base area}\times\text{height}\Rightarrow\text{height}=\frac{6\sqrt{\sqrt{6}-2}}{2}=3\sqrt{\left(\sqrt{6}-2\right)}.$$

22.(B) $[(\overline{a}\times\overline{b})\times(\overline{c}\times\overline{d})]\cdot(\overline{c}\times\overline{b})=0$ $\Rightarrow$ $([\overline{a}\,\overline{b}\,\overline{d}]\overline{c}-[\overline{a}\,\overline{b}\,\overline{c}]d)\cdot(\overline{c}\times\overline{b})=0$ $\Rightarrow$ $[\overline{a}\,\overline{b}\,\overline{c}]\,[\overline{d}\,\overline{c}\,\overline{b}]=0$

23.(C) Take cross product with $\hat{\alpha}$, we get : $\hat{\alpha}\times\overline{x}=\hat{\alpha}\times\hat{\beta}-\hat{\alpha}\times(\hat{\alpha}\times\overline{x})$ $\Rightarrow$ $\hat{\alpha}\times\overline{x}=\hat{\alpha}\times\hat{\beta}+\overline{x}-(\hat{\alpha}\cdot\overline{x})\hat{\alpha}$

But if we take dot product with $\hat{\alpha}$ we get $\hat{\alpha}\cdot\overline{x}=0$ So $2\overline{x}=\hat{\beta}-(\hat{\alpha}\times\hat{\beta})$

Now $4|\overline{x}|^2=|\hat{\beta}|^2+|\hat{\alpha}\times\hat{\beta}|^2-2\hat{\alpha}\cdot(\hat{\alpha}\times\hat{\beta})=2\Rightarrow|x|=\dfrac{1}{\sqrt{2}}$.



24.(D) $|\vec{d}-\vec{a}|^2+|\vec{b}-\vec{c}|^2=(\vec{d}-\vec{a})\cdot(\vec{d}-\vec{a})+(\vec{b}-\vec{c})\cdot(\vec{b}-\vec{c})$

$=|\vec{d}|^2+|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2-2\vec{d}\cdot\vec{a}-2\vec{b}\cdot\vec{c}$

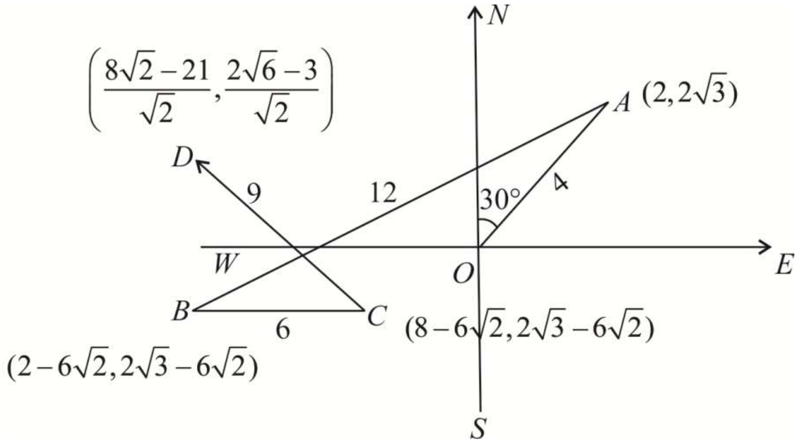
$=4R^2-2(\vec{d}\cdot\vec{a}+\vec{b}\cdot\vec{c})$

$=4R^2-2R^2(\cos\angle AOB+\cos\angle BOC)$

$=4R^2\;(\because\angle AOB=\pi-\angle BOC)$

25.(A) The internal bisector will be parallel to $\hat{a}+\hat{b}$

26-30. **26.(C)** **27.(D)** **28.(B)** **29.(B)** **30.(A)**



31.(A) The vector along the bisector of the given angle is $\vec{c}=t\left(\frac{7\hat{i}-4\hat{j}-4\hat{k}}{\sqrt{81}}+\frac{-2\hat{i}-\hat{j}+2\hat{k}}{\sqrt{9}}\right)=t\left(\frac{\hat{i}-7\hat{j}+2\hat{k}}{9}\right)$

Since $|c|=5\sqrt{6}, 5\sqrt{6}=t\frac{\sqrt{54}}{9} \Rightarrow t=15.$

32.(B) Vectors along AB, BC and CA are $\vec{b}-\vec{a}, \vec{c}-\vec{b}$ and $\vec{a}-\vec{c}$ respectively. Hence the bisectors of the angle B and C respectively are

$\vec{r}=\vec{b}+\mu\left(\frac{\vec{b}-\vec{a}}{c}+\frac{\vec{c}-\vec{b}}{a}\right)$ and $\vec{r}=\vec{c}+t\left(\frac{\vec{b}-\vec{c}}{a}+\frac{\vec{c}-\vec{a}}{b}\right)$ so that for the point P

$\vec{b}+\mu\left(\frac{\vec{b}-\vec{a}}{c}+\frac{\vec{c}-\vec{b}}{a}\right)=\vec{c}+t\left(\frac{\vec{b}-\vec{c}}{a}+\frac{\vec{c}-\vec{a}}{b}\right) \Rightarrow t=\frac{b\mu}{c}$ and $\mu=\frac{ac}{b+c+a} \Rightarrow$ the p.v. of P is $\frac{b\vec{b}+c\vec{c}-a\vec{a}}{b+c-a}$.

33.(C) Let A be the initial point and the p.v. of B and C be $\vec{b}$ and $\vec{c}$ respectively. Hence, let $AB=c$ and $AC=b$. The internal bisector is $r=t\left(\frac{\vec{b}}{c}+\frac{\vec{c}}{b}\right)$ and the equation of the line BC is $\vec{r}=\vec{b}+k(\vec{c}-\vec{b})$ and hence the p.v. of D is $\frac{b\vec{b}+c\vec{c}}{b+c}$

$$\Rightarrow \quad \overrightarrow{BD}=\frac{b\vec{b}+c\vec{c}}{b+c}-\vec{b}=\frac{c(\vec{c}-\vec{b})}{b+c} \quad \Rightarrow \quad BD=\frac{c|\vec{c}-\vec{b}|}{b+c}=\frac{ca}{b+c}.$$

Similarly $BE=\frac{c|\vec{c}-\vec{b}|}{c-b}=\frac{ca}{c-b}$ and hence $\frac{1}{BE}+\frac{1}{BD}=\frac{c-b}{ac}+\frac{c+b}{ac}=\frac{2c}{ac}=\frac{2}{a}=\frac{2}{BC}$.

34.(C) Incentre is the point of intersection of the internal bisectors of angles A, B and C. Equation of the bisectors of angles A and B are $\vec{r}=\vec{a}+t\left(\frac{\vec{c}-\vec{a}}{b}+\frac{\vec{b}-\vec{a}}{c}\right)$ and $\vec{r}=\vec{b}+\mu\left(\frac{\vec{a}-\vec{c}}{c}+\frac{\vec{c}-\vec{b}}{a}\right)$

Equating the coefficient of $\vec{a}, \vec{b}$ and $\vec{c}$, we find that the *p.v* of the incentre is $\frac{a\vec{a}+b\vec{b}+c\vec{c}}{a+b+c}$.

35.(A) Since the three vertices *A*, *B*, *C* of the triangle are coplanar, all the points lying as the linear *AB*, *BC*, *CD* are naturally coplanar.

36.(B) $(\vec{a}\cdot\vec{c})\vec{b}-(\vec{b}\cdot\vec{c})\vec{a}=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}-\vec{b})\vec{c} \Rightarrow \quad \vec{b}\cdot\vec{c}=0$ and $\vec{a}\cdot\vec{b}=0$ or $\vec{a}=\lambda\vec{c}$

37.(A)

38.(B) $(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=\dfrac{\vec{b}}{2} \Rightarrow \vec{a}\cdot\vec{c}=\dfrac{1}{2}$ & $\vec{a}\cdot\vec{b}=0 \Rightarrow$ angle between $\vec{a}$ & $\vec{c}=\dfrac{\pi}{3}$

39.(D) **40.(C)**

39-40 $\vec{r}\cdot(10\hat{j}-8\hat{i}-\vec{r})=40$ **;** $\vec{r}=x\hat{i}+y\hat{j}$

Curve is circle $x^2+y^2+8x-10y+40=0$; Centre $=(-4, 5)$; $r=1$, We need to find distance from $(-2, 3)$

41.(CD) $|\vec{a}+\vec{b}+\vec{c}+\vec{d}|^2=\Sigma|\vec{a}|^2+2\Sigma\,\vec{a}\cdot\vec{b}=4+2\vec{d}\,.(\vec{a}+\vec{b}+\vec{c}\,)$

Let $\vec{d}=\lambda\vec{a}+\mu\vec{b}+\nu\vec{c}$, then $\vec{d}\cdot\vec{a}=\vec{d}\cdot\vec{b}=\vec{d}\cdot\vec{c}=\cos\theta \quad\Rightarrow\quad \lambda=\mu=\nu=\cos\theta$

Also $\lambda^2+\mu^2+\nu^2=1 \Rightarrow 3\cos^2\theta=1 \Rightarrow \cos\theta=\pm\dfrac{1}{\sqrt{3}} \qquad \therefore \qquad |\vec{a}+\vec{b}+\vec{c}+\vec{d}|^2=4\pm\dfrac{2\cdot 3}{\sqrt{3}}=4\pm 2\sqrt{3}$.

42.(AB) As $\bar{d}$ is perpendicular to $\bar{a}+\bar{b}+\bar{c}$

$\Rightarrow \quad \bar{d}\cdot\bar{a}+\bar{d}\cdot\bar{b}+\bar{d}\cdot\bar{c}=0$ **... (1)**

Given $\bar{d}=x(\bar{a}\times\bar{b})+y(\bar{b}\times\bar{c})+z(\bar{c}\times\bar{a})$ **... (2)**

$\Rightarrow \quad y[\bar{b}\,\bar{c}\,\bar{a}]+z[\bar{c}\,\bar{a}\,\bar{b}]+x[\bar{a}\,\bar{b}\,\bar{c}]=0 \quad\Rightarrow\quad (x+y+z)[\bar{a}\,\bar{b}\,\bar{c}]=0$

As $\bar{a},\bar{b}\,\&\,\bar{c}$ are non coplanar $\quad x+y+z=0$ **... (3)**

$\Rightarrow \quad x^3+y^3+z^3-3xyz=(x+y+z)(x^2+y^2+z^2-xy-yz-zx)=0$

$(x+y+z)^2=0$; $x^2+y^2+z^2=-2(xy+yz+zx) \quad\Rightarrow\quad (xy+yz+zx)=\dfrac{x^2+y^2+z^2}{-2}$

43.(AC) $\bar{u}\cdot\bar{b}=0$ **... (1)**

$\bar{u}=\bar{a}-(\bar{a}\cdot\bar{b})\bar{b}\;=(\bar{b}\cdot\bar{b})\bar{a}-(\bar{a}\cdot\bar{b})\bar{b}$

$\Rightarrow \quad |\bar{u}|=\bar{b}\times(\bar{a}\times\bar{b}) \quad\Rightarrow\quad |\bar{u}|=|\bar{b}||\bar{a}\times\bar{b}|\sin\dfrac{\pi}{2}=|\bar{a}\times\bar{b}| \;\Rightarrow\; |\bar{u}|=|\bar{v}|$ **... (2)**

44.(AC) $\bar{a}\times(\bar{b}\times\bar{c})=(x^2-2x+6)\bar{b}+\sin y\,\bar{c}$

$\Rightarrow \quad (\bar{a}\cdot\bar{c})\bar{b}-(\bar{a}\cdot\bar{b})\bar{c}=(x^2-2x+6)\bar{b}+\sin y\bar{c} \quad\Rightarrow\quad \bar{a}\cdot\bar{c}=x^2-2x+6$ and $\bar{a}\cdot\bar{b}=-\sin y$

As $\bar{a}\cdot\bar{b}+\bar{a}\cdot\bar{c}=4 \;\Rightarrow\; x^2-2x+6-\sin y=4$, $(x-1)^2+1=\sin y \;\Rightarrow\; x=1\;\&\; y=(4n+1)\dfrac{\pi}{2}$

45.(ABC) As $\bar{\alpha},\bar{\beta}\,\&\,\bar{\gamma}$ are coplanar $\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$

or $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)=\dfrac{(a+b+c)}{2}((a-b)^2+(b-c)^2+(c-a)^2)=0$

$\Rightarrow$ Either $a+b+c=0$ or $a=b=c$

46.(ABCD) Given $\bar{\alpha}\cdot\bar{\beta}=0,\ \bar{\alpha}\cdot\bar{\gamma}=\bar{\beta}\cdot\bar{\gamma}=\cos\theta$ **... (1)**

By taking dot product by $\hat{\alpha}\,\&\,\hat{\beta}$ we get $x=y=\cos\theta$ **... (2)**

$z(\hat{\alpha}\times\hat{\beta})=(\hat{\gamma}-x\hat{\alpha}-y\hat{\beta}) \;\Rightarrow\; z^2|\hat{\alpha}\times\hat{\beta}|^2=|(\hat{\gamma}-x\hat{\alpha}-y\hat{\beta})|^2 \;\Rightarrow\; z^2=1-2\cos^2\theta$ **... (3)**

47.(AB) $\hat{a}\cdot\hat{b}=\cos 2\theta$ and $|\hat{a}-\hat{b}|<1 \Rightarrow 1+1-2\hat{a}\cdot\hat{b}<1$; $2\hat{a}\cdot\hat{b}>1 \Rightarrow \cos 2\theta>1/2 \Rightarrow 1/2<\cos 2\theta\le 1$

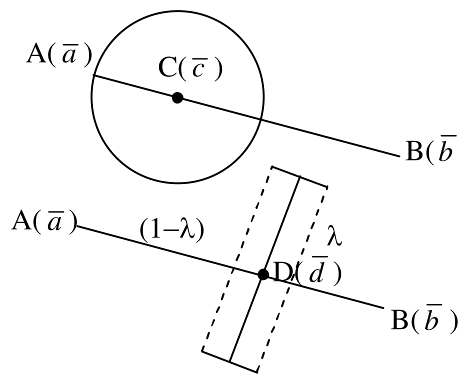
48.(BC) Let $\vec{r}=\overrightarrow{AB}+\lambda\overrightarrow{AC}=\hat{k}-\hat{i}+\lambda\left(\hat{k}-\hat{j}\right)=-\hat{i}-\lambda\hat{j}+(1+\lambda)\hat{k}$ **....(i)**

Again since $|AB|=|BC|=|CA|$,

Incentre is same as circumcentre, and hence IA is perpendicular to $BC \Rightarrow \vec{r}$ is parallel to $BC \Rightarrow \vec{r}=\lambda\left(\hat{i}-\hat{j}\right)$.

Hence unit vector $\vec{r}=\pm\dfrac{1}{\sqrt{2}}\left(\hat{i}-\hat{j}\right)$

49.(ABC) Take $\bar{c}=\dfrac{2\bar{a}+\bar{b}}{3}$ **... (1)**



then first equation becomes $|\bar{r}-\bar{c}|=\dfrac{1}{3}|\bar{a}-\bar{b}|$ **... (2)**

Locus of variable point P is a sphere whose centre is at point C ($\bar{c}$)

and radius $\dfrac{1}{3}|\bar{a}-\bar{b}|=\dfrac{1}{3}AB$

Taking $\lambda\bar{a}+(1-\lambda)\bar{b}=\bar{d}$. D ($\bar{d}$) lie on the line joining A ($\bar{a}$)

and B ($\bar{b}$) and the equation becomes $(\bar{r}-\bar{d})\cdot(\bar{a}-\bar{b})=0$

$\Rightarrow$ In this case locus of the variable point is a plane passing through point $D(\bar{d})$ and normal to the plane containing $(\bar{a}-\bar{b})$

Obviously the system of equation will not have any solution if point D ($\bar{d}$) lies outside the sphere

$\Rightarrow$ $|3\bar{d}-2\bar{a}-\bar{b}|-|\bar{a}-\bar{b}|>0$ ... (3)

$\Rightarrow$ $|3\lambda-2|>1$ $\Rightarrow$ $\lambda<\dfrac{1}{3}$ or $\lambda>1 \Rightarrow \lambda\in(-\infty, 1/3)\cup(1,\infty)$.

50.(AC) Let vector $\bar{d}$ be $\bar{d}=\dfrac{\bar{a}+\bar{b}+a\times\bar{b}}{3}$ then

$\bar{d}.\bar{a}=\dfrac{|\bar{a}|^2+\bar{a}.\bar{b}}{3}=|\bar{d}||\bar{a}|\cos\alpha$ **... (1)**

$\bar{d}.\bar{b}=\dfrac{|\bar{b}|^2+\bar{a}.\bar{b}}{3}=|\bar{d}||\bar{b}|\cos\alpha$ **... (2)**

$\bar{d}.(\bar{a}\times\bar{b})=\dfrac{|\bar{a}\times\bar{b}|^2}{3}=|\bar{d}||\bar{a}\times\bar{b}|\cos\alpha$ **... (3)**

(i) From equation (1) and (2) we get $\dfrac{|\bar{a}|^2+\bar{a}.\bar{b}}{3|\bar{a}|}=\dfrac{|\bar{b}|^2+\bar{a}.\bar{b}}{3|\bar{b}|}$

$\Rightarrow$ $(|\bar{a}|-|\bar{b}|)\left(1-\dfrac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|}\right)=0$ $\Rightarrow$ $|\bar{a}|=|\bar{b}|$, as $\bar{a}$ and $\bar{b}$ are non-collinear

(ii) From equation (2) and (3)

$|\bar{b}|=\dfrac{|\bar{b}|^2+\bar{a}.\bar{b}}{|\bar{a}\times\bar{b}|}\Rightarrow|\bar{b}|^3\sin\theta=|\bar{b}|^2(1+\cos\theta)$ (where θ is the angle between $\bar{a}$ and $\bar{b}$)

$\Rightarrow$ $|\bar{b}|=|\bar{a}|=\dfrac{1+\cos\theta}{\sin\theta}$

Now $|d|^2 = \dfrac{(\bar{a}+\bar{b}+\bar{a}\times\bar{b})}{3}\cdot\dfrac{(\bar{a}+\bar{b}+\bar{a}\times\bar{b})}{3}$

$\Rightarrow \quad |d|^2 = \dfrac{|\bar{a}|^2+|\bar{b}|^2+2\bar{a}.\bar{b}+|\bar{a}\times\bar{b}|^2}{9} \quad \Rightarrow \quad 9|\bar{d}|^2 = |\bar{a}|^2(2+2\cos\theta+|\bar{a}|^2\sin^2\theta)$

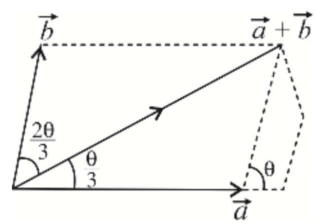
$\Rightarrow \quad 9|\bar{d}|^2 = |\bar{a}|^2(2+2\cos\theta+(1+\cos\theta)^2) \quad \Rightarrow \quad 9|\bar{d}|^2 = |\bar{a}|^2(1+\cos\theta)(3+\cos\theta)$

Now $|\bar{d}|\cos\alpha = \dfrac{|\bar{a}\times\bar{b}|}{3} \Rightarrow |\bar{d}|\cos\alpha = \dfrac{|\bar{a}||\bar{b}|\sin\theta}{3}$

$\Rightarrow \quad \cos\alpha = \dfrac{|\bar{a}|^2\sin\theta}{|\bar{a}|\sqrt{(1+\cos\theta)(3+\cos\theta)}} \quad \Rightarrow \quad \cos\alpha = \sqrt{\dfrac{1-\cos\theta}{3+\cos\theta}}$

$\Rightarrow \quad \cos\alpha = \dfrac{1}{\sqrt{3}}$ or $\alpha = \cos^{-1}\dfrac{1}{\sqrt{3}}$ when $\bar{a}.\bar{b} = 0$.

51.(BD) $\tan\dfrac{\theta}{3} = \dfrac{b\sin\theta}{a+b\cos\theta} \Rightarrow a\sin\dfrac{\theta}{3} + b\sin\dfrac{\theta}{3}\cos\theta = b\sin\theta\cos\dfrac{\theta}{3}$



$\Rightarrow \quad a\sin\dfrac{\theta}{3} = b\left[\sin\theta\cos\dfrac{\theta}{3} - \cos\theta\sin\dfrac{\theta}{3}\right] \Rightarrow a\sin\dfrac{\theta}{3} = b\sin\dfrac{2\theta}{3}$

$\Rightarrow \quad a\sin\dfrac{\theta}{3} = 2b\sin\dfrac{\theta}{3}\cos\dfrac{\theta}{3} \Rightarrow \cos\dfrac{\theta}{3} = \dfrac{a}{2b} \Rightarrow \theta = 3\cos^{-1}\left(\dfrac{a}{2b}\right)$

$\left|\vec{a}+\vec{b}\right| = \sqrt{a^2+b^2+2ab\cos\theta} = \sqrt{a^2+b^2+2ab\left(4\cos^3\dfrac{\theta}{3} - 3\cos\dfrac{\theta}{3}\right)}$

$= \sqrt{a^2+b^2+2ab\left(\dfrac{4a^3}{8b^3}\right) - 6ab\left(\dfrac{a}{2b}\right)} = \sqrt{b^2+\dfrac{a^4}{b^2}-2a^2} = \dfrac{a^2-b^2}{b}$

52.(BD) $\bar{a}\times(\bar{b}\times\bar{c})$ is coplanar with $\bar{b}$ & $\bar{c}$ and $\perp \bar{a}$.

53.(AB) Equation of line in paramagnetic form is $\bar{r} = (\hat{i}+\hat{j}-2\hat{k}) + \lambda\left(\dfrac{4}{5}\hat{j} - \dfrac{3}{5}\hat{k}\right)$ Place $\lambda = \pm 1$

54.(AD) Knowing the volume of the prism we find its altitude $H = (AA_1) = \sqrt{6}$ and

Let Vertex $A_1(x, y, z)$

$\vec{AA_1} = (x-1, y, z-1)$. We get the other equations and from the condition $\vec{AA_1}$

Perpendicular to $\vec{AC}$ or compute $\pm\dfrac{\overline{AB}\times\overline{AC}}{|\overline{AB}\times\overline{AC}|} = \hat{n}$

$\sqrt{6}\hat{n} = \pm\left(\hat{i}+2\hat{j}+\hat{k}\right) = (x-1)\hat{i} + (y)\hat{j} + (z-1)\hat{k}$ Compare to get A & D

55.(ABCD) $|(\bar{a}\times\bar{b})\cdot\bar{c}| = |\bar{a}||\bar{b}|\sin\alpha(\hat{n}\cdot\bar{c}) = |\bar{a}||\bar{b}||\bar{c}|\sin\alpha\cos\beta$

Where α is angle between $\bar{a}$ & $\bar{b}$, & β is angle between $\bar{c}$ & normal to $\bar{a}$ & $\bar{b}$.

If $|(\bar{a}\times\bar{b})\cdot\bar{c}| = |\bar{a}||\bar{b}||\bar{c}| \Rightarrow \alpha = \dfrac{\pi}{2}, \beta = 0$

56.AC (A) Section Formula (C) $(\vec{b}-\vec{a})\times(\vec{c}-\vec{a}) = 0 \Rightarrow \vec{b}\times\vec{c}+\vec{a}\times\vec{b}+\vec{c}\times\vec{a} = 0$

57.BC (B) Line of intersection of 2 planes (C) $\perp$ to both $\vec{a}\times\vec{b}$ and $\vec{c}\times\vec{d}$

58.(CD) $\left|\vec a+\vec b\right|<1 \Rightarrow \left|\vec a+\vec b\right|^2<1 \Rightarrow 1+1+2\cos\alpha<1 \Rightarrow \cos\alpha<-\frac{1}{2}$

59.CD Let vector be $l\vec b+m\vec c$; $\left|\left(l\vec b+m\vec c\right)\cdot\vec a\right|=\frac{\sqrt2}{\sqrt3}$

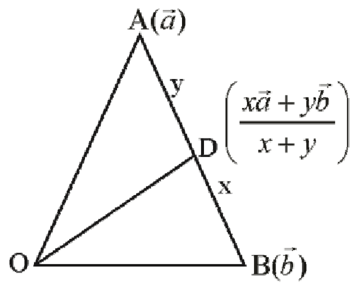
60.AC use angle bisector formula $\frac{\vec a+\vec b}{\left|\vec a+\vec b\right|}\times 2$

61.BC $\vec\lambda+\vec\mu+\vec v=0 \Rightarrow$ coplanar

62.AC

63. **[A-q, r, s] [B-p, s] [C-p, r] [D-p, q]**

$\vec c=(x+y)\overline{OD}$



64. **[A-r] [B-p] [C-s] [D-q]**

(B) $[\vec a\ \ \vec b+\vec a\times\vec b\ \ \vec b]=[\vec a\ \vec b\ \vec b]+[\vec a\ \ \vec a\times\vec b\ \ \vec b]=[\vec a\ \ \vec a\times\vec b\ \ \vec b]=\vec a\times(\vec a\times\vec b).\vec b]=((\vec a.\vec b)\vec a-(\vec a.\vec a)\vec b).\vec b$

$(\vec a.\vec b)^2-(\vec a.\vec a)(\vec b.\vec b)$

(C) $[\vec a+\vec b+\vec c\quad \vec a+\vec b\quad \vec b+\vec c]=[\vec a\ \vec b\ \vec c]=|a|^2=1$

(D) As $\vec x$ & $\vec y$ are in the plane of $\vec a$ & $\vec b$ & $\vec a$, $\vec b$ & $\vec c$ are coplanar $\Rightarrow[\vec x\ \vec y\ \vec c]=0$

65. **[A-p, q, r, s] [B-p, q] C-p, r] [D-r]**

(A) $\begin{vmatrix}1 & 2 & -1\\ -1 & 1 & a\\ 0 & 1 & 0\end{vmatrix}\neq 0$

(B) $\vec a.\vec b>0$ **... (1)**

As $\vec b$ makes obtuse angle $\Rightarrow\lambda<0$ **... (2)**

(C) $\begin{vmatrix}2 & -1 & 1\\ 1 & 2 & 1+a\\ 3 & a & 5\end{vmatrix}=0$

66. **[A-r, s] [B-q, r] [C-q, r] [D-p, q, r]**

(A) $\vec a\cdot\vec c=\frac{1}{4}$ Also $\vec a\cdot\vec b-2\vec a\cdot\vec c=\lambda\vec a\cdot\vec a\ \ \vec a\cdot\vec b=\lambda+\frac{1}{2}$

Also $\vec b\cdot\vec b-2\vec b\cdot\vec c=\lambda\vec a\cdot\vec b\ \ \vec b\cdot\vec c=8-\frac{\lambda^2}{2}-\frac{\lambda}{4}$

Also $\vec b\cdot\vec c-2\vec c\cdot\vec c=\lambda\vec a\cdot\vec c\ \ 8-\frac{\lambda^2}{2}-\frac{\lambda}{4}-2=\frac{\lambda}{4} \Rightarrow \lambda^2+\lambda-12=0\Rightarrow\lambda=3,-4$

(B) $xy-2zy+3zx=2zx+3xy-zy$

$\Rightarrow \quad 2xy + yz - zx = 0 \quad$ ......(i) and $x - y + 2z = 0 \quad$ ......(ii)

$x^2 + y^2 + z^2 = 12 \quad$ .....(iii) and $y < 0 \quad$ ....(iv)

On solving (i) and (ii) we get $x + y = 0$

Solving with $x - y = 2z$ [from (ii)]

$2x = -2z \Rightarrow x = -z$

$\therefore \quad x^2 + y^2 + z^2 = 3z^2 = 12 \quad \Rightarrow \quad z = 2, -2$

Given $y < 0 \Rightarrow x > 0$ and $z < 0 \quad \therefore \quad z = -2 \quad \therefore \quad x + y - z = 2 \quad \therefore \quad q, r$ are correct option

(C) $G = \dfrac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{H} = \vec{a} + \vec{b} + \vec{c}$

Now $\overrightarrow{AD} + \overrightarrow{BD} + \overrightarrow{CH} + 3\overrightarrow{HG} = \lambda\overrightarrow{HD}$

$\vec{\alpha} - \vec{a} + \vec{\alpha} - \vec{b} + \vec{a} + \vec{b} - 3\left(\dfrac{2}{3}(\vec{a} + \vec{b} + \vec{c})\right) = \lambda(\vec{\alpha} - (\vec{a} + \vec{b} + \vec{c}))$

$\Rightarrow \quad 2(\vec{d} - \vec{a} + \vec{b} + \vec{c}) = \lambda(\vec{d} - (\vec{a} + \vec{b} + \vec{c})) \quad \Rightarrow \quad \lambda = 2 \quad \therefore \quad q, r$ are correct option

(D) If angle between $\vec{b}$ and $\vec{c}$ is α, then $\left|\vec{b} \times \vec{c}\right| = \sqrt{15}$

$\Rightarrow \quad \left|\vec{b}\right|\left|\vec{c}\right| \sin \pm = \sqrt{15} \Rightarrow \sin \pm = \dfrac{\sqrt{15}}{4}$; $\cos \pm = \dfrac{1}{4}$

$\Rightarrow \quad \vec{b} - 2\vec{c} = 4\lambda\vec{a} \Rightarrow \left(\vec{b} - 2\vec{c}\right)^2 = 16\lambda^2\left(\vec{a}\right)^2 \Rightarrow \left|\vec{b}\right|^2 + 4\left|\vec{c}\right|^2 - 4\vec{b}.\vec{c} = 16\lambda^2\left|\vec{a}\right|^2$

$\Rightarrow \quad 16 + 4 - 4\left|\vec{b}\right|\left|\vec{c}\right| \cos\alpha = 16\lambda^2 \Rightarrow \lambda = \pm 1$

67. **[A-r] [B-p] [C-p, q] [D-s]**

(A) $\because \left|\vec{a} \times \vec{b}\right| = \left|\vec{b} \times \vec{c}\right| = \left|\vec{c} \times \vec{a}\right|$ = twice the area of ΔABC having same direction. $\therefore \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

(B) As all the edges of a regular tetrahedron are of equal lengths.

$\therefore \quad \left|\vec{a}\right| = \left|\vec{b}\right| = \left|\vec{c}\right|$ and $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a}$

(C) $\vec{a} \times \vec{b} = \vec{c} \Rightarrow \vec{a} \perp \vec{c}$ and $\vec{b} \perp \vec{c}$; $\vec{b} \times \vec{c} = \vec{a} \Rightarrow \vec{b} \perp \vec{a}$ and $\vec{c} \perp \vec{a}$

$\Rightarrow \quad \vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular $\Rightarrow \vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$

Further $\left|\vec{a} \times \vec{b}\right| = \left|\vec{c}\right| \Rightarrow \left|\vec{a}\right|\left|\vec{b}\right| = \left|\vec{c}\right|$ and $\left|\vec{b} \times \vec{c}\right| = \left|\vec{a}\right| \Rightarrow \left|\vec{b}\right|\left|\vec{c}\right| = \left|\vec{a}\right| \Rightarrow \left|\vec{c}\right| = \left|\vec{a}\right|$ and $\left|\vec{b}\right| = 1$

(D) $\vec{a} + \vec{b} + \vec{c} = 0 \Rightarrow a^2 + b^2 + c^2 + 2\left(\vec{a}.\vec{b} + \vec{b}\vec{c} + \vec{c}\vec{a}\right) = 0 \Rightarrow \vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a} = -\dfrac{3}{2}$

Also $\left|\vec{a}\right| = \left|\vec{b}\right| = \left|\vec{c}\right| = 1$

68.(5) Let $P(x_1, y_1), Q(x_2, y_2)$ be the two points on $y = 2^{x+2}$

$\overrightarrow{OP}.\hat{i}$ = projection of $\overrightarrow{OP}$ on x – axis, so $x_1 = -1 \Rightarrow y_1 = 2$

$\overrightarrow{OQ}.\hat{i}$ = projection of $\overrightarrow{OQ}$ on x – axis, so $x_2 = 2 \Rightarrow y_2 = 16$

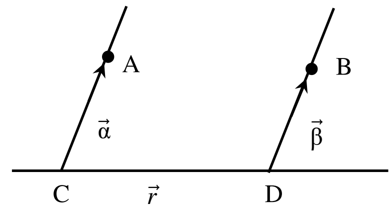
If $\hat{j}$ is a unit vector along y-axis, then $\overrightarrow{OP} = -\hat{i} + 2\hat{j}$, $\overrightarrow{OQ} = 2\hat{i} + 16\hat{j}$

$\Rightarrow \quad \overrightarrow{OQ} - 4\,\overrightarrow{OP} = 6\hat{i} + 8\hat{j} \Rightarrow |\overrightarrow{OQ} - 4\,\overrightarrow{OP}| = \sqrt{36 + 64} = 10$

69.(9) $P.V.$ of C

$\vec{r}_1 = (7\hat{i}+6\hat{j}+2\hat{k}) + a(3\hat{i}+2\hat{j}+4\hat{k}),\ a \in R$

P.V. of D, $\vec{r}_2 = (5\hat{i}+3\hat{j}+4\hat{k}) + b(2\hat{i}+\hat{j}+3\hat{k}),\ b \in R$



$\overrightarrow{CD} = \vec{r}_2 - \vec{r}_1$ and we know that $\overrightarrow{CD} \parallel \vec{r} \Rightarrow \overrightarrow{CD} = c(2\hat{i}+2\hat{j}+\hat{k})$

Hence by comparing both $\overrightarrow{CD}$ we get $3a - 2b - 2c + 2 = 0,\ 2a - b - 2c + 3 = 0$

and $4a - 3b - c - 2 = 0 \Rightarrow a = 2, b = 1, c = 3 \Rightarrow \left|\overrightarrow{CD}\right| = 3\sqrt{2^2+2^2+1^2} = 9$

70.(3) $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right| = \left|\vec{a}\times\vec{b}\right| |\vec{c}| \sin 30^0$

$$\vec{a}\times\vec{b} = \begin{vmatrix} i & j & k \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2i - 2j + k \Rightarrow |\vec{a}\times\vec{b}| = 3$$

$|\vec{c}-\vec{a}|^2 = 8 \Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c}.\vec{a} = 8 \Rightarrow |\vec{c}|^2 - 2|\vec{c}| + 1 = 0 \Rightarrow |\vec{c}| = 1 \Rightarrow \left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right| = 1.\ 3.\ \frac{1}{2} = \frac{3}{2}$

71.(1) $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ **... (1)**

$\Rightarrow (\vec{a}.\vec{b}+\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$

$\Rightarrow \vec{a}.\vec{b}+\vec{a}.\vec{c} = 4-2\beta-\sin\alpha$ **... (2)** $\quad \vec{a}.\vec{b} = 1-\beta^2$ **... (3)** $\quad$ As $(\vec{c}.\vec{c})\vec{a} = \vec{c} \Rightarrow \vec{a}.\vec{c} = 1$ **... (4)**

$\Rightarrow 2-\beta^2 = 4-2\beta-\sin\alpha \Rightarrow \sin\alpha = \beta^2-2\beta+2 \Rightarrow \sin\alpha = (\beta-1)^2+1 \Rightarrow \beta = 1, \alpha = (4n+1)\frac{\pi}{2}$

72.(6) $\vec{w} + (\vec{w}\times\vec{u}) = \vec{v}$ **... (1)**

Taking dot product with $\vec{v}$ & $\vec{w}$ we get :

$[\vec{u}\,\vec{v}\,\vec{w}] = 1 - \vec{w}.\vec{v}$ **... (2)** $\quad |w|^2 = \vec{w}.\vec{v}$ **... (3)** $\quad \Rightarrow [\vec{u}\,\vec{v}\,\vec{w}] = 1 - |w|^2$ **... (4)**

Now $|\vec{w}\times\vec{u}|^2 = |\vec{u}-\vec{w}|^2 \Rightarrow |w|^2 = \dfrac{1}{1+\sin^2\theta} \Rightarrow$ Max $[\vec{u}\ \vec{v}\ \vec{w}] = \dfrac{1}{2}$

73.(7) Given $\vec{v}.\vec{u} = \hat{w}.\hat{u}$ & $\vec{v}.\vec{w} = 0 \Rightarrow |\vec{u}-\vec{v}+\vec{w}|^2 = 14$

74.(5) $\vec{a}.\vec{b} = 0\ \ x_1 + x_2 + x_3 = 0$

Thus, we have to obtain the number of integral solutions of this equation.

Coefficient of x^0 *in* $(x^{-3}+x^{-2}+x^{-1}+x^0+x+x^2)^3$

$$= x^0 \left| \left(\frac{1+x+x^2+x^3+x^4+x^5}{x^3}\right)^3 = x^9 \right| (1-x^6)^3 (1-x)^{-3} = {}^{11}C_9 - 3.{}^5C_3 = 25.$$

75.(0) $\overrightarrow{BC} = \dfrac{\vec{u}}{|\vec{u}|} - \dfrac{\vec{v}}{|\vec{v}|},\ \overrightarrow{AC} = \dfrac{2\vec{u}}{|\vec{u}|} \Rightarrow \overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC} = \dfrac{\vec{u}}{|\vec{u}|} + \dfrac{\vec{v}}{|\vec{v}|}$

$$\Rightarrow \quad \overrightarrow{AB}\cdot\overrightarrow{BC} = \left(\frac{\vec{u}}{|\vec{u}|}+\frac{\vec{v}}{|\vec{v}|}\right)\cdot\left(\frac{\vec{u}}{|\vec{u}|}-\frac{\vec{v}}{|\vec{v}|}\right) = \left(\frac{\vec{u}\cdot\vec{u}}{|\vec{u}|^2}-\frac{\vec{v}\cdot\vec{v}}{|\vec{v}|^2}\right) = \frac{|\vec{u}|^2}{|\vec{u}|^2}-\frac{|\vec{v}|^2}{|\vec{v}|^2} = 1 - 1 = 0$$

$$\Rightarrow \quad B = \frac{\pi}{2} \quad \Rightarrow \quad 1+\cos 2A+\cos 2B+\cos 2C = 1-1-4\cos A\cos B\cos C = 0.$$

76.(1) Given that $\overline{u}\times\overline{v}+\overline{u}=\overline{w}$ and $\overline{w}\times\overline{u}=\overline{v}$

$\Rightarrow(\overline{u}\times\overline{v}+\overline{u})\times\overline{u}=\overline{v}$ $\Rightarrow$ $(\overline{u}\times\overline{v})\times\overline{u}=\overline{v}$ $\Rightarrow$ $\overline{v}-(\overline{u}.\overline{v})\overline{u}=\overline{v}$ $\Rightarrow$ $(\overline{u}.\overline{v})\overline{u}=0\Rightarrow(\overline{u}.\overline{v})=0$

Now, $[u\ v\ w]=u.\ (v\times w)=u.\ (\overline{v}\times(\overline{u}\times\overline{v}+\overline{u})=u.(\overline{v}\times(\overline{u}\times\overline{v})+\overline{v}\times\overline{u})=u.\ (\overline{v}^2\overline{u}-(\overline{u}.\overline{v})\overline{v}+\overline{v}\times\overline{u})=v^2u^2=1.$

77.(4) $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]=[\vec{a}\ \vec{b}\ \vec{c}]^2=\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}=16$

78.(0) $\vec{r}_1=4\vec{a}+m_1\vec{b};\ \vec{r}_2=l_2\vec{a}+m_2\vec{b}\ ;\ \vec{r}_3=l_3\vec{a}+m_3\vec{b}$

Since $\vec{r}_1$, $\vec{r}_2$ and $\vec{r}_2$ are co-linear $(\vec{r}_1-\vec{r}_2)\times(\vec{r}_2-\vec{r}_3)=0$

$\left((l_1-l_2)\vec{a}+(m_1-m_2)\vec{b}\right)\times\left((l_2-l_3)\vec{a}+(m_2-m_3)\vec{b}\right)=0$

$(l_1-l_2)(m_2-m_3)(\vec{a}\times\vec{b})-(m_1-m_2)(l_2-l_3)(\vec{a}\times\vec{b})=0$ or $(l_1-l_2)(m_2-m_3)-(m_1-m_2)(l_2-l_3)=0$

$$\begin{vmatrix}1 & 1 & 1\\ l_1 & l_2 & l_3\\ m_1 & m_2 & m_3\end{vmatrix}=\begin{vmatrix}1 & 1 & 1\\ l_1-l_2 & l_2 & l_3\\ m_1-m_2 & m_2 & m_3\end{vmatrix}=\begin{vmatrix}0 & 1 & 1\\ l_1-l_2 & l_2-l_3 & l_3\\ m_1-m_2 & m_2-m_3 & m_3\end{vmatrix}$$

$(l_1-l_2)(m_2-m_3)-(m_1-m_2)(l_2-l_3)=0$

79.(2) The p.v. of mid-point of BC is $\left(\frac{\lambda-1}{2}\right)\hat{i}+4\hat{j}+\left(\frac{\lambda+2}{2}\right)\hat{k}$

Since it is equally inclined to the axes, therefore its d.c. are $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

80.($2\sqrt{\frac{5}{7}}$) The equation of plane OAC

$(\vec{r}-\vec{0})=\vec{r}_1\times\vec{r}_3-(say\ \pi)$ Now, find the perpendicular distance of point B from the plane π

81.(7) $\vec{r}_1=\hat{i}-6\hat{j}+10\hat{k}$, $\vec{r}_2=-\hat{i}-3\hat{j}+7\hat{k}$, $\vec{r}_3=5\hat{i}-\hat{j}+h\hat{k}$, $\vec{r}_4=7\hat{i}-4\hat{j}+7\hat{k}$

The vectors corresponding to the edges are $\vec{a}=\vec{r}_1-\vec{r}_2=2\hat{i}-3\hat{j}+3\hat{k}$

$\vec{b}=\vec{r}_3-\vec{r}_2=6\hat{i}+2\hat{j}+(h-7)\hat{k}$; $\vec{c}=\vec{r}_4-\vec{r}_1=6\hat{i}+2\hat{j}-3\hat{k}$ Volume of the tetrahedron $=\left|\frac{1}{2}[\vec{a}\ \vec{b}\ \vec{c}]\right|=11$

82.(1) $\vec{a}\cdot\vec{b}=0=\vec{a}\cdot\vec{c}$; $\left|\vec{a}\times\vec{b}-\vec{a}\times\vec{c}\right|=\left|\vec{a}\times(\vec{b}-\vec{c})\right|=|\vec{a}|\cdot\left|\vec{b}-\vec{c}\right|$

Since $\vec{a}$ is perpendicular to $\vec{b}-\vec{c}$ as $\vec{a}\cdot(\vec{b}-\vec{c})=0$

$|\vec{a}|=1$ as $\left|\vec{b}-\vec{c}\right|=\sqrt{|b|^2+|c|^2-2\vec{b}\cdot\vec{c}}$

83.(3) $\vec{a}+\vec{b}=\lambda\vec{c}$ and $\vec{b}+\vec{c}=\mu\vec{a}$

$(\vec{a}-\vec{c})=\lambda\vec{c}-\mu\vec{a}$ $\Rightarrow\vec{a}(1+\mu)=\vec{c}(\lambda+1)$ $\Rightarrow$ $\mu=1,\ \lambda=-1$ $\Rightarrow$ $\vec{a}+\vec{b}+\vec{c}=0$

$|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$; $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-3$

84.(2) Shortest distance $=\left|(\vec{a}-\vec{c})\cdot(\vec{a}\times\vec{d})\right|$

THREE DIMENSIONAL GEOMETRY

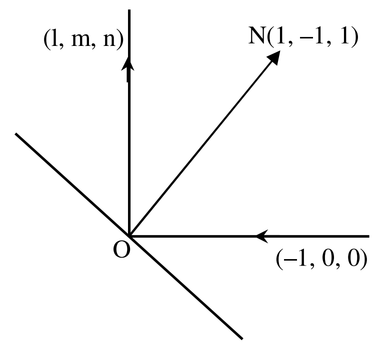
1.(D) Let $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}=\lambda$

$x = 2\lambda,\ y = 2 + 3\lambda,\ z = 3 + 4\lambda$

Now $(2\lambda + 2).2 + (7 + 3\lambda).3 + (-4 + 4\lambda).4 = 0$; $4\lambda + 4 + 21 + 9\lambda - 16 + 16\lambda = 0$; $29\lambda + 9 = 0$; $\lambda = -\frac{9}{29}$.

2.(A) The line is $\frac{x-1}{2}=\frac{y}{-1}=\frac{z+2}{1}=r$. There is no point of intersection of line and the plane hence line is parallel to the plane so direction cosines of the projected line remain same.

3.(D) The *d.c*'*s* of incident ray are (–1, 0, 0). Let the *d.c*'*s* of reflected ray be (l, m, n), then the direction ratios of the normal to the plane of mirror will be (l + 1, m , n).



So, $\frac{l+1}{1}=\frac{m}{-1}=\frac{n}{1}=k$ (say)

$\Rightarrow \quad 1 = k - 1,\ m = -k,\ n = k$

$\Rightarrow \quad (k - 1)^2 + (-k)^2 + (k)^2 = l^2 + m^2 + n^2 = 1 \Rightarrow k = 2/3.$

So, (l, m, n) is $\left(\frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}\right)$.

4.(C) Since $\begin{vmatrix} 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{vmatrix} \neq 0$, the given equations has a unique solution. Therefore the given planes meet in a unique point.

5.(A) Let (λ, λ, λ) be the point of intersection of two lines.

$\Rightarrow \quad \lambda(\sin A + \sin B + \sin C) = 2d^2$ and $\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

$\Rightarrow \quad \frac{\sin 2A+\sin 2B+\sin 2C}{\sin A+\sin B+\sin C}=\frac{1}{2} \Rightarrow \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=\frac{1}{16}$.

6.(D) Point A is (a, b, c)

$\Rightarrow$ Points P, Q, R are (a, b, $-c$), ($-a$, b, c) and (a, $-b$, c) respectively.

$\Rightarrow$ Centroid of triangle PQR is $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right) \quad \Rightarrow \quad G \equiv \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$

$\Rightarrow$ A, O, G are collinear $\Rightarrow$ area of triangle AOG is zero.

7.(A) Let the equation to the plane be $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$

$\Rightarrow \quad \frac{a}{\alpha}+\frac{b}{\beta}+\frac{c}{\gamma}=1$ ($\because$ The plane passes through a, b, c)

Now the points of intersection of the plane with the coordinate axes are $A(\alpha, 0, 0)$, $B(0, \beta, 0)$ and $C(0, 0, \gamma)$

$\Rightarrow$ Equation to planes parallel to the coordinate planes and passing through A, B and C are $x = \alpha$, $y = \beta$ and $z = \gamma$.

$\therefore$ The locus of the common point is $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$ (by eliminating α, β, γ from above equation)

Hence **(A)** is the correct answer.

8.(A) $a_1a_1 + b_1b_2 + c_1c_2 = 6 + 2 + 12 =$ positive

So acute angle bisector is $\dfrac{2x-y+2z+3}{3} = -\dfrac{3x-2y+6z+8}{7}$ $\Rightarrow$ $23x - 13y + 32z + 45 = 0$

Hence **(A)** is the correct answer.

9.(A) Let l, m, n be direction ratio of the normal

$l + m + n = 0$

$l + m + nd = 0 \quad \Rightarrow \quad n(1 - d) = 0 \Rightarrow n = 0$

$\left(\dfrac{1}{\sqrt{2}}, -\dfrac{1}{\sqrt{2}}, 0\right), \left(-\dfrac{1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}, 0\right).$

10.(C) Equation of straight line through origin is $\dfrac{x-0}{l} = \dfrac{y-0}{m} = \dfrac{z-0}{n}$

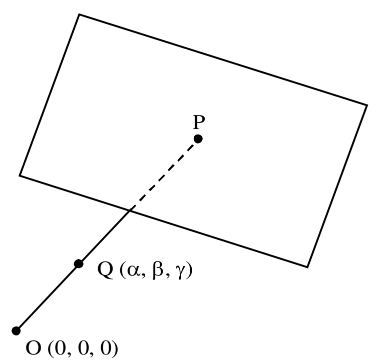
Where $l((b + c) + m(c + a) + n(a + b) = 0$ and $l(b - c) + m(c - a) + n(a - b) = 0$

On solving, $\dfrac{l}{2\left(a^2-bc\right)} = \dfrac{m}{2\left(b^2-ca\right)} = \dfrac{n}{2\left(c^2-ab\right)}$

$\therefore$ Equation of the straight line is $\dfrac{x}{a^2-bc} = \dfrac{y}{b^2-ca} = \dfrac{z}{c^2-ab}$

Hence **(C)** is the correct answer.

11.(D) Let P (α, β, γ) be the foot of perpendicular from origin to the plane.



Equation of plane is $\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$

$$\frac{x}{\frac{\alpha^2+\beta^2+\gamma^2}{\alpha}} + \frac{y}{\frac{\alpha^2+\beta^2+\gamma^2}{\beta}} + \frac{z}{\frac{\alpha^2+\beta^2+\gamma^2}{\gamma}} = 1.$$

$(\alpha^2 + \beta^2 + \gamma^2)^2(\alpha^{-2} + \beta^{-2} + \gamma^{-2}) = k^2$

So locus is $(x^2 + y^2 + z^2)^2 (x^{-2} + y^{-2} + z^{-2}) = k^2$.

12.(A) Let $OP : PQ = \lambda : 1$

$P \equiv \left(\dfrac{\lambda\alpha}{\lambda-1}, \dfrac{\lambda\beta}{\lambda-1}, \dfrac{\lambda\gamma}{\lambda-1}\right)$

P lies on plane $lx + my + nz = p$

$\Rightarrow \quad \dfrac{\lambda}{\lambda-1}(\alpha l + \beta m + \gamma n) = p \quad \Rightarrow \quad \dfrac{\lambda}{\lambda-1} = \dfrac{p}{l\alpha + m\beta + n\gamma}$

Also $OP \,.\, OQ = p^2 \;\Rightarrow\; \sqrt{\alpha^2+\beta^2+\gamma^2}\left(\dfrac{\lambda}{\lambda-1}\right)\sqrt{\alpha^2+\beta^2+\gamma^2} = p^2 \quad \Rightarrow \quad \dfrac{p(\alpha^2+\beta^2+\gamma^2)}{l\alpha+m\beta+n\gamma} = p^2$

So locus is $x^2 + y^2 + z^2 = (lx + my + nz)p$.

13.(A) Equation of the plane passing through the given line is $3x - 2y + 1 + \lambda(2x - z + 1) = 0$.

Since the plane is orthogonal to the given plane,

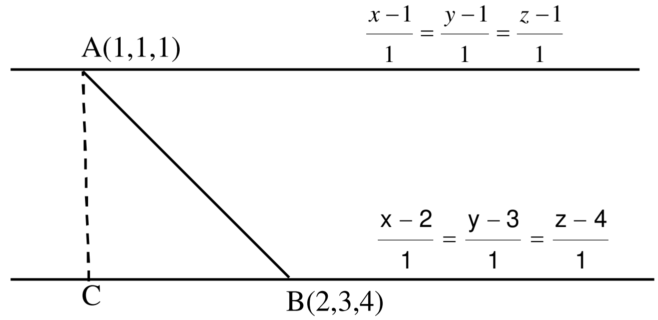
$3(3 + 2\lambda) + 4(-2) + 5(-\lambda) = 0$ or $\lambda = -1$.

Hence the plane thorough the given line and orthogonal to the given plane is $x - 2y + z = 0$.

Thus the equation to the required line is $3x + 4y + 5z = 0 = x - 2y + z$.

Writing in symmetrical form, we find that the orthogonal projection of the given line is $\dfrac{x}{7} = \dfrac{y}{1} = \dfrac{z}{-5}$.

14.(C) Since the given lines are parallel. From figure,



$$BC=\frac{(2-1)\cdot 1}{\sqrt{3}}+\frac{(3-1)\cdot 1}{\sqrt{3}}+\frac{(4-1)\cdot 1}{\sqrt{3}}$$

$$=\frac{1+2+3}{\sqrt{3}}=2\sqrt{3}.$$

$$AB=\sqrt{1+4+9}=\sqrt{14}$$

Shortest Distance $=AC=\sqrt{14-12}=\sqrt{2}$.

15.(A) Equation of line $x + 2y + z - 1 + \lambda(-x + y - 2z - 2) = 0$. **... (1)**

$x + y - 2 + \mu(x + z - 2) = 0$ **...(2)**

(0, 0, 1) lies on it $\Rightarrow \lambda = 0,\ \mu = -2$

For point of intersection point $z = 0$ and solve (1) and (2).

16.(ABC) Mid-point of object and image lies on mirror. Line joining object and image is $\perp$ to mirror.

17.(A) Any point on line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}=\lambda$ is $P(\lambda, 2\lambda+1, 3\lambda+2)$, $A(1, 6, 3)$

d.r. of AP $\equiv <\lambda-1, 2\lambda-5, 3\lambda-1>$

AP 1 to the line

$\Rightarrow \quad 1(\lambda-1)+2(2\lambda-5)+3(3\lambda-1)=0 \quad \Rightarrow \quad 14\lambda-14=0 \quad \Rightarrow \quad \lambda=1 \quad \therefore \quad P\equiv(1, 3, 5)$

18.(B) Family of planes containing the given line is $x + y + z - 4 + \lambda\,[\,2x + y + 3z - 1] = 0$. If it passes through origin then $\lambda = -4$.

$\Rightarrow$ Equation of plane is $7x + 3y + 11z = 0$ **... (i)**

Equation of plane containing the given line and perpendicular to plane (i) is $x + y + z - 4 + \lambda_1\,(2x + y + 3z - 1) = 0$**... (ii)**

where $7(1 + 2\lambda_1) + 3(1 + \lambda_1) + 11(1 + 3\lambda_1) = 0$

$\Rightarrow \quad \lambda_1=\frac{-21}{50}=$ Line OA is perpendicular to plane (ii)

Hence direction ratios of OA are (8, 29, –13) $\quad\Rightarrow\quad$ Equation of line OA is $\frac{x}{8}=\frac{y}{29}=\frac{z}{-13}$.

19.(D) Let direction cosines of the line be l, m and $\cos 60^0$. Then its equation is

$$\frac{x}{l}=\frac{y}{m}=\frac{z-2}{1/2}=\lambda \qquad \textbf{... (1)}$$

$\Rightarrow \quad x = \ell\lambda,\ y = m\lambda$ and $z = 2+\frac{\lambda}{2}$ putting in plane

$\ell\lambda+m\lambda+2+\frac{\lambda}{2}=1 \Rightarrow \lambda\,(l+m+1/2)=-1 \quad\Rightarrow\quad \lambda=\frac{-2}{2l+2m+1}$

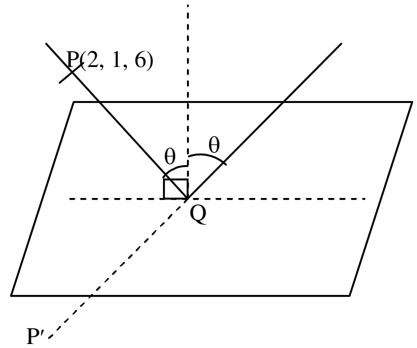
$x=\frac{-2l}{2l+2m+1},\ y=\frac{-2m}{2l+2m+1}$ and $z=2+\frac{-1}{2l+2m+1}$

Are the point of intersection of the line and plane

$\Rightarrow \quad 2l+2m+1=\frac{-1}{z-2}$ putting in x and y we get :

$$x = 2l(z-2) \Rightarrow l = \frac{x}{2(z-2)} \text{ and } m = \frac{y}{2(z-2)} \quad \Rightarrow \quad \left(\frac{x}{2(z-2)}\right)^2 + \left(\frac{y}{2(z-2)}\right)^2 + \frac{1}{4} = 1$$

$\Rightarrow \quad x^2 + y^2 - 3(z-2)^2 = 0 \qquad \Rightarrow \quad x^2 + y^2 = 3(z-2)^2$ is the locus



20.(A) Equation of plane passing though (2, 1, 0), (5, 0, 1) and (4, 1, 1) is

$$\begin{vmatrix} x-2 & y-1 & z-0 \\ 5-2 & 0-1 & 1-0 \\ 4-2 & 1-1 & 1-0 \end{vmatrix} = 0$$

$\Rightarrow (x-2)(-1) - (y-1)(3-2) + z(0+2) = 0$

or, $\quad x + y - 2z = 3 \qquad$ **...(i)**

A point on the line $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5}$ is (2, 1, 6). Now the reflection point of this point (2, 1, 6) w. r. t mirror kept along the plane $x + y - 2z = 3$ is $\frac{\alpha'-2}{1} = \frac{\beta'-1}{1} = \frac{\gamma'-6}{-2} = \frac{-2(2+1-2\times6-3)}{1^2+1^2+(-2)^2}$

$\Rightarrow \quad \alpha' = 2+4 = 6, \quad \beta' = 1+4 = 5 \quad \& \quad \gamma' = -8+6 = -2$.

Also the line $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5} = r$,

Let it meets the plane $x + y - 2z = 3$ at $(3r + 2, 4r + 1, 5r + 6)$

$\Rightarrow \quad 3r + 2 + 4r + 1 - 2(5r + 6) = 3 \qquad \Rightarrow \quad -3r = 12 \Rightarrow r = -4$

$\therefore \quad$ Point $Q = (-10, -15, -14)$

$\therefore \quad$ Equation of reflected ray is line joining P' and Q.

$\Rightarrow \quad \frac{x+10}{16} = \frac{y+15}{20} = \frac{z+14}{12}$

$\therefore \quad$ Reflected line $\frac{x+10}{4} = \frac{y+15}{5} = \frac{z+14}{3}$.

21.(C) Let any point on the given line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-1}{1} = r$ is $(2r + 1, r + 1, r + 1)$

On putting this point on the plane $x + y + z = 1$

we get $r = -\frac{1}{2}$

$\Rightarrow \quad$ Given line cuts the plane $x + y + z = 1$ at the point $Q = \left(0, \frac{1}{2}, \frac{1}{2}\right)$

Now the line from point (1, 1, 1) and perpendicular to the given plane is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1} = \lambda$ **... (1)**

$\Rightarrow \quad$ Any point on the line (1) is $(\lambda+1, \lambda+1, \lambda+1)$.

Therefore line (1) cuts the given plane at point R when $\lambda = \frac{1}{3} \Rightarrow \qquad$ point R is $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

Hence the equation of required line QR is $\dfrac{x-\frac{1}{3}}{\frac{1}{3}-0}=\dfrac{y-\frac{1}{3}}{\frac{1}{3}-\frac{1}{2}}=\dfrac{z-\frac{1}{3}}{\frac{1}{3}-\frac{1}{2}}$

or $\dfrac{x-\frac{1}{3}}{\frac{1}{3}}=\dfrac{y-\frac{1}{3}}{-\frac{1}{6}}=\dfrac{z-\frac{1}{3}}{-\frac{1}{6}}\Rightarrow\dfrac{3x-1}{2}=\dfrac{3y-1}{-1}=\dfrac{3z-1}{-1}$.

22.(B) Any point on the given line is $2r+1, -r-1, 4r+3$.

If this point lies on the planes, then $2r+1-2r-2+4r+3=12\Rightarrow r=\dfrac{5}{2}$.

Hence the point of intersection of the given line and that of the plane is $\left(6, -\dfrac{7}{2}, 13\right)$. Also a point on the line is $(1, -1, 3)$.

Let (α, β, γ) be its image in the given plane. In such a case $\dfrac{\alpha-1}{1}=\dfrac{\beta+1}{2}=\dfrac{\gamma-3}{1}=\lambda$

$\Rightarrow$ $\alpha=\lambda+1, \beta=2\lambda-1, \gamma=\lambda+3$. Now the midpoint of the image and the point $(1, -1, 3)$ lies on the plane i.e. $\left(1+\dfrac{\lambda}{2}, \lambda-1, 3+\dfrac{\lambda}{2}\right)$ lies in the plane

$\Rightarrow$ $\lambda=\dfrac{10}{3}$. Hence the image of $(1, -1, 3)$ is $\left(\dfrac{8}{3}, \dfrac{7}{3}, \dfrac{14}{3}\right)$.

Hence the equation of the required line is $\dfrac{x-6}{\frac{10}{3}}=\dfrac{y+\frac{7}{2}}{\frac{-35}{6}}=\dfrac{z-13}{\frac{25}{3}}$ or $\dfrac{x-6}{4}=\dfrac{y+\frac{7}{2}}{-7}=\dfrac{z-13}{10}$.

23.(A) x-axis and line of intersection of two plane must be coplanar which means point $(x_1, 0, 0)$ will satisfy both the planes

i.e. $a_1x_1=d_1$ and $a_2x_1=d_2$ $\Rightarrow$ $\dfrac{d_1}{a_1}=\dfrac{d_2}{a_2}\Rightarrow d_1a_2=d_2a_1$.

24.(D) The equation of plane is $x+\lambda z=0$

As it is making an angle α from $x=0$ $\Rightarrow$ $\dfrac{1}{\sqrt{1+\lambda^2}}=\cos\alpha$ $\Rightarrow$ $\lambda=\pm\sqrt{\sec^2\alpha-1}$

25.(C) $P(x, y, z)$ lies on line through $(0, 0, 2)$ making $60°$ with z-axis

i.e. $\dfrac{(xi+yj+(z-2)k)\cdot k}{\sqrt{x^2+y^2+(z-2)^2}}=\cos 60°$ $\Rightarrow$ $x^2+y^2-3(z-2)^2=0.$

26.(B) The plane does not pass through $(0, 0, 2)$ and its inclination with z-axis is θ, where $\sin\theta=\dfrac{1}{\sqrt{3}}$

i.e. $\theta<60°\Rightarrow$ it represents a hyperbola.

27.(A) The plane passes through $(0, 0, 2)$ and its inclination with z-axis is θ, where $\sin\theta=\dfrac{1}{\sqrt{3}}$

i.e. $\theta<60°\Rightarrow$ it represents a pair of straight lines.

28.(B) The plane is parallel to z-axis $\Rightarrow$ locus is a hyperbola.

29.(A) The plane passes through (0, 0, 2) and its inclination with z-axis is θ, where $\sin\theta = \frac{1}{\sqrt{6}}$

i.e. $\theta < 60° \Rightarrow$ it represents a pair of straight lines.

30.(D) In the new definition $l = \frac{x}{|x|+|y|+|z|}$, etc

31.(B) $d(O,P) = k \Rightarrow |x|+|y|+|z| = k$

which represents a set of 8 planes making intercepts of lengths k on positive as well as negative sides of all three axes.

32.(D) $d(O,P) = d(A,P)$

$\Rightarrow |x|+|y|+|z| = |x-1|+|y-2|+|z-3| \Rightarrow x+y+z = |x-1|+|y-2|+|z-3|$.

Now the sign-scheme is

$x-1$	$y-2$	$z-3$	Resulting equation	Inference
+	+	+	$0=-6$	No solution
−	+	+	$x=-2$	No solution
+	−	+	$y=-1$	No solution
+	+	−	$z=0$	xy-plane
−	−	+	$x+y=0$	$x=0, y=0 \Rightarrow z$-axis
−	+	−	$x+z=1$	Plane || to y-axis
+	−	−	$y+z=2$	Plane || to x-axis
−	−	−	$x+y+z=3$	Part of an oblique plane

33.(D) Side length of triangle = 2

So, the coordinates of vertices are $(\sqrt{2}, 0, 0), (0, \sqrt{2}, 0), (0, 0, \sqrt{2})$ and hence the centroid is $\left(\frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}\right)$.

34.(D) Both lines are coplanar

$$\begin{vmatrix} 2 & 3 & \lambda \\ 3 & 2 & 3 \\ 1 & -1 & -1 \end{vmatrix} = 0 \Rightarrow 2(-2+3)+3(3+3)+\lambda(-3-2) = 0 \Rightarrow \lambda = 4$$

$\sin^{-1}\sin 4 = \sin^{-1}\sin(\pi-4) = \pi - 4$

35.(B) Let $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{4} = r_1$

$\Rightarrow x = 3+2r_1,\ y = 2+3r_1,\ z = 1+4r_1$ it will line on $\frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3} \Rightarrow r_1 = 1$

So, point of intersection is (5, 5, 5)

36.(C) Equation of plane contains both lines $\begin{vmatrix} x-2 & y-2 & z-1 \\ 2 & 3 & 4 \\ 3 & 2 & 3 \end{vmatrix} = 0$

$(x-3)(1)+(y-2)(12-6)+(z-1)(4-9)=0$

$x+6y-5z=10$

37.(B) The equation of the line of intersection of the planes $y+z=0$ and $z+x=0$ are $\frac{x}{1}=\frac{y}{1}=\frac{z}{-1}$ ......(1)

Similarly the equation of edge of intersection of the planes $x+y=0$ and $x+y+z=a$ are $\frac{x}{1}=\frac{y}{-1}=\frac{z-a}{0}$ ......(2)

If l, m, n be the d.c's of the shortest distance between the lines (1) and (2)

$\therefore \quad 1\cdot l+1\cdot m-1\cdot n=0$ and $1\cdot l-1\cdot m+0\cdot n=0$

Solve these we get $l=-\frac{1}{\sqrt{6}},\ m=-\frac{1}{\sqrt{6}},\ n=-\frac{1}{\sqrt{6}}$

38.(A) It is clear from (1) and (2) that $A(0, 0, 0)$ is a point on lien (1) and $B(0, 0, a)$ is a point on line (2)

$\therefore \quad S.D=\left|(0-0)+m(0-0)+n(0-a)\right|=-\frac{2}{\sqrt{6}}(-a)=\frac{2}{\sqrt{6}}a$

39.(BC) The locus is the pair of planes passing through the point of intersection of the lines and having their bisectors as the normals.

40.(AB) The equation of the required plane is $(x-y+1)+\lambda\,(2y+z-6)=0$

$\Rightarrow \quad x+(2\lambda-1)\,y+\lambda z+1-6\lambda=0$

Since it makes an angle of 30^0 with $x+y+z=5$

$\Rightarrow \quad \frac{|1+(2\lambda-1)+\lambda|}{\sqrt{3}\cdot\sqrt{1+\lambda^2+(2\lambda-1)^2}}=\frac{\sqrt{3}}{2} \quad \Rightarrow |6\lambda|=3\sqrt{5\lambda^2-4\lambda+2}$

$\Rightarrow \quad 4\lambda^2=5\lambda^2-4\lambda+2 \Rightarrow \lambda^2-4\lambda+2=0 \Rightarrow \lambda=(2\pm\sqrt{2})$

$\Rightarrow \quad (x-y+1)+(2\pm\sqrt{2})\,(2y+z-6)=0$ are two required planes.

41.(A) Equation of planes $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

Three mutually perpendicular planes are

$x=0,\ y=0$ and $z=0$.

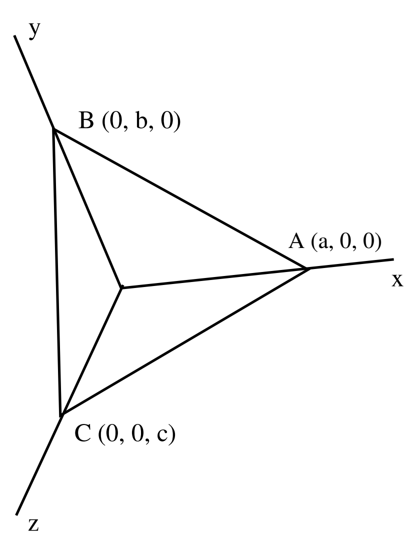
$d.r.$ of AB is $(a, -b, 0)$

$d.r.$ of AC is $(a, 0, -c)$

So, $\cos A=\frac{a^2}{\sqrt{a^4+\sum a^2b^2}} \Rightarrow \cot A=\frac{a^2}{\sqrt{\sum a^2b^2}}$

similarly $\cot B=\frac{b^2}{\sqrt{\sum a^2b^2}}$, $\cot C=\frac{c^2}{\sqrt{\sum a^2b^2}}$.

Let α be the angle between the planes $z=0$ and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ then



$$\cos\alpha = \frac{\frac{1}{c}}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}} = \frac{ab}{\sqrt{\sum a^2b^2}} \Rightarrow \cos^2\alpha = \frac{a^2b^2}{\sum a^2b^2} = \cot A \cot B.$$

Similarly other results can be proved.

42.(A) $l = \left(\frac{-bm-cn}{a}\right) \Rightarrow fmn + (gn+hm)\left(\frac{-bm-cn}{a}\right) = 0 \quad \Rightarrow \quad -\frac{hb}{a}\left(\frac{m}{n}\right)^2 + \left(f - \frac{gb}{a} - \frac{ch}{a}\right)\frac{m}{n} - \frac{gc}{a} = 0$.

If $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$ are the roots then $\frac{m_1}{n_1}\cdot\frac{m_2}{n_2} = \frac{gc}{hb}$.

Similarly $\frac{l_1}{n_1}\cdot\frac{l_2}{n_2} = \frac{fc}{ha}$.

Two lines are perpendicular then

$$\Rightarrow \quad l_1l_2 + m_1m_2 + n_1n_2 = 0 \Rightarrow \frac{l_1l_2}{n_1n_2} + \frac{m_1m_2}{n_1n_2} + 1 = 0 \Rightarrow \quad \frac{gc}{hb} + \frac{fc}{ha} + 1 = 0 \Rightarrow \frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0.$$

43.(ABCD)

As perpendicular distances are in *A.P.*

Let $|a-d|, |a|, |a+d|$ be these distances.

So, the coordinates of point P are $(\pm(a-d), \pm a, \pm(a+d))$

Now, distances from P of z–axis, y–axis and x–axis are respectively,

$2a^2 + d^2 - 2ad = 5$ **...(1)**

$2a^2 + 2d^2 = 10$ **... (2)**

$2a^2 + d^2 + 2ad = 13$ **...(3)**

Solving (1) and (3), we get : $4ad = 8$

Substituting in (2) : $\frac{4}{d^2} + d^2 = 5 \Rightarrow d^2 = 1$ or 4

$\Rightarrow \quad d = \pm 1, \ a = \pm 2$

So, Points are $(\pm 1, \pm 2, \pm 3)$. (There will be 8 such points)

44.(AB) The equation of a plane through the line of intersection of the plane $3y + 4z = 0$ and $x = 0$ is $3y + 4z + \lambda x = 0$.

This plane makes an angle 60^0 with the plane $3y + 4z = 0$

$$\therefore \quad \cos 60^0 = \frac{3^2+4^2}{\sqrt{3^2+4^2}\sqrt{3^2+4^2+\lambda^2}} \Rightarrow \cos^2 60^0 = \frac{3^2+4^2}{3^2+4^2+\lambda^2} \quad \Rightarrow \quad \frac{1}{4} = \frac{25}{25+\lambda^2} \Rightarrow \lambda = \pm 5\sqrt{3}$$

Hence required plane is $3y + 4z \pm 5\sqrt{3}\,x = 0$.

45.(CD) For the points where the line intersects the curve, we have $z = 0$. Therefore $\frac{x-2}{3} = \frac{y+1}{2} = \frac{0-1}{-1}$

$\Rightarrow \frac{x-2}{3} = 1$ and $\frac{y+1}{2} = 1 \Rightarrow x = 5$ and $y = 1$

Put these values in $xy = c^2$, we get $c = \pm\sqrt{5}$.

46.(C) If the two lines have direction cosines l_1, m_1, n_1 and l_2, m_2, n_2 then the direction ratio of their bisectors will be $l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2$

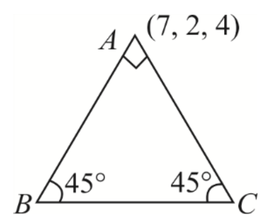
47.(AD) Let the equation of plane be $Ax + By + Cz = 0$ **... (1)**

As line is parallel $\Rightarrow 2A - B - 2C = 0$ **... (2)**

As distance is $\frac{5}{3}$ and the point (1, – 3, –1) is lying on the line $\Rightarrow |A - 3B - C| = \frac{5}{3}$ **... (3)**

& $A^2 + B^2 + C^2 = 1$ **... (4)**

Solve (2), (3) and (4) for *A, B, C*.



48.(BD) Let the point *B* be $(-6 + 5\lambda, -10 + 3\lambda, -14 + 8\lambda)$

$\Rightarrow$ direction ratio of AB $\quad 5\lambda - 13, 3\lambda - 12, 8\lambda - 18$

The value of $\lambda(\lambda = 2,\ \lambda = 3)$ can be found as angle between *AB* and *BC* is 45°

49.(AB) Let the point on the line be $(2\lambda, 2 + 2\lambda, 1)$

Now as it strikes $x = 0 \Rightarrow \lambda = 0 \quad \Rightarrow \quad A(0, 2, 1)$

Find the reflection of any point *P* on the line along $x = 0$ let it be *Q* then *AQ* will be reflected ray.

50.(AB) Let the required line passes through the point (*h*, 0, 0). Then its equation is

$$\frac{x-h}{1} = \frac{y-0}{0} = \frac{z-0}{1} \text{ or, } \frac{x-h}{1} = \frac{y-0}{0} = \frac{z-0}{-1}$$

Let MN be a line perpendicular to given line and required line and its direction cosines be *l* : *m* : *n*, then

$$l \times 1 + m \times 1 + n \times 1 = 0$$
$$l \times 1 + m \times 0 + n \times 1 = 0$$

Solving we get $l = -n$ and $m = 0 \quad \Rightarrow \quad l = \frac{1}{\sqrt{2}}, n = -\frac{1}{\sqrt{2}}, m = 0$

Let point *A* be (*h*, 0, 0) and *B* be (1, 1, 0) then projection of *AB* on the line *MN* is $l(h - 1) + m(0 - 1) + n(0 - 0) = 1$

$$\Rightarrow \quad \frac{1}{\sqrt{2}}(h - 1) + 0 - \left(\frac{1}{\sqrt{2}} \times 0\right) = 1 \Rightarrow h - 1 = \sqrt{2} \Rightarrow h = 1 + \sqrt{2}$$

$$\Rightarrow \quad \frac{x - 1 - \sqrt{2}}{1} = \frac{y-0}{0} = \frac{z-0}{1} \text{ or, } \frac{x-h}{1} = \frac{y-0}{0} = \frac{z-0}{-1} \quad \Rightarrow \quad l + m + n = 0 \text{ and } -l + 0 + n = 0$$

Solving, we get $n = \frac{1}{\sqrt{6}}, 1 = \frac{1}{\sqrt{6}}$ and $m = -\frac{2}{\sqrt{6}}$

$$\Rightarrow \quad \frac{1}{\sqrt{6}}(h - 1) - \frac{2}{\sqrt{6}}(0 - 1) + \frac{1}{\sqrt{6}}(0 - 0) = 1 \Rightarrow h = \sqrt{6} - 1$$

$$\therefore \quad \frac{x - \sqrt{6} + 1}{1} = \frac{y-0}{0} = \frac{z}{-1}.$$

Hence $\frac{x - \sqrt{6} + 1}{1} = \frac{y-0}{0} = \frac{z}{-1}$ and $\frac{x - 1 - \sqrt{2}}{1} = \frac{y-0}{0} = \frac{z-0}{1}$ are the required lines.

51.(AC) Find the point of intersection of line and plane let it be *A*.

Find the foot of normal of point (–1, –1, –3) along the plane let it be *B* then *AB* is required line.

52.(ABCD) Three planes meet at two points it means they have infinite many solution, so $\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0$ on solving $\alpha = 4$

$P_1 : 2x + y + z = 1$ **;** $P_2 : x - y + z = 2$ **;** $P_3 : 4x - y + 3z = 5$

P on XOY plane $\equiv (1, -1, 0)$ (Which can be obtained by putting $z = 0$ in any two equations)

Q on YOZ plane $\equiv \left(0, -\frac{1}{2}, \frac{3}{2}\right)$ (By putting $x = 0$ in any two equations)

53.(AC) Any point on the line $(3\lambda - 2, 2\lambda - 1, 2\lambda + 3)$, then

$(3\lambda - 3)^2 + (2\lambda - 3)^2 + (2\lambda)^2 = 18 \quad \Rightarrow \quad \lambda = 0, \frac{30}{17} \Rightarrow$ points are $(-2, -1, 3)$ and $\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$.

54.(ABC) Let $P(2r_1, -3r_1, r_1)$ and $Q(3r_2 + 2, -5r_2 + 1, 2r_2 - 2)$ be the points on the given lines so that PQ is the line of shortest distance

d.r.s of PQ $2r_1 - 3r_2 - 2, -3r_1 + 5r_2 - 1, r_1 - 2r_2 + 2$

Since it is perpendicular to given lines $2(2r_1 - 3r_2 - 2) - 3(3r_1 + 5r_2 - 1) + (r_1 - 2r_2 + 2) = 0$

and $(2r_1 - 3r_2 - 2) - 5(-3r_1 + 5r_2 - 1) + 2(r_1 - 2r_2 + 2) = 0$

$\Rightarrow \quad r_1 = 31/3, \ r_2 = 19/3 \quad \Rightarrow$ P is $\left(\frac{62}{3}, -31, \frac{31}{3}\right)$ and Q is $\left(21, \frac{-92}{3}, \frac{32}{3}\right)$ d. r. s of PQ is $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

55.(BC) Equation of planes bisecting the angle between given planes are

$$\frac{2x - y + 2z + 3}{\sqrt{4+1+4}} = \pm \frac{3x - 2y + 6z + 8}{\sqrt{9+4+36}}$$

$\Rightarrow \quad 5x - y - 4z - 3 = 0$ and $23x - 13y + 32z + 45 = 0$ are required planes.

56.(BC)

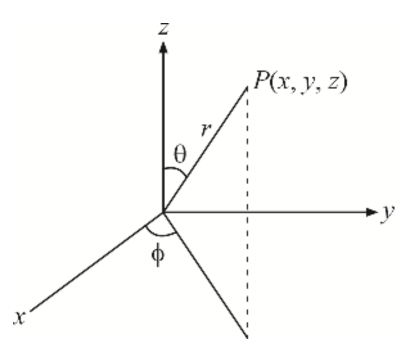
57.(AC) Since point of intersection of the given lines is (0, 0, 0). It must lie on the angle bisector

So, $\dfrac{0-2}{-1} = \dfrac{0+2}{1} = \dfrac{0+k^2-1}{4} \Rightarrow k \pm 3$

58.(ABC) If P be (x, y, z) then from the figure, $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$

$\Rightarrow \quad 1 = r\sin\theta\cos\phi, \ 2 = r\sin\theta\sin\phi, \ 3 = r\cos\theta \quad \Rightarrow \quad 1^2 + 2^2 + 3^2 = r^2 = \pm\sqrt{14}$

$\therefore \quad \sin\theta\cos\phi = \dfrac{1}{\sqrt{14}}, \ \sin\theta\sin\phi = \dfrac{1}{\sqrt{14}}$



$\sin\theta\sin\phi = \dfrac{2}{\sqrt{14}}, \ \cos\theta = \dfrac{3}{\sqrt{14}}$

(neglecting negative sign as θ and ϕ are acute)

$\therefore \quad \dfrac{\sin\theta\sin\phi}{\sin\theta\cos\phi} = \dfrac{2}{1} \Rightarrow \tan\phi = 2$

Also, $\tan\theta = \dfrac{\sqrt{5}}{3}$

59.(AB)

60. **[A-p, q] [B-p, q] [C-r, s] [D-q]**

(A) Let the point on the line be $(-1-\lambda, 12+5\lambda, 7+2\lambda)$, place it on the surface.

(B) Let the points on the two lines be $A(\lambda_1, \lambda_1-5, \lambda_1-1)$ and $B(\lambda_2-5, \frac{\lambda_2}{3}, \frac{\lambda_2}{2})$ respectively the variable line will be parallel to AB.

(C) Any plane through the intersection of the two planes is $(2x-y)+\lambda(3z-y)=0$.

Find λ using the condition that it is perpendicular to third plane

61. **[A-q] [B-p] [C-s] [D-r]**

(B) The line will be parallel to normal of the plane

(C) Let the points of shortest distance on the two lines be

$A(2\lambda_1, -3\lambda_1, \lambda_1)$ & $B(2+3\lambda_2, 1-5\lambda_2, -2+2\lambda_2)$

Now use the condition that AB will be perpendicular to the two lines to find the value of λ_1 & λ_2.

(D) The direction ratio of line of intersection will be cross product of $(3i-j+k)$ & $(i+4j-2k)$

62. **[A-t] [B-p] [C-s] [D-r]**

(A) The line is $\dfrac{x-5}{\frac{4}{\sqrt{26}}}=\dfrac{y}{\frac{1}{\sqrt{26}}}=\dfrac{z+6}{\frac{3}{\sqrt{26}}}=\approx$. Place $\lambda=\pm 3$

(B) The equation of plane is $\begin{vmatrix} x-2 & y-4 & z-6 \\ 3 & 5 & 7 \\ 1 & 4 & 7 \end{vmatrix}=0$

(C) The equation of line in parametric form is $\dfrac{x-2}{\frac{6}{7}}=\dfrac{y+3}{\frac{2}{7}}=\dfrac{z+1}{\frac{3}{7}}=\approx$

Now place $\lambda=\pm 14$ (Check which is nearer to origin)

(D) Let foot of perpendicular of the point $A(3,-1,11)$ on line be $B(2\lambda, 2+3\lambda, 3+4\lambda)$.

Now as AB is perpendicular to line, find the value of λ.

63. **[A-q, s] [B-s] [C-r] [D-p]**

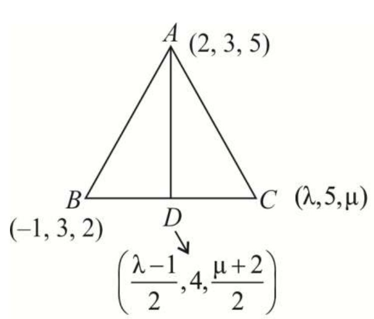
(A) Let d.c of the line be l, m, n then

$2l-3m-n=0$ **... (1)**, $-l+2m-2n=0$ **... (2)** $l^2+m^2+n^2=1$ **... (3)**

64. **[A-q] [B-p] [C-p] [D-r]**

(A) Find the point of intersection of the line & plane. Let it be A. Find the foot of perpendicular of the point (1, 3, 4) on the given plane. Let it be B. AB will be the line $L=0$

(B)



As AD is equally inclined from axes its direction ratio will be 1, 1, 1.

(D) As $A(a,b,c)$ is lying on the plane $\Rightarrow 3a+2b+c=7$

If $\vec{p}\ \&\ \vec{q}$ be two vectors then $\vec{p}\cdot\vec{q}\le|\vec{p}||\vec{q}|$

$$\Rightarrow \quad (3i+2j+k)(ai+bj+ck)\le\sqrt{14}\sqrt{a^2+b^2+c^2}$$

$$\sqrt{a^2+b^2+c^2}\ge\frac{7}{\sqrt{14}}$$

65.(2) Any plane passing through first line $2x + y + z - 1 + \lambda(3x + y + 2z - 2) = 0$. If it is parallel to second line

$(2 + 3\lambda)1 + (1 + \lambda)1 + (1 + 2\lambda)1 = 0 \Rightarrow \lambda = -2/3$.

Plane is $y - z + 1 = 0$

Distance from (0, 0, 0) $=\frac{1}{\sqrt{2}}$.

66.(1) Intersection of line $y - z - 1 = 0$, $x = 0$ with xy plane gives $x = 0$, $y = 1$ and $z = 0$ so point A is (0, 1, 0). Similarly point B is (– 1, 0, 0) and point C is (1, 0, 0). Here $AB^2 + AC^2 = BC^2$.So triangle ABC is right angled at A. Hence orthocentre is point A (0, 1, 0).

67.(4) Equation of plane passing through (1, 1, 1) and perpendicular to the given line is $x + 2y + 3z = 6$.

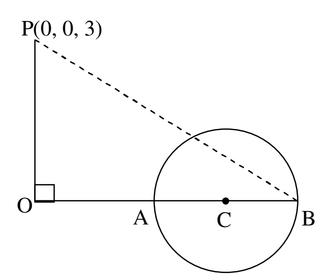
The minimum distance of the point (1, 1, 1) from the plane $x + y + z = 1$ measured perpendicular to $\frac{x-x_1}{1}=\frac{y-y_1}{2}=\frac{z-z_1}{3}$ is the perpendicular distance of the point (1, 1, 1) from the line of intersection of the above two planes.

The equation of line of intersection of these planes is

$$\frac{x+4}{1}=\frac{y-5}{-2}=\frac{z-0}{1}=r \qquad \textbf{... (1)}$$

The minimum distance of (1, 1, 1) from the line (1) is $\frac{2}{3}\sqrt{21}$ units.

68.(5) Let O be the foot of perpendicular from the point P(0, 0, 3) to the $x - y$ plane. Hence it is evident that PB is the maximum distance of point P from the circle



$$\Rightarrow \quad PB^2=OP^2+OB^2=25 \ \Rightarrow PB=5$$

69.(6) Equation of plane which contains the line $\frac{x-2}{3}=\frac{y-3}{1}=\frac{z-1}{2}$ is

$(x - 2 - 3(y - 3)) + \lambda(2(y - 3) - (z - 1)) = 0$

$x + (2\lambda - 3)y - \lambda z + 7 - 5\lambda = 0$

If the plane is parallel to $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$

$\Rightarrow 1 + (2\lambda - 3)2 - 3\lambda = 0 \Rightarrow \lambda = 5$.

So, equation of plane which contains second line and parallel to first is $x + 7y - 5z - 18 = 0$.

Similarly, equation of plane which contains first line and parallel to second is $(x - 1) + 7(y - 2) - 5(z - 3) = 0$

$\Rightarrow \quad x + 7y - 5z = 0$

Hence equidistant plane is $\frac{(x+7y-5z-18)+(x+7y-5z)}{2}=0$

$\Rightarrow \quad x + 7y - 5z - 9 = 0$.

Alternate Solution:

General points on lines are $P(\lambda + 1, 2\lambda + 2, 3\lambda + 3)$ and $Q(3\mu + 2, \mu + 3, 2\mu + 1)$

Let the required plane be $ax + by + cz = 1$

Then mid point of PQ will always lie on the plane for all values of λ and μ

$$\Rightarrow \quad a\left(\frac{3\mu+\lambda+3}{2}\right)+b\left(\frac{2\lambda+\mu+5}{2}\right)+c\left(\frac{3\lambda+2\mu+4}{2}\right)=1$$

$$\Rightarrow \quad \mu(3a+b+2c)+\lambda(a+2b+3c)+3a+5b+4c-2=0 \quad \Rightarrow \quad 3a+b+2c=0,\ \ a+2b+3c=0$$

$$3a + 5b + 4c = 2 \quad \Rightarrow \quad a = \frac{1}{9},\ b = \frac{7}{9},\ c = -\frac{5}{9}.$$

So, plane is $x + 7y - 5z = 9$.

70.(3) New plane will pass through the line of intersection of the planes $2x + 3y + 4z = 5$ and $z = 0$. This plane can be written as $2x + 3y + 4z - 5 + \lambda z = 0$. Now $(1, 1, 1)$ lies on it

So, $\lambda = -4$

So, the equation of plane in new position is $2x + 3y - 5 = 0$

Angle between these planes is given by $\cos\alpha = \dfrac{2^2+3^2}{\sqrt{29}\sqrt{13}} \Rightarrow \alpha = \cos^{-1}\sqrt{\dfrac{13}{29}}$.

71.(4) Let the equation of plane be $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$ **... (1)**

$$\Rightarrow \quad \left|\frac{-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}\right| = p \quad \Rightarrow \quad \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{p^2} \qquad \textbf{... (2)}$$

The coordinate of centroid `is $\left(\dfrac{a}{4},\dfrac{b}{4},\dfrac{c}{4}\right) \Rightarrow$ locus is $\dfrac{1}{x^2}+\dfrac{1}{y^2}+\dfrac{1}{z^2}=\dfrac{16}{p^2}$

72.(1) The given planes are $x-cy-bz=0,\ cx-y+az=0,\ bx+ay-z=0$

$\therefore$ The rectangular array is $\left\|\begin{matrix}1 & -c & -b & 0\\ c & -1 & a & 0\\ b & a & -1 & 0\end{matrix}\right\|$ $\quad \therefore \quad \Delta_4 = \begin{vmatrix}1 & -c & -b\\ c & -1 & a\\ b & a & -1\end{vmatrix} = \begin{vmatrix}1 & -c & -b\\ 0 & c^2-1 & bc+a\\ 0 & a+bc & b^2-1\end{vmatrix}$

(Subtracting c times 1st row from 2nd and b times from third row.)

$$= \begin{vmatrix}c^2-1 & bc+a\\ a+bc & b^2-1\end{vmatrix} = (c^2-1)(b^2-1)-(bc+a)(bc+a)$$

$$= (c^2b^2-c^2-b^2+1)-(b^2c^2+a^2+2abc) = 1-a^2-b^2-c^2-2abc$$

and
$$\Delta_1 = \begin{vmatrix}-c & -b & 0\\ -1 & a & 0\\ a & -1 & 0\end{vmatrix} = 0$$

Now, if the given planes intersect in a line, then Δ_4 must be zero.

i.e., $\quad 1-a^2-b^2-c^2-2abc=0 \qquad$ i.e., $\quad a^2+b^2+c^2+2abc=1$

73.(4) The equations of the given planes can be written as

$$x-y+z+1=0$$
$$\lambda x+3y+2z-3=0$$
$$3x+\lambda y+z-2=0$$

$\therefore$ The rectangular array is $\begin{vmatrix} 1 & -1 & 1 & 1 \\ \lambda & 3 & 2 & -3 \\ 3 & \lambda & 1 & -2 \end{vmatrix}=0$

$\therefore$ $\Delta_4=\begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix}=(\lambda-4)(\lambda+3)$ **... (i)**

$\Delta_1=\begin{vmatrix} -1 & 0 & 0 \\ 3 & 5 & 0 \\ \lambda & \lambda+1 & \lambda-2 \end{vmatrix}=-5(\lambda-2)$ **... (ii)**

$\Delta_2=\begin{vmatrix} 1\ldots & 1\ldots & 0 \\ \lambda & 2-\lambda & -3-\lambda \\ 3 & -2 & -5 \end{vmatrix}=3\lambda-16$ **... (iii)**

$\Delta_3=\begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & -3 \\ 3 & \lambda & -2 \end{vmatrix}=(\lambda+3)(\lambda-2)$ **... (iv)**

If, the given planes form a triangular prism, then we know that $\Delta_4=0$ and none of $\Delta_1,\Delta_2,\Delta_3$ is zero. Here from equations (i), (ii), (iii) and (iv) we find that if $\lambda=4,$ then $\Delta_4=0$ and none of $\Delta_1,\Delta_2,\Delta_3$ is zero.

Consequently for $\lambda=4,$ the given planes form a triangular prism.

74.(7) Equation of the required plane is $2x+y+2z-9+\lambda(4x-5y-4z-1)=0$

It passes through (3, 2, 1)

$\therefore$ $6+2-9+\lambda(12-10-4-1)=0 \quad\Rightarrow\quad 1-3\lambda=0 \Rightarrow \lambda=\frac{1}{3}$

$\therefore$ Equation of the required plane is $2x+y+2z-9+\frac{1}{3}(4x-5y-4z-1)=0$

$\Rightarrow$ $10x-2y+2z=28 \Rightarrow \frac{x}{14/15}+\frac{y}{14}+\frac{z}{14}=1 \Rightarrow a=\frac{14}{5}, b=-14, c=14 \Rightarrow (14+14-14)^2=196$

75.(4) The given line passes through $A(5,-2,6)$

$(PQ)^2=(AP)^2-(AQ)^2$

$AP^2=19$

Now AQ is the projection of AP on the given line

$\therefore$ $AQ=(5-4)\frac{3}{\sqrt{50}}+(-2+5)\left(\frac{-4}{\sqrt{50}}\right)+(6-3)\left(\frac{-5}{\sqrt{50}}\right)=\frac{6}{\sqrt{50}} \Rightarrow PQ^2=19-\frac{36}{50}=\frac{427}{25}$

$\therefore$ $100(PQ)^2=1828$

76.(5) The plane passes through the point of intersection of lines (0, 1, 1)

The point (2, 3, 2) lines on the line, whose projection on the second line is (1, 2, 0)

$\therefore$ the required plane has normal (1, 1, 2)

$\therefore$ Equation of the plane is $1(x-0)+1(y-1)+2(z-1)=0$

$x+y+2z-3=0$; $(k,-2,0)$ lies on the plane $\therefore$ $k=5$

77.(2) Let $x\hat{i}+y\hat{j}+z\hat{k}=\vec{r}$

$\vec{r}\cdot(10\hat{j}-8\hat{j}-\vec{r})=41 \Rightarrow -\vec{r}\cdot(10\hat{i}-8\hat{i})+r^2+40=0$ ... (1)

$P_1>\max\left\{\vec{r}+2\hat{i}-3\hat{j}\right\}, P_2=\max\left\{\vec{r}+2\hat{i}-3\hat{j}\right\}$

Equation (1) is a sphere of centre $\dfrac{10\hat{j}-8\hat{i}}{2}$ and radius 1 unit $P_1-P_2=2\times1=2$.

78.(4) $\hat{i}(x+3y-4z)+\hat{j}(x-3y+5z)+\hat{k}(3x+y)=\lambda(x\hat{i}+y\hat{j}+z\hat{k})$

$(1-\lambda)x+3y-4z=0$; $x-(3+\lambda)y+5z=0$; $3x+y-\lambda z=0$

$$\begin{vmatrix}1-\lambda & 3 & -4\\ 1 & -(\lambda+3) & 5\\ 3 & 1 & -\lambda\end{vmatrix}=0 \Rightarrow \lambda=-1$$

$2x+3y=4z$ ... (1)

$x-2y=-5z$ ... (2)

$3x+y=-z$

Solving equation (1) and (2), we get

$y=\dfrac{4z-2(-5z)}{3-2(-2)}=\dfrac{14z}{7}=2z$; $x=-z,\ y=2z$; $\dfrac{x-y-z}{x}=\dfrac{-z-2z-z}{-z}=4.$

79.(6) Let the equation of the plane containing the lines $\dfrac{x-1}{2}=\dfrac{y-2}{3}=\dfrac{z-3}{4}$ and $\dfrac{x-2}{3}=\dfrac{y-3}{4}=\dfrac{z-4}{5}$ be

$a(x-1)+b(y-2)+c(z-3)=0$ ... (1)

where $2a+3b+4c=0$ and $3a+4b+5c=0$

Solving last two equations, we get $\dfrac{a}{1}=\dfrac{b}{-2}=\dfrac{c}{1}$

$\therefore$ From equation (1), equation of the plane is $(x-1)-2(y-2)+(z-3)=0 \Rightarrow$ $x-2y+z=0$

Now the distance between the planes $Ax-2y+z=d$ and $x-2y+z=0$

(Clearly two planes must be parallel so $A=1$) ; $\dfrac{|d|}{\sqrt{1^2+2^2+1^2}}=\sqrt{6} \Rightarrow |d|=6$

80.(2) $4+\lambda_1=1+2\lambda_2 \Rightarrow \lambda_1-2\lambda_2=-3$ ... (1)

$-3-4\lambda_1=-1-3\lambda_2 \Rightarrow 4\lambda_1-3\lambda_2=-2$ ... (2)

From (1) and (2), $-5\times2=-10$

$\lambda_2=2,\ \lambda_1=1 \Rightarrow$ point of intersection $(4+1,-3-4,-1+7)$ i.e. $(5,-7,6)$.

Distance of (5, –7, 6) from (1, –4, 7)

$=\sqrt{16+9+1}=\sqrt{26}$.

PROBABILITY

1.(C) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc = 0 \Rightarrow$ whether $ad = 1$, $bc = 1$ or $ad = -1$, $bc = -1$

Which occur in eight ways. Total number of 2×2 determinants from $\{-1, 1\}$ is 16. Thus required probability is $\frac{8}{16} = \frac{1}{2}$.

2.(D) There can be four such numbers i.e. 43, 34, 62, 26.
Whose product of digit is 12.

$\Rightarrow$ Probability that the man will laugh by seeing the chosen numbers $= \frac{4}{90} = \frac{2}{45}$

$\Rightarrow$ Required probability $= 1 - \left(1 - \frac{2}{45}\right)^3 = 1 - \left(\frac{43}{45}\right)^3$

3.(D) Let A be the event of P_1 winning in 3rd round and B be the event of P_2 winning in first but loosing in second.

To find $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$

$P(A) = \frac{{}^{8n-1}C_{n-1}}{{}^{8n}C_n} = \frac{1}{8}$ and $P(B \cap A)$

= Probability of both P_1 and P_2 winning in first round $\times$ probability of P_1 winning and P_2 loosing in second round $\times$ probability of P_1 winning in 3rd round.

$= \frac{{}^{8n-2}C_{4n-2}}{{}^{8n}C_{4n}} \times \frac{{}^{4n-2}C_{2n-1}}{{}^{4n}C_{2n}} \times \frac{{}^{2n-1}C_{n-1}}{{}^{2n}C_n} = \frac{n}{4(8n-1)}$. Hence, $P\left(\frac{B}{A}\right) = \frac{2n}{8n-1}$.

Alternate:

Probability that P_2 wins in first round given, P_1 wins $= \frac{{}^{8n-2}C_{4n-2}}{{}^{8n-1}C_{4n-1}} = \frac{4n-1}{8n-1}$.

In second round probability that P_2 looses in second round given P_1 wins $= 1 - \frac{2n-1}{4n-1} = \frac{2n}{4n-1}$.

Hence, probability that P_2 looses in second round. Given P_1 wins in 3rd round $\frac{4n-1}{8n-1} \times \frac{2n}{4n-1} = \frac{2n}{8n-1}$.

4.(D) The number of favorable cases is ${}^nC_0 \times 2^n + {}^nC_1\, 2^{n-1} + \ldots + {}^nC_n 2^0 = 3^n$. Total number of cases is $2^n \times 2^n = 4^n$.

Thus required probability is $\left(\frac{3}{4}\right)^n$.

5.(D) Probability that 3 different dices shows different faces is $\frac{{}^6P_3}{6^3} = \frac{5}{9}$, so required probability

${}^3C_2\left(\frac{5}{9}\right)^2 \times \left(\frac{4}{9}\right)^1 = \frac{100}{243}$.

6.(A) 4 players will reach in semi final. Since all the players are of equal strength so this is equivalent to select 4 out of 2^n players

Total number of ways = ${}^{2^n}C_4$

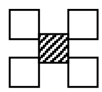
Favourable number of ways = ${}^{2^n-2}C_3 \times {}^2C_1$

Therefore required probability $= \dfrac{{}^{(2^n-2)}C_3 \times 2}{{}^{2^n}C_4} = \dfrac{(2^n-2)\times(2^n-3)\times(2^n-4)\times 8}{2^n(2^n-1)(2^n-2)(2^n-3)} = \dfrac{8\times(2^n-4)}{2^n(2^n-1)}$

7.(B) They will say same thing in two ways i.e. either both speak truth or both tell a lie.
So probability = (0.6 × 0.7) + (1 – 0.6) (1 – 0.7) = 0.42 + 0.12 = 0.54.

8.(C) If the shaded square is chosen we can chose 2 squares out of 4 other at the corner in ${}^4C_2$ ways such shaded squares are 6^2.

Hence probability $= \dfrac{{}^4C_2 6^2}{{}^{64}C_3}$.



9.(C) Given that 5 and 6 have appeared on two of the dice the sample space reduces to $6^4 - 2\times 5^4 + 4^4$ (inclusion exclusion principle) also favourable cases are 4! = 24.

So, the required probability is $\dfrac{24}{302} = \dfrac{12}{151}$.

10.(C) Let a_n be the number of strings of H and T of length n with no two adjacent H's, then $a_1 = 2, a_2 = 3$.

Also, $a_{n+2} = a_{n+1} + a_n$ ($\because$ the string must begin with T or HT)

So, $a_3 = 5, a_4 = 8, a_5 = 8 + 5 = 13 \Rightarrow$ the required probability $= \dfrac{13}{2^5} = \dfrac{13}{32}$.

11.(B) A number has exactly 3 factors if the number is square of a prime number.
Squares of 11, 13, 17, 19, 23, 29, 31 are 3-digit numbers.

Required probability = $\dfrac{7}{900}$

12.(A) Favorable cases $m = {}^6C_2(2\,{}^4C_4 + 2\,{}^4C_3\,{}^1C_1 + {}^4C_2\,{}^2C_2)$

Total cases $\rightarrow n = {}^{12}C_8$

Required probability = $\dfrac{m}{n} = \dfrac{16}{33}$

13.(B) The problem is of conditional probability. Total cases in which at least one of the cubes is red painted is 125 – 27 = 98 out of which 8 are painted on three sides.

$\Rightarrow$ Probability = $\dfrac{8}{98} = \dfrac{4}{49}$

14.(C) Either first ball is black or first three balls are black or first five and so on…. all being equally likely.

Total probability of both balls being odd is $\frac{1}{k}\left[\frac{1}{1!(2n-1)!}+\frac{1}{3!(2n-3)!}+.........\right]$ where k is number of ways both can be odd. Probability of exactly one black ball is

$$\frac{\frac{1}{k}\left(\frac{1}{1!(2n-1)!}\right)}{\frac{1}{k}\left(\frac{1}{1!(2n-1)!}+\frac{1}{3!(2n-3)!}+.........\right)}=\frac{2n}{{}^{2n}C_1+{}^{2n}C_3+.........{}^{2n}C_{2n-1}}=\frac{2n}{2^{2n-1}}=\frac{n}{2^{2n-2}}$$

15.(D) Total ways in which they can seated in the buses are 3^{10}. Applying the principle of Inclusion-exclusion, we get number of favourable ways as $3^{10}-3.2^{10}+3$. Hence, requires probability $=\frac{3^{10}-3.2^{10}+3}{3^{10}}$.

16.(C) Let us assume that 'A' wins after n deuces, $n \in [(0, \infty)$ Probability of a deuce $=\frac{2}{3}.\frac{2}{3}+\frac{1}{3}.\frac{1}{3}=\frac{5}{9}$ (A wins his serve then B wins his serve or A loses his serve then B also loses his serve)

Now probability of 'A' winning the game $=\sum_{n=0}^{\infty}(5/9)^n.\left(\frac{2}{3}\right)\frac{1}{3}=\frac{1}{1-(5/9)}.\frac{2}{9}=\frac{1}{2}$.

17.(B) Total number of ways to distribute the balls so that no box is empty are [(1, 1, 4), (2, 1, 3), (2, 2, 2)]

$$=\frac{3!}{2!}\left[\left({}^6C_1.\ {}^5C_1.\ {}^4C_4\right)\right]+3!\left({}^6C_2.\ {}^4C_1.\ {}^3C_3\right)+\left({}^6C_2.\ {}^4C_2.\ {}^2C_2\right)=90+6.60+90=540$$

Required probability $=\frac{90}{540}=\frac{1}{6}$.

18.(D) First we select r rows and r columns which these squares will be consisting of by ${}^mC_r\,{}^nC_r$ ways.

Now for any one row any one out of r columns can be selected to get the first square, then for second row one column out of $(r-1)$ columns can be selected and so on in $r!$ ways.

Total number of ways are ${}^mC_r\,{}^nC_r\,r!$. Hence probability is $\frac{{}^mC_r\,{}^nC_r\,r!}{{}^{mn}C_r}$.

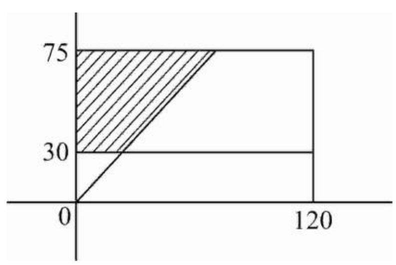
19.(D) Mr. Walia calls at 2 hrs x minutes

Mr. Sharma calls at 2 hr y minutes

then $x\in(0,120)$, $y\in(30,75)$

Favourable case $\equiv x<y$

Probability $=\frac{\frac{1}{2}(30+75)\times 45}{45\times 120}=\frac{7}{16}$



20.(C) $P(I\ II \cup I\ III)=P(I\ II)+P(I\ III)-P(I\ II\ III)$

$$=pq+\frac{p}{2}-\frac{pq}{2}=\frac{p}{2}(q+1)=\frac{1}{2} \quad\Rightarrow\quad p(q+1)=1$$

21.(C) Total number of ways $={}^6C_3=20$

Out of which only two equilateral triangle is possible, So required probability $=\frac{2}{20}=\frac{1}{10}$.

22.(A) $P(E_r) =$ Probability of getting r number of balls

$P(E) =$ The particular ball is first balls to pop up.

$$P(E) = \sum_{r=1}^{n} P(E_r) P\left(\frac{E}{E_r}\right)$$

$$P(E_r) = Kr \qquad \sum_{r=1}^{n} P(E_r) = 1 \quad \Rightarrow \sum_{r=1}^{n} kr = 1 \quad \Rightarrow \quad k = \frac{2}{n(n+1)}$$

$$P\left(\frac{E}{E_r}\right) = \frac{{}^{n-1}C_{r-1}\,(r-1)!}{{}^{n}C_r\; r!} = \frac{1}{n}$$

$$\Rightarrow \quad P(E) = \sum_{r=1}^{n} \frac{1}{n} \times \frac{2r}{n(n+1)} = \frac{1}{n} \quad \Rightarrow \quad P\left(\frac{E_n}{E}\right) = \frac{P(E_n)\cdot P\left(\frac{E}{E_n}\right)}{P(E)} = \frac{\frac{2n}{n(n+1)} \times \frac{1}{n}}{\frac{1}{n}} = \frac{2}{n+1}$$

23.(D) Let d be the common difference

If $d = 2$, we can take 3 numbers as (1, 3, 5), (2, 4, 6), … $(4m - 3, 4m - 1, 4m + 1)$.

In this case we have $4m - 3$ triplets.

If $d = 4$, we can take 3 numbers as

(1, 5, 9), (2, 6, 10), … $(4m - 7, 4m - 3, 4m + 1)$ and in this case we have $4m - 7$ triplets.

If $d = 2m$, we have only one triplet $(1, 2m + 1, 4m + 1)$

Total number of ways of selecting such triplet is $4m - 3 + 4m - 7 + \ldots + 1 = \frac{m}{2}(2 + (m - 1)4) = m(2m - 1)$

Total number of ways of selecting 3 numbers out of 4m + 1 numbers is ${}^{4m+1}C_3$.

Required probability $= \dfrac{m(2m-1)}{{}^{4m+1}C_3} = \dfrac{6m(2m-1)}{(4m+1)4m(4m-1)} = \dfrac{3(2m-1)}{2(16m^2-1)}$.

24.(A) $P(A \cap B') = P(A) - P(A \cap B) = 0.20$

Also $\quad P(A' \cap B) = P(B) - P(A \cap B) = 0.15, \quad \Rightarrow \quad P(A) + P(B) - 2P(A \cap B) = 0.35$

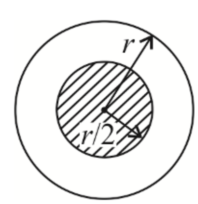
Now, $\quad P(A' \cap B') = 1 - P(A \cup B)\,; \quad 0.1 = 1 - P(A) - P(B) + P(A \cap B)$

$\Rightarrow \quad P(A) + P(B) - P(A \cap B) = 0.9 \quad \Rightarrow \quad P(A \cap B) = 0.9 - 0.35 = 0.55$

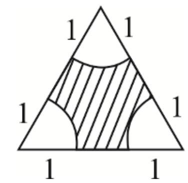
and $\quad P(A) = 0.75P(B) = 0.70 \qquad$ Now $\quad P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.55}{0.70}$

25.(C) For the favourable case, the points should lie inside the concentric circle of radius $\frac{r}{2}$.

So desired probability $= \dfrac{\text{area of smaller circle}}{\text{area of larger circle}} = \dfrac{\pi\left(\frac{r}{2}\right)^2}{\pi r^2} = \dfrac{1}{4}$



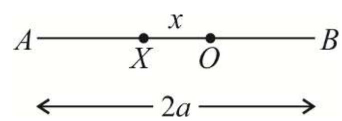
26.(B) The area of the triangle $=\dfrac{\sqrt{3}}{4}(3)^2=\dfrac{9}{4}\sqrt{3}$



The points should lie in the shaded region.

Area of any of the circular segment $=\left(\dfrac{\pi}{6}\right)(1)^2$ $\quad\therefore$ Desired probability $=1-\dfrac{3\times\dfrac{\pi}{6}}{\dfrac{9}{4}\sqrt{3}}=1-\dfrac{2\pi}{9\sqrt{3}}$

27.(A) AX, XB and AO can form a triangle if X lies in OA then



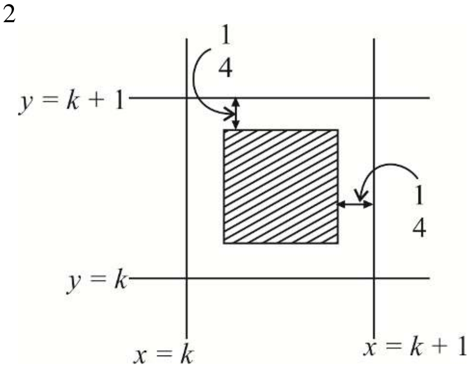
$AX+AO>XB\Rightarrow a-x+a>a+x$

$\therefore \quad x<\dfrac{a}{2}$. Similarly if x lies in OB section, then again $x<\dfrac{a}{2}$

So, for favourable condition X should lie symmetrically about O, such that $OX<\dfrac{a}{2}$.

Hence the desired probability $=\dfrac{\text{Distance in which } X \text{ can be lie}}{\text{Entire length } AB}=\dfrac{a}{2a}=\dfrac{1}{2}$

28.(C) Adjacent line $x=k, x=k+1$



$y=k, y=k+1$ from a unit square

For the favourable event, the coin should fall into the shaded square of side length $\dfrac{1}{2}$

$\therefore \quad$ Desired probability $=\dfrac{\text{area of shaded square}}{\text{area of larger square}}=\dfrac{1}{4}$

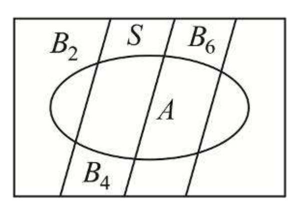
29.(D) If A got at least one head during the first 10 rounds, he will not be drained out at the 11th round.

30.(C) To finish at the 12th round he must have exactly 1 head in the first 10 rounds, and a tail at the 11th and the 12th round. The probability of this is ${}^{10}C_1\, pq^{11}$.

31.(A) Again A cannot be drained at the 13th round if A gets at least 2 heads in the first 10 rounds. To finish at the 14th round, A get exactly 2 heads in the first 10 rounds and a tail on all the round from 11th to 14th. This has a probability ${}^{10}C_2\, p^2q^{12}$ or if A get 1 head in first 10 & 1 head in 11$^{\text{th}}$ or 12$^{\text{th}}$ round i.e. ${}^{10}C_1 q^9 p\left(2pq^3\right)$

Therefore required probability $=q^{10}(1+10pq+65p^2q^2)$

32.(C) $P(A/B_E)=\dfrac{P(A\cap B_E)}{P(B_E)}$, where A = sum is 4.



$\therefore \quad P(A/B_E)=\dfrac{P(A\cap B_2)+P(A\cap B_4)}{P(B_2)+P(B_4)+P(B_6)+\ldots\infty}$

$$=\frac{P(B_2)P(A/B_2)+P(B_4)(A/B_4)}{\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\ldots\infty}=\frac{\frac{1}{4}\left(\frac{3}{36}\right)+\frac{1}{16}\left(\frac{1}{6^4}\right)}{\left(\frac{1}{4}\right)/\left(1-\frac{1}{4}\right)}=\left(\frac{3}{144}+\frac{1}{16\times 1296}\right)\left(\frac{3}{1}\right)\cong\frac{1}{16}.$$

33.(A) $n(S) = 6.6.6 = 216$

Now greatest number is 4, no atleat one of the dice shown up 4;

$\therefore \quad n(A) = 4^3 - 3^3 = 37$. Hence $P(A) = \dfrac{37}{216}$.

34.(B) $P(B_2/S) = \dfrac{P(B_2 \cap A)}{P(A)} \Rightarrow P(B_2/S) = \dfrac{\frac{1}{4}\left(\frac{2}{36}\right)}{\frac{1}{2}\left(\frac{1}{6}\right) + \frac{1}{4}\left(\frac{2}{36}\right) + \frac{1}{8}\left(\frac{1}{216}\right)} = \dfrac{24}{169}$.

35.(ABCD) We have $P(X \ge a) = \sum_{r=a}^{\infty} pq^r = \dfrac{pq^a}{1-q} = q^a$

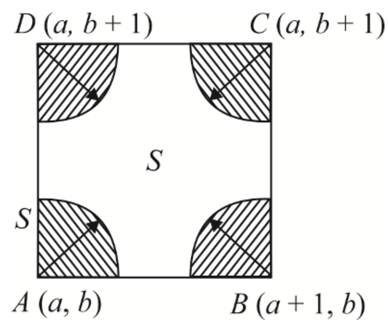
Next $P(X \ge a+b \mid X \ge a) = \dfrac{P[(X \ge a+b) \cap (X \ge a)]}{P(X \ge a)} = \dfrac{P(X \ge a+b)}{P(X \ge a)} = \dfrac{q^{a+b}}{q^a} = q^b = P(X \ge b)$

Similarly $P(X \ge a+b \mid x \ge b) = P(X \ge a)$

Again $P(X = a+b \mid X \ge a) = \dfrac{P[(X = a+b) \cap (X \ge a)]}{P(X \ge a)} = \dfrac{P(X = a+b)}{P(X \ge a)} = \dfrac{pq^{a+b}}{q^a} = pq^b = P(X = b)$

36.(ABC) Let S denote the set of points inside a square with corners $(a,b),(a,b+1),(a+1,b),(a+1,b+1), a$ and b are integers. Clearly each of the four points belong to the set X.

Let P denote the set of points in S with distance less than $\dfrac{1}{4}$ from any corner point. P consists of four quarter circles each of radius $\dfrac{1}{4}$.



A coin, whose centre falls in S, will cover a point of X if and only if its centre falls in P.

Hence, the required probability, $p = \dfrac{\text{area of } P}{\text{area of } S} = \dfrac{\pi(1/4)^2}{1 \times 1} = \dfrac{\pi}{16}$

37.(BD) $p_1 = \dfrac{2}{\pi}, \ p_2 = 1 - \dfrac{2}{\pi}$

38.(BC) $0 \le \dfrac{1+4p}{4} \le 1 \Rightarrow -\dfrac{1}{4} \le p \le \dfrac{3}{4}$ … (1) and $0 \le \dfrac{1-p}{3} \le 1 \Rightarrow -2 \le p \le 1$ … (2)

$0 \le \dfrac{1-2p}{2} \le 1 \Rightarrow -\dfrac{1}{2} \le p \le \dfrac{1}{2}$ … (3)

Also, $0 \le P(A \cup B \cup C) \le 1$

$\Rightarrow 0 \le \dfrac{1+4p}{4} + \dfrac{1-p}{3} + \dfrac{1-2p}{2} \le 1 \Rightarrow 0 \le \dfrac{13-4p}{12} \le 1 \Rightarrow \dfrac{1}{4} \le p \le \dfrac{13}{4}$ … (4)

From (1), (2), (3) and (4) : $\dfrac{1}{4} \le p \le \dfrac{1}{2}$.

39.(AD) If the match is won at the r^{th} game then the result of $(r-1)^{th}$ and r^{th} game will be a win for a particular player. The favourable outcomes of different games would look like this (starting from first and ending at r^{th} game).

	I	II	III	IV		$(r-2)^{th}$	$(r-1)^{th}$	r^{th}
	LW,	LW,	LW,	LW	.......,	LW,	LW,	WL
Or,	WL,	LW,	LW,	LW	…….,	LW,	LW,	WL
Or,	WL,	WL,	LW,	LW	…….,	LW,	LW,	WL
Or,	WL,	WL,	WL,	LW	…….,	LW,	LW,	WL.
	---	----	---	---	---	----	----	
	---	----	---	---	---	----	----	
Or,	WL,	WL,	WL,	WL	……..,	WL,	LW,	WL

Here 'L' stands for loss and 'W' for win 'LW' stands for loss of P_k and winof P_{k+1}, WL stands for win of P_k and loss of P_{k+1}.

That means there are $(r-1)$ favourable cases in which the match ends at r^{th} game. Also probability corresponding to each case is $\frac{1}{2}\frac{1}{2}\frac{1}{2}....'r'\text{ times} = \frac{1}{2^r}$ $\Rightarrow$ Required probability $= \frac{r-1}{2^r}$.

40.(AD) Let λ, μ and λ′, μ′, be the numbers of heads and tails by A and B respectively, so that $\lambda + \lambda' = n + 1$ and $\mu + \mu' = n$

The required probability P is the probability of inequality $\lambda > \mu$

The probability $1 - P$ of the opposite even $\lambda \le \cup\ \mu$ is at the same time the probability of the inequality $\lambda' > \mu'$ that is $1 - P$ is the probability that A will throw more tails than B.

As $\lambda \le \mu \Rightarrow n + 1 - \lambda' < n - \mu' \Rightarrow 1 - \lambda' \le -\mu' \Rightarrow \lambda' - 1 \ge \mu' \Rightarrow \lambda' \ge \mu + 1 > \mu'$

So by reason of symmetry $1 - p = p$ or $p = \frac{1}{2}$.

41.(ABCD) We have $P(A \cup B) \ge \max\{P(A), P(B)\} = \frac{2}{3}$,

Next, $P(A \cap B) = P(A) + P(B) - P(A \cup B) \ge P(A) + P(B) - 1 = 1/6$

And $P(A \cap B) \le \min\{P(A), P(B)\} = \frac{1}{2} \Rightarrow \frac{1}{6} \le P(A \cap B) \le \frac{1}{2}$

Also $P(A \cap B') = P(A) - P(A \cap B) \le \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$

Lastly $P(A' \cap B) = P(B) - P(A \cap B)$

$$2/3 - 1/2 \le P(A' \cap B) \le \frac{2}{3} - \frac{1}{6} \Rightarrow 1/6 \le P(A' \cap B) \le ½$$

42.(AC) $\frac{6}{n(n+1)(2n+1)}$

Let E_i be the event that the bag contains exactly i white balls $(0 \le i \le n)$

$\Rightarrow \quad P(E_i) \propto i \Rightarrow P(E_1) = ki \Rightarrow P(E_0) = 0.$

Since $\{E_i\}$ is the set of mutually exclusive events.

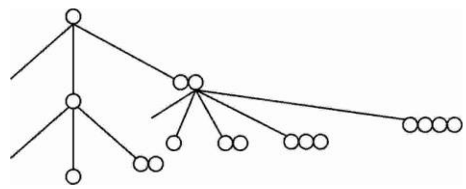
$$\sum_{i=1}^{n} P(E_i) = 1 \Rightarrow \sum_{i=1}^{n} ki = 1 \Rightarrow k\frac{n(n+1)}{2} = 1 \Rightarrow k = \frac{2}{n(n+1)}. \Rightarrow P(E_i) = \frac{2i}{n(n+1)}$$

Let B be the event that the ball, drawn out, is white

$$\Rightarrow P(E_1/B)=\frac{P(E_1).P(B/E_1)}{\sum_{j=1}^{n}P(E_j).P(B/E_j)}=\frac{\frac{2}{n(n+1)}.\frac{1}{n}}{\sum_{j=1}^{n}\frac{2j}{n(n+1)}.\frac{j}{n}}=\frac{1}{\sum_{j=1}^{n}.j^2}=\frac{6}{n(n+1)(2n+1)}.$$

43.(AC) $p=\dfrac{({}^nC_0)^2+({}^nC_1)^2+....+({}^nC_n)^2}{4^n}=\dfrac{{}^{2n}C_n}{2^{2n}}$

44.(AC) $P(X_2>0)=1-\dfrac{1}{4}-\dfrac{1}{2}\times\dfrac{1}{4}-\dfrac{1}{4}\times\dfrac{1}{4}\times\dfrac{1}{4}=\dfrac{64-16-8-1}{64}=\dfrac{39}{64}$



$$P(X_1=2\,|\,X_2=1)=\frac{P(X_2=1|\,X_1=2)\,.\,P(X_1=2)}{P(X_2=1)}$$

$$=\frac{\left(\frac{1}{2}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{4}\right).\frac{1}{4}}{\frac{1}{2}\times\frac{1}{2}+\frac{1}{4}\left(\frac{1}{2}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{4}\right)}=\frac{1}{5}$$

45.(ABCD) $A\subseteq A\cup B \quad\Rightarrow\quad P(A\cup B)\ge P(A)\Rightarrow P(A\cup B)\ge\dfrac{3}{4}$

$P(A\cap B)=P(A)+P(B)-P(A\cup B)\ge P(A)+P(B)-1\ge\dfrac{3}{4}+\dfrac{5}{8}-1\ge\dfrac{3}{8}$

$(A\cap B)\subseteq B \quad\Rightarrow\quad P(A\cap B)\le P(B)\le\dfrac{5}{8}$

$P(A\cap B')=P(A)-P(A\cap B)\;\therefore\quad \dfrac{3}{4}-\dfrac{5}{8}\le P(A\cap B')\le\dfrac{3}{4}-\dfrac{3}{8}\quad\Rightarrow\quad \dfrac{1}{8}\le P(A\cap B')\le\dfrac{3}{8}$

Now $P(A'\cap B)=P(B)-P(A\cap B)\quad\Rightarrow\quad P(A\cap B)=P(B)-(A'\cap B)$

$\Rightarrow\quad \dfrac{3}{8}\le P(B)-P(A'\cap B)\le\dfrac{5}{8}\quad\Rightarrow\quad 0\le P(A'\cap B)\le\dfrac{1}{4}$

46.(ABCD) $P(E_0)=\dfrac{2^3}{3^3}$

$P(E_1)=\dfrac{{}^3C_1\cdot1\cdot2\cdot2}{3^3}$; $\quad P(E_2)=\dfrac{{}^3C_2\cdot1\cdot1\cdot2}{3^3}$; $\;P(E_3)=\dfrac{{}^3C_3\cdot1\cdot1\cdot1}{3^3}$

47.(BC) Let the number of green marbles be x.

Probability of drawing green values in one try = $\dfrac{1}{y}=\dfrac{x}{x+32}\quad\Rightarrow\quad x+32=yx\quad\Rightarrow\quad x=\dfrac{32}{y-1}$

Since x is integer, possible values of y are {33, 17, 9, 5, 3, 2}. Corresponding value of x are {1, 2, 4, 8, 16, 32.}

48.(ABCD) $P(E_1)=\dfrac{{}^6C_3}{216}=\dfrac{5}{54}$; $\quad P(E_2)=\dfrac{4+2}{216}=\dfrac{1}{36}$

[(1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 6), (1, 3, 5), (2, 4, 6)]

$E_1 \cap E_2 = E_2 \quad \therefore \qquad P(E_1 \cap E_2) = P(E_2) = \frac{1}{36}$

$P(E_2 / E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{1/36}{5/54} = \frac{54}{180} = \frac{3}{10}$

49.(AC) $P(A') = \frac{39}{52} = \frac{3}{4}$ (Event A' is that missing card is not a spade)

$P(S/A) = \frac{{}^{12}C_2}{{}^{51}C_2}$ (only 12 spades are left if missing card is spade)

$$P(A/S) = \frac{P(A \cap S)}{P(S)} = \frac{P(A)P(S/A)}{P(S)} = \frac{\frac{1}{4}\cdot\frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4}\cdot\frac{{}^{12}C_2}{{}^{51}C_2} + \frac{3}{4}\cdot\frac{{}^{13}C_2}{{}^{51}C_2}} = \frac{11}{50}$$

50.(BD) If a is the radius of the circle, the area of the inscribed square = $2a^2$

$\therefore \quad p_1 = \frac{2a^2}{\pi a^2} = 2/\pi$ and $p_2 = 1 - p = \frac{\pi - 2}{\pi} \qquad \therefore \qquad \pi - 2 < 2 \Rightarrow \frac{\pi-2}{\pi} < \frac{2}{\pi} \Rightarrow p_2 < p_1$

$p_1^2 - p_2^2 = (p_1 + p_2)(p_1 - p_2) = \frac{4-\pi}{\pi} < \frac{1}{3}$

51.(ABCD) Let A, B and C be the events that the student is successful in tests I, II and III, respectively.

Then P(the student is successful)

$= P[(A \cap B \cap C') \cup (A \cap B' \cap C') \cup (A \cap B \cap C)] = P(A \cap B \cap C') + p(A \cap B' \cap C') + P(A \cap B \cap C')$

$= P(A)P(B)P(C') + P(A)P(B')P(C') + P(A)P(B)P(C) \qquad [\because$ A, B and C are independent]

$= pq(1 - 1/2) + p(1-q)(1/2) + (pq)(1/2) = \frac{1}{2}[pq + p(1-q) + pq] = \frac{1}{2}p(1+q)$

$\therefore \quad \frac{1}{2} = \frac{1}{2}p(1+q) \quad \Rightarrow \qquad p(1+q) = 1$

This equation is satisfied for all pairs of values in (A), (B) and (C). Also, it is satisfied for infinitely many values of p and q. For instance, when $p = n/(n+1)$ the $nq = 1/n$, where n is any positive integer.

52.(BCD) $P(E_1) = P(E_2) = \frac{\frac{10!}{2!}}{\frac{11!}{2!2!}} = \frac{2}{11}$

$P(E_1 \cap E_2)$ = probability that two I's are together, and two B's are together = $\frac{9!}{\frac{11!}{2!2!}} = \frac{2}{55}$

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{2}{11} + \frac{2}{11} - \frac{2}{55} = \frac{18}{55}$

$P(E_1 / E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{2}{55}}{\frac{2}{11}} = 1/5$

53.(ACD) We have $a^2+b^2+c^2=2$ where a, b, c are integers. $\Rightarrow$ $a=0,\ b=\pm1,\ c=\pm1$

Or $a=\pm1,\ b=0,\ c=\pm1$ Or $a=\pm1,\ b=\pm1,\ c=0$

There are 12 possibilities.

(a) Now $|A|\neq 0 \Rightarrow a+b+c\neq 0$

So, there are 6 possibilities. Probability = 1/2

(b) $|A|=0 \Rightarrow a+b+c=0$

So, there are 6 possibilities

But in no case the system has infinitely many solution

(c) If $a=0,\ b=1,\ c=-1,\ a=0,\ b=-1,\ c=1$

So there are two possibilities

(d) Range of $ab+bc+ca$ is $\{-1,\ 1\}$

54.(AC) (a) The probability of S_1 to be among the eight winners is equal to the probability of S_1 winning in the group, which is given by ½.

(b) If S_1 and S_2 are in the same pair-then exactly one wins.

If S_1 and S_2 are in two separate pairs, then for exactly one of S_1 and S_2 to be among the eight winners, S_1 wins and S_2 loses of S_1 lose and S_2 wins.

Now the probability of S_1, S_2 being in the same pair and one wins is (Probability of S_1, S_2 being in the same pair) $\times$ (Probability of any one winning in the pair). And the probability of S_1, S_2 being in the same pair is $\dfrac{n(E)}{n(S)}$

The number of ways 16 players are divided into 8 pairs is $n(S)=\dfrac{16!}{(2!)^8\times 8!}$

The number of ways in which 16 persons can be divided in 9 pairs so that S_1 and S_2 are in same pair is $n(E)=\dfrac{14!}{(2!)^7\times 7!}$

Therefore, the probability of S_1 and S_2 being in the same pair is $\dfrac{\dfrac{14!}{(2!)^7\times 7!}}{\dfrac{16}{(2!)^8\times 8!}}=\dfrac{2!\times 8}{16\times 15}=\dfrac{1}{15}$

The probability of any one winning in the pair of S_1, S_2 is P(certain event) = 1.

Hence, the probability that the pair of S_1, S_2 being in two pairs separately and any one of S_1, S_2 wins is given by the sum the probability of S_1, S_2 being in two pairs separately and S_1 wins, S_2 loses and the probability of S_1, S_2 being in two pairs separately and S_1 loses, S_2 wins. It is given by $\left[1-\dfrac{1}{15}\right]\times\dfrac{1}{2}\times\dfrac{1}{2}+\left[1-\dfrac{1}{15}\right]\times\dfrac{1}{2}\times\dfrac{1}{2}=\dfrac{1}{2}\times\dfrac{14}{15}=\dfrac{7}{15}$

Therefore, the required probability is $(1/15)+(7/15)=(8/15)$.

55.(AB) Given that A and B independent events.

$\therefore \quad p(A\cap B)=P(A)P(B)$ ...(i)

Also given that

$P(A\cap B)=\frac{1}{6}$ ...(ii)

And $P(\bar{A}\cap\bar{B})=\frac{1}{3}$

Also, $P(\bar{A}\cap\bar{B})=1-P(A\cup B)$ or $P(\bar{A}\cap\bar{B})=1-P(A)-P(B)+P(A\cap B)$

or $\frac{1}{3}=1-P(A)-P(B)+\frac{1}{6}$ or $P(A)+P(B)=\frac{5}{6}$

from eqs. (i) and (ii), we get $P(A)P(B)=\frac{1}{6}$

Let $P(A)=x$ and $P(B)=y$. Then eqs (iv) and (v) become $x+y=\frac{5}{6}$; $xy=\frac{1}{6}$

Solving we get $x = 1/2$ and $y = 1/3$ or $x = 1/3$ and $y = ½$. Thus, $P(A) = ½$ or 1/3.

56.(AD) $P(A\cap B)=a,\ P(A)=a+d,\ P(B)=a+2d$ and $P(A\cup B)=a+3d$ also $a+d=d$

$\Rightarrow \quad a=0 \quad \Rightarrow \quad P(A\cap B)=0,\ P(A)=d,\ P(B)=2d$ and $P(A\cup B)=3d$

57.(ACD) $P(A)=\frac{{}^5C_1 2^6}{2^{10}}=\frac{5}{16},\ P(B)=\frac{{}^5C_1 2^6}{2^{10}}=\frac{5}{16}$

$P(A\cap B)=\frac{{}^5C_2\times 2\times 2^3}{2^{10}}=\frac{5}{32} \quad \Rightarrow \quad P(A\cup B)=\frac{15}{32}$

58.(ABCD) A, B are exhaustive events $\Rightarrow \quad A\cup B=S$

$\Rightarrow \quad P(A\cup B)=1$

$2P(B)=P(A)=P(A\cup B)=K+1$

$\Rightarrow \quad P(B)=\frac{K+1}{2}-P(A\cup B)-P(A)-PB\ =1-K-\frac{(K+1)}{2}=\frac{2-2K-K-1}{2}$

$\therefore \quad P(A\cap B)=\frac{3K-1}{2} \qquad P(A'\cup B')=P\left((A\cap B)'\right)=1-P(A\cap B)=1-\frac{3K-1}{2}=\frac{3(1-K)}{2}$

59.(ABC) Let the no. of blue marbles be a and the no. of green marbles be b.

$\therefore \quad \frac{ab}{{}^{a+b}C_2}=\frac{1}{2} \quad \Rightarrow \quad (a+b)(a+b-1)=4ab \quad \Rightarrow \quad b^2-(2a+1)b+a^2-a=0$

But $b\in R\Rightarrow D=(2a+1)^2-4\left(a^2-a\right)=8a+1 \quad \therefore \quad 8a+1$ must be a perfect square

Hence possible values of a are 3, 6, 10

60.(AD) From the given condition, it follows that ${}^nC_4,\ {}^nC_5$ and ${}^nC_6$ are in A.P.

i.e, $2.\ {}^nC_5={}^nC_4+{}^nC_6$

$$\frac{2n(n-1)(n-2)(n-3)(n-4)}{5!}=\frac{n(n-1)(n-2)(n-3)}{4!}+\frac{n(n-1)(n-2)(n-3)(n-4)(n-5)}{6!}$$

$\Rightarrow \quad \frac{2(n-4)}{5}=1+\frac{(n-4)(n-5)}{5.6} \quad \Rightarrow \quad n^2-21n+98=0 \quad \Rightarrow \quad (n-7)(n-14)=0 \quad \Rightarrow \quad n=7,\ n=14$

61.(BC) (a) Is false since $A_1, A_2, \ldots, A_n$ may be overlapping.

(b) If $A_1, A_2, \ldots, A_n$ are disjoint and exhaustive both then $\sum P(A_i)=1$ if they are only exclusive then $\sum P(A_i)\le 1$

If exhaustive then $\sum P(A_i)\ge 1$ (choice (c) follows)

62.(AD) Clearly $P(A)=P(C)$

Event A is no boy or exactly one boy in family $\Rightarrow$ $P(A)=\left(\frac{1}{2}\right)^3+{}^3C_1\left(\frac{1}{2}\right)^3=\frac{1}{8}+\frac{3}{8}=\frac{1}{2}$

Event B is 2 boys, 1 girl or 1 boy, 2 girls $P(B)={}^3C_1\left(\frac{1}{2}\right)^3+{}^3C_2\left(\frac{1}{2}\right)^3=\frac{3}{4}$

Event C is no girl or exactly one girl $P(C)=\left(\frac{1}{2}\right)^3+{}^3C_1\left(\frac{1}{2}\right)^3=\frac{1}{2}$

Event $A\cap B$ is one boy and two girls $P(A\cap B)={}^3C_1\left(\frac{1}{2}\right)^3=\frac{3}{8}$

Event $B\cap C$ is one girl and two boys $P(B\cap C)=\frac{3}{8}$

$A\cap C\to\phi \quad\Rightarrow\quad A\cap B\cap C$ is also ϕ

$P(A\cap B)=\frac{3}{8}=P(A)\times P(B)$, so, A and B are independent neither $P(A\cap B\cap C)=P(A)\times P(B)\times P(B)$ nor $P(A\cap C)=P(A)\times P(C)$ So, ABC is not independent

63.(AC) We have $q = P(OH \text{ or } 2H \text{ or } 4H)$

$\therefore\quad q=p^4+{}^4C_2p^2(1-p)^2+(1-p)^4=8p^4-16p^3+12p^2-4p+1=\frac{(2p-1)^4+1}{2}$

Now put the values and check.

64.(ABC) Let event E_1 is first digit is '2' $\Rightarrow$ 211 or 222

$P(E_1)=\frac{2}{4}=\frac{1}{2}$

Let event E_2 is second digit is '2' $\Rightarrow$ 121, 222

$P(E_2)=\frac{1}{2}$

Let event E_3 is third digit is '2' $\Rightarrow$ 222 and 112

$P(E_3)=\frac{1}{2}$

$(E_1\cap E_2)=222 \Rightarrow P(E_1\cap E_2)=\frac{1}{4}==P(E_1)P(E_2)$

Similarly E_2 and E_3, E_1 and E_3

Also, $E_1\cap E_2\cap E_3\to 222 \quad\Rightarrow\quad P(E_1\cap E_2\cap E_3)=\frac{1}{4}\ne P(E_1)P(E_2)P(E_3)$

65. **[A-q, s] [B-p, t] [C-r, t]**

(A) $\lambda = 1 -$ Problem will not be solved

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) = 1 - P(\bar{A})P(\bar{B})P(\bar{C})$$

$$= 1 - \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right) = 1 - \frac{1}{2}\times\frac{2}{3}\times\frac{3}{4} = 1-\frac{1}{4} = \frac{3}{4}$$

and $\mu = P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)$

$$= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C)$$

$$= \frac{1}{2}\cdot\left(1-\frac{1}{3}\right)\cdot\left(1-\frac{1}{4}\right) + \left(1-\frac{1}{2}\right)\cdot\frac{1}{3}\cdot\left(1-\frac{1}{4}\right) + \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\cdot\frac{1}{4} = \frac{6}{24}+\frac{3}{24}+\frac{2}{24} = \frac{11}{24}$$

$\therefore \lambda + \mu = \frac{3}{4} + \frac{11}{24} = \frac{29}{24}$ and $\lambda - \mu = \frac{7}{24} (Q, S)$

(B) Here, $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(C) = \frac{1}{4}$

$\therefore \lambda = P(A \cap B \cap \bar{C}) + P(\bar{A} \cap B \cap C) + P(A \cap \bar{B} \cap C)$

$$= P(A)P(B)P(\bar{C}) + P(\bar{A})P(B)P(C) + P(A)P(\bar{B})P(C) = \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{3}{4} + \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{1}{4} = \frac{6}{24}$$

and $\mu = \lambda + P(A)\cdot P(B)\cdot P(C) \quad = \lambda + \frac{1}{24} = \frac{7}{24}$

$\therefore$ and $\mu - \lambda = \frac{1}{24}$ and $\lambda + \mu = \frac{13}{24} (P, T)$

(C) $\lambda = \frac{2}{6}\times\frac{5}{8} = \frac{5}{24}$ and $\mu = \frac{4}{6}\times\frac{3}{8} = \frac{6}{24}$

$\therefore \quad \mu - \lambda = \frac{1}{24} (T)$ and $\mu + \lambda = \frac{11}{24} (R)$

66. **[A-t]** **[B-q]** **[C-s]** **[D-r]**

(A) A_1 : missing card is red, A_2 : missing card is black.

B: 1st 13 cards drawn are all red.

$$P\left(\frac{A_2}{B}\right) = \frac{P(A_2)\cdot P\left(\frac{B}{A_2}\right)}{P(A_1)\cdot P\left(\frac{B}{A_1}\right) + P(A_2)\cdot\left(\frac{B}{A_2}\right)}$$ 51 [$25^{\text{red}} + 26^{\text{black}}$]

$$P\left(\frac{B}{A_1}\right) = \frac{{}^{25}C_{13}}{{}^{51}C_{13}}$$ 51 [$26^{\text{red}} + 25^{\text{black}}$]

$$P\left(\frac{B}{A_2}\right) = \frac{{}^{26}C_{13}}{{}^{51}C_{13}} \qquad = P\left(\frac{A_2}{B}\right) = \frac{\frac{1}{2}\times\frac{{}^{26}C_{13}}{{}^{51}C_{13}}}{\frac{1}{2}\left[\frac{{}^{25}C_{13}}{{}^{51}C_{13}} + \frac{{}^{26}C_{13}}{{}^{51}C_{13}}\right]} = \frac{{}^{26}C_{13}}{{}^{25}C_{13} + {}^{26}C_{13}}.$$

(B) $\frac{25}{51}$

Let A be the event of drawing a red card when one card is drawn out of 51 cards (excluding missing card).

Let A_1 be the event that the missing card is red and A_2 be the event that the missing card is black.

Now by Bayes's theorem, required probability,

$$P(A_1/A) = \frac{P(A_1).\ P(A/A_1)}{P(A_1).\ P(A/A_1) + P(A_2).P(A/A_2)} \quad \ldots (1)$$

In a pack of 52 cards 26 are red and 26 are black.

Now $P(A_1) =$ probability that the missing card is red $= \frac{{}^{26}C_1}{{}^{52}C_1} = \frac{26}{52} = \frac{1}{2}$

$P(A_2) =$ probability that the missing card is black $= \frac{26}{52} = \frac{1}{2}$

$P(A/A_1) =$ probability of drawing a red card when the missing card is red.

$= \frac{25}{51}$ [$\because$ Total number of cards left is 51 out of which 25 are red and 26 are blacks as the missing card is red]

Again $P(A/A_2) =$ Probability of drawing a red card when the missing card is black $= \frac{26}{51}$. Now from (1), required probability, $P(A_1/A) = \frac{\frac{1}{2}.\frac{25}{51}}{\frac{1}{2}.\frac{25}{51} + \frac{1}{2}.\frac{26}{51}} = \frac{25}{51}$.

(C) $E_1 =$ Event that A has drawn a white ball, $E_2 =$ Event that A has drawn a black ball.

$E_3 =$ event that A has drawn a green ball.

$P(E_1) = \frac{2}{12}$, $P(E_2) = \frac{4}{12}$, $P(E_3) = \frac{6}{12}$.

$E =$ event that B has drawn two green balls.

Hence $P\left(\frac{E}{E_1}\right) = \frac{{}^6C_2}{{}^{11}C_2} = P\left(\frac{E}{E_2}\right)$, and $P\left(\frac{E}{E_3}\right) = \frac{{}^5C_2}{{}^{11}C_2}$

$$\Rightarrow \quad P\left(\frac{E_2}{E}\right) = \frac{P(E_2)P\left(\frac{E}{E_2}\right)}{\sum_{i=1}^{3} P(E_i)P\left(\frac{E}{E_i}\right)} = \frac{\frac{4}{12} \times \frac{{}^6C_2}{{}^{11}C_2}}{\frac{2}{12} \times \frac{{}^6C_2}{{}^{11}C_2} + \frac{4}{12} \times \frac{{}^6C_2}{{}^{11}C_2} + \frac{6}{12} \times \frac{{}^5C_2}{{}^{11}C_2}} = \frac{2}{5}.$$

(D) Suppose, there exist 3 rational points or more on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$

Therefore if $(x_1, y_1), (x_2, y_2)$ and (x_2, y_3) be those 3 points, then

$$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0 \quad \ldots (1)$$

$$x_2^2 + y_2^2 + 2gx_2 + 2fy_1 + c = 0 \quad \ldots (2)$$

$$x_3^2 + y_3^2 + 2gx_3 + 2fy_3 + c = 0 \quad \ldots (3)$$

Solving (1), (2) and (3), we will get g, f, c as rational

Thus, centre of the circle $(-g, -f)$ is a rational point

$\Rightarrow$ both the coordinates of the centre are rational numbers.

Obviously, the possible values of p are 1, 2

Similarly, the possible values of q are 1, 2

$\Rightarrow$ for this case (p, q) may be in (2×2) i.e. 4 ways.

Now, (p, q) can be, without restriction, in (6×6) i.e. 36 ways.

Hence, the probability that at the most two rational points exist on the circle $= \frac{36-4}{36} = \frac{32}{36} = \frac{8}{9}$.

67. **[A-r]** **[B-s]** **[C-p]** **[D-q]**

(A) $\frac{{}^{11}C_5}{{}^{12}C_6} = \frac{1}{2}$.

(B) Let E_1 be the event that S_3 and S_4 are in same group

Let E_2 be the event that S_3 and S_4 are in different group

$P(E_1) = \frac{1}{11}$, $\quad P(E_2) = \frac{10}{11}$

Let E be the event that exactly one of S_3 and S_4 is among the losers, then

$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2)$

$= \frac{1}{11} \times 1 + \frac{10}{11} \times \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}\right) = \frac{6}{11}$.

(C) S_2 and S_4 should be in different groups for both being winner

Required probability $= \frac{10}{11}\left(\frac{1}{2} \cdot \frac{1}{2}\right) = \frac{5}{22}$ or $\frac{{}^{10}C_4}{{}^{12}C_6} = \frac{5}{22}$.

68. **[A-r]** **[B-p]** **[C-q]** **[D-s]**

(A) The req. event will occur if last digit in all the chosen numbers is 1, 3, 7 or 9.

Therefor the req. prob. $= \left(\frac{4}{10}\right)^n$

(B) P(last digit is 1, 2, 3, 4, 6, 7, 8, 9) – P(last digit is 1, 3, 7, 9) $= \frac{8^n - 4^n}{10^n}$

(C) $P(1,3,5,7,9) - P(1,3,7,9) = \frac{5^n - 4^n}{10^n}$

(D) $P(0,5) - P(5) = \frac{(10^n - 8^n) - (5^n - 4^n)}{10^n}$

69. **[A-q]** **[B-p]** **[C-r]** **[D-q]**

(A) If E_i denotes the event that the bag contains i black balls and $12-i$ white balls $(i=0,1,2,....,12)$ and if A denotes the event that the four balls drawn are black,

$$P(E_i)=\frac{1}{13},\ \text{for } i=0,1,2,.......,12$$

$$P(A/E_i)=0,\ \text{for } i=0,1,2,3 \qquad \ldots (1)$$

$$P(A/E_i)=\frac{{}^iC_4}{{}^{12}C_4},\ \text{for } 4\le i\le 12$$

$$\therefore \quad P(A)=\sum_{i=0}^{12}P(E_i)P(A/E_i)=\frac{1}{13}\sum_{i=4}^{12}P(A/E_i)$$

$$=\frac{1}{13}\times\frac{1}{{}^{12}C_4}\left[{}^4C_4+{}^5C_4+{}^6C_4+.....+{}^{12}C_4\right]=\frac{1}{13\times{}^{12}C_4}{}^{13}C_5=\frac{13!\,4!\,8!}{13(12)!\,5!\,8!}=\frac{1}{5}$$

(B) $$P(A/E_{10})=\frac{{}^{10}C_4}{{}^{12}C_4}=\frac{10!\,4!\,8!}{12!\,4!\,6!}=\frac{7\times 8}{11\times 12}=\frac{14}{33}$$

(C) By Baye's theorem, $$P(E_{10}/A)=\frac{P(E_{10})P(A/E_{10})}{P(A)}=\frac{\frac{1}{13}\times\frac{14}{33}}{\frac{1}{5}}=\frac{70}{429}$$

(D) Let B denote the probability of drawing 2 white and 2 black balls.

$$P(B/E_i)=\frac{{}^iC_2\times{}^{(12-i)}C_2}{{}^{12}C_4},\ \text{for } i=2,3,4,......,10 \quad \text{and} \quad P(B/E_i)=0,\ \text{for } i=0,1,11,12$$

$$\therefore \quad P(B)=\sum_{i=0}^{12}P(E_i)P(B/E_i)=\frac{1}{13}\times\frac{1}{{}^{12}C_4}\left[{}^2C_2\ {}^{10}C_2+{}^3C_2\ {}^9C_2+{}^4C_2\ {}^8C_2+.......+{}^{10}C_2\ {}^2C_2\right]$$

$$=\frac{1}{13\times 495}\left[2\left({}^2C_2\ {}^{10}C_2+{}^3C_2\ {}^9C_2+{}^4C_2\ {}^8C_2+{}^5C_2\ {}^7C_2\right)+{}^6C_2\ {}^6C_2\right]$$

$$=\frac{1}{13\times 495}\left[2(45+108+168+210)+225\right]=\frac{2(531)+225}{13\times 495}=\frac{1287}{13\times 495}=\frac{13\times 99}{13\times 99\times 5}=\frac{1}{5}$$

70.(7) Last digit of a^2 and b^2

Frequency	a^2	b^2
1	0	0
2	1	1
2	4	4
1	5	5
2	6	6
2	9	9

Favorable case = total number of cases when sum of last digit is 0 or 5.

= (0, 0), (0, 5), (1, 4), (1, 9), (4, 6), (5, 5), (6, 9)

Hence required probability = (1 × 1 × 1 + 1 × 1 × 2 + 2 × 2 × 2 + 2 × 2 × 2 + 2 × 2 × 2 + 1 × 1 × 1 + 2 × 2 × 2) / (10 × 10) = $\frac{9}{25}$.

71.(8) $\lim\limits_{x\to 0}\left(\dfrac{a^x+b^x}{2}\right)^{2/x}$ $\qquad (1^{\infty}$ form)

$ln\ k = \lim\limits_{x\to 0}\dfrac{2}{x}\left(\dfrac{a^x+b^x}{2}-1\right) \quad \Rightarrow \quad k = ab$. So favorable case = (6, 1), (1, 6), (3,), (3, 2)

Hence required probability $=\dfrac{4}{36}=\dfrac{1}{9}$.

72.(3) Let E_1, E_2, E_3, E_4 be the events that first two drawn book are (math, math) (math, phy) (phy, math) and (phy, phy) respectively and A be the event that third drawn book is of math.

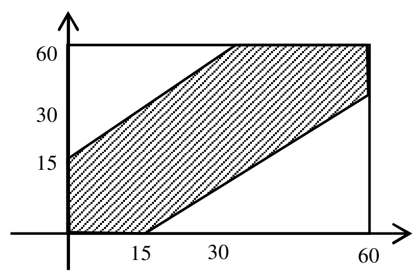
Here $\quad P(E_1)=\dfrac{3}{5}\times\dfrac{5}{7}=\dfrac{3}{7}$, $\quad P(E_2)=\dfrac{3}{5}\times\dfrac{2}{7}=\dfrac{6}{35}$, $\quad P(E_3)=\dfrac{2}{5}\times\dfrac{3}{4}=\dfrac{3}{10}$, $\quad P(E_4)=\dfrac{2}{5}\times\dfrac{1}{4}=\dfrac{1}{10}$

Also $P(A/E_1)=\dfrac{7}{9}; P(A/E_2)=5/6; P(A/E_3)=\dfrac{5}{6};\ P(A/E_4)=1$

Now by Baye's theorem $P(E_4/A)=\dfrac{P(E_4).P(A/E_4)}{P(A)}$

$$=\frac{\frac{1}{10}\times 1}{\frac{3}{7}\times\frac{7}{9}+\frac{6}{35}\times\frac{5}{6}+\frac{3}{10}\times\frac{5}{6}+\frac{1}{10}\times 1}=\frac{1}{\frac{10}{3}+\frac{10}{7}+\frac{5}{2}+1}=\frac{42}{140+60+105+42}=\frac{42}{347}.$$

73.(9) Let line of arrival of A be taken along x–axis and that of B along y–axis meeting between A and B taken place if and only if $|x-y|\le 15$.



Let x and y be the coordinates in the plane and scale be taken in minutes.

Therefore all possible out comes will be a square with side 60 the favourable out come will be in shaded region to the whole area $=\dfrac{60^2-45^2}{60^2}=\dfrac{7}{16}$.

74.(8) Let E_1 be the event that balls drawn from bag I are of different colours.

$E_2 \to$ both are white ; $E_3 \to$ both are red ; $A \to$ third drawn ball is red

$P(E_2)=\dfrac{{}^5C_2}{{}^9C_2}=\dfrac{10}{36}=\dfrac{5}{18}$, $P(E_3)=\dfrac{{}^4C_2}{{}^9C_2}=\dfrac{6}{36}=\dfrac{1}{6}$

$P(E_1)=1-(P(E_2)+P(E_3))=1-\left[\dfrac{5}{18}+\dfrac{1}{6}\right]=1-\dfrac{8}{18}=\dfrac{5}{9}$

Now by Baye's Theorem $P(E_2/A)=\dfrac{P(E_2)\ P(A/E_2)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)+\quad P(E_3)P(A/E_3)}$

$$=\frac{\frac{5}{18}\times\frac{4}{7}}{\frac{5}{9}\times\frac{5}{9}+\frac{5}{18}\times\frac{4}{7}+\frac{1}{6}\times\frac{2}{7}}=\frac{45}{146}.$$

75.(9) Let E_1 = the event that one rupee coin is transferred from first purse to the second purse when 9 coins are transferred ; E_2 the event that one rupee coins is not transferred from first purse to the second purse when 9 coins are transferred ; E = the event that one rupee coin is found in the first purse after the second transfer.

Let $A_1 = E_1 \cap E, A_2 = E_2 \cap E$; Let $A = A_1 \cap A_2$

Since E_1 & E_2 are mutually exclusive events, therefore A_1 & A_2 are also mutually exclusive. Therefore required probability, $P(A) = P(A_1) + P(A_2) = P(E_1 \cap E) + P(E_2 \cap E) = P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)\cdot P\left(\frac{E}{E_2}\right) = P\left(\frac{E}{E_1}\right)$

= probability that one rupee coin will be transferred from second purse to the first purse when one rupee coin and 8 fifty paise coins are transferred from first purse to the second purse $= \frac{{}^1C_1 \cdot {}^{18}C_8}{{}^{19}C_9} = \frac{9}{19} =$ $P\left(\frac{E}{E_1}\right) =$

probability that one rupee coin will be in the first purse when it has not been transferred to the second purse = 1

So $\boldsymbol{P}$ **(A)** $= \frac{{}^1C_1 \cdot {}^9C_8}{{}^{10}C_9} \cdot \frac{9}{19} + \frac{{}^9C_9}{{}^{10}C_9} 1 = \frac{9}{10} \cdot \frac{9}{19} + \frac{1}{10} = \frac{81+19}{190} = \frac{10}{19}$.

76.(8) The queens will be facing each other in case both are placed on the same column, the same row or the same diagonal of the chess board. There are eight rows, eight columns, two principal diagonals and four sets each containing diagonal lines of 7, 6, 5, 4, 3, 2, 1 squares. Hence the probability of the queens taking on each other is

$$P(\bar{A}) = \frac{18\,{}^8C_2 + 4\left({}^7C_2 + {}^6C_2 + {}^5C_2 + {}^4C_2 + {}^3C_2 + {}^2C_2\right)}{{}^{64}C_2}$$

$$= \frac{18\times 8\times 7 + 4[7\times 6 + 6\times 5 + 5\times 4 + 4\times 3 + 3\times 2 + 2\times 1]}{64\times 63}$$

$$= \frac{18\times 7 + [7\times 3 + 3\times 5 + 5\times 2 + 2\times 3 + 3 + 1]}{8\times 63} = \frac{18\times 7 + [7\times 3 + 3\times 7 + 14]}{8\times 63} = \frac{18+6+2}{8\times 9} = \frac{26}{72} = \frac{13}{36}.$$

Hence the probability that the two queen to not take on each other is $P(A) = 1 - \frac{13}{36} = \frac{23}{36}$.

77.(3) Let p and q denote probability of things going to man and woman respectively.

Therefore $p = \frac{1}{1+\mu}$ and $q = \frac{\mu}{1+\mu}$.

Probability of men receiving r things is given by $P_r = {}^{\alpha}C_r . q^{\alpha - r} . p^r$.

So required probability is given by $P_1 + P_3 + P_5 + \ldots\ldots$ i.e. ${}^{\alpha}C_1 q^{\alpha-1} p + {}^{\alpha}C_3 q^{\alpha-3} p^3 + {}^{\alpha}C_5 q^{\alpha-5} p^5 + \ldots\ldots$

$$= \frac{1}{2}\left[(q+p)^{\alpha} - (q-p)^{\alpha}\right] = \frac{1}{2}\left[1 - \left(\frac{\mu-1}{\mu+1}\right)^{\alpha}\right] = \frac{1}{2} - \frac{1}{2}\left(\frac{\mu-1}{\mu+1}\right)^{\alpha}.$$

By comparison, we have $\left(\frac{\mu-1}{\mu+1}\right) = \frac{1}{2} \Rightarrow 2\mu - 2 = \mu + 1$. Thus $\mu = 3$.

78.(1.44) Total number of outcomes = 6^6 . Number of ways of choosing 4 other different numbers is ${}^6C_2$ and choosing 2 out of remaining 4 can be lone in ${}^4C_2$ ways. Also, number of ways of arranging 6 numbers of which 2 are alike and 2 are alike is $\frac{6!}{212!}$

$$\therefore \quad \text{Required probability} = \frac{{}^6C_2 \times {}^4C_2 \times \frac{6}{2!2!}}{6^6} = \frac{26}{72}$$

79.(2.27) Probability of getting a six is 1/6. If Sanchita starts the same then the probability that she wins is

$$\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{1}{6}\right)+\ldots\infty=\frac{\frac{1}{6}}{1-\frac{25}{56}}=\frac{6}{11}$$ and if Raj starts the game then probability that Sanchita, wins the game is

$1-6/11=5/11$

80.(0.40) Probability of getting 5 is 4/36, = 1/9, Probability of getting 7 is 6/36 = 1/6 probability that neither 5 nor 7 will appear is $1-10/36=13/18$

Required probability is $$\left(\frac{1}{9}\right)+\left(\frac{13}{18}\right)\left(\frac{1}{9}\right)+\left(\frac{13}{18}\right)\left(\frac{13}{18}\right)\left(\frac{1}{9}\right)+\ldots=(1/9)\left\{\frac{1}{1-\frac{13}{18}}\right\}=\frac{2}{5}=0.40$$

81.(0.75) He will fail in exam in two cases:
Case (i) He studied and failed, probability of this case is (1/3)(1/2) = 1/6
Case (ii) He didn't studied and failed, probability of this case is (2/3)(3/4) = 1/2
So total probability is $1/6+1/2=4/6=2/3$
Then required probability = (1/2/(2/3) = 3/4

82.(0.45) Claim: on a rod of length $a+b+c$, lengths a, b are measured at random. The probability that no point of the measured lines will coincide is $\dfrac{c^2}{(a+c)\,(b+c)}$.

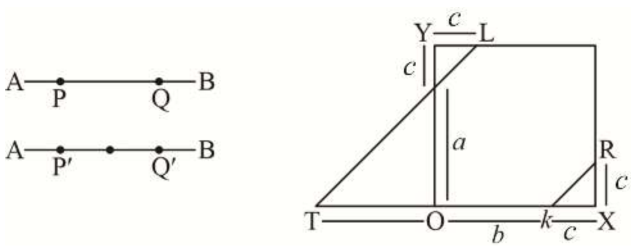
Let AB be the line, and suppose $AP = x$ and $PQ = a$; also let a be measured from P towards B, so that x must be less than $b + c$. Again let $AP' = y$, $P'Q' = b$, and suppose $P'Q'$ measured from P' towards B, then y must be less than $a + c$. Now in favorable case we must have $AP' > AQ$ or else $AP > AQ'$

Hence $y > a + x$ or $x > b + y$ … (i)

We must have $x > 0$ and $< b+c$

$y > 0$ and $a+c$ … (ii)

Take a pair of rectangular axes and make OX equal to $b + c$, and OY equal to $a + c$.
Draw the line $y = a + x$, respectively by TML in the figure; and $x = b + y$ represented by kR.



Then YM, KX are each equal to c, OM, OT are equal to a. The conditions (i) are only satisfied by points in the triangles MYL and KXR, while condition (2) are satisfied by any points within the rectangle OX, OY satisfied by any points within the rectangle OX, OY required probability $=\dfrac{c^2}{(a+c)\,(b+c)}$.

83.(0.9523) Here sample space is ${}^{10}C_6=(10!)/(6!)(4!)=(7\times8\times9\times10)/(24)=210$

Required answer is 200/210 = 20/21 = 0.9523

84.(1.25) $E: B_1B_2B_3\times\times\times$, where $\times$ means B or G $\quad\therefore\quad P(E)=\frac{1}{2}.\frac{1}{2}.\frac{1}{2}=\frac{1}{8}=\frac{8}{64}$

85.(0.0130) In a throw of two dice probability that one will get 12 is 1/36 and will not get 12 is 35/36. As per the given condition in 1st 19 throw outcomes is not 12 and 20th throw outcome is 12 so required probability is $\left(\frac{35}{36}\right)^{19}\left(\frac{1}{36}\right)=0.0130$

86.(0.299) Required probability is $\left({}^{4}C_1\right)\left({}^{48}C_4\right)/\left({}^{52}C_5\right)=0.299$ (approx).

87.(0.1054) $\dfrac{\dfrac{(48!)}{(12!)^4}}{\dfrac{(52!)}{(13!)^4 4!}}=0.1054$

88.(0.422) Total number of ways of selecting 2 cards (6×13) and that for 3rd, 4th and 5th card is 48, 44 and 40 ways, so total number of favorable cases is $(78\times 48\times 44\times 40)$. Sample space is ${}^{52}C_5=2,598,560$

Required probability = $\frac{78\times 48\times 44\times 40}{2598560}=0.422$ (approx).

89.(0.018181) $=\frac{1}{55}$

90.(0.142857) Required probability is 720/(7!) = 1/7

0.142857

91.(1.35) This question is based on principle of inclusion and exclusion

Let X, Y and Z be the events that the student passes in Maths, Physics, Chemistry.

$P(X)=m$, $P(Y)=p$ and $P(Z)=c$ and P(passing in at least one) = $P(X\cup B\cup C)=0.75$ [given]

Now, $1-P(X'\cap Y'\cap Z')=0.75$, $P(X)=1-P(X')$ and $P(X\cup B\cup C)'=P(X\cap Y\cap Z)$

$\Rightarrow\quad 1-P(X')P(Y')P(Z')=0.75$

X, Y and Z are independent event therefore X', Y' and Z' are also independent.

$1-(1-m)(1-p)(1-c)=0.75 \quad\Rightarrow\quad (1-m)(1-p)(1-c)=0.25$ $\quad\ldots$(i)

Also P(passing exactly in one subject) = 0.4 $\Rightarrow\quad P(X\cap Y\cap Z'\cup X\cap Y'\cap Z\cup X'\cap Y\cap Z)=0.4$

$\Rightarrow\quad P(X\cap Y\cap Z')\cup P(X\cap Y'\cap Z)\cup P(X'\cap Y\cap Z)=0.4$

$pm-pmc+pc-pmc+mc-pmc=0.4$ $\quad\ldots$(ii)

Again P(passing at least in two subjects) = 0.5

$\Rightarrow\quad P(X\cap Y\cap Z')\cup P(X\cap Y'\cap Z)\cup P(X'\cap Y\cap Z)\cup P(X\cap Y\cap Z)=0.5$

$\Rightarrow\quad (pm+pc+mc)-pcm=0.5$ $\quad\ldots$(iii)

From (ii) we get $(pm+pc+mc)-3pcm=0.4$ $\quad\ldots$(iv)

From (i) we get

$1-(m+p+c)+(pm+pc+cm)-pcm=0.25$ $\quad\ldots$(v)

Now from (iii), (iv) and (v) we get,

$p+m+c=1.35=27/20$ and pmc = 1/100.10

92.(0.1142) As $P_1, P_2, \ldots P_8$ (the eight players) the number of ways they can be paired in four pairs

$= 1/4!\left[\left({}^8C_2\right)\left({}^8C_2\right)\left({}^4C_2\right)\left({}^2C_2\right)\right]$ ways = 105 ways.

Now, at least two players certainly reach the second round in between P_1, P_2 and P_3, P_4 can reach in the final, if exactly two players play against each other in between P_1, P_2 and P_3 and remaining player will play against one of players from P_5, P_6, P_7, P_8 and P_4 plays against one of the remaining three from P_5 to P_8. Now this can be possible in ${}^3C_2 \times {}^4C_1 \times {}^2C_1 = 36$ ways. Required probability = 4/35

93.(1.33) $P\left(\left(\dfrac{A^C}{B^C}\right)^C\right) = 1 - P\left(\dfrac{A^C}{B^C}\right) = 1 - \dfrac{P\left(A^C \cap B^C\right)}{P\left(B^C\right)} = 1 - \dfrac{1 - P(A \cup B)}{1 - P(B)} = 1 - \dfrac{1 - P(A) - P(B) + P(A \cap B)}{1 - P(B)}$

$= 1 - \dfrac{0.65}{0.75} = 1 - \dfrac{13}{15} = \dfrac{2}{15}$

94.(0.20) Let C = the event of the selected number being composite,

E = the event of the there being no remainder.

$P(C) = \dfrac{n(C)}{n(S)} = \dfrac{15}{25} = \dfrac{3}{5},\ P(\bar{C}) = \dfrac{2}{5}$

$P(E/C) = \dfrac{4}{15},\ P(E/\bar{C}) = \dfrac{1}{10}$

$P(E) = P(C).\ P(E/C) + P(\bar{C}).P(E/\bar{C}) = \dfrac{3}{5}.\dfrac{4}{15} + \dfrac{2}{5}.\dfrac{1}{10} = \dfrac{4}{25} + \dfrac{1}{25} = \dfrac{1}{5} = 0.2$

95.(0.3488) Of the five bags, 3 belong to the first group and 2 to the second group. Hence $p_1 = \dfrac{3}{5},\ p_2 = \dfrac{2}{5}$

If a bag is selected from the first group, the chance of drawing a black ball is $\dfrac{2}{7}$. If from the second group, the chance is $\dfrac{4}{5}$. Thus $p, = \dfrac{2}{7},\ p_2 = \dfrac{4}{5}$, $p_1p_1 = \dfrac{6}{35},\ p_2p_2 = \dfrac{8}{25}$.

Hence the chances that the black ball came from one of the first group is $\dfrac{6}{35} - \left(\dfrac{6}{35} + \dfrac{8}{25}\right) = \dfrac{15}{43}$.

96.(0.9722) There are two hypotheses, (i) their coincident testimony is true, (ii) it is false.

Hence, $p_1 = \dfrac{1}{6},\ p_2 = \dfrac{5}{6}$, $p_1 = \dfrac{3}{4} \times \dfrac{7}{10},\ p_2 = \dfrac{1}{25} \times \dfrac{1}{4} \times \dfrac{3}{10}$.

For in estimating p_2 we must take into account the chance that As B will both select the while ball when it has not been drawn; this choice is $\dfrac{1}{5} \times \dfrac{1}{5}$ or $\dfrac{1}{25}$.

Now the probabilities of the two hypotheses are as 35:1; thus probability $= \dfrac{35}{36}$.

97.(4.17) The number of ways of drawing 7 balls (second drawn) = ${}^{10}C_7$. For each set of 7 balls of the second draw 3 must be common to the set of 5 balls of the first draw, i.e., 2 other balls can be drawn in ${}^3C_2$ ways. Thus, of making the first draw so that there are 3 balls common. Hence, the probability of having 3 balls in common = $\dfrac{{}^7C_3 \times {}^3C_2}{{}^{10}C_7} = \dfrac{5}{12}$

MATRICES & DETERMINANTS

1.(C) $A=\begin{pmatrix}1 & -1\\ 1 & -2\end{pmatrix}$ satisfies $A^2+A-I=0$

$A^2=I-A \quad \Rightarrow \quad A^3=A-A^2=2A-I$

$A^4=2A^2-A=2I-2A-A=2I-3A \quad \Rightarrow \quad A^5=2A-3A^2=2A-3I+3A=5A-3I$

$A^6=5A^2-3A=5I-5A-3A=5I-8A \Rightarrow n=6$

2.(D) $a_{rs}=(r-s)2^{i(r-s)} \quad \Rightarrow \quad A=\begin{bmatrix}0 & -2^{-i} & -2.2^{-2i} & -3.2^{-3i}\ldots\ldots\\ 2^{i} & 0 & -2^{-i} & -2.2^{-2i}\ldots\ldots\\ 2.2^{2i} & 2^{i} & 0 & \ldots\ldots\ldots\\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots\end{bmatrix}$

$\overline{A}=\left[(r-s)2^{-i(r-s)}\right] \Rightarrow A=-(\overline{A})^T$

3.(C) A_1 is cofactor matrix of $A \quad \Rightarrow A_1=(Adj(A))' \Rightarrow |A_1|=|Adj(A)|$

A_2 is cofactor matrix of $A_1 \quad \Rightarrow |A_2|=|Adj(A_1)|=|Adj(Adj(A))|$

$\Rightarrow |A_n|=\left|\underbrace{Adj(Adj(\ldots\ldots(Adj(A))\ldots\ldots)}_{n\text{ times}}\right|=K=|A|^{(n-1)^n} \Rightarrow |A|=K^{\frac{1}{(n-1)^n}}$

4.(B) $A^2=A \;\;\&\;\; B=I-A$

$\Rightarrow \quad AB=A(I-A)=A-A^2=A-A=0$ and $BA=(I-A)A=A-A^2=A-A=0$

$B^2=(I-A).(I-A)=I-A-A+A^2=I-A=B$

5.(C) Trace $(A)=\sum_{i=1}^{n}a_{ii}=\sum_{i=1}^{n}(-1)^i({}^nC_i)^2=-({}^nC_1)^2+({}^nC_2)^2-({}^nC_3)^2+({}^nC_4)^2+\ldots+(-1)^n({}^nC_n)^2=-1$ (as n is odd)

6.(A) $A=\begin{pmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{pmatrix}$; $B=\begin{pmatrix}a_{11} & \frac{1}{3}a_{12} & \frac{1}{9}a_{13}\\ 3a_{21} & a_{22} & \frac{1}{3}a_{23}\\ 9a_{31} & 3a_{32} & a_{33}\end{pmatrix} \Rightarrow |B|=|A|$ Similarly $|C|=|B|$

7.(D) $\dfrac{f(x)}{x}=\begin{vmatrix}x & 1+x^2 & x^2\\ \ln(1+x^2) & e^x & \dfrac{\sin x}{x}\\ \cos x & \tan x & \dfrac{\sin^2 x}{x}\end{vmatrix}$; $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=\begin{vmatrix}0 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 0\end{vmatrix}=1$

8.(D) $ax+4y+z=0$; $2y+3z=1$; $3x-bz=-2$

$|A|=\begin{vmatrix}a & 4 & 1\\ 0 & 2 & 3\\ 3 & 0 & -b\end{vmatrix}=a(-2b)+3(10)=-2ab+30$

$\Rightarrow$ Unique solution if $ab \neq 15$

$$adj.(A).B = \begin{pmatrix} -2b & 4b & 10 \\ 9 & -ab-3 & -3a \\ -6 & 12 & 2a \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 4b-20 \\ -ab-3+6a \\ 12-4a \end{pmatrix}$$

Infinitely many solutions if $ab = 15,\ b = 5,\ a = 3$

No solution if $ab = 15,\ a \neq 3$ or $b \neq 5$

9.(A) $\begin{vmatrix} 1 & 2\omega & 3\omega^2 \\ 2 & 3\omega & \omega^2 \\ 3 & \omega & 2\omega^2 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 5 - 2(4-3) + 3(2-9) = 5 - 2 - 21 = -18$

10.(C) 3×3 symmetric matrix

$\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$, let all diagonal elements be one

$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

Diagonal two zero's, one 1

$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ or $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ or $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

By permuting diagonal elements, will get 6 more such matrices

Hence Total 12 symmetric matrices, of these nonsingular matrices are

$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ or $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ or $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ or $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ Hence probability = $\frac{1}{2}$

11.(C) $f(x) = \begin{vmatrix} x\cos x & 2x\sin x & x\tan x \\ 1 & x & 1 \\ 1 & 2x & 1 \end{vmatrix}$, $\frac{f(x)}{x^2} = \begin{vmatrix} \cos x & \frac{2\sin x}{x} & \tan x \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix}$ $\lim\limits_{x \to 0} \frac{f(x)}{x^2} = \begin{vmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = -1$

12.(A) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3 = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 8$

As $a+b+c=-2$; $ab+bc+ca=0$; $abc=-1$; $a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+bc+ca) = 4-2(0) = 4$

13.(A) $A^2 = 2A - I \Rightarrow A^3 = 2A^2 - A = 4A - 2I - A = 3A - 2I$

$A^4 = 3A^2 - 2A = 6A - 3I - 2A = 4A - 3I$; $A^n = nA - (n-1)I$

14.(C) $\begin{vmatrix} \lambda & \lambda+1 & \lambda-1 \\ \lambda+1 & \lambda & \lambda+2 \\ \lambda-1 & \lambda+2 & \lambda \end{vmatrix} = 0 = \begin{vmatrix} \lambda & \lambda+1 & \lambda-1 \\ 1 & -1 & 3 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} \lambda & \lambda+1 & \lambda-1 \\ 1 & -1 & 3 \\ 0 & 0 & 4 \end{vmatrix} = 4(-\lambda-\lambda-1) = 0 \Rightarrow \lambda = -\dfrac{1}{2}$

15.(A) $\begin{vmatrix} \cos X & \sin X & 0 \\ \cos Y & \sin Y & 0 \\ \cos Z & \sin Z & 0 \end{vmatrix} . \begin{vmatrix} \cos L & \cos M & \cos N \\ \sin L & \sin M & \sin N \\ 0 & 0 & 0 \end{vmatrix} = 0$

16.(B) $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $P = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, $Q = p^T AP$

$P\,Q^{2014} P^T = P(P^T AP)(P^T AP)\text{.........}(P^T AP)P^T = A^{2014}$

Note $P\,P^T = I$

$A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$; $A^3 = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} \Rightarrow A^{2014} = \begin{pmatrix} 1 & 4028 \\ 0 & 1 \end{pmatrix}$

17.(C) $adj(adjA) = |A|^{n-2} A = |A|A$

18.(D) $\left|(xA)^{-1}\right| = \dfrac{1}{|xA|} = \dfrac{1}{x^3 K}$

19.(C) $A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \\ 1 & 5 & 10 \end{pmatrix}$ $\qquad |A| = 0$

$Adj(A).(B) = \begin{pmatrix} 10 & -15 & 5 \\ -6 & 9 & -3 \\ 2 & -3 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ K \\ K^2 \end{pmatrix} \Rightarrow K^2 - 3K + 2 = 0 \Rightarrow K = 1 \text{ or } 2$

20.(C) **21.(B)** **22.(C)**

$A = \begin{bmatrix} 4 & 1 \\ -9 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix}$

$\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix}^2 = \begin{bmatrix} 3 & -1 \\ -9 & -3 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow A^{100} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 100\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} = \begin{bmatrix} 301 & 100 \\ -900 & -299 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = B$

$\Rightarrow \quad a + d = 2$

$\dfrac{c}{b} = -\dfrac{900}{100} = -9$; $\qquad A^{100}$ satisfies $B^2 - 2B + I = 0$ i.e. $A^{200} - 2A^{100} + I = 0$

23.(A) Characteristic equation of A is given by $\begin{vmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 4 \\ -1 & -1 & -2-\lambda \end{vmatrix} = 0$

$\Rightarrow \lambda^3 - 3\lambda^2 - \lambda + 3 = 0 \quad \Rightarrow \lambda^2(\lambda-3) - (\lambda-3) = 0 \quad \Rightarrow \lambda = \pm 1, 3$

24.(C) Matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ have eigen values 1, 1

25.(B) Matrix A satisfies its characteristic equation given by $|A-\lambda I|=0$

i.e. $A^3-5A^2+11A-7I=0$

$A^7-4A^6+6A^5=(A^7-5A^6+11A^5-7A^4)+(A^6-5A^5+11A^4-7A^3)-4A^4+7A^3=-4A^4+7A^3$

$=-4A^4+20A^3-44A^2+28A-13A^3+44A^2-28A$

$=-13A^3+65A^2-143A+91I-21A^2+115A-91I=-21A^2+115A-91I \Rightarrow \alpha=-21,\ \beta=115,\ \gamma=-91$

26.(BCD) $A^5=B^5$ & $A^4B=B^4A$, $A\neq B$

$A^5-A^4B=B^5-B^4A \Rightarrow A^4(A-B)=B^4(B-A)$

$(A^4+B^4)(A-B)=0$ & $A^5+A^4B=B^5+B^4A \Rightarrow A^4(A+B)=B^4(B+A) \Rightarrow (A^4-B^4).(A+B)=0$

Now if $|A^4+B^4|\neq 0$ then $(A^4+B^4)^{-1}$ exist hence

$A-B=0 \Rightarrow A=B$ (contradicts $A\neq B$) Hence $|A^4+B^4|=0$

27.(ABC) $X^n-Y^n-X^{n+1}+Y^{n+1}=0$

$X^{n+1}-XY^n-X^{n+2}+XY^{n+1}=0$ **;** $X^nY-Y^{n+1}-X^{n+1}Y+Y^{n+2}=0$

$X^{n+1}-XY^n-X^{n+2}+XY^{n+1}+X^nY-Y^{n+1}-X^{n+1}Y+Y^{n+2}=0$

$\Rightarrow XY^n-XY^{n+1}-X^nY+X^{n+1}Y=0 \Rightarrow (X^n-Y^n)(I-X)(I-Y)=0$

As X^n-Y^n is invertible $\Rightarrow (I-X)(I-Y)=0$. If $|I-X|\neq 0$ then $Y=I$ contradiction $\therefore |I-X|=|I-Y|=0$

28.(AB) $YZ=\begin{bmatrix}4&5\\3&4\end{bmatrix}\begin{bmatrix}4&-5\\-3&4\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I$; $(YZ)^2=I^2=\begin{bmatrix}1&0\\0&1\end{bmatrix}.\begin{bmatrix}1&0\\0&1\end{bmatrix}=I \Rightarrow (YZ)^k=I$

$tr(X(YZ)^k)=tr(X)=3 \Rightarrow \sum_{k=0}^{\infty}\frac{tr(X(YZ)^k)}{2^k}=3\sum_{k=0}^{\infty}\frac{1}{2^k}=\frac{3}{1-\frac{1}{2}}=6$ and $\sum_{k=1}^{\infty}\frac{tr(X(YZ)^k)}{2^k}=\frac{3\times 1/2}{1-1/2}=3$

29.(CD) $Adj(A)=\begin{pmatrix}1&4&4\\2&1&7\\1&1&3\end{pmatrix}$; $|Adj(A)|=\begin{vmatrix}1&4&4\\2&1&7\\1&1&3\end{vmatrix}=-4-4(6-7)+4(2-1)=4=|A|^2 \Rightarrow |A|=\pm 2$

30.(AD) In $f(x)$ $C_2\to C_2-C_1-4C_3$

$C_2=0 \Rightarrow f(4)=0$

In $g(x)$, $C_2\to C_2-C_3$

$g(x)=\begin{vmatrix}2x-2&0&x-1\\3x-4&x-2&x-1\\3x-5&0&2x-4\end{vmatrix}=(x-2)(4x^2-8x-4x+8-3x^2+3x+5x-5)$

$=(x-2)(x^2-4x+3)=(x-2)(x-1)(x-3)$

$f(x)=g(x) \Rightarrow x=1,2,3$

31.(AC) $C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} x^2-3 & 2 & x^2-12 \\ x^2-3 & x^2-12 & 3 \\ x^2-3 & 2 & 7 \end{vmatrix} = (x^2-3)\begin{vmatrix} 1 & 2 & x^2-12 \\ 1 & x^2-12 & 3 \\ 1 & 2 & 7 \end{vmatrix}$$

$$R_3 \to R_3 - R_1 = (x^2-3)\begin{vmatrix} 1 & 2 & x^2-12 \\ 1 & x^2-12 & 3 \\ 0 & 0 & 19-x^2 \end{vmatrix} = (x^2-3)(19-x^2)(x^2-14)$$

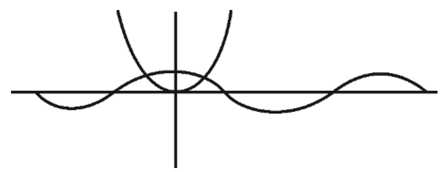
$f(x)=0$ has 6 real roots given by $\pm\sqrt{3}, \pm\sqrt{14}, \pm\sqrt{19}$

32.(ABC) $X \odot (Y+Z) = X \odot Y + X \odot Z$

33.(CD) $Adj(AB) = Adj(B).Adj(A) = |B|B^{-1}.|A|A^{-1} = |A||B|B^{-1}.A^{-1} = |A||B|(AB)^{-1}$

34.(AB) $f(0)=0 \Rightarrow (a)$; $f'(0)=0 \Rightarrow (b)$

35.(BCD) $\begin{vmatrix} x^2+\sin x\cos x & x(1+\sin x) \\ x+\cos x & x+1 \end{vmatrix} = 0 \Rightarrow x^3 + x\sin x\cos x + x^2 + \sin x\cos x - x^2 - x\cos x - x^2\sin x - x\sin x\cos x = 0$



$\Rightarrow x(x^2-\cos x) - \sin x(x^2-\cos x) = 0 \Rightarrow x = \sin x$ or $x^2 = \cos x$

$\Rightarrow x = 0$ or $\alpha, -\alpha$ st. $\alpha^2 = \cos\alpha$

$\cos\frac{\pi}{6} > \left(\frac{\pi}{6}\right)^2$; $\cos\frac{\pi}{4} > \left(\frac{\pi}{4}\right)^2$ & $\cos\left(\frac{\pi}{3}\right) < \left(\frac{\pi}{3}\right)^2$

36.(AB) $M_1 = n^2 - n$, $M_2 = \frac{n^2-n}{2} \Rightarrow \frac{M_1}{M_2} = 2$; $M_1 = 2\sum_{k=1}^{n-1} k$

37.(AB) $a_{ij} = i^n - j^n = -(j^n - i^n) = -a_{ji} \Rightarrow$ A is skew symmetric matrix

38.(ACD) $f(x) = \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix}$; $f(-x) = -f(x) \Rightarrow f(x)$ is an odd function

$\ln\left(\frac{a-x}{a+x}\right)$ is odd function

39.(ABCD) $A^m B^n = B^n A^m$

$(A-\lambda I).(B+\mu I) = (B+\mu I).(A-\lambda I)$; $(A+\lambda I).(\mu I - B) = (\mu I - B).(A+\lambda I)$

$(A+B)^n = A^n + {}^nC_1 A^{n-1}B + ... + B^n$

40.(ABC) $adj(adj\ A) = |A|A = -A$

$|adj(AB)| = |AB|^2 = |A|^2|B|^2 = 576$; $|adj(adj(adj(adj(A))))| = (-1)^{2^4} = 1$; $|adj(B^{-1})| = |B^{-1}|^2 = \frac{1}{|B|^2} = \frac{1}{576}$

41.(ABD) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3 \neq 0$ for only trivial solution

$\Rightarrow a \neq b$ or $b \neq c$ or $c \neq a$ as $a, b, c \in R^+ \Rightarrow \dfrac{a^3 + b^3 + c^3}{3} > abc \Rightarrow a^3 + b^3 + c^3 > 3abc$

For non trivial solution $a = b = c$

For $a = b = c$ all 3 equations becomes $x + y + z = 0$

Distance of this plane from (1, 2, 3) is $\dfrac{1+2+3}{\sqrt{1+1+1}} = \dfrac{6}{\sqrt{3}}$

Hence minimum value of $(x-1)^2 + (y-2)^2 + (z-3)^2 = \dfrac{36}{3} = 12$

42.(ABCD) For each place we have 3 possibilities hence $O(A) = 3^9$

$\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$ for $a + b + c = 0,\ a, b, c \in \{0, 1, -1\}$

Let $a = b = c = 0 \Rightarrow 3^3$ such matrices

Let $a = 1, b = -1,\ c = 0 \Rightarrow 3^3$ such matrices

$\Rightarrow 6 \times 3^3$ such matrices (by permuting a, b, c) $\Rightarrow 7 \times 3^3$ matrices with trace zero

$\begin{pmatrix} 0 & d & e \\ -d & 0 & f \\ -e & -f & 0 \end{pmatrix}$ skew symmetric matrices = 3^3

Number of matrices in A such that each of 0,1,–1 occurs at least once

$$= 3^9 - {}^3C_1 \times 2^9 + {}^3C_2 \times 1^9 = 19683 - 3 \times 512 + 3 = 18150$$

Skew symmetric matrix of odd order has determinant zero.

43.(ABC) For symmetric $x = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \Rightarrow 5^2$ matrices

For skew symmetric $\begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}$ i.e. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ only possibility, already included in symmetric matrices.

$|X| = a^2 - b^2 = (a+b)(a-b),\ a, b \in \{0, 1, 2, 3, 4\}$

For it to be divisible by 5 either $a - b = 0$ *i.e.* $a = b$

$\begin{pmatrix} a & a \\ a & a \end{pmatrix}$ i.e. 5 matrices or $a + b = 5$ *i.e.* $\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$ or $\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$ or $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ or $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

Trace of $\begin{pmatrix} a & b \\ c & a \end{pmatrix}$ is $2a$, not divisible by 5 $\Rightarrow a \neq 0$

Determinant is $a^2 - bc$ for it to be divisible by 5 $b \neq 0, c \neq 0 \Rightarrow a,b,c \in \{1,2,3,4\}$

For any set of values of $a,b \in \{1,2,3,4\}$ we get a unique value of c $\Rightarrow 16$ such matrices.

$$\begin{pmatrix}1&1\\1&1\end{pmatrix},\begin{pmatrix}2&2\\2&2\end{pmatrix},\begin{pmatrix}3&3\\3&3\end{pmatrix},\begin{pmatrix}4&4\\4&4\end{pmatrix},\begin{pmatrix}1&4\\4&1\end{pmatrix},\begin{pmatrix}2&3\\3&2\end{pmatrix},\begin{pmatrix}3&2\\2&3\end{pmatrix},\begin{pmatrix}4&1\\1&4\end{pmatrix}$$

$$\begin{pmatrix}1&2\\3&1\end{pmatrix},\begin{pmatrix}1&3\\2&1\end{pmatrix},\begin{pmatrix}2&4\\1&2\end{pmatrix},\begin{pmatrix}4&2\\3&4\end{pmatrix},\begin{pmatrix}2&1\\4&2\end{pmatrix},\begin{pmatrix}4&3\\2&4\end{pmatrix},\begin{pmatrix}3&1\\4&3\end{pmatrix},\begin{pmatrix}3&4\\1&3\end{pmatrix}$$

Number of matrices in A whose determinant is not divisible by 5 = total – those which have determinant divisible by 5 $= 5^3 - (16+9) = 100$

9 comprises of $a=0,\ b=0,\ c\neq 0$ i.e. 4 ; $a=0, b\neq 0, c=0$ *i.e.* 4 ; $a=0, b=0, c=0$ *i.e.* 1

44.(AB) $x^a y^b = e^m$

$$x^c y^d = e^n \ ; \ \frac{x^{ac}y^{bc}}{x^{ca}y^{da}} = \frac{e^{mc}}{e^{na}} \Rightarrow y^{bc-da} = e^{mc-na} \Rightarrow y^R = e^Q \Rightarrow y = e^{Q/R} \Rightarrow R = Q\log_y e$$

$$\frac{x^{ad}y^{bd}}{x^{cb}y^{db}} = \frac{e^{md}}{e^{nb}} \Rightarrow x^{ad-bc} = e^{md-bn} \Rightarrow x^R = e^P \Rightarrow x = e^{P/R} \Rightarrow R = P\log_x e \ ; \ P\log_x e = Q\log_y e$$

45.(ACD) $AB = I \Rightarrow B = A^{-1} = \frac{1}{2}\begin{pmatrix}7&9&-1\\-19&-23&3\\2&2&0\end{pmatrix}$; $|Adj(B)| = |B|^2 = \frac{1}{|A|^2} = \frac{1}{4}$

46.(AC) $f(x) = \begin{vmatrix}n&n+1&n+2\\n!&(n+1)!&(n+2)!\\1&1&1\end{vmatrix} = \begin{vmatrix}n&1&1\\n!&n.n!&(n+1)(n+1)!\\1&6&0\end{vmatrix} = (n+1)(n+1)! - nn! = n!\left((n+1)^2-n\right) = n!\left(n^2+n+1\right)$

47.(ABC) $|A(\theta)| = 1$ Hence $A(\theta)$ is invertible

$$A(\pi+\theta) = \begin{bmatrix}\sin(\pi+\theta) & i\cos(\pi+\theta)\\ i\cos(\pi+\theta) & \sin(\pi+\theta)\end{bmatrix} \ ; \ = \begin{bmatrix}-\sin\theta & -i\cos\theta\\ -i\cos\theta & -\sin\theta\end{bmatrix} = -A(\theta)$$

$$\text{adj } A(\theta) = \begin{bmatrix}\sin\theta & -i\cos\theta\\ -i\cos\theta & \sin\theta\end{bmatrix} \ ; \ A(\theta)^{-1} = \begin{bmatrix}\sin\theta & -i\cos\theta\\ -i\cos\theta & \sin\theta\end{bmatrix} = A(\pi-\theta)$$

48.(BD) Applying $C_1 \to c_1 - (\cot\phi)C_2$. we get

$$\Delta = \begin{vmatrix}0 & \sin\theta\sin\phi & \cos\theta\\ 0 & \cos\theta\sin\phi & -\sin\theta\\ -\frac{\sin\theta}{\sin\phi} & \sin\theta\cos\phi & 0\end{vmatrix} = -\frac{\sin\theta}{\sin\phi}\left[-\sin\phi\sin^2\theta - \cos^2\theta\sin\phi\right]$$

Expanding along $C_1 = \sin\theta$; Which is independent of ϕ, also

$$\frac{d\Delta}{d\theta} = \cos\theta \Rightarrow \left[\frac{d\Delta}{d\theta}\right]_{\theta=\frac{\pi}{2}} = \cos\frac{N}{2} = 0$$

49.(AB) Let $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Since $\Delta^c = \Delta^2$, where Δ^c is determinant of cofactors of Δ

$$\Rightarrow \quad \begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$$

$$\Rightarrow \quad \begin{vmatrix} a^2+b^2+c^2 & ab+bc+ca & ac+ba+bc \\ ab+bc+ca & b^2+c^2+a^2 & bc+ac+ab \\ ca+ab+bc & bc+ca+ab & c^2+a^2+b^2 \end{vmatrix} = \begin{vmatrix} \alpha^2 & \beta^2 & \beta^2 \\ \beta^2 & \alpha^2 & \beta^2 \\ \beta^2 & \beta^2 & \alpha^2 \end{vmatrix} \text{ where } \alpha^2 = a^2+b^2+c^2, \ \beta^2 = ab+bc+ca$$

50.(BCD) $ax + 2y = \lambda$

$3x - 2y = \mu$

(A) $a = -3$ gives ; $\lambda = -\mu$ or $\lambda + \mu = 0$ not for all λ, μ

(B) $a \neq -3 \Rightarrow \Delta \neq 0$ where $\Delta = \begin{vmatrix} a & 2 \\ 3 & -2 \end{vmatrix} = -2a - 6$ $\quad \therefore \quad$ (B) is correct

(C) correct

(D) If $\lambda + \mu \neq 0 \Rightarrow -3x + 2y = \lambda$ and $3x - 2y = \mu$ Inconsistent (D) correct

51.(AC) $\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & a+b \\ 0 & 1 & 2a+2b \end{vmatrix} = a \begin{vmatrix} 1 & a & 0 \\ 1 & 2a+b & a+b \\ 0 & 1 & 2a+3b \end{vmatrix} = a(a+b) \begin{vmatrix} 1 & a & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2a+3b \end{vmatrix} = a(a+b)(2a+3b-1)$

52.(BC) $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$

$$\begin{vmatrix} 1+\alpha^2+2\alpha & 1+4\alpha^2+4\alpha & 1+9\alpha^2+6\alpha \\ 4+\alpha^2+4\alpha & 4+4\alpha^2+8\alpha & 4+9\alpha^2+12\alpha \\ 9+\alpha^2+6\alpha & 9+4\alpha^2+12\alpha & 9+9\alpha^2+18\alpha \end{vmatrix} = -684\alpha$$

$$R_2 \to R_2 - R_1 ; \ R_2 \to R_3 - R_2$$

$$\begin{vmatrix} 1+\alpha^2+2\alpha & 1+4\alpha^2+4\alpha & 1+4\alpha^2+6\alpha \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix} = -648\alpha \ ; \ \begin{vmatrix} \alpha^2-4 & 4\alpha^2-4 & 9\alpha^2-4 \\ -2 & -2 & -2 \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix} = -648\alpha$$

$$-2\begin{vmatrix} \alpha^2-4 & 4\alpha^2-4 & 9\alpha^2-4 \\ 1 & 1 & 1 \\ 5+2\alpha & 2\alpha & 4\alpha \end{vmatrix} = -648\alpha \ ; \ -2\begin{vmatrix} \alpha^2-4 & 3\alpha^2 & 8\alpha^2 \\ 1 & 0 & 0 \\ 5+2\alpha & 2\alpha & 4\alpha \end{vmatrix} = -648\alpha$$

$$\Rightarrow \quad 2\left(12\alpha^3 - 16\alpha^3\right) = -648\alpha \Rightarrow 2\left(-4\alpha^3\right) = -648\alpha \Rightarrow 8\alpha^3 = 648\alpha \Rightarrow \alpha\left(\alpha^2 - 81\right) = 0 \ ; \ \alpha = 0, 9, -9$$

53.(AC) A is an orthogonal matrix

$\therefore \quad AA^T = I$

or $\frac{1}{3}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}, \frac{1}{3}\begin{bmatrix}1 & 2 & a\\ 2 & 1 & 2\\ 2 & -2 & b\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ or $\frac{1}{9}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}\begin{bmatrix}1 & 2 & a\\ 2 & 1 & 2\\ 2 & -2 & b\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

$\Rightarrow \begin{bmatrix}9 & 0 & a+4+2b\\ 0 & 9 & 2a+2-2b\\ a+4+2b & 2a+b-2b & a^2+4+b^2\end{bmatrix} = \begin{bmatrix}9 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 9\end{bmatrix} \Rightarrow a+4+2b=0,\ 2a+2-2b=0\ \&\ a^2+4+b^2=9$

$\Rightarrow \quad a+2b+4=0,\ a-b+1=0\ \ a^2+b^2=5 \quad \Rightarrow \quad a=-2,\ b=-1$

54.(A) $\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}\begin{bmatrix}a & b & g\\ b & e & h\\ c & f & i\end{bmatrix}$

$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2=5$

Case-I. Five $(1's)$ and four $(0's)$

$^9C_5=126$

Case-II. One (2) and one $^9C_2.2!=72$ $\quad\therefore\quad$ total 198

55.(BCD) $n=1$

$p=\left[w^2\right];\ p^2=\left[w^4\right]\neq 0$

$n=2$

$p=\begin{bmatrix}w^2 & w^3\\ w^3 & w^4\end{bmatrix}=\begin{bmatrix}w^2 & 1\\ 1 & w\end{bmatrix};\ p^2=\begin{bmatrix}w^4+1 & ..\\ .. & ..\end{bmatrix}\neq 0$

$n=3$

$P=\begin{bmatrix}w^2 & 1 & w\\ 1 & w & w^2\\ w & w^2 & 1\end{bmatrix}\begin{bmatrix}w^2 & 1 & w\\ 1 & w & w^2\\ w & w^2 & 1\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$ Similarly $p^2\neq 0$ when n is not multiple of 3.

56.(AC) $\begin{vmatrix}1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta\\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta\\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta\end{vmatrix}=0 \quad\Rightarrow\quad \begin{vmatrix}2 & \cos^2\theta & 4\sin 4\theta\\ 2 & 1+\cos^2\theta & 4\sin 4\theta\\ 1 & \cos^2\theta & 1+4\sin 4\theta\end{vmatrix}=0$ Applying $C_1\to C_1+C_2$

$\Rightarrow \begin{vmatrix}2 & \cos^2\theta & 4\sin 4\theta\\ 0 & 1 & 0\\ -1 & 0 & 0\end{vmatrix}=0$ Applying $R_2\to R_2-R_1$, $R_3\to R_3-R_1$

$\Rightarrow \quad 2+4\sin 4\theta=0 \quad\Rightarrow\quad \sin 4\theta=-\frac{1}{2}=-\sin\frac{\pi}{6}=\sin\left(-\frac{\pi}{6}\right)$

The value of θ lying between 0 and $\frac{\pi}{2}$ are $\frac{7\pi}{24}$ and $\frac{11\pi}{24}$ for $n=1\ \&\ 2$

57.(ABC) $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$

$= \begin{vmatrix} 2 & \cos^2 x & \sin 2x \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$ apply $C_1 \to C_1 + C_2$ $= \begin{vmatrix} 2 & \cos^2 x & \sin^2 x \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$ $\begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$

$=$ $2+\sin 2x$, max $\sin 2x = 1$; $\alpha = 3, \beta = 1$, $\min \sin 2x = -1$

Now, $\alpha - \beta = 2,\ \alpha+\beta = 4,\ \alpha + 3\beta = 6$; Thus, $(\alpha-\beta)+(\alpha+\beta) = \alpha + 3\beta$ so, $\alpha-\beta,\ \alpha+\beta,\ \alpha+3\beta$ cannot for a triangle.

58.(CD) (A) $\left(Y^3Z^4 - Z^4Y^3\right)^T$

$= \left(Y^3Z^4\right)^T - \left(Z^4Y^3\right)^T = \left(Z^4\right)^T.\left(Y^3\right)^T - \left(Y^3\right)^T\left(Z^4\right)^T = -Z^4Y^3 + Y^3Z^4 = Y^3Z^4 - Z^4Y^3$ $\therefore$ symmetric

(B) $\left(X^{44}+Y^{44}\right)^T = \left(X^{44}\right)^T + \left(Y^{44}\right)^T = X^{44}+Y^{44}$ $\therefore$ symmetric

(C) $\left(X^4Z^3 - Z^3 \times 4\right)^T$

$= \left(X^4Z^3\right)^T - \left(Z^3\times 4\right)^T = \left(Z^3\right)^T\left(X^4\right)^T - \left(X^4\right)^T\left(Z^3\right)^T = -\left(X^4Z^3 - Z^3\times 4\right)$ $\therefore$ skew symmetric

(D) $\left(X^{23}+Y^{23}\right)^T = -X^{23} - Y^{23}$ $\therefore$ skew symmetric

59.(AC) $A = B^2 \Rightarrow |A| = |B|^2 =$ positive

(A) $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = 1(-1)$ = negative Matrix (a) can not be possible

(B) $\begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = 1(1-0)$ = positive Matrix (b) can be possible

Ex. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \end{vmatrix}\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix}$

(C) $\begin{vmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = -1$ = negative Matrix (c) can not be possible

(D) $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$ = positive Matrix (D) can be possible

60.(CD) $M=\begin{bmatrix} a & b \\ b & c \end{bmatrix}$

(A) $\begin{bmatrix} a \\ b \end{bmatrix}$ & [b, c] are transpose So $\begin{bmatrix} a \\ b \end{bmatrix}=\begin{bmatrix} b \\ c \end{bmatrix}$ is given $a=b=c$

$M=\begin{bmatrix} a & a \\ a & a \end{bmatrix} \Rightarrow |M|=0$

(B) [b, c] and $\begin{bmatrix} a \\ b \end{bmatrix}$ are transpose So $a=b=c$

(C) $M=\begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \Rightarrow |M|=ac\neq 0$

(D) $M=\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ given $=ac\neq\lambda^2$

61.(BD) $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}=0$

$\Rightarrow \begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ 0 & 0 & -\alpha(a\alpha+b)-b\alpha-c \end{vmatrix}=0 \quad [R_3\to R_3-\alpha R_1-R_2] \quad \Rightarrow \quad -\left(a\alpha^2+2b\alpha+c\right)\left(ac-b^2\right)=0$

So the determinant va if $ac-b^2=0$ or $a\alpha^2+2b\alpha+c=0$, i,e. if a, b, c are in G.P or α is a root of $ax^2+2bx+c=0$

62.(ABC) $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}=0$

or $(ab)^3+(bc)^3+(ca)^3-3(ab)(bc)(ca)=0$

or $\left(ab+bcw^2+caw\right)\left(abw+bcw^2+ca\right)\left(abw^2+bcw+ca\right)=0$

or $ab+bcw^2+caw=0$; $abw+bcw^2+ca=0$; $abw^2+bcw+ca=0$

$\Rightarrow \dfrac{1}{cw^2}+\dfrac{1}{a}+\dfrac{1}{b\omega}=0,\ \dfrac{1}{cw}+\dfrac{1}{a}+\dfrac{1}{bw^2}=0$; $\dfrac{1}{c}+\dfrac{1}{aw}+\dfrac{1}{bw^2}=0$

63.(AB) The number of third order determinants = the number of arrangements of nine different number is nine places = 9! Corresponding to each determinant made there is a determinant obtained by interchanging, two consecutive rows (or columns) So, the sum of this pair will be 0. $\therefore$ the sum of all determinants = 0

64.(CD) (A) $\left(N^TMN\right)^T=N^TM^TN$ is symmetric if M is symmetric and skew symmetric if M is skew- symmetric

(B) $(MN-NM)^T=(MN)^T-(NM)^T=NM-MN$

$-(MN-NM)$ is skew symmetric

(C) $(MN)^T=N^TMT=NM\neq MN$

(D) standard result is $\operatorname{adj}(MN)=\left[(\operatorname{adj}N)(\operatorname{adj}M)\right]$

$\neq(\operatorname{adj}M)(\operatorname{adj}N)$

65.(CD) For existence of non-trivial solution $\begin{vmatrix} 6 & 5 & \lambda \\ 3 & -1 & 4 \\ 1 & 2 & -3 \end{vmatrix} = 0 \Rightarrow \lambda = -5$

If $\lambda \neq -5,\ x = 0,\ y = 0,\ z = 0$ is the only solutions.

66. **[A-q, r]** **[B-p, s]** **[C-s]** **[D-r]**

A is idempotent $\Rightarrow A^2 = A$

$$\begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix};\ \begin{bmatrix} x^2 & xy+y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow \quad x^2 = x\ \&\ xy + y = y \quad \Rightarrow x = 0$ or $1\ \&\ xy = 0 \Rightarrow x = 0,\ y \in R\ or\ x = 1,\ y = 0 \quad (A) \to Q, R$

A is involuntary

$\Rightarrow \quad A^2 = I$

$$\begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix};\ \begin{bmatrix} x^2 & xy+y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow x^2 = 1 \quad \& \quad xy + y = 0 \ \Rightarrow x = 1,\ y = 0\ or\ x = -1,\ y \in R \qquad (B) \to P, S$

A is orthogonal $\Rightarrow AA' = I$

$\Rightarrow \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x & 0 \\ y & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x^2+y^2 & y \\ y & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow x^2 + y^2 = 1,\ y = 0 \ \Rightarrow x = \pm 1,\ y = 0 \qquad (C) \to S$

A is singular $\Rightarrow |A| = 0 \Rightarrow x = 0,\ y \in R \qquad (D) \to R$

67. **[A-q]** **[B-r, s]** **[C-p]** **[D-p]**

$$\frac{f(x)}{x} = \begin{vmatrix} (\alpha x+1)\cos^2 x & 1 & 1-x \\ \beta \sin x & x & 2x \\ (\gamma x^2+1)\tan x & 1 & 1-x^2 \end{vmatrix} \Rightarrow \frac{f(x)}{x^2} = \begin{vmatrix} (\alpha x+1)\cos^2 x & 1 & 1-x \\ \dfrac{\beta \sin x}{x} & 1 & 2 \\ (\gamma x^2+1)\tan x & 1 & 1-x^2 \end{vmatrix}$$

$\lim\limits_{x\to 0} \dfrac{f(x)}{x} = 0,\ \lim\limits_{x\to 0} \dfrac{f(x)}{x^2} = \begin{vmatrix} 1 & 1 & 1 \\ \beta & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -1 \qquad (A) \to Q$

$f(0) = f'(0) = 0 \Rightarrow \lim\limits_{x\to 0} \dfrac{f'(x)}{x} = \lim\limits_{x\to 0} f''(x) = f''(0)$

$f''(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \end{vmatrix} = -2 \qquad (B) \to R, S$

$$A = \lim_{x\to 0} \frac{1}{x^6} \int_{x^2}^{x^3} f(x)dx = \lim_{x\to 0} \frac{f(x^3).3x^2 - f(x^2).2x}{6x^5}$$

$$= \lim_{x\to 0} \frac{3x f(x^3) - 2f(x^2)}{6x^4} = \lim_{x\to 0} \frac{3f(x^3) + 9x^3 f'(x^3) - 4x f'(x^2)}{24x^3}$$

$$= \lim_{x\to 0} \frac{3f(x^3)-4xf'(x^2)}{24x^3} + \frac{9}{24}f'(x^3) \;\; = \lim_{x\to 0} \frac{9x^2 f'(x^3)-4f'(x^2)-8x^2 f''(x^2)}{72x^2} + \frac{9}{24}f'(x^3)$$

$$= \lim_{x\to 0} \frac{-4f'(x^2)-8x^2 f''(x^2)}{72x^2} + \frac{1}{2}f'(x^3)$$

$$= \lim_{x\to 0} \frac{-8xf''(x^2)-16xf''(x^2)-16x^3 f'''(x^2)}{144x} + \frac{1}{2}f'(x^3) = \frac{48}{144} = \frac{1}{3} \Rightarrow [A]=0 \qquad (C)\to P$$

$$g(x) = \begin{vmatrix} \cos^2 x & 1 & 1-x \\ 0 & 1 & 2 \\ \tan x & 1 & 1-x^2 \end{vmatrix} = \cos^2 x(1-x^2-2) + \tan x(2-1+x) = -\cos^2 x(1+x^2) + \tan x(1+x)$$

$$g\left(\frac{\pi}{4}\right) = -\frac{1}{2}\left(1+\left(\frac{\pi}{4}\right)^2\right)+1+\frac{\pi}{4} = \frac{1}{2}+\frac{\pi}{4}-\frac{1}{2}\left(\frac{\pi}{4}\right)^2 = 0.97 \text{ (Approx.)} \qquad \left[g\left(\frac{\pi}{4}\right)\right]=0 \qquad (D)\to P$$

68. **[A-s] [B-q] [C-r] [D-p]**

Symmetric A

$$\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \text{ possible matrices } = 2^6 = 64; \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix} \text{ maximum value of } |A| \text{ is } 4$$

By interchanging any 2 rows or columns minimum value of $|A|$ is -4 ; Maximum value of trace of A is 3

69. **[A-p, q]** **[B-r]** **[C-s]** **[D-p, q]**

$A.Adj(A) = |A|I$

$\Rightarrow (Adj(A))^{-1} = \dfrac{A}{|A|} \qquad (A)\to P,Q$

$(Adj(KA)) = K^{n-1}Adj(A) \qquad (B)\to R$

$Adj(Adj(KA)) = Adj(K^{n-1}Adj(A)) = K^{(n-1)^2}Adj(Adj(A)) = K^{(n-1)^2}|A|^{n-2}A \qquad (C)\to S$

$A^{-1}.Adj(A^{-1}) = |A^{-1}|I = \dfrac{1}{|A|}I \Rightarrow adj(A^{-1}) = \dfrac{A}{|A|} \qquad (D)\to P,Q$

70. **[A-q] [B-p, s] [C-p, r] [D-p, r]**

$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \alpha \end{vmatrix} = (\alpha-3)\neq 0$ *i.e.* $\alpha\neq 3$ unique solution $\qquad (A)\to Q$

$\alpha=3, \beta\neq 10$ no solution $\qquad (B)\to P,S$

$\alpha=3, \beta=10$ infinitely many solution $\qquad (C)\to P,R$

Atleast 2 solution $\Rightarrow$ infinitely many solutions $\qquad (D)\to P,R$

71.(0) $(A-A^T)^{2015}$ is skew symmetric matrix of odd order $\Rightarrow |(A-A^T)^{2015}| = 0$

72.(0) $z^6-1=0,\ z^{21}-1=0$ common roots are $1,\omega,\omega^2$

$R_1 \to R_1+R_2+R_3$

$$\begin{vmatrix} x+y+z & x+y+z & x+y+z \\ 2y & y-x-z & 2y \\ 2z & 2z & z-x-y \end{vmatrix} = 0$$

73.(7) $P^T P = I = PP^T$

$PQ^{2014}P^T = P(P^T AP)(P^T AP)..........(P^T AP).P^T$

$$= A^{2014} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{2014} = \begin{bmatrix} 1 & 2014 \\ 0 & 1 \end{bmatrix} \Rightarrow b = 2014 \Rightarrow \text{sum of digits of } b \text{ is } 7$$

74.(2) $S = \begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$

$|S| = (1 - c\omega) - \omega(a - b\omega) + \omega^2(ac - b) = 1 - c\omega - a\omega + b\omega^2 + ac\omega^2 - b\omega^2 = 1 - a\omega - c\omega + ac\omega^2$

$a = \omega, c = \omega \qquad |S| = -3\omega^2 \neq 0, \quad b = \omega \text{ or } \omega^2$

$a = \omega, c = \omega^2 \qquad |S| = 0$

$a = \omega^2, c = \omega \qquad |S| = 0$

$a = \omega^2, c = \omega^2 \qquad |S| = 0 \qquad \Rightarrow$ 2 non singular matrices

$a = \omega, b = \omega, c = \omega$ & $a = \omega,\ b = \omega^2, c = \omega$

75.(9) $f(x) = \begin{vmatrix} x^2 + 3x & x - 1 & x + 3 \\ x + 1 & -2x & x - 4 \\ x - 3 & x + 4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + e$

$a + b + c + d + e = f(1) = \begin{vmatrix} 4 & 0 & 4 \\ 2 & -2 & -3 \\ -2 & 5 & 3 \end{vmatrix} = 4(-6 + 15) + 4(10 - 4) = 36 + 24 = 60 \quad ; \ e = f(0) = \begin{vmatrix} 0 & -1 & 3 \\ 1 & 0 & -4 \\ -3 & 4 & 0 \end{vmatrix} = 0$

$a =$ coefficient of $x^4 = -7$

$-\left[\dfrac{a + b + c + d + e}{a + e}\right] = -\left[\dfrac{60}{-7}\right] = 9$

76.(0) Use ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$

77.(6) $\frac{1}{2}|[x]| = \{x\} \quad x \in [-2, 2] \Rightarrow x = 0, -\frac{1}{2}, \frac{3}{2}$ using graph

$|\sin y| = \{y\} \quad y \in (-\pi, \pi) \Rightarrow y$ can take 7 values, using graph

$\cos z = \{z\} \quad z \in (-\pi, \pi) \Rightarrow z$ can take 2 values, using graph $\quad \Rightarrow (x, y, z)$ can take $3 \times 7 \times 2$ i.e. 42 triplets

78.(7) $A^2 - 5A - 2I = 0$; $A^2 = 5A + 2I$; $A^{2014} = 5A^{2013} + 2A^{2012} \Rightarrow \lambda + \mu = 7$

79.(2) $\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$

Symmetric matrices

Diagonal all 'a', or 1 'a', 2 'b'

$\begin{pmatrix} a & a & b \\ a & a & b \\ b & b & a \end{pmatrix}$ or $\begin{pmatrix} a & b & a \\ b & a & b \\ a & b & a \end{pmatrix}$ or $\begin{pmatrix} a & b & b \\ b & a & a \\ b & a & a \end{pmatrix} \equiv 3$

$$\begin{pmatrix} a & b & a \\ b & b & a \\ a & a & b \end{pmatrix} \text{ or } \begin{pmatrix} a & a & b \\ a & b & a \\ b & a & b \end{pmatrix} \text{ or } \begin{pmatrix} a & a & a \\ a & b & b \\ a & b & b \end{pmatrix} \equiv 3\times 3 \text{ (permuting } a,b,b\text{)}$$

$\Rightarrow$ 12 symmetric matrices $\Rightarrow k = 12$, number of zeroes at end of 12! is 2

80.(1) $A^2 = 3A - 2I \;;\; A^3 = 3A^2 - 2A = 3(3A-2I) - 2A = 7A - 6I$

$$A = 3I - 2A^{-1} \Rightarrow A^{-1} = \frac{3I - A}{2} = \frac{3I - \left(\frac{A^3+6I}{7}\right)}{2} = \frac{21I - A^3 - 6I}{14} = \frac{15I - A^3}{14} \;;\; \lambda + k\mu = 15 - 14 = 1$$

81.(2.25) In ΔABC $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$

$\Rightarrow \quad 1\left(c^2 - ab\right) - a\left(c - a\right) + b\left(b - c\right) = 0 \;\Rightarrow\; a^2 + b^2 + c^2 - ab - bc - ca = 0$

$\Rightarrow \quad 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0 \;\Rightarrow\; (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$

$\Rightarrow \quad a = b = c \;\Rightarrow\; \Delta ABC$ is equilateral $\Rightarrow \angle A = \angle B = \angle C = 60°$

$\therefore \quad \sin^2 A + \sin^2 B + \sin^2 C = \sin^2 60 + \sin^2 60 + \sin^2 60 = 3\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{9}{4}$

82.(3) Using $C_1 \to C_1 + C_3$, we get $\begin{vmatrix} 1 & \tan\theta + \sec^2\theta & 3 \\ 0 & \cos\theta & \sin\theta \\ 0 & -4 & 3 \end{vmatrix} = 3\cos\theta + 4\sin\theta$

$\frac{d\Delta}{d\theta} = -3\sin\theta + 4\cos\theta \;;\; \frac{d\Delta}{d\theta} = 0 \Rightarrow 3\sin\theta - 4\cos\theta = 0$

$\tan\theta = \frac{4}{3} \Rightarrow \sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$ min $\Delta = 3$

83.(103) $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$

$$P^2 = \begin{vmatrix} 1 & 0 & 0 \\ 8 & 1 & 0 \\ 16(1+2) & 8 & 1 \end{vmatrix};\; P^3 = \begin{vmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 16(1+2+3) & 12 & 1 \end{vmatrix};\; P^{50} = \begin{vmatrix} 1 & 0 & 0 \\ 4\times 50 & 1 & 0 \\ 16(1+2+...+50) & 4\times 50 & 1 \end{vmatrix};\; P^{50} = \begin{vmatrix} 1 & 0 & 0 \\ 200 & 1 & 0 \\ 20400 & 200 & 1 \end{vmatrix}$$

$\therefore \quad P^{50} - Q = I$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 200 & 1 & 0 \\ 20400 & 200 & 1 \end{bmatrix} - \begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \quad 200 - q_{21} = 0 \Rightarrow q_{21} = 200$

$20400 - q_{31} = 0 \Rightarrow q_{31} = 20400$ and $200 - q_{32} = 0 \Rightarrow q_{32} = 200 \quad \therefore$

$\frac{q_{31} + q_{32}}{q_{21}} = \frac{20400 + 200}{200} = 103$

84.(3) We have $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = \begin{vmatrix} x^3 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} + \begin{vmatrix} 1 & x^2y & x^2z \\ 0 & y^3+1 & y^2z \\ 0 & yz^2 & z^3+1 \end{vmatrix}$

$$= x\begin{vmatrix} x^2 & x^2y & x^2z \\ y^2 & y^3+1 & y^2z \\ z^2 & yz^2 & z^3+1 \end{vmatrix} + \left(y^3+1\right)\left(z^3+1\right) - y^3z^3$$

$C_2 \to C_2 - yC_1;\ C_3 \to C_3 - zC_1$ $\quad = x\begin{vmatrix} x^2 & 0 & 0 \\ y^2 & 1 & 0 \\ z^2 & 0 & 1 \end{vmatrix} + y^3z^3 + z^3 + y^3 + 1 - y^3z^3 = x^3 + y^3 + z^3 + 1$

Given $x^3 + y^3 + z^3 + 1 = 30$; $x^3 + y^3 + z^3 = 29$ $\therefore$ solutions are (3, 1, 1), (1, 3, 1) (1, 1, 3)

85.(99) $A = \begin{bmatrix} 1 & \omega^2 \\ \omega & 1 \end{bmatrix}$ $\quad \therefore\ A^2 = \begin{bmatrix} 1 & \omega^2 \\ \omega & 1 \end{bmatrix}\begin{bmatrix} 1 & \omega^2 \\ \omega & 1 \end{bmatrix} = \begin{bmatrix} 1+\omega^3 & 2\omega^2 \\ 2\omega & \omega^3+1 \end{bmatrix} = \begin{bmatrix} 2 & 2\omega^2 \\ 2\omega & 2 \end{bmatrix} = 2\begin{bmatrix} 1 & \omega^2 \\ \omega & 1 \end{bmatrix} = 2^1.\ A$

Similarly $A^3 = A^2 \times A$ $= 2A \times A$ $= 2A^2 = 4A = 2^2A$ $\quad \therefore\ A^{100} = 2^{99}.A$ $\quad \therefore$ k = 99

86.(10) Determenant D = 0

$\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 4 & -3 \end{vmatrix} = 0 \Rightarrow -3k + 8 - k(-9+4) + 3(12-2k) = 0 \Rightarrow -3k + 8 + 5k + 36 - 6k = 0 \Rightarrow -4k = -44 \Rightarrow k = 11$

$\therefore$ we have, $x + 11y + 3z = 0$; $3x + 11y - 2z = 0$; $2x + 4y - 3z = 0$, Put $z = t$; $x + 11y = -3t$; $3x + 11y = 2t$

Solving, $x = \dfrac{5t}{2},\ y = -\dfrac{t}{2}$ $\quad \therefore\ \dfrac{xz}{y^2} = \dfrac{\frac{5t}{2}\times t}{\frac{t^2}{4}} = \dfrac{5}{2}\times 4 = 10$

87.(2) $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = \begin{vmatrix} 3 & \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 \\ \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 & \alpha^3+\beta^3+\gamma^3 \\ \alpha^2+\beta^2+\gamma^2 & \alpha^2+\beta^3+\gamma^3 & \alpha^4+\beta^4+\gamma^4 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} . \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix}^2 = (\alpha-\beta)^2(\beta-\gamma)^2(\gamma-\alpha)^2$$

88.(6) $B.C = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 9-8 & -12+12 \\ 6-6 & -8+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$BC = I$; $\quad tr(A) + tr\left(\dfrac{A.BC}{2}\right) + tr\left(\dfrac{A(BC)^2}{4}\right) + \ldots.\infty$

$= tr(A) + tr\left(\dfrac{A}{2}\right) + tr\left(\dfrac{A}{4}\right) + \ldots.$ $\quad = tr\left(A + \dfrac{A}{2} + \dfrac{A}{4} + \dfrac{A}{8} + \ldots\right)$ $\quad = tr\left(\dfrac{A}{1-\frac{1}{2}}\right) = tr(2A) = 2tr(A) = 2\times 3 = 6$

89.(6) $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2 \Rightarrow \begin{vmatrix} 5x-4 & 2x & 2x \\ 5x-4 & x-4 & 2x \\ 5x-4 & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$

$C_1 \to C_1 + C_2 + C_3$

$\Rightarrow \quad (5x-4)\begin{vmatrix} 1 & 2x & 2x \\ 1 & x-4 & 2x \\ 1 & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2 \Rightarrow \quad (5x-4)\begin{vmatrix} 1 & 2x & 2x \\ 0 & -x-4 & 0 \\ 0 & 0 & -x-4 \end{vmatrix} = (A+Bx)(x-A)^2$

$\therefore \quad R_2 \to R_2 - R_1\ ;\ R_3 \to R_3 - R_1$

$(5x-4)(x+4)^2 = (A+Bx)(x-A)^2\ ; \quad A=-4,\ B=5\ ;\ A+2B = 10-4 = 6$

90.(9) The number of third order determinants = the number of arrangements of nine different numbers in places = 9!

$\therefore \quad (a+b)! = 9! \quad ; \quad a+b = 9$

91.(4) Let $A = \begin{vmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{vmatrix}$

$= \begin{vmatrix} 2k-1 & 0 & 2\sqrt{k} \\ 2\sqrt{k} & 1+2k & -2k \\ -2\sqrt{k} & 2k+1 & -1 \end{vmatrix} C_2 \to C_2 - C_3 \ = \begin{vmatrix} 2k-1 & 0 & 2\sqrt{k} \\ 4\sqrt{k} & 0 & 1-2k \\ -2\sqrt{k} & 2k+1 & -1 \end{vmatrix} R_2 \to R_2 - R_3 \ = (2k+1)3$

$\because$ B is a skew-symmetric matrix of odd order therefore det $B = 0$

Now, det $(\text{adj} A)$ + det$(\text{adj} B) = 10^6$

$\left\{(2k+1)^3\right\}^2 + 0 = 10^6 \quad \Rightarrow \quad 2k+1 = 10$ as $k > 0$; k = 4.5 ; $[k] = 4$

92.(1) $D = 0$

$\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{vmatrix} = 0\ ; \quad \begin{vmatrix} 1+\alpha+\alpha^2 & 2\alpha+1 & \alpha^2+\alpha+1 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{vmatrix} = 0$

$\begin{vmatrix} 1+\alpha+\alpha^2 & 2\alpha+1 & 0 \\ \alpha & 1 & 0 \\ \alpha^2 & \alpha & 1-\alpha^2 \end{vmatrix} = 0 \Rightarrow (1-\alpha^2)(1+\alpha+\alpha^2-2\alpha^2-\alpha) = 0 \Rightarrow (1-\alpha^2)^2 = 0\ ;\ \alpha = -1$ or 1

For $\alpha = 1$, system of linear equation has no solutions $\qquad \therefore \qquad \alpha = -1 \ ; \ 1+\alpha+\alpha^2 = 1$

93.(4) $\Delta = \begin{vmatrix} 1 & x+y & x+y+z \\ 2 & 3x+2y & 4x+3y+2z \\ 3 & 6x+3y & 10x+6y+3z \end{vmatrix} \quad = x^2 \begin{vmatrix} 1 & 1 & x+y \\ 2 & 3 & 4x+3y \\ 3 & 6 & 10x+6y \end{vmatrix}$

$C_3 \to C_3 - zC_1\ ;\ C_2 \to C_2 - yC_1$

$= x^3 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 6 & 10 \end{vmatrix} \quad = x^3(6-8+3) = 64 \ = x^3 = 64 \Rightarrow \ x = 4$

94.(2) Using $C_3 \to C_3 - (C_1 + C_2)$ in D_1 and D_2, we have, $\left|\dfrac{D_1}{D_2}\right| = \left|\dfrac{+2b(ad-bc)}{b(ad-bc)}\right| = +2$

95.(9) $M=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Then $a_{12}=-1$

$a_{11}-a_{12}=1 \Rightarrow a_{11}=0$; $a_{11}+a_{12}+a_{13}=0 \Rightarrow a_{13}=1$

$a_{22}=2,\ a_{21}-a_{22}=1 \Rightarrow a_{21}=3$; $a_{21}+a_{22}+a_{23}=0 \Rightarrow a_{21}=3$; $a_{21}+a_{22}+a_{23}=0 \Rightarrow a_{23}=-5$

$a_{32}=3,\ a_{31}-a_{32}=1 \Rightarrow a_{31}=2$; $a_{31}+a_{32}+a_{33}=12 \Rightarrow a_{33}=7$

Hence sum of diagonal of M is $a_{11}+a_{22}+a_{23}=0+2+7=9$

96.(1) $\omega=\cos\frac{2N}{3}+i\sin\frac{2\pi}{3}$

By $C_1+C_1+C_2+C_3$

$$\begin{vmatrix} z & \omega & \omega^2 \\ z & z+\omega^2 & 1 \\ z & 1 & z+\omega \end{vmatrix}=0 \Rightarrow \begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & z+\omega^2 & 1 \\ 1 & 1 & z+\omega \end{vmatrix}=0 \Rightarrow \begin{vmatrix} 1 & \omega & \omega^2 \\ 0 & z+\omega^2-\omega & 1-\omega^2 \\ 0 & 1-\omega & z+\omega-\omega^2 \end{vmatrix}=0$$

$R_2 \to R_2-R_1$; $R_3 \to R_3-R_1$

$\Rightarrow \left(z-\omega^2-\omega\right)\left(z+\omega-\omega^2\right)-(1-\omega)\left(1-\omega^2\right)=0 \Rightarrow z^2=0$ $\therefore$ only one solution

97.(4) $|\text{adj } A|=|A|^{n-1}=(-2)^2=4$ where $|A|=-2$

98.(1) $$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}=1$$

99.(0) Taking e^{x_1} common from R_1, e^{x_4} common from R_2 and e^{x_1} from R_3, we get,

$\Delta=e^{x_1+x_2+x_3}\Delta_1$

Where $\Delta_1=\begin{vmatrix} 1 & e^{3d} & e^{6d} \\ 1 & e^{3d} & e^{6d} \\ 1 & e^{3d} & e^{6d} \end{vmatrix}=0$ Where d is common difference of AP $\therefore$ $\Delta=0$

100.(1) We can write the following Δ_1 as product of two determinants as follows:

$$\Delta_1=\begin{vmatrix} x^2 & -2x & 1 \\ y^2 & -2y & 1 \\ z^2 & -2z & 1 \end{vmatrix}\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

We interchange C_1 and C_3 in the first determinant and absorb the resulting negative sign in C_2 to get

$$\Delta_1=\begin{vmatrix} 1 & 2x & x^2 \\ 1 & 2y & y^2 \\ 1 & 2z & z^2 \end{vmatrix}\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}=\begin{vmatrix} (1+ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix}=\Delta_2$$